Engineering Physics: Top Down

A conceptual descent from fundamental fields to engineering systems

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ENGINEERING PHYSICS: TOP DOWN

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Engineering Physics: Top Down

A conceptual descent from fundamental fields to engineering systems

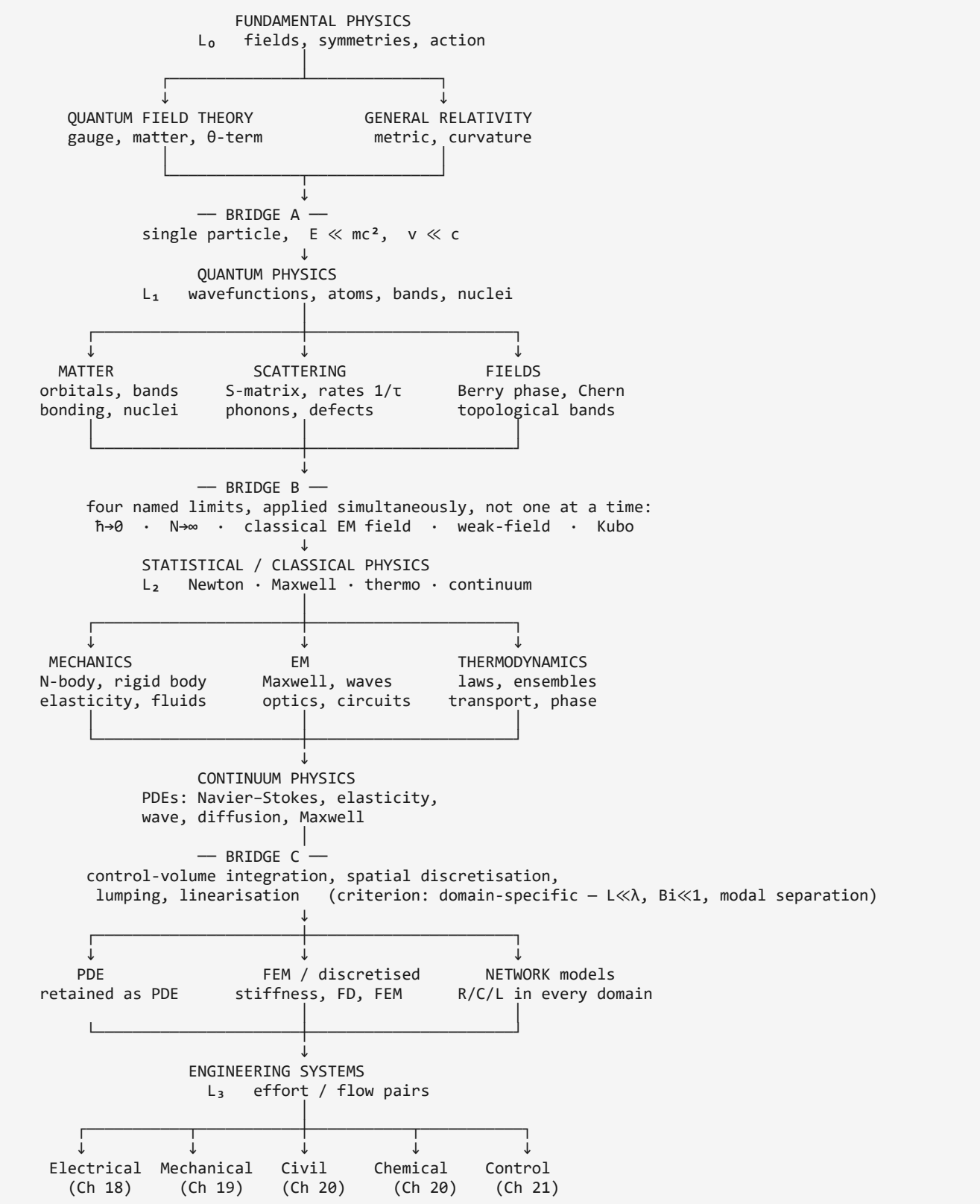
> **How far can you descend from fundamental physical principles before you reach the equations engineers use every day?**

This repository is a personal academic knowledge system that answers that question by following one connected chain of theories, approximations, and reductions: from the Standard Model and the Einstein–Hilbert action, through quantum, classical, and continuum physics, to the lumped models, circuits, machines, structures, reactors, and control systems that four engineering disciplines actually use. Its subject is not physics *or* engineering. It is the **chain of named reductions between them**, and the exact point at which each reduction stops being valid.

The hierarchy represents a hierarchy of physical descriptions and approximations, not a claim that every higher-level theory has been quantitatively derived from fundamental physics. Some connections in this book are exact mathematical reductions. Many are controlled approximations. Some are structural correspondences — two different systems sharing one piece of mathematics without one producing the other. A few are honest analogies, and some are phenomenological models fit to experiment. Every connection that carries weight is tagged so you always know which kind you are looking at.

Core architecture

The descent is not a single vertical stack. It branches, runs in parallel, and reconverges — and the reconvergence points are the interesting part.



Two features of this diagram carry the argument:

- **The reconvergence is real.** Several apparently independent fundamental threads — gauge structure, gravity, quantum statistics, scattering, and field topology — converge on a small set of classical continuum descriptions used throughout engineering. The Generalized Transport Law (Ch. 13) is where this is stated most sharply: Ohm, Fourier, Fick and Newton viscosity are instances of one linear-response framework with different operators and closures, while Hooke's law enters only as a static limit. This is a shared *framework*, not one calculation producing five results.

- **The four engineering branches reconverge too.** They are not four subjects that happen to share methods. They are four choices of effort/flow pair, instantiated on one R/C/L template — a statement about representations rather than about shared physics, and strongest exactly where the lumping criterion holds. Chapters 18–21 are the proof of concept for everything Chapters 0–17 set up.

The full tagged version of this graph — every arrow labelled, every discarded term named — is in the Master Map.

Layers and bridges

	Name	What happens
L0	Framework	$S_{\text{eff}} = S_{EH} + S_{SM} + S_{\text{unknown}}$, with $S_{\text{unknown}} \equiv \int d^4x\,\sqrt{-g}\,\mathcal{F}[\ldots]$ — the best available framework, with an explicit placeholder for what we do not know. Canonical form: Ch. 0 §0.10
A	Bridge A	Isolate matter fields; take $E \ll mc^2$, $v \ll c$. Bridge zone: the Dirac equation and the Foldy–Wouthuysen expansion
L1	Quantum / atomic	Wavefunctions, spin, exchange, atoms, molecules, bands, nuclei, quantum scattering, topology
B	Bridge B	Four simultaneous descents: $\hbar \to 0$ (classical mechanics), $N \to \infty$ (statistical mechanics), classical $U(1)$ (Maxwell), weak-field metric (Newton). Plus Kubo linear response (transport)
L2	Classical continuum & statistical	Lagrangian and Hamiltonian mechanics, thermodynamics, electromagnetism, fluid mechanics, elasticity, heat and mass transfer, wave and diffusion equations
C	Bridge C	Integrate the L2 PDEs over control volumes; discretise; lump. Bridge zone: transmission lines, Euler–Bernoulli beams
L3	Engineering systems	Electrical, mechanical/thermal, civil, chemical — and the control abstraction that unifies them

---

Chapter map

Chapter	Subject	Layer
00	The equation; the whole architecture; the five cross-layer threads	L0
01	The action principle and the mathematical machinery	L0
02	Phenomena catalogue — the index of everything the book explains, and where it is explained	—
03	**Bridge A** — Dirac → Schrödinger → Pauli, via the Foldy–Wouthuysen expansion	A
04	Schrödinger equation; hydrogen; the ladder of bound states	L1
05	Spin, exchange statistics, many-electron atoms, the periodic table	L1
06	Bonding, band theory, nuclear physics, fission and fusion	L1
07	Topology — Berry phase, Chern numbers, quantum Hall, superconductivity	L1
08	**Bridge B.d** — GR → Newton's law; the measurable corrections	B
09	**Bridge B.a** — quantum mechanics → classical mechanics	B

Chapter	Subject	Layer
10	**Bridge B.b** — quantum statistics → statistical mechanics and thermodynamics	B
11	**Bridge B.c** — classical $U(1)$ → Maxwell's equations	B
12	Electromagnetic waves, optics, waveguides, photonics	L2
13	**Bridge B.e** — Kubo → the Generalized Transport Law *(the convergence chapter)*	B
14	Sensors, semiconductor devices, thermal and electrical metrology	L2
15	Fluid mechanics: Navier–Stokes, viscous flow, dimensional analysis, turbulence	L2
16	Solid and continuum mechanics: elasticity, waves, FEM discretisation, PDE forms	L2
17	**Bridge C** — the R/C/L template, effort/flow pairs, distributed-parameter models	C
18	Electrical engineering systems: circuits, AC, power, machines, signals	L3
19	Mechanical and thermal systems: dynamics, machines, heat transfer, fluids, acoustics	L3
20	Civil and chemical systems: structures, hydraulics, geotechnics, reactors, separations	L3
21	Feedback, control, and the cross-branch capstone	L3
—	Epilogue — the unfinished equation, and what the unknown sector $\mathcal{F}[\ldots]$ still hides	—

Chapters 18–21 are the load-bearing part. Without them this is an unconventional physics textbook; with them it is a physics-to-engineering abstraction framework.

Worked demonstrations of single descents, end to end, are in `EXAMPLES/` — each one takes a starting theory, names the approximation, shows the reduction, lands on the engineering equation, and states the validity limits.

Full derivations and proofs that the chapters state or sketch are in `APPENDIX/`. The appendices support the 21 chapters; they are not new chapters.

Appendix	Subject	Supports
A	Variational calculus, Noether's theorems, conservation laws	Ch 0, 1, 9, 11, 17
B	Dirac → Foldy–Wouthuysen → Pauli; hydrogen fine structure	Ch 3, 4, 5
C	Quantum → classical (stationary phase, Ehrenfest, WKB); GR → Newton	Ch 1, 4, 8, 9
D	Ensembles, quantum statistics, Kubo, Green–Kubo, Drude	Ch 10, 13, 14
E	Continuum balance laws and their reduction to lumped networks	Ch 11, 15, 16, 17, 19, 20
F	Linear systems and feedback: Laplace, Nyquist, Routh, root locus	Ch 17, 18, 21

Figures live in `FIGURES/`, one folder per chapter or example. Every figure is an SVG drawn by the `make_figures.py` script beside it, from the equations and parameters in the text. Run `python FIGURES/build_all.py` to regenerate them all.

AI tooling. `MCP.md` is the context manifest for AI agents. `mcp-server/` is an MCP server with structure audits, Model Ledger extraction, model tracing and a figure index. The repo's `.mcp.json` registers it with Claude Code.

How connections are marked

Every load-bearing arrow in this book carries one of five tags. This is the mechanism that prevents the central failure mode of interdisciplinary writing — confusing *"these equations have the same mathematical structure"* with *"these phenomena have the same physical origin."*

Tag	Meaning
`[DERIVATION]`	An exact or controlled mathematical reduction. Apply calculus under the stated assumptions and this follows.
`[APPROXIMATION]`	Obtained by dropping terms under an explicit, named limit. The small parameter and the regime are stated.
`[STRUCTURAL CONNECTION]`	Two different physical systems obey the same mathematics because both are instances of one underlying mathematical object — not because one causes the other.
`[ANALOGY]`	A conceptual correspondence for intuition or pedagogy. Not identity.
`[PHENOMENOLOGICAL]`	Form or parameters come from experiment or an effective theory, not from a controlled derivation.

Applied inline, a tag is a blockquote attached to the equation it qualifies:

> **[APPROXIMATION]** Dirac equation → nonrelativistic expansion in $(v/c)^2$ → Pauli equation, retaining the Zeeman and spin–orbit terms to order $(v/c)^2$.

> **[STRUCTURAL CONNECTION]** An electrical RC network and a thermal RC network obey the same lumped first-order form, $C_x\,dx/dt = (x_{in} - x)/R_x$, with $x = V$ and $C_x = C$ for the electrical case and $x = T$ and $C_x = \rho c_p V$ for the thermal case. Same differential-equation structure, different physical variables, constitutive parameters, and energy-storage mechanisms.

Model validity

The book's central rule:

> **Every useful engineering model has a domain of validity.**

A model is never presented as an isolated equation. At every major model transition the book records a **Model Ledger**:

Property	Description
Parent theory	The deeper model this came from
Reduction	The named operation or limit applied
Assumptions	What must hold for the reduction to be legal
Retained	The physics that survives

Property	Description
Neglected	The physics that was discarded
Valid when	The regime where the model is trustworthy
Fails when	The regime where it must be replaced
Next model	What to reach for when it fails

```
Parent theory       3D elastodynamics (Ch. 15, 16)
Reduction           slender-beam assumption,  L / r ≫ 1
Model               Euler–Bernoulli beam,  EI ∂⁴w/∂x⁴ = q
Retained            bending deformation
Neglected          transverse shear deformation, rotary inertia
Valid when          slender, low-frequency, small-deflection
Fails when          thick beams (L/r ≲ 10), or high-frequency
Next model          Timošenko beam theory
```

The goal is to make **model selection** part of learning physics and engineering: not *what is the equation* but *what was assumed, what was thrown away, and when does it stop working*.

How to navigate

Sequential — the intended descent:

```
Master Map  →  Chapter 0  →  Chapter 1  →  …  →  Chapter 21  →  Epilogue
```

> **Note: Two different "0" documents** The **Master Map** (`METADATA/00 Map.md`) is navigation and provenance — the dependency graph, tagged arrows, and break points. It is not part of the conceptual sequence. **Chapter 0** (`CHAPTERS/CHAPTER 0.md`) is the first chapter of the descent, where the action and the Mexican hat are introduced. They are numbered in different systems and are easy to confuse; "Chapter 0" always means the chapter.

Reverse — start from an equation you use at work and walk it backwards to its origin. This is the more useful direction for a working engineer:

```
PID gain tuning
      ↓
transfer function
      ↓
lumped ODE
      ↓
PDE / conservation law
      ↓
Noether's theorem
      ↓
U(1) gauge invariance
```

Single descent — one chain, fully worked, with validity limits: see `EXAMPLES/`.

Conventions used throughout: SI units; standard mathematical notation; approximation limits stated where relevant; dimensional consistency maintained; sign conventions stated where they affect the result. There is no equation numbering — cross-references are by chapter and section, e.g. *"Ch. 16 §16.7.2"*.

Status

Physics Core — Chapters 0–21 + Epilogue. The architecture is frozen at 21 chapters.

The chapter count will not grow. What the project needs now is rigour, not breadth:

- [x] Four-layer architecture, three named bridges, five cross-layer threads
- [x] Chapters 0–21 and the Epilogue
- [x] Connection-type tagging applied at load-bearing transitions
- [x] Model Ledgers on the major model transitions
- [x] `REFERENCES.md` — load-bearing claims traced to sources, with unverified items flagged rather than guessed
- [x] `EXAMPLES/` — worked single-descent demonstrations
- [x] `APPENDIX/` — full derivations and proofs behind the chapters (A–F)
- [x] `FIGURES/` — script-generated figures embedded across chapters, examples, Map and Epilogue
- [ ] Extend tagging coverage to the remaining heuristic sections
- [ ] Pin CODATA / PDG citations to fixed dataset editions
- [ ] Second-pass verification of Chapters 16, 17, 19, 20 against primary sources

Read it as a **research and study map**, not as an authoritative replacement for specialist textbooks. The detailed conceptual argument — every layer expanded, every discarded term named, every arrow tagged — lives in `00 Map.md`. Writing rules for extending it are in `CONTRIBUTING.md`.

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ENGINEERING PHYSICS: TOP DOWN

A Detailed Layer Map with Smooth Transitions

v5 — four layers, three bridges, master dependency graph, tagged connections, Model Ledger convention

This document is the **conceptual argument** of the repository: how the whole system works, layer by layer, arrow by arrow. The `README.md` answers *what this repository is*; this file answers *how the intellectual system it describes actually operates*. Read this before the chapters in sequence.

Contents

- §M — The Master Dependency Graph — the full branch-and-reconvergence map, with every arrow tagged
- §0 — The Framework Scale — the action, Noether's theorem, symmetry breaking, topology
- Bridge A — Layer 0 → Layer 1
- Layer 1 — quantum / atomic
- Bridge B — Layer 1 → Layer 2
- Layer 2 — classical continuum & statistical
- Bridge C — Layer 2 → Layer 3
- Layer 3 — engineering systems
- Notation · Model Ledger · Cross-Layer Threads

Companion files. `README.md` — what the repository is. `REFERENCES.md` — sources for load-bearing claims. `EXAMPLES/` — single descents worked end to end, each with its validity limits. `CONTRIBUTING.md` — the rules for extending any of this. `FIGURES/` — every figure, with the script that regenerates it. `APPENDIX/` — full derivations and proofs behind the chapters (Appendices A–F).

The Organizational Claim (Stated Once, Up Front)

Most fundamental physical theories are naturally formulated in terms of an action, and this has proved to be an extraordinarily powerful organizing framework. Whether *every* fundamental theory must admit an action formulation is an open question — the amplitudes program and AdS/CFT both hint that the Lagrangian may be a convenient organizing language rather than the only possible one. Nevertheless: **if a Theory of Everything exists, we have overwhelming reason to expect it will appear in this form:**

$$S_{\text{eff}} = S_{EH} + S_{SM} + S_{\text{unknown}} = \int d^4x\,\sqrt{-g}\left[\frac{R}{16\pi G} + \mathcal{L}_{SM} + \mathcal{F}[g,\,\phi_?,\,\text{new fields},\,\text{topology},\,\ldots]\right]$$

Notation. $\mathcal{F}[\ldots]$ is a *placeholder for an unknown sector*, not a specific functional form. It is deliberately **not** written as $f(\phi)R$: that form would assert the missing physics is a scalar times the Ricci scalar, which nothing supports. The canonical statement is Ch. 0 §0.10.

Term	What it captures	Status
$R/16\pi G$	Curvature of spacetime (gravity)	Confirmed, tested to high precision
$\mathcal{L}_{SM}$	Quantum fields for all known matter and three forces	Confirmed, tested to extraordinary precision
$\mathcal{F}[\ldots]$	Everything we don't know yet	Unknown; active research

What we know about $\mathcal{F}$, even without knowing $\mathcal{F}$:

- It must be a scalar (more precisely, a scalar density assembled from diffeomorphism- and local-Lorentz-invariant ingredients), so that the action is coordinate-independent
- It must reduce to zero correction in all regimes that $\mathcal{L}_{SM} + R/16\pi G$ already handles correctly
- It *may* include further topological terms. The SM already carries one (the QCD θ-term, $\theta \frac{g^2}{32\pi^2} G^a_{\mu\nu} \tilde{G}^{a\mu\nu}$), and the anomaly structure strongly constrains which additional topological terms are allowed — whether any beyond the SM are genuinely required is an open question, not a premise
- Known candidates: dark matter fields, dark energy beyond Λ, quantum gravity corrections (R^2, Gauss-Bonnet), explanation of why the QCD θ-parameter ≈ 0 (the strong CP problem), possible extension of the gauge group to SU(5)/SO(10) (GUTs), supersymmetric partners

The pedagogical stance: this equation is the best framework humanity has. Every chapter of this book is a controlled, named approximation to it. Students who finish this book will know which approximations they are standing on at every moment of their career — and will know exactly when to distrust those approximations.

> **This is a hierarchy of physical descriptions, not a claim that every higher-level theory can currently be derived quantitatively from the Standard Model.** Some arrows in this map are exact derivations. Many are controlled approximations. Some are structural correspondences — two different systems sharing one piece of mathematics without one causing the other. A few are honest analogies. The notation below tells you, at every arrow, which kind you are looking at.

Notation: How to Read Every Connection

Dense chains of the form A → B → C → D appear throughout this book. Each arrow can mean something different, and collapsing them into one visual weight is the single most common way this kind of map overstates itself. Every non-trivial arrow in this document is tagged with one of five labels (defined in full in `README.md`, "Notation and Conventions"):

> **[DERIVATION]** An exact or controlled mathematical reduction: apply calculus or algebra to the prior result under the stated assumptions, and this follows.

> **[APPROXIMATION]** A theory obtained by dropping terms under an explicit, named limiting assumption (a small parameter going to zero, a scale separation, etc.).

> **[STRUCTURAL CONNECTION]** Two different physical systems obey the same mathematics because both are instances of the same underlying mathematical object (a conserved current, a topological invariant, a free-energy expansion) — not because one causes or produces the other.

> **[ANALOGY]** A conceptual correspondence, useful for intuition or pedagogy, that should not be mistaken for identity.

> **[PHENOMENOLOGICAL]** A model whose parameters or functional form come from experiment or an effective theory, not from a controlled derivation.

Where a chain in this map does not carry a tag, treat it as informal narrative connective tissue, not a claim of derivation.

Model Ledger: the Convention

Tags classify a *connection*. The **Model Ledger** classifies a *model*. Wherever a chapter performs a major model transition — a PDE becoming an ODE, a field equation becoming a lumped circuit, a full model becoming a linearisation — it records the transition in a fixed nine-row table. The Ledger is the operational form of the book's central rule: **every useful engineering model has a domain of validity.**

```
### Model Ledger — <name of the reduced model>

| Property | Description |
|---|---|
| Parent theory | The deeper model this was obtained from |
| Reduction | The named operation or limit applied |
| Model | The reduced model itself |
| Assumptions | What must hold for the reduction to be legal |
| Retained | The physics that survives the reduction |
| Neglected | The physics that was discarded |
| Valid when | The regime in which the model is trustworthy |
| Fails when | The regime in which it must be abandoned |
| Next model | What to reach for when it fails |
```

Worked instance — the Euler–Bernoulli beam, which is the reference example for the whole Layer 2 → Layer 3 transition:

```
### Model Ledger — Euler–Bernoulli beam

| Property | Description |
|---|---|
| Parent theory | 3D linear elastodynamics, Navier's equation (Ch. 15, 16) |
| Reduction | Slender-beam assumption, L / r ≫ 1 |
| Model | EI ∂⁴w/∂x⁴ = q(x,t)  (Ch. 16 §16.7.2, Ch. 20 §20.2) |
| Assumptions | Slender beam, small deflection, linear elasticity, Euler–Bernoulli kinematics (cross-sections stay normal) |
| Retained | Bending deformation; axial force; distributed inertia |
| Neglected | Transverse shear deformation, rotary inertia, warping torsion |
| Valid when | L/r ≳ 10, excitation wavelengths ≫ L, small deflection |
| Fails when | Thick beams (shear-dominated), high-frequency excitation, large deflection (geometric nonlinearity) |
| Next model | Timošenko beam (adds shear + rotary inertia); shell theory for wide/thin members |

Second instance — the lumped thermal RC node, which is the reference example for Bridge C:

### Model Ledger — Lumped thermal RC node

| Property | Description |
|---|---|
| Parent theory | Heat equation, ρc ∂T/∂t = ∇·(k∇T) (Ch. 16) |
| Reduction | Control-volume integration + lumped-capacitance assumption |
| Model | C dT/dt = (T_source − T)/R_th  (Ch. 17, 19) |
| Assumptions | Bi ≪ 1 (lumped capacitance); short conduction path (small k/ρ thermal resistance); single dominant storage mode; boundaries at fixed T_source |
| Retained | Total energy stored; net heat flow into the node |
| Neglected | Internal temperature gradients; conduction delays; 3D spreading resistance; radiation (unless added as a parallel R) |
| Valid when | Characteristic diffusion time across the body ≪ process timescale — typically true for compact bodies (Biot number ≪ 0.1) |
| Fails when | Large or slow-heated bodies, phase change, or a genuinely distributed thermal mass; then the internal gradient *is* the answer, and the heat equation is required |
| Next model | Rod/slab/fin models; distributed-parameter model (Ch. 19 §19.6) |
```

Placement rule. One Ledger per major model transition — not one per equation, and not one per chapter. Most chapters contain several distinct models, and a chapter with no transition (for example Ch. 2, the phenomena catalogue) carries none. Chapters carrying Ledgers in this edition: **3, 7, 13, 16, 17, 19, 20, 21**.

Why these chapters. Each performs a transition a reader could otherwise mistake for a definition rather than a reduction, and each is a place where the book's central claim is under most pressure:

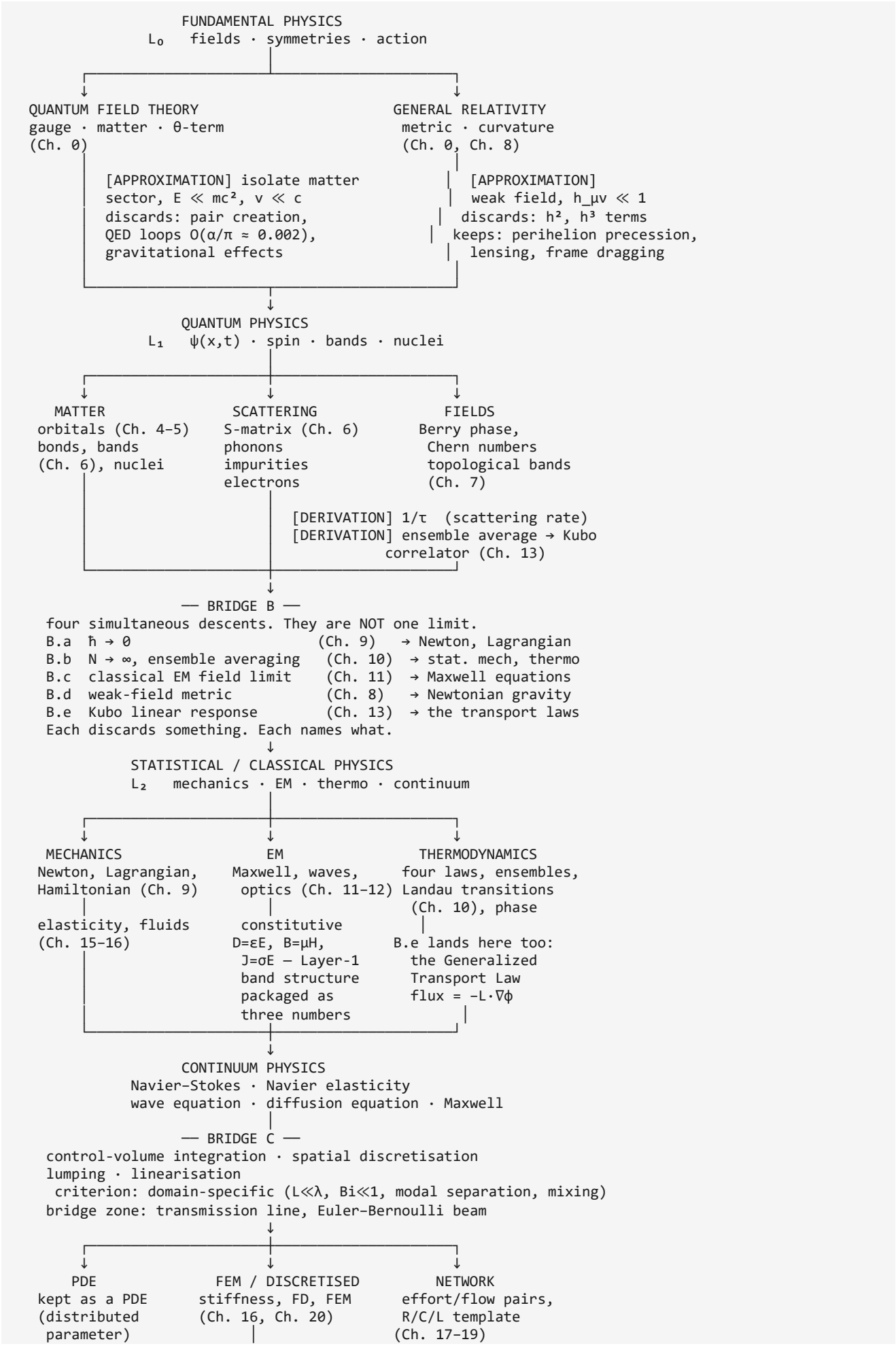
Ch.	Transition recorded	Claim at risk
3	Dirac → Foldy–Wouthuysen → Pauli → Schrödinger	that each is a theorem rather than a truncation
7	Topological invariant → quantised transport coefficient	that topology "proves" a measurement
13	Quantum Hamiltonian → linear response coefficient	that Ohm's law is fundamental
16	Distributed PDE → lumped ODE	that the cross-branch result is an identity
17	Control-volume integral → R/C element	the same, from the engineering side
19	Thermofluid bond graph → linear loop gain	that bond graphs remove the need for models
20	Physics → CE/ChE design law	that "same mathematics" means "same equation"
21	Nonlinear plant → LTI transfer function → PID	that a transfer function describes every plant

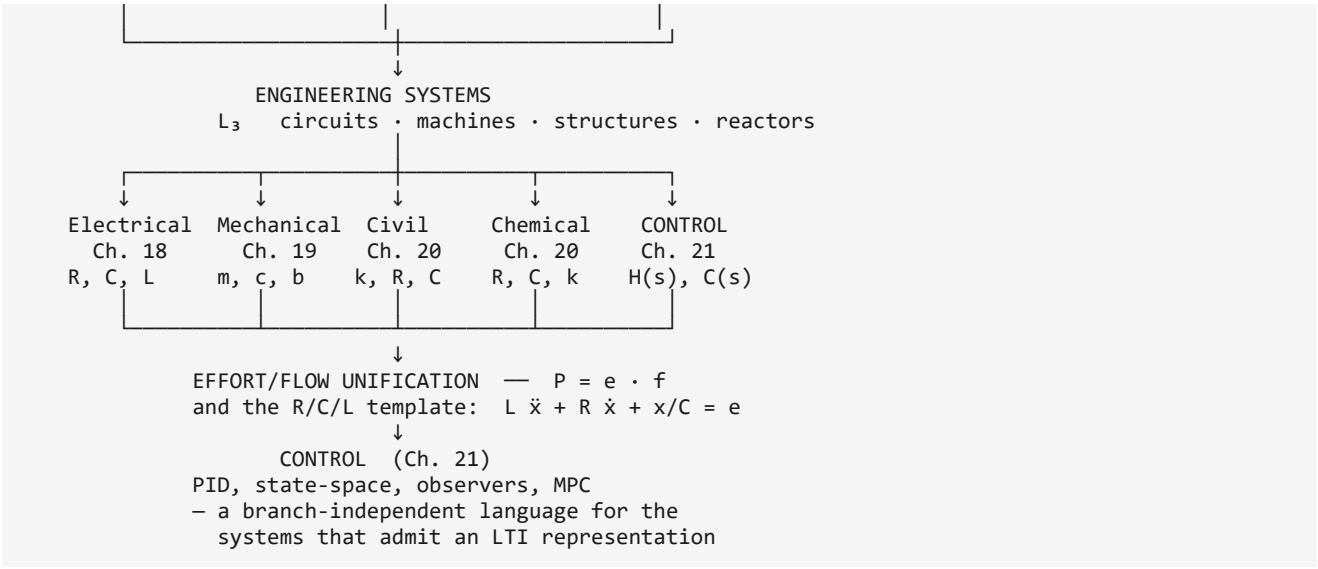
The remaining chapters carry none, and that is a deliberate statement: a Ledger records a *reduction*, and a chapter that establishes, catalogues, or surveys has not reduced anything yet. Adding Ledgers elsewhere to make the count look uniform would dilute the convention.

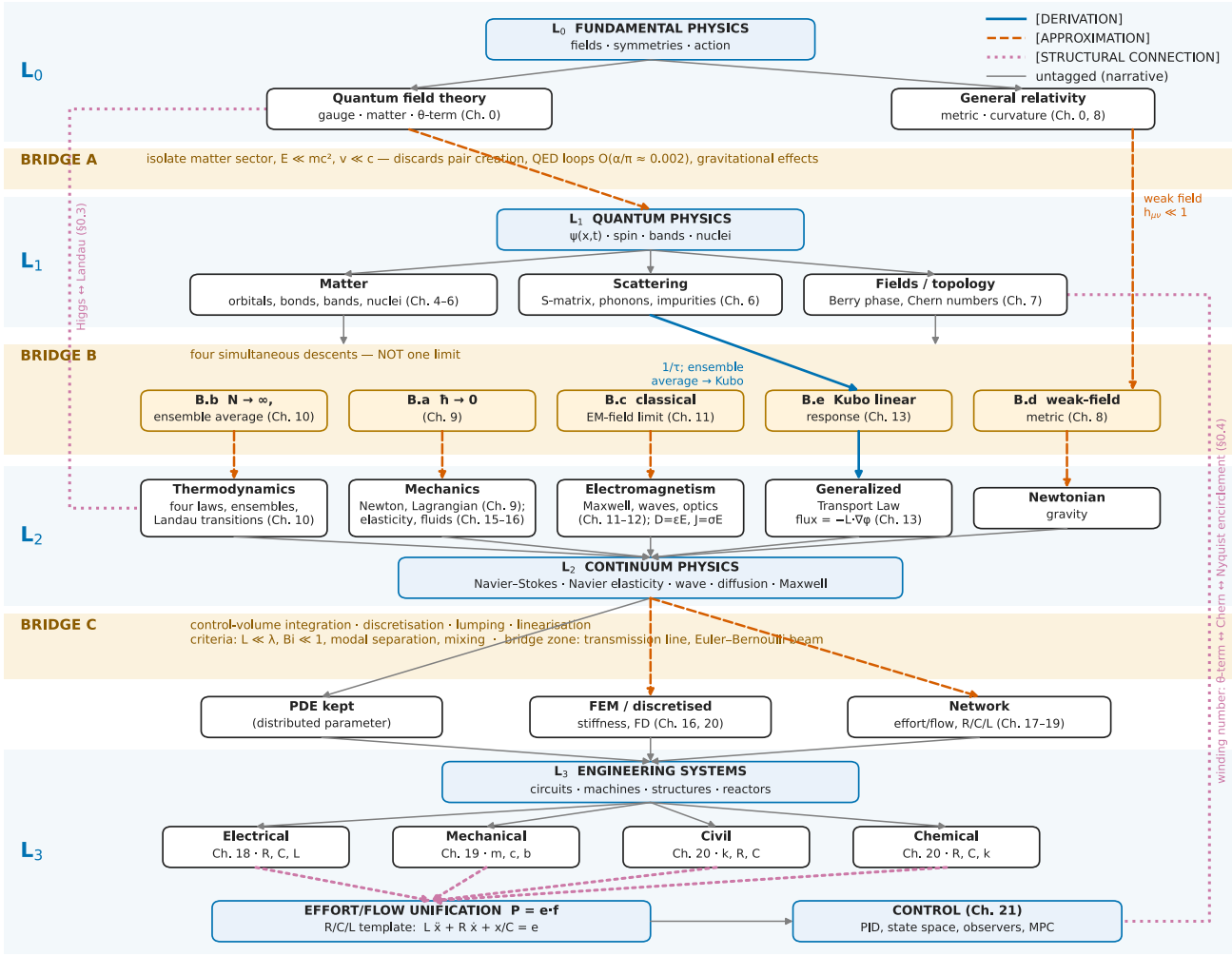
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§M — The Master Dependency Graph

This is the whole book in one diagram, with **every arrow tagged**. It is the same picture as the README's core architecture, at full resolution. Read it top to bottom as a descent; read it bottom to top as a reverse derivation from an engineering equation back to the action.


FUNDAMENTAL PHYSICS
L₀ fields · symmetries · action
QUANTUM FIELD THEORY
gauge · matter · θ-term
(Ch. 0)
GENERAL RELATIVITY
metric · curvature
(Ch. 0, Ch. 8)
[APPROXIMATION] isolate matter sector, E ≪ mc², v ≪ c
discards: pair creation, QED loops O(α/π ≈ 0.002), gravitational effects
[APPROXIMATION] weak field, h_μν ≪ 1
discards: h², h³ terms
keeps: perihelion precession, lensing, frame dragging
QUANTUM PHYSICS
L₁ ψ(x,t) · spin · bands · nuclei
MATTER
orbitals (Ch. 4–5)
bonds, bands
(Ch. 6), nuclei
SCATTERING
S-matrix (Ch. 6)
phonons
impurities
electrons
FIELDS
Berry phase,
Chern numbers
topological bands
(Ch. 7)
[DERIVATION] 1/τ (scattering rate)
[DERIVATION] ensemble average → Kubo correlator (Ch. 13)
── BRIDGE B ──
four simultaneous descents. They are NOT one limit.
B.a ħ → 0 (Ch. 9) → Newton, Lagrangian
B.b N → ∞, ensemble averaging (Ch. 10) → stat. mech, thermo
B.c classical EM field limit (Ch. 11) → Maxwell equations
B.d weak-field metric (Ch. 8) → Newtonian gravity
B.e Kubo linear response (Ch. 13) → the transport laws
Each discards something. Each names what.
STATISTICAL / CLASSICAL PHYSICS
L₂ mechanics · EM · thermo · continuum
MECHANICS
Newton, Lagrangian, Hamiltonian (Ch. 9)
elasticity, fluids (Ch. 15–16)
EM
Maxwell, waves, optics (Ch. 11–12)
constitutive D=εE, B=μH, J=σE — Layer-1 band structure packaged as three numbers
THERMODYNAMICS
four laws, ensembles, Landau transitions (Ch. 10), phase
B.e lands here too: the Generalized Transport Law flux = −L·∇ϕ
CONTINUUM PHYSICS
Navier–Stokes · Navier elasticity
wave equation · diffusion equation · Maxwell
── BRIDGE C ──
control-volume integration · spatial discretisation
lumping · linearisation
criterion: domain-specific (L≪λ, Bi≪1, modal separation, mixing)
bridge zone: transmission line, Euler–Bernoulli beam
PDE
kept as a PDE (distributed parameter)
FEM / DISCRETISED
stiffness, FD, FEM
(Ch. 16, Ch. 20)
NETWORK
effort/flow pairs, R/C/L template
(Ch. 17–19)





The ASCII graph above redrawn as a layered network, L0 → L1 → L2 → L3, with Bridges A, B (B.a–B.e) and C as bands and every edge styled by its tag (solid = DERIVATION, dashed = APPROXIMATION, dotted = STRUCTURAL CONNECTION, thin grey = untagged narrative); the only DERIVATION edges are the scattering → Kubo → transport-law chain (B.e), general relativity's weak-field limit is routed straight to B.d where Bridge B lists it, and the cross-domain links (Higgs ↔ Landau, the winding-number thread, effort/flow) are structural.

The three things this graph is claiming

1. The reconvergence at Layer 2 is the book's central technical result. Five apparently independent L0/L1 threads — gauge structure, gravity, quantum statistics, scattering, field topology — converge on one small set of classical continuum equations. The sharpest statement is the Generalized Transport Law (Bridge B.e / Ch. 13):

$\hat{A}$, $\hat{B}$	L_{AB}	**Law**	**Domain**
$\hat{J}$, $\hat{J}$	σ	Ohm	Electrical
$\hat{J}_Q$, $\hat{J}_Q$	κ	Fourier	Thermal
$\hat{J}_N$, $\hat{J}_N$	D	Fick	Mass / species
$\hat{\Pi}$, $\hat{\Pi}$	η	Newton viscosity	Fluid momentum
$\hat{\sigma}$, $\hat{\sigma}$	C	Hooke	Structural

One linear-response framework, four dissipative coefficients and one static susceptibility, each with its own operators and closure. This is a `[STRUCTURAL CONNECTION]` across domains — *not* a claim that the five domains are physically the same thing, and not a claim that one calculation emits all five (Ch. 13 §13.0, §13.2.2).

2. The reconvergence at Layer 3 is a statement about representations, not about physics. EEE, ME, CE and ChE do not share physics; they share an effort/flow pair and an R/C/L template. `P = e · f` in every domain because work is work. The unification is in the *mathematics of the lumped model*, and it is exactly as strong as — and exactly as weak as — the lumping assumption that produced it. The effort/flow construction is moreover available for almost any conjugate variable pair, so its generality is a property of the representation rather than evidence of deep shared structure; and not every system is naturally effort/flow — distributed, multi-port, or strongly constrained systems are often better served by Hamiltonian or port-based descriptions (Ch. 17 §17.1.1).

3. The break points map backwards, and that is the point. Every Layer-3 model has a documented failure regime and a documented destination:

L3 failure mode	**Physical cause**	**Go back to**
Transistor gate tunnelling	Oxide $\lesssim$ few nm	L1 — band structure, WKB tunnelling (Ch. 4 §4.4)
Quantum Hall, topological insulators	Non-trivial band topology	L1 — Chern number, edge states (Ch. 7)
Fracture, fatigue	Bond-level failure mechanics	L1 — molecular dynamics, fracture mechanics
EM behaviour above ~GHz	Lumping criterion $L \ll \lambda$ violated	L2 — transmission line, full Maxwell (Ch. 17)
Turbulence	Navier–Stokes → chaos	Stays in L2; there is no closed L3 model. This is an honest dead end.
Irradiation, plasma, high-T materials	Damage physics beyond equilibrium transport	L1 or L0

The last column is the argument for building the map bottom-up in the first place: a working engineer who knows where their model's failure regime lives knows *which* deeper theory to go and learn.

Architecture Overview

```
LAYER 0    S = ∫ √-g [ (R−2Λ)/16πG + L_SM + F[…] ]
           The Framework Scale — the universe unresolved
                        |
              ====== BRIDGE A ======
              Isolate matter fields
              Single particle, E ≪ mc²
              (Bridge zone: Dirac equation)
                        |
LAYER 1    Quantum / Atomic Scale
           Wavefunctions, atoms, bands, nuclei
                        |
              ====== BRIDGE B ======
              Four simultaneous descents:
              ħ→0, N→∞, classical EM field→Maxwell, g→η
              (Bridge zones: WKB, quantum stat mech,
               classical coherent fields)
                        |
LAYER 2    Classical Continuum & Statistical Scale
           Newton, Thermo, Maxwell, Continuum,
           + the Generalized Transport Law
                        |
              ====== BRIDGE C ======
              Discretize: integrate PDEs over
              control volumes (lumped assumption)
              (Bridge zone: distributed-parameter models)
                        |
LAYER 3    Engineering Systems Scale
           EEE | ME | CE | ChE — same template, different units
```

LAYER 0 — The Framework Scale

What this layer is: The universe before we resolve it into separate forces, particles, or scales. Everything is a field. Spacetime is dynamic. Forces are geometrical consequences of symmetry requirements.

0.1 — Anatomy of the Action

The Einstein–Hilbert sector $\frac{R}{16\pi G}$:

R is the Ricci scalar — a single number at each spacetime point that measures how much spacetime is curved there. $\sqrt{-g}$ is the spacetime volume element (the "size" of a patch of spacetime in curved coordinates). Together they encode gravity not as a force but as geometry.

The Standard Model Lagrangian $\mathcal{L}_{SM}$:

$$\mathcal{L}_{SM} = \underbrace{-\frac{1}{4}F^a_{\mu\nu}F^{a\mu\nu}}_{\text{gauge fields (forces)}} + \underbrace{\bar{\psi}(i\gamma^\mu D_\mu - m)\psi}_{\text{fermion fields (matter)}} + \underbrace{|D_\mu H|^2 - V(H)}_{\text{Higgs field (mass)}} + \underbrace{\mathcal{L}_{Yukawa}}_{\text{matter-Higgs coupling}} + \underbrace{\theta\frac{g^2}{32\pi^2}G^a_{\mu\nu}\tilde{G}^{a\mu\nu}}_{\text{topological θ-term}}$$

Piece	Symmetry generating it	Physics it produces
Gauge fields $F^a_{\mu\nu}$	U(1)×SU(2)×SU(3) gauge invariance	EM, weak force, strong force
Fermion fields $\bar{\psi}\psi$	Poincaré invariance + spin-statistics	Electrons, quarks, neutrinos
Higgs H	Spontaneous breaking of SU(2)×U(1)	Why anything has mass
Yukawa coupling	Coupling of H to ψ	Specific fermion masses
θ-term	Topology of the SU(3) gauge field	Vacuum structure; strong CP problem

0.2 — Noether's Theorem: The One Idea That Threads All Four Layers

State it once here, recognize its shadow everywhere below.

> For every continuous symmetry of the action, there exists a corresponding conserved current and conserved charge.

Symmetry of S	Conserved quantity	Engineering manifestation (Layer 3)
Time translation	Energy	First law of thermodynamics; energy balance
Space translation	Linear momentum	Newton's 2nd law; force balance
Rotation	Angular momentum	Torque balance; shaft power
Global phase symmetry of charged matter	Electric charge	Kirchhoff's current law
SU(3) gauge	Color charge	(Confined inside nuclei; never reaches Layer 3)

KCL is not one step from Noether's theorem — it is the endpoint of a six-link chain, each link a separate, named operation:

> **[DERIVATION]** Within a theory possessing local $U(1)$ gauge structure, the matter fields carry a conserved charge. The charge is associated with the **global** phase symmetry $\psi \to e^{i\alpha}\psi$ (a continuous one-parameter group), so Noether's *first* theorem applies and yields a conserved current j^μ with $\partial_\mu j^\mu = 0$.

> **Warning: Precise attribution: which symmetry, and which Noether** It is worth being exact, because the Noether → KCL thread is load-bearing for this book. Two distinct things are easily conflated:
>
> - The **global** phase symmetry of charged matter is a continuous one-parameter symmetry group. *This* is what Noether's first theorem acts on, and it is what produces the conserved charge and the current j^μ.
> - **Local** gauge invariance — invariance of the action under $\psi \to e^{ig\alpha(x)}\psi$, $A_\mu \to A_\mu + \partial_\mu \alpha$ with α an arbitrary *function of spacetime* — is not a symmetry in the Noether-first-theorem sense at all. Because the parameter is a function rather than a number, the usual charge-conserving variation does not follow. Local gauge structure is instead associated with Noether's **second** theorem: constraints among the equations of motion, with the "charges" being the identity among those EOMs (the Noether identities $D_\mu J^\mu \equiv 0$).
>
> So the honest statement of link 1 is: *gauge structure is what makes the electromagnetic field a connection A_μ at all*, and the *global* residual phase symmetry of charged matter carries the Noether charge. Writing "U(1) gauge symmetry ⟹ Noether ⟹ conserved current" is a compressed shorthand for this, and it is used as shorthand in a few places below — but the distinction matters, because the gauge redundancy is a redundancy (a description choice) whereas the global symmetry is a physical symmetry.

> **[DERIVATION]** That conservation law is exactly the continuity equation $\partial_\mu j^\mu = 0$ — worked out explicitly in Bridge B.c below.

> **[DERIVATION]** Integrating $\partial_\mu j^\mu = 0$ over a finite spatial volume and applying the divergence theorem gives an integral charge-balance statement: the rate of charge accumulation inside any closed surface equals the net current crossing that surface.

> **[APPROXIMATION]** Bridge C's lumped-node approximation (the appropriate domain criterion — $L/\lambda \ll 1$ for EM, $Bi \ll 1$ for thermal, modal separation for structures, mixing time for reactors; defined in full in Bridge C below) shrinks that closed surface down to a single circuit node, discarding spatial variation inside it.

> **[DERIVATION]** In steady state, with no charge accumulating at the node, the integral balance reduces algebraically to $\sum I_{node} = 0$ — Kirchhoff's current law.

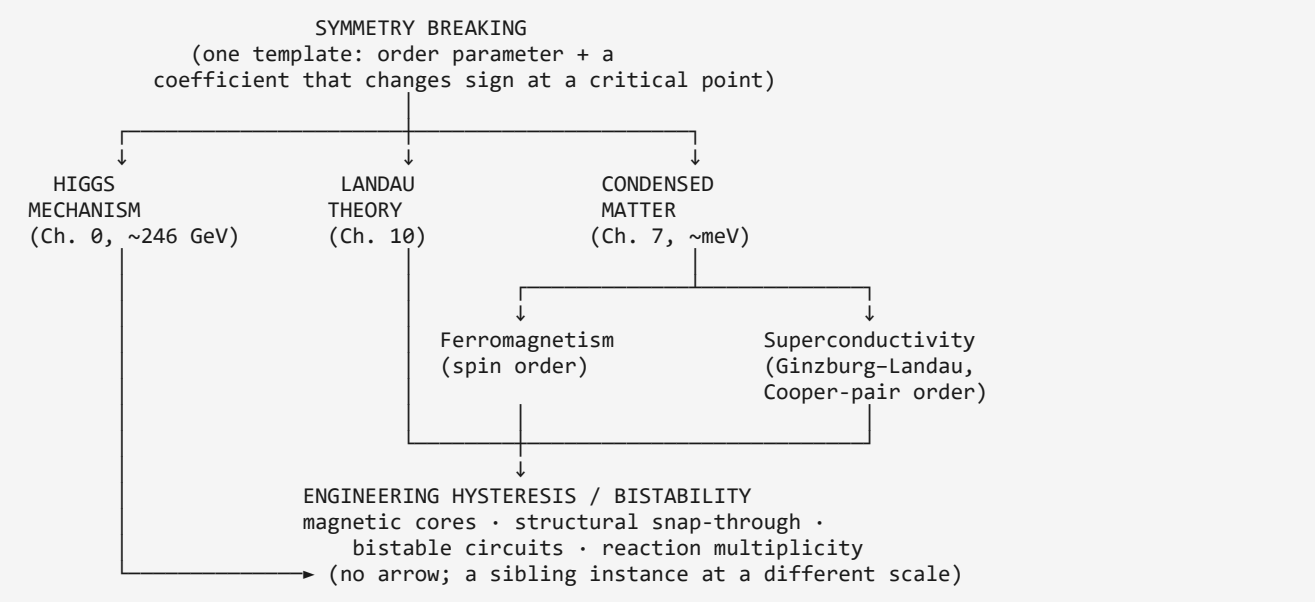
The compressed slogan — "KCL is Noether's theorem, seen from far away" — is a useful thing to say to a student *after* they have walked this chain once. It is not, by itself, a derivation, and this book does not present it as one. The chain is made visible at every layer it passes through: the continuity-equation step lives in Bridge B.c, the lumping step lives in Bridge C, and the final algebraic form is assembled in §3.1.

0.3 — Symmetry Breaking and the Higgs: A Template to Recognize Later

The Higgs potential $V(H) = -\mu^2|H|^2 + \lambda|H|^4$ has a "Mexican hat" shape, so the vacuum is degenerate rather than a single point. The familiar picture shows a circle of degenerate minima; for the SM Higgs doublet the literal minimum manifold in four-real-dimensional field space is S^3, with gauge fixing selecting a representative. The system "picks one" — breaking the SU(2)×U(1) symmetry. This gives masses. (See Ch. 0 §0.3 for the full S^3 statement and why the naive circle picture undercounts the Goldstone directions.)

Mark this mechanism, not this instance. [STRUCTURAL CONNECTION] The Higgs mechanism is *one example* of spontaneous symmetry breaking in field theory — it is not the physical ancestor of ferromagnetism or superconductivity. Landau theory (Layer 2, §2.3) is the general mathematical template underneath all of them: a free energy expanded in an order parameter, with a coefficient that changes sign at a critical point.

The relationship is a **fan out from one template into independent instances**, not a chain. Nothing below the template is upstream of anything else in the fan:



Per-arrow classification. The tree shape above is the whole point; the tags say exactly how much each edge carries.

Edge	Tag	What it actually means
Higgs mechanism →	`[DERIVATION]`	The Higgs potential is obtained within the Standard Model, not imported. SU(2)×U(1) → U(1) at $v \approx 246$ GeV.
Landau theory →	`[DERIVATION]`	Landau's expansion follows from analyticity of the free energy near a critical point, plus symmetry constraints on the allowed terms.
Landau → ferromagnetism	`[APPROXIMATION]`	Landau's *mean-field* theory applied to a spin order parameter, with a coefficient $a(T)$ that crosses zero at T_c. Valid near T_c and above T_c; quantitatively wrong far below it because it neglects critical fluctuations.
Landau → superconductivity	`[APPROXIMATION]`	Ginzburg–Landau, with the order parameter being the Cooper-pair condensate and $a(T) = a_0(T - T_c)$. Valid near T_c; the microscopic mechanism (BCS pairing) is *added*, not derived, from the L1 side.
Any of the above → engineering hysteresis	`[ANALOGY]`	Hysteresis needs a bistable free-energy landscape with a barrier. That is a structural feature, not a derivation. A Schmitt trigger is not a superconductor.
Higgs ↔ Landau	`[STRUCTURAL CONNECTION]`	Same template, independently applied: $\mu^2 \leftrightarrow \mu^2(T_c - T)$. The dictionary is exact; the physics is not connected.

The scales differ by roughly **twelve orders of magnitude** ($\sim 10^{11}$ eV versus $\sim 10^{-3}$ eV), which is the cleanest available demonstration that energy scale is irrelevant to the template's validity and everything to the instantiation.

What a student should take away: **symmetry breaking is a scale-independent mathematical phenomenon with multiple independent physical realizations** — not a single lineage running outward from the Higgs field. Reading the tree above as a genealogy is the specific error this section exists to prevent.

0.4 — Topology in the Action: Already There, Already Relevant

The θ-term is a topological term — it integrates to a topological invariant (the Pontryagin number) over closed spacetime regions. It doesn't contribute to the classical equations of motion (it's a total derivative) but profoundly affects the quantum vacuum structure.

Why tell engineering students this? Because **topology has already reached semiconductor labs.** The quantum Hall effect, topological insulators, and Weyl semimetals all derive their exotic properties from topological terms in their effective Layer-1 Hamiltonians — which are shadows of Layer-0 topology. Note the epistemic status: the link between fundamental topological terms and effective topological phases is a **structural correspondence**, not a claim that these materials require new physics. Most of topological condensed matter is already accounted for by established quantum mechanics and quantum field theory. Whether genuinely new fundamental input is required in that sector is an open question, not a premise — and the unknown sector $\mathcal{F}[\ldots]$ is where such new physics would enter *if* it turned out to be needed.

[STRUCTURAL CONNECTION] Note in advance, because this is the thread most often over-read: the θ-term, the Chern numbers of Layer 1, and the encirclement count that appears in Layer-3 control theory (Ch. 21) share a common branch of mathematics — the classification of maps by winding number or degree — **without one being derived from another.** This is a *shared mathematical thread*, not a physical descent.

The interesting claim is not that Nyquist control came from QCD. It is this:

> **Topological invariants, winding numbers, phase, and counting arguments appear independently in physically unrelated areas — particle physics, condensed matter, and control engineering — because classifying maps by degree is a general mathematical tool that any field theory, any band structure, and any complex transfer function can end up needing.**

Three settings, three genuinely different physical questions, one shared skeleton:

Setting	The invariant	What it counts	Physical question it answers
Ch. 0 — QCD θ-term	Pontryagin number $\in \mathbb{Z}$	Windings of the gauge field on a closed 4-manifold	Why is $\theta_{QCD} < 10^{-10}$? (strong CP)
Ch. 7 — band structure	Chern number / $\mathbb{Z}_2$ invariant	Winding of the Bloch states over the Brillouin torus	Why is the Hall conductance quantised? Why do edge states exist?
Ch. 21 — Nyquist criterion	Encirclement number of $(-1, 0)$	Windings of $L(j\omega)$ in the complex plane	How many closed-loop poles are unstable?

The structures are analogous in the strong sense — integer-valued, invariant under smooth deformation, changing only across a genuine qualitative transition. The *physics* is not analogous at all: a Pontryagin number constrains the vacuum structure of a gauge field; a Chern number produces a measurable Hall conductance; an encirclement count is a stability test applied to an engineering transfer function. Where the chain is completed, in Ch. 21 §21.11.2, this is stated explicitly and the analogy is explicitly declined.

BRIDGE A — Descent from Layer 0 to Layer 1

The operation: Isolate the matter (fermion) sector. Take the limit $E \ll mc^2$, $v \ll c$, and reduce to single-particle (or few-particle) quantum mechanics.

What you are assuming small: v/c and $E_{kinetic}/m_e c^2$. For electrons in atoms: $v/c \approx \alpha \approx 1/137$ — excellent approximation.

Bridge A.1 — What Survives the Descent

Layer-0 object	Survives as
Fermion field $\psi(x,t)$	Wavefunction $\Psi(\mathbf{r},t)$
Gauge coupling $D_\mu = \partial_\mu + ieA_\mu$	Minimal coupling: $\mathbf{p} \to \mathbf{p} - e\mathbf{A}$
Lorentz group representations	Spin-1/2; Pauli matrices
U(1) symmetry (Noether: charge conservation)	Continuity equation $\partial_t \rho + \nabla \cdot \mathbf{J} = 0$
SU(3) (strong force)	Reduced to nuclear binding (residual force); not discarded, just confined

Bridge A.2 — What Gets Discarded and Why It Is Allowed

Discarded	Why allowed at $E \ll mc^2$, $v \ll c$
Virtual particle creation/annihilation	Requires $E \sim 2mc^2$ to produce pairs
Renormalization corrections	Of order $\alpha/\pi \approx 0.002$ — tiny
Gravitational effects on electrons	Gm_e^2/r is 10^{42} times smaller than Coulomb force at atomic scales
Higher-order gauge corrections (QED loops)	Suppressed by α^n

Bridge A.3 — The Bridge Zone: The Dirac Equation

The Dirac equation sits *between* QFT and Schrödinger QM. It is still relativistic and first-order in time, but it is a single-particle equation rather than a full field theory.

$$i\hbar\frac{\partial \psi}{\partial t} = \left(c\boldsymbol{\alpha}\cdot(\mathbf{p} - e\mathbf{A}) + \beta m_e c^2 + eV\right)\psi$$

The Dirac equation, **without any additional assumptions**, predicts:

- Spin-1/2 — not postulated, derived
- Magnetic moment with g ≈ 2 — not fitted, derived
- Fine structure of hydrogen — not patched in, derived
- The existence of antimatter — it falls out of the algebra

The Foldy-Wouthuysen transformation then block-diagonalizes the Dirac equation in powers of $(v/c)^2$, producing in order:

1. The Schrödinger equation (leading order)
2. The Pauli equation with spin-orbit coupling (next order)

3. Darwin term corrections (next order)

Each step is a named, controlled expansion. Students see the Schrödinger equation emerge as a first term in a series — not as an axiom handed down from authority.

LAYER 1 — The Quantum / Atomic Scale

What this layer is: The world of individual atoms and small collections of them. Matter has identity here — this is where carbon differs from silicon, conductor differs from insulator, and an atom can absorb exactly one photon at a time. Every engineering branch's materials science sits on this layer.

1.1 — Single-Particle Quantum Mechanics

The Schrödinger equation — derived from Bridge A, not postulated:

$$i\hbar\frac{\partial\Psi}{\partial t} = \hat{H}\Psi = \left(-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r})\right)\Psi$$

Key structures:

- Superposition: not a strange axiom, but a consequence of linearity in the field equation that survived from Layer 0
- Probability current: U(1) symmetry and Noether's theorem give a conserved current of $|\Psi|^2$ `[DERIVATION]`. The step from *conserved density* to *probability* is the Born rule, which is a **postulate** of standard QM, not a Noether consequence. See Ch. 3 §3.9.2
- Quantization of energy: not an axiom, but the result of imposing boundary conditions on the wavefunction (same reason a guitar string has discrete harmonics)

1.2 — Spin and Exchange Statistics

From the Pauli equation (one step below Dirac in Bridge A):

- Spin-1/2 operators $\hat{S} = \frac{\hbar}{2}\boldsymbol{\sigma}$ fall directly out
- The **spin-statistics theorem** from QFT: half-integer spin fields *must* anticommute → Pauli exclusion principle

The Pauli exclusion principle is *not* an empirical rule added to explain the periodic table. It is a mathematical consequence of the anticommutation relations of the fermion field ψ in Layer 0. This is one of the clearest demonstrations of why starting at Layer 0 gives deeper insight.

1.3 — The Hydrogen Atom and Atomic Structure

The Schrödinger equation with $V = -e^2/4\pi\epsilon_0 r$:

- Exact solution gives quantized energy levels $E_n = -13.6/n^2$ eV
- Orbital angular momentum and magnetic quantum numbers fall out of the spherical symmetry (Noether's theorem for rotations: symmetry → conserved L)

- Shell structure → the periodic table — not memorized but derived from: (a) energy-level ordering, (b) Pauli exclusion, (c) spin degeneracy

The periodic table is Layer-1 output. Every "materials property" that EEE, ME, CE, and ChE engineers work with — conductivity, thermal expansion, yield strength, reaction enthalpy — ultimately traces back to where an atom sits in this table and why.

1.4 — Molecular Bonding: The Bridge to Materials

Linear Combination of Atomic Orbitals (LCAO) and Molecular Orbital (MO) theory:

- Bonding and antibonding orbitals arise from symmetric/antisymmetric combinations of atomic wavefunctions
- This is why H_2O has the bond angle it has, why graphene is a conductor while diamond is not, why polymer chains behave the way they do

Branch connections from this section:

- ChE: reaction thermodynamics, reaction kinetics (activation energy = energy barrier in the molecular potential)
- ME: material phases and alloy behavior
- CE: cement hydration chemistry (Layer-1 bond formation)
- EEE: semiconductor dopant energy levels

1.5 — Periodic Lattice and Band Theory

This is the most important Layer-1 chapter for working engineers.

Take many atoms. Arrange them periodically. Apply the Schrödinger equation with a periodic potential $V(\mathbf{r}) = V(\mathbf{r} + \mathbf{a})$.

Bloch's theorem follows from the *discrete* lattice translation symmetry — not from Noether's theorem, which requires a continuous one-parameter group. Since $\hat{H}$ commutes with every lattice translation, the states may be chosen as joint eigenstates, and the one-dimensional irreducible representations of a discrete translation group are labelled by a phase $e^{i\mathbf{k}\cdot\mathbf{a}}$:

$$\Psi_{n\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_{n\mathbf{k}}(\mathbf{r})$$

- Allowed energies form **bands** separated by **gaps** — not put in by hand but caused by Bragg reflection at the Brillouin zone boundary
- Where the Fermi level sits relative to bands determines:

Fermi level position	Material type	Branch relevance
Inside a band	Metal/conductor	EEE (wiring, contacts)
In a gap, gap ≫ kT	Insulator	EEE, ME (dielectrics)
In a gap, gap ~ kT–3eV	Semiconductor	EEE (devices)
In a gap, gap ~ 1–10eV	Optical material	EEE (photonics, LEDs)

Topological bands (forward pointer to the unknown sector $\mathcal{F}[\ldots]$): Some band structures have a non-trivial topological invariant (Chern number, Z2 invariant). These give quantum Hall conductance, topological insulator surface states, and Weyl semimetal behavior — effects that survive disorder and temperature in a way ordinary band theory can't explain. This is where Layer-0 topology (the θ-term and anomaly structure) has already reached the lab bench. [STRUCTURAL CONNECTION — same mathematical object as the θ-term (a topological invariant), independently applied to a different physical system.]

1.6 — Quantum Scattering Theory: The Foundation of All Transport

This is the Layer-1 chapter that Bridge B will build on directly.

When a quantum particle (electron, phonon, neutron) encounters a potential, it scatters. The **S-matrix** and differential cross-section $d\sigma/d\Omega$ give the probability of scattering into each direction.

For an electron in a crystal, the relevant scatterers are:

- **Phonons** (lattice vibrations): temperature-dependent scattering
- **Impurities/defects**: disorder-dependent scattering
- **Other electrons** (electron-electron): correlation effects

Each of these gives a scattering rate $1/\tau$ (inverse relaxation time).

Why this matters: the transport coefficients in Layer 2 (conductivity, thermal conductivity, viscosity, diffusivity) are all **averages of these scattering rates over the appropriate ensemble.** Bridge B will make this explicit.

BRIDGE B — Descent from Layer 1 to Layer 2

The operation: four named limits, applied to different parts of Layer 1 simultaneously. They are not the same limit. They produce different parts of Layer 2. Each is independent and can be understood on its own.

```
Layer 1
    │
    ├─(B.a) ħ→0 ──────────────────────────► Classical mechanics (§2.1, §2.2)
    ├─(B.b) N→∞, ensemble avg ────────────► Thermodynamics (§2.3)
    ├─(B.c) Classical limit of U(1) ──────► Maxwell's equations (§2.4)
    ├─(B.d) Weak-field metric ────────────► Newtonian gravity (§2.5)
    └─(B.e) Scattering avg (Kubo) ────────► Generalized Transport Law (§2.7)
```

Bridge B.a — ħ → 0: Quantum to Classical Mechanics

What you are assuming small: $\hbar/S_{action}$ where S_{action} is the characteristic classical action of the system. For a macroscopic object, $S_{action}/\hbar \sim 10^{34}$ — the approximation is essentially exact.

The Ehrenfest theorem gives the first sign that this limit works:

$$\frac{d\langle \mathbf{p} \rangle}{dt} = -\langle \nabla V \rangle$$

This is Newton's second law for the expectation value of momentum.

The path integral perspective: Feynman's path integral says the amplitude for going from A to B sums over *all* paths, each weighted by $e^{iS/\hbar}$. When $\hbar \to 0$, the integrand oscillates wildly for all paths *except* the one where $\delta S = 0$ — the stationary phase path. The **classical trajectory** emerges from the stationary phase of the quantum path integral. This is the most transparent connection between Layer 0's action principle and Layer 2's classical mechanics.

Bridge zone: WKB approximation. When $\hbar$ is small but not zero, the WKB wavefunction $\Psi \approx A(\mathbf{r})e^{iS_{cl}(\mathbf{r})/\hbar}$ sits between quantum and classical. It governs:

- Quantum tunneling through barriers (relevant for tunnel diodes, STM)
- Semiclassical quantization (Bohr-Sommerfeld, WKB energy levels)
- High-frequency wave optics (geometric optics is WKB for EM waves)

What survives into Layer 2: Newton's 2nd law, Lagrangian formalism (the action principle persists in classical form), conservation laws (Noether's theorem survives the ħ→0 limit exactly).

Bridge B.b — Thermodynamic Limit + Statistical Description: QM to Thermodynamics

What you are assuming: that you cannot track all $N \sim 10^{23}$ microstates — only macroscopic averages matter.

The density matrix $\hat{\rho}$ describes a statistical mixture of quantum states:

$$S = -k_B \mathrm{Tr}(\hat{\rho} \ln \hat{\rho}) \quad \text{(von Neumann entropy)}$$

In the $N \to \infty$ limit with maximum entropy (least-biased) reasoning:

- Canonical ensemble → Boltzmann distribution $P_i \propto e^{-E_i/k_B T}$
- Partition function $Z = \mathrm{Tr}(e^{-\hat{H}/k_B T})$ encodes all thermodynamics
- Statistical mechanics provides microscopic foundations for the thermodynamic laws, with the **second** law acquiring its characteristic statistical interpretation. The four are not equivalent in this respect: the first law is essentially energy conservation (a dynamical symmetry, not a statistical one), the zeroth is explained via transitivity of $\beta = 1/k_B T$, and the third needs extra assumptions about low-temperature spectra. Ch. 10 §10.5 gives the per-law accounting. Note also that Bridge B.b bundles *two* separate steps — the thermodynamic limit ($N, V \to \infty$ at fixed density) and the statistical/ensemble description — and ensemble averaging is not itself a limit.

Phase transitions via Landau theory: Near a phase transition, write the free energy as a power series in an order parameter ϕ:

$$F = F_0 + a(T)\phi^2 + b\phi^4 + \cdots$$

When $a(T) = a_0(T - T_c)$ changes sign, the minimum shifts from $\phi = 0$ to $\phi \neq 0$ — spontaneous symmetry breaking.

This is the same free-energy template as §0.3 [STRUCTURAL CONNECTION] — the Higgs mechanism (~246 GeV) and the ferromagnetic transition (~10^{-3} eV) are independent instances of identical mathematics, not one causing the other. The student has already seen this template. Tell them.

Bridge zone: Quantum statistical mechanics. The Fermi-Dirac and Bose-Einstein distributions sit between quantum (Layer 1) and classical thermodynamics (Layer 2). At $k_B T \gg \hbar\omega$, they reduce to the classical Maxwell-Boltzmann distribution. At $k_B T \ll E_F$ (the Fermi energy), electron behavior is dominated by quantum statistics — this is why metals have the electronic specific heat they do, and why the Layer-2 classical equipartition theorem fails for electrons.

Bridge B.c — Classical Electromagnetic-Field Limit of the U(1) Gauge Theory

What the name means. This is **not** a limit in which the group $U(1)$ itself becomes Maxwell's equations. It denotes the *classical-field limit of the electromagnetic $U(1)$ gauge theory*, in which the field behaves as a coherent classical configuration and quantum fluctuations of the field are negligible. The gauge group is unchanged by the limit; what is discarded is the quantisation of the field.

What you are assuming: Many photons in coherent states, so quantum fluctuations are negligible; fields are smooth and classical.

The U(1) piece of $\mathcal{L}_{SM}$ is:

$$\mathcal{L}_{EM} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + j^\mu A_\mu$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

Applying the Euler-Lagrange equations to this action:

$$\partial_\mu F^{\mu\nu} = j^\nu$$

In 3+1 notation, this is exactly $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$ and $\nabla \times \mathbf{B} - \partial_t \mathbf{E}/c^2 = \mu_0 \mathbf{J}$. The other two Maxwell equations ($\nabla \cdot \mathbf{B} = 0$, $\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$) follow from the antisymmetry of $F_{\mu\nu}$ — they are mathematical identities (Bianchi identity), not independent physical laws.

Four Maxwell equations = one covariant equation + one algebraic identity. Both fall directly out of the Layer-0 action.

The KCL thread, link 2 of 6 (see §0.2): [DERIVATION] Charge conservation — carried by the **global** phase symmetry of charged matter, via Noether's first theorem — survives the classical limit as the continuity equation

$$\partial_\mu j^\mu = 0$$

This is the exact statement that will be integrated over a control volume and lumped in Bridge C to produce Kirchhoff's current law in §3.1. Nothing about circuits has been assumed yet — this is still pure field theory.

Bridge B.d — Weak-Field Metric: GR to Newtonian Gravity

What you are assuming: $h_{\mu\nu}$ is small, where $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$. For Earth's surface: $h_{00} \approx 2GM/rc^2 \approx 10^{-9}$ — extremely small.

Varying the Einstein-Hilbert action w.r.t. $g_{\mu\nu}$ gives:

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}$$

In the weak-field, non-relativistic limit, the 00-component reduces to:

$$\nabla^2 \Phi = 4\pi G\rho \quad \Longrightarrow \quad F = -\frac{GMm}{r^2}$$

Newton's law of gravitation is a first-order expansion of general relativity. The corrections (perihelion precession, gravitational lensing, frame dragging) are higher-order terms in $h_{\mu\nu}$ — small but measurable, and engineers running GPS systems must account for them (general relativistic time dilation of ~45 μs/day is corrected for in GPS). This is Layer-0 physics quietly inside everyday technology.

Bridge B.e — Quantum Scattering → Kubo → Generalized Transport Law

What you are assuming: Linear response (perturbative E field), statistical averaging over the ensemble, and that the current-current correlator decays (i.e., the system has some scattering — a real material at finite T or with disorder).

The **Green-Kubo formula** for any transport coefficient L_{AB}:

$$L_{AB} = \frac{1}{Vk_BT}\int_0^\infty \langle \hat{A}(0)\hat{B}(t)\rangle_0 \, dt$$

This single *template*, with different operators $\hat{A}$, $\hat{B}$ and different closure assumptions, gives:

$\hat{A}$, $\hat{B}$	L_{AB}	Law	Equation	Extra input required
Current $\hat{J}$, $\hat{J}$	Electrical conductivity σ	Ohm	$\mathbf{J} = \sigma\mathbf{E}$	Drude / memory closure
Heat current $\hat{J}_Q$, $\hat{J}_Q$	Thermal conductivity κ	Fourier	$\mathbf{q} = -\kappa\nabla T$	closure; Wiedemann–Franz needs elastic scattering
Particle current $\hat{J}_N$, $\hat{J}_N$	Diffusivity D	Fick	$\mathbf{J}_N = -D\nabla c$	Einstein relation
Momentum flux $\hat{\Pi}$, $\hat{\Pi}$	Viscosity η	Newton viscosity	$\tau = -\eta\, dv/dy$	transverse projection, τ_v
Stress $\hat{\sigma}$, $\hat{\sigma}$	Elastic moduli	Hooke	$\sigma = C : \varepsilon$	**free-energy second derivative** — a static susceptibility, not a transport coefficient

The convergence chapter: all five laws instantiate one response-theory framework. For the four dissipative transport coefficients (σ, κ, D, η), the coefficient is a time-integrated autocorrelation of the corresponding flux operator in the equilibrium quantum state, and each requires its own closure. The fifth, C, sits in the framework only as a zero-frequency nondissipative limit, not as a transport coefficient.

Precise scope of the claim. Ohm's law and Fourier's law are *not* the same computation — they use different operators, different couplings, and Wiedemann–Franz requires an extra physical assumption (quasi-elastic scattering). What they share is a template. Ch. 13 §13.2.2 states exactly what "exact" does and does not attach to: the Kubo *relation* is exact given linear response; any particular closed form for a coefficient additionally requires the correct operator definitions, equilibrium state, order of limits, thermodynamic limit, boundary conditions, and separation of nondissipative/contact terms.

The role of scattering (Bridge A.1.6 paying off here): If the current-current correlator never decays (perfect crystal, zero temperature, no disorder), the integral diverges → σ → ∞ → the material is a perfect conductor or superconductor. Ohm's law requires finite scattering. The irreversibility of $\mathbf{J} = \sigma\mathbf{E}$ comes from this decay — quantum decoherence injected by the environment.

LAYER 2 — The Classical Continuum & Statistical Scale

What this layer is: The world of continuous fields, average densities, and macroscopic forces. The vast majority of classical engineering science already exists here, completely assembled, before a student has chosen a branch.

2.1 — Classical Mechanics and the Lagrangian

Newton's 2nd law, derived in Bridge B.a, now treated as a working tool:

$$m\ddot{\mathbf{r}} = \mathbf{F}$$

But the more powerful form — which carries the Layer-0 action principle into this layer — is the **Lagrangian**:

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = 0, \quad L = T - V$$

This is Hamilton's principle $\delta S = 0$ applied to a classical, particle system. Students who started at Layer 0 already know this is the stationary-phase condition of a path integral. Students starting here for the first time can verify it reduces to Newton's law — but should be told: this is not a coincidence. The variational principle is the same one from Layer 0, just in different notation.

Noether thread at Layer 2: Every continuous symmetry of L gives a conserved quantity:

- Time-translation symmetry → Energy conservation (1st law, later)
- Spatial translation → Momentum conservation
- Rotation symmetry → Angular momentum conservation

These are not separate postulates. They are Noether's theorem, living here.

2.2 — Hamiltonian Mechanics and Phase Space

The Hamiltonian $H(q, p) = T + V$ reformulates mechanics in phase space. Its value: classical mechanics is the *grammar* of much of Layer-3 dynamics — many engineering dynamical models admit a variational, Hamiltonian, port-Hamiltonian, or contact/DAE formulation. But this is a strong tendency, not a universal theorem, and the exceptions are not edge cases. Plenty of standard engineering models are

dissipative (ordinary differential equations with friction, drag, and irreversibility), **stochastic** (random walks, Markov chains, stochastic differential equations in state-space or Fokker–Planck form), **empirical** (curve fits and lookup tables), **constrained** (describing functions, complementarity), **hybrid** (mode switching with logic), **non-variational**, or **reduced-order closures** standing in for unresolved microstructure. Dissipative and stochastic models in particular need extra structure — a friction/metric term, a noise intensity, a dissipative potential — before any variational principle is available. So the honest formulation is: *variational structure is a powerful and widely applicable way to organise dynamics, not a property that every engineering differential equation possesses.*

Canonical transformations and action-angle variables: worth introducing here because they explain why rotating machinery, vibrating structures, and oscillating circuits all share mathematical form (they're all harmonic oscillators in appropriate coordinates).

2.3 — Thermodynamics and Statistical Mechanics

The four laws, given microscopic foundations rather than postulated (Bridge B.b) — with the per-law qualifications in Ch. 10 §10.5:

- Zeroth: equilibrium is the maximum-entropy state
- First: energy conservation (Noether: time-translation symmetry survives Bridge B.b)
- Second: entropy never decreases for isolated systems — falls out of the statistical definition, because the volume of high-entropy macrostates is astronomically larger than low-entropy ones
- Third: entropy → 0 as T → 0 (the ground state is unique or finitely degenerate)

Phase transitions with Landau theory: the symmetry-breaking template from §0.3 and Bridge B.b, now applied to:

- Liquid-gas transitions (ChE, ME)
- Solid-liquid (solidification in ME/CE)
- Magnetic transitions (EEE magnetics)
- Superconducting transition (EEE)

The critical point: all these transitions share the same universality class mathematics. This is one of the deepest results of 20th-century physics, and it's entirely within Layer 2.

2.4 — Maxwell's Equations in Working Form

Derived in Bridge B.c, now written in the forms engineers actually use:

Differential form (for field problems):

$$\nabla \cdot \mathbf{D} = \rho_f, \quad \nabla \times \mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t}$$

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Integral form (for Kirchhoff's laws and circuit theory — pointing forward to Layer 3):

$$\oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d\Phi_B}{dt}, \qquad \oint \mathbf{H} \cdot d\mathbf{l} = I_{enc}$$

Neither integral form maps onto a Kirchhoff law by itself. **KCL and KVL come from two different parts of the Maxwell/continuity structure, taken through two different limits — they are not mirror images of each other via Faraday and Ampère.**

```
MAXWELL'S EQUATIONS
      │
      ├── Faraday:  ∮E·dl = −dΦ_B/dt
      │      [STRUCTURAL CONNECTION] ──► circuit voltage relations
      │      (supplies the inductor relation V_L = L di/dt once the
      │       changing flux is confined to inductors — not neglected)
      │
      ├── Ampère–Maxwell:  ∮H·dl = I_enc + dΦ_D/dt
      │      [STRUCTURAL CONNECTION] ──► field/current relations
      │      (this is where current *sources*; it is not a conservation law)
      │
      └── Charge continuity:  ∂_μ j^μ = 0
             [DERIVATION, Bridge B.c] ──► KCL
             (the actual conservation statement, after Bridge C lumping)
```

KCL comes from the charge-continuity branch above — already derived in Bridge B.c and carried through Bridge C's lumped-node approximation (§0.2, §3.1). It is not read directly off Ampère's law; Ampère's law is where the *field* equation that current sources into lives, but the *conservation statement* that becomes KCL is continuity, a separate (Bianchi-adjacent) piece of structure.

KVL requires a second, independent step — the quasi-static / lumped approximation that defines Bridge C:

```
Maxwell (full, with flux terms)
       │
       │  [APPROXIMATION — lumped / quasi-static:
       │     changing flux confined to inductors,
       │     none linking the interconnect; L_loop ≪ λ_EM]
      ↓
lumped circuit theory (branch voltages, incl. L di/dt)
       │
       │  [DERIVATION — Bridge C control-volume
       │     integration of Faraday around a closed loop]
      ↓
KVL
```

Under the lumped assumption, Faraday's law integrated around a loop becomes $\sum_k V_k = 0$ — KVL — with each inductor's induced EMF entering as its branch voltage $L\,di/dt$: the changing flux is *accounted for* in the inductor, not dropped (dropping it would delete the inductor). KVL is a loop statement about $\mathbf{E}$, not energy conservation; the energy balance is Poynting's theorem (Example 03, step 4). Away from the lumped assumption (high frequency, electrically large loops), flux links the interconnect itself and cannot be assigned to any one element, KVL stops being exact, and the full Faraday integral must be used instead — which is exactly why transmission-line theory (Bridge C.3) exists as a separate model.

Constitutive relations (bringing in Layer-1 band theory output):

$$\mathbf{D} = \epsilon\mathbf{E}, \quad \mathbf{B} = \mu\mathbf{H}, \quad \mathbf{J} = \sigma\mathbf{E}$$

These ε, μ, σ are not phenomenological constants — they are the Layer-1 band structure and scattering physics, packaged into single numbers valid at low frequency and moderate field strength. The student should know where these numbers come from and when they break.

2.5 — Continuum Mechanics

The continuum limit of $N \to \infty$ particles (Bridge B.b, spatial version). Note that the thermodynamic limit alone does not produce a continuum field — coarse-graining additionally requires scale separation, $\ell_{\text{micro}} \ll L_{\text{variation}}$ (Ch. 15 §15.1):

Fluid mechanics:

- Conservation of mass: $\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$
- Conservation of momentum: Navier-Stokes equation

$$\rho \left(\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} \right) = -\nabla p + \eta \nabla^2 \mathbf{v} + \mathbf{f}$$

- The viscosity η here is the Kubo transport coefficient from Bridge B.e — not an empirical constant but a Layer-1 momentum scattering rate

Solid mechanics:

- Displacement field $\mathbf{u}(\mathbf{r})$ — the continuum limit of atomic positions
- Strain tensor: $\varepsilon_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i)$
- Stress tensor: $\sigma_{ij} = C_{ijkl}\varepsilon_{kl}$ (generalized Hooke — a static, nondissipative response, not a dissipative Green–Kubo coefficient)
- Equilibrium: $\nabla \cdot \boldsymbol{\sigma} + \mathbf{f} = 0$

2.6 — The Unified PDE Perspective

A striking fact that Layer 2 reveals: three different physical situations share one equation form.

The wave equation $\frac{\partial^2 \phi}{\partial t^2} = c^2 \nabla^2 \phi$:

- EM waves (c = speed of light)
- Acoustic waves (c = speed of sound)
- Elastic waves in solids (c = √(E/ρ))
- All analyzed with the same mathematics. The branch only changes what c is.

The diffusion equation $\frac{\partial \phi}{\partial t} = D \nabla^2 \phi$:

- Heat conduction (φ = temperature, D = thermal diffusivity)
- Mass diffusion (φ = concentration, D = diffusivity)
- Charge diffusion in semiconductors (φ = carrier density)
- All analyzed with the same mathematics. The branch only changes what D is.

This is the Layer-2 version of the Generalized Transport Law: not just the constitutive relation flux $= -L \cdot \nabla \phi$, but the full time-dependent PDE that falls out when you combine the constitutive relation with a continuity equation (which is itself Noether's conservation law).

2.7 — The Generalized Transport Law (The Convergence Chapter)

Already introduced in Bridge B.e, now written in its full engineering form:

$$\underbrace{\text{Flux}}_{\text{what flows}} = - \underbrace{L}_{\text{material coeff}} \cdot \nabla \underbrace{\phi}_{\text{driving potential}}$$

Domain	Flux	L	φ (potential)	Named law	Branch
Electrical	Current density **J**	σ (conductivity)	Electric potential V	Ohm	EEE
Thermal	Heat flux **q**	κ (thermal cond.)	Temperature T	Fourier	ME, ChE
Mass/species	Molar flux **J**_N	D (diffusivity)	Concentration c	Fick	ChE, CE (env.)
Fluid momentum	Shear stress τ	η (viscosity)	Velocity u	Newton viscosity	ME, CE (fluid)
Structural	Stress σ	E (modulus)	Strain ε	Hooke	CE, ME
Chemical	Reaction rate r	k (rate const.)	Concentration c	Arrhenius (modified) [PHENOMENOLOGICAL]	ChE

Many of the linear-response transport coefficients in this table can be represented through Kubo/Green–Kubo relations, and for those the Layer-1 route is the one to remember. But **not every row is a Kubo coefficient**, and the distinction is worth keeping: Arrhenius rate constants are chemical-kinetic parameters fixed by reaction energetics and collision theory, not by a current-response correlator; elastic moduli are static elastic constants (strain energy second derivatives), which enter linear response only in the trivial static limit and are set by the full interatomic potential, not by a dissipative transport correlator. Each needs its own microscopic or phenomenological framework. When any of these coefficients breaks down (high field, quantum regime, exotic material), the student knows exactly where to go: back to Layer 1, sometimes Layer 0.

BRIDGE C — Descent from Layer 2 to Layer 3

The operation: Integrate the Layer-2 PDEs over **control volumes** (finite regions of space), discarding spatial variation *within* each element. The PDE becomes an ODE or algebraic equation.

What you are assuming small: *the relevant within-element scale* — and this depends on the domain. There is no single lumping criterion for all of physics. For wave-carrying domains it is the shortest relevant wavelength; for diffusive domains it is a diffusion length and the criterion is time-dependent; for structural dynamics it is modal separation; for chemical reactors it is a mixing time. Treating $L_{element} \ll \lambda_{field}$ as *the* criterion is a wave-domain statement misapplied as a general one.

Domain	Lumping valid when	Relevant scale	Notes
Electrical (RF circuits)	$Lf/v_{em} \ll 1$	EM wavelength	Valid below ~300 MHz for 10 cm components; interconnect delay matters
Thermal	$Bi = hL/\kappa \ll 1$	Diffusion length $\sqrt{\alpha t}$	**Time-dependent** — validity window in time, not a bandwidth
Acoustic	$Lf/c_s \ll 1$	Sound wavelength	Also needs absorption layer thick vs λ; $c_s = 343$ m/s
Fluid / hydraulic	$Lf/c_{water} \ll 1$	Acoustic wavelength	$c \approx 1480$ m/s; also fails on water hammer

Domain	Lumping valid when	Relevant scale	Notes
Structural	$Lf/v_P \ll 1$	Elastic stress-wave wavelength	Quasi-static limit; first bending/shear mode sets the edge ($L/\lambda \gtrsim 1/10$)
Chemical (CSTR)	Mixing time $\ll$ reaction time	Residence time	Perfect-mixing assumption, not a spatial scale at all

`[APPROXIMATION]` — the criterion is a claim about the *input*, not just the component: a lumped model is valid near one frequency or timescale, and drops every spatial mode above the first. Full derivation and the three separate things lumping assumes: Ch. 16 §16.10.1, Ch. 17 §17.1.2.

This is the operation referenced in §0.2 as link 4 of the KCL chain, and in §2.4 as the second stage of the KVL derivation.

Bridge C.1 — The Effort/Flow Variable Pair: The Formal Unification

When you lump any continuous domain, two types of variable emerge:

- **Effort variable (e):** the "potential," the thing that drives flow (does work *per unit* of flow)
- **Flow variable (f):** the "current," the thing that flows

Power in every domain is simply $P = e \cdot f$ — provided the flow is the true conjugate of the effort.

Domain	Effort e	Flow f	Power e·f
Electrical	Voltage V [V]	Current I [A]	Watts [W]
Translational mechanical	Force F [N]	Velocity v [m/s]	Watts [W]
Rotational mechanical	Torque τ [Nm]	Angular vel. ω [rad/s]	Watts [W]
Thermal	Temperature T [K]	Entropy flow Ṡ = Q̇/T [W/K]	Watts: T·Ṡ = Q̇ [W]
Hydraulic/pneumatic	Pressure P [Pa]	Volume flow Q [m³/s]	Watts [W]
Chemical	Chem. potential μ [J/mol]	Molar flow ṅ [mol/s]	Watts [W]

> **Warning: The thermal row needs the entropy flow, not the heat flow** $T\dot{Q}$ has units K·W and is **not** power. Temperature is power-conjugate to the **entropy rate** $\dot{S}$ [W/K], so that $T\dot{S} = \dot{Q}$ [W]. Engineering texts instead quote $R_{th} = \Delta T/\dot{Q}$ [K/W] (Ch. 14, 17, 19). The two are related only by a **small-signal linearisation** about a reference temperature T_0: $\dot{S} \approx \dot{Q}/T_0$, giving $R_S \approx T_0 R_{th}$. This is not an exact identity — across a finite ΔT the relation is nonlinear, and the transfer generates entropy $\dot{Q}(1/T_{cold} - 1/T_{hot})$. Full treatment, plus the caveat on the "conserved quantity" column: Ch. 17 §17.2.

Bridge C.2 — The Universal R/C/L Template

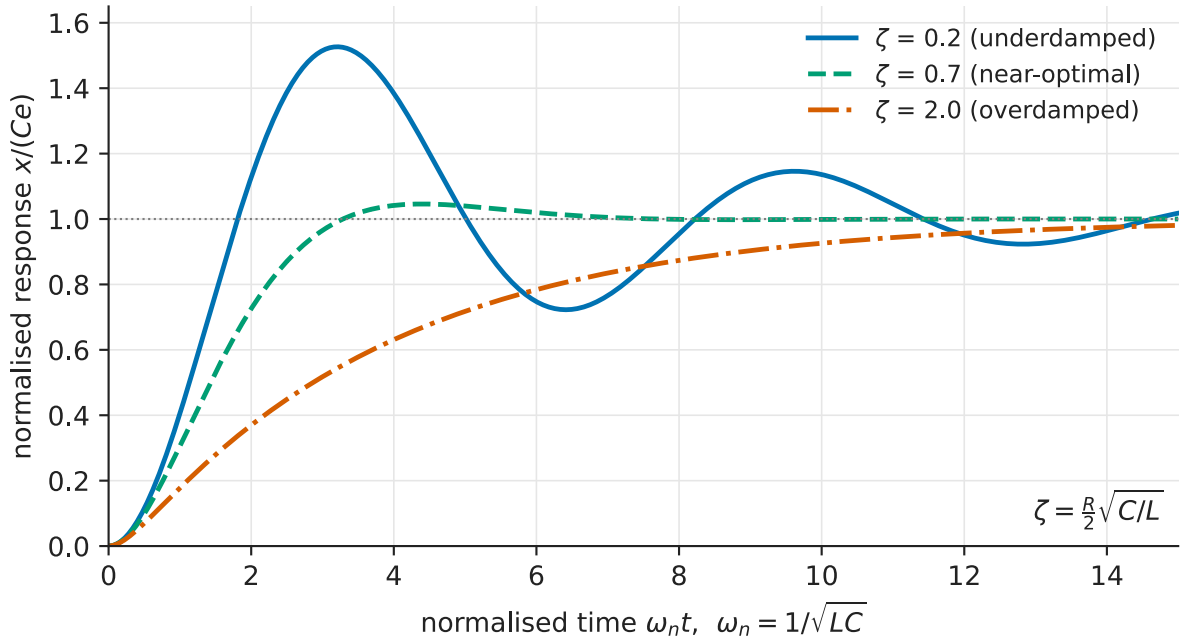
Many lumped physical systems can be represented using storage, dissipation, and inertial elements — the R/C/L template — arising from the three ways energy can appear in the transport equations. This is a **powerful template, not a theorem**: the enumeration below is a guide to what to look for, not a claim that every physical domain possesses exactly three passive element types and nothing else.

Element	Electrical	Mechanical (trans.)	Thermal	Hydraulic
R (dissipation)	Resistor	Damper/dashpot	Thermal resistance	Pipe friction
C (potential E. storage)	Capacitor	Spring	Thermal capacitance	Accumulator
L (kinetic E. storage)	Inductor	Mass/inertia	(rare)	Fluid inertia

The governing equation in every domain:

$$L\ddot{x} + R\dot{x} + \frac{x}{C} = e_{source}$$

An RLC circuit, a damped mass-spring, a hydraulic pipe-with-inertia, and a first-order thermal system: all described by this ODE. Only the labels change.



Unit-step response of $L\ddot{x} + R\dot{x} + x/C = e$, normalised to $\omega_n = 1/\sqrt{LC}$ and $\zeta = \frac{R}{2}\sqrt{C/L}$, for ζ = 0.2, 0.7 and 2: the same three curves describe every domain in the table (the thermal row is the $L = 0$ first-order case), and the four-domain schematic is in Ch. 17.

Why is this not an analogy? Because all of these are derived from the same Layer-2 conservation law (continuity of a Noether charge) + the same constitutive relation (the Generalized Transport Law), just integrated over different control volumes with different units. The structural identity is exact — this is [STRUCTURAL CONNECTION] in the strongest sense the tag allows, one step short of a literal derivation across domains.

Bridge C.3 — The Bridge Zone: Distributed-Parameter Models

The bridge zone between Layer 2 and Layer 3 is the set of models that are neither full PDEs nor fully lumped:

The transmission line (EEE):

$$\frac{\partial V}{\partial x} = -L'\frac{\partial I}{\partial t} - R'I, \quad \frac{\partial I}{\partial x} = -C'\frac{\partial V}{\partial t} - G'V$$

This retains spatial variation along x (Layer 2 PDE) while lumping in the transverse directions (Layer 3 approximation). It is exactly the right model when the component is long compared to its cross-section but not short compared to the wavelength — precisely the regime where §2.4's KVL derivation stops applying.

The Euler-Bernoulli beam (CE/ME):

$$EI\frac{\partial^4 w}{\partial x^4} + \rho A\frac{\partial^2 w}{\partial t^2} = q(x,t)$$

Again: PDE along the beam axis, lumped in cross-section.

Both of these models will be shown to be living between Layer 2 and Layer 3 — students who understand this will know exactly when to use each.

LAYER 3 — The Engineering Systems Scale

What this layer is: The world of systems, circuits, machines, and processes. The fundamental variables are effort and flow. The governing equations are ODEs and algebraic balance equations. This is where discipline identity emerges — not because the physics is different, but because engineers chose different effort/flow pairs in different physical domains.

3.1 — The Balance Laws: Noether's Theorem at Layer 3

The conservation laws from Layer 0 (Noether's theorem) survive all the way here. They are the **Kirchhoff laws** of every domain:

Conservation law	Layer 3 form	Name
Charge conservation (U(1) Noether)	$\sum I_{node} = 0$	Kirchhoff's Current Law
Faraday's law, lumped (not a conservation law — energy balance is Poynting's theorem)	$\sum V_{loop} = 0$	Kirchhoff's Voltage Law
Mass conservation	$\sum \dot{m}_{node} = 0$	Continuity (hydraulic, ChE)
Momentum conservation	$\sum F_{node} = 0$	Equilibrium (structural)
Momentum conservation	$\sum F = m\ddot{x}$	Newton's 2nd (dynamic)

The student should feel the weight of this: KCL has a full derivation chain, not a one-line pedigree — global phase symmetry of charged matter (§0.2) → conserved current (Noether's first theorem) → continuity equation $\partial_\mu j^\mu = 0$ (Bridge B.c) → integral charge balance → lumped-node approximation (Bridge C) → $\sum I = 0$. At Layer 3, it's written in two seconds. Both are true. The short form is what you use every day. The long form is what you reach for when the short form stops working — at nanoscale, at high frequency, in a quantum device. (The common shorthand "U(1) gauge symmetry ⟹ Noether ⟹ conserved current" compresses link 1; §0.2 gives the precise version and explains why the global symmetry, not the gauge redundancy, carries the Noether charge.)

KVL follows a parallel but distinct chain (§2.4): Faraday's law → lumped approximation (changing flux confined to inductors, entering as $L\,di/dt$) → Bridge C lumping → $\sum V = 0$. It is not the same chain as KCL, and does not reduce to the same physics.

3.2 — EEE: Circuits, Devices, Signals

Built from R/C/L in the electrical domain:

- Circuit analysis (node/mesh — these are KCL/KVL, i.e., Noether plus the quasi-static approximation)
- AC analysis and impedance (Fourier decomposition of the Layer-2 Maxwell equations)
- Semiconductor devices (built on Layer-1 band structure + Layer-2 drift-diffusion)
- Signal processing (Fourier/Laplace — mathematical tools inherited from Layer-2 wave equations)

3.3 — ME: Machines, Structures, Thermofluid Systems

Built from R/C/L in translational, rotational, and thermal domains:

- Dynamics: mass-spring-damper (L-C-R in mechanical domain)
- Structures: statics is Noether (momentum conservation), dynamics adds inertia
- Thermofluids: heat exchangers, engines, HVAC (Layer-2 thermo + fluid, lumped into control volumes)
- Vibrations: same ODE as RLC circuit — modal analysis is the mechanical frequency domain

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3.4 — CE: Structural and Environmental Systems

Built from R/C/L in structural and hydraulic domains:

- Structural analysis: FEM as a systematic lumping of the Layer-2 elasticity PDE
- Hydraulics: pipe networks (Darcy-Weisbach as the R in the hydraulic circuit)
- Geotechnical: soil consolidation as a diffusion equation (from Bridge B.e, Layer 2 diffusion → Layer 3 lumped drain)
- Environmental: pollutant transport is Fick's law → lumped-compartment models

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3.5 — ChE: Reactors, Separations, Process Control

Built from R/C/L in chemical and thermal domains:

- Reactors: material and energy balance = continuity + 1st law, over a control volume
- Heat exchangers: thermal R/C in a distributed-parameter model (Bridge C zone)
- Distillation/absorption: Fick's law (Layer-2 mass transport) → stage-by-stage (lumped)
- Reaction kinetics: Layer-1 activation energy → Arrhenius rate law [PHENOMENOLOGICAL, structurally matched to the Boltzmann factor] → Layer-2 reaction-diffusion PDE → Layer-3 reactor ODE

3.6 — The Cross-Branch Capstone: Feedback and Control

A PID controller is:

$$u(t) = K_p e(t) + K_i \int e\,dt + K_d \frac{de}{dt}$$

It doesn't know what domain it's in. It doesn't know if e is a voltage error, a temperature error, a position error, or a flow error. It doesn't know if u is a current, a valve opening, or a throttle position.

This is possible because at Layer 3, all domains share the same R/C/L template, which shares the same second-order ODE, which shares the same transfer function. Control theory operates at the level of the ODE — entirely above the physics. This capstone chapter should make students feel what the whole book has been building toward: a *domain-general* mathematical language for engineering that applies in every branch because every branch was always solving the same underlying equations.

3.7 — Where Layer 3 Breaks

Every chapter in Layer 3 should end with one paragraph answering: **when does this model fail, and where do you go?**

Layer-3 failure mode	Physical cause	Where to go
Transistor quantum tunneling	Gate oxide too thin (~nm)	Back to Layer 1 (band structure, WKB tunneling)
Topological insulators / quantum Hall	Non-trivial band topology	Back to Layer 1 (Chern number, edge states)
Fracture and fatigue	Bond-level failure mechanics	Back to Layer 1 (molecular dynamics, fracture mechanics)
High-frequency EM (above ~GHz)	Lumping criterion violated	Back to Layer 2 (transmission line, full Maxwell)
Turbulence (high Re)	Navier-Stokes → chaotic	Stays in Layer 2, no closed Layer-3 model
Extreme temperature materials (nuclear, aerospace)	Radiation damage, plasma	Back to Layer 1 or Layer 0
Novel materials (graphene, Weyl semimetals)	Topology, Dirac-like dispersion	Back to Layer 0 for the unknown-sector corrections

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Cross-Layer Threads

These five threads run through every layer. Every chapter should briefly identify which thread(s) it participates in.

```
Thread 1: NOETHER'S THEOREM
  Layer 0: Formal theorem — every symmetry ⇔ conservation law
  Layer 1: Charge conservation → continuity equation for |Ψ|²
  Layer 2: Conservation of energy/momentum/charge as PDEs
  Layer 3: Kirchhoff's laws, force balance, mass balance — all Noether,
           via the explicit chain worked in §0.2 and §3.1 (KCL) and
           §2.4 and §3.1 (KVL) — these are two related but separate chains.

Thread 2: SYMMETRY BREAKING (The Mexican Hat)
  All four layers below are independent instances of one mathematical
  template (order parameter + sign-changing coefficient), not a single
  causal lineage running through the Higgs field — see §0.3.
  Layer 0: Higgs potential — SU(2)×U(1) → U(1) at ~246 GeV
  Layer 1: Crystal symmetry breaking → band gaps, piezoelectricity
  Layer 2: Landau theory — phase transitions at ~meV–eV scale
  Layer 3: Hysteresis in magnetics, snap-through in structures, bistable circuits

Thread 3: THE ACTION PRINCIPLE (Stationary S)
  Layer 0: S = ∫√-g[R/16πG + L_SM + f]
  Layer 2: Classical action S = ∫L dt; Hamilton's principle
  Layer 3: Virtual work / minimum potential energy in structures;
           minimum dissipation in thermal networks (Onsager)

Thread 4: THE WAVE / DIFFUSION DICHOTOMY
  Layer 0: Field equations are hyperbolic (waves) or parabolic (diffusion)
  Layer 1: Schrödinger is parabolic; Klein-Gordon/Dirac hyperbolic
  Layer 2: Wave equation (EM, acoustic, elastic); diffusion equation (heat, mass)
  Layer 3: Filter design; thermal transient; signal propagation

Thread 5: TOPOLOGY  (shared mathematical thread — NOT a physical descent)
  Layer 0: θ-term in QCD; anomaly cancellation constrains particle charges
           → Pontryagin number (integer classifying a map)
  Layer 1: Chern number, Berry phase → quantum Hall, topological insulators
           → integer classifying a map over the Brillouin torus
  Layer 2: topological defects in ordered media (vortices, domain walls)
  Layer 3: (Mostly invisible at the engineering level — until it breaks
           something at Layer 1, which is why the unknown sector $\mathcal{F}[\ldots]$
           matters. Where it does appear explicitly — the Nyquist
           encirclement count, Ch. 21 §21.3.4 and §21.11.2 — it shares
           only the *mathematics* of integer invariants under smooth
           deformation, on a completely different space. Not a
           physical lineage, and explicitly not derived from Ch. 0.)
```

Final Visual Map (Complete)



The Equation

A complete term-by-term account of the framework from which all physics in this book descends

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> *This chapter is not meant to be fully understood on first reading. It is meant to be fully understood by the last reading. Return to it after every major chapter. Something new will have clicked.*

0.0 — The Full Action

"The best-established fundamental theories used in this book can be organized into an action principle combining general relativity, the Standard Model, and an explicitly unresolved sector."

$$S = \int d^4x\,\sqrt{-g}\left[\underbrace{\frac{R}{16\pi G}}_{\S 0.3} + \underbrace{-\frac{1}{4}B_{\mu\nu}B^{\mu\nu} - \frac{1}{4}W^a_{\mu\nu}W^{a\mu\nu} - \frac{1}{4}G^a_{\mu\nu}G^{a\mu\nu}}_{\S 0.4\text{ — gauge forces}} + \underbrace{i\bar{\psi}_f\gamma^\mu D_\mu\psi_f}_{\S 0.5\text{ — matter}} + \underbrace{|D_\mu H|^2 - V(H)}_{\S 0.7\text{ — Higgs}} + \underbrace{Y_i}\right.$$

The symmetry group of everything except gravity in this action is:

$$G_{SM} = \mathrm{SU}(3)_C \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$$

and the symmetry of the geometry is **diffeomorphism invariance** — the action is the same in any coordinate system.

From this point through the relativistic action discussion, c=ħ=1 unless SI units are explicitly restored.

The subscripts on each term tell you which section explains it. The rest of this chapter unpacks every symbol.

0.1 — How to Read This Chapter

Each section follows the same structure:

1. **The term** — written explicitly
2. **The objects** — what each symbol is
3. **The symmetry** — what invariance forces this term to take this form
4. **The physics** — what phenomenon this term is responsible for
5. **Equation of motion** — what you get when you vary the action with respect to this term's field
6. **If removed** — what breaks if you delete this term
7. **Engineering thread** — how far the consequence reaches into later chapters

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0.2 — The Stage: $\int d^4x\,\sqrt{-g}$

Before any physics, we must specify *where* the action is evaluated.

0.2.1 — The Integration Measure d^4x

$$d^4x = dx^0\,dx^1\,dx^2\,dx^3$$

This is the four-dimensional volume element over spacetime. $x^0 = ct$ is the time coordinate; x^1, x^2, x^3 are the three spatial coordinates. The action integrates the Lagrangian density $\mathcal{L}$ over all of spacetime — summing contributions from every point in space at every moment in time.

This is why $\mathcal{L}$ is called a *density*: it carries units per unit volume, and the action is the spacetime integral of that density.

> **Warning: Dimensional bookkeeping in the measure — and why natural units are used**
> This trips up almost everyone at first reading, so it is worth doing carefully.
>
> **In SI, with $x^0 = ct$:** all four coordinates have dimensions of length, so $[d^4x] = L^4$. An energy density has $[\mathcal{L}] = \mathrm{J/m^3}$, and therefore
>
> $$[d^4x\,\mathcal{L}] = m^4 \cdot \frac{\mathrm{J}}{m^3} = \mathrm{J\cdot m}$$
>
> which is energy × **length**, not energy × time. To recover the familiar "action has units of $\hbar$" statement one must either divide the measure by c (equivalently, use $x^0 = t$ and write d^4x/c), or restore the dimension-carrying constants explicitly.
>
> **In the natural units declared above ($\hbar = c = 1$):** time and length carry the same dimension, and energy carries dimensions of *inverse* length ($E = pc$, with $p \sim 1/L$). So
>
> $$[S] = [E]\cdot[L] = \frac{1}{L}\cdot L = 1$$
>
> — **the action is dimensionless**, and the scale that would otherwise be carried by $\hbar$ is simply set to 1. The measure, the Lagrangian, and the statement "S has units of $\hbar$" are then all mutually consistent.
>
> Everything in §§0.2–0.6 is written in these natural units. SI is restored explicitly wherever a number is quoted (G, the Higgs mass, engineering quantities), and the conversion is flagged at the point of use.

0.2.2 — The Factor $\sqrt{-g}$

g is the **determinant of the metric tensor** $g_{\mu\nu}$.

The metric tensor $g_{\mu\nu}$ is a 4×4 symmetric matrix that encodes:

- How distances are measured at each point in spacetime
- How the coordinate axes are oriented relative to the geometry
- The local gravitational field (in GR, these are the same thing)

In flat spacetime with Cartesian coordinates:

$$g_{\mu\nu} = \eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1)$$

and $\det(\eta_{\mu\nu}) = -1$, so $\sqrt{-g} = 1$ — the factor disappears.

In curved spacetime (near a mass, for instance), $g_{\mu\nu}$ is a nontrivial matrix and $\sqrt{-g} \neq 1$. The factor $\sqrt{-g}$ corrects the volume element so that the integral measures the actual geometric volume of a spacetime region, not just the coordinate volume. Without it, the action would change value when you changed coordinates — violating the core principle of general covariance.

Engineering relevance: In every practical engineering calculation, spacetime is flat, $\sqrt{-g} = 1$, and you never see this factor again. It matters for GPS corrections, gravitational wave detectors, and any device near an extreme gravitational field.

0.3 — Gravity: $\frac{R}{16\pi G}$

The Term

$$S_{EH} = \frac{1}{16\pi G} \int d^4x \, \sqrt{-g} \, (R - 2\Lambda)$$

This is the **Einstein–Hilbert action**, the simplest possible action for gravity. The Λ term is the cosmological constant; §0.10 returns to why its *physical origin and magnitude* remain one of the deepest open problems, but it belongs here rather than in the unknown sector.

The Objects

G — Newton's gravitational constant:

$$G = 6.674 \times 10^{-11} \, \text{N} \cdot \text{m}^2/\text{kg}^2$$

The same G that appears in $F = Gm_1m_2/r^2$. It sets the strength of gravity relative to the other forces.

$g_{\mu\nu}$ — the metric tensor (already introduced). This is the **dynamical variable** of gravity — the field that "is" gravity in GR. In classical GR, it is a smooth classical field; how to quantize it is unknown and is one of the primary targets of the $\mathcal{F}$ placeholder in §0.10.

R — the **Ricci scalar**. Built from the metric and its derivatives:

$$R = g^{\mu\nu} R_{\mu\nu}$$

where $R_{\mu\nu}$ is the Ricci tensor:

$$R_{\mu\nu} = \partial_\rho \Gamma^\rho_{\mu\nu} - \partial_\nu \Gamma^\rho_{\mu\rho} + \Gamma^\rho_{\rho\lambda}\Gamma^\lambda_{\mu\nu} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\rho}$$

and $\Gamma^\rho_{\mu\nu}$ are the **Christoffel symbols** — functions of $g_{\mu\nu}$ and its first derivatives that encode how the geometry curves. R at a point is a single number that summarizes "how much spacetime is curved here." But it is only **one scalar contraction** of the full Riemann tensor, so its sign and value must not be read off as a verdict on curvature:

- $R = 0$ does **not** by itself mean flat spacetime
- $R > 0$: net positive scalar curvature
- $R < 0$: net negative scalar curvature

> **Warning: The sign of R does not identify a spacetime** Two cautions on the bullet list above. First, the sign of the Ricci scalar is **not** a signature of the geometry taken as a whole. A Schwarzschild black hole — the most striking example of curved spacetime — has $R_{\mu\nu} = 0$ and hence $R = 0$ *everywhere outside the horizon*, despite carrying nonzero **Weyl** curvature. So "near a mass" is not associated with $R > 0$; in vacuum, mass sources Weyl curvature, not Ricci curvature. Positive R arises in genuinely matter- or vacuum-energy-dominated regions, such as the positive-curvature interior solution in the Newtonian/stellar regime, not from the mere presence of a gravitating source.
>
> Second, "$R < 0$ is anti-de Sitter" names a common *example*, not an identification. Anti-de Sitter space is a maximally symmetric vacuum solution with constant negative R, but negative R also occurs in a great many other spacetimes — matter-dominated cosmological solutions, gravitational-wave regions, generic inhomogeneous curvature. The sign alone never selects AdS.

> **Warning: $R = 0$ does not imply flat spacetime** This is one of the most common misconceptions in the subject, and it is worth stating precisely. **Flat spacetime requires the entire Riemann tensor to vanish:**
>
> $$R^{\rho}{}_{\sigma\mu\nu} = 0 \quad \text{(all components)}$$
>
> The Ricci scalar $R = g^{\mu\nu}R_{\mu\nu}$ is a single number obtained by contracting that tensor twice. A nonzero tensor can easily have all its contractions cancel.
>
> **The standard counterexample is the Schwarzschild exterior.** Outside the source the field is vacuum, so
>
> $$R_{\mu\nu} = 0 \;\Rightarrow\; R = 0$$
>
> yet spacetime there is genuinely curved: the Riemann tensor is nonzero, tidal forces are measurable, and the geometry is not Minkowski. A geodesic deviation equation gives $\xi^{\mu''} = -\frac{1}{2}R^{\mu}{}_{\nu\rho\sigma}u^{\nu}u^{\rho}\xi^{\sigma} \neq 0$.
>
> The correct summary is: **vacuum implies $R_{\mu\nu} = 0$, not $R^{\rho}{}_{\sigma\mu\nu} = 0$.** Weyl curvature (tidal effects, the part not determined by the matter distribution) survives in vacuum. The distinction reappears in action design (§0.3) and in the Riemann-based expansion around flat spacetime used in Chapter 8.

The Symmetry

Setting $\Lambda = 0$, the R term is the **unique** local scalar, linear in the Riemann tensor, that is at most second order in the metric derivatives and whose variation yields equations of motion at most second order in $g_{\mu\nu}$:

- A scalar built from the metric and at most its second derivatives
- Invariant under all diffeomorphisms (coordinate reparametrizations)
- At most second order in the equations of motion

Two qualifications keep this from being overstated. It is unique **within linear-in-curvature, second-order, and no-ghost** choices: adding R^2 or $R_{\mu\nu}R^{\mu\nu}$ also yields a scalar of the same symmetry class, but produces fourth-order equations and is excluded by the requirement of no unphysical (ghost) degrees of freedom. And the cosmological constant Λ is *not* excluded by any of this — it is the unique cosmological term, which is why it sits inside S_{EH} above rather than in $\mathcal{F}$.

Diffeomorphism invariance is the gauge symmetry of gravity: physics cannot depend on which coordinate system you use to describe spacetime.

Equation of Motion

Varying S_{EH} with respect to $g^{\mu\nu}$ gives the **Einstein field equations**:

$$\boxed{G_{\mu\nu} + \Lambda g_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R + \Lambda g_{\mu\nu} = 8\pi G\, T_{\mu\nu}}$$

$G_{\mu\nu}$ is the **Einstein tensor** — geometric information about curvature. $T_{\mu\nu}$ is the **stress-energy tensor** — energy, momentum, and stress content of matter and fields. The $\Lambda g_{\mu\nu}$ term is the vacuum energy contribution. This equation says:

> **"Spacetime curvature = (constant) × energy-momentum content + vacuum energy"**

Setting $\Lambda = 0$ recovers the vacuum form $R_{\mu\nu} = 0$ of §0.3 — which, as noted above, constrains only the Ricci tensor, not the full Riemann tensor.

Limits and Descendants

Limit	Result	Appears in
$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, $h \ll 1$	Linearized gravity	Ch. 8
Further: $v \ll c$, non-relativistic matter	Poisson equation $\nabla^2\Phi = 4\pi G\rho$	Ch. 8
Integrate Poisson: point source	$F = GMm/r^2$ (Newton's law)	Ch. 8
Propagating perturbations of $h_{\mu\nu}$	Gravitational waves	Ch. 8

If Removed

Without this term, spacetime is a fixed, flat background — special relativity but not general relativity. Gravity would have to be added as a force by hand (Newton's way), contradicting every precision gravitational test above $v/c \sim 10^{-5}$.

Engineering Thread

→ Ch. 8 (Newtonian gravity) → Ch. 20 (GPS timing corrections)

0.4 — The Forces: Gauge Kinetic Terms

0.4.1 — General Structure

Every fundamental force in the Standard Model is a **gauge force**: it exists because the action must be invariant under a local (spacetime-dependent) symmetry transformation. The kinetic energy of the gauge field — the "cost" of having the force field vary in space and time — appears as:

$$\mathcal{L}_{gauge} = -\frac{1}{4}F^a_{\mu\nu}F^{a\mu\nu}$$

where $F^a_{\mu\nu}$ is the **field strength tensor** — the gauge-invariant measure of how the gauge field varies. The factor $-\frac{1}{4}$ is a convention that gives the correct normalization.

The field strength is built from the **gauge potential** A^a_μ:

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g\, f^{abc} A^b_\mu A^c_\nu$$

- The first two terms: the "curl" of the gauge potential (as in EM: $\mathbf{B} = \nabla \times \mathbf{A}$)
- The last term: **only present for non-Abelian gauge groups** (SU(2), SU(3)). f^{abc} are the **structure constants** of the group. This term means the force field can interact with itself — gluons interact with gluons; W bosons interact with each other. Photons do not (QED is Abelian).

0.4.2 — The $U(1)_Y$ Term: Hypercharge / Proto-Electromagnetism

$$\mathcal{L}_B = -\frac{1}{4} B_{\mu\nu} B^{\mu\nu}$$

B_μ — the **hypercharge gauge field**. A single four-vector field (one real component per spacetime direction). Its field strength:

$$B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu$$

This is the same structure as the electromagnetic field tensor $F_{\mu\nu}$ — because after the Higgs mechanism (§0.7), B_μ mixes with W^3_μ to produce the physical photon A_μ and the Z boson.

Symmetry: Invariance under $\psi \to e^{ig_1 Y \alpha(x)} \psi$ where Y is the **hypercharge** of the field and $\alpha(x)$ is a local phase. This is a U(1) group — just like rotating complex numbers by an angle.

Gauge boson: After symmetry breaking, B_μ and W^3_μ mix:

$$A_\mu = B_\mu \cos\theta_W + W^3_\mu \sin\theta_W \quad \text{(photon)}$$
$$Z_\mu = -B_\mu \sin\theta_W + W^3_\mu \cos\theta_W \quad \text{(Z boson)}$$

where $\theta_W \approx 28.7^\circ$ is the **Weinberg angle** — measured experimentally.

Physical consequence: Together with SU(2), this term is responsible for electromagnetism and the weak force — after the Higgs mechanism disentangles them.

0.4.3 — The $SU(2)_L$ Term: Weak Isospin

$$\mathcal{L}_W = -\frac{1}{4} W^a_{\mu\nu} W^{a\mu\nu}, \quad a = 1, 2, 3$$

W^a_μ — three gauge fields (a = 1, 2, 3), one for each generator of SU(2). Each is a four-vector. Their field strength:

$$W^a_{\mu\nu} = \partial_\mu W^a_\nu - \partial_\nu W^a_\mu + g_2\, \epsilon^{abc} W^b_\mu W^c_\nu$$

The $\epsilon^{abc} W^b W^c$ term: the W bosons carry weak charge themselves, so they interact with each other. This self-interaction makes the weak force structurally different from electromagnetism.

Symmetry: SU(2) — 2×2 unitary matrices with determinant 1. The subscript L means this symmetry acts **only on left-handed fermions** (§0.5.3). This is the only force in the Standard Model that violates parity — left and right are not the same under the weak force.

Gauge bosons: After Higgs mechanism:

- W_μ^1 and W_μ^2 combine into the charged $W^\pm$ bosons (mass ~80 GeV)
- W_μ^3 mixes with B_μ to form Z^0 (~91 GeV) and the photon

Physical consequences:

- Radioactive β-decay: $n \to p + e^- + \bar{\nu}_e$ (mediated by W^-)
- Nuclear fusion in stars (crucial for solar energy)
- Neutrino interactions (neutrinos feel only this force, plus gravity)

If removed: No radioactivity. No nuclear burning in stars. No nuclear synthesis of heavy elements. No neutrino interactions.

0.4.4 — The SU(3)$_C$ Term: Color / Strong Force (QCD)

$$\mathcal{L}_G = -\frac{1}{4} G^a_{\mu\nu} G^{a\mu\nu}, \quad a = 1, \ldots, 8$$

G^a_μ — **eight gluon fields**, one for each generator of SU(3). SU(3) is the group of 3×3 unitary matrices with determinant 1; it has 8 independent generators (the Gell-Mann matrices λ^a). Their field strength:

$$G^a_{\mu\nu} = \partial_\mu G^a_\nu - \partial_\nu G^a_\mu + g_3 \, f^{abc} G^b_\mu G^c_\nu$$

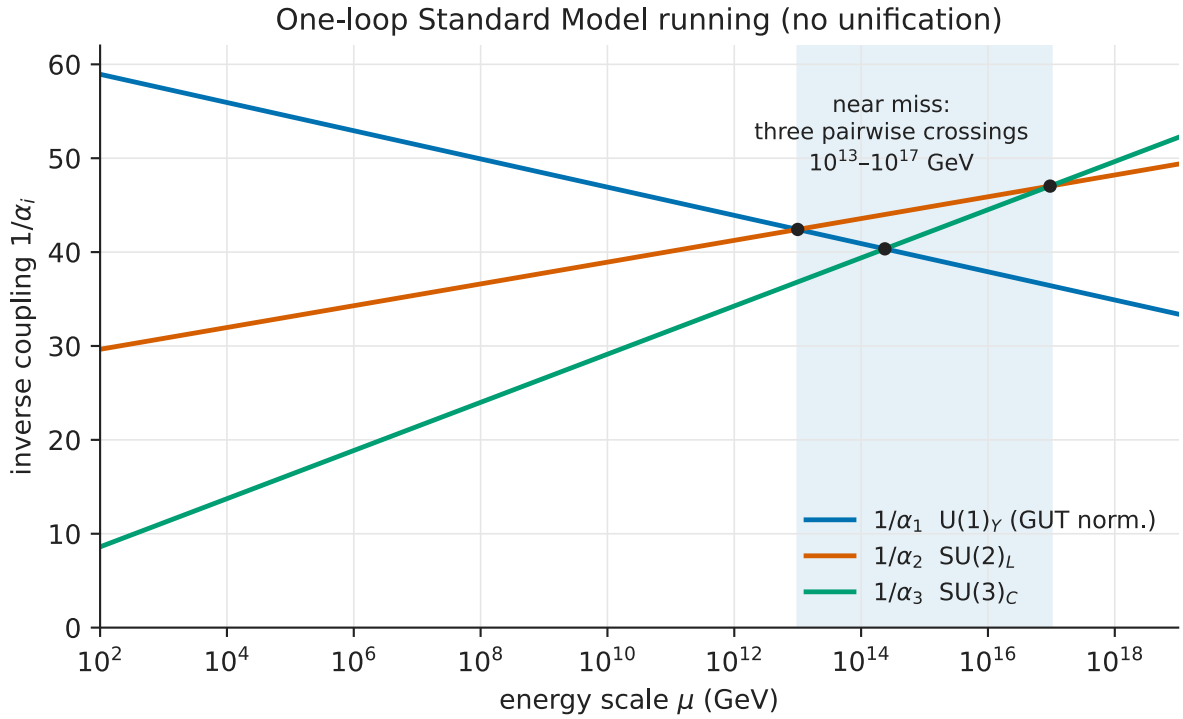
f^{abc} are the SU(3) structure constants — completely determined by the algebra of the Gell-Mann matrices.

Symmetry: SU(3)$_C$ — the subscript C stands for "color." Quarks come in three "colors" (red, green, blue — labels, not actual colors). The SU(3) symmetry rotates these color labels. Gluons carry color charge themselves, which is why the $f^{abc} G^b G^c$ self-interaction is large.

Physical consequences:

- **Confinement:** The strong force grows with distance (unlike EM, which weakens). No isolated quark has ever been observed; they are permanently bound into color-neutral hadrons (protons, neutrons, pions).
- **Asymptotic freedom:** At very short distances (high energy), the coupling g_3 *decreases* — quarks inside a proton behave almost freely. This is why perturbation theory works at high energy in QCD.
- **Nuclear binding:** The residual strong force between color-neutral protons and neutrons (mediated by pion exchange — an effective, not fundamental, description) holds nuclei together.

Coupling constant: g_3 (or equivalently $\alpha_s = g_3^2/4\pi \approx 0.12$ at the Z boson mass scale) is the largest coupling constant of all three forces at everyday energies. The strong force is literally strong.



One-loop Standard Model running $1/\alpha_i(\mu) = 1/\alpha_i(M_Z) - (b_i/2\pi)\ln(\mu/M_Z)$ *with GUT-normalised* $b_i = (41/10,\ -19/6,\ -7)$ *and standard inputs* $1/\alpha_i(M_Z) \approx 59.0,\ 29.6,\ 8.5$ *(not taken from the text):* $1/\alpha_3$ *rises with energy (asymptotic freedom), and the three lines cross pairwise between about* 10^{13} *and* 10^{17} *GeV — a near miss, not a single unification point.*

If removed: No protons, no neutrons, no nuclei, no atoms. No chemistry, no materials, no engineering.

Engineering thread: → Ch. 6 (nuclear binding energy, fission, fusion)

0.5 — Matter: The Fermion Term $i\bar{\psi}_f\gamma^\mu D_\mu\psi_f$

The Term

$$\mathcal{L}_{fermion} = \sum_f i\bar{\psi}_f\gamma^\mu D_\mu\psi_f$$

The sum runs over all fermion fields f in the Standard Model. This is the kinetic energy and gauge-force coupling of all matter particles.

The Objects

ψ_f — a **Dirac spinor field** for each fermion species f. A Dirac spinor has four complex components — not because we need four numbers to specify position in spacetime, but because the Lorentz group acts on spinors in a four-dimensional representation that combines spin-up, spin-down, particle, and antiparticle into one object. This will be derived in Bridge A (Ch. 3).

$\bar{\psi}_f = \psi_f^\dagger\gamma^0$ — the **Dirac adjoint**, the appropriate "conjugate" of ψ that makes bilinear expressions Lorentz scalars.

γ^μ — the four **Dirac gamma matrices** ($\mu = 0, 1, 2, 3$), each a 4×4 matrix satisfying the **Clifford algebra**:

$$\{\gamma^\mu, \gamma^\nu\} \equiv \gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = -2g^{\mu\nu}\mathbf{1}_{4\times4}$$

The minus sign goes with this book's signature $g = \mathrm{diag}(-1,+1,+1,+1)$ (§0.2.2): it gives $(\gamma^0)^2 = +\mathbf{1}$ and $(\gamma^i)^2 = -\mathbf{1}$, the same matrices as the particle-physics convention $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$ with $\eta = \mathrm{diag}(+1,-1,-1,-1)$, so that the Dirac equation below squares to the Klein–Gordon equation with the correct sign of m^2. This relation is the algebraic encoding of the Lorentz metric into the spinor structure. The gamma matrices make $\bar{\psi}\gamma^\mu\partial_\mu\psi$ a Lorentz scalar — ensuring the action does not depend on the observer's frame.

D_μ — the **gauge covariant derivative**, which is the ordinary partial derivative ∂_μ plus terms that couple the fermion to each gauge field according to the fermion's charges:

$$D_\mu = \partial_\mu - ig_1 Y B_\mu - ig_2\frac{\tau^a}{2}W_\mu^a - ig_3\frac{\lambda^a}{2}G_\mu^a$$

- g_1, g_2, g_3: the three gauge coupling constants (measured, not predicted)
- Y: hypercharge of the specific fermion
- τ^a: SU(2) generators (2×2 matrices; Pauli matrices divided by 2)
- λ^a: SU(3) generators (3×3 Gell-Mann matrices divided by 2)

Only the pieces that act on a given fermion's representation are included. For example, right-handed electrons have no SU(2) coupling — the τ^a term drops out completely.

0.5.1 — The Complete Fermion Content

The Standard Model has exactly **three generations** of matter, each identical in structure but differing in mass (via Yukawa couplings, §0.8). Each generation contains:

Field	Symbol	SU(3)	SU(2)	Y	Particles
Left-handed quark doublet	Q_L^i	**3**	**2**	+1/6	(u_L, d_L), (c_L, s_L), (t_L, b_L)
Right-handed up quark	u_R^i	**3**	**1**	+2/3	u_R, c_R, t_R
Right-handed down quark	d_R^i	**3**	**1**	−1/3	d_R, s_R, b_R
Left-handed lepton doublet	L_L^i	**1**	**2**	−1/2	(ν_{eL}, e_L), $(\nu_{\mu L}, \mu_L)$, $(\nu_{\tau L}, \tau_L)$
Right-handed charged lepton	e_R^i	**1**	**1**	−1	e_R, μ_R, τ_R

Notation: **3** in SU(3) column means it transforms as a color triplet (a quark); **1** means it is colorless (a lepton). **2** in SU(2) means it is a weak doublet (left-handed); **1** means it is a singlet (right-handed). $i = 1, 2, 3$ labels the generation.

Written out in full, the fermion Lagrangian is:

$$\mathcal{L}_{fermion} = \sum_{i=1}^{3}\Big[i\bar{Q}_L^i\gamma^\mu D_\mu Q_L^i + i\bar{u}_R^i\gamma^\mu D_\mu u_R^i + i\bar{d}_R^i\gamma^\mu D_\mu d_R^i + i\bar{L}_L^i\gamma^\mu D_\mu L_L^i + i\bar{e}_R^i\gamma^\mu D_\mu e_R^i\Big]$$

0.5.2 — Chirality: Why Left and Right Are Different

The **chiral projection operators**:

$$P_L = \frac{1-\gamma^5}{2}, \qquad P_R = \frac{1+\gamma^5}{2}$$

where $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ is the fifth gamma matrix.

For a massless fermion: $\psi_L = P_L\psi$ and $\psi_R = P_R\psi$ are independent — left-handed and right-handed fields decouple completely. The crucial asymmetry of the Standard Model is that SU(2) acts **only on** ψ_L — left-handed fields are doublets, right-handed fields are singlets. This is **parity violation** built directly into the structure of the action.

Experimental confirmation: Wu's experiment (1956) showed beta decay violates parity. The cobalt-60 nucleus emitted electrons preferentially opposite to its spin — a right-handed universe would give the opposite result.

0.5.3 — Equation of Motion

Varying the action with respect to $\bar{\psi}_f$:

$$\boxed{i\gamma^\mu D_\mu \psi_f = m_f \psi_f}$$

This is the **Dirac equation** for a massive fermion in a gauge field. For $m = 0$ and no gauge fields ($D_\mu = \partial_\mu$), it reduces to the free massless Dirac equation. In the non-relativistic limit, it becomes the Pauli equation, then the Schrödinger equation (Bridge A, Ch. 3).

0.5.4 — The Antiparticles

The four-component Dirac spinor ψ automatically contains both particle and antiparticle solutions. Negative-energy solutions of the Dirac equation are reinterpreted as positive-energy antiparticles (Dirac sea / Feynman-Stückelberg interpretation). This is a prediction of the formalism, not an assumption.

Particle	Antiparticle
Electron e^-	Positron e^+
Up quark u	Anti-up quark $\bar{u}$
Neutrino ν_e	Antineutrino $\bar{\nu}_e$

If Removed

Without the fermion term, there are no matter particles — no electrons, quarks, or neutrinos. There is only a universe of self-interacting gauge bosons. No atoms. No chemistry. No engineers.

Engineering thread: → Bridge A (Ch. 3–4): Dirac→Schrödinger→atomic structure→band theory→every material property in every branch

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0.6 — The Mass Term: Why There Isn't One

You might expect a mass term $-m\bar{\psi}\psi$ in the action. It is not there as a standalone term. Here is why.

A Dirac mass term can be rewritten as:

$$m\bar{\psi}\psi = m(\bar{\psi}_R\psi_L + \bar{\psi}_L\psi_R)$$

This **mixes left-handed and right-handed fields**. But ψ_L transforms as an SU(2) doublet and ψ_R as an SU(2) singlet. The combination $\bar{\psi}_R\psi_L$ is **not invariant under SU(2) gauge transformations** — it would break gauge invariance, which would make the weak force non-renormalizable and the theory inconsistent.

Therefore: **bare mass terms for quarks and charged leptons are forbidden by gauge invariance.** Masses must come from a different mechanism — the Yukawa coupling to the Higgs field (§0.8), which generates an effective mass after spontaneous symmetry breaking.

The one exception worth noting: **Majorana mass terms** $m\bar{\psi}^c\psi$ (where ψ^c is the charge conjugate) are gauge-invariant for gauge-singlet fermions. This is the dominant hypothesis for neutrino masses, which is part of why neutrino masses point toward physics beyond the Standard Model (part of $\mathcal{F}$, §0.10).

0.7 — Mass from Symmetry Breaking: The Higgs Sector $|D_\mu H|^2 - V(H)$

The Term

$$\mathcal{L}_{Higgs} = (D_\mu H)^\dagger (D^\mu H) - V(H)$$
$$V(H) = -\mu^2 (H^\dagger H) + \lambda (H^\dagger H)^2$$

The Objects

H — the **Higgs field**, a complex scalar SU(2) doublet:

$$H = \begin{pmatrix} H^+ \\ H^0 \end{pmatrix}$$

It has four real degrees of freedom. H^+ is the charged component; H^0 is the neutral component. The superscripts indicate electric charge. The Higgs carries hypercharge $Y = +\frac{1}{2}$.

μ^2 — a mass-squared parameter. Despite the notation, $\mu^2 > 0$ in the Standard Model — the negative sign in front of it is what creates the Mexican hat shape. (Do not confuse this with the muon μ.)

λ — the Higgs self-coupling constant. Must be positive ($\lambda > 0$) for the potential to be bounded below (stable vacuum). Measured value: $\lambda \approx 0.13$.

$D_\mu H$ — the gauge-covariant derivative acting on H:

$$D_\mu H = \left(\partial_\mu - ig_2 \frac{\tau^a}{2} W_\mu^a - ig_1 \frac{1}{2} B_\mu \right) H$$

The $|D_\mu H|^2$ term is what gives the W and Z bosons their masses after symmetry breaking — when $\langle H \rangle \neq 0$, the $(D_\mu H)^\dagger (D^\mu H)$ term generates mass terms for W_μ^a and B_μ directly.

The Potential in Detail

$$V(H) = -\mu^2 |H|^2 + \lambda |H|^4$$

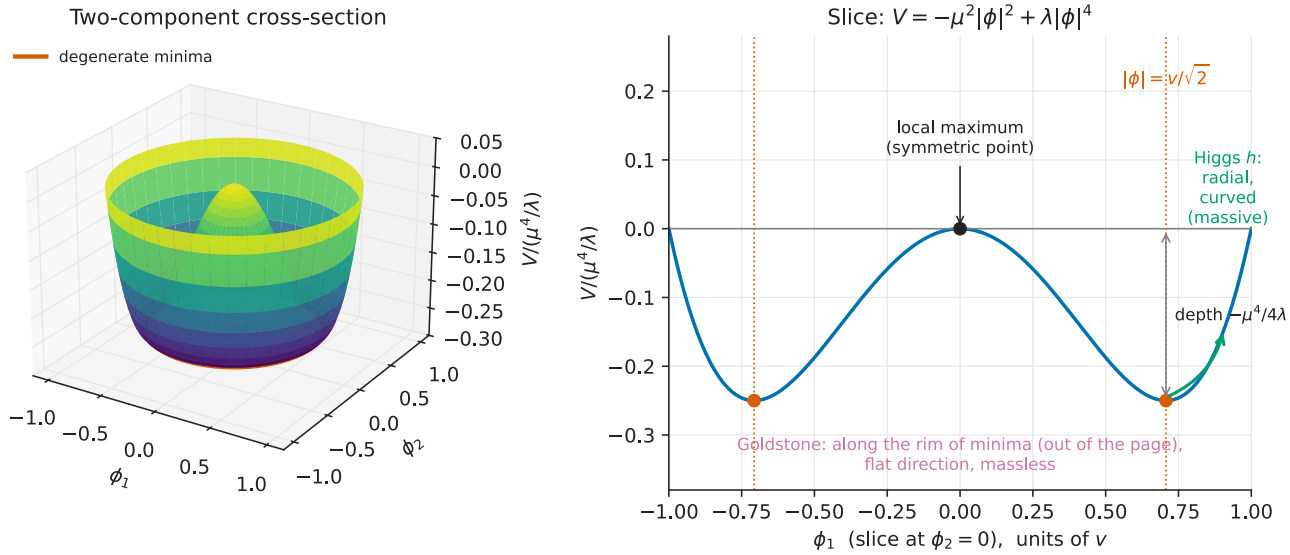
Region	Shape	Meaning
$\|H\| = 0$	Origin of the potential — a local maximum	Symmetry unbroken; the field sits at the centre

Region	Shape	Meaning
$\lvert H\rvert = v/\sqrt{2}$	Degenerate manifold of minima in field space (the "Mexican hat"); after gauge fixing, a single representative is chosen	Spontaneous symmetry breaking; the vacuum settles into the degenerate set

The minimum occurs at:

$$|H|_{\min} = \frac{v}{\sqrt{2}}, \quad v = \sqrt{\frac{\mu^2}{\lambda}} \approx 246\,\text{GeV}$$

v is the **vacuum expectation value (VEV)** of the Higgs field. In "empty" space, the Higgs field is not zero — it sits in the degenerate set of vacuum minima at $|H| = v/\sqrt{2}$. This nonzero background value permeates all of spacetime.



The potential $V = -\mu^2|\phi|^2 + \lambda|\phi|^4$ in dimensionless units ($\mu = \lambda = 1$) for a two-component field — a cross-section only, since the true minimum manifold of the four-component Higgs doublet is S^3, not a circle: the origin is a local maximum, the minima sit at $|\phi| = v/\sqrt{2}$ with depth $-\mu^4/4\lambda$, the radial (Higgs) direction is curved and the direction along the rim (Goldstone) is flat.

> **Warning: "Circle of degenerate minima" is a picture, not the geometry** The familiar Mexican-hat image shows a **circle of degenerate minima**. Strictly, H is a complex $SU(2)$ doublet, i.e. **four real components**, and the minimum condition is
>
> $$H^\dagger H = \frac{v^2}{2}$$
>
> In four-dimensional real field space this level set is an S^3, not a circle S^1. The 2D circle picture is the correct picture for a *complex scalar* (two real components), and it is what almost every textbook draws. It is a legitimate pedagogical representation of the *pattern* — a flat degenerate manifold of minima — but it is not the literal geometry, and the distinction matters as soon as one asks how many Goldstone directions there are. (There are three. The manifold itself gives $\dim S^3 = 3$, and the group structure agrees: $\dim\left[SU(2)_L \times U(1)_Y\right] - \dim U(1)_{EM} = (3+1) - 1 = 3$ broken generators. The naive S^1 picture would wrongly suggest one.)

> **A second subtlety: this is a gauge symmetry.** Calling $SU(2)_L \times U(1)_Y \to U(1)_{EM}$ "spontaneous symmetry breaking" is the standard and useful shorthand, and this book uses it. But the electroweak symmetry is a *gauge redundancy*, not a physical global symmetry, so strictly the gauge-fixed vacuum is a **choice of representative**, not a selection of a preferred physical state. The physical content is the pattern of symmetry of the vacuum and the masses it produces; the particular point chosen is a gauge convention. The conventional statement is retained throughout, with that understood.

Spontaneous Symmetry Breaking

The vacuum state breaks $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ down to $\mathrm{U}(1)_{EM}$. We choose:

$$\langle H \rangle = \begin{pmatrix} 0 \\ v/\sqrt{2} \end{pmatrix}$$

(Any point in the degenerate set of minima is equivalent; we fix one representative by gauge choice.)

Writing $H = \langle H \rangle +$ perturbations, four degrees of freedom split:

- **Three Goldstone bosons** — would be massless, but are absorbed ("eaten") by $W^{\pm}$ and Z^0 as their longitudinal polarizations, giving them mass
- **One physical Higgs boson** h — the remaining real scalar, observed at CERN in 2012 with mass $m_h = 125.09$ GeV

The Physical Masses from $|D_\mu H|^2$

Inserting $\langle H \rangle$ into $(D_\mu H)^\dagger (D^\mu H)$:

$$m_W = \frac{1}{2} g_2 v \approx 80.4\,\mathrm{GeV}$$

$$m_Z = \frac{v}{2}\sqrt{g_1^2 + g_2^2} \approx 91.2\,\mathrm{GeV}$$

$$m_\gamma = 0 \quad \text{(photon remains exactly massless)}$$

The photon is massless because the unbroken $\mathrm{U}(1)_{EM}$ symmetry forbids a photon mass term — and this exact masslessness is why the electromagnetic force has infinite range.

If Removed

Without the Higgs term, all fermions and the W, Z bosons are exactly massless. The $\mathrm{SU}(2) \times \mathrm{U}(1)$ symmetry is unbroken. The weak force has infinite range like electromagnetism. Atoms as we know them cannot exist. No chemistry.

Engineering thread: → Ch. 6 (atomic masses) → Ch. 11 (phase transitions — same mathematical structure) → Ch. 7 (superconductivity — same broken-symmetry structure at lower energy) → all of materials science

---

0.8 — Specific Masses: Yukawa Couplings $Y_{ij}\bar{\psi}_i H \psi_j$

The Term

$$\mathcal{L}_{Yukawa} = -\left(Y^u_{ij}\bar{Q}^i_L \tilde{H} u^j_R + Y^d_{ij}\bar{Q}^i_L H\, d^j_R + Y^e_{ij}\bar{L}^i_L H\, e^j_R\right) + \text{h.c.}$$

where $\tilde{H} = i\sigma^2 H^*$ is the **charge-conjugated Higgs doublet** (has hypercharge $Y = -\frac{1}{2}$, needed for up-type quarks).

The Objects

$Y^u_{ij}, Y^d_{ij}, Y^e_{ij}$ — the **Yukawa coupling matrices**, each a 3×3 complex matrix in generation space ($i\,j = 1, 2, 3$). These are dimensionless numbers — measured, not predicted.

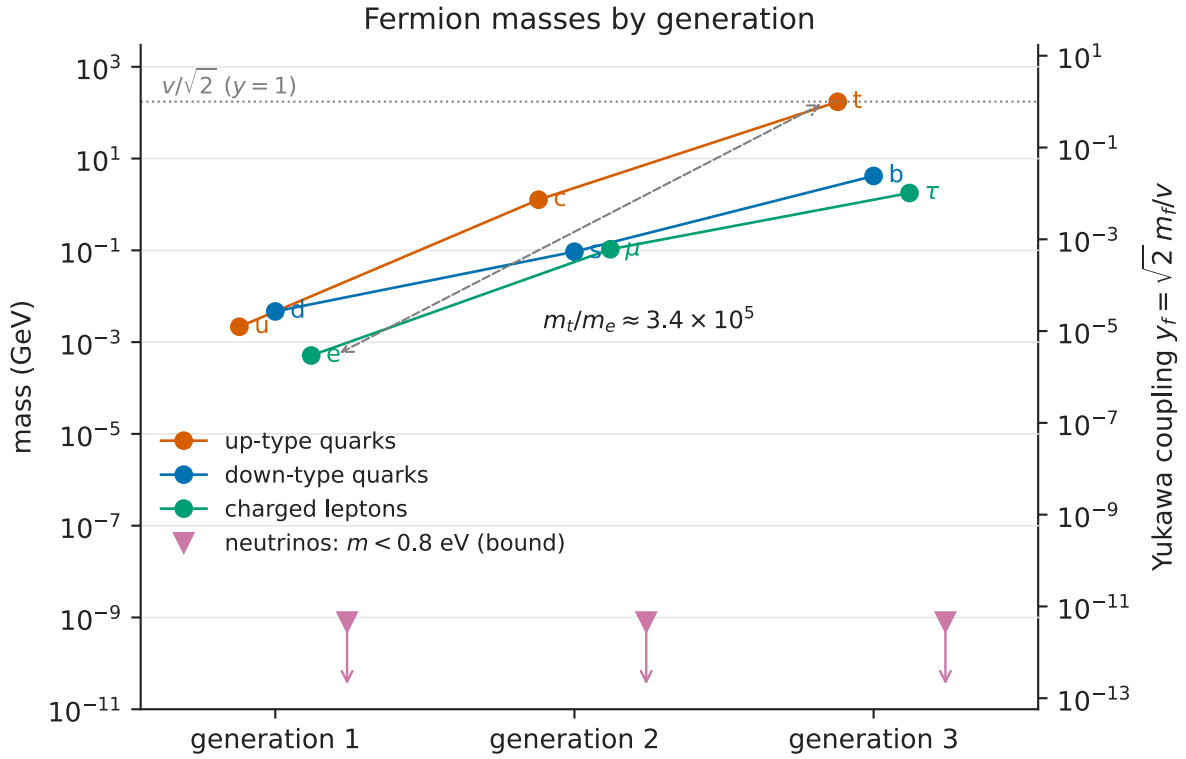
After symmetry breaking, insert $\langle H \rangle = (0,\, v/\sqrt{2})^T$:

$$\mathcal{L}_{Yukawa} \supset -\frac{v}{\sqrt{2}}\left(Y^u_{ij}\bar{u}^i_L u^j_R + Y^d_{ij}\bar{d}^i_L d^j_R + Y^e_{ij}\bar{e}^i_L e^j_R\right) + \text{h.c.}$$

Comparing with the Dirac mass term $m\bar{\psi}\psi$:

$$m_f = Y_f \cdot \frac{v}{\sqrt{2}}$$

Every fermion mass is the Yukawa coupling times the Higgs VEV. The Higgs mechanism gives the framework for mass; the Yukawa coupling provides the specific value.



Masses of the up-type quarks, down-type quarks and charged leptons by generation (PDG central values), with the equivalent Yukawa coupling $y_f = \sqrt{2}\, m_f/v$ ($v = 246$ GeV) on the right axis and neutrinos shown only as the direct bound $m < 0.8$ eV: only the top quark has $y \approx 1$, and $m_t/m_e \approx 3.4 \times 10^5$ is unexplained by the Lagrangian.

The CKM Matrix and Generation Mixing

The Yukawa matrices Y^u_{ij} and Y^d_{ij} are generally not diagonal. Diagonalizing them by unitary rotations in generation space:

- Produces the physical quark mass eigenstates
- The mismatch between the up-type and down-type rotation matrices gives the **Cabibbo-Kobayashi-Maskawa (CKM) matrix** V_{CKM}, a 3×3 unitary matrix with one complex phase

This complex phase is the source of **CP violation** in the quark sector — the slight asymmetry between matter and antimatter in certain decays. (It is insufficient alone to explain the matter-antimatter asymmetry of the observable universe — another entry in the $\mathcal{F}$ placeholder.)

For **leptons**, a corresponding **PMNS matrix** U_{PMNS} describes neutrino mixing — the phenomenon of neutrino oscillations (see §0.10). Two precision points, consistent with the parameter count in §0.12: this is a matrix over *neutrino* weak eigenstates and mass eigenstates, not over charged leptons (charged-lepton mixing is unphysical and rotated away). And it exists **only if neutrinos have masses** — in the minimal massless-neutrino Standard Model there is no PMNS matrix, because massless neutrinos of different flavour do not oscillate. Its appearance is therefore direct evidence that the minimal SM needs extension.

What Yukawa Couplings Do Not Explain

Question	Status
Why is $m_e = 0.511$ MeV?	Y_e is measured, not derived
Why is $m_t/m_e \approx 3.5 \times 10^5$?	Unknown — no theory predicts the hierarchy
Why exactly three generations?	Unknown

These are genuine open questions. The $\mathcal{F}$ term is expected to eventually provide a mechanism that predicts the Yukawa couplings from deeper principles.

Engineering thread: → Ch. 6 (specific atomic masses that determine chemical properties and nuclear binding)

0.9 — Topology Already in the Action: The QCD θ-Term

The Term

$$\mathcal{L}_\theta = \frac{\theta, g_3^2}{32\pi^2} G^a_{\mu\nu} \tilde{G}^{a\mu\nu}$$

where $\tilde{G}^{a\mu\nu} = \frac{1}{2}\varepsilon^{\mu\nu\rho\sigma} G^a_{\rho\sigma}$ is the **dual field strength** of the gluon field ($\varepsilon^{\mu\nu\rho\sigma}$ is the completely antisymmetric Levi-Civita tensor in four dimensions).

What This Term Is

$G^a_{\mu\nu}\tilde{G}^{a\mu\nu}$ is the **QCD Pontryagin density** — a Lorentz scalar built from the gluon field that is a **total derivative**:

$$G^a_{\mu\nu}\tilde{G}^{a\mu\nu} = \partial_\mu K^\mu$$

where K^μ is the Chern-Simons current. Since it is a total derivative, it does not contribute to the classical equations of motion. Integrating it over a closed spacetime region gives a **topological invariant** — the winding number n — which is always an integer and does not change under smooth deformations of the gauge field.

Why It Matters Despite Not Affecting Classical EOM

In quantum mechanics, the path integral sums over *all* field configurations. Different topological sectors (different winding numbers) give separate contributions that interfere. The θ-term weights each sector by $e^{i\theta n}$, so its value matters for quantum processes even though $\delta S_\theta = 0$ classically.

Physical consequence: If $\theta \neq 0$, the strong force violates **CP symmetry** — the combined symmetry of charge conjugation (C) and parity (P). This would give the neutron a nonzero **electric dipole moment (EDM)**.

Experimental bound: $d_n < 1.8 \times 10^{-26}\, e \cdot \text{cm}$ (measured). This constrains:

$$|\theta| < 10^{-10}$$

The puzzle: why is θ so small? Nothing in the theory forces it to be zero. This is the **strong CP problem** — one of the major open questions in physics. A leading solution is the **Peccei-Quinn mechanism**, which promotes θ to a dynamical field (the **axion**) that relaxes to zero dynamically. The axion is a candidate for dark matter — and therefore a candidate for part of $\mathcal{F}$.

Engineering Connection

Topology is already inside everyday condensed matter devices. The Berry phase (a geometric/topological phase in quantum mechanics) and the Chern number (a topological invariant of electron band structures) directly determine the conductance of quantum Hall devices and topological insulators. These are Layer-1 manifestations of the same mathematical topology that appears here in the QCD θ-term.

Engineering thread: → Ch. 7 (Berry phase, quantum Hall effect, topological insulators)

0.10 — The Honest Placeholder: S_{unknown}

The Canonical Notation

This book uses **one** schematic form of the action, stated here once and used everywhere else:

$$S_{\text{eff}} = S_{EH} + S_{SM} + S_{\text{unknown}}$$

$$S_{EH} \equiv \int d^4x\,\sqrt{-g}\,\frac{R}{16\pi G}, \qquad S_{SM} \equiv \int d^4x\,\sqrt{-g}\,\mathcal{L}_{SM}, \qquad \boxed{S_{\text{unknown}} \equiv \int d^4x\,\sqrt{-g}\,\mathcal{F}[g,\,\phi_?,\,\text{new fields},}$$

Why this form and not $f(\phi)R$. Earlier drafts of this material wrote the unknown sector as $f(\phi)\,R$ — a *nonminimal coupling* multiplying the Ricci scalar. That was a mistake, and a consequential one: it silently asserted that whatever physics is missing must be a scalar function times gravity. Nothing supports that. The missing sector could be fermions, gauge fields, topological terms, curvature-squared corrections,

modified dynamics, or something with no precedent. Writing $\mathcal{F}[\ldots]$ as a **placeholder for an unknown sector**, rather than a specific functional form, keeps the claim exactly as strong as the evidence — no stronger.

So the single expanded form, used in the README, the Map, and the Epilogue, is:

$$\boxed{S_{\text{eff}} = \int d^4x\,\sqrt{-g}\left[\frac{R}{16\pi G} + \mathcal{L}_{SM} + \mathcal{F}[g,\,\phi_?,\,\text{new fields},\,\text{topology},\,\ldots]\right]}$$

Notation discipline. $\mathcal{F}[g,\ldots]$ with square brackets means "a yet-unspecified functional of these inputs." Write it as a functional, never as a specific product, and never drop it. The master action is *always* written with all three sectors present; abbreviating to $S_{EH} + S_{SM}$ without a statement about $\mathcal{F}$ is a silent claim that the unknown sector is empty.

What This Term Is

$\mathcal{F}$ is **explicitly unknown**. It is not a gap we are filling lazily — it is a gap that represents the genuine frontier of human knowledge. We include it in the action because the action principle is the framework within which any future physics will almost certainly be expressed.

What We Know About $\mathcal{F}$, Even Without Knowing $\mathcal{F}$

Structural constraints:

- $\mathcal{F}$ must be a Lorentz scalar (coordinate independence)
- $\mathcal{F}$ must be consistent with all symmetries not already broken at accessible energy scales
- $\mathcal{F}$ must vanish (or be negligible) in all regimes where the known action already agrees with experiment
- $\mathcal{F}$ must be diffeomorphism-covariant (couple consistently to gravity)

Required content:

Missing physics	Why $\mathcal{F}$ must contain it
Dark matter	27% of the universe's mass-energy is gravitationally confirmed but has no particle in $\mathcal{L}_{SM}$
Dark energy	68% of mass-energy; drives accelerating expansion. The minimal $\Lambda g_{\mu\nu}$ term is *already in* S_{EH} — what $\mathcal{F}$ must supply is an explanation of **why** $\Lambda \approx 10^{-52}\,\text{m}^{-2}$ (naturalness, see the Epilogue), or a dynamical alternative to a bare constant
Quantum gravity	The Einstein-Hilbert term is classical; at Planck scale ($E \sim 10^{19}$ GeV), $R/16\pi G$ must be replaced or corrected
Strong CP resolution	Why $\theta \approx 0$? Possibly an axion field $a(x)$ in $\mathcal{F}$
Neutrino masses	Measured nonzero (from oscillations); require Majorana or Dirac mass terms beyond minimal SM
Matter-antimatter asymmetry	The universe has matter; additional CP violation beyond the CKM phase is required
Gauge coupling unification	At $\sim 10^{16}$ GeV, the three coupling constants g_1, g_2, g_3 nearly converge — suggesting a larger symmetry group (GUT) in $\mathcal{F}$

Topological and Anomaly Terms in $\mathcal{F}$

Anomalies are quantum mechanical violations of classical symmetries. When you quantize a classically symmetric theory, some symmetries can be destroyed by quantum corrections (loop diagrams). This is catastrophic if it happens to a *gauge* symmetry — it makes the theory inconsistent (non-unitary).

Anomaly cancellation in the SM is a remarkable non-trivial fact: the anomalies from all the fermions in one generation cancel exactly. For example, the U(1) gauge anomaly requires:

$$\sum_{\text{fermions}} Y^3 = 0$$

> **Warning: Convention first: the sum runs over left-handed Weyl fermions** The $U(1)_Y^3$ anomaly is a sum over left-handed Weyl fields only. Right-handed fields are not counted separately — each is replaced by its left-handed charge conjugate, and that conjugation **reverses the hypercharge**, $Y \to -Y$. Since the anomaly is cubic, this sign flip is essential, and getting it wrong produces a plausible-looking sum that does not vanish. The hypercharges that actually enter the sum are therefore:

Field	Type	Y	Multiplicity
Q_L	left-handed	$+\frac{1}{6}$	6 (3 colours × 2 $SU(2)$ components)
u_R^c	left-handed charge conjugate of u_R	$-\frac{2}{3}$	3 (3 colours)
d_R^c	left-handed charge conjugate of d_R	$+\frac{1}{3}$	3 (3 colours)
L_L	left-handed	$-\frac{1}{2}$	2 ($SU(2)$ doublet)
e_R^c	left-handed charge conjugate of e_R	$+1$	1

Note especially that d_R^c carries $+\frac{1}{3}$ (not $-\frac{1}{3}$) and e_R^c carries $+1$ (not -1), because both are charge conjugates. With these values the sum is:

$$\mathcal{A}_{U(1)_Y^3} = 3\left[2\left(\tfrac{1}{6}\right)^3 + \left(-\tfrac{2}{3}\right)^3 + \left(\tfrac{1}{3}\right)^3\right] + \left[2\left(-\tfrac{1}{2}\right)^3 + (1)^3\right] = 0$$

The arithmetic checks term by term: the quark bracket is $\frac{1}{108} - \frac{8}{27} + \frac{1}{27} = -\frac{1}{4}$, and the lepton bracket is $-\frac{1}{4} + 1 = \frac{3}{4}$, so the total is $3\left(-\frac{1}{4}\right) + \frac{3}{4} = -\frac{3}{4} + \frac{3}{4} = 0$. ✓

This works **only because quarks come in 3 colors and have those specific hypercharges, and only when leptons are included.** Change the particle content and the anomaly does not cancel — the theory becomes inconsistent. This is a profound constraint: the structure of $\mathcal{F}$ must be anomaly-free. It is worth registering how strong this is: the cancellation is not approximate and not fitted. No free parameter is adjusted to achieve it, and it holds generation by generation.

Global anomalies (Witten anomaly, etc.) provide additional constraints on what $\mathcal{F}$ can contain.

Known topological extensions of $\mathcal{F}$:

- Higher-derivative gravity terms: R^2, $R_{\mu\nu}R^{\mu\nu}$, Gauss-Bonnet (arise naturally in string theory effective actions)
- Chern-Simons terms (important in 2+1 dimensions; describe quantum Hall effect effective field theory)

- Pontryagin density for gravity: $R_{\mu\nu\rho\sigma}\tilde{R}^{\mu\nu\rho\sigma}$

The Pedagogical Statement

> If a Theory of Everything exists, we have compelling reason to believe it will be expressible as:
>
> $$\delta\left(\int d^4x\,\sqrt{-g}\left[\frac{R}{16\pi G}+\mathcal{L}_{SM}+\mathcal{F}\right]\right)=0$$
>
> The known terms $R/16\pi G+\mathcal{L}_{SM}$ are not wrong — they are correct limiting cases that any future $\mathcal{F}$ must reproduce. The student who understands the known part completely understands what any future theory must reduce to.

0.11 — The Complete Equation: Everything Together

$$\boxed{S=\int d^4x\sqrt{-g}\left[\frac{R-2\Lambda}{16\pi G}+\mathcal{L}_{SM}+\mathcal{F}[\ldots]\right]}$$

with the Standard Model piece expanded as

$$\mathcal{L}_{SM}=-\frac{1}{4}B_{\mu\nu}B^{\mu\nu}-\frac{1}{4}W^a_{\mu\nu}W^{a\mu\nu}-\frac{1}{4}G^a_{\mu\nu}G^{a\mu\nu}+\sum_f i\bar{\psi}_f\gamma^\mu D_\mu\psi_f+|D_\mu H|^2-V(H)+Y_{ij}\bar{\psi}_i H\psi_j+\text{h.c.}+$$

Note the structure: the cosmological constant sits in the **known** gravitational sector alongside R, the Standard Model is known, and only $\mathcal{F}$ is explicitly unknown.

Varying with respect to each field gives:

Vary w.r.t.	Equation of motion	Name
$g^{\mu\nu}$	$G_{\mu\nu}+\Lambda g_{\mu\nu}=8\pi G\,T_{\mu\nu}$	Einstein field equations
B_μ	$\partial^\nu B_{\nu\mu}=g_1 j^\mu_Y$	Hypercharge Maxwell equation
W^a_μ	$D^\nu W^a_{\nu\mu}=g_2 j^{a\mu}_W$	Weak field equations
G^a_μ	$D^\nu G^a_{\nu\mu}=g_3 j^{a\mu}_C$	QCD field equations
$\bar{\psi}_f$	$i\gamma^\mu D_\mu\psi_f=m_f\psi_f$	Dirac equation (with Higgs-generated mass)
$H^\dagger$	$D^2H=\frac{\partial V}{\partial H^\dagger}+Y^f\psi_f$	Higgs field equation

Together, these equations encode the fundamental dynamics included in the GR + minimal-SM framework used in this book.

0.12 — The Counting: How Complex Is This Equation?

The minimal Standard Model, with neutrinos massless, has 19 free parameters. The count is:

Source	Count	Detail
Gauge couplings g_1, g_2, g_3	3	one per gauge group

Source	Count	Detail
Higgs potential μ^2, λ	2	the only free parameters in the scalar sector
Charged-fermion Yukawa eigenvalues	9	$3+3+3$ for up, down, and charged leptons
CKM matrix	4	3 mixing angles + 1 CP-violating phase
QCD θ	1	strong-CP angle
Total	**19**	$3+2+9+4+1=19$

There is no physical PMNS matrix in the minimal massless-neutrino Standard Model. Without neutrino masses there is nothing for neutrinos to mix *with*, so the PMNS matrix is not a free parameter of that model — including one in the count of 19 would be a bookkeeping error. This is exactly why neutrino masses are experimentally significant: they prove the minimal SM is incomplete.

Extending to Dirac neutrino masses adds 3 neutrino Yukawa eigenvalues, 3 additional mixing angles, and 1 Dirac CP-violating phase:

$$19+3+3+1=\boxed{26}$$

For **Majorana** neutrinos the count differs again, because Majorana phases add further physical parameters (two extra phases per generation for three generations, with one combination removable). The clean statement: **the minimal Standard Model with massless neutrinos has 19 free parameters; neutrino masses and mixing require an extension of the minimal SM and introduce additional parameters.**

A true Theory of Everything would predict these numbers. The fact that we cannot yet derive them is a measure of how much of $\mathcal{F}$ remains unknown.

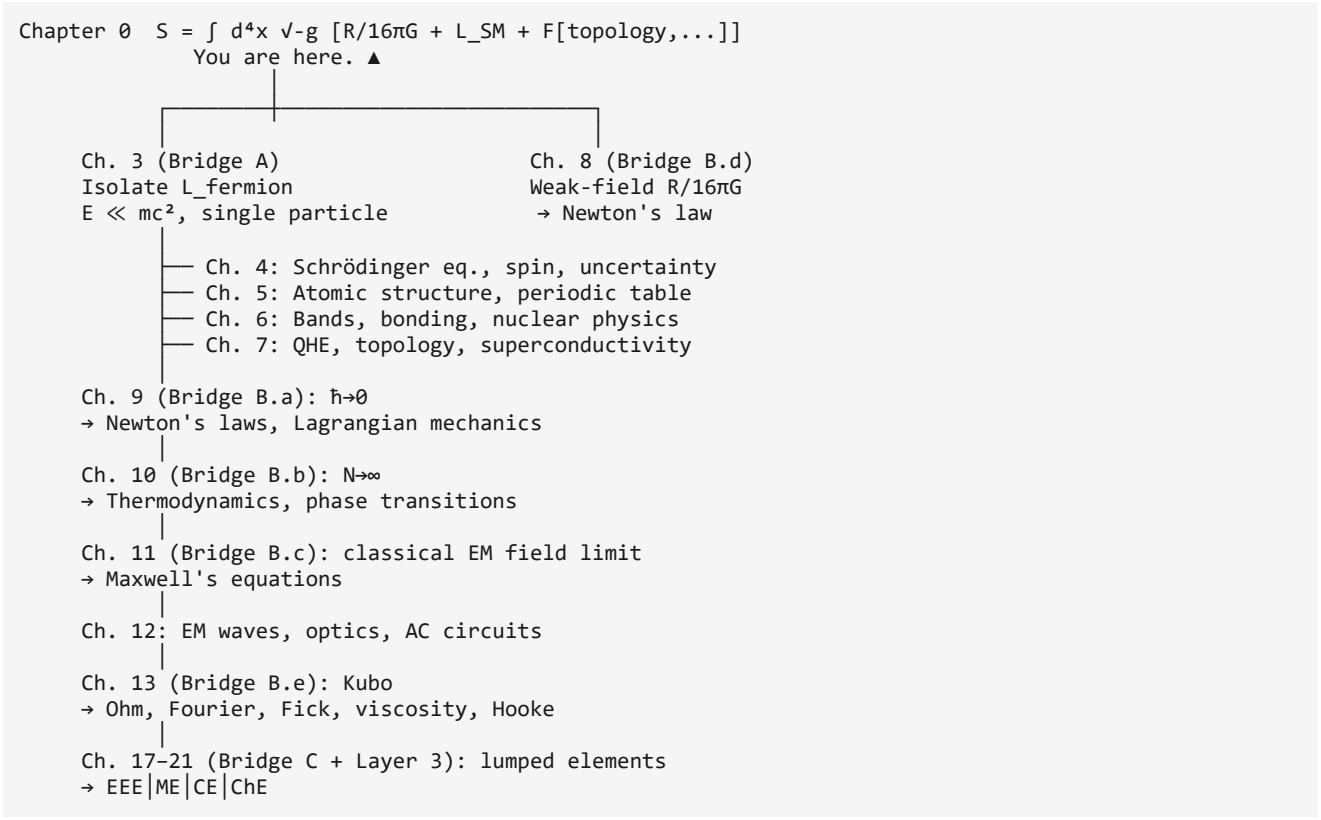
0.13 — The Symmetry Group: A Summary

$$G_{total}=\underbrace{\mathrm{Diff}(M^4)}_{\text{gravity/GR}}\times\underbrace{\mathrm{SU}(3)_C}_{\text{strong}}\times\underbrace{\mathrm{SU}(2)_L}_{\text{weak}}\times\underbrace{\mathrm{U}(1)_Y}_{\text{hypercharge}}$$

Group	Dimension	Force	Generators	Gauge bosons
$\mathrm{Diff}(M^4)$	∞-dim	Gravity	Diffeomorphisms	Graviton (unquantized)
$\mathrm{SU}(3)_C$	8	Strong	Gell-Mann matrices λ^a	8 gluons
$\mathrm{SU}(2)_L$	3	Weak	Pauli matrices τ^a	W^+, W^-, Z^0
$\mathrm{U}(1)_Y$	1	Hyper-charge → EM	Phase rotation	Photon γ

After the Higgs mechanism: $\mathrm{SU}(2)_L\times\mathrm{U}(1)_Y\to\mathrm{U}(1)_{EM}$, giving the physical photon, $W^\pm$, and Z^0.

0.14 — Where Each Term Goes in This Book



0.15 — One Last Note Before the Descent

This equation is not a decoration at the front of the book. Every derivation in every subsequent chapter will appeal to one or more terms in it.

When you derive Ohm's law in Chapter 13, you are working in the $U(1)$ gauge sector, with the linear-response (Kubo) formalism applied to the fermion kinetic term, in the limit of weak electric fields, finite temperature, and finite disorder.

When you write KCL at a circuit node in **Chapter 18**, you are using charge conservation. The conceptual chain has several links, and it is worth keeping them distinct rather than compressing them:

```
U(1) gauge structure
   → conserved charge current (Noether)
   → local continuity equation  ∂μjμ = 0
   → integral charge balance over a closed surface (divergence theorem)
   → lumped-node approximation (Ch. 17, Bridge C)
   → KCL:  Σ I = 0 at a node
```

Note what is *not* being claimed: KCL is not a direct application of the hypercharge symmetry $U(1)_Y$ to a circuit node. The electromagnetic $U(1)_{em}$ that couples to charge is a *combination* of $U(1)_Y$ and $SU(2)_L$ fixed by the Higgs vacuum (Ch. 7 §7.4), and $U(1)_Y$ by itself does not commute with $SU(2)_L$ after symmetry breaking. KCL is a **macroscopic conservation statement** — charge does not accumulate in a node faster than it leaves — obtained by integrating the continuity equation and then *approximating* the spatially distributed current by a single value per node. The gauge-symmetry reasoning establishes the conservation law; the lumping approximation turns it into a circuit equation. The two steps are separate, and the second is an approximation with a validity criterion (Ch. 17 §17.1.1).

The equation is the map. Every chapter is a destination on that map.

End of Chapter 0.

Next: Chapter 1 — The Principle: What the Action Means, and Why It Governs Everything

The Action Principle

Why Nature Extremizes, and Why That Explains Everything

> *The historical development of mechanics went from Newton to Lagrange to Hamilton. The logical development goes the other way.* — Cornelius Lanczos, *The Variational Principles of Mechanics*

1.0 — The Problem This Chapter Solves

In Chapter 0 you saw the equation:

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G} + \mathcal{L}_{SM} + \mathcal{F} \right]$$

and you were told: *vary this, set $\delta S = 0$, and you get all of physics.*

Two questions must be bothering you.

First: Why does writing $\delta S = 0$ give the equations of motion at all? What is the action S, and why should nature care whether it is stationary?

Second: Even if it works for one example, why does the same framework — varying an action — correctly describe gravity, electromagnetism, nuclear forces, and quantum mechanics? Why is this one mathematical idea so unreasonably effective?

This chapter answers both questions. By the end, you will understand the action principle not as a trick that happens to work, but as the only self-consistent language for writing down a physical law that is coordinate-independent, respects symmetry, and generalizes cleanly from particles to fields to quantum mechanics.

1.1 — What Is Wrong with Newton?

Newton's second law $\mathbf{F} = m\ddot{\mathbf{r}}$ works. It correctly predicts the motion of planets, projectiles, and charged particles. So what is wrong with it?

1.1.1 — It Is Coordinate-Dependent

Newton's law is written in Cartesian coordinates. Apply it in polar coordinates:

$$m\ddot{r} - mr\dot{\theta}^2 = F_r$$
$$mr\ddot{\theta} + 2m\dot{r}\dot{\theta} = F_\theta$$

These look nothing like $F = ma$. The "extra" terms ($mr\dot{\theta}^2$, $2m\dot{r}\dot{\theta}$) are not real forces — they are artifacts of the coordinate system (centrifugal and Coriolis terms). In Cartesian coordinates they do not appear. In polar coordinates they do. The law changes its form depending on your coordinate choice.

This is a serious deficiency. The laws of physics should not depend on the human decision of how to label points in space.

1.1.2 — It Requires All Forces to Be Known Explicitly

If a ball rolls on a surface, the surface exerts a normal force on the ball. Newton requires you to find this constraint force explicitly — but it does not do any work (it is perpendicular to the motion) and you may not even know its direction in advance.

Constraint forces are a computational nuisance. They appear on the right-hand side of Newton's equations, must be solved for, and then discarded. There should be a way to formulate mechanics that makes constraint forces invisible.

1.1.3 — It Does Not Reveal Symmetry

If a system is symmetric under rotation (a central-force problem), Newton's law does not make this transparent. You solve three coupled differential equations in x, y, z and then discover after the fact that angular momentum is conserved. The symmetry should have told you this *before* you did any calculation.

1.1.4 — It Does Not Generalize to Fields or Quantum Mechanics

$\mathbf{F} = m\ddot{\mathbf{r}}$ describes the motion of a point particle. The electromagnetic field, the Higgs field, and the wavefunction of an electron are not particles. They are fields — functions defined over all of spacetime. There is no direct generalization of $\mathbf{F} = m\ddot{\mathbf{r}}$ to fields.

The action principle has a clean generalization. That is why $\mathcal{L}_{SM}$ looks the way it does in Chapter 0.

1.2 — Generalized Coordinates and Configuration Space

1.2.1 — Degrees of Freedom

The first step is to count how many numbers you actually need to completely specify the state of a mechanical system, ignoring constraints.

System	Naive coordinates	Constraints	True DOF
Point particle in 3D	x, y, z	None	3
Pendulum (2D)	x, y	$x^2 + y^2 = L^2$	1
Double pendulum	$x_1\, y_1\, x_2\, y_2$	Two length constraints	2
Rigid body	$x\, y\, z\, \theta\, \phi\, \psi$	None (rigid body has 6 DOF)	6
N particles	$3N$ Cartesian	k constraints	$3N - k$

The number of true degrees of freedom is $n = 3N - k$. These are the **generalized coordinates** $q_1, q_2, \ldots, q_n$ — any n independent parameters that completely describe the configuration of the system.

1.2.2 — Configuration Space

The set of all possible values $(q_1, \ldots, q_n)$ forms an n-dimensional space called **configuration space** $\mathcal{Q}$. The motion of the system is a **curve** in configuration space:

$$\gamma : [t_1, t_2] \to \mathcal{Q}, \quad t \mapsto (q_1(t), \ldots, q_n(t))$$

Newton's law, reformulated, becomes: *find the curve γ that the system actually follows*. The Lagrangian approach gives a beautiful answer to this question.

1.2.3 — Generalized Velocities

The time derivatives $\dot{q}_i = dq_i/dt$ are the **generalized velocities**. Together, $(q_i, \dot{q}_i)$ completely specify the instantaneous state and its rate of change — the "position and velocity" in generalized language.

1.3 — The Lagrangian

1.3.1 — Definition

For a mechanical system with generalized coordinates q_i, the **Lagrangian** is:

$$L(q_i, \dot{q}_i, t) = T(q_i, \dot{q}_i) - V(q_i, t)$$

where T is the kinetic energy and V is the potential energy.

This combination $T - V$ (not $T + V$!) is what enters the variational principle. The reason for the minus sign will become clear in §1.4 — it is what makes the stationary path correspond to the actual physical motion.

1.3.2 — Examples

Free particle in 3D:

$$T = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2), \quad V = 0$$
$$L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2)$$

Harmonic oscillator:

$$L = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2$$

Simple pendulum (angle θ as generalized coordinate, length ℓ):

$$T = \frac{1}{2}m\ell^2\dot{\theta}^2, \quad V = -mg\ell\cos\theta$$
$$L = \frac{1}{2}m\ell^2\dot{\theta}^2 + mg\ell\cos\theta$$

The constraint force (string tension) does not appear anywhere. The coordinate θ automatically respects the constraint — this is the advantage of generalized coordinates.

Charged particle in electromagnetic field (recovering what Chapter 3 needs):

$$L = \frac{1}{2}m|\dot{\mathbf{r}}|^2 - eV(\mathbf{r}, t) + \frac{e}{c}\dot{\mathbf{r}} \cdot \mathbf{A}(\mathbf{r}, t)$$

The gauge potential $\mathbf{A}$ enters the Lagrangian directly — and the resulting equations of motion will give the Lorentz force law $\mathbf{F} = e(\mathbf{E} + \dot{\mathbf{r}}/c \times \mathbf{B})$. Verify: this is exactly the minimal coupling $\mathbf{p} \to \mathbf{p} - e\mathbf{A}/c$ seen in Chapter 3 (§3.9.4). The Lagrangian already knows about gauge coupling.

1.4 — Hamilton's Principle and the Action

1.4.1 — The Action

Given a curve γ in configuration space parameterized by time, the **action** is the real number:

$$S[\gamma] = \int_{t_1}^{t_2} L(q_i(t), \dot{q}_i(t), t)\, dt$$

The action is a **functional** — it assigns a number to each possible path, not to a point. The notation $S[\gamma]$ (square brackets) indicates this: γ is a function, and S is a function of that function.

Different paths between the same endpoints $(q_i(t_1), q_i(t_2))$ give different values of S.

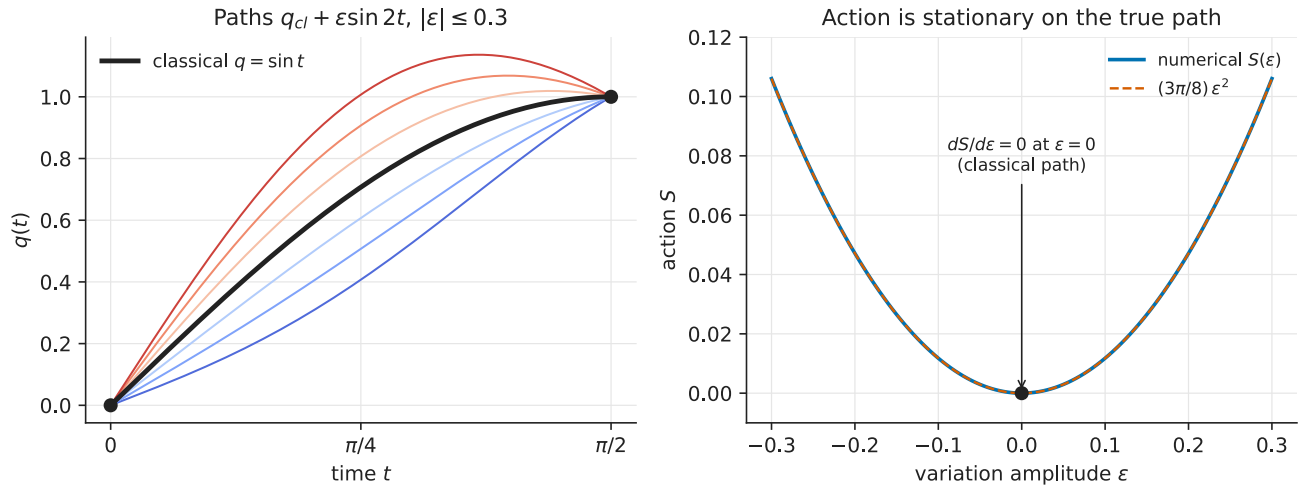
1.4.2 — Hamilton's Principle (Principle of Stationary Action)

The physical path is the one for which the action is stationary with respect to all variations that fix the endpoints.

That is: among all curves connecting $(q_i(t_1))$ to $(q_i(t_2))$, the actual motion of the system is the one for which $\delta S = 0$.

"Stationary" means: if you perturb the path slightly (while keeping the endpoints fixed), the action does not change to first order in the perturbation. It is analogous to finding the minimum (or maximum, or saddle point) of an ordinary function by setting its derivative to zero.

This is sometimes called the "principle of least action" — but "stationary" is more accurate. The action is not always minimum; it is always stationary.



Harmonic oscillator with $m = k = 1$ and fixed endpoints $q(0) = 0$, $q(\pi/2) = 1$: the classical path $q = \sin t$ and varied paths $\sin t + \epsilon \sin 2t$ for $|\epsilon| \le 0.3$ (left), and the numerically integrated action $S(\epsilon) = \int (\frac{1}{2}\dot{q}^2 - \frac{1}{2}q^2)\, dt$ (right), which matches $(3\pi/8)\epsilon^2$ and is stationary — here a minimum — exactly at $\epsilon = 0$.

1.4.3 — Why Should $\delta S = 0$?

This is the deep question. Several answers exist:

The Feynman path integral answer (deepest): In quantum mechanics, a particle does not take one path — it takes *all* paths simultaneously. The probability amplitude for going from A to B is:

$$\langle B | e^{-i\hat{H}t/\hbar} | A \rangle = \int_{\text{all paths}} e^{iS[\gamma]/\hbar} \, \mathcal{D}\gamma$$

Each path contributes with equal magnitude but a phase $e^{iS/\hbar}$. In the limit $\hbar \to 0$, paths with rapidly oscillating phases cancel each other out — *except* near the path where S is stationary, where neighboring paths have nearly the same phase and add constructively.

The classical trajectory is the stationary-phase path of the quantum path integral.

Classical mechanics is not wrong — it is the $\hbar \to 0$ limit of quantum mechanics, and the action principle is the classical shadow of the path integral. We will make this exact in §1.8.

The Newtonian answer (verification): Hamilton's principle is equivalent to Newton's second law for conservative systems. The action principle is not replacing Newton — it is a reformulation that happens to be far more powerful and more fundamental. The equivalence is shown in §1.5.

1.5 — The Euler-Lagrange Equations

1.5.1 — Derivation

We derive the condition $\delta S = 0$ explicitly. Consider a single generalized coordinate $q(t)$ (the multi-coordinate case is the same, applied independently to each q_i). The physical path is $q(t)$; a nearby perturbed path is:

$$q_\epsilon(t) = q(t) + \epsilon\, \eta(t)$$

where $\eta(t)$ is an arbitrary smooth function and ϵ is a small number. The boundary conditions require:

$$\eta(t_1) = \eta(t_2) = 0$$

(the endpoints are fixed; we are only varying the middle of the path).

The action of the perturbed path:

$$S[q_\epsilon] = \int_{t_1}^{t_2} L(q + \epsilon\eta,\, \dot{q} + \epsilon\dot{\eta},\, t)\, dt$$

Expand to first order in ϵ using the Taylor expansion of L:

$$S[q_\epsilon] = S[q] + \epsilon \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q}\,\eta + \frac{\partial L}{\partial \dot{q}}\,\dot{\eta} \right) dt + \mathcal{O}(\epsilon^2)$$

For stationarity, $dS[q_\epsilon]/d\epsilon\big|_{\epsilon=0} = 0$:

$$\delta S \equiv \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q}\,\eta + \frac{\partial L}{\partial \dot{q}}\,\dot{\eta} \right) dt = 0$$

Integrate the second term by parts, using $\int u\,dv = uv - \int v\,du$ with $u = \partial L/\partial\dot{q}$ and $dv = \dot{\eta}\,dt$:

$$\int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{q}}\,\dot{\eta}\,dt = \underbrace{\left[\frac{\partial L}{\partial \dot{q}}\,\eta\right]_{t_1}^{t_2}}_{=\,0\ \text{(endpoints fixed)}} - \int_{t_1}^{t_2} \frac{d}{dt}\frac{\partial L}{\partial \dot{q}}\,\eta\,dt$$

Substituting back:

$$\delta S = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q} - \frac{d}{dt}\frac{\partial L}{\partial \dot{q}}\right)\eta\,dt = 0$$

Since this must hold for **all** smooth η vanishing at the endpoints, the integrand itself must be identically zero (by the fundamental lemma of variational calculus):

$$\boxed{\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = 0 \qquad i = 1, 2, \ldots, n}$$

These are the **Euler-Lagrange equations** — the equations of motion that follow from Hamilton's principle.

1.5.2 — Recovering Newton's Second Law

For a particle in 3D with $L = \frac{1}{2}m\dot{\mathbf{r}}^2 - V(\mathbf{r})$, apply the Euler-Lagrange equation to the coordinate x:

$$\frac{\partial L}{\partial x} = -\frac{\partial V}{\partial x} = F_x, \qquad \frac{\partial L}{\partial \dot{x}} = m\dot{x}, \qquad \frac{d}{dt}(m\dot{x}) = m\ddot{x}$$

Therefore:

$$m\ddot{x} = -\frac{\partial V}{\partial x} = F_x$$

This is exactly $F = ma$ in the x-direction. The Euler-Lagrange equations contain Newton's second law as a special case.

1.5.3 — The Pendulum (Euler-Lagrange in Action)

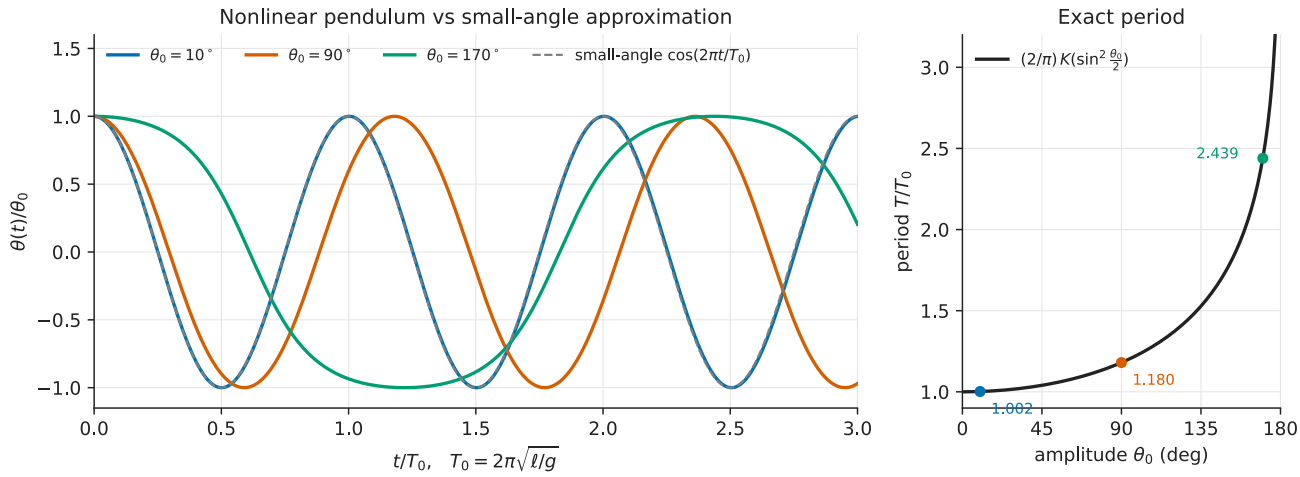
Using $L = \frac{1}{2}m\ell^2\dot{\theta}^2 + mg\ell\cos\theta$:

$$\frac{\partial L}{\partial \theta} = -mg\ell\sin\theta, \qquad \frac{\partial L}{\partial \dot{\theta}} = m\ell^2\dot{\theta}$$

$$\frac{d}{dt}(m\ell^2\dot{\theta}) - (-mg\ell\sin\theta) = 0$$

$$m\ell^2\ddot{\theta} + mg\ell\sin\theta = 0 \quad\Longrightarrow\quad \ddot{\theta} + \frac{g}{\ell}\sin\theta = 0$$

The pendulum equation of motion — derived in one line, with no forces to resolve, no constraint to handle, and valid in any coordinate.



Numerical solutions of $\ddot{\theta} + (g/\ell)\sin\theta = 0$ released from rest at $\theta_0 = 10^\circ, 90^\circ, 170^\circ$, normalised to θ_0 and plotted against t/T_0 with $T_0 = 2\pi\sqrt{\ell/g}$, against the small-angle $\cos(2\pi t/T_0)$ (left); the exact period $T/T_0 = (2/\pi)K(\sin^2\frac{\theta_0}{2})$ (right) is 1.002, 1.180 and 2.439 for the three amplitudes, so only the 10° swing is well described by the linear approximation.

1.6 — The Canonical Momentum and the Hamiltonian

1.6.1 — Canonical Momentum

Define the **canonical momentum** conjugate to q_i:

$$p_i = \frac{\partial L}{\partial \dot{q}_i}$$

For a particle in Cartesian coordinates, $p_x = m\dot{x}$ — the ordinary momentum. For a charged particle, $p_x = m\dot{x} + eA_x/c$ — momentum plus a gauge contribution. The canonical momentum is not always mv; it depends on the Lagrangian.

1.6.2 — The Legendre Transform

The Hamiltonian is obtained from the Lagrangian by a **Legendre transform**:

$$H(q_i, p_i, t) = \sum_i p_i \dot{q}_i - L(q_i, \dot{q}_i, t)$$

where $\dot{q}_i$ on the right must be expressed in terms of q_i and p_i by inverting $p_i = \partial L/\partial \dot{q}_i$.

For a particle with $L = \frac{1}{2}m\dot{q}^2 - V(q)$:

$$p = m\dot{q} \quad \Longrightarrow \quad \dot{q} = p/m$$

$$H = p \cdot \frac{p}{m} - \left(\frac{p^2}{2m} - V\right) = \frac{p^2}{2m} + V(q) = T + V$$

The Hamiltonian is the total energy — kinetic plus potential.

1.6.3 — Hamilton's Equations

Differentiating $H(q_i, p_i, t)$:

$$\boxed{\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i}}$$

These are **Hamilton's equations** — 2n first-order ODEs replacing the n second-order Euler-Lagrange equations. They treat positions q_i and momenta p_i on equal footing.

Phase space: The $2n$-dimensional space $(q_1, \ldots, q_n, p_1, \ldots, p_n)$ is **phase space**. Hamilton's equations define a flow on phase space: starting from any initial point $(q_i(0), p_i(0))$, the equations uniquely determine the entire future trajectory.

1.6.4 — Poisson Brackets

For any two functions $f(q, p, t)$ and $g(q, p, t)$ on phase space, define:

$$\{f, g\} = \sum_i \left(\frac{\partial f}{\partial q_i}\frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i}\frac{\partial g}{\partial q_i} \right)$$

Hamilton's equations become: $\dot{f} = \{f, H\} + \partial f/\partial t$.

The fundamental Poisson brackets are:

$$\{q_i, p_j\} = \delta_{ij}, \qquad \{q_i, q_j\} = 0, \qquad \{p_i, p_j\} = 0$$

Why this matters: In quantum mechanics, Poisson brackets become commutators:

$$\{A, B\}_{classical} \quad \longrightarrow \quad \frac{1}{i\hbar}[\hat{A}, \hat{B}]_{quantum}$$

The canonical commutation relation $[\hat{x}, \hat{p}] = i\hbar$ is the quantum version of $\{x, p\} = 1$. The entire structure of quantum mechanics — operators, commutators, the Heisenberg picture — is the Poisson bracket structure of classical Hamiltonian mechanics, promoted to operators. This is called **canonical quantization**, and it is one of the bridges to quantum mechanics (Ch. 3).

1.7 — Noether's Theorem: Symmetry and Conservation

This is the theorem that Chapter 1 exists to prove. Everything else in this book is downstream of it.

1.7.1 — Statement

> **Noether's Theorem (Emmy Noether, 1915):** For every continuous one-parameter symmetry of the action S, there exists a corresponding conserved quantity J (a constant of motion).

A **continuous symmetry** is a transformation $q_i \to q_i + \epsilon K_i(q, \dot{q}, t)$ parameterized by a continuous parameter ϵ, such that the action is invariant: $\delta S = 0$.

1.7.2 — Proof

Suppose the action is unchanged under $q_i \to q_i + \epsilon K_i$. Then $\delta S = 0$, which means (by the same integration-by-parts calculation as §1.5.1, but now with $\eta_i = K_i$):

$$\delta S = \int_{t_1}^{t_2} \sum_i \left(\frac{\partial L}{\partial q_i} - \frac{d}{dt}\frac{\partial L}{\partial \dot{q}_i} \right) K_i \, dt + \left[\sum_i \frac{\partial L}{\partial \dot{q}_i} K_i \right]_{t_1}^{t_2} = 0$$

The first integral is zero by the Euler-Lagrange equations (the system is on-shell, i.e., following its actual trajectory). Therefore:

$$\left[\sum_i \frac{\partial L}{\partial \dot{q}_i} K_i \right]_{t_1}^{t_2} = 0$$

This holds for any t_1, t_2. Therefore the quantity:

$$\boxed{J = \sum_i p_i K_i = \sum_i \frac{\partial L}{\partial \dot{q}_i} K_i = \text{constant}}$$

is conserved along the physical trajectory. □

Symmetry up to a total derivative. The proof assumed L itself is unchanged. The action is still a symmetry if L changes by a total time derivative, $\delta L = \epsilon \, dF/dt$, since that adds only a boundary term; repeating the calculation then gives the conserved quantity $J = \sum_i p_i K_i - F$. Time translation needs this form: for $\partial L/\partial t = 0$, the shift $K_i = \dot{q}_i$ changes L by $\epsilon \, dL/dt$, so $F = L$ and $J = \sum_i p_i \dot{q}_i - L = H$ (§1.7.3).

1.7.3 — The Major Conservation Laws, Derived

Time translation symmetry → Conservation of energy:

Consider $t \to t + \epsilon$ (shift everything forward in time). For a Lagrangian with no explicit time dependence ($\partial L/\partial t = 0$):

$$\frac{dL}{dt} = \sum_i \frac{\partial L}{\partial q_i}\dot{q}_i + \frac{\partial L}{\partial \dot{q}_i}\ddot{q}_i = \frac{d}{dt}\left(\sum_i \frac{\partial L}{\partial \dot{q}_i}\dot{q}_i \right)$$

(using the Euler-Lagrange equations). Therefore:

$$\frac{d}{dt}\left(\sum_i p_i \dot{q}_i - L \right) = 0 \quad \Longrightarrow \quad \frac{dH}{dt} = 0$$

Energy is conserved because the laws of physics do not change with time.

Space translation symmetry → Conservation of momentum:

Consider $\mathbf{r} \to \mathbf{r} + \boldsymbol{\epsilon}$ (shift the entire system by a constant vector). If V depends only on relative positions, L is unchanged. The generator is $K_i = 1$ for each coordinate:

$$J = \sum_i p_i = \mathbf{p}_{total} = \text{const}$$

Momentum is conserved because empty space looks the same everywhere.

Rotation symmetry → Conservation of angular momentum:

Consider rotation by ϵ around the z-axis: $(x, y) \to (x - \epsilon y, y + \epsilon x)$. The generators are $K_x = -y$, $K_y = x$, $K_z = 0$. The Noether charge:

$$J_z = p_x(-y) + p_y(x) = xp_y - yp_x = L_z$$

Angular momentum is conserved because space looks the same in all directions.

U(1) phase symmetry → Conservation of electric charge:

For the electron field ψ, consider $\psi \to e^{i\epsilon}\psi$ (a global phase rotation). The Lagrangian $\bar{\psi}(i\gamma^\mu D_\mu - m)\psi$ is unchanged because $\bar{\psi} \to e^{-i\epsilon}\bar{\psi}$ cancels the $e^{i\epsilon}$ from ψ. The Noether current:

$$j^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi)} \cdot (i\psi) = -e\bar{\psi}\gamma^\mu\psi$$

and the conserved charge $Q = \int j^0 \, d^3r = -e \int |\psi|^2 \, d^3r = -e$ (for one electron). **Electric charge is conserved because the action is invariant under phase rotations of the electron field.**

Summary of Noether's theorem through the full descent:

Symmetry	Generator K_i	Conserved quantity	Layer 3 form
Time translation	$\dot{q}_i$ (with $F = L$, §1.7.2)	H (energy)	Energy balance, 1st law of thermodynamics
Space translation	$(1,0,0)$, $(0,1,0)$, $(0,0,1)$	p_x, p_y, p_z	Force balance ($\sum F = 0$)
Rotation	$(-y, x, 0)$ etc.	L_x, L_y, L_z	Torque balance; shaft power
U(1) phase	$i\psi$	Electric charge Q	Kirchhoff's Current Law

The entire foundation of circuit theory, structural mechanics, and process engineering rests on these four rows.

1.8 — Hamilton-Jacobi Theory: The Bridge to Quantum Mechanics

1.8.1 — Hamilton's Principal Function

We seek a canonical transformation $(q_i, p_i) \to (Q_i, P_i)$ such that the new Hamiltonian $K(Q_i, P_i, t) = 0$. If such a transformation exists, then Hamilton's equations in the new coordinates are trivially:

$$\dot{Q}_i = \frac{\partial K}{\partial P_i} = 0, \qquad \dot{P}_i = -\frac{\partial K}{\partial Q_i} = 0$$

meaning $Q_i = \text{const}$ and $P_i = \text{const}$ for all time. The problem is solved — we need only transform back.

A canonical transformation can be generated by a function $S(q_i, P_i, t)$ (Hamilton's **principal function**), with:

$$p_i = \frac{\partial S}{\partial q_i}, \qquad Q_i = \frac{\partial S}{\partial P_i}, \qquad K = H + \frac{\partial S}{\partial t}$$

Setting $K = 0$:

$$\boxed{H\left(q_i, \frac{\partial S}{\partial q_i}, t\right) + \frac{\partial S}{\partial t} = 0}$$

This is the **Hamilton-Jacobi (HJ) equation** — a single first-order PDE for the function $S(q_i, t)$.

1.8.2 — The Time-Independent Case

For a time-independent Hamiltonian, write:

$$S(q_i, t) = W(q_i) - Et$$

where E is the (conserved) total energy. Substituting:

$$\boxed{H\left(q_i, \frac{\partial W}{\partial q_i}\right) = E}$$

This is the **time-independent Hamilton-Jacobi equation**. $W(q_i)$ is called **Hamilton's characteristic function**.

Example — Free particle in 1D:

$$H = \frac{p^2}{2m} = \frac{1}{2m}\left(\frac{dW}{dq}\right)^2 = E$$

$$\frac{dW}{dq} = \sqrt{2mE} = p \quad \Longrightarrow \quad W = pq$$

$$S = pq - Et = pq - \frac{p^2}{2m}t$$

The constant-S surfaces are the **wavefronts** of a wave with wavenumber $k = p/\hbar$ and frequency $\omega = E/\hbar$ — exactly a plane wave $e^{i(kq - \omega t)}$. We are already seeing the quantum wavefunction.

1.8.3 — The Key Connection: HJ Equation → Schrödinger Equation

This is the most important result in classical mechanics. It directly shows why quantum mechanics takes the form it does.

Write the wavefunction as:

$$\psi(\mathbf{r}, t) = A(\mathbf{r}, t)\, e^{iS(\mathbf{r},t)/\hbar}$$

where A is a real amplitude and S is Hamilton's principal function. Substitute into the Schrödinger equation $i\hbar\,\partial_t\psi = \left(-\frac{\hbar^2}{2m}\nabla^2 + V\right)\psi$:

Left side:

$$i\hbar\,\partial_t\psi = i\hbar\left(\dot{A} + \frac{iA\dot{S}}{\hbar}\right)e^{iS/\hbar} = \left(i\hbar\dot{A} - A\dot{S}\right)e^{iS/\hbar}$$

Right side: (using $\nabla\psi = (\nabla A + \frac{iA}{\hbar}\nabla S)e^{iS/\hbar}$)

$$\nabla^2\psi = \left(\nabla^2 A + \frac{2i}{\hbar}\nabla A\cdot\nabla S + \frac{iA}{\hbar}\nabla^2 S - \frac{A}{\hbar^2}|\nabla S|^2\right)e^{iS/\hbar}$$

Collecting terms order by order in $\hbar$ (after dividing through by $Ae^{iS/\hbar}$):

Order $\hbar^0$ (leading order):

$$\boxed{-\frac{\partial S}{\partial t} = \frac{|\nabla S|^2}{2m} + V = H(\mathbf{r}, \nabla S)}$$

This is the **Hamilton-Jacobi equation** — appearing directly as the $\hbar \to 0$ limit of the Schrödinger equation. Classical mechanics is exactly the leading-order term of quantum mechanics in a $\hbar$-expansion.

Order $\hbar^1$ (next order):

$$\frac{\partial A^2}{\partial t} + \nabla \cdot \left(A^2 \frac{\nabla S}{m} \right) = 0$$

This is the **continuity equation for** $\rho = A^2 = |\psi|^2$ — the conservation of probability. It arises automatically at next order in $\hbar$.

Order $\hbar^2$ (quantum corrections):

$$\text{Quantum potential: } Q = -\frac{\hbar^2}{2m} \frac{\nabla^2 A}{A}$$

This is the purely quantum term that has no classical analog. It is responsible for tunneling, zero-point energy, and interference effects.

The hierarchy:

$$\begin{gathered}
\text{Full quantum: } \psi = Ae^{iS/\hbar} \\
\downarrow \quad \hbar \to 0 \\
\text{HJ equation (classical mechanics): } H + \partial_t S = 0 \\
\downarrow \quad \text{simple geometry} \\
\text{Newton's 2nd law: } m\ddot{\mathbf{r}} = -\nabla V
\end{gathered}$$

Classical mechanics is quantum mechanics with quantum corrections suppressed by $\hbar$. The Hamilton-Jacobi equation is the hinge between them.

1.8.4 — WKB Approximation: The Bridge Zone

The **WKB approximation** (Wentzel-Kramers-Brillouin) keeps $\hbar$ small but nonzero:

$$\psi(x) \approx \frac{C}{\sqrt{p(x)}} \exp\left(\pm \frac{i}{\hbar} \int p(x)\, dx \right), \quad p(x) = \sqrt{2m(E - V(x))}$$

This is valid when the potential varies slowly over a de Broglie wavelength ($|dp/dx| \ll p^2/\hbar$). It describes:

- Quantum tunneling through a barrier (p becomes imaginary inside the barrier)
- Semiclassical quantization: $\oint p\, dq = (n + \frac{1}{2})h$ (Bohr-Sommerfeld)
- Geometric optics as the WKB limit of wave optics

WKB sits in the bridge zone between classical mechanics (Layer 2) and quantum mechanics (Layer 1). It is the correct equation to use when $\hbar$ is small but cannot be set to zero.

1.9 — From Particles to Fields: The Extension to $\mathcal{L}_{SM}$

Everything in §§1.2–1.8 was for a finite number of degrees of freedom $q_1, \ldots, q_n$. The Standard Model Lagrangian involves **fields** — quantities defined at every point in spacetime. This is an infinite number of degrees of freedom.

The extension is straightforward and the structure is identical.

1.9.1 — The Field Lagrangian Density

Replace:

Particle mechanics	Field theory
Generalized coordinate $q_i(t)$	Field $\phi_a(\mathbf{r}, t)$
Index i labeling DOF	Continuous label $(\mathbf{r}, a)$
Velocity $\dot{q}_i(t)$	Gradient $\partial_\mu \phi_a$ (space-time derivative)
Lagrangian $L(q_i, \dot{q}_i, t)$	Lagrangian density $\mathcal{L}(\phi_a, \partial_\mu \phi_a)$
Action $S = \int L\, dt$	Action $S = \int d^4x\, \mathcal{L}$

1.9.2 — Field Euler-Lagrange Equations

Varying $S[\phi_a] = \int d^4x\, \mathcal{L}(\phi_a, \partial_\mu \phi_a)$ with respect to ϕ_a (same integration-by-parts procedure as §1.5.1, now over 4D spacetime):

$$\boxed{\frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} = 0}$$

This is the **field Euler-Lagrange equation**. It is the equation of motion for the field ϕ_a.

Verification on the simplest field: For $\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$ (Maxwell, from §0.4.3 of Chapter 0):

$$\frac{\partial \mathcal{L}}{\partial A_\nu} = 0, \qquad \frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\nu)} = -F^{\mu\nu}$$

The field equation: $\partial_\mu F^{\mu\nu} = 0$ — Maxwell's equations in vacuum.

1.9.3 — Noether's Theorem for Fields

For a field transformation $\phi_a \to \phi_a + \epsilon\, \delta\phi_a$ that leaves $\mathcal{L}$ invariant, the conserved current is:

$$j^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \delta\phi_a$$

satisfying $\partial_\mu j^\mu = 0$ (a continuity equation in 4D).

The conserved charge:

$$Q = \int j^0\, d^3x = \text{const}$$

The energy-momentum tensor comes from invariance under spacetime translations $x^\mu \to x^\mu + \epsilon^\mu$:

$$T^{\mu\nu}_{\text{can}} = -\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)}\partial^\nu \phi_a + g^{\mu\nu}\mathcal{L}$$

(signs for the book's $(-,+,+,+)$ signature, so that $T^{00} = +\mathcal{H}$; in the $(+,-,-,-)$ signature both terms flip — Appendix A §A.7.1)

- T^{00} = energy density
- T^{0i} = momentum density
- $\partial_\mu T^{\mu\nu} = 0$ encodes energy-momentum conservation in field theory

This is the **canonical** Noether tensor, and the label matters. The object that sources Einstein's equations is the **Hilbert** stress-energy tensor,

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta S_{\text{matter}}}{\delta g^{\mu\nu}}$$

which is defined by a *functional derivative with respect to the metric* — a different construction, and one that requires no preferred flat background. In flat spacetime, with fields taken to be independent of the metric, the two agree. In general relativity they do not coincide: the canonical tensor is typically neither symmetric nor gauge invariant, so it is not by itself a well-behaved source for a diffeomorphism-invariant theory.

The two are nonetheless **related**. The canonical tensor can be *improved* — the Belinfante-Rosenfeld procedure adds a total divergence,

$$T^{\mu\nu} = T^{\mu\nu}_{\text{can}} + \partial_\lambda B^{\lambda\mu\nu}, \qquad B^{\lambda\mu\nu} = -B^{\mu\lambda\nu}$$

which leaves the conserved four-momentum unchanged, and under suitable conditions (scalar, spin-0 and spin-1/2 matter with the standard improvement) the result matches the Hilbert tensor up to the flat-background limit. So the correct statement is: the Hilbert tensor is the general-relativistic object, and for restricted matter content it coincides with an improved version of the canonical Noether tensor. **Not** "exactly this object."

What survives without qualification is the structural point that follows the display above: gravity couples to energy-momentum because spacetime-translation symmetry is what *defines* energy-momentum, so the stress-energy tensor is the natural source of Einstein's equations. Identifying it with the flat-space Noether current is what fails, not the symmetry argument.

1.10 — Why This Framework Is Unreasonably Effective

We can now answer the question from §1.0.

Coordinate independence: The action $S = \int \mathcal{L}\, d^4x$ is a single number — it does not depend on coordinates. The field equations derived from it are automatically covariant (they transform correctly under any coordinate change). This is why the same framework works in Cartesian, polar, spherical, and curved coordinates without modification.

Symmetry is primary: By writing physics as an action, symmetries are directly visible as transformations that leave S unchanged. Noether's theorem then guarantees conserved quantities without calculation. In Newton's framework, discovering a conservation law requires solving the equations first and observing the pattern. In the action framework, symmetry $\Rightarrow$ conservation *by construction*.

Constrained generalization: There is a *tight* way to write a Lorentz-invariant, gauge-invariant, renormalizable quantum field theory in 3+1 dimensions — an action of the form $\int d^4x\, \mathcal{L}$ whose Lagrangian density is dimensionless (power-counting renormalizability), local, and restricted by the internal symmetry group. Those requirements are strong, but they are not a uniqueness theorem. Many such theories exist, and they are not related by cosmetic choices: scalar QED, non-Abelian gauge theories with different groups, four-fermion theories, and — within the Standard Model gauge group itself — many admissible field contents and parameter choices. Even fixing the gauge group, the particle content and Yukawa structure are inputs, not outputs.

What is genuinely remarkable is the **convergence from data**, not a deduction from the framework. The observed Standard Model is pinned down from many directions at once: the measured gauge group, the observed particle spectrum and its quantum numbers, charge quantisation, anomaly cancellation across each generation, the measured fermion masses and mixing angles, and the absence of any observed extension at collider energies. A large space of consistent theories is reduced to one *empirically*. The framework narrows the possibilities a great deal; experiment selects among what survives. Those are two different acts, and conflating them is what makes "the framework almost forces the answer" sound stronger than it is.

Classical-quantum bridge: As shown in §1.8.3, classical mechanics is the $\hbar \to 0$ limit of quantum mechanics, and this limit is made precise by the HJ equation. The action S is literally the phase of the quantum wavefunction. Writing physics as an action is writing it in the language that is maximally transparent about this relationship.

1.11 — Summary

Concept	Key equation	Engineering shadow
Lagrangian	$L = T - V$	Energy bookkeeping in every control volume
Euler-Lagrange	$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = 0$	Equations of motion for any mechanical system
Canonical momentum	$p_i = \partial L / \partial \dot{q}_i$	Impulse, momentum flux
Hamiltonian	$H = \sum p_i \dot{q}_i - L$	Total energy; the thing that is conserved
Poisson brackets	$\{q_i, p_j\} = \delta_{ij}$	Become commutators in QM
Noether's theorem	Symmetry $\Rightarrow$ conserved $J = p_i K_i$	KCL, KVL, force balance, mass balance
Hamilton-Jacobi	$H(q, \nabla S) + \partial_t S = 0$	Classical limit of Schrödinger equation
WKB	$\psi \sim e^{iS/\hbar}/\sqrt{p}$	Bridge zone between classical and quantum
Field EL equation	$\partial_\mu(\partial\mathcal{L}/\partial(\partial_\mu\phi)) - \partial\mathcal{L}/\partial\phi = 0$	Maxwell, Dirac, Klein-Gordon from one formula
Field Noether	$j^\mu = (\partial\mathcal{L}/\partial(\partial_\mu\phi))\delta\phi$	$T^{\mu\nu}$ in GR; charge current in EM

1.12 — Looking Ahead

The next chapter (Chapter 2) is the phenomena catalogue — a map of every named effect in this book, placed on the layer diagram. Consult it as a reference.

Chapter 3 then performs the first actual descent: from the fermion sector of $\mathcal{L}_{SM}$ (Chapter 0), using the field Euler-Lagrange equation (this chapter, §1.9.2), down to the Dirac equation and from there to the Schrödinger equation. The minimal coupling in the Schrödinger equation ($\hat{p} \to \hat{p} - e\mathbf{A}/c$) comes from the covariant derivative; the probability interpretation comes from Noether's theorem applied to U(1) phase symmetry; and the uncertainty principle comes from the Fourier structure of the wavefunction — all traceable to ideas introduced in this chapter.

End of Chapter 1.

Next: Chapter 2 — A Map of Physical Phenomena

A Map of Physical Phenomena

Every named physical effect in this book — located, described, and connected

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2.0 — How to Use This Chapter

This chapter is a **reference map**, not a reading chapter. Every named physical effect that appears anywhere in this book is listed here, briefly described, and placed on the four-layer framework from Chapters 0 and 1.

Read it now once, lightly. You will not understand every entry. That is expected and fine. The goal of the first read is to get a sense of the territory — to see how many different phenomena trace back to the same few mathematical structures.

Return to it often. Whenever a subsequent chapter introduces a named effect, find it here. The layer code and cross-references will tell you where it sits in the framework and what it connects to.

Layer Codes

Code	Scale	What lives here
L0	Framework scale	Particle physics, QFT, general relativity
L0→L1	Bridge A	Effects with QFT origin that manifest in single-particle QM
L1	Quantum / atomic scale	Wavefunctions, atoms, bands, nuclear structure
L1→L2	Bridge B	Effects quantum in origin but classical/continuum in manifestation
L2	Classical continuum scale	Maxwell, Newton, thermodynamics, continuum mechanics
L2→L3	Bridge C	Distributed-to-lumped transition
L3	Engineering systems scale	Circuits, machines, structures, processes

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Format of Each Entry

Effect Name *What it is, in one or two sentences.* **Layer:** code | **Revealed by:** chapter/section | **Branch:** disciplines | **→ Ch. X**

2.1 — Quantum and Atomic Phenomena

Photoelectric Effect Monochromatic light above a threshold frequency ejects electrons from a metal with kinetic energy $K = h\nu - W$; below the threshold no electrons are emitted regardless of intensity. **Layer:** L0→L1 | **Revealed by:** Ch. 3 Bridge A: photon is a quantum of the U(1) gauge field; electron absorbs one quantum | **Branch:** EEE (photovoltaics, photodetectors, photomultipliers) | **→ Ch. 4**

Compton Scattering A photon scattered by a free electron shifts to longer wavelength $\Delta\lambda = (h/m_e c)(1 - \cos\theta)$; both momentum and energy are conserved as if the photon were a particle. **Layer:** L0→L1 | **Revealed by:** Ch. 3: relativistic electron-photon kinematics in QED | **Branch:** EEE (X-ray imaging, radiation dosimetry) | → **Ch. 3**

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de Broglie Matter Waves Every particle with momentum p has wavelength $\lambda = h/p$; matter produces interference and diffraction patterns. **Layer:** L1 | **Revealed by:** Ch. 3 Bridge A: momentum eigenstates of the Schrödinger equation are plane waves e^(ik·r) with $p = \hbar k$ | **Branch:** EEE (electron microscopy, neutron diffractometry) | → **Ch. 4**

Heisenberg Uncertainty Principle Conjugate observables satisfy $\Delta x \Delta p \geq \hbar/2$ and $\Delta E \Delta t \geq \hbar/2$; the more precisely one is known the less precisely the other can be. **Layer:** L1 | **Revealed by:** Ch. 1 §1.8 + Ch. 3 §3.11: the Fourier bandwidth theorem applied to wavefunctions; $\hat{p} = -i\hbar\nabla$ makes position and momentum Fourier conjugates | **Branch:** All (fundamental measurement limit; transistor scaling limits) | → **Ch. 4**

Quantum Tunneling A quantum particle has nonzero probability of crossing a classically forbidden barrier; amplitude decays as e^(−2κd) inside the barrier. **Layer:** L1 | **Revealed by:** Ch. 3 Bridge A: Schrödinger equation has exponentially decaying solutions in forbidden regions; WKB from Ch. 1 §1.8.4 | **Branch:** EEE (tunnel diodes, Josephson junctions, STM, MOSFET gate leakage) | → **Ch. 4**

Zero-Point Energy A quantum oscillator in its ground state has energy $E_0 = \hbar\omega/2 > 0$; quantum systems cannot be at rest even at absolute zero. **Layer:** L1 | **Revealed by:** Ch. 3: the uncertainty principle forbids $x = p = 0$ simultaneously | **Branch:** EEE (laser threshold; squeezed light); ME (helium never solidifies at 1 atm) | → **Ch. 4**

Zeeman Effect (Normal and Anomalous) An external magnetic field splits degenerate atomic energy levels; the anomalous splitting pattern requires spin-1/2. **Layer:** L1 | **Revealed by:** Ch. 3 Bridge A: spin-1/2 falls from the Dirac equation; Zeeman Hamiltonian $-\mu \cdot B$ from minimal coupling | **Branch:** EEE (NMR, MRI, magnetometers, atomic clocks) | → **Ch. 5**

Stark Effect An external electric field shifts and splits atomic energy levels; linear for degenerate states, quadratic for non-degenerate. **Layer:** L1 | **Revealed by:** Ch. 3 + perturbation theory: add eEz to the Coulomb potential | **Branch:** EEE (electro-optic modulators, electric-field sensing, Stark-shift clocks) | → **Ch. 5**

Fine Structure Spectral lines split into closely spaced doublets from spin-orbit coupling (L·S interaction) and the relativistic kinetic energy correction; splitting of order $\alpha^2 m_e c^2 \times \alpha^2$. **Layer:** L1 | **Revealed by:** Ch. 3 §3.5 Foldy-Wouthuysen at order $(v/c)^2$ | **Branch:** Metrology (fine structure constant α; precision spectroscopy) | → **Ch. 5**

Hyperfine Structure Further splitting from the interaction between the electron and nuclear magnetic moments; the 21-cm hydrogen line (1420 MHz) is the most famous instance. **Layer:** L1 | **Revealed by:** Ch. 3 Bridge A: contact interaction of electron wavefunction with nuclear spin | **Branch:** EEE (atomic

hydrogen maser; GPS atomic clocks; radio astronomy) | → **Ch. 5**

Lamb Shift The $2s_{1/2}$ and $2p_{1/2}$ hydrogen levels, predicted degenerate by the Dirac equation, are split by 1057.8 MHz due to QED vacuum fluctuations. **Layer:** L0 (requires full QED) | **Revealed by:** One-loop QED; beyond Bridge A | **Branch:** Metrology (Rydberg constant definition) | → **Epilogue**

Spontaneous Emission An excited atom decays to its ground state by emitting a photon even with no external radiation; rate given by Einstein's A coefficient. **Layer:** L0→L1 | **Revealed by:** Ch. 3 + quantized EM field: vacuum fluctuations drive decay; Fermi's Golden Rule gives the rate | **Branch:** EEE (LED efficiency; laser threshold; amplifier noise) | → **Ch. 5**

Stimulated Emission An incident photon triggers a second identical photon from an excited atom, enabling coherent amplification. **Layer:** L0→L1 | **Revealed by:** Ch. 3 Bridge A: Einstein B coefficient from Fermi's Golden Rule; bosonic stimulation factor (n+1) | **Branch:** EEE (operating principle of every laser and optical amplifier) | → **Ch. 5**

Rabi Oscillations A two-level atom driven at resonance coherently oscillates between ground and excited states at the Rabi frequency $\Omega R = d\varepsilon/\hbar$. **Layer:** L1 | **Revealed by:** Ch. 3 + time-dependent perturbation theory on two-level Schrödinger equation | **Branch:** EEE (qubit control; pulsed NMR π-pulses; atomic clocks) | → **Ch. 5**

Fermi's Golden Rule The transition rate between states under a weak perturbation is $\Gamma = (2\pi/\hbar)|\langle f|V|i\rangle|^2\rho(Ef)$; proportional to matrix element squared times density of final states. **Layer:** L1 | **Revealed by:** Ch. 3 Bridge A: first-order time-dependent perturbation theory | **Branch:** All (scattering rates → transport coefficients; optical rates; chemical reaction rates) | → **Ch. 6**

Pauli Exclusion Principle No two identical fermions can occupy the same quantum state; the many-body wavefunction is antisymmetric under particle exchange. **Layer:** L0→L1 | **Revealed by:** Ch. 0 §0.5: anticommutation relations of fermion fields in L_SM | **Branch:** All (periodic table; why matter is rigid; Fermi-Dirac statistics for metals and semiconductors) | → **Ch. 5**

Spin-Statistics Theorem Half-integer spin → Fermi-Dirac statistics + Pauli exclusion; integer spin → Bose-Einstein statistics + stimulation. **Layer:** L0 | **Revealed by:** Ch. 0 §0.5: requires relativistic QFT; proven from Lorentz invariance and locality | **Branch:** All (Fermi sea → metals/semiconductors; Bose enhancement → lasers, BEC, superfluids) | → **Ch. 5**

Quantum Entanglement Two particles in a joint quantum state show measurement correlations stronger than any local hidden-variable theory allows (Bell's theorem). **Layer:** L0→L1 | **Revealed by:** Ch. 3: tensor product structure of multi-particle Hilbert space | **Branch:** EEE (quantum key distribution, quantum repeaters, quantum error correction) | → **Ch. 4**

Quantum Decoherence A quantum superposition loses phase coherence from interaction with environmental degrees of freedom; off-diagonal density matrix elements decay exponentially. **Layer:** L1→L2 | **Revealed by:** Bridge B.b (Ch. 10): tracing out environment from the density matrix; same

operation as the statistical limit | **Branch:** EEE (limits quantum computing; explains classical behavior of macroscopic objects) | → **Ch. 10**

Quantum Zeno Effect Frequent measurement freezes a quantum system in its initial state; survival probability → 1 as measurement interval τ → 0. **Layer:** L1 | **Revealed by:** Ch. 3: wavefunction collapse resets the short-time quadratic decay | **Branch:** EEE (decoherence engineering in qubits; error-correction protocols) | → **Ch. 4**

Born-Oppenheimer Approximation Electrons adjust instantaneously to slowly-moving nuclei ($m_e/M \ll 1$); electronic and nuclear motions decouple; nuclear dynamics happen on an effective potential surface. **Layer:** L1 | **Revealed by:** Ch. 3 Bridge A: expand Schrödinger equation in m_e/M | **Branch:** ChE (quantum chemistry, DFT, molecular simulation); ME/CE (first-principles material properties) | → **Ch. 6**

2.2 — Condensed Matter and Band Phenomena

Bloch's Theorem Electron wavefunctions in a periodic crystal take the form ψ_nk(r) = e^(ik·r) u_nk(r), a plane wave modulated by a lattice-periodic function. **Layer:** L1 | **Revealed by:** Ch. 6: commuting with the *discrete* lattice translations gives joint eigenstates labelled by the irreducible representations of that discrete group (crystal momentum ħk). *Not* Noether — that theorem needs a continuous symmetry; see Ch. 6 §6.1.2 | **Branch:** EEE/ME/CE (foundation of all band-structure-based materials analysis) | → **Ch. 6**

Band Gaps Allowed electron energies cluster into bands separated by forbidden gaps; gaps open from Bragg reflection at Brillouin zone boundaries. **Layer:** L1 | **Revealed by:** Ch. 6: nearly-free-electron + degenerate perturbation theory at zone boundary | **Branch:** EEE (conductor/semiconductor/insulator distinction; all device physics); ME (optical and electrical material properties) | → **Ch. 6**

Hall Effect (Classical) A transverse voltage develops across a current-carrying conductor in a magnetic field; R_H = 1/ne gives carrier density and sign directly. **Layer:** L2 | **Revealed by:** Bridge B.c (Ch. 11): Lorentz force on classical charge carrier | **Branch:** EEE (Hall sensors, current probes, material characterization) | → **Ch. 14**

Quantum Hall Effect (Integer) Hall conductance in a 2D electron gas at low T and high B is precisely quantized: $\sigma_{xy} = n e^2/h$ for integer n, regardless of disorder. **Layer:** L1→L2 | **Revealed by:** Ch. 7: Landau levels + Chern number of occupied bands; topology protects the quantization | **Branch:** EEE (primary resistance standard; $R_K = h/e^2 \approx 25812.807\ \Omega$ by SI definition since 2019) | → **Ch. 7**

Fractional Quantum Hall Effect Hall conductance at $\sigma_{xy} = (p/q)e^2/h$ for fractional p/q; quasiparticles carry fractional charge e/q; non-Abelian anyons emerge. **Layer:** L0→L1 | **Revealed by:** Beyond single-particle theory; many-body Laughlin wavefunction; topological order; relevant to the unknown-sector discussion in Ch. 0 | **Branch:** EEE (topological quantum computing platform) | → **Epilogue**

Landau Levels Electrons in a uniform B-field occupy discrete energy levels $E_n = \hbar\omega_c(n + 1/2)$ where $\omega_c = eB/m_e$ is the cyclotron frequency. **Layer:** L1 | **Revealed by:** Ch. 4: Schrödinger equation with minimal coupling $\hat{p} \to \hat{p} - eA/c$ maps to a harmonic oscillator | **Branch:** EEE (basis of QHE; magneto-oscillations; magnetoresistance) | → **Ch. 7**

Aharonov-Bohm Effect The wavefunction phase is shifted by the vector potential A even where $B = \nabla \times A = 0$; the topological phase produces observable interference in ring geometries. **Layer:** L1 | **Revealed by:** Ch. 3 §3.9.4: minimal coupling in Schrödinger equation; A (not B) is the fundamental object | **Branch:** EEE (mesoscopic physics; flux qubit; SQUID interference) | → **Ch. 5**

Berry Phase Adiabatic evolution around a closed loop in parameter space gives a geometric phase $\gamma_n = i\oint \langle n|\nabla_R|n\rangle \cdot dR$ independent of the evolution rate. **Layer:** L1 | **Revealed by:** Ch. 7: adiabatic theorem; Berry connection $A_n = i\langle n|\nabla_k|n\rangle$ is the gauge field of band topology | **Branch:** EEE (anomalous velocity; topological insulators; Weyl semimetals; quantum gate operations) | → **Ch. 7**

Superconductivity (BCS) Below T_c, phonon-mediated attraction forms Cooper pairs that condense into a macroscopic quantum state carrying current with zero resistance. **Layer:** L1→L2 | **Revealed by:** Ch. 7: phonon-mediated pairing from Fermi's Golden Rule; spontaneous U(1) breaking (same Mexican hat as Higgs in Ch. 0, at meV scale) | **Branch:** EEE (MRI magnets, accelerators, power transmission R&D, quantum computing) | → **Ch. 7**

Meissner Effect A superconductor expels all magnetic flux on cooling below T_c ($B = 0$ inside), distinguishing it from a perfect conductor. **Layer:** L1→L2 | **Revealed by:** Ch. 7: London equations from broken U(1) symmetry of BCS state; photon acquires effective mass inside — Higgs mechanism analog | **Branch:** EEE/ME (MRI shielding; Maglev trains; flux exclusion in SQUID design) | → **Ch. 7**

Josephson Effect (DC and AC) A supercurrent tunnels through an insulating barrier with no voltage (DC); applied voltage V drives AC supercurrent at $f = 2eV/h$ (AC). **Layer:** L1 | **Revealed by:** Ch. 7: quantum tunneling of Cooper pairs; macroscopic BCS wavefunction phase difference is the dynamical variable | **Branch:** EEE (voltage standard $1\ V \equiv f/483597.8$ GHz exact; SQUID magnetometers; superconducting qubits) | → **Ch. 7**

Flux Quantization Magnetic flux through a superconducting loop is quantized in units $\Phi_0 = h/2e = 2.067 \times 10^{-15}$ Wb; the 2e reflects Cooper pair charge. **Layer:** L1→L2 | **Revealed by:** Ch. 7: single-valuedness of BCS macroscopic wavefunction around the loop | **Branch:** EEE (SQUID sensing; flux qubit; type-II vortex physics) | → **Ch. 7**

Casimir Effect Two uncharged parallel conductors in vacuum attract each other due to suppression of vacuum EM fluctuations in the gap; $F/A = -\hbar c\pi^2/240d^4$. **Layer:** L0→L1 | **Revealed by:** Ch. 3 Bridge A: quantized photon field zero-point energy; plates change mode spectrum in gap relative to outside | **Branch:** EEE/ME (stiction in MEMS/NEMS at sub-micron gaps; nanotechnology design) | → **Ch. 5**

Kondo Effect A magnetic impurity in a metal creates a resistance minimum below the Kondo temperature T_K; resistance shows logarithmic upturn before saturating as T → 0. **Layer:** L1→L2 | **Revealed by:** Bridge B.e (Ch. 13): many-body scattering resonance at Fermi level; requires non-perturbative renormalization group | **Branch:** EEE (limits quantum device performance at millikelvin; heavy-fermion materials) | **→ Ch. 13**

Anderson Localization Sufficient disorder spatially localizes all electron wavefunctions; a disordered metal becomes an insulator through quantum interference of multiply-scattered waves. **Layer:** L1 | **Revealed by:** Ch. 6/13: backscattered paths interfere constructively, suppressing diffusion; weak localization correction to conductivity | **Branch:** EEE (thin-film resistance; mesoscopic conductance fluctuations; weak localization in 2DEG) | **→ Ch. 13**

Spin Hall Effect A longitudinal charge current generates a transverse spin current in materials with strong spin-orbit coupling, without a magnetic field. **Layer:** L1 | **Revealed by:** Ch. 7: intrinsic spin Hall from Berry curvature (topological); extrinsic from spin-dependent impurity scattering (Fermi's Golden Rule) | **Branch:** EEE (spintronics; spin-orbit torque MRAM; spin current generation) | **→ Ch. 7**

Giant Magnetoresistance (GMR) Resistance of a ferromagnet/metal/ferromagnet trilayer is much lower when magnetic layers are parallel vs. antiparallel; effect up to 80%. **Layer:** L1→L2 | **Revealed by:** Ch. 7: spin-dependent scattering rates from Fermi's Golden Rule differ for majority and minority spin channels | **Branch:** EEE (HDD read heads since 1997; magnetic field sensors) | **→ Ch. 7**

Topological Insulator Surface States Bulk is insulating but time-reversal-protected metallic surface states with spin-momentum locking cannot be backscattered by non-magnetic perturbations. **Layer:** L1 | **Revealed by:** Ch. 7: non-trivial Z_2 topological invariant of bulk bands; same topology thread as QCD θ-term in Ch. 0 §0.9 | **Branch:** EEE (spin-polarized currents; topological transistors; quantum computing) | **→ Ch. 7**

Weyl Semimetal Band-touching Weyl nodes in the Brillouin zone host electrons behaving as massless relativistic Weyl fermions with definite chirality; each node is a Berry curvature monopole. **Layer:** L1 | **Revealed by:** Ch. 7: Weyl node carries Chern number ±1; chiral anomaly (from Ch. 0 ABJ anomaly) manifests as negative magnetoresistance | **Branch:** EEE (ultralow dissipation transport; potential quantum devices) | **→ Ch. 7**

Bose-Einstein Condensation (BEC) Below T_c a macroscopic fraction of bosons occupies the single-particle ground state, forming a coherent macroscopic quantum state. **Layer:** L1→L2 | **Revealed by:** Bridge B.b (Ch. 10): Bose-Einstein statistics + thermodynamic limit; same U(1) breaking as Higgs and BCS, at μeV scale | **Branch:** EEE (atom lasers; quantum simulation); ME (cryogenic systems) | **→ Ch. 10**

Superfluidity Liquid He-4 below 2.17 K flows with exactly zero viscosity; persistent current; quantized vortices. **Layer:** L1→L2 | **Revealed by:** Bridge B.b (Ch. 10): BEC + weak interactions; broken U(1); Landau two-fluid model | **Branch:** ME (dilution refrigerators; cryogenic technology for quantum computing) | **→ Ch. 10**

Piezoelectric Effect A crystal without inversion symmetry develops electric polarization under stress (direct: $P_i = d_{ijk}\sigma_{jk}$) or deforms under applied field (converse). **Layer:** L1→L2 | **Revealed by:** Bridge B (Ch. 13): broken inversion symmetry in crystal + quantum-mechanical coupling of strain to charge distribution | **Branch:** EEE/ME (quartz oscillators, ultrasound transducers, MEMS, accelerometers, inkjet actuators) | → **Ch. 13**

Ferroelectric Effect Spontaneous switchable electric polarization with hysteresis; analogous to ferromagnetism but for electric dipoles. **Layer:** L1→L2 | **Revealed by:** Bridge B.b (Ch. 10): Landau double-well free energy for electric polarization order parameter — same Mexican hat as Ch. 0 §0.7 | **Branch:** EEE (FeRAM non-volatile memory; electro-optic devices) | → **Ch. 10**

Magnetostriction A ferromagnetic material changes dimensions when magnetized (Joule magnetostriction); stress changes its magnetization (Villari effect). **Layer:** L1→L2 | **Revealed by:** Bridge B (Ch. 13): spin-orbit coupling couples magnetic order to elastic strain | **Branch:** ME/EEE (sonar transducers — Terfenol-D; vibration energy harvesting; magnetomechanical sensors) | → **Ch. 13**

Shape Memory Effect Deformation at low temperature is fully recovered on heating through a reversible martensitic phase transformation. **Layer:** L2 | **Revealed by:** Bridge B.b (Ch. 10): first-order structural phase transition; Landau theory with shear strain as order parameter | **Branch:** ME/CE/Biomedical (arterial stents; actuators; orthodontic wires; pipe couplings) | → **Ch. 10**

Coulomb Blockade In a nanoscale conductor with small capacitance C, charging energy $E_C = e^2/2C$ suppresses tunneling unless exactly compensated by gate voltage; conductance shows Coulomb oscillations. **Layer:** L1→L2 | **Revealed by:** Ch. 7: classical charging energy (L2 capacitance) + quantum tunneling (L1 Ch. 4) meet in the mesoscopic regime | **Branch:** EEE (single-electron transistors; charge sensing; qubit implementations) | → **Ch. 7**

2.3 — Electromagnetic Phenomena

Faraday's Law of Induction A changing magnetic flux through a loop induces EMF $\varepsilon = -d\Phi_B/dt$; Lenz's law says the induced current opposes the change. **Layer:** L2 | **Revealed by:** Bridge B.c (Ch. 11): Bianchi identity $\partial[\mu F\nu\rho] = 0$ gives $\nabla\times E = -\partial B/\partial t$ directly from the U(1) sector of Ch. 0 | **Branch:** EEE (transformers, generators, motors, inductive charging, wireless power) | → **Ch. 11**

Skin Effect AC current concentrates near the conductor surface; penetration depth $\delta = \sqrt{(2/\mu\sigma\omega)}$ decreasing as $1/\sqrt{f}$. **Layer:** L2 | **Revealed by:** Bridge B.c + Ohm's law: diffusion equation $\nabla^2 E = \mu\sigma\, \partial_t E$ has exponentially decaying solutions | **Branch:** EEE (RF conductor design, transformer iron losses, induction heating, EMI shielding) | → **Ch. 11**

Eddy Currents Closed current loops induced in a bulk conductor by changing magnetic flux; dissipate energy and oppose the inducing change. **Layer:** L2 | **Revealed by:** Bridge B.c (Ch. 11): Faraday's law + Ohm's law inside a bulk conductor | **Branch:** EEE/ME (transformer losses; induction motors; eddy-current brakes; non-destructive testing) | → **Ch. 11**

Faraday Rotation Plane of polarization rotates in a magnetized medium by $\theta_F = VBd$; sign is independent of propagation direction (non-reciprocal). **Layer:** L2 | **Revealed by:** Bridge B.c (Ch. 12): B breaks time-reversal symmetry; off-diagonal dielectric tensor elements from Zeeman splitting | **Branch:** EEE (optical isolators; fiber-optic current sensors; gyroscopes) | **→ Ch. 12**

Pockels Effect Refractive index changes linearly with electric field ($\Delta n \propto E$) in non-centrosymmetric crystals; enables phase modulation at GHz rates. **Layer:** L1→L2 | **Revealed by:** Bridge B (Ch. 13): broken inversion + field perturbation of electronic states gives $\chi^{(2)} \neq 0$ | **Branch:** EEE (electro-optic modulators in fiber-optic communications; $LiNbO_3$ devices) | **→ Ch. 13**

Kerr Electro-Optic Effect Refractive index changes quadratically with field ($\Delta n \propto E^2$); exists in all media including centrosymmetric. **Layer:** L1→L2 | **Revealed by:** Bridge B (Ch. 13): third-order susceptibility $\chi^{(3)}$ always nonzero | **Branch:** EEE (ultrafast optical switching; Q-switching; Kerr-lens mode-locking) | **→ Ch. 13**

Birefringence Anisotropic medium has different refractive indices for different polarizations; incident beam splits into ordinary and extraordinary rays. **Layer:** L2 | **Revealed by:** Bridge B.c (Ch. 12): anisotropic dielectric tensor ε_{ij} from crystal symmetry | **Branch:** EEE (waveplates, polarizing beam splitters, LCD displays, optical coherence tomography) | **→ Ch. 12**

Thermionic Emission Electrons heated above the work function escape from a metal surface; Richardson-Dushman: $J = AT^2\exp(-W/k_BT)$. **Layer:** L1→L2 | **Revealed by:** Bridge B.b (Ch. 14): high-energy tail of Fermi-Dirac distribution at elevated T | **Branch:** EEE (electron guns, X-ray tubes, vacuum tubes, thermionic energy converters) | **→ Ch. 14**

Seebeck Effect A temperature gradient across dissimilar materials drives EMF $V = S\alpha\beta\Delta T$; the operating principle of thermocouples. **Layer:** L1→L2 | **Revealed by:** Bridge B.e (Ch. 13): off-diagonal Kubo coefficient L_{QE} relates temperature gradient to charge current | **Branch:** ME/EEE (thermocouples; thermoelectric generators for waste-heat recovery) | **→ Ch. 13**

Peltier Effect Current through a thermoelectric junction absorbs/releases heat at rate $\dot{Q} = \Pi I$ (Peltier coefficient $\Pi = ST$ by Kelvin's relation). **Layer:** L1→L2 | **Revealed by:** Bridge B.e (Ch. 13): same Onsager off-diagonal element as Seebeck | **Branch:** EEE/ME (solid-state cooling; laser diode stabilization; precision calorimetry) | **→ Ch. 13**

Thomson Effect Current in a temperature-gradient conductor absorbs/releases heat at rate $\dot{q} = \tau J\, dT/dx$ (Thomson coefficient τ); distinct from Joule heating ($\propto J^2$). **Layer:** L1→L2 | **Revealed by:** Bridge B.e (Ch. 13): third Onsager coefficient; $\tau = T\, dS/dT$ | **Branch:** EEE/ME (complete thermoelectric module efficiency requires all three effects) | **→ Ch. 13**

Joule Heating Current I through resistance R dissipates power $P = I^2R = V^2/R$ regardless of current direction. **Layer:** L2→L3 | **Revealed by:** Bridge B.e (Ch. 13): diagonal Kubo → Ohm's law → $P = J \cdot E$ per unit volume | **Branch:** EEE/ME (conductor sizing; fuse design; electric heating; thermal management in electronics) | **→ Ch. 13, 15**

2.4 — Optical and Wave Effects

Wave Interference Coherent waves superpose; amplitude enhancement where phases align, cancellation where they oppose. **Layer:** L2 | **Revealed by:** Bridge B.c (Ch. 11): linearity of Maxwell's equations → superposition; same in acoustics from linear Navier-Stokes | **Branch:** EEE (antenna arrays, interferometric sensors, thin-film coatings, holography) | **→ Ch. 12**

Diffraction A wave at an aperture spreads beyond geometric shadow; far-field pattern is the Fourier transform of the aperture function. **Layer:** L2 | **Revealed by:** Bridge B.c (Ch. 11): Huygens-Fresnel from Maxwell; Kirchhoff diffraction integral | **Branch:** EEE (imaging resolution limits; X-ray crystal structure determination) | **→ Ch. 12**

Snell's Law (Refraction) At an interface: $n_1 \sin\theta_1 = n_2 \sin\theta_2$; wave bends toward normal when entering denser medium. **Layer:** L2 | **Revealed by:** Bridge B.c (Ch. 11): tangential E, H boundary conditions; equivalent to Fermat's principle (Ch. 1) | **Branch:** EEE (lens design, fiber optics, waveguides); CE (acoustic refraction in layered soils) | **→ Ch. 12**

Total Internal Reflection Above critical angle $\theta_c = \arcsin(n_2/n_1)$, the wave in the denser medium is completely reflected; an evanescent field decays into the less dense medium. **Layer:** L2 | **Revealed by:** Bridge B.c (Ch. 11): Snell → $\sin\theta_t > 1$ has no real solution; $|r|^2 = 1$ | **Branch:** EEE (optical fiber waveguiding; integrated photonics; frustrated TIR biosensors) | **→ Ch. 12**

Brewster's Angle At $\theta_B = \arctan(n_2/n_1)$, reflected p-polarization vanishes; reflected beam is completely s-polarized. **Layer:** L2 | **Revealed by:** Bridge B.c (Ch. 11): Fresnel $r_p = 0$ when reflected and refracted rays are orthogonal | **Branch:** EEE (anti-reflection for laser systems; polarizing beam splitters) | **→ Ch. 12**

Rayleigh Scattering Scattered intensity $\propto \lambda^{-4}$ from particles much smaller than wavelength; sky is blue, sunsets are red. **Layer:** L2 | **Revealed by:** Bridge B.c (Ch. 11): oscillating dipole radiation power $\propto \omega^4$ | **Branch:** EEE (dominant fiber-optic attenuation below 1550 nm); CE/ME (atmospheric optics, LIDAR) | **→ Ch. 12**

Raman Scattering Inelastically scattered light shifts in frequency by a molecular/crystal vibration frequency; Stokes (photon loses energy) and anti-Stokes (photon gains energy). **Layer:** L1→L2 | **Revealed by:** Ch. 6 + nonlinear optics: photon-phonon interaction via $\chi^{(2)}$ or $\chi^{(3)}$ | **Branch:** ChE/ME (material identification; semiconductor stress measurement; in-situ reaction monitoring) | **→ Ch. 6**

Brillouin Scattering Light scattered by acoustic phonons (thermal sound waves) shifts by the phonon frequency; used for distributed structural sensing. **Layer:** L1→L2 | **Revealed by:** Bridge B (Ch. 13): quantized elastic waves + photoelastic coupling | **Branch:** EEE/CE (Brillouin OTDR for structural health monitoring in pipelines, bridges, dams) | **→ Ch. 13**

Cherenkov Radiation Charged particle traveling faster than c/n emits a cone of radiation; $\cos\theta = c/nv$. **Layer:** L2 | **Revealed by:** Bridge B.c (Ch. 11): constructive interference of the classical radiation cone; nuclear reactor pool glows blue from this | **Branch:** EEE (particle detectors; Cherenkov counters for particle ID) | → **Ch. 12**

Bremsstrahlung Decelerated charged particle emits a broad photon spectrum; high-energy cutoff at $h\nu_{max} = eV$ (Duane-Hunt). **Layer:** L0→L2 | **Revealed by:** Bridge B.c (Ch. 11) classically; quantum correction gives the spectrum | **Branch:** EEE (X-ray tubes; synchrotrons; radiation shielding design) | → **Ch. 12**

Doppler Effect Wave frequency shifts with relative motion of source and observer; EM: $f_{obs} = f_0\sqrt{((1-\beta)/(1+\beta))}$; acoustic formula differs because sound has a medium. **Layer:** L2 | **Revealed by:** Bridge B.c (Ch. 11 EM): Lorentz transform of wave 4-vector; classical waves for acoustics | **Branch:** EEE (radar, lidar, Doppler ultrasound); ME/CE (acoustic flow meters, vibration analysis) | → **Ch. 12**

Blackbody Radiation Thermal equilibrium emission follows Planck's law $B_\nu = (2h\nu^3/c^2)/(e^{(h\nu/k_BT)}-1)$; total power $\propto T^4$ (Stefan-Boltzmann); peak wavelength $\propto 1/T$ (Wien). **Layer:** L0→L2 | **Revealed by:** Bridge B.b (Ch. 10): quantized EM field energy (L0) + Bose-Einstein distribution (Bridge B.b); classical Rayleigh-Jeans gives the UV catastrophe | **Branch:** ME/CE/EEE (furnace design, thermal imaging, solar cell spectrum, satellite thermal control) | → **Ch. 10**

Optical Activity A chiral medium rotates the polarization plane of light; opposite enantiomers rotate in opposite directions. **Layer:** L1→L2 | **Revealed by:** Bridge A+B (Ch. 12): molecular chirality at L1 → non-symmetric susceptibility tensor at L2 | **Branch:** ChE (pharmaceutical purity; sugar concentration); Biomedical (circular dichroism spectroscopy) | → **Ch. 12**

Second Harmonic Generation (SHG) High-intensity light at ω in a non-centrosymmetric medium generates output at 2ω; requires phase matching. **Layer:** L1→L2 | **Revealed by:** Bridge B (Ch. 13): $\chi^{(2)}$ from broken inversion symmetry; nonlinear polarization $P^{(2)} = \varepsilon_0\chi^{(2)}E^2$ at 2ω | **Branch:** EEE (green laser pointers from 1064 nm YAG; frequency doubling in photonics labs) | → **Ch. 13**

2.5 — Mechanical, Fluid, and Acoustic Effects

Bernoulli Effect Steady inviscid flow: $p + \frac{1}{2}\rho v^2 + \rho g z = \text{const}$ along a streamline; pressure decreases where flow speed increases. **Layer:** L2 | **Revealed by:** Bridge B.b (Ch. 16): Euler equation (Navier-Stokes, $\eta = 0$) integrated along a streamline | **Branch:** ME/CE (aerodynamics, hydraulics, Venturi meters, Pitot tubes, carburetors) | → **Ch. 16**

Magnus Effect Spinning body in cross-flow experiences lift $F_L = \rho\Gamma L$ perpendicular to both velocity and spin axis (Kutta-Joukowski). **Layer:** L2 | **Revealed by:** Bridge B.b (Ch. 16): Navier-Stokes + rotation creates asymmetric pressure distribution | **Branch:** ME/CE (sports aerodynamics; Flettner rotor ships; gyroplane rotors) | → **Ch. 16**

Coanda Effect A fluid jet attaches to and follows an adjacent curved surface due to entrainment and resulting low pressure. **Layer:** L2 | **Revealed by:** Bridge B.b (Ch. 16): viscous entrainment creates sub-ambient pressure between jet and surface | **Branch:** ME/CE (fluidic devices; VTOL aircraft; high-lift airfoils; industrial burners) | → **Ch. 16**

Reynolds Number Transition / Turbulence Flow is laminar below critical Re = ρvL/η; nonlinear inertial terms dominate above it and flow becomes chaotic turbulence; no closed-form Layer-3 model exists. **Layer:** L2 | **Revealed by:** Bridge B.b (Ch. 16): nonlinear $(v\cdot\nabla)v$ term in Navier-Stokes; where L2 refuses to lump cleanly | **Branch:** ME/CE/ChE (pipe flow, aerodynamics, reactor mixing, heat exchanger performance) | → **Ch. 16**

Kelvin-Helmholtz Instability Small perturbations at a shear interface between fluids of different velocities grow exponentially into vortices. **Layer:** L2 | **Revealed by:** Bridge B.b (Ch. 16): linear stability of Navier-Stokes at a shear interface | **Branch:** ME/CE (ocean-atmosphere interface; fuel atomization; wind-driven waves; cloud streaks) | → **Ch. 16**

Rayleigh-Taylor Instability Dense fluid on top of light fluid is unstable; perturbations grow with rate $\gamma = \sqrt{(Akg)}$ where A is the Atwood number. **Layer:** L2 | **Revealed by:** Bridge B.b + B.d (Ch. 16): Navier-Stokes + gravity; buoyancy on density perturbations | **Branch:** CE/ME (geophysical flows; nuclear fusion implosion; ocean overturning; salt diapirs) | → **Ch. 16**

Rayleigh-Bénard Convection Organized convection cells form above critical Rayleigh number $Ra = g\alpha\Delta T d^3/\nu\kappa$ in a fluid heated from below; canonical pattern-formation example. **Layer:** L2 | **Revealed by:** Bridge B.b (Ch. 16): Navier-Stokes + thermal diffusion + buoyancy; linear stability → critical Ra | **Branch:** ME/CE/ChE (building HVAC, electronics cooling, geophysical convection, industrial furnaces) | → **Ch. 16**

Acoustic Cavitation Large rarefaction pressures nucleate bubbles that collapse violently, producing ~5000 K and ~1000 atm locally. **Layer:** L2 | **Revealed by:** Bridge B.b (Ch. 16): Rayleigh-Plesset bubble dynamics from Navier-Stokes | **Branch:** ME/ChE (ultrasonic cleaning, sonochemistry, food processing); CE (pump and propeller erosion) | → **Ch. 16**

Resonance A system driven at natural frequency $\omega_0 = 1/\sqrt{(LC)}$ responds maximally; Q factor quantifies sharpness; all domains share the same second-order ODE. **Layer:** L2→L3 | **Revealed by:** Bridge C (Ch. 15/18): $L\ddot{x} + R\dot{x} + x/C = F$ has a resonant peak identical in every physical domain | **Branch:** All (RLC circuits, mechanical vibration, acoustic cavities, MEMS, optical resonators) | → **Ch. 18**

Surface Acoustic Waves (SAW) Acoustic waves guided along a solid surface, decaying exponentially with depth; Rayleigh waves (elliptical motion) and Love waves (transverse) are primary modes. **Layer:** L2 | **Revealed by:** Bridge B (Ch. 9/16): elastic wave equation + free-surface boundary condition | **Branch:** EEE/ME/CE (SAW filters in mobile phones; chemical sensors; structural health monitoring) | → **Ch. 16**

Thermal Expansion Most materials expand on heating because thermal vibrations in an anharmonic interatomic potential give a positive mean displacement. **Layer:** L1→L2 | **Revealed by:** Bridge B (Ch. 10): anharmonic interatomic potential (L1) → positive Grüneisen parameter → macroscopic $\alpha = (1/V)(\partial V/\partial T)_P$ | **Branch:** ME/CE (expansion joints, bimetallic actuators, precision machining, bridge joints) | → **Ch. 10**

2.6 — Gravitational and Relativistic Effects

Equivalence Principle Gravitational and inertial mass are exactly equal; free fall is locally indistinguishable from absence of gravity. **Layer:** L0 | **Revealed by:** Ch. 0 §0.3 + Ch. 8: geodesic equation describes free fall with no force; gravity is geometry | **Branch:** ME (inertial navigation; GPS/INS fusion needed because accelerometers cannot separate gravity from acceleration) | → **Ch. 8**

Gravitational Time Dilation Clocks deeper in a gravitational well run slower; $\Delta f/f = \Delta\Phi/c^2$. **Layer:** L0 | **Revealed by:** Ch. 8 Bridge B.d: metric $g_{00} = 1 + 2\Phi/c^2$; proper time $d\tau^2 = g_{00}\, dt^2$ | **Branch:** EEE (GPS: without +45 μs/day correction positional error is ~11 km/day) | → **Ch. 8**

Gravitational Redshift Photons climbing out of a gravitational well lose energy and shift to lower frequency; $\Delta\nu/\nu = -\Delta\Phi/c^2$. **Layer:** L0 | **Revealed by:** Ch. 8: energy conservation for photon $E = h\nu$ in gravitational field; confirmed by Pound-Rebka 1959 | **Branch:** EEE (precision atomic clocks; gravitational wave detector calibration) | → **Ch. 8**

Gravitational Waves Accelerating masses produce spacetime ripples propagating at c; strain $h = \Delta L/L$; first detected by LIGO in 2015. **Layer:** L0 | **Revealed by:** Ch. 8: linearized Einstein equations give wave equation for $h_{\mu\nu}$; two transverse-traceless polarizations | **Branch:** EEE (gravitational wave detectors are precision optical interferometers at Layer 3) | → **Ch. 8**

Gravitational Lensing Massive objects deflect light by $\alpha = 4GM/c^2b$ (twice Newtonian); confirmed by Eddington 1919. **Layer:** L0 | **Revealed by:** Ch. 8: photon geodesic in Schwarzschild metric; factor 2 over Newton because both spatial and temporal metric components contribute | **Branch:** Astronomy (dark matter mapping; gravitational telescopes) | → **Ch. 8**

Frame Dragging (Lense-Thirring) Rotating massive body drags surrounding spacetime; satellite orbital plane precesses at $\Omega_{LT} = 2GJ/c^2r^3$. **Layer:** L0 | **Revealed by:** Ch. 8: off-diagonal g_{0i} components of Kerr metric; confirmed Gravity Probe B 2011 | **Branch:** EEE (precise orbit determination for navigation satellites) | → **Ch. 8**

2.7 — Nuclear and Particle Effects

Radioactive Decay (α, β, γ) Unstable nuclei transform spontaneously: α-decay (He nucleus tunnels through Coulomb barrier); β-decay (n↔p via weak SU(2) force); γ-emission (nuclear excitation photon). **Layer:** L0→L1 | **Revealed by:** Ch. 0 §0.4.3 (SU(2) → β); Ch. 4 tunneling → α; Ch. 5 nuclear levels → γ |

Branch: ME/CE/ChE (nuclear reactors, radiation shielding, medical imaging, geological dating) | → **Ch. 6**

Nuclear Fission A fissile nucleus absorbs a slow neutron, splits into two lighter nuclei, releases ~2.4 neutrons and ~200 MeV per event; chain reaction possible. **Layer:** L0→L1 | **Revealed by:** Ch. 6: nuclear binding energy curve (SU(3) residual) + fission barrier tunneling | **Branch:** ME/ChE (nuclear power plants; naval propulsion; weapon physics) | → **Ch. 6**

Nuclear Fusion Light nuclei fuse under extreme conditions; stellar fusion requires quantum tunneling through Coulomb barrier at thermal velocities. **Layer:** L0→L1 | **Revealed by:** Ch. 6: strong nuclear force + Gamow tunneling peak; plasma confinement physics | **Branch:** ME (fusion reactor engineering — ITER; plasma engineering) | → **Ch. 6**

Pair Production and Annihilation Photon near nucleus converts to e^+e^- pair ($h\nu \geq 2m_ec^2 = 1.022$ MeV); pair annihilates into two back-to-back 511-keV photons. **Layer:** L0 | **Revealed by:** Ch. 0 §0.5: antimatter solutions of Dirac equation; fundamental QED process | **Branch:** EEE (PET scanners; radiation detectors; radiotherapy) | → **Ch. 3**

Mössbauer Effect A nucleus in a crystal emits or absorbs a γ-ray with no recoil (whole crystal recoils); linewidth $\Delta E/E \sim 10^{-13}$. **Layer:** L1 | **Revealed by:** Ch. 6: zero-phonon transition; Lamb-Mössbauer factor gives recoil-free fraction | **Branch:** ME/CE (residual stress in steel; geological dating; Pound-Rebka gravitational redshift test) | → **Ch. 6**

Neutrino Oscillations Neutrinos produced in flavor eigenstates (ν_e, $\nu\mu$, $\nu\tau$) propagate as mass eigenstate superpositions; flavor oscillates with distance $L_{osc} \propto E/\Delta m^2$. **Layer:** L0 | **Revealed by:** Ch. 0 §0.8 (PMNS matrix); requires neutrino masses → extension of L_{SM} → in the $\mathcal{F}[\ldots]$ unknown sector; direct evidence the action in Ch. 0 is incomplete | **Branch:** Metrology (reactor neutrino monitoring); physics frontier | → **Epilogue**

2.8 — Surface and Interface Phenomena

Surface Tension / Young-Laplace Liquid-vapor interface has surface energy γ per area; pressure difference across curved interface $\Delta p = \gamma(1/R_1 + 1/R_2)$. **Layer:** L1→L2 | **Revealed by:** Bridge B.b (Ch. 10): broken bonds at surface raise free energy; thermodynamic minimization gives Young-Laplace | **Branch:** ChE/ME/CE (bubble dynamics, droplet coalescence, capillary pressure in porous media, coating flows) | → **Ch. 16**

Capillary Action Liquid rises in a narrow tube to $h = 2\gamma\cos\theta/(\rho g r)$; balance of surface tension and hydrostatic pressure. **Layer:** L2 | **Revealed by:** Bridge B.b + B.d (Ch. 10 + Ch. 8): Young-Laplace pressure + hydrostatic equilibrium | **Branch:** CE/ChE (groundwater, inkjet, microfluidics, concrete moisture, lab-on-chip) | → **Ch. 16**

Wetting and Contact Angle Contact angle θ satisfies Young's equation $\gamma_{SV} = \gamma_{SL} + \gamma_{LV}\cos\theta$; determines whether a liquid spreads on a surface. **Layer:** L1→L2 | **Revealed by:** Bridge B.b (Ch. 10): free energy minimization over three interface energies; molecular origins (L1) determine γ values | **Branch:** ME/ChE/CE (waterproofing; adhesion; coating; microfluidic valves; anti-icing surfaces) | **→ Ch. 16**

Osmotic Pressure Solution separated from pure solvent by semipermeable membrane develops pressure $\Pi = cRT$ (van't Hoff, dilute limit). **Layer:** L2 | **Revealed by:** Bridge B.b (Ch. 10): solute lowers chemical potential of solvent (entropy of mixing); thermodynamic equilibrium gives van't Hoff | **Branch:** ChE/CE/Biomedical (desalination, drug delivery, cell mechanics, wastewater treatment) | **→ Ch. 10**

Marangoni Effect (Thermocapillary Flow) Gradients in surface tension (from T or concentration gradients) drive surface flow from low to high surface tension; causes "tears of wine." **Layer:** L2 | **Revealed by:** Bridge B.b + B.c (Ch. 16): temperature/concentration-dependent γ → tangential stress boundary condition in Navier-Stokes | **Branch:** ME/ChE (crystal growth, thin-film uniformity, welding pool dynamics, inkjet) | **→ Ch. 16**

Leidenfrost Effect A droplet on a surface far above boiling point levitates on a vapor cushion, dramatically reducing heat transfer and extending lifetime. **Layer:** L2 | **Revealed by:** Bridge B.b (Ch. 16): vapor pressure exceeds ambient; vapor film insulates droplet by buoyancy and pressure | **Branch:** ME/ChE (spray cooling; nuclear safety — departure from nucleate boiling; quenching; droplet manipulation) | **→ Ch. 16**

Electrowetting Applied voltage modifies contact angle of a conducting droplet: $\cos\theta(V) = \cos\theta_0 + \varepsilon V^2/2\gamma d$; contact angle decreases with voltage. **Layer:** L1→L2 | **Revealed by:** Bridge B.c + B.b (Ch. 11, 10): electric field energy in dielectric modifies effective solid-liquid surface energy | **Branch:** EEE/ChE (lab-on-chip; digital microfluidics; electrowetting displays; adaptive lenses) | **→ Ch. 16**

Triboelectric Effect Contact and separation of dissimilar materials transfers charge via surface electron transfer; sign depends on triboelectric series. **Layer:** L1→L2 | **Revealed by:** Bridge A+B (Ch. 13): work function difference (L1) drives charge transfer; detailed mechanism for insulators still debated | **Branch:** EEE (triboelectric nanogenerators for energy harvesting; ESD hazard in manufacturing) | **→ Ch. 13**

2.9 — Cross-Domain and Emerging Phenomena

Johnson-Nyquist Noise (Thermal Noise) A resistor R at temperature T generates voltage noise $S_V = 4k_BTR$ (white spectrum); unavoidable floor on electronic signal detection. **Layer:** L1→L2 | **Revealed by:** Bridge B.e (Ch. 13): fluctuation-dissipation theorem connects Kubo conductivity to equilibrium fluctuations; $S_V = 4k_BT\,\mathrm{Re}[Z(\omega)]$ | **Branch:** EEE (noise floor of all analog systems; antenna temperature; precision measurement) | **→ Ch. 13**

Shot Noise Discreteness of electron charge gives current noise $S_I = 2eI$ (Poissonian); Fano factor $F = S_I/2eI$ measures deviation from Poissonian due to correlations. **Layer:** L1→L2 | **Revealed by:** Bridge B.e (Ch. 13): quantized charge (Noether U(1) from Ch. 0) + Poisson statistics of independent tunnel events

| **Branch:** EEE (detector sensitivity limits; qubit readout noise; characterizing quantum transport) | → **Ch. 13**

Surface Plasmon Resonance (SPR) Collective electron oscillations at metal-dielectric interface couple resonantly to evanescent EM waves; resonance is exquisitely sensitive to dielectric environment. **Layer:** L1→L2 | **Revealed by:** Bridge B.c (Ch. 11): plasma frequency $\omega_p = \sqrt{(ne^2/\varepsilon_0 m)}$ (L1 electron density) + evanescent wave phase-matching condition | **Branch:** EEE/ChE (label-free biosensors for antigen-antibody binding; nanophotonic circuits) | → **Ch. 13**

Photovoltaic Effect Photons absorbed in a p-n junction generate electron-hole pairs; built-in field separates them; quasi-Fermi level splitting gives open-circuit voltage. **Layer:** L1→L2 | **Revealed by:** Bridge A+B: photoelectric effect (L0→L1, Ch. 4) + band structure (L1, Ch. 6) + drift-diffusion from Kubo (Bridge B.e) | **Branch:** EEE (solar cells; photodetectors; image sensors) | → **Ch. 13**

Phonons Quantized collective lattice vibrations; acoustic (gapless, in-phase) and optical (gapped, out-of-phase) branches; carry most heat in dielectrics. **Layer:** L1 | **Revealed by:** Ch. 6: many-body quantization of lattice vibrations; phonon is the boson of the elastic field, parallel to photon in EM | **Branch:** ME/EEE (thermal conductivity; electron-phonon scattering → R vs. T in metals; thermoelectric optimization) | → **Ch. 6**

Plasma Oscillations (Langmuir Waves) Free electrons displaced from equilibrium oscillate collectively at $\omega_p = \sqrt{(ne^2/\varepsilon_0 m_e)}$; EM waves cannot propagate for $\omega < \omega_p$. **Layer:** L1→L2 | **Revealed by:** Bridge B.c (Ch. 11): Kubo conductivity → complex $\varepsilon(\omega)$ → $\mathrm{Re}[\varepsilon] = 0$ at ω_p | **Branch:** EEE (RF shielding; waveguide cutoff; surface plasmonics; UV reflectance edge of metals) | → **Ch. 13**

Electrophoresis Charged particles migrate in an applied electric field at velocity $v = \mu_E E$ where $\mu_E = \varepsilon\zeta/\eta$ (zeta potential ζ, viscosity η); electric and drag forces balance. **Layer:** L2 | **Revealed by:** Bridge B.c + B.b (Ch. 11, 16): Lorentz force on charged particle + Stokes drag from Navier-Stokes | **Branch:** ChE/Biomedical (gel electrophoresis for DNA; capillary electrophoresis; colloid separation) | → **Ch. 16**

Fluorescence Photon absorption excites a molecule; after rapid vibrational relaxation, emission occurs at longer wavelength (Stokes shift); lifetime 1–10 ns. **Layer:** L1 | **Revealed by:** Ch. 5: electronic transitions between molecular orbitals; Franck-Condon (Born-Oppenheimer); Fermi's Golden Rule gives the rate | **Branch:** EEE/ChE/Biomedical (fluorescence microscopy; FRET sensors; LEDs; laser media; flow cytometry) | → **Ch. 5**

NMR and MRI Nuclei with nonzero spin precess at Larmor frequency $\omega_L = \gamma_N B_0$ in static field B_0; resonant RF pulse tips magnetization; free induction decay encodes structural information. **Layer:** L1 | **Revealed by:** Ch. 5: nuclear Zeeman effect; nuclear spin precession from the Pauli equation | **Branch:** EEE/Biomedical (MRI for medical imaging; NMR spectroscopy; NMR quantum computing) | → **Ch. 5**

2.10 — Master Index (Alphabetical)

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End of Chapter 2. Return to it whenever a named effect appears in a later chapter.

Next: Chapter 3 — Bridge A: The Quantum Descent

Bridge A — The Quantum Descent

From the Standard Model Lagrangian to the Schrödinger Equation

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> *The Dirac equation is the most beautiful equation in physics. It predicted spin, antimatter, and the fine structure of hydrogen before anyone had asked it to.* — attributed to Paul Dirac (paraphrased)

3.0 — Where We Are and Where We Are Going

In Chapter 0 you saw the full action. Every term was labeled and every field was named. Now we begin the actual descent.

This chapter performs the first controlled approximation — the one that takes you from quantum field theory (the full action) to single-particle quantum mechanics (the Schrödinger equation). This transition is called **Bridge A** in the layer map of this book.

At the end of this chapter you will have the Schrödinger equation written on your page. But unlike every other physics course, you will know exactly where it came from, what was thrown away to get it, and precisely when you are allowed to use it. That knowledge — not the equation itself — is what this chapter delivers.

The map of this chapter:

```
L_SM (Chapter 0)
      │
      │  Step 1: Isolate the U(1)-coupled fermion sector (§3.1)
      │  [discard SU(3) and SU(2): not relevant for electrons in atoms]
      ▼
The Dirac Lagrangian: L = ψ̄(iγᵘDμ - m)ψ
      │
      │  Step 2: Euler-Lagrange equations (§3.2)
      ▼
The Dirac Equation: iγᵘDμψ = mψ        ← Bridge zone (§3.3–3.5)
      │
      │  Step 3: Foldy-Wouthuysen transformation, order (v/c)¹ (§3.6)
      ▼
The Pauli Equation: spin included, relativistic corrections at O(v/c)²
      │
      │  Step 4: Drop spin-field coupling for spinless or B=0 case (§3.7)
      ▼
The Schrödinger Equation: iħ ∂ψ/∂t = [-ħ²∇²/2m + V]ψ    ← Layer 1
```

3.1 — Isolating the Right Sector

3.1.1 — Which Part of $\mathcal{L}_{SM}$ We Are Using

The full Standard Model Lagrangian (Chapter 0) contains three gauge sectors: $\mathrm{SU}(3)_C$, $\mathrm{SU}(2)_L$, and $\mathrm{U}(1)_Y$. For electrons in atoms at energies $E \ll m_W c^2 \approx 80$ GeV:

- **SU(3)$_C$:** Acts only on quarks. Electrons carry no color charge. This sector is completely invisible to electrons. Discard it.

- **SU(2)$_L$:** Acts on left-handed fermions and is mediated by $W^{\pm}$ and Z^0 bosons. The dimensionless weak coupling at energy E is $G_F E^2$ (with $G_F \approx 1.17 \times 10^{-5}\,\text{GeV}^{-2}$, so E must be expressed in GeV). For an atomic electron, $E \sim 10\,\text{eV} = 10^{-8}\,\text{GeV}$, giving $G_F E^2 \sim 1.17 \times 10^{-5} \times 10^{-16} \sim 10^{-21}$. Measured against the electromagnetic coupling $\alpha \approx 7.3 \times 10^{-3}$, the ratio is

$$\frac{G_F E^2}{\alpha} \sim \frac{1.2 \times 10^{-21}}{7.3 \times 10^{-3}} \sim 10^{-19}$$

small enough to discard for atomic structure — but it is three orders of magnitude larger than the value a careless estimate would suggest, and the order-of-magnitude is not the interesting part. What matters is the reasoning. Two traps are worth flagging. G_F carries units of GeV^{-2}, so $G_F E^2$ is dimensionless *only* if E is in GeV; writing $E \sim 10$ without converting silently gains or loses nine orders of magnitude. And there is no $\hbar c$ to divide out, because the combination is already dimensionless in natural units. The ratio is also energy-dependent: at $E \sim 1$ GeV the same calculation gives $\sim 10^{-3}$, which is precisely why the weak interaction dominates at collider energies while being irrelevant in atoms. Discard it.

- **U(1)$_{EM}$:** Electromagnetism. After the Higgs mechanism (§0.7), the physical photon A_μ couples to electrons with charge $e = -|e|$ (throughout this chapter e denotes the particle's *signed* charge, so every formula below holds for any charge and is evaluated for the electron with $e < 0$). This is the only force relevant for atomic structure.

What we keep:

$$\mathcal{L}_{electron} = \bar{\psi}\left(i\gamma^\mu \partial_\mu - m_e - e\gamma^\mu A_\mu\right)\psi$$

where A_μ is the electromagnetic four-potential ($A_0 = V/c$ is the scalar potential, $\mathbf{A}$ is the vector potential).

Using the shorthand $\not{\partial} \equiv \gamma^\mu \partial_\mu$ and $\not{A} \equiv \gamma^\mu A_\mu$:

$$\mathcal{L}_{electron} = \bar{\psi}\left(i\ \not{\partial} - m_e - e\ \not{A}\right)\psi$$

This is the **QED Lagrangian for the electron** — quantum electrodynamics reduced to its matter content. The photon's own kinetic term $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ is still present (it governs the electromagnetic field), but for atomic physics we treat A_μ as an external, given field rather than a dynamical one.

3.1.2 — What We Are Discarding and Why

By treating A_μ as a classical, external field and working at low energies, we are discarding:

Discarded	Physical content	Validity
Full QED vertex corrections	Lamb shift, $g-2$ correction to magnetic moment	Corrections of order $\alpha/\pi \approx 0.002$
Virtual pair creation loops	Vacuum polarization, Uehling potential	Order α^2 — measurable in precision spectroscopy, negligible in engineering
Pair creation processes	$\gamma\gamma \to e^+e^-$, $e^- \to e^-\gamma$	Requires $E \geq 2m_e c^2 = 1.022$ MeV — far above atomic energies

Discarded	Physical content	Validity
Quantization of the EM field	Spontaneous emission, Casimir effect	Included approximately via Einstein A coefficient; full treatment needs QFT

The validity criterion for this entire chapter:

$$\boxed{E_{kinetic} \ll m_e c^2 = 0.511\,\text{MeV}}$$

For ground-state hydrogen: $E_1 = 13.6\,\text{eV} \ll 511{,}000\,\text{eV}$. The ratio is $\sim 10^{-5}$. The approximation is excellent.

3.2 — Euler-Lagrange Descent: The Dirac Equation

3.2.1 — Applying the Euler-Lagrange Equations

The equation of motion for ψ is found by varying $\mathcal{L}_{electron}$ with respect to $\bar{\psi}$ (the Dirac adjoint). The Euler-Lagrange equation for a field is:

$$\frac{\partial \mathcal{L}}{\partial \bar{\psi}} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \bar{\psi})} = 0$$

Since $\mathcal{L}_{electron}$ depends on $\bar{\psi}$ but not on $\partial_\mu \bar{\psi}$ (the kinetic term $i\bar{\psi}\gamma^\mu \partial_\mu \psi$ is first-order in ψ, not $\bar{\psi}$), the second term vanishes. The first term gives directly:

$$\boxed{i\gamma^\mu \partial_\mu \psi - m_e \psi - e\gamma^\mu A_\mu \psi = 0}$$

Rearranging, and writing $D_\mu = \partial_\mu + ieA_\mu$ for the covariant derivative:

$$\boxed{i\gamma^\mu D_\mu \psi = m_e \psi}$$

This is the **Dirac equation** — the equation of motion of a spin-1/2 particle in an electromagnetic field, derived directly from the Lagrangian by the Euler-Lagrange procedure. It was not guessed or postulated here; it fell out of the variational principle applied to the fermion sector of the Standard Model.

3.2.2 — Restoring $\hbar$ and c

In the derivation above, we used natural units ($\hbar = c = 1$). Restoring physical units:

$$\left(i\hbar\gamma^\mu \partial_\mu - \frac{e}{c}\gamma^\mu A_\mu - m_e c\right)\psi = 0$$

Or, separating time and space components ($\mu = 0$: time; $\mu = 1, 2, 3$: space):

$$\boxed{i\hbar\frac{\partial \psi}{\partial t} = \left[c\boldsymbol{\alpha} \cdot \left(\hat{\mathbf{p}} - \frac{e}{c}\mathbf{A}\right) + \beta m_e c^2 + eV\right]\psi}$$

This is the **Dirac equation in Hamiltonian form**, where:

- $\hat{\mathbf{p}} = -i\hbar\nabla$ is the momentum operator
- $V = A_0$ is the electric scalar potential, and eV the potential energy (e is the signed charge, $e = -|e|$ for the electron)

- $\boldsymbol{\alpha}$ and β are 4×4 matrices defined by the gamma matrices

---

3.3 — Anatomy of the Dirac Equation

3.3.1 — The Gamma Matrices and the Clifford Algebra

The four gamma matrices γ^μ ($\mu = 0, 1, 2, 3$) are 4×4 complex matrices satisfying the **Clifford algebra**:

$$\{\gamma^\mu, \gamma^\nu\} = \gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2g^{\mu\nu}\mathbf{1}_{4\times 4}$$

Signature in this chapter. The Dirac algebra of this chapter is written in the particle-physics signature $g^{\mu\nu} = \text{diag}(+1, -1, -1, -1)$, so that $(\gamma^0)^2 = +\mathbf{1}$ (consistent with $\gamma^0 = \beta$ below), $(\gamma^i)^2 = -\mathbf{1}$, and $A_0 = +V/c$ as in §3.1. In the book's $(-,+,+,+)$ signature of Ch. 0 §0.2.2 the same matrices satisfy $\{\gamma^\mu, \gamma^\nu\} = -2g^{\mu\nu}\mathbf{1}_{4\times 4}$ (Ch. 0 §0.5).

In the **Dirac representation** (standard representation, most transparent for the non-relativistic limit):

$$\gamma^0 = \beta = \begin{pmatrix} \mathbf{1} & 0 \\ 0 & -\mathbf{1} \end{pmatrix}, \qquad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}$$

where $\mathbf{1}$ is the 2×2 identity matrix and σ^i are the **Pauli matrices**:

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

The **alpha matrices** $\boldsymbol{\alpha}$ appearing in the Hamiltonian form are:

$$\boldsymbol{\alpha} = \gamma^0\boldsymbol{\gamma} = \begin{pmatrix} 0 & \boldsymbol{\sigma} \\ \boldsymbol{\sigma} & 0 \end{pmatrix}$$

3.3.2 — The Four-Component Spinor

The field ψ is a **Dirac spinor** — a four-component complex column vector:

$$\psi = \begin{pmatrix} \phi \\ \chi \end{pmatrix}$$

In the non-relativistic limit, it will be shown (§3.5) that:

- ϕ: two-component **large component** — survives as the physical electron wavefunction
- χ: two-component **small component** — magnitude suppressed by (v/c) relative to ϕ; encodes relativistic corrections and the positron (antimatter) degree of freedom

The four components are not four spatial dimensions. They encode spin (up/down — two components) and particle/antiparticle (electron/positron — two more). The Lorentz group requires four dimensions for a consistent relativistic description of spin-1/2.

3.3.3 — The Dirac Hamiltonian

Write the Dirac equation in the form $i\hbar\,\partial_t\psi = \hat{H}_{Dirac}\psi$:

$$\hat{H}_{Dirac} = \begin{pmatrix} m_e c^2 + eV & c\,\boldsymbol{\sigma}\cdot\hat{\boldsymbol{\pi}} \\ c\,\boldsymbol{\sigma}\cdot\hat{\boldsymbol{\pi}} & -m_e c^2 + eV \end{pmatrix}$$

where $\hat{\boldsymbol{\pi}} = \hat{\mathbf{p}} - \frac{e}{c}\mathbf{A}$ is the **kinetic momentum** (canonical momentum minus gauge contribution).

The diagonal blocks:

- Upper-left $m_e c^2 + eV$: rest mass energy plus potential — the electron block
- Lower-right $-m_e c^2 + eV$: negative rest mass energy — the positron block

The off-diagonal blocks $c\,\boldsymbol{\sigma} \cdot \hat{\boldsymbol{\pi}}$ couple electrons and positrons. This coupling is what makes the Dirac equation an inherently relativistic, multi-particle object — and what must be systematically removed to reach the Schrödinger equation.

3.4 — What the Dirac Equation Predicts Without Additional Assumptions

Before we descend to the Schrödinger equation, it is worth pausing to appreciate what the Dirac equation gives us **automatically** — with no extra postulates:

3.4.1 — Spin-1/2

The Dirac spinor automatically has two spin degrees of freedom per particle (the two components of ϕ). The spin operator:

$$\hat{\mathbf{S}} = \frac{\hbar}{2}\begin{pmatrix} \boldsymbol{\sigma} & 0 \\ 0 & \boldsymbol{\sigma} \end{pmatrix}$$

falls directly from the spinor structure. It satisfies $\hat{S}^2 = \frac{3}{4}\hbar^2$, confirming spin-$\frac{1}{2}$.

Nobody put spin into the Dirac equation. It is there because the Lorentz group, when acting on a four-component field, has spin-$\frac{1}{2}$ built into its representation theory. The Clifford algebra $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$ (signature as in §3.3.1) is the minimal algebraic structure consistent with a first-order, Lorentz-covariant wave equation — and it forces spin-$\frac{1}{2}$ as a consequence.

3.4.2 — The Magnetic Moment: $g = 2$

For an electron in a magnetic field $\mathbf{B}$, the Dirac equation contains an interaction term:

$$\hat{H}_{mag} = -\frac{e\hbar}{2m_e c}\boldsymbol{\sigma} \cdot \mathbf{B} = -g_e \frac{e}{2m_e c}\hat{\mathbf{S}} \cdot \mathbf{B}$$

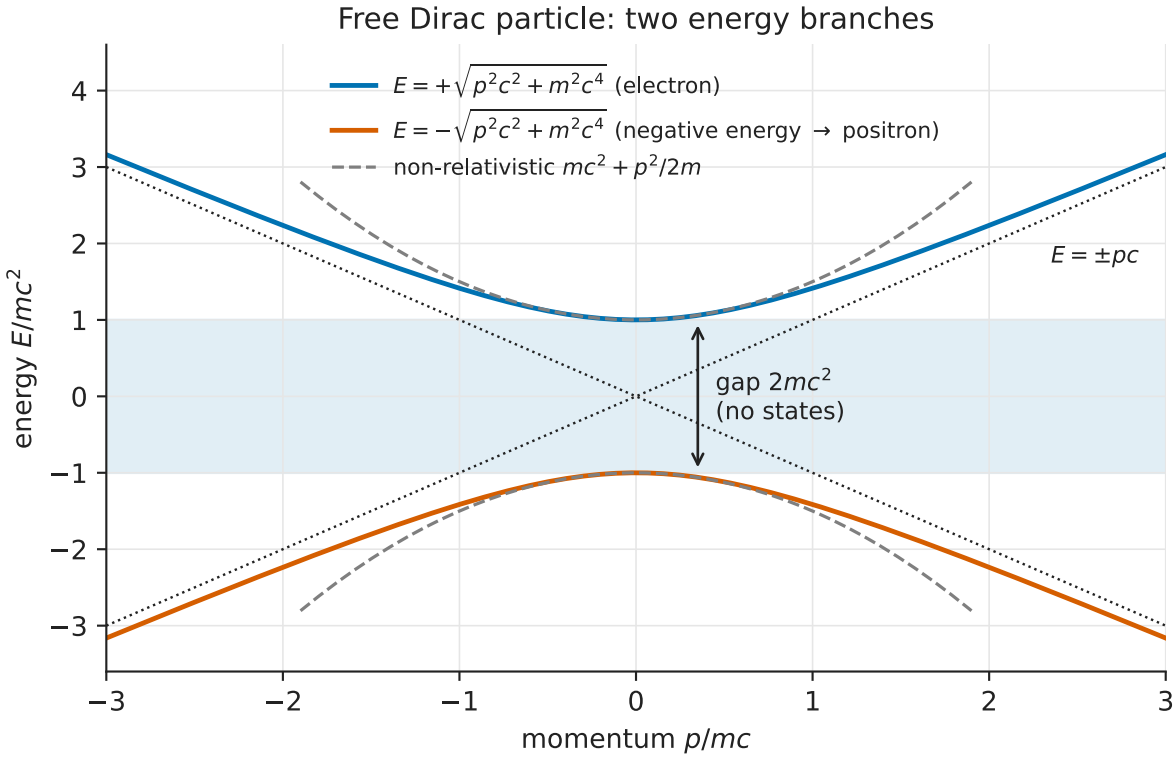
This gives $g_e = 2$ exactly — the electron has twice the magnetic moment you would expect from classical orbital motion. This was a mysterious experimental result before 1928. The Dirac equation explains it with no fitting: $g = 2$ is a mathematical consequence of the Clifford algebra together with minimal coupling $D_\mu = \partial_\mu + ieA_\mu$ (a non-minimal Pauli term $\propto \bar{\psi}\sigma^{\mu\nu}F_{\mu\nu}\psi$ is equally Lorentz- and gauge-invariant and would shift g freely).

(The small departure from exactly 2, $g_e = 2.00231930\ldots$, is the QED correction — the Lamb-shift-era legacy, discarded in §3.1.2 as α/π.)

3.4.3 — Antimatter

The negative-energy solutions of the Dirac equation — the lower two components becoming large for solutions with energy $E \approx -m_e c^2$ — were initially puzzling. Dirac interpreted them in 1930 as **positrons**: the antiparticle of the electron with the same mass and opposite charge.

The positron was discovered by Anderson in 1932. It was the first antiparticle ever found — predicted by the formalism, not hypothesized by experiment.



The two branches $E = \pm\sqrt{p^2c^2 + m^2c^4}$ of the free Dirac equation in units of mc^2 against p/mc, with the non-relativistic parabola $mc^2 + p^2/2m$ (dashed) and the light-cone asymptotes $E = \pm pc$ (dotted): no states exist in the shaded gap of width $2mc^2$, and the negative-energy branch is the one reinterpreted as the positron.

3.4.4 — Fine Structure of Hydrogen

Without any modification, the Dirac equation applied to the hydrogen atom (Coulomb potential energy $eV = -e^2/r$) gives energy levels:

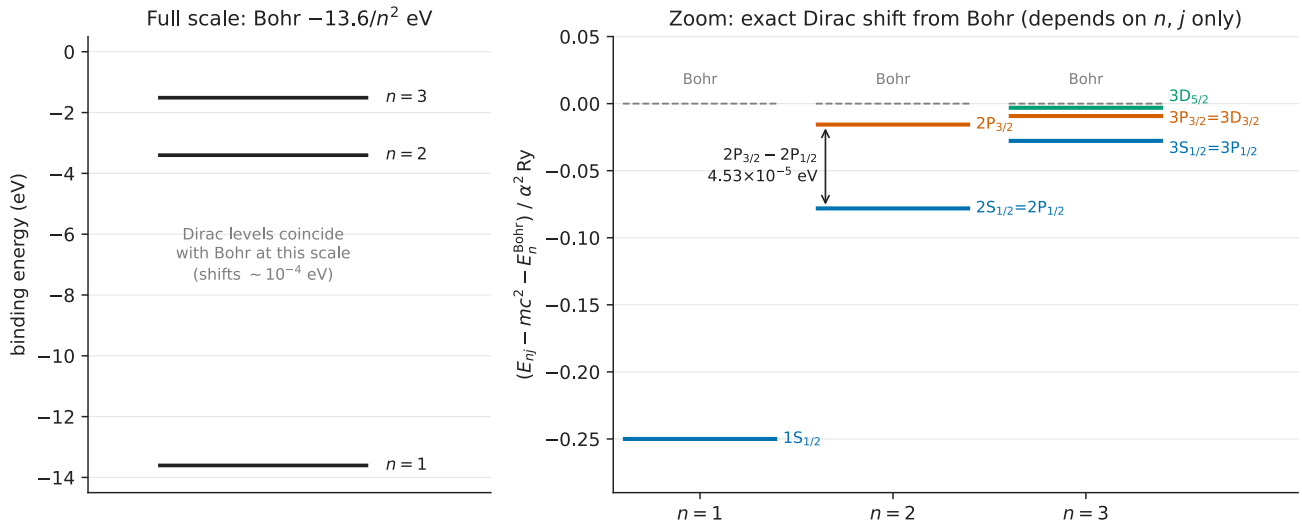
$$E_{nj} = m_e c^2 \left(1 + \left(\frac{\alpha}{n - j - \frac{1}{2} + \sqrt{(j+\frac{1}{2})^2 - \alpha^2}}\right)^2\right)^{-1/2}$$

Expanding in α^2 (where $\alpha = e^2/\hbar c \approx 1/137$ is the fine structure constant):

$$E_{nj} - m_e c^2 \approx -\frac{m_e c^2 \alpha^2}{2n^2}\left(1 + \frac{\alpha^2}{n}\left(\frac{1}{j+\frac{1}{2}} - \frac{3}{4n}\right)\right)$$

The first term is the Bohr energy $-13.6/n^2$ eV. The second term — the **fine structure correction** — splits energy levels with the same n but different j (total angular momentum). This splitting is observed in the hydrogen spectrum to exactly the precision the Dirac equation predicts.

No additional assumptions. No spin-orbit coupling inserted by hand. Just the Dirac equation and a Coulomb potential.



Hydrogen $n = 1, 2, 3$ from the exact Dirac formula: at full scale (left) the levels are indistinguishable from Bohr's $-13.6/n^2$ eV, while the zoom (right) shows $E_{nj} - m_e c^2$ minus the Bohr energy in units of α^2 Ry — every level is pushed down, levels depend only on n and j (so $2S_{1/2}$ and $2P_{1/2}$ stay degenerate), and $2P_{3/2}$ lies 4.53×10^{-5} eV above them.

3.5 — The Foldy-Wouthuysen Transformation: The Systematic Descent

To reach the Schrödinger equation from the Dirac equation, we need to **decouple the large and small components** of ψ order by order in v/c. The systematic procedure for this is the **Foldy-Wouthuysen (FW) transformation**.

3.5.1 — The Strategy

The Dirac Hamiltonian has **even** operators (block-diagonal: couple ϕ to ϕ and χ to χ) and **odd** operators (block-off-diagonal: couple ϕ to χ):

$$\hat{H}_{Dirac} = \underbrace{\beta m_e c^2 + eV}_{\text{even: } \mathcal{E}} + \underbrace{c\boldsymbol{\alpha} \cdot \hat{\pi}}_{\text{odd: } \mathcal{O}}$$

In the non-relativistic limit, the odd part $\mathcal{O}$ is small compared to the rest mass energy $m_e c^2$. We perform a sequence of unitary transformations $\psi \to e^{iS}\psi$ to eliminate odd operators order by order.

3.5.2 — First Transformation: Eliminating $\mathcal{O}$ at Leading Order

Choose the generator:

$$S = -\frac{i\beta\mathcal{O}}{2m_e c^2} = -\frac{i\beta\, c\boldsymbol{\alpha} \cdot \hat{\boldsymbol{\pi}}}{2m_e c^2}$$

The transformed Hamiltonian (to order $1/m_e c^2$):

$$\hat{H}' = e^{iS}\hat{H}_{Dirac}e^{-iS} = \beta m_e c^2 + eV + \frac{\hat{\pi}^2}{2m_e} - \frac{e\hbar}{2m_e c}\beta\boldsymbol{\sigma} \cdot \mathbf{B} + \text{remaining odd terms of order } (v/c)^3$$

The remaining odd terms are of order $(v/c)^3 \times m_e c^2$ — two orders smaller than the original odd term. A second FW transformation eliminates these.

3.5.3 — The Result: The Foldy-Wouthuysen Hamiltonian

After completing the transformation to order $(v/c)^2$, the full FW Hamiltonian acting on the **large component ϕ only** is:

$$\hat{H}_{FW} = m_e c^2 + \frac{\hat{\pi}^2}{2m_e} + eV \underbrace{- \frac{e\hbar}{2m_e c}\boldsymbol{\sigma}\cdot\mathbf{B}}_{\text{Zeeman term}} \underbrace{- \frac{\hat{\pi}^4}{8m_e^3 c^2}}_{\text{rel. mass corr.}} \underbrace{- \frac{e\hbar}{4m_e^2 c^2}\boldsymbol{\sigma}\cdot(\mathbf{E}\times\hat{\mathbf{p}})}_{\text{spin-orbit coupling}} \underbrace{- \frac{e\hbar^2}{8m_e^2 c^2}\nabla\cdot\mathbf{E}}_{\text{Darwin term}}$$

Each term has a name and a physics:

Term	Size	Physics
$m_e c^2$	Constant offset	Rest mass energy (set to zero in non-rel. limit)
$\hat{\pi}^2/2m_e + eV$	$\sim E_{atom}$	**Schrödinger Hamiltonian**
$-\frac{e\hbar}{2m_e c}\boldsymbol{\sigma}\cdot\mathbf{B}$	$\sim \alpha^2 E_{atom}$	**Zeeman effect** (spin in magnetic field)
$-\hat{\pi}^4/8m_e^3c^2$	$\sim \alpha^4 m_e c^2$	Relativistic kinetic energy correction
$-\frac{e\hbar}{4m_e^2c^2}\boldsymbol{\sigma}\cdot(\mathbf{E}\times\hat{\mathbf{p}})$	$\sim \alpha^4 m_e c^2$	**Spin-orbit coupling** (fine structure)
$-\frac{e\hbar^2}{8m_e^2c^2}\nabla\cdot\mathbf{E}$	$\sim \alpha^4 m_e c^2$	**Darwin term** (contact interaction at $r=0$; positive for hydrogen, since $e<0$)

The last three terms together account for the fine structure of hydrogen — matching the exact Dirac result to order α^4.

`[APPROXIMATION]` — **two reductions happen here, and they are different in kind.** The Foldy-Wouthuysen transformation is a **unitary change of basis constructed perturbatively, order by order** — a sequence $\psi \to e^{iS}\psi$ with S fixed at each order to cancel the odd terms then present (§3.5.1–3.5.2). It does not *diagonalise* $\mathcal{O}$; it **annihilates** it, removing the $\boldsymbol{\alpha}\cdot\hat{\boldsymbol{\pi}}$ mixing between the large and small components. Unitarity is exact at every order, since the exponential of a Hermitian generator is unitary. But there is no closed-form generator for a general external field: S is an asymptotic series in $1/m_e$, so the *truncation* of that series at order $(v/c)^2$ is `[APPROXIMATION]`, the odd terms being squared once and dropped. The boxed Hamiltonian is therefore a **truncated expansion, valid through the order displayed** — not an exact closed-form result. It acts on the 4×4 spinor at the accuracy retained, and as a 2×2 operator on ϕ it is the expansion, accurate to $\mathcal{O}(\alpha^4)$ corrections, and it stops being adequate once α^4 terms compete with the structure being resolved.

Model Ledger — Dirac Hamiltonian to Foldy-Wouthuysen Hamiltonian

Field	Content
Parent theory	Relativistic quantum mechanics: $i\hbar c\gamma^\mu D_\mu\psi - m_e c^2\psi = 0$ (Ch. 0, Ch. 3 §3.2)
Reduction	Exact unitary transformation to the even/odd (FW) basis; square the odd operator once; truncate at $(v/c)^2$
Model	2×2 operator on the large component: $\hat{\pi}^2/2m_e + eV - \frac{e\hbar}{2m_e c}\boldsymbol{\sigma}\cdot\mathbf{B} - \frac{\hat{\pi}^4}{8m_e^3c^2} - \frac{e\hbar}{4m_e^2c^2}\boldsymbol{\sigma}\cdot(\mathbf{E}\times\hat{\mathbf{p}}) - \frac{e\hbar^2}{8m_e^2c^2}\nabla\cdot\mathbf{E}$

Field	Content
Assumptions	one-particle (no pair creation); external potential with $\mathbf{E}$, $\mathbf{B}$ treated as given; expansion valid where $v/c \ll 1$; spin-½
Physics retained	kinetic energy, Zeeman splitting, relativistic mass correction, spin-orbit coupling, Darwin contact term
Physics neglected	vacuum polarisation, pair creation, radiative self-energy (Lamb shift), higher-order α^6 terms, negative-energy states
Validity	$E \ll m_e c^2$ — i.e. bound atomic states, not free-particle scattering at relativistic energy
Fails when	$E \gtrsim m_e c^2$; a strong external field makes the expansion parameter non-uniform; radiation shifts exceed the retained terms
Next model	full Klein-Gordon/Dirac with antiparticles; QED radiative corrections; two-particle Breit equation

The discarded terms are not small by accident. Each corresponds to something Chapter 0's $\mathcal{L}_{SM}$ contains and this equation does not, and the size table above is what justifies dropping them rather than declaring them irrelevant.

3.6 — First Stop: The Pauli Equation

Keeping only the Schrödinger term and the Zeeman term, and dropping the rest-mass constant $m_e c^2$ (a trivial energy offset), we arrive at the **Pauli equation**:

$$\boxed{i\hbar\frac{\partial\phi}{\partial t} = \left[\frac{\left(\hat{\mathbf{p}} - \frac{e}{c}\mathbf{A}\right)^2}{2m_e} - \frac{e\hbar}{2m_e c}\boldsymbol{\sigma}\cdot\mathbf{B} + eV\right]\phi}$$

where ϕ is now a **two-component Pauli spinor**:

$$\phi = \begin{pmatrix}\phi_\uparrow \\ \phi_\downarrow\end{pmatrix}$$

$\phi_\uparrow$ and $\phi_\downarrow$ are the spin-up and spin-down components.

`[APPROXIMATION]` **— the Pauli equation is a truncation of the FW Hamiltonian, not an independent theory.** Three terms were removed (the relativistic mass correction, spin-orbit coupling, and the Darwin term) and three component degrees of freedom were discarded (the small component χ). What is paid for that truncation is not only accuracy: it is the ability to reason at all. Spin now lives in a 2×2 matrix acting on two components, so spin-dependent phenomena become algebra instead of a 4×4 operator on a four-component field.

Model Ledger — FW Hamiltonian to the Pauli equation

Field	Content
Parent theory	$\hat{H}_{FW}$ acting on the large component (§3.5.3)
Reduction	drop the three α^4 terms; drop the rest-mass constant; retain the Zeeman term
Model	$i\hbar\,\partial_t\phi = \left[\frac{(\hat{\mathbf{p}}-e\mathbf{A}/c)^2}{2m_e} - \frac{e\hbar}{2m_e c}\boldsymbol{\sigma}\cdot\mathbf{B} + eV\right]\phi$
Assumptions	$E \ll m_e c^2$; spin-½; $\hbar$-expanded in $\mathbf{A}$; no radiation; fields externally imposed
Physics retained	orbital motion, magnetic coupling to spin, electromagnetic potential

Field	Content
Physics neglected	relativistic dispersion, spin-orbit interaction, Darwin contact, Lamb shift, antiparticles
Validity	atoms and solids in laboratory fields; magnetic fine structure is the leading neglected effect
Fails when	fine structure must be resolved (then use the full Dirac or FW operator); strong fields; heavy nuclei where v/c is not small
Next model	Dirac (§3.3); spin-orbit-corrected Pauli; Bloch equation for solids (Ch. 18)

What the Pauli Equation Adds Over Schrödinger

The Pauli equation is the correct equation for a non-relativistic electron **including its spin**. The Zeeman term $-\frac{e\hbar}{2m_e c}\boldsymbol{\sigma}\cdot\mathbf{B}$ gives:

- Splitting of energy levels in magnetic fields (Zeeman effect)
- Spin precession in magnetic fields (the basis of NMR and MRI)
- Pauli paramagnetism in metals

The Pauli equation is what you should use whenever:

- Spin matters (magnetic fields, spin-orbit coupling)
- But energies are still much less than $m_e c^2$

Expanding $(\hat{p} - eA/c)^2$

$$\left(\hat{\mathbf{p}} - \frac{e}{c}\mathbf{A}\right)^2 = \hat{p}^2 - \frac{e}{c}(\hat{\mathbf{p}}\cdot\mathbf{A} + \mathbf{A}\cdot\hat{\mathbf{p}}) + \frac{e^2}{c^2}A^2$$

In the Coulomb gauge $\nabla\cdot\mathbf{A} = 0$, the operators commute ($\hat{\mathbf{p}}\cdot\mathbf{A} = \mathbf{A}\cdot\hat{\mathbf{p}}$):

$$= \hat{p}^2 - \frac{2e}{c}\mathbf{A}\cdot\hat{\mathbf{p}} + \frac{e^2}{c^2}A^2$$

For weak fields, drop the A^2 term. Using $\mathbf{B} = \nabla\times\mathbf{A}$:

$$\hat{H}_{Pauli} = \frac{\hat{p}^2}{2m_e} + eV - \frac{e}{2m_e c}(\hat{\mathbf{L}} + 2\hat{\mathbf{S}})\cdot\mathbf{B}$$

where $\hat{\mathbf{L}} = \hat{\mathbf{r}}\times\hat{\mathbf{p}}$ is the orbital angular momentum. The factor of 2 in front of $\hat{\mathbf{S}}$ is the $g_e = 2$ we already saw in §3.4.2 — it appears here automatically, as it must.

3.7 — Second Stop: The Schrödinger Equation

When the magnetic field is absent or spin is irrelevant (e.g., we are studying spatial wavefunctions for spinless situations), each component of ϕ satisfies the same equation independently. Setting $\mathbf{B} = 0$ and writing ψ for a single spin component:

$$\boxed{i\hbar\frac{\partial\psi}{\partial t} = \hat{H}\psi = \left(-\frac{\hbar^2}{2m_e}\nabla^2 + V(\mathbf{r})\right)\psi}$$

This is the **Schrödinger equation** — the foundation of all of Layer 1.

`[APPROXIMATION]` — **this is the loosest step in the chapter, and the most productive.** Setting $\mathbf{B} = 0$ and ignoring spin is a much larger simplification than the previous one: it discards an entire observable degree of freedom, not a correction term. Two-component spinor becomes one-component scalar; the $\boldsymbol{\sigma}$ operator disappears; the equation becomes the one that admits closed-form solutions.

Model Ledger — Pauli equation to the Schrödinger equation

Field	Content
Parent theory	Pauli equation for the two-component spinor (§3.6)
Reduction	set $\mathbf{B} = 0$; retain one spin component; replace $\hat{\mathbf{p}} \to -i\hbar\nabla$ for a scalar field
Model	$i\hbar\,\partial_t\psi = \left(-\frac{\hbar^2}{2m_e}\nabla^2 + V\right)\psi$
Assumptions	$E \ll m_e c^2$; spin decoupled and irrelevant; scalar (single-valued) wavefunction; time-independent mass
Physics retained	wave nature, quantised bound states, interference, tunnelling
Physics neglected	spin, magnetic coupling, relativistic dispersion, particle creation, particle indistinguishability
Validity	the entire non-relativistic single-particle regime — the working domain of Chapters 4–6
Fails when	spin is observable (Zeeman, spin-orbit); $v \gtrsim c$; exchange matters (then use the many-body Schrödinger equation, Ch. 6); fields are time-dependent and strong
Next model	Pauli (spin back in); Dirac (relativity back in); many-body Schrödinger (exchange, Ch. 6)

The trade is worth stating plainly: what is discarded is exactly what makes the equation *solvable*. Scalar one-body Schrödinger admits analytic solutions for the hydrogen atom; the corresponding Pauli or Dirac problems do not, and are attacked numerically or perturbatively instead. Solvability is a property of the approximation, not a hint that the approximation is right.

What This Derivation Reveals

Every introductory quantum mechanics course treats the Schrödinger equation as an axiom — a starting postulate handed down without justification. From the perspective of this book, it is not an axiom. It is:

1. The non-relativistic, single-particle, spin-ignored limit of the Dirac equation
2. Which is itself the Euler-Lagrange equation of the electron Lagrangian
3. Which is the U(1)-coupled fermion sector of $\mathcal{L}_{SM}$
4. Which follows from requiring the action to be invariant under U(1) gauge transformations

Every symbol in the Schrödinger equation has a traceable origin:

Symbol	Origin
i	Required by unitarity of time evolution (unitary representation of the Poincaré group)
$\hbar$	Sets the scale between the action and quantum phenomena; appears in the path integral weight $e^{iS/\hbar}$
$\partial/\partial t$	Time translation; by Noether → energy; operator $\hat{H} = i\hbar\partial_t$
$-\hbar^2\nabla^2/2m$	Non-relativistic kinetic energy; from $\hat{p}^2/2m$ with $\hat{p} = -i\hbar\nabla$
$V(\mathbf{r})$	The electromagnetic potential eA_0; from the U(1) gauge coupling in D_μ

Symbol	Origin
ψ	One component $\phi_\uparrow$ or $\phi_\downarrow$ of the large component of the Dirac spinor

The Time-Independent Schrödinger Equation

For a stationary state $\psi(\mathbf{r}, t) = \phi(\mathbf{r})e^{-iEt/\hbar}$:

$$\boxed{\hat{H}\phi = E\phi \qquad \text{i.e.,} \qquad \left(-\frac{\hbar^2}{2m}\nabla^2 + V\right)\phi = E\phi}$$

This is an eigenvalue equation: ϕ are the **energy eigenstates** and E are the **energy eigenvalues**. Its solutions for various potentials $V(\mathbf{r})$ are the substance of Chapters 4 and 5.

3.8 — The Bridge Zone: When to Use Which Equation

The three equations of this chapter are not competitors — they are a nested hierarchy. The appropriate equation depends on the regime:

```
Regime                        Equation to use          Chapters
──────────────────────────────────────────────────────────────
E ~ mc², pair creation        Full QED / L_SM          Ch. 0
E ≲ mc², single particle      Dirac equation           §3.3–3.4
E ≪ mc², spin matters         Pauli equation           §3.6, Ch. 5
E ≪ mc², spin irrelevant      Schrödinger equation     §3.7, Ch. 4–6
Classical: S ≫ ħ              Newtonian mechanics      Ch. 9 (Bridge B.a)
```

The bridge zone is the Dirac equation — it lives between the full QFT of Chapter 0 and the non-relativistic quantum mechanics of Chapters 4–7. The Dirac equation is the correct equation to use for:

- Relativistic electrons in heavy atoms (large nuclear charge Z; inner electrons have $v/c \sim Z\alpha$, which becomes significant for $Z \gtrsim 50$)
- Electron transport in graphene (the electrons in graphene mimic massless Dirac fermions because the band structure near the Dirac point is linear, not parabolic — a Layer-1 accident with Layer-0 mathematical structure)
- Positron emission tomography (PET): pair annihilation, requires the antiparticle solutions
- Spintronic devices where spin-orbit coupling is designed in deliberately

---

3.9 — What Survived the Descent

After the descent from $\mathcal{L}_{SM}$ to the Schrödinger equation, the following structures survive intact. Each one is traceable to a Layer-0 origin:

3.9.1 — The Wavefunction $\psi(\mathbf{r}, t)$

A complex-valued function of position and time. It is not a classical field you can directly observe — it is a **probability amplitude**. Its origin is the large component of the Dirac spinor, which is itself the fermion field ψ of $\mathcal{L}_{SM}$ in the single-particle, non-relativistic limit.

3.9.2 — The Probability Current: What Noether's Theorem Does and Does Not Give

From the Dirac Lagrangian, the U(1) symmetry $\psi \to e^{i\alpha}\psi$ gives, via Noether's theorem, a conserved current:

$$j^\mu = e\bar{\psi}\gamma^\mu\psi = \left(e|\psi|^2,\, e\bar{\psi}\gamma^i\psi\right)$$

The conservation law $\partial_\mu j^\mu = 0$ becomes, in 3+1 notation:

$$\frac{\partial(e|\psi|^2)}{\partial t} + \nabla \cdot \mathbf{j} = 0$$

Dividing by e ($= -|e|$ for the electron):

$$\frac{\partial|\psi|^2}{\partial t} + \nabla \cdot \left(\frac{\mathbf{j}}{e}\right) = 0$$

This is the **continuity equation for** $|\psi|^2$: the quantity $|\psi|^2$ is a conserved density, and current flows so that its integral over any closed region is time-independent.

> **Warning: What Noether's theorem does not establish** The step from "$|\psi|^2$ is conserved" to "$|\psi|^2\, d^3x$ is the *probability* of finding the particle there" is **not** a consequence of Noether's theorem. Noether's theorem delivers a conserved current from a continuous symmetry. It does not deliver a probability interpretation.
>
> The two claims are logically distinct, and conflating them is a standard error in "everything is just symmetry" treatments of QM:
>
> | Step | What supplies it | Status |
> |---|---|---|
> | Global U(1) symmetry ⇒ conserved current j^μ | Noether's theorem (Ch. 1) | `[DERIVATION]` |
> | Normalization $\int | \psi | ^2 = 1$ |
> | $P(x) = | \psi(x) | ^2$ is a probability |
> | Non-negativity, $\sum_i P_i = 1$ | positivity + normalization of $|\psi|^2$ | follows once Born is assumed |
>
> **Where the Born rule stands in this book.** The Born rule is retained as a **foundational probabilistic postulate of quantum mechanics**. It is *not* derived here, and the book's own tagging rules forbid claiming otherwise. What the descent *does* earn is real and worth stating precisely: the continuity equation above fixes the *functional form* of the conserved quantity to be $|\psi|^2$, so once the Born postulate is adopted, probability conservation follows from charge conservation. The interpretive step is assumed; the conservation arithmetic is derived.
>
> There is a substantial literature attempting to derive or motivate Born-type rules from deeper principles — envariance, Wigner-friendliness and related no-go results, decision-theoretic arguments, and axiomatic approaches. None is accepted as a derivation within mainstream practice, and this book does not reproduce or rely on them. Claiming a settled derivation would be exactly the kind of overclaim the `[SYNTHESIS]` / `[DERIVATION]` distinction exists to prevent. See `REFERENCES.md` for how the book's other interpretive commitments are tiered.

3.9.3 — Superposition

The Schrödinger equation is linear in ψ. If ψ_1 and ψ_2 are solutions, so is $c_1\psi_1 + c_2\psi_2$ for any complex constants. This linearity is inherited from the linearity of the Dirac equation, which is itself linear because the Lagrangian $\bar{\psi}(i\gamma^\mu D_\mu - m)\psi$ is quadratic in the fields (first power of ψ and first power of $\bar{\psi}$).

But the two claims must be kept apart, and the second sentence of this section used to run them together. The quadratic action gives a *linear field equation*. The superposition principle for states rests on something else: the **linear structure of the state space itself**, $\mathcal{H} = \{c_1|\psi_1\rangle + c_2|\psi_2\rangle\}$ for any $|\psi_1\rangle, |\psi_2\rangle \in \mathcal{H}$ and any complex c_i. That is a statement about the vector space of physical states — what quantum theory assumes, along with the Born rule that makes the amplitudes physical — and it is not deduced from the form of any particular Lagrangian.

The clean separation:

- **quadratic free-field action ⇒ linear field equation.** Real, and demonstrated by the Dirac equation above.
- **linear quantum state space ⇒ superposition principle.** Structural, and independent of the free-field action.
- **particular interacting or effective equations may be nonlinear.** Interacting QFTs have higher-order interaction terms — $\lambda\phi^4$, $g^2\phi^4$, electroweak terms — and states still superpose. Conversely, nonlinear equations such as the Gross–Pitaevskii equation describe a *mean-field* or classical-field limit in which the many-body state has already been packaged, and it is the quantised fluctuations about that solution that again obey a linear equation and again superpose.

So the free Dirac equation is *consistent* with superposition because the state space is linear, and its linearity is *visible* as a linear equation — not because the quadratic action caused the superposition principle to exist.

3.9.4 — Minimal Coupling: $\hat{\mathbf{p}} \to \hat{\mathbf{p}} - \frac{e}{c}\mathbf{A}$

The replacement of the canonical momentum $\hat{\mathbf{p}}$ by the kinetic momentum $\hat{\mathbf{p}} - \frac{e}{c}\mathbf{A}$ — the standard way of introducing electromagnetic fields into quantum mechanics — comes directly from the covariant derivative $D_\mu = \partial_\mu + ieA_\mu$ in the original Lagrangian. Minimal coupling is not a separate rule; it is what covariant derivative means, evaluated in the non-relativistic limit.

3.9.5 — Spin (in the Pauli equation)

The Pauli matrices $\boldsymbol{\sigma}$ in the Pauli equation are the 2×2 blocks of the 4×4 gamma matrices — a direct inheritance from the Dirac spinor structure. Spin is not added to the Schrödinger equation from outside; it lives in the Pauli equation, which is one order higher in v/c from Schrödinger.

3.10 — What Was Discarded, and Why It Was Allowed

Discarded	Physical content	Smallness parameter	When it matters
Antiparticle components (χ)	Positron; pair creation	$\sim (v/c)^2$ relative to ϕ	$E \sim m_e c^2$ — not in atoms

Discarded	Physical content	Smallness parameter	When it matters
Relativistic kinetic correction $-p^4/8m^3c^2$	Relativistic mass increase	$\alpha^4 m_e c^2 / E_{Bohr} \sim \alpha^2$	Fine structure (Ch. 5)
Spin-orbit coupling	Fine structure splitting	$\sim \alpha^2$	Fine structure (Ch. 5)
Darwin term	s-state contact interaction	$\sim \alpha^2$	Fine structure (Ch. 5)
QED loop corrections	Lamb shift, $g-2$	$\sim \alpha/\pi \approx 0.002$	Precision spectroscopy
Virtual pair loops	Vacuum polarization	$\sim \alpha^2$	Precision spectroscopy

The key point: every discarded term has a named smallness parameter. None are thrown away without justification. When you need more precision (spectroscopy, metrology, exotic atoms), you add terms back in the appropriate order.

3.11 — The Uncertainty Principle: A Structural Consequence

Before moving on, one of the most famous results in quantum mechanics deserves to be placed in its proper context.

The **Heisenberg uncertainty principle** $\Delta x\, \Delta p \geq \hbar/2$ is often presented as a mysterious feature of quantum mechanics — a limit on measurement or a consequence of disturbing the system. Its actual origin is simpler and more structural.

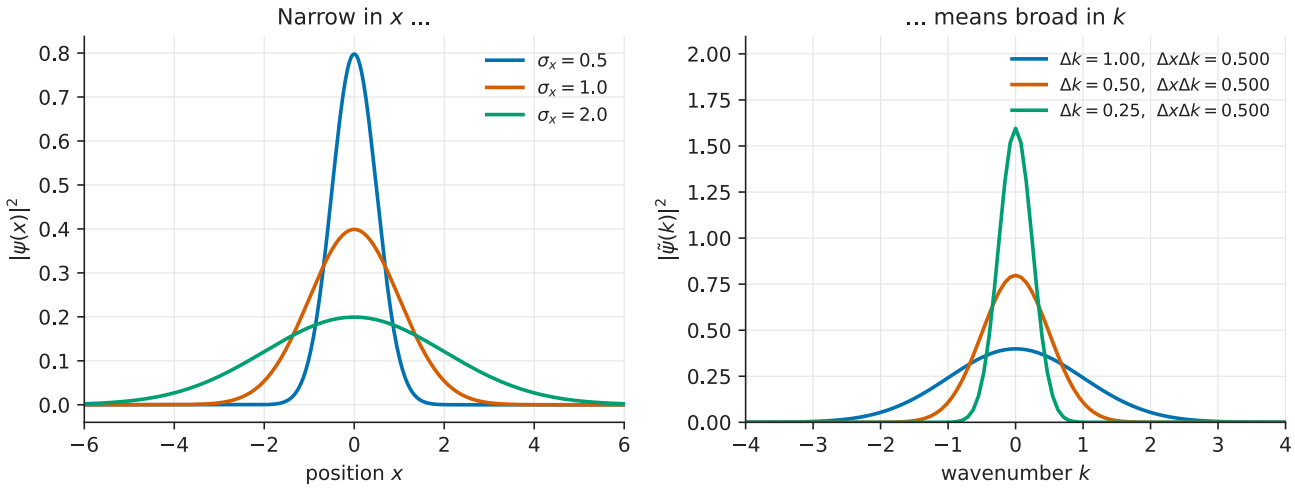
The momentum operator in the Schrödinger equation is $\hat{p} = -i\hbar\nabla$. The position operator is $\hat{x} = x$ (multiplication by x). Their commutator:

$$[\hat{x}, \hat{p}] = \hat{x}\hat{p} - \hat{p}\hat{x} = i\hbar$$

This commutation relation is a **mathematical consequence of** $\hat{p} = -i\hbar\nabla$ — which is itself a consequence of the field ψ having the Fourier decomposition $\psi(\mathbf{r}) = \int \tilde{\psi}(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{r}} d^3k$ with $\hat{p}\psi = \hbar\mathbf{k}\psi$ for a definite-k component.

A standard result from Fourier analysis then gives:

$$\Delta x\, \Delta k \geq \frac{1}{2} \qquad \Longrightarrow \qquad \Delta x\, \Delta p \geq \frac{\hbar}{2}$$



Gaussian packets $|\psi(x)|^2$ with $\sigma_x = 0.5, 1, 2$ (left) and their numerically computed Fourier transforms $|\tilde{\psi}(k)|^2$ (right): halving the width in x doubles it in k, and every Gaussian sits exactly on the bound, $\Delta x\,\Delta k = 1/2$.

The uncertainty principle is the Fourier bandwidth theorem, applied to the wavefunction. It is structural — it would hold for any wave-like description, classical or quantum. What makes it quantum is that $p = \hbar k$ — that momentum is proportional to wavenumber, which is itself a consequence of $\hat{p} = -i\hbar\nabla$ in the Schrödinger equation, which traces back to the covariant derivative in $\mathcal{L}_{SM}$.

3.12 — Summary of Bridge A

What we started with:

$$\mathcal{L}_{SM} = \text{full Standard Model} + \text{gravity}$$

The three steps:

Step	Operation	Result	Connection
§3.1	Isolate U(1)-coupled electron sector; treat A_μ as classical	Electron QED Lagrangian	`[APPROXIMATION]` — sector isolation, mean field
§3.2	Apply Euler-Lagrange	**Dirac equation** (bridge zone)	`[DERIVATION]` — exact within the Lagrangian
§3.5–3.6	Foldy-Wouthuysen to order $(v/c)^2$	**Pauli equation** (with spin)	`[APPROXIMATION]` — unitary at each order, series truncated at $(v/c)^2$
§3.7	Drop spin terms, $B \to 0$	**Schrödinger equation**	`[APPROXIMATION]` — discards a degree of freedom

The transformation in §3.5 is a derivation; the truncation that follows it is an approximation; the drop in §3.7 is an approximation that costs an entire observable. Only one row in this table is lossless, and the Model Ledgers in §3.5.3, §3.6, and §3.7 record exactly what each row pays.

What was preserved: wavefunction, probability conservation (Noether/U(1)), superposition, minimal coupling, spin (in Pauli), uncertainty principle (Fourier structure)

What was discarded: antiparticles, pair creation, QED loops, relativistic corrections (available on demand: add FW terms back for fine structure)

The validity criterion:

$$E_{kinetic} \ll m_e c^2 = 511\,\text{keV} \qquad \Leftrightarrow \qquad v \ll c$$

For hydrogen ground state: $v/c = \alpha \approx 1/137$; the relative error in the kinetic energy is $v^2/c^2 = \alpha^2 \approx 5 \times 10^{-5}$, i.e. 0.005%. The criterion is not a matter of degree — it is the boundary at which pair creation becomes energetically available and the reduction loses its meaning.

3.13 — Looking Ahead: Layer 1

With the Schrödinger equation in hand, Layer 1 opens up. The next four chapters use it to build the atomic world from scratch:

Chapter 4 — The Schrödinger equation in free space and in simple potentials. The infinite well, the harmonic oscillator, the hydrogen atom. Quantum numbers, energy levels, wavefunctions. The periodic table begins.

Chapter 5 — Spin, multi-electron systems, and the periodic table completed. Spin-orbit coupling (FW term) gives fine structure. The Pauli equation gives the Zeeman effect. Many-electron Schrödinger + Pauli exclusion (which, remember, came from the anticommutation of ψ in $\mathcal{L}_{SM}$) gives the shell structure of atoms.

Chapter 6 — The Schrödinger equation in a periodic lattice. Bloch's theorem. Band structure. Metals, semiconductors, insulators. Molecular bonding. Nuclear binding energy. These are the material properties every engineer works with.

Chapter 7 — Where Layer-1 physics becomes topological. Quantum Hall effect, Berry phase, topological insulators. The echo of the θ-term from Chapter 0, heard at the scale of centimeter-sized condensed matter devices.

End of Chapter 3.

Next: Chapter 4 — Layer 1 Begins: The Schrödinger Equation, Energy Quantization, and the Hydrogen Atom

The Schrödinger Equation and the Hydrogen Atom

Layer 1 Begins: Quantization, Wavefunctions, and the Architecture of Atoms

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> *The more I think about the physical portion of Schrödinger's theory, the more repulsive I find it. What does it mean for an electron to be a wave?* — Werner Heisenberg, 1926
>
> *He was wrong to be repulsed. Schrödinger was right.* — The rest of physics, 1927–present

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4.0 — Where We Are

In Chapter 3, you watched the Schrödinger equation fall out of the Standard Model Lagrangian through a sequence of named, controlled approximations. The equation arrived at your door already carrying its justification.

Now we use it.

This chapter does two things. First, it builds physical intuition through three exactly-solvable quantum systems — the infinite well, the harmonic oscillator, and the finite barrier. Each one teaches something about quantization that the next system confirms. Second, it solves the hydrogen atom completely — the most important calculation in all of chemistry and materials science.

By the end of this chapter, you will understand not just the energy levels of hydrogen but *why* they are discrete, *why* three quantum numbers appear, and *why* atomic structure is the way it is. The entire periodic table follows from what happens here.

4.1 — Operators and the Language of Quantum Mechanics

4.1.1 — Physical Quantities as Operators

In classical mechanics, physical quantities are just numbers: the position of a particle is a vector $\mathbf{r}(t)$, its momentum is $\mathbf{p}(t) = m\dot{\mathbf{r}}$, and its energy is $H = p^2/2m + V$.

In quantum mechanics, physical quantities are **operators** — mathematical objects that act on wavefunctions:

Classical quantity	Quantum operator	Action on ψ
Position x	$\hat{x}$	Multiply by x
Momentum p_x	$\hat{p}_x = -i\hbar\,\partial/\partial x$	Differentiate, multiply by $-i\hbar$
Kinetic energy $p^2/2m$	$\hat{T} = -(\hbar^2/2m)\nabla^2$	Apply Laplacian, multiply by $-\hbar^2/2m$
Potential energy $V(\mathbf{r})$	$\hat{V} = V(\mathbf{r})$	Multiply by $V(\mathbf{r})$
Total energy H	$\hat{H} = \hat{T} + \hat{V}$	Apply both above

The momentum operator $\hat{p}_x = -i\hbar\,\partial/\partial x$ is not an arbitrary choice — it follows directly from the plane wave $\psi = e^{ikx}$ being an eigenstate with $\hat{p}_x\psi = \hbar k\,\psi$, and $p = \hbar k$ being the de Broglie relation from Ch. 3.

4.1.2 — The Eigenvalue Equation

The Schrödinger equation in its time-independent form (§3.7):

$$\hat{H}\psi = E\psi \qquad \Longrightarrow \qquad \left(-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r})\right)\psi = E\psi$$

is an **eigenvalue equation**: find the functions ψ_n (eigenfunctions) and numbers E_n (eigenvalues) such that $\hat{H}$ acting on ψ_n returns E_n times ψ_n.

Why eigenvalues are what you measure: When you perform an energy measurement on a quantum system, the result is always one of the eigenvalues E_n. You cannot get an intermediate value. This **is** a postulate of the standard measurement formalism — the spectral rule, under which measuring an observable $\hat{A}$ returns an eigenvalue of $\hat{A}$, with probabilities given by the Born rule. It is not derived here, and **Noether's theorem does not derive it**: Noether's first theorem maps a continuous symmetry to a conserved current, and says nothing about which outcomes a measurement is allowed to return. The connection to conservation runs the other way and is much weaker. Once the Born postulate is adopted, the conserved $\hat{\psi}^\dagger\hat{\psi}$ current of Ch. 1 §1.7 implies conservation of probability under the resulting unitary evolution. The discreteness of measured values is then a mathematical consequence of the operator structure together with the spectral rule — not a consequence of probability conservation, and not a consequence of any symmetry principle. [STRUCTURAL CONNECTION] The operator structure of Ch. 3 produces the discrete spectrum; the spectral rule connects that spectrum to measurement. Neither step is a Noether statement.

4.1.3 — Expectation Values

You will not always find the system in an energy eigenstate. For a general wavefunction ψ, the **expectation value** of an observable $\hat{A}$ is:

$$\langle A \rangle = \int_{-\infty}^{\infty} \psi^* \hat{A}\psi \, d^3r$$

This is the statistical average of many repeated measurements on identically prepared systems. For the position:

$$\langle x \rangle = \int x, |\psi|^2 \, dx$$

The probability of finding the particle between x and $x + dx$ is $|\psi(x)|^2 dx$. This is why normalization is required:

$$\int_{-\infty}^{\infty} |\psi|^2 \, d^3r = 1$$

Total probability must be 1 — the particle is somewhere. This is the continuity equation from Ch. 3 §3.9.2, integrated over all space.

4.2 — The Infinite Square Well: Where Quantization Is Born

4.2.1 — The Setup

The simplest quantum system that exhibits quantization: a particle of mass m confined to a one-dimensional box of length L, with impenetrable walls:

$$V(x) = \begin{cases} 0 & 0 \le x \le L \\ \infty & \text{otherwise} \end{cases}$$

The infinite potential walls mean the particle cannot exist outside $[0, L]$:

$$\psi(x) = 0 \quad \text{for } x \le 0 \text{ and } x \ge L$$

4.2.2 — The Solution

Inside the well ($V = 0$), the Schrödinger equation is:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi \qquad \Longrightarrow \qquad \frac{d^2\psi}{dx^2} = -k^2\psi, \quad k = \frac{\sqrt{2mE}}{\hbar}$$

The general solution is $\psi(x) = A\sin(kx) + B\cos(kx)$.

Applying boundary conditions:

- $\psi(0) = 0$: gives $B = 0$, so $\psi = A\sin(kx)$
- $\psi(L) = 0$: requires $\sin(kL) = 0$, which means $kL = n\pi$ for integer n

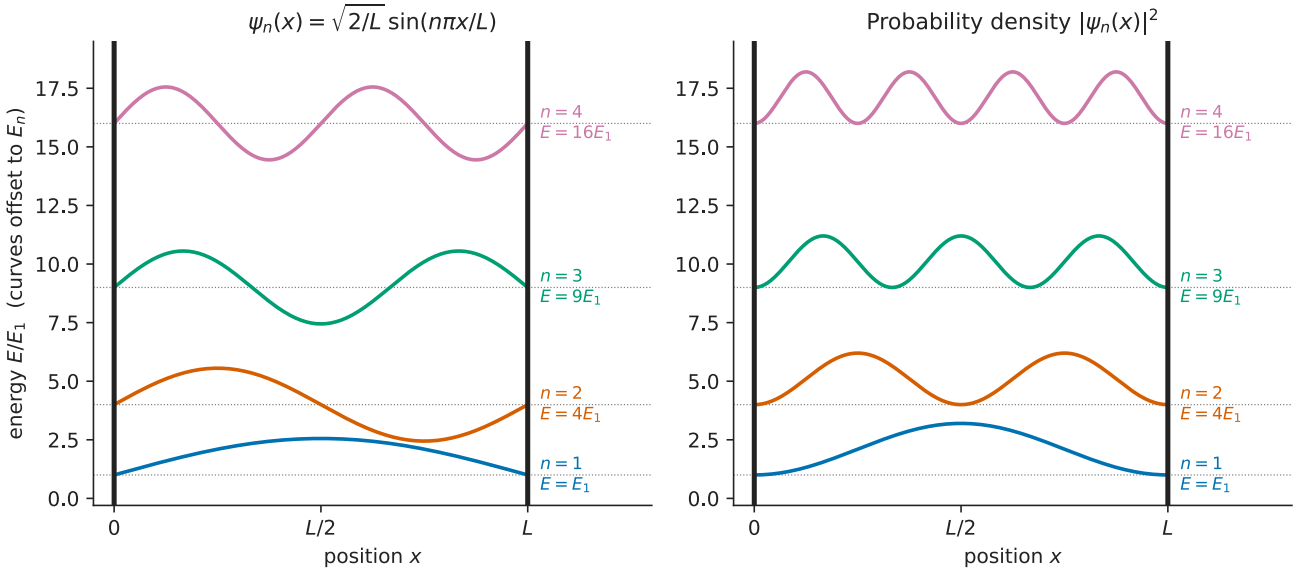
$$\boxed{k_n = \frac{n\pi}{L}, \qquad n = 1, 2, 3, \ldots}$$

The quantization condition emerges from the boundary conditions — not from any additional axiom. The allowed wavefunctions and energies are:

$$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right)$$

$$\boxed{E_n = \frac{n^2\pi^2\hbar^2}{2mL^2} = n^2 E_1, \qquad E_1 = \frac{\pi^2\hbar^2}{2mL^2}}$$

The $\sqrt{2/L}$ normalization constant comes from requiring $\int_0^L |\psi_n|^2 dx = 1$.



Wavefunctions $\psi_n = \sqrt{2/L}\sin(n\pi x/L)$ (left) and densities $|\psi_n|^2$ (right) for $n = 1$–4, each offset to its energy $E_n = n^2 E_1$: every state vanishes at both walls, ψ_n has $n - 1$ interior nodes, and the level spacing grows as $(2n + 1)E_1$.

4.2.3 — What This Teaches

Lesson 1: Quantization comes from boundary conditions. The integer n is not put in by hand — it is forced by requiring the wavefunction to vanish at both walls. This is exactly how a guitar string has discrete harmonics: n nodes fit in the length L. Quantum mechanics is wave mechanics.

Lesson 2: Zero-point energy is not zero. The lowest allowed energy is $E_1 = \pi^2\hbar^2/2mL^2 > 0$, not zero. The particle cannot be at rest. From the uncertainty principle (Ch. 3 §3.11): confining a particle to length $\Delta x \approx L$ forces $\Delta p \geq \hbar/2L$, which requires nonzero kinetic energy. Zero-point energy is the uncertainty principle made concrete.

Lesson 3: Quantum numbers label orthogonal states. Different eigenfunctions are orthogonal: $\int_0^L \psi_m^* \psi_n \, dx = \delta_{mn}$. States with different n are completely distinguishable. The quantum number n is an integer label for a standing-wave mode.

Lesson 4: The correspondence principle. For large n, the probability density $|\psi_n|^2$ becomes nearly uniform over the well — matching what classical mechanics predicts (equal probability of being anywhere in the well). Quantum mechanics recovers classical mechanics at large quantum numbers.

4.2.4 — Uncertainty Principle Check

For the ground state $\psi_1 = \sqrt{2/L}\sin(\pi x/L)$:

$$\langle x \rangle = \frac{L}{2}, \quad \Delta x = \frac{L}{\sqrt{12}}\sqrt{1 - \frac{6}{\pi^2}}$$

$$\langle p \rangle = 0, \quad \Delta p = \frac{\pi\hbar}{L}$$

$$\Delta x \, \Delta p = \frac{\pi\hbar}{L} \cdot \frac{L}{\sqrt{12}}\sqrt{1 - \frac{6}{\pi^2}} \approx 0.568\,\hbar \geq \frac{\hbar}{2} \checkmark$$

The inequality is satisfied — as it must be for any quantum state.

4.3 — The Quantum Harmonic Oscillator

The harmonic oscillator $V = \frac{1}{2}m\omega^2 x^2$ is the most important potential in all of physics. Every stable potential near equilibrium is approximately harmonic (Taylor expansion around the minimum). Atoms in a crystal, electrons in a magnetic field, electromagnetic modes in a cavity — all are harmonic oscillators in disguise.

4.3.1 — The Ladder Operator Method

Rather than grinding through the differential equation, introduce the **ladder operators**:

$$\hat{a}_\pm = \frac{1}{\sqrt{2\hbar m\omega}}\left(m\omega\hat{x} \mp i\hat{p}\right)$$

They satisfy $[\hat{a}_-, \hat{a}_+] = 1$. The Hamiltonian becomes:

$$\hat{H} = \hbar\omega\left(\hat{a}_+\hat{a}_- + \frac{1}{2}\right)$$

If ψ is an eigenstate with energy E, then $\hat{a}_+\psi$ is an eigenstate with energy $E + \hbar\omega$, and $\hat{a}_-\psi$ is an eigenstate with energy $E - \hbar\omega$. They "climb" and "descend" the energy ladder.

Since energy must be bounded below, there exists a ground state ψ_0 with $\hat{a}_-\psi_0 = 0$. Solving this:

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \exp\left(-\frac{m\omega x^2}{2\hbar}\right)$$

The ground state is a Gaussian. All higher states are generated by applying $\hat{a}_+$ repeatedly:

$$\psi_n = \frac{(\hat{a}_+)^n}{\sqrt{n!}}\psi_0$$

4.3.2 — Energy Levels

$$\boxed{E_n = \hbar\omega\left(n + \frac{1}{2}\right), \qquad n = 0, 1, 2, \ldots}$$

Every energy level is separated by exactly $\hbar\omega$. The zero-point energy is $E_0 = \hbar\omega/2$.

Why this matters beyond quantum mechanics: The quantum field in Chapter 0 is a collection of quantum harmonic oscillators — one per mode per field. When you quantize a mode of the electromagnetic field (a photon of frequency ω), you get $E = \hbar\omega(n + \frac{1}{2})$ where n is the photon number. The $\hbar\omega/2$ zero-point energy summed over all modes gives the vacuum energy — and the Casimir effect (Ch. 2 entry) is the measurable consequence of this sum being different between two plates than in free space. The harmonic oscillator is the engine of quantum field theory.

4.3.3 — Wavefunctions

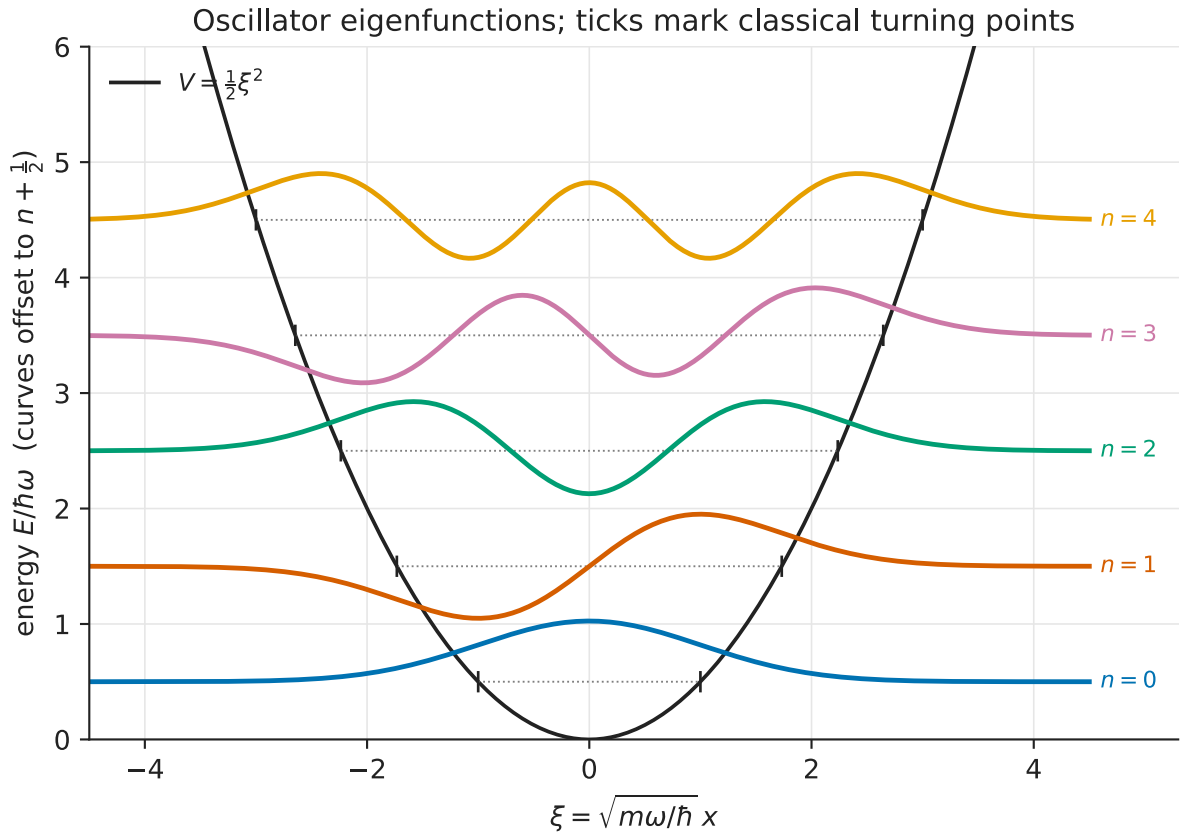
The n-th eigenfunction involves the n-th Hermite polynomial H_n:

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2^n n!}} H_n\left(\sqrt{\frac{m\omega}{\hbar}}x\right) \exp\left(-\frac{m\omega x^2}{2\hbar}\right)$$

The first few:

- $\psi_0 \propto e^{-m\omega x^2/2\hbar}$ — Gaussian, no nodes
- $\psi_1 \propto x\, e^{-m\omega x^2/2\hbar}$ — one node at $x = 0$
- $\psi_2 \propto (2m\omega x^2/\hbar - 1)e^{-m\omega x^2/2\hbar}$ — two nodes

The n-th state has exactly n nodes — the same node-counting rule as the infinite square well. This is not a coincidence: for any confining potential, the n-th eigenstate has $n - 1$ (or n) nodes, by the oscillation theorem.



Eigenfunctions $\psi_n(\xi) \propto H_n(\xi)e^{-\xi^2/2}$ for $n = 0$–4, each drawn on its level $E_n = (n + \frac{1}{2})\hbar\omega$ inside $V = \frac{1}{2}\xi^2$, with ticks at the classical turning points $\xi = \pm\sqrt{2n+1}$: the levels are equally spaced, ψ_n has n nodes, and every state leaks a little beyond its turning points into the classically forbidden region.

4.4 — Quantum Tunneling: The Finite Barrier

4.4.1 — The Setup

For a rectangular potential barrier of height V_0 and width d, with particle energy $E < V_0$:

$$V(x) = \begin{cases} 0 & x < 0 \text{ (region I)} \\ V_0 & 0 \le x \le d \text{ (region II)} \\ 0 & x > d \text{ (region III)} \end{cases}$$

4.4.2 — The Solution Inside the Barrier

Inside the barrier, the Schrödinger equation becomes:

$$\frac{d^2\psi}{dx^2} = \kappa^2\psi, \qquad \kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$$

Since $E < V_0$, κ is real and positive. The solution inside is:

$$\psi_{II}(x) = Ce^{\kappa x} + De^{-\kappa x}$$

This is not oscillatory — it decays (and grows) exponentially. Classically, the particle simply reflects at $x = 0$. Quantum mechanically, the wavefunction extends into the classically forbidden region.

4.4.3 — Transmission Coefficient

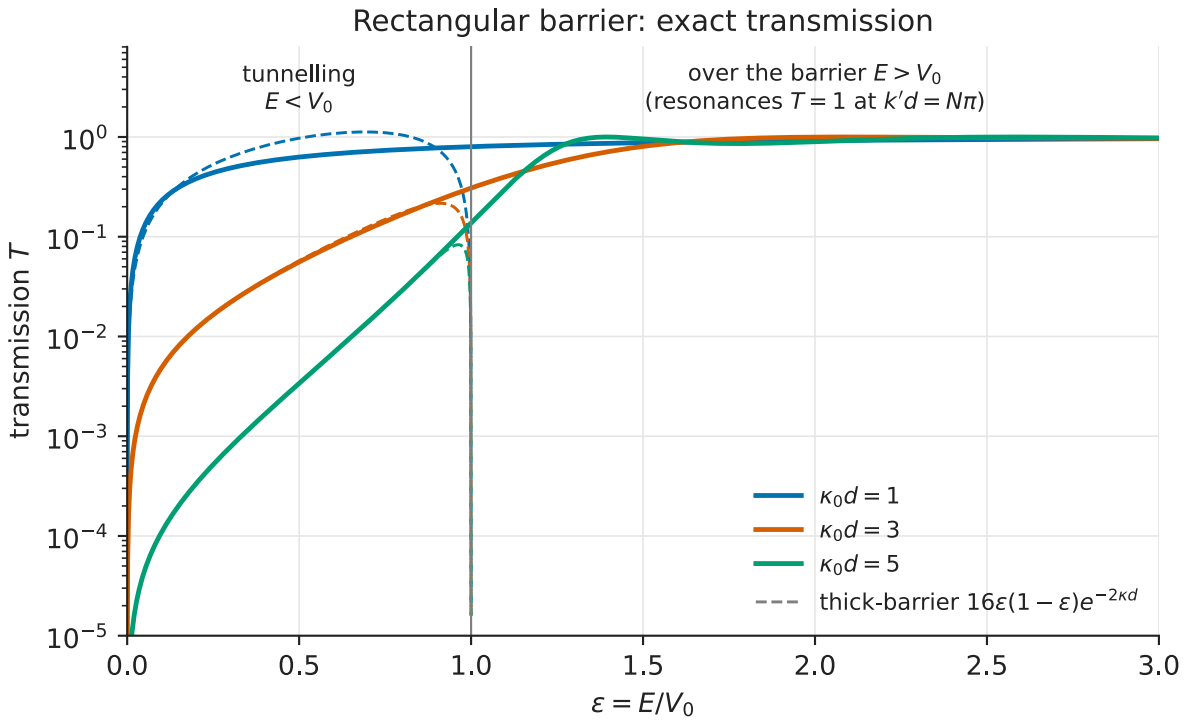
Matching the wavefunction and its derivative at both boundaries ($x = 0$ and $x = d$) gives the transmission coefficient — the probability that the particle passes through:

$$\boxed{T \approx 16\frac{E}{V_0}\left(1 - \frac{E}{V_0}\right)e^{-2\kappa d}}$$

(valid for $\kappa d \gg 1$, the thick barrier limit)

Key physics:

- T depends **exponentially** on the barrier width d and on $\kappa \propto \sqrt{V_0 - E}$
- Doubling the barrier width squares the exponential factor: $e^{-2\kappa d} \to e^{-4\kappa d}$ (the prefactor is unchanged)
- This extreme sensitivity to d and $V_0 - E$ makes tunneling both measurable and controllable



Exact transmission $T(E/V_0)$ for a rectangular barrier with $\kappa_0 d = \sqrt{2mV_0}\,d/\hbar = 1, 3, 5$ (solid), using $T = [1 + V_0^2 \sinh^2(\kappa d)/4E(V_0 - E)]^{-1}$ below the top and the $\sin^2(k'd)$ form above it, with the thick-barrier approximation $16\epsilon(1-\epsilon)e^{-2\kappa d}$ dashed: the approximation is excellent for $\kappa_0 d = 5$, fails for $\kappa_0 d = 1$, and even above the barrier $T < 1$ except at the resonances $k'd = N\pi$.

4.4.4 — Engineering Applications

Application	What tunnels	Barrier	Engineering use
Tunnel diode	Electrons	Semiconductor junction	Oscillators, fast switches

Application	What tunnels	Barrier	Engineering use
STM (Scanning Tunneling Microscope)	Electrons	Vacuum gap 1–10 Å	Atomic-resolution imaging
Flash memory	Electrons	Thin oxide	Non-volatile data storage
α-decay (Ch. 6)	α particle	Coulomb barrier	Nuclear dating
MOSFET gate leakage	Electrons	Gate oxide	Limits transistor scaling

The STM tunnel current $I \propto e^{-2\kappa z}$ where z is tip-sample distance; at $\kappa \approx 10\,\text{nm}^{-1}$, a 1 Å change in z changes the current by a factor of $e^2 \approx 7.4$. This extraordinary sensitivity is what gives STM atomic resolution.

4.5 — The Hydrogen Atom

This is the most important calculation in atomic physics. Everything that follows in Chapters 5, 6, and 7 is either an extension or an approximation of what we derive here.

4.5.1 — The Setup

One electron of mass m_e and charge $-e$ orbiting a proton of charge $+e$. The potential energy is the Coulomb attraction:

$$V(r) = -\frac{e^2}{4\pi\epsilon_0 r}$$

This depends only on r — the distance between electron and proton. The Schrödinger equation is:

$$-\frac{\hbar^2}{2m_e}\nabla^2\psi + V(r)\psi = E\psi$$

4.5.2 — Why Spherical Coordinates

The Coulomb potential has **spherical symmetry**: V depends only on r, not on the angles θ and ϕ. Working in spherical coordinates (r, θ, ϕ) lets us exploit this symmetry directly.

The Laplacian in spherical coordinates:

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2}$$

This looks terrifying. The key observation: **the angular part is a known operator**. The expression in angular brackets is proportional to $\hat{L}^2$ — the squared angular momentum operator — which has known eigenfunctions (spherical harmonics).

4.5.3 — Separation of Variables: The Core Strategy

Try a solution of the form:

$$\psi(r, \theta, \phi) = R(r)\,Y(\theta, \phi)$$

Substituting into the Schrödinger equation and dividing through by $R(r)Y(\theta,\phi)$:

$$\underbrace{\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) - \frac{2m_e r^2}{\hbar^2}(V(r)-E)}_{\text{function of } r \text{ only}} = -\underbrace{\frac{1}{Y}\hat{\Lambda}^2 Y}_{\text{function of } \theta,\phi \text{ only}}$$

where $\hat{\Lambda}^2$ is the angular part of ∇^2. Since the left side depends only on r and the right only on θ, ϕ, **both sides must equal the same constant** — call it $\ell(\ell+1)$. This gives two separate equations:

Angular equation:

$$\hat{\Lambda}^2 Y = -\ell(\ell+1)Y$$

Radial equation:

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \left[\frac{2m_e}{\hbar^2}(E - V(r)) - \frac{\ell(\ell+1)}{r^2}\right]R = 0$$

The constant $\ell(\ell+1)$ is called the **separation constant**. It will turn out to be related to the orbital angular momentum.

4.6 — The Angular Equations: Quantum Numbers ℓ and m

4.6.1 — The ϕ Equation

Further separate $Y(\theta,\phi) = \Theta(\theta)\Phi(\phi)$. The ϕ equation is:

$$\frac{d^2\Phi}{d\phi^2} = -m^2\Phi$$

Solution: $\Phi(\phi) = e^{im\phi}$

Quantization of m: The wavefunction must be single-valued — physically, rotating by 2π must return you to the same state:

$$\Phi(\phi + 2\pi) = \Phi(\phi) \qquad \Longrightarrow \qquad e^{im\cdot 2\pi} = 1$$

This forces m to be an integer: $m = 0, \pm 1, \pm 2, \ldots$

The origin of quantized angular momentum: m is the magnetic quantum number — it is forced to be an integer by the simple requirement that the wavefunction is single-valued around the z-axis. This is the same topological argument that gives flux quantization in a superconducting ring (Ch. 7).

4.6.2 — The θ Equation: Associated Legendre Polynomials

The θ equation becomes:

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) + \left[\ell(\ell+1) - \frac{m^2}{\sin^2\theta}\right]\Theta = 0$$

The solutions are the **associated Legendre polynomials** $P_\ell^m(\cos\theta)$.

Quantization of ℓ: For $\Theta(\theta)$ to be finite at $\theta = 0$ and $\theta = \pi$ (the poles), the separation constant must satisfy:

$$\ell = 0, 1, 2, 3, \ldots \quad \text{and} \quad |m| \leq \ell$$

The constraint $|m| \leq \ell$ has a clear physical meaning: the z-component of angular momentum cannot exceed the total angular momentum.

4.6.3 — Spherical Harmonics

Combining the Θ and Φ solutions gives the **spherical harmonics** $Y_\ell^m(\theta, \phi)$:

$$Y_\ell^m(\theta, \phi) = \sqrt{\frac{(2\ell+1)}{4\pi}\frac{(\ell-|m|)!}{(\ell+|m|)!}} P_\ell^m(\cos\theta)\, e^{im\phi}$$

These are the angular eigenfunctions of the hydrogen atom — and of every spherically symmetric potential. The first few:

ℓ	m	Label	Y_ℓ^m
0	0	s	$1/\sqrt{4\pi}$
1	0	p_z	$\sqrt{3/4\pi}\cos\theta$
1	±1	$p_{x,y}$	$\mp\sqrt{3/8\pi}\sin\theta\, e^{\pm i\phi}$
2	0	d_{z^2}	$\sqrt{5/16\pi}(3\cos^2\theta - 1)$

The shapes of these functions — spherical, dumbbell, cloverleaf — are the shapes of the atomic orbitals that govern all of chemistry.

Key properties:

- Orthonormal: $\int Y_\ell^{m*} Y_{\ell'}^{m'}\, d\Omega = \delta_{\ell\ell'}\delta_{mm'}$
- Angular momentum: $\hat{L}^2 Y_\ell^m = \ell(\ell+1)\hbar^2 Y_\ell^m$ and $\hat{L}_z Y_\ell^m = m\hbar Y_\ell^m$

4.7 — The Radial Equation: Quantum Number n and Energy

4.7.1 — Substitution and Simplification

With $u(r) \equiv rR(r)$, the radial equation becomes:

$$-\frac{\hbar^2}{2m_e}\frac{d^2u}{dr^2} + \left[V(r) + \frac{\hbar^2\ell(\ell+1)}{2m_e r^2}\right]u = Eu$$

This looks like the one-dimensional Schrödinger equation with an **effective potential**:

$$V_{eff}(r) = -\frac{e^2}{4\pi\epsilon_0 r} + \underbrace{\frac{\hbar^2\ell(\ell+1)}{2m_e r^2}}_{\text{centrifugal barrier}}$$

The centrifugal term $\ell(\ell+1)\hbar^2/2m_e r^2$ pushes the electron away from the nucleus for $\ell > 0$ — the quantum analog of the classical centrifugal effect.

4.7.2 — Boundary Conditions and the Quantization of Energy

At $r \to \infty$: For bound states ($E < 0$), the wavefunction must vanish: $u(r) \to 0$.

At $r \to 0$: $u(0) = 0$ (since $R(r) = u/r$ must be finite at the origin).

Solving the radial equation with the Coulomb potential $V = -e^2/4\pi\epsilon_0 r$, the mathematical requirement that $R(r)$ remains finite at both boundaries forces the energy to take discrete values. The solutions are the **associated Laguerre polynomials** $L_{n-\ell-1}^{2\ell+1}$, and they only exist when:

$$n = 1, 2, 3, \ldots \qquad \text{and} \qquad \ell \leq n - 1$$

where n is called the **principal quantum number**.

4.7.3 — The Energy Levels of Hydrogen

$$\boxed{E_n = -\frac{m_e e^4}{2(4\pi\epsilon_0)^2\hbar^2} \cdot \frac{1}{n^2} = -\frac{13.6\,\text{eV}}{n^2}, \qquad n = 1, 2, 3, \ldots}$$

This is the **Bohr formula** — but now derived rigorously from the Schrödinger equation, not guessed from Bohr's circular orbit model. Bohr guessed correctly; Schrödinger explained why.

The **Bohr radius** a_0 sets the natural length scale:

$$a_0 = \frac{4\pi\epsilon_0\hbar^2}{m_e e^2} = 0.529\,\text{Å} = 0.0529\,\text{nm}$$

In terms of a_0, the energy becomes $E_n = -e^2/8\pi\epsilon_0 a_0 n^2$.

4.8 — The Complete Solution: Three Quantum Numbers

Assembling the pieces:

$$\boxed{\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r)\, Y_\ell^m(\theta, \phi)}$$

The three quantum numbers and their origin:

Quantum number	Name	Values	Origin	Physical meaning
n	Principal	$1, 2, 3, \ldots$	Normalizability of $R_{n\ell}$ at $r \to \infty$	Sets the energy: $E_n = -13.6/n^2$ eV
ℓ	Orbital/azimuthal	$0, 1, \ldots, n-1$	Finiteness of $R_{n\ell}$ at $r = 0$	Total orbital angular momentum: $L = \sqrt{\ell(\ell+1)}\hbar$
m	Magnetic	$-\ell, \ldots, 0, \ldots, +\ell$	Single-valuedness of $\Phi(\phi)$	z-component: $L_z = m\hbar$

The constraint chain: Solving one equation forces the next quantum number. Solving the ϕ equation gives integer m. The θ equation then requires $\ell \geq |m|$. The radial equation requires $n \geq \ell + 1$. The three quantum numbers are not three independent choices — they are three consecutive constraints on one wavefunction.

4.8.1 — Explicit Wavefunctions

Ground state ($n = 1$, $\ell = 0$, $m = 0$): the 1s state

$$\psi_{100} = \frac{1}{\sqrt{\pi}} \left(\frac{1}{a_0}\right)^{3/2} e^{-r/a_0}$$

Spherically symmetric (no angular dependence). Electron most likely found at $r = a_0$ (the Bohr radius). Energy: $E_1 = -13.6$ eV.

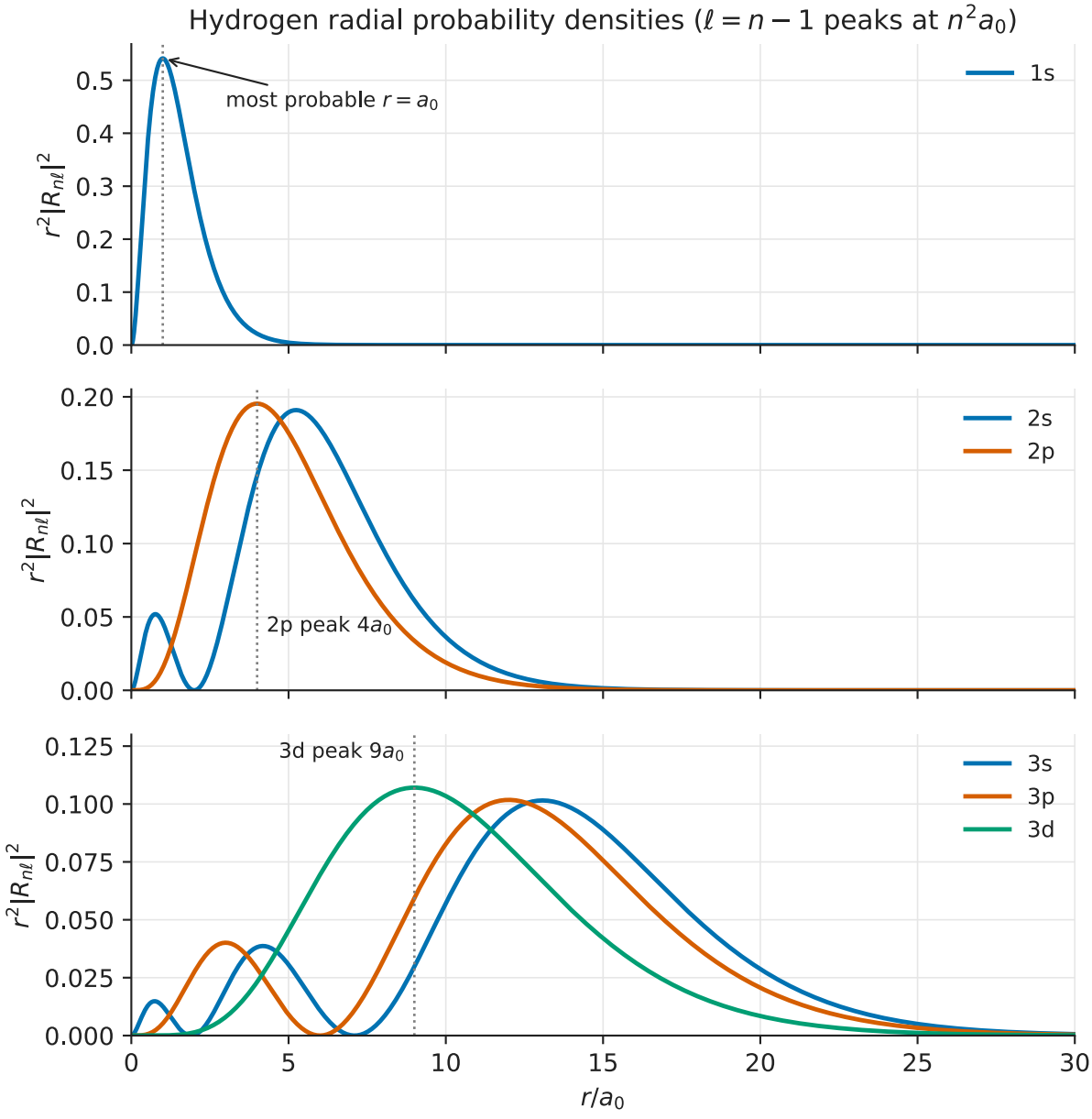
First excited states ($n = 2$): four degenerate states

$$\psi_{200} = \frac{1}{4\sqrt{2\pi}} \left(\frac{1}{a_0}\right)^{3/2} \left(2 - \frac{r}{a_0}\right) e^{-r/2a_0}$$

$$\psi_{210} = \frac{1}{4\sqrt{2\pi}} \left(\frac{1}{a_0}\right)^{3/2} \frac{r}{a_0} e^{-r/2a_0} \cos\theta$$

$$\psi_{21\pm1} = \mp\frac{1}{8\sqrt{\pi}} \left(\frac{1}{a_0}\right)^{3/2} \frac{r}{a_0} e^{-r/2a_0} \sin\theta\, e^{\pm i\phi}$$

All have energy $E_2 = -13.6/4 = -3.4$ eV.



Radial probability density $r^2|R_{n\ell}(r)|^2$ against r/a_0 for the $n = 1, 2, 3$ states (each normalised numerically to 1): the 1s peak is exactly at a_0, each state has $n - \ell - 1$ radial nodes, and the nodeless $\ell = n - 1$ states peak at $n^2 a_0$ (a_0, $4a_0$, $9a_0$), while low-ℓ states put a small inner lobe close to the nucleus.

4.9 — Degeneracy: Why Multiple States Have the Same Energy

For hydrogen, the energy $E_n = -13.6/n^2$ eV depends **only on** n, not on ℓ or m. For a given n:

- ℓ can take values $0, 1, \ldots, n-1$ (n values)
- For each ℓ, m takes $2\ell + 1$ values

Total number of degenerate states at energy E_n:

$$\sum_{\ell=0}^{n-1}(2\ell + 1) = n^2$$

So the ground state is non-degenerate (1 state), the first excited level has 4 degenerate states, the second has 9, and so on.

Why does this degeneracy exist? The Coulomb potential $V \propto 1/r$ has an extra, accidental symmetry beyond rotational symmetry — the **Laplace-Runge-Lenz vector** is conserved, giving an additional conserved quantity. By Noether's theorem (Ch. 1 §1.7), this additional symmetry produces the extra degeneracy. This is special to the $1/r$ potential; a perturbed Coulomb potential (e.g., relativistic corrections, spin-orbit coupling) breaks this degeneracy and splits the levels.

4.10 — Spectroscopy: The Rydberg Formula

When an electron transitions from a higher energy state n_i to a lower state n_f, a photon is emitted with energy:

$$h\nu = E_{n_i} - E_{n_f} = 13.6\,\text{eV}\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

The corresponding wavelength:

$$\boxed{\frac{1}{\lambda} = R_\infty\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right), \qquad R_\infty = \frac{m_e e^4}{8\epsilon_0^2 h^3 c} = 1.097 \times 10^7\,\text{m}^{-1}}$$

This is the **Rydberg formula**, empirically discovered in 1888, theoretically derived here. The **Rydberg constant** R_∞ is one of the most precisely measured constants in physics ($R_\infty = 1.0973731568539 \times 10^7\,\text{m}^{-1}$) and is used to define the hydrogen energy scale in the SI system.

Spectral series:

Series	n_f	Spectral region	Engineering relevance
Lyman	1	Ultraviolet	UV lithography, astrophysics
Balmer	2	Visible	The red H-α line at 656 nm; plasma diagnostics

Series	n_f	Spectral region	Engineering relevance
Paschen	3	Near infrared	IR spectroscopy
Brackett	4	Infrared	Fiber-optic window

4.11 — Orbital Notation and Chemists' Language

The quantum numbers ($\ell = 0, 1, 2, 3$) have spectroscopic labels (s, p, d, f) — a historical accident from "sharp, principal, diffuse, fundamental" series of spectral lines, before quantum mechanics existed.

A state with $n = 2$, $\ell = 1$ is called "2p". The labels you see in chemistry:

ℓ	Label	Number of m values	States
0	s	1	$m = 0$
1	p	3	$m = -1, 0, +1$
2	d	5	$m = -2, -1, 0, +1, +2$
3	f	7	$m = -3, \ldots, +3$

At $n = 1$: only $1s$ (1 state). At $n = 2$: $2s$ and $2p$ (4 states). At $n = 3$: $3s$, $3p$, $3d$ (9 states).

Including spin (from Ch. 3 Bridge A, the Pauli equation): each orbital state splits into spin-up ($m_s = +\frac{1}{2}$) and spin-down ($m_s = -\frac{1}{2}$), doubling the count:

$$\text{Capacity of shell } n = 2n^2$$

Shell $n = 1$ holds 2 electrons, $n = 2$ holds 8, $n = 3$ holds 18. This is **why** the rows of the periodic table have lengths 2, 8, 18, 32...

4.12 — The Correspondence Principle Revisited

At large n, the classical orbit of hydrogen has radius $r_n = n^2 a_0$ and frequency $\nu_n = e^2/(4\pi\epsilon_0 2\hbar n^3)$. The photon frequency emitted in a transition from n to $n-1$:

$$\nu = R_\infty c \left(\frac{1}{(n-1)^2} - \frac{1}{n^2} \right) \xrightarrow{n \to \infty} \frac{2R_\infty c}{n^3} = \nu_n^{classical}$$

At large n, the quantum orbital frequency matches the classical orbit frequency exactly. **Quantum mechanics reduces to classical mechanics at large quantum numbers** — the correspondence principle of Ch. 1 §1.8, seen in the hydrogen spectrum.

4.13 — What This Chapter Has Built

The hydrogen atom solution rests on three pillars, each of which you now have:

Pillar 1: The Schrödinger equation — derived in Ch. 3 from $\mathcal{L}_{SM}$

Pillar 2: Separation of variables — a mathematical technique for PDEs with symmetric potentials; valid here because $V = V(r)$ only

Pillar 3: Boundary conditions imposing quantization — the same principle as the infinite square well, now in 3D with three separate conditions giving three separate quantum numbers

Together they give a **complete and exact solution of the nonrelativistic Coulomb Schrödinger model** for hydrogen.

That claim needs its two halves kept apart, because they are different claims. *Exact* is true of the model: for a point nucleus of infinite mass and no relativistic or radiative corrections, the separation-of-variables solution above is the whole answer, with no residual equation left to solve. *Accurate for real hydrogen* is not, and the gap is far larger than the model's elegance suggests. The physical spectrum carries, among others:

- **finite proton mass** — the reduced-mass correction, worth $\approx 5 \times 10^{-4}$ on its own;
- **relativistic** fine structure (spin–orbit, relativistic mass, Darwin);
- the **Lamb shift** and other radiative corrections;
- **hyperfine** structure, recoil, and finite proton size.

The reduced-mass correction alone is some 10^4 times larger than one part in 10^8, so the model cannot be said to reproduce spectroscopy to that accuracy under any reading. The honest summary is: **exact solution of the nonrelativistic Coulomb model**, accurate to about one part in 10^4 for real hydrogen, with the rest supplied by the corrections catalogued in §4.13 and developed in Ch. 5.

4.14 — Summary

Result	Equation	Origin
Schrödinger equation	$\hat{H}\psi = E\psi$	Ch. 3 Bridge A
Particle in box: energies	$E_n = n^2\pi^2\hbar^2/2mL^2$	Boundary conditions
Harmonic oscillator: energies	$E_n = \hbar\omega(n + \frac{1}{2})$	Ladder operators
Tunneling transmission	$T \approx e^{-2\kappa d}$	Exponential decay in barrier
Hydrogen energies	$E_n = -13.6\,\text{eV}/n^2$	Radial equation + normalizability
Quantum numbers	$n \geq 1;\ \ell \leq n-1;\ \lvert m\rvert \leq \ell$	Three quantum numbers label a state
Rydberg formula	$1/\lambda = R_\infty(1/n_f^2 - 1/n_i^2)$	Energy conservation for photon emission
Shell capacity (with spin)	$2n^2$ electrons per shell	Degeneracy × 2 for spin

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4.15 — Looking Ahead: Chapter 5

The hydrogen atom has one electron — the problem is exactly solvable. Real atoms have many electrons. Adding them one by one using three facts:

1. The quantum numbers (n, ℓ, m, m_s) from this chapter
2. The Pauli exclusion principle from Ch. 0 (anticommutation of fermion fields)
3. The spin from Ch. 3 (Foldy-Wouthuysen, Pauli equation)

...the periodic table falls out completely, without any additional assumptions.

The Zeeman effect, fine structure, and the spectroscopy of real atoms come from the FW correction terms (spin-orbit, Darwin, relativistic mass) that were systematically set aside in §3.5 of Bridge A. Chapter 5 puts them back in.

End of Chapter 4.

Next: Chapter 5 — Spin, Many-Electron Systems, and the Periodic Table

Spin, Many-Electron Systems, and the Periodic Table

From One Electron to All of Chemistry

> *Pauli's exclusion principle is what prevents your hand from going through a table. It is not the electromagnetic repulsion of electrons — it is the antisymmetry of their wavefunction.* — common observation among condensed matter physicists

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5.0 — Where We Are

Chapter 4 solved the hydrogen atom exactly and derived three quantum numbers (n, ℓ, m). The solution depended on one simplification: a single electron.

Two things were left aside in Chapter 4 that this chapter now restores:

First: The electron has spin. It was derived from the Dirac equation in Ch. 3 and placed in the Pauli equation, but then suppressed when we descended further to the Schrödinger equation. Spin generates a fourth quantum number m_s, is responsible for fine structure, the Zeeman effect, and magnetic resonance.

Second: Real atoms have more than one electron. Adding a second electron immediately makes the Schrödinger equation impossible to solve exactly — but a single additional principle (the Pauli exclusion principle, traceable to Ch. 0's anticommutation relations) is enough to determine the structure of every atom in the periodic table without solving anything exactly.

By the end of this chapter, the periodic table is not a memorization exercise. It is a theorem.

5.1 — Spin: The Fourth Quantum Number

5.1.1 — Where Spin Comes From

In Ch. 3 §3.4.1, spin fell out of the Dirac equation automatically. The four-component Dirac spinor contains two spin degrees of freedom per particle as a mathematical consequence of requiring a first-order, Lorentz-covariant wave equation consistent with the Clifford algebra $\gamma^\mu, \gamma^\nu = 2g^{\mu\nu}$.

Spin is not the electron physically rotating — an electron is a point particle with no size, so "spinning" is meaningless classically. Spin is an intrinsic angular momentum with no classical analog, forced into existence by relativistic quantum mechanics.

5.1.2 — Spin Operators

The spin operator $\hat{\mathbf{S}}$ satisfies the same algebra as orbital angular momentum $\hat{\mathbf{L}}$:

$$[\hat{S}_x, \hat{S}_y] = i\hbar\hat{S}_z, \quad [\hat{S}_y, \hat{S}_z] = i\hbar\hat{S}_x, \quad [\hat{S}_z, \hat{S}_x] = i\hbar\hat{S}_y$$

The eigenvalues of $\hat{S}^2$ and $\hat{S}_z$:

$$\hat{S}^2\,\chi = s(s+1)\hbar^2\,\chi, \qquad \hat{S}_z\,\chi = m_s\hbar\,\chi$$

For the electron: $s = \frac{1}{2}$ (a fixed intrinsic property), so:

$$\hat{S}^2\,\chi = \frac{3}{4}\hbar^2\,\chi, \qquad m_s = \pm\frac{1}{2}$$

There are exactly two spin states. In the Pauli equation (Ch. 3 §3.6), they are represented by two-component **spinors**:

$$\chi_+ = \begin{pmatrix}1\\0\end{pmatrix} \equiv\uparrow \quad \text{(spin-up, } m_s = +\tfrac{1}{2}\text{)}$$

$$\chi_- = \begin{pmatrix}0\\1\end{pmatrix} \equiv\downarrow \quad \text{(spin-down, } m_s = -\tfrac{1}{2}\text{)}$$

In terms of the Pauli matrices (which appeared in the FW transformation of Ch. 3):

$$\hat{\mathbf{S}} = \frac{\hbar}{2}\boldsymbol{\sigma} = \frac{\hbar}{2}\begin{pmatrix}\sigma_x\\ sigma_y\\ sigma_z\end{pmatrix}$$

5.1.3 — The Stern-Gerlach Experiment: Spin Made Visible

In 1922, Otto Stern and Walther Gerlach sent a beam of silver atoms through an inhomogeneous magnetic field. Classically, a magnetic dipole in a field gradient experiences a force $F_z = \mu_z\,\partial B_z/\partial z$; if the orientation of the magnetic moment is continuous, the beam should spread into a continuous smear on the detector.

Instead: two discrete spots appeared.

Silver has the electronic configuration $[\text{Kr}], 4d^{10}, 5s^1$ — one unpaired electron with zero orbital angular momentum ($\ell = 0$) and spin $s = 1/2$. The magnetic moment is purely from spin. The two spots correspond to $m_s = +1/2$ and $m_s = -1/2$, separated by:

$$\Delta z = \frac{\mu_B}{m_{\text{atom}} v^2}\frac{\partial B_z}{\partial z} L_{\text{field}}$$

This experiment directly measures spin quantization. The discreteness is not a resolution limit — it is forced by the eigenvalue equation $\hat{S}_z\chi = m_s\hbar\chi$.

Engineering connection: The Stern-Gerlach principle is the basis of atomic beam frequency standards. The cesium atomic clock — which defines the SI second — uses a magnetic state selector that works exactly like a Stern-Gerlach apparatus, selecting the $m_s = 0$ hyperfine transition to define $1\,\text{s} \equiv 9{,}192{,}631{,}770$ periods.

5.1.4 — The Complete Hydrogen Quantum Numbers

Adding spin, the complete state of the electron in hydrogen is:

$$\psi_{n\ell m m_s}(\mathbf{r}) = \psi_{n\ell m}(\mathbf{r})\,\chi_{m_s}$$

Four quantum numbers, four constraints:

Quantum number	Symbol	Values	Constraint origin
Principal	n	$1, 2, 3, \ldots$	Normalizability of radial equation
Orbital	ℓ	$0, 1, \ldots, n-1$	Finiteness at $r = 0$
Magnetic	m	$-\ell, \ldots, +\ell$	Single-valuedness in ϕ
Spin	m_s	$+\frac{1}{2},, -\frac{1}{2}$	Dirac equation (Ch. 3 §3.4.1)

The degeneracy of level n including spin: $2n^2$.

5.2 — Many-Electron Atoms: The Antisymmetry Requirement

5.2.1 — Why the Two-Electron Problem Has No Exact Solution

For helium (two electrons at positions $\mathbf{r}_1$ and $\mathbf{r}_2$):

$$\hat{H} = \underbrace{-\frac{\hbar^2}{2m_e}\nabla_1^2 - \frac{Ze^2}{4\pi\epsilon_0 r_1}}_{\text{electron 1 alone}} + \underbrace{-\frac{\hbar^2}{2m_e}\nabla_2^2 - \frac{Ze^2}{4\pi\epsilon_0 r_2}}_{\text{electron 2 alone}} + \underbrace{\frac{e^2}{4\pi\epsilon_0 |\mathbf{r}_1 - \mathbf{r}_2|}}_{\text{electron-electron repulsion}}$$

The first two terms would be exactly solvable — two independent hydrogen-like problems. The last term, the electron-electron repulsion, couples the two coordinates and makes the problem analytically intractable. This is the quantum mechanical three-body problem.

There is no exact closed-form solution for helium or any atom with $Z > 1$.

Yet we can determine the structure of every atom in the periodic table. The key is a principle, not a solution.

5.2.2 — The Pauli Exclusion Principle: Its True Origin

From Ch. 0 §0.5: the fermion field ψ satisfies **anticommutation relations**:

$$\hat{\psi}(\mathbf{r}), \hat{\psi}^\dagger(\mathbf{r}') = \delta^3(\mathbf{r} - \mathbf{r}')$$

This is not put in by hand. It is required by Lorentz invariance and causality for half-integer spin fields (the spin-statistics theorem, Ch. 0 §0.5).

The consequence for many-electron wavefunctions: swapping any two identical electrons must change the sign of the wavefunction:

$$\Psi(\ldots, \mathbf{x}_i, \ldots, \mathbf{x}_j, \ldots) = -\Psi(\ldots, \mathbf{x}_j, \ldots, \mathbf{x}_i, \ldots)$$

where $\mathbf{x} = (\mathbf{r}, m_s)$ denotes both spatial and spin coordinates. This is **antisymmetry** — the full meaning of the Pauli exclusion principle.

Immediate corollary: If two electrons have identical quantum numbers ($\mathbf{x}_i = \mathbf{x}_j$), then swapping them gives:

$$\Psi = -\Psi \qquad \Longrightarrow \qquad \Psi = 0$$

A wavefunction that is identically zero means the probability of that configuration is exactly zero. **Two electrons cannot occupy the same quantum state.** This is the Pauli exclusion principle — a theorem, not a postulate.

5.2.3 — The Slater Determinant

The correct antisymmetric wavefunction for N electrons, each occupying a single-particle state ϕ_i, is:

$$\Psi(\mathbf{x}_1, \ldots, \mathbf{x}_N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \phi_1(\mathbf{x}_1) & \phi_1(\mathbf{x}_2) & \cdots & \phi_1(\mathbf{x}_N) \\ \phi_2(\mathbf{x}_1) & \phi_2(\mathbf{x}_2) & \cdots & \phi_2(\mathbf{x}_N) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_N(\mathbf{x}_1) & \phi_N(\mathbf{x}_2) & \cdots & \phi_N(\mathbf{x}_N) \end{vmatrix}$$

This is the **Slater determinant**. Its key properties:

- Swapping two columns (= two electrons) changes the sign of the determinant ✓
- If two rows are identical (= two electrons in the same state), the determinant is zero ✓

The Slater determinant is not the exact wavefunction of a real atom — it neglects correlation between electron positions. But it is the correct starting point for the Hartree-Fock mean-field theory and for density functional theory (DFT), both of which are the workhorses of computational chemistry and materials simulation.

5.3 — Screening and the Lifting of Hydrogen's Degeneracy

In hydrogen, the energy depends only on n — all states with the same n are degenerate regardless of ℓ. In multi-electron atoms, this degeneracy is lifted. The reason is **screening**.

5.3.1 — The Effective Nuclear Charge

Inner electrons partially shield the outer electrons from the full nuclear charge Z. An electron in orbital $n\ell$ sees an **effective nuclear charge**:

$$Z_{\text{eff}} = Z - \sigma_{n\ell}$$

where $\sigma_{n\ell}$ is the **screening constant** — the amount of nuclear charge shielded by inner electrons. Slater's rules give approximate values.

The energy of a state in a multi-electron atom is approximately:

$$E_{n\ell} \approx -\frac{Z_{\text{eff}}^2 \times 13.6\,\text{eV}}{n^2}$$

5.3.2 — Why $2s$ Has Lower Energy Than $2p$

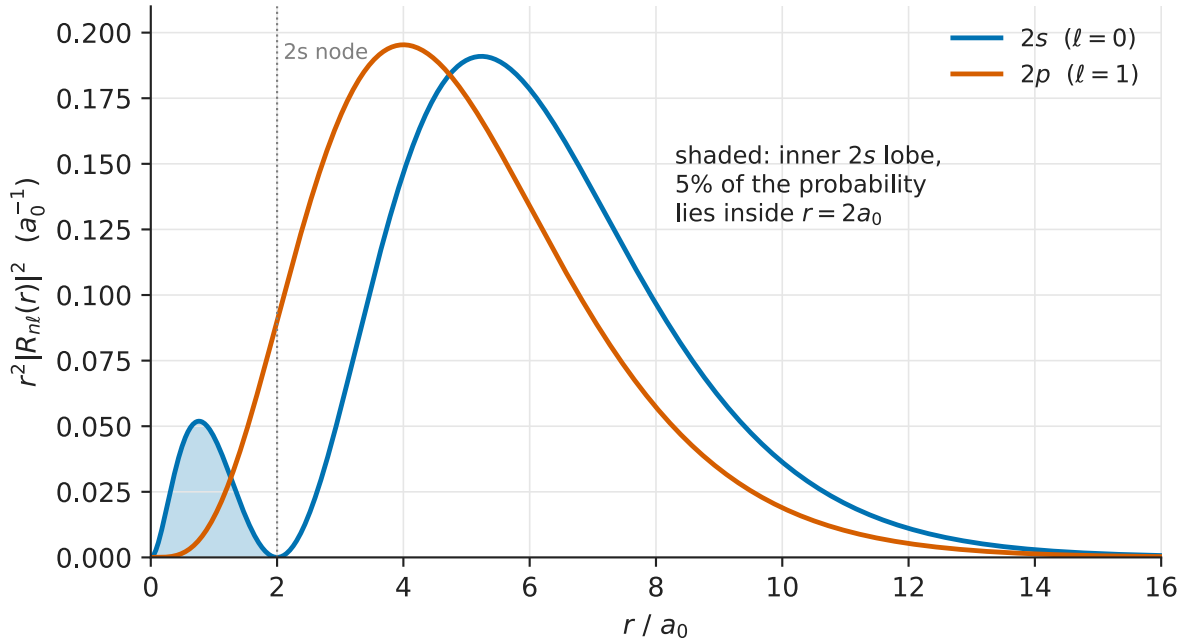
For $\ell = 0$ (s-states), the radial wavefunction has nonzero probability at the nucleus ($|\psi_{n00}(0)|^2 > 0$). An s-electron "penetrates" to the nucleus and is not fully shielded — it sees a larger Z_{eff} and has lower (more negative) energy.

For $\ell > 0$ (p, d, f-states), the centrifugal barrier $\hbar^2\ell(\ell+1)/2m_e r^2$ in the effective potential keeps the electron further from the nucleus. A p-electron is more effectively screened and has higher energy.

The energy ordering in multi-electron atoms:

$$E_{ns} < E_{np} < E_{nd} < E_{nf} \quad \text{(for the same } n\text{)}$$

This splitting of energies within a shell is what makes the periodic table non-trivial. In hydrogen, $2s$ and $2p$ are degenerate. In lithium, $2s$ is significantly lower than $2p$ — which is why lithium's third electron goes into $2s$, not $2p$.



Radial probability $r^2|R_{n\ell}|^2$ for hydrogenic $2s$ and $2p$ ($Z = 1$): the $2p$ peak lies closer in, but the $2s$ orbital has a small inner lobe (shaded, about 5% of its probability inside its node at $r = 2a_0$) that penetrates the core electrons and feels the nearly unscreened nucleus.

5.4 — Building the Periodic Table: Aufbau and Hund

5.4.1 — The Aufbau Principle

Fill orbitals from lowest to highest energy, subject to:

- Pauli exclusion: at most 2 electrons per orbital (one spin-up, one spin-down)
- Hund's first rule (below)

The energy ordering of orbitals in multi-electron atoms:

$$1s \to 2s \to 2p \to 3s \to 3p \to 4s \to 3d \to 4p \to 5s \to 4d \to 5p \to \cdots$$

Why does $4s$ fill before $3d$? Although $n = 3$ is closer to the nucleus, the $3d$ wavefunction has high angular momentum ($\ell = 2$), is pushed outward by the centrifugal barrier, and is well-screened. The $4s$ wavefunction ($\ell = 0$) penetrates the core and sees a large Z_{eff}, making its energy lower than $3d$ for elements from K to Zn.

5.4.2 — Hund's Rules

For a partially filled subshell (multiple electrons in orbitals with the same n and ℓ):

Rule 1 (most important): Maximize total spin S. Electrons singly occupy orbitals before doubling up, with parallel spins.

Rule 2: For the same S, maximize total orbital angular momentum L.

Rule 3: For less-than-half-filled subshells, $J = |L - S|$; for more-than-half, $J = L + S$.

Why Rule 1? Two electrons with parallel spins must have different spatial wavefunctions (Pauli exclusion). This spatial antisymmetry keeps them further apart, reducing their Coulomb repulsion energy — so parallel spins are energetically preferred. This is the **exchange interaction** — a purely quantum effect with no classical analog.

Example: Carbon ($Z = 6$, configuration $1s^2 2s^2 2p^2$): The two $2p$ electrons have $\ell = 1$ with three m values $(-1, 0, +1)$. By Hund's Rule 1, both occupy different m values with parallel spins: $m = 0, m_s = +\frac{1}{2}$ and $m = +1, m_s = +\frac{1}{2}$. Ground state: ${}^3P_0$.

5.4.3 — Period-by-Period Construction

Period 1 ($n = 1$ shell, capacity 2):

Z	Element	Configuration	Notes
1	H	$1s^1$	One unpaired electron
2	He	$1s^2$	First noble gas; $1s$ shell complete

Period 2 ($n = 2$ shell, capacity 8):

Z	Element	Configuration	Chemical character
3	Li	[He]$2s^1$	Alkali metal; easily loses $2s$ electron
4	Be	[He]$2s^2$	Alkaline earth
5	B	[He]$2s^2 2p^1$	First p-block element
6	C	[He]$2s^2 2p^2$	Four valence electrons; basis of organic chemistry
7	N	[He]$2s^2 2p^3$	Half-filled $2p$ (Hund maximum); very stable
8	O	[He]$2s^2 2p^4$	Two unpaired p electrons; highly reactive
9	F	[He]$2s^2 2p^5$	One electron short of noble gas; most electronegative
10	Ne	[He]$2s^2 2p^6$	Noble gas; $2p$ complete

Period 3 ($n = 3$ shell, $3s$ and $3p$ only — $3d$ doesn't fill yet):

Z	Element	Config	Notes
11	Na	[Ne]$3s^1$	Alkali; analogue of Li
14	Si	[Ne]$3s^2 3p^2$	Semiconductor; same valence structure as C
18	Ar	[Ne]$3s^2 3p^6$	Noble gas

Period 4 ($4s$ fills first, then $3d$ transition metals, then $4p$):

Range	Elements	Notes
$Z = 19, 20$	K, Ca	$4s^1$, $4s^2$ — fill $4s$ before $3d$
$Z = 21$–30	Sc–Zn	$3d$ transition metals; magnetic properties from partially filled $3d$
$Z = 31$–36	Ga–Kr	Fill $4p$; Kr is next noble gas

The 10 transition metals (Sc through Zn) arise from filling the 5 $3d$ orbitals. Their chemical and magnetic properties (ferromagnetism in Fe, Co, Ni; catalytic activity of transition metals) all trace back to unpaired $3d$ electrons and the Hund's Rule exchange interaction.

5.4.4 — Periodic Trends

Three key properties vary systematically across the table, all explained by screening and orbital filling:

Atomic radius: Decreases across a period (increasing Z_{eff} pulls electrons inward), increases down a group (larger $n \to$ larger orbitals). This determines bond lengths, ionic sizes, and crystal structure.

Ionization energy: Energy to remove the outermost electron. Increases across a period (larger $Z_{\text{eff}} \to$ harder to remove). Noble gases have the highest ionization energies; alkali metals the lowest. This determines electrochemical series, corrosion behavior, and semiconductor doping character.

Electronegativity: Tendency to attract electrons in a bond. Increases across a period and up a group. F is the most electronegative; Cs the least. Electronegativity differences determine bond polarity, solubility, and surface chemistry.

Engineering consequence: Every materials selection decision in EEE, ME, CE, and ChE ultimately traces back to these three properties and the electronic configurations that produce them. Copper's high conductivity comes from its $3d^{10}4s^1$ configuration giving a half-filled $4s$ band. Silicon's semiconductor behavior comes from its four-valence-electron $3s^2 3p^2$ configuration. Steel's properties are those of iron ($3d^6 4s^2$) modified by carbon occupying interstitial sites.

5.5 — Fine Structure: Spin-Orbit Coupling

In Ch. 3 §3.5.3, the Foldy-Wouthuysen transformation at order $(v/c)^2$ gave — among other terms — the spin-orbit coupling:

$$\hat{H}_{SO} = -\frac{e\hbar}{4m_e^2 c^2}\boldsymbol{\sigma} \cdot (\mathbf{E} \times \hat{\mathbf{p}})$$

For the Coulomb field of the nucleus, $\mathbf{E} = (e/4\pi\epsilon_0 r^3)\mathbf{r}$, this becomes:

$$\boxed{\hat{H}_{SO} = \frac{1}{2m_e^2 c^2}\frac{1}{r}\frac{dV}{dr}\hat{\mathbf{L}} \cdot \hat{\mathbf{S}} \equiv \xi(r)\hat{\mathbf{L}} \cdot \hat{\mathbf{S}}}$$

where the factor of $\frac{1}{2}$ is the **Thomas precession correction** (a relativistic kinematic effect).

5.5.1 — Physical Picture

An electron moving through the electric field $\mathbf{E}$ of the nucleus sees, in its own rest frame, a magnetic field:

$$\mathbf{B}_{\text{eff}} \approx -\frac{\mathbf{v} \times \mathbf{E}}{c^2}$$

The electron's spin magnetic moment $\boldsymbol{\mu}_S = -g_e \mu_B \mathbf{S}/\hbar$ (with $g_e = 2$ from the Dirac equation) interacts with this effective field. **The spin-orbit coupling is the magnetic moment of the electron interacting with the magnetic field it experiences due to its own orbital motion.** The coupling strength is $\propto (v/c)^2 \propto \alpha^2$, which is why it is a fine structure correction.

5.5.2 — Total Angular Momentum J

Because $\hat{H}_{SO} = \xi(r)\hat{\mathbf{L}} \cdot \hat{\mathbf{S}}$ couples $\mathbf{L}$ and $\mathbf{S}$, neither L_z nor S_z is individually conserved. What is conserved is the **total angular momentum**:

$$\hat{\mathbf{J}} = \hat{\mathbf{L}} + \hat{\mathbf{S}}$$

Using $\hat{\mathbf{L}} \cdot \hat{\mathbf{S}} = \frac{1}{2}(\hat{J}^2 - \hat{L}^2 - \hat{S}^2)$:

$$\hat{H}_{SO} = \frac{\xi(r)}{2}\left(\hat{J}^2 - \hat{L}^2 - \hat{S}^2\right)$$

The energy shift from spin-orbit coupling in hydrogen (first-order perturbation theory):

$$\boxed{E_{SO} = \frac{\xi_{n\ell}}{2}\hbar^2 \left[j(j+1) - \ell(\ell+1) - s(s+1)\right]}$$

where $\xi_{n\ell} = \langle n\ell | \xi(r) | n\ell \rangle$ and $j = \ell \pm \frac{1}{2}$.

5.5.3 — The j Quantum Number

For an electron with orbital quantum number ℓ and spin $s = \frac{1}{2}$:

$$j = \ell + \frac{1}{2} \quad \text{or} \quad j = \ell - \frac{1}{2} \quad (\text{for } \ell \geq 1)$$

For $\ell = 0$: only $j = \frac{1}{2}$ (spin and orbital don't couple when there's no orbital angular momentum).

Each j value gives a different energy — the spin-orbit coupling **splits the $\ell > 0$ levels into doublets**:

$$\Delta E_{SO} = E_{j=\ell+1/2} - E_{j=\ell-1/2} = \frac{\xi_{n\ell}}{2}\hbar^2(2\ell+1)$$

This is the **fine structure** of the spectral lines.

5.5.4 — Spectroscopic Term Symbols

The state of a single electron is specified by (n, ℓ, j) using the notation:

$$n,^{2s+1}L_j$$

where $L = s, p, d, f, \ldots$ for $\ell = 0, 1, 2, 3, \ldots$ and $2s+1$ is the spin multiplicity (2 for one electron, since $s = 1/2$).

State	n	ℓ	j	Term	Energy relative to ground
Hydrogen 2p	2	1	1/2	$2,^2P_{1/2}$	$E_2 + E_{SO}(j=1/2)$
Hydrogen 2p	2	1	3/2	$2,^2P_{3/2}$	$E_2 + E_{SO}(j=3/2)$

The splitting between $2,^2P_{1/2}$ and $2,^2P_{3/2}$ in hydrogen is $\Delta E = 4.53 \times 10^{-5}$ eV — tiny but precisely measurable and exactly matching the FW prediction from Ch. 3.

5.6 — The Zeeman Effect: Spin Included

5.6.1 — Interaction with an External Field

Adding an external magnetic field $\mathbf{B}$ along $\hat{z}$, the perturbation Hamiltonian is:

$$\hat{H}_Z = -\hat{\boldsymbol{\mu}} \cdot \mathbf{B} = \frac{e}{2m_e}\left(\hat{\mathbf{L}} + 2\hat{\mathbf{S}}\right) \cdot \mathbf{B} = \frac{eB}{2m_e}\left(\hat{L}_z + 2\hat{S}_z\right)$$

The factor of 2 in front of $\hat{S}_z$ is $g_e = 2$ from the Dirac equation (Ch. 3 §3.4.2) — not put in by hand.

5.6.2 — Weak Field: The Anomalous Zeeman Effect

For B weak enough that $\hat{H}_Z \ll \hat{H}_{SO}$, the spin-orbit coupling dominates and $\mathbf{J} = \mathbf{L} + \mathbf{S}$ is a good quantum number. The energy shift is:

$$\Delta E = g_J \mu_B B m_J$$

where $\mu_B = e\hbar/2m_e = 9.274 \times 10^{-24}$ J/T is the **Bohr magneton**, and g_J is the **Landé g-factor**:

$$\boxed{g_J = 1 + \frac{j(j+1) + s(s+1) - \ell(\ell+1)}{2j(j+1)}}$$

This gives $g_J = 1$ for purely orbital states (S = 0) and $g_J = 2$ for purely spin states ($\ell = 0$), consistent with Ch. 3.

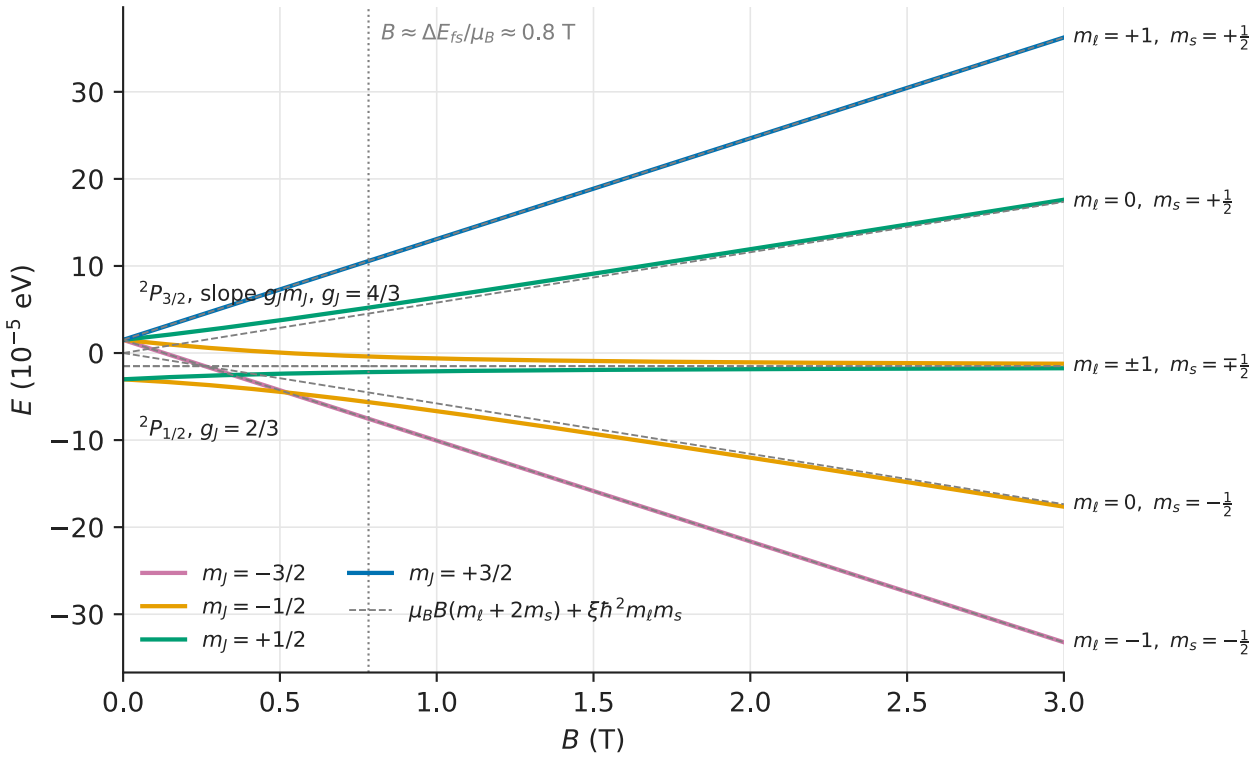
Why it was called "anomalous": Before spin was understood, the factor of 2 for spin was mysterious. It seemed as if the electron's magnetism was "anomalous." After the Dirac equation explained $g_e = 2$, it became simply the Landé formula. Nothing about it is anomalous — it is an exact consequence of quantum mechanics.

5.6.3 — Strong Field: The Paschen-Back Effect

For B large enough that $\hat{H}_Z \gg \hat{H}_{SO}$, the external field decouples $\mathbf{L}$ and $\mathbf{S}$ from each other; both couple independently to $\mathbf{B}$:

$$\Delta E = \mu_B B (m_\ell + 2m_s)$$

The Paschen-Back regime shows that $\mathbf{J}$ is not fundamental — it is the good quantum number only when spin-orbit coupling dominates. In strong fields, m_ℓ and m_s are separately conserved (separately good quantum numbers).



Exact eigenvalues of $\xi\,\mathbf{L}\cdot\mathbf{S} + \mu_B B(L_z + 2S_z)$ for hydrogen $2p$ (six $|m_\ell, m_s\rangle$ states, ξ fixed by the zero-field $^2P_{1/2}$–$^2P_{3/2}$ splitting of 4.53×10^{-5} eV): at low field the levels fan out with slopes $g_J m_J \mu_B$ ($g_J = 4/3$ and $2/3$), and above $B \approx \Delta E_{fs}/\mu_B \approx 0.8$ T they approach the dashed Paschen-Back lines $\mu_B B(m_\ell + 2m_s)$; levels of equal m_J never cross.

Engineering connection — NMR and MRI: The Zeeman effect is the foundation of NMR and MRI. A nucleus with spin I (e.g., ^{1}H has $I = 1/2$, ^{13}C has $I = 1/2$) in a magnetic field B_0 has energy levels split by:

$$\Delta E = \gamma_N \hbar B_0$$

where γ_N is the **nuclear gyromagnetic ratio**. An RF pulse at the Larmor frequency $\omega_L = \gamma_N B_0$ drives Rabi oscillations between the m_I levels. The free induction decay after the pulse contains structural and chemical information. MRI at 3 T uses $\omega_L = 128$ MHz for protons.

5.7 — Many-Electron Term Symbols

For atoms with multiple valence electrons, the total L, S, and J are formed by vector addition of individual angular momenta.

LS coupling (Russell-Saunders): Valid for light atoms ($Z \lesssim 30$). Individual spins couple to total S; individual orbital moments to total L; then $\mathbf{J} = \mathbf{L} + \mathbf{S}$.

jj coupling: Valid for heavy atoms ($Z \gtrsim 60$). Each electron's spin and orbital angular momentum couple first (giving individual j_i), then these couple to total J. Spin-orbit coupling is stronger than electron-electron repulsion.

The ground state term symbol for LS coupling:

$$^{2S+1}L_J$$

where $L = 0, 1, 2, 3, 4, \ldots$ is written as $S, P, D, F, G, \ldots$

Examples of ground state terms:

Atom	Config	S	L	J	Term	Why
H	$1s^1$	1/2	0	1/2	$^2S_{1/2}$	Single s electron
He	$1s^2$	0	0	0	$^1S_0$	Paired spins; closed shell
C	$2p^2$	1	1	0	$^3P_0$	Hund: max S, max L, $J = L - S$ (< half-filled)
N	$2p^3$	3/2	0	3/2	$^4S_{3/2}$	Half-filled p: max S; $L = 0$ by symmetry
O	$2p^4$	1	1	2	$^3P_2$	More than half-filled: $J = L + S$
Fe	$3d^6$	2	2	4	$^5D_4$	More than half-filled d; $J = L + S$

Iron ($Z = 26$, $^5D_4$) has 4 unpaired $3d$ electrons giving $S = 2$ — a large magnetic moment of $4\mu_B$. This is why iron is ferromagnetic: sufficiently strong exchange interactions align these large moments macroscopically.

5.8 — Molecular Bonding: Where Chemistry Begins

The periodic table tells you what each atom wants to do. Molecular bonding tells you what happens when atoms find each other. This section is a preview of Ch. 6, where bonding is treated fully.

5.8.1 — Linear Combination of Atomic Orbitals (LCAO)

When two hydrogen atoms approach, their $1s$ wavefunctions overlap. Form linear combinations (the molecular orbital approach):

$$\psi_\pm = \frac{1}{\sqrt{2(1 \pm S_{AB})}}(\phi_{1s}^A \pm \phi_{1s}^B)$$

where $S_{AB} = \int \phi_{1s}^A \phi_{1s}^B \, d^3r$ is the **overlap integral**.

- ψ_+ (symmetric, bonding): probability density **between** the nuclei is enhanced — reduces the electron-nuclear distance and lowers energy
- ψ_- (antisymmetric, antibonding): node between the nuclei — electrons are excluded from the internuclear region, raising energy

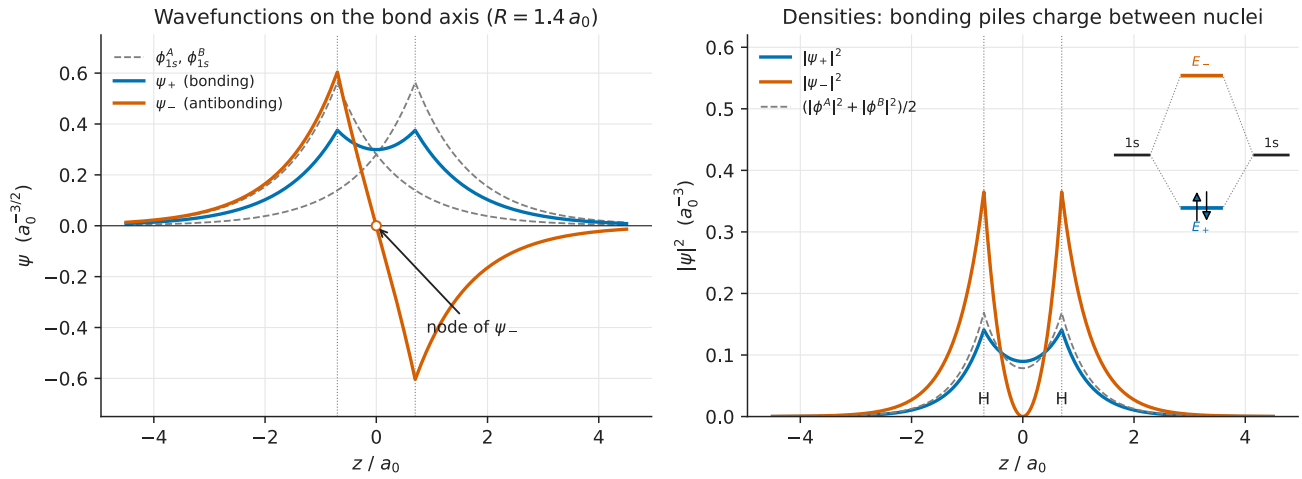
Energy of the two molecular orbitals:

$$E_\pm = E_{1s} \pm \frac{H_{AB} - E_{1s} S_{AB}}{1 \pm S_{AB}}$$

where $H_{AB} = \int \phi_{1s}^A \hat{H} \phi_{1s}^B \, d^3r < 0$ is the resonance integral (negative, so E_+ is below E_{1s}).

Two electrons (one from each H atom, with opposite spins by Pauli exclusion) both occupy ψ_+: total energy is $2E_+ < 2E_{1s}$. The molecule is stable.

This is the covalent bond in H_2 — derived from the same quantum mechanics that gave us the hydrogen atom, using only the Pauli exclusion principle and the variational principle.



Hydrogen 1s orbitals $e^{-|z \mp R/2|/a_0}$ at $R = 1.4\,a_0$ (the H₂ bond length) combined into $\psi_\pm$ with the $1/\sqrt{2(1 \pm S_{AB})}$ normalisation: ψ_+ adds charge between the nuclei, ψ_- has a node at the midpoint; the inset (schematic) shows both electrons in ψ_+ and the antibonding level raised more than the bonding level is lowered.

5.8.2 — Bond Types from Electronic Configuration

The type of molecular orbital formed depends on the symmetry of the atomic orbitals that overlap:

Overlap type	Orbital	Symmetry	Bond	Example
Head-on (s-s, s-p, p_z-p_z)	σ	Cylindrical around bond axis	Single bond	H-H, C-H
Side-on (p_x-p_x, p_y-p_y)	π	Two lobes above/below axis	Part of double/triple bond	C=C, C≡C
d-d	δ	Four-lobed	Metal-metal bonds	Transition metal clusters

The double bond (e.g., ethylene C_2H_4): one σ bond (head-on p_z overlap) + one π bond (side-on p_x overlap). The π bond is weaker (less overlap) and more reactive.

The triple bond (e.g., nitrogen N_2): one σ + two π bonds. Exceptionally strong — N_2 is the most stable diatomic molecule at room temperature, dissociation energy 945 kJ/mol.

5.8.3 — Hybridization: Carbon's Flexibility

Carbon ($2s^2 2p^2$) forms four bonds. But why are all four equivalent in methane (CH_4) if carbon has one $2s$ and two $2p$ electrons? Answer: the $2s$ and three $2p$ orbitals mix to form four equivalent **hybrid orbitals**:

sp^3 **hybridization** (methane, diamond):

$$\phi_i = \frac{1}{2}(\phi_{2s} \pm \phi_{2p_x} \pm \phi_{2p_y} \pm \phi_{2p_z})$$

Four equivalent orbitals pointing to the corners of a tetrahedron, 109.5° apart. This is why diamond (all sp^3 C-C bonds) is the hardest natural material.

sp^2 **hybridization** (graphene, graphite): Three sp^2 hybrids in a plane (120° apart) + one remaining unhybridized $2p_z$. The $2p_z$ orbitals from all carbons overlap to form a delocalized π network. This delocalized π system gives graphene its extraordinary electrical conductivity (the electrons in the π band are nearly free) — and makes it a Dirac semimetal with linear band dispersion (Ch. 7).

sp **hybridization** (acetylene, carbon nanotubes): Two sp hybrids in a line (180° apart) + two unhybridized p orbitals forming two π bonds.

Engineering consequence: The hybridization of carbon determines everything:

- sp^3 → diamond (insulator), polymer chains, organic molecules
- sp^2 → graphene, graphite, carbon nanotubes, fullerenes
- Mixed → amorphous carbon, diamond-like carbon (DLC) for hard coatings

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5.9 — Summary

Starting from three things established in earlier chapters:

1. The quantum numbers (n, ℓ, m) from Ch. 4
2. Spin ($m_s = \pm\frac{1}{2}$) from the Dirac equation (Ch. 3)
3. Pauli exclusion from anticommutation of fermion fields (Ch. 0)

This chapter derived:

Result	Derived from
Spin quantization	Dirac equation; Stern-Gerlach confirms
Pauli exclusion principle	Antisymmetry of fermion wavefunction (Slater determinant)
Aufbau orbital filling	Pauli exclusion + energy ordering from screening
Hund's rules	Exchange interaction (Pauli + Coulomb repulsion)
The periodic table	Aufbau + Hund + 4 quantum numbers
Fine structure	FW spin-orbit term (Ch. 3 §3.5.3)
Zeeman effect	Minimal coupling to external B; $g_e = 2$ from Dirac
Landé g-factor	LS coupling + vector addition of J
Covalent bond	LCAO: bonding/antibonding MOs from overlapping wavefunctions
Hybridization	Linear combinations of atomic orbitals from same atom

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5.10 — Engineering Thread

Physics from this chapter	Engineering application
Spin quantization	NMR spectroscopy; MRI imaging; atomic clocks
Zeeman effect	Magnetometers; electron spin resonance (ESR) sensors
Pauli exclusion	Why metals are stiff; why matter has volume
Electronic configurations	Conductors vs. insulators vs. semiconductors (Ch. 6)

Physics from this chapter	Engineering application
Transition metal 3*d* electrons	Ferromagnetism in Fe, Co, Ni; hard drives; inductors
Electronegativity (from periodic trends)	Battery electrode selection; corrosion engineering; semiconductor doping polarity
sp^3 carbon bonding	Diamond coatings; polymer chemistry; organic semiconductor design
sp^2 carbon bonding	Graphene electronics; carbon nanotube interconnects; carbon fiber composites
Molecular orbitals	Computational chemistry (DFT) for materials design

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5.11 — Looking Ahead: Chapter 6

The periodic table gives us the identity of atoms. Chapter 6 asks: what happens when you assemble $\sim 10^{23}$ of them into a solid?

The answer — Bloch's theorem, band theory, and the metal-semiconductor-insulator distinction — is Layer 1's most direct contribution to engineering practice. Every transistor, every conductor, every solar cell depends on what happens when 10^{23} atomic wavefunctions from this chapter are made to overlap.

End of Chapter 5.

Next: Chapter 6 — Band Theory, Molecular Bonding, and Nuclear Physics

Band Theory, Molecular Bonding, and Nuclear Physics

Assembling Matter: From Atoms to Solids to Nuclei

> *The electron does not belong to any particular atom. In a metal, every electron belongs to every atom simultaneously. This is why metals conduct electricity and why they are shiny.* — Richard Feynman, *The Feynman Lectures*, Vol. III

6.0 — Where We Are

Chapters 4 and 5 built the quantum mechanical description of isolated atoms. Chapter 6 asks: what happens when you take 10^{23} atoms and pack them together?

The answer to that question is **band theory** — the quantum mechanics of electrons in a periodic crystal lattice. It is the single most engineering-relevant result in all of quantum mechanics:

- It explains why copper conducts and glass insulates
- It explains why silicon can be a transistor
- It explains why GaAs emits light and silicon does not
- It is the quantum foundation of every semiconductor device ever built

This chapter also completes two other branches of Layer 1: the full taxonomy of molecular bonds that hold materials together, and the nuclear physics that governs radioactivity, fission, and fusion. By the end of this chapter, Layer 1 is complete — every material property that engineering depends on has a quantum mechanical origin that you can trace.

6.1 — From One Atom to 10^{23}: Bloch's Theorem

6.1.1 — The Problem

In Chapter 4, the potential was $V(r) = -e^2/4\pi\epsilon_0 r$ — a single nucleus. In a crystal with lattice constant a, there is a nucleus at every lattice point $\mathbf{R} = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + n_3\mathbf{a}_3$, so the potential is:

$$V(\mathbf{r}) = \sum_{\mathbf{R}} V_{atom}(\mathbf{r} - \mathbf{R})$$

This is a **periodic potential**: $V(\mathbf{r} + \mathbf{a}) = V(\mathbf{r})$.

The Schrödinger equation is now:

$$\left(-\frac{\hbar^2}{2m_e}\nabla^2 + V(\mathbf{r})\right)\psi = E\psi$$

with $V(\mathbf{r})$ having the periodicity of the crystal lattice.

6.1.2 — Bloch's Theorem from the Discrete Translation Group

A discrete translation by lattice vector $\mathbf{a}$ is a symmetry of the Hamiltonian: $[\hat{H}, \hat{T}_{\mathbf{a}}] = 0$ where $\hat{T}_{\mathbf{a}}$ is the translation operator.

Since $\hat{H}$ commutes with every lattice translation, its eigenstates can be simultaneously chosen as eigenstates of $\hat{T}_{\mathbf{a}}$.

> **Warning: This is representation theory of a *discrete* group, not Noether's theorem** Noether's first theorem requires a **continuous one-parameter** symmetry group: it manufactures a conserved charge by differentiating the action with respect to a continuous parameter. The set of lattice translations is a **discrete** group ($\mathbb{Z}^3$ in three dimensions) with no continuous parameter to differentiate, so Noether's theorem does not apply to it.
>
> What does the work here is that the one-dimensional **irreducible representations** of a discrete translation group are labelled by a phase. Because translations are unitary, the eigenvalue of $\hat{T}_{\mathbf{a}}$ must have unit modulus ($|\hat{T}_{\mathbf{a}}\psi|^2 = |\psi|^2$), so
>
> $$\hat{T}_{\mathbf{a}}\psi = e^{i\mathbf{k}\cdot\mathbf{a}}\psi$$
>
> for some wavevector $\mathbf{k}$ — fixed **only modulo reciprocal lattice vectors** (§6.1.3), precisely because it indexes a representation of a discrete group.

Carrying the translation through gives the Bloch form

$$\boxed{\psi_{n\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}\, u_{n\mathbf{k}}(\mathbf{r})}$$

where $u_{n\mathbf{k}}(\mathbf{r}+\mathbf{a}) = u_{n\mathbf{k}}(\mathbf{r})$ is a function with the same periodicity as the lattice. This is **Bloch's theorem**.

The quantum number $\mathbf{k}$ is the **crystal momentum**, conventionally quoted as $\hbar\mathbf{k}$: the label of the irreducible representation of the discrete translation group, equivalently fixed by the phase $e^{i\mathbf{k}\cdot\mathbf{a}}$ that is the eigenvalue of $\hat{T}_{\mathbf{a}}$. It is a *quasimomentum*, not a Noether charge — it is defined only modulo $\mathbf{G}$ and need not equal the mechanical momentum. The band index n labels which periodic function $u_{n\mathbf{k}}$ the electron occupies.

> **Note: Where Noether language *is* correct here** A free particle in featureless space enjoys a **continuous** translation symmetry, and Noether's theorem then gives conservation of ordinary momentum $\mathbf{p}$. A crystal lattice breaks that continuous symmetry down to a discrete subgroup, so mechanical momentum is *not* conserved — the lattice can absorb momentum — and the label that behaves conservatively is the quasimomentum instead. So: Noether for continuous translations and ordinary momentum; irreducible representations of a discrete group for Bloch's theorem and crystal momentum. Conflating the two is a common and consequential slip, because only the genuinely continuous symmetries come with a Noether charge in the usual sense.

6.1.3 — The Brillouin Zone

The crystal momentum $\mathbf{k}$ is not unique: adding a **reciprocal lattice vector** $\mathbf{G}$ (defined by $e^{i\mathbf{G}\cdot\mathbf{a}} = 1$) gives the same Bloch state. This means $\mathbf{k}$ and $\mathbf{k} + \mathbf{G}$ describe the same physical state.

We therefore restrict $\mathbf{k}$ to the **first Brillouin zone** — the region of $\mathbf{k}$-space closest to the origin, bounded by the planes where $|\mathbf{k}| = |\mathbf{k} - \mathbf{G}|$ (the perpendicular bisectors of reciprocal lattice vectors).

For a 1D crystal with lattice constant a: the first Brillouin zone is $k \in [-\pi/a, +\pi/a]$. The zone boundaries at $k = \pm\pi/a$ are where the electron wavelength $\lambda = 2\pi/k = 2a$ equals twice the lattice constant — the condition for Bragg reflection.

6.2 — The Nearly Free Electron Model: Where Band Gaps Come From

6.2.1 — Free Electrons as the Starting Point

For $V = 0$, the solutions are plane waves $\psi_k = e^{ikx}/\sqrt{L}$ with energy $E = \hbar^2 k^2 / 2m_e$. This is the free electron parabola.

Now add a weak periodic potential. Write $V(x) = \sum_G V_G e^{iGx}$ (Fourier series), where $G = 2\pi n/a$ are reciprocal lattice vectors.

6.2.2 — Bragg Reflection and the Band Gap

Most $\mathbf{k}$ values are barely affected by the weak perturbation. But at the Brillouin zone boundary $k = \pm\pi/a$, two degenerate states $|k\rangle$ and $|k - G\rangle = |-\pi/a\rangle$ have equal energy. Degenerate perturbation theory gives two new eigenstates:

$$\psi_\pm = \frac{1}{\sqrt{2}}\left(e^{i\pi x/a} \pm e^{-i\pi x/a}\right) = \begin{cases} \sqrt{2}\cos(\pi x/a) \\ i\sqrt{2}\sin(\pi x/a) \end{cases}$$

These have different spatial distributions:

- ψ_+: $|\psi_+|^2 \propto \cos^2(\pi x/a)$ — maxima at atomic sites (low V: the attractive ionic potential is deepest there) → **lower energy**, $E_+ = \varepsilon_{\pi/a} + V_G$ with $V_G < 0$
- ψ_-: $|\psi_-|^2 \propto \sin^2(\pi x/a)$ — maxima between atoms (high V) → **higher energy**, $E_- = \varepsilon_{\pi/a} - V_G$

The energy splitting is:

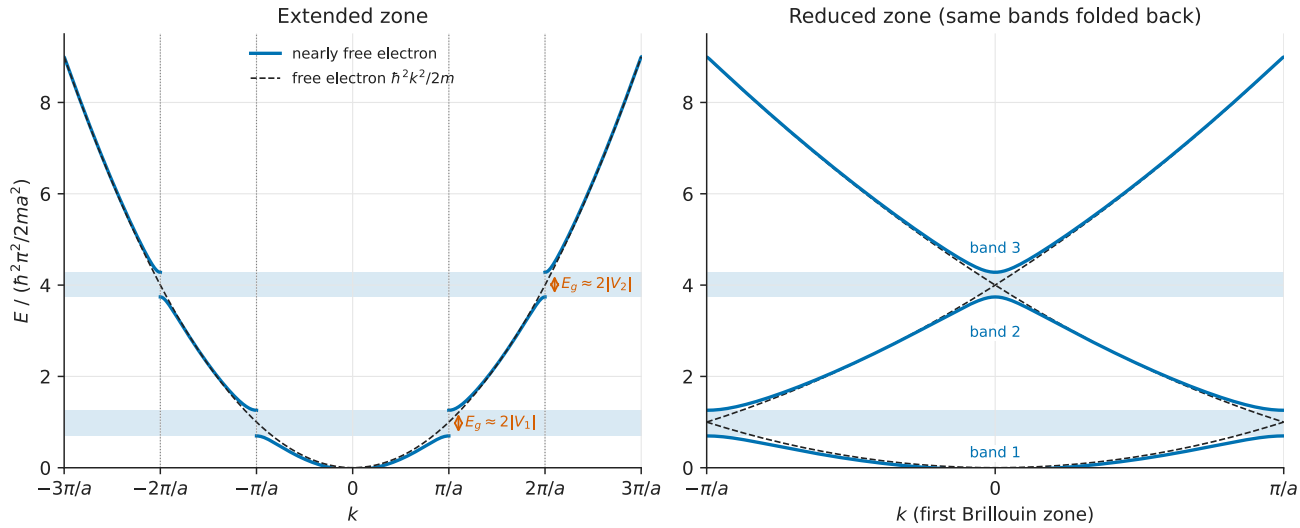
$$E_g = E_- - E_+ = 2|V_G| = 2\left|\int_0^a V(x) e^{-i2\pi x/a} \frac{dx}{a}\right|$$

This is the **band gap** — it is twice the first Fourier coefficient of the periodic potential. **Band gaps are Bragg diffraction of electron waves.**

The electron wavelength $\lambda = 2a$ at the zone boundary satisfies the Bragg condition for reflection from planes separated by a: $2a\sin(90^\circ) = \lambda$. The electron standing waves are Bragg-reflected, and the two standing waves have different energies — the gap.

6.2.3 — The Band Structure Picture

The result: energy $E(k)$ is not a single parabola but a sequence of **bands** separated by **gaps**, with the parabolic shape modified near zone boundaries:



Nearly-free-electron bands from diagonalising the plane-wave Hamiltonian with Fourier components $|V_1| = 0.3$ and $|V_2| = 0.25$ (in units of $\hbar^2\pi^2/2ma^2$, exaggerated for visibility): away from the zone boundaries the bands follow the free parabola (dashed), while at $k = \pm\pi/a, \pm 2\pi/a$ they split by $E_g \approx 2|V_G|$ (shaded gaps), with the upper and lower bands given by $E = \frac{1}{2}(\varepsilon_k + \varepsilon_{k-G}) \pm \sqrt{\frac{1}{4}(\varepsilon_k - \varepsilon_{k-G})^2 + |V_G|^2}$; the right panel folds the same bands into the first Brillouin zone.

Each band holds exactly $2N$ electrons (factor of 2 for spin), where N is the number of unit cells. The first band holds $2N$ states, the second holds $2N$ states, and so on.

6.3 — The Tight-Binding Model: Band Theory from the Bottom Up

The nearly-free-electron model works when the potential is weak and electrons are nearly free. The **tight-binding model** works from the opposite limit: electrons are nearly localized on atoms, with small overlap between neighbors.

6.3.1 — The 1D Tight-Binding Chain

Consider N atoms in a line with spacing a, each contributing one atomic orbital $\phi(\mathbf{r} - \mathbf{R}_n)$. The Bloch wavefunction:

$$\psi_k(x) = \frac{1}{\sqrt{N}} \sum_{n=1}^{N} e^{ikna} \phi(x - na)$$

Computing the energy $\langle \psi_k | \hat{H} | \psi_k \rangle$:

$$E(k) = E_0 + \langle \phi_0 | \hat{H} - \hat{H}_{atom} | \phi_0 \rangle - 2t\cos(ka) + \cdots$$

where $t = -\langle \phi_0 | \hat{H} | \phi_1 \rangle > 0$ is the **hopping integral** — the matrix element for an electron to tunnel from one atom to its neighbor.

Dropping constants:

$$\boxed{E(k) = E_0 - 2t\cos(ka)}$$

This is the 1D tight-binding band. Its width is $4t$ (from $E_0 - 2t$ at $k = 0$ to $E_0 + 2t$ at $k = \pi/a$). The **bandwidth** is set by the overlap between neighboring orbitals:

- Large overlap (e.g., metallic conductors): large t → wide band → electrons nearly free → good conductor
- Small overlap (e.g., molecular crystals): small t → narrow band → electrons nearly localized → poor conductor or insulator

6.3.2 — Graphene: The 2D Tight-Binding Model

Graphene is a single atomic layer of carbon with a hexagonal lattice. Each carbon is sp^2 hybridized (Ch. 5 §5.8.3), contributing one unhybridized p_z orbital per site to the π-band.

The honeycomb lattice has two atoms per unit cell (sublattice A and B), giving two tight-binding bands. Solving the 2×2 Hamiltonian near the K-point (corner of the hexagonal Brillouin zone):

$$E(\mathbf{q}) \approx \pm\hbar v_F |\mathbf{q}|, \qquad v_F = \frac{3ta}{2\hbar} \approx 10^6 \text{ m/s}$$

where $\mathbf{q} = \mathbf{k} - \mathbf{K}$ is measured from the K-point.

This is a linear dispersion relation — the same as for a massless relativistic particle ($E = \hbar c|\mathbf{k}|$ with v_F replacing c). The electrons in graphene near the K-point behave as massless Dirac fermions. The Schrödinger equation is not sufficient — a 2D Dirac equation governs them.

This is exactly why graphene appears as a Weyl semimetal case study in Ch. 7: a Layer-0 concept (Dirac/Weyl fermions from $\mathcal{L}_{SM}$) reappears in a carbon atom arrangement — a remarkable example of the top-down thread.

6.4 — Effective Mass: How Band Electrons Behave Like Classical Particles

6.4.1 — Deriving the Effective Mass

An electron in a band with dispersion $E_n(\mathbf{k})$ responds to an external force $\mathbf{F}$ (from electric or magnetic fields). Its equation of motion:

$$\mathbf{F} = \hbar\frac{d\mathbf{k}}{dt}$$

The group velocity (velocity of the wavepacket):

$$\mathbf{v}_g = \frac{1}{\hbar}\frac{\partial E_n}{\partial \mathbf{k}}$$

Differentiating:

$$\frac{d\mathbf{v}_g}{dt} = \frac{1}{\hbar}\frac{\partial^2 E_n}{\partial \mathbf{k}^2}\frac{d\mathbf{k}}{dt} = \frac{1}{\hbar^2}\frac{\partial^2 E_n}{\partial \mathbf{k}^2}\cdot\mathbf{F}$$

Comparing with $\mathbf{F} = m^*\mathbf{a}$:

$$\boxed{m^* = \hbar^2 \left(\frac{\partial^2 E}{\partial k^2} \right)^{-1}}$$

The **effective mass** m^* is the reciprocal of the band curvature. It tells you how the band electron responds to forces — replacing the bare electron mass m_e in all macroscopic equations.

Key physical consequences:

Region of band	$\partial^2 E/\partial k^2$	m^*	Physical meaning
Bottom of band	> 0 (concave up)	$m^* > 0$, often $< m_e$	Light electrons, easy to accelerate
Top of band	< 0 (concave down)	$m^* < 0$	Electron accelerates opposite to force
Inflection point	$= 0$	$m^* \to \infty$	Electron doesn't respond to force

Holes: Near the top of a filled band, it is convenient to describe the missing electrons (vacancies) as **holes** with:

- Positive charge $+e$
- Positive effective mass $m_h^* = -m_e^* > 0$
- Energy measured downward from the band top

A hole in the valence band acts exactly like a positively charged particle — it moves opposite to an electron under the same field. This is not a classical analog; it is a consequence of the antisymmetry of the many-electron Fermi sea.

6.5 — Metals, Semiconductors, and Insulators

6.5.1 — The Fermi Level and Band Filling

At $T = 0$, electrons fill all available states up to the **Fermi energy** E_F. The key question is: **where does E_F fall relative to the band structure?**

The **Fermi-Dirac distribution** (derived rigorously in Ch. 10, Bridge B.b):

$$f(E) = \frac{1}{e^{(E - E_F)/k_B T} + 1}$$

gives the probability that a state at energy E is occupied at temperature T. At $T = 0$: $f = 1$ for $E < E_F$, $f = 0$ for $E > E_F$. At finite T, the step broadens over a range $\sim k_B T$ around E_F.

6.5.2 — The Three Cases

Metals: E_F lies inside a band — the band is partially filled. States immediately above E_F are available for electrons to scatter into. Any infinitesimal applied field produces a current. $\sigma \sim 10^6$–10^8 S/m.

Example: copper. Configuration $[\text{Ar}]3d^{10}4s^1$ — the half-filled $4s$ band gives one conduction electron per atom. The $3d$ band is full and below E_F, contributing to high density of states but not to conduction.

Semiconductors: E_F lies in a band gap $E_g \sim 0.1$–3 eV. At $T = 0$, the valence band is completely full and the conduction band is completely empty — the material is an insulator. At room temperature ($k_B T \approx 0.026$ eV), a small fraction of electrons is thermally excited across the gap, producing both

conduction electrons and valence band holes. Conductivity is intermediate and strongly temperature-dependent. $\sigma \sim 10^{-4}$–10^{3} S/m.

Insulators: Same physics as semiconductors but $E_g > 3$–4 eV. Thermal excitation across the gap is negligible at room temperature. $\sigma \lesssim 10^{-10}$ S/m.

The quantitative boundary is not sharp — "semiconductor" and "insulator" differ in degree, not kind. Diamond ($E_g = 5.5$ eV) is an insulator; GaN ($E_g = 3.4$ eV) is a semiconductor used in blue LEDs.

6.5.3 — Why Filled Bands Don't Conduct

If a band is completely filled, for every electron at $+k$ there is one at $-k$ with opposite velocity. The net current is exactly zero regardless of field, because applying a field cannot promote any electron — all states are occupied. A small field shifts the k-distribution slightly, but a completely filled band has no available states to shift into.

This is a topological statement: a completely filled band's contribution to current is identically zero by crystal symmetry. This will reappear in Ch. 7 when we discuss the quantum Hall effect.

6.6 — Semiconductor Physics

6.6.1 — The Carrier Concentrations

In an intrinsic (undoped) semiconductor with band gap E_g:

$$n = p = n_i = \sqrt{N_c N_v}\, \exp\left(-\frac{E_g}{2k_B T}\right)$$

where $N_c = 2(m_e^* k_B T / 2\pi\hbar^2)^{3/2}$ and $N_v = 2(m_h^* k_B T / 2\pi\hbar^2)^{3/2}$ are the **effective densities of states** for the conduction and valence bands.

For silicon at 300 K: $n_i = 1.5 \times 10^{10}\ \text{cm}^{-3}$ compared to the total atomic density $5 \times 10^{22}\ \text{cm}^{-3}$. Only 1 in 3×10^{12} silicon atoms contributes a free electron at room temperature.

The exponential factor $e^{-E_g/2k_BT}$ is the Boltzmann weight for exciting an electron across the gap. The same factor appears in reaction kinetics (Arrhenius law, Ch. 2 catalogue) — both reflect thermal activation over an energy barrier.

6.6.2 — Doping: Engineering the Fermi Level

The power of semiconductors is that n and p can be controlled over many orders of magnitude by adding **impurity atoms** — atoms with one more or one fewer valence electron than the host.

n-type doping: Replace Si (4 valence electrons) with phosphorus (5 valence electrons). The extra electron has no available bond — it sits in a hydrogen-like orbital bound to the P^+ ion with binding energy:

$$E_d = \frac{m_e^* e^4}{2(4\pi\epsilon_0\epsilon_r)^2\hbar^2} = E_H \frac{m_e^*}{m_e \epsilon_r^2}$$

For Si ($m_e^* \approx 0.33 m_e$, $\epsilon_r \approx 11.7$): $E_d \approx 45$ meV. The donor level sits just 45 meV below the conduction band — easily ionized at room temperature, donating a free electron to the conduction band.

At room temperature, essentially all donors are ionized: $n \approx N_D$.

p-type doping: Replace Si with boron (3 valence electrons). One bond is incomplete — a hole is "bound" to the B^- ion with acceptor energy $E_a \approx 45$ meV above the valence band. At room temperature: $p \approx N_A$.

The mass action law (valid for non-degenerate semiconductors):

$$n \cdot p = n_i^2 \qquad \text{(always, regardless of doping)}$$

This follows from the chemical equilibrium condition for electron-hole pair generation: $e^- + h^+ \rightleftharpoons 0$ with equilibrium constant n_i^2.

6.6.3 — Common Semiconductors and Their Properties

Material	E_g (eV)	Gap type	m_e^*/m_e	Application
Ge	0.67	Indirect	0.22	Infrared detectors
Si	1.12	Indirect	0.36	MOSFETs, solar cells, ICs
GaAs	1.42	Direct	0.067	Lasers, LEDs, HEMTs, solar cells
InP	1.34	Direct	0.077	Telecom lasers (1550 nm)
GaN	3.40	Direct	0.20	Blue LEDs, power electronics
SiC	3.26	Indirect	0.42	High-power, high-temperature electronics
diamond	5.50	Indirect	0.36	Potential future ultra-wide-bandgap devices

Direct vs. indirect gap (Ch. 2 catalogue entry made explicit):

- **Direct gap** (E_g at same **k** for valence and conduction band extrema): An electron can recombine with a hole by emitting a photon directly — momentum is conserved with no phonon required. High radiative efficiency. ⇒ LEDs and semiconductor lasers.
- **Indirect gap** (extrema at different **k**): Recombination requires a phonon to conserve crystal momentum. Very low radiative efficiency. Silicon does not emit light efficiently. ⇒ Transistors and solar cells (where carrier collection, not emission, matters).

The laser condition requires a direct-gap material and population inversion ($E_F^n - E_F^p > E_g$, where E_F^n, E_F^p are the quasi-Fermi levels under forward bias). All practical semiconductor lasers use III-V direct-gap materials.

6.7 — Phonons: The Quantum Mechanics of Heat

6.7.1 — Quantizing the Lattice Vibrations

The atoms in a crystal are not fixed — they vibrate around their equilibrium positions. For small displacements, every vibration mode is a harmonic oscillator (Ch. 4 §4.3). Quantizing it gives a **phonon** with energy $\hbar\omega$.

A crystal with N atoms and p atoms per unit cell has $3pN$ vibrational modes. These divide into branches:

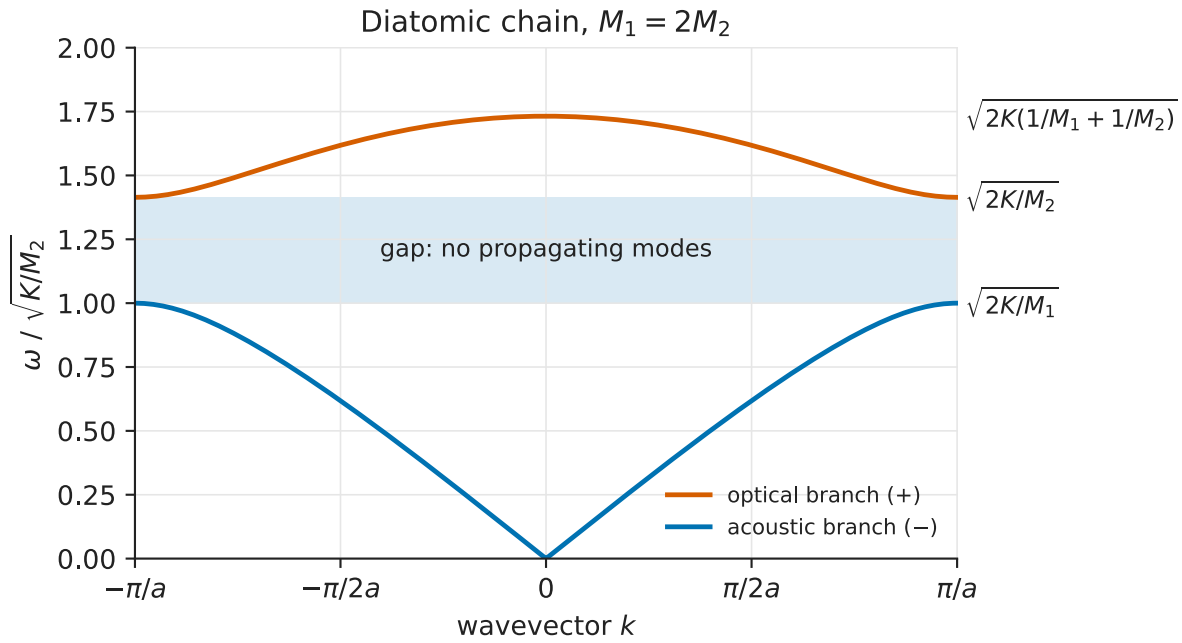
- **Acoustic branches** (3 branches): all atoms in a unit cell move in phase; $\omega \to 0$ as $k \to 0$ (long-wavelength sound waves)

- **Optical branches** ($3p - 3$ branches): atoms in the unit cell move out of phase; finite frequency even at $k = 0$ (can interact with light — hence "optical")

For a diatomic chain with masses M_1 and M_2 and spring constant K:

$$\omega^2 = K\left(\frac{1}{M_1} + \frac{1}{M_2}\right) \pm K\sqrt{\left(\frac{1}{M_1} + \frac{1}{M_2}\right)^2 - \frac{4\sin^2(ka/2)}{M_1 M_2}}$$

$+$ sign: optical branch; $-$ sign: acoustic branch. The gap between them is $\sqrt{2K/M_1}$ to $\sqrt{2K/M_2}$ — no phonon modes exist in this gap.



The diatomic-chain dispersion above for $M_1 = 2M_2$: the acoustic branch rises linearly from $\omega = 0$ (sound), the optical branch starts at $\sqrt{2K(1/M_1 + 1/M_2)}$ at $k = 0$, and at the zone boundary they end at $\sqrt{2K/M_1}$ and $\sqrt{2K/M_2}$, bounding a frequency gap (shaded) with no propagating modes.

6.7.2 — Phonons Carry Heat

Thermal conductivity in a crystal comes from phonon transport (in insulators and semiconductors) and electron transport (in metals). For phonons:

$$\kappa_{phonon} = \frac{1}{3} C_v v_s \ell$$

where C_v is the heat capacity, v_s is the phonon group velocity (speed of sound), and ℓ is the phonon mean free path (limited by phonon-phonon scattering, phonon-boundary scattering, and phonon-impurity scattering).

Diamond has an exceptionally high thermal conductivity ($\kappa \approx 2000$ W/m·K, five times copper) because: (1) light carbon atoms → high v_s, (2) strong covalent bonds → high ω → sparse phonon-phonon scattering (fewer phonons to scatter from at room temperature), (3) no heavy atoms to scatter phonons.

6.7.3 — Electron-Phonon Scattering: Why Metals Resist

In metals, electrons near E_F scatter from phonons. Phonon occupation increases with temperature as $n_{ph} \propto k_B T/\hbar\omega$ (Bose-Einstein distribution in the classical regime). Scattering rate $\propto$ phonon occupation $\propto T$. Since $\sigma = ne^2\tau/m^*$ (Bridge B.e result), and $\tau \propto 1/T$:

$$\rho_{metal} = 1/\sigma \propto T \quad \text{(at room temperature, Bloch-Grüneisen)}$$

This is the measured linear resistivity of metals at room temperature — aluminum, copper, gold, all show $\rho \propto T$ above ~100 K.

At low T: phonon occupation $\propto T^5$ (few phonons; only small-angle scattering), giving $\rho \propto T^5$ (Bloch-Grüneisen). At very low T: residual resistance from impurities and defects (temperature-independent). The Mathiessen rule adds these: $\rho = \rho_0 + AT^5 + BT$.

Engineering thread: The temperature dependence of resistance is what makes RTDs (Resistance Temperature Detectors) work. Platinum has a precisely characterized $\rho(T)$ from this physics, enabling temperature measurement to $\pm 0.01\,^\circ$C.

6.8 — Bonding in Solids: A Complete Taxonomy

Chapter 5 gave covalent bonding from LCAO. Here we complete the picture with every bond type and the crystal structures they produce.

6.8.1 — Ionic Bonding

Origin: Transfer of electrons from electropositive to electronegative atoms creates charged ions; Coulomb attraction holds them together.

Example: NaCl. Na gives its $3s^1$ electron to Cl (which wants one to fill $3p^6$). The lattice energy (per ion pair):

$$U_{Madelung} = -\frac{Me^2}{4\pi\epsilon_0 r_0}\left(1 - \frac{1}{n}\right)$$

where $M \approx 1.748$ is the **Madelung constant** for the NaCl structure (from summing the alternating +/- Coulomb interactions over the entire lattice), r_0 is the equilibrium spacing, and $n \approx 9$ is the Born repulsion exponent.

Properties: High melting point (strong Coulomb attraction), brittle (cleavage on ionic planes), dissolve in polar solvents, conduct electricity when molten or dissolved (free ions).

Engineering: Ceramic insulators (Al_2O_3, MgO), ionic conductors for fuel cells (ZrO_2), piezoelectrics ($BaTiO_3$).

6.8.2 — Covalent Bonding

Origin: Shared electrons in bonding molecular orbitals (LCAO, Ch. 5 §5.8). Bond strength from electron density accumulation between nuclei.

Properties: Directional (bond angles determined by orbital geometry), hard, brittle (breaking a bond requires disrupting the electron sharing), covalent crystals: diamond, Si, GaAs.

Engineering: Semiconductor devices (Si, GaAs, SiC), hard coatings (diamond, SiC), ceramics (SiC, Si_3N_4).

6.8.3 — Metallic Bonding

Origin: Valence electrons delocalize from their parent atoms and form a "sea" that bonds all ions together. Band theory is the quantum description of this: the electrons occupy wide, partially filled bands (low effective mass, high mobility).

Properties: Non-directional (no preferred bond angles → ductile, malleable), high electrical and thermal conductivity (delocalized electrons), opaque and lustrous (plasma frequency in UV → reflects all visible light, Ch. 2).

Why metals are ductile but ionic crystals are brittle: In a metal, shifting atomic planes relative to each other merely rearranges which electrons are between which ions — the bond doesn't break. In an ionic crystal, shifting planes brings like-charge ions adjacent, creating a repulsive plane — catastrophic fracture.

6.8.4 — Van der Waals (London Dispersion) Bonding

Origin: Quantum fluctuations in the electron distribution of neutral atoms create instantaneous dipoles; these instantaneous dipoles polarize neighboring atoms. The interaction energy between induced dipoles:

$$U_{vdW} = -\frac{C}{r^6}$$

where C depends on the atomic polarizabilities and ionization energies. This is the **London dispersion force** — it has a purely quantum origin (zero for classical atoms with no fluctuations) and is always attractive.

Magnitude: Much weaker than ionic or covalent bonds ($\sim$ 1–10 kJ/mol vs. $\sim$ 100–1000 kJ/mol). But present between *all* atoms.

Engineering: Holds graphite layers together (weak interlayer vdW → easy sliding → graphite is a lubricant); adhesion of geckos to surfaces (van der Waals contact area); 2D material exfoliation (Scotch tape method for graphene exploits weak vdW between layers); molecular recognition in drug design.

6.8.5 — Hydrogen Bonding

Origin: An H atom covalently bonded to an electronegative atom (F, O, N) has its electron pulled away, exposing the bare proton. This positive proton attracts the lone pairs of neighboring electronegative atoms. Energy: $\sim$ 10–40 kJ/mol.

Engineering consequence: Water's anomalously high boiling point (100°C vs. -60°C for H_2S, the next heavier group-16 hydride), its high heat capacity and latent heat — all from hydrogen bonds. These make water a uniquely effective heat transfer fluid and give it its role in biology and industrial cooling. Concrete strength depends on H-bonding in C-S-H gel. Polymer properties (nylon, proteins) depend critically on H-bond geometry.

6.9 — Nuclear Physics: The Strong Force's Engineering Consequences

6.9.1 — The Nuclear Force

In Ch. 0, the strong force was SU(3) color — confined to quark interactions inside hadrons. The **residual strong force** between nucleons (protons and neutrons) is an effective force mediated by pion exchange — an analog of the van der Waals force between atoms (which is a residual electromagnetic interaction between neutral atoms).

The nucleon-nucleon potential is approximately:

$$V_{NN}(r) \approx -\frac{g^2}{4\pi}\frac{e^{-m_\pi c r/\hbar}}{r}$$

(Yukawa potential, where $m_\pi c^2 \approx 135$ MeV is the pion mass). The range of the strong force:

$$r_0 = \frac{\hbar}{m_\pi c} \approx 1.4\,\text{fm} \approx 1.4 \times 10^{-15}\,\text{m}$$

This sets the scale of nuclear physics. At $r < r_0$: strong attraction. At $r > 2\text{–}3\,r_0$: negligible. At $r \lesssim 0.5$ fm: short-range repulsion (hard core). The strong force is short-range, much stronger than EM at close range, and saturates (each nucleon bonds only to its immediate neighbors).

6.9.2 — Nuclear Binding Energy

The binding energy $B(Z, A)$ is the energy required to completely separate a nucleus of Z protons and $N = A - Z$ neutrons into free nucleons:

$$B(Z,A) = (Zm_p + Nm_n - m_{nucleus})c^2$$

The **semi-empirical mass formula** (Bethe-Weizsäcker):

$$\boxed{B(Z,A) = a_V A - a_S A^{2/3} - a_C \frac{Z(Z-1)}{A^{1/3}} - a_A \frac{(A-2Z)^2}{A} + \delta(A,Z)}$$

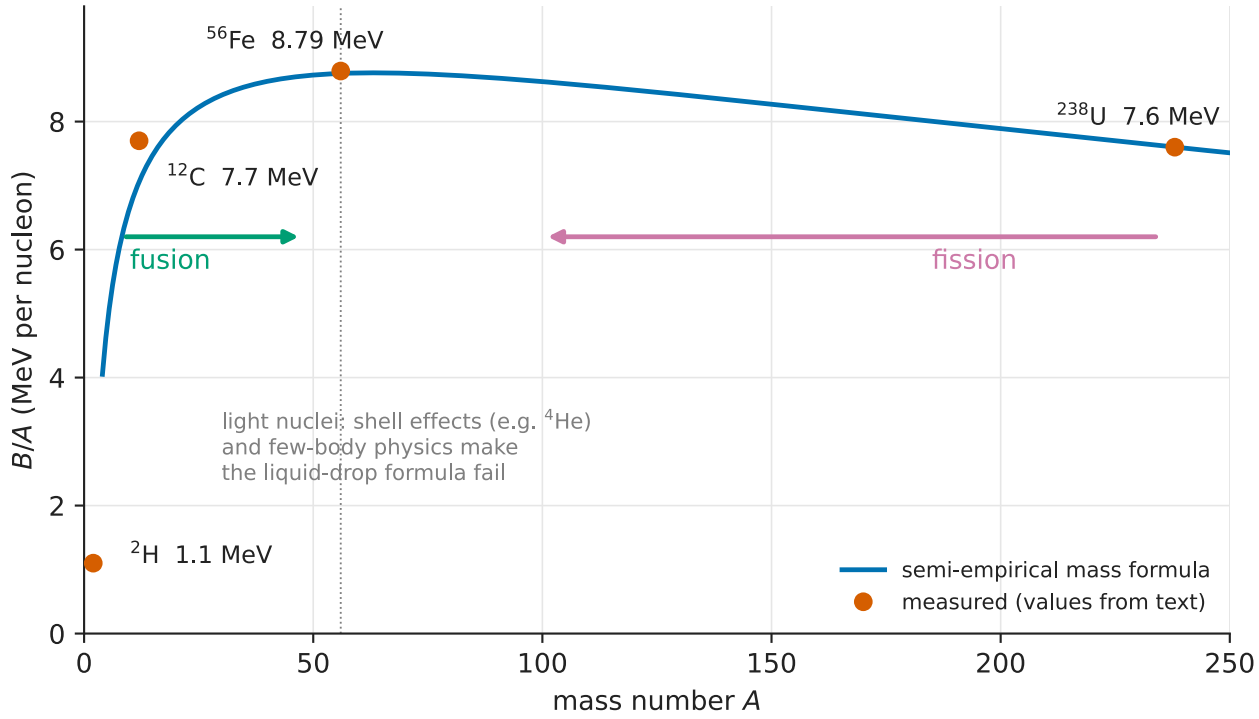
Each term has a transparent physical origin:

Term	Value	Physical origin	Analogy
Volume: $a_V A$	$a_V = 15.8$ MeV	Each nucleon bonds to its neighbors; binding energy $\propto A$	Bulk cohesion energy
Surface: $-a_S A^{2/3}$	$a_S = 18.3$ MeV	Surface nucleons have fewer bonds; correction $\propto$ surface area $\propto A^{2/3}$	Surface tension
Coulomb: $-a_C Z(Z-1)/A^{1/3}$	$a_C = 0.714$ MeV	Proton-proton repulsion; all pairs interact; $R \propto A^{1/3}$	Electrostatic energy
Asymmetry: $-a_A(A-2Z)^2/A$	$a_A = 23.2$ MeV	Pauli exclusion: symmetric filling of n and p levels minimizes energy	Fermi pressure
Pairing: $+\delta/\sqrt{A}$	$\delta = \pm 12$ MeV	Paired nucleons are extra stable (like Cooper pairs); $+$ for even-even, $-$ for odd-odd, 0 for odd A	Pairing gap in BCS

6.9.3 — The Binding Energy per Nucleon Curve

Dividing $B(Z, A)$ by A gives binding energy per nucleon, which:

- Rises steeply from ^{2}H ($B/A = 1.1$ MeV) to carbon ($B/A \approx 7.7$ MeV)
- Peaks near ^{56}Fe ($B/A = 8.79$ MeV) — the most stable nucleus
- Decreases slowly from Fe to ^{238}U ($B/A = 7.6$ MeV)



Semi-empirical mass formula with the coefficients in the table above ($a_V = 15.8$, $a_S = 18.3$, $a_C = 0.714$, $a_A = 23.2$ MeV; pairing omitted), evaluated along the valley of stability $Z \approx A/(2 + 0.0153A^{2/3})$, with the measured values quoted here: the curve peaks near $A \approx 56$ and reproduces Fe and U well, while light nuclei such as ^{2}H and ^{12}C deviate because a liquid-drop formula cannot capture few-body and shell effects.

This single curve determines all nuclear energy release:

- **Fusion** ($A < 56$, building toward Fe): joining two light nuclei increases B/A → releases energy. Sun: $4\text{H} \rightarrow {}^{4}\text{He}$, releases 26.7 MeV per reaction.
- **Fission** ($A > 56$, moving toward Fe): splitting a heavy nucleus increases B/A → releases energy. $^{235}\text{U} + n \rightarrow {}^{141}\text{Ba} + {}^{92}\text{Kr} + 3n$, releases ≈ 200 MeV per fission.

Both processes maximize binding energy per nucleon, approaching Fe from their respective sides of the curve.

6.10 — Radioactive Decay

6.10.1 — Alpha Decay: Tunneling Through the Coulomb Barrier

An α particle (^{4}He nucleus, $Z = 2$, $A = 4$) pre-forms inside a heavy nucleus and tunnels through the Coulomb barrier between the nuclear surface ($r = R$) and the classical turning point ($r = r_c$).

The Gamow tunneling factor (Ch. 4 §4.4, now applied to the Coulomb potential):

$$T_\alpha \propto \exp\left(-2\int_R^{r_c} \kappa(r)\,dr\right), \qquad \kappa(r) = \frac{\sqrt{2m_\alpha(V_C(r) - E)}}{\hbar}$$

where $V_C(r) = 2Ze^2/4\pi\epsilon_0 r$ is the daughter-nucleus Coulomb potential. The integral gives the **Gamow factor** G:

$$G = \frac{2\pi Z e^2}{4\pi\epsilon_0 \hbar v_\alpha}\left(1 - \frac{4}{\pi}\sqrt{\frac{E}{V_C(R)}}\right) \approx \frac{2\pi Z e^2}{4\pi\epsilon_0 \hbar v_\alpha}$$

Decay rate: $\lambda \propto f \cdot e^{-2G}$ where $f \approx 10^{21}\ \text{s}^{-1}$ is the assault frequency.

The Geiger-Nuttall law: $\log \lambda = C_1 Z/\sqrt{E_\alpha} + C_2$. Alpha emitters spanning 24 orders of magnitude in decay rate obey this relation — from ^{212}Po ($t_{1/2} = 0.3\,\mu$s) to ^{232}Th ($t_{1/2} = 14$ Gyr). The enormous range comes from the exponential sensitivity of tunneling probability to E_α — a change of 1 MeV in α energy changes $t_{1/2}$ by ~ 20 orders of magnitude.

6.10.2 — Beta Decay: The Weak Force at Work

β^- decay: $n \to p + e^- + \bar{\nu}_e$ A down quark converts to an up quark via W^- boson emission (Ch. 0 §0.4.3): $d \to u + W^-$, then $W^- \to e^- + \bar{\nu}_e$.

β^+ decay: $p \to n + e^+ + \nu_e$ (requires $Q > 2m_e c^2$)

Electron capture: $p + e^- \to n + \nu_e$

The Q-value (energy released):

$$Q = (m_{\text{parent}} - m_{\text{daughter}} - m_{\text{emitted}})c^2$$

The electron energy spectrum is continuous (not discrete as in α decay), because the Q is shared between the electron and the neutrino. The maximum electron energy equals Q (when ν carries no energy).

Fermi's Golden Rule applied to β decay: The decay rate:

$$\lambda = \frac{G_F^2 |M_{fi}|^2}{2\pi^3\hbar^7}(m_e c^2)^5 f(Z, Q)$$

where G_F is the Fermi constant (weak coupling, from g_2 in Ch. 0), $|M_{fi}|$ is the nuclear matrix element, and $f(Z, Q)$ is the Fermi function accounting for Coulomb effects on the outgoing electron.

6.10.3 — Gamma Decay and the Decay Law

Gamma decay: A nucleus in an excited state transitions to a lower state, emitting a γ photon. Energies: 0.1–10 MeV. Selection rules from angular momentum conservation determine which transitions are allowed and their rates.

The radioactive decay law (any mode):

$$N(t) = N_0\, e^{-\lambda t}, \qquad t_{1/2} = \frac{\ln 2}{\lambda}$$

Activity: $A = \lambda N$ (decays per second, unit: Becquerel, 1 Bq = 1 decay/s).

Decay chains: A parent decays to a daughter which may itself be unstable, forming a chain. The Bateman equations govern the chain:

$$\frac{dN_i}{dt} = \lambda_{i-1} N_{i-1} - \lambda_i N_i$$

Engineering: Geological dating uses isotope ratios (U-Pb, Rb-Sr, K-Ar) measured against the decay law to determine rock ages with precision of $\pm 1\%$ over billions of years. Medical nuclear imaging uses short-lived isotopes (^{99m}Tc, $t_{1/2} = 6$ h; ^{18}F, $t_{1/2} = 110$ min).

6.11 — Nuclear Fission and Fusion

6.11.1 — Fission: The Liquid Drop Model

The binding energy formula (§6.9.2) predicts the **fission barrier**. When a ^{235}U nucleus deforms from a sphere to a prolate ellipsoid:

- Surface energy increases (larger area → less stable)
- Coulomb energy decreases (charges further apart → more stable)

At small deformation: surface term wins → nucleus is stable (barrier exists). At large deformation: Coulomb term wins → nucleus splits.

The **fission barrier height** E_b:

$$E_b \approx a_S A^{2/3} - \frac{3}{5}\frac{e^2 Z^2}{4\pi\epsilon_0 R}$$

The fissility parameter $x = E_C / 2E_S = Z^2/50.9A$ determines whether spontaneous fission can occur. For $x > 1$ ($Z^2/A > 50.9$): no barrier, spontaneous fission. ^{235}U has $x \approx 0.77$ — a barrier of ≈ 6 MeV.

A slow neutron absorbed by ^{235}U provides ~ 6.5 MeV of excitation (neutron binding energy in ^{236}U) — just enough to cross the barrier. This is why ^{235}U is fissile with thermal neutrons while ^{238}U is not (its fission barrier is ~ 1 MeV higher than the excitation energy from neutron capture).

Chain reaction criticality: Each fission releases $\nu \approx 2.4$ neutrons. Define:

- k_{eff}: average number of neutrons from one fission that cause the next fission
- $k_{eff} < 1$: subcritical (dies out)
- $k_{eff} = 1$: critical (sustained chain reaction — nuclear reactor)
- $k_{eff} > 1$: supercritical (exponentially growing — weapon)

The four-factor formula for a large reactor: $k_{eff} = \eta f p \varepsilon$ where:

- η: neutrons produced per neutron absorbed in fuel
- f: fraction absorbed in fuel vs. moderator
- p: resonance escape probability
- ε: fast fission factor

6.11.2 — Fusion: The Gamow Peak

For two nuclei to fuse, they must overcome the Coulomb barrier: $V_C = Z_1 Z_2 e^2 / 4\pi\epsilon_0 r$ (at nuclear contact radius $r \approx 2$ fm for D-T: $V_C \approx 1.44\,\text{MeV fm}/2\,\text{fm} \approx 0.72$ MeV).

The thermal energy in a plasma at T: $\langle E \rangle = \frac{3}{2} k_B T$. Achieving $\langle E \rangle > V_C$ would require $T \gtrsim 6 \times 10^9$ K — far above stellar core temperatures ($\sim 10^7$ K for the Sun).

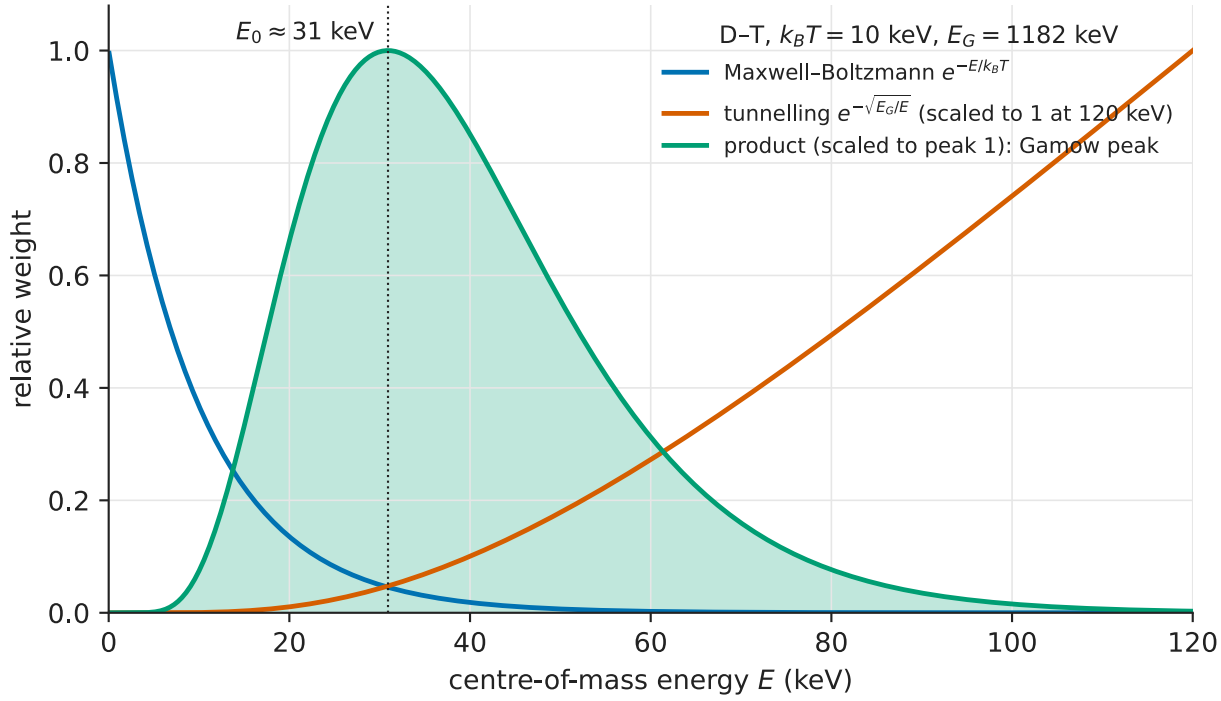
The resolution is quantum tunneling (Ch. 4 §4.4): Even at $kT \ll V_C$, particles in the high-energy tail of the Maxwell-Boltzmann distribution can tunnel. The reaction rate involves:

$$\sigma(E) v \propto e^{-2\pi\eta} \times e^{-E/k_B T}$$

where $e^{-2\pi\eta}$ is the Gamow tunneling factor and $e^{-E/k_B T}$ is the Maxwell-Boltzmann weight. The product peaks at the **Gamow peak energy**:

$$E_0 = \left[\frac{(\pi\alpha Z_1 Z_2\, k_B T)^2\, m_r c^2}{2} \right]^{1/3} = \left(\frac{E_G\, (k_B T)^2}{4} \right)^{1/3}, \qquad E_G = 2 m_r c^2 (\pi\alpha Z_1 Z_2)^2$$

For D-T fusion at $k_B T = 10$ keV ($T \approx 10^8$ K): $E_0 \approx 31$ keV — in the tail of the Maxwell-Boltzmann distribution but exponentially enhanced by tunneling.



D-T at $k_B T = 10$ keV: the falling Maxwell-Boltzmann weight $e^{-E/k_B T}$ times the rising tunnelling factor $e^{-\sqrt{E_G/E}}$ ($E_G = 2 m_r c^2 (\pi\alpha Z_1 Z_2)^2 \approx 1.18$ MeV) gives a narrow window of reacting energies peaked at $E_0 \approx 31$ keV, three times $k_B T$ yet far below the Coulomb barrier (curves scaled as labelled).

The D-T reaction (^{2}H + ^{3}H $\rightarrow$ ^{4}He + n + 17.6 MeV) is the most practical fusion reaction due to its low Gamow peak energy and high cross-section. ITER (International Thermonuclear Experimental Reactor) targets $Q \geq 10$ (energy out/energy in ≥ 10).

6.12 — Summary

Layer 1 is now complete. The tools assembled:

From Ch. 4	From Ch. 5	From Ch. 6
Wavefunctions $\psi_{n\ell m}$	Spin m_s	Bloch states $\psi_{n\mathbf{k}}$
Quantization by BCs	Pauli exclusion from Ch. 0	Band structure $E_n(\mathbf{k})$
Hydrogen atom solution	Periodic table	Metal/SC/insulator distinction
Tunneling coefficient $T \propto e^{-2\kappa d}$	Fine structure (FW)	Fission barrier; Gamow peak
Uncertainty principle	Zeeman effect	Phonons; $\rho \propto T$ in metals

Engineering threads from Ch. 6:

Physics	Application
Band structure + Fermi level	Every semiconductor device (transistors, LEDs, solar cells)
Effective mass m^*	Carrier mobility → device speed
Direct vs. indirect gap	LEDs and lasers vs. transistors; material selection
Doping (n-type, p-type)	p-n junction, MOSFET, BJT
Phonon thermal conductivity	Heat sink design; thermoelectrics; CPU cooling
$\rho \propto T$ (electron-phonon)	RTD temperature sensors; conductor loss
Ionic bonding	Ceramic insulators; fuel cell electrolytes
vdW bonding	Graphene exfoliation; 2D materials; MEMS stiction
Nuclear binding energy curve	Nuclear reactor fuel selection; fusion energy
Radioactive decay law	Nuclear waste management; medical imaging; geological dating
Gamow tunneling	α-decay rates; fusion cross-sections; tunnel diode design

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6.13 — Looking Ahead: Chapter 7

Chapter 6 treated electrons as if their quantum numbers were $(n, \mathbf{k}, s)$ with well-defined energies $E_n(\mathbf{k})$ — a single-particle picture. Chapter 7 asks what happens when the bands themselves carry topological invariants, when electron-electron interactions are strong, or when quantum phase coherence extends over macroscopic distances.

The results: the quantum Hall effect, topological insulators, Weyl semimetals, and superconductivity. These are not exotic additions to band theory — they are what band theory becomes when you stop ignoring topology.

End of Chapter 6.

Next: Chapter 7 — Topology, Quantum Hall Effect, and Superconductivity

Topology, Quantum Hall Effect, and Superconductivity

When Band Theory Becomes Topological

> *It is remarkable that the Hall conductance is quantized even in the presence of impurities and disorder. The reason is topology.* — David Thouless, Nobel Lecture, 2016

7.0 — The Question This Chapter Answers

Chapter 6 built band theory — the quantum mechanics of electrons in periodic potentials. It explained conductors, semiconductors, and insulators. It gave you the tools to design transistors and solar cells.

But band theory as presented in Ch. 6 misses something profound. Two insulators can have identical band gaps, identical carrier concentrations, and completely identical local physics — and yet be fundamentally different. They cannot be continuously deformed into each other without closing the gap, passing through a metallic state.

The difference is **topology** — a global property of the band structure that cannot be detected by looking at any single **k**-point, only by integrating a geometric quantity over the entire Brillouin zone.

This chapter develops the topological perspective on bands and shows that:

1. The integer quantum Hall conductance is quantized because it equals a topological integer (the Chern number) — immune to any disorder or imperfection
2. Topological insulators have surface states protected by symmetry — they cannot be removed without breaking that symmetry or closing the bulk gap
3. Weyl semimetals host low-energy quasiparticles described by an effective Weyl Hamiltonian - the same *mathematical form* as the Weyl fermion fields of the Standard Model Lagrangian (Ch. 0). These are emergent quasiparticles of a condensed-matter band structure, not the fundamental SM fields, so this is a `[STRUCTURAL CONNECTION]` rather than an identification: topology has brought particle-physics form language into a material you can hold in your hand
4. Superconductivity is broken U(1) symmetry — the same mathematical template as the Higgs mechanism (Ch. 0 §0.7), now at millielectronvolt energies

The topology thread promised in Ch. 0 §0.9 is collected here.

7.1 — Berry Phase: Geometry Hidden in Quantum Mechanics

7.1.1 — The Adiabatic Theorem

Consider a quantum system with a Hamiltonian $\hat{H}(\mathbf{R})$ that depends on a set of slowly-varying external parameters $\mathbf{R}(t)$ — which could be a slowly-rotating magnetic field, or the crystal momentum $\mathbf{k}$ varying as an electron moves through the Brillouin zone.

Adiabatic theorem (Born-Fock, 1928): If the system starts in eigenstate $|n(\mathbf{R}(0))\rangle$ and the parameters change slowly (adiabatically), the system remains in the instantaneous eigenstate $|n(\mathbf{R}(t))\rangle$ — it tracks the eigenstate as $\mathbf{R}$ evolves.

The state acquires two phases:

$$|\psi(t)\rangle = e^{i\gamma_n(t)}\, e^{-\frac{i}{\hbar}\int_0^t E_n(t')dt'}\, |n(\mathbf{R}(t))\rangle$$

- The second exponential: the familiar **dynamical phase** $-\int E_n\, dt/\hbar$
- The first exponential: a **geometric phase** $\gamma_n(t)$ discovered by Berry in 1984 — it depends only on the geometry of the path in parameter space, not on how fast you traverse it

7.1.2 — The Berry Connection and Berry Phase

Substituting into the Schrödinger equation and solving for γ_n:

$$\gamma_n = i\int_{\mathbf{R}(0)}^{\mathbf{R}(t)} \langle n(\mathbf{R})|\nabla_{\mathbf{R}}|n(\mathbf{R})\rangle \cdot d\mathbf{R}$$

Define the **Berry connection** (vector potential in parameter space):

$$\mathbf{A}_n(\mathbf{R}) = i\langle n(\mathbf{R})|\nabla_{\mathbf{R}}|n(\mathbf{R})\rangle$$

For a **closed loop** $\mathcal{C}$ in parameter space (returning to the starting point), the Berry phase is:

$$\boxed{\gamma_n = \oint_{\mathcal{C}} \mathbf{A}_n(\mathbf{R}) \cdot d\mathbf{R}}$$

This is gauge-dependent (like any vector potential) — but for a closed loop, the gauge-invariant **Berry curvature** gives a gauge-independent result. By Stokes' theorem:

$$\gamma_n = \iint_{\mathcal{S}} \boldsymbol{\Omega}_n(\mathbf{R}) \cdot d\mathbf{S}$$

where the **Berry curvature** is:

$$\boldsymbol{\Omega}_n(\mathbf{R}) = \nabla_{\mathbf{R}} \times \mathbf{A}_n = i\langle \nabla_{\mathbf{R}} n| \times |\nabla_{\mathbf{R}} n\rangle$$

The Berry curvature is the "magnetic field" in parameter space — just as $\mathbf{B} = \nabla \times \mathbf{A}$ in real space. The Berry phase accumulated around a loop is the "flux" of $\boldsymbol{\Omega}$ through the surface bounded by that loop.

7.1.3 — The Chern Number: Topology from Integration

For electrons in a crystal, the natural parameter space is the Brillouin zone (BZ) — a torus (periodic in all directions of $\mathbf{k}$-space). The Berry curvature for the n-th band is:

$$\Omega_n(\mathbf{k}) = i \sum_{m \neq n} \frac{\langle u_{n\mathbf{k}} | \partial_{k_x} \hat{H} | u_{m\mathbf{k}} \rangle \langle u_{m\mathbf{k}} | \partial_{k_y} \hat{H} | u_{n\mathbf{k}} \rangle - (x \leftrightarrow y)}{(E_n - E_m)^2}$$

The **Chern number** (first Chern number) of the n-th band:

$$\boxed{C_n = \frac{1}{2\pi} \iint_{\text{BZ}} \Omega_n(\mathbf{k}) \, d^2k \in \mathbb{Z}}$$

This is always an integer. It cannot take fractional values. It cannot change continuously. It can only change by jumping to another integer — which requires closing the band gap (making $E_n = E_m$ somewhere, making Ω_n singular).

The reason it is an integer is the Chern-Gauss-Bonnet theorem from differential geometry: the integral of a curvature over a closed surface is 4π times the Euler characteristic, which for a torus equals zero — but for a band over the BZ torus, the Chern number counts how many times the eigenvector $|u_{n\mathbf{k}}\rangle$ "wraps" around the BZ, and wrapping numbers are always integers.

This is the same mathematics that makes flux quantization an integer: the winding number of the superconducting phase around a loop (§7.8) is an integer for the same reason.

7.2 — Anomalous Velocity: How Topology Produces a Hall Effect

7.2.1 — The Semiclassical Equations of Motion, Corrected

The standard semiclassical equations of motion for a Bloch electron:

$$\dot{\mathbf{r}} = \frac{1}{\hbar} \frac{\partial E_n}{\partial \mathbf{k}}, \qquad \hbar \dot{\mathbf{k}} = -e\mathbf{E} - e\dot{\mathbf{r}} \times \mathbf{B}$$

are incomplete. Including the Berry curvature correction (derivable from the Schrödinger equation itself):

$$\dot{\mathbf{r}} = \frac{1}{\hbar} \frac{\partial E_n}{\partial \mathbf{k}} - \frac{e}{\hbar} \mathbf{E} \times \boldsymbol{\Omega}_n(\mathbf{k})$$

The second term is the **anomalous velocity** — it is **transverse** to the applied electric field $\mathbf{E}$. It produces a current perpendicular to the field without any magnetic field:

$$\mathbf{J}_{anomalous} = \frac{e^2}{\hbar} \sum_{n, \in, \text{filled}} \int \frac{d^2k}{(2\pi)^2} \boldsymbol{\Omega}_n(\mathbf{k}) \times \mathbf{E}$$

The Hall conductivity:

$$\sigma_{xy} = \frac{e^2}{\hbar} \sum_{n, \in, \text{filled}} \frac{1}{2\pi} \iint_{\text{BZ}} \Omega_n \, d^2k = \frac{e^2}{h} \sum_n C_n$$

The Hall conductivity equals the sum of Chern numbers of all filled bands, in units of e^2/h. This is the TKNN formula (Thouless, Kohmoto, Nightingale, den Nijs, 1982) — one of the deepest results in condensed matter physics.

7.3 — The Integer Quantum Hall Effect

7.3.1 — The Setup

A 2D electron gas (2DEG) — electrons confined to a plane (a GaAs/AlGaAs heterostructure, for example) — in a strong perpendicular magnetic field $B \sim 1$–30 T at temperatures $T \sim 1$–4 K.

Landau levels (from Ch. 6): the 2D kinetic energy in a magnetic field is quantized into discrete levels:

$$E_n = \hbar\omega_c \left(n + \frac{1}{2}\right), \quad \omega_c = \frac{eB}{m^*}$$

Each Landau level has a macroscopic degeneracy:

$$N_{LL} = \frac{eB}{h} \times (\text{area}) = \frac{\Phi}{\Phi_0}$$

where Φ is the total magnetic flux and $\Phi_0 = h/e$ is the (single-particle) flux quantum. The degeneracy equals the number of flux quanta threading the sample.

7.3.2 — Integer Hall Conductance

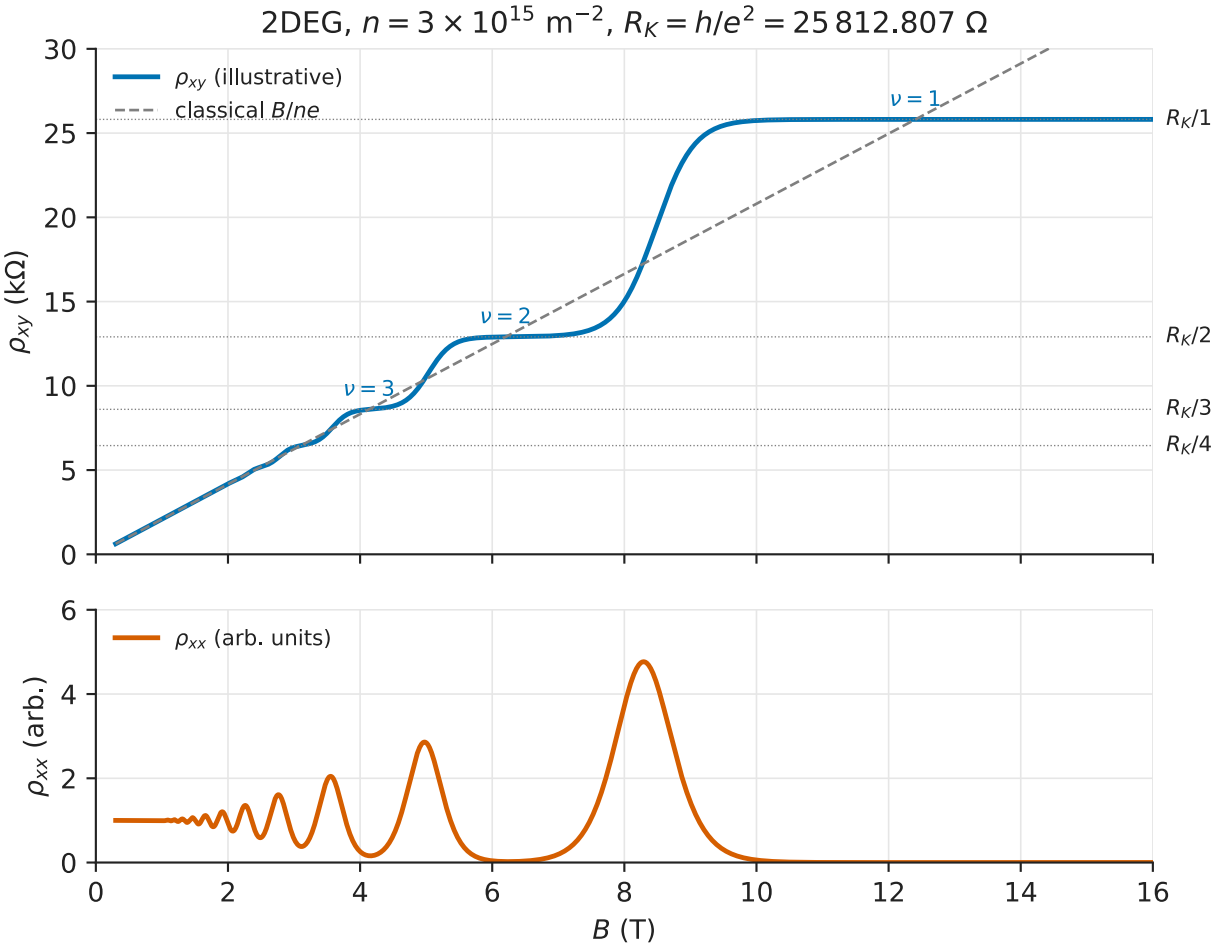
When exactly ν Landau levels are filled (filling factor $\nu = nh/eB$ where n is the electron density):

$$\sigma_{xy} = \nu \frac{e^2}{h}, \quad \sigma_{xx} = 0$$

The measurements (von Klitzing, 1980):

- σ_{xy} is quantized to 1 part in 10^9 — regardless of sample imperfections, impurities, or edge geometry
- $\sigma_{xx} = 0$ simultaneously — no energy dissipation at plateaus

The resistance standard: $R_K = h/e^2 \approx 25812.807\,\Omega$ is now exact by SI definition (2019).



Phenomenological illustration (not a microscopic calculation) for a 2DEG with $n = 3 \times 10^{15}$ m^{-2}: ρ_{xy} follows the classical line B/ne at low field but locks onto plateaus R_K/ν ($R_K = 25\,812.807\ \Omega$) centred where $\nu = nh/eB$ is an integer (e.g. $\nu = 1$ at 12.4 T), while ρ_{xx} vanishes on each plateau and peaks only in the transitions between them; disorder broadening is put in by hand.

7.3.3 — Why Topology Protects It

The TKNN result §7.2 says $\sigma_{xy} = (e^2/h) \sum_n C_n$. The Chern number is:

1. **Quantized by mathematics** (always an integer)
2. **Robust against perturbations** — adding disorder, impurities, or changing sample shape does not change C_n as long as the gap remains open
3. **Changed only by a phase transition** — closing and reopening the gap

This is why a "fractional" Hall conductance never arises in integer QHE: integers cannot be fractional. Disorder smears the Landau levels into bands but does not change the Chern numbers.

7.3.4 — Edge States: The Bulk-Edge Correspondence

A topologically non-trivial insulator (filled Landau levels) in contact with vacuum (topologically trivial, Chern number 0) must have conducting states at the boundary. The gap must close at some point as you go from bulk to vacuum — and it closes by having electronic states that cross the Fermi level.

Bulk-edge correspondence: Number of chiral edge channels = $|C_{total}|$.

For ν filled Landau levels: ν chiral edge channels, each carrying current $e^2/h \times V$ in the same direction. These channels are:

- **Chiral**: electrons move in one direction only (set by $\mathbf{B}$)
- **Backscattering-free**: an electron moving right cannot scatter to move left without crossing the bulk (there is no left-mover at the same edge)

The quantization is protected by chirality. No amount of disorder can cause backscattering in a chiral edge channel — there are no backward states to scatter into. This is the physical reason for the extraordinary precision of the QH plateau.

7.4 — Topological Insulators

7.4.1 — Time-Reversal Symmetry and the Z_2 Invariant

The QHE requires a magnetic field — time-reversal symmetry ($\mathcal{T}$) is broken. A different class of topological phases exists in systems with **preserved** time-reversal symmetry, characterized by a $\mathbb{Z}_2$ topological invariant $\nu_0 \in 0, 1$ (not a Chern number).

Time-reversal for spin-1/2 systems: $\hat{\mathcal{T}}^2 = -1$.

The Kramers theorem: for $\hat{\mathcal{T}}^2 = -1$, every eigenstate is at least doubly degenerate. At high-symmetry $\mathbf{k}$-points in the BZ (where $\mathbf{k} \equiv -\mathbf{k}$, called time-reversal invariant momenta TRIM), these Kramers pairs are pinned at the same energy.

The Z2 invariant counts the parity of the number of times the surface band crosses the Fermi energy between two TRIM points. Informally:

$$(-1)^{\nu_0} = \prod_{\text{TRIM}} \xi_{2m}(\Gamma_i)$$

where ξ_{2m} are parity eigenvalues of occupied bands at TRIM points.

- $\nu_0 = 0$: topologically trivial (ordinary insulator)
- $\nu_0 = 1$: topological insulator (non-trivial)

7.4.2 — 2D Topological Insulator: Quantum Spin Hall Effect

HgTe/CdTe quantum wells (König et al., 2007): HgTe has inverted band order (the Γ_8 light-hole band is above the Γ_6 conduction band) while CdTe is normal. At a critical well thickness, band inversion occurs — the Z_2 invariant changes from 0 to 1.

Result: **helical edge states** with **spin-momentum locking**:

- Spin-up electrons move right
- Spin-down electrons move left
- These are time-reversal partners (Kramers pair)

Why they cannot backscatter: Backscattering would require spin-up (moving right) to become spin-down (moving left). Time reversal takes ↑right to ↓left. Since $\hat{\mathcal{T}}^2 = -1$, the backscattering matrix element $\langle \downarrow \text{left} | \hat{V} | \uparrow \text{right} \rangle$ satisfies:

$$\langle \downarrow \text{left} | \hat{V} | \uparrow \text{right} \rangle = -\langle \downarrow \text{left} | \hat{V} | \uparrow \text{right} \rangle = 0$$

A time-reversal invariant, non-magnetic impurity cannot cause elastic single-particle backscattering between Kramers partners on an isolated helical edge. A magnetic impurity (which breaks $\mathcal{T}$) can gap the edge states — a feature used in quantum anomalous Hall devices.

> **Warning: The protected-channel statement has a domain of validity** "Zero backscattering" is a statement about a *specific idealisation*, and it is easy to over-apply. Precisely what the cancellation above rules out is **elastic, single-particle backscattering between the Kramers pair, by a perturbation that respects $\hat{\mathcal{T}}$, with no other mechanism available**. Outside that, protection fails:
>
> - **Interactions.** Electron–electron interactions can redistribute momentum between edge modes; while strict single-particle backscattering remains forbidden, the current is no longer carried by independent quasiparticles and the ideal conductance picture needs modification.
> - **Inelastic processes.** Phonon absorption/emission introduces energy dissipation and a non-elastic path; the $\mathcal{T}$-selection argument applies to elastic scattering only.
> - **Multiple-particle / higher-order processes.** Sequential two-impurity events involving an intermediate bulk or virtual state can in effect reverse the propagation direction.
> - **Magnetic or otherwise symmetry-breaking perturbations.** These open a backscattering channel directly.
> - **Coupling to bulk states.** Once the edge is coupled to bulk conduction or additional edge channels, a reverse-propagating state exists and backscattering is possible.
>
> So the defensible statement is the narrow one above; "any time-reversal invariant impurity produces exactly zero backscattering" is too strong as written.

7.4.3 — 3D Topological Insulators

Materials: Bi_2Se_3, Bi_2Te_3, $Bi_2Sb_3Te_3$ (and alloys).

These have strong spin-orbit coupling (from heavy Bi, Se, Te atoms) that inverts the band structure at the Γ-point. The result: $\nu_0 = 1$.

Surface states: A topological insulator with $\nu_0 = 1$ has a Dirac cone on every surface — a single cone (odd number — this is what $\nu_0 = 1$ means). The Hamiltonian near the cone:

$$H_{surface}(\mathbf{k}) = \hbar v_F (\hat{z} \times \boldsymbol{\sigma}) \cdot \mathbf{k}$$

The spin is locked perpendicular to $\mathbf{k}$ (helical spin texture). ARPES (angle-resolved photoemission spectroscopy) directly images this cone and spin texture — it is one of the clearest demonstrations that a quantum-mechanical prediction is correct.

Key properties:

- Bulk is insulating: $E_g \approx 0.3$ eV for Bi_2Se_3
- Surface is metallic: single Dirac cone with $v_F \approx 5 \times 10^5$ m/s
- Surface conductance is topologically protected: cannot be gapped by any non-magnetic surface perturbation (contamination, reconstruction, gentle disorder)

7.4.4 — The Axion Connection to Chapter 0

The effective electromagnetic action of a 3D TI contains a topological term:

$$S_\theta = \frac{\theta e^2}{2\pi h} \int d^3r\, dt\, \mathbf{E} \cdot \mathbf{B}$$

For a trivial insulator: $\theta = 0$. For a strong TI: $\theta = \pi$.

The **mathematical structure is the same class of object** as the QCD θ-term from Ch. 0 §0.9: $\theta \frac{g^2}{32\pi^2} G_{\mu\nu}\tilde{G}^{\mu\nu}$, with EM fields in place of gluon fields — a total-derivative term $\theta\, F\tilde{F}$ whose coefficient is **defined only modulo** 2π and which time-reversal symmetry pins to a discrete set of values. This is a `[STRUCTURAL CONNECTION]`, and the structural language is deliberate: the systems, the effective theories, and the physics are **not** the same. In QCD, θ is a single number whose smallness ($\lesssim 10^{-10}$) is the strong-CP problem, measured in a fundamental theory of the Standard Model. In a three-dimensional time-reversal-invariant topological insulator, time reversal sends $\theta \to -\theta$, so the only values consistent with the symmetry are $\theta = 0$ or $\pi \pmod{2\pi}$. Two distinctions matter here. First, θ is **not** an integer-valued coefficient: it is a continuous variable, defined modulo 2π, that the symmetry restricts to two classes. The integers in this story live in the *band-structure invariant* — the Chern number or the $\mathbb{Z}_2$ index — not in the coefficient of $F\tilde{F}$. Second, θ is a genuine physical response coefficient: it is proportional to the half-quantized magnetoelectric polarizability P_3, measurable as a surface Hall conductance. The QCD θ, by contrast, is a free parameter of a fundamental theory, and nothing in the Standard Model fixes it. Sharing a mathematical form is not sharing an explanation, and the analogy earns its keep by illuminating the *form* of the strong-CP problem, not by equating the two θ's.

The Peccei-Quinn mechanism (Ch. 0) promotes θ to a dynamical field — the **axion**. Proposed axion dark matter could be detected by its coupling $\mathbf{E} \cdot \mathbf{B}$ — using topological insulator heterostructures as detectors. **This is a direct link from the unknown sector $\mathcal{F}[\ldots]$ of Ch. 0 to a table-top condensed matter experiment.**

7.5 — Weyl Semimetals

7.5.1 — Weyl Nodes in the Brillouin Zone

A Weyl semimetal has pairs of band-touching points (Weyl nodes) in the Brillouin zone where the Hamiltonian near each node takes the form:

$$H_{Weyl}(\mathbf{q}) = \pm \hbar v_F \boldsymbol{\sigma} \cdot \mathbf{q}$$

where $\mathbf{q} = \mathbf{k} - \mathbf{k}_{Weyl}$ is measured from the node. This is the **Weyl equation** — the equation for a massless relativistic fermion of definite chirality from Ch. 0. The $+/-$ sign is the chirality.

The condensed matter Weyl fermion is not an analogy — the mathematical structure $H = \hbar v_F \boldsymbol{\sigma} \cdot \mathbf{k}$ is identical to the Weyl spinor in $\mathcal{L}_{SM}$, with v_F replacing c. A Weyl semimetal is a material in which Standard Model Lagrangian physics emerges at low energies.

7.5.2 — Topological Protection and Berry Monopoles

The Berry curvature of a Weyl node:

$$\boldsymbol{\Omega}_+(\mathbf{q}) = \frac{\mathbf{q}}{2|\mathbf{q}|^3} \quad \text{(for chirality } +1\text{)}$$

This is the field of a magnetic monopole at $\mathbf{q} = 0$ in $\mathbf{k}$-space, with strength (Chern number) $C = +1$. The opposite-chirality node has $C = -1$.

Why Weyl nodes are robust: A single Weyl node cannot be removed by any perturbation — it is a monopole in Berry curvature, and monopoles cannot be continuously annihilated. Two nodes of opposite chirality can annihilate (same as a magnetic monopole and antimonopole meeting). This is why Weyl nodes come in pairs and can only disappear by merging in $\mathbf{k}$-space.

Nielsen-Ninomiya theorem: Any lattice regularization of a theory with Weyl fermions must have equal numbers of left- and right-handed Weyl nodes. This is a lattice version of the anomaly cancellation condition in Ch. 0 §0.10.

7.5.3 — Fermi Arcs: Open Surface Fermi Surfaces

Between Weyl nodes of opposite chirality projected onto the surface BZ, there are **Fermi arc surface states** — open arcs of Fermi surface connecting the projected positions of the two nodes.

This is forbidden in ordinary band theory (Fermi surfaces must be closed) but allowed here because the surface states are terminated by the bulk Weyl nodes — a purely topological consequence.

Measurement: ARPES directly measures the Fermi surface; Fermi arcs have been observed in TaAs, NbP, and WTe_2 exactly as predicted.

7.5.4 — Chiral Anomaly: A Quantum Field Theory Effect in a Material

In the presence of parallel electric and magnetic fields ($\mathbf{E} \parallel \mathbf{B}$), charge is pumped between Weyl nodes of opposite chirality at rate:

$$\frac{d(n_R - n_L)}{dt} = \frac{e^2}{2\pi^2\hbar^2}\mathbf{E}\cdot\mathbf{B}$$

This is the **ABJ chiral anomaly** from quantum field theory — the breaking of classical chiral symmetry by quantum effects, discovered in particle physics in 1969. It appears directly in the transport properties of Weyl semimetals as:

Negative longitudinal magnetoresistance: When $\mathbf{B} \parallel \mathbf{J}$, the chiral charge pumping increases conductivity (more carriers contribute to current). σ increases with B — **negative** magnetoresistance. This is the experimental signature of the chiral anomaly, measured in TaAs in 2015.

7.6 — Superconductivity: Broken U(1) Symmetry

7.6.1 — The Cooper Instability

In a normal metal at $T = 0$, the Fermi sea is filled. Adding two electrons above E_F — even with a vanishingly small attractive interaction — leads to a bound state. This is the **Cooper instability** (1956), proven by showing that the two-electron Schrödinger equation above a Fermi sea has a bound-state solution for any $V < 0$ (attractive interaction), no matter how weak.

The binding energy of a Cooper pair:

$$E_{binding} \approx -2\hbar\omega_D \exp\left(-\frac{2}{N(0)V}\right)$$

where ω_D is the Debye frequency (maximum phonon frequency), $N(0)$ is the density of states at the Fermi level, and $V > 0$ is the dimensionless pairing strength.

Why phonon-mediated attraction: An electron moving through the lattice slightly polarizes the surrounding ions (they are attracted toward the electron's path). This positive ion polarization lingers for a time $\sim 1/\omega_D$ after the electron has passed. A second electron arriving later is attracted to this positive polarization — experiencing an effective attractive electron-electron interaction mediated by the lattice. The net effect: the retarded phonon interaction can overcome the instantaneous Coulomb repulsion.

7.6.2 — The BCS Ground State and Energy Gap

The full BCS theory (Bardeen-Cooper-Schrieffer, 1957) treats all pairs simultaneously. The BCS ground state is a coherent superposition:

$$|BCS\rangle = \prod_{\mathbf{k}} (u_{\mathbf{k}} + v_{\mathbf{k}} \hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}^\dagger_{-\mathbf{k}\downarrow}) |0\rangle$$

where $|v_{\mathbf{k}}|^2 = \frac{1}{2}\left(1 - \frac{\xi_{\mathbf{k}}}{\sqrt{\xi_{\mathbf{k}}^2 + |\Delta|^2}}\right)$ is the probability that the pair $(\mathbf{k}\uparrow, -\mathbf{k}\downarrow)$ is occupied, and $\xi_{\mathbf{k}} = E_{\mathbf{k}} - E_F$ is the energy measured from the Fermi level.

The **BCS gap equation** determines Δ self-consistently:

$$\boxed{\Delta = V \sum_{\mathbf{k}} \frac{\Delta}{2\sqrt{\xi_{\mathbf{k}}^2 + |\Delta|^2}}}$$

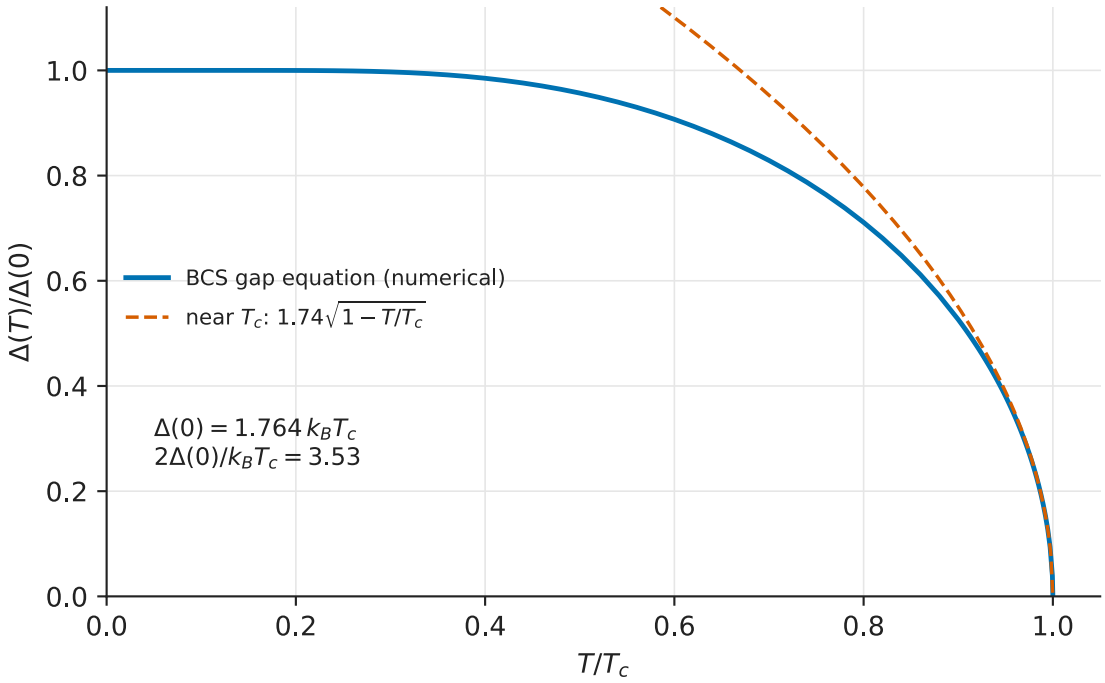
At $T = 0$, the solution:

$$\Delta(0) = 2\hbar\omega_D \exp\left(-\frac{1}{N(0)V}\right)$$

At temperature T:

$$\Delta(T) \approx 1.74\,\Delta(0)\sqrt{1 - \frac{T}{T_c}} \;\; (T \to T_c), \qquad \Delta(0) = 1.764\, k_B T_c$$

The ratio $2\Delta(0)/k_B T_c = 3.528$ is a universal BCS prediction — one of the first great successes of the theory, confirmed in many conventional superconductors.



$\Delta(T)/\Delta(0)$ from solving the weak-coupling BCS gap equation numerically: the gap is essentially flat below $T \approx 0.4\,T_c$ and closes as $1.74\,\Delta(0)\sqrt{1 - T/T_c}$ (dashed) near T_c, with $\Delta(0) = 1.764\,k_B T_c$.

7.6.3 — The Mexican Hat: Superconductivity as Symmetry Breaking

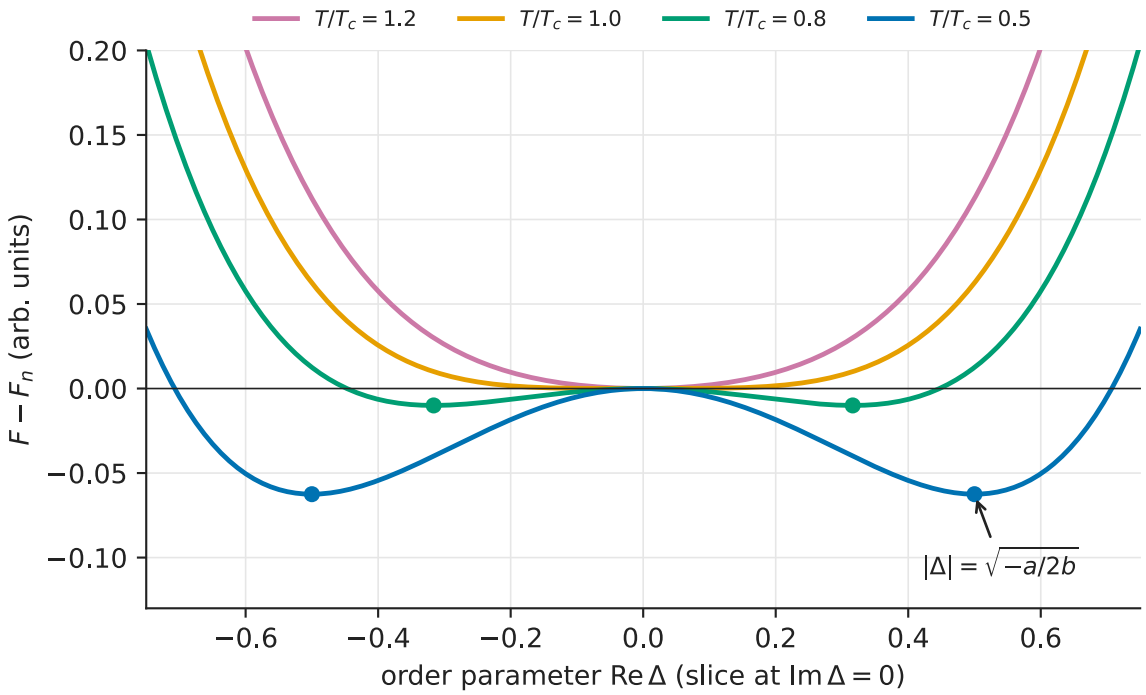
The BCS ground state has a definite macroscopic phase ϕ. The order parameter:

$$\Delta = |\Delta|, e^{i\phi}$$

The free energy (Ginzburg-Landau):

$$F = a(T)|\Delta|^2 + b|\Delta|^4 + \frac{\hbar^2}{2m^*}|\nabla\Delta|^2 + \cdots$$

with $a(T) = a_0(T - T_c)$. Below T_c: $a < 0$, the minimum is at $|\Delta| = \sqrt{-a/2b} \neq 0$.



$F = a_0(T - T_c)|\Delta|^2 + b|\Delta|^4$ (with $a_0 = b = 1$) along a slice through the complex Δ plane, for $T/T_c = 1.2, 1.0, 0.8, 0.5$: above T_c the only minimum is $\Delta = 0$, at T_c the bottom is flat (quartic), and below T_c the minima (dots) move out to $|\Delta| = \sqrt{-a/2b}$; rotating the curve about the vertical axis gives the Mexican hat (3D picture: Ch. 0 §0.7).

This is the Mexican hat potential from Ch. 0 §0.7 — identical mathematics, ten orders of magnitude different in energy scale:

System	Order parameter	Symmetry broken	Energy scale
Higgs field (Ch. 0)	$H = \|H\|e^{i\phi}$	electroweak $SU(2) \times U(1) \to U(1)_{em}$	~100 GeV
BEC (Ch. 10)	$\Psi = \|\Psi\|e^{i\phi}$	global $U(1)$	~nK
Superconductor	$\Delta = \|\Delta\|e^{i\phi}$	global $U(1)$ (gauge-invariant; see below)	~meV
Ferromagnet	M — a real vector, **not** $\|M\|e^{i\phi}$	time-reversal, not $U(1)$	~eV

Same free-energy form for the first three — but not the same symmetry story. The superconducting and BEC order parameters both carry a global $U(1)$ phase, so the free energy is identical in form. The ferromagnet does *not*: its order parameter is a real vector and it breaks time-reversal symmetry, not $U(1)$. Putting it in this table as $|M|e^{i\phi}$ was wrong and has been removed.

On the gauge case, precisely. For a superconductor the broken $U(1)$ is *global*, and the condensate phase is a gauge-invariant physical observable, not a gauge artefact. Electromagnetic $U(1)$ is **not** physically broken — by Elitzur's theorem a local gauge symmetry cannot be spontaneously broken, because gauge-dependent expectation values are not gauge invariant. What happens is the **Anderson–Higgs mechanism**: the phase stiffness of the condensate supplies an effective mass to the electromagnetic field within the penetration depth, $\lambda_L^{-2} = \mu_0 n_s e^{*2}/m^*$, which is dynamically equivalent to the Higgs mechanism. So "the photon acquires a mass inside the superconductor" is a correct and useful statement; "U(1) is broken" is shorthand that needs the qualification above. The observable consequence is the Meissner effect — magnetic field excluded from the bulk, decaying over λ_L (§7.7).

7.7 — London Equations and the Meissner Effect

7.7.1 — The London Equations

The BCS condensate has a current density (from minimal coupling, Ch. 3):

$$\mathbf{J}_s = -\frac{n_s e^2}{m_e}\mathbf{A} - \frac{n_s e\hbar}{m_e}\nabla\phi$$

where n_s is the superfluid density, ϕ is the condensate phase, and $\mathbf{A}$ is the vector potential (in the London gauge $\nabla\phi = 0$):

$$\mathbf{J}_s = -\frac{n_s e^2}{m_e}\mathbf{A}$$

This is **London's second equation**. Taking the curl and using $\mathbf{B} = \nabla \times \mathbf{A}$:

$$\nabla \times \mathbf{J}_s = -\frac{n_s e^2}{m_e}\mathbf{B}$$

Combined with Ampere's law $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}_s$:

$$\nabla^2 \mathbf{B} = \frac{\mathbf{B}}{\lambda_L^2}, \qquad \lambda_L = \sqrt{\frac{m_e}{\mu_0 n_s e^2}}$$

The solution inside a superconductor occupying $x > 0$:

$$B(x) = B_0\, e^{-x/\lambda_L}$$

The magnetic field **decays exponentially** from the surface with the **London penetration depth** λ_L. For most conventional SCs: $\lambda_L \sim 10$–100 nm. For $x \gg \lambda_L$: $\mathbf{B} = 0$ — the Meissner effect.

The Higgs mechanism made explicit: The photon inside the superconductor acquires effective mass $m_\gamma = \hbar/c\lambda_L$ — exactly the Higgs mechanism where the W and Z bosons acquire mass from the Higgs VEV. In the SC, the Higgs VEV is replaced by $|\Delta|$.

7.7.2 — Type I and Type II Superconductors

Compare the London penetration depth λ_L (field expulsion) with the **Ginzburg-Landau coherence length** ξ (spatial variation of $|\Delta|$, the distance over which the order parameter recovers its bulk value):

$$\kappa_{GL} = \frac{\lambda_L}{\xi}$$

- **Type I** ($\kappa_{GL} < 1/\sqrt{2}$): perfect Meissner up to H_c; then normal. Rare in practice.
- **Type II** ($\kappa_{GL} > 1/\sqrt{2}$): Meissner up to H_{c1}; then **mixed (Abrikosov vortex) state** from H_{c1} to H_{c2}; then normal.

Abrikosov vortices (Nobel Prize 2003): In the mixed state, quantized tubes of magnetic flux penetrate the superconductor. Each vortex carries exactly one flux quantum $\Phi_0 = h/2e$ and has:

- A **normal core** of radius ξ (where $|\Delta| \to 0$)
- Circulating supercurrents in a ring of radius λ_L

High-T_c superconductors (YBCO, BSCCO) are Type II with $H_{c2} > 100$ T — the reason they can be used for high-field magnets.

7.8 — Flux Quantization and the Josephson Effect

7.8.1 — Flux Quantization

The single-valuedness of the superconducting order parameter $\Delta = |\Delta| e^{i\phi}$ around a closed loop requires:

$$\oint \nabla\phi \cdot d\mathbf{l} = 2\pi n, \quad n \in \mathbb{Z}$$

Using the expression for the supercurrent (London equation), the total magnetic flux threading the loop:

$$\boxed{\Phi = n\Phi_0, \qquad \Phi_0 = \frac{h}{2e} = 2.067 \times 10^{-15}\ \text{Wb}}$$

The $2e$ denominator confirms that the supercurrent is carried by Cooper pairs (charge $2e$), not individual electrons (e). This was experimentally confirmed in 1961 — two years before BCS theory was fully accepted.

7.8.2 — The Josephson Effect

Two superconductors with order parameters $\Delta_1 = |\Delta| e^{i\phi_1}$ and $\Delta_2 = |\Delta| e^{i\phi_2}$ separated by a thin insulating barrier (a Josephson junction):

DC Josephson effect: A supercurrent flows with no applied voltage:

$$\boxed{I = I_c \sin(\phi_1 - \phi_2)}$$

where I_c is the critical current (maximum Cooper pair tunneling rate) and $\phi_1 - \phi_2$ is the phase difference across the junction.

AC Josephson effect: Apply a DC voltage V across the junction. The phase evolves as $d(\phi_1 - \phi_2)/dt = 2eV/\hbar$, giving an AC supercurrent:

$$I(t) = I_c \sin\left(\phi_0 + \frac{2eV}{\hbar} t\right) \quad \Longrightarrow \quad f_{AC} = \frac{2eV}{h}$$

The Josephson frequency-voltage relation $f = 2eV/h = 483597.8\ \text{GHz/V}$:

- It depends only on fundamental constants (e and h), not on the material
- Since 2019, it defines the volt in the SI system: $1\ \text{V} \equiv f/483597.8\ \text{GHz}$ (exact by definition, traceable to quantum mechanics)

7.8.3 — The SQUID: Most Sensitive Magnetometer

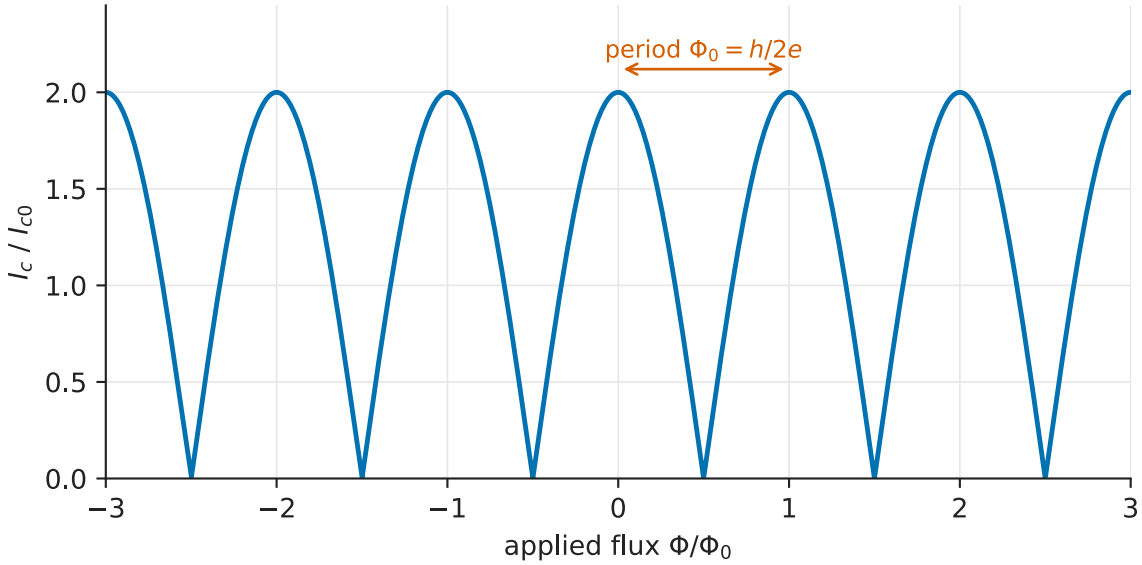
A superconducting quantum interference device (SQUID) consists of a superconducting loop with two Josephson junctions. The critical current:

$$I_c(\Phi) = 2I_{c0} \left| \cos\left(\frac{\pi\Phi}{\Phi_0}\right) \right|$$

oscillates with the magnetic flux Φ through the loop with period $\Phi_0 = h/2e$. Measuring I_c to 1 part in 10^3 resolves flux changes of:

$$\delta\Phi \sim 10^{-3}\Phi_0 = 2 \times 10^{-18}\,\text{Wb}$$

For a loop area of $1\,\text{mm}^2$: $\delta B \sim 2 \times 10^{-12}$ T — far below Earth's field ($\sim 50\,\mu$T), though still $\sim 20\times$ the magnetic field of the human brain ($\sim 10^{-13}$ T at the scalp). Practical magnetometers close that gap with finer flux resolution ($\sim 10^{-6}\Phi_0/\sqrt{\text{Hz}}$) and a cm-scale superconducting pickup coil (flux transformer), reaching ~ 1–$5\,\text{fT}/\sqrt{\text{Hz}}$.



The dc-SQUID critical current $I_c(\Phi) = 2I_{c0}|\cos(\pi\Phi/\Phi_0)|$ for $-3 \le \Phi/\Phi_0 \le 3$: it repeats every flux quantum $\Phi_0 = h/2e$, so counting periods measures flux in units of Φ_0 and interpolating within one period resolves small fractions of it.

Engineering applications: Magnetoencephalography (MEG) for brain imaging; non-destructive testing; geodetic surveys; detection of unexploded ordnance; qubit readout in superconducting quantum computers.

7.9 — Coulomb Blockade and Mesoscopic Physics

At the boundary between Layer 1 (quantum) and Layer 2 (classical continuum), sits a regime where systems are large enough to have continuous energy spectra but small enough that single-electron charging matters.

7.9.1 — The Quantum Dot

A **quantum dot** — a small conductor (semiconductor or metallic island) of capacitance C — has a charging energy:

$$E_C = \frac{e^2}{2C}$$

For a dot with $C \sim 1$ aF: $E_C \sim 80$ meV $\gg k_B T$ at low temperature. Adding one electron to the dot costs energy E_C — but if the gate voltage exactly compensates this, the electron tunnels freely.

The result: **Coulomb oscillations** — the conductance through the dot oscillates as a function of gate voltage, with period $\Delta V_g = e/C_g$. Each peak corresponds to adding exactly one electron. In a DC bias, single electrons tunnel one at a time: a **single-electron transistor**.

The quantum dot level spectrum: For very small dots, the discrete energy levels from quantum confinement (Ch. 4 particle-in-box) become relevant. The dot becomes an artificial atom — with shell structure, spin states, and exchange interactions controlled by gate voltages.

Engineering: Quantum dots are the leading platform for semiconductor spin qubits; the Coulomb blockade is used for charge sensing with $\sim 10^{-4} e/\sqrt{\text{Hz}}$ sensitivity; single-photon emitters for quantum communication.

7.10 — The Topology Thread: From Ch. 0 to Engineering

Five connections between the topology of Ch. 0 and the physics of this chapter:

Ch. 0 structure	Condensed matter analog	Ch. 7 section	Connection
θ-term in QCD	Axion electrodynamics in TI; $\theta = \pi$	§7.4.4	`[STRUCTURAL CONNECTION]` — same topological term, different field content
Weyl fermion field ψ_L in $\mathcal{L}_{SM}$	Weyl quasiparticle in TaAs, NbAs	§7.5.1	`[STRUCTURAL CONNECTION]` — same Dirac algebra, emergent spectrum
ABJ chiral anomaly	Negative magnetoresistance in Weyl metals	§7.5.4	`[DERIVATION]` — the anomaly is universal and the calculation is the same
Higgs mechanism: $SU(2)_L \times U(1)_Y \to U(1)_{\text{EM}}$; $W^\pm$ and Z acquire mass, **the photon does not** ($m_\gamma = 0$, since $U(1)_{\text{EM}}$ is unbroken)	Meissner: broken global $U(1)$ condensate + Anderson–Higgs → effective photon mass *inside the superconductor*	§7.7.1	`[STRUCTURAL CONNECTION]` — same field-theoretic mechanism (a condensate makes the gauge field massive), different symmetry status (§7.6.3)
Anomaly cancellation in SM requires specific particle content	Nielsen–Ninomiya: Weyl nodes come in equal and opposite pairs	§7.5.2	`[STRUCTURAL CONNECTION]` — *not* the same theorem; related through the fermion-doubling obstruction, but one is a continuum gauge-anomaly constraint and the other a lattice no-go result (§7.10.2)

7.10.1 — What this thread is, and what it is not

`[STRUCTURAL CONNECTION]` — **a shared mathematical and topological thread, not a descent.** The book's claim is that later physics is a *limit or projection* of earlier action. That claim is true of Chapter 3 (Dirac → Pauli → Schrödinger) and true of Chapter 16 (continuum → lumped). It is **not** true here, and the distinction is worth being precise about rather than quietly rounding up.

What is genuinely shared:

- The **Berry phase** of §7.1 and the θ-term of Chapter 0 are both integrals of a curvature over a closed parameter manifold, and both are integer-quantised for topological indices. That is a real structural identity, and it is why the axion angle θ of QCD and the magnetoelectric coupling of a topological insulator can be discussed in the same language.

- The **chiral anomaly** and the **ABJ anomaly** are the same theorem. The calculation is not analogous to itself; it is one result, applied once to quark fields and once to Weyl quasiparticles.

The Nielsen–Ninomiya theorem is *not* on this list, and the distinction is worth making explicitly — see §7.10.2.

What is *not* shared, and what "the same action" would wrongly imply:

- A Weyl semimetal is **not** a sector of $\mathcal{L}_{SM}$ with a different parameter value. It is a low-energy **effective** theory in which Weyl nodes are *emergent* — they are not present in the microscopic crystal Hamiltonian and appear only after integrating out the full band structure. The quasiparticle fields are not the Standard Model's ψ_L; they are constructed from Bloch states.
- The relevant energy scales are unrelated. Chapter 0's fermions sit at $E \gg m_W$; a Weyl node at k in TaAs sits at meV–100 meV. There is no limit operation relating them.
- The symmetries are analogies, not identifications: QED's $\mathrm{U}(1)_A$ is an approximate flavour symmetry; a condensed-matter system's particle-hole symmetry is a lattice property with no Standard Model counterpart.

Model Ledger — topological classification to a measured transport coefficient

Field	Content
Parent theory	Berry connection and Chern number from Bloch band structure (§7.1)
Reduction	Semiclassical equations of motion for a Weyl node pair; linear response to E and B
Model	$\sigma_{xy} = e^2/h$ per chiral channel, set by the integer ν
Assumptions	clean band structure; Fermi level in the gap; no phonon or disorder scattering; $k_B T \ll \Delta$; semiclassical regime $\hbar\omega_c \ll E_F$
Physics retained	Berry-curvature pumping; integer topological invariant
Physics neglected	disorder and localisation physics, phonon-assisted scattering, interactions, thermal activation, edge reconstruction
Validity	quantised plateau observed; $T \lesssim 1$ K in high-mobility 2DEGs
Fails when	disorder broadening exceeds the gap; interactions localise; sample edges are reconstructed
Next model	non-interacting edge theory (Landauer, Ch. 13 §13.11); K-matrix/fractional QH; magnetotransport with disorder (Ch. 18)

The honest summary statement: topology in Chapter 0 and topology in condensed matter are **the same mathematics applied to different systems**. That is a real and useful connection — it is why techniques transfer — but it is a statement about mathematics, not about one theory descending from the other, and the two should not be conflated.

7.10.2 — Anomaly cancellation and Nielsen–Ninomiya are not the same theorem

`[STRUCTURAL CONNECTION]` — These two are constantly conflated because they are both about anomalies and chiral fermions, and because a real mathematical relationship connects them. They are nevertheless different theorems with different hypotheses, different conclusions, and different provenances, and labelling the second `[DERIVATION]`-equivalent to the first overstates the case.

	SM anomaly cancellation	Nielsen–Ninomiya
Setting	continuum relativistic chiral gauge theory	lattice-regularized fermion theory
Statement	gauge and gravitational anomalies must cancel for the theory to be consistent, fixing the required fermion representation content of $SU(2) \times U(1)$	a single Weyl fermion cannot be put on a lattice with locality, translational invariance, and Hermiticity: nodes of opposite chirality must occur, with total chirality vanishing on the Brillouin torus
Origin	Adler–Bell–Jackiw anomaly and its descendants	fermion doubling / topological charge in the Brillouin zone
What it constrains	particle content of a fundamental theory	existence of isolated Weyl nodes in a band structure

The relationship is real but indirect, and it runs through the **fermion-doubling obstruction**: the Nielsen–Ninomiya theorem is one of the obstructions that make it difficult to formulate a consistent purely-chiral continuum gauge theory without additional structure, and anomaly cancellation is part of the price of making such a theory work anyway. But the logical directions differ — Nielsen– Ninomiya says a *lattice* chiral theory must double; anomaly cancellation says a *continuum* chiral gauge theory needs matching representations. Neither is obtained from the other by relabelling ψ or by changing the sign of a momentum, and the Brillouin zone is not a place where gauge anomalies are evaluated.

The honest label is `[STRUCTURAL CONNECTION]`: a shared root concern (what blocks a consistent chiral fermion description) reached by different arguments. The connection earns its keep as a warning about doubling, not as a license to transfer SM particle-content arguments into band theory.

7.11 — Summary

Layer 1 closes with the topological perspective:

Result	Key formula	Protection
Berry phase	$\gamma = \oint \mathbf{A}_n \cdot d\mathbf{k}$	Geometric (path-dependent)
Chern number	$C = (1/2\pi) \iint \Omega \, d^2k \in \mathbb{Z}$	Topological (always integer)
Hall conductivity	$\sigma_{xy} = (e^2/h) \sum_n C_n$	Topological (quantized)
TI surface states	$H = \hbar v_F (\hat{z} \times \boldsymbol{\sigma}) \cdot \mathbf{k}$	Symmetry ($\mathcal{T}$)
Weyl node	$H = \pm \hbar v_F \boldsymbol{\sigma} \cdot \mathbf{k}$; $C = \pm 1$	Topological (Berry monopole)
BCS gap	$\Delta = 2\hbar\omega_D e^{-1/N(0)V}$	Symmetry breaking
Meissner	$B = B_0 e^{-x/\lambda_L}$; $B_{bulk} = 0$	Anderson–Higgs / phase stiffness
Flux quant.	$\Phi = nh/2e$	Phase winding (integer)
Josephson	$I = I_c \sin\Delta\phi$; $f = 2eV/h$	Phase coherence

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7.12 — Engineering Thread

Physics	Application
Quantum Hall effect	Primary resistance standard $R_K = h/e^2$ (exact); calibration
Topological surface states	Spin-polarized current sources; future low-dissipation transistors

Physics	Application
Weyl semimetal chiral anomaly	Ultra-high mobility; negative MR sensors
Meissner effect	MRI magnets (YBCO, NbTi); magnetic shielding; Maglev
Flux quantization + SQUID	Magnetoencephalography; brain imaging; geophysical surveys
Josephson voltage standard	SI volt definition; precision metrology
BCS superconductors	Particle accelerator magnets (LHC uses 8.3 T NbTi); power cables
High-T_c superconductors	Future power grids; FCL (fault current limiters)
Superconducting qubits	IBM, Google, IQM quantum computers all use Josephson junctions
Coulomb blockade / quantum dots	Spin qubits; single-electron transistors; single-photon emitters

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7.13 — Looking Ahead: Bridge B

Layer 1 is complete. Starting with the Standard Model Lagrangian (Ch. 0) and the action principle (Ch. 1), we derived:

- The Schrödinger equation (Ch. 3 Bridge A)
- Quantization and the hydrogen atom (Ch. 4)
- Spin, Pauli exclusion, the periodic table (Ch. 5)
- Band theory, bonding, nuclear physics (Ch. 6)
- Topology, quantum Hall effect, superconductivity (Ch. 7)

Everything above is quantum mechanical. Chapter 8 begins **Bridge B** — the descent to the classical world. Four simultaneous limits (§1.9 of the layer map):

- $\hbar \to 0$: quantum expectation values → Newton's laws (Ch. 9)
- $N \to \infty$, ensemble average: quantum states → thermodynamics (Ch. 10)
- Classical EM field limit: quantum EM → Maxwell's equations (Ch. 11)
- Weak metric field: GR → Newtonian gravity (Ch. 8)
- Kubo averaging: quantum scattering → Ohm, Fourier, Fick, viscosity, Hooke (Ch. 13)

The entire landscape of classical engineering science emerges from these five limiting operations applied to the quantum world of Layer 1.

End of Chapter 7. End of Layer 1.

Next: Chapter 8 — Bridge B.d: From General Relativity to Newton's Law of Gravitation

Bridge B.d — From General Relativity to Newton's Law of Gravitation

How Spacetime Curvature Becomes an Inverse-Square Force

> *Before Einstein, space was just the stage on which physics happened. After Einstein, space became a participant. It curves. It waves. And in the limit where nothing moves fast and everything is weak, it pretends to be Newton's force.* — paraphrased from Misner, Thorne, Wheeler, *Gravitation*

8.0 — Bridge B.d in Context

Chapters 3–7 descended through Layer 1: quantum mechanics from the Standard Model Lagrangian. Bridge B now takes us from Layer 1 to Layer 2 — the classical continuum world.

Bridge B is not one transition but five simultaneous limiting operations (from the layer map, Ch. 1 §1.9):

Bridge	Limit	Produces	Chapter
B.d	Weak field + slow motion + static	Newton's gravity	**8 (this chapter)**
B.a	$\hbar \to 0$	Classical mechanics	9
B.b	$N \to \infty$, ensemble average	Thermodynamics	10
B.c	Classical EM field limit	Maxwell's equations	11
B.e	Kubo + quantum scattering	Generalized transport law	13

This chapter extracts the gravitational sector from Ch. 0's action:

$$S_{EH} = \frac{1}{16\pi G}\int d^4x\,\sqrt{-g}\,R$$

The destination is Newton's law $F = -GMm/r^2$ — but we will pass through general relativity properly, collecting every correction that engineering currently cares about along the way.

8.1 — The Starting Point: Einstein Field Equations

From Ch. 0 §0.3: varying S_{EH} with respect to the inverse metric $g^{\mu\nu}$ gives the **Einstein field equations** (in SI units):

$$\boxed{G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = \frac{8\pi G}{c^4}\,T_{\mu\nu}}$$

Everything in this equation has appeared before, but let us restate it clearly:

Left side — geometry:

- $g_{\mu\nu}$: the metric tensor — the dynamical field of gravity, encoding distances and angles in curved spacetime
- $R_{\mu\nu}$: the Ricci tensor — a contraction of the Riemann curvature tensor, measuring how much parallel transport around a loop rotates a vector
- $R = g^{\mu\nu}R_{\mu\nu}$: the Ricci scalar — a scalar contraction of the Riemann tensor giving one particular local curvature invariant. It does **not** measure the total local curvature: a nonzero R does not imply a curved spacetime (the Schwarzschild vacuum has $R = 0$ yet non-zero Weyl curvature), and a zero R does not imply flat spacetime (Ch. 0 §0.3).
- $G_{\mu\nu}$: the Einstein tensor — the unique symmetric, divergence-free ($\nabla^{\mu}G_{\mu\nu} = 0$), second-order combination of the metric

Right side — matter:

- $T_{\mu\nu}$: the stress-energy tensor — matter's energy-momentum content, encoding:
  - T^{00} = energy density (including rest mass energy ρc^2)
  - T^{0i} = momentum density (energy flux)
  - T^{ij} = stress tensor (pressure and shear)

  Its connection to Noether's theorem needs a caveat, because in general relativity the naive statement is not correct. In **flat-space** field theory, translation invariance is a continuous symmetry and the (canonical, or symmetrised Belinfante) stress-energy tensor is indeed the Noether current of spacetime translations (Ch. 1 §1.9.3). In **GR** this becomes delicate: the diffeomorphism group is a *gauge* redundancy rather than a physical global symmetry, so translations are not an ordinary Noether symmetry of general covariance. Gravitational energy cannot be assigned a local gauge-invariant density at all (it lives in the pseudotensor or in the ADM/Ashtekar variables at infinity), and the quantity appearing on the right of Einstein's equation is the **matter** stress tensor, which is covariantly conserved ($\nabla_{\mu}T^{\mu\nu} = 0$) rather than being a plain Noether current. The safe statement, used here: $T_{\mu\nu}$ **is the stress-energy tensor associated with matter's energy-momentum.** The definition actually used in this chapter is the metric variation given immediately after this list. In flat-spacetime field theory the same tensor is instead connected to translation symmetry through Noether's theorem; the two constructions agree in the flat limit, but only the metric-variation one is valid in a curved geometry.

The covariant definition. In general relativity the stress-energy tensor of matter is defined by varying the matter action with respect to the metric:

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta S_{\text{matter}}}{\delta g^{\mu\nu}}$$

Its covariant conservation, $\nabla_{\mu}T^{\mu\nu} = 0$, then follows from the diffeomorphism invariance of S_{matter} rather than from a Noether current (Ch. 1 §1.9.3). This ordering matters: the flat-space statement "the stress-energy tensor is the Noether current of translations" and the GR statement "the stress-energy tensor is the metric variation of the matter action" are **different constructions**, not two ways of writing one definition. They coincide in the flat limit, where the metric is fixed and translations are a genuine global symmetry; in curved spacetime the translation generators are part of a gauge redundancy, and only the metric-variation definition is available. The symbol S_{matter} denotes the matter action alone, so $T_{\mu\nu}$ here is the source on the right-hand side of Einstein's equation and carries no description of gravitational energy itself.

The equation in words:

> Spacetime curvature = $\frac{8\pi G}{c^4}\times$ energy-momentum content

The factor $8\pi G/c^4 \approx 2.07 \times 10^{-43}$ m/J is extraordinarily small — this is why you need planetary masses to produce measurable curvature.

8.2 — The Weak-Field Expansion

8.2.1 — The Perturbation

For weak gravitational fields (everywhere in the solar system, in every engineering context except near black holes), the metric is nearly flat:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \qquad |h_{\mu\nu}| \ll 1$$

where $\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1)$ is the flat Minkowski metric and $h_{\mu\nu}$ is a small perturbation. We work to **first order in** $h_{\mu\nu}$.

Quantitative justification:

Environment	$\lvert h_{00}\rvert = 2\lvert\Phi\rvert/c^2$	Where
Earth's surface	1.4×10^{-9}	Engineering on Earth
GPS satellite orbit	3.3×10^{-10}	GPS receivers
Solar surface	4.2×10^{-6}	Solar physics
Neutron star surface	~ 0.3	Not weak! GR required exactly
Black hole horizon	~ 1	Full GR, no perturbation theory

The perturbative treatment is valid from a laboratory balance through the GPS network, and in the **weak-field wave zone** of systems such as neutron-star binaries during portions of their inspiral. That last case needs care: linearised gravity handles the early inspiral and the far-field wave zone well, but the **late inspiral and merger involve strong-field, nonlinear general relativity** where the first-order $h_{\mu\nu}$ expansion fails and post-Newtonian or numerical methods are required. One does not compute merger with $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ and $|h_{\mu\nu}| \ll 1$.

8.2.2 — Linearized Ricci Tensor

Expanding the Christoffel symbols to first order in h:

$$\Gamma^\rho_{\mu\nu} = \frac{1}{2}\eta^{\rho\sigma}\left(\partial_\mu h_{\nu\sigma} + \partial_\nu h_{\mu\sigma} - \partial_\sigma h_{\mu\nu}\right) + \mathcal{O}(h^2)$$

The linearized Ricci tensor:

$$R^{(1)}_{\mu\nu} = \frac{1}{2}\left(-\Box h_{\mu\nu} - \partial_\mu\partial_\nu h + \partial^\rho\partial_\mu h_{\nu\rho} + \partial^\rho\partial_\nu h_{\mu\rho}\right)$$

where $\Box = -\partial_t^2/c^2 + \nabla^2$ is the d'Alembertian and $h = \eta^{\mu\nu}h_{\mu\nu}$ is the trace.

8.2.3 — Lorenz Gauge and the Trace-Reversed Perturbation

Define the **trace-reversed perturbation**:

$$\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h$$

In the **Lorenz gauge** (the gravitational analog of the electromagnetic Lorenz gauge $\partial^\mu A_\mu = 0$):

$$\partial^\mu \bar{h}_{\mu\nu} = 0$$

the linearized Einstein equations simplify dramatically to:

$$\boxed{\Box \bar{h}_{\mu\nu} = -\frac{16\pi G}{c^4} T_{\mu\nu}}$$

This is a set of ten wave equations — one for each independent component of $\bar{h}_{\mu\nu}$. The source is the stress-energy tensor; the retarded solution is a superposition of gravitational waves propagating at c.

8.3 — The Newtonian Limit: Three Conditions

To recover Newton's law, apply three simultaneous limits:

Condition 1: Slow Motion ($v \ll c$)

The stress-energy tensor for a pressureless dust of density ρ and velocity $\mathbf{v}$:

$$T^{\mu\nu} = \rho\, u^\mu u^\nu$$

where $u^\mu = \gamma(c, \mathbf{v})$ is the four-velocity. For $v \ll c$: $\gamma \approx 1$, and:

$$T^{00} \approx \rho c^2 \gg T^{0i} \approx \rho c v^i \gg T^{ij} \approx \rho v^i v^j$$

Only $T^{00} = \rho c^2$ is significant. All spatial components of $T_{\mu\nu}$ are negligible.

Condition 2: Weak Field ($|h_{\mu\nu}| \ll 1$)

Already imposed in §8.2. The linearized equations are valid.

Condition 3: Static Field ($\partial_t h_{\mu\nu} \approx 0$)

For a slowly-varying source ($v \ll c$, quasi-static), time derivatives of $\bar{h}_{\mu\nu}$ are negligible compared to spatial derivatives: $\Box \approx \nabla^2$.

8.3.1 — The Poisson Equation

With all three conditions, the $(\mu, \nu) = (0, 0)$ component of $\Box \bar{h}_{\mu\nu} = -16\pi G T_{\mu\nu}/c^4$ becomes:

$$\nabla^2 \bar{h}_{00} = -\frac{16\pi G}{c^4} \cdot \rho c^2 = -\frac{16\pi G \rho}{c^2}$$

From the gauge conditions and slow-motion limit: $\bar{h}_{00} = -4\Phi/c^2$ where Φ is to be determined, and all other $\bar{h}_{\mu\nu} \approx 0$.

Substituting:

$$\boxed{\nabla^2\Phi = 4\pi G\rho}$$

This is **Poisson's equation for the Newtonian gravitational potential** — the exact equation Newton would have written.

For a point mass M at the origin ($\rho = M\delta^3(\mathbf{r})$):

$$\nabla^2\Phi = 4\pi GM\delta^3(\mathbf{r}) \quad \Longrightarrow \quad \Phi(r) = -\frac{GM}{r}$$

8.3.2 — Newton's Law from the Geodesic Equation

The geodesic equation (the equation of motion for a freely falling particle in curved spacetime):

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\nu\rho}\frac{dx^\nu}{d\tau}\frac{dx^\rho}{d\tau} = 0$$

For slow motion ($dx^i/d\tau \approx v^i \ll c$, $dx^0/d\tau \approx c$):

$$\frac{d^2x^i}{dt^2} \approx -c^2\Gamma^i_{00} = +c^2 \cdot \frac{1}{2}\eta^{ij}\partial_j h_{00} = -\partial_i\Phi$$

Therefore:

$$\boxed{\mathbf{a} = -\nabla\Phi \quad \Longrightarrow \quad \mathbf{F} = m\mathbf{a} = -\frac{GMm}{r^2}\hat{r}}$$

Newton's law of gravitation is recovered in the weak-field, slow-motion, quasi-static limit of general relativity — three conditions taken *together*, each with a stated scale. The passage from the geodesic equation to $\mathbf{a} = -\nabla\Phi$ requires:

1. $|h_{\mu\nu}| \ll 1$ — the field is weak, $r \gg r_s$;
2. $v \ll c$ — slow motion, so g_{0i} terms are negligible and the geodesic equation's leading spatial term is just acceleration;
3. **quasi-static** — the metric is effectively time-independent over the interval used, so $\partial_0 g_{00}$ terms can be neglected and the potential can be integrated once, globally.

Dropping any one of the three leaves a different equation. Without (1) the field strength itself is not perturbative. Without (2) velocity-dependent and gravitomagnetic terms survive. Without (3) the potential is not static and Φ cannot be defined as a single-valued function of position — which is why this is a *quasi-static* limit and not merely a "weak-field, slow-motion" one.

The key insight: there is no gravitational *force* in GR. A freely falling particle follows a **geodesic** — the straightest possible path in curved spacetime. What appears as a gravitational force to an accelerating observer (standing on the ground) is the Christoffel symbol term — the correction from the curvature of spacetime.

8.4 — The Metric in the Newtonian Limit

The full spacetime metric in the Newtonian limit (with $\Phi/c^2 \ll 1$):

$$\boxed{ds^2 = -\left(1 + \frac{2\Phi}{c^2}\right) c^2\,dt^2 + \left(1 - \frac{2\Phi}{c^2}\right)(dx^2 + dy^2 + dz^2)}$$

This is the **linearized Schwarzschild metric** — valid wherever $|\Phi|/c^2 \ll 1$.

Physical consequences read directly from the metric:

8.4.1 — Gravitational Time Dilation

For a clock at rest ($d\mathbf{x} = 0$), the proper time interval:

$$d\tau = \sqrt{-g_{00}/c^2}\,dt = \sqrt{1 + \frac{2\Phi}{c^2}}\,dt \approx \left(1 + \frac{\Phi}{c^2}\right) dt$$

A clock deeper in a gravitational well (Φ more negative) ticks more slowly relative to coordinate time. The fractional rate difference between clocks at potentials Φ_1 and Φ_2:

$$\frac{\Delta\tau_1 - \Delta\tau_2}{\Delta\tau_2} = \frac{\Phi_1 - \Phi_2}{c^2} = \frac{\Delta\Phi}{c^2}$$

8.4.2 — Gravitational Redshift

A photon emitted at potential Φ_e and received at $\Phi_r > \Phi_e$:

$$\frac{\nu_r}{\nu_e} = \sqrt{\frac{1 + 2\Phi_e/c^2}{1 + 2\Phi_r/c^2}} \approx 1 + \frac{\Phi_e - \Phi_r}{c^2} = 1 - \frac{|\Delta\Phi|}{c^2}$$

The photon climbing out of the gravitational well loses energy and redshifts. **Pound and Rebka (1959)** verified this at Harvard, measuring the redshift of ^{57}Fe gamma rays falling 22.5 m: $\Delta\nu/\nu = -2.5 \times 10^{-15}$, confirmed to 1%.

8.5 — The Schwarzschild Solution: Gravity of a Spherical Mass

For a static, spherically symmetric mass M (the Sun, Earth, a star), the exact solution to the Einstein equations is the **Schwarzschild metric** (Schwarzschild, 1916):

$$\boxed{ds^2 = -\left(1 - \frac{r_s}{r}\right) c^2\,dt^2 + \frac{dr^2}{1 - r_s/r} + r^2\,d\Omega^2}$$

where $d\Omega^2 = d\theta^2 + \sin^2\theta\,d\phi^2$ and the **Schwarzschild radius**:

$$r_s = \frac{2GM}{c^2}$$

Object	Mass M	Schwarzschild radius r_s	Actual radius
Earth	5.97×10^{24} kg	8.87 mm	6371 km
Sun	1.99×10^{30} kg	2.95 km	696,000 km
MW central BH (Sgr A*)	$4 \times 10^6 M_\odot$	1.2×10^7 km	$\approx r_s$
GW150914 final BH	$62 M_\odot$	183 km	$\approx r_s$

For Earth and the Sun: the actual radius is enormously larger than r_s, so we are always in the $r \gg r_s$ regime — the Newtonian limit applies.

8.5.1 — Gravitational Time Dilation from Schwarzschild

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{r_s}{r}} \approx 1 - \frac{GM}{rc^2}$$

A clock at radius r runs slow compared to a clock at infinity by factor $\sqrt{1 - r_s/r}$. At Earth's surface ($r = R_\oplus = 6.371 \times 10^6$ m):

$$\frac{d\tau_{surface}}{dt} = 1 - \frac{GM_\oplus}{R_\oplus c^2} = 1 - 6.95 \times 10^{-10}$$

The surface clock loses $60.1\ \mu$s per day relative to a clock at infinity.

8.5.2 — Circular Orbits and Orbital Velocity

The geodesic equation for a circular orbit at radius r gives:

$$v_{orbit} = \sqrt{\frac{GM}{r}}$$

This is identical to the Newtonian result — circular orbits at $r \gg r_s$ are indistinguishable from Newtonian ones to leading order.

The orbital period: $T = 2\pi r / v_{orbit} = 2\pi\sqrt{r^3/GM}$ (Kepler's third law).

8.6 — Gravitational Waves

8.6.1 — Propagating Solutions

In vacuum ($T_{\mu\nu} = 0$), the linearized equations reduce to:

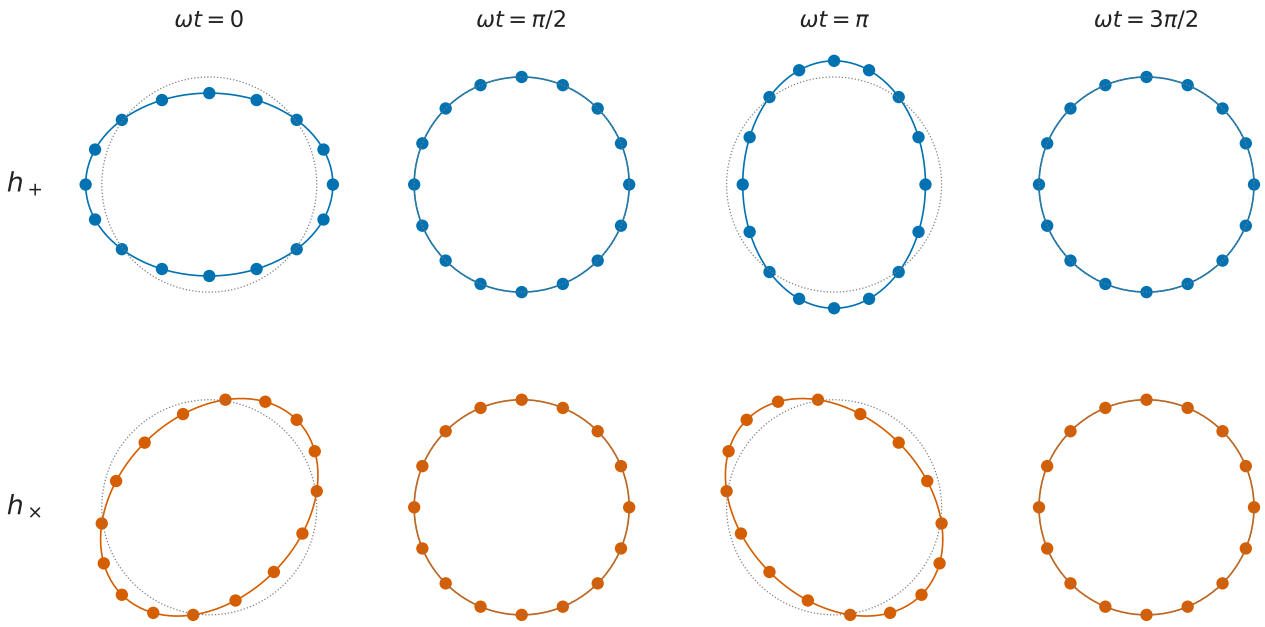
$$\Box \bar{h}_{\mu\nu} = 0$$

This has wave solutions propagating at c. In the **transverse-traceless (TT) gauge** (imposing both $\partial^\mu \bar{h}_{\mu\nu} = 0$ and $\bar{h} = 0$ and choosing the wave propagating in the z-direction), only two independent components survive:

$$h^{TT}_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & h_+ & h_\times & 0 \\ 0 & h_\times & -h_+ & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \cos(\omega t - kz)$$

The plus polarization (h_+): stretches space in x, compresses in y simultaneously, then reverses. Ring of test particles: oscillates between ◯ → ⇒ → ◯ → ⇑ → ◯.

The cross polarization ($h_\times$): same as h_+ but rotated $45°$.



A ring of free test particles in the plane transverse to the wave, at $\omega t = 0, \pi/2, \pi, 3\pi/2$, with an exaggerated strain $h = 0.3$ (real signals are $\sim 10^{-21}$): h_+ stretches x while squeezing y and then the reverse, $h_\times$ does the same along the diagonals, and the ring is momentarily circular between the extremes.

Gravitational waves have exactly 2 independent polarizations — as predicted for a massless spin-2 field. (Compare: photons have 2 polarizations for a massless spin-1 field.)

8.6.2 — The Quadrupole Formula

A gravitational wave is generated by any mass distribution with a time-varying quadrupole moment:

$$I_{ij}(t) = \int \rho(\mathbf{r}, t) \left(x_i x_j - \frac{1}{3}\delta_{ij} r^2 \right) d^3 r$$

The power radiated:

$$P = -\frac{G}{5c^5} \left\langle \dddot{I}_{ij} \dddot{I}^{ij} \right\rangle$$

A spherically symmetric pulsation emits no gravitational waves — $I_{ij} = 0$. A uniformly rotating perfect sphere emits no GWs. You need asymmetry: binary stars, merging black holes, rotating neutron stars with bumps.

8.6.3 — LIGO and the First Detection

GW150914 (14 September 2015): merger of two black holes of masses $36 M_\odot$ and $29 M_\odot$ at $d \approx 1.3$ billion light-years.

The measured strain:

$$h = \frac{\Delta L}{L} \approx 10^{-21}$$

For LIGO's 4 km arm: $\Delta L \approx 4 \times 10^{-18}$ m — one-thousandth the diameter of a proton.

Why LIGO can measure this: The Michelson interferometer measures the differential phase shift between two arms perpendicular to each other. A GW with + polarization stretches one arm and compresses the other — doubling the phase difference. With 300 kW of laser power recycled in the 4 km arms, shot noise and thermal noise can be controlled below the gravitational wave signal.

The engineering: LIGO is the most sensitive instrument ever built. Its suspension systems achieve seismic isolation of 10^{-12} at 100 Hz. Its mirror coatings have thermal noise levels of 10^{-20} m/$\sqrt{\text{Hz}}$. Layer-2 classical physics (interferometry, noise analysis) serves a Layer-0 measurement.

8.7 — Post-Newtonian Corrections: Where GR Matters in the Solar System

The expansion of GR beyond leading-order Newtonian gravity gives **post-Newtonian (PN) corrections** of order $(v/c)^2$ and Φ/c^2. These are small but measurable.

8.7.1 — Perihelion Precession of Mercury

The Newtonian orbit of a planet around the Sun is a closed ellipse (Kepler). In GR, there is an additional term in the effective potential:

$$V_{eff}(r) = -\frac{GM}{r} + \frac{L^2}{2m^2r^2} - \underbrace{\frac{GML^2}{m^2c^2r^3}}_{\text{GR correction}}$$

This causes the ellipse to **precess** — the perihelion advances by:

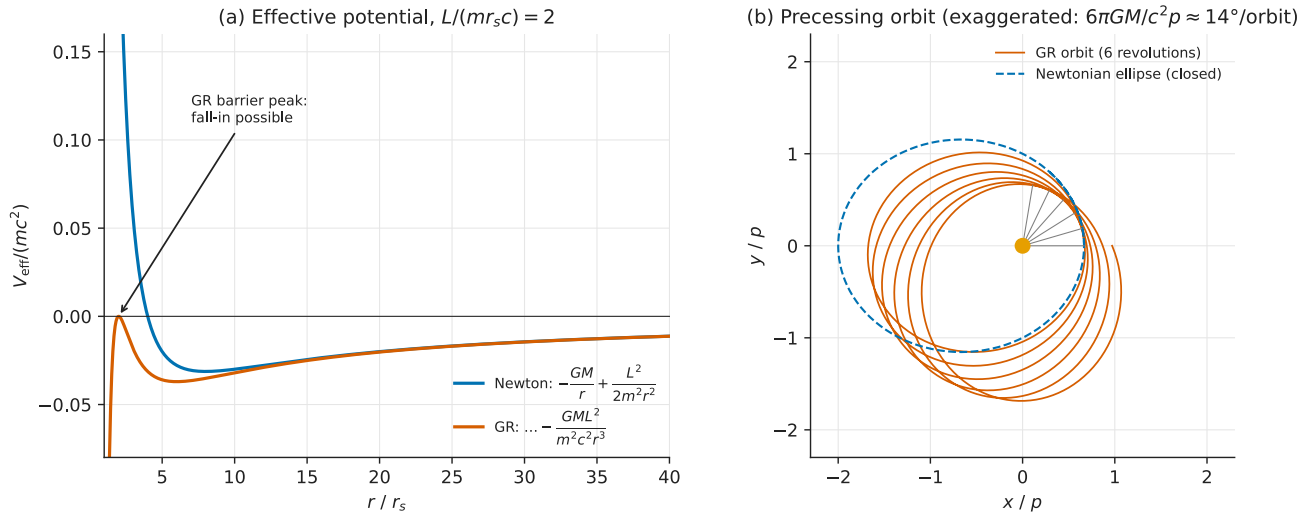
$$\Delta\phi_{\text{per orbit}} = \frac{6\pi GM}{a(1-e^2)c^2}$$

For Mercury: $a = 5.79 \times 10^{10}$ m, $e = 0.206$:

$$\Delta\phi = \frac{6\pi \times (6.67\times10^{-11})(1.99\times10^{30})}{(5.79\times10^{10})(1-0.206^2)(9\times10^{16})} = 5.02 \times 10^{-7}\,\text{rad/orbit}$$

Mercury completes 415 orbits per century: $\Delta\phi_{century} = 2.08 \times 10^{-4}$ rad/century $= 43.0''$/century.

This matches the observed anomalous precession of Mercury (after accounting for perturbations from other planets) to better than 0.1% — one of the first and most famous confirmations of GR.



(a) V_{eff}/m in units of c^2 versus r/r_s for $L/(mr_sc) = 2$: far out the curves agree, but the $-GML^2/m^2c^2r^3$ term turns the Newtonian centrifugal wall into a finite barrier near $r = 2r_s$. (b) Orbit from $u'' + u = GMm^2/L^2 + 3GMu^2/c^2$ with the GR term exaggerated to about 14° of perihelion advance per orbit (Mercury: 5×10^{-7} rad), against the closed Newtonian ellipse.

8.7.2 — Gravitational Deflection of Light

A photon passing a massive body at impact parameter b is deflected by:

$$\alpha = \frac{4GM}{bc^2}$$

This is **twice the Newtonian prediction** ($2GM/bc^2$) because in GR both the temporal and spatial components of the metric are perturbed — light is deflected by time curvature AND space curvature equally, whereas Newton's law only captures the time-curvature analog.

For light grazing the Sun ($b = R_\odot$):

$$\alpha = \frac{4 \times (6.67 \times 10^{-11}) \times (1.99 \times 10^{30})}{(6.96 \times 10^{8}) \times (9 \times 10^{16})} = 8.48 \times 10^{-6}\,\text{rad} = 1.75''$$

Eddington's 1919 expedition to measure stellar positions during a solar eclipse confirmed $\alpha = 1.75'' \pm 0.09''$ — the first empirical test of GR and the observation that made Einstein world-famous.

Modern measurements (Very Long Baseline Interferometry, VLBI) confirm the prediction to 10^{-4} precision.

8.7.3 — Frame Dragging: The Lense-Thirring Effect

A rotating mass (M, angular momentum J) drags spacetime around it, adding an off-diagonal term to the metric:

$$g_{0i} = -\frac{2G}{c^3 r^3}\epsilon_{ijk}J^j x^k$$

A gyroscope in orbit around Earth precesses at the Lense-Thirring rate:

$$\boldsymbol{\Omega}_{LT} = \frac{G}{c^2 r^3}\left[3(\mathbf{J}\cdot\hat{r})\hat{r} - \mathbf{J}\right]$$

For a circular polar orbit at $r = R_\oplus + h$: $\Omega_{LT} \approx 39$ mas/yr (milli-arcseconds per year).

Gravity Probe B (Stanford / NASA, 2004–2011): Four cryogenic gyroscopes in polar orbit at 642 km altitude. Measured:

- Geodetic precession (mass-only, from g_{00} and g_{ij} terms): 6606 mas/yr (predicted 6606.1 mas/yr) ✓
- Lense-Thirring (from g_{0i} terms): 37.2 mas/yr (predicted 39.2 mas/yr) ✓ (to 15%)

8.8 — GPS: General Relativity in an Engineering System

GPS is the most commercially important application of general relativity. The calculation below is a worked example of the theory in this chapter applied to a real engineering system.

8.8.1 — GPS Orbital Parameters

- Orbital altitude: $h = 20{,}200$ km above Earth's surface
- Orbital radius: $r_{sat} = R_\oplus + h = 2.657 \times 10^7$ m
- Orbital velocity: $v_{sat} = \sqrt{GM_\oplus / r_{sat}} = 3.87$ km/s

8.8.2 — Gravitational Time Dilation (GR Effect)

Clock rate relative to infinity (from §8.5.1):

$$\left.\frac{d\tau}{dt}\right|_{surface} = 1 - \frac{GM_\oplus}{R_\oplus c^2} = 1 - 6.95 \times 10^{-10}$$

$$\left.\frac{d\tau}{dt}\right|_{satellite} = 1 - \frac{GM_\oplus}{r_{sat} c^2} = 1 - 1.66 \times 10^{-10}$$

Satellite clock runs faster than surface clock due to weaker gravity by:

$$\Delta\left.\frac{d\tau}{dt}\right|_{GR} = (6.95 - 1.66) \times 10^{-10} = 5.29 \times 10^{-10}$$

Per day: $5.29 \times 10^{-10} \times 86{,}400\,\text{s} = +45.7\,\mu\text{s/day}$ (satellite runs fast)

8.8.3 — Velocity Time Dilation (Special Relativity Effect)

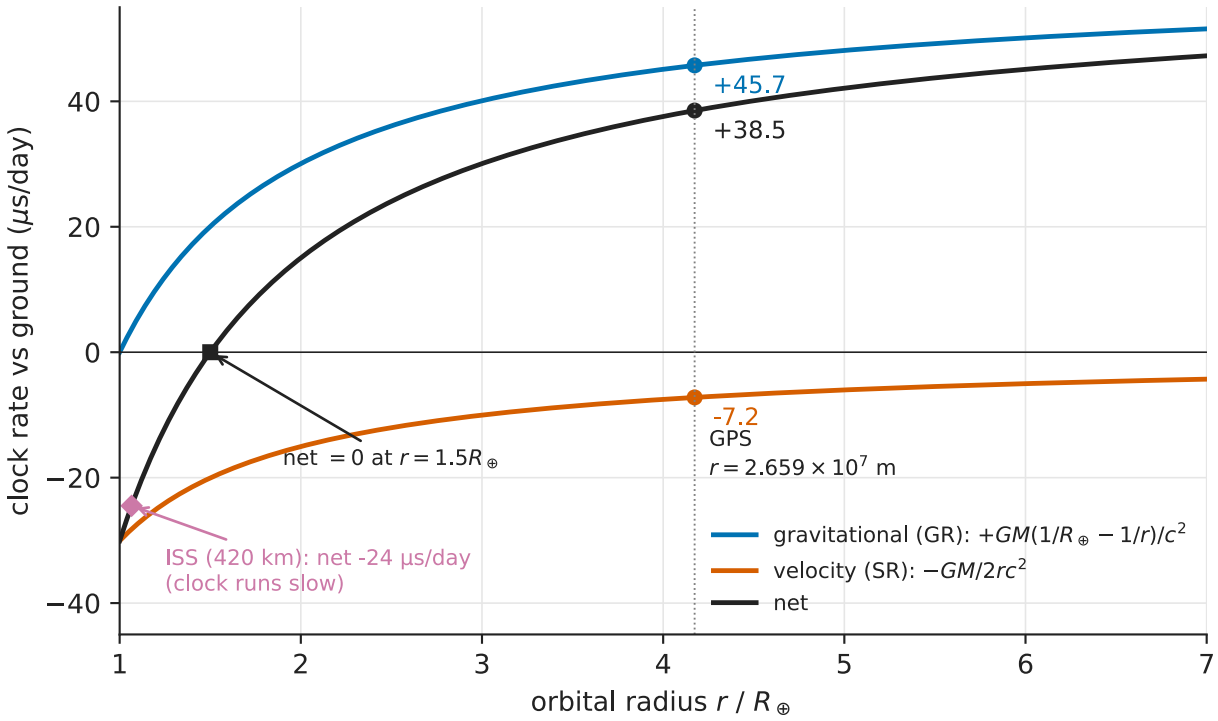
From SR (time dilation due to velocity):

$$\left.\frac{d\tau}{dt}\right|_{SR} = \sqrt{1 - \frac{v_{sat}^2}{c^2}} \approx 1 - \frac{v_{sat}^2}{2c^2} = 1 - 8.32 \times 10^{-11}$$

Per day: $-8.32 \times 10^{-11} \times 86{,}400\,\text{s} = -7.2\,\mu\text{s/day}$ (satellite runs slow)

8.8.4 — Net Correction

$$\Delta t_{net} = +45.7 - 7.2 = +38.5\,\mu\text{s/day}$$



Rate of a circular-orbit clock relative to a ground clock, in μs/day, versus $r/R_\oplus$: the gravitational term $GM(1/R_\oplus - 1/r)/c^2$ (blue) and the velocity term $-GM/2rc^2$ (orange) sum to a net that crosses zero at $r = 1.5\ R_\oplus$; GPS at $r = 2.657 \times 10^7$ m gets $+45.7 - 7.2 = +38.5$ μs/day, while the ISS clock runs slow (Earth's rotation neglected).

The satellite clock runs fast by 38.5 μs/day. Since GPS positioning requires timing accuracy $\delta t < 20$ ns for $\delta x < 6$ m accuracy, and the uncorrected error would be 38,500 ns/day:

$$\delta x_{uncorrected} \approx c \times 38.5\,\mu\text{s} \approx 11.5\,\text{km/day}$$

Without the GR and SR corrections, the dominant error would be a range error of roughly 11.5 km per day — this is a light-travel-time-equivalent displacement $\delta x \approx c\,\delta t$, i.e. the range error a receiver would attribute to a satellite if it trusted the uncorrected onboard clock. It is *not* a statement that a receiver's reported position drifts by exactly 11.5 km per day: in a four-satellite solution the clock bias is estimated as a fourth unknown, so an uncorrected satellite clock bias is largely absorbed into the receiver's own clock estimate, and the positional error that actually appears is much smaller and geometry-dependent. The number is retained because it is the standard order-of-magnitude figure for *why* relativistic clock effects must be corrected at all. The correction is implemented by pre-adjusting the satellite clock frequency before launch:

$$f_{corrected} = f_0 \times \left(1 - \frac{38.5 \times 10^{-6}}{86{,}400}\right) = f_0 \times \left(1 - 4.46 \times 10^{-10}\right)$$

The satellite clock is made to tick slightly slow before launch (f reduced by 4.46×10^{-10}), so that after it reaches orbit and both relativistic effects apply, it ticks at the correct rate relative to ground clocks.

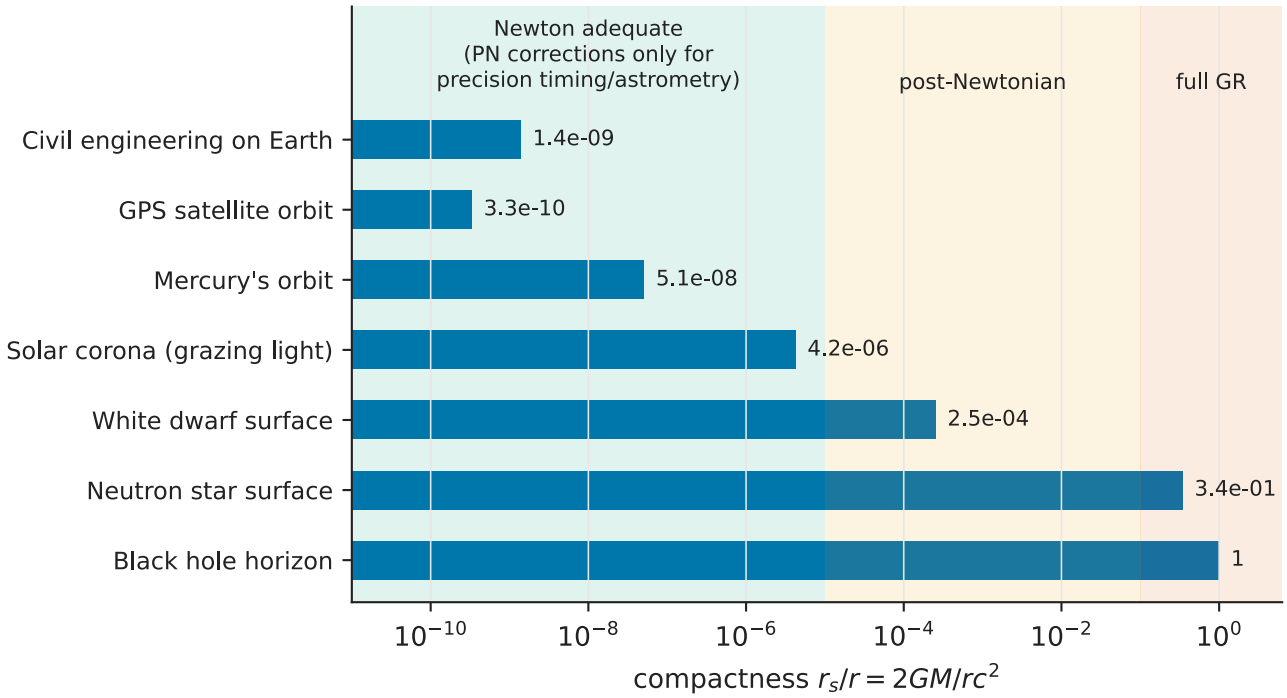
This is GR in commercial engineering practice, used by every smartphone, aircraft, and ship navigation system on Earth.

8.9 — The Validity Criterion: When to Use GR vs. Newton

The weak-field expansion is valid when $r_s/r \ll 1$ and $v \ll c$. More precisely:

$$\text{GR correction} \sim \frac{v^2}{c^2} \sim \frac{r_s}{r} = \frac{2GM}{rc^2}$$

Context	r_s/r	GR correction needed?
Civil engineering on Earth	$\sim 10^{-9}$	Never (except GPS receivers)
GPS satellite orbit	$\sim 10^{-9}$	Yes — 38 μs/day matters
Mercury's orbit	$\sim 10^{-8}$	Yes — 43"/century
Solar corona	$\sim 10^{-6}$	Yes — light deflection
White dwarf surface	$\sim 10^{-4}$	Yes — spectral line shifts
Neutron star surface	~ 0.3	Full GR
Black hole horizon	$= 1$	Full GR (beyond perturbation theory)



Compactness $r_s/r = 2GM/rc^2$ computed for each row of the table (Earth's surface, GPS orbit, Mercury, the solar limb, a $0.6\,M_\odot$ white dwarf, a $1.4\,M_\odot$ neutron star, a horizon) on a log scale: the contexts span ten decades, and only the last two lie where the weak-field expansion fails; the band edges are indicative, not sharp.

The rule of thumb: **Newton's law is sufficient when** $GM/rc^2 \ll 1$ **and** $v \ll c$. For engineering on Earth or in near-Earth orbit, the only GR correction that matters in practice is the gravitational time dilation for precision timekeeping.

8.10 — Summary: The Descent from GR to Newton

Three conditions applied to the Einstein field equations:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

Condition	Mathematical operation	What it removes
Weak field	$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, linearize	Nonlinear metric self-interaction
Slow motion	$v \ll c \to T_{00} \approx \rho c^2$ dominates	Momentum and stress as sources
Static	$\partial_t h_{\mu\nu} = 0$	Retardation, gravitational waves

Output: $\nabla^2 \Phi = 4\pi G\rho \to F = -GMm/r^2 \hat{r}$

What was discarded and when it matters:

Discarded	GR effect	Magnitude	Engineering relevance
Wave term $\Box$	Gravitational waves	$h \sim 10^{-21}$	LIGO, LISA
T_{ij} terms	Gravitational waves from pressure	Small	Neutron stars
h_{ij} spatial metric	Gravitational lensing factor of 2	$1.75''$	VLBI astrometry
Time-dep. metric	Frame dragging	39 mas/yr	Gravity Probe B
Higher-order h^2	Perihelion precession	43"/century	Mercury, binary pulsars
Velocity of source	Gravitomagnetic effects	$\sim v/c$ times GR	Active research

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8.11 — Looking Ahead: Bridge B Continues

With Newton's law in hand, Chapter 9 takes the next Bridge B limit: $\hbar \to 0$, deriving classical mechanics from quantum mechanics. Newton's law (this chapter) and Lagrangian mechanics (Ch. 9) combine to give the full toolkit of classical particle and rigid-body dynamics — the spine of mechanical engineering.

Gravity enters engineering practice overwhelmingly through Newtonian approximation. The systematic position is clear: Newton is GR's leading-order term, fully justified in any context where $GM/rc^2 \ll 1$ and $v \ll c$.

End of Chapter 8.

Next: Chapter 9 — Bridge B.a: From Quantum Mechanics to Classical Mechanics ($\hbar \to 0$)

Bridge B.a — From Quantum Mechanics to Classical Mechanics

The $\hbar \to 0$ Descent: How Newton Emerges from Schrödinger

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> *Classical mechanics is not wrong. It is quantum mechanics seen from far away, in the limit where the action is much larger than ħ. Every classical trajectory is the stationary phase of a quantum path integral.* — Richard Feynman, *QED: The Strange Theory of Light and Matter*

9.0 — Bridge B.a in Context

Chapter 8 performed Bridge B.d: the weak-field, slow-motion limit of general relativity giving Newton's gravitational law. This chapter performs Bridge B.a: the $\hbar \to 0$ limit of quantum mechanics giving classical mechanics.

These are **separate, independent limits** — both are part of Bridge B, running simultaneously. The complete Layer-2 picture requires all five Bridge B operations (B.a through B.e) to run in parallel. Chapter 9 handles B.a alone; the others follow in Chapters 10, 11, and 13.

The descent in this chapter:

```
Layer 1: Quantum mechanics
         ψ satisfies iħ ∂ψ/∂t = Ĥψ
                │
                │  ħ → 0
                │  (action S ≫ ħ)
                │
                ▼
Bridge zone: WKB approximation (§9.3)
         ψ ≈ A(r,t) exp(iS(r,t)/ħ)
         leading order → Hamilton-Jacobi equation
                │
                │  wavepacket width Δx → 0
                │  (V varies slowly over Δx)
                │
                ▼
Layer 2: Classical mechanics
         Euler-Lagrange, Hamilton's equations, Newton's 2nd law
```

A critical point: the classical mechanics recovered here is not a separate subject patched onto quantum mechanics. It is **the same physical law**, expressed in the language appropriate to the regime $S \gg \hbar$.

9.1 — Ehrenfest's Theorem: The First Sign of Newton

9.1.1 — The General Theorem

For any quantum observable $\hat{A}$ (not explicitly time-dependent):

$$\frac{d\langle \hat{A} \rangle}{dt} = \frac{d}{dt} \int \psi^* \hat{A} \psi \, d^3r$$

Using the Schrödinger equation $i\hbar \, \partial_t \psi = \hat{H}\psi$ and its conjugate:

$$\boxed{\frac{d\langle \hat{A} \rangle}{dt} = \frac{i}{\hbar} \langle [\hat{H}, \hat{A}] \rangle + \left\langle \frac{\partial \hat{A}}{\partial t} \right\rangle}$$

This is **Ehrenfest's theorem** — the quantum analog of Hamilton's equation of motion $\dot{f} = \{f, H\}_{Poisson} + \partial f/\partial t$. Replacing the Poisson bracket with $(i/\hbar)\times$commutator is canonical quantization (Ch. 1 §1.6.4) operating in reverse: here we see the quantum equation reducing to the classical one.

9.1.2 — Recovering Newton's Second Law

Apply Ehrenfest's theorem to position and momentum, with $\hat{H} = \hat{p}^2/2m + V(\mathbf{r})$:

For position $\hat{x}$:

$$[\hat{H}, \hat{x}] = [\hat{p}^2/2m, \hat{x}] = \frac{1}{2m}[\hat{p}^2, \hat{x}] = \frac{1}{2m}(\hat{p}[\hat{p}, \hat{x}] + [\hat{p}, \hat{x}]\hat{p}) = \frac{-i\hbar\hat{p}}{m}$$

Therefore:

$$\frac{d\langle \hat{x} \rangle}{dt} = \frac{i}{\hbar} \left\langle \frac{-i\hbar\hat{p}}{m} \right\rangle = \frac{\langle \hat{p} \rangle}{m}$$

This is exactly Newton's kinematic relation $v = p/m$, now holding for expectation values. **A quantum wavepacket moves like a classical particle, at least in this sense.**

For momentum $\hat{p}$:

$$[\hat{H}, \hat{p}] = [V(\mathbf{r}), \hat{p}] = V(\mathbf{r})(-i\hbar\nabla) - (-i\hbar\nabla)V(\mathbf{r}) = i\hbar\nabla V$$

Therefore:

$$\boxed{\frac{d\langle \hat{p} \rangle}{dt} = -\langle \nabla V(\hat{\mathbf{r}}) \rangle}$$

This is almost Newton's second law $\dot{p} = -\nabla V = F$, but not quite: the right side is $-\langle \nabla V \rangle$, not $-\nabla V(\langle \mathbf{r} \rangle)$.

9.1.3 — The Classical Limit Condition

When do these become identical?

Taylor expand $\nabla V(\mathbf{r})$ around $\langle \mathbf{r} \rangle$:

$$\langle \nabla V \rangle = \nabla V\big|_{\langle \mathbf{r} \rangle} + \frac{1}{2}\langle (\Delta \mathbf{r})^2 \rangle \nabla^3 V\big|_{\langle \mathbf{r} \rangle} + \cdots$$

The correction term involves $(\Delta x)^2 V''' \sim (\Delta x)^2 / L_V^3$ where L_V is the scale over which V varies. This is negligible when:

$$\boxed{\Delta x \ll L_V}$$

The wavepacket must be narrow compared to the scale over which the potential varies. When this holds:

$$\frac{d\langle \mathbf{p} \rangle}{dt} \approx -\nabla V(\langle \mathbf{r} \rangle) = \mathbf{F}(\langle \mathbf{r} \rangle)$$

This is Newton's second law — the expectation values of position and momentum of a narrow quantum wavepacket obey classical equations of motion.

What controls the wavepacket width? The uncertainty principle (Ch. 3 §3.11) requires $\Delta x \, \Delta p \geq \hbar/2$. As $\hbar \to 0$, the minimum wavepacket width goes to zero — a perfectly sharp classical trajectory. For macroscopic objects, $\hbar / S_{action} \sim 10^{-34}$ J·s / (1 kg × 1 m/s × 1 m) $= 10^{-34}$ — the wavepacket is effectively a point.

9.2 — The Path Integral and Stationary Phase

9.2.1 — The Feynman Path Integral

From quantum mechanics (the rigorous formulation), the amplitude for a particle to travel from $(\mathbf{r}_a, t_a)$ to $(\mathbf{r}_b, t_b)$ is:

$$K(\mathbf{r}_b, t_b;\, \mathbf{r}_a, t_a) = \int_{\text{all paths}} \mathcal{D}\mathbf{r}(t)\, \exp\left(\frac{i}{\hbar} S[\mathbf{r}(t)] \right)$$

where $S[\mathbf{r}(t)] = \int_{t_a}^{t_b} L(\mathbf{r}, \dot{\mathbf{r}}, t)\, dt$ is the classical action (Ch. 1 §1.4) evaluated along each possible path, and the integral runs over **all continuous paths** connecting the endpoints — not just the classical one.

Every path contributes equally in magnitude but with a phase $e^{iS/\hbar}$. Paths of very different action have rapidly oscillating phases that cancel.

9.2.2 — Stationary Phase: $\hbar \to 0$

As $\hbar \to 0$, the phase $S/\hbar$ oscillates infinitely rapidly for any variation of the path. Neighboring paths cancel by destructive interference — **except near the path where S is stationary**, where neighboring paths have nearly the same phase and interfere constructively.

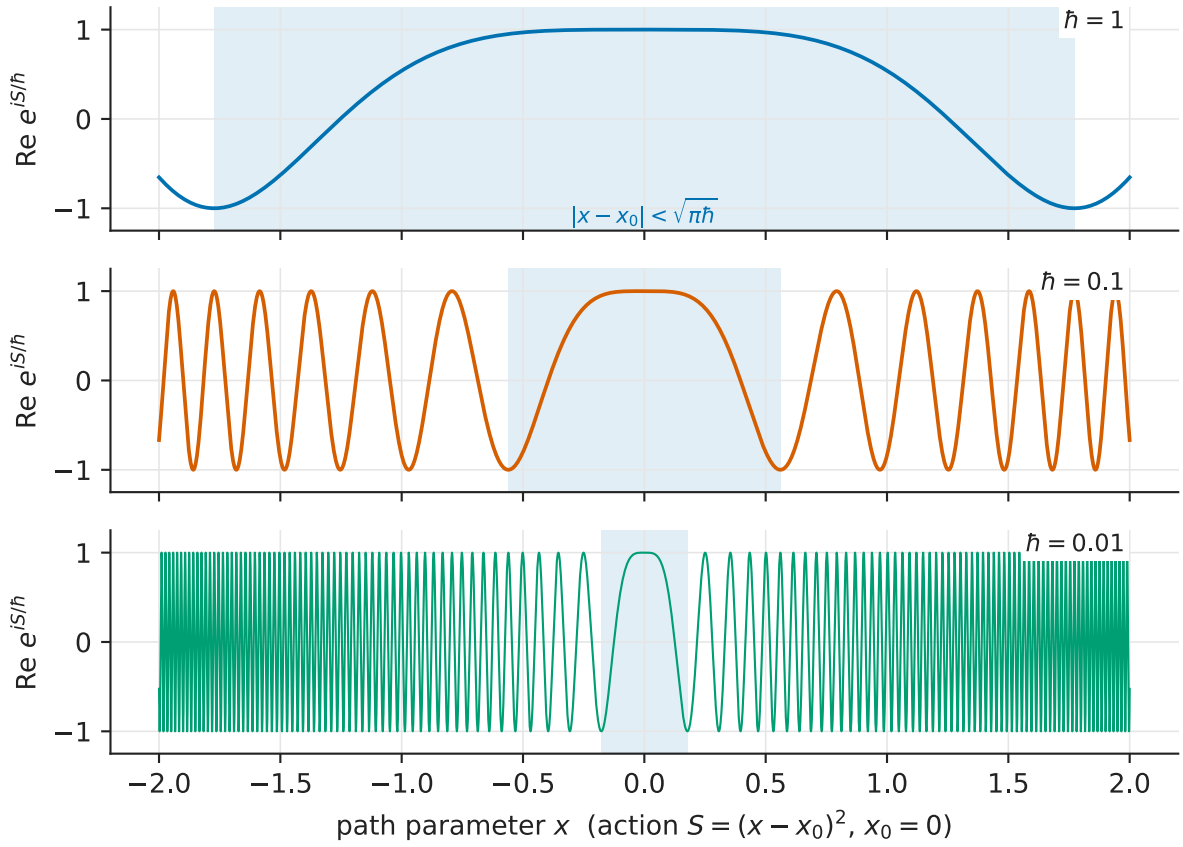
The condition $\delta S = 0$ — stationary action — is exactly the **Euler-Lagrange equation** (Ch. 1 §1.5). The dominant contribution to K comes from the classical trajectory $\mathbf{r}_{cl}(t)$:

$$K \xrightarrow{\hbar \to 0} A(\mathbf{r}_b, \mathbf{r}_a, t)\, \exp\left(\frac{i}{\hbar} S_{cl}[\mathbf{r}_{cl}(t)] \right)$$

where A is a slowly-varying prefactor from the Gaussian integral over fluctuations around the classical path.

The physical statement: In the classical limit, a particle takes the single path of stationary action. In the quantum world, it takes all paths. Classical mechanics is the stationary-phase approximation of quantum mechanics.

This is the answer to the question raised in Ch. 1 §1.4.2: *Why should $\delta S = 0$?* Because nature is quantum. The classical trajectory is the path that survives when all other paths cancel by interference as $\hbar / S \to 0$.



$\mathrm{Re}\, e^{iS/\hbar}$ *for the model action* $S = (x - x_0)^2$ *with* $x_0 = 0$ *at* $\hbar = 1, 0.1, 0.01$*: away from the stationary point the phase oscillates ever faster and neighbouring contributions cancel, so only a window of width* $\sim \sqrt{\pi\hbar}$ *(shaded) around* $\delta S = 0$ *survives as* $\hbar \to 0$.

9.3 — The WKB Approximation: The Bridge Zone

9.3.1 — The WKB Ansatz

Write the wavefunction as:

$$\psi(\mathbf{r}, t) = A(\mathbf{r}, t)\, \exp\left(\frac{i}{\hbar} S(\mathbf{r}, t)\right)$$

where $A(\mathbf{r}, t)$ is a real amplitude and $S(\mathbf{r}, t)$ is a real phase. Substituting into the Schrödinger equation and separating real and imaginary parts, collecting terms by power of $\hbar$:

Order $\hbar^0$ (leading order):

$$\frac{\partial S}{\partial t} + \frac{|\nabla S|^2}{2m} + V = 0$$

This is the **Hamilton-Jacobi equation** (Ch. 1 §1.8.1). The quantum phase S satisfies the classical equation for Hamilton's principal function. The HJ equation is exactly what the Schrödinger equation reduces to in the limit $\hbar \to 0$ — confirming that classical mechanics is the ħ→0 limit.

Order $\hbar^1$ (next order):

$$\frac{\partial A^2}{\partial t} + \nabla \cdot \left(A^2 \frac{\nabla S}{m}\right) = 0$$

This is the **continuity equation** for the probability density $\rho = A^2 = |\psi|^2$, with current density $\mathbf{J} = A^2 \nabla S/m = |\psi|^2 \mathbf{v}_{cl}$. Probability is conserved — as it must be by Noether's theorem (Ch. 1 §1.7).

Order $\hbar^2$ (purely quantum correction):

$$\text{Quantum potential: } Q = -\frac{\hbar^2}{2m}\frac{\nabla^2 A}{A}$$

This term — sometimes called the **Bohm potential** — vanishes as $\hbar \to 0$. It is responsible for tunneling (the wavefunction penetrates barriers where $E < V$, giving imaginary p and an evanescent wave), interference, and zero-point energy. Setting $Q = 0$ is the WKB approximation.

9.3.2 — The WKB Wavefunction

In one dimension, the HJ equation gives $p(x) = \partial S/\partial x = \sqrt{2m(E - V(x))}$. The WKB wavefunction in a classically allowed region ($E > V$):

$$\psi_{WKB}(x) = \frac{C}{\sqrt{p(x)}} \exp\left(\pm \frac{i}{\hbar}\int^x p(x')\,dx'\right)$$

In a classically forbidden region ($E < V$), $p \to i\kappa$ where $\kappa = \sqrt{2m(V-E)}/\hbar$:

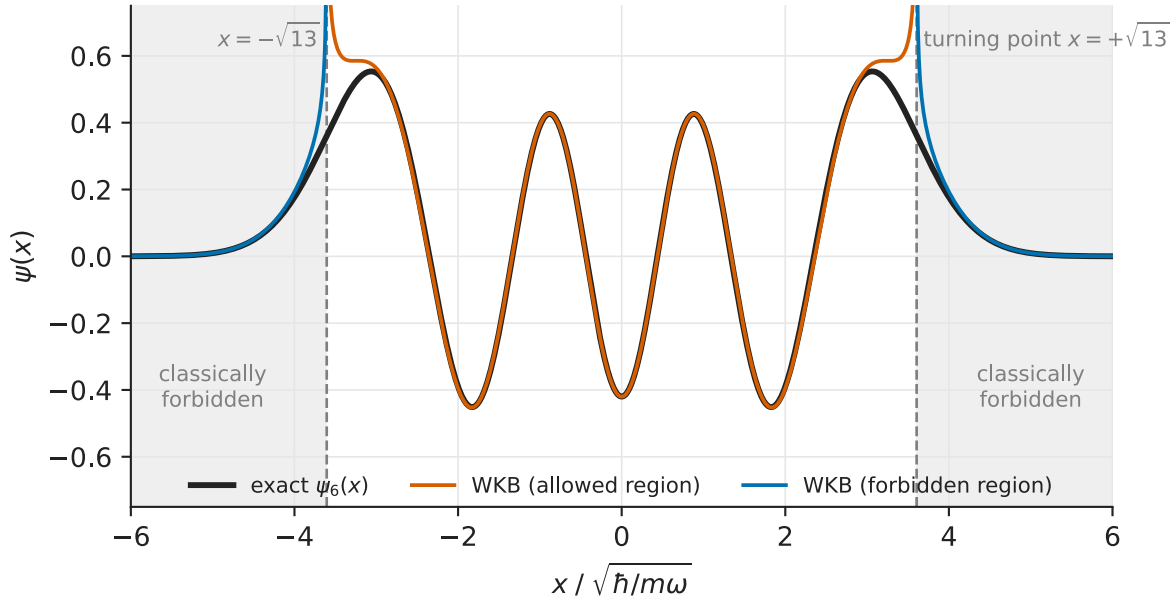
$$\psi_{WKB}(x) = \frac{C}{\sqrt{\kappa(x)}} \exp\left(\mp \int^x \kappa(x')\,dx'\right)$$

9.3.3 — Validity Criterion

WKB is valid when the de Broglie wavelength $\lambda = h/p$ changes slowly over one wavelength — i.e., when the potential is nearly constant locally:

$$\left|\frac{d\bar{\lambda}}{dx}\right| \ll 1, \quad \bar{\lambda} = \frac{\lambda}{2\pi} = \frac{\hbar}{p} \qquad \Longleftrightarrow \qquad \frac{\hbar}{p^2}\left|\frac{dp}{dx}\right| \ll 1$$

This fails at **classical turning points** ($p = 0$) where $\lambda \to \infty$. Connection formulas (derived by matching to exact Airy function solutions near the turning point) bridge the oscillatory and evanescent WKB solutions.



Exact $\psi_6(x)$ of the harmonic oscillator ($\hbar = m = \omega = 1$, $E = 6.5$) against the WKB form $C\,p^{-1/2}\cos(\int_{-a}^{x} p\,dx'/\hbar - \pi/4)$ inside and the decaying WKB exponential outside: the two agree to within line width away from the turning points $x = \pm\sqrt{13}$ (dashed), and the WKB curves diverge exactly where $p \to 0$ and the validity criterion fails.

9.3.4 — WKB Applications

Bohr-Sommerfeld quantization (the semiclassical bridge): For a particle in a potential well, matching the WKB solutions at both turning points:

$$\oint p\,dq = \left(n + \frac{1}{2}\right) h, \qquad n = 0, 1, 2, \ldots$$

The $\frac{1}{2}$ is the **Maslov correction** from the phase shift at turning points — it gives the zero-point energy automatically. For the harmonic oscillator: $\oint p\,dq = 2\pi m\omega E/\omega^2 = 2\pi E/\omega = (n+\frac{1}{2})h$, recovering $E_n = \hbar\omega(n+\frac{1}{2})$ exactly.

Geometric optics as WKB: In the limit of short wavelength, electromagnetic waves (governed by Maxwell's equations in Layer 2) satisfy a WKB-like equation. The eikonal equation $|\nabla S|^2 = n^2(\mathbf{r})$ is the HJ equation for photons with $p = n(\mathbf{r})\hbar\omega/c$. Fermat's principle of least time is the stationary phase of the wave equation. **Optics is the $\lambda \to 0$ limit of electromagnetism, just as classical mechanics is the $\hbar \to 0$ limit of quantum mechanics — the mathematical structure is identical.**

9.4 — Coherent States: Quantum States That Look Classical

9.4.1 — The Most Classical Quantum State

A **coherent state** $|\alpha\rangle$ of the harmonic oscillator is the eigenstate of the lowering operator $\hat{a}_-|\alpha\rangle = \alpha|\alpha\rangle$ for complex $\alpha = |\alpha|e^{i\phi}$.

Properties:

- $\langle\hat{x}\rangle(t) = x_0\cos(\omega t + \phi)$ — classical oscillatory motion
- $\langle\hat{p}\rangle(t) = -m\omega x_0\sin(\omega t + \phi)$ — classical momentum
- $\Delta x = \sqrt{\hbar/2m\omega}$ and $\Delta p = \sqrt{m\omega\hbar/2}$ — constant, minimum uncertainty
- $\Delta x\,\Delta p = \hbar/2$ — saturates the uncertainty principle at all times

A coherent state is a **minimum-uncertainty Gaussian wavepacket** that follows the classical trajectory without spreading. It is the quantum state that most closely resembles a classical particle in a harmonic potential.

9.4.2 — Decoherence: Why Macroscopic Objects Are Classical

A quantum superposition of macroscopically distinct states (the "Schrödinger's cat" state) rapidly loses its quantum coherence due to entanglement with environmental degrees of freedom.

For a system interacting with an environment at temperature T:

$$\tau_{decoherence} \sim \frac{\hbar}{k_B T}\left(\frac{\lambda_{dB}}{\Delta x}\right)^2$$

where Δx is the spatial separation between the superposed states and $\lambda_{dB} = h/\sqrt{2mkT}$ is the thermal de Broglie wavelength.

For a 1 g dust particle with $\Delta x = 1$ μm in air at 300 K:

$$\tau_{decoherence} \sim 10^{-31}\ \text{s}$$

The coherence is destroyed in a time far shorter than any observable timescale. This is why macroscopic objects always appear classical — their quantum superpositions decohere before they can be measured. **Classical mechanics is not just $\hbar \to 0$; it is also the open quantum system limit**, where the environment continuously measures and collapses the quantum state.

This decoherence picture will be developed fully in Ch. 10 (Bridge B.b), where tracing out environmental degrees of freedom is the operation that produces statistical mechanics from quantum mechanics.

9.5 — Lagrangian Mechanics as a Classical Tool

The Lagrangian was derived in Ch. 1 and used to demonstrate Newton's law in Ch. 1 §1.5.2. In this chapter it becomes a working tool for engineering dynamics problems.

9.5.1 — The Method

For any mechanical system:

1. Choose n generalized coordinates $q_1, \ldots, q_n$ (fewest independent parameters that specify the configuration)
2. Write T and V in these coordinates
3. Form $L = T - V$
4. Apply Euler-Lagrange: $\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = 0$

Constraint forces never appear. The approach is coordinate-free: it produces the same equations of motion regardless of which valid q_i you choose.

9.5.2 — The Charged Particle in an EM Field

The Lagrangian for a particle of charge q and mass m in fields $(\phi, \mathbf{A})$ (from Ch. 3 §3.9.4 — minimal coupling in the Schrödinger equation, now in the classical limit):

$$L = \frac{1}{2}m|\dot{\mathbf{r}}|^2 - q\phi(\mathbf{r}, t) + q\dot{\mathbf{r}} \cdot \mathbf{A}(\mathbf{r}, t)$$

Applying Euler-Lagrange to coordinate x:

$$\frac{d}{dt}(m\dot{x} + qA_x) = q\dot{\mathbf{r}} \cdot \frac{\partial \mathbf{A}}{\partial x} - q\frac{\partial \phi}{\partial x}$$

Expanding using $\mathbf{E} = -\nabla\phi - \partial_t \mathbf{A}$ and $\mathbf{B} = \nabla \times \mathbf{A}$:

$$m\ddot{x} = q(E_x + \dot{y}B_z - \dot{z}B_y) = q(\mathbf{E} + \dot{\mathbf{r}} \times \mathbf{B})_x$$

This is the **Lorentz force law** — recovered from the Lagrangian with no additional input. The same Lagrangian term $q\dot{\mathbf{r}} \cdot \mathbf{A}$ that appeared as minimal coupling in quantum mechanics (Ch. 3) gives the Lorentz force in the classical limit. The physics is continuous across Bridge B.a.

9.5.3 — The Double Pendulum

Two masses m_1, m_2 on strings of length ℓ_1, ℓ_2, angles θ_1, θ_2:

$$T = \frac{1}{2}(m_1 + m_2)\ell_1^2\dot{\theta}_1^2 + \frac{1}{2}m_2\ell_2^2\dot{\theta}_2^2 + m_2\ell_1\ell_2\dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2)$$

$$V = -(m_1 + m_2)g\ell_1\cos\theta_1 - m_2 g\ell_2\cos\theta_2$$

Applying Euler-Lagrange gives two coupled nonlinear ODEs — the equations of motion for the double pendulum. These cannot be derived from Newton's law without first computing the tension in both strings (constraint forces). The Lagrangian approach is not just elegant; it is simpler.

For small oscillations (§9.8 below), the double pendulum has two normal modes. For large oscillations, it is **chaotic** — sensitive to initial conditions, with adjacent trajectories diverging exponentially. This is the first appearance of chaos in the book. We will return to it briefly in §9.10.

9.6 — Hamiltonian Mechanics and Phase Space

9.6.1 — Phase Space

The state of a mechanical system with n degrees of freedom is a point in **phase space** $\Gamma = \{(q_1, \ldots, q_n, p_1, \ldots, p_n)\}$ — a $2n$-dimensional space. Hamilton's equations (Ch. 1 §1.6.3):

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i}$$

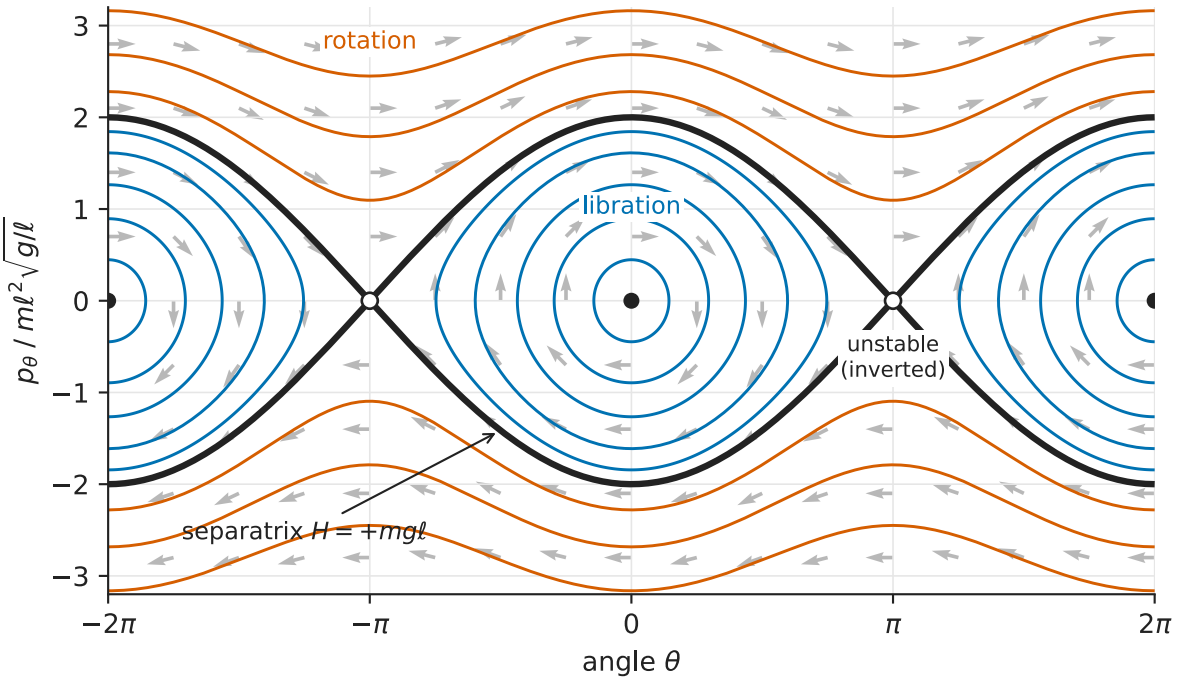
define a **flow** on phase space. Every initial condition maps to a unique trajectory. Trajectories never cross (uniqueness of solutions to ODEs with smooth H).

Phase space portraits for the harmonic oscillator ($H = p^2/2m + \frac{1}{2}m\omega^2 q^2$):

- Trajectories are ellipses $E = \text{const}$ in the (q, p) plane
- All motion is periodic; no chaos possible for one degree of freedom
- Trajectory frequency: ω (independent of amplitude — special to harmonic oscillator)

For the pendulum ($H = p^2/2m\ell^2 - mg\ell\cos\theta$):

- Small energy: ellipses (harmonic approximation)
- Energy $= 2mg\ell$ above the lowest point ($H = +mg\ell$ with the zero of V at the pivot): **separatrix** — the figure-eight curve separating libration (bounded oscillation) from rotation (complete loops)
- Large energy: rotation trajectories (pendulum spins over the top)
- The separatrix itself is a **homoclinic orbit** — asymptotically approaches the unstable equilibrium as $t \to \pm\infty$; infinite period



Contours of $H = p^2/2m\ell^2 - mg\ell\cos\theta$ (zero of energy at the pivot height; units $m = \ell = g = 1$): closed libration orbits (blue) around the stable points $\theta = 0, \pm 2\pi$, open rotation orbits (orange), and the separatrix $H = +mg\ell$ (black, $2mg\ell$ above the bottom of the well) through the unstable inverted points $\theta = \pm\pi$; grey arrows show the Hamiltonian flow $(\dot\theta, \dot p) = (p/m\ell^2, -mg\ell\sin\theta)$.

9.6.2 — Liouville's Theorem

The phase space flow is **incompressible**: the volume occupied by any ensemble of initial conditions is conserved under Hamilton's equations.

$$\frac{d}{dt}\int_\Omega d^{2n}(q,p) = 0 \qquad \Longleftrightarrow \qquad \frac{\partial \rho}{\partial t} + H, \rho = 0$$

where $\rho(q,p,t)$ is the phase space density and $\cdot,\cdot$ is the Poisson bracket. This is the **classical Liouville equation**.

Why this matters for Ch. 10: The quantum analog of the Liouville equation is the **von Neumann equation** for the density matrix:

$$\frac{\partial \hat\rho}{\partial t} = -\frac{i}{\hbar}[\hat H, \hat\rho]$$

This is the quantum Liouville equation — the Poisson bracket replaced by the commutator (same canonical quantization rule as always). Bridge B.b (Ch. 10) takes the quantum Liouville equation, applies $N \to \infty$ and traces over unobserved degrees of freedom, to get thermodynamics.

Engineering application: Liouville's theorem governs the performance of optical and particle beam systems. A beam of particles (or photons) occupies a volume of phase space (x, p_x, y, p_y). This volume — called **emittance** or **étendue** — is conserved by any lossless optical element (lens, mirror, waveguide). You cannot focus a beam to have smaller emittance than it started with; you can only redistribute phase space volume, not reduce it.

9.6.3 — Action-Angle Variables

For an integrable system (one with as many conserved quantities as degrees of freedom), a canonical transformation $(q_i, p_i) \to (J_i, \theta_i)$ exists where:

- $J_i = \frac{1}{2\pi}\oint p_i\, dq_i$ are the **action variables** — adiabatic invariants
- $H = H(J_1, \ldots, J_n)$ — the Hamiltonian depends only on J_i
- $\dot{\theta}_i = \partial H/\partial J_i = \omega_i(J) = \text{const}$ — all angles evolve uniformly
- $\dot{J}_i = -\partial H/\partial \theta_i = 0$ — all action variables are constants of motion

The motion is **quasi-periodic** (superposition of oscillations at the fundamental frequencies ω_i). For the harmonic oscillator: $J = E/\omega$ (the adiabatic invariant) and $\theta = \omega t$.

Adiabatic invariance: If parameters of the Hamiltonian change slowly (over many oscillation periods), the action J is conserved even though E changes. This governs:

- Plasma confinement (charged particle mirrors in magnetic bottles)
- Adiabatic demagnetization (reaching millikelvin temperatures)
- Quantum connection: $J = (n + \frac{1}{2})\hbar$ — the Bohr-Sommerfeld condition from §9.3.4 is adiabatic invariance at the quantum level

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9.7 — Rigid Body Dynamics

9.7.1 — The Inertia Tensor

A rigid body with density $\rho(\mathbf{r})$ rotating with angular velocity $\boldsymbol{\omega}$ has angular momentum:

$$L_i = \sum_j I_{ij}\omega_j, \qquad I_{ij} = \int \rho(\mathbf{r})\left(|\mathbf{r}|^2\delta_{ij} - r_i r_j\right) d^3r$$

The **inertia tensor** I_{ij} is a 3×3 symmetric matrix — a generalized version of the scalar moment of inertia $I = mr^2$ for rotation about a fixed axis.

In the **principal axis frame** (the frame in which I_{ij} is diagonal):

$$I_{ij} = \text{diag}(I_1, I_2, I_3)$$

The principal moments I_1, I_2, I_3 are the eigenvalues of I_{ij}; the eigenvectors are the principal axes. Every rigid body has a unique set of principal axes through its center of mass.

Computing principal moments:

Body	Symmetry	$I_1 = I_2$	I_3
Uniform sphere (radius R)	$I_1 = I_2 = I_3$	$\frac{2}{5}mR^2$	$\frac{2}{5}mR^2$
Thin rod (length ℓ, axis = 3)	$I_1 = I_2 \gg I_3$	$\frac{1}{12}m\ell^2$	≈ 0
Disk (radius R, axis = 3)	$I_1 = I_2 = \frac{1}{2}I_3$	$\frac{1}{4}mR^2$	$\frac{1}{2}mR^2$
Cylinder (radius R, length ℓ)	$I_1 = I_2$	$\frac{m}{12}(3R^2 + \ell^2)$	$\frac{1}{2}mR^2$

Parallel axis theorem: If I_{cm} is the moment about the center of mass, the moment about a parallel axis at distance d is:

$$I = I_{cm} + md^2$$

9.7.2 — Euler's Equations

In the **body frame** (rotating with the body), Newton's law for rotational motion:

$$\boxed{I_1\dot{\omega}_1 - (I_2 - I_3)\omega_2\omega_3 = \tau_1}$$

$$\boxed{I_2\dot{\omega}_2 - (I_3 - I_1)\omega_3\omega_1 = \tau_2}$$

$$\boxed{I_3\dot{\omega}_3 - (I_1 - I_2)\omega_1\omega_2 = \tau_3}$$

These are **Euler's equations** — the equations of motion for rotational dynamics in the principal axis (body) frame. They are three coupled, nonlinear ODEs in general.

Torque-free symmetric body ($\tau = 0$, $I_1 = I_2 \neq I_3$, e.g., a football):

From Euler's equations: $\omega_3 = \text{const}$ (spin about symmetry axis is constant). The transverse components satisfy:

$$\dot{\omega}_1 = -\Omega_b \omega_2, \qquad \dot{\omega}_2 = +\Omega_b \omega_1$$

where $\Omega_b = \omega_3(I_3 - I_1)/I_1$ is the **body precession frequency**.

Solution: $\omega_1 = A\sin(\Omega_b t)$, $\omega_2 = A\cos(\Omega_b t)$ — the angular velocity vector traces a **body cone** at rate Ω_b around the symmetry axis.

In the **space frame**, the symmetry axis simultaneously precesses around the fixed angular momentum vector $\mathbf{L}$ at the **space precession frequency** $\Omega_s = L/I_1$. This is **torque-free precession** — the wobble of Earth's rotation axis relative to its geographic pole (the Chandler wobble, period ~14 months).

9.7.3 — The Gyroscope

A spinning gyroscope with large angular momentum $\mathbf{L} = I\boldsymbol{\omega}$ ($I\omega \gg$ other angular momenta) subject to a torque $\boldsymbol{\tau}$:

$$\boldsymbol{\tau} = \frac{d\mathbf{L}}{dt}$$

For a gyroscope with spin axis making angle θ with the vertical, weight mg acting at distance r from the pivot:

$$|\boldsymbol{\tau}| = mgr\sin\theta$$

The angular momentum is large and nearly fixed in direction, so $d\mathbf{L}/dt$ is perpendicular to $\mathbf{L}$. The spin axis **precesses** around the vertical:

$$\boxed{\Omega_{prec} = \frac{mgr}{I\omega}}$$

Engineering applications:

- Gyroscopic stabilization: the faster the spin, the slower the precession, the more stable the platform
- Gyrocompasses: precess to align with Earth's rotation axis (geographic North)
- MEMS gyroscopes (ADXRS, ITG-3200): vibrating ring or fork structures measuring Coriolis force on oscillating mass — in every smartphone
- Attitude control on spacecraft: control moment gyroscopes (CMGs) exert torques to orient the spacecraft

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9.8 — Small Oscillations and Normal Modes

9.8.1 — The General Setup

Any stable mechanical equilibrium can be treated as a collection of coupled harmonic oscillators for small displacements. Near equilibrium $q_i^{(0)}$, expand T and V to second order in displacements $\eta_i = q_i - q_i^{(0)}$:

$$T = \frac{1}{2}\sum_{i,j} T_{ij}\dot{\eta}_i\dot{\eta}_j, \qquad V = V_0 + \frac{1}{2}\sum_{i,j} V_{ij}\eta_i\eta_j$$

where T_{ij} and V_{ij} are the kinetic and potential energy matrices (symmetric, positive-definite for a stable equilibrium).

The Euler-Lagrange equations:

$$\sum_j (T_{ij}\ddot{\eta}_j + V_{ij}\eta_j) = 0$$

9.8.2 — Normal Mode Eigenvalue Problem

Try $\boldsymbol{\eta}(t) = \mathbf{A}\cos(\omega t + \phi)$:

$$\left(\mathbf{V} - \omega^2 \mathbf{T}\right)\mathbf{A} = \mathbf{0}$$

Non-trivial solution requires:

$$\boxed{\det\left(\mathbf{V} - \omega^2\mathbf{T}\right) = 0}$$

This gives n normal frequencies ω_k^2 (eigenvalues) and n normal mode vectors $\mathbf{A}^{(k)}$ (eigenvectors). Each normal mode is an independent oscillation in which all coordinates move at the same frequency ω_k.

Properties of normal modes:

- Modes are orthogonal with respect to T: $\mathbf{A}^{(j)T}\mathbf{T}\mathbf{A}^{(k)} = \delta_{jk}$
- A zero frequency $\omega = 0$ corresponds to a **zero mode** — a rigid body translation or rotation
- In normal coordinates Q_k, the system decouples into n independent harmonic oscillators

9.8.3 — Example: Coupled Pendulums

Two identical pendulums of length ℓ, mass m, connected by a spring of constant k at their midpoints. Generalized coordinates: θ_1, θ_2.

$$T_{ij} = m\ell^2\delta_{ij}, \qquad V_{ij} = \begin{pmatrix} mg\ell + k\ell^2/4 & -k\ell^2/4 \\ -k\ell^2/4 & mg\ell + k\ell^2/4 \end{pmatrix}$$

Normal frequencies:

$$\omega_1^2 = \frac{g}{\ell} \quad \text{(in-phase: spring doesn't stretch)}$$
$$\omega_2^2 = \frac{g}{\ell} + \frac{k}{2m} \quad \text{(out-of-phase: spring fully engaged)}$$

Normal modes:

- Mode 1: $(\theta_1, \theta_2) \propto (1, 1)$ — pendulums swing together
- Mode 2: $(\theta_1, \theta_2) \propto (1, -1)$ — pendulums swing opposite

This is the classical analog of bonding/antibonding molecular orbitals (Ch. 5 §5.8.1):

- In-phase combination = lower frequency = bonding orbital
- Out-of-phase combination = higher frequency = antibonding orbital

The mathematical structure is identical.

9.8.4 — Connection to Phonons (Ch. 6)

For N atoms in a crystal, each atom is displaced from equilibrium and interacts with its neighbors through a potential expanded to second order in displacements. The result is exactly the normal mode eigenvalue problem with $3N$ coordinates.

The $3N$ normal mode frequencies give the **phonon dispersion relation** $\omega_k(\mathbf{q})$ (Ch. 6 §6.7): the $3N$ normal modes of the classical crystal become $3N$ quantum harmonic oscillators upon quantization, each with energy $\hbar\omega_k(n_k + \frac{1}{2})$. **Phonons are the quantum particles of the normal mode vibration field.**

The chain from Ch. 0 → Ch. 6 → Ch. 9 is explicit: the SM Lagrangian's matter fields give atomic bonding (Ch. 5); atomic bonding gives the crystal potential; the crystal potential quantized gives phonons (Ch. 6); the classical normal mode analysis of the crystal gives the phonon dispersion (this chapter).

9.9 — Non-Inertial Reference Frames

9.9.1 — The Rotating Frame

In a frame rotating with angular velocity $\boldsymbol{\Omega}$ relative to an inertial frame, the time derivative of any vector $\mathbf{A}$ transforms as:

$$\left(\frac{d\mathbf{A}}{dt}\right)_{inertial} = \left(\frac{d\mathbf{A}}{dt}\right)_{rot} + \boldsymbol{\Omega} \times \mathbf{A}$$

Applying this to the position and velocity, Newton's second law in the rotating frame:

$$\boxed{m\mathbf{a}_{rot} = \mathbf{F} \underbrace{-2m\boldsymbol{\Omega} \times \mathbf{v}_{rot}}_{\text{Coriolis}} \underbrace{-m\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r})}_{\text{centrifugal}} \underbrace{-m\dot{\boldsymbol{\Omega}} \times \mathbf{r}}_{\text{Euler force}}}$$

Coriolis force $-2m\boldsymbol{\Omega} \times \mathbf{v}_{rot}$:

- Acts perpendicular to velocity in the rotating frame

- Deflects moving objects to the right (Northern hemisphere) or left (Southern)
- Responsible for the rotation direction of cyclones and anticyclones
- Zero for stationary objects

Centrifugal force $-m\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}) = m\Omega^2 \mathbf{r}_\perp$:

- Acts outward, perpendicular to rotation axis
- Modifies effective gravity: $\mathbf{g}_{eff} = \mathbf{g} - \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r})$
- Makes Earth oblate at the equator: equatorial radius 21 km larger than polar

9.9.2 — Connection to Chapter 8

These fictitious forces are not mysterious — they are Christoffel symbol terms in the geodesic equation (Ch. 8 §8.1) evaluated in a non-inertial coordinate system. Specifically, in a uniformly rotating coordinate system:

$$\Gamma^i_{0j} \sim \epsilon_{ijk}\Omega^k \quad \Longrightarrow \quad -c^2\Gamma^i_{0j}\frac{dx^j}{dt} = -2\boldsymbol{\Omega} \times \mathbf{v}_{rot}$$

The Coriolis force is the gravitomagnetic term in the metric. For slow rotation ($\Omega \ll c/R$), it is purely a coordinate artifact — exactly what "fictitious force" means. In GR, fictitious forces and gravity are unified as curvature.

9.9.3 — The Foucault Pendulum

A pendulum free to swing in any direction, at latitude λ:

The effective angular velocity vector in the local vertical frame: $\boldsymbol{\Omega}_{vert} = \Omega_{Earth}\sin\lambda\,\hat{z}$

The Coriolis force on the pendulum causes the plane of oscillation to precess:

$$\Omega_{prec} = -\Omega_{Earth}\sin\lambda$$

- At the North Pole ($\lambda = 90^\circ$): full rotation in 24 hours (sidereal day)
- At the equator ($\lambda = 0^\circ$): no precession
- At latitude 45°N (Paris, where Foucault demonstrated in 1851): rotation period $= 24/\sin(45^\circ) = 33.9$ hours
- At latitude 53°N (Hamburg): period $= 24/\sin(53^\circ) = 30.1$ hours

Engineering applications of Coriolis:

- Flow meters (Coriolis mass flowmeters): fluid forced to rotate deflects due to Coriolis; deflection ∝ mass flow rate. Precision: 0.1% of reading.
- MEMS gyroscopes: vibrating proof mass deflected by Coriolis when device rotates; capacitive sensing of deflection gives angular rate. Used in phones, cars, aircraft, missiles.
- Ballistic trajectory correction: long-range projectiles deflected $\sim$ meters by Coriolis over km-scale trajectories.

9.10 — Chaos: The Boundary of Classical Predictability

9.10.1 — Integrable vs. Non-Integrable Systems

An n-degree-of-freedom system is **integrable** if it has n independent conserved quantities in involution ($F_i, F_j = 0$). By the Liouville-Arnold theorem, such systems can be transformed to action-angle variables (§9.6.3), and all motion is quasi-periodic on n-dimensional tori in phase space.

A system with fewer than n independent conserved quantities is **non-integrable**. For these systems, the KAM (Kolmogorov-Arnold-Moser) theorem says:

- Most phase space tori survive small perturbations (KAM tori)
- Some tori (those with rational frequency ratios $\omega_1/\omega_2 = p/q$) are destroyed and replaced by chains of islands and chaotic regions
- As perturbation grows, more tori are destroyed; eventually large regions become chaotic

9.10.2 — Sensitive Dependence and Lyapunov Exponents

In a chaotic region, nearby trajectories diverge exponentially:

$$|\delta\mathbf{x}(t)| \approx |\delta\mathbf{x}(0)|, e^{\lambda t}$$

where $\lambda > 0$ is the **largest Lyapunov exponent**. For the double pendulum: $\lambda^{-1} \sim 1/\omega$ for large-angle motion — predictability horizon is only a few oscillation periods.

The weather: Earth's atmosphere is a non-integrable fluid system. Lyapunov exponent corresponds to predictability limit of $\sim$ 2 weeks. No amount of computing power overcomes this; it is a fundamental feature of nonlinear dynamics.

9.10.3 — Engineering Relevance

- **Mechanical systems:** Fatigue failure can be triggered by chaotic vibration in structures near homoclinic bifurcations
- **Power electronics:** DC-DC converters with fast switching can exhibit chaotic oscillations — undesirable in precision applications, studied in circuit design
- **Chemical reactors:** Belousov-Zhabotinsky oscillating reactions; chaotic mixing enhances reaction rates
- **Quantum chaos:** Statistical properties of quantum energy spectra in classically chaotic systems follow random matrix theory — relevant to quantum transport in mesoscopic devices

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9.11 — Summary

Bridge B.a is complete. The descent from quantum mechanics to classical mechanics:

Step	Mathematical operation	Result
Ehrenfest theorem	Commutator → Poisson bracket	$d\langle p\rangle/dt = -\langle\nabla V\rangle$
Classical limit condition	$\Delta x \ll L_V$	$d\langle p\rangle/dt = -\nabla V(\langle x\rangle) = F$
Path integral, $\hbar \to 0$	Stationary phase	Classical trajectory = $\delta S = 0$ path

Step	Mathematical operation	Result
WKB, order $\hbar^0$	Substitute $\psi = Ae^{iS/\hbar}$	Hamilton-Jacobi equation
WKB, order $\hbar^1$	Next order	Continuity equation for probability current
Normal modes	T, V matrices → eigenvalue problem	Classical phonon dispersion
Rigid body	Euler-Lagrange for rotation	Euler's equations, gyroscope precession
Rotating frame	Coordinate transformation	Coriolis + centrifugal forces

What was discarded and when it matters:

Discarded	Physical content	Reinstated when
$Q = -\hbar^2\nabla^2 A/2mA$	Tunneling, zero-point energy, interference	$\Delta x \sim \lambda_{dB}$: quantum devices, atoms
Wavepacket spread	Uncertainty principle, quantum diffraction	Sub-nm scales, cold atoms
Quantum decoherence dynamics	Quantum-to-classical transition	Quantum computing, mesoscopic physics
Non-classical correlations	Entanglement, Bell inequality violations	Quantum information, precision sensing

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9.12 — Engineering Thread

Physics from this chapter	Engineering application
Euler-Lagrange mechanics	Robotics dynamics, cable systems, spacecraft attitude
Hamilton's equations + phase space	Nonlinear dynamics, control system design
Liouville's theorem	Étendue conservation in optical systems; particle beam emittance
Action-angle + adiabatic invariance	Plasma magnetic mirror; adiabatic demagnetization refrigeration
Rigid body inertia tensor	Rotating machinery balancing; spacecraft moment of inertia
Euler's equations	Attitude control; gyroscope design
Gyroscope precession $\Omega_{prec} = mgr/I\omega$	Gyrocompasses; MEMS rate gyroscopes; CMGs
Normal modes	Vibration analysis; structural resonance; noise suppression
Coupled normal modes → phonons	Connects to thermal conductivity, specific heat (Ch. 10)
Coriolis force	Coriolis flow meters; MEMS gyroscopes; ballistics
Chaos / Lyapunov exponent	Predictability limits; robust control design; fatigue

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9.13 — Looking Ahead

Chapter 9 has descended from quantum mechanics to classical mechanics via $\hbar \to 0$. The resulting tools — Lagrangian, Hamiltonian, rigid body, normal modes — are the core of classical mechanical engineering science.

Chapter 10 performs Bridge B.b: the same quantum mechanics but with $N \to \infty$ and ensemble averaging, producing statistical mechanics and thermodynamics. The quantum Liouville equation from this chapter becomes the von Neumann equation; tracing over unobserved degrees of freedom produces the Boltzmann distribution and the four laws of thermodynamics.

The connection between Chapters 9 and 10 is important: classical mechanics (Ch. 9) gives individual deterministic trajectories; statistical mechanics (Ch. 10) gives ensemble-averaged behavior. The two descriptions are complementary — neither replaces the other.

End of Chapter 9.

Next: Chapter 10 — Bridge B.b: Statistical Mechanics and Thermodynamics ($N \to \infty$)

CHAPTER 10

Bridge B.b — Statistical Mechanics and Thermodynamics

The Thermodynamic Limit and the Statistical Description: What Counting Can and Cannot Derive

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> *The second law of thermodynamics is, without a doubt, one of the most perfect laws in physics. Any reproducible violation of it, however small, would bring the discoverer great fame and probably a Nobel Prize. Yet all we have is a deep sense that the law is correct.* — Josef Loschmidt (paraphrased), 1876
>
> *The reason is simply that there are far more ways for a system to be disordered than ordered. Entropy increases because the universe is doing statistics.* — modern response

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10.0 — Bridge B.b in Context

Chapter 9 performed Bridge B.a: taking $\hbar \to 0$ on quantum mechanics to recover classical mechanics. This chapter performs Bridge B.b: a completely different operation on the same quantum mechanics — taking $N \to \infty$ and averaging over microscopic degrees of freedom.

The two operations are independent and produce different physics:

Bridge	Operation	Produces	Chapter
B.a	$\hbar \to 0$	Classical mechanics	9
B.b	Thermodynamic limit ($N, V \to \infty$ at fixed density) + statistical/ensemble description	**Statistical mechanics + thermodynamics**	**10 (this)**
B.c	Classical EM field limit	Maxwell's equations	11
B.d	Weak-field metric	Newtonian gravity	8
B.e	Kubo + quantum scattering	Generalized transport law	13

> **Warning: "Bridge B.b" bundles two logically separate steps — do not conflate them** The shorthand "$N \to \infty$" that appears throughout this book as the label for Bridge B.b compresses two distinct operations, and precision is worth the extra words:
>
> 1. **The thermodynamic limit** is a genuine limiting procedure: $N, V \to \infty$ with $N/V = n$ held fixed, and with intensive quantities (density n, temperature T, pressure p, energy per particle u) held at their limiting values. What it buys you is that fluctuations become negligible relative to mean values, so that averages $\langle X \rangle$ are sharply peaked and extensive quantities scale linearly. Legitimately: $N, V \to \infty$ at fixed density.

2. **The statistical / ensemble description** is a *logical and probabilistic* step, not a limit at all. It says a system of N particles should be described by a probability distribution over microstates rather than a point in phase space. Nothing about this requires N to be large — for $N = 1$ the microcanonical ensemble is perfectly well defined, and for a few particles one still needs the ensemble. It is what licenses Gibbs's $1/N!$ and replaces an intractable 10^{23}-dimensional phase space by a tractable distribution over energy levels.

Conversely, averaging over an ensemble does **not** substitute for the thermodynamic limit: ensemble averaging without $N \to \infty$ leaves you with large fluctuations, and the "thermodynamic" description of a small system is legitimately different (fluctuation theorems, negative-entropy microstates). Both steps are needed for ordinary thermodynamics, and they do different work. The old label "$N \to \infty \to$ thermodynamics" suggests one magical limit; the accurate label is *thermodynamic limit + statistical description*.

The same caveat applies to the continuum hypothesis and other uses of the symbol elsewhere in the book: a limit in which N grows while a length scale is simultaneously introduced is a *coarse-graining*, and should be read as such.

The deep point: **you can take these limits in any order.** Classical statistical mechanics applies Bridge B.a first (to get classical trajectories), then Bridge B.b (to average over them). Quantum statistical mechanics applies Bridge B.b directly to the quantum system, without Bridge B.a. This chapter does the latter — the more fundamental approach — and recovers classical thermodynamics as a further limiting case.

The organizing question: When you have $N \sim 10^{23}$ quantum particles and you cannot possibly track every quantum number, what can you say? Answer: exactly as much as an engineer needs.

10.1 — The Density Matrix: Describing What We Don't Know

10.1.1 — Pure States and Mixed States

A **pure state** $|\psi\rangle$ is a complete quantum mechanical description — every property of the system is determined (up to measurement). The density matrix of a pure state:

$$\hat{\rho} = |\psi\rangle\langle\psi|$$

has the property $\hat{\rho}^2 = \hat{\rho}$ (it is a projector).

A **mixed state** describes a statistical mixture of pure states — it is what you use when you know the system is in one of several states $|n\rangle$, with probabilities p_n, but you don't know which one:

$$\boxed{\hat{\rho} = \sum_n p_n |n\rangle\langle n|, \qquad p_n \geq 0, \qquad \sum_n p_n = 1}$$

A mixed state has $\hat{\rho}^2 \neq \hat{\rho}$; equivalently $\mathrm{Tr}(\hat{\rho}^2) < 1$.

Why we need mixed states for thermodynamics: A hot gas in a box is not in a pure state. It is in a statistical ensemble of energy eigenstates with Boltzmann probabilities. You don't know (and couldn't measure) which exact configuration the 10^{23} atoms are in. The density matrix encodes that ignorance.

10.1.2 — The Von Neumann Equation

The equation of motion for the density matrix (the quantum Liouville equation):

$$\boxed{\frac{\partial \hat{\rho}}{\partial t} = -\frac{i}{\hbar}[\hat{H}, \hat{\rho}]}$$

This is the quantum analog of the classical Liouville equation $\partial\rho/\partial t + \{H, \rho\}_{Poisson} = 0$ from Ch. 9 §9.6.2. The Poisson bracket has been replaced by $(-i/\hbar) \times$ commutator — the canonical quantization rule (Ch. 1 §1.6.4) applied consistently.

Expectation values of any observable $\hat{A}$:

$$\langle \hat{A} \rangle = \mathrm{Tr}(\hat{\rho}\hat{A})$$

This works for both pure and mixed states. Thermodynamics is the study of expectation values of macroscopic observables (energy, pressure, magnetization, etc.) in the thermodynamic limit — $N, V \to \infty$ at fixed density, together with the statistical/ensemble description (see the Bridge B.b table above).

10.1.3 — The Reduced Density Matrix and Decoherence

Consider a system S coupled to an environment E (thermal reservoir). The total density matrix $\hat{\rho}_{SE}$ evolves under the combined Hamiltonian $\hat{H} = \hat{H}_S + \hat{H}_E + \hat{H}_{int}$.

The **reduced density matrix** of S alone:

$$\hat{\rho}_S = \mathrm{Tr}_E(\hat{\rho}_{SE})$$

is obtained by tracing out the environment's degrees of freedom. This operation:

- Destroys the off-diagonal elements of $\hat{\rho}_S$ in the energy eigenbasis → decoherence (Ch. 9 §9.4.2)
- Produces a mixed state even if $\hat{\rho}_{SE}$ was initially pure
- Converts the quantum Liouville equation into the **Lindblad master equation** (introduced in Ch. 0 §0.2 context) with dissipation terms from the environment

This is the mathematical operation of Bridge B.b: tracing out environmental degrees of freedom converts the pure, reversible quantum dynamics into the irreversible statistical mechanics of the reduced system. Irreversibility is not put in by hand — it emerges from coarse-graining over many ($N \sim 10^{23}$) environmental degrees of freedom.

10.2 — Von Neumann Entropy: Quantifying Ignorance

10.2.1 — Definition

The **von Neumann entropy** of a density matrix $\hat{\rho}$:

$$\boxed{S = -k_B \mathrm{Tr}(\hat{\rho} \ln \hat{\rho}) = -k_B \sum_n p_n \ln p_n}$$

where the last form uses the eigenvalues p_n of $\hat{\rho}$.

Properties:

- $S \geq 0$, with $S = 0$ iff $\hat{\rho}$ is a pure state ($p_n = 1$ for one n, zero otherwise)
- S is maximized for the uniform distribution: $p_n = 1/\mathcal{N}$ for all $\mathcal{N}$ states, giving $S_{max} = k_B \ln \mathcal{N}$
- S is invariant under unitary evolution: $dS/dt = 0$ under the von Neumann equation
- This last property means entropy **cannot increase** under closed quantum dynamics — irreversibility requires the coupling to an environment (§10.1.3)

10.2.2 — Connection to Boltzmann

For a classical system of N particles with Ω accessible microstates:

$$S = k_B \ln \Omega$$

This is **Boltzmann's formula** (carved on his tombstone). It is the classical limit of the von Neumann entropy for a uniform distribution over Ω states.

What is a microstate? In quantum mechanics: a specific eigenstate of the full Hamiltonian. In classical statistical mechanics: a point in phase space. The number of accessible microstates Ω grows astronomically with energy and particle number — the central fact underlying the second law.

10.3 — The Maximum Entropy Principle: Temperature Emerges

10.3.1 — The Constrained Optimization

Problem: An isolated system has fixed total energy $\langle \hat{H} \rangle = U$. What is the equilibrium density matrix $\hat{\rho}_{eq}$?

Answer: The one that maximizes $S = -k_B \mathrm{Tr}(\hat{\rho} \ln \hat{\rho})$ subject to:

- $\mathrm{Tr}(\hat{\rho}) = 1$ (normalization)
- $\mathrm{Tr}(\hat{\rho}\hat{H}) = U$ (fixed average energy)

Use Lagrange multipliers λ_0 (for normalization) and β (for energy):

$$\frac{\partial}{\partial \hat{\rho}} \left[-k_B \mathrm{Tr}(\hat{\rho} \ln \hat{\rho}) - \lambda_0 \mathrm{Tr}(\hat{\rho}) - \beta \mathrm{Tr}(\hat{\rho}\hat{H}) \right] = 0$$

Result: $-k_B(\ln \hat{\rho} + \mathbf{1}) - \lambda_0 \mathbf{1} - \beta \hat{H} = 0$

$$\boxed{\hat{\rho}_{eq} = \frac{1}{Z} e^{-\beta \hat{H}}, \qquad Z = \mathrm{Tr}(e^{-\beta \hat{H}})}$$

This is the **canonical ensemble density matrix**. The normalization factor Z is the **partition function** — one of the most powerful objects in physics.

10.3.2 — Temperature as a Lagrange Multiplier

The Lagrange multiplier β enforces the energy constraint. Comparing with thermodynamics (to be derived in §10.4), β equals:

$$\beta = \frac{1}{k_B T}$$

Temperature is the Lagrange multiplier for energy in the maximum-entropy optimization. It is not a fundamental concept — it is the parameter that controls the average energy of a system in contact with a reservoir. Two systems in thermal equilibrium share the same β — which is the thermodynamic zeroth law, derived rather than postulated.

10.3.3 — The Boltzmann Distribution

In the energy eigenbasis $\hat{H}|n\rangle = E_n|n\rangle$:

$$p_n = \frac{e^{-\beta E_n}}{Z}, \qquad Z = \sum_n e^{-\beta E_n}$$

The probability of finding the system in state n decreases exponentially with energy — the **Boltzmann distribution**. This is not a postulate; it is the maximum-entropy distribution subject to fixed average energy.

The Boltzmann factor e^{-E/k_BT} is the most important factor in statistical mechanics. It appears in:

- Chemical reaction rates (Arrhenius: $k \propto e^{-E_a/k_BT}$)
- Carrier concentrations in semiconductors ($n \propto e^{-E_g/2k_BT}$, Ch. 6)
- Vacancy concentrations in crystals ($c_v \propto e^{-E_f/k_BT}$)
- Rate of radioactive decay products (nuclear systematics)
- Every thermally activated process in materials science

All of these are the same Boltzmann factor, applied to different activation energies. A student who understands §10.3 understands all of them at once.

10.4 — Thermodynamic Quantities from the Partition Function

All thermodynamic quantities follow from $Z = \mathrm{Tr}(e^{-\beta\hat{H}})$:

Helmholtz free energy:

$$\boxed{F = -k_BT \ln Z}$$

This is the fundamental relation — everything else is a derivative of F.

Internal energy:

$$U = \langle \hat{H} \rangle = -\frac{\partial \ln Z}{\partial \beta} = F + TS$$

Entropy:

$$S = -\left.\frac{\partial F}{\partial T}\right|_V = k_B\left(\ln Z + \beta U\right) = -k_B \mathrm{Tr}(\hat{\rho}\ln\hat{\rho})$$

(Consistent with the von Neumann entropy — as it must be.)

Pressure (from the volume-dependence of energy levels $E_n(V)$):

$$P = -\left.\frac{\partial F}{\partial V}\right|_T = k_BT\left.\frac{\partial \ln Z}{\partial V}\right|_T$$

Chemical potential (for a system where particle number N can vary):

$$\mu = \left.\frac{\partial F}{\partial N}\right|_{T,V}$$

Heat capacity:

$$C_V = \left.\frac{\partial U}{\partial T}\right|_V = k_B\beta^2\frac{\partial^2 \ln Z}{\partial \beta^2} = \frac{\langle \hat{H}^2\rangle - \langle \hat{H}\rangle^2}{k_B T^2}$$

The last form shows C_V is proportional to the energy **variance** — how much the energy fluctuates around its mean. Larger fluctuations → larger heat capacity.

10.5 — The Four Laws: Statistical Foundations and Their Limits

> **Warning: What "deriving the four laws" does and does not mean** Statistical mechanics provides genuine microscopic foundations for thermodynamics, and for one of the four laws it provides something remarkable: a *statistical explanation of why the law holds at all*, rather than a restatement of it. But the four laws are not equivalent in this respect, and presenting them as uniformly "derived from counting" overstates the case.
>
> | Law | What statistical mechanics contributes |
> |---|---|
> | **Zeroth** | Explains thermal equilibrium and the *transitivity* of temperature via the equality of $\beta = 1/k_B T$ across equilibrium systems. A structural account, not a counting result. |
> | **First** | Essentially **energy conservation** — Noether's theorem for time-translation symmetry, a dynamical symmetry, not a statistical one. Statistical mechanics shows *how* energy divides into work and heat (the $\mathrm{Tr}(d\hat{\rho}\,\hat{H})$ / $\mathrm{Tr}(\hat{\rho}\,d\hat{H})$ split below). It does not derive energy conservation. |
> | **Second** | The one law that receives a true *statistical* explanation: entropy increase is overwhelmingly probable because disordered macrostates have exponentially more microstates. This is the characteristic success of the approach, and the reason the second law is usually singled out. |
> | **Third** | Requires assumptions *beyond* counting: non-degenerate ground states, bounded or well-behaved low-temperature spectra, and a gap or finite ground-state degeneracy. $S/N \to 0$ is a statement about the low-energy density of states, which pure counting of accessible states does not by itself deliver. |
>
> The defensible summary: **statistical mechanics provides microscopic foundations for the thermodynamic laws, with the second law acquiring its characteristic statistical interpretation.** The others are clarified, structured, or given a mechanism — but they are not all derived from counting, and a derivation that claims otherwise is usually smuggling in an extra assumption.

Law 0 — Thermal Equilibrium (The Zeroth Law)

Statement: If two systems are each in thermal equilibrium with a third, they are in equilibrium with each other.

Derivation: At equilibrium, each system maximizes its own entropy subject to its energy. The Lagrange multiplier $\beta = 1/k_B T$ is the same for all systems in equilibrium with each other (since energy flows until β equalizes). The zeroth law is the statement that β is a transitive equality — a mathematical fact about Lagrange multipliers.

Law 1 — Conservation of Energy (The First Law)

Statement: $dU = \delta Q - \delta W$ (energy is conserved; work and heat are both forms of energy transfer).

Derivation: From $U = \mathrm{Tr}(\hat{\rho}\hat{H})$:

$$dU = \underbrace{\mathrm{Tr}(d\hat{\rho} \cdot \hat{H})}_{\delta Q} + \underbrace{\mathrm{Tr}(\hat{\rho} \cdot d\hat{H})}_{\delta W}$$

- $\delta W = \mathrm{Tr}(\hat{\rho}\, d\hat{H}) = -P\, dV$: work done by the system changes the energy levels $E_n(V)$ by changing the volume — the Hamiltonian itself changes
- $\delta Q = \mathrm{Tr}(d\hat{\rho} \cdot \hat{H})$: heat changes the occupation probabilities p_n — the density matrix changes while energy levels stay fixed

The first law is the statement that total energy is conserved (Noether's theorem for time-translation symmetry, Ch. 1 §1.7) applied to a system where energy can be transferred via two different mechanisms: rearranging the Hamiltonian (work) or rearranging the probability distribution (heat).

Law 2 — Entropy Never Decreases (The Second Law)

Statement: The entropy of an isolated system is non-decreasing; it is constant for reversible processes and increases for irreversible ones.

Derivation sketch: Consider all microstates of a gas in a $V/2$-box, then remove the partition. The number of accessible microstates goes from $\Omega_{1/2}$ to $\Omega_{full} \sim 2^N \Omega_{1/2}$. The entropy change:

$$\Delta S = k_B \ln \frac{\Omega_{full}}{\Omega_{1/2}} = N k_B \ln 2 > 0$$

For $N = 10^{23}$: $\Delta S = 10^{23} k_B \ln 2 > 0$ — overwhelmingly positive.

Why does entropy increase? The volume of phase space accessible to high-entropy macrostates is astronomically larger than that for low-entropy macrostates. A system that starts in a low-entropy state will move toward higher-entropy states simply because there are so many more of them — not because of any arrow-of-time in the microscopic laws.

The **Clausius inequality** for any cycle:

$$\oint \frac{\delta Q}{T} \leq 0$$

with equality for reversible cycles. This follows from the second law and is the quantitative form used in engineering analysis.

Law 3 — Zero Entropy at Zero Temperature (The Third Law)

Statement: The entropy of a system approaches a constant (usually zero) as $T \to 0$.

Derivation: As $T \to 0$ ($\beta \to \infty$), the Boltzmann distribution concentrates entirely on the ground state:

$$p_0 \to 1, \quad p_{n \neq 0} \to e^{-\beta(E_n - E_0)} \to 0$$

If the ground state is non-degenerate: $\hat{\rho} \to |0\rangle\langle 0|$, a pure state, and $S = -k_B \mathrm{Tr}(\hat{\rho} \ln \hat{\rho}) \to 0$.

If the ground state is g-fold degenerate: $S \to k_B \ln g$. For macroscopic systems, $g \ll e^N$ so $S/N \to 0$ as $N, V \to \infty$ at fixed density — the extensive entropy per particle still vanishes.

Engineering consequence: As $T \to 0$, all specific heats $C \to 0$ (consistent with $S \to 0$ since $\int_0^T C\, dT'/T' = S(T)$ must converge). This means you cannot remove a finite amount of heat to reach $T = 0$ in a finite number of steps — it is asymptotically approachable but never reached.

10.6 — The Carnot Efficiency

From the first and second laws, the maximum efficiency of any heat engine operating between a hot reservoir at T_h and a cold reservoir at T_c:

$$\boxed{\eta_{Carnot} = 1 - \frac{T_c}{T_h}}$$

Derivation: In a reversible (Carnot) cycle:

- Heat absorbed from hot reservoir: Q_h
- Heat rejected to cold reservoir: Q_c
- Work output: $W = Q_h - Q_c$ (first law)
- Entropy balance for reversible cycle: $Q_h/T_h = Q_c/T_c$ (second law, equality)
- Efficiency: $\eta = W/Q_h = 1 - Q_c/Q_h = 1 - T_c/T_h$

No engine can exceed Carnot efficiency — if it did, entropy would decrease, violating the second law. This is the fundamental limit on all heat engines: steam turbines, internal combustion, Stirling, thermoelectric generators.

Engineering reality: Real engines achieve 40–60% of Carnot efficiency. A coal power plant at $T_h = 600^\circ C$ and $T_c = 30^\circ C$: $\eta_{Carnot} = 1 - 303/873 = 65.3\%$. Actual efficiency: ~42%. The gap is irreversibility (friction, heat leakage, finite-time operation) and is the subject of finite-time thermodynamics.

10.7 — Ideal Gas and the Equipartition Theorem

10.7.1 — The Ideal Gas Partition Function

For a monatomic ideal gas of N identical particles in volume V, each with only translational kinetic energy $\hat{H}_i = \hat{p}_i^2/2m$:

$$Z_1 = \mathrm{Tr}(e^{-\beta \hat{p}^2/2m}) = \frac{V}{\lambda_{th}^3}$$

where the **thermal de Broglie wavelength**:

$$\lambda_{th} = h/\sqrt{2\pi m k_B T}$$

sets the length scale at which quantum effects become important (λ_{th} comparable to interparticle spacing $n^{-1/3}$ → quantum statistics matter).

For N distinguishable particles: $Z = Z_1^N$. For identical particles (indistinguishable):

$$Z = \frac{Z_1^N}{N!} = \frac{V^N}{N!\lambda_{th}^{3N}}$$

The $N!$ is the **Gibbs correction factor** — without it, entropy is non-extensive (Gibbs paradox). Quantum mechanics resolves this: identical particles are truly indistinguishable.

From $F = -k_B T \ln Z$:

$$PV = Nk_BT, \qquad U = \frac{3}{2}Nk_BT, \qquad C_V = \frac{3}{2}Nk_B$$

The ideal gas law and monatomic heat capacity, derived from the partition function.

10.7.2 — The Equipartition Theorem

For any quadratic term in the Hamiltonian $H = \ldots + \frac{1}{2}\alpha q^2 + \ldots$:

$$\left\langle \frac{1}{2}\alpha q^2 \right\rangle = \frac{1}{2}k_BT$$

Every quadratic degree of freedom contributes $\frac{1}{2}k_BT$ to the average energy.

System	Quadratic DOF	C_V	Prediction
Monatomic gas	3 translational	$\frac{3}{2}Nk_B$	Exact (He, Ar, Ne)
Diatomic gas (rigid)	3 trans + 2 rot	$\frac{5}{2}Nk_B$	Good at room T (N_2, O_2)
Diatomic gas (vibrating)	+ 2 vib	$\frac{7}{2}Nk_B$	Fails!
Monatomic solid (classical)	3 pos + 3 mom	$3Nk_B$	Dulong-Petit law

10.7.3 — Failure of Equipartition and the Quantum Correction

The problem: Equipartition predicts $C_V = \frac{7}{2}Nk_B$ for vibrating diatomic molecules and $C_V = 3Nk_B$ for all solids at all temperatures. Experiment shows $C_V \to 0$ as $T \to 0$.

The fix: A quantum harmonic oscillator with frequency ω only contributes to the heat capacity if $k_BT \gtrsim \hbar\omega$. For $k_BT \ll \hbar\omega$, the mode is **frozen out** — it remains in its ground state $E_0 = \hbar\omega/2$ regardless of temperature.

Einstein model for a solid (all modes at same frequency ω_E):

$$C_V = 3Nk_B \left(\frac{\hbar\omega_E}{k_BT}\right)^2 \frac{e^{\hbar\omega_E/k_BT}}{(e^{\hbar\omega_E/k_BT}-1)^2}$$

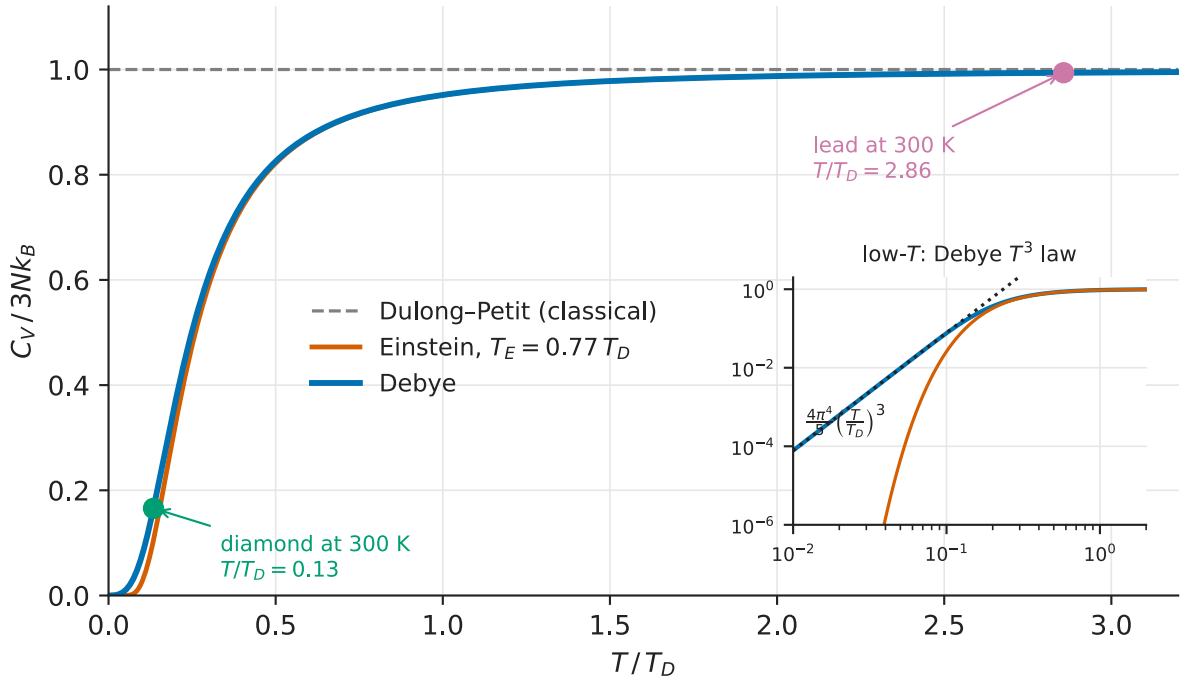
- At high T ($k_BT \gg \hbar\omega_E$): $C_V \to 3Nk_B$ (Dulong-Petit) ✓
- At low T ($k_BT \ll \hbar\omega_E$): $C_V \to 0$ exponentially ✓

Debye model (realistic phonon spectrum, ω up to Debye cutoff ω_D):

$$C_V = 9Nk_B\left(\frac{T}{T_D}\right)^3\int_0^{T_D/T}\frac{x^4e^x}{(e^x-1)^2}dx$$

At low T: $C_V = \frac{12\pi^4}{5}Nk_B\left(\frac{T}{T_D}\right)^3$ — the **Debye T^3 law** ✓

The Debye temperature $T_D = \hbar\omega_D/k_B$ characterizes when quantum effects matter: for diamond $T_D = 2230$ K; for lead $T_D = 105$ K. This is why diamond remains quantum (low C_V) at room temperature while lead is classical.



$C_V/3Nk_B$ versus T/T_D for the Debye model, the Einstein model with $T_E = \sqrt{3/5}\,T_D \approx 0.77\,T_D$, and the classical Dulong–Petit value; at 300 K diamond ($T_D = 2230$ K, $T/T_D = 0.13$) sits on the steep quantum rise at $C_V \approx 0.17 \times 3Nk_B$ while lead ($T_D = 105$ K, $T/T_D = 2.86$) is at the classical plateau; the log–log inset shows the Debye T^3 law against the far faster exponential freeze-out of the Einstein model.

10.8 — Quantum Statistics: Bose-Einstein and Fermi-Dirac

10.8.1 — The Grand Canonical Ensemble

When particle number N can fluctuate (the system is coupled to a particle reservoir), maximize entropy subject to both fixed $\langle\hat{H}\rangle = U$ and fixed $\langle\hat{N}\rangle = N$:

$$\hat{\rho} = \frac{1}{\Xi}e^{-\beta(\hat{H}-\mu\hat{N})}, \qquad \Xi = \mathrm{Tr}(e^{-\beta(\hat{H}-\mu\hat{N})})$$

where μ is the **chemical potential** — the Lagrange multiplier for particle number. The grand potential: $\Omega = -k_BT\ln\Xi = F - \mu N$.

10.8.2 — Mean Occupation Numbers

For a system of non-interacting particles, each single-particle state ϵ can be occupied by n particles. The mean occupation:

Bose-Einstein (bosons, integer spin, $n = 0, 1, 2, \ldots$):

$$\boxed{\bar{n}_{BE}(\epsilon) = \frac{1}{e^{(\epsilon-\mu)/k_BT} - 1}}$$

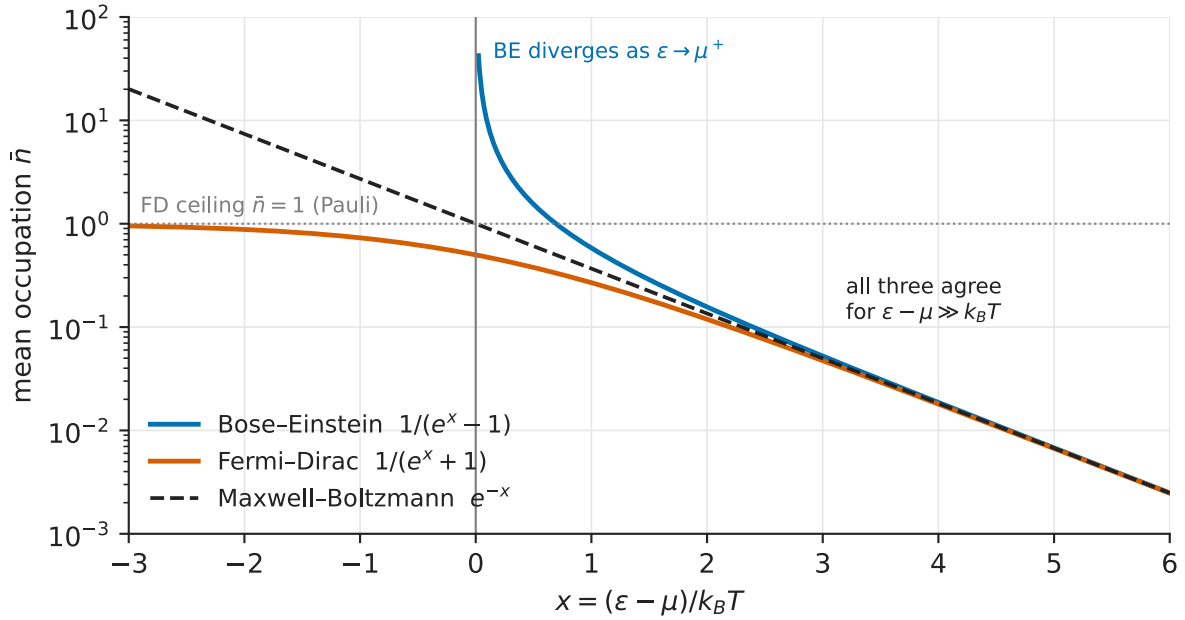
Fermi-Dirac (fermions, half-integer spin, $n = 0$ or 1 by Pauli exclusion):

$$\boxed{\bar{n}_{FD}(\epsilon) = \frac{1}{e^{(\epsilon-\mu)/k_BT} + 1}}$$

Maxwell-Boltzmann (classical limit, $e^{(\epsilon-\mu)/k_BT} \gg 1$):

$$\bar{n}_{MB}(\epsilon) = e^{-(\epsilon-\mu)/k_BT}$$

At high T (or low density), all three converge to Maxwell-Boltzmann. The quantum corrections matter when $\lambda_{th}^3 n \gtrsim 1$ — when the thermal de Broglie wavelength becomes comparable to the interparticle spacing.



Mean occupation versus $x = (\epsilon - \mu)/k_BT$: Bose–Einstein diverges as $\epsilon \to \mu^+$, Fermi–Dirac saturates at the Pauli ceiling $\bar{n} = 1$ for $\epsilon < \mu$, and both merge with the Maxwell–Boltzmann exponential once $\epsilon - \mu \gtrsim 3k_BT$, where $\bar{n} \ll 1$.

10.8.3 — The Fermi-Dirac Distribution and Free Electrons

For electrons (fermions, $\mu \to E_F$ at $T = 0$):

At $T = 0$: step function — all states below E_F filled, all above empty.

The Fermi energy for a free electron gas of density n:

$$E_F = \frac{\hbar^2}{2m_e}(3\pi^2 n)^{2/3}$$

For copper ($n = 8.5 \times 10^{28}$ m^{-3}): $E_F = 7.0$ eV, $T_F = E_F/k_B = 81{,}000$ K. At room temperature ($T = 300$ K $\ll T_F$): the Fermi-Dirac distribution is nearly a step function. Electrons are **degenerate** — their quantum statistics dominate over thermal effects.

Electronic specific heat: Only electrons within k_BT of E_F can be thermally excited. The fraction $\sim k_BT/E_F \ll 1$:

$$C_V^{electron} = \frac{\pi^2}{2} N k_B \frac{T}{T_F} = \gamma T$$

where $\gamma = \pi^2 N k_B / 2T_F$ is the **Sommerfeld coefficient**. At low T, $C_V^{electron} = \gamma T$ (linear in T) while $C_V^{phonon} = AT^3$ (Debye). Total low-T specific heat: $C_V = \gamma T + AT^3$ — observed in all metals, confirming the Fermi-Dirac picture.

Connection to Ch. 6: The Fermi-Dirac distribution with μ adjusted by doping is exactly the occupation function used in semiconductor device physics (Ch. 6 §6.6). The connection is direct: the grand canonical ensemble here produces the Fermi-Dirac function used there.

10.8.4 — Blackbody Radiation: Photons Are Bosons with $\mu = 0$

Photons have integer spin (spin-1) → bosons. They are not conserved (can be created and destroyed freely) → $\mu = 0$.

The mean number of photons in a mode of frequency ν:

$$\bar{n} = \frac{1}{e^{h\nu/k_BT} - 1}$$

Energy per mode: $\bar{E} = h\nu\,\bar{n} = \frac{h\nu}{e^{h\nu/k_BT}-1}$

Spectral energy density (summing over all modes in volume V in frequency interval $d\nu$):

$$\boxed{u(\nu, T) = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/k_BT} - 1}}$$

This is **Planck's law** for blackbody radiation — derived from Bose-Einstein statistics for photons.

Limits:

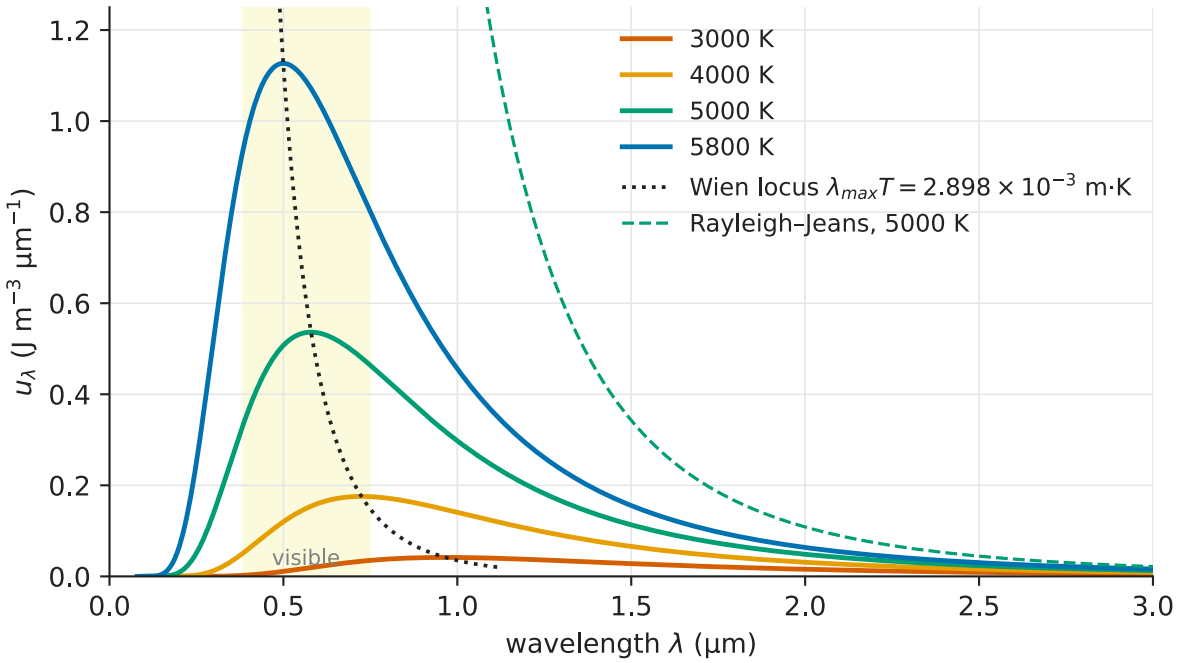
- Low frequency ($h\nu \ll k_BT$): $u \to 8\pi\nu^2 k_BT/c^3$ — the classical Rayleigh-Jeans law. This diverges as $\nu \to \infty$ (the ultraviolet catastrophe), the failure that motivated Planck to quantize energy in 1900.
- High frequency ($h\nu \gg k_BT$): $u \to 8\pi h\nu^3/c^3 \cdot e^{-h\nu/k_BT}$ — Wien's approximation; exponential suppression of high-frequency modes.

Wien's displacement law: Peak of $u(\nu, T)$ at $h\nu_{max} = 2.82\,k_BT$; the peak of the per-wavelength density u_λ is a different point, at $hc/\lambda_{max} = 4.965\,k_BT$:

$$\lambda_{max} T = b = 2.898 \times 10^{-3}\ \text{m} \cdot \text{K}$$

Stefan-Boltzmann law: Total power radiated per unit area:

$$j = \sigma T^4, \qquad \sigma = \frac{2\pi^5 k_B^4}{15 c^2 h^3} = 5.67 \times 10^{-8}\ \text{W/m}^2\text{K}^4$$



Planck spectral energy density per unit wavelength, $u_\lambda = (8\pi hc/\lambda^5)/(e^{hc/\lambda k_B T} - 1)$, at $T = 3000$, 4000, 5000 and 5800 K: the peak slides along the Wien locus $\lambda_{max}T = 2.898 \times 10^{-3}$ m·K (dotted) into the visible band for the Sun's 5800 K, the area grows as T^4, and the classical Rayleigh–Jeans law (dashed, 5000 K) matches only at long wavelength before diverging.

10.9 — Phase Transitions and Landau Theory

10.9.1 — Order Parameters and Broken Symmetry

A **phase transition** is a discontinuous (or singular) change in the equilibrium state as a control parameter (temperature, pressure, field) passes through a critical value. The key concept is the **order parameter** ϕ: a quantity that is zero in the high-symmetry (disordered) phase and nonzero in the low-symmetry (ordered) phase.

10.9.2 — The Landau Free Energy

Near a continuous (second-order) phase transition, the free energy as a functional of the order parameter ϕ:

$$\boxed{F = F_0 + a(T)\phi^2 + b\phi^4 + c|\nabla\phi|^2 + \cdots, \qquad b > 0}$$

with $a(T) = a_0(T - T_c)$ changing sign at the transition.

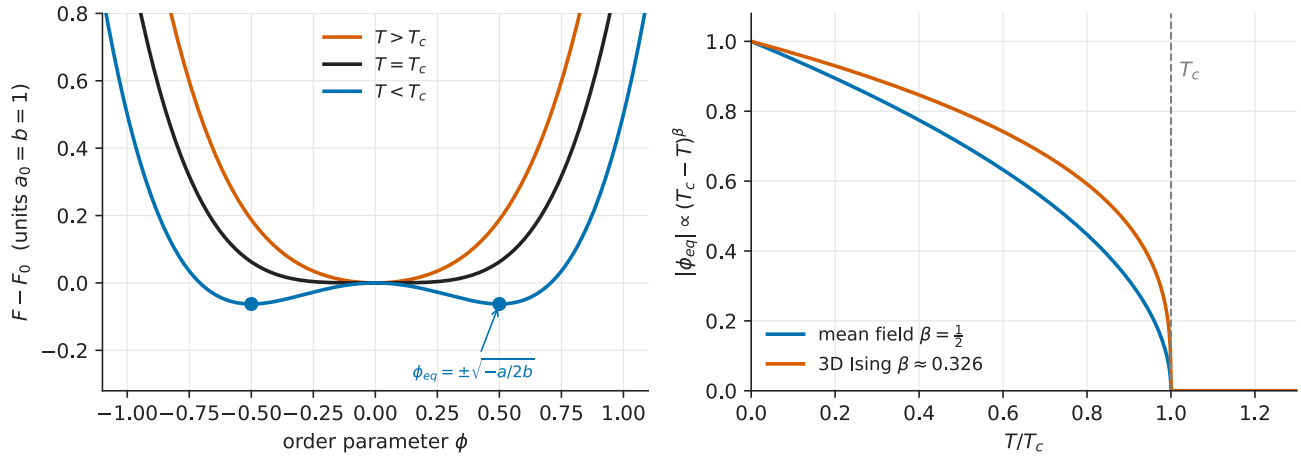
This is the Mexican hat, appearing for the fourth time:

System	ϕ	Symmetry broken	T_c	Chapter
Higgs field	Complex scalar H	SU(2)×U(1) → U(1)	≈160 GeV (crossover; VEV $v = 246$ GeV)	0, §0.7
BCS superconductor	Gap $\Delta = \|\Delta\|e^{i\theta}$	U(1)	$\sim$ meV	7, §7.6

System	ϕ	Symmetry broken	T_c	Chapter
BEC	Condensate $\Psi = \lvert\Psi\rvert e^{i\theta}$	U(1)	$\sim$ μeV	7, 10
Ferromagnet	Magnetization **M**	SO(3)	Curie T	10
Ferroelectric	Polarization **P**	Inversion	Curie T	10
Liquid-gas	Density ρ-ρ_c	Z2	Critical T	10
Antiferromagnet	Staggered mag.	Translation + SO(3)	Néel T	10

The universal result: For $T > T_c$ ($a > 0$): minimum at $\phi = 0$ (disordered phase). For $T < T_c$ ($a < 0$): minimum at $|\phi|^2 = -a/2b = a_0(T_c - T)/2b \neq 0$ (ordered phase, spontaneous symmetry breaking).

Order parameter near T_c: $|\phi| \propto (T_c - T)^\beta$ with $\beta = \frac{1}{2}$ (mean-field value; corrected by fluctuations to $\beta \approx 0.326$ in 3D Ising universality class).



Left: $F - F_0 = a_0(T - T_c)\phi^2 + b\phi^4$ (units $a_0 = b = 1$, $T - T_c = \pm 0.5$ and 0) — a single minimum at $\phi = 0$ above T_c, a flat quartic bottom at T_c, and the double well with minima at $\phi_{eq} = \pm\sqrt{-a/2b}$ below T_c (the 1D cut of the Mexican hat of Ch. 0 §0.7); right: $|\phi_{eq}| \propto (T_c - T)^\beta$ for the mean-field $\beta = \frac{1}{2}$ and the 3D Ising $\beta \approx 0.326$, both normalised to 1 at $T = 0$ — the fluctuation-corrected order parameter rises much more steeply just below T_c.

10.9.3 — First-Order vs. Second-Order Transitions

Second-order (continuous): ϕ grows continuously from zero below T_c. No latent heat. Diverging correlation length $\xi \propto |T - T_c|^{-\nu}$. Examples: ferromagnetic transition, λ-transition in helium, superconducting transition in zero field.

First-order (discontinuous): ϕ jumps discontinuously. Latent heat $L = T_c \Delta S$. Phase coexistence. Requires a cubic term $c\phi^3$ in Landau free energy (making the energy landscape asymmetric) or a $b < 0$ term. Examples: melting/freezing, liquid-gas transition away from critical point, first-order ferromagnetic transitions.

Clausius-Clapeyron equation for first-order phase boundaries:

$$\frac{dP}{dT} = \frac{L}{T\Delta V}$$

where L is latent heat and ΔV is volume change. Applications:

- Ice: negative dP/dT (water has larger ΔV than ice — anomalous)
- Most solids: positive dP/dT
- Boiling water at altitude: lower T_{boil} due to lower pressure (same equation)

10.9.4 — Gibbs Phase Rule

For a system of C chemical components in P phases:

$$\boxed{F = C - P + 2}$$

where F is the number of **degrees of freedom** (intensive variables that can be independently varied without changing the number of phases).

Example	C	P	F
Water, one phase	1	1	2 (choose T and p)
Water+ice	1	2	1 (only one T for each p)
Water+ice+vapor (triple point)	1	3	0 (unique T, P)
Salt water, one phase	2	1	3 (T, P, concentration)

The Gibbs phase rule tells ChE engineers how many independent variables can be specified in a separation process.

10.10 — Entropy, Information, and the Arrow of Time

10.10.1 — Boltzmann's H-Theorem and the Approach to Equilibrium

Boltzmann proved (for a dilute gas obeying his kinetic equation) that the quantity:

$$H = \int f(\mathbf{v}) \ln f(\mathbf{v}) \, d^3v$$

(where f is the single-particle distribution function) satisfies $dH/dt \leq 0$. Since $S \propto -H$, entropy increases: $dS/dt \geq 0$.

The proof assumes **molecular chaos** (Stosszahlansatz): the velocities of two particles about to collide are uncorrelated. This assumption breaks time-reversal symmetry (the future-pointing assumption) — and is what injects the arrow of time.

10.10.2 — Maxwell's Demon and Landauer's Principle

Maxwell's demon can, in principle, sort molecules and decrease entropy — seemingly violating the second law. The resolution (Szilard 1929, Landauer 1961):

Erasing one bit of information costs at least $k_B T \ln 2$ of free energy.

The demon must remember each measurement result in order to operate the door. When the demon's memory is eventually erased (as it must be, since it has finite memory), the entropy increase from erasure exactly compensates the entropy decrease from sorting. Information is physical — it has thermodynamic

cost.

Landauer's principle: $W_{erasure} \geq k_B T \ln 2$ per bit.

At 300 K: $k_B T \ln 2 \approx 2.87 \times 10^{-21}$ J ≈ 0.018 meV per bit. Modern CMOS logic dissipates $\sim 10^{-16}$ J per bit — still 10^5 times above the Landauer limit, leaving enormous room for efficiency improvement.

Engineering consequence: The Landauer limit is the ultimate lower bound on computing energy dissipation. It is currently unreachable in practice but provides a target for reversible computing research.

10.10.3 — Shannon Entropy and the Unified Picture

The Shannon information entropy:

$$H = -\sum_i p_i \log_2 p_i \text{ bits}$$

is identical in form to the von Neumann entropy (with $k_B \log_e$ replaced by $\log_2$). This is not a coincidence. Statistical mechanics is the physics of information: entropy measures how much we *don't know* about the microscopic state. Temperature is the currency of that ignorance.

The arrow of time is the arrow of decreasing information: The universe started in an extremely low-entropy (high information, low probability) state (the Big Bang). It has been evolving toward higher entropy (lower information, higher probability) ever since — not because the microscopic laws prefer the future, but because high-entropy states are so overwhelmingly more numerous.

10.11 — Summary

Bridge B.b is complete. The descent from quantum mechanics to thermodynamics:

Step	Operation	Result
Density matrix	$\hat{\rho} = \Sigma p_n \lvert n\rangle\langle n\rvert$	Encodes statistical ignorance
Trace environment	$\hat{\rho}_S = \mathrm{Tr}_E(\hat{\rho}_{SE})$	Irreversibility from many DOF
Von Neumann entropy	$S = -k_B \mathrm{Tr}(\hat{\rho} \ln \hat{\rho})$	Quantifies ignorance
Max entropy, fixed $\langle \hat{H} \rangle$	Lagrange multiplier β	Boltzmann distribution; temperature
Partition function	$Z = \mathrm{Tr}(e^{-\beta \hat{H}})$	All thermodynamics from one function
Zeroth law	β equalizes at equilibrium	Temperature is transitive
First law	$dU = \delta Q - \delta W$	Energy conservation (Noether, time symmetry)
Second law	$S = k_B \ln \Omega$ increases	High-entropy states vastly outnumber low
Third law	$S \to 0$ as $T \to 0$	Ground state is pure
Carnot efficiency	$\eta = 1 - T_c/T_h$	Maximum from first + second law
Equipartition	$\frac{1}{2} k_B T$ per quadratic DOF	Fails when $\hbar\omega \gg k_B T$
Fermi-Dirac	$n_{FD} = 1/(e^{(\epsilon-\mu)/kT} + 1)$	Electrons in metals and semiconductors
Bose-Einstein	$n_{BE} = 1/(e^{(\epsilon-\mu)/kT} - 1)$	Phonons, photons, BEC
Planck radiation	$u(\nu, T) = (8\pi h\nu^3/c^3)/(e^{h\nu/kT} - 1)$	Blackbody spectrum

Step	Operation	Result
Landau theory	$F = a(T)\phi^2 + b\phi^4$	Phase transitions (Mexican hat, 4th appearance)
Clausius-Clapeyron	$dP/dT = L/T\Delta V$	Phase boundaries
Gibbs phase rule	$F = C - P + 2$	ChE separation processes
Landauer limit	$W_{min} = k_B T \ln 2$ per bit	Ultimate computing energy limit

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10.12 — Engineering Thread

Physics	Application
Boltzmann factor $e^{-E/kT}$	Arrhenius reaction kinetics (ChE); semiconductor carrier density (EEE); diffusion activation (ME/CE)
Carnot efficiency $\eta = 1 - T_c/T_h$	Design limit for steam turbines, engines, refrigerators, heat pumps
Clausius-Clapeyron $dP/dT = L/T\Delta V$	Boiling points vs. altitude; steam tables; vapor pressure calculations
Gibbs phase rule $F = C - P + 2$	Distillation design; alloy phase diagram reading; ChE process control
Equipartition + quantum freezeout	Specific heat of materials; thermal mass in HVAC; heat storage
Fermi-Dirac distribution	Electronic specific heat in metals; semiconductor device physics; Fermi level engineering (Ch. 6)
Bose-Einstein statistics (photons)	Blackbody furnace design; solar cell efficiency limits; thermal imaging
Planck/Stefan-Boltzmann	Furnace radiation heat transfer; spacecraft thermal control; IR sensor design
Debye T^3 law	Low-T cryogenic specific heat; thermal mass at millikelvin for quantum computers
Landau theory / phase diagrams	Materials processing (heat treatment, solidification); alloy design
Landauer limit	Energy-efficient computing; reversible logic design; fundamental sensor noise
Von Neumann entropy + decoherence	Quantum computing error budgets; qubit coherence time engineering

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10.13 — Looking Ahead

Chapters 9 and 10 have recovered classical mechanics and thermodynamics from quantum mechanics via the $\hbar \to 0$ and $N \to \infty$ limits respectively.

Chapter 11 performs Bridge B.c: the classical field limit of the U(1) gauge sector of $\mathcal{L}_{SM}$ (Ch. 0 §0.4.2). This gives Maxwell's equations directly from the Lagrangian — showing that every field equation in classical electromagnetism is a consequence of the Euler-Lagrange equations applied to one term in $\mathcal{L}_{SM}$.

The entire Layer 2 landscape is now taking shape: Newtonian gravity (Ch. 8), classical mechanics (Ch. 9), thermodynamics (Ch. 10), and next — Maxwell's equations (Ch. 11) and the transport laws (Ch. 13). These five, together, constitute the complete classical engineering science that Chapters 15–21 will lump into the engineering systems of Layer 3.

End of Chapter 10.

Next: Chapter 11 — Bridge B.c: Maxwell's Equations from the U(1) Gauge Sector

CHAPTER 11

Bridge B.c — Maxwell's Equations from the U(1) Gauge Sector

Four Equations from One Lagrangian Term

> *From the long view of the history of mankind, seen from, say, ten thousand years from now, there can be little doubt that the most significant event of the 19th century will be judged as Maxwell's discovery of the laws of electrodynamics.* — Richard Feynman, *The Feynman Lectures on Physics*, Vol. II

11.0 — Bridge B.c in Context

Bridges B.a and B.b ($\hbar \to 0$ and $N \to \infty$) recovered classical mechanics and thermodynamics from quantum mechanics. Bridge B.c is a different operation entirely: it takes the **classical electromagnetic-field limit of the $U(1)$ gauge theory** — the classical limit of the $U(1)$ gauge sector of $\mathcal{L}_{SM}$.

> **Warning: What the name of this bridge does and does not mean** This is **not** a limit in which the group $U(1)$ itself becomes Maxwell's equations. It denotes the *classical-field limit of the electromagnetic $U(1)$ gauge theory*, in which quantum fluctuations of the field are negligible. The gauge group is unchanged by the limit; what is discarded is the quantisation of the field. The arrow in "B.c → Maxwell" is a statement that Maxwell's equations are *derived from* the classical $U(1)$ gauge theory, not that the two objects are the same thing.

The operation: The quantum U(1) gauge field A_μ is, at the quantum level, the photon field — its excitations are photons. In the classical limit (many photons in a coherent state, field amplitudes large compared to $\hbar\omega$), the quantum field operator $\hat{A}_\mu$ becomes a classical field $A_\mu(\mathbf{r}, t)$. The Euler-Lagrange equations for this classical field are **Maxwell's equations**.

What this chapter shows:

The four Maxwell equations — which every EEE student learns as four separate, empirically-derived postulates — are in fact:

- **Two equations:** from a single covariant Euler-Lagrange equation
- **Two more:** from a mathematical identity on the field tensor

Neither set is postulated. Both follow from the U(1) gauge sector of Ch. 0 by the field Euler-Lagrange procedure of Ch. 1 §1.9.

Bridge B sub-path	Limit	Result	Chapter
B.a	$\hbar \to 0$	Classical mechanics	9
B.b	$N \to \infty$	Thermodynamics	10
B.c	Classical EM field	**Maxwell's equations**	**11**
B.d	Weak-field metric	Newton's gravity	8
B.e	Kubo averaging	Transport laws	13

11.1 — The Starting Point: The Electromagnetic Lagrangian Density

From Ch. 0 §0.4.2, the U(1) gauge sector of $\mathcal{L}_{SM}$ in the classical field limit:

$$\boxed{\mathcal{L}_{EM} = -\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} + A_\mu j^\mu}$$

Every symbol:

$A^\mu = (\phi/c,\ \mathbf{A})$ — the **electromagnetic four-potential**. The scalar potential ϕ (volts) and the vector potential $\mathbf{A}$ (V·s/m) are unified into a single four-vector. The physical fields $\mathbf{E}$ and $\mathbf{B}$ are both encoded in A^μ.

$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ — the **field strength tensor** (or Faraday tensor). It is antisymmetric: $F_{\mu\nu} = -F_{\nu\mu}$. It is the gauge-invariant content of the theory — every physical observable is expressed through $F_{\mu\nu}$, not through A_μ directly.

$j^\mu = (c\rho,\ \mathbf{J})$ — the **electromagnetic four-current**: $j^0 = c\rho$ (charge density scaled to have units consistent with current density), and $j^i = J^i$ (current density components). This is the Noether current of U(1) symmetry (Ch. 1 §1.7.3) — charge conservation is built into its very definition.

$\mu_0 = 4\pi \times 10^{-7}$ **H/m** — the permeability of free space (in SI units). The factor $1/\mu_0$ gives the correct normalization for SI electromagnetic fields.

$A_\mu j^\mu = -\phi\rho + \mathbf{A} \cdot \mathbf{J}$ — the coupling between the gauge field and the current: the source term that drives the fields.

11.2 — The Field Strength Tensor: E and B Unified

11.2.1 — Constructing $F_{\mu\nu}$

Using the spacetime metric $\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1)$ and the four-potential $A^\mu = (\phi/c, A_x, A_y, A_z)$:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

Computing each independent component using $\mathbf{E} = -\nabla\phi - \partial_t \mathbf{A}$ and $\mathbf{B} = \nabla \times \mathbf{A}$:

$$F_{0i} = \partial_0 A_i - \partial_i A_0 = -\frac{E_i}{c}$$

$$F_{12} = \partial_1 A_2 - \partial_2 A_1 = B_z, \qquad F_{23} = B_x, \qquad F_{31} = B_y$$

The complete field tensor in matrix form:

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix}$$

The contravariant version $F^{\mu\nu} = \eta^{\mu\alpha}\eta^{\nu\beta}F_{\alpha\beta}$ differs only in the sign of the E/c entries:

$$F^{\mu\nu} = \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & -B_z & B_y \\ -E_y/c & B_z & 0 & -B_x \\ -E_z/c & -B_y & B_x & 0 \end{pmatrix}$$

Physical meaning: A single antisymmetric tensor $F_{\mu\nu}$ encodes all six components of the electromagnetic field: three electric (E_x, E_y, E_z) and three magnetic (B_x, B_y, B_z). What one observer sees as purely electric, another observer moving relative to the first sees as a mixture of electric and magnetic. $F_{\mu\nu}$ is the observer-independent object; **E** and **B** separately are observer-dependent projections.

11.2.2 — The Lagrangian in Terms of E and B

$$-\frac{1}{4\mu_0}F_{\mu\nu}F^{\mu\nu} = \frac{1}{2}\left(\frac{\epsilon_0 E^2}{1} - \frac{B^2}{\mu_0}\right) = \frac{\epsilon_0}{2}\left(E^2 - c^2B^2\right)$$

using $\mu_0\epsilon_0 = 1/c^2$.

The electromagnetic Lagrangian density is proportional to $E^2 - c^2B^2$ — the Lorentz-invariant combination of the field magnitudes. This is not $E^2 + c^2B^2$ (the energy density); the Lagrangian is $T - V$, not $T + V$.

11.3 — Euler-Lagrange Descent: Covariant Maxwell Equations

11.3.1 — Applying the Field Euler-Lagrange Equation

From Ch. 1 §1.9.2, the field Euler-Lagrange equation for dynamical field A_ν:

$$\frac{\partial \mathcal{L}_{EM}}{\partial A_\nu} - \partial_\mu \frac{\partial \mathcal{L}_{EM}}{\partial(\partial_\mu A_\nu)} = 0$$

First term: $\frac{\partial \mathcal{L}_{EM}}{\partial A_\nu} = j^\nu$ (from the source term $A_\mu j^\mu$)

Second term: Using $\frac{\partial F_{\alpha\beta}}{\partial(\partial_\mu A_\nu)} = \delta^\mu_\alpha\delta^\nu_\beta - \delta^\mu_\beta\delta^\nu_\alpha$:

$$\frac{\partial \mathcal{L}_{EM}}{\partial(\partial_\mu A_\nu)} = -\frac{1}{\mu_0}F^{\mu\nu}$$

Substituting:

$$j^\nu - \partial_\mu\left(-\frac{1}{\mu_0}F^{\mu\nu}\right) = 0$$

$$\boxed{\partial_\mu F^{\mu\nu} = -\mu_0 j^\nu}$$

This single covariant equation contains two of Maxwell's four equations.

11.3.2 — The Bianchi Identity: Two More Equations for Free

Since $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, direct computation shows:

$$\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} = 0$$

This is the **Bianchi identity** — not an equation of motion, but a **mathematical identity** that holds for any antisymmetric tensor derived from a four-potential. It requires no variation, no Lagrangian — it is automatic from the definition $F = \partial \wedge A$.

The Bianchi identity contains the other two Maxwell equations.

The structure: one covariant equation from dynamics, one identity from geometry. Maxwell's four equations split exactly 2+2.

11.4 — The Four Maxwell Equations in 3D Form

11.4.1 — From $\partial_\mu F^{\mu\nu} = -\mu_0 j^\nu$

Component $\nu = 0$ (time component, using $j^0 = c\rho$):

$$\partial_i F^{i0} = -\mu_0 j^0 = -\mu_0 c\rho$$

$$-\frac{1}{c}\partial_i E^i = -\mu_0 c\rho \quad \Longrightarrow \quad \boxed{\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}}$$

Gauss's law for electricity. The divergence of $\mathbf{E}$ is sourced by free charge density.

Component $\nu = i$ (spatial components, $j^i = J^i$):

Expanding $\partial_0 F^{0i} + \partial_j F^{ji} = -\mu_0 J^i$:

$$\boxed{\nabla \times \mathbf{B} - \mu_0\epsilon_0 \frac{\partial \mathbf{E}}{\partial t} = \mu_0 \mathbf{J}}$$

The Ampere-Maxwell law. The curl of $\mathbf{B}$ is sourced by current density plus Maxwell's displacement current $\epsilon_0 \partial \mathbf{E}/\partial t$ — the term Maxwell added in 1865. This single addition made electromagnetism predict electromagnetic waves at speed $c = 1/\sqrt{\mu_0\epsilon_0}$, confirmed by Hertz in 1887.

11.4.2 — From the Bianchi Identity

Components $(\lambda, \mu, \nu) = (1, 2, 3)$**:**

$$\partial_1 F_{23} + \partial_2 F_{31} + \partial_3 F_{12} = 0 \quad \Longrightarrow \quad \boxed{\nabla \cdot \mathbf{B} = 0}$$

Gauss's law for magnetism. There are no magnetic monopoles in the Standard Model — and this is why. The equation $\nabla \cdot \mathbf{B} = 0$ is a mathematical consequence of writing $\mathbf{B} = \nabla \times \mathbf{A}$, which is guaranteed by the field tensor structure. (Note: if magnetic monopoles are discovered — possibly in the $\mathcal{F}[\ldots]$ sector — $\nabla \cdot \mathbf{B} = \mu_0 \rho_m$ would need to be added.)

Components $(\lambda, \mu, \nu) = (0, i, j)$**:**

$$\partial_0 F_{ij} + \partial_i F_{j0} + \partial_j F_{0i} = 0$$

In 3D vector form:

$$\boxed{\nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = 0}$$

Faraday's law. A time-varying magnetic field induces a circulating electric field. The source of electromotive force in every transformer, generator, and inductive sensor.

11.4.3 — All Four Maxwell Equations, Assembled

$$\begin{aligned}
\nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0} && \text{(Gauss — electric)} \\
\nabla \times \mathbf{B} &= \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} && \text{(Ampere-Maxwell)} \\
\nabla \cdot \mathbf{B} &= 0 && \text{(Gauss — magnetic)} \\
\nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} && \text{(Faraday)}
\end{aligned}$$

The counting: Four vector equations in 3D — but in covariant form, just one dynamical equation and one identity. The apparent complexity of Maxwell's equations in 3D is a coordinate artifact. In four-dimensional spacetime notation:

$$\partial_\mu F^{\mu\nu} = -\mu_0 j^\nu \qquad \partial_{[\lambda} F_{\mu\nu]} = 0$$

Two lines. All of classical electrodynamics.

11.5 — Charge Conservation: Noether's Theorem Completing the Circle

Taking the four-divergence of both sides of the dynamical equation:

$$\partial_\nu \partial_\mu F^{\mu\nu} = -\mu_0 \partial_\nu j^\nu$$

The left side vanishes identically because $\partial_\nu \partial_\mu$ is symmetric in (μ, ν) while $F^{\mu\nu}$ is antisymmetric: $\partial_\nu \partial_\mu F^{\mu\nu} = 0$.

Therefore:

$$\partial_\mu j^\mu = 0 \qquad \Longrightarrow \qquad \frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0$$

Charge conservation is built into Maxwell's equations. It is not an additional postulate. It follows from the antisymmetry of $F^{\mu\nu}$, which follows from $F = \partial \wedge A$, which follows from U(1) gauge invariance.

The chain from Ch. 0 to KCL:

1. Ch. 0: U(1) gauge symmetry in $\mathcal{L}_{SM}$
2. Ch. 1 Noether: U(1) symmetry → conserved charge, $\partial_\mu j^\mu = 0$
3. This chapter: $\partial_\mu j^\mu = 0 \to$ continuity equation $\partial_t \rho + \nabla \cdot \mathbf{J} = 0$
4. Quasi-static limit (§11.12): $\partial_t \rho \to 0 \to \nabla \cdot \mathbf{J} = 0 \to \sum I_{node} = 0$
5. Ch. 18 (Layer 3): Kirchhoff's Current Law

KCL is Noether's theorem for the **global** phase symmetry of charged matter, seen from the other end of a five-step descent.

11.6 — Gauge Invariance and the Electromagnetic Potentials

11.6.1 — Gauge Freedom

The field tensor $F_{\mu\nu}$ is invariant under:

$$A_\mu \to A_\mu + \partial_\mu \chi(\mathbf{r}, t)$$

for any smooth scalar function χ. Since $\mathbf{E}$ and $\mathbf{B}$ are determined by $F_{\mu\nu}$, they are also gauge-invariant. Physical observables cannot depend on the choice of χ.

This is the electromagnetic manifestation of U(1) gauge invariance from Ch. 0. The redundancy in A_μ (the gauge freedom) is the price of writing the theory in a way that makes Lorentz invariance and locality manifest.

11.6.2 — Gauge Choices

Lorenz gauge $\partial_\mu A^\mu = 0$ (Lorentz-covariant):

$$\Box A^\mu = -\mu_0 j^\mu, \qquad \Box = -\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \nabla^2$$

Each component of A^μ satisfies an independent wave equation driven by the corresponding component of j^μ. Manifest covariance; retarded potentials.

Coulomb gauge $\nabla \cdot \mathbf{A} = 0$ (non-covariant but practical):

$$\nabla^2 \phi = -\frac{\rho}{\epsilon_0} \quad \text{(instantaneous Coulomb potential)}$$

$$\Box \mathbf{A} = -\mu_0 \mathbf{J} + \frac{1}{c^2}\nabla\frac{\partial \phi}{\partial t}$$

The scalar potential is determined instantaneously by the charge distribution (no retardation); retardation enters only through $\mathbf{A}$. Useful for static and low-frequency problems.

11.7 — Electromagnetic Waves

11.7.1 — The Wave Equation in Vacuum

In free space ($\mathbf{J} = 0$, $\rho = 0$), the Lorenz gauge gives:

$$\Box A^\mu = 0 \quad \Longrightarrow \quad \left(-\frac{1}{c^2}\frac{\partial^2}{\partial t^2} + \nabla^2\right) A^\mu = 0$$

Plane wave solutions: $A^\mu = \epsilon^\mu \, e^{i(k_\nu x^\nu)} = \epsilon^\mu \, e^{i(\mathbf{k}\cdot\mathbf{r} - \omega t)}$

Dispersion relation: $k_\mu k^\mu = 0 \Rightarrow \omega^2 = c^2 |\mathbf{k}|^2$, so:

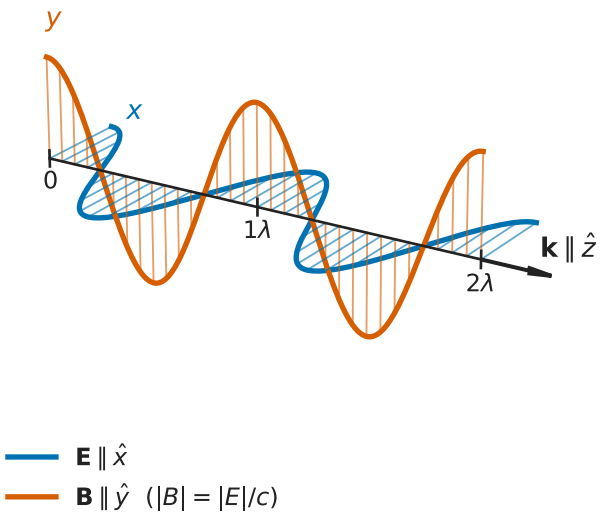
$$\boxed{c = \frac{\omega}{|\mathbf{k}|} = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 2.998 \times 10^8 \ \text{m/s}}$$

The speed of light falls out automatically — it is not put in by hand. Maxwell noticed this in 1865: the ratio $\sqrt{1/\mu_0\epsilon_0}$ equals the measured speed of light, and concluded that light must be an electromagnetic wave.

11.7.2 — Polarization: Two Physical Degrees of Freedom

The Lorenz gauge condition $k_\mu \epsilon^\mu = 0$ constrains ϵ^μ to be orthogonal to the wavevector — eliminating the time-like component. A residual gauge freedom $\epsilon^\mu \to \epsilon^\mu + k^\mu \Lambda$ removes one more degree of freedom, leaving exactly **two independent transverse polarization states**.

For a wave propagating in $\hat{z}$: ϵ_x (linear x-polarization) and ϵ_y (linear y-polarization). Or equivalently, right and left circular polarization. No longitudinal photon in vacuum — this is a consequence of U(1) gauge invariance.



Instantaneous fields of a linearly x-polarized vacuum plane wave over two wavelengths: $\mathbf{E} = E_0 \cos(kz - \omega t)\,\hat{x}$ and $\mathbf{B} = (E_0/c)\cos(kz - \omega t)\,\hat{y}$ oscillate in phase, both transverse to $\mathbf{k} \parallel \hat{z}$, with $\mathbf{E} \times \mathbf{B}$ along $\mathbf{k}$ — no longitudinal component appears.

Parallel: gravitational waves (Ch. 8) also have exactly two transverse polarizations ($+$ and $\times$), now from the diffeomorphism invariance of the metric. The number of physical polarizations is $d - (1 + 1) = d - 2$ (gauge condition plus residual gauge freedom) for a massless spin-1 field in d spacetime dimensions: 2 in 3+1D.

11.7.3 — The Electromagnetic Spectrum

The wave equation gives $\omega = ck$ for any frequency. The physical spectrum:

Band	Frequency	Wavelength	Engineering use
Radio (LF–UHF)	30 kHz–3 GHz	10 km–10 cm	Broadcast, radar, GPS
Microwave	3–300 GHz	10 cm–1 mm	WiFi, 5G, satellite, radar
Infrared	300 GHz–400 THz	1 mm–750 nm	Thermal imaging, fiber comms
Visible	400–750 THz	750–400 nm	LEDs, lasers, solar cells

Band	Frequency	Wavelength	Engineering use
UV	750 THz–30 PHz	400–10 nm	Lithography, sterilization
X-ray	30 PHz–30 EHz	10 nm–10 pm	Medical imaging, crystallography
Gamma	$>$ 30 EHz	$<$ 10 pm	Nuclear medicine, radiation

All governed by the same wave equation $\Box A^\mu = 0$ from one Lagrangian term.

11.8 — Energy and Momentum: Poynting's Theorem

11.8.1 — The Electromagnetic Energy-Momentum Tensor

From the Noether procedure (Ch. 1 §1.9.3), the energy-momentum tensor of the EM field:

$$T^{\mu\nu}_{EM} = \frac{1}{\mu_0}\left(F^{\mu\alpha}F^{\nu}{}_{\alpha} - \frac{1}{4}\eta^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}\right)$$

Components in 3D:

Energy density:

$$u = T^{00} = \frac{1}{2}\left(\epsilon_0 E^2 + \frac{B^2}{\mu_0}\right)$$

Poynting vector (energy flux density):

$$\mathbf{S} = \frac{1}{\mu_0}\mathbf{E}\times\mathbf{B} = c^2\epsilon_0\mathbf{E}\times\mathbf{B}$$

Maxwell stress tensor T^{ij} (momentum flux):

$$T^{ij} = \epsilon_0\left(E^iE^j - \frac{\delta^{ij}}{2}E^2\right) + \frac{1}{\mu_0}\left(B^iB^j - \frac{\delta^{ij}}{2}B^2\right)$$

11.8.2 — Poynting's Theorem

Conservation of EM energy (from $\partial_\mu T^{\mu 0} = -j^\mu F_{\mu 0}$):

$$\boxed{\frac{\partial u}{\partial t} + \nabla\cdot\mathbf{S} = -\mathbf{J}\cdot\mathbf{E}}$$

- Left side: rate of change of EM energy density + outward energy flux
- Right side: $-\mathbf{J}\cdot\mathbf{E}$ = power delivered to charges per unit volume (negative = EM field does work on charges; positive if charges do work on field)

For a conductor with $\mathbf{J} = \sigma\mathbf{E}$ (Ohm's law, from Ch. 13):

$$-\mathbf{J}\cdot\mathbf{E} = -\sigma E^2 < 0$$

The EM field continuously loses energy to the conductor as Joule heat σE^2 per unit volume.

The Poynting vector S tells you where the energy flows:

For a DC-carrying wire: $\mathbf{E}$ is parallel to the wire (driving the current); $\mathbf{B}$ circles around it. $\mathbf{S} = \mathbf{E} \times \mathbf{B}/\mu_0$ points **radially inward** toward the wire — the electrical power flows through the space *around* the wire, not through the wire itself. This counterintuitive but correct picture follows directly from Maxwell's equations.

Momentum density:

$$\mathbf{g} = \frac{\mathbf{S}}{c^2} = \epsilon_0(\mathbf{E} \times \mathbf{B})$$

Radiation pressure on a perfectly absorbing surface: $P_{rad} = u = (\epsilon_0 E^2 + B^2/\mu_0)/2$ — the energy density equals the radiation pressure. For sunlight at Earth's surface (~ 1400 W/m²): $P_{rad} \approx 4.7$ μPa — small but measurable, and crucial for spacecraft attitude control (solar sails, thermal radiation torque).

11.9 — Constitutive Relations: Connecting Layer 1 to Layer 2

Maxwell's equations are exact but describe vacuum. In matter, the response of charge distributions to applied fields must be added. In the macroscopic limit, this is done through **constitutive relations** — relationships between the macroscopic fields.

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} = \epsilon \mathbf{E} = \epsilon_0(1 + \chi_e)\mathbf{E}$$

$$\mathbf{H} = \frac{\mathbf{B}}{\mu_0} - \mathbf{M} = \frac{\mathbf{B}}{\mu} = \frac{\mathbf{B}}{\mu_0(1 + \chi_m)}$$

$$\mathbf{J}_f = \sigma \mathbf{E}$$

The permittivity ϵ, permeability μ, and conductivity σ are **not fundamental constants** — they are Layer-1 properties of the material that emerge from the quantum mechanics of electrons in the crystal lattice.

Quantity	Layer-1 origin	Engineering implication
$\epsilon(\omega)$	Electronic polarizability (Ch. 4 atoms, Ch. 6 bands)	Determines refractive index, capacitance, optical dispersion
$\mu(\omega)$	Spin magnetization (Ch. 5 Zeeman), orbital currents	Determines inductance, magnetic shielding
$\sigma(\omega)$	Kubo formula (Ch. 13): $\sigma = (e^2\tau/m)n$	Determines resistance, skin depth, loss

The Drude model gives $\sigma(\omega)$ for metals:

$$\sigma(\omega) = \frac{\sigma_0}{1 - i\omega\tau} = \frac{ne^2\tau/m}{1 - i\omega\tau}$$

At $\omega = 0$: DC conductivity $\sigma_0 = ne^2\tau/m$. At $\omega\tau \gg 1$ (high frequency): $\sigma \to -ine^2/(m\omega)$ — purely imaginary, giving reactive behavior. The plasma frequency $\omega_p = \sqrt{ne^2/m\epsilon_0}$ marks where $\mathrm{Re}[\epsilon(\omega)] = 0$ and the metal becomes transparent.

Frequency dispersion — Kramers-Kronig relations: The real and imaginary parts of any causal response function $\epsilon(\omega)$ are related by:

$$\mathrm{Re}[\epsilon(\omega)] - 1 = \frac{2}{\pi}\mathcal{P}\int_0^\infty \frac{\omega' \,\mathrm{Im}[\epsilon(\omega')]}{\omega'^2 - \omega^2} d\omega'$$

These follow from causality alone (the response cannot precede the stimulus) and are universal — satisfied by any physical $\epsilon(\omega)$, $\mu(\omega)$, or $\sigma(\omega)$. They are used to extract absorption spectra from reflectance measurements and to verify self-consistency of optical models.

11.10 — Electrostatics: The Static Limit of Two of Maxwell's Equations

When $\partial/\partial t = 0$ and no magnetic fields are present:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \qquad \nabla \times \mathbf{E} = 0$$

Since $\nabla \times \mathbf{E} = 0$, write $\mathbf{E} = -\nabla\phi$ (always possible for a curl-free vector field). Substituting into Gauss's law:

$$\boxed{\nabla^2 \phi = -\frac{\rho}{\epsilon_0}}$$

Poisson's equation for the electrostatic potential. For $\rho = 0$: Laplace's equation $\nabla^2\phi = 0$.

Solution via Green's function:

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0}\int \frac{\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}d^3r'$$

Boundary value problems: Unique solution to Poisson/Laplace with boundary conditions (Dirichlet: ϕ specified; Neumann: $\partial_n\phi$ specified). Standard methods: separation of variables (spherical harmonics from Ch. 4), image charges, conformal mapping.

Capacitance:

For a conductor system at potential V holding charge Q:

$$C = \frac{Q}{V}, \qquad U = \frac{Q^2}{2C} = \frac{1}{2}CV^2 = \frac{\epsilon_0}{2}\int E^2\, d^3r$$

The energy stored equals the volume integral of $\epsilon_0 E^2/2$ — consistent with the Poynting energy density formula.

11.11 — Magnetostatics: The Static Limit of the Other Two

When $\partial/\partial t = 0$ and no electric fields:

$$\nabla \cdot \mathbf{B} = 0, \qquad \nabla \times \mathbf{B} = \mu_0 \mathbf{J}$$

Since $\nabla \cdot \mathbf{B} = 0$, write $\mathbf{B} = \nabla \times \mathbf{A}$. In Coulomb gauge ($\nabla \cdot \mathbf{A} = 0$):

$$\nabla \times (\nabla \times \mathbf{A}) = \mu_0 \mathbf{J} \quad \Longrightarrow \quad -\nabla^2 \mathbf{A} = \mu_0 \mathbf{J}$$

Solution via Green's function:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi}\int \frac{\mathbf{J}(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}d^3r' \quad \Longrightarrow \quad \mathbf{B} = \frac{\mu_0}{4\pi}\int \frac{\mathbf{J}(\mathbf{r}')\times(\mathbf{r}-\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|^3}d^3r'$$

This is the **Biot-Savart law** — derived, not postulated.

Inductance:

For a circuit carrying current I with total flux linkage $\Lambda = N\Phi$:

$$L = \frac{\Lambda}{I} = \frac{N\Phi}{I}, \qquad U = \frac{1}{2}LI^2 = \frac{1}{2\mu_0}\int B^2\, d^3r$$

The energy stored equals the volume integral of $B^2/2\mu_0$.

11.12 — The Quasi-Static Limit: Bridge to Circuit Theory

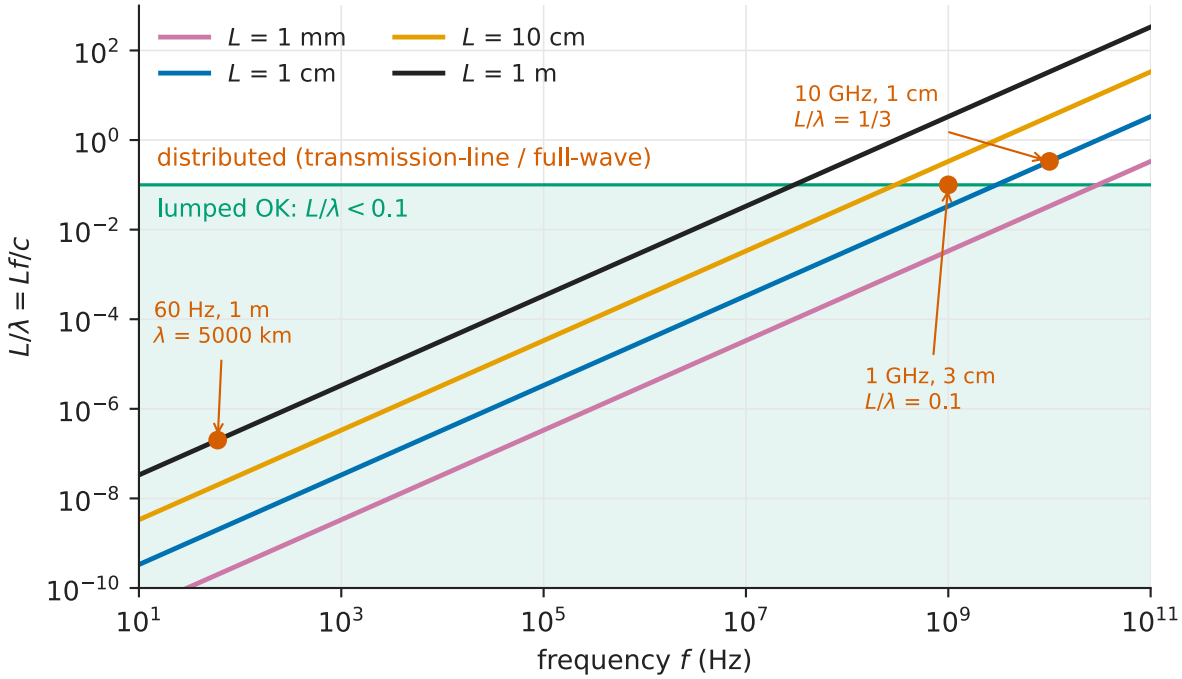
11.12.1 — The Lumping Criterion for Electromagnetism

The full Maxwell equations are PDEs. To use lumped-element circuit theory (Ch. 18), we need the **quasi-static approximation**: the system is small compared to the electromagnetic wavelength.

For a circuit of characteristic size L operating at frequency f:

$$\text{Quasi-static valid when: } \frac{L}{\lambda_{EM}} = \frac{Lf}{c} \ll 1$$

At 60 Hz (power systems), $\lambda = 5000$ km → valid for all practical circuits. At 1 GHz (RF), $\lambda = 30$ cm → a 3 cm device satisfies $L/\lambda = 0.1$ — marginally valid. At 10 GHz, $L = 1$ cm → $L/\lambda = 1/3$ — distributed effects essential.



Electrical size $L/\lambda = Lf/c$ from 10 Hz to 100 GHz for $L = 1$ mm, 1 cm, 10 cm and 1 m, with the lumped-element region $L/\lambda < 0.1$ shaded: a 1 m circuit at 60 Hz sits seven orders of magnitude inside it, a 3 cm device at 1 GHz sits exactly on the boundary, and a 1 cm structure at 10 GHz ($L/\lambda = 1/3$) is firmly distributed.

11.12.2 — Kirchhoff's Voltage Law from Faraday

In the quasi-static limit, consider a closed loop $\mathcal{C}$. Integrate Faraday's law over a surface $\mathcal{S}$ bounded by $\mathcal{C}$:

$$\oint_{\mathcal{C}} \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt} \iint_{\mathcal{S}} \mathbf{B} \cdot d\mathbf{S} = -\frac{d\Phi_B}{dt}$$

For a circuit with lumped inductors (where flux is concentrated in the inductors):

$$\sum_{elements} \int \mathbf{E} \cdot d\mathbf{l} = -\frac{d\Phi_B}{dt}$$

The flux term does not vanish; it is relocated. With the changing flux confined to the inductors (and negligible in the "empty" space between components), each inductor's share of $-d\Phi_B/dt$ is booked as its branch voltage $V_L = L\, di/dt$. With that bookkeeping, the loop sum of branch voltages, inductors included, is zero:

$$\sum_{\text{branches}} V_k = 0$$

This is **Kirchhoff's Voltage Law** — the integral form of Faraday's law in the quasi-static limit with lumped flux.

At DC ($d\Phi_B/dt = 0$): the right side vanishes exactly, giving the simple form $\sum V_k = 0$ without qualification.

11.12.3 — Kirchhoff's Current Law from Charge Conservation

From the continuity equation $\partial_t \rho + \nabla \cdot \mathbf{J} = 0$, integrate over the volume of a node:

$$\frac{dQ_{node}}{dt} = -\oint \mathbf{J} \cdot d\mathbf{S} = -\sum_k I_k$$

In the quasi-static limit, charge does not accumulate at a node on timescales of interest ($dQ_{node}/dt \to 0$ in steady state or low-frequency AC):

$$\sum_k I_k = 0$$

This is **Kirchhoff's Current Law** — charge conservation (Noether U(1)) applied to a lumped node.

The two Kirchhoff laws are the lumped-circuit balance relations obtained from charge conservation and Faraday's law under the appropriate quasi-static and lumped assumptions — not "exactly Maxwell's equations." The distinction is worth holding onto, because the circuit model has *already discarded something*: the spatial degrees of freedom of the field. The continuous $\nabla \cdot \mathbf{E}$ and $\nabla \times \mathbf{E}$ statements have been integrated over vanishing volumes, so what survives is a set of algebraic constraints on a finite set of node voltages and branch currents. That reduction is a real approximation with a real criterion (§11.12.1), and the word "exactly" erases it.

What *is* exact is each Kirchhoff law relative to the conservation law it comes from — KCL from charge conservation, KVL from Faraday's law (§11.12.2–3). What is approximate is the lumping that produces them. Every circuit calculation you will ever perform in Chapter 18 does trace back to the two lines of covariant EM theory in §11.3, but through an approximation rather than without one.

11.12.4 — Impedance from Maxwell

In phasor (frequency-domain) form, the relation between voltage and current for each passive element follows from Maxwell:

Resistor: $\mathbf{J} = \sigma \mathbf{E} \to V = IR$ (Ohm's law from Ch. 13 Kubo formula)

Capacitor: Gauss's law $\to Q = CV$, differentiated $\to I = C\, dV/dt \to Z_C = 1/i\omega C$

Inductor: Faraday's law $\to V = L\, dI/dt \to Z_L = i\omega L$

The impedance concept is the frequency-domain projection of Maxwell's equations onto a lumped circuit. AC circuit analysis (impedance, transfer functions, filters, resonance) is Layer-3 Maxwell, Noether, and Ohm packaged for daily engineering use.

11.13 — Electromagnetic Duality

Maxwell's equations in vacuum ($\rho = 0$, $\mathbf{J} = 0$) are symmetric under the **duality transformation**:

$$\mathbf{E} \to c\mathbf{B}, \qquad c\mathbf{B} \to -\mathbf{E}$$

or equivalently $F_{\mu\nu} \to \tilde{F}_{\mu\nu} = \frac{1}{2}\epsilon_{\mu\nu\rho\sigma}F^{\rho\sigma}$.

This symmetry is broken by the presence of electric charges (source for $\nabla \cdot \mathbf{E}$) but not magnetic charges (since $\nabla \cdot \mathbf{B} = 0$ — no magnetic monopoles). If magnetic monopoles exist (they appear in some GUT extensions in $\mathcal{F}$), the symmetry would be restored: $\nabla \cdot \mathbf{B} = \mu_0 \rho_m$, $\nabla \times \mathbf{E} = -\partial_t \mathbf{B} - \mu_0 \mathbf{J}_m$.

This is another instance where $\mathcal{F}$ of Ch. 0 would modify our equations — and where the phenomenology of Layer 2 (the presence or absence of magnetic monopoles) would directly reflect Layer-0 physics.

11.14 — Summary

Bridge B.c complete. Maxwell's equations descended from one Lagrangian term:

Result	Origin
$\mathcal{L}_{EM} = -\frac{1}{4\mu_0}F_{\mu\nu}F^{\mu\nu} + A_\mu j^\mu$	U(1) sector of $\mathcal{L}_{SM}$, classical field limit
$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$	Definition of field tensor
$\partial_\mu F^{\mu\nu} = -\mu_0 j^\nu$	Euler-Lagrange applied to A_ν
$\nabla \cdot \mathbf{E} = \rho/\epsilon_0$	$\nu = 0$ component
$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0\epsilon_0 \partial_t \mathbf{E}$	$\nu = i$ components
$\partial_{[\lambda} F_{\mu\nu]} = 0$	Bianchi identity (automatic from $F = \partial A$)
$\nabla \cdot \mathbf{B} = 0$	Bianchi, all-spatial
$\nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0$	Bianchi, one time index
$\partial_\mu j^\mu = 0$	Antisymmetry of $F^{\mu\nu}$ → automatic charge conservation
$\omega = ck$ (EM waves)	Wave equation in vacuum

Result	Origin
$c = 1/\sqrt{\mu_0 \epsilon_0}$	Speed of light from Lagrangian parameters
KVL: $\sum V_k = 0$	Faraday's law, quasi-static, lumped
KCL: $\sum I_k = 0$	Charge conservation, quasi-static, lumped node
$Z_C = 1/i\omega C$	Gauss's law → $Q = CV$ → differentiated
$Z_L = i\omega L$	Faraday's law → $V = L\, dI/dt$

11.15 — Engineering Thread

Physics	Application
Wave equation $\omega = ck$	Antenna design; waveguides; optical fiber; radar
Faraday's law	Transformers; generators; electric motors; wireless charging
Ampere-Maxwell (displacement current)	Why capacitors "conduct" at AC; RF circuit behavior
Poynting vector $\mathbf{S} = \mathbf{E} \times \mathbf{B}/\mu_0$	Power flow in transmission lines; electromagnetic power delivery
Poisson equation $\nabla^2 \phi = -\rho/\epsilon_0$	Capacitor design; electrostatic actuators; MEMS
Biot-Savart	Inductor and transformer design; MRI coil design
KVL (from Faraday, quasi-static)	All circuit analysis
KCL (from charge conservation)	All node analysis
$\epsilon(\omega), \mu(\omega)$	Dielectric material selection; antenna substrate; optical coatings
$\sigma(\omega)$ Drude + Kubo	Conductor sizing; skin depth; RF shielding design
Kramers-Kronig	Optical material characterization; loss estimation
Radiation pressure	Laser-based positioning; solar sail mission design

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11.16 — Looking Ahead

Chapters 8–11 have now completed four of the five Bridge B paths:

- B.a (Ch. 9): $\hbar \to 0$ → classical mechanics
- B.b (Ch. 10): $N \to \infty$ → thermodynamics
- B.c (Ch. 11): classical EM field limit of the $U(1)$ gauge theory → Maxwell's equations
- B.d (Ch. 8): weak-field GR → Newton's gravity

Chapter 12 uses the Maxwell equations derived here to develop electromagnetic wave propagation, optics, and photonics — applying the wave equation to the full EM spectrum and building toward the optical phenomena catalogued in Ch. 2.

Chapter 13 then performs Bridge B.e: applying the Kubo formula (quantum scattering + statistical averaging) to derive the constitutive relations $\mathbf{J} = \sigma\mathbf{E}$, $\mathbf{q} = -\kappa\nabla T$, and the full table of transport laws. This completes Bridge B and assembles the complete Layer-2 toolkit.

End of Chapter 11.

Next: Chapter 12 — Electromagnetic Waves, Optics, and Photonics

Electromagnetic Waves, Optics, and Photonics

One Wave Equation, the Entire Electromagnetic Spectrum

> *We have reason to conclude that light itself, including radiant heat and other radiations, is an electromagnetic disturbance in the form of waves propagated through the electromagnetic field according to electromagnetic laws.* — James Clerk Maxwell, *A Dynamical Theory of the Electromagnetic Field*, 1865

12.0 — Overview

Chapter 11 derived Maxwell's equations from the U(1) gauge sector of $\mathcal{L}_{SM}$. The vacuum wave equation fell out immediately: $\Box\mathbf{E} = 0$, predicting wave solutions propagating at $c = 1/\sqrt{\mu_0\epsilon_0}$.

This chapter does four things with that result:

1. **Extends** the wave equation to material media — showing how $\epsilon(\omega)$ and $\mu(\omega)$ modify propagation, giving dispersion, attenuation, and the refractive index
2. **Applies** Maxwell's boundary conditions to derive, not postulate, the laws of reflection, refraction (Snell's law), Brewster's angle, total internal reflection, and the Fresnel equations
3. **Identifies** waveguides as the bridge zone between Layer 2 (Maxwell's PDEs) and Layer 3 (lumped circuit theory) — the transmission line is the distributed- parameter model that sits between them
4. **Connects** to engineering photonics: optical fiber, lasers, integrated photonics, and nonlinear optics

Every named optical phenomenon from the Chapter 2 catalogue appears here with its derivation. Students who return to the catalogue after this chapter will find every optical entry fully grounded.

12.1 — Wave Equations in Material Media

12.1.1 — Maxwell in a Linear Isotropic Medium

In a homogeneous, isotropic, linear medium with constitutive relations $\mathbf{D} = \epsilon\mathbf{E}$, $\mathbf{B} = \mu\mathbf{H}$, $\mathbf{J}_f = \sigma\mathbf{E}$ (no free charges, $\rho_f = 0$), Maxwell's equations become:

$$\nabla \cdot \mathbf{E} = 0, \qquad \nabla \times \mathbf{H} = \sigma\mathbf{E} + \epsilon\frac{\partial \mathbf{E}}{\partial t}$$

$$\nabla \cdot \mathbf{H} = 0, \qquad \nabla \times \mathbf{E} = -\mu\frac{\partial \mathbf{H}}{\partial t}$$

Taking the curl of Faraday's law and using Ampere-Maxwell:

$$\boxed{\nabla^2 \mathbf{E} = \mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} + \mu\sigma \frac{\partial \mathbf{E}}{\partial t}}$$

This is the **electromagnetic wave equation in a lossy medium**:

- $\mu\epsilon\, \partial_t^2 \mathbf{E}$: wave propagation term (speed $v = 1/\sqrt{\mu\epsilon}$)
- $\mu\sigma\, \partial_t \mathbf{E}$: damping term from Ohmic loss ($\sigma > 0$)

12.1.2 — The Complex Wavenumber

For a monochromatic plane wave $\mathbf{E} = \mathbf{E}_0 e^{i(\tilde{k}z - \omega t)}$:

$$\tilde{k}^2 = \mu\epsilon\omega^2 + i\mu\sigma\omega = \mu\omega^2\left(\epsilon + i\frac{\sigma}{\omega}\right) \equiv \mu\omega^2\tilde{\epsilon}$$

Define the **complex permittivity** $\tilde{\epsilon} = \epsilon' - i\epsilon''$:

$$\tilde{\epsilon} = \epsilon + i\frac{\sigma}{\omega} = \epsilon_0\tilde{\epsilon}_r, \qquad \tilde{\epsilon}_r = \epsilon_r + i\frac{\sigma}{\omega\epsilon_0}$$

The complex wavenumber $\tilde{k} = \beta + i\alpha$ gives:

$$\mathbf{E}(z,t) = \mathbf{E}_0\, e^{-\alpha z}\, e^{i(\beta z - \omega t)}$$

- β **(phase constant):** determines the phase velocity $v_p = \omega/\beta$ and wavelength $\lambda = 2\pi/\beta$ in the medium
- α **(attenuation constant):** the field decays as $e^{-\alpha z}$; intensity decays as $e^{-2\alpha z}$
- **Skin depth:** $\delta = 1/\alpha$ — the depth at which the field amplitude falls to $1/e$ of its surface value

12.1.3 — The Complex Refractive Index

$$\tilde{n} = \sqrt{\tilde{\epsilon}_r \mu_r} = n + i\kappa$$

where n is the **refractive index** (determines phase velocity) and κ is the **extinction coefficient** (determines absorption):

$$v_p = \frac{c}{n}, \qquad \alpha = \frac{\omega\kappa}{c} = \frac{2\pi\kappa}{\lambda_0}$$

The **Beer-Lambert law** for intensity: $I = I_0 e^{-2\alpha z} = I_0 e^{-\mu_a z}$ where $\mu_a = 2\alpha$ is the absorption coefficient (m^{-1}).

Material classification by $\tilde{n}$:

Material type	n	κ	Example
Transparent dielectric	$n > 1$	$\kappa \approx 0$	Glass, water at optical freq.
Weakly absorbing	$n > 1$	$\kappa \ll n$	Optical glass with defects
Metal (optical freq.)	$n \sim \kappa \sim 1$–10	large	Cu, Al, Au at visible
Good conductor (RF)	$n = \kappa = \sqrt{\sigma/2\omega\epsilon_0}$	$n = \kappa \gg 1$	Copper at GHz

For a good conductor ($\sigma \gg \omega\epsilon$):

$$\alpha = \beta = \sqrt{\frac{\omega\mu\sigma}{2}} = \frac{1}{\delta}, \qquad \delta = \sqrt{\frac{2}{\omega\mu\sigma}}$$

This is the **skin depth** (derived again in Ch. 16 §16.8.1 as the magnetic-diffusion length) — confirming that the skin effect is a direct consequence of the complex refractive index of conductors.

12.2 — Dispersion: Phase and Group Velocity

12.2.1 — The Two Velocities

A monochromatic wave at frequency ω propagates at the **phase velocity** $v_p = \omega/k(\omega) = c/n(\omega)$.

A pulse — a superposition of frequencies — travels at the **group velocity**:

$$\boxed{v_g = \frac{d\omega}{dk} = \frac{c}{n + \omega\, dn/d\omega} = \frac{c}{n_g}}$$

where $n_g = n + \omega\, dn/d\omega$ is the **group index**.

When n is frequency-independent: $v_g = v_p = c/n$ (no dispersion). When $n = n(\omega)$: $v_g \neq v_p$ and different frequency components travel at different speeds — **dispersion**.

12.2.2 — Types of Dispersion

Normal dispersion ($dn/d\omega > 0$, $v_g < v_p$): occurs in most transparent materials at optical frequencies. A prism separates white light because the refractive index is higher at shorter wavelengths (blue light bends more than red).

Anomalous dispersion ($dn/d\omega < 0$, $v_g > v_p$): occurs near absorption resonances. The Sellmeier equation for glass:

$$n^2(\lambda) = 1 + \sum_i \frac{B_i \lambda^2}{\lambda^2 - C_i}$$

where (B_i, C_i) are material-specific constants fitted to absorption resonances.

12.2.3 — Pulse Spreading in Optical Fibers

A pulse of spectral width $\Delta\omega$ propagates through a fiber of length L. Different frequency components arrive at different times. The temporal spread:

$$\Delta\tau = L \left| \frac{d^2 k}{d\omega^2} \right| \Delta\omega = L|\beta_2|\Delta\omega$$

where $\beta_2 = d^2k/d\omega^2$ is the **group velocity dispersion** (GVD) parameter.

For standard SMF-28 fiber at 1550 nm: $\beta_2 = -21.7$ ps²/km. For a 10 Gb/s signal ($\Delta\omega \sim 60$ GHz), pulse spreads by ~ 13 ps/km — limiting the transmission distance before regeneration.

Dispersion management: Alternating sections of normal and anomalous dispersion fiber (dispersion-shifted or compensating fiber) can reduce β_2^{eff} to near zero over long spans.

12.3 — Boundary Conditions and the Laws of Optics

12.3.1 — Maxwell Boundary Conditions

At an interface between medium 1 and medium 2, integrating Maxwell's equations over a pillbox (normal to surface) and a loop (tangential to surface):

$$\boxed{\hat{n} \times (\mathbf{E}_2 - \mathbf{E}_1) = 0} \quad \text{(tangential E continuous)}$$

$$\boxed{\hat{n} \times (\mathbf{H}_2 - \mathbf{H}_1) = \mathbf{K}} \quad \text{(tangential H discontinuous by surface current)}$$

$$\boxed{\hat{n} \cdot (\mathbf{D}_2 - \mathbf{D}_1) = \sigma_f} \quad \text{(normal D discontinuous by surface charge)}$$

$$\boxed{\hat{n} \cdot (\mathbf{B}_2 - \mathbf{B}_1) = 0} \quad \text{(normal B always continuous)}$$

For non-conducting dielectric interfaces: $\mathbf{K} = 0$, $\sigma_f = 0$, and both tangential components of $\mathbf{E}$ and $\mathbf{H}$ are continuous.

12.3.2 — Snell's Law from Phase Matching

At a planar interface, the boundary conditions must be satisfied for all positions on the interface and for all time. This forces all three waves (incident, reflected, transmitted) to have the **same frequency** (obvious) and the **same tangential component of k** (phase matching):

$$k_1 \sin\theta_i = k_r \sin\theta_r = k_2 \sin\theta_t$$

Since $k = n\omega/c$ and $k_1 = k_r$ (same medium):

$$\theta_r = \theta_i \qquad \text{(law of reflection)}$$

$$\boxed{n_1 \sin\theta_i = n_2 \sin\theta_t} \qquad \text{(Snell's law)}$$

Snell's law follows from translational invariance parallel to the interface: the tangential component of wavevector is conserved. The section title says it — phase matching — and that is the direct route, because the boundary conditions have to hold at *every* point along the interface and at *every* time, which admits only a common tangential $\mathbf{k}$.

This *is* an optical analogue of a symmetry-conservation relation, and the connection to Noether's theorem is worth preserving in that weaker form. Continuous translation symmetry parallel to a planar interface is precisely what forces the phase-matching condition, so the structural shape is Noether-like: a continuous spatial symmetry $\Rightarrow$ a conserved quantity. But the conserved quantity carried here is the **tangential wavevector**, not a Noether current of a Lagrangian, and the derivation that produced the boxed equation above is the boundary-condition argument, not an application of Noether's theorem to an action. Invoking Noether directly is stronger than the derivation supports: there is no field-translation symmetry of the optical action being used to get this, and the wave amplitude in each medium is governed by a different wave equation.

So: the tangential $\mathbf{k}$ is conserved because the interface is translation-invariant along itself, and that conservation is what Snell's law encodes. The result is not empirical and it is not a coincidence — but the honest label is *phase matching arising from a geometric symmetry*, not *Noether's theorem*.

12.3.3 — Fresnel Equations

Applying the tangential boundary conditions with the phase-matching constraint, we get the **Fresnel amplitude coefficients** for each polarization:

s-polarization (TE: E field ⊥ to plane of incidence):

$$r_s = \frac{n_1 \cos\theta_i - n_2 \cos\theta_t}{n_1 \cos\theta_i + n_2 \cos\theta_t}, \qquad t_s = \frac{2n_1 \cos\theta_i}{n_1 \cos\theta_i + n_2 \cos\theta_t}$$

p-polarization (TM: E field in plane of incidence):

$$r_p = \frac{n_2 \cos\theta_i - n_1 \cos\theta_t}{n_2 \cos\theta_i + n_1 \cos\theta_t}, \qquad t_p = \frac{2n_1 \cos\theta_i}{n_2 \cos\theta_i + n_1 \cos\theta_t}$$

Reflectance and transmittance (power, accounting for beam cross-section change):

$$R_s = |r_s|^2, \quad R_p = |r_p|^2$$
$$T_s = \frac{n_2 \cos\theta_t}{n_1 \cos\theta_i}|t_s|^2, \quad T_p = \frac{n_2 \cos\theta_t}{n_1 \cos\theta_i}|t_p|^2$$

Energy conservation: $R + T = 1$ for each polarization (lossless interface).

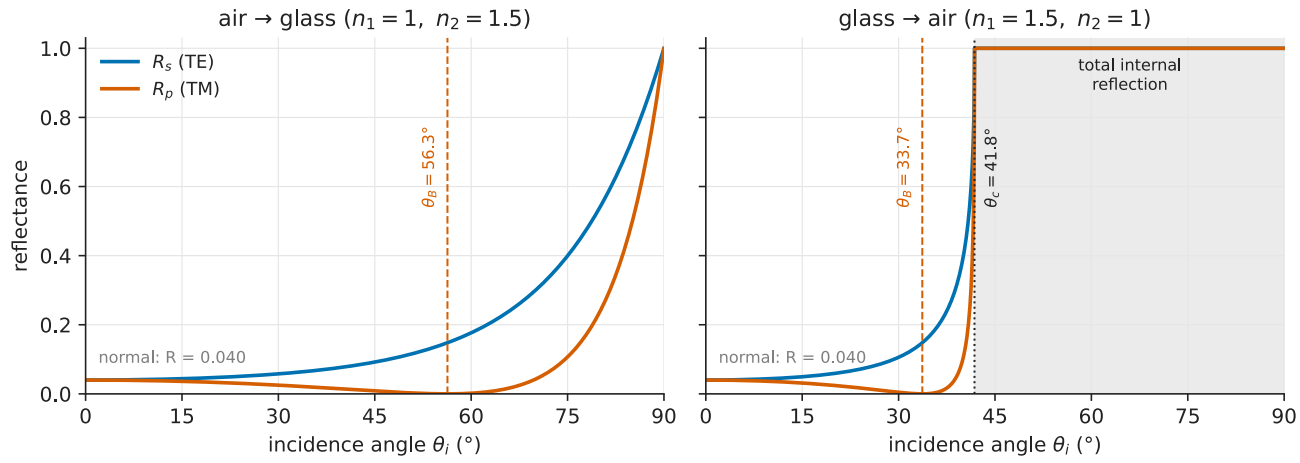
12.3.4 — Brewster's Angle

When $r_p = 0$: the numerator $n_2 \cos\theta_i - n_1 \cos\theta_t = 0$. Using Snell's law: this occurs when $\theta_i + \theta_t = 90^\circ$, giving:

$$\tan\theta_B = \frac{n_2}{n_1}$$

At Brewster's angle, the p-polarization is completely transmitted — reflected light is purely s-polarized. **Physical origin:** The refracted wave tries to radiate back as the reflected wave, but the oscillating dipoles in the medium cannot radiate in the direction of their own oscillation (a transverse dipole does not radiate along its axis). When the reflected and refracted beams are perpendicular, the dipoles cannot radiate into the reflection direction.

Engineering applications: Brewster windows in laser cavities eliminate reflection losses for one polarization; polarizing beam splitter cubes use Brewster-angle stacks; anti-glare coatings exploit Brewster's angle for horizontal polarization.



Power reflectances $R_s = |r_s|^2$ and $R_p = |r_p|^2$ versus incidence angle for $n = 1.5$ glass: from air (left) R_p vanishes at Brewster's angle $\theta_B = 56.3^\circ$ while both rise to 1 at grazing incidence; from inside the glass (right) $R_p = 0$ at $\theta_B = 33.7^\circ$ and both reach 1 at the critical angle $\theta_c = 41.8^\circ$, beyond which reflection is total (shaded); both panels start from the normal-incidence value $R = 0.04$.

12.3.5 — Total Internal Reflection

For $n_1 > n_2$ (denser medium), at incidence angle $\theta_i > \theta_c$:

$$\theta_c = \arcsin\left(\frac{n_2}{n_1}\right)$$

Snell's law gives $\sin\theta_t > 1$ — no real solution. The transmitted wave becomes **evanescent**: it decays exponentially into medium 2 with no time-averaged power transmitted.

The evanescent field:

$$\mathbf{E}_t \propto \exp\left(-\frac{z}{\delta_{ev}}\right)\exp(ik_x x), \qquad \delta_{ev} = \frac{\lambda_0}{2\pi\sqrt{n_1^2\sin^2\theta_i - n_2^2}}$$

- At $\theta_i = \theta_c$: $\delta_{ev} \to \infty$ (grazing transmission)
- For typical glass-air TIR at $\theta_i = 60^\circ$, $n_1 = 1.5$: $\delta_{ev} \approx 100$ nm

Frustrated TIR: Bringing a second medium within the evanescent decay length couples the evanescent field back into propagating waves — tunneling of light (cf. quantum tunneling, Ch. 4 §4.4, same mathematics). Used in ATR (attenuated total reflectance) spectroscopy and fiber coupling.

12.4 — Polarization

12.4.1 — The Jones Vector

The polarization state of a plane wave is described by the **Jones vector**:

$$\mathbf{J} = \begin{pmatrix} E_x \\ E_y \end{pmatrix}$$

where E_x and E_y are complex amplitudes. Standard polarization states:

Polarization	Jones vector
Linear (horizontal)	$(1,\ 0)^T$
Linear (vertical)	$(0,\ 1)^T$
Linear (45°)	$(1,\ 1)^T/\sqrt{2}$
Right circular	$(1,\ -i)^T/\sqrt{2}$
Left circular	$(1,\ +i)^T/\sqrt{2}$

12.4.2 — Waveplates and Birefringence

A birefringent medium has different refractive indices for orthogonal polarizations: $n_x \neq n_y$ (ordinary and extraordinary axes). A wave propagating through thickness d accumulates a **phase retardation**:

$$\delta = \frac{2\pi d}{\lambda_0}(n_e - n_o)$$

Quarter-wave plate ($\delta = \pi/2$): converts linear to circular polarization. Jones matrix: QWP $= e^{i\pi/4}\begin{pmatrix}1 & 0\\ 0 & -i\end{pmatrix}$

Half-wave plate ($\delta = \pi$): rotates linear polarization by 2α, where α is the angle between the fast axis and the input polarization. Jones matrix: HWP $= \begin{pmatrix}\cos 2\alpha & \sin 2\alpha\\ \sin 2\alpha & -\cos 2\alpha\end{pmatrix}$

Malus's law: Intensity transmitted through an analyzer at angle θ to a linearly polarized input:

$$I = I_0 \cos^2\theta$$

This follows from the Jones vector projection: $I \propto |E_x|^2 = |E_0 \cos\theta|^2$.

Engineering applications: LCD displays (electrically controlled birefringence to rotate polarization); ellipsometry (polarization analysis for thin film measurement); optical isolators (Faraday rotation + polarizers).

12.5 — Interference

12.5.1 — Two-Beam Interference

The superposition of two coherent waves with intensities I_1, I_2, and phase difference δ:

$$I_{total} = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\delta$$

Maximum ($\delta = 2m\pi$): $I_{max} = (\sqrt{I_1} + \sqrt{I_2})^2$ Minimum ($\delta = (2m+1)\pi$): $I_{min} = (\sqrt{I_1} - \sqrt{I_2})^2$

Visibility: $\mathcal{V} = (I_{max} - I_{min})/(I_{max} + I_{min})$ — measures contrast. For $I_1 = I_2$: $\mathcal{V} = 1$ (perfect contrast); $I_{min} = 0$.

The phase difference from an optical path difference ΔL:

$$\delta = \frac{2\pi}{\lambda_0}\Delta L = k_0 \Delta L$$

12.5.2 — Young's Double Slit

Two slits separated by distance d, at distance L from screen:

Bright fringes at: $y_m = m\lambda L/d$ (fringe spacing $\Delta y = \lambda L/d$)

The intensity pattern is the Fourier transform of the two-slit aperture:

$$I(\theta) = 4I_0 \cos^2\left(\frac{\pi d \sin\theta}{\lambda}\right)$$

From linearity of Maxwell: The superposition principle — which makes interference possible — is a direct consequence of Maxwell's equations being linear. Nonlinear media ($\chi^{(2)}, \chi^{(3)}$) modify this.

12.5.3 — Thin Film Interference

For a film of refractive index n, thickness t, at near-normal incidence:

$$\delta = \frac{4\pi n t}{\lambda_0} + \delta_{phase}$$

Phase shifts at reflection: π at low-to-high n interface; 0 at high-to-low. For a film (air–film–substrate with $n_{film} < n_{sub}$): both reflections have π phase shift, net phase shift = 0:

- **Destructive reflection** (anti-reflection): $2nt = (m + \frac{1}{2})\lambda_0$
- **Constructive reflection:** $2nt = m\lambda_0$

Quarter-wave anti-reflection coating: $n_f = \sqrt{n_1 n_2}$, $t = \lambda/4n_f$. For glass ($n = 1.5$) in air: optimal coating index $n_f = 1.22$ (MgF_2 has $n = 1.38$ — close enough for broadband AR).

High-reflectance stack: Alternating quarter-wave layers of high and low index: $R \to 1$ as the number of pairs increases. Distributed Bragg reflectors (DBR) in VCSELs use this principle.

12.5.4 — Fabry-Perot Etalon

Two partially reflecting mirrors of reflectance R separated by spacing d:

$$I_T = \frac{I_0}{1 + F\sin^2(\delta/2)}, \qquad F = \frac{4R}{(1-R)^2}$$

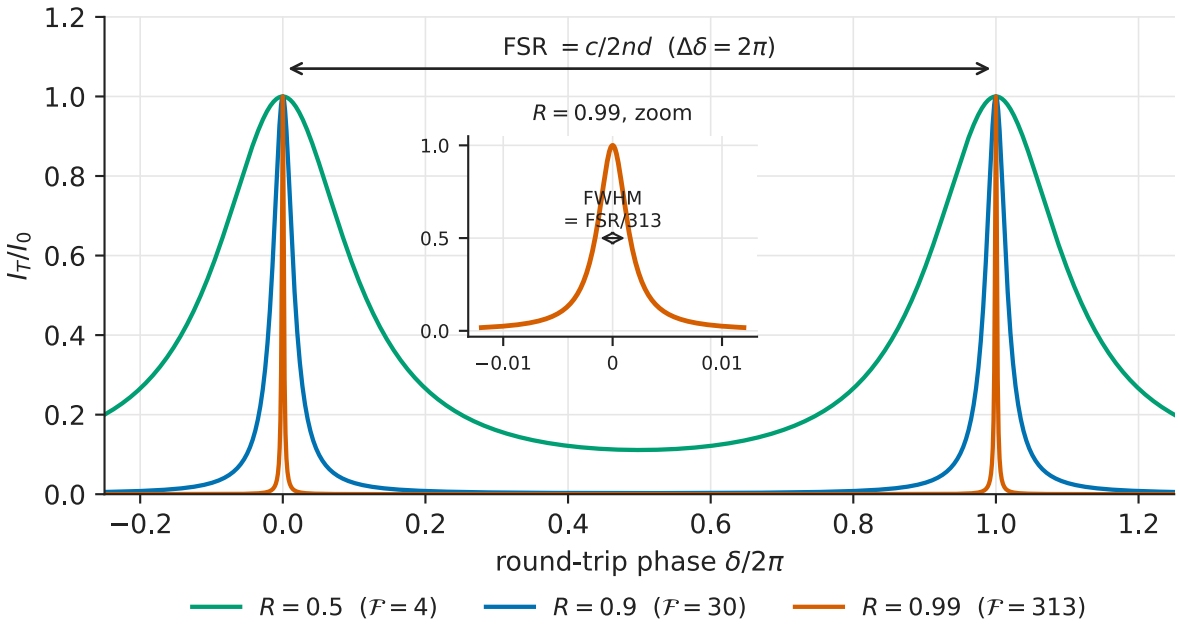
where $\delta = 4\pi n d\cos\theta/\lambda_0$ is the round-trip phase and F is the **coefficient of finesse**.

The resonances (full transmission) occur at $\delta = 2m\pi$:

$$\text{FSR} = \frac{c}{2nd} \quad \text{(free spectral range)}$$

Finesse: $\mathcal{F} = \pi\sqrt{F}/2 = \pi\sqrt{R}/(1-R)$ — the ratio of FSR to linewidth. For $R = 0.99$: $\mathcal{F} \approx 313$.

Resolving power: $\mathcal{R} = m\mathcal{F}$ — the Fabry-Perot can resolve wavelength differences $\Delta\lambda = \lambda/\mathcal{R}$.



Airy transmission $I_T/I_0 = 1/[1 + F\sin^2(\delta/2)]$ with $F = 4R/(1-R)^2$ for $R = 0.5$, 0.9 and 0.99 over one free spectral range ($\Delta\delta = 2\pi$, FSR $= c/2nd$): raising R leaves the peaks at full transmission but narrows them, and the inset shows that for $R = 0.99$ the FWHM is FSR/313, i.e. finesse $\mathcal{F} = \pi\sqrt{R}/(1-R) \approx 313$.

Engineering: Fabry-Perot cavities are the resonators in all lasers; optical spectrum analyzers use scanning FP etalons; wavelength-selective filters in DWDM fiber systems.

12.6 — Diffraction and Fourier Optics

12.6.1 — The Huygens-Fresnel Principle

Every point on a wavefront is the source of a secondary spherical wavelet. The field at any subsequent point is the coherent superposition of all wavelets — the Kirchhoff diffraction integral:

$$U(\mathbf{r}) = \frac{-i}{\lambda}\iint_{\Sigma} U(\mathbf{r}')\frac{e^{ik|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|}\cos\theta\, dS'$$

This follows from Green's second identity applied to the wave equation from Ch. 11 — not a new assumption.

12.6.2 — Fraunhofer Diffraction: The Fourier Transform

In the far field ($z \gg a^2/\lambda$, Fraunhofer regime), the Kirchhoff integral simplifies to a **Fourier transform** of the aperture function $t(x, y)$:

$$U(f_x, f_y) \propto \iint t(x,y)\, e^{-i2\pi(f_x x + f_y y)}dx\, dy = \mathcal{F}\{t\}$$

where $f_x = \sin\theta_x/\lambda$, $f_y = \sin\theta_y/\lambda$ are spatial frequencies.

Single slit (width a): $t(x) = \text{rect}(x/a)$

$$I(\theta) = I_0\left(\frac{\sin(\pi a\sin\theta/\lambda)}{\pi a\sin\theta/\lambda}\right)^2 = I_0\,\text{sinc}^2\left(\frac{a\sin\theta}{\lambda}\right)$$

First zero at $\sin\theta = \lambda/a$.

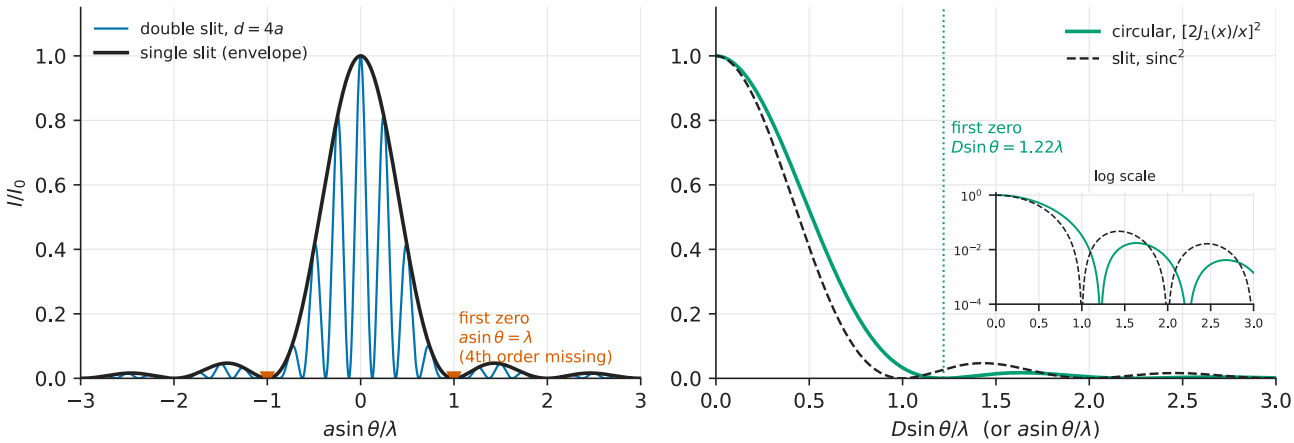
Circular aperture (diameter D): Fourier transform is an Airy function.

First zero at $\sin\theta = 1.22\lambda/D$ → **Rayleigh criterion** for angular resolution:

$$\boxed{\theta_{min} = 1.22\frac{\lambda}{D}}$$

A 1 m telescope at $\lambda = 550$ nm: $\theta_{min} = 0.14''$ of arc. A microwave radar dish of 3 m at 10 GHz ($\lambda = 3$ cm): $\theta_{min} = 1.2^\circ$.

Diffraction grating (N slits, spacing d): principal maxima at $d\sin\theta = m\lambda$, spectral resolving power $\mathcal{R} = mN$.



Left: single-slit pattern $\text{sinc}^2(a\sin\theta/\lambda)$ *with its first zeros at* $a\sin\theta = \pm\lambda$*, and the double slit with* $d = 4a$*, whose* $\cos^2(\pi d\sin\theta/\lambda)$ *fringes fill the sinc*^{2} *envelope with every fourth order missing; right: the circular-aperture Airy pattern* $[2J_1(x)/x]^2$*,* $x = \pi D\sin\theta/\lambda$*, whose first zero moves out to* $D\sin\theta = 1.22\lambda$ *(Rayleigh criterion), with weaker side lobes than the slit (log-scale inset).*

12.6.3 — Fourier Optics: A Lens as an Analog Computer

A converging lens of focal length f **computes the Fourier transform** of the field in its front focal plane, producing the transform in its back focal plane:

$$U_{back}(x', y') \propto \mathcal{F}\{U_{front}(x, y)\}_{f_x = x'/\lambda f,\, f_y = y'/\lambda f}$$

Two lenses in succession perform the inverse Fourier transform — restoring the original image. Between the two lenses, the **Fourier plane** contains the spatial frequency spectrum of the object: low frequencies near the center, high frequencies (fine detail) far from the center.

Spatial filtering (optical image processing):

- Block high spatial frequencies (low-pass filter) → blur the image
- Block low spatial frequencies (high-pass filter) → edge enhancement
- Block specific spatial frequencies → remove periodic noise
- All implemented with physical apertures in the Fourier plane

This is a direct connection to signal processing (Ch. 18): the same Fourier mathematics that describes digital filters here operates in analog light.

12.7 — Waveguides: The Bridge Zone

Waveguides sit in the bridge zone between Layer 2 (full Maxwell PDEs) and Layer 3 (lumped circuit elements). They retain spatial variation along the propagation direction while being "lumped" in the transverse directions.

12.7.1 — The Rectangular Metallic Waveguide

A metallic rectangular waveguide (width $a > b$) has perfect-conductor walls where $\mathbf{E}_{tangential} = 0$ (boundary condition from Maxwell).

The wave equation separates: $E_z = X(x)Y(y)e^{i(\beta z - \omega t)}$. Applying BCs:

$$\beta_{mn}^2 = \left(\frac{\omega}{c}\right)^2 - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2$$

A mode propagates (β real) only above its **cutoff frequency**:

$$f_{c,mn} = \frac{c}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

Dominant mode (TE₁₀): $m = 1, n = 0$; cutoff at $f_c = c/2a$.

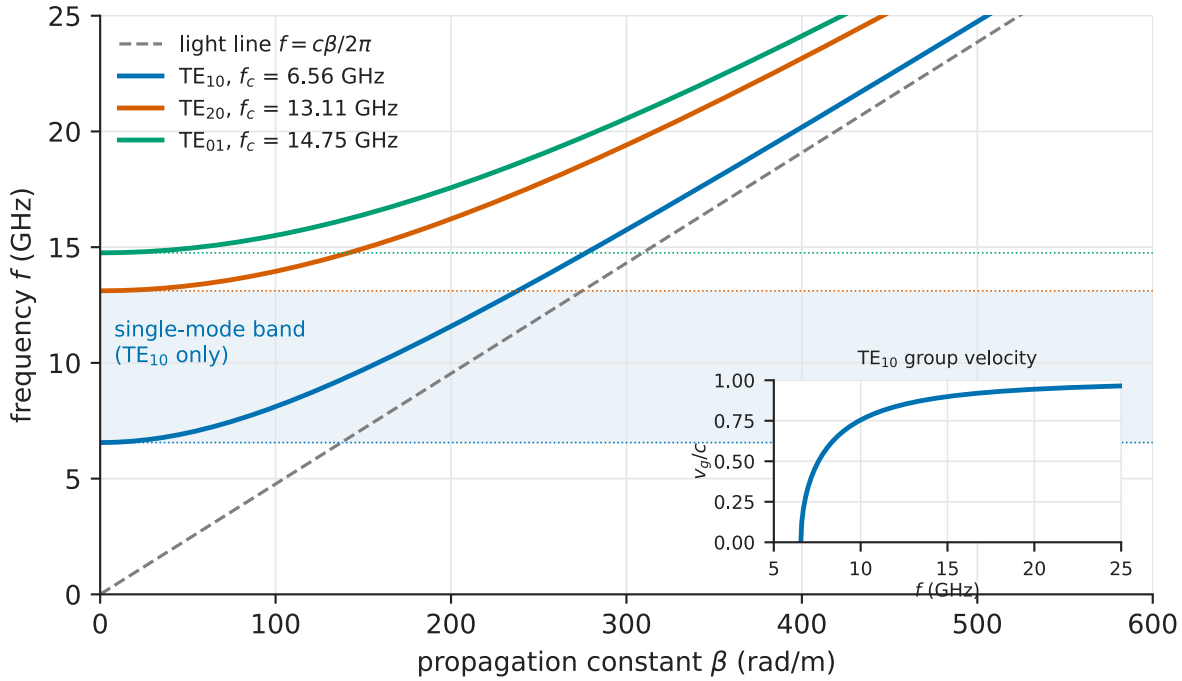
Below cutoff: β imaginary → exponential decay (evanescent). This is the waveguide analog of quantum tunneling (Ch. 4): the wave decays through the cutoff region as $e^{-|\beta|z}$ exactly as $e^{-\kappa d}$ in §4.4.

The Brillouin zone boundary in a crystal (Ch. 6) is the k-space analog of the cutoff: at the BZ boundary, Bragg reflection opens a gap just as the waveguide cutoff prevents propagation below f_c.

Group velocity in the waveguide:

$$v_g = c\sqrt{1 - (f_c/f)^2} < c$$

At $f \gg f_c$: $v_g \to c$. Near cutoff $f \to f_c^+$: $v_g \to 0$ (energy "stagnates").



Dispersion $f = (c/2\pi)\sqrt{\beta^2 + (m\pi/a)^2 + (n\pi/b)^2}$ for WR-90 ($a = 22.86$ mm, $b = 10.16$ mm): TE$_{10}$ cuts off at 6.56 GHz, TE$_{20}$ at 13.11 GHz and TE$_{01}$ at 14.75 GHz, so only TE$_{10}$ propagates between 6.56 and 13.11 GHz (shaded); every mode lies above the light line and approaches it at high frequency, and the inset shows the TE$_{10}$ group velocity $v_g/c = \sqrt{1 - (f_c/f)^2}$ falling to zero at cutoff.

12.7.2 — Optical Fiber: Guided Waves by Total Internal Reflection

A step-index optical fiber has a core (index n_1, radius a) surrounded by cladding ($n_2 < n_1$). Light launched at angle $\theta < \theta_{acc}$ from the axis undergoes TIR at the core-cladding interface and is guided.

Numerical aperture:

$$\text{NA} = n_0 \sin\theta_{acc} = \sqrt{n_1^2 - n_2^2}$$

V-number (normalized frequency):

$$V = \frac{2\pi a}{\lambda}\text{NA} = \frac{2\pi a}{\lambda}\sqrt{n_1^2 - n_2^2}$$

- $V < 2.405$: **single-mode fiber** (one guided mode, HE$_{11}$)
- $V \gg 2.405$: **multimode fiber** ($\sim V^2/2$ modes)

Single-mode fiber (SMF-28) at 1550 nm: $a = 4.1\,\mu$m, $\Delta n = 0.0036$, $V = 2.1$. One mode only — no modal dispersion.

Why 1550 nm? Silica fiber has minimum attenuation at 1550 nm ($\alpha \approx 0.2$ dB/km from Rayleigh scattering $\propto \lambda^{-4}$ plus infrared absorption). The entire global internet runs on 1550 nm light.

12.7.3 — Transmission Lines: EM Bridge Between Layers 2 and 3

A transmission line (coaxial cable, microstrip, twin-wire) is the most important bridge-zone model in electrical engineering. Derived by integrating Maxwell's equations over the cross-section, it retains spatial variation along z:

$$\frac{\partial V}{\partial z} + L'\frac{\partial I}{\partial t} + R'I = 0$$
$$\frac{\partial I}{\partial z} + C'\frac{\partial V}{\partial t} + G'V = 0$$

where L', C', R', G' are per-unit-length inductance, capacitance, resistance, and conductance — obtained from the static field solutions of Maxwell.

For a lossless line ($R' = G' = 0$):

$$v = \frac{1}{\sqrt{L'C'}} = \frac{c}{\sqrt{\epsilon_r \mu_r}}, \qquad Z_0 = \sqrt{\frac{L'}{C'}}$$

The transmission line equations are **Maxwell's equations in one dimension**: V plays the role of E and I plays the role of H. In the limit $z \to 0$ (lumped element), the partial derivatives become ordinary derivatives, and the transmission line equations collapse to KVL and KCL (Ch. 11 §11.12).

The bridge zone explicitly: A short line ($\ell \ll \lambda$): lumped. A long line ($\ell \sim \lambda$): distributed (transmission line equations). A very long line ($\ell \gg \lambda$): must be treated as a continuous medium (full Maxwell).

12.8 — Lasers and Coherent Light

12.8.1 — Population Inversion and the Laser Condition

From Ch. 5 §5 (Einstein A and B coefficients): in a medium with N_2 atoms in the upper level and N_1 in the lower, the **gain coefficient** for a wave at frequency $\nu_{21} = (E_2 - E_1)/h$:

$$g(\nu) = \frac{c^2}{8\pi\nu^2 n^2} A_{21}(N_2 - N_1) g(\nu)$$

For $N_2 > N_1$ (**population inversion**): $g > 0$ — the medium amplifies. For $N_2 < N_1$ (thermal equilibrium): $g < 0$ — the medium absorbs.

Population inversion requires pumping: A two-level system cannot achieve inversion (Einstein showed this in 1917). Practical lasers use three or four energy levels.

Threshold condition for lasing (round-trip gain equals round-trip loss):

$$2gL = 2\alpha_i L + \ln\frac{1}{R_1 R_2}$$

where L is cavity length, α_i is internal loss, $R_1 R_2$ is the product of mirror reflectances. Below threshold: spontaneous emission only. Above threshold: stimulated emission dominates and the output power rises sharply.

12.8.2 — Laser Properties

Spectral coherence: A laser oscillates on one or a few longitudinal cavity modes, separated by $\Delta\nu = c/2nL$ (FSR of the Fabry-Perot cavity from §12.5.4). Linewidth can be < 1 Hz for stabilized lasers — 10^{14} times narrower than thermal emission at the same frequency.

Spatial coherence: The cavity selects a transverse mode (usually TEM$_{00}$ — a Gaussian beam). The beam propagates with well-defined phase across its full diameter.

Directionality: Diffraction-limited beam divergence $\theta_{div} = \lambda/\pi w_0$, where w_0 is the beam waist. For a 1 mm beam at 633 nm: $\theta_{div} = 0.2$ mrad.

12.8.3 — Semiconductor Lasers

From Ch. 6 §6.6.3: a direct-gap semiconductor (GaAs, InP, GaN) under forward-bias injection has:

- Electrons pumped into the conduction band
- Holes pumped into the valence band
- Population inversion for frequencies near E_g/h

Double heterostructure: Sandwiching a thin active layer (e.g., GaAs, $E_g = 1.42$ eV) between wider-gap cladding layers (AlGaAs) confines both carriers (via potential wells) and the optical mode (via index step → waveguiding). This reduces the threshold current density by $\sim 1000\times$ compared to bulk homojunction lasers.

Quantum well lasers: Active layer ~ 10 nm thick — quantum confinement (Ch. 4 particle-in-a-box) shifts the energy levels and narrows the gain spectrum. Threshold current densities < 100 A/cm² routinely achieved.

Edge-emitting vs. VCSEL:

- Edge-emitting: cleaved facets form the FP cavity; beam from the edge
- VCSEL (Vertical-Cavity Surface-Emitting Laser): DBR mirrors top and bottom; beam from the surface; easier to test, cheaper, used in data centers

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12.9 — Nonlinear Optics

At high intensities, the polarization response becomes nonlinear:

$$\mathbf{P} = \epsilon_0 \left(\chi^{(1)}\mathbf{E} + \chi^{(2)}\mathbf{E}^2 + \chi^{(3)}\mathbf{E}^3 + \cdots \right)$$

12.9.1 — Second-Order Effects ($\chi^{(2)}$)

$\chi^{(2)} \neq 0$ only in **non-centrosymmetric crystals** (broken inversion symmetry, from Ch. 5 §5.8.3 and Ch. 6 bonding). The crystal symmetry requirement is exactly the same as for piezoelectricity.

Second harmonic generation (SHG): Input at ω, output at 2ω. Requires **phase matching** $k(2\omega) = 2k(\omega)$ for efficient conversion. Since $n(2\omega) \neq n(\omega)$ (dispersion), phase matching uses birefringence.

Power conversion (phase-matched, length L):

$$P_{2\omega} = \frac{\omega^2 d^2 L^2}{n^3 \epsilon_0 c^3 A} P_\omega^2$$

Proportional to P_ω^2 — a quadratic process. Efficiency increases with intensity and crystal length.

Engineering: 532 nm green lasers (used in laser pointers, projectors) are SHG of 1064 nm Nd:YAG lasers in KTP or LBO crystals.

12.9.2 — Third-Order Effects ($\chi^{(3)}$)

Present in **all materials** (including isotropic media, glasses, fibers — no symmetry requirement).

Kerr effect: $n = n_0 + n_2 I$ (intensity-dependent refractive index). For silica fiber: $n_2 = 2.6 \times 10^{-20}$ m²/W.

Self-phase modulation (SPM): A pulse propagating in a Kerr medium acquires an intensity-dependent phase $\phi(t) = -n_2 I(t) \omega L / c$. This broadens the spectrum of the pulse.

Optical soliton: When SPM (spectral broadening) exactly cancels the GVD (temporal broadening), the pulse shape is preserved indefinitely:

$$|E(t)| \propto \text{sech}(t/T_0)$$

Soliton communication systems exploit this balance for lossless (amplifier-compensated) dispersion-free propagation. They are solutions to the **Nonlinear Schrödinger Equation (NLSE)**:

$$\frac{\partial A}{\partial z} = -\frac{\alpha}{2}A - \frac{i\beta_2}{2}\frac{\partial^2 A}{\partial t^2} + i\gamma|A|^2 A$$

The NLSE is itself an engineering approximation to Maxwell's equations in a slowly-varying envelope approximation — a bridge zone between the full Maxwell PDE (Layer 2) and the simplified propagation model used in system design.

12.10 — Scattering

12.10.1 — Rayleigh Scattering

A small dielectric sphere ($a \ll \lambda$) in an electric field develops a dipole moment $\mathbf{p} = \alpha_{pol}\mathbf{E}$, where α_{pol} is the polarizability (from Ch. 4: the quantum-mechanical response of the electron cloud to an applied field).

The oscillating dipole radiates (Hertz dipole radiation):

$$\frac{dP_{rad}}{d\Omega} = \frac{\alpha_{pol}^2 \omega^4}{32\pi^2 \epsilon_0 c^3} \sin^2 \Phi$$

Total scattered power $\propto \omega^4 \propto \lambda^{-4}$: **Rayleigh scattering.**

- Blue light (450 nm): scattered 5.5× more than red (700 nm)
- Sky is blue: sunlight scattered by air molecules; N_2 and O_2 preferentially scatter short wavelengths toward the observer
- Sunsets are red: long path through atmosphere removes blue, leaving red
- Fiber attenuation: Rayleigh scattering from density fluctuations in silica limits attenuation to 0.2 dB/km floor at 1550 nm

12.10.2 — Raman Scattering

Inelastic scattering with energy exchange with a molecular vibration (phonon):

Stokes Raman: Photon loses energy $\hbar\omega_{vib}$ → phonon created $\nu_{Stokes} = \nu_{inc} - \nu_{vib}$

Anti-Stokes Raman: Photon gains energy $\hbar\omega_{vib}$ from a thermal phonon $\nu_{anti-Stokes} = \nu_{inc} + \nu_{vib}$

The Stokes-to-anti-Stokes ratio:

$$\frac{I_{aS}}{I_S} = \exp\left(-\frac{\hbar\omega_{vib}}{k_B T}\right)$$

This is directly the Boltzmann factor from Ch. 10 §10.3 — measuring the ratio directly gives the sample temperature (Raman thermometry).

Engineering: Raman spectroscopy identifies molecular species through vibrational fingerprints; distributed Raman sensing measures temperature along optical fibers by backscattered Stokes/anti-Stokes ratio; Raman amplifiers in fiber systems use stimulated Raman scattering to amplify signals at 1550 nm.

12.11 — The Doppler Effect for Electromagnetic Waves

For a source moving with velocity v relative to the observer, the relativistic Doppler shift (from Ch. 8 — special relativity, $g_{\mu\nu} = \eta_{\mu\nu}$):

$$\nu_{obs} = \nu_0 \sqrt{\frac{1 - v/c}{1 + v/c}} \approx \nu_0 \left(1 - \frac{v}{c}\right) \quad \text{(recession, } v \ll c\text{)}$$

The EM Doppler formula differs from the acoustic one (Ch. 9 §9.9) because light has no preferred medium — only the relative velocity between source and observer matters.

Radar Doppler: Transmitted frequency f_0, received frequency from target moving at v_r (radial component):

$$\Delta f = f_{Doppler} = \frac{2 v_r f_0}{c}$$

For X-band radar ($f_0 = 10$ GHz) and $v_r = 30$ m/s: $\Delta f = 2$ kHz. **Engineering:** Speed cameras, police radar, weather Doppler radar (measures precipitation velocity), air traffic control, automotive adaptive cruise control.

12.12 — Summary

One wave equation from Ch. 11 — extended to matter and systematically applied:

Result	Derivation
Wave equation in matter	Maxwell + constitutive relations: $\nabla^2 \mathbf{E} = \mu\epsilon \ddot{\mathbf{E}} + \mu\sigma \dot{\mathbf{E}}$
Complex refractive index $\tilde{n} = n + i\kappa$	Complex $\tilde{\epsilon} = \epsilon + i\sigma/\omega$
Skin depth $\delta = 1/\alpha$	Imaginary part of $\tilde{k}$ in good conductor
Phase vs. group velocity	$v_p = \omega/\beta$; $v_g = d\omega/dk$
Snell's law $n_1 \sin\theta_i = n_2 \sin\theta_t$	Phase matching at boundary; tangential $\mathbf{k}$ conserved by interface translation symmetry
Fresnel equations	Maxwell boundary conditions applied
Brewster's angle $\tan\theta_B = n_2/n_1$	$r_p = 0$ condition on Fresnel equation
TIR, evanescent field	$\sin\theta_t > 1$; field decays as $e^{-z/\delta_{ev}}$
Interference $I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\delta$	Superposition (linearity of Maxwell)
Fraunhofer diffraction = Fourier transform	Kirchhoff integral in far field
Rayleigh criterion $\theta_{min} = 1.22\lambda/D$	Airy disk first zero
Waveguide cutoff $f_c = c/2a$	TM/TE mode BCs in rectangular guide
TIR fiber, V-number	Total internal reflection + waveguide mode analysis
Transmission line equations	Maxwell equations averaged over cross-section
Laser threshold	Round-trip gain = round-trip loss
Semiconductor laser	Direct-gap + population inversion + FP or DBR cavity
SHG $\propto P_\omega^2$	$\chi^{(2)}$ nonlinear polarization, phase matching
Optical soliton	SPM balanced by GVD → NLSE
Rayleigh scattering $\propto \lambda^{-4}$	Hertz dipole radiation from α_{pol}
Raman: Stokes/anti-Stokes ratio $= e^{-\hbar\omega/kT}$	Boltzmann factor (Ch. 10)
EM Doppler $\Delta f = 2v_r f_0/c$	Relativistic phase shift

12.13 — Engineering Thread

Physics	Application
$v_p = c/n(\omega)$, dispersion	Fiber optic system design; pulse spreading; WDM channel spacing
Skin depth $\delta = \sqrt{2/\omega\mu\sigma}$	RF conductor sizing; shielding; induction heating
Fresnel equations + anti-reflection coatings	Camera lenses; solar panels; display screens
TIR in optical fiber	Global internet; endoscopes; optical interconnects
Rayleigh criterion $1.22\lambda/D$	Telescope, microscope, camera, satellite resolution
Waveguide cutoff	Microwave oven shielding (holes < cutoff); radar waveguides; MRI coils
Fabry-Perot cavity	All laser design; optical spectrum analyzers; DWDM filters
Semiconductor laser + fiber	Fiber communications; LIDAR; barcode scanners; laser machining
SHG	Green/blue lasers; OCT medical imaging; nonlinear characterization
Raman scattering	Material identification (pharmacy, security); fiber temperature sensing
Doppler radar	Traffic enforcement; weather forecasting; automotive safety
Transmission line $Z_0 = \sqrt{L'/C'}$	PCB trace impedance matching; RF amplifier design; coaxial systems
Soliton / NLSE	Long-haul undersea fiber transmission; ultrafast pulse compression

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12.14 — Looking Ahead

Chapter 12 has deployed Maxwell's equations across the electromagnetic spectrum, from waveguides to lasers to nonlinear photonics.

Chapter 13 performs Bridge B.e: the final Bridge B path. Starting from the quantum scattering theory of Ch. 6, applying the Kubo formula, and taking the statistical average gives the **generalized transport law** — Ohm's law, Fourier's law, Fick's law, and Newton's viscosity, all from one Green–Kubo linear-response scheme. Hooke's law enters Ch. 13 as well, but on a different footing: the elastic modulus is a static, nondissipative response coefficient relating stress to strain, so Hooke's law is a constitutive relation arising from static response rather than a fifth dissipative Green–Kubo transport coefficient. Ch. 13 draws that distinction explicitly. The conductivity σ that appeared in this chapter as a constitutive parameter will be derived from first principles in Ch. 13.

After Ch. 13, all five Bridge B paths are complete, and Layer 2 is fully assembled. Bridge C (Ch. 17) then begins the lumping operation that converts the PDEs of Layer 2 into the ODE systems of Layer 3.

End of Chapter 12.

Next: Chapter 13 — Bridge B.e: The Kubo Formula and the Generalized Transport Law

Bridge B.e — The Kubo Formula and the Generalized Transport Law

Five Constitutive Laws Through the Kubo / Linear-Response Framework

> **Note: Why not "one calculation"?** An earlier title claimed *Five Transport Laws from One Quantum Mechanical Calculation*. That is false as stated, and it is false in a way the chapter itself demonstrates. The five laws share one **response-theory structure**, but producing each one requires a different generalized force, a different current operator, a different constitutive closure, and — for C — a different limiting procedure entirely (§13.9.2). What this chapter delivers is a *framework* in which all five sit, not a single calculation that emits all five.

> *The Green-Kubo relations are among the most profound results in statistical mechanics. They say that the dissipation of a system driven out of equilibrium is completely determined by the spontaneous fluctuations of the same system at equilibrium.* — David Chandler, *Introduction to Modern Statistical Mechanics*

13.0 — Bridge B.e and the Convergence Chapter

This is the final Bridge B path — and the most important chapter in the book for making it discipline-agnostic.

The claim of Chapter 1 was that every engineering branch shares one mathematical template at Layer 2:

$$\text{flux} = -L \cdot \nabla\phi$$

The claim was that Ohm's law, Fourier's law, Fick's law, Newton's law of viscosity, and Hooke's law are the same formula with different labels.

This chapter delivers the framework. Four of the five response coefficients (σ, κ, D, η) are genuine **Green–Kubo transport coefficients**; the fifth, C, is a static susceptibility taken from the curvature of the free energy rather than from a current autocorrelation. Each coefficient requires its own generalized force, its own current operator, and its own closure. What is shared is the response structure, not the calculation:

Law	Generalized force $\hat{A}$	Current operator $\hat{B}$	Extra step needed
Ohm	$\hat{\mathbf{E}}$	$\hat{\mathbf{J}}$ (charge)	Drude/memory closure
Fourier	$\hat{\mathbf{T}}$ or $-\nabla T$	$\hat{\mathbf{J}}_Q$	same closure; Wiedemann–Franz adds an assumption
Fick	$\nabla\mu$	$\hat{\mathbf{J}}^N$	Einstein relation

Law	Generalized force $\hat{A}$	Current operator $\hat{B}$	Extra step needed
Newton viscosity	$\partial_j u_i$	$\hat{\sigma}_{ij}$ (transverse)	transverse projection
Hooke	ε_{kl}	$\delta\hat{\sigma}_{ij}$	**free-energy second derivative** — not a transport coefficient at all

`[STRUCTURAL CONNECTION]` — the unification is real and it is the book's organizing insight, but it is a shared *framework* claim, not a claim that one substitution yields five laws. The honest formulation: **linear-response theory gives a single template for dissipative response, and these five laws are instances of that template applied to five different observables — with the fifth, elasticity, only partially an instance.**

`[DERIVATION]` — *within* that framework, the Kubo relation itself is derived rather than assumed, but it carries boundary conditions that must be stated, because "delivers the proof" is doing more work than it should. Three assumptions carry the whole chapter:

1. **Linear response.** The system is perturbed weakly, $\hat{A} = \hat{A}_0 + \delta\hat{A}$ with $|\delta\hat{A}| \ll |\hat{A}_0|$, and the response is taken to first order in $\delta\hat{A}$. This is what produces the linear constitutive laws at all. Finite-amplitude transport is not covered (§13.12).
2. **Equilibrium initial condition.** The correlator $\langle\cdots\rangle_0$ is an *equilibrium* average. This is what produces the time integral, and hence the dissipative arrow of time described below.
3. **Time-translation invariance of the unperturbed state.** Without it, the $\langle J(t)J(0)\rangle_0$ in the Kubo formula is not a function of t alone.

Note also that the five results are not equally "the same". **Four** of them (σ, κ, D, η) are dissipative transport coefficients, obtained from the same dissipative part of the correlator — but η is built on a *transverse* current, the off-diagonal stress component $\hat{\sigma}_{xy}$, rather than on a conserved total current, which is why its Kubo formula is written with a factor of $1/V$ and a different relaxation time τ_v. The fifth, C in Hooke's law, is not a transport coefficient at all: it is a *zero-frequency, non-dissipative* static susceptibility, a second derivative of the free energy rather than an integral over a dissipative correlator. The unified template holds, but the members of the family differ in which part of the Kubo expression they use — a distinction that matters when the relaxation time goes to zero or the response becomes non-Markovian, and that is why "Hooke's law falls out of Kubo alongside Ohm's law" is a `[STRUCTURAL CONNECTION]` in the strict sense rather than a fifth instance of the same calculation.

Bridge B.e is also the bridge that injects irreversibility. Chapters 9 and 11 (classical mechanics and Maxwell's equations) are time-reversible: run them backward and you get equally valid physics. Ohm's law is not time-reversible — current flows from high to low potential, not randomly. The irreversibility comes from this chapter: quantum scattering and decoherence, averaged over an equilibrium ensemble, produce a finite relaxation time and a finite, dissipative conductivity.

Model Ledger — microscopic Hamiltonian to linear transport coefficient

Field	Content
Parent theory	Quantum many-body Hamiltonian $\hat{H}_0$ in thermal equilibrium (Ch. 5, Ch. 6)
Reduction	Weak perturbation; first order in the field; equilibrium initial state; replace the bath with a relaxation time τ (Drude closure)

Field	Content
Model	$L_{\alpha\beta} = \frac{1}{Vk_BT}\int_0^\infty \langle \hat{J}^\alpha(0)\hat{J}^\beta(t)\rangle_0\, e^{i\omega t}\, dt$; then $L = \frac{nq^2\tau}{m}$
Assumptions	linear response; equilibrium; ergodicity (so τ is a state variable, not history-dependent); elastic scattering only; **single relaxation time** (the Drude closure — the only modelling input); Boltzmann equation valid in the local, diffusive regime (fails for $L \lesssim \ell$, §13.4.4). Note: the result is derived via the linearized Boltzmann equation and the Fermi-surface δ-function, so it holds for degenerate electrons ($k_BT \ll E_F$) without a classical-equipartition assumption — see §13.4.2
Physics retained	dissipation, the arrow of time, the relation between charge/heat/momentum diffusion coefficients; Fermi-surface restriction of the current-carrying states
Physics neglected	inelastic and memory-dependent scattering, phonon drag, interactions (in the elastic limit), non-equilibrium distributions, band-structure anisotropy. The **free** Fermi gas is exact and gives $\sigma \to \infty$; all finite resistivity comes from the closure
Validity	diffusive regime $\ell \ll L_{\text{device}}$; $\omega\tau \ll 1$; well-defined τ
Fails when	$\ell \gtrsim L$ (ballistic/Landauer, §13.11), $\omega\tau \gtrsim 1$, strong correlations, or non-equilibrium driving
Next model	memory kernel / non-Markovian response; Landauer (ballistic); hydrodynamic transport; BTE (Ch. 19 for phonons)

Bridge	Path	Result
B.a	$\hbar \to 0$	Classical mechanics (Ch. 9)
B.b	$N \to \infty$	Thermodynamics (Ch. 10)
B.c	Classical EM field limit	Maxwell's equations (Ch. 11)
B.d	Weak-field GR	Newton's gravity (Ch. 8)
B.e	**Kubo averaging**	**Generalized Transport Law (Ch. 13)**

13.1 — Linear Response Theory: The Setup

13.1.1 — The Perturbation

Consider a system in equilibrium, described by density matrix $\hat{\rho}_0 = e^{-\beta\hat{H}_0}/Z$ (Ch. 10 §10.3). At $t = 0$, a weak time-dependent perturbation is switched on:

$$\hat{H}(t) = \hat{H}_0 - \hat{A}\,F(t)$$

where $\hat{A}$ is a quantum observable (the operator that couples to the perturbation) and $F(t)$ is the classical driving force (an electric field, temperature gradient, pressure difference, etc.).

Examples of coupling:

- Electric field: $\hat{A} = -e\hat{\mathbf{r}}$ (dipole coupling), $F = \mathbf{E}(t)$
- Temperature gradient: $\hat{A} = \hat{H}_0$ (energy), $F = \nabla T/T^2$
- Chemical potential gradient: $\hat{A} = \hat{N}$ (particle number), $F = -\nabla\mu/T$

13.1.2 — The Linear Response

To first order in F, the expectation value of a second observable $\hat{B}$:

$$\langle \hat{B}(t) \rangle = \langle \hat{B} \rangle_0 + \int_{-\infty}^{t} \chi_{BA}(t - t')\, F(t')\, dt'$$

where χ_{BA} is the **retarded Green's function** (generalized susceptibility):

$$\boxed{\chi_{BA}(t - t') = \frac{i}{\hbar} \Theta(t - t') \langle [\hat{B}(t), \hat{A}(t')] \rangle_0}$$

Θ is the Heaviside step function (enforcing **causality**: response cannot precede the perturbation). The commutator $[\hat{B}, \hat{A}]$ is computed in the Heisenberg picture with respect to $\hat{H}_0$, and $\langle \cdot \rangle_0$ is the equilibrium average.

In frequency space (Fourier transform):

$$\langle \hat{B}(\omega) \rangle = \tilde{\chi}_{BA}(\omega)\, F(\omega)$$

The response at frequency ω is proportional to the driving at the same frequency — this is linear response.

13.2 — The Kubo Formula

13.2.1 — Derivation Sketch

Starting from the von Neumann equation (Ch. 10 §10.1.2) for the perturbed system:

$$i\hbar \frac{\partial \hat{\rho}}{\partial t} = [\hat{H}_0 + \hat{H}', \hat{\rho}]$$

Write $\hat{\rho} = \hat{\rho}_0 + \delta\hat{\rho}$ and linearize in $\hat{H}'$:

$$i\hbar \frac{\partial\, \delta\hat{\rho}}{\partial t} = [\hat{H}_0, \delta\hat{\rho}] + [\hat{H}', \hat{\rho}_0]$$

Solving this first-order equation using the interaction picture and integrating:

$$\delta\hat{\rho}(t) = -\frac{i}{\hbar} \int_{-\infty}^{t} e^{-i\hat{H}_0(t-t')/\hbar} [\hat{H}'(t'), \hat{\rho}_0] e^{i\hat{H}_0(t-t')/\hbar} dt'$$

Substituting $\hat{H}' = -\hat{A}\, F(t')$ (which flips the overall sign to $+i/\hbar$), taking the trace with $\hat{B}$ and using the cyclic property of the trace and the equilibrium density matrix $[\hat{\rho}_0, e^{-\beta \hat{H}_0}] = 0$:

$$\chi_{BA}(\tau) = \frac{i}{\hbar} \Theta(\tau) \langle [\hat{B}(\tau), \hat{A}(0)] \rangle_0$$

This confirms §13.1.2. The transport coefficient L_{BA} (the DC, $\omega \to 0$ limit):

$$\boxed{L_{BA} = \frac{1}{V k_B T} \int_0^\infty \langle \hat{J}_B(0) \hat{J}_A(t) \rangle_0\, dt}$$

where $\hat{J}_A$, $\hat{J}_B$ are the current operators associated with observables $\hat{A}$, $\hat{B}$ (related by continuity equations), and V is the system volume. This is the **Green-Kubo formula** for transport coefficients.

13.2.2 — What the Formula Says

The transport coefficient L_{BA} — which tells you how much of current B flows in response to a force on A — equals the **time-integrated autocorrelation** of the equilibrium current fluctuations.

Three things to notice:

1. No perturbation appears in the formula. The right side is an equilibrium average — no external force, no out-of-equilibrium state. The coefficient that describes the dissipative response to a perturbation is entirely determined by the system's behavior at equilibrium. Dissipation and equilibrium fluctuations are two faces of the same physics.

2. The integral must converge for a finite transport coefficient to exist. If the current-current correlator decays to zero at long times (as it does in any real material with scattering), L_{BA} is finite. If the correlator never decays (perfect crystal at $T = 0$ with no defects), $L_{BA} \to \infty$ — the material is a perfect conductor. **Ohm's law requires scattering.**

3. The framework is exact; a given formula for a given coefficient is not automatically so. The derivation above made exactly one approximation — linear response. Within that, the Kubo relation is an identity. But "exact" attaches to the *relation*, not to any particular closed form for σ, κ, D, or η. Getting a usable expression for a specific coefficient additionally requires:

- choosing the correct observable–force pair $\hat{A}$, $\hat{B}$ and defining the current operator consistently (§13.4.1 shows this is where factors of e and V come from);
- an equilibrium state and a well-defined order of limits — $\omega \to 0$, $V \to \infty$, and $t \to \infty$ **do not always commute**, and dc conductivity is a case where the order matters;
- the thermodynamic limit, or an explicit finite-size treatment;
- boundary conditions, and for charged systems the treatment of **diamagnetic and contact terms**, which contribute a nondissipative (δ-function in ω) piece that must be separated from the dissipative part;
- handling **conservation laws** — a conserved current has a Drude weight, and a nondecaying correlator is a statement about a conserved quantity, not a divergence;
- deciding whether the coefficient wanted is dissipative or nondissipative, since only the dissipative part is positive and only that part is a "transport coefficient" in the Green–Kubo sense.

So the defensible statement is: **the Kubo linear-response formalism is exact within its domain of assumptions; specific transport-coefficient formulas require the appropriate equilibrium state, operator definitions, limiting procedure, and treatment of nondissipative/contact contributions.** "Valid for any quantum system, any temperature, any material" was an overclaim and has been removed.

13.3 — The Fluctuation-Dissipation Theorem

13.3.1 — Statement

The imaginary part of the susceptibility (which governs energy absorption from a periodic perturbation) is directly related to the spectral density of equilibrium fluctuations:

$$\boxed{\mathrm{Im}[\tilde{\chi}_{AA}(\omega)] = \frac{\omega}{2k_B T} S_A(\omega)}$$

where the **spectral density** (power spectrum of fluctuations):

$$S_A(\omega) = \int_{-\infty}^{\infty} \langle \hat{A}(t)\hat{A}(0)\rangle_0 \, e^{i\omega t}\, dt$$

> **Warning: This is the *classical* limit, not the general quantum relation** `[APPROXIMATION]` The boxed formula is valid when $\hbar\omega \ll k_B T$ — the **classical (high-temperature) regime**, where the quantum energy scale is negligible compared with thermal energy.
>
> The general quantum fluctuation-dissipation relation depends on whether the spectrum S_A is **symmetrized** or **unsymmetrized**, and the two normalizations differ. The S_A defined above is unsymmetrized (it carries the ordering $\langle \hat{A}(t)\hat{A}(0)\rangle$, not a commutator average), so the matching statement is
>
> $$\boxed{\mathrm{Im}[\tilde{\chi}_{AA}(\omega)] = \frac{1 - e^{-\hbar\omega/k_B T}}{2\hbar}\, S_A(\omega)}$$
>
> which is equivalently $S_A(\omega) = \dfrac{2\hbar}{1 - e^{-\beta\hbar\omega}} \mathrm{Im}[\tilde{\chi}_{AA}(\omega)]$. The **detailed-balance** relation that the equilibrium spectrum must satisfy is
>
> $$S_A(-\omega) = e^{-\hbar\omega/k_B T}\, S_A(\omega)$$
>
> so the same result can be written as the antisymmetric combination $\mathrm{Im}[\tilde{\chi}_{AA}] = \big[S_A(\omega) - S_A(-\omega)\big]/2\hbar$ — note the $1/2\hbar$, and note that the spectral factor belongs in the *numerator*.
>
> **Check the classical limit explicitly.** For $x = \hbar\omega/k_B T \ll 1$, $1 - e^{-x} \to x$, and the boxed general form gives $\frac{\hbar\omega}{2\hbar k_B T} S_A = \frac{\omega}{2k_B T} S_A$ — exactly the classical result above. In the deep quantum regime $x \gtrsim 1$ the factor $1 - e^{-x} \to 1$ saturates, so the classical form must not be used: zero-point fluctuations, cryogenic electronics, and any genuinely quantum regime all fall here. Since this chapter presents itself as *quantum* linear-response theory, the distinction is flagged rather than glossed.
>
> If one prefers the **symmetrized** spectrum $S_{sym} = \frac{1}{2}\int e^{i\omega t}\langle \hat{A}(0)\hat{A}(t) + \hat{A}(t)\hat{A}(0)\rangle\, dt$, the compact statement is $S_{sym}(\omega) = \hbar \coth\!\big(\frac{\hbar\omega}{2k_B T}\big)\mathrm{Im}[\tilde{\chi}_{AA}(\omega)]$, whose classical limit is again $\mathrm{Im}[\tilde{\chi}_{AA}] = \frac{\omega}{2k_B T} S_{sym}$: $\coth(x) \approx 1/x$. Both forms are standard; the only thing that matters is that S and the prefactor are quoted for the *same* convention.

Physical interpretation: The rate at which the system absorbs energy from an external drive at frequency ω is exactly determined by how strongly the system spontaneously fluctuates at the same frequency at equilibrium. Noise and dissipation are the same phenomenon.

13.3.2 — Johnson-Nyquist Noise

Applied to a resistor R at temperature T:

$$S_V(\omega) = 4k_B T\, \mathrm{Re}[Z(\omega)] \xrightarrow{\omega \to 0} 4k_B T R$$

The **open-circuit voltage noise** of a resistor: $\langle V^2 \rangle = 4k_B T R \Delta f$ in bandwidth Δf.

For $R = 1\,\text{M}\Omega$ at $T = 300\,\text{K}$ in $B = 10\,\text{kHz}$: $V_{rms} = \sqrt{4k_B TR\Delta f} = \sqrt{4 \times 1.38 \times 10^{-23} \times 300 \times 10^6 \times 10^4} = 12.9\,\mu\text{V}$

This is the **floor on voltage measurement** — regardless of amplifier quality, the resistor itself generates this noise. It sets the sensitivity limit of every resistive sensor, every voltmeter, every impedance measurement.

13.3.3 — Shot Noise

When current is carried by discrete particles (electrons tunneling one at a time), the current fluctuations follow Poisson statistics:

$$S_I = 2eI \qquad \text{(full shot noise)}$$

The **Fano factor** $F = S_I/2eI$ measures the deviation from Poissonian statistics. $F = 1$: independent tunnel events. $F < 1$: sub-Poissonian (fermion antibunching; Pauli exclusion supresses fluctuations). $F > 1$: super-Poissonian (bunching; correlated transport).

Engineering: Shot noise limits the SNR of photodetectors and PIN diodes. The Fano factor is used to characterize quantum transport regimes — a direct measurement of whether transport is diffusive, ballistic ($F = 0$), or correlated. The famous $F = 1/3$ suppression is the standard result for **coherent, elastic, single-particle diffusive transport in a many-channel quasi-one-dimensional conductor at zero temperature**; it is not a generic property of anything informally called "a 1D diffusive wire." It requires dephasing to be negligible over the wire, contacts to be fully equilibrating, spin degeneracy to be present, and the channel number large enough for the average to be self-averaging. Depart from those conditions and the number moves: finite temperature, spin-active or strongly correlated channels, long-range disorder, or a short wire with few channels all shift F away from $1/3$. Read F as a diagnostic whose value depends on the regime, not as a label attached to "diffusive."

13.4 — Deriving Ohm's Law

13.4.1 — The Kubo Conductivity

Operator convention first, because the factors of e and V depend on it. Two different objects are both called "current" in the literature, and conflating them is the usual source of a Drude derivation that loses an e^2:

Symbol	Object	Units	Definition
$\hat{\mathbf{J}}$	**total** charge current	C·m/s	$\hat{\mathbf{J}} = \sum_k \mathbf{j}_k$
$\hat{\mathbf{j}}$	**current density**	A/m²	$\hat{\mathbf{j}} = \hat{\mathbf{J}}/V$

For a free electron in band state k, with velocity $\mathbf{v}_k = \hbar\mathbf{k}/m_e$ and charge $-e$, the single-particle current is

$$\mathbf{j}_k = -e\,\mathbf{v}_k = -\frac{e\hbar}{m_e}\,\mathbf{k}$$

so the total current operator is

$$\hat{\mathbf{J}} = -\frac{e\hbar}{m_e}\sum_k \mathbf{k}\,\hat{c}_k^\dagger \hat{c}_k = -\frac{e}{m_e}\hat{\mathbf{P}}$$

with $\hat{\mathbf{P}} = \sum_k \hbar\mathbf{k}\,\hat{c}_k^\dagger \hat{c}_k$ the total momentum operator. **Note the sign** — the electron charge is $-e$ with $e > 0$, and it survives into the correlator as a positive e^2.

The conductivity **tensor** follows from the full retarded response. Writing the current–current correlator in spectral (imaginary-time) form, with $\beta = 1/k_B T$:

$$\sigma_{\alpha\beta}(\omega) = \frac{1}{V}\int_0^\infty dt\, e^{i\omega t}\int_0^\beta d\lambda\, \langle \hat{J}_\beta(-i\hbar\lambda)\hat{J}_\alpha(t)\rangle_0$$

Note the $1/V$: the prefactor converts a **total** current into a current **density**. That single factor is what keeps the conductivity's units at $(\mathrm{S/m})$.

For an isotropic system in the DC limit ($\omega \to 0$), the standard Green–Kubo reduction replaces the λ integral by a factor β (exact only if $\hat{\mathbf{J}}$ commutes with $\hat{H}_0$, i.e. a conserved current, or in the classical limit; otherwise an approximation valid when the current decays slowly on the scale $\hbar/k_B T$) and contracts the tensor with $\frac{1}{3}\delta_{\alpha\beta}$:

$$\sigma_{DC} = \frac{1}{3Vk_BT}\int_0^\infty \langle \hat{\mathbf{J}}(0)\cdot\hat{\mathbf{J}}(t)\rangle_0\, dt$$

The $\frac{1}{3}$ is the **isotropic average** $\langle k_\alpha k_\beta\rangle \to \frac{1}{3}k^2\delta_{\alpha\beta}$ — it is required because a conductor has no preferred direction. It is *not* a fudge factor, and the next step shows exactly what it does.

13.4.2 — The Drude Result

The honest structure of this result is a chain, and it is worth writing the chain down before filling in any of its links:

```
Kubo                          exact linear-response relation
  ↓
current-current correlator exact many-body object (not elementary)
  ↓
Boltzmann / memory function microscopic transport equation
  ↓
relaxation-time closure     ← the one modelling assumption
  ↓
Drude form                    σ = nₑe²τ/mₑ
```

Each arrow is a different kind of step. Only the first is "the framework".

Why the correlator cannot be skipped, and why it cannot be guessed

An earlier version of this section tried to evaluate $\langle \hat{J}^\alpha \hat{J}^\beta\rangle$ directly and reached the right answer through a wrong correlator. The failure is instructive, so it is worth recording what actually blocks the direct route.

Start from the one occupation relation that *is* elementary. Since a fermionic mode satisfies $n_k^2 = n_k$,

$$\langle \hat{n}_k\hat{n}_{k'}\rangle_0 = f_k f_{k'} + \delta_{kk'}\, f_k(1-f_k), \qquad f_k = \frac{1}{e^{\beta(\epsilon_k-\mu)}+1}$$

(Note what this is *not*: the variance $\mathrm{Var}(n_k) = f_k(1-f_k)$ governs number and charge fluctuations and shot noise. It is not the dc conductivity.)

But the correct number correlator does not deliver the current correlator. Two obstructions:

1. **A non-interacting gas has no resistivity at all.** For quadratic dispersion, $[\hat{\mathbf{P}}, \hat{H}_0] = 0$, so $\hat{\mathbf{J}} = -e\hat{\mathbf{P}}/m_e$ is conserved and $\langle \hat{\mathbf{J}}(0) \cdot \hat{\mathbf{J}}(t) \rangle_0$ is time-independent. The Green–Kubo integral of a constant **diverges**: $\sigma_{DC} \to \infty$, the perfect conductor. *That is the correct exact answer for a collision-free gas.* All finite resistivity is a statement about collisions, so the decay cannot be derived from the free gas.
2. **The classical shortcut is invalid here.** Conduction electrons in a metal are degenerate, $k_B T \ll E_F$, with typical speed v_F — *not* $\sqrt{k_B T/m_e}$. The familiar classical velocity autocorrelation $C_{vv}(t) = \frac{k_B T}{m_e} e^{-t/\tau}$ is built on equipartition and is simply **not a valid microscopic description of these electrons**.

> **Warning: A retraction worth stating explicitly** It is tempting to write the classical correlator, notice that its $k_B T$ cancels against the $1/k_B T$ in the Green–Kubo prefactor, and conclude the classical form must be fine. **That inference is invalid.** A factor that happens to cancel in the final answer does not repair an incorrect intermediate object. The correlator is a physical quantity in its own right — it governs shot noise, current fluctuations, and the full optical response — and it is wrong for degenerate electrons regardless of what survives division. Where the cancellation *is* genuinely useful is in justifying the classical *Boltzmann* treatment of transport (§13.6), not in licensing a classical correlator for a quantum gas.

So the correlator is a genuine many-body object: Fermi-surface geometry, occupation factors, band structure, and scattering matrix elements all enter. The efficient way to handle it is not to write it down, but to compute the transport coefficient from the kinetic equation that governs the distribution function.

The legitimate route: linearized Boltzmann transport

> **Note: Kubo does not imply Boltzmann — they are two branches** This is a **change of framework**, not a step inside Kubo. Kubo is a many-body response theory; the Boltzmann equation is a *kinetic* description of a distribution function, valid under additional assumptions (well-defined quasiparticles, weak correlations, separation of microscopic collision scales from macroscopic transport scales). The relationship is:
>
> ```
> Quantum many-body theory
>        │
>        ├── Kubo linear response          (exact, given linear response)
>        │
>        └── kinetic / semiclassical limit (requires the assumptions above)
>                ↓
>        Boltzmann equation
>                ↓
>        relaxation-time closure
>                ↓
>           Drude conductivity
> ```
>
> So `[DERIVATION]` below is scoped: it is a derivation *within the Boltzmann framework*, and that framework's own validity conditions are part of the claim.

`[DERIVATION]` **Charge convention for the rest of this subsection:** the electron charge is $-e$ with $e > 0$, so the semiclassical equation of motion is

$$\hbar\dot{\mathbf{k}} = -e\mathbf{E} \qquad \Longleftrightarrow \qquad \dot{\mathbf{k}} = -\frac{e\mathbf{E}}{\hbar}$$

Note this is a $1/\hbar$ times a **k-gradient**. (The familiar $1/m_e$ form, $\dot{\mathbf{v}} = -e\mathbf{E}/m_e$, is equivalent but must be paired with a **v-gradient**; mixing the $1/m_e$ coefficient with a $\nabla_{\mathbf{k}}$ is a units error.) Since $\mathbf{v}_{\mathbf{k}} = \frac{1}{\hbar}\nabla_{\mathbf{k}}\epsilon_{\mathbf{k}}$, the Boltzmann equation is

$$\frac{\partial f}{\partial t} - \frac{e\mathbf{E}}{\hbar}\cdot\nabla_{\mathbf{k}} f(\mathbf{k}) = \left(\frac{\partial f}{\partial t}\right)_{\text{coll}}$$

`[PHENOMENOLOGICAL]` The collision term is the **relaxation-time closure**: all scattering — electron–phonon, electron–impurity, electron–electron — is replaced by relaxation toward the *unperturbed* equilibrium f_0 with a single time constant τ (Ch. 6 §6.7.3),

$$\left(\frac{\partial f}{\partial t}\right)_{\text{coll}} = -\frac{f - f_0}{\tau} = -\frac{\delta f}{\tau}$$

(More refined collision models relax toward a *displaced* local equilibrium δf_{eq} — the displaced-Dirac-distribution or shift model of hot-electron transport. That sophistication is not needed here and is deliberately omitted rather than half-introduced.)

Linearizing, $f = f_0 + \delta f$ with $|\delta f| \ll f_0$, and taking the steady state $\partial_t \delta f = 0$:

$$-\frac{e\mathbf{E}}{\hbar}\cdot\nabla_{\mathbf{k}} f_0 = -\frac{\delta f}{\tau}$$

Using $\nabla_{\mathbf{k}} f_0 = \frac{\partial f_0}{\partial \epsilon_{\mathbf{k}}}\nabla_{\mathbf{k}}\epsilon_{\mathbf{k}} = \hbar\,\mathbf{v}_{\mathbf{k}}\,\frac{\partial f_0}{\partial \epsilon_{\mathbf{k}}}$:

$$\delta f(\mathbf{k}) = e\tau\,(\mathbf{E}\cdot\mathbf{v}_{\mathbf{k}})\,\frac{\partial f_0}{\partial \epsilon_{\mathbf{k}}}$$

Since $\partial f_0/\partial\epsilon_{\mathbf{k}} < 0$, this correctly *depletes* states moving along $\mathbf{E}$ — which is the right sign for electrons, whose drift is opposite to $\mathbf{E}$.

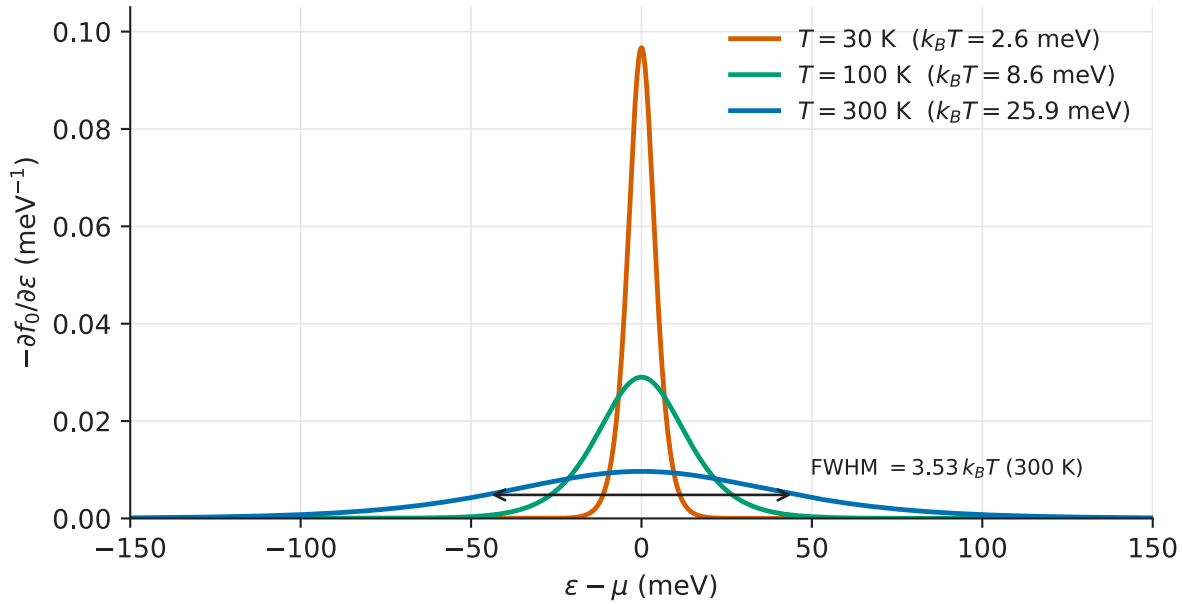
The current density carries the electron charge $-e$:

$$\mathbf{J} = -2e\int \mathbf{v}_{\mathbf{k}}\,\delta f(\mathbf{k})\,\frac{d^3k}{(2\pi)^3} = 2e^2\tau\int \frac{d^3k}{(2\pi)^3}\,(\mathbf{E}\cdot\mathbf{v}_{\mathbf{k}})\left(-\frac{\partial f_0}{\partial\epsilon_{\mathbf{k}}}\right)\mathbf{v}_{\mathbf{k}}$$

Both signs are now correct and they work together: $-\partial f_0/\partial\epsilon_{\mathbf{k}} > 0$ and the angular average of $(\mathbf{E}\cdot\mathbf{v})\mathbf{v}$ is along $+\mathbf{E}$, so $\mathbf{J} \parallel +\mathbf{E}$, as Ohm's law requires.

This is where degeneracy is handled correctly. The Fermi–Dirac derivative becomes a surface delta function as $T \to 0$:

$$-\frac{\partial f_0}{\partial\epsilon_{\mathbf{k}}} = \frac{1}{4k_BT}\operatorname{sech}^2\!\left(\frac{\epsilon_{\mathbf{k}} - \mu}{2k_BT}\right) \xrightarrow[T\to 0]{} \delta(\epsilon_{\mathbf{k}} - \mu)$$



The Fermi window $-\partial f_0/\partial\epsilon = (4k_BT)^{-1}\,\mathrm{sech}^2[(\epsilon-\mu)/2k_BT]$ at $T = 30$, 100 and 300 K: each curve has unit area and a full width of $3.53\,k_BT$ (about 91 meV at room temperature), so only electrons within a few k_BT of μ — a sliver of a metal's several-eV Fermi sea — carry current, and the window sharpens toward $\delta(\epsilon-\mu)$ as $T \to 0$.

So the current is carried by states
on the Fermi surface
— which is the actual physical content of conduction in a degenerate metal, and something the classical correlator cannot express at all. Performing the angular average,

$$2\int \frac{d^3k}{(2\pi)^3}\,\delta(\epsilon_{\mathbf{k}}-\mu)\,v^\alpha v^\beta = \frac{n_e}{m_e}\,\delta_{\alpha\beta}$$

(The factor 2 counts spin, since n_e includes both spin states. Angular average gives $2\int(\mathbf{E}\cdot\mathbf{v})\mathbf{v}(-\partial f_0/\partial\epsilon)\frac{d^3k}{(2\pi)^3} = \frac{\mathbf{E}}{3}\,2\int v^2\delta(\epsilon_{\mathbf{k}}-\mu)\frac{d^3k}{(2\pi)^3}$, and the radial integral is $2\int\frac{d^3k}{(2\pi)^3}\delta(\epsilon_{\mathbf{k}}-\mu)\,v^2 = \frac{3n_e}{m_e}$ using $v_F^2 = \hbar^2k_F^2/m_e^2$ and $n_e = k_F^3/3\pi^2$ — so the angular $1/3$ and the radial factor of 3 cancel exactly.) Therefore

$$\mathbf{J} = \frac{n_e e^2\tau}{m_e}\,\mathbf{E} \quad\Longrightarrow\quad \boxed{\sigma_{DC} = \frac{n_e e^2\tau}{m_e}}$$

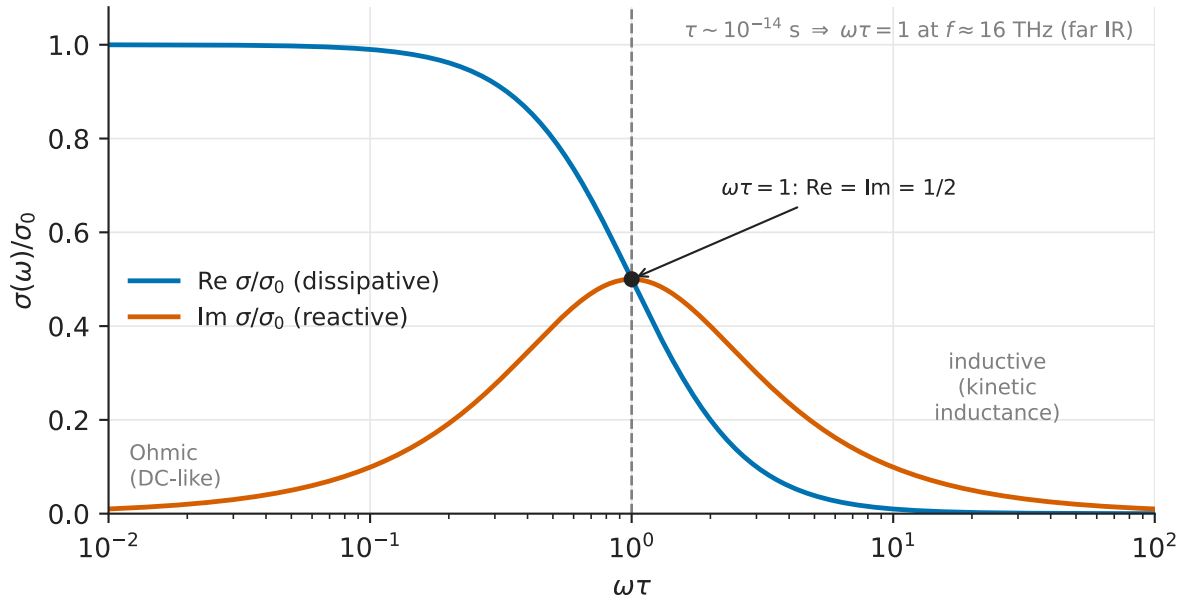
The e^2 because the current is *charge* current; n_e because only states in a shell at the Fermi surface respond; τ because it is the closure parameter. The mean free path is $\ell = v_F\tau$, so

$$\rho = \frac{1}{\sigma} = \frac{m_e}{n_e e^2\tau} = \frac{m_e v_F}{n_e e^2\ell}$$

Frequency dependence. Retaining $\partial_t\delta f$ gives the Drude form

$$\sigma(\omega) = \frac{n_e e^2\tau}{m_e}\cdot\frac{1}{1-i\omega\tau}, \qquad \sigma_0 = \frac{n_e e^2\tau}{m_e}$$

At $\omega\tau \ll 1$: purely real, Ohmic. At $\omega\tau \gg 1$: purely imaginary, reactive. The crossover at $\omega \sim 1/\tau$ therefore marks where the Drude response stops being predominantly **dissipative** and becomes predominantly **reactive**. For typical metals $\tau \sim 10^{-14}$ s, so this sits in the far infrared.



Drude conductivity $\sigma(\omega)/\sigma_0 = 1/(1 - i\omega\tau)$: the dissipative part $\mathrm{Re}\,\sigma/\sigma_0 = 1/(1+\omega^2\tau^2)$ is flat (Ohmic) for $\omega\tau \ll 1$ and falls as $(\omega\tau)^{-2}$ beyond, the reactive part $\mathrm{Im}\,\sigma/\sigma_0 = \omega\tau/(1+\omega^2\tau^2)$ peaks at the crossover $\omega\tau = 1$ where the two are equal at $1/2$ — about 16 THz for $\tau = 10^{-14}$ s, well below the plasma frequency.

> **Warning: $1/\tau$ is *not* the optical transparency frequency** It is tempting to read the Drude crossover as the frequency at which a metal stops reflecting and becomes transparent. That is wrong, and conflating the two is a common error. Reflectivity and transparency are governed principally by the **plasma frequency**
>
> $$\omega_p = \sqrt{\frac{n_e e^2}{\varepsilon_0 m_e^*}}$$
>
> together with **interband** transitions. For typical metals $\omega_p \sim 10^{15}$–10^{16} Hz (visible to UV), roughly one to two orders of magnitude *above* $1/\tau$. Below ω_p the free-electron gas cannot respond fast enough to screen the field, so metals are strongly reflective — which is why metals look shiny well past the infrared Drude crossover. Metals only become transparent in the thin-film or narrow-gap sense when the film is thinner than the skin depth, or when interband absorption closes the gap. **The correct statement:** $\omega\tau \sim 1$ marks the crossover from predominantly dissipative to predominantly reactive Drude response. Optical reflectivity and transparency additionally depend on ω_p and on interband contributions.

The classical Drude model, and why it agrees

The 1900-era Drude model treats electrons as classical billiard balls between collisions. It gets the same answer — and it is *not* the same argument. In that picture one writes $\mathbf{v}(t) = \mathbf{v}(0)e^{-t/\tau}$ for a tagged particle and uses equipartition for $\langle v^2 \rangle$, which is legitimate *for that model*. The two routes converge because at $T \to 0$ the only thing τ needs to encode is momentum relaxation, and both the classical and the quantum descriptions deliver it.

The agreement is the reassuring part: the Drude formula is robust to the choice of description, because the Fermi-surface δ-function of the quantum treatment and the momentum-relaxation assumption of the classical treatment encode the same physics. What is *not* legitimate is borrowing the classical model's equipartition correlator and presenting it as the quantum result.

> **Warning: Epistemic summary**
>
> | Step | Status |
> |---|---|
> | Kubo relates correlator to response | `[DERIVATION]` — exact given linear response |
> | Free Fermi gas gives $\sigma \to \infty$ | `[DERIVATION]` — exact, and the reason a closure is needed |
> | Current correlator from Fermi-surface integrals | `[DERIVATION]` via linearized Boltzmann |
> | Single relaxation time τ | `[PHENOMENOLOGICAL]` — **the closure**; the only modelling input |
> | Classical $C_{vv} = \frac{k_B T}{m} e^{-t/\tau}$ | **not used** — invalid for degenerate electrons |
> | $\sigma = n_e e^2 \tau / m_e$ | `[DERIVATION]` within the closure |
>
> The framework is exact, the closure is a model, and the result does not depend on a classical approximation that fails for the electrons in question.

13.4.3 — Ohm's Law as an Emergent, Not Fundamental, Law

Why Ohm's law has an arrow of time: The current-current correlator decays because electrons relax momentum into phonons, defects, and other environmental degrees of freedom. Careful about what this claim is: the underlying electron–phonon or electron–impurity interaction is *reversible quantum dynamics* — closed-system unitary evolution cannot produce genuine irreversibility. What produces the decaying correlator is a **coarse-graining**: tracing over the phonons and defects, averaging over ensembles, and adopting a kinetic approximation that discards the full dynamics of the environment. Within that coarse-grained description the correlator decays, and that decay is what makes σ finite and real. Irreversibility is a feature of the effective description, not of the microscopic laws. (The mechanism is the same partial trace that produces decoherence in Ch. 10 §10.1.3 — with the important difference that here we trace over *bath* degrees of freedom, not a measuring apparatus.)

In an ideal translationally invariant system at $T = 0$: With no momentum-relaxing mechanism, the current correlation function has a non-decaying component, so $\tau \to \infty$ and $\sigma_{DC} \to \infty$. Note precisely what this does and does not say: it means the DC conductivity contains a **persistent (Drude) contribution** — the system sustains a current against arbitrarily small dissipation. It does *not* mean a current spontaneously appears in the absence of an applied field. There is still no current unless something drives one; what is unbounded is the *response* to a drive, and the absence of resistivity. A finite resistivity requires momentum relaxation, whether from scattering, from boundaries, or from an explicit momentum sink.

This should also not be identified with superconductivity. A clean normal metal with exactly conserved momentum has infinite Drude weight while remaining an ordinary metal: $\omega_p \neq 0$, no gap, no phase-coherent supercurrent. Superconductivity is a distinct phenomenon — it adds a many-body state (Cooper pairing, energy gap, phase rigidity) on top of the electrons. Infinite Drude weight and superconductivity are related but not equivalent.

Ohm's law is not fundamental. It is an emergent law valid in the regime of diffusive, decoherent transport. Its emergence from the Kubo formula tells you exactly when it fails: when mean free path exceeds the device size (ballistic transport), when topology protects the current from backscattering, or when strong correlations destroy the quasiparticle picture.

13.5 — Deriving Fourier's Law of Heat Conduction

13.5.1 — The Heat Current Operator

For electrons, the heat current is the energy current minus the chemical potential times the particle current:

$$\hat{\mathbf{J}}_Q = \hat{\mathbf{J}}_E - \mu \hat{\mathbf{J}}_N = \sum_k (E_k - \mu) \mathbf{v}_k \hat{c}_k^\dagger \hat{c}_k$$

The Kubo formula for thermal conductivity (at zero electric field):

$$\kappa_{\alpha\beta} = \frac{1}{V k_B T^2} \int_0^\infty \langle \hat{J}_Q^\alpha(0) \hat{J}_Q^\beta(t) \rangle_0 \, dt$$

13.5.2 — The Result and the Wiedemann-Franz Law

For metals (electronic heat conduction dominates), the same relaxation time τ governs both charge and heat transport. Taking the ratio:

$$\frac{\kappa}{\sigma T} = \frac{\pi^2}{3} \left(\frac{k_B}{e} \right)^2 = L_0 = 2.44 \times 10^{-8} \, \mathrm{W} \cdot \Omega \cdot \mathrm{K}^{-2}$$

The *number* is universal; the *law* is not unconditional. L_0 is a **low-temperature Fermi-liquid** result, valid in the degenerate limit where electronic quasiparticle transport dominates.

`[APPROXIMATION]` **The universality of L_0 is a statement about a scattering model, not about metals.** The ratio is constant precisely because the *same* τ cancels between σ and κ — which requires elastic scattering off a static bath (the same $\tau(\epsilon)$ for charge and heat), a degenerate electron gas, and heat carried by electrons alone; no particular dispersion is needed. Whenever heat and charge relax by *different* mechanisms, the ratio drifts:

Regime	What happens to $\kappa/\sigma T$
Elastic scattering, low T	approaches L_0
Electron–phonon scattering, high T	$\tau_Q \neq \tau_N$ — ratio below L_0
Momentum-conserving electron–electron scattering	relaxes heat but not charge — ratio below L_0; a key diagnostic of strange metals
Transition metals, strong spin–orbit	s-wave scattering suppressed — ratio above L_0
Strongly correlated (heavy fermions, cuprates)	quasiparticle picture fails — no constant ratio
High T (classical regime)	degenerate-gas assumption breaks; equipartition applies instead

So "a constant depending only on fundamental constants, independent of material" accurately describes the *low-temperature Fermi-liquid limit* and is false in general — including in exactly the materials where the interesting physics lives. Reading the deviations diagnostically is more useful than treating L_0 as a law:

the size and sign of the departure identifies which additional relaxation channel is active.

This is a good illustration of the book's method. The *universal number* is `[DERIVATION]`; the *scope of validity* is `[APPROXIMATION]`; and the failure cases are where the new physics becomes visible.

Physical origin: Both charge and heat are carried by electrons near the Fermi level. The energy E_F per electron is the same whether you are measuring electrical or thermal transport. The factor k_B^2/e^2 converts energy² (thermal) to energy/charge (electrical) squared.

Engineering consequence: Wiedemann-Franz allows you to estimate thermal conductivity of a metal from its electrical resistivity (easier to measure): $\kappa = L_0 \sigma T$. For copper at 300 K: $\sigma = 6 \times 10^7$ S/m → $\kappa = 2.44 \times 10^{-8} \times 6 \times 10^7 \times 300 = 439$ W/m·K (measured: 401 W/m·K — 10% error from non-Drude effects).

13.5.3 — Phonon Thermal Conductivity

For insulators and semiconductors (no free electrons), heat is carried by phonons. The Green-Kubo formula:

$$\kappa = \frac{1}{V k_B T^2} \int_0^\infty \langle \hat{\mathbf{J}}_Q^{ph}(0) \cdot \hat{\mathbf{J}}_Q^{ph}(t) \rangle_0 \, dt$$

In kinetic theory language (the result of integrating the correlator):

$$\kappa_{phonon} = \frac{1}{3} C_v v_s^2 \tau_{ph} = \frac{1}{3} C_v v_s \ell_{ph}$$

where C_v is specific heat (per unit volume, Ch. 10 §10.7.3), v_s is the phonon group velocity, and $\ell_{ph} = v_s \tau_{ph}$ is the phonon mean free path.

High thermal conductivity requires: large C_v (many phonon modes active) × large v_s (stiff material, strong bonds) × large ℓ_{ph} (few scattering events).

Material	κ (W/m·K)	Primary mechanism	Why
Diamond	2000	Phonon	Light C atoms, stiff bonds, few defects
Silver	429	Electron	High σ, Wiedemann-Franz
Copper	401	Electron	High σ, Wiedemann-Franz
Silicon	148	Phonon	Moderate v_s, moderate ℓ_{ph}
Glass	1.0	Phonon	Amorphous → very short ℓ_{ph}
Air	0.026	Gas kinetics	Very low C_v per unit volume

13.6 — Deriving Fick's Law of Diffusion

13.6.1 — The Velocity Autocorrelation Function

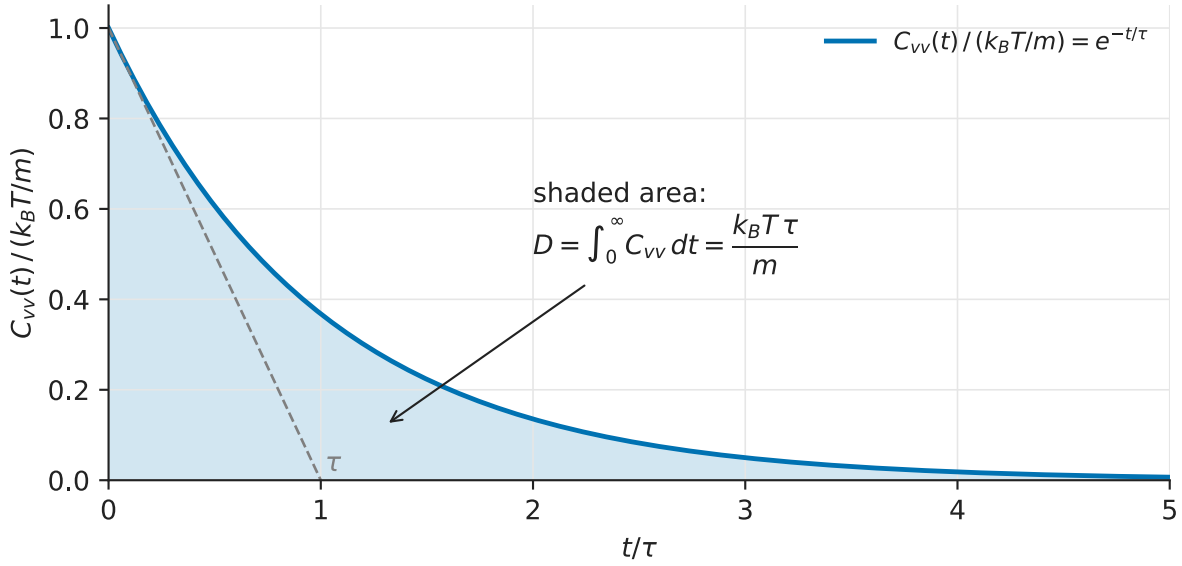
The diffusivity of a species in a medium is given by the **Green-Kubo formula for diffusion** — the integral of the velocity autocorrelation function (VACF):

$$D = \frac{1}{3} \int_0^\infty \langle \mathbf{v}(0) \cdot \mathbf{v}(t) \rangle_0 \, dt$$

For a particle undergoing Brownian-like motion with momentum relaxation time τ: $\langle v_\alpha(0) v_\alpha(t)\rangle = \frac{k_B T}{m} e^{-t/\tau}$

$$D = \frac{1}{3} \cdot 3 \cdot \frac{k_B T}{m}\tau = \frac{k_B T \tau}{m} = \frac{k_B T}{m} \cdot \frac{\ell}{v_{th}}$$

where ℓ is the mean free path and $v_{th} = \sqrt{k_B T/m}$ is the thermal velocity.



Normalised velocity autocorrelation $C_{vv}(t)/(k_BT/m) = e^{-t/\tau}$ for a Brownian tracer: the tangent at $t = 0$ reaches zero at $t = \tau$, and the shaded area under the curve is the Green–Kubo diffusivity $D = \int_0^\infty C_{vv}\,dt = k_BT\tau/m$.

> **Note: When the classical correlator *is* legitimate — and when it is not** The velocity autocorrelation $C_{vv}(t) = \frac{k_BT}{m}e^{-t/\tau}$ used above is **legitimate here**, because a Brownian tracer in a classical thermal medium is non-degenerate: $v_{th} = \sqrt{k_BT/m}$ really is the relevant speed, and equipartition really does apply.
>
> The same expression is **not** legitimate for conduction electrons in a metal, which are degenerate with $v \sim v_F$. That is why §13.4.2 does not use it — it derives the Drude result from the Fermi-surface δ-function instead. The contrast is worth keeping straight: the two cases share a formula but not its justification, and transferring the classical result to degenerate electrons without re-examining the correlator is a common and consequential error.

13.6.2 — The Einstein-Smoluchowski Relation

For a charged particle (charge q, mobility $\mu_{mob} = e\tau/m$):

$$D = \frac{k_B T}{m}\tau = \frac{k_B T}{q} \cdot \frac{q\tau}{m} = \frac{k_B T}{q}\mu_{mob}$$

$$\boxed{D = \frac{k_B T}{q}\mu_{mob}}$$

This is the **Einstein-Smoluchowski (or Einstein) relation** — a direct consequence of the fluctuation-dissipation theorem. It connects:

- **Diffusion** D: random spreading driven by thermal fluctuations (Fick's law)
- **Drift** μ_{mob}: directed motion under an applied force (Ohm's law)

Both are caused by the same scattering events. They cannot be independently specified — knowing one gives the other through k_BT/q.

Engineering: In semiconductors, $D_n = \mu_n k_B T/e$ and $D_p = \mu_p k_B T/e$ (thermal voltage $V_T = k_BT/e =$ 26 mV at 300 K). The p-n junction equations (Shockley diode equation), minority carrier diffusion lengths, and transistor gain all trace back to this relation.

13.6.3 — Fick's First and Second Laws

With the Green-Kubo diffusivity, the constitutive relation (Fick's first law):

$$\mathbf{J}_N = -D\nabla c$$

Combined with the continuity equation (mass conservation, Noether for particle number, Ch. 1 §1.7):

$$\frac{\partial c}{\partial t} + \nabla \cdot \mathbf{J}_N = 0 \quad \Longrightarrow \quad \boxed{\frac{\partial c}{\partial t} = D\nabla^2 c}$$

Fick's second law — the diffusion equation. Same mathematical structure as the heat equation (Ch. 12 §12.1.1 with $\sigma = 0$), as the charge diffusion equation in semiconductors, and as the Schrödinger equation in imaginary time. One equation, four domains.

13.7 — Deriving Newton's Law of Viscosity

13.7.1 — The Stress Tensor Autocorrelation

The viscosity of a fluid is given by the Green-Kubo formula applied to the off-diagonal stress tensor component $\hat{\sigma}_{xy}$ (the momentum flux in the y-direction from flow in the x-direction):

$$\eta = \frac{V}{k_BT}\int_0^\infty \langle \hat{\sigma}_{xy}(0)\hat{\sigma}_{xy}(t)\rangle_0 \, dt$$

For a simple fluid with a single structural relaxation time τ_v:

$$\langle \hat{\sigma}_{xy}(0)\hat{\sigma}_{xy}(t)\rangle_0 = \frac{n(k_BT)^2}{V}e^{-t/\tau_v}$$

$$\eta = nk_BT\tau_v$$

The viscosity equals the thermal energy density times the structural relaxation time.

Engineering interpretation: Viscosity measures how long a fluid "remembers" a stress. A gas has $\tau_v \sim \ell/\bar{v} \sim$ ps (very short memory, low η); a polymer melt has $\tau_v \sim$ ms–s (long memory, very high η); glass at room temperature has $\tau_v > 10^{12}$ s — effectively infinite (solid-like).

13.7.2 — Temperature Dependence of Viscosity

Gases (kinetic theory): $\eta \propto \sqrt{T}$ (more collisions at higher T, but higher mean free velocity; they nearly cancel — viscosity of gases **increases** with temperature).

Liquids (thermally-activated flow, Arrhenius): $\eta = \eta_0 e^{E_a/k_B T}$ — viscosity **decreases** with temperature (activation energy E_a for a molecule to squeeze past neighbors).

Polymers (WLF equation — empirical): $\log(\eta)$ varies with $T - T_g$ where T_g is the glass transition temperature.

This difference (gas vs. liquid viscosity-temperature behavior) is a key practical indicator for the ChE engineer selecting a working fluid.

13.8 — Deriving Hooke's Law: Elastic Moduli

13.8.1 — The Stress-Strain Correlation

The elastic modulus tensor C_{ijkl} can be derived from two routes:

Route 1: Green-Kubo (isothermal modulus):

$$C_{ijkl} = \frac{V}{k_B T} \int_0^\infty \langle \hat{\sigma}_{ij}(0) \hat{\sigma}_{kl}(t) \rangle_0 \, dt \Big|_{t \to 0}$$

For an elastic solid (no viscous relaxation at short times), the correlator does not decay to zero at $t \to 0^+$ but retains its elastic value — giving a finite modulus.

Route 2: Second derivative of free energy (more practical for static moduli):

$$C_{ijkl} = \frac{1}{V} \frac{\partial^2 F}{\partial \varepsilon_{ij} \partial \varepsilon_{kl}} \bigg|_{\varepsilon = 0}$$

where ε_{ij} is the strain tensor. This can be computed from first-principles DFT (density functional theory, using the band structure machinery of Ch. 6) by calculating how the total energy changes with strain.

13.8.2 — Hooke's Law for Engineering

The constitutive relation for a linear elastic solid:

$$\sigma_{ij} = C_{ijkl} \varepsilon_{kl}$$

For an isotropic material, C_{ijkl} has only two independent parameters (Lamé constants λ, μ_L):

$$\sigma_{ij} = \lambda \varepsilon_{kk} \delta_{ij} + 2\mu_L \varepsilon_{ij}$$

Equivalently, in terms of Young's modulus E and Poisson's ratio ν:

$$E = \frac{\mu_L(3\lambda + 2\mu_L)}{\lambda + \mu_L}, \qquad \nu = \frac{\lambda}{2(\lambda + \mu_L)}$$

The same Kubo framework shows that Hooke's law has a frequency-dependent generalization. For a **viscoelastic** material:

$$\tilde{C}(\omega) = \frac{V}{k_B T} \int_0^\infty \langle \hat{\sigma}_{xy}(0) \hat{\sigma}_{xy}(t) \rangle_0 \, e^{i\omega t} \, dt$$

At $\omega \to 0$: recovers the static elastic modulus (material responds slowly and fully). At $\omega \to \infty$: recovers the unrelaxed (glassy) modulus (material doesn't have time to rearrange). At intermediate ω: complex modulus $C' + iC''$ — energy storage and loss.

13.9 — The Onsager Matrix: All Transport Unified

13.9.1 — Coupled Transport

When multiple driving forces act simultaneously, the currents become coupled. Define thermodynamic forces:

$$\mathbf{X}_E = \frac{1}{T}\left(\mathbf{E} + \frac{\nabla \mu}{e}\right), \quad \mathbf{X}_Q = -\frac{\nabla T}{T^2}$$

(for carriers of charge $-e$; these are the forces conjugate to $\mathbf{J}^e$ and $\mathbf{J}^Q$ in the entropy production).

The Onsager transport equations:

$$\begin{pmatrix} \mathbf{J}^e \\ \mathbf{J}^Q \end{pmatrix} = \begin{pmatrix} L_{EE} & L_{EQ} \\ L_{QE} & L_{QQ} \end{pmatrix} \begin{pmatrix} \mathbf{X}_E \\ \mathbf{X}_Q \end{pmatrix}$$

Diagonal elements (from Kubo):

- L_{EE}: charge current from electric force → conductivity $\sigma = L_{EE}/T$
- L_{QQ}: heat current from temperature gradient → thermal conductivity $\kappa = (L_{QQ} - L_{QE}^2/L_{EE})/T^2$

Off-diagonal elements (from Kubo — cross-correlators):

- $L_{EQ} = L_{QE}$: charge current from temperature gradient (Seebeck), or heat current from electric field (Peltier)

13.9.2 — Onsager Reciprocal Relations

$$\boxed{L_{AB} = L_{BA}}$$

Proof: Time-reversal symmetry of the equilibrium correlation functions. Under time reversal, $\langle J_A(0) J_B(t)\rangle_0 = \langle J_B(0) J_A(t)\rangle_0$ for currents with the same sign under time reversal (both odd or both even). Therefore the Kubo integrals are equal: $L_{AB} = L_{BA}$. In general (Onsager–Casimir), with time-reversal parities $\varepsilon_A, \varepsilon_B = \pm 1$ and a magnetic field $\mathbf{B}$, which is odd under time reversal, the statement is $L_{AB}(\mathbf{B}) = \varepsilon_A \varepsilon_B \, L_{BA}(-\mathbf{B})$; the boxed form is the $\mathbf{B} = 0$, equal-parity case. For the thermal force ∇T, which is not a term in the Hamiltonian, writing L_{QQ} and L_{EQ} as Kubo correlators itself needs an extra step (Luttinger's fictitious gravitational potential).

Physical consequence: If a temperature gradient drives a charge current (Seebeck effect), then an electric field drives an equal heat current (Peltier effect). These must be related — you cannot have one without the other.

13.9.3 — Thermoelectric Coefficients

Seebeck coefficient (thermopower) S — open-circuit voltage per degree:

$$S = \frac{L_{EQ}}{T L_{EE}} = \left.\frac{\Delta V}{\Delta T}\right|_{J^e=0}$$

From the Mott formula $S = -\frac{\pi^2 k_B^2 T}{3e}\frac{d\ln\sigma}{dE}\Big|_{E_F}$, a free electron gas with constant mean free path ($\sigma \propto E$) gives $S = -\frac{\pi^2 k_B^2 T}{3eE_F}$; constant τ ($\sigma \propto E^{3/2}$) gives $-\frac{\pi^2 k_B^2 T}{2eE_F}$ instead. At 300 K for copper ($E_F = 7$ eV) the first gives $S \approx -1.05\,\mu$V/K. Measured copper is $\approx +1.8\,\mu$V/K: the free-electron model gets the magnitude roughly right but the **sign wrong** — a known failure, attributed to energy-dependent scattering and the non-spherical Fermi surface of Cu (necks touching the zone boundary).

Peltier coefficient Π — heat current per unit charge current:

$$\Pi = \frac{L_{QE}}{L_{EE}}$$

Kelvin's relation (from Onsager reciprocity):

$$\Pi = ST$$

This is not an independent empirical law — it is the Onsager reciprocal relation $L_{QE} = L_{EQ}$ written in terms of S and Π.

Thermoelectric figure of merit:

$$ZT = \frac{S^2\sigma T}{\kappa} = \frac{T^2 S^2 L_{EE}}{L_{QQ} - L_{QE}^2/L_{EE}}$$

$ZT > 1$ is the threshold for useful thermoelectric devices. Best materials (Bi_2Te_3 alloys, SnSe single crystals): $ZT \approx 2$–3. The Onsager matrix shows exactly what needs to be optimized — maximize $S^2\sigma$ (power factor) while minimizing κ (thermal conductivity). These are coupled through the underlying band structure and phonon spectrum.

13.10 — The Generalized Transport Law: The Complete Table

Five constitutive laws sharing one linear-response template. Read the table as *five instances of one template*, not five outputs of one calculation — and note the distinction in the last column: the first four rows are Green–Kubo transport coefficients, whereas C is a static curvature of the free energy, not a transport coefficient (§13.9.2):

$$\boxed{\mathbf{J}_X = -L_{XX}\nabla\phi_X}$$

Current $\mathbf{J}_X$	**Coefficient L_{XX}**	**Potential ϕ_X**	**Named Law**	**Branch**	**Kubo integrand**
Charge density $\mathbf{J}^e$	σ (conductivity)	Electric potential V	**Ohm's law**	EEE	$\langle \hat{J}^e(0)\hat{J}^e(t)\rangle$
Heat flux $\mathbf{J}^Q$	κ (thermal cond.)	Temperature T	**Fourier's law**	ME, ChE	$\langle \hat{J}^Q(0)\hat{J}^Q(t)\rangle$
Particle flux $\mathbf{J}^N$	D (diffusivity)	Concentration c	**Fick's law**	ChE, CE	$\langle \mathbf{v}(0)\mathbf{v}(t)\rangle$
Momentum flux τ_{xy}	η (viscosity)	Velocity u	**Newton's viscosity**	ME, CE	$\langle \hat{\sigma}_{xy}(0)\hat{\sigma}_{xy}(t)\rangle$

Current $\mathbf{J}_X$	Coefficient L_{XX}	Potential ϕ_X	Named Law	Branch	Kubo integrand
Stress σ_{ij}	C_{ijkl} (elastic moduli)	Strain ε_{kl}	**Hooke's law**	CE, ME	$\partial^2 F/\partial\varepsilon^2$

Off-diagonal (Onsager) transport also from Kubo cross-correlators:

Effect	From	Coefficient	Branch
Seebeck	$\langle \hat{J}^e(0)\hat{J}^Q(t)\rangle$	$S = L_{EQ}/TL_{EE}$	EEE, ME
Peltier	(Onsager: $L_{QE} = L_{EQ}$)	$\Pi = ST$	EEE, ME
Soret (thermodiffusion)	$\langle \hat{J}^N(0)\hat{J}^Q(t)\rangle$	S_T	ChE
Dufour (diffusion-thermo)	(Onsager)	D^Q	ChE

This table is the architectural backbone of Layer 2 engineering science. It collects the constitutive relations used across electrical, mechanical, civil, and chemical engineering — though not every engineering constitutive relation is a row here, and the table is organised by the *shared linear-response structure* rather than by exhaustiveness. On the coefficients themselves, precision matters: **many** of the transport coefficients above are Green–Kubo integrals of an equilibrium correlation function, but **not all of them are**, and the exceptions are not oversights. Chemical reaction rate constants are kinetic parameters set by reaction energetics and collision theory, not by a current-response correlator; elastic moduli are static elastic constants — second derivatives of the strain energy — fixed by the full interatomic potential and entering a linear-response framework only in the trivial static limit. Each needs its own microscopic or phenomenological framework, and labelling them uniformly `[DERIVATION]` would be wrong.

13.11 — Beyond Kubo: Landauer and Topology

13.11.1 — Ballistic Transport and the Limits of the Bulk Formulation

Kubo linear response is **not** restricted to diffusive transport — it is the general framework. What is restricted is the *bulk conductivity* formulation used in §13.4, in which a local constitutive law $\mathbf{J} = \sigma\mathbf{E}$ with a single material constant is assumed to be meaningful. That assumption requires the sample to be large compared with the mean free path, so that scattering is frequent and a local σ exists:

```
Linear response
      ├── bulk / diffusive  (L ≫ ℓ)  → Kubo conductivity, local σ
      └── open ballistic mesoscopic (L ≲ ℓ) → Landauer / scattering, mode-resolved
```

For $L \lesssim \ell_{mfp}$, an electron crosses the device without scattering, so no local bulk conductivity is defined. The Drude estimate then *overestimates* the resistance, because it assumes a scattering probability that the geometry never gives the electron the chance to take. The right object is then the transmission of individual modes, not a material constant.

For a **ballistic conductor** (negligible scattering inside the channel), each transverse mode contributes a conductance:

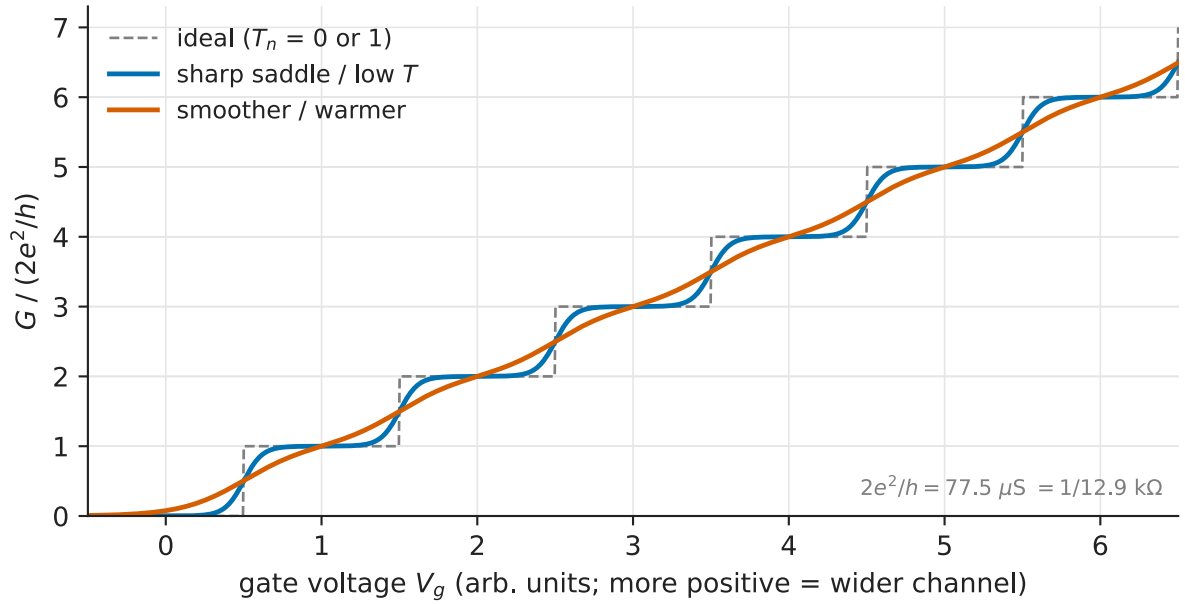
$$G_n = \frac{2e^2}{h} T_n$$

where $T_n \in [0, 1]$ is the **transmission probability** of mode n. Total conductance:

$$\boxed{G = \frac{2e^2}{h} \sum_n T_n}$$

This is the **Landauer formula** (1957). For a perfect conductor ($T_n = 1$) with N modes: $G = 2Ne^2/h$.

The quantum of conductance $G_0 = 2e^2/h \approx 7.75 \times 10^{-5}\ \text{S} = 1/(12.9\,\text{k}\Omega)$: the conductance of a single perfectly transmitting channel. First measured directly in quantum point contacts (1988).



Landauer conductance $G = (2e^2/h)\sum_n T_n$ of a quantum point contact as the gate widens the channel, with saddle-point transmissions $T_n = 1/[1 + e^{-(V_g - V_n)/w}]$ for evenly spaced mode thresholds V_n (illustrative gate units): each newly opened mode adds one quantum $2e^2/h = 77.5\ \mu$S, and the plateaus round off as the transmission steps broaden (larger w, e.g. higher temperature or a smoother constriction).

The hierarchy of transport theories:

```
Quantum field theory (Ch. 0)
        ↓ Kubo formula (this chapter)
Diffusive: G = σA/L     (Ohm's law, L ≫ ℓ_mfp)
        ↓ Landauer
Ballistic: G = (2e²/h)ΣTₙ  (L < ℓ_mfp)
        ↓ Topology (Ch. 7)
Topological: G = νe²/h  (Tₙ = 1, protected by topology)
```

13.11.2 — The Quantum Hall Conductance from Topology

In the integer quantum Hall effect (Ch. 7 §7.3), the fundamental statement is about a **conductivity**, not a conductance:

$$\boxed{\sigma_{xy} = \nu \frac{e^2}{h}}$$

the quantized **Hall conductivity**, with ν the number of filled Landau levels and the longitudinal conductivity vanishing ($\sigma_{xx} = 0$) on the plateau.

The microscopic content, from Kubo applied to the full 2D system, is the Chern-number integral over the Brillouin zone:

$$\sigma_{xy} = \frac{e^2}{h} \sum_{n \in \text{filled}} \frac{1}{2\pi} \iint_{\text{BZ}} \Omega_n(\mathbf{k})\, d^2k = \nu \frac{e^2}{h}$$

where Ω_n is the Berry curvature of band n. This is the TKNN result (Ch. 7 §7.2.1) seen from the transport-theory side: the Kubo integral *counts Chern number*, which is why the value is topologically protected and independent of disorder, carrier density, and sample geometry.

> **Note: Conductivity vs. conductance — a distinction worth keeping** σ_{xy} and G are different quantities, and identifying them is a common shortcut. σ_{xy} is an intrinsic **material/response** property (conductivity, S), whereas G is a **device** property (conductance, S) that depends on geometry and contacts.
>
> In a suitable ideal Hall-bar geometry with well-separated edge channels and equilibrating contacts, the two-terminal conductance of the same sample also takes the quantized value $G = \nu e^2/h$, because the contact resistance is negligible and $\sigma_{xx} = 0$ forces the current to be carried entirely by the edges. That is a *consequence* of the conductivity quantization plus the geometry, not the primary statement — and in a two-terminal device without equilibrating contacts the Hall voltage and the two-terminal conductance are in general *not* quantized, because of contact and fringe effects.
>
> For this book's purposes the clean formulation is: the Hall **conductivity** is quantized; suitable geometries then exhibit the corresponding quantized conductance.

13.12 — When Transport Laws Break Down (and What to Do)

Failure	Regime	Alternative	Return to
Ohm's law (high E)	$eE\ell \gtrsim E_F$: hot carriers	Nonlinear Boltzmann equation; impact ionization	L1 (band structure, Ch. 6)
Ohm's law (small L)	$L \lesssim \ell_{mfp}$: ballistic	Landauer formula	L1 (quantum transport, Ch. 7)
Fourier's law (nanoscale)	$L \lesssim \ell_{phonon}$: ballistic phonons	Boltzmann transport equation for phonons	L1 (phonon dispersion, Ch. 6)
Fick's law (crowded medium)	High concentration: interactions matter	Generalized diffusion equation; activity coefficients	L2 (thermodynamics, Ch. 10)
Newton's viscosity (non-Newtonian)	Polymers, colloids: τ_{flow} spans many decades	Generalized Maxwell/Oldroyd equations	L2 (viscoelasticity)
Hooke's law (large strain)	$\varepsilon > 0.01$: nonlinear elasticity	Hyperelastic models (Mooney-Rivlin, Ogden)	L2 (continuum mechanics, Ch. 15)
All laws (strongly correlated)	Mott insulators, strange metals	DMFT, slave-boson, RVB theories	L0/L1 (beyond quasiparticles)

The engineer who knows this table knows exactly when to trust their formula and exactly where to go when it fails.

13.13 — Summary

Bridge B.e is complete. The five transport laws, unified:

Step	Operation	Result
Linear response theory	Perturbation on $\hat{\rho}_0$	Response $\propto$ equilibrium correlator
Kubo formula	Time-integrate current correlator	Transport coefficient L_{AB}
FDT	Imaginary $\chi \leftrightarrow$ noise spectrum	$S_V = 4k_BTR$ (Johnson-Nyquist)
Ohm's law	Charge current-current Kubo	$\sigma = ne^2\tau/m$; $\rho \propto T$
Fourier's law	Heat current-current Kubo	$\kappa = L_0\sigma T$ (Wiedemann-Franz)
Fick's law	Velocity autocorrelation	$D = k_BT\mu/q$ (Einstein relation)
Newton viscosity	Stress autocorrelation	$\eta = nk_BT\tau_v$
Hooke's law	Free energy second derivative	C_{ijkl} from DFT or Kubo
Onsager matrix	Cross-correlators + reciprocity	Seebeck-Peltier-Thomson relations
Landauer	Ballistic transport	$G = (2e^2/h)\sum T_n$
Topology	$T_n = 1$ protected	$\sigma_{xy} = \nu e^2/h$ (QHE); G quantized in ideal Hall-bar geometry

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13.14 — Engineering Thread

Physics	Application
$\sigma = ne^2\tau/m$	Conductor sizing; resistor design; contact resistance minimization
$\sigma(\omega) = \sigma_0/(1 - i\omega\tau)$	Skin effect; RF conductor loss; plasma frequency
Wiedemann-Franz $\kappa = L_0\sigma T$	Thermal management: copper heat spreaders; Peltier coolers
Phonon $\kappa = C_v v_s \ell/3$	Thermal interface materials; diamond heat sinks; thermoelectric leg design
Einstein relation $D = \mu k_B T/e$	Semiconductor diffusion length $L_D = \sqrt{D\tau}$; p-n junction analysis
Fick's equation $\partial_t c = D\nabla^2 c$	Semiconductor doping profiles; ChE mass transfer; concrete carbonation
Newton viscosity $\eta = nk_BT\tau_v$	Pipe flow ($\Delta P = 8\eta LQ/\pi r^4$); lubrication; rheology
Viscoelastic $\tilde{C}(\omega)$	Polymer processing; damping materials; tire hysteresis
Seebeck S, Peltier $\Pi = ST$	Thermocouples; Peltier coolers; thermoelectric generators
$ZT = S^2\sigma T/\kappa$	Thermoelectric device efficiency optimization
Johnson-Nyquist $S_V = 4k_BTR$	Amplifier noise floor; sensor sensitivity limit; low-noise design
Landauer $G = (2e^2/h)\sum T_n$	Nanoscale transistor resistance; quantum point contact sensors
QHE $\sigma_{xy} = \nu e^2/h$	Primary resistance standard; metrological applications (Hall-bar geometry)

13.15 — Looking Ahead: Bridge B Complete, Layer 2 Assembled

With Chapter 13, all five Bridge B paths are complete:

```
Layer 1 (Quantum)
   │
   ├─ B.a (ħ→0) ──────────────► Classical mechanics (Ch. 9)
   ├─ B.b (N→∞) ──────────────► Thermodynamics (Ch. 10)
   ├─ B.c (EM field limit) ──► Maxwell's equations (Ch. 11)
   │                              └─ EM waves, optics (Ch. 12)
   ├─ B.d (weak-field GR) ──► Newton's gravity (Ch. 8)
   └─ B.e (Kubo) ─────────────► Generalized transport law (Ch. 13)
                                  └─ Ohm, Fourier, Fick, viscosity, Hooke
```

Layer 2 is now complete. Every classical engineering science equation you will use in Chapters 15–21 is assembled. The inventory:

- Equations of motion: Newton (Ch. 8, 9), Euler-Lagrange (Ch. 9)
- Thermodynamics: four laws, partition function, phase diagrams (Ch. 10)
- Electromagnetism: Maxwell's equations, wave equation, KVL/KCL (Ch. 11, 12)
- Transport: Ohm, Fourier, Fick, viscosity, Hooke (Ch. 13)
- Continuum mechanics: stress-strain, Navier-Stokes (Ch. 15)

Chapter 15 covers continuum mechanics (fluid and solid) — deriving Navier-Stokes from Newton + statistical mechanics and the stress-strain equation from Hooke's law in the continuum limit.

Chapter 16 then shows that the wave equation and diffusion equation appear identically in every domain — the final unification of Layer 2 before the descent to engineering systems begins in Chapter 17 (Bridge C).

End of Chapter 13. End of Bridge B.

Next: Chapter 14 — Transport Phenomena and Thermoelectrics (Layer 2 Applied)

Transport Phenomena and Thermoelectrics

The Kubo Coefficients in Engineering Practice

> *In science, there is no distinction between theory and application. The equations that describe the flow of electrons in a wire are the same ones that describe the flow of heat through a wall — because they are both the same equation, applied to the same phenomenon.* — paraphrased from Lars Onsager

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14.0 — Overview

Chapter 13 places the transport laws in a common Green–Kubo linear-response framework, with no additional input beyond equilibrium quantum mechanics *beyond the linear-response and closure assumptions stated there*. The four dissipative coefficients σ, κ, D, η share that framework: each is expressible through an equilibrium current-correlation integral, subject to the relevant quantum/classical and closure assumptions. The elastic modulus C_{ijkl} is different — it is a static, nondissipative response coefficient relating stress to strain, so Hooke's law is a constitutive relation arising from static response, not a fifth dissipative Green–Kubo transport coefficient. The Onsager matrix gives the off-diagonal effects.

This chapter applies those results to concrete engineering problems. It is **Layer 2 applied** — the transport laws in their full engineering detail, with quantitative worked results, before Bridge C lumps them into circuit elements and control volumes in Chapter 17.

The chapter divides into three groups:

Electrical transport (§§14.1–14.2): resistivity vs. temperature, Hall sensors, contact resistance.

Thermoelectric transport (§§14.3–14.5): the full Onsager matrix in engineering form — thermocouples, Peltier coolers, thermoelectric generators, the ZT optimization problem.

Coupled transport (§§14.6–14.8): semiconductor drift-diffusion, thermionic emission, the thermal resistance network as bridge to Layer 3.

14.1 — Electrical Resistivity in Practice

14.1.1 — Matthiessen's Rule

From Ch. 13 §13.4.2, the conductivity $\sigma = ne^2\tau/m$ requires knowing τ. In real metals, electrons scatter from multiple sources simultaneously. Each scattering mechanism contributes an independent rate $1/\tau_i$:

$$\frac{1}{\tau} = \frac{1}{\tau_{phonon}} + \frac{1}{\tau_{impurity}} + \frac{1}{\tau_{grain}} + \cdots$$

Since $\rho = m/ne^2\tau$, this gives **Matthiessen's rule**:

$$\rho(T) = \rho_0 + \rho_{phonon}(T) + \rho_{other}$$

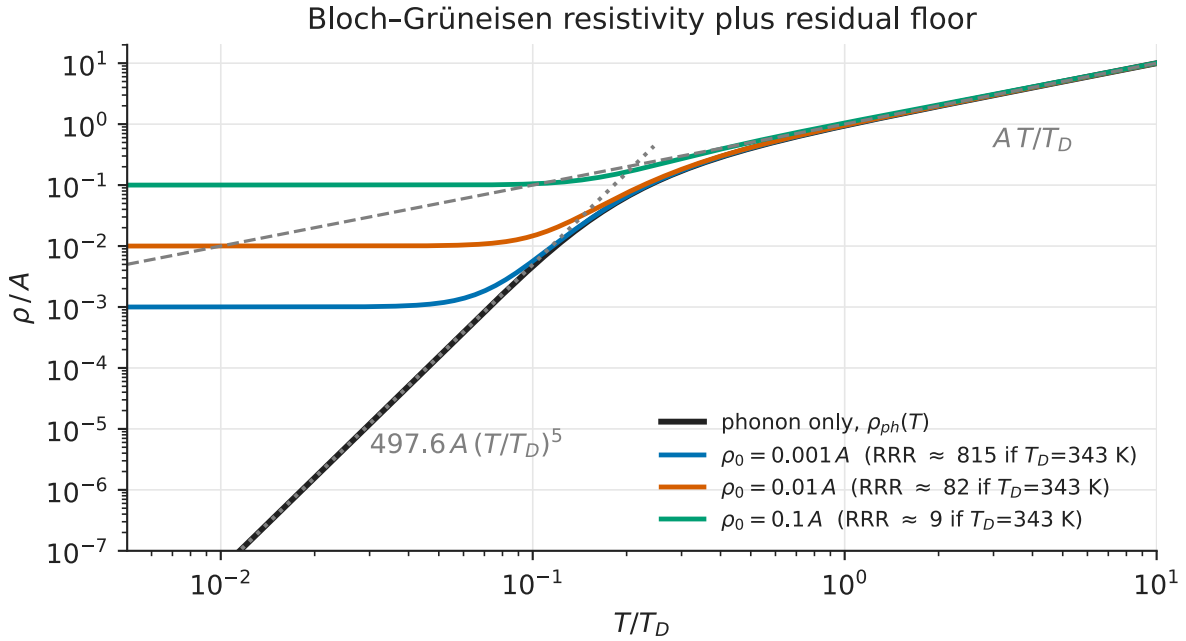
- ρ_0 (residual resistivity): from impurities and defects; temperature-independent. Sets the $T \to 0$ floor.
- $\rho_{phonon}(T) \propto T$: at room temperature (Bloch-Grüneisen, Ch. 6 §6.7.3). At high T (above Debye temperature): linear. At low T: $\rho_{phonon} \propto T^5$ (only small-angle phonon scattering survives).
- ρ_{grain}: grain boundary scattering in polycrystalline materials; roughly temperature-independent.

Bloch-Grüneisen formula for phonon contribution:

$$\rho_{phonon}(T) = 4A \left(\frac{T}{T_D}\right)^5 \int_0^{T_D/T} \frac{x^5}{(e^x - 1)(1 - e^{-x})} dx$$

Limits:

- $T \gg T_D$: $\rho_{phonon} \to AT/T_D$ (linear, observed in room-temperature metals)
- $T \ll T_D$: $\rho_{phonon} \to 4 \cdot 5!\,\zeta(5)\, A(T/T_D)^5 \approx 497.6A(T/T_D)^5$ (steep T^5 power law)



$\rho(T) = \rho_0 + \rho_{phonon}(T)$ from the Bloch–Grüneisen integral for $\rho_0 = 0.001A$, $0.01A$ and $0.1A$, with the $497.6A(T/T_D)^5$ and AT/T_D asymptotes (grey); the residual floor hides the T^5 regime at low T, and the purer the metal the larger the RRR.

14.1.2 — The Residual Resistivity Ratio

The **residual resistivity ratio (RRR)** is the single most informative measure of metal purity for cryogenic and superconducting applications:

$$RRR = \frac{\rho(300\,\mathrm{K})}{\rho(4\,\mathrm{K})}$$

For copper: commercial purity (99.9%) gives $RRR \sim 40$–100. Ultra-pure copper (99.9999%) gives $RRR > 10{,}000$.

Higher RRR means lower residual resistivity, which means lower heat load in superconducting magnet coil leads, better signal integrity in cryogenic detectors, and higher purity for electron beam lithography.

14.1.3 — Resistance Temperature Detectors (RTDs)

The nearly linear $\rho(T)$ of platinum is exploited in RTDs. The Callendar-Van Dusen equation (empirical fit to the Bloch-Grüneisen result):

$$R(T) = R_0\left(1 + A\,T + B\,T^2 + C\,T^3(T - 100)\right)$$

For Pt100 ($R_0 = 100\,\Omega$ at 0°C): $A = 3.9083 \times 10^{-3}$, $B = -5.775 \times 10^{-7}$, $C = -4.183 \times 10^{-12}$ (IEC 60751 standard).

Sensitivity: $dR/dT \approx 0.385\,\Omega/^\circ\text{C}$ for Pt100.

Why platinum? Stable crystal structure, reproducible RRR, chemically inert over a wide temperature range, well-characterized impurity scattering. The NIST Pt resistance standard is the most precise thermometer from 14 K to 962°C.

14.1.4 — Contact Resistance and the Ballistic Limit

At a metal contact, resistance arises from three sources:

Spreading resistance (current flowing from a small contact area into a large bulk — current funnels through the contact):

$$R_{spread} = \frac{\rho}{4a}$$

for a circular contact of radius a.

Interface/contact resistance (imperfect bonding, oxide layer, phonon mismatch at metal-semiconductor interfaces): characterized by contact resistivity ρ_c ($\Omega\cdot\text{m}^2$).

Sharvin resistance (ballistic limit when contact size $a < \ell_{mfp}$):

$$R_{Sharvin} = \frac{h}{2e^2}\frac{1}{\langle T\rangle N_{modes}} = \frac{4\rho\ell_{mfp}}{3\pi a^2}$$

Sharvin resistance is geometry-limited, not scattering-limited — it represents the quantum resistance of the few ballistic modes that fit through the contact. For an aluminum nanocontact ($a = 1$ nm, $\ell_{mfp} = 18$ nm): $R_{Sharvin} \approx 4\,\text{k}\Omega$.

Engineering implication: As interconnect widths in semiconductor ICs approach ℓ_{mfp} (currently ~ 10–20 nm), Matthiessen's rule breaks down, resistivity increases beyond bulk values (sidewall scattering), and quantum resistance becomes significant. This is the physical wall constraining Moore's Law.

14.2 — The Hall Effect in Engineering

14.2.1 — Classical Hall Effect Revisited

From Ch. 13 §13.10 (Ohm's law): a current I in a conductor of thickness t in transverse field B:

$$R_H = \frac{1}{nq}, \qquad V_H = \frac{IB}{nqt} = R_H\frac{IB}{t}$$

The Hall coefficient $R_H = 1/nq$ (positive for holes, negative for electrons) simultaneously gives:

- **Carrier density n:** from the magnitude of R_H
- **Carrier type:** from the sign of R_H
- **Mobility:** $\mu_H = |R_H|\sigma = |R_H|/\rho$

For copper at 300 K: $R_H = -5.5 \times 10^{-11}$ m³/C → $n = 1/|R_H|e = 1.14 \times 10^{29}$ m^{-3} (one conduction electron per atom). ✓

For p-type silicon doped at $N_A = 10^{23}$ m^{-3}: $R_H = +1/ep = +6.25 \times 10^{-5}$ m³/C (positive — holes dominate). ✓

14.2.2 — Hall Sensors

Hall sensors output a voltage proportional to the product $I \times B$. Keep I constant → $V_H \propto B$: a linear magnetic field sensor. Keep B constant → $V_H \propto I$: a current sensor (no galvanic connection needed).

Key figures of merit:

- Sensitivity: $S_A = V_H/IB = 1/nqt$ (higher for thin films, low-n semiconductors)
- Offset voltage: zero-field output from geometry imperfections (trimmed with laser)
- Temperature coefficient: from $\partial R_H/\partial T$ via Fermi-Dirac statistics

InAs and InSb (high mobility $\mu \sim 10^4$ cm²/Vs) give high sensitivity. Si Hall sensors are more stable and CMOS-compatible.

Applications: Brushless DC motor commutation, automotive angle/speed sensors, current measurement in EV battery management systems, anti-lock braking systems, compass and positioning sensors in smartphones.

14.2.3 — The Quantum Hall Resistance Standard

From Ch. 7 §7.3 and Ch. 13 §13.11.2: at filling factor ν:

$$R_{xy} = \frac{h}{\nu e^2} = \frac{R_K}{\nu}, \qquad R_K = \frac{h}{e^2} = 25812.807\,\Omega$$

Since 2019 (SI redefinition), h and e are exact, so R_K is exact. A 2DEG in a cryostat at $T < 1$ K, $B \sim 10$ T holds this value to better than 10^{-9} — the most precisely reproducible resistance in physics.

Metrological use: National measurement institutes maintain GaAs/AlGaAs QH samples as the primary resistance standard. All secondary resistors (decade boxes, etc.) are calibrated against these. Ultimately, every ohmmeter you use traces back to the topology of Landau levels.

14.3 — Thermoelectric Effects: Complete Engineering Treatment

14.3.1 — The Three Effects, Unified

From Ch. 13 §13.9 (Onsager matrix), the three thermoelectric effects:

Seebeck effect — open-circuit EMF from temperature gradient:

$$V_{OC} = S \cdot \Delta T \qquad (S\text{: Seebeck coefficient, V/K})$$

Peltier effect — heat pumped by current:

$$\dot{Q}_{Peltier} = \Pi \cdot I = ST \cdot I \qquad (S\text{: same coefficient, by Kelvin's relation})$$

Thomson effect — heat generated/absorbed in a current-carrying conductor with a temperature gradient:

$$\dot{q}_{Thomson} = \tau_{Th} \cdot J \cdot \nabla T \qquad (\tau_{Th} = T, dS/dT\text{: Thomson coefficient})$$

The three are not independent — they are all encoded in the single Seebeck function $S(T)$:

$$\Pi(T) = S(T) \cdot T, \qquad \tau_{Th}(T) = T\frac{dS}{dT}$$

This is the Onsager reciprocal relation (Ch. 13 §13.9.2) made explicit: knowing $S(T)$ completely determines all thermoelectric behavior.

14.3.2 — Thermocouples

Two dissimilar conductors A and B, junction at temperature T, reference at T_0:

$$V = \int_{T_0}^{T} [S_A(T') - S_B(T')]\, dT' = S_{AB}\Delta T \quad \text{(for small } \Delta T\text{)}$$

The relative Seebeck coefficient $S_{AB} = S_A - S_B$ (in μV/K) defines the thermocouple type. IEC standard thermocouple types:

Type	Materials	Range	Sensitivity	Note
K	NiCr / NiAl	-200 to $1260\,^\circ$C	$\approx 41\,\mu$V/°C	Most common; general purpose
J	Fe / CuNi	-210 to $760\,^\circ$C	$\approx 52\,\mu$V/°C	Oxidises above 760°C
T	Cu / CuNi	-270 to $370\,^\circ$C	$\approx 43\,\mu$V/°C	Best for cryogenic
E	NiCr / CuNi	-270 to $1000\,^\circ$C	$\approx 68\,\mu$V/°C	Highest sensitivity
N	NiCrSi / NiSi	-270 to $1300\,^\circ$C	$\approx 39\,\mu$V/°C	Stable at high T
R	Pt13Rh / Pt	0 to $1600\,^\circ$C	$\approx 12\,\mu$V/°C	Precision high-T
B	Pt30Rh / Pt6Rh	100 to $1820\,^\circ$C	$\approx 10\,\mu$V/°C	Very high T; self-compensating
S	Pt10Rh / Pt	0 to $1600\,^\circ$C	$\approx 10\,\mu$V/°C	Primary standard

Cold junction compensation: The reference junction must be at a known temperature T_0. In practice, an isothermal terminal block with an on-board temperature sensor (thermistor or RTD) measures T_0; the microcontroller adds $V(T_0)$ from a lookup table to the measured voltage.

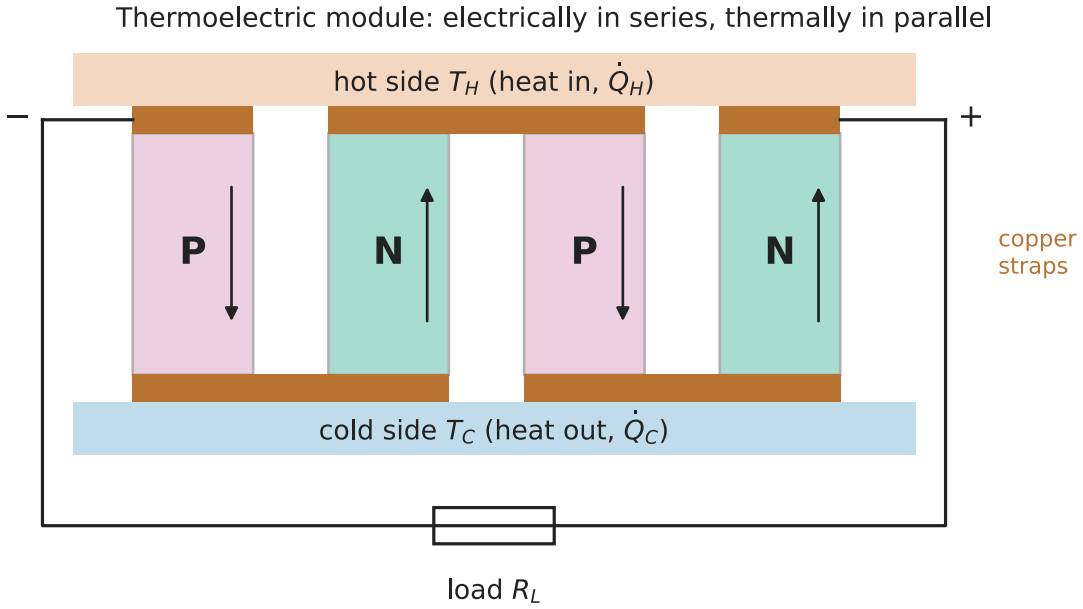
Seebeck coefficient origin: From the Mott formula (Ch. 13 §13.9.3): $S = -\pi^2 k_B^2 T / 3eE_F \times (d\ln\sigma/dE)|_{E_F}$. Materials with strongly energy-dependent $\sigma(E)$ near E_F (e.g., from a sharp band edge or resonant level) have large $|S|$.

14.4 — Thermoelectric Device Design

14.4.1 — The Thermoelectric Module

A thermoelectric module consists of N thermocouple pairs (P-type and N-type semiconductor legs) electrically in series and thermally in parallel:

```
Hot side (T_H)
├── [P-leg] [N-leg] [P-leg] [N-leg] ...
└── Cold side (T_C)
       ↑                ↑
    (+) current    current flows through external circuit
```



Generator-mode module: P and N legs joined by copper straps so current runs down each P leg and up each N leg (electrically in series), while every leg spans the same hot-to-cold gap (thermally in parallel).

For N couples with matched resistance $R_{total} = NR_0$ and thermal conductance $K_{total} = NK_0$:

Thermoelectric generator (TEG):

Open-circuit voltage: $V_{OC} = N\,S_{PN}\Delta T$

Maximum power: $P_{max} = N\frac{(S_{PN}\Delta T)^2}{4R_0}$ (matched load $R_L = R_{total}$)

Efficiency at maximum power:

$$\eta_{max} = \frac{\Delta T}{T_H}\cdot\frac{\sqrt{1+ZT_M}-1}{\sqrt{1+ZT_M}+T_C/T_H}$$

where $T_M = (T_H + T_C)/2$ is the mean temperature and:

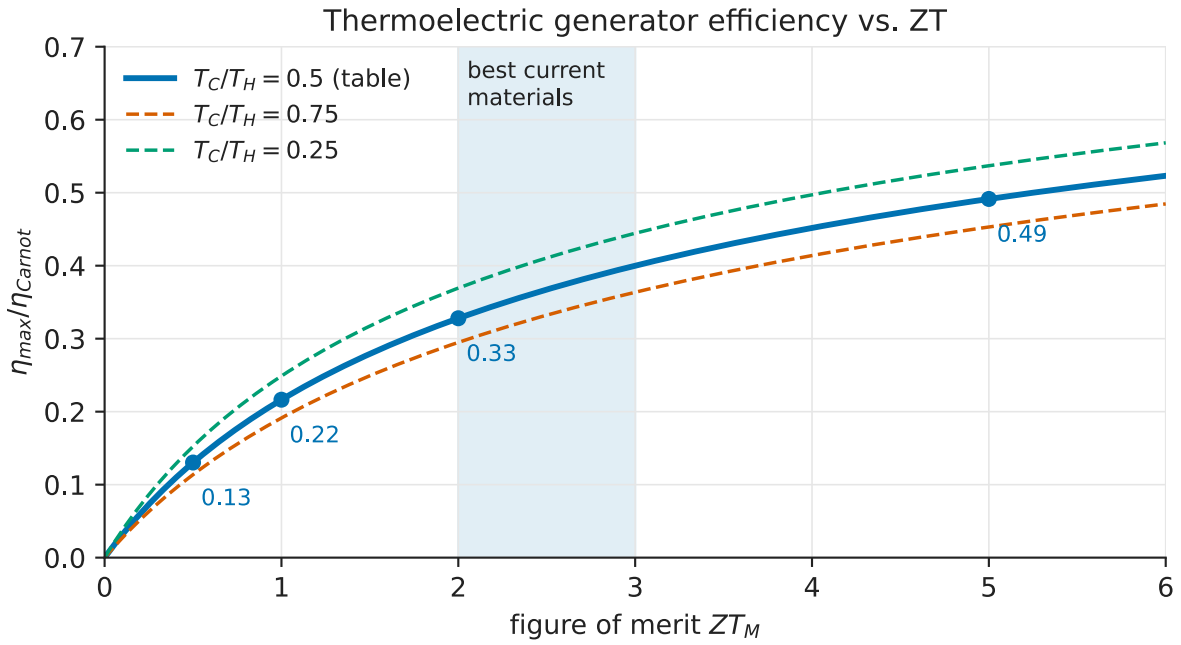
$$\boxed{ZT = \frac{S^2\sigma}{\kappa}T = \frac{S^2}{\rho\kappa}T}$$

is the **dimensionless figure of merit**. The Carnot efficiency $\eta_C = \Delta T/T_H$ is recovered in the limit $ZT \to \infty$.

Efficiency at matched load vs. Carnot:

ZT	η_{max}/η_{Carnot} **(for** $T_C/T_H = 300/600$ **K)**
0.5	0.13
1.0	0.22
2.0	0.33
5.0	0.49
∞	1.00 (Carnot)

Current best materials ($ZT \sim 2$–3): ~33–40% of Carnot efficiency.



$\eta_{max}/\eta_{Carnot} = (\sqrt{1+ZT_M}-1)/(\sqrt{1+ZT_M}+T_C/T_H)$ *for* $T_C/T_H = 0.5$ *(solid; dots are the table values) and 0.25, 0.75 (dashed); the approach to Carnot is slow — even* $ZT = 5$ *reaches only about half of Carnot.*

Thermoelectric cooler (TEC/Peltier module):

Peltier cooling power (heat pumped from cold side):

$$\dot{Q}_C = S_{PN}\, T_C\, I - \frac{1}{2} I^2 R_{total} - K_{total}\Delta T$$

Maximum ΔT (at $\dot{Q}_C = 0$, optimizing over I):

$$\boxed{\Delta T_{max} = \frac{1}{2} Z\, T_C^2}$$

For Bi_2Te_3 module ($ZT = 1$ at 300 K, $T_C = 300$ K, so $Z = 1/300$ K^{-1}): the ideal formula gives $\Delta T_{max} = \frac{1}{2}(1/300)(300)^2 = 150$ K; real modules achieve ~ 70 K because of contact resistance, parasitic heat leaks and temperature-dependent properties.

Coefficient of performance at maximum ΔT:

$$COP_{max} = \frac{\dot{Q}_C}{P_{input}}\bigg|_{I_{opt}} = \frac{T_C}{\Delta T} \cdot \frac{\sqrt{1+ZT_M} - T_H/T_C}{\sqrt{1+ZT_M}+1}$$

The first factor is the Carnot COP; the second factor is the ZT correction.

14.4.2 — Materials Engineering for High ZT

$ZT = S^2\sigma T/\kappa$: maximize power factor $S^2\sigma$, minimize κ.

The conflict: High σ → high κ (Wiedemann-Franz, Ch. 13 §13.5.2). Good thermoelectrics need high S and σ (like a metal) but low κ (like a glass). The ideal: a "phonon glass, electron crystal" (PGEC).

Strategies to achieve high ZT:

Strategy	Mechanism	Example
Resonant levels	Sharp feature in DOS near E_F → large $\lvert dS/dE\rvert$	HgTe quantum wells
Band convergence	Multiple valleys contributing to transport → high S with high σ	PbTe (high-T), SnSe
Nanostructuring	Phonon scattering at grain boundaries without electron scattering	BiSbTe nanocomposite
Lattice softening	Rattler atoms in cages scatter phonons (PGEC)	Skutterudites, clathrates
2D/layered	Quantum confinement enhances DOS asymmetry	Bi_2Te_3 thin films

Current champions:

- Room temperature ($\sim$ 300 K): BiSbTe alloys, $ZT \sim 2.0$–2.5
- Mid-range ($\sim$ 700 K): PbTe:Na/Se, $ZT \sim 2.5$
- High temperature ($\sim$ 900 K): SiGe alloys for RTG, $ZT \sim 1.0$; SnSe single crystal $ZT \sim 2.6$
- Cryogenic ($<$ 200 K): BiSb, $ZT \sim 0.6$

14.4.3 — Applications of Thermoelectric Devices

Radioisotope thermoelectric generators (RTGs): The Voyager 1 and 2 spacecraft (launched 1977) are still transmitting — powered by ^{238}Pu RTGs delivering $\sim$ 40 W at launch. PbTe thermoelectric couples convert decay heat directly to electricity with $\eta \sim 6\%$, no moving parts, 100% reliability over decades.

Automotive waste heat recovery: A typical gasoline engine wastes $\sim 60\%$ of fuel energy as heat. TEG modules on the exhaust pipe can recover 1–2 kW, reducing alternator load. Currently limited by ZT and cost.

Precision temperature control: Peltier coolers maintain laser diode junctions to $\pm 0.001\,^\circ$C — critical for wavelength stability in fiber-optic communications (DWDM channels are 0.8 nm apart).

Medical cooling: Portable vaccine refrigerators, DNA analysis devices, personal cooling vests for MS patients (heat sensitivity).

14.5 — Semiconductor Drift-Diffusion Equations

14.5.1 — Combining Ohm and Fick for Carriers

From Ch. 13, charge transport in a semiconductor involves two mechanisms:

- **Drift**: carriers move in response to electric field (Ohm's law)
- **Diffusion**: carriers move from high to low concentration (Fick's law)

The **total current densities** (using Einstein relation $D = \mu k_B T/e$):

$$\boxed{J_n = e\mu_n n\mathbf{E} + eD_n \nabla n = e\mu_n \left(n\mathbf{E} + \frac{k_B T}{e}\nabla n \right)}$$

$$\boxed{J_p = e\mu_p p\mathbf{E} - eD_p \nabla p = e\mu_p \left(p\mathbf{E} - \frac{k_B T}{e}\nabla p \right)}$$

Signs: Electrons drift opposite to $\mathbf{E}$ (negative charge) but diffuse down their concentration gradient. Holes drift with $\mathbf{E}$ but diffuse down their gradient. Both contribute to current in the same direction for most practical configurations.

14.5.2 — Continuity Equations

Conservation of electron and hole densities (charge conservation, Noether U(1)):

$$\frac{\partial n}{\partial t} = \frac{1}{e}\nabla \cdot \mathbf{J}_n + (G - R)$$
$$\frac{\partial p}{\partial t} = -\frac{1}{e}\nabla \cdot \mathbf{J}_p + (G - R)$$

where G is the carrier generation rate (thermal or optical) and R is the recombination rate. For low-level injection (minority carrier analysis):

$$R - G = \frac{\Delta n}{\tau_n} \quad \text{(electrons in p-type)}$$

where τ_n is the minority carrier lifetime (typically μs–ms in silicon).

14.5.3 — Minority Carrier Diffusion and Diffusion Length

For minority electrons in p-type material (no applied field, $\mathbf{E} \approx 0$ in neutral bulk), in steady state:

$$D_n \frac{d^2 \Delta n}{dx^2} - \frac{\Delta n}{\tau_n} = 0$$

Solution: $\Delta n(x) = \Delta n_0 \, e^{-x/L_n}$

Electron diffusion length: $L_n = \sqrt{D_n \tau_n}$

This is the average distance a minority carrier diffuses before recombining. It sets the length scale for p-n junction and transistor behavior.

For silicon at 300 K: $D_n \approx 25$ cm²/s, $\tau_n \approx 1\,\mu$s → $L_n = \sqrt{25 \times 10^{-4} \times 10^{-6}} = 50\,\mu$m.

14.5.4 — The p-n Junction and Shockley Equation

At the p-n junction, minority carriers diffuse across the depletion region. The boundary conditions (from Boltzmann factor e^{eV/k_BT}):

$$\Delta n_p(0) = n_{p0}(e^{eV/k_BT} - 1), \qquad \Delta p_n(0) = p_{n0}(e^{eV/k_BT} - 1)$$

Solving the diffusion equation in each neutral region:

$$\boxed{I = I_0\left(e^{eV/k_BT} - 1\right), \qquad I_0 = Ae^2\left(\frac{D_n n_{p0}}{L_n} + \frac{D_p p_{n0}}{L_p}\right)}$$

Shockley's ideal diode equation — derived from drift-diffusion (Ch. 13) applied to a p-n junction (Ch. 6 §6.6.2). Every diode model in circuit simulators (SPICE) starts here.

14.5.5 — Solar Cell Physics

Under illumination, minority carriers are generated at rate G_{opt}. The short-circuit current:

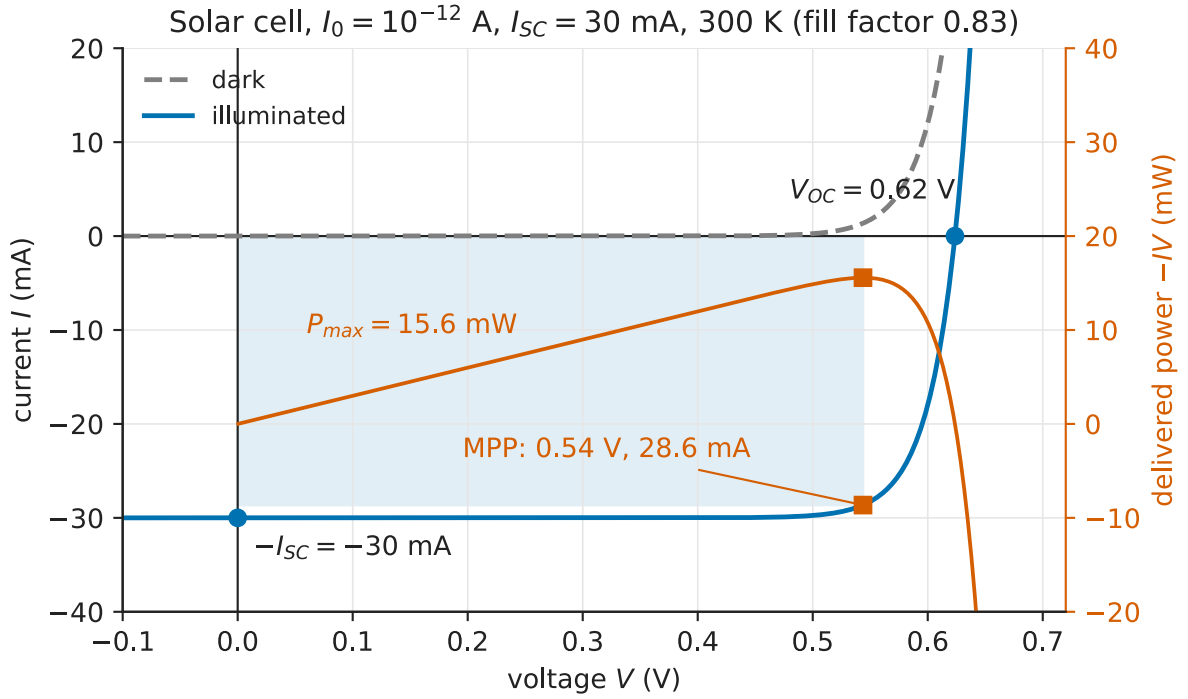
$$I_{SC} = Ae(G_{opt}L_n + G_{opt}L_p) = Ae\,G_{opt}(L_n + L_p)$$

Open-circuit voltage (from $I = 0$):

$$V_{OC} = \frac{k_BT}{e}\ln\left(\frac{I_{SC}}{I_0} + 1\right) \approx \frac{k_BT}{e}\ln\frac{I_{SC}}{I_0}$$

For a 1 cm² silicon solar cell: $I_0 \approx 10^{-12}$ A, $I_{SC} \approx 30$ mA → $V_{OC} \approx 0.62$ V at 300 K (typical cells: 0.6–0.7 V).

Maximum power point: Efficiency limited by the Shockley-Queisser limit ($\eta_{max} \approx 33\%$ for single-junction under AM1.5 spectrum) from thermodynamic analysis of blackbody radiation (Ch. 10 §10.8.4) + Carnot considerations at the bandgap.



Ideal-diode solar cell $I = I_0(e^{eV/k_BT} - 1) - I_{SC}$ with $I_0 = 10^{-12}$ A, $I_{SC} = 30$ mA at 300 K, dark (dashed) and illuminated, with delivered power $-IV$ on the right axis; illumination shifts the curve down by I_{SC}, giving $V_{OC} \approx 0.62$ V and a maximum power point near 0.54 V, 28.6 mA (15.6 mW, fill factor 0.83).

14.6 — Thermionic Emission

14.6.1 — The Richardson-Dushman Equation

From Ch. 10 §10.8.2 (Fermi-Dirac distribution): electrons in a metal at temperature T have an energy distribution $f(E) = 1/(e^{(E-E_F)/k_BT}+1)$. The fraction with enough energy to escape the surface (energy $> E_F + W$, where W is the work function):

$$J = A_R\, T^2\, e^{-W/k_BT}$$

$$\boxed{A_R = \frac{4\pi m_e k_B^2 e}{h^3} = 1.20 \times 10^6\ \text{A/(m}^2\text{K}^2\text{)}}$$

This is the **Richardson-Dushman equation**. A_R is the Richardson constant; experimentally it ranges from 0.5×10^6 to 1.5×10^6 depending on crystal orientation and surface condition.

14.6.2 — Work Functions and Schottky Effect

Work functions of common metals:

Metal	W (eV)	Notes
Cs	1.95	Lowest; used in photocathodes
Ba	2.52	Oxide cathodes
W	4.55	High T stability; electron guns
Mo	4.36	Vacuum tubes
Ni	5.15	Catalyst surfaces
Pt	5.65	Highest; inert contacts
Au	5.10	Ohmic contacts to p-GaAs

Schottky effect: An applied electric field E_{field} lowers the work function by the image-charge potential:

$$\Delta W = \sqrt{\frac{e^3 E_{field}}{4\pi\epsilon_0}}$$

For $E_{field} = 10^7$ V/m: $\Delta W \approx 0.38$ eV — significant enhancement of emission current at high fields.

14.6.3 — Engineering Applications

X-ray tubes: Tungsten filament heated to ~ 2500 K emits electrons; accelerated to ~ 100 kV; decelerated by target (Au or W) → bremsstrahlung X-rays.

SEM/TEM electron guns: Thermionic (W or LaB_6) or field emission (cold) sources. LaB_6 has a lower work function ($W = 2.66$ eV) and higher brightness than W, widely used in analytical SEMs.

Vacuum tubes (historical and niche): Thermionic triode/pentode — once the backbone of electronics; now confined to audio amplifiers, high-power RF transmitters, and CRT displays.

Thermionic energy converters: A hot cathode emits electrons that travel to a cooler anode across a vacuum gap, doing work. Efficiency limited by space-charge effects but potentially higher than Carnot (uses non-equilibrium electron distribution). Research-stage technology for concentrated solar power.

14.7 — The Thermal Resistance Network: Bridge to Layer 3

14.7.1 — Fourier → Thermal Resistance

For steady-state heat flow through a slab (Fourier's law, Ch. 13 §13.5):

$$\dot{Q} = -\kappa A \frac{dT}{dx}$$

For a uniform slab of length L, area A, in steady state ($\nabla^2 T = 0$): T is linear, and:

$$\dot{Q} = \frac{\kappa A}{L} \Delta T = \frac{\Delta T}{R_{th}}$$

Thermal resistance:

$$\boxed{R_{th} = \frac{L}{\kappa A}} \qquad \text{(units: K/W)}$$

This is exactly Ohm's law with $\Delta T \leftrightarrow \Delta V$ and $\dot{Q} \leftrightarrow I$ and $R_{th} \leftrightarrow R$. The analogy is exact because Fourier's law and Ohm's law are instances of the same linear-response (Kubo) template, applied to different currents with different closures (Ch. 13 §13.10).

14.7.2 — Thermal Resistance Networks

Series (heat flows through each layer sequentially):

$$R_{th,total} = \sum_i R_{th,i} = \sum_i \frac{L_i}{\kappa_i A}$$

Example: CPU die → TIM (thermal interface material) → heatspreader → TIM → heatsink:

Layer	κ (W/mK)	L (mm)	A (cm²)	R_{th} (K/W)
Si die	150	0.5	1	0.033
TIM1 (indium)	80	0.1	1	0.013
Cu spreader	400	3.0	100	0.075
TIM2 (paste)	8	0.2	100	0.025
Al heatsink	200	20	1000	0.010
Convection	—	—	—	0.50
Total				**0.66 K/W**

At 100 W TDP: $\Delta T = 66\,^\circ\text{C}$ from junction to ambient. Junction at $T_j = 66 + 25 = 91\,^\circ\text{C}$ ✓

Parallel (heat flows through multiple paths):

$$\frac{1}{R_{th,total}} = \sum_i \frac{1}{R_{th,i}}$$

Contact thermal resistance (Kapitza resistance): At an interface between two materials, phonon transmission is imperfect. Contact resistance $R_c = 1/hA$ where h is the interface conductance ($\sim 10^7$–10^9 W/m²K). Dominant effect at nanoscale (where $L < \ell_{phonon}$).

14.7.3 — Transient Thermal: The RC Thermal Network

Adding thermal mass (the thermal capacitance):

$$C_{th} = \rho c_p V \qquad (\text{J/K})$$

The thermal RC time constant:

$$\tau_{th} = R_{th} \cdot C_{th}$$

For an LED package ($R_{th} = 5$ K/W, volume $= 1$ cm³, $\rho c_p = 3 \times 10^6$ J/m³K): $C_{th} = 3 \times 10^6 \times 10^{-6} = 3$ J/K → $\tau_{th} = 5 \times 3 = 15$ s.

A pulsed LED can handle much higher peak power than continuous — the pulse must be shorter than τ_{th} for the junction temperature not to exceed its limit.

The thermal RC network is Bridge C applied to heat flow — the Fourier PDE lumped into a first-order ODE for each thermal node. This is exactly the same lumping operation as Ch. 17 (Bridge C) will perform for all domains. Thermal engineering students are already doing Layer 3 analysis; they just call it "thermal resistance modeling" instead of "control volume lumping."

14.8 — Diffusion in Engineering: The Error Function Solution

14.8.1 — Semi-Infinite Solid with Constant Surface Concentration

Fick's second law: $\partial c/\partial t = D\nabla^2 c$

For a semi-infinite solid ($x > 0$) with surface concentration c_s (constant) and initial uniform concentration c_0:

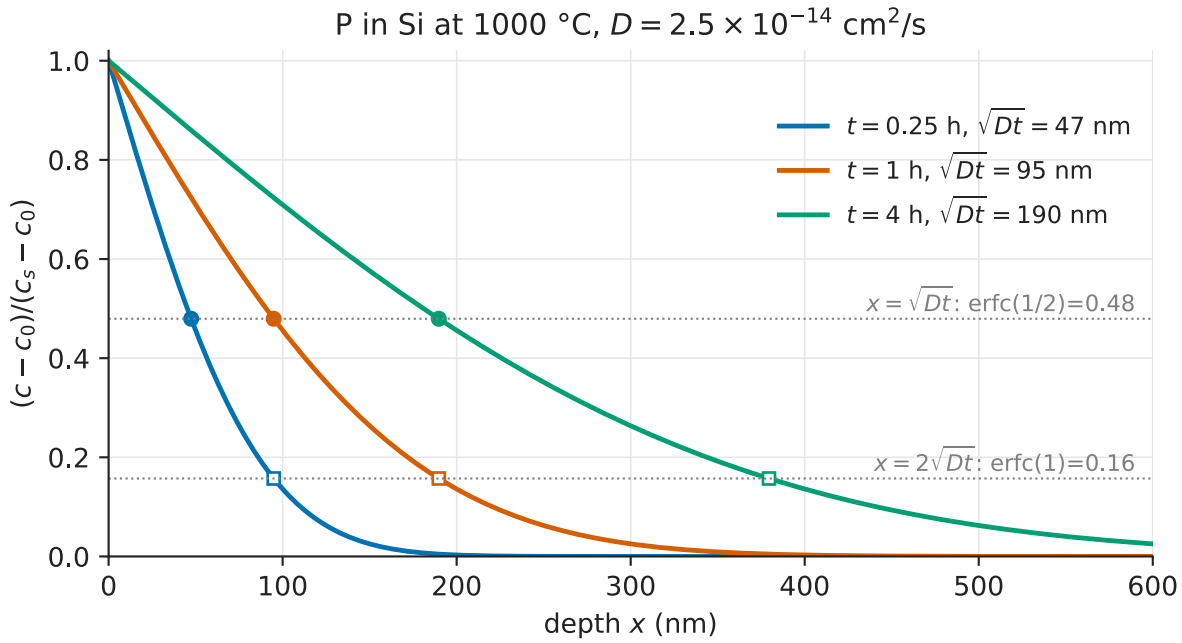
$$\boxed{c(x,t) = c_s + (c_0 - c_s)\,\text{erf}\left(\frac{x}{2\sqrt{Dt}}\right)}$$

where $\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-u^2}\,du$.

Junction depth in semiconductor doping: After a diffusion anneal at temperature T for time t, the dopant profile is the complementary error function above (constant-source predeposition; a Gaussian profile arises instead for a limited-source drive-in). The junction depth x_j (where $c = c_{background}$):

$$x_j \approx 2\sqrt{Dt}\,\text{erfc}^{-1}\left(\frac{c_{background}}{c_s}\right)$$

For phosphorus in silicon at 1000°C ($D = 2.5 \times 10^{-14}$ cm²/s) for $t = 1$ h: $\sqrt{Dt} = \sqrt{2.5 \times 10^{-14} \times 3600} = 9.5 \times 10^{-6}$ cm $= 95$ nm.



Normalised concentration $(c - c_0)/(c_s - c_0) = \operatorname{erfc}(x/2\sqrt{Dt})$ for P in Si with $D = 2.5 \times 10^{-14}$ cm²/s at $t = 0.25$, 1 and 4 h; dots mark $x = \sqrt{Dt}$ (47, 95, 190 nm) and squares $x = 2\sqrt{Dt}$ — quadrupling the time only doubles the depth.

14.8.2 — Arrhenius Diffusivity in Solids

Solid-state diffusion requires atoms to jump between lattice sites, overcoming an energy barrier Q (activation energy, from Ch. 6 §6.8.1 — bond breaking):

$$D = D_0\, e^{-Q/RT} = D_0\, e^{-Q/k_B T N_A}$$

Same Boltzmann factor as reaction kinetics. For self-diffusion in copper: $D_0 = 0.78$ cm²/s, $Q = 211$ kJ/mol.

At 800°C (1073 K): $D = 0.78\, e^{-211000/8.314/1073} = 5.2 \times 10^{-11}$ cm²/s. At 600°C: $D = 2.4 \times 10^{-13}$ cm²/s — 200× slower. Activated processes are highly sensitive to temperature.

Engineering applications:

- **Steel carburization**: carbon diffuses into iron surface; $Q = 135$ kJ/mol for C in γ-Fe
- **Semiconductor doping**: B, P, As in Si; $Q \sim 3$–4 eV
- **Oxidation of metals**: O^{2-} diffuses through oxide scale (Pilling-Bedworth ratio)
- **Grain growth**: grain boundary diffusion; $Q < Q_{bulk}$ (less barrier at boundaries)

14.9 — Magnetoresistance Effects

14.9.1 — Ordinary Magnetoresistance

For free electrons, a magnetic field curves the trajectories (Lorentz force) and increases the effective path length between collisions. The resistance increase:

$$\frac{\Delta\rho}{\rho_0} = (\mu B)^2 \quad \text{for } \mu B \ll 1$$

For copper ($\mu = 44$ cm²/Vs) at $B = 1$ T: $\mu B = 0.044 \to \Delta\rho/\rho_0 \approx 0.002$ — tiny.

For high-mobility semiconductors (InSb, $\mu = 7.7$ m²/Vs) at $B = 0.1$ T: $\mu B = 0.77 \to$ significant magnetoresistance. This is why Hall sensors need low-μ materials for linear response.

14.9.2 — Anisotropic Magnetoresistance (AMR)

In ferromagnetic metals (Ni, Fe, permalloy), the resistivity depends on the angle θ between the current direction and the magnetization direction:

$$\rho(\theta) = \rho_\perp + (\rho_\parallel - \rho_\perp)\cos^2\theta$$

AMR ratio: $(\rho_\parallel - \rho_\perp)/\rho_\perp \approx 2$–$5\%$ for Ni-Fe alloys.

Application: Hard disk drive read heads (1990s–early 2000s) used AMR sensors detecting the fringing field of magnetic bits. Superseded by GMR.

14.9.3 — Giant Magnetoresistance (GMR) and Tunnel Magnetoresistance (TMR)

From Ch. 7 §7.2 (spin-dependent transport):

GMR (spin-valve): $\Delta R/R \sim 5$–80% for metallic spacer. Current HDD read heads use TMR for still higher ratios.

TMR (magnetic tunnel junction, MTJ): Two ferromagnetic electrodes separated by a thin (~ 1 nm) insulating barrier. Tunnel current depends on relative magnetization orientation:

$$\text{TMR} = \frac{R_{AP} - R_P}{R_P} = \frac{2P_1P_2}{1 - P_1P_2}$$

where P_i is the spin polarization of electrode i. For CoFeB/MgO/CoFeB: TMR $> 600\%$ at room temperature — the spin-polarized tunneling through crystalline MgO is symmetry-filtered.

MRAM (Magnetic RAM): Each MTJ is a non-volatile memory bit — high resistance = "1", low resistance = "0". Written by spin-transfer torque (STT), read by measuring resistance. STT-MRAM offers SRAM-like speed, flash-like non-volatility, and essentially unlimited endurance. In production at 22–28 nm nodes (Everspin, Samsung, GlobalFoundries).

14.10 — Summary

Effect	Key equation	Engineering use
Matthiessen's rule	$\rho = \rho_0 + \rho_{ph}(T) + \ldots$	RTD calibration; purity assessment
RTD (Pt100)	Callendar-Van Dusen	Temperature measurement ±0.01°C
Hall effect	$V_H = IB/nqt$	Current sensors; carrier density measurement
QH resistance	$R_K = h/e^2$ (exact)	SI resistance standard
Seebeck	$V = S\Delta T$	Thermocouples; all 8 standard types
Peltier	$\dot{Q} = ST \cdot I$	Laser cooling; solid-state refrigeration
TEG efficiency	$\eta = \eta_C[\sqrt{1+ZT}-1]/[\sqrt{1+ZT}+T_C/T_H]$	Waste heat recovery; RTGs

Effect	Key equation	Engineering use
TEC ΔT_{max}	$\Delta T_{max} = ZT_C^2/2$	Minimum achievable cold-side temperature
Figure of merit	$ZT = S^2\sigma T/\kappa$	Materials optimization target
Drift-diffusion	$J = en\mu E + eD\nabla n$	All semiconductor device modeling
Shockley equation	$I = I_0(e^{eV/kT} - 1)$	Diode, BJT, solar cell analysis
Solar cell V_{OC}	$V_{OC} = (kT/e)\ln(I_{SC}/I_0)$	PV cell characterization
Richardson-Dushman	$J = A_R T^2 e^{-W/kT}$	Electron guns; thermionic converters
Thermal resistance	$R_{th} = L/\kappa A$	Junction temperature; heatsink design
Transient thermal	$\tau_{th} = R_{th}C_{th}$	Pulsed power; thermal shock analysis
Error function diffusion	$c = c_s\text{erfc}(x/2\sqrt{Dt})$	Doping profiles; case hardening
Arrhenius diffusion	$D = D_0 e^{-Q/RT}$	Process time-temperature control
AMR	$\Delta\rho/\rho \sim 2$–5%	Magnetic field sensors
GMR/TMR	$\Delta R/R$ up to 600%	HDD read heads; MRAM

14.11 — Engineering Thread

Physics	Engineering system
Matthiessen's rule + Pt100	Industrial process temperature measurement
Hall sensors	Brushless DC motor control; current measurement in EVs
Seebeck + thermocouples	Furnace control; gas turbine exhaust monitoring
Peltier coolers	Laser diode temperature control; CCD sensor cooling
TEG	Automotive exhaust recovery; spacecraft power (RTGs)
ZT optimization	Next-generation thermoelectric materials research
Drift-diffusion	MOSFET design; BJT gain; solar cell efficiency
Shockley equation	SPICE device models; analog circuit design
Thermal resistance	CPU/GPU thermal design; LED luminaire design; power module packaging
Arrhenius diffusion	IC process recipe development; steel heat treatment
STT-MRAM	Embedded non-volatile memory in advanced CMOS nodes

14.12 — Looking Ahead

Chapter 14 completed the "Layer 2 Applied" section — taking the transport laws derived in Chapters 8–13 and working through their engineering applications in detail. The next two chapters complete Layer 2 itself:

Chapter 15 covers **continuum mechanics** — deriving the Navier-Stokes equations for fluids and the elasticity equations for solids. These are the remaining Layer-2 PDEs that, alongside Maxwell (Ch. 11) and the transport equations (Ch. 13), complete the classical continuum description.

Chapter 16 reveals the **unified PDE structure** — showing that the wave equation and diffusion equation appear identically across acoustics, electromagnetism, elasticity, heat, and mass transport. This is the last unification before Bridge C begins the descent to Layer 3.

End of Chapter 14.

Next: Chapter 15 — Continuum Mechanics: Fluids and Solids

Continuum Mechanics: Fluids and Solids

Newton's Second Law Applied to 10^{23} Particles Simultaneously

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> *The equations of fluid motion are Newton's second law for a fluid parcel. Everything else is bookkeeping.* — commonly attributed to G. I. Taylor

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15.0 — Overview

Chapters 8–14 completed the five Bridge B paths, bringing all of classical physics into Layer 2. But Layer 2 still has two important members unwritten:

- **Continuum mechanics** (this chapter): the mechanics of deformable media — fluids and solids — treated as continuous fields rather than discrete particles
- **The unified PDE structure** (Ch. 16): showing that the wave equation and diffusion equation appear identically across all physical domains

This chapter derives the governing equations of fluid mechanics (Euler and Navier-Stokes) and solid mechanics (Navier's equation of linear elasticity) from first principles. The derivation has a single organizing strategy: **apply Newton's second law to a fluid or solid parcel** and use the stress tensor to represent internal forces. Every subsequent complication — viscosity, elasticity, compressibility, non-linearity — is a refinement of this one idea.

Connections to previous chapters:

Concept	Origin
Continuum hypothesis	Coarse-graining with $\ell_{\text{micro}} \ll L_{\text{variation}}$, built on the $N \to \infty$ limit of Ch. 10 (Bridge B.b)
Strain tensor ε_{ij}	Generalization of Ch. 9 spring extension to 3D
Viscous stress	Newton's viscosity from Ch. 13 §13.7 (Kubo)
Elastic stress	Hooke's law from Ch. 13 §13.8
Conservation laws	Noether's theorem (Ch. 1 §1.7): momentum, energy; particle number — hence mass for a single nonrelativistic species
Elastic waves	Classical limit of phonons (Ch. 6 §6.7)
Turbulence	Where the PDE refuses to lump — Layer 3 has no exact model

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15.1 — The Continuum Hypothesis

15.1.1 — From Particles to Fields

The coarse-grained continuum description of Ch. 10, with many microscopic degrees of freedom inside each representative volume, gave thermodynamic quantities as ensemble averages. Applied to a spatially inhomogeneous system, it gives **field variables** — quantities that vary continuously in space:

- **Density field:** $\rho(\mathbf{r}, t) = \langle \hat{\rho} \rangle$ — mass per unit volume
- **Velocity field:** $\mathbf{v}(\mathbf{r}, t) = \langle \hat{\mathbf{v}} \rangle$ — momentum per unit mass
- **Pressure field:** $p(\mathbf{r}, t)$ — isotropic stress
- **Temperature field:** $T(\mathbf{r}, t)$ — local thermodynamic temperature
- **Displacement field:** $\mathbf{u}(\mathbf{r}, t)$ — displacement from equilibrium (solids)

The averaging volume must be:

- Large enough to contain many molecules (thermodynamic limit valid)
- Small enough to resolve the spatial variation of interest

> **Warning: $N \to \infty$ alone does not give a continuum field** The thermodynamic limit of Ch. 10 is a statement about *particle number*; it says nothing about length scales. A continuum description additionally requires **scale separation** — a microscopic length ℓ_{micro} and a scale of variation $L_{\text{variation}}$ with $\ell_{\text{micro}} \ll L_{\text{variation}}$. A system with $N \to \infty$ but no such separation, such as a cluster small enough that every molecule is still individually resolved, has no continuum field to coarse-grain onto. This is the same reason the Knudsen criterion below is the practical test, and the two must not be conflated: the thermodynamic limit supplies the many-body averaging, the Knudsen number supplies the spatial validity.

Validity criterion — Knudsen number:

$$Kn = \frac{\ell_{mfp}}{L} \ll 1$$

where ℓ_{mfp} is the mean free path and L is the characteristic length of the problem. Continuum mechanics is valid for $Kn < 0.01$.

Medium	ℓ_{mfp}	Continuum breaks at
Air at 1 atm, 300 K	68 nm	Micro/nano-channels $L < 7\,\mu$m
Water at 300 K	~ 0.3 nm	Near-molecular systems
Liquid metals	~ 0.1 nm	Essentially always valid
Air at 30 km altitude	~ 1 mm	Small aircraft structures
Re-entry vehicle wake	several cm	Kinetic theory needed

15.1.2 — Eulerian vs. Lagrangian Description

Lagrangian (tracking particles): follow each fluid parcel as it moves. Natural for solid mechanics (track each material point).

Eulerian (fixed spatial grid): observe what passes through a fixed point. Natural for fluid mechanics (the flow field at fixed locations).

The connection is the **material derivative** (substantial derivative) — the rate of change of a quantity as seen by a parcel moving with the fluid:

$$\boxed{\frac{D}{Dt} = \frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla}$$

- $\partial/\partial t$: local rate of change at fixed position (Eulerian)
- $\mathbf{v} \cdot \nabla$: change due to moving to a new location (advection)

Example: The temperature of a fluid parcel changes because (a) the temperature at that location changes with time, and (b) the parcel moves to a region of different temperature. Both contribute to DT/Dt.

The advection term $(\mathbf{v} \cdot \nabla)\mathbf{v}$ in Newton's law for a fluid parcel is what makes Navier-Stokes **nonlinear** — and what makes turbulence so difficult.

15.2 — Kinematics: Describing Deformation

15.2.1 — The Strain Tensor

When a solid body deforms, each point $\mathbf{r}$ moves to $\mathbf{r} + \mathbf{u}(\mathbf{r})$. For small deformations, the infinitesimal **strain tensor**:

$$\boxed{\varepsilon_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)}$$

Symmetric: $\varepsilon_{ij} = \varepsilon_{ji}$ (6 independent components in 3D).

Physical interpretation of components:

- **Diagonal** $\varepsilon_{11}, \varepsilon_{22}, \varepsilon_{33}$: normal strains (fractional extension along each axis). $\varepsilon_{kk} = \nabla \cdot \mathbf{u}$ = volumetric strain.
- **Off-diagonal** $\varepsilon_{12}, \varepsilon_{13}, \varepsilon_{23}$: shear strains (change in angle between originally perpendicular line elements). Factor of 2 compared to engineering shear strain $\gamma_{ij} = 2\varepsilon_{ij}$.

Decomposition: Any displacement gradient decomposes into:

$$\frac{\partial u_i}{\partial x_j} = \underbrace{\varepsilon_{ij}}_{\text{strain}} + \underbrace{\omega_{ij}}_{\text{rotation}}$$

where the antisymmetric part $\omega_{ij} = \frac{1}{2}(\partial_i u_j - \partial_j u_i)$ is a rigid body rotation — it causes no stress and is not related to deformation.

15.2.2 — The Strain Rate Tensor (Fluids)

For a flowing fluid, it is not the displacement but the **velocity gradient** that causes viscous stress. The strain rate tensor:

$$\dot{\varepsilon}_{ij} = \frac{1}{2}\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}\right)$$

Same structure as ε_{ij} but with velocities instead of displacements. This is Newton's law of viscosity generalized to three dimensions.

Vorticity: The antisymmetric part of the velocity gradient:

$$\omega_{ij} = \frac{1}{2}\left(\frac{\partial v_i}{\partial x_j} - \frac{\partial v_j}{\partial x_i}\right), \qquad \boldsymbol{\omega} = \nabla \times \mathbf{v}$$

Irrotational flow: $\boldsymbol{\omega} = 0 \rightarrow$ potential flow ($\mathbf{v} = \nabla\phi$).

15.3 — The Stress Tensor: Internal Forces

15.3.1 — Cauchy's Stress Tensor

Consider a material cut by an imaginary plane with unit normal $\hat{n}$. The force per unit area exerted by the material on the positive side on the negative side is the **traction vector** $\mathbf{t}^{(\hat{n})}$.

Cauchy's theorem: The traction vector is linear in $\hat{n}$:

$$t_i^{(\hat{n})} = \sigma_{ij} n_j$$

where σ_{ij} is the **Cauchy stress tensor** — a 3×3 matrix (9 components, reduced to 6 by symmetry) that completely characterizes the internal force state at a point.

Physical interpretation:

- σ_{11}, σ_{22}, σ_{33}: **normal stresses** — tension (positive) or compression (negative)
- $\sigma_{12} = \sigma_{21}$, etc.: **shear stresses** — tangential forces on each face
- **Pressure:** $p = -\frac{1}{3}\sigma_{kk}$ (hydrostatic stress, sign convention: compression is positive pressure)

Symmetry of σ_{ij}: Angular momentum balance for a fluid parcel requires $\sigma_{ij} = \sigma_{ji}$ — the stress tensor is symmetric (6 independent components, not 9).

15.3.2 — Principal Stresses

The stress tensor can always be diagonalized by rotating to principal axes — the three orthogonal directions in which only normal stresses act (no shear). The eigenvalues $\sigma_1 \geq \sigma_2 \geq \sigma_3$ are the **principal stresses**.

The **von Mises equivalent stress** (used in failure criteria, §15.11):

$$\sigma_{vM} = \sqrt{\frac{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2}{2}}$$

15.4 — Conservation Laws: Noether's Theorem in Continuum Form

The conservation laws from Ch. 1 (Noether's theorem) apply to continua as **partial differential equations** — one for each conserved density.

15.4.1 — Mass Conservation (Continuity Equation)

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0}$$

or equivalently: $\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} = 0$

For incompressible flow ($D\rho/Dt = 0$, constant density following a parcel):

$$\nabla \cdot \mathbf{v} = 0$$

Every divergence-free velocity field is volume-preserving. This is the continuum form of the conservation of **particle number**. For a single nonrelativistic species the relevant symmetry is the *global* phase $U(1)$ of the matter field, whose Noether charge is the particle number $N = \int n \, d^3x$ (Ch. 1 §1.7); the mass density $\rho = mn$ is proportional to that charge for fixed m, so mass conservation follows for that species as a *consequence*. It should not be claimed, however, that mass conservation in general descends from a fundamental $U(1)$. For relativistic fields the rest mass is part of the four-momentum rather than a separate conserved charge, and for a multi-component system the total mass is not fixed by any one conserved charge at all. The particle-number statement is the primitive one; mass conservation follows only where the proportionality $\rho = mn$ holds.

15.4.2 — Momentum Conservation (Cauchy's Equation of Motion)

Newton's second law for a material parcel of density ρ, acted on by body force $\mathbf{f}$ per unit volume and surface force from the stress tensor:

$$\boxed{\rho \frac{D\mathbf{v}}{Dt} = \rho \left(\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} \right) = \nabla \cdot \boldsymbol{\sigma} + \mathbf{f}}$$

This is **Cauchy's equation of motion** — the universal momentum equation for any continuum. It is Newton's second law in continuum form. The specific form of $\boldsymbol{\sigma}$ (the constitutive relation) determines whether this is fluid mechanics or solid mechanics.

- **Body force $\mathbf{f}$**: gravity $\rho\mathbf{g}$, electromagnetic (Lorentz force on a charged continuum), centrifugal and Coriolis (in a rotating frame, Ch. 9 §9.9)
- **Surface force $\nabla \cdot \boldsymbol{\sigma}$**: from the divergence of the stress tensor — a net stress gradient gives a net force

This is Noether's theorem (space translation → momentum conservation, Ch. 1 §1.7) in PDE form: $\partial_t(\rho v_i) + \partial_j(\rho v_i v_j - \sigma_{ij}) = f_i$.

15.4.3 — Energy Conservation

The energy equation (first law of thermodynamics per unit volume in motion):

$$\rho \frac{De}{Dt} = \sigma_{ij}\dot{\varepsilon}_{ij} - \nabla \cdot \mathbf{q} + \dot{Q}_{vol}$$

where e is the specific internal energy, $\sigma_{ij}\dot{\varepsilon}_{ij} = -p\,\nabla \cdot \mathbf{v} + \tau_{ij}\dot{\varepsilon}_{ij}$ is the stress power — the reversible compression work $-p\,\nabla \cdot \mathbf{v}$ (either sign) plus the viscous dissipation rate $\tau_{ij}\dot{\varepsilon}_{ij}$ (always non-negative, second law), $\mathbf{q}$ is the heat flux (from Fourier's law, Ch. 13 §13.5), and $\dot{Q}_{vol}$ is any volumetric heat source.

15.5 — The Ideal Fluid: Euler Equations

15.5.1 — The Constitutive Relation for an Ideal Fluid

An **ideal** (inviscid) fluid has no viscosity: no tangential stress between fluid layers. The only stress is isotropic pressure:

$$\sigma_{ij} = -p\,\delta_{ij}$$

Substituting into Cauchy's equation:

$$\boxed{\rho \frac{D\mathbf{v}}{Dt} = -\nabla p + \rho \mathbf{g}}$$

These are the **Euler equations** for an inviscid fluid. Together with the continuity equation and an equation of state $p = p(\rho)$, they form a closed system.

15.5.2 — Bernoulli's Equation

For **steady** ($\partial/\partial t = 0$), **incompressible** ($\rho = \text{const}$), **irrotational** ($\boldsymbol{\omega} = \nabla \times \mathbf{v} = 0$) flow, the Euler equation integrates along a streamline to:

$$\boxed{p + \frac{1}{2}\rho v^2 + \rho g z = \text{const along a streamline}}$$

Bernoulli's equation — derived from Newton's law for an ideal fluid parcel.

The three terms:

- p: static pressure (energy density stored as pressure)
- $\frac{1}{2}\rho v^2$: dynamic pressure (kinetic energy density)
- $\rho g z$: hydrostatic pressure (potential energy density)

Applications:

- **Pitot tube:** measures dynamic pressure $\frac{1}{2}\rho v^2 = p_{stagnation} - p_{static}$ → airspeed
- **Venturi meter:** pressure drop at constriction $\Delta p = \frac{1}{2}\rho(v_2^2 - v_1^2)$ → flow rate
- **Lift on an airfoil:** faster flow over upper surface → lower pressure → net upward force (though Kutta-Joukowski is more precise)
- **Cavitation onset:** when local p drops below vapor pressure p_v

15.5.3 — Hydrostatics

For a static fluid ($\mathbf{v} = 0$), Euler reduces to:

$$\nabla p = \rho \mathbf{g} \quad \Longrightarrow \quad \frac{dp}{dz} = -\rho g$$

For incompressible fluid: $p(z) = p_0 - \rho g z$ (linear pressure increase with depth).

Archimedes' principle follows directly: integrate pressure over the surface of a submerged body → net upward force $= \rho_f g V_{displaced}$.

15.6 — Viscous Flow: The Navier-Stokes Equation

15.6.1 — The Constitutive Relation for a Newtonian Fluid

A **Newtonian fluid** has a linear relationship between the stress tensor and the strain rate tensor. The most general isotropic, linear, symmetric relation:

$$\sigma_{ij} = -p\,\delta_{ij} + 2\mu\dot{\varepsilon}_{ij} + \lambda\dot{\varepsilon}_{kk}\delta_{ij}$$

where:

- $-p\delta_{ij}$: isotropic pressure (as in ideal fluid)
- $2\mu\dot{\varepsilon}_{ij}$: deviatoric (shear) viscous stress (μ = dynamic viscosity)
- $\lambda\dot{\varepsilon}_{kk}\delta_{ij}$: bulk viscous stress from volume rate of change

Stokes' hypothesis ($\lambda = -\frac{2}{3}\mu$). This is a **constitutive closure, not a derivation.** It does not follow from the Navier–Stokes equations, and nothing in the preceding steps forces it. The Navier–Stokes statement above says only that λ is *some* coefficient; the particular value $\lambda = -\frac{2}{3}\mu$ is imported, on the argument that monatomic gases have no internal degrees of freedom that could store energy in a uniform compression. That is a physical argument about molecular structure, not a theorem, and it is the sort of step this book otherwise labels honestly.

The status of the claim that follows — "the bulk viscosity is zero" — is correspondingly weaker than "well-verified" suggests. For dilute monatomic gases the bulk viscosity is *very small*, which is why the hypothesis works so well in practice; but it is not identically zero, and modern Green–Kubo determinations — equilibrium molecular dynamics and their quantum-statistical refinements — resolve a small but nonzero value for dilute argon. (The same Green–Kubo machinery used in Ch. 13 to *derive* μ is what detects this: a nice illustration that a coefficient thought to be a postulate turns out to be measurable.) It is also simply false for many ordinary fluids — λ is substantial in liquids, and large in melts, emulsions and other complex media, which is one reason they need a constitutive model rather than this one.

With Stokes' hypothesis adopted and incompressibility imposed ($\dot{\varepsilon}_{kk} = \nabla\cdot\mathbf{v} = 0$), the bulk term drops out and

$$\sigma_{ij} = -p\,\delta_{ij} + 2\mu\dot{\varepsilon}_{ij}$$

Note the ordering of the two moves: the *assumption* kills the term, and incompressibility would have done so anyway. Keeping them distinct is what lets you see which assumption is load-bearing.

15.6.2 — The Navier-Stokes Equation

Substituting the Newtonian constitutive relation into Cauchy's equation (using $\nabla\cdot\boldsymbol{\sigma} = -\nabla p + \mu\nabla^2\mathbf{v}$ for incompressible flow):

$$\boxed{\rho\frac{D\mathbf{v}}{Dt} = \rho\underbrace{\left(\frac{\partial\mathbf{v}}{\partial t} + (\mathbf{v}\cdot\nabla)\mathbf{v}\right)}_{\text{inertia}} = \underbrace{-\nabla p}_{\text{pressure}} + \underbrace{\mu\nabla^2\mathbf{v}}_{\text{viscosity}} + \underbrace{\rho\mathbf{g}}_{\text{gravity}}}$$

together with the incompressibility constraint:

$$\nabla\cdot\mathbf{v} = 0$$

These are the **Navier-Stokes equations for incompressible Newtonian flow** — four scalar equations (3 momentum + 1 continuity) for four unknowns (v_x, v_y, v_z, p).

The nonlinear term $(\mathbf{v}\cdot\nabla)\mathbf{v}$: advection of momentum by the flow itself. This is the source of all complexity in fluid mechanics — instabilities, turbulence, chaotic behavior. Without this term, NS is linear and easily solvable.

Origin of the viscous term: The term $\mu\nabla^2\mathbf{v}$ is Newton's law of viscosity (Ch. 13 §13.7) applied at every point and every direction simultaneously. The viscosity μ is the Kubo Green function of §13.7.1.

15.6.3 — Vorticity Transport

Taking the curl of NS for incompressible flow:

$$\frac{D\boldsymbol{\omega}}{Dt} = (\boldsymbol{\omega}\cdot\nabla)\mathbf{v} + \nu\nabla^2\boldsymbol{\omega}$$

where $\nu = \mu/\rho$ is the **kinematic viscosity** and $\boldsymbol{\omega} = \nabla\times\mathbf{v}$.

- $(\boldsymbol{\omega}\cdot\nabla)\mathbf{v}$: vortex stretching — the term responsible for the cascade to smaller scales in turbulence (absent in 2D!)
- $\nu\nabla^2\boldsymbol{\omega}$: viscous diffusion of vorticity

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15.7 — Exact Solutions of Navier-Stokes

15.7.1 — Hagen-Poiseuille Flow (Pipe Flow)

Steady, fully-developed, incompressible flow through a circular pipe of radius R. Assume: $\mathbf{v} = v_z(r)\hat{z}$, $\partial/\partial t = 0$, $\partial/\partial z = 0$.

NS simplifies to: $\mu\nabla^2 v_z = \frac{dp}{dz} = \text{const} \equiv -G$

In cylindrical coordinates: $\frac{\mu}{r}\frac{d}{dr}\left(r\frac{dv_z}{dr}\right) = -G$

Boundary conditions: $v_z(R) = 0$ (no-slip), $dv_z/dr|_{r=0} = 0$ (symmetry):

$$\boxed{v_z(r) = \frac{G}{4\mu}(R^2 - r^2)} \quad \text{(parabolic profile)}$$

Hagen-Poiseuille law (volume flow rate Q and pressure drop Δp):

$$Q = \int_0^R v_z(r)\cdot 2\pi r\,dr = \frac{\pi R^4}{8\mu}\frac{\Delta p}{L} \quad\Longrightarrow\quad \Delta p = \frac{8\mu L Q}{\pi R^4}$$

The R^4 dependence is dramatic: doubling the pipe radius increases flow by 16× at the same pressure drop. Halving the radius increases resistance by 16×.

The hydraulic resistance:

$$R_{hyd} = \frac{\Delta p}{Q} = \frac{8\mu L}{\pi R^4}$$

This is the fluid analog of electrical resistance ($R = \rho L/A$) — the same linear effort–flow structure, though R_{hyd} comes from solving Navier–Stokes in the pipe while electrical R comes from Ohm's law; what they share is the R/C/L representation [STRUCTURAL CONNECTION]. In Bridge C (Ch. 17), this becomes the "R" element in the hydraulic circuit.

Validity: Laminar flow only. Turbulence onset at $Re_c \approx 2300$.

15.7.2 — Couette Flow (Flow Between Plates)

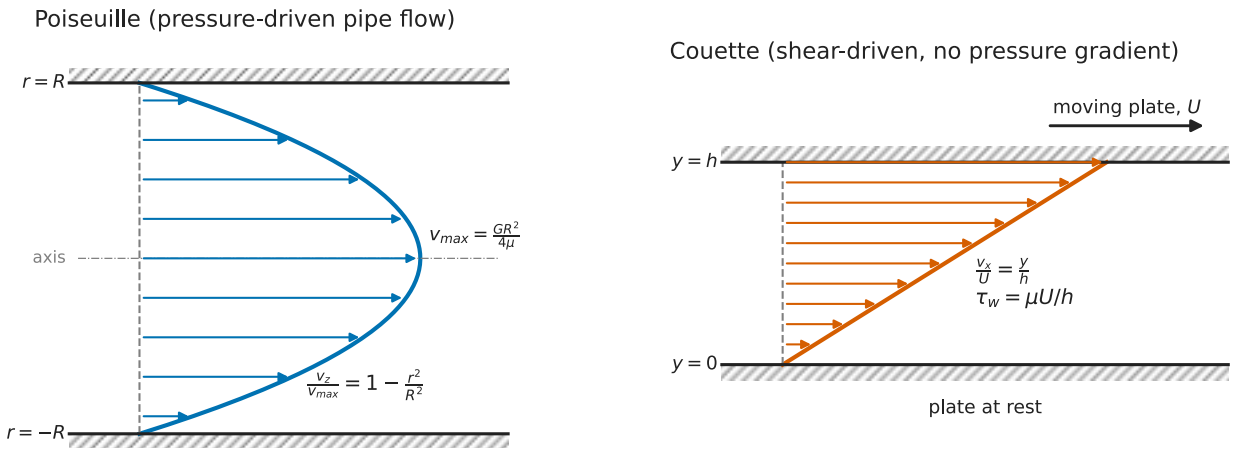
Steady flow between two parallel plates, lower at rest, upper moving at velocity U. Only $v_x(y)$ nonzero, no pressure gradient:

$$\mu \frac{d^2 v_x}{dy^2} = 0 \quad \Longrightarrow \quad v_x = U\frac{y}{h}$$

Linear velocity profile. Wall shear stress:

$$\tau_w = \mu \frac{dv_x}{dy} = \frac{\mu U}{h}$$

This is Newton's law of viscosity from Ch. 13 §13.7 — now derived as an exact solution of NS with the appropriate boundary conditions. Couette flow is the canonical laboratory setup for measuring viscosity (the cone-and-plate rheometer is a variant).



Left: Poiseuille profile $v_z/v_{max} = 1 - (r/R)^2$ across a pipe, zero at the no-slip walls and maximal on the axis. Right: Couette profile $v_x/U = y/h$ between a fixed and a moving plate, with uniform shear stress $\tau_w = \mu U/h$.

15.7.3 — Stokes Flow (Very Low Reynolds Number)

When $Re \ll 1$, the inertia term $\rho(\mathbf{v}\cdot\nabla)\mathbf{v}$ is negligible compared to viscous forces. NS becomes linear (Stokes equations):

$$\mu \nabla^2 \mathbf{v} = \nabla p, \qquad \nabla \cdot \mathbf{v} = 0$$

Stokes drag on a sphere of radius a moving at velocity U:

$$\mathbf{F}_{drag} = -6\pi\mu a \mathbf{U}$$

Applications: Sedimentation of small particles; blood cell motion; swimming of microorganisms; MEMS fluid channels; aerosol particle dynamics.

The Stokes drag formula is also the basis for the Einstein relation (Ch. 13 §13.6.2). The derivation has to get the *driving force* right, because that is where the usual shortcut slips. The thermodynamic force on a dilute solute comes from the chemical potential, and in the ideal dilute limit

$$\mathbf{F}_{th} = -k_B T \nabla \ln c = -k_B T \frac{\nabla c}{c}$$

This has units of energy per length — a force ✓. (The tempting shorthand $\mathbf{F} = D\nabla c/c$ is dimensionally wrong: D is m²/s and $\nabla c/c$ is 1/m, so the product is m/s, a **velocity**, not a force. It is easy to write because it *looks* like it will cancel into something sensible, but it silently mixes the transport coefficient into the driving term.)

With the force correct, the two routes agree. Balancing drift against drag and converting to a flux, $k_B T|\nabla c|/c = 6\pi\mu a\,\mathbf{U}$ together with $\mathbf{J} = -c\mathbf{U} = -D\nabla c$, gives

$$D = \frac{k_B T}{6\pi\mu a}$$

Equivalently, and more transparently as a statement about *mobility*: Stokes drag fixes the hydrodynamic mobility of the sphere, $\mu_{mech} = 1/(6\pi\mu a)$, and Einstein's relation is $D = \mu_{mech} k_B T$ — same answer, but now the force, the mobility, and the diffusion coefficient each have their own dimension and the balance is visible rather than assumed. (Stokes–Einstein).

15.8 — Dimensional Analysis and the Reynolds Number

15.8.1 — Non-Dimensionalizing the Navier-Stokes Equation

Scale all variables by characteristic values: velocity U_0, length L_0, time L_0/U_0, pressure ρU_0^2. The dimensionless NS equation:

$$\frac{D\mathbf{v}^*}{Dt^*} = -\nabla^* p^* + \frac{1}{Re}\nabla^{*2}\mathbf{v}^* + \frac{1}{Fr^2}\hat{g}$$

where the **Reynolds number** Re and **Froude number** Fr:

$$\boxed{Re = \frac{\rho U_0 L_0}{\mu} = \frac{U_0 L_0}{\nu}} \qquad Fr = \frac{U_0}{\sqrt{g L_0}}$$

The Reynolds number is the single most important parameter in fluid mechanics. It measures the ratio of inertial forces to viscous forces.

Re	Flow regime	Example
$Re \ll 1$	Stokes (creeping) flow	Pollen in air, microfluidics
$Re \sim 1$	Transitional (still laminar)	Blood in capillaries
$Re \sim 10^2$	Laminar with separation	Flow past a cylinder, streamlined bodies
$Re \sim 10^3$	Wake instabilities, vortex shedding	Kármán vortex street
$Re \gtrsim 2300$	Transition in pipes	Pipe flow becomes turbulent

Re	Flow regime	Example
$Re \sim 10^6$	Fully turbulent	Aircraft at cruise, water in rivers
$Re \sim 10^9$	Atmospheric flows	Ocean circulation, hurricanes

15.8.2 — Turbulence: Where Layer 3 Has No Exact Model

At high Re, the inertial term $(\mathbf{v} \cdot \nabla)\mathbf{v}$ dominates. The flow becomes unstable to perturbations. Small eddies cascade into ever smaller eddies (Kolmogorov energy cascade) until viscous dissipation absorbs the energy at the Kolmogorov length scale:

$$\eta_K = \left(\frac{\nu^3}{\varepsilon_{diss}} \right)^{1/4}$$

For air at $Re = 10^6$: $\eta_K \sim 0.1$ mm — nine orders of magnitude smaller than the flight vehicle. Direct numerical simulation (DNS) of the NS equations at these Re is computationally intractable.

The engineering consequence: Turbulence is the single most important unsolved problem in classical physics for engineering practice. Engineers use:

- **RANS** (Reynolds-Averaged NS): solve for time-averaged flow, model turbulence stresses (k-ε, k-ω models)
- **LES** (Large-Eddy Simulation): resolve large eddies, model small ones
- **DNS** (Direct Numerical Simulation): resolve all scales; feasible only for moderate Re

None of these is a closed-form Layer 3 model. Turbulence is where the Navier-Stokes PDE refuses to be lumped — one of the most honest demonstrations of why this book maintains all four layers rather than jumping straight to engineering formulas.

15.9 — Linear Elasticity: The Navier Equation

15.9.1 — Constitutive Relation for a Linear Elastic Solid

From Ch. 13 §13.8.2 (Hooke's law, general form):

$$\sigma_{ij} = C_{ijkl} \varepsilon_{kl}$$

For an **isotropic** material (same properties in all directions), C_{ijkl} has only two independent constants — the **Lamé constants** λ and μ_L:

$$\boxed{\sigma_{ij} = \lambda \varepsilon_{kk} \delta_{ij} + 2\mu_L \varepsilon_{ij}}$$

Equivalently, in terms of Young's modulus E (stiffness) and Poisson's ratio ν_P (lateral contraction):

$$\lambda = \frac{\nu_P E}{(1+\nu_P)(1-2\nu_P)}, \qquad \mu_L = G = \frac{E}{2(1+\nu_P)}$$

Quantity	Definition	Typical range
Young's modulus E	Uniaxial stress/strain	Steel: 200 GPa; Al: 70 GPa; rubber: 0.01–0.1 GPa
Shear modulus G	Shear stress/strain	Steel: 80 GPa; Al: 26 GPa

Quantity	Definition	Typical range
Poisson's ratio ν_P	$-\varepsilon_{yy}/\varepsilon_{xx}$ for uniaxial σ_{xx}	Metals: 0.25–0.35; rubber: ≈ 0.5; foam: ≈ 0
Bulk modulus K	Pressure/volumetric strain	Steel: 160 GPa; water: 2.2 GPa

Relations: $K = \lambda + \frac{2}{3}\mu_L = \frac{E}{3(1-2\nu_P)}$

15.9.2 — The Navier Equation of Motion

Substituting the isotropic Hooke's law into Cauchy's equation:

$$\rho\frac{\partial^2\mathbf{u}}{\partial t^2} = (\lambda+\mu_L)\nabla(\nabla\cdot\mathbf{u}) + \mu_L\nabla^2\mathbf{u} + \mathbf{f}$$

This is the **Navier equation** (or **Lamé-Navier equation**) of linear elasticity — the governing equation for displacement in a deformable elastic solid.

Comparison with Navier-Stokes:

- In NS: $\rho D\mathbf{v}/Dt = -\nabla p + \mu\nabla^2\mathbf{v} + \mathbf{f}$ (velocities)
- In Navier: $\rho\ddot{\mathbf{u}} = (\lambda+\mu_L)\nabla(\nabla\cdot\mathbf{u}) + \mu_L\nabla^2\mathbf{u} + \mathbf{f}$ (displacements)

Same mathematical structure, different physical variables. The elastic equation is linear; NS is nonlinear (the $(\mathbf{v}\cdot\nabla)\mathbf{v}$ term). This is why solid mechanics is mathematically much more tractable than turbulent fluid mechanics.

15.9.3 — Static Equilibrium

In statics ($\ddot{\mathbf{u}} = 0$):

$$(\lambda+\mu_L)\nabla(\nabla\cdot\mathbf{u}) + \mu_L\nabla^2\mathbf{u} + \mathbf{f} = 0$$

For incompressible material ($\nabla\cdot\mathbf{u} = 0$, $\nu_P \to \frac{1}{2}$):

$$\mu_L\nabla^2\mathbf{u} + \mathbf{f} = 0 \quad\Longrightarrow\quad \nabla^2\mathbf{u} = -\mathbf{f}/\mu_L$$

This is a vector Poisson equation — same structure as electrostatics (Ch. 11 §11.10). The Green's function for the elastic displacement field due to a point force is the **Kelvin fundamental solution** — the elastic analog of the Coulomb potential.

15.10 — Elastic Waves: P-waves and S-waves

15.10.1 — Wave Solutions to the Navier Equation

In a homogeneous elastic medium with no body forces ($\mathbf{f} = 0$), try plane wave solutions $\mathbf{u} = \mathbf{A}e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}$.

Helmholtz decomposition: $\mathbf{u} = \nabla\phi + \nabla\times\boldsymbol{\psi}$ separates into a scalar potential (irrotational) and vector potential (solenoidal) part.

P-waves (Primary, Pressure, Longitudinal): $\nabla\times\mathbf{u} = 0$, $\mathbf{k} \parallel \mathbf{A}$

$$\rho\omega^2\mathbf{A} = (\lambda + 2\mu_L)k^2\mathbf{A} \quad \Longrightarrow \quad \boxed{v_P = \sqrt{\frac{\lambda + 2\mu_L}{\rho}} = \sqrt{\frac{K + \frac{4}{3}G}{\rho}}}$$

P-waves: particles oscillate parallel to propagation direction (compression and rarefaction); transmitted through solids, liquids, and gases.

S-waves (Secondary, Shear, Transverse): $\nabla \cdot \mathbf{u} = 0$, $\mathbf{k} \perp \mathbf{A}$

$$\rho\omega^2\mathbf{A} = \mu_L k^2\mathbf{A} \quad \Longrightarrow \quad \boxed{v_S = \sqrt{\frac{\mu_L}{\rho}} = \sqrt{\frac{G}{\rho}}}$$

S-waves: particles oscillate perpendicular to propagation; transmitted only through solids (liquids and gases have $G = 0$, so $v_S = 0$).

Since $\lambda + 2\mu_L > \mu_L$: **P-waves always travel faster than S-waves.**

Material	v_P (m/s)	v_S (m/s)	v_P/v_S
Steel	5960	3235	1.84
Aluminum	6420	3040	2.11
Granite	5950	3640	1.63
Water	1480	0	∞
Air (1 atm)	343	0	∞

Connection to phonons (Ch. 6 §6.7.1): The elastic wave equations derived here are the classical ($\hbar \to 0$) limit of quantum phonon dispersion. Acoustic phonons (Ch. 6) → acoustic elastic waves in the long-wavelength limit. The speed of sound in a solid is the group velocity of acoustic phonons at $k \to 0$.

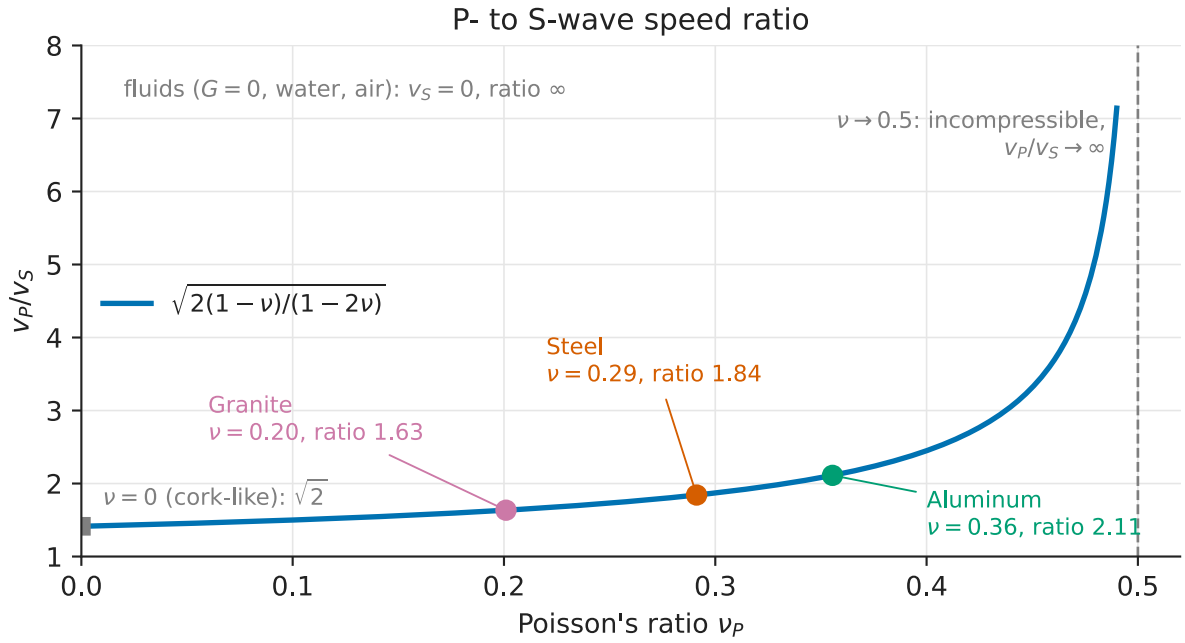
15.10.2 — The Poisson's Ratio Constraint

The wave speed ratio:

$$\frac{v_P}{v_S} = \sqrt{\frac{\lambda + 2\mu_L}{\mu_L}} = \sqrt{\frac{2(1 - \nu_P)}{1 - 2\nu_P}}$$

For most metals ($\nu_P \approx 0.3$): $v_P/v_S \approx 1.87$. For incompressible materials ($\nu_P \to 0.5$): $v_P \to \infty$ (pressure waves travel instantaneously — bulk modulus $K \to \infty$). For materials with $\nu_P = 0$ (cork-like): $v_P/v_S = \sqrt{2}$.

Seismology: The time delay between P-wave and S-wave arrivals at a seismometer directly gives the distance to an earthquake epicenter. Three stations triangulate the location. The ratio v_P/v_S also reveals subsurface geology — different for water-saturated rock vs. dry rock.



$v_P/v_S = \sqrt{2(1-\nu_P)/(1-2\nu_P)}$ for $0 \le \nu_P \le 0.49$, with steel, aluminum and granite placed at the Poisson's ratio implied by their tabulated speeds (0.29, 0.36, 0.20); the ratio starts at $\sqrt{2}$ and diverges as $\nu_P \to 0.5$.

15.11 — Failure Criteria

15.11.1 — The von Mises Criterion (Ductile Materials)

Ductile materials (metals) yield when the **elastic strain energy of distortion** reaches a critical value. In terms of principal stresses:

$$\sigma_{vM} = \sqrt{\frac{(\sigma_1-\sigma_2)^2 + (\sigma_2-\sigma_3)^2 + (\sigma_3-\sigma_1)^2}{2}} \geq \sigma_y$$

where σ_y is the uniaxial yield strength. The von Mises surface is a cylinder in principal stress space — yielding is insensitive to hydrostatic pressure.

For plane stress ($\sigma_3 = 0$):

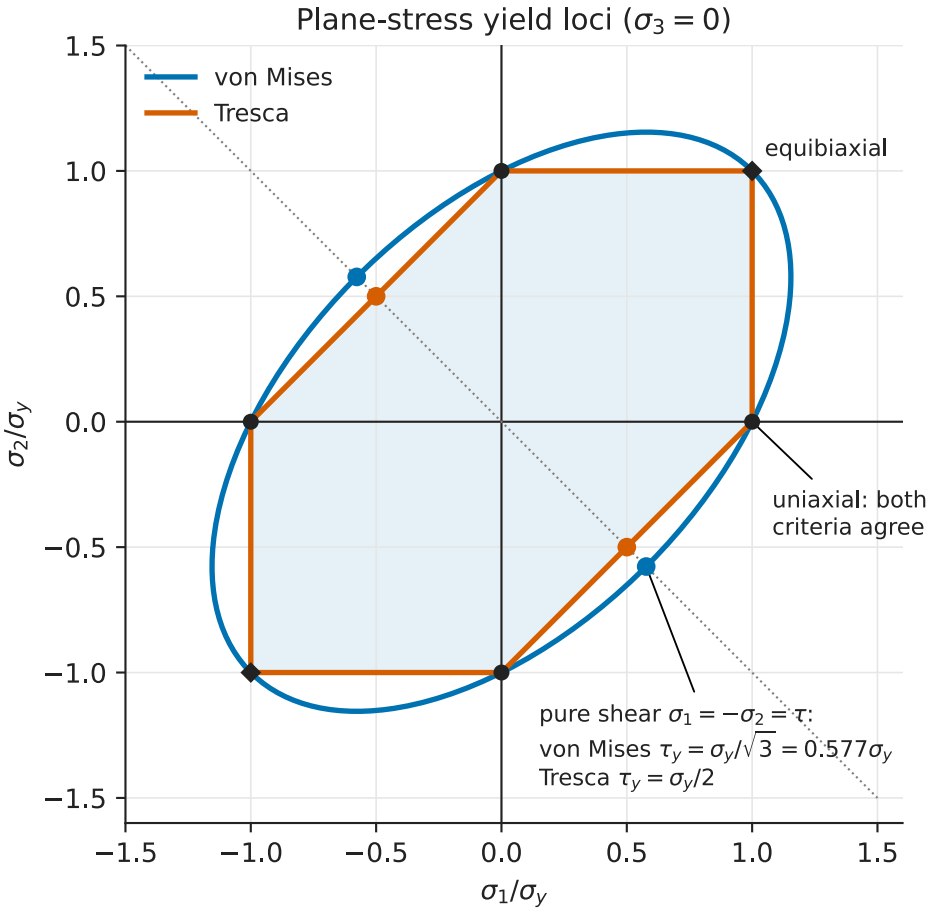
$$\sigma_{vM} = \sqrt{\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2} \geq \sigma_y$$

15.11.2 — The Tresca Criterion (Maximum Shear Stress)

Simpler but slightly more conservative: yielding occurs when the maximum shear stress exceeds the shear yield strength $\tau_y = \sigma_y/2$:

$$\tau_{max} = \frac{\sigma_1 - \sigma_3}{2} \geq \frac{\sigma_y}{2}$$

Von Mises is more accurate for ductile metals; Tresca is simpler and safe (conservative). Both give the same result for uniaxial stress.



Plane-stress yield loci in $(\sigma_1, \sigma_2)/\sigma_y$: the von Mises ellipse $\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2 = \sigma_y^2$ circumscribes the Tresca hexagon; they touch at uniaxial and equibiaxial states and differ most in pure shear ($\tau_y = 0.577\sigma_y$ vs. $0.5\sigma_y$), which is why Tresca is the conservative choice.

15.11.3 — Fracture Mechanics (Brittle Materials)

Brittle materials (ceramics, glass, rock) fail by crack propagation before yielding. Griffith's criterion: a crack of half-length a propagates when the stress intensity factor exceeds the fracture toughness:

$$K_I = \sigma\sqrt{\pi a} \geq K_{Ic}$$

K_{Ic} (MPa√m): steel $\sim$ 50–200; aluminum $\sim$ 20–45; glass $\sim$ 0.7; concrete $\sim$ 0.2–1.4; Si wafer $\sim$ 0.9.

The fracture mechanics approach is not covered in full depth here — it is the subject of an entire CE/ME course. The key point: **the failure criterion is as important as the stress distribution.** Solving the Navier equation gives you stresses; the failure criterion tells you what those stresses mean.

15.12 — Beyond Newtonian: Viscoelasticity and Non-Newtonian Fluids

15.12.1 — Viscoelastic Materials

From Ch. 13 §13.8.1: at finite frequency, the elastic modulus becomes complex: $\tilde{C}(\omega) = C'(\omega) + iC''(\omega)$. Materials with significant C'' are **viscoelastic**:

- Elastic behavior at short times (behave like solids)

- Viscous behavior at long times (flow like liquids)
- Maxwell model: $\frac{d\sigma}{dt} + \frac{\sigma}{\tau_{rel}} = E\dot{\varepsilon}$ (series spring + dashpot)
- Kelvin-Voigt model: $\sigma = E\varepsilon + \eta\dot{\varepsilon}$ (parallel spring + dashpot)

Engineering materials: Polymers, biological tissues, asphalt, concrete (under creep), metals at elevated temperature. The loss tangent $\tan\delta = C''/C'$ characterizes energy dissipation per cycle — critical for vibration damping, acoustic absorption, and tire hysteresis.

15.12.2 — Non-Newtonian Fluids

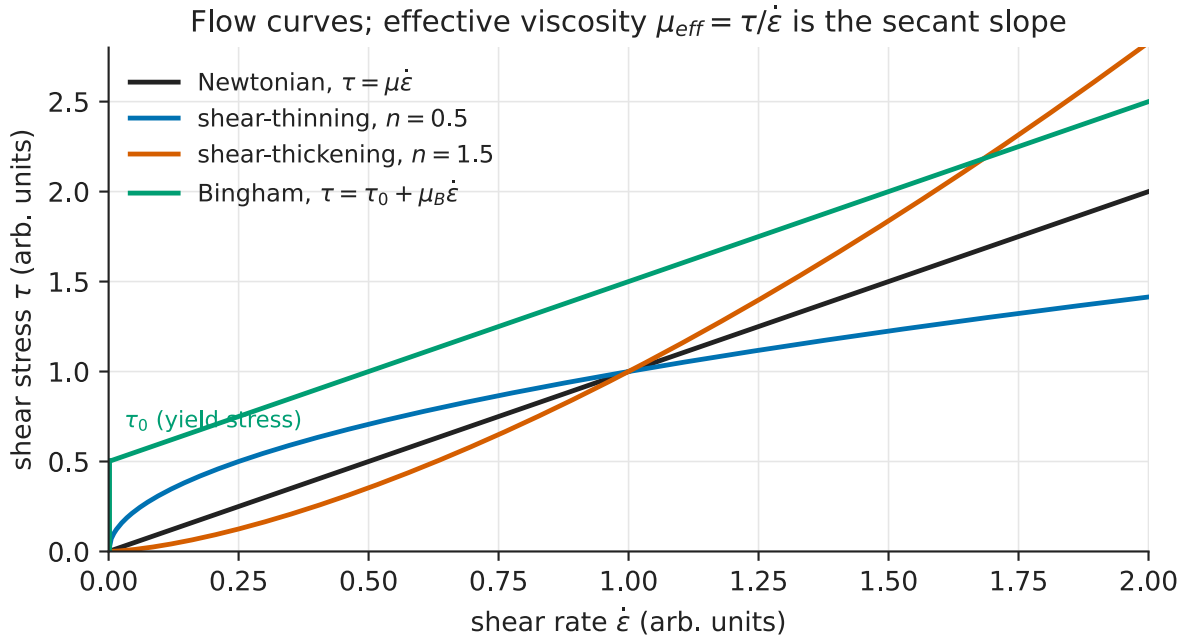
The Newtonian constitutive relation $\tau = \mu\dot{\varepsilon}$ (constant μ) is an approximation. For complex fluids:

Power-law fluids: $\mu_{eff} = K\dot{\varepsilon}^{n-1}$

- $n < 1$: **shear-thinning** (ketchup, blood, polymer melts) — viscosity decreases with shear rate
- $n > 1$: **shear-thickening** (cornstarch in water, some slurries)
- $n = 1$: Newtonian

Bingham plastics: require a yield stress τ_0 before flowing: $\tau = \tau_0 + \mu_B\dot{\varepsilon}$ for $|\tau| > \tau_0$, $\dot{\varepsilon} = 0$ otherwise. Examples: toothpaste, certain drilling muds, fresh concrete.

Thixotropy: viscosity is history-dependent (decreased by recent shearing and recovers over time). Paints, gels, some clays.



Flow curves $\tau(\dot{\varepsilon})$ for a Newtonian fluid, power-law fluids with $n = 0.5$ and 1.5 ($K = 1$), and a Bingham plastic with $\tau_0 = 0.5$, $\mu_B = 1$ (arbitrary units); the effective viscosity is the secant slope $\tau/\dot{\varepsilon}$, falling with rate for $n < 1$ and rising for $n > 1$.

Engineering importance: Non-Newtonian rheology governs polymer processing (injection molding, extrusion), food processing, pharmaceutical formulation, concrete placement, drilling mud design, and biological fluid dynamics (blood is shear-thinning; this matters in arterial flow).

15.13 — Summary

All major equations of continuum mechanics derived from Cauchy's equation + constitutive relations:

Result	Constitutive input	Application
Euler equation	$\sigma_{ij} = -p\delta_{ij}$ (ideal fluid)	Aerodynamics, hydraulics
Bernoulli equation	Euler + steady + irrotational	Pitot, Venturi, airfoil
Navier-Stokes	Newtonian: $\sigma_{ij} = -p\delta_{ij} + 2\mu\dot{\varepsilon}_{ij}$	All viscous flow
Hagen-Poiseuille	NS + pipe geometry	Pipe flow, blood vessels
Couette flow	NS + shear geometry	Viscometry, lubrication
Stokes drag	NS + $Re \ll 1$	Particle settling, microfluidics
Reynolds number	Non-dimensionalized NS	Flow regime classification
Navier equation	Hooke: $\sigma_{ij} = \lambda\varepsilon_{kk}\delta_{ij} + 2\mu_L\varepsilon_{ij}$	Solid deformation
P-wave speed	$v_P = \sqrt{(\lambda + 2\mu_L)/\rho}$	Seismology, NDT, geophysics
S-wave speed	$v_S = \sqrt{\mu_L/\rho}$	Seismology, shear wave imaging
Von Mises criterion	Yield when $\sigma_{vM} \geq \sigma_y$	Ductile metal design
Stokes-Einstein	$D = k_B T/6\pi\mu a$	Particle diffusion, Einstein relation

15.14 — Engineering Thread

Physics	Application
Continuity equation $\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0$	Mass balance in all pipe and duct systems; conservation checks in CFD
Bernoulli $p + \frac{1}{2}\rho v^2 + \rho g z = \text{const}$	Pitot tubes; Venturi meters; nozzle design; pump head calculations
Hagen-Poiseuille $\Delta p = 8\mu L Q/\pi R^4$	Pipe sizing; hydraulic system design; microfluidic channel design
Reynolds number $Re = \rho v L/\mu$	Flow regime prediction; scale model testing (similarity); turbulence onset
Stokes drag $F = 6\pi\mu a U$	Particle settling in ChE; filtration; aerosol dynamics; electrophoresis (Ch. 13)
Navier equation + elastic waves	NDT (ultrasonic testing); geophysical exploration; medical ultrasound
Von Mises criterion $\sigma_{vM} \geq \sigma_y$	Structural design of pressure vessels, beams, frames
Fracture mechanics $K_I = \sigma\sqrt{\pi a}$	Crack growth prediction; fatigue life; brittle fracture of glass/ceramic components
Non-Newtonian rheology	Polymer processing; concrete mix design; food and pharmaceutical manufacturing
Viscoelastic $\tan\delta$	Vibration damper selection; acoustic panel design; polymer material selection
Stokes-Einstein $D = k_B T/6\pi\mu a$	Drug diffusion in tissue; membrane separation processes; colloid stability

15.15 — Looking Ahead

Chapter 15 completes the governing equations of continuum mechanics. Layer 2 now contains its full set of PDEs:

- Newton's gravity: $\nabla^2\Phi = 4\pi G\rho$ (Ch. 8)
- Classical mechanics: Hamilton's equations (Ch. 9)
- Thermodynamics: 4 laws + equations of state (Ch. 10)
- Maxwell's equations (Ch. 11)
- EM wave equation: $\Box\mathbf{E} = 0$ (Ch. 12)
- Transport laws: Ohm, Fourier, Fick, viscosity, Hooke (Ch. 13)
- Navier-Stokes: fluid mechanics (Ch. 15, this chapter)
- Navier equation: solid mechanics (Ch. 15, this chapter)

Chapter 16 now shows the unifying PDE structure: the wave equation $\partial^2\phi/\partial t^2 = v^2\nabla^2\phi$ and the diffusion equation $\partial\phi/\partial t = D\nabla^2\phi$ appear identically across acoustics, electromagnetism, elasticity, heat, and mass transport — with only v, D, and the identity of ϕ changing. This is the final Layer-2 unification before Bridge C begins the descent to Layer 3.

End of Chapter 15.

Next: Chapter 16 — The Unified PDE: Waves and Diffusion Across All Domains

The Unified PDE: Waves and Diffusion Across All Domains

Shared PDE Structures: Waves and Diffusion Across Domains

> *The unreasonable effectiveness of mathematics in the natural sciences is a gift we neither understand nor deserve — but we should at least notice when the same equation shows up in ten different places.* — after Eugene Wigner

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16.0 — The Point of This Chapter

Look at the governing equations assembled across Chapters 8–15:

$$\Box \mathbf{E} = 0 \quad \text{(EM waves, Ch. 11)}$$

$$\rho \ddot{\mathbf{u}} = \mu_L \nabla^2 \mathbf{u} + \ldots \quad \text{(elastic waves, Ch. 15)}$$

$$\frac{\partial T}{\partial t} = \alpha \nabla^2 T \quad \text{(heat diffusion, Ch. 13)}$$

$$\frac{\partial c}{\partial t} = D \nabla^2 c \quad \text{(mass diffusion, Ch. 13)}$$

$$\frac{\partial^2 V}{\partial z^2} = L'C' \frac{\partial^2 V}{\partial t^2} \quad \text{(transmission line, Ch. 12)}$$

These look like five separate equations from five separate branches of physics. They are not. They are all the same equation.

This chapter makes that statement precise — showing the universal PDE structure that all Layer-2 physics shares, deriving the general solution methods that work for all of them simultaneously, and identifying exactly when each PDE reduces to the lumped-element description of Layer 3.

This is the final unification of Layer 2 before Bridge C begins.

16.1 — The Wave Equation

16.1.1 — The Universal Form

The wave equation in d spatial dimensions:

$$\boxed{\frac{\partial^2 \phi}{\partial t^2} = v^2 \nabla^2 \phi}$$

where $\phi(\mathbf{r}, t)$ is any field variable and v is the wave speed.

This is a **hyperbolic** PDE. The ratio v has units of speed. The equation admits solutions that propagate without changing shape at speed v in any direction.

Where it comes from in this book:

Physical domain	ϕ	v	Origin
EM waves in vacuum	$\mathbf{E}$ or $\mathbf{B}$	$c = 1/\sqrt{\mu_0 \epsilon_0}$	Maxwell (Ch. 11): $\Box \mathbf{E} = 0$
EM waves in dielectric	$\mathbf{E}$	c/n	Maxwell + constitutive (Ch. 12)
Acoustic waves in fluid	p or ρ'	$c_s = \sqrt{\partial p/\partial \rho}$	Linearized Euler (Ch. 15 §15.5)
Elastic P-waves	$\nabla \cdot \mathbf{u}$	$\sqrt{(\lambda + 2\mu_L)/\rho}$	Navier equation (Ch. 15 §15.10)
Elastic S-waves	$\nabla \times \mathbf{u}$	$\sqrt{\mu_L/\rho}$	Navier equation (Ch. 15 §15.10)
Shallow water waves	η (surface height)	$\sqrt{gh}$	Linearized shallow-water Euler
Lossless transmission line	V or I	$1/\sqrt{L'C'}$	Telegraph equations (Ch. 12 §12.7)
de Broglie waves	ψ (in WKB)	$v_g = p/m$	Schrödinger → HJ → WKB (Ch. 1, Ch. 3)

All eight are the same equation with different labels.

16.1.2 — Linearization: How the Wave Equation Arises

The wave equation always arises by **linearizing** a nonlinear PDE around equilibrium. The two-step procedure:

1. Let $\phi = \phi_0 + \phi'$ where ϕ_0 is the equilibrium and ϕ' is a small perturbation
2. Substitute and drop terms quadratic in ϕ' → linear PDE

Example: Acoustic wave from Navier-Stokes

Full equations: continuity $\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0$; Euler $\rho D\mathbf{v}/Dt = -\nabla p$.

Linearize: $\rho = \rho_0 + \rho'$, $p = p_0 + p'$, $\mathbf{v} = \mathbf{v}'$ (all primes small):

$$\frac{\partial \rho'}{\partial t} + \rho_0 \nabla \cdot \mathbf{v}' = 0, \qquad \rho_0 \frac{\partial \mathbf{v}'}{\partial t} = -\nabla p'$$

Using $p' = c_s^2 \rho'$ (linearized equation of state, $c_s^2 = \partial p/\partial \rho$):

$$\frac{\partial^2 p'}{\partial t^2} = c_s^2 \nabla^2 p'$$

The acoustic wave equation. The nonlinear term $(\mathbf{v} \cdot \nabla)\mathbf{v}$ from Navier-Stokes vanished because it is quadratic in $\mathbf{v}'$.

The same linearization procedure applied to the Navier equation (Ch. 15 §15.9.2), Maxwell's equations (Ch. 11 §11.7), and the shallow-water equations gives the wave equation in each domain. The common structure is: a linear restoring force (pressure, stress, electromagnetic field tension) plus an inertia term.

16.2 — The Diffusion Equation

16.2.1 — The Universal Form

$$\boxed{\frac{\partial \phi}{\partial t} = D \nabla^2 \phi}$$

where D is the **diffusivity** (units m²/s). This is a **parabolic** PDE. It admits no propagating wave solutions — disturbances spread and decay.

Where it comes from:

Physical domain	ϕ	D	Origin
Heat conduction	Temperature T	$\alpha = \kappa/\rho c_p$ (thermal diffusivity)	Fourier's law + continuity (Ch. 13, 14)
Mass diffusion	Concentration c	Diffusivity D (Fick)	Fick's law + continuity (Ch. 13)
Carrier diffusion (semiconductors)	Carrier density n or p	$D_n = \mu_n k_B T/e$	Einstein relation (Ch. 14)
Vorticity diffusion	Vorticity ω	Kinematic viscosity $\nu = \mu/\rho$	Linearized NS (Ch. 15)
Magnetic flux in conductor	**B** field	$1/\mu\sigma$	Maxwell + Ohm (Ch. 11)
Brownian motion	Probability density P	$k_B T/\gamma$ (friction coeff.)	Fokker-Planck (Ch. 10)
Option price (Black-Scholes)	Option price V	$\sigma^2 S^2/2$	Mathematical finance

All six are the same equation. A financial engineer solving the Black-Scholes equation for option pricing is solving the same PDE as a process engineer computing diffusion in a reactor.

16.2.2 — Derivation: Conservation Law + Flux Law = Diffusion Equation

Every diffusion equation has the same two-step derivation:

Step 1: Conservation law (Noether, Ch. 1):

$$\frac{\partial \phi}{\partial t} + \nabla \cdot \mathbf{J} = S$$

Step 2: Constitutive relation (flux proportional to gradient):

$$\mathbf{J} = -D \nabla \phi$$

Combining:

$$\frac{\partial \phi}{\partial t} = D \nabla^2 \phi + S$$

The source term S accounts for generation (heat sources, chemical reactions, carrier recombination-generation). This two-step procedure — Noether + Kubo — is the foundation of every transport PDE in engineering.

16.3 — The General Unified PDE

16.3.1 — The Telegraph Equation

The wave equation and diffusion equation are limiting cases of a more general equation:

$$\boxed{\frac{\partial^2 \phi}{\partial t^2} + \beta \frac{\partial \phi}{\partial t} = v^2 \nabla^2 \phi - \omega_c^2 \phi}$$

where:

- β: damping coefficient (relaxation rate)
- v: propagation speed (undamped)
- ω_c: cutoff frequency (restoring force)

Special cases:

Limit	Equation	Physics
$\beta = 0, \omega_c = 0$	$\partial_{tt}\phi = v^2\nabla^2\phi$	Wave equation (lossless)
$\beta \to \infty, \omega_c = 0$ (overdamped)	$\beta\partial_t\phi = v^2\nabla^2\phi \to \partial_t\phi = D\nabla^2\phi$	Diffusion equation
$\beta \neq 0, \omega_c = 0$	Damped wave equation	Lossy EM waves; viscous acoustic waves
$\beta = 0, \omega_c \neq 0$	Klein-Gordon equation	Massive relativistic field (Ch. 0); waveguide modes (Ch. 12)
General $\beta \neq 0, \omega_c = 0$	Telegraph equation	Transmission line (Ch. 12 §12.7.3)

16.3.2 — The Transmission Line as the Archetype

The transmission line equations (Ch. 12 §12.7.3) with loss:

$$\frac{\partial V}{\partial z} = -L'\frac{\partial I}{\partial t} - R'I$$
$$\frac{\partial I}{\partial z} = -C'\frac{\partial V}{\partial t} - G'V$$

Combining into a single equation for V:

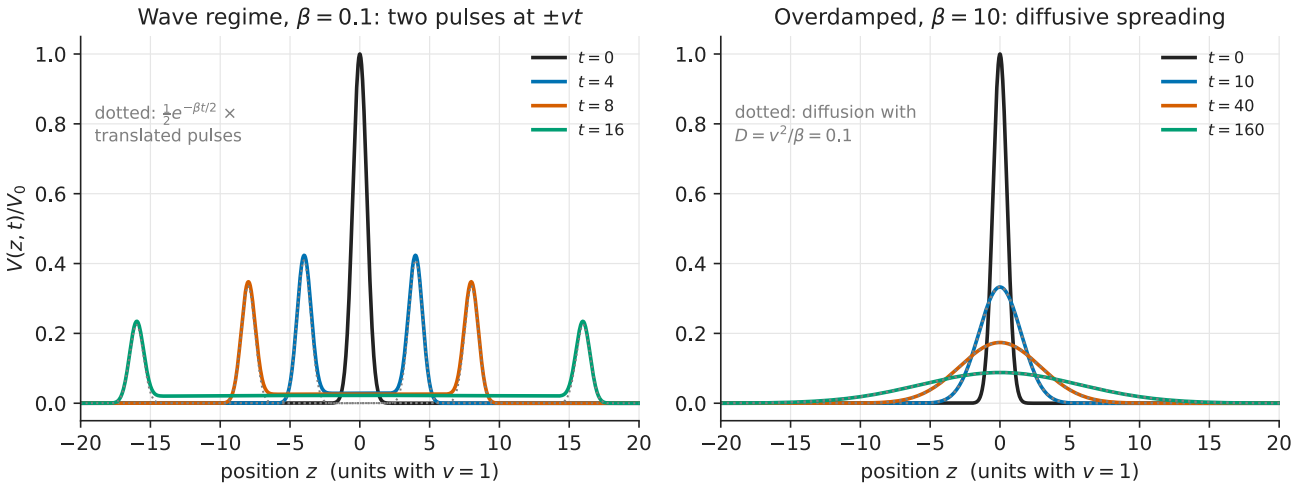
$$\frac{\partial^2 V}{\partial z^2} = L'C'\frac{\partial^2 V}{\partial t^2} + (R'C' + G'L')\frac{\partial V}{\partial t} + R'G'V$$

Identify: $v = 1/\sqrt{L'C'}$, $\beta = R'/L' + G'/C'$, $\omega_c^2 = R'G'/L'C'$.

Two limiting regimes:

- **High frequency** ($\omega \gg R'/L', \omega \gg G'/C'$): $\beta \to 0, \omega_c \to 0 \to$ wave equation: $\partial_{tt}V = v^2\partial_{zz}V \to$ signals propagate at speed $v = 1/\sqrt{L'C'} = c/\sqrt{\epsilon_r}$
- **DC/low frequency** (R' dominates, $G' = 0$): overdamped → diffusion equation: $\partial_t V = (1/R'C')\partial_{zz}V \to$ signals "diffuse" down the line (relevant for very resistive cables, or underground power transmission)

The transmission line is the bridge zone (Ch. 1 §1.9, Bridge C) — it retains spatial variation that the lumped circuit drops, but can be analyzed with the same mathematics as wave/diffusion problems.



Finite-difference solutions of $\partial_{tt}V + \beta\,\partial_t V = v^2\partial_{zz}V$ for a Gaussian initial pulse ($v = 1$): with $\beta = 0.1$ (left) it splits into two pulses moving at $\pm v$ that shrink as $e^{-\beta t/2}$ and leave a low wake, while with $\beta = 10$ (right) it stays put and spreads exactly like diffusion with $D = v^2/\beta$ (dotted).

16.4 — Plane Wave Solutions and the Dispersion Relation

16.4.1 — The Dispersion Relation

Try a plane wave solution $\phi = \phi_0 e^{i(\mathbf{k}\cdot\mathbf{r} - \omega t)}$. Substituting into the general unified PDE:

$$-\omega^2 - i\beta\omega = -v^2k^2 - \omega_c^2$$

$$\boxed{\omega^2 - i\beta\omega - v^2k^2 + \omega_c^2 = 0}$$

This is the **dispersion relation** — the algebraic relationship between frequency ω and wavenumber k for the given PDE. It encodes all wave physics for that system.

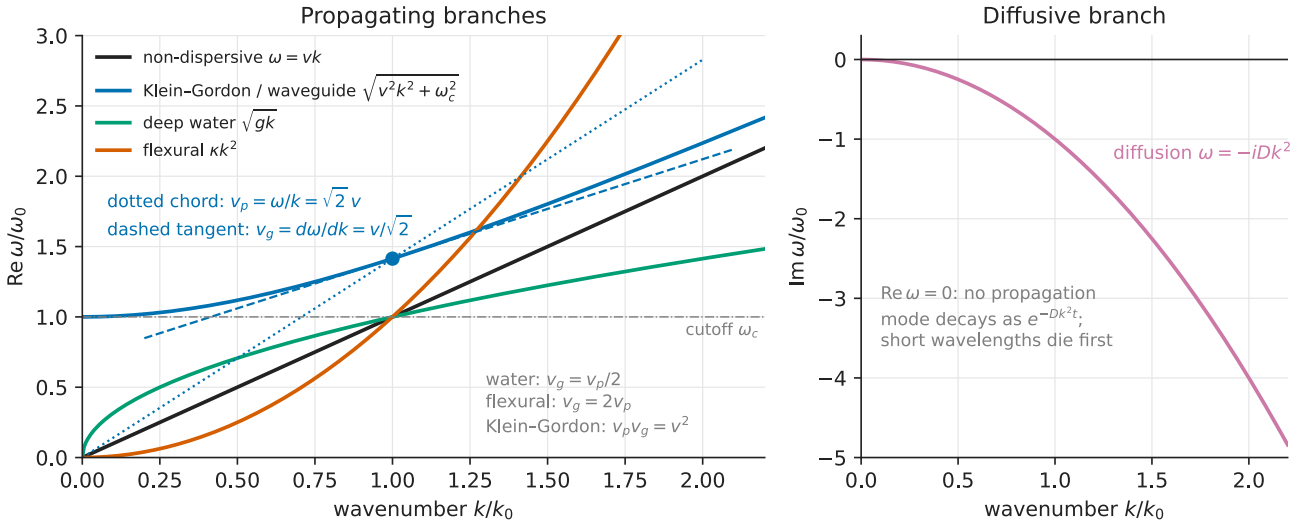
Reading the dispersion relation:

Feature of $\omega(k)$	Physical meaning
$d\omega/dk = v_g$	Group velocity (speed of signal/energy)
$\omega/k = v_p$	Phase velocity (speed of wavefronts)
$\omega(k=0) = \omega_c$	Cutoff frequency (minimum propagation frequency)
$\mathrm{Im}(\omega) < 0$	Temporal damping (amplitude decays in time)
$\mathrm{Im}(k) > 0$	Spatial attenuation (evanescent wave)
$\omega = \omega(k)$ is nonlinear	Dispersive system (pulse spreading)
$\omega = vk$ exactly	Non-dispersive (ideal wave, pulse shape preserved)

16.4.2 — Dispersion Relations Across All Domains

System	Dispersion relation	Notes
EM wave (vacuum)	$\omega = ck$	Non-dispersive; all frequencies same $v_p = c$
EM wave (dielectric)	$\omega = (c/n)k$	Non-dispersive within medium
EM wave (conductor)	$\omega = ck/\tilde{n}$	$\tilde{n}$ complex → attenuation
Waveguide TE$_{mn}$	$\omega^2 = c^2k^2 + \omega_{c,mn}^2$	Dispersive; cutoff → Klein-Gordon form
Acoustic (fluid)	$\omega = c_s k$	Non-dispersive for linear acoustics
Elastic P-wave	$\omega = v_P k$	Non-dispersive
Elastic S-wave	$\omega = v_S k$	Non-dispersive
Phonon (acoustic, long λ)	$\omega \approx v_s k$	Linear near $k = 0$ → non-dispersive
Phonon (optical)	$\omega \approx \omega_0 - Ak^2$	Dispersive near zone center
Deep water gravity waves	$\omega = \sqrt{gk}$	Strongly dispersive ($v_p = g/\omega$)
Flexural wave (beam)	$\omega = \kappa k^2$	Highly dispersive (structural waves)
Heat diffusion	$\omega = -iDk^2$	Purely imaginary: decay, no propagation
Schrödinger	$\omega = \hbar k^2/2m$	Same as diffusion! (imaginary time)
Transmission line (lossless)	$\omega = k/\sqrt{L'C'}$	Non-dispersive

The Schrödinger equation is a diffusion equation in imaginary time: replacing $t \to it$ in the diffusion equation gives $-i\partial_t\phi = D\nabla^2\phi$, which is $i\hbar\partial_t\psi = -\hbar^2\nabla^2\psi/2m$ with $D = \hbar/2m$. This mathematical connection between quantum mechanics and diffusion is not a coincidence — it underlies the path integral formulation (Ch. 1 §1.8.2): the Euclidean (imaginary-time) path integral is literally a diffusion problem.



Normalised dispersion relations: non-dispersive $\omega = vk$, Klein–Gordon/waveguide $\sqrt{v^2k^2 + \omega_c^2}$ (cutoff at ω_c), deep water $\sqrt{gk}$ and flexural κk^2, with the phase velocity (chord from the origin) and group velocity (tangent) marked on the Klein–Gordon branch at $k = k_0$; the right panel shows diffusion, whose $\omega = -iDk^2$ is purely imaginary, so every mode decays without propagating.

16.5 — General Solutions: Separation of Variables and Normal Modes

16.5.1 — Separation of Variables

For a PDE in a finite domain with appropriate boundary conditions, try:

$$\phi(\mathbf{r}, t) = X(\mathbf{r})\, T(t)$$

Wave equation: Substituting $\phi = XT$ into $\partial_{tt}\phi = v^2\nabla^2\phi$:

$$X\ddot{T} = v^2 T \nabla^2 X \quad \Longrightarrow \quad \frac{\ddot{T}}{T} = v^2 \frac{\nabla^2 X}{X} = -\omega_n^2 = \text{const}$$

Two ODEs:

- **Spatial:** $\nabla^2 X_n + k_n^2 X_n = 0$ (Helmholtz equation) with $k_n = \omega_n / v$
- **Temporal:** $\ddot{T}_n + \omega_n^2 T_n = 0 \to T_n(t) = A_n \cos\omega_n t + B_n \sin\omega_n t$

Diffusion equation: Substituting into $\partial_t\phi = D\nabla^2\phi$:

$$\dot{T} = DT\frac{\nabla^2 X}{X} = -\lambda_n D \quad \Longrightarrow \quad T_n(t) = e^{-\lambda_n D t}$$

- **Spatial:** $\nabla^2 X_n + \lambda_n X_n = 0$ (same Helmholtz equation!)
- **Temporal:** $T_n(t) = e^{-D\lambda_n t}$ (exponential decay)

The spatial problem is identical for both equations. The boundary conditions determine the eigenvalues $k_n^2 = \lambda_n$ and eigenfunctions X_n. Only the temporal behavior differs:

PDE	Temporal behavior
Wave	$\cos(\omega_n t)$, $\sin(\omega_n t)$ — oscillatory
Diffusion	$e^{-D\lambda_n t}$ — decaying exponential

16.5.2 — The Sturm-Liouville Problem

The spatial equation $\nabla^2 X + \lambda X = 0$ with homogeneous boundary conditions is a **Sturm-Liouville eigenvalue problem**. Its solutions:

- **Eigenvalues** $\lambda_n \geq 0$: determined by the geometry and boundary conditions
- **Eigenfunctions** X_n: an orthogonal set, complete in the function space

The eigenfunction expansion:

$$\phi(\mathbf{r}, t) = \sum_n c_n(t)\, X_n(\mathbf{r})$$

The coefficients $c_n(t)$ are the **normal mode amplitudes** — each evolves independently (for linear PDEs). This is the field-theoretic version of normal mode analysis from Ch. 9 §9.8.

Examples of eigenfunction sets:

Domain	Boundary conditions	Eigenfunctions X_n	Eigenvalues λ_n
1D box $[0, L]$	$X(0) = X(L) = 0$	$\sin(n\pi x/L)$	$(n\pi/L)^2$

Domain	Boundary conditions	Eigenfunctions X_n	Eigenvalues λ_n
1D box $[0, L]$	$X'(0) = X'(L) = 0$	$\cos(n\pi x/L)$	$(n\pi/L)^2$
Circular disk radius R	$X(R) = 0$	$J_m(j_{mn} r/R) e^{im\phi}$	$(j_{mn}/R)^2$
Sphere radius R	$X(R) = 0$	$j_\ell(k_{n\ell} r) Y_\ell^m(\hat{r})$	$k_{n\ell}^2$
Rectangular box $a \times b \times c$	Dirichlet	$\sin\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right)\sin\left(\frac{p\pi z}{c}\right)$	$\pi^2(m^2/a^2 + n^2/b^2 + p^2/c^2)$

Connection to quantum mechanics (Ch. 4): The spherical eigenfunctions $j_\ell(kr)Y_\ell^m(\hat{r})$ are exactly the wavefunctions of the hydrogen atom in free space. The Helmholtz equation $\nabla^2 X + \lambda X = 0$ is the spatial part of the Schrödinger equation for a free particle — the same eigenfunctions appear in acoustics (resonant cavities), EM (microwave cavities), and quantum mechanics. Different physics, identical mathematics.

16.5.3 — The General Solution (Superposition)

For the wave equation:

$$\phi(\mathbf{r}, t) = \sum_n \left[A_n \cos(\omega_n t) + B_n \sin(\omega_n t)\right] X_n(\mathbf{r})$$

Coefficients A_n, B_n from initial conditions $\phi(\mathbf{r}, 0)$ and $\dot{\phi}(\mathbf{r}, 0)$ using orthogonality of X_n.

For the diffusion equation:

$$\phi(\mathbf{r}, t) = \sum_n c_n \, e^{-D\lambda_n t} \, X_n(\mathbf{r})$$

Coefficients c_n from initial condition $\phi(\mathbf{r}, 0)$.

Key observation: The $n = 0$ (lowest) mode decays slowest (smallest λ_0) for diffusion, or oscillates at lowest frequency for waves. All higher modes decay faster or oscillate faster. At long times, only the fundamental mode survives.

16.6 — Green's Functions: The Impulse Response

16.6.1 — The Green's Function Concept

The **Green's function** $G(\mathbf{r}, t\,\mathbf{r}', t')$ is the response of the system to an impulse at position $\mathbf{r}'$ and time t':

$$\mathcal{L} G(\mathbf{r}, t\,\mathbf{r}', t') = \delta^3(\mathbf{r} - \mathbf{r}')\delta(t - t')$$

where $\mathcal{L}$ is the differential operator of the PDE. The general solution for any source distribution $S(\mathbf{r}', t')$:

$$\phi(\mathbf{r}, t) = \int G(\mathbf{r}, t\,\mathbf{r}', t') \, S(\mathbf{r}', t') \, d^3r' \, dt' + \text{boundary terms}$$

Systems thinking: The Green's function is the **impulse response** — exactly the same concept as in control theory and signal processing (Ch. 21). The convolution integral $\phi = G * S$ is the same operation as filtering a signal through a linear system. **Layer 2 PDEs and Layer 3 transfer functions are the same mathematics operating at different spatial scales.**

16.6.2 — Green's Functions for the Wave and Diffusion Equations

Diffusion equation in free 3D space ($t > t'$):

$$\boxed{G_{diff}(\mathbf{r}, t\,\mathbf{r}', t') = \frac{1}{[4\pi D(t-t')]^{3/2}} \exp\left(-\frac{|\mathbf{r}-\mathbf{r}'|^2}{4D(t-t')}\right)}$$

A **Gaussian** centered at $\mathbf{r}'$, spreading with time as $\sigma^2 = 2D(t-t')$. This is the fundamental solution — any initial condition spreads according to convolution with this Gaussian.

Key property: $G_{diff}(\mathbf{r}, t\,\mathbf{r}', t') \neq 0$ for any $|\mathbf{r}-\mathbf{r}'|$ at $t > t'$ — information travels at **infinite speed** in the diffusion equation. This is not a physical paradox; it is an approximation that breaks down for $|\mathbf{r}-\mathbf{r}'| \gtrsim c_s(t-t')$ (the thermal phonon speed).

Wave equation in free 3D space:

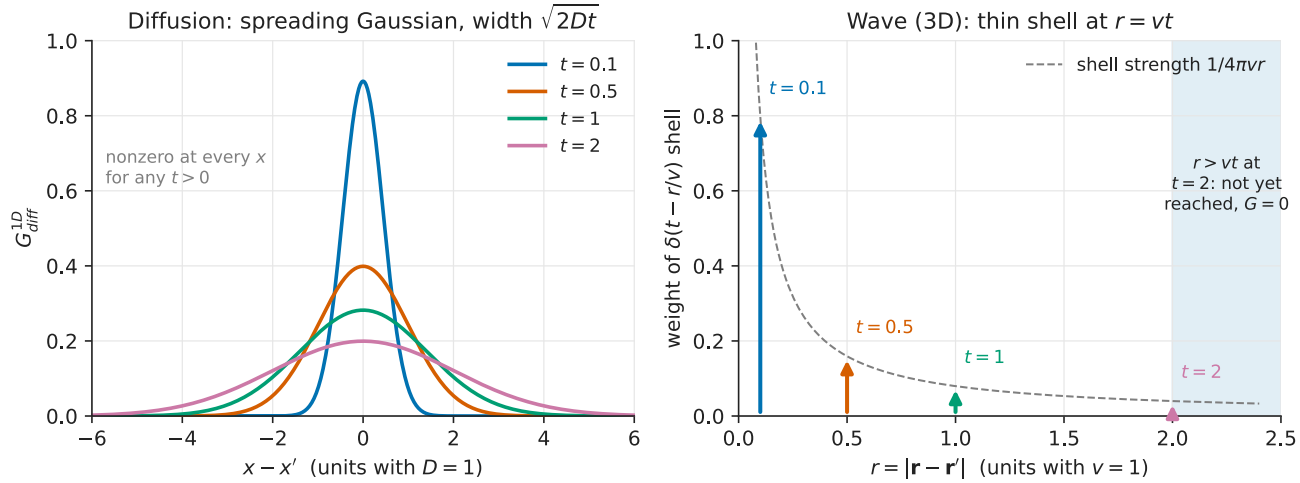
$$G_{wave}(\mathbf{r}, t\,\mathbf{r}', t') = \frac{\delta(t-t'-|\mathbf{r}-\mathbf{r}'|/v)}{4\pi v|\mathbf{r}-\mathbf{r}'|}$$

An **expanding spherical shell** at radius $v(t-t')$ from the source. Information travels at exactly speed v — the retarded potential. This is the Green's function used in antenna radiation theory (Ch. 12).

1D diffusion (heat or mass):

$$G^{1D}_{diff}(x, t\,x', t') = \frac{1}{\sqrt{4\pi D(t-t')}} \exp\left(-\frac{(x-x')^2}{4D(t-t')}\right)$$

This gives the solution to the semi-infinite solid problem of Ch. 14 §14.8.1 directly by convolution with the initial condition.



Left: 1D diffusion Green's function $e^{-x^2/4Dt}/\sqrt{4\pi Dt}$ ($D = 1$) at $t = 0.1$, 0.5, 1, 2 — it widens as $\sqrt{2Dt}$ and is nonzero everywhere at once. Right: the 3D wave Green's function at the same times ($v = 1$) is a sharp shell at $r = vt$ of strength $1/4\pi vr$, zero both inside and outside — signals arrive exactly at speed v.

16.6.3 — The Connection Between Wave and Diffusion Green's Functions

Replace $t \to it$ (Wick rotation to imaginary time) in the wave equation $\partial_{tt}G = v^2\nabla^2 G$:

$$-\partial^2_\tau G = v^2\nabla^2 G \quad \Longrightarrow \quad \partial_\tau G = v^2\nabla^2 G/2\tau \ldots$$

More precisely: writing $\tau = it$ converts the oscillatory wave solutions $e^{-i\omega t}$ into decaying exponentials $e^{-\omega\tau}$ — exactly the diffusion form. **The wave equation in imaginary time IS the diffusion equation.**

This is not just a mathematical trick. The path integral for quantum mechanics (Ch. 1 §1.8.2) is evaluated in Euclidean (imaginary) time precisely because the oscillatory path integral is hard to evaluate while the diffusion-like Euclidean path integral is well-defined. Monte Carlo simulations of quantum field theories exploit this connection.

16.7 — The Wave Equation Across All Domains: A Complete Picture

16.7.1 — Acoustic Waves

From §16.1.2: the wave equation for pressure p:

$$\frac{\partial^2 p}{\partial t^2} = c_s^2 \nabla^2 p, \qquad c_s = \sqrt{\left.\frac{\partial p}{\partial \rho}\right|_s} = \sqrt{\frac{\gamma p_0}{\rho_0}}$$

For air at 20°C: $c_s = 343$ m/s. For water: 1480 m/s. For steel: $v_P = 5960$ m/s.

Acoustic impedance: $Z_{ac} = \rho_0 c_s$ (analogous to wave impedance $Z_0 = \sqrt{\mu_0/\epsilon_0}$ in EM). Reflection at interfaces from impedance mismatch: $r = (Z_2 - Z_1)/(Z_2 + Z_1)$ — same Fresnel formula as optics (Ch. 12 §12.3.3) because the same wave equation governs both.

Resonant modes of acoustic cavities: Same eigenvalue problem as EM cavities. A rectangular room of dimensions $a \times b \times c$ has resonant frequencies:

$$f_{mnp} = \frac{c_s}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{c}\right)^2}$$

Engineering: Room acoustics design, noise cancellation, ultrasonic sensors, sonar, SONAR bathymetry.

16.7.2 — Flexural (Bending) Waves in Beams

From the Euler-Bernoulli beam equation (the bridge zone between Ch. 15 continuum elasticity and Ch. 17 lumped structures):

$$EI\frac{\partial^4 w}{\partial x^4} + \rho A \frac{\partial^2 w}{\partial t^2} = q(x,t)$$

where $w(x,t)$ is the lateral deflection, EI is the bending stiffness, ρA is the mass per unit length.

For plane wave solutions $w \propto e^{i(kx-\omega t)}$:

$$\omega^2 = \frac{EI}{\rho A}k^4 \quad \Longrightarrow \quad v_{flexural} = \sqrt[4]{\frac{EI}{\rho A}}\,\sqrt{\omega}$$

The flexural wave speed increases as $\sqrt{\omega}$ — it is **highly dispersive**. A sharp hammer blow on a rail arrives as a "chirp" (high frequencies first, low frequencies later) at a distant microphone.

This is **not** the standard wave equation. It is $\partial_{tt} w = -\kappa^2 \partial_{xxxx} w$ (fourth-order in space, second in time) — the beam equation. It gives the bridge zone between full 3D elasticity (Navier equation, Ch. 15) and the lumped beam element (Bridge C, Ch. 17).

16.8 — The Diffusion Equation Across All Domains: A Complete Picture

16.8.1 — Magnetic Flux Diffusion

In a conducting medium ($\sigma \neq 0$), from Maxwell's equations (Ch. 11):

$$\nabla \times \mathbf{B} = \mu\sigma \mathbf{E}, \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Combining:

$$\frac{\partial \mathbf{B}}{\partial t} = \frac{1}{\mu\sigma} \nabla^2 \mathbf{B}$$

The **magnetic diffusivity** is $\eta_m = 1/\mu\sigma$. Magnetic field diffuses into a conductor at rate $\partial_t \mathbf{B} = \eta_m \nabla^2 \mathbf{B}$.

Skin depth from the diffusion equation: at frequency ω, the field penetrates to depth $\delta = \sqrt{2\eta_m/\omega} = \sqrt{2/\mu\sigma\omega}$ — exactly the skin depth formula from Ch. 12 §12.1.2, now derived as the characteristic length of the magnetic diffusion equation.

Magnetic Reynolds number: $Rm = \mu\sigma v L$ compares convective transport of magnetic field (by conducting fluid flow) to diffusive transport. $Rm \gg 1$: magnetic field is "frozen into" the fluid (MHD limit, important for plasma physics, stellar interiors, liquid metal reactors).

16.8.2 — Vorticity Diffusion

From the vorticity transport equation (Ch. 15 §15.6.3) at low Re (no vortex stretching):

$$\frac{\partial \boldsymbol{\omega}}{\partial t} = \nu \nabla^2 \boldsymbol{\omega}$$

Vorticity diffuses away from solid boundaries with diffusivity $\nu = \mu/\rho$ (kinematic viscosity). The thickness of a viscous boundary layer growing in time:

$$\delta_{BL} \sim \sqrt{\nu t}$$

This is the same $\sqrt{Dt}$ scaling as all diffusion problems — the boundary layer thickness is the characteristic length of the vorticity diffusion equation.

Stokes' second problem: Flat plate oscillating at frequency ω. Vorticity diffuses away to depth $\delta = \sqrt{2\nu/\omega}$ — same formula as the electromagnetic skin depth, with ν replacing $1/\mu\sigma$.

Quantity	EM skin depth	Viscous Stokes depth
Governing PDE	$\partial_t \mathbf{B} = (1/\mu\sigma)\nabla^2 \mathbf{B}$	$\partial_t \boldsymbol{\omega} = \nu \nabla^2 \boldsymbol{\omega}$
Diffusivity	$\eta_m = 1/\mu\sigma$	$\nu = \mu/\rho$
Penetration depth	$\delta_{EM} = \sqrt{2\eta_m/\omega}$	$\delta_{Stokes} = \sqrt{2\nu/\omega}$
Same formula:	$\delta = \sqrt{2D_{eff}/\omega}$	

Same equation, different physics, same formula.

16.9 — Boundary and Initial Conditions: What Makes Physics Unique

The PDE alone does not determine the solution — it is the PDE **plus** the boundary and initial conditions that give a unique physical prediction.

16.9.1 — Types of Boundary Conditions

Type	Form	Physical meaning	Example
Dirichlet	$\phi = f$ on boundary	Prescribed value	Temperature of a wall; potential on a conductor
Neumann	$\nabla\phi \cdot \hat{n} = g$ on boundary	Prescribed flux	Insulating wall ($g = 0$); heat flux into surface
Robin (mixed)	$a\phi + b\nabla\phi \cdot \hat{n} = h$	Combination	Newton cooling: $-\kappa \partial T/\partial n = h_c(T - T_\infty)$
Radiation (non-reflecting)	Outgoing wave only	Absorbing boundary	Open domains, infinite media
Periodic	$\phi(0) = \phi(L)$	Repeating media	Crystal unit cell, periodic waveguides

The same PDE with different boundary conditions gives physically distinct solutions:

- Dirichlet heat equation → prescribed wall temperature → Fourier series solution
- Neumann heat equation → insulated wall → different Fourier series (cosines instead of sines)
- Robin → convective cooling → mixed series

This is why different engineering problems look different even though they obey the same PDE: the geometry and boundary conditions encode the engineering specifics.

16.9.2 — Uniqueness Theorems

For elliptic problems (steady-state, $\partial/\partial t = 0$):

- **Dirichlet problem:** Unique solution (max/min principle prevents two different solutions)
- **Neumann problem:** Unique up to an additive constant (only the gradient is fixed)

For parabolic problems (diffusion):

- Unique solution given initial condition $\phi(\mathbf{r}, 0)$ and boundary conditions

For hyperbolic problems (wave equation):

- Unique solution given initial conditions $\phi(\mathbf{r}, 0)$ and $\dot{\phi}(\mathbf{r}, 0)$

16.10 — The Lumping Criterion: Bridge to Layer 3

16.10.1 — When Does a PDE Become an ODE?

The wave equation $\partial_{tt}\phi = v^2\nabla^2\phi$ in a finite domain of size L has eigenfrequencies $\omega_n \sim nv\pi/L$.

If the driving frequency $\omega_{drive} \ll \omega_1 = v\pi/L$ (lowest mode):

- The wavelength $\lambda = 2\pi v/\omega_{drive} \gg 2L$
- The spatial variation of ϕ within the domain is negligible
- The PDE reduces to an ODE: $\ddot{\phi} \approx \text{const} \times \phi$

The lumping criterion:

$$\boxed{Kn_{EM} = \frac{L}{\lambda} = \frac{Lf}{v} \ll 1}$$

When this condition holds, the system behaves as a lumped element. When it fails, spatial variation matters and the PDE must be retained.

`[APPROXIMATION]` — **lumping is the single most consequential approximation in this book, because every engineering model in Chapters 17–21 is downstream of it.** Three things are being assumed at once when $\lambda \gg 2L$, and they are worth separating because they fail differently:

1. **Spatial uniformity.** The field is replaced by a single value. This is the $L/\lambda \ll 1$ criterion.
2. **Locality of the constitutive law.** The flux–gradient relation is evaluated at one point. This is usually *not* an approximation at all — it is exact — but it becomes one for media whose transport coefficient varies in space.
3. **A single timescale.** By dropping ∇^2, the model also drops the spatial eigenmodes. So the ODE describes only the first mode. A lumped model with a Q of 10^6 is a claim that no higher mode is excited — which is a statement about the *input*, not just the system.

Domain	PDE variable	Speed v	Lumping valid when	Fails when
EM (circuit)	V, I	$c/\sqrt{\epsilon_r}$	$f \ll v/L$ (e.g. $f \ll 300$ MHz for 10 cm)	rise time $< 2L/v$; interconnect delay matters
Acoustic (Helmholtz)	p	$c_s = 343$ m/s	$f \ll c_s/2L$	room modes excited; absorption layer thin vs λ
Elastic (lumped mass–spring)	u	v_P, v_S	$f \ll v_P/2L$	first bending or shear mode reached; $L/\lambda \gtrsim 1/10$
Thermal (lumped capacitance)	T	diffusive, not wave	$Bi = hL/\kappa \ll 1$	$Bi \gtrsim 0.1$; internal gradients matter
Hydraulic (pipe sections)	p	$c_{water} = 1480$ m/s	$f \ll c_{water}/L$	water hammer; valve closure time $< 2L/c$

The Biot number for thermal systems is the analog of the spatial Knudsen number: it measures whether the temperature is uniform within the body (small Bi, lumped valid) or has significant spatial variation (large Bi, distributed model needed). Note that it is the *only* row in this table with no wave speed — because thermal transport is diffusive, its relevant scale is the diffusion length $\sqrt{\alpha t}$, not a wavelength. The

lumping criterion for a diffusive system is $L \ll \sqrt{\alpha t}$, which is time-dependent, and that is why a lumped thermal model has a validity *window in time* rather than a bandwidth. This is worth keeping in mind in Chapter 19, where lumped thermal capacitance is used as a fixed parameter.

Model Ledger — distributed PDE to lumped ODE

Field	Content
Parent theory	Wave / diffusion PDE on a continuum domain (Ch. 15, this chapter)
Reduction	$L/\lambda \ll 1$; drop ∇^2; integrate the conservation law over the domain volume; lump all energy storage into one state variable per mode
Model	$C\dot{\phi} + R\phi = u(t)$ — first-order in time, zero order in space
Assumptions	only the fundamental spatial mode participates; uniform material; linear; no distributed source; state variable single-valued over the domain
Physics retained	total energy, total entropy production, and their balance
Physics neglected	all spatial gradients; every mode except the first; shape effects (a lumped model cannot predict resonance or standing waves)
Validity	a point operating point, not a range — a lumped model is valid near one frequency/timescale, not across a band
Fails when	excitation approaches ω_1; distributed sources act; the geometry's shape is itself the physics (acoustics, waveguides)
Next model	modal expansion; distributed model (Ch. 17 §17.10); finite-volume discretisation (§16.11.3)

16.10.2 — The Lumped Equivalents

When the lumping criterion is satisfied, the PDE terms reduce to:

PDE term	Lumped equivalent	Physical meaning
$C\partial_t\phi$	$C\dot{\phi}$ (capacitance × rate)	Energy storage (potential)
$(1/L)\int \phi\, dt$	$(1/L)\phi$ (inductance inverse)	Energy storage (kinetic)
$R\phi$ (or $G\nabla\phi$)	$R\phi$ (resistance)	Dissipation
$\nabla^2\phi$	$\Delta\phi/L^2$ (finite difference)	Spatial coupling

The transmission line equations (§16.3.2) become the lumped L-C ladder network when spatial variation is dropped — each segment becomes an inductor and capacitor in the electrical analogy. This is exactly Bridge C (Ch. 17).

16.11 — Numerical Methods: Approximating the Continuum

When exact analytic solutions don't exist (complex geometry, nonlinear PDEs, non-standard boundary conditions), numerical methods discretize the continuum:

16.11.1 — Finite Difference Method (FDM)

Replace derivatives with finite differences on a regular grid:

$$\left.\frac{\partial^2 \phi}{\partial x^2}\right|_{x=x_i} \approx \frac{\phi_{i+1} - 2\phi_i + \phi_{i-1}}{\Delta x^2}$$

The PDE becomes a large system of algebraic equations. Simple to implement; handles irregular boundaries poorly.

16.11.2 — Finite Element Method (FEM)

Divide the domain into elements; approximate ϕ within each element by polynomial interpolation (shape functions). Use the Galerkin weighted residual method to convert the PDE into a matrix equation:

$$\mathbf{K}\mathbf{u} = \mathbf{f} \quad \text{(statics)} \qquad \mathbf{M}\ddot{\mathbf{u}} + \mathbf{C}\dot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{f} \quad \text{(dynamics)}$$

where $\mathbf{K}$ is the stiffness matrix, $\mathbf{M}$ is the mass matrix, $\mathbf{C}$ is the damping matrix.

FEM for structural mechanics: The Navier equation (Ch. 15 §15.9.2) is solved by discretizing the structure into elements (bars, beams, shells, solids). The global stiffness matrix is assembled from element stiffness matrices. **This is the connection between the continuum Navier equation and structural engineering practice.**

The FEM stiffness matrix is the **discretized continuum** — a lumped model of the distributed elasticity, valid in each element separately.

16.11.3 — Finite Volume Method (FVM)

Integrate the PDE over control volumes; apply flux conservation across faces. Naturally conserves mass, momentum, and energy. Standard for fluid dynamics (CFD).

The hierarchy: FVM is the rigorous version of the lumped control volume analysis that Bridge C performs — the difference is that FVM keeps many small volumes while Bridge C uses one large one.

16.12 — The Unified Picture: Summary Table

Domain	Wave eq. var	v or D	Diffusion eq. var	D_{eff}
EM (vacuum)	$\mathbf{E}$, $\mathbf{B}$	c	—	—
EM (conductor)	—	—	$\mathbf{B}$	$1/\mu\sigma$
EM (circuit, lossless)	V, I	$1/\sqrt{L'C'}$	—	—
Acoustic	pressure p	$c_s = \sqrt{\gamma p/\rho}$	—	—
Elastic solid	$\nabla\cdot\mathbf{u}$, $\nabla\times\mathbf{u}$	v_P, v_S	—	—
Flexural beam	deflection w	ω-dependent	—	—
Heat conduction	—	—	T	$\alpha = \kappa/\rho c_p$
Mass diffusion	—	—	c	D
Vorticity (viscous)	—	—	$\boldsymbol{\omega}$	$\nu = \mu/\rho$
Carrier (semiconductor)	—	—	n, p	$D_n = \mu_n k_B T/e$
Quantum (WKB)	ψ	p/m	ψ (imaginary t)	$\hbar/2m$

The master equation governing all of them:

$$\frac{\partial^2 \phi}{\partial t^2} + \beta \frac{\partial \phi}{\partial t} = v^2 \nabla^2 \phi - \omega_c^2 \phi + S(\mathbf{r}, t)$$

One canonical form. Many domains. Often only the labels change — but not always.

> **Warning: This is a canonical form, not a universal law of physics** Read the table above as: *many* important engineering PDEs can be brought to this form. That is a strong and useful statement, and it is worth knowing exactly *why* it works, because the reasons are what tell you when it will fail.
>
> Getting a system into this form requires several restrictive steps, each of which is a real approximation:
>
> - **Linearity.** The coefficients (v, β, ω_c) must be constant. Nonlinear elasticity, compressible flow with density-dependent coefficients, and most reaction-diffusion systems are genuinely nonlinear and do not reduce to it.
> - **A single scalar field ϕ.** Systems needing several coupled fields (Navier–Stokes velocity and pressure, Maxwell **E** and **B**, multi-phase flow) carry extra structure this form has no slot for.
> - **A single characteristic speed.** Wave-like transport with genuinely different speeds in different directions (flexural waves, where $v \propto \sqrt{\omega}$) or genuine anisotropy needs a tensorial operator, not $v^2 \nabla^2$.
> - **Diffusion-dominated regimes.** Pure diffusion is the $\omega_c = \beta = 0$ limit, which is why the table shows "—" for the inertial column. Steady incompressible flow, pure reaction kinetics, and algebraic constraints are not described by this equation at all.
> - **Time derivatives of order two.** First-order-in-time systems, delay equations, and fractional-order systems need different forms.
>
> So the accurate formulation is: **many engineering PDEs admit canonical wave/diffusion/storage forms after appropriate linearisation, scalar reduction, and limiting.** That is more valuable than a universal claim, because it tells the reader both *why* the same equation keeps appearing and *when* it will not.

16.13 — Engineering Thread

Physics	Engineering application
Acoustic resonance (wave eq. eigenmodes)	Noise control; concert hall acoustics; ultrasonic transducer design
Flexural wave dispersion $v \propto \sqrt{\omega}$	Rail NDE via dispersion of hammer pulse; guided wave inspection
Skin depth $\delta = \sqrt{2/\mu\sigma\omega}$	Cable shielding; transformer core lamination; RF conductor loss
Thermal diffusivity $\alpha = \kappa/\rho c_p$	Transient heat conduction; pulsed laser processing; thermal barrier coatings
Biot number $Bi = hL/\kappa$	Lumped capacitance validity; heat treatment uniformity; food safety cooking models
Diffusion length $L = \sqrt{D\tau}$	Minority carrier device physics; carburization case depth
Green's function (Gaussian diffusion)	Pollution plume modeling; drug release profiles; doping profile prediction
FEM stiffness matrix	Structural analysis of all engineered structures; ANSYS, ABAQUS, OpenFOAM

Physics	Engineering application
FVM for CFD	Aerodynamic drag; heat exchanger performance; combustion simulation
Lumping criterion $L/\lambda \ll 1$	Circuit vs. transmission line model; acoustic duct vs. Helmholtz; structural modal limit

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16.14 — Looking Ahead: Bridge C

This chapter completes **Layer 2**. The full inventory:

- Newton's gravity (Ch. 8) → Poisson equation $\nabla^2\Phi = 4\pi G\rho$
- Classical mechanics (Ch. 9) → Hamilton's equations, Lagrangian dynamics
- Thermodynamics (Ch. 10) → Four laws, partition function, Landau theory
- Maxwell's equations (Ch. 11) → Two covariant equations
- EM waves and optics (Ch. 12) → Wave equation in various media
- Transport laws (Ch. 13) → Kubo → Ohm, Fourier, Fick, viscosity, Hooke
- Transport applications (Ch. 14) → Thermoelectrics, drift-diffusion, RTDs
- Continuum mechanics (Ch. 15) → Navier-Stokes, Navier elasticity, elastic waves
- Unified PDE (Ch. 16) → Wave equation and diffusion equation across all domains

Chapter 17 begins Bridge C: the descent from Layer 2 PDEs to Layer 3 ODEs. The operation: integrate the PDE over a control volume, invoke the lumping criterion ($L/\lambda \ll 1$, $Bi \ll 1$), and the PDE becomes a discrete algebraic or ODE system. The result is the effort-flow template, R/C/L in every domain, and the engineering systems of Chapters 18–21.

End of Chapter 16. End of Layer 2.

Next: Chapter 17 — Bridge C: Lumping and the Effort-Flow Template

Bridge C — Lumping and the Effort-Flow Template

The Descent from PDEs to Engineering Systems

> *An electrical engineer, a mechanical engineer, and a chemical engineer are all solving the same equation. The only difference is what they call the variables.* — teaching tradition in engineering mathematics

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17.0 — Bridge C in Context

Chapters 8–16 built Layer 2 completely. The full set of classical continuum physics — Newton, thermodynamics, Maxwell, transport laws, fluid mechanics, solid mechanics, and the unified PDE — is now assembled.

Bridge C performs one operation: integrate the Layer-2 PDEs over finite control volumes, invoke the lumping criterion ($L/\lambda \ll 1$ or $Bi \ll 1$), and the PDEs collapse into ordinary differential equations and algebraic balance equations.

The result is **Layer 3** — the engineering systems scale. Its defining feature: all physical domains share one mathematical template, with the only difference being the names of variables.

This chapter is the book's central structural claim, made explicit: EEE, ME, CE, and ChE are not four different subjects. They are four different *labelings* of one mathematical system, valid when the appropriate lumping criterion holds.

`[STRUCTURAL CONNECTION]` — the claim is about **representations, not physics**, and it holds strongly for the domains in this book while not being a theorem about all of engineering. Three qualifications keep it honest:

1. **The unification is a modeling choice, not a discovered identity.** An effort/flow pair can be *constructed* for any two conjugate variables, which means the template is available essentially anywhere. That generality is a feature of the representation and a warning against reading physics into it.
2. **Not every system is naturally effort/flow.** Systems with multiple distributed inputs and outputs, with strong spatial structure, or with constraints that are not captured by a single conjugate pair, are better served by other representations — Hamiltonian, port-Hamiltonian, passivity- based, or a full network. The template is most valuable where the lumping criterion holds, because then the system genuinely reduces to a small number of states.
3. **"Four branches" is illustrative, not exhaustive.** These four are the ones this book develops. Their formal similarities extend to further domains, and their internal differences are far larger than the similarity suggests.

Bridge C in the layer map:

```
Layer 2 (PDEs)
   ∂²φ/∂t² = v²∇²φ         (wave equation)
   ∂φ/∂t = D∇²φ            (diffusion equation)
   ρ Dv/Dt = -∇p + μ∇²v    (Navier-Stokes)
   ρü = (λ+μ)∇(∇·u) + μ∇²u (Navier elasticity)
                │
                │  Integrate over control volume
                │  Invoke L/λ ≪ 1 (or Bi ≪ 1)
                │
                ▼
Layer 3 (ODEs + algebraic balance equations)
   L ẍ + R ẋ + x/C = e_source    (universal second-order ODE)
   Σ flows at node = 0            (Kirchhoff-type laws)
   Σ efforts around loop = 0      (Kirchhoff-type laws)
```

17.1 — The Lumping Operation: PDE to ODE

17.1.1 — The Formal Procedure

Consider the diffusion equation on an element of size L:

$$\frac{\partial \phi}{\partial t} = D\nabla^2\phi + S$$

Step 1: Integrate over the control volume $V = L^3$:

$$\int_V \frac{\partial \phi}{\partial t}\, dV = D\int_V \nabla^2\phi\, dV + \int_V S\, dV$$

Step 2: Apply the divergence theorem to the Laplacian:

$$\frac{d}{dt}\int_V \phi\, dV = D\oint_{\partial V} \nabla\phi \cdot \hat{n}\, dA + \dot{Q}_{source}$$

Step 3: Invoke the lumping assumption — ϕ is spatially uniform inside V, so:

$$\frac{d\phi}{dt}\cdot V = D\cdot\frac{\phi_{out} - \phi_{in}}{L}\cdot A + \dot{Q}_{source}$$

Step 4: Identify the lumped parameters:

$$C_{lump}\frac{d\phi}{dt} = \frac{\phi_{out} - \phi_{in}}{R_{lump}} + S_{lump}$$

where $C_{lump} = V$ (capacitance = volume for mass storage, $\rho c_p V$ for heat) and $R_{lump} = L/DA$ (resistance).

This is an ODE — the PDE has become a lumped circuit element.

The same procedure applied to the wave equation gives an LC oscillator. Applied to Navier-Stokes (linearized) gives a mass-spring-damper.

But the identity is not exact, and the reason is on line 58. **Steps 1 and 2 are exact** — the divergence theorem is an identity and the integral form is a restatement. **Step 3 is the approximation**: replacing $\phi(\mathbf{r}, t)$ inside the volume by a single $\phi(t)$ is valid only under the lumping criterion of Ch. 16 §16.10 ($L/\lambda \ll 1$, or $Bi \ll 0.1$ for thermal). **Step 4 is a definition**, not a derivation: C_{lump} and R_{lump} are *defined* by identifying volume with capacitance and L/DA with resistance, which makes them easy to remember and easy to misapply.

`[APPROXIMATION]` — so the cross-branch result is a **shared procedure applied to different physics**, not a mathematical identity between the domains. What is identical across domains is Steps 1, 2, and 4. The physics is entirely in Step 3's justification and in the values that Step 4 assigns.

Model Ledger — control-volume integral to lumped element

Field	Content
Parent theory	Continuum conservation law + constitutive flux law (Ch. 15, Ch. 16)
Reduction	Integrate over V (exact); apply the divergence theorem (exact); impose spatial uniformity inside V (approximate); define C_{lump}, R_{lump} by correspondence
Model	$C\dot{\phi} = \frac{\phi_{out} - \phi_{in}}{R} + \dot{Q}_{source}$
Assumptions	$L/\lambda \ll 1$ or $Bi < 0.1$; linear constitutive law; time-invariant parameters; no distributed source inside V; a single state variable per element
Physics retained	total storage, total flow, and the balance between them
Physics neglected	every internal gradient; internal modes; spatial distribution of the source
Validity	a point in parameter space, not a region — valid at one frequency or timescale
Fails when	the criterion is violated; the constitutive law is nonlinear; parameters drift with state
Next model	retain $\phi(\mathbf{r}, t)$ and solve the PDE (Ch. 17 §17.10); or resolve modes (Ch. 19 §19.3)

17.1.2 — The Lumping Criterion, Domain by Domain

From Ch. 16 §16.10.1, the lumping criterion for each domain:

Domain	Criterion	Physical meaning
EM (circuits)	$L/\lambda = Lf/v_{em} \ll 1$	Component smaller than EM wavelength
Acoustic	$L/\lambda = Lf/c_s \ll 1$	Component smaller than sound wavelength
Elastic structural	$L/\lambda = Lf/v_P \ll 1$	Component smaller than stress wave wavelength
Thermal	$Bi = hL/\kappa \ll 1$	Temperature uniform within body
Hydraulic (pipe network)	$Lf/c_{water} \ll 1$	No water hammer effects
Chemical (CSTR)	Perfect mixing assumed	Concentration uniform in vessel

When the criterion fails: do not use the lumped model. Use the distributed-parameter bridge zone model (transmission line, Euler-Bernoulli beam, heat equation) or the full PDE. The chapter map tells you exactly where to go.

17.2 — Effort and Flow: A Cross-Domain Variable Pair

17.2.1 — Definitions

Across the domains modelled in this book, a consistent description of a two-terminal lumped element can be built on a pair of conjugate variable types:

Effort variable e: The "potential" — the quantity that drives the flow, does work per unit of flow passing through. Measured across an element.

Flow variable f: The "through-variable" — the quantity that flows, whose product with the effort gives instantaneous power. Measured through an element. In conservation-based domains f is often the current (rate of flux) of a conserved quantity and is then connected to a Noether current (Ch. 1 §1.7); but the effort–flow abstraction is **broader than Noether's theorem**, and the connection is not automatic. Velocity paired with force is a perfectly good effort–flow pair without being a Noether current; likewise entropy flow, molar flow, volume flow, and flux rate. What all of them share is the power identity $e \cdot f = P$, not a shared conservation law.

Power: $P = e \times f$ in every domain.

17.2.2 — The Complete Effort-Flow Table

Domain	Effort e	Flow f	Power $e \times f$	Conserved quantity
Electrical	Voltage V [V]	Current I [A]	Watts [W]	Charge Q
Translational mech.	Force F [N]	Velocity v [m/s]	Watts [W]	Momentum p
Rotational mech.	Torque τ [N·m]	Angular velocity ω [rad/s]	Watts [W]	Angular momentum L
Thermal	Temperature T [K]	Entropy flow $\dot{S} = \dot{Q}/T$ [W/K]	Watts: $T\dot{S} = \dot{Q}$ [W]	Energy U (transferred; entropy *generated*)
Hydraulic	Pressure P [Pa]	Volume flow Q [m³/s]	Watts [W]	Volume/mass
Pneumatic	Pressure P [Pa]	Mass flow $\dot{m}$ [kg/s]	Watts [W]	Mass
Chemical	Chem. potential μ [J/mol]	Molar flow $\dot{n}$ [mol/s]	Watts [W]	Moles
Magnetic	Magnetomotive force $\mathcal{F}$ [A]	Flux rate $\dot{\Phi}$ [Wb/s = V]	Watts [W]	Flux linkage Φ (inductive state variable, *not* a Noether charge)

> **Warning: The thermal row is a genuine special case — the units are not optional**
> Temperature is **not** the power-conjugate of heat flow: $T\dot{Q}$ has units K·W, which is not power. That is why the naive row "Thermal: $T \times \dot{Q}$ = watts" is wrong. The thermodynamically conjugate flow is the **entropy rate** $\dot{S}$ [W/K], and only then is $e \cdot f = T\dot{S} = \dot{Q}$ [W] exact.
>
> **Which convention is this book using?** The engineering convention throughout Ch. 14, 17 and 19 is $R_{th} = \Delta T/\dot{Q}$ [K/W], because that is what appears in datasheets and heat-sink specifications. In the entropy-flow convention the same element obeys $\Delta T = R \cdot \dot{S}$, and the two coefficients are related **only approximately**:
>
> $$\Delta T = R_{th}\,\dot{Q} \;\simeq\; T_0\,R_{th}\,\dot{S} = R\,\dot{S} \qquad \Longrightarrow \qquad R \simeq T_0\,R_{th}$$
>
> This is a **small-signal linearisation about a reference temperature** T_0, not an identity. The step is $\dot{S} \approx \dot{Q}/T_0$, which is valid only when the temperature variation across the element is small compared with T_0. For a finite temperature difference the relation is nonlinear, and the entropy generated by a transfer from T_{hot} to T_{cold} is

$$\dot{S}_{gen} = \dot{Q}\left(\frac{1}{T_{cold}} - \frac{1}{T_{hot}}\right)$$

so no single constant R relates ΔT to $\dot{S}$ across the whole range. **Do not read $R = T\,R_{th}$ as an exact transformation for arbitrary ΔT.** The tables below quote K/W; the two conventions must never be mixed silently, and the conversion is a linearisation.

Warning: Not every flow in this table is a Noether current For electrical rows the flow is the conserved charge current itself, which is what makes KCL a statement of charge conservation. For mechanical rows, F and v are a *power-conjugate* pair — a representation choice, not a conservation law. The Noether current of spatial translations is momentum density and its flux, not velocity: in a continuum, $\partial_t(\rho v_i) + \partial_j \Pi_{ij} = f_i$ expresses momentum balance, and the transported quantity is Π_{ij}. A lumped F–v pair therefore encodes momentum transport without v being a conserved current. Two further rows are weaker still, and the table should not pretend otherwise:

- **Thermal.** $\int \dot{S}\,dt$ is entropy **production**, not a conserved quantity. What is conserved is energy U; heat flow *moves* energy between bodies. The Noether framing applies to the energy transfer, not to the entropy current.
- **Magnetic.** Flux Φ is not a Noether charge of any ordinary continuous symmetry. It is the state variable of an inductive element; its *quantization* in superconductors comes from the winding of the condensate phase, not from Noether's theorem applied to a magnetic symmetry.

The generalized momentum (integral of effort): $p_e = \int e\,dt$ (charge in electrical, momentum in mechanical, flux linkage in magnetic)

The generalized displacement (integral of flow): $q_f = \int f\,dt$ (charge $Q = \int I\,dt$ in electrical, displacement $x = \int v\,dt$ in mechanical)

17.2.3 — Why a Pair, and Not More?

From the network perspective (Ch. 1 §1.7): many conserved physical quantities participate in a pair of power-conjugate variables whenever a suitable lumped representation exists. Conservation is a statement about a Noether current; the effort–flow pair is a modelling choice that corresponds to one in the electrical case and only indirectly in the mechanical case. In the electrical case the conserved charge becomes the flow variable and its conjugate potential (from the Hamiltonian formulation, Ch. 1 §1.6) becomes the effort variable; in the mechanical case the power-conjugate pair is velocity and force, and what is transported is the momentum flux Π_{ij} rather than either member of the pair. (Two rows of the table above are weaker than this picture — thermal and magnetic — and the caveats are recorded there.)

Power = effort × flow in every domain because this product equals the rate of energy transfer, and energy is the Hamiltonian — the conserved quantity of time translation (Ch. 1 §1.7.3). The product is invariant across domains because energy is invariant across domains. (This holds for the thermal domain only once the flow is the entropy rate $\dot{S}$ rather than $\dot{Q}$ — see the warning above. Fixing the flow variable is what makes the slogan literally true instead of approximately true.)

The electrical case is the clearest. In a theory with local $U(1)$ gauge structure, the **global** phase symmetry of charged matter gives a Noether current j^μ with $\partial_\mu j^\mu = 0$; integrating over a volume yields $\dot{Q} = -\oint \mathbf{j} \cdot d\mathbf{A}$, which is KCL (Ch. 11 §11.12). The gauge potential becomes the voltage as effort. Note the distinction:

the *global* symmetry is what Noether's first theorem acts on, while the *local* gauge redundancy is what makes the potential a gauge field and relates to Noether's second theorem — they are not the same statement.

For mechanical domains the chain is a continuous symmetry, so Noether applies directly: space-translation symmetry → momentum conservation → momentum flux as the flow; force as effort.

17.3 — The Three Passive Elements: An R/C/L Template for Lumped Systems

> **Note: A template, not a theorem** Storage, dissipation, and inertia are the three things energy can do in a lumped system, so it is unsurprising — and very useful — that R, C, and L organise a great deal of engineering practice. But this is a **template**, not a universal result. Plenty of legitimate lumped elements do not fit the three, and the repository already contains examples: nonlinear hydraulic inertance, pumps and valves (active, not passive), chemical reaction elements, hysteretic and material-memory elements, and distributed delays. Thermal systems frequently have a capacitance and a resistance but **no** inertial element at all, so the L column is simply empty. Expect to add sources, nonlinearities, constraints, and domain-specific elements as the problem demands. The template earns its keep by telling you what to look for first, not by claiming to be exhaustive.

17.3.1 — The Resistor R: Dissipation

A resistor relates effort to flow **instantaneously** (no memory):

$$e = R \cdot f \quad \text{or} \quad f = \frac{e}{R} = G \cdot e$$

where $G = 1/R$ is the conductance. Power dissipated: $P = e \cdot f = Rf^2 = e^2/R > 0$.

The resistor is the lumped form of the **constitutive transport relation** (Kubo, Ch. 13): flux = conductivity × gradient. Lumped: $\Delta\phi/L \to \Delta\phi$, $DA/L \to 1/R$.

Domain	Resistor R	Physical origin	Value
Electrical	R [Ω]	Ohm's law ($\sigma = ne^2\tau/m$, Ch. 13)	$\rho L/A$
Translational mech.	Damper b [N·s/m]	Newton's viscosity (Ch. 13)	$6\pi\mu a$ (Stokes)
Rotational mech.	Rotational damper b_r [N·m·s/rad]	Viscous torque	
Thermal	R_{th} [K/W]	Fourier's law (Ch. 13)	$L/\kappa A$
Hydraulic	R_{hyd} [Pa·s/m³]	Hagen-Poiseuille (Ch. 15)	$8\mu L/\pi r^4$
Chemical	R_{chem} [J·s/mol²]	Fick's law (Ch. 13)	L/DA

17.3.2 — The Capacitor C: Potential Energy Storage

A capacitor relates effort to the **integral of flow** (has memory of past flow):

$$e = \frac{1}{C}\int f\,dt \quad \Longleftrightarrow \quad f = C\frac{de}{dt}$$

Energy stored: $W_C = \frac{1}{2}Ce^2$ (effort-squared storage).

The capacitor is the lumped form of the **∂φ/∂t storage term** in the PDE.

Domain	Capacitor C	Physical origin	Value
Electrical	C [F]	Electric field energy ($\frac{1}{2}\epsilon_0 E^2$, Ch. 11)	$\epsilon A/d$
Translational mech.	Compliance $1/k$ [m/N]	Spring potential energy	$1/k$
Rotational mech.	Torsional compliance $1/k_r$ [rad/N·m]	Torsional spring	$1/k_r$
Thermal	C_{th} [J/K]	Thermal energy storage (Ch. 10)	$\rho c_p V$
Hydraulic	C_{hyd} [m³/Pa]	Fluid compressibility or vessel compliance	$V/\rho c_s^2$ or $A\Delta h/\Delta P$
Chemical	C_{chem} [mol²/J]	Vessel/tank volume	V/RT (ideal gas)

17.3.3 — The Inductor L: Kinetic Energy Storage

An inductor relates flow to the **integral of effort** (has memory of past effort):

$$f = \frac{1}{L}\int e\,dt \quad \Longleftrightarrow \quad e = L\frac{df}{dt}$$

Energy stored: $W_L = \frac{1}{2}Lf^2$ (flow-squared storage).

The inductor is the lumped form of the **∂²φ/∂t² inertia term** in the PDE.

Domain	Inductor L	Physical origin	Value
Electrical	L [H]	Magnetic field energy ($\frac{1}{2}\mu_0 H^2$, Ch. 11)	$\mu N^2 A/\ell$
Translational mech.	Mass m [kg]	Kinetic energy $\frac{1}{2}mv^2$ (Ch. 9)	m
Rotational mech.	Moment of inertia I_r [kg·m²]	Rotational KE $\frac{1}{2}I_r\omega^2$ (Ch. 9)	$\sum mr^2$
Hydraulic	Fluid inertance L_{hyd} [kg/m⁴]	Kinetic energy of flowing fluid	$\rho L/A$
Pneumatic	Same	Fluid inertia	$\rho L/A$

Why no thermal inductor? Pure heat conduction has no "inertia" term — the thermal diffusion equation is first-order in time, not second. Temperature does not overshoot past equilibrium in a simple thermal system (unlike mechanical oscillators). When thermal wave phenomena appear (very fast laser heating, pulsed electron beams), a "thermal inertance" can be defined — but it is rarely relevant in engineering.

17.4 — Conservation Laws at Nodes: The Kirchhoff Generalizations

17.4.1 — The Flow Conservation Law (KCL Generalization)

At any node (junction point), the net inflow equals the rate of storage at that node — a balance law from the continuity equation (Ch. 1 §1.7). When storage is drawn as its own element branch (a mass to the inertial reference, C_{th} to ambient), the storage flow is one of the f_k and the node law takes the form:

$$\boxed{\sum_k f_k = 0 \quad \text{at every node}}$$

Domain	Statement	Physical origin
Electrical	$\sum I_k = 0$ — KCL	Charge conservation (Noether U(1), Ch. 11 §11.12)
Translational mech.	$\sum F_k = m\ddot{x}$ — Newton's law	Momentum conservation (Noether, translation)
Rotational mech.	$\sum \tau_k = I_r\dot{\omega}$ — rotational Newton	Angular momentum conservation
Thermal	$\sum \dot{Q}_k = C_{th}\dot{T}$ — energy balance	Energy balance (first law); Noether time-translation applies only to closed Lagrangian systems
Hydraulic	$\sum Q_k = 0$ — continuity	Mass conservation (Noether, particle number)
Chemical	$\sum \dot{n}_k = 0$ — mole balance	Particle conservation

For static cases (no storage: $C = 0$, $L = 0$): every conservation law becomes simply $\sum f_k = 0$ — the flows sum to zero at every node.

17.4.2 — The Effort Conservation Law (KVL Generalization)

Around any closed loop (cycle in the graph), the sum of all effort drops must be zero:

$$\boxed{\sum_k e_k = 0 \quad \text{around every loop}}$$

Domain	Statement	Physical origin
Electrical	$\sum V_k = 0$ — KVL	Faraday's law, quasi-static (Ch. 11 §11.12)
Translational mech.	$\sum F_k = 0$ — force loop	D'Alembert's principle (inertia as a "force")
Rotational mech.	$\sum \tau_k = 0$	Torque loop balance
Thermal	$\sum \Delta T_k = 0$	Temperature is a potential (no thermal circulation)
Hydraulic	$\sum \Delta P_k = 0$	Pressure is a potential (no spontaneous pressure circuits)

Why does the loop law hold? Effort variables are **potentials** — they derive from scalar potential functions. For electrical: $\mathbf{E} = -\nabla\phi - \partial_t \mathbf{A}$ and $\oint \mathbf{E} \cdot d\mathbf{l} = -d\Phi/dt$, which is not zero in general. The lumped approximation confines changing flux to the inductors: outside them $\mathbf{E}$ is (approximately) a gradient, so the loop integral there reduces to a sum of potential differences, and each inductor's Faraday EMF is booked as its branch voltage $L\,di/dt$. For mechanical: the loop law expresses the compatibility of relative velocities (D'Alembert form). The loop sum vanishes because, under these lumping assumptions, the off-element field is a gradient and the line integral of a gradient over a closed loop is zero.

17.5 — Bond Graphs: The Formal Multi-Domain Language

17.5.1 — Bond Graph Notation

A **bond** is a line connecting two subsystems, with a **half-arrow** indicating the positive flow direction. Every bond carries both an effort and a flow:

```
  e
──────→ (half-arrow shows positive flow convention)
  f
```

Power flows in the direction of the half-arrow when $P = ef > 0$.

Bond graph elements:

Symbol	Name	Law	Domain example
R	Resistor	$e = Rf$	Resistor, damper, pipe
C	Capacitor	$f = C\dot{e}$	Capacitor, spring, thermal mass
I	Inductor	$e = L\dot{f}$	Inductor, mass, fluid inertance
Se	Effort source	$e = e_s(t)$	Voltage source, force source, pressure source
Sf	Flow source	$f = f_s(t)$	Current source, velocity source, flow source
TF	Transformer	$e_1 = n \cdot e_2$, $f_2 = n \cdot f_1$	Ideal transformer, gear, lever, piston
GY	Gyrator	$e_1 = r \cdot f_2$, $e_2 = r \cdot f_1$	Motor/generator, gyroscope, Hall sensor

17.5.2 — Junctions: The Topology

Two junction types define how bonds connect:

0-junction (common effort): All connected bonds share the same effort. Flows sum to zero ($\sum f_k = 0$). Analog: nodes in parallel in circuits.

$$e_1 = e_2 = \cdots = e_n, \qquad \sum_k f_k = 0$$

1-junction (common flow): All connected bonds share the same flow. Efforts sum to zero ($\sum e_k = 0$). Analog: elements in series in circuits.

$$f_1 = f_2 = \cdots = f_n, \qquad \sum_k e_k = 0$$

17.5.3 — Bond Graph Examples: Identical Topology, Different Physics

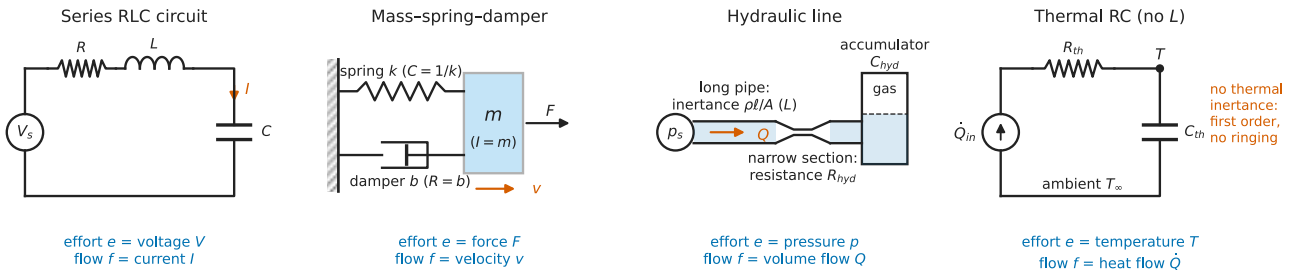
Series RLC circuit and **mass-spring-damper** have identical bond graphs:

```
Series RLC:
Se ── 1 ── R
          ├─ C
          └─ I

Mass-spring-damper:
Se ── 1 ── R (damper b)
          ├─ C (spring 1/k)
          └─ I (mass m)
```

The 1-junction says: same current through all elements (series circuit) = same velocity for all elements (everything moves at the same velocity). The effort equation: $V_{source} = V_R + V_C + V_L$ = $F_{applied} = F_{damper} + F_{spring} + F_{mass}$.

These are **the same equation with different labels.** Every analytical result derived for one (step response, frequency response, resonance) applies directly to the other.



The R/C/L template in four domains, each labelled with its effort and flow: series RLC circuit, mass–spring–damper ($I = m$, $R = b$, $C = 1/k$), hydraulic line (pipe inertance $\rho\ell/A$, constriction resistance, accumulator compliance) and thermal RC, which has no inductive element and so can only respond in first order.

Transformer and lever:

```
Hydraulic cylinder → mechanical:
P_in ─── TF(A) ─── F_out
Q_in                v_out

Gear train:
τ_in ─── TF(N) ─── τ_out
ω_in                ω_out
```

Both are transformers with effort/flow transformation ratio $n = A$ (area) or $n = N$ (gear ratio): $e_1 = n \cdot e_2$, $f_2 = n \cdot f_1$, power conserved: $e_1 f_1 = e_2 f_2$.

Motor/generator as gyrator:

An electric motor converts electrical effort (voltage V) and flow (current I) to mechanical effort (torque τ) and flow (angular velocity ω):

$$V = K_e \omega, \qquad \tau = K_t I$$

For an ideal motor: $K_e = K_t = K$ (back-EMF constant = torque constant). This is a gyrator with $r = K$: electrical effort drives mechanical flow, and mechanical effort drives electrical flow. The gyrator is non-reciprocal — it is what makes motors and generators directional.

17.6 — Complex Impedance and Transfer Functions

17.6.1 — Impedance in Every Domain

For sinusoidal steady state ($e = \hat{e}e^{i\omega t}$, $f = \hat{f}e^{i\omega t}$), the **impedance** is:

$$Z(\omega) = \frac{\hat{e}}{\hat{f}}$$

For each element:

Element	Impedance $Z(\omega)$	At DC ($\omega \to 0$)	At high ω
R	R	R (finite)	R (finite)
C	$1/i\omega C$	∞ (blocks DC)	0 (short)
L (I)	$i\omega L$	0 (short)	∞ (blocks HF)

Admittance: $Y(\omega) = 1/Z(\omega)$.

Combination rules (same in every domain):

- Series: $Z_{total} = Z_1 + Z_2 + \cdots$ (impedances add)
- Parallel: $Y_{total} = Y_1 + Y_2 + \cdots$ (admittances add)

17.6.2 — The Transfer Function

The **transfer function** $H(s) = Y(s)/X(s)$ in the Laplace domain ($s = i\omega$ for sinusoidal; $s = \sigma + i\omega$ for general transients) relates any output to any input for a linear system.

For a series RLC circuit with input voltage V_{in} and output voltage across C:

$$H(s) = \frac{V_C}{V_{in}} = \frac{1/Cs}{Ls^2 + Rs + 1/C} = \frac{\omega_0^2}{s^2 + 2\zeta\omega_0 s + \omega_0^2}$$

where $\omega_0 = 1/\sqrt{LC}$ and $\zeta = R/2\sqrt{L/C}$.

The transfer function encodes:

- **Poles:** values of s where $H \to \infty$ — system's natural frequencies
- **Zeros:** values of s where $H \to 0$ — frequencies blocked by the system
- **Frequency response:** $H(i\omega)$ gives amplitude and phase vs. frequency (Bode plot)

The **Bode plot** (log magnitude vs. log frequency):

- Each pole at ω_p: slope changes by -20 dB/decade
- Each zero at ω_z: slope changes by $+20$ dB/decade
- Reading off poles and zeros directly from the Bode plot is a core engineering skill

This machinery works in every domain. A thermal system, a hydraulic network, and a mechanical vibration problem all have transfer functions of the same form — and all use the same Bode plot analysis.

17.7 — The Second-Order System: The Universal Template

17.7.1 — The Governing ODE

The second-order system (L-R-C, mass-spring-damper, hydraulic RCL):

$$\boxed{L\ddot{q} + R\dot{q} + \frac{q}{C} = e_{source}(t)}$$

where $q = \int f\, dt$ is the generalized displacement (charge, position, volume). In standard form, dividing by L:

$$\ddot{q} + 2\zeta\omega_0\dot{q} + \omega_0^2 q = \frac{e_{source}}{L}$$

Natural frequency:

$$\omega_0 = \frac{1}{\sqrt{LC}} \qquad [\text{rad/s}]$$

Damping ratio:

$$\zeta = \frac{R}{2}\sqrt{\frac{C}{L}} = \frac{R}{2\omega_0 L} = \frac{1}{2Q}$$

Quality factor:

$$Q = \frac{\omega_0 L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{1}{2\zeta}$$

17.7.2 — Free Response (Homogeneous Solution)

Characteristic equation: $s^2 + 2\zeta\omega_0 s + \omega_0^2 = 0$

Roots: $s_{1,2} = -\zeta\omega_0 \pm \omega_0\sqrt{\zeta^2 - 1}$

Three regimes:

Underdamped ($\zeta < 1$): $s_{1,2} = -\zeta\omega_0 \pm i\omega_d$ where $\omega_d = \omega_0\sqrt{1-\zeta^2}$

$$q(t) = e^{-\zeta\omega_0 t}\left(A\cos\omega_d t + B\sin\omega_d t\right)$$

Oscillates at ω_d with exponentially decaying envelope. Time to decay: $\tau = 1/\zeta\omega_0$.

Critically damped ($\zeta = 1$): repeated root $s = -\omega_0$

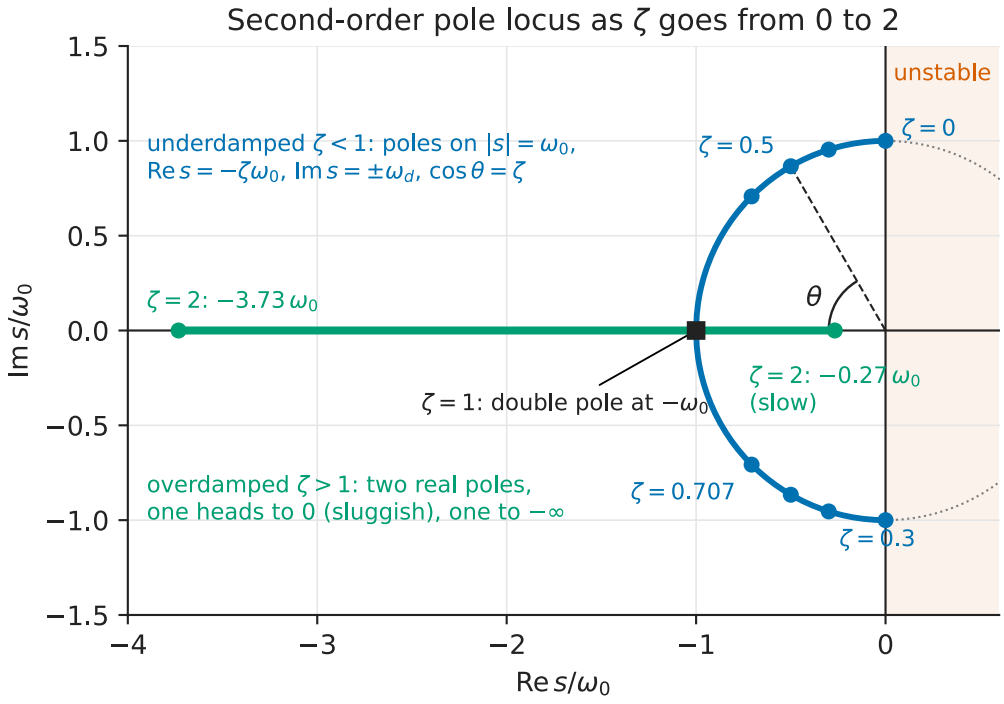
$$q(t) = (A + Bt)\,e^{-\omega_0 t}$$

Fastest approach to equilibrium without overshoot.

Overdamped ($\zeta > 1$): two real roots $s_1 < s_2 < 0$

$$q(t) = A\,e^{s_1 t} + B\,e^{s_2 t}$$

Slow exponential approach; no oscillation.



Roots $s = -\zeta\omega_0 \pm \omega_0\sqrt{\zeta^2 - 1}$ as ζ rises from 0 to 2: underdamped poles travel along the circle $|s| = \omega_0$ at angle θ with $\cos\theta = \zeta$, meet at $-\omega_0$ when $\zeta = 1$, then split along the real axis — at $\zeta = 2$ one pole sits at $-0.27\omega_0$, which is why overdamped systems are sluggish.

17.7.3 — Step Response

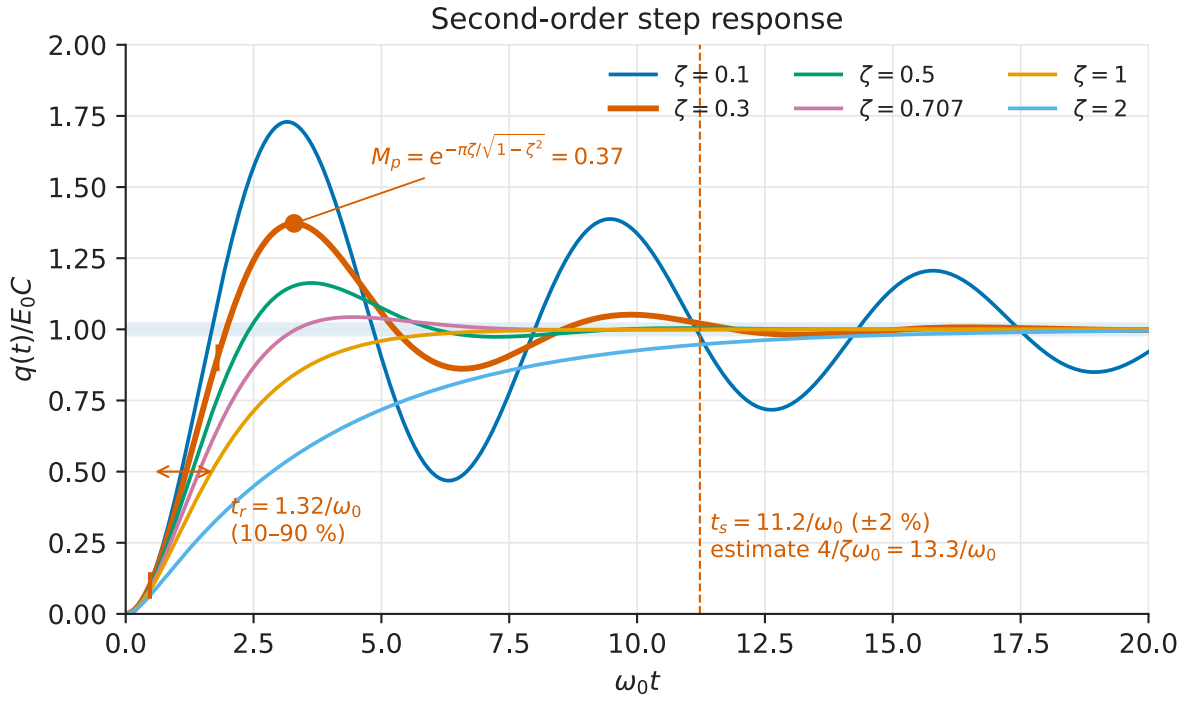
For a unit step input $e_{source} = E_0\, u(t)$:

Underdamped ($\zeta < 1$):

$$q(t) = E_0 C\left[1 - e^{-\zeta\omega_0 t}\left(\cos\omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_d t\right)\right]$$

- Overshoot: $M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}$
- Rise time: $t_r \approx (1.8)/\omega_0$ (10% to 90%, for $\zeta \approx 0.5$)
- Settling time: $t_s \approx 4/\zeta\omega_0$ (±2% band)

The same metrics apply to every domain: overshoot in a mass-spring system (mechanical), ringing in an RLC circuit (electrical), thermal overshoot in a temperature controller (thermal), pressure oscillation in a hydraulic ram (hydraulic).



Step response $q(t)/E_0C$ versus $\omega_0 t$ for $\zeta = 0.1, 0.3, 0.5, 0.707, 1, 2$, with the $\zeta = 0.3$ curve annotated: overshoot $M_p = 0.37$, 10–90% rise time $1.32/\omega_0$, and ±2% settling at $11.2/\omega_0$ against the $4/\zeta\omega_0 = 13.3/\omega_0$ estimate; $\zeta = 1$ is the fastest response without overshoot.

17.7.4 — Frequency Response

For sinusoidal input $e_{source} = E_0 \cos\omega t$:

$$|H(i\omega)| = \frac{\omega_0^2}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\zeta^2\omega_0^2\omega^2}}$$

$$\angle H(i\omega) = -\arctan\left(\frac{2\zeta\omega_0\omega}{\omega_0^2 - \omega^2}\right)$$

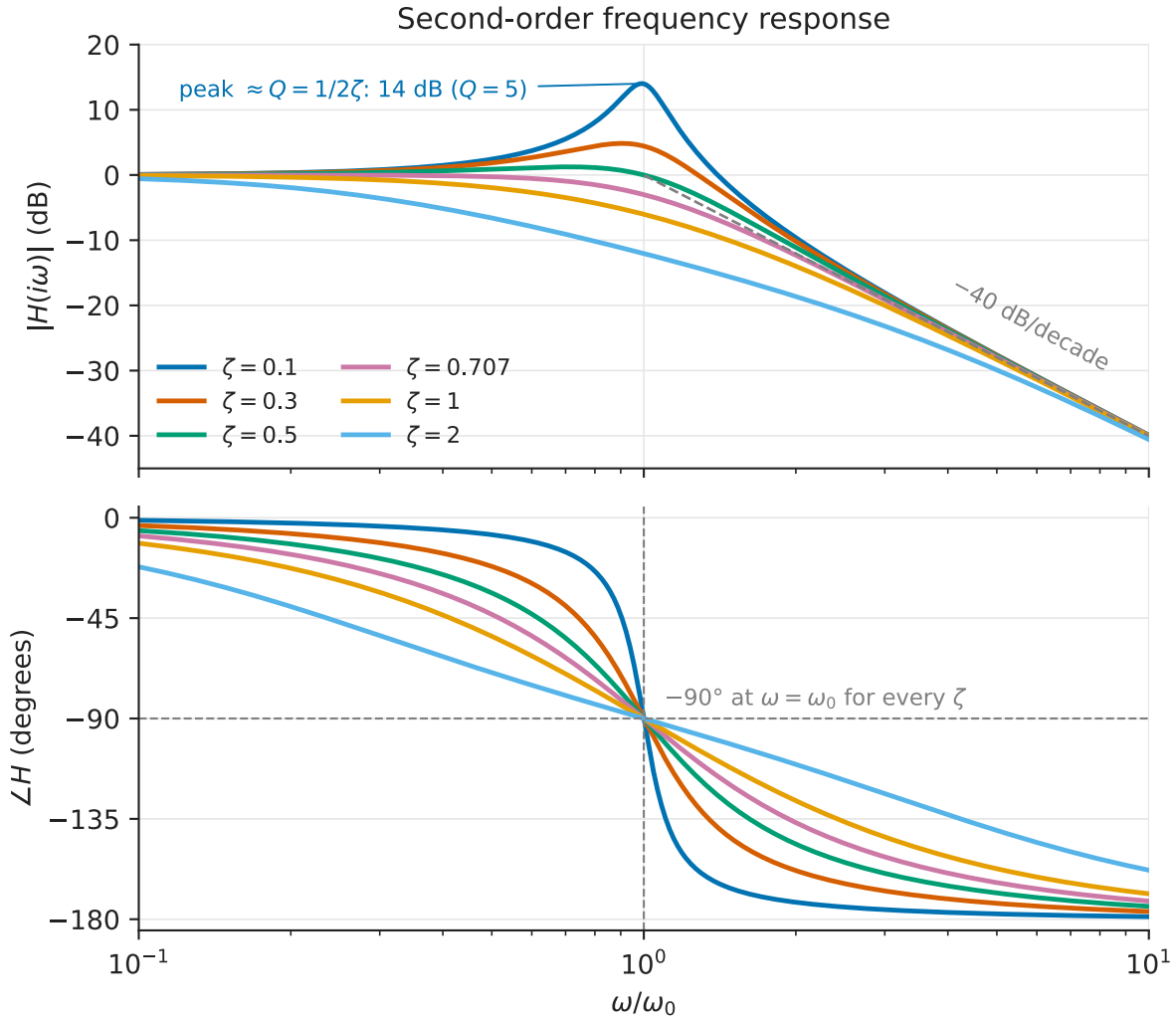
(four-quadrant arctangent, $-\operatorname{atan2}(2\zeta\omega_0\omega,\ \omega_0^2 - \omega^2)$; the phase passes -90° at ω_0 and tends to -180° as $\omega \to \infty$)

Resonance peak: At $\omega_p = \omega_0\sqrt{1-2\zeta^2}$ (for $\zeta < 1/\sqrt{2}$):

$$|H_{peak}| = \frac{1}{2\zeta\sqrt{1-\zeta^2}} \approx \frac{1}{2\zeta} = Q \quad \text{(for small } \zeta\text{)}$$

The **Q factor physically means:** the resonance amplitude is Q times the DC amplitude. High-Q systems are highly resonant and selective (narrow bandpass).

Half-power bandwidth: $\Delta\omega = \omega_0/Q$ — the frequency range around resonance where power is above $P_{max}/2$.



Magnitude (dB) and phase of $H = \omega_0^2/(\omega_0^2 - \omega^2 + 2i\zeta\omega_0\omega)$ for $\zeta = 0.1, 0.3, 0.5, 0.707, 1, 2$: light damping gives a resonant peak $\approx Q = 1/2\zeta$ (14 dB for $\zeta = 0.1$), every curve rolls off at −40 dB/decade above ω_0, and the phase passes −90° at $\omega = \omega_0$ regardless of ζ.

17.7.5 — The Universal Template Table

Physical system	L (inertance)	R (resistance)	C (compliance)	ω_0
Series RLC circuit	Inductance L	Resistance R	Capacitance C	$1/\sqrt{LC}$
Mass-spring-damper	Mass m	Damping b	$1/k$ (spring)	$\sqrt{k/m}$
Pendulum (small angle)	$m\ell^2$	Air drag	$1/(mg\ell)$	$\sqrt{g/\ell}$
Torsional oscillator	Inertia I_r	Bearing friction	$1/k_r$	$\sqrt{k_r/I_r}$
Hydraulic L-R-C	Fluid inertance $\rho L/A$	Pipe resistance	Accumulator C_{hyd}	$1/\sqrt{L_{hyd}C_{hyd}}$
Helmholtz resonator	Air inertance in neck	Air viscosity	Volume compliance	$c_s\sqrt{A/(V\ell)}$
Thermal R-C	(no inductor)	R_{th}	C_{th}	(first-order only)

Design insight: Every system in this table can be analyzed with identical mathematics. The engineer chooses which row to work in; the equations are the same.

17.8 — Thévenin and Norton Equivalents

17.8.1 — The Theorems

Thévenin's theorem: Any linear network seen from two terminals is equivalent to:

- An effort source e_{Th} = open-circuit effort (measure effort with no load)
- A series impedance Z_{Th} = impedance seen looking into the network with all sources killed

Norton's theorem: The same network is also equivalent to:

- A flow source $f_N = e_{Th}/Z_{Th}$ = short-circuit flow
- A parallel impedance $Z_N = Z_{Th}$

Maximum power transfer: Power delivered to load Z_L is maximized when:

$$Z_L = Z_{Th}^*$$

For resistive networks: $R_L = R_{Th}$, and $P_{max} = e_{Th}^2/4R_{Th}$.

17.8.2 — Thévenin/Norton in Every Domain

Domain	Thévenin source	Thévenin impedance	Maximum power transfer
Electrical	V_{OC} [V]	R_{Th} [Ω]	$R_L = R_{Th}$
Mechanical	F_{free} (free deflection force)	k_{Th} (stiffness)	$k_L = k_{Th}$
Thermal	T_{OC} [K]	$R_{th,Th}$ [K/W]	$R_{th,L} = R_{th,Th}$
Hydraulic	P_{OC} [Pa]	$R_{hyd,Th}$ [Pa·s/m³]	$R_{hyd,L} = R_{hyd,Th}$
Acoustic	p_{OC} [Pa]	$Z_{ac,Th}$ [Pa·s/m³]	Impedance match

Engineering use: Every subnetwork can be reduced to its Thévenin/Norton equivalent before connecting to the next subsystem — modularity. The impedance matching principle (maximum power transfer) governs:

- RF amplifier matching networks (EEE)
- Acoustic transducer design (ME/EEE)
- Hydraulic actuator sizing (ME)
- Heat exchanger coupling to thermal loads (ME/ChE)

17.9 — Energy and Power Accounting

17.9.1 — Power Balance at Every Instant

For any closed system, the power balance:

$$\underbrace{P_{source}}_{\text{input}} = \underbrace{\frac{d}{dt}\left(\frac{1}{2}Ce^2 + \frac{1}{2}Lf^2\right)}_{\text{stored}} + \underbrace{Rf^2}_{\text{dissipated}}$$

This is the first law of thermodynamics (Noether energy conservation, Ch. 1 §1.7.3) expressed in lumped-element form. It applies identically in every domain — because it follows from the same Noether theorem in every domain.

17.9.2 — Reactive Power and Power Factor

For sinusoidal steady state at frequency ω:

Complex power: $\tilde{S} = \frac{1}{2}\hat{e}\,\hat{f}^* = P + iQ_{reac}$

- $P = \mathrm{Re}[\tilde{S}]$: **active power** (actually dissipated, time-averaged)
- $Q_{reac} = \mathrm{Im}[\tilde{S}]$: **reactive power** (stored/returned each cycle)
- $|\tilde{S}|$: **apparent power**

Power factor: $\cos\phi = P/|\tilde{S}| = \mathrm{Re}[Z]/|Z|$

For a pure R: $\phi = 0$, power factor = 1 (all power dissipated). For a pure L or C: $\phi = \pm 90^\circ$, power factor = 0 (power oscillates, none dissipated).

Engineering: Electrical utilities charge for apparent power (kVA) but only deliver active power (kW). Low power factor means the utility must supply large currents for small useful power → line losses. Power factor correction capacitors reduce reactive current in inductive loads (motors) — a direct application of the impedance concepts above.

17.10 — The Bridge Zone: When Lumping Fails

17.10.1 — The Distributed-Parameter Models

When the lumping criterion is violated, the element must be modeled as a **distributed parameter system** — a PDE retaining spatial variation. The bridge zone models that sit between the Layer-2 PDEs (Ch. 15–16) and Layer-3 lumped elements:

Lumped fails when	Bridge zone model	Governing equation
Electrical: $f > v_{em}/10L$	Transmission line	$\partial_{zz}V = L'C'\partial_{tt}V + R'C'\partial_t V$

Lumped fails when	Bridge zone model	Governing equation
Structural: $f > v_P/10L$	Euler-Bernoulli beam	$EI\partial_{xxxx}w + \rho A\partial_{tt}w = q$
Acoustic: $f > c_s/10L$	1D wave equation	$\partial_{tt}p = c_s^2\partial_{xx}p$
Thermal: $Bi > 0.1$	1D heat equation	$\partial_t T = \alpha\partial_{xx}T$
Hydraulic: $f > c_{water}/10L$	Water hammer equations	$\partial_{tt}p = c_w^2\partial_{xx}p$

The transmission line and Euler-Bernoulli beam are the most important bridge zone models in engineering practice. Both retain spatial variation along one axis while lumping in the transverse directions.

17.10.2 — The Hierarchy: Layer 2 → Bridge Zone → Layer 3

```
Layer 2: Full PDE
(full 3D, all frequencies)
        │
        │ if L >> characteristic dimension (thin beam, long cable)
        ▼
Bridge zone: 1D PDE
(full spatial variation along primary direction)
        │
        │ if L/λ << 1 (lumping criterion satisfied)
        ▼
Layer 3: ODE / algebraic
(lumped element: R, L, C)
```

The engineer always works at the lowest layer valid for the problem. This chapter gives the explicit criteria for when to move up.

17.11 — The Complete Layer-3 Template

17.11.1 — The Four Engineering Disciplines as Instances

The effort-flow template, R/C/L elements, KCL/KVL analogs, and Thévenin/Norton equivalents constitute a complete mathematical framework. Each engineering discipline is one instantiation:

EEE (Chapter 18): Effort = voltage V; Flow = current I; R = resistor; C = capacitor; L = inductor; KCL: $\sum I = 0$; KVL: $\sum V = 0$

ME (Chapter 19): Translational: Effort = force F; Flow = velocity v; R = damper b; C = spring $1/k$; L = mass m Rotational: Effort = torque τ; Flow = angular velocity ω; R = friction; C = torsional spring; L = moment of inertia

CE (Chapter 20): Hydraulic: Effort = pressure P; Flow = volume flow Q; R = pipe; C = accumulator; L = inertance Structural: Effort = force F; Flow = velocity v; R = damping; C = compliance; L = mass

ChE (Chapter 20): Effort = chemical potential μ; Flow = molar flow $\dot{n}$; R = diffusion resistance; C = vessel; energy balances close the system

17.11.2 — The Cross-Branch Identity

Every second-order system in every domain satisfies:

$$L\ddot{q} + R\dot{q} + q/C = e_{source}$$

with the same ω_0, ζ, Q, step response, Bode plot, and Thévenin/Norton equivalents.

`[STRUCTURAL CONNECTION]` — **the differential equation is genuinely the same; the R, C, L are genuinely different quantities with different units and different values.** This is the honest limit of the cross-branch claim. What unifies is the *order* of the ODE and the algebra of its solution, not the physics that fixes the coefficients.

Domain	R	C	L	Sets the timescale
Electrical	resistance [Ω]	capacitance [F]	inductance [H]	$1/RC$
Mechanical (translational)	damping [N·s/m]	compliance [m/N]	mass [kg]	$\sqrt{m/k}$
Thermal	thermal resistance [K/W]	heat capacity [J/K]	— (no inertia)	RC only
Hydraulic	hydraulic resistance [Pa·s/m³]	compliance [m³/Pa]	inertance [Pa·s²/m³]	RC or LC

Two things this table is for. First, **thermal systems have no L**: heat has no inertial term, so a lumped thermal system is first-order and has no resonance, no overshoot, and no ζ. Any PID tuning rule imported from mechanical or electrical practice has to be re-derived for thermal loops. Second, **the dimensionless ratios are what actually transfer**: ω_0, ζ, and Q are unit-free, so a normalised step response is comparable across domains even though R and C are not.

A PID controller (Ch. 21 preview) doesn't know what domain it's in:

$$u(t) = K_p e(t) + K_i \int e\,dt + K_d \dot{e}$$

The gains K_p, K_i, K_d are tuned the same way whether the controlled variable is voltage, temperature, position, pressure, or concentration — **provided the loop is adequately described by one linear time-invariant transfer function $H(s)$.** That proviso is a real restriction, not a formality: Ch. 21 §21.0.2 is explicit that nonlinear, time-varying, hybrid, distributed, stochastic, and delayed systems each need a different representation first. A controller that "does not know the domain" is a controller that *assumes* the domain is linear, time-invariant, and describable by a single rational function of s.

This is possible only because Bridge C has reduced all domains to the same mathematical object — and that reduction is an approximation whose validity is established in Ch. 16 §16.10, not a free-standing fact about nature.

17.12 — Summary

Bridge C performed three operations:

Step	Mathematical operation	Result	Connection
1. Integrate PDE over control volume	$\int_V (\partial_t \phi = D\nabla^2\phi) dV$	Flux in = rate of change	`[DERIVATION]` — exact

Step	Mathematical operation	Result	Connection
2. Invoke lumping criterion	$L/\lambda \ll 1$ or $Bi \ll 1$	Spatial uniformity assumed	`[APPROXIMATION]` — the load-bearing step
3. Identify lumped parameters	$C = V$, $R = L/DA$	ODE with lumped R, C, L	`[STRUCTURAL CONNECTION]` — same form, different coefficients

The three laws of Layer 3:

- Flow sum at every node = 0 (Noether conservation, KCL-type)
- Effort sum around every loop = 0 (potential structure, KVL-type)
- Element constitutive laws: $e = Rf$, $f = C\dot{e}$, $e = L\dot{f}$

The result for this class of systems: $L\ddot{q} + R\dot{q} + q/C = e_{source}(t)$ — the governing equation for every second-order linear RLC-type lumped system in any physical domain where lumping is valid. The equation is a consequence of step 1, is conditional on step 2, and is merely re-labelled in step 3. §17.10 is where it stops working.

Note the scope: this is the canonical **second-order linear** member of a family, not the governing equation of every system. Engineering models are also routinely first-order, higher-order, algebraic, differential-algebraic, stochastic, nonlinear, hybrid, delayed, fractional-order, or constrained — and the second-order oscillator, important as it is, is one subset. §17.10 works through the cases where this form is the wrong tool.

17.13 — Engineering Thread

Physics	Application
Lumping criterion $L/\lambda \ll 1$	When to use lumped circuit vs. transmission line; structural lumped mass vs. continuous beam
Effort-flow pairs	Hydraulic-electric analogy for system simulation; bond graph modeling
R, C, L in every domain	Cross-domain system modeling; multi-physics simulation (COMSOL, Simulink)
KCL/KVL analogs	Node and mesh analysis in all domains
Second-order ODE	Vibration design (ω_0, ζ spec); filter design; control system stability
Overshoot $M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}$	Structure settling time; amplifier stability margin; hydraulic ram design
Q factor and bandwidth	Resonator design; selectivity of filters; structural damping specification
Thévenin/Norton	Source-load matching; impedance matching in RF, audio, mechanical, acoustic
Power factor $\cos\phi$	Utility billing; power factor correction; reactive power management
Bond graphs	Multi-domain system modeling without domain-specific knowledge
Transfer function $H(s)$	Control system design; filter synthesis; structural FRF measurement

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17.14 — Looking Ahead: Layer 3 Chapters

Bridge C is complete. Layer 3 opens with the same template instantiated in each engineering discipline:

Chapter 18 — EEE Systems: Circuits, semiconductor devices, signals and systems. The electrical instantiation of the effort-flow template, with additional richness from nonlinear devices (diodes, transistors) built on Layer-1 physics (Ch. 6).

Chapter 19 — ME Systems: Machines, structures, thermofluid systems. The mechanical and thermal instantiation. Structural analysis using FEM (the discretized Navier equation, Ch. 15) and vibration design using the second-order template.

Chapter 20 — CE and ChE Systems: Structural systems, hydraulic networks, reactors, separations. The hydraulic and chemical instantiations.

Chapter 21 — Feedback, Control, and the Cross-Branch Capstone: PID control, transfer functions, stability criteria — the branch-agnostic mathematics that operates entirely at the level of the ODE, applying equally to all domains.

End of Chapter 17. Bridge C complete. Layer 3 begins.

Next: Chapter 18 — EEE Systems: Circuits, Devices, and Signals

EEE Systems: Circuits, Devices, and Signals

The Electrical Instantiation of Layer 3

> *A circuit is Maxwell's equations with the spatial variation averaged out. Everything else is bookkeeping — elegant, powerful bookkeeping, but bookkeeping nonetheless.* — teaching tradition in electrical engineering

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18.0 — Overview

Chapter 17 established the Layer-3 template: effort and flow, the R/C/L storage-dissipation-inertia pattern, KCL-type and KVL-type conservation at nodes and loops, and the canonical second-order R/C/L ODE that appears in many lumped linear systems. It also established where that template stops: first-order, higher-order, algebraic, differential-algebraic, stochastic, nonlinear, hybrid, delayed, and constrained systems all need other forms, and lumping is conditional on a stated criterion. This chapter instantiates the template in the electrical domain, where the RLC second-order case is realised most completely.

This chapter instantiates that template in the electrical domain. Every concept introduced here already appeared in some form in earlier chapters:

Ch. 18 concept	Physics origin
KCL, KVL	Maxwell quasi-static (Ch. 11 §11.12); Noether U(1) (Ch. 1 §1.7)
Impedance $Z(\omega)$	Ch. 17 §17.6 universal definition
Diode IV characteristic	p-n junction drift-diffusion (Ch. 6, Ch. 14)
MOSFET operation	Band bending, inversion layer (Ch. 6 §6.6)
Filter transfer function	Dispersion relation (Ch. 16 §16.4); second-order system (Ch. 17 §17.7)
Fourier/Laplace analysis	Plane wave decomposition (Ch. 16 §16.4); eigenmodes (Ch. 16 §16.5)
Sampling theorem	Nyquist from Ch. 16 (spatial frequencies → bandwidth)
Power electronics	Ohm + Faraday + Kirchhoff (Ch. 11) + control (Ch. 21 preview)

This chapter is not a replacement for a full circuits course. It is the systematic connection of that course's material to the physics that underlies it.

18.1 — Circuit Analysis: Maxwell's Equations in Lumped Form

18.1.1 — The Two Laws, Their Origins

Kirchhoff's Current Law (KCL):

At any node, $\sum_k I_k = 0$, where positive convention is defined per branch.

Origin (Ch. 11 §11.12): The continuity equation $\partial_t \rho + \nabla \cdot \mathbf{J} = 0$ integrated over the node volume, with quasi-static assumption $\partial_t \rho \to 0$:

$$\oint_{\partial V} \mathbf{J} \cdot d\mathbf{A} = 0 \quad \Longrightarrow \quad \sum_k I_k = 0$$

KCL is charge conservation (Noether U(1)) applied to a lumped node.

Kirchhoff's Voltage Law (KVL):

Around any closed loop, $\sum_k V_k = 0$.

Origin (Ch. 11 §11.12): Faraday's law $\oint \mathbf{E} \cdot d\mathbf{l} = -d\Phi_B/dt$ with quasi-static assumption ($d\Phi_B/dt \approx 0$ for circuits much smaller than the electromagnetic wavelength):

$$\sum_k V_k = -\frac{d\Phi_B}{dt} \approx 0$$

KVL fails when magnetic flux through the loop changes significantly — i.e., when inductors are not lumped or at high frequencies where the distributed transmission line model (Ch. 12 §12.7) must be used.

18.1.2 — Nodal Analysis

Apply KCL to each non-reference node. Express all currents in terms of node voltages using Ohm's law:

$$I = \frac{V_a - V_b}{R} = (V_a - V_b)G$$

For n nodes (one chosen as reference/ground), this gives $(n-1)$ equations in $(n-1)$ unknowns. In matrix form:

$$\mathbf{G}\mathbf{v} = \mathbf{i}_s$$

where $\mathbf{G}$ is the **conductance matrix** (nodal admittance matrix):

- $G_{kk} = \sum$ (all conductances connected to node k)
- $G_{jk} = G_{kj} = -\sum$ (all conductances between nodes j and k)

Nodal analysis with impedances (AC): Replace G with admittance $Y = 1/Z$ and work in phasors.

18.1.3 — Mesh Analysis

Apply KVL to each independent mesh (closed loop). Express all voltages using Ohm's law with mesh currents. For m independent meshes:

$$\mathbf{Z}\mathbf{i}_{mesh} = \mathbf{v}_s$$

where $\mathbf{Z}$ is the **impedance matrix** (mesh impedance matrix):

- $Z_{kk} = \sum$ (all impedances in mesh k)
- $Z_{jk} = Z_{kj} = -\sum$ (all impedances shared between meshes j and k)

When to use which: Nodal analysis works for any topology; mesh analysis is easier when there are more nodes than meshes or for planar circuits. For systematic computer-aided analysis: modified nodal analysis (MNA) handles voltage sources, op-amps, and other elements naturally.

18.1.4 — Superposition and Linearity

From the linearity of Maxwell's equations (Ch. 11): the response of a linear circuit to multiple independent sources equals the sum of responses to each source individually (all others killed: voltage sources → short circuits, current sources → open circuits).

Superposition fails for: nonlinear elements (diodes, transistors operating in nonlinear region), circuits with dependent sources that depend on the full combined response.

18.1.5 — Thévenin and Norton: Engineering Application

From Ch. 17 §17.8:

Finding V_{Th}: Remove the load; calculate the open-circuit voltage at the terminals.

Finding R_{Th} (or Z_{Th} in AC): Kill all independent sources (voltage sources → short, current sources → open); calculate the resistance (or impedance) seen at the terminals.

Systematic method (with dependent sources): Apply a test voltage V_t, calculate the resulting current $I_t \to R_{Th} = V_t / I_t$.

Maximum power transfer:

$$R_L = R_{Th} \quad \Longrightarrow \quad P_{max} = \frac{V_{Th}^2}{4R_{Th}}$$

Engineering use: Every measurement instrument (voltmeter, oscilloscope, current probe) has an input impedance. The Thévenin/Norton framework tells you exactly how much the instrument loads the circuit being measured.

18.2 — AC Analysis: Phasors, Impedance, Power

18.2.1 — Phasor Representation

For a linear circuit driven by sinusoidal sources at frequency ω, all voltages and currents are sinusoidal at the same frequency (by superposition and linearity):

$$v(t) = |V|\cos(\omega t + \phi_V) = \mathrm{Re}[\underbrace{|V|e^{j\phi_V}}_{\tilde{V}} \cdot e^{j\omega t}]$$

The **phasor** $\tilde{V} = |V|e^{j\phi_V} = V_R + jV_I$ is a complex number encoding amplitude and phase. All circuit analysis reduces to complex algebra:

- KCL: $\sum_k \tilde{I}_k = 0$ (complex equation, equivalent to 2 real equations)
- KVL: $\sum_k \tilde{V}_k = 0$
- Ohm's law: $\tilde{V} = Z(\omega)\tilde{I}$ (complex impedance)

18.2.2 — Complex Power

For phasors $\tilde{V}$ and $\tilde{I}$:

$$\tilde{S} = \frac{1}{2}\tilde{V}\tilde{I}^* = P + jQ_{reac}$$

- **Active power** $P = \frac{1}{2}|\tilde{V}||\tilde{I}|\cos\phi$ [W] — time-averaged power dissipated
- **Reactive power** $Q_{reac} = \frac{1}{2}|\tilde{V}||\tilde{I}|\sin\phi$ [VAr] — power oscillating between source and reactive elements
- **Apparent power** $|\tilde{S}|$ [VA] — magnitude of complex power
- **Power factor** $\text{PF} = \cos\phi = P/|\tilde{S}|$

Power factor correction: An inductive load (motor, transformer) has $\phi > 0$ (lagging current), $Q_{reac} > 0$. Adding shunt capacitance $C_{pf} = Q_{reac}/\omega V_{rms}^2$ brings $\phi \to 0$, reducing the current the utility must supply for the same useful power.

18.2.3 — Resonance

Series RLC (from Ch. 17 §17.7.4):

$$Z(\omega) = R + j\omega L + \frac{1}{j\omega C} = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

At resonance $\omega = \omega_0 = 1/\sqrt{LC}$: $\text{Im}[Z] = 0$, $|Z| = R$ (minimum). Maximum current flows; voltage across R equals source voltage.

Voltage magnification across L or C: $|V_L| = |V_C| = Q \cdot |V_{source}|$

Parallel RLC: Dual of series. At resonance: maximum impedance $= R_p$, minimum current from source. Used in bandpass filters, tank circuits.

18.3 — Semiconductor Devices: Physics to Circuit Models

18.3.1 — The p-n Junction Diode

From Ch. 14 §14.5.4 (drift-diffusion) and Ch. 6 §6.6 (doping):

IV characteristic:

$$I_D = I_0\left(e^{V_D/\eta V_T} - 1\right)$$

where I_0 is the reverse saturation current, $\eta = 1$–2 is the ideality factor (1 for diffusion-dominated, 2 for recombination-dominated), and $V_T = k_B T/e \approx 26$ mV at 300 K.

Operating regions:

Region	Condition	I_D	Application
Forward bias	$V_D > 0$	Exponentially increasing	Rectification, LED, laser
Reverse bias	$V_D < 0$	$\approx -I_0$ (tiny)	Blocking
Breakdown	$V_D < -V_{BR}$	Large (controlled for Zener)	Voltage reference, protection

Small-signal model (for AC analysis around DC bias point I_{DQ}):

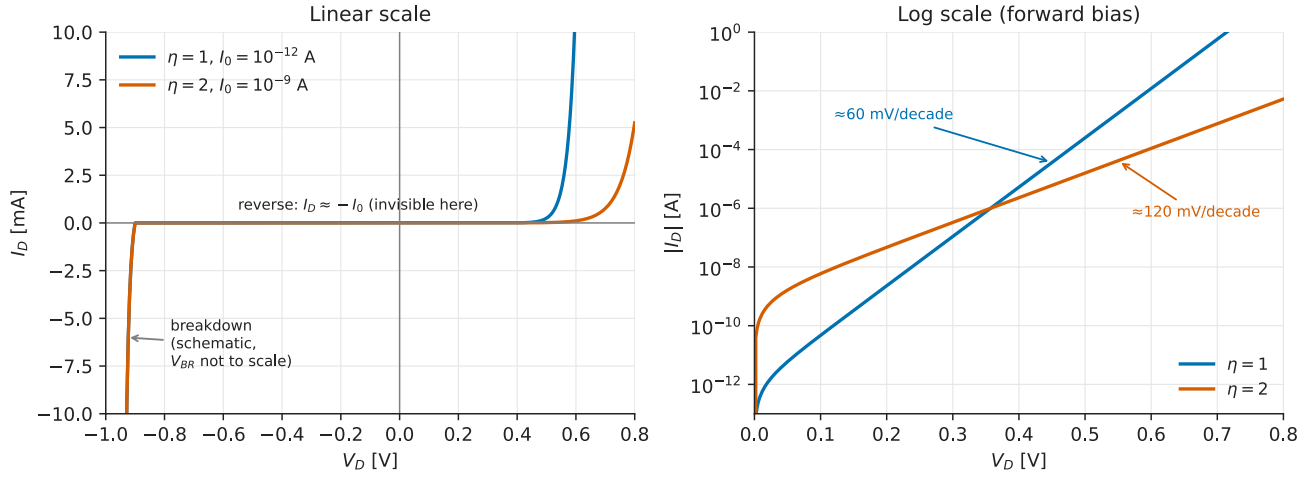
$$r_d = \left.\frac{\partial V_D}{\partial I_D}\right|_{I_{DQ}} = \frac{\eta V_T}{I_{DQ}}$$

The diode becomes a resistor r_d for small AC signals. At $I_{DQ} = 1$ mA: $r_d = 26\,\Omega$ (η=1). At 10 mA: $r_d = 2.6\,\Omega$.

Junction capacitance: The depletion region is a capacitor:

$$C_j = C_{j0}\left(1 - \frac{V_D}{\phi_B}\right)^{-m}$$

where $\phi_B \approx 0.7$ V is the built-in potential and $m \approx 0.33$–0.5. Reverse bias increases depletion width → decreases C_j (varactor diode, used in voltage-controlled oscillators and RF tuners).



$I_D = I_0(e^{V_D/\eta V_T} - 1)$ at 300 K for $\eta = 1$ ($I_0 = 10^{-12}$ A) and $\eta = 2$ ($I_0 = 10^{-9}$ A): the linear panel shows the forward "turn-on" knee, the negligible reverse current and a schematic breakdown knee (not to scale), while the log panel shows straight lines of slope $\eta V_T \ln 10 \approx 60$ and 120 mV per decade.

18.3.2 — The Bipolar Junction Transistor (BJT)

An NPN BJT is two back-to-back p-n junctions (Emitter-Base-Collector).

Active region operation ($V_{BE} \approx 0.7$ V, $V_{CE} > V_{CE,sat}$):

$$I_C = \beta I_B = I_S e^{V_{BE}/V_T}$$

where $\beta = h_{FE}$ is the current gain (50–500 typical) and I_S is the saturation current (from the drift-diffusion physics, Ch. 14 §14.5.3).

Ebers-Moll model (physics-based): connects directly to Ch. 14 minority carrier diffusion — the collector current is the minority carrier current injected from the emitter base junction and collected at the collector junction.

Hybrid-π small-signal model:



Key parameters:

- $g_m = I_C/V_T$ (transconductance, from diode equation)
- $r_\pi = \beta/g_m = \beta V_T/I_C$ (input resistance)
- $r_o = V_A/I_C$ (Early voltage)
- $C_\pi + C_\mu$ (junction capacitances, from depletion region physics)

Common-emitter amplifier:

$$A_v = -g_m R_C \approx -\frac{R_C}{r_d} = -\frac{I_C R_C}{V_T}$$

Voltage gain is proportional to $I_C R_C/V_T$ — the ratio of the DC voltage drop across R_C to the thermal voltage. At $I_C R_C = 5$ V: $A_v = -192$.

18.3.3 — The MOSFET

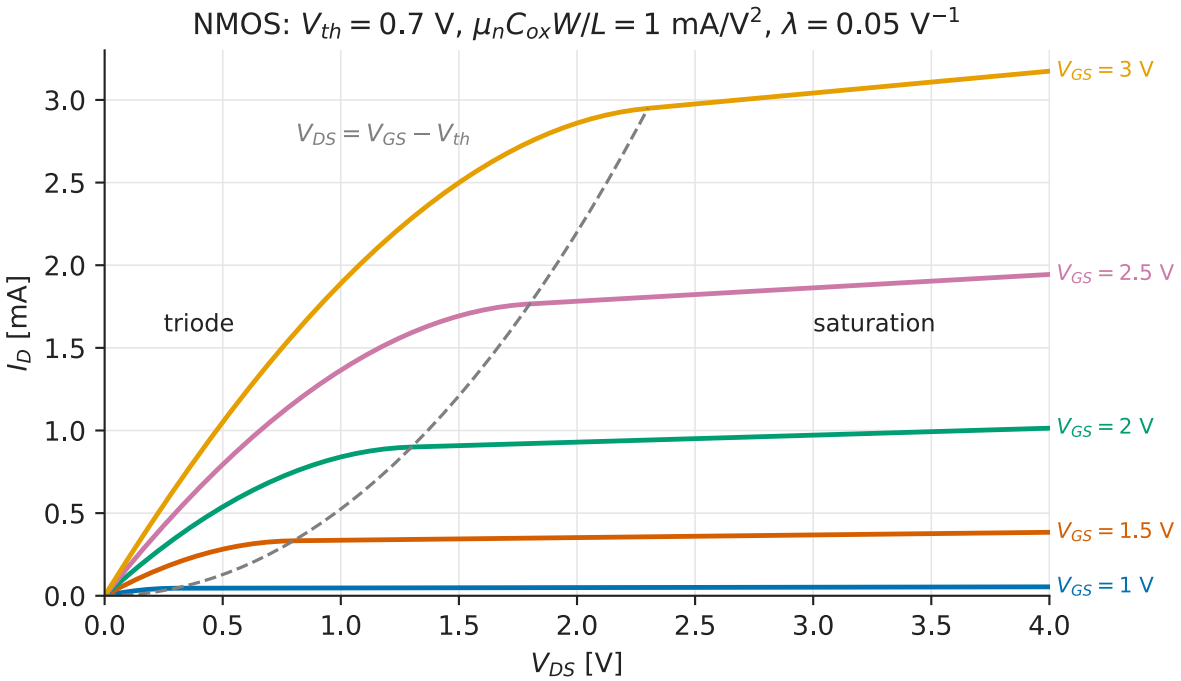
An enhancement-mode NMOS transistor has a p-type substrate with an oxide layer (gate dielectric) and n+ source/drain regions. Applying $V_{GS} > V_{th}$ inverts the surface, creating an n-type channel (inversion layer, from Ch. 6 §6.6 Fermi level above conduction band edge at the surface).

Saturation region ($V_{GS} > V_{th}$, $V_{DS} > V_{GS} - V_{th}$):

$$I_D = \frac{\mu_n C_{ox} W}{2L}(V_{GS} - V_{th})^2 (1 + \lambda V_{DS})$$

where $C_{ox} = \epsilon_{ox}/t_{ox}$ is the gate oxide capacitance per area (from t_{ox}: oxide thickness), W/L is the width-to-length ratio, and λ is the channel length modulation parameter.

Triode region ($V_{GS} > V_{th}$, $V_{DS} < V_{GS} - V_{th}$): $I_D = \mu_n C_{ox}\frac{W}{L}\left[(V_{GS} - V_{th})V_{DS} - \frac{1}{2}V_{DS}^2\right]$, which meets the saturation expression at $V_{DS} = V_{GS} - V_{th}$.



Drain current I_D versus V_{DS} for $V_{GS} = 1$–3 V with $V_{th} = 0.7$ V, $\mu_n C_{ox} W/L = 1$ mA/V² and $\lambda = 0.05$ V⁻¹ (the $(1 + \lambda V_{DS})$ factor applied in both regions for continuity); the dashed parabola $V_{DS} = V_{GS} - V_{th}$ separates triode from saturation, and the saturated current grows as $(V_{GS} - V_{th})^2$ with a slight λ slope.

Connection to physics: $\mu_n C_{ox}$ traces directly to:

- μ_n: electron mobility in the inversion layer (from scattering, Ch. 6 §6.4)
- C_{ox}: parallel-plate capacitance of gate oxide (ϵ_{ox}/t_{ox})
- V_{th}: threshold voltage from Fermi level alignment (Ch. 6 §6.5.2)

Small-signal model:

$$g_m = \frac{\partial I_D}{\partial V_{GS}} = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D} = \frac{2I_D}{V_{GS} - V_{th}}$$

Common-source amplifier:

$$A_v = -g_m R_D$$

CMOS inverter (complementary NMOS + PMOS):

- Input HIGH ($V_{in} = V_{DD}$): NMOS ON, PMOS OFF → output LOW
- Input LOW ($V_{in} = 0$): NMOS OFF, PMOS ON → output HIGH
- Static power dissipation: near zero (only leakage current)
- Dynamic power dissipation: $P_{dyn} = \alpha C_L V_{DD}^2 f$ (switching only) where α is the activity factor, C_L is the load capacitance

Scaling (Moore's Law physics): As L decreases, $\mu_n C_{ox}/L$ increases → faster switching. But: short-channel effects, gate leakage (tunneling through thin oxide, Ch. 4 §4.4), drain-induced barrier lowering (DIBL) — all quantum effects that break the simple model. This is where Layer-1 physics (Ch. 6, 7) re-enters EEE practice.

18.4 — Operational Amplifiers

18.4.1 — The Ideal Op-Amp

A differential amplifier with:

- Open-loop gain: $A_{OL} \to \infty$
- Input impedance: $R_{in} \to \infty$
- Output impedance: $R_{out} \to 0$
- Infinite bandwidth (for ideal)

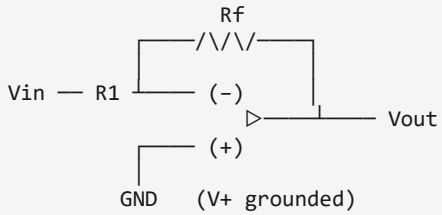
The two golden rules (valid with negative feedback):

1. $V_+ = V_-$ (virtual short: the feedback keeps the input differential voltage negligible)
2. $I_+ = I_- = 0$ (no current into the input terminals)

These rules follow from $A_{OL} \to \infty$ and negative feedback — they are not approximations when both conditions hold.

18.4.2 — Standard Op-Amp Configurations

Inverting amplifier:



Applying golden rules: $V_- = V_+ = 0$ (virtual ground); $I_{Rf} = I_{R1}$:

$$\frac{0 - V_{out}}{R_f} = \frac{V_{in} - 0}{R_1} \quad \Longrightarrow \quad \boxed{A_v = -\frac{R_f}{R_1}}$$

Non-inverting amplifier: $A_v = 1 + R_f/R_1$

Voltage follower: $A_v = 1$ (Rf = 0, R1 = ∞); presents high-Z input to source, low-Z to load — the buffer.

Summing amplifier:

$$V_{out} = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \cdots \right)$$

Integrator (C replaces Rf):

$$V_{out} = -\frac{1}{RC} \int V_{in}\, dt \quad \Longrightarrow \quad H(s) = -\frac{1}{sRC}$$

Differentiator (C replaces R1):

$$V_{out} = -RC \frac{dV_{in}}{dt} \quad \Longrightarrow \quad H(s) = -sRC$$

These directly implement the integration/differentiation operations from Ch. 16 (PDE solutions) and Ch. 21 (PID control).

18.4.3 — Real Op-Amp Limitations

Gain-bandwidth product (GBW): For a single-pole op-amp:

$$A_{OL}(f) = \frac{A_{DC}}{1 + jf/f_p} \quad \Longrightarrow \quad |A_v|_{closed} \cdot B_{3dB} = \text{GBW}$$

Closed-loop gain × bandwidth = constant. Gain of 100 → bandwidth = GBW/100. Typical GBW: 1 MHz (general purpose), 100 MHz (high speed), GHz (RF op-amps).

Slew rate: Maximum dV_{out}/dt (limits large-signal performance): For $V_{out} = V_{pk} \sin(2\pi f t)$: maximum $dV/dt = 2\pi f V_{pk}$. Full power bandwidth: $f_{FP} = SR/(2\pi V_{pk})$.

Offset voltage V_{OS}: Small DC voltage at input (mismatch in input pair); amplified by closed-loop gain. Nulled by external trim or digital calibration.

18.5 — Filters: Frequency-Selective Systems

18.5.1 — Filter Specifications

From Ch. 17 §17.6.2: the transfer function $H(s)$ characterizes any linear system. A filter is a system designed to have a specific $|H(j\omega)|$ shape:

- **Low-pass (LPF):** Pass $\omega < \omega_c$, attenuate $\omega > \omega_c$
- **High-pass (HPF):** Pass $\omega > \omega_c$, attenuate $\omega < \omega_c$
- **Band-pass (BPF):** Pass $\omega_1 < \omega < \omega_2$
- **Band-stop (BSF/Notch):** Block $\omega_1 < \omega < \omega_2$

Specifications:

- Passband ripple: how much $|H|$ varies in the passband
- Stopband attenuation: minimum rejection in the stopband
- Transition band: slope of the transition from pass to stop

18.5.2 — First-Order RC Filter

The simplest LPF (RC with output across C):

$$H(s) = \frac{1/sC}{R + 1/sC} = \frac{1}{1 + sRC} = \frac{1}{1 + s/\omega_c}$$

where $\omega_c = 1/RC$ is the -3 dB cutoff frequency.

Bode plot: Flat at 0 dB below ω_c; rolls off at -20 dB/decade above. Phase: -45° at ω_c; approaches -90° asymptotically.

Physical origin: The RC time constant $\tau = RC$ is the thermal RC from Ch. 14 §14.7.3 in the electrical domain. The same equation, the same time constant, the same first-order step response $V_C(t) = V_{in}(1 - e^{-t/RC})$.

18.5.3 — Classical Filter Approximations

Butterworth (maximally flat magnitude):

$$|H_n(j\omega)|^2 = \frac{1}{1 + (\omega/\omega_c)^{2n}}$$

- Monotonically decreasing $|H|$; no ripple anywhere
- $-20n$ dB/decade rolloff in stopband
- Poles equally spaced on left half-circle of radius ω_c

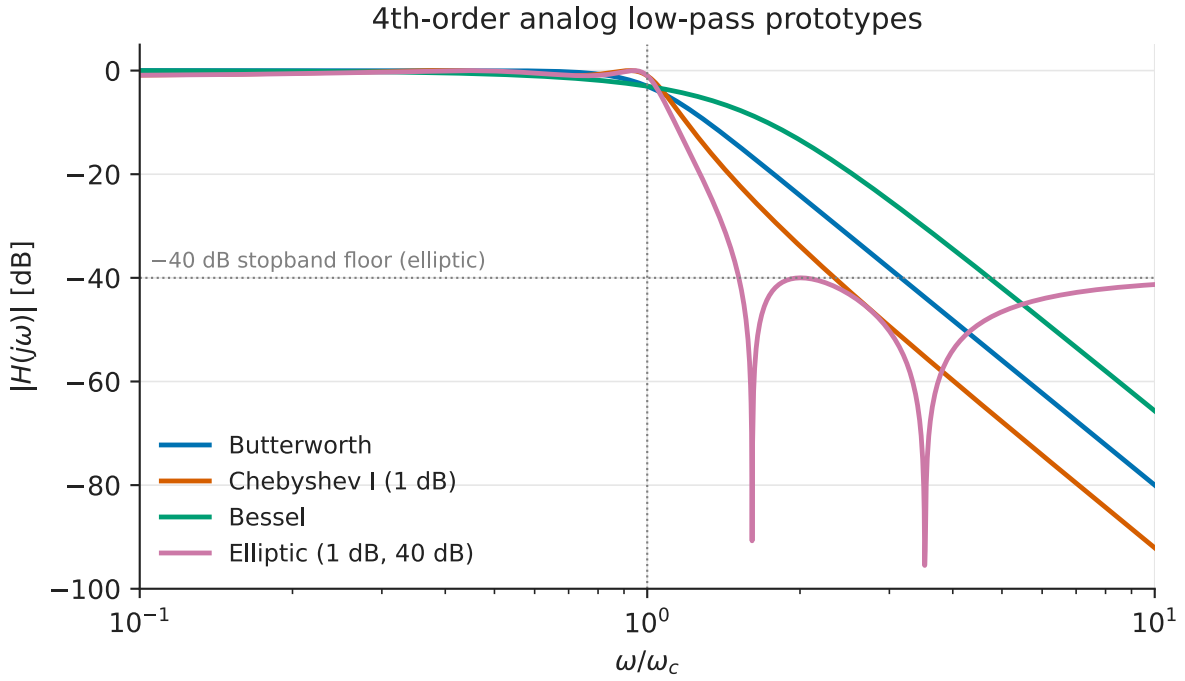
Chebyshev Type I (equiripple passband):

- Steeper rolloff than Butterworth for same order
- Ripple δ_p dB in passband; monotonic in stopband
- Better selectivity; worse transient response (phase nonlinearity)

Bessel (maximally flat group delay):

- Group delay $\tau_g = -d\angle H/d\omega$ = const in passband
- Preserves pulse shape (no dispersion)
- Poorest amplitude selectivity
- Used in audio, radar pulse shaping, medical waveform preservation

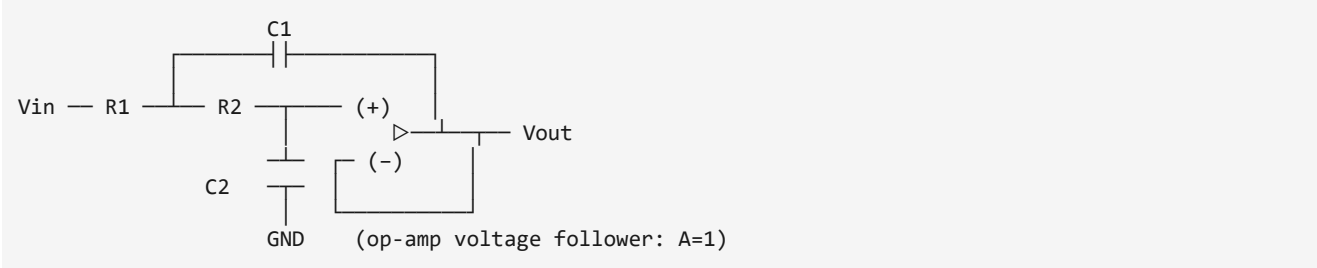
Elliptic (Cauer): Ripple in both passband and stopband; most efficient (narrowest transition band for given order) — used when component count matters.



$|H(j\omega)|$ in dB versus ω/ω_c for fourth-order analog prototypes: Butterworth and Bessel (-3 dB at ω_c), Chebyshev I (1 dB ripple) and elliptic (1 dB ripple, 40 dB stopband); notice the trade from Bessel (gentlest roll-off) through Butterworth and Chebyshev to the elliptic's sharp edge, bought with stopband nulls and a 40 dB floor.

18.5.4 — Active Filters: Op-Amp Implementation

Sallen-Key second-order LPF:



Transfer function:

$$H(s) = \frac{\omega_0^2}{s^2 + (\omega_0/Q)s + \omega_0^2}$$

with $\omega_0 = 1/\sqrt{R_1 R_2 C_1 C_2}$ ($= 1/RC$ for equal components) and $Q = \sqrt{R_1 R_2 C_1 C_2}/[C_2(R_1 + R_2)]$ adjustable by varying component values. No inductors needed — inductors are bulky, lossy, and non-integrable at audio and signal processing frequencies. Op-amp active filters replace them with R, C, and

controlled sources.

Higher-order filters: Cascade second-order sections. An n-th order Butterworth is cascaded from $n/2$ Sallen-Key sections.

Connection to Ch. 16: A filter is a frequency-selective wave propagation device — it implements a chosen dispersion relation $\omega(k)$ in the frequency domain. The poles of $H(s)$ are the complex natural frequencies of the system — exactly the eigenfrequencies from Ch. 16 §16.5.

18.6 — Fourier Analysis: Signals in Frequency Domain

18.6.1 — Fourier Series

A periodic signal $x(t)$ with period T_0 (fundamental frequency $f_0 = 1/T_0$):

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}, \qquad c_n = \frac{1}{T_0}\int_0^{T_0} x(t) e^{-jn\omega_0 t}\, dt$$

Connection to Ch. 16 §16.5: The Fourier series is the eigenfunction expansion for the 1D wave equation with periodic boundary conditions. The modes $e^{jn\omega_0 t}$ are eigenfunctions of d/dt, and the expansion is the same superposition as in Ch. 16 §16.5.2.

Parseval's theorem (power conservation):

$$\frac{1}{T_0}\int_0^{T_0} |x(t)|^2\, dt = \sum_{n=-\infty}^{\infty} |c_n|^2$$

Average power in time = sum of powers at each harmonic.

18.6.2 — Fourier Transform

For aperiodic signals:

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft}\, dt, \qquad x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi ft}\, df$$

Key transform pairs:

$x(t)$	$X(f)$	**Engineering use**
$\delta(t)$	1	Impulse response, white spectrum
1	$\delta(f)$	DC signal
$e^{-t/\tau}u(t)$	$\tau/(1+j2\pi f\tau)$	RC filter step response
$\text{rect}(t/T)$	$T\,\text{sinc}(fT)$	Ideal pulse, brick-wall filter
$e^{-\pi t^2}$	$e^{-\pi f^2}$	Gaussian pulse (minimum time-bandwidth)
$\cos(2\pi f_0 t)$	$\frac{1}{2}[\delta(f-f_0)+\delta(f+f_0)]$	Single tone

Convolution theorem:

$$y(t) = h(t) * x(t) \quad \Longleftrightarrow \quad Y(f) = H(f)\cdot X(f)$$

Convolution in time = multiplication in frequency. The output of any LTI system is the input spectrum multiplied by the transfer function — the entire filtering operation reduces to pointwise multiplication after the Fourier transform.

Time-bandwidth product: $\Delta t \cdot \Delta f \geq 1/(4\pi)$ (Heisenberg uncertainty principle, Ch. 3 §3.11, applied to signals instead of wavefunctions). A short pulse has a wide spectrum; a narrowband signal lasts long in time. This is the fundamental trade-off in every communications and radar system.

18.6.3 — Laplace Transform and System Analysis

The bilateral Laplace transform:

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st}\, dt, \quad s = \sigma + j\omega$$

For causal systems (unilateral Laplace, $t \geq 0$): generalizes the Fourier transform to include exponentially growing/decaying signals.

Key transform pairs:

$x(t)$	$X(s)$
$\delta(t)$	1
$u(t)$ (step)	$1/s$
$e^{-at}u(t)$	$1/(s+a)$
$\sin(\omega_0 t)u(t)$	$\omega_0/(s^2+\omega_0^2)$
$t^n e^{-at}u(t)$	$n!/(s+a)^{n+1}$

Transfer function from the circuit: Replace each element with its Laplace impedance ($Z_R = R$, $Z_C = 1/sC$, $Z_L = sL$), apply KVL/KCL in the s-domain, and solve algebraically.

Poles and stability:

- All poles in the left half-plane ($\text{Re}[s_k] < 0$): stable (response decays)
- Simple (non-repeated) poles on the imaginary axis: marginally stable (sustained oscillation) — but not BIBO stable: a bounded input at the pole frequency produces an unbounded response
- Any pole in the right half-plane: unstable (growing response)

Initial and final value theorems:

$$x(0^+) = \lim_{s\to\infty} sX(s), \qquad x(\infty) = \lim_{s\to 0} sX(s)$$

The final value theorem holds only if all poles of $sX(s)$ lie in the open left half-plane (i.e. $x(t)$ actually settles); otherwise the limit can return a finite number for a signal that oscillates or diverges. Useful for checking step response endpoints without full inverse Laplace.

18.7 — Sampling and Digital Signals

18.7.1 — The Nyquist-Shannon Sampling Theorem

Statement: A continuous signal $x(t)$ bandlimited to f_{max} Hz can be perfectly reconstructed from samples taken at rate $f_s \geq 2f_{max}$ (the Nyquist rate).

Derivation from Fourier theory: Sampling at interval $T_s = 1/f_s$ multiplies the signal by a train of impulses $\sum_n \delta(t - nT_s)$. In the frequency domain, this convolution creates periodic copies of $X(f)$ centered at multiples of f_s. If $f_s > 2f_{max}$, the copies do not overlap → perfect reconstruction by a lowpass filter.

Aliasing: If $f_s < 2f_{max}$, the copies overlap and frequency components above $f_s/2$ are "aliased" — they appear as false lower-frequency components. An anti-aliasing filter (LPF at $f_{max} = f_s/2$) before the ADC prevents this.

Connection to Ch. 16: The Nyquist criterion is the sampling version of the spatial Nyquist criterion in optics ($\theta_{min} = \lambda/D$, Ch. 12 §12.6.2) — both limit how finely a signal can be resolved given a finite aperture or sampling rate.

18.7.2 — Analog-to-Digital and Digital-to-Analog Conversion

ADC quantization: Mapping a continuous amplitude to 2^N discrete levels.

- **Resolution:** Voltage per bit = $V_{FSR}/2^N$
- **Dynamic range:** DR $= 20\log_{10}(2^N) = 6.02N$ dB
- **SNR (ideal ADC):** SNR $= 6.02N + 1.76$ dB (for sinusoidal input)

For 16-bit audio: DR = 96 dB (vs. human hearing ~120 dB dynamic range → 20-bit ideal). For 12-bit industrial ADC: DR = 72 dB.

ADC architectures: Flash (fastest, 2^N comparators), successive approximation (SAR, most common at moderate speeds), sigma-delta ($\Delta\Sigma$, highest resolution via oversampling + noise shaping).

18.7.3 — Discrete Fourier Transform (DFT) and FFT

The DFT of N samples $x[n]$:

$$X[k] = \sum_{n=0}^{N-1} x[n], e^{-j2\pi kn/N}, \qquad k = 0, 1, \ldots, N-1$$

Direct computation: $O(N^2)$ operations. **Fast Fourier Transform (FFT):** Cooley-Tukey algorithm exploiting the twiddle factor symmetry → $O(N \log_2 N)$ operations.

For $N = 10^6$: DFT needs 10^{12} operations; FFT needs 2×10^7 — 50,000× faster. The FFT is one of the most important algorithms in engineering computing.

Frequency resolution: $\Delta f = f_s/N$ (Hz per bin). To resolve two frequencies Δf_{sep}, need $N \geq f_s/\Delta f_{sep}$ samples.

18.8 — Power Electronics

18.8.1 — Rectifiers

Half-wave rectifier: One diode; passes only positive half-cycles. Average output: $V_{avg} = V_{pk}/\pi$. Large ripple.

Full-wave bridge rectifier: Four diodes; both half-cycles used. Average output: $V_{avg} = 2V_{pk}/\pi$.

Capacitor filter: A capacitor C across the output holds the voltage between charging pulses. Ripple voltage (approximate):

$$V_{ripple} \approx \frac{I_{load}}{2fC}$$

For $I_{load} = 100$ mA, $f = 60$ Hz, $C = 1000\,\mu$F: $V_{ripple} \approx 0.83$ V. Larger C → smaller ripple → larger peak current demand.

18.8.2 — Switching Converters

DC-DC converters use switching (transistor on/off at high frequency f_{sw}) plus energy storage (L and C) to efficiently change voltage levels.

Buck converter (step-down):

When switch ON: current through inductor ramps up; when OFF: inductor maintains current through diode. In steady state:

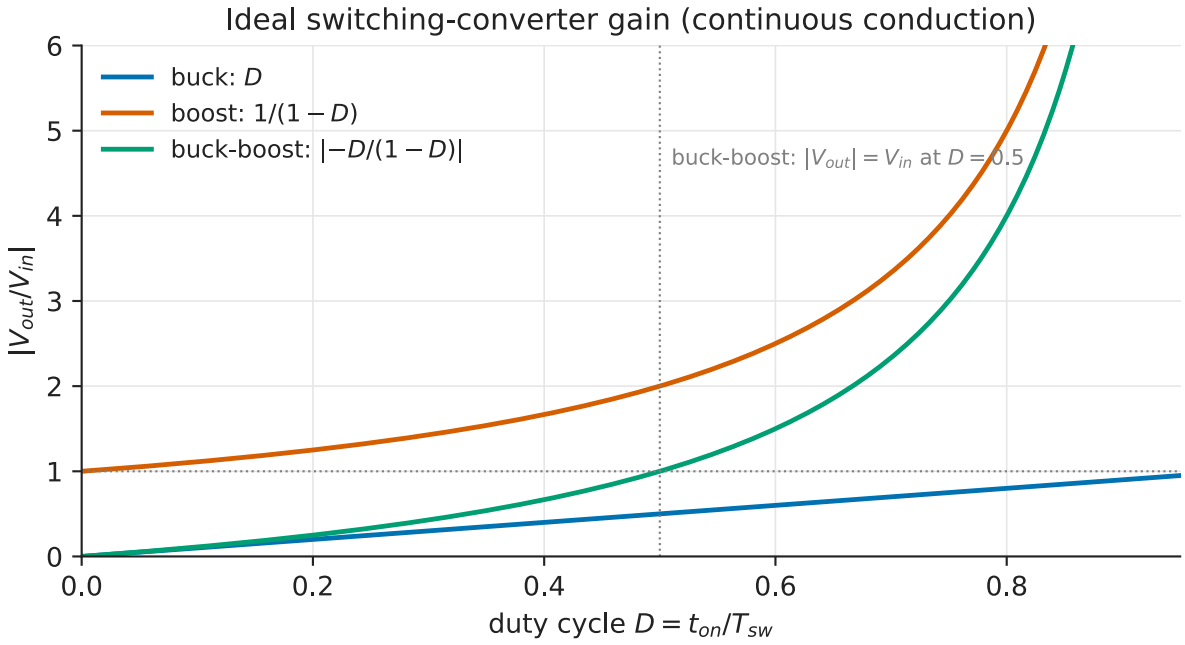
$$V_{out} = D \cdot V_{in}, \qquad I_{in} = D \cdot I_{out}, \qquad P_{in} = P_{out}$$

where $D = t_{on}/T_{sw}$ is the duty cycle. Efficiency: 90–99% (vs. linear regulator: $\eta = V_{out}/V_{in}$, as low as 20% for large step-down).

Connection to physics: The energy stored in the inductor during ON time ($W = \frac{1}{2}LI^2$, from Ch. 17 §17.9.1) is released during OFF time. The inductor "stores" kinetic energy of current, exactly as a flywheel stores mechanical kinetic energy. Faraday's law (Ch. 11) governs the inductor voltage.

Boost converter (step-up): $V_{out} = V_{in}/(1 - D)$

Buck-boost: $V_{out} = -DV_{in}/(1 - D)$ (inverts polarity)



Ideal continuous-conduction gain $|V_{out}/V_{in}|$ versus duty cycle D for the buck (D), boost ($1/(1-D)$) and buck-boost ($|D/(1-D)|$) converters; note the buck-boost crosses unity at $D = 0.5$ and both step-up gains diverge as $D \to 1$, where real losses cap them.

Switching frequency trade-off:

- Higher f_{sw} → smaller L, C → smaller converter
- Higher f_{sw} → more switching losses → lower efficiency
- Typical: 100 kHz–10 MHz for power ICs; 20–100 kHz for large converters

Engineering applications:

- Laptop power adapters (AC→DC, then DC-DC regulation)
- Electric vehicle battery management (bidirectional DC-DC for cell balancing)
- Solar inverters (DC-DC boost + DC-AC inverter)
- LED drivers (constant-current buck converter)

18.9 — Where EEE Layer 3 Fails

18.9.1 — High-Frequency Effects

At frequencies where the lumping criterion $Lf/v_{em} \ll 1$ fails:

- **Parasitic inductance and capacitance** of component leads and PCB traces dominate
- **Transmission line effects** — signal reflections on PCB traces longer than $\lambda/10$ (at 1 GHz: $\lambda = 30$ cm; $\lambda/10 = 3$ cm — every long trace is a transmission line)
- **Radiation** — antennas radiate; circuits become radiators (EMC problems)
- **Skin effect** — current concentrates in thin surface layer, increasing effective resistance

At these frequencies, full Maxwell's equations (Ch. 11–12) or distributed models (Ch. 12 §12.7, Ch. 17 §17.10) are required.

18.9.2 — Quantum Device Limits

As transistor gate lengths approach $\sim$ 3–5 nm:

- **Gate oxide tunneling** (Ch. 4 §4.4): $\sim$ 1 nm SiO_2 gates have unacceptable leakage → replaced by high-κ dielectrics (HfO_2)
- **Ballistic transport** (Ch. 13 §13.11.1): mean free path exceeds channel length → Landauer formula, not Drude
- **Random dopant fluctuations**: at 5 nm gate lengths, the depletion region may contain 0 or 1 dopant atoms → statistical variation in V_{th}
- **Quantum confinement**: 2D electron gas at the interface behaves quantum mechanically → need Ch. 6–7 band theory, not bulk MOSFET model

The simple MOSFET equations from §18.3.3 break down. The physicist who derived them from Ch. 6 physics will know which Layer-1 effects are responsible and what the corrected model should be.

18.9.3 — Topological and Quantum Devices

Some devices are inherently beyond Layer-3 description:

- **Josephson junction** (Ch. 7 §7.8): superconducting quantum switch; the current-phase relation $I = I_c \sin(\Delta\phi)$ cannot be captured by any R, C, L model
- **Quantum dot single-electron transistor** (Ch. 7 §7.9): Coulomb blockade gives a conductance that oscillates with gate voltage — no DC operating point
- **Topological qubit** (Ch. 7 §7.4): information encoded in Majorana modes, non-local and topologically protected — no equivalent classical circuit

These devices require returning to Layer 1 (Chs. 6–7) for design.

18.10 — Summary

Layer-3 concept	Physical origin	Chapter
KCL $\sum I = 0$	Noether U(1) / charge conservation	Ch. 1, 11, 17
KVL $\sum V = 0$	Faraday's law, quasi-static	Ch. 11, 17
Complex impedance $Z(\omega)$	Lumped L, R, C from Wave/diffusion PDE	Ch. 16, 17
Diode $I = I_0(e^{V/V_T} - 1)$	p-n junction drift-diffusion	Ch. 6, 14
MOSFET $I_D \propto (V_{GS} - V_{th})^2$	Inversion layer, surface charge	Ch. 6
Op-amp golden rules	$A_{OL} \to \infty$ + negative feedback	Ch. 17 (Thévenin)
Filter $H(s)$ poles	Natural frequencies of LC network	Ch. 16, 17
Butterworth $\lvert H\rvert^2 = 1/(1 + (\omega/\omega_c)^{2n})$	Optimal flat passband	Ch. 17 (2nd-order)
Fourier transform $X(f) = \mathcal{F}x(t)$	Eigenfunction expansion	Ch. 16 §16.5
Nyquist $f_s \geq 2f_{max}$	Spectral aliasing / Fourier sampling	Ch. 16
Buck converter $V_{out} = DV_{in}$	Faraday (inductor) + KVL	Ch. 11, 17
GBW product	Single-pole op-amp rolloff	Ch. 17 (Bode)

Layer-3 concept	Physical origin	Chapter
ADC dynamic range $6.02N$ dB	2^N quantization levels	Ch. 10 (information)

18.11 — Engineering Thread

Concept	Engineering system
KVL/KCL + Thévenin	All circuit simulation (SPICE, LTspice, Cadence)
Phasors + power factor	Grid-connected inverters, motor drives, utility metering
Diode model	Rectifiers; photodetectors; LED drivers; solar cell characterization
BJT small-signal	RF amplifier design; audio amplifiers; discrete analog design
MOSFET saturation	Digital logic (CMOS); analog VLSI; power switching
Op-amp golden rules	Instrumentation amplifiers; ADC input stages; active filters
Butterworth/Chebyshev	Anti-aliasing filters; audio equalizers; IF filters in receivers
Fourier/FFT	Spectrum analyzers; software-defined radio; vibration analysis
Nyquist + ADC specs	Data acquisition system design; audio codec selection
Buck/boost converters	Power management ICs in phones, laptops, EVs
Laplace poles/zeros	Control system design (Ch. 21); stability analysis; loop shaping

18.12 — Looking Ahead

Chapter 18 established EEE as the electrical instantiation of the Layer-3 template from Ch. 17. The same template — different labels — drives Chapters 19 and 20.

Chapter 19 covers **ME Systems**: the mechanical and thermal instantiation. Mass replaces inductance; spring compliance replaces capacitance; damping replaces resistance. The same ω_0, ζ, Q, Bode plots, and Thévenin equivalents appear, now governing vibrating structures, rotating machinery, and thermofluid systems.

Chapter 20 covers **CE and ChE Systems**: the hydraulic, structural, and chemical process instantiations.

Chapter 21 then closes the book with feedback and control — the branch-agnostic mathematics that sits above all domains and applies equally to every engineering system.

End of Chapter 18.

Next: Chapter 19 — ME Systems: Machines, Structures, and Thermofluid Systems

CHAPTER 19

ME Systems: Machines, Structures, and Thermofluid Systems

The Mechanical and Thermal Instantiation of Layer 3

> *All mechanical engineering reduces to three questions: Where does the energy come from? Where does it go? How fast? The effort-flow template answers all three, in every sub-domain.*

19.0 — Overview

Chapter 18 established EEE as the electrical instantiation of the Layer-3 template: effort = voltage, flow = current, $R/C/L$ = resistor/capacitor/inductor.

This chapter does the same for **mechanical engineering**. The template (Ch. 17) applies to translational mechanics, rotational mechanics, thermodynamic cycles, heat transfer, and fluid systems — each with its own effort-flow pair, its own R/C/L elements, and its own KCL-type and KVL-type conservation laws.

Physics connections:

Ch. 19 concept	Physics origin
Mass-spring-damper ODE	Navier equation (Ch. 15), lumped by Bridge C (Ch. 17)
Thermodynamic cycles	Four laws of thermodynamics (Ch. 10)
Heat transfer modes	Fourier's law + radiation (Ch. 13, 10)
Pipe flow, pump curves	Hagen-Poiseuille (Ch. 15), hydraulic R/C/L (Ch. 17)
Acoustic systems	Wave equation (Ch. 15–16)
Vibration isolation	Second-order transfer function (Ch. 17 §17.7.4)
Shaft critical speed	Rotor dynamics = mass-spring eigenfrequency
Heat exchanger NTU	Thermal circuit + dimensionless groups (Ch. 13–14)

19.1 — Translational Mechanical Systems

19.1.1 — The Effort-Flow Pair

From Ch. 17 §17.2.2:

- **Effort:** Force F [N]
- **Flow:** Velocity v [m/s]
- **Power:** $P = Fv$ [W]

- **Generalized momentum:** $p = \int F\,dt$ [N·s] = linear momentum
- **Generalized displacement:** $x = \int v\,dt$ [m]

The three passive elements:

Element	Symbol	Constitutive law	Energy	Physical meaning
Mass m	$L_{mech} = m$	$F = m\dot{v} = m\ddot{x}$	$\frac{1}{2}mv^2$	Kinetic energy storage (inertia)
Spring k	$C_{mech} = 1/k$	$F = kx$, $v = \dot{x} = \frac{1}{k}\dot{F}$	$\frac{1}{2}kx^2$	Potential energy storage
Damper b	$R_{mech} = b$	$F = bv$	$P_{diss} = bv^2$	Viscous dissipation

19.1.2 — The Single-DOF Mass-Spring-Damper

Newton's second law for a mass m on a spring k with damping b, driven by force $F_{ext}(t)$:

$$\boxed{m\ddot{x} + b\dot{x} + kx = F_{ext}(t)}$$

This is exactly the universal second-order ODE from Ch. 17 §17.7.1, with: $L = m$, $R = b$, $C = 1/k$, $q = x$, $e_{source} = F_{ext}$.

Natural frequency and damping:

$$\omega_0 = \sqrt{\frac{k}{m}}, \qquad \zeta = \frac{b}{2\sqrt{km}} = \frac{b}{2m\omega_0}, \qquad Q = \frac{\sqrt{km}}{b} = \frac{m\omega_0}{b}$$

Common engineering systems and their ω_0:

System	ω_0 (rad/s)	f_0 (Hz)
Automotive suspension	6–12	1–2
Building floor (occupied)	13–20	2–3
Typical machine tool	60–600	10–100
MEMS accelerometer proof mass	10^4–10^5	10^3–10^4
Quartz crystal resonator	10^7–10^8	10^6–10^7
Atomic force microscope cantilever	10^4–10^6	10^3–10^5

19.1.3 — Vibration Isolation

Problem: A rotating machine on a rigid floor vibrates at frequency ω. Isolate the floor from the vibration force.

The transmissibility $T(\omega)$ is the ratio of force transmitted to the floor to the force generated by the machine:

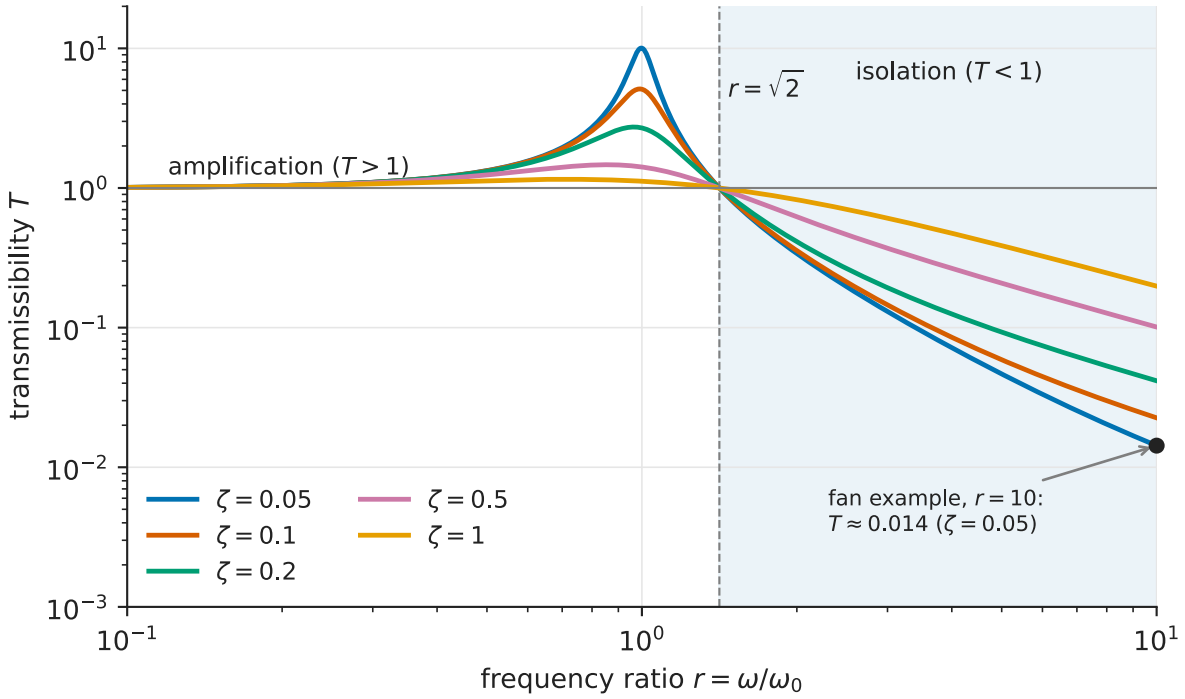
$$T(\omega) = \left|\frac{F_{transmitted}}{F_{machine}}\right| = \sqrt{\frac{1 + (2\zeta r)^2}{(1 - r^2)^2 + (2\zeta r)^2}}$$

where $r = \omega/\omega_0$.

Key result: For $r > \sqrt{2}$, $T < 1$ (isolation achieved). For $r < \sqrt{2}$, $T > 1$ (amplification).

Design rule: Mount the machine on springs such that $\omega_0 < \omega_{machine}/\sqrt{2}$. For a fan at 3000 rpm ($\omega = 314$ rad/s): need $\omega_0 < 222$ rad/s → $f_0 < 35$ Hz. For $f_0 = 5$ Hz → $r = 314/31.4 = 10$ → $T \approx 0.01$ (99% isolation).

Damping trade-off: High ζ reduces resonance peak during startup (when machine passes through ω_0) but worsens isolation at operating frequency. Optimal: $\zeta \approx 0.05$–0.15 for continuous rotating machinery.



Transmissibility $T(r)$ on log–log axes for $\zeta = 0.05, 0.1, 0.2, 0.5, 1$: every curve passes through $T = 1$ at $r = \sqrt{2}$, damping tames the resonance peak but raises T throughout the isolation region (shaded), and the fan example at $r = 10$ gives $T \approx 0.014$ at $\zeta = 0.05$ (0.010 undamped).

19.1.4 — Base Excitation

When the base (floor/ground) moves with displacement $y(t) = Y\cos\omega t$, the absolute displacement of the mass x and relative displacement $z = x - y$:

$$m\ddot{z} + b\dot{z} + kz = -m\ddot{y} = m\omega^2 Y\cos\omega t$$

Transmissibility (displacement ratio $|X/Y|$): same formula as force transmissibility.

Engineering: Seismic isolation of buildings and precision instruments. A building on base isolators with $f_0 = 0.3$ Hz ($\omega_0 = 1.9$ rad/s): at earthquake frequency $f = 1$–10 Hz ($r = 3$–30): $T \approx 0.001$–0.1.

19.2 — Rotational Mechanical Systems

19.2.1 — The Rotational Effort-Flow Pair

- **Effort:** Torque τ [N·m]
- **Flow:** Angular velocity ω [rad/s]
- **Power:** $P = \tau\omega$ [W]

Elements:

Element	Law	Energy	Example
Moment of inertia I_r	$\tau = I_r\dot{\omega}$	$\frac{1}{2}I_r\omega^2$	Flywheel, rotor, disk
Torsional spring k_r	$\tau = k_r\theta$	$\frac{1}{2}k_r\theta^2$	Shaft, coupling
Rotational damper b_r	$\tau = b_r\omega$	$P = b_r\omega^2$	Bearing friction, fluid damping

Governing equation for a single-DOF torsional system:

$$I_r\ddot{\theta} + b_r\dot{\theta} + k_r\theta = \tau_{ext}(t)$$

Same second-order template; $\omega_0 = \sqrt{k_r/I_r}$, $\zeta = b_r/2\sqrt{k_r I_r}$.

19.2.2 — Gear Trains as Transformers

A gear pair with N_1 and N_2 teeth is a **mechanical transformer** (TF in bond graph):

$$\frac{\tau_2}{\tau_1} = \frac{N_2}{N_1} = n, \qquad \frac{\omega_1}{\omega_2} = \frac{N_2}{N_1} = n$$

Power conserved (ideal gear): $\tau_1\omega_1 = \tau_2\omega_2$.

Reflected inertia: A load inertia I_{load} on the output shaft appears at the input shaft as I_{load}/n^2 — the gear ratio squared. This is the mechanical analog of the transformer's impedance transformation:

$$Z_1 = n^2 Z_2 \quad \text{(electrical)} \qquad I_{1,eff} = I_{load}/n^2 \quad \text{(mechanical)}$$

Gear efficiency: Real gears have friction losses. Efficiency varies from $\eta_{spur} \approx 0.97$–0.99 per stage (helical gears); worm gears can be as low as $\eta \approx 0.4$–0.7 (high reduction ratio but lossy).

19.2.3 — Shaft Critical Speed

A rotating shaft with an out-of-balance mass m_u at eccentricity e_u from the geometric center behaves as a forced harmonic oscillator. At the **critical speed** $\omega_c = \sqrt{k_{shaft}/m_{rotor}}$, the shaft deflects resonantly — potentially to failure.

For a simply-supported shaft with a central disk of mass m on a shaft of stiffness $k = 48EI/L^3$ (from beam theory):

$$\omega_c = \sqrt{\frac{48EI}{mL^3}}$$

Design rules:

- Operating speed well above or below ω_c (separation factor > 1.3)
- Supercritical design (above ω_c): cross through critical speed quickly during startup; add damping to limit amplitude at ω_c
- Used in: turbines, centrifuges, grinding spindles

Flywheel energy storage: $E = \frac{1}{2}I_r\omega^2 = \frac{1}{2}mr_{gyr}^2\omega^2$. For a steel disk ($r = 0.5$ m, $m = 100$ kg) at $\omega = 3000$ rpm: $I_r = \frac{1}{2}(100)(0.5)^2 = 12.5$ kg·m²; $E = \frac{1}{2}(12.5)(314)^2 = 616$ kJ.

19.3 — Multi-DOF Systems and Modal Analysis

19.3.1 — The Equations of Motion

For N masses with displacements $\mathbf{q} = (q_1, \ldots, q_N)^T$, the equations of motion in matrix form (from Ch. 9 §9.8 normal modes):

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{F}(t)$$

where $\mathbf{M}$, $\mathbf{C}$, $\mathbf{K}$ are the mass, damping, and stiffness matrices.

Free undamped vibration ($\mathbf{C} = 0$, $\mathbf{F} = 0$):

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{0}$$

Try $\mathbf{q} = \mathbf{v}\, e^{i\omega t}$ → generalized eigenvalue problem:

$$(\mathbf{K} - \omega^2 \mathbf{M})\mathbf{v} = \mathbf{0}$$

The N eigenvalues ω_n^2 are the natural frequencies squared; the N eigenvectors $\mathbf{v}^{(n)}$ are the **mode shapes**.

19.3.2 — Modal Decomposition

Transform to modal coordinates $\mathbf{q} = \mathbf{\Phi}\boldsymbol{\eta}$ where $\mathbf{\Phi} = [\mathbf{v}^{(1)} | \cdots | \mathbf{v}^{(N)}]$:

$$\ddot{\eta}_n + 2\zeta_n \omega_n \dot{\eta}_n + \omega_n^2 \eta_n = f_n(t)/m_n^*$$

Each mode is an independent single-DOF oscillator — with its own ω_n, ζ_n, and modal mass $m_n^* = \mathbf{v}^{(n)T}\mathbf{M}\mathbf{v}^{(n)}$.

This is the engineering application of the eigenfunction expansion from Ch. 16 §16.5: the normal mode decomposition is the same operation whether applied to a lumped mass-spring system (this chapter) or a continuous elastic body (Ch. 15–16).

19.3.3 — Vibration Absorber (Frahm Dynamic Absorber)

Add a small mass-spring subsystem (m_a, k_a) tuned to the excitation frequency ω_{exc}: set $\omega_a = \sqrt{k_a/m_a} = \omega_{exc}$.

At $\omega = \omega_a$: the absorber mass oscillates 180° out of phase, creating a force that exactly cancels the excitation on the main mass. The main mass becomes perfectly stationary.

Design: $k_a = m_a \omega_{exc}^2$. Typical mass ratio $\mu = m_a/m_{main} = 0.05$–$0.20$.

Applications: Building sway dampers (Taipei 101: 660-tonne pendulum absorber at $f_0 = 0.15$ Hz); bridges (Millennium Bridge London: TMDs after pedestrian resonance event); machining chatter suppression.

19.4 — Power Transmission and Machine Elements

19.4.1 — Power Flow in Mechanical Transmissions

Any power transmission chain: prime mover (engine/motor) → transmission (gears, belt, chain) → load (pump, generator, wheel).

Power balance: $P_{output} = \eta_{total}\, P_{input}$

Efficiency chain: For n elements: $\eta_{total} = \eta_1 \times \eta_2 \times \cdots \times \eta_n$

Typical efficiencies:

- Helical gear stage: 98–99%
- V-belt: 93–97%
- Chain drive: 97–99%
- Worm gear: 40–95% (depends on lead angle)
- Ball screw (rotation→translation): 90–95%

A drivetrain with five elements at 97% each: $\eta_{total} = 0.97^5 = 0.86$ — 14% loss.

19.4.2 — The Ball Screw as a Transformer

A ball screw converts rotation (effort = torque τ, flow = ω) to translation (effort = force F, flow = velocity v):

$$v = \frac{p}{2\pi}\omega, \qquad F = \frac{2\pi\tau\eta}{p}$$

where p is the lead (m/revolution) and η is the efficiency.

This is a **mechanical transformer** with ratio $n = p/2\pi$ [m/rad]: $e_2 = e_1/n$ (force from torque), $f_2 = n \cdot f_1$ (velocity from angular velocity).

Application: CNC machine axes, industrial robots, hydraulic press actuators. Typical ball screw: lead = 5–20 mm/rev, efficiency 90–95%, dynamic load rating determines life (L10 life from Hertz contact stress at ball-raceway interface).

19.5 — Thermodynamic Cycles in Engineering

19.5.1 — The Framework: Carnot and Real Cycles

From Ch. 10 §10.6: any heat engine operating between T_H and T_C:

$$\eta_{Carnot} = 1 - \frac{T_C}{T_H}$$

All real cycles are less efficient. The gap measures irreversibility from: friction, heat transfer across finite temperature differences, incomplete combustion, non-quasi-static processes.

The thermodynamic cycle in effort-flow language: The working fluid carries energy. Effort = pressure P (or temperature T for heat exchange), flow = volume flow rate (or entropy flow $\dot{S} = \dot{Q}/T$). The cycle traces a closed path in the P-v (pressure-volume) or T-s (temperature-entropy) diagram.

19.5.2 — Rankine Cycle (Steam Power)

The workhorse of electrical power generation (coal, nuclear, concentrated solar):

Process	Device	Change
1→2	Pump	Isentropic compression of liquid
2→3	Boiler	Constant-pressure heat addition (evaporation + superheat)
3→4	Turbine	Isentropic expansion (work output)
4→1	Condenser	Constant-pressure heat rejection (condensation)

Thermal efficiency:

$$\eta_{Rankine} = \frac{W_{turbine} - W_{pump}}{Q_{boiler}} = 1 - \frac{Q_{condenser}}{Q_{boiler}}$$

Improvements:

- **Superheat:** Raise T_3 → higher η, drier steam in turbine
- **Reheat:** Expand partway, reheat, expand again → higher T_{avg} in cycle
- **Regeneration:** Extract steam from turbine to preheat feedwater → less heat needed from boiler

Modern supercritical coal plant: $T_H = 600^\circ\text{C}$, $T_C = 40^\circ\text{C}$, $\eta_{Carnot} = 64\%$, actual $\eta \approx 40$–47%.

19.5.3 — Brayton Cycle (Gas Turbine, Jet Engine)

Process	Device	
1→2	Compressor	Isentropic compression
2→3	Combustor	Constant pressure heat addition
3→4	Turbine	Isentropic expansion
4→1	Exhaust	Constant pressure heat rejection

Efficiency as a function of pressure ratio $r_p = P_2/P_1$:

$$\boxed{\eta_{Brayton} = 1 - \frac{1}{r_p^{(\gamma-1)/\gamma}}}$$

For $r_p = 30$, $\gamma = 1.4$: $\eta = 62\%$ (ideal). Actual: 35–45%.

Why gas turbines can achieve high T_H: Turbine inlet temperature (TIT) now exceeds 1700°C in modern aero-engines — well above the melting point of the blade alloy. Possible through: thermal barrier coatings, internal cooling channels with compressed air, single-crystal Ni superalloy blades (no grain boundaries → no creep at high temperature). This is materials physics (Ch. 6 bonding + diffusion, Ch. 13 Arrhenius) in direct engineering service.

19.5.4 — Otto and Diesel Cycles (Internal Combustion)

Otto cycle (gasoline, spark ignition):

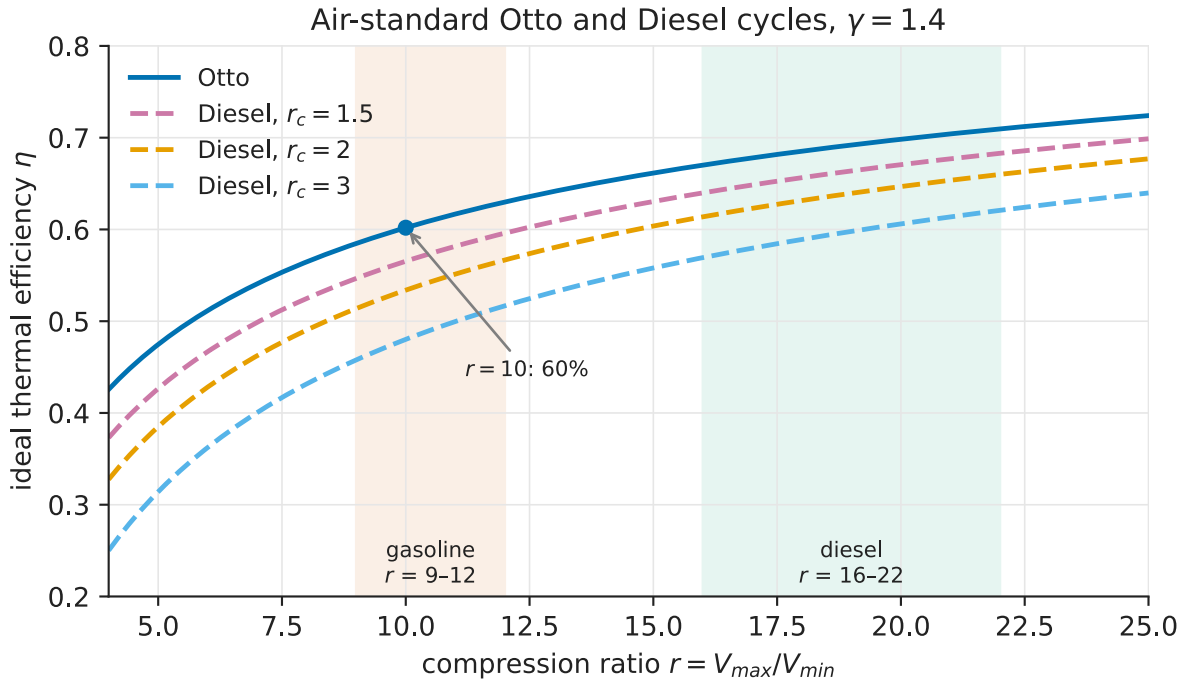
$$\boxed{\eta_{Otto} = 1 - \frac{1}{r^{\gamma-1}}}$$

where $r = V_{max}/V_{min}$ is the compression ratio. For $r = 10$, $\gamma = 1.4$: $\eta_{Otto} = 60\%$ (ideal). Actual: 25–35%.

Diesel cycle (compression ignition, constant-pressure combustion):

$$\eta_{Diesel} = 1 - \frac{1}{r^{\gamma-1}} \cdot \frac{r_c^{\gamma} - 1}{\gamma(r_c - 1)}$$

where $r_c = V_3/V_2$ is the cutoff ratio. For $r_c <$ corresponding Otto ratio: $\eta_{Diesel} < \eta_{Otto}$ at same compression ratio — but diesel engines run at higher r (16–22 vs. 9–12 for gasoline) → typically higher actual η.



Air-standard efficiency versus compression ratio for $\gamma = 1.4$: Otto $1 - r^{1-\gamma}$ (60% at $r = 10$) and Diesel for cutoff ratios $r_c = 1.5, 2, 3$, with typical gasoline (r = 9–12) and diesel (r = 16–22) ranges shaded; at equal r the Diesel is always below the Otto, but a diesel at $r \approx 18$ still beats a gasoline engine at $r \approx 10$.

19.5.5 — Vapor Compression Refrigeration Cycle

Inverse of the Rankine cycle: work input pumps heat from cold to hot.

COP (Coefficient of Performance):

$$COP_{refrigerator} = \frac{Q_C}{W_{compressor}} = \frac{T_C}{T_H - T_C}$$

(Carnot limit)

For a room air conditioner ($T_C = 20^\circ$C, $T_H = 35^\circ$C): $COP_{Carnot} = 293/15 = 19.5$. Actual: 3–6.

Heat pump COP:

$$COP_{heat\,pump} = \frac{Q_H}{W_{compressor}} = COP_{refrigerator} + 1$$

A heat pump with $COP = 4$ delivers 4 kW of heat for 1 kW of electrical input — 4× more efficient than a direct electric heater. Governed by the same Carnot limit as everything else.

Refrigerants: The thermodynamic cycle efficiency depends on the refrigerant's T-s diagram shape (specific heat, latent heat, saturation curve). HFCs (R-134a, R-410A): high GWP → replacing with low-GWP alternatives (R-32, R-1234yf, CO_2).

19.6 — Heat Transfer in Engineering Practice

19.6.1 — The Three Modes and Their Resistances

From Ch. 13 §13.5 and Ch. 14 §14.7, all three heat transfer modes have thermal resistance representations:

Conduction:

$$\dot{Q} = \frac{\kappa A}{L}\Delta T = \frac{\Delta T}{R_{cond}}, \qquad R_{cond} = \frac{L}{\kappa A}$$

Convection:

$$\dot{Q} = hA(T_{wall} - T_{fluid}) = \frac{\Delta T}{R_{conv}}, \qquad R_{conv} = \frac{1}{hA}$$

The convection coefficient h is not a material property — it depends on flow conditions (Re, Pr) and must be found from correlations.

Radiation:

$$\dot{Q} = \sigma\epsilon A(T_1^4 - T_2^4) \approx h_{rad}A(T_1 - T_2)$$

where $h_{rad} \approx 4\sigma\epsilon T_m^3$ linearized around the mean temperature T_m. For $T_m = 350$ K, $\epsilon = 0.9$: $h_{rad} \approx 7.5$ W/m²K — comparable to natural convection.

19.6.2 — Dimensionless Numbers for Convection

Reynolds number $Re = \rho v L/\mu$ (from Ch. 15 §15.8.1): determines flow regime (laminar/turbulent) in the convecting fluid.

Prandtl number $Pr = \mu c_p/\kappa = \nu/\alpha$: ratio of momentum diffusivity to thermal diffusivity.

- $Pr \ll 1$ (liquid metals): thermal diffusion much faster than momentum → wide thermal BL
- $Pr \approx 1$ (gases): thermal and velocity BLs similar thickness
- $Pr \gg 1$ (oils, viscous fluids): thermal BL much thinner than velocity BL

Nusselt number $Nu = hL/\kappa$: ratio of convective to conductive heat transfer. All convection correlations express Nu as a function of Re and Pr:

Dittus-Boelter (turbulent pipe flow, $Re > 10^4$, $0.6 < Pr < 160$):

$$Nu_D = 0.023\, Re_D^{0.8}\, Pr^n, \qquad n = 0.4 \text{ (heating)}\ \ 0.3 \text{ (cooling)}$$

Churchill-Chu (natural convection on vertical plate):

$$Nu_L = \left[0.825 + \frac{0.387\, Ra_L^{1/6}}{[1 + (0.492/Pr)^{9/16}]^{8/27}}\right]^2$$

where $Ra = Gr \cdot Pr = g\beta \Delta T L^3/\nu\alpha$ is the Rayleigh number.

Biot number (from Ch. 16 §16.10.1): $Bi = hL/\kappa_{solid}$

$Bi \ll 1$ (≤ 0.1): temperature uniform in solid → lumped capacitance valid $Bi \gg 1$: significant temperature gradient → solve heat equation

19.6.3 — Heat Exchangers: LMTD and NTU Methods

A heat exchanger transfers heat from a hot fluid (inlet $T_{H,in}$, outlet $T_{H,out}$) to a cold fluid (inlet $T_{C,in}$, outlet $T_{C,out}$).

Overall heat transfer coefficient U:

$$\frac{1}{UA} = \frac{1}{h_H A_H} + \frac{\ln(r_o/r_i)}{2\pi\kappa L} + R_{fouling} + \frac{1}{h_C A_C}$$

(thermal resistances in series: hot-side convection + wall conduction + fouling + cold-side convection)

LMTD method (when inlet and outlet temperatures are known):

$$\dot{Q} = UA \cdot \text{LMTD}, \qquad \text{LMTD} = \frac{(T_{H,in} - T_{C,out}) - (T_{H,out} - T_{C,in})}{\ln\frac{T_{H,in} - T_{C,out}}{T_{H,out} - T_{C,in}}}$$

NTU-effectiveness method (when outlet temperatures are unknown — design problem):

$$\varepsilon = \frac{\dot{Q}}{\dot{Q}_{max}} = \frac{\dot{Q}}{C_{min}(T_{H,in} - T_{C,in})}$$
$$NTU = \frac{UA}{C_{min}}, \qquad C_r = \frac{C_{min}}{C_{max}}$$

For a counterflow heat exchanger:

$$\varepsilon = \frac{1 - e^{-NTU(1-C_r)}}{1 - C_r e^{-NTU(1-C_r)}} \quad (C_r < 1)$$

Engineering: A shell-and-tube heat exchanger designed with $NTU = 2$, $C_r = 0.5$ achieves $\varepsilon \approx 0.77$ (77% of maximum possible heat transfer). The NTU-effectiveness framework connects directly to the thermal circuit (R-network) of Ch. 14 — $NTU = UA/C_{min}$ is the ratio of thermal conductance to heat capacity flow rate.

19.7 — Fluid Systems as Hydraulic Circuits

19.7.1 — The Hydraulic Effort-Flow Pair

From Ch. 17 §17.2.2 and Ch. 15:

- **Effort:** Pressure P [Pa] (or pressure difference ΔP)
- **Flow:** Volume flow rate Q [m³/s]
- **Power:** $P = \Delta P \cdot Q$ [W]

Hydraulic R/C/L elements:

Element	Law	Physical
Pipe resistance R_{hyd}	$\Delta P = R_{hyd} Q$	Hagen-Poiseuille: $R = 8\mu L/\pi r^4$
Accumulator C_{hyd}	$Q = C_{hyd}\dot{P}$	Fluid compressibility or elastic vessel: $C = V/\rho c_s^2$
Fluid inertance L_{hyd}	$\Delta P = L_{hyd}\dot{Q}$	Accelerating/decelerating fluid column: $L = \rho L/A$

KCL analog (mass conservation): $\sum Q_k = 0$ at every junction. **KVL analog (energy/pressure):** $\sum \Delta P_k = 0$ around every loop (modified by elevation: $\Delta P + \rho g \Delta h = 0$).

19.7.2 — Pipe Network Analysis

Pipe networks (water distribution, HVAC, hydraulic machinery) are analyzed exactly as electrical resistor networks:

1. Apply KCL (flow balance) at each junction node
2. Apply KVL (pressure balance) around each loop: $\sum_k \Delta P_k = 0$ where $\Delta P_k = R_k Q_k$ for each pipe
3. **Nonlinearity:** For turbulent flow, $\Delta P = f L \rho Q^2/(2DA^2) \propto Q^2$ (Darcy-Weisbach); the resistance is not constant → iterative solution (Hardy Cross method)

Minor losses: Fittings, valves, bends add equivalent length $L_{eq} = KD/f$:

Fitting	K
Gate valve (fully open)	0.1
Globe valve (fully open)	6–10
90° elbow (standard)	0.9
90° elbow (long radius)	0.4
T-junction (branch)	1.5

19.7.3 — Pump and System Curves

A pump's **characteristic curve** $H(Q)$ (head vs. flow rate) and a pipe system's **system curve** $H_{sys}(Q)$ (pressure drop vs. flow) intersect at the operating point.

Pump curve: $H_{pump} = H_0 - AQ^2$ (centrifugal pump, parabolic)

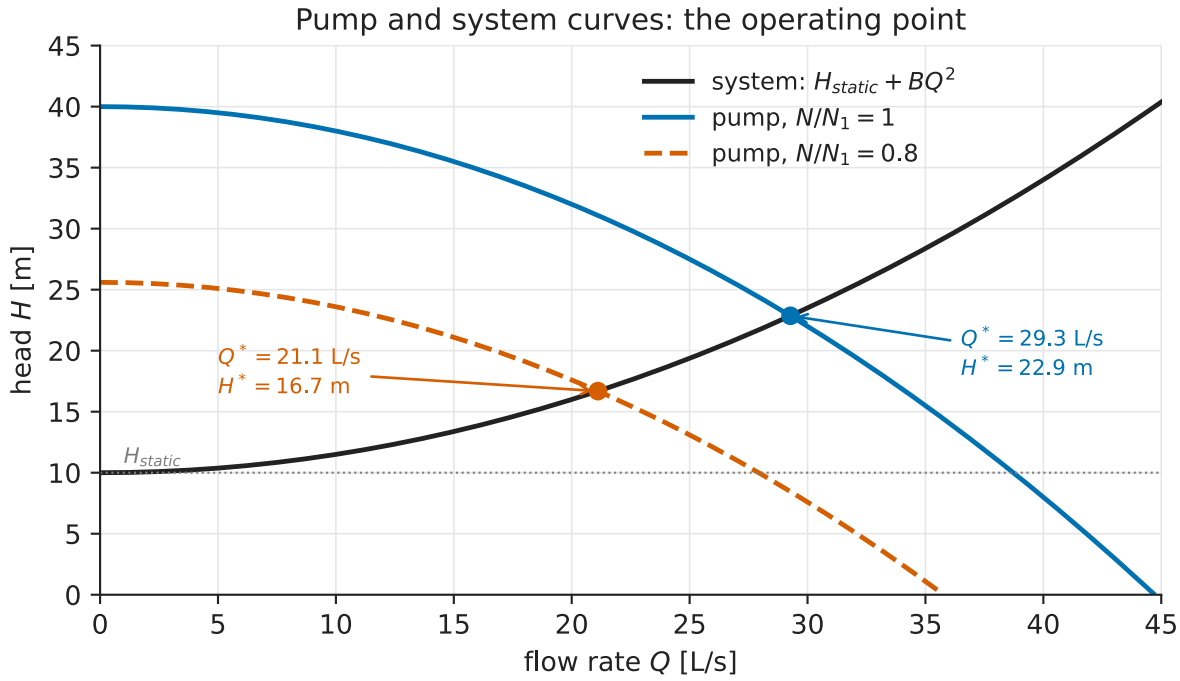
System curve: $H_{sys} = H_{static} + BQ^2$ (static head + friction losses)

Operating point: $H_{pump} = H_{sys} \rightarrow$ solve for Q^*, H^*.

Pump similarity laws (for geometrically similar pumps of different size or speed):

$$\frac{Q_2}{Q_1} = \frac{N_2}{N_1}\left(\frac{D_2}{D_1}\right)^3, \qquad \frac{H_2}{H_1} = \left(\frac{N_2}{N_1}\right)^2\left(\frac{D_2}{D_1}\right)^2, \qquad \frac{P_2}{P_1} = \left(\frac{N_2}{N_1}\right)^3\left(\frac{D_2}{D_1}\right)^5$$

These follow from dimensional analysis (Ch. 15 §15.8) — the Reynolds similarity and Froude similarity conditions applied to turbomachinery.



Illustrative pump curve $H = H_0 - AQ^2$ ($H_0 = 40$ m, $A = 0.020$ m/(L/s)²) against the system curve $H = H_{static} + BQ^2$ ($H_{static} = 10$ m, $B = 0.015$ m/(L/s)²), with the same pump slowed to $N/N_1 = 0.8$ by the affinity laws ($H_0 \rightarrow 0.64H_0$); because of the static head, a 20% speed cut drops the operating flow by about 28%, not 20%.

19.8 — Acoustics: Sound and Vibration

19.8.1 — Sound as a Wave

From Ch. 15–16: acoustic pressure p' satisfies:

$$\frac{\partial^2 p'}{\partial t^2} = c_s^2 \nabla^2 p'$$

with $c_s = \sqrt{\gamma P_0/\rho_0}$ (speed of sound in a gas).

Sound pressure level (SPL):

$$L_p = 20\log_{10}\left(\frac{p_{rms}}{p_{ref}}\right) \text{ dB}, \qquad p_{ref} = 20\ \mu\text{Pa}$$

Source	SPL (dB)
Threshold of hearing	0
Library	30–40
Conversation	60
Factory floor	80–95
Threshold of pain	130
Jet engine (1 m)	150

Acoustic impedance $Z_{ac} = \rho c_s$: analogous to electromagnetic wave impedance $Z_{EM} = \sqrt{\mu/\epsilon}$ (Ch. 11 §11.8). Reflection at interfaces from impedance mismatch — the same Fresnel formula as optics (Ch. 12 §12.3.3).

19.8.2 — Room Acoustics: Reverberation

A room of volume V and total sound absorption A_{room} (Sabins):

$$RT_{60} = 0.161 \frac{V}{A_{room}}$$

Sabine formula for reverberation time (time for sound to decay 60 dB): $A_{room} = \sum S_i \alpha_i$ where S_i is the area and α_i the absorption coefficient of each surface.

Absorption coefficients α at 500 Hz (typical):

- Concrete: 0.02 (nearly perfect reflector)
- Carpeted floor: 0.30
- Upholstered seats: 0.60
- Acoustic foam: 0.80–0.99

Optimal RT60: Concert halls: 1.8–2.2 s; speech/lecture: 0.5–1.0 s; recording studios: 0.2–0.5 s.

19.8.3 — Vibration Isolation vs. Noise Control

Vibration isolator (mechanical): see §19.1.3. Targets: prevent machine vibration from radiating as sound, or prevent ground vibration from reaching sensitive equipment.

Active noise cancellation: An additional sound source (loudspeaker) generates the anti-phase signal to cancel the primary noise at the observer's ears. Works for correlated, low-frequency noise (up to $\sim$ 500 Hz):

- Aircraft cabin: 8–12 dB reduction in engine noise
- In-ear headphones (ANC): 20–30 dB reduction of low-frequency ambient noise
- Requires real-time digital signal processing (DSP, from Ch. 18) with feed-forward or feedback control (Ch. 21 preview)

19.9 — Thermofluid Systems as Coupled Networks

19.9.1 — Multi-Domain Bond Graph: HVAC

An HVAC system couples hydraulic (refrigerant circuit), thermal (heat transfer), and mechanical (compressor, fans) domains:

Refrigerant circuit (hydraulic domain):

- Compressor: effort source (pressure head) driving refrigerant flow
- Condenser: hydraulic resistance + thermal interface to air
- Expansion valve: controlled variable resistance (throttle)
- Evaporator: hydraulic resistance + thermal interface to air

Thermal domain:

- Condenser: TF (transformer) coupling hydraulic to thermal: $\dot{Q} = \dot{m}_{ref} \Delta h_{cond}$
- Evaporator: TF coupling hydraulic to thermal: $\dot{Q}_{cooling} = \dot{m}_{ref} \Delta h_{evap}$

The coupling: the refrigerant mass flow rate $\dot{m}$ connects the hydraulic circuit power ($\dot{m}\Delta P$) to the thermal circuit power ($\dot{m}\Delta h$) through enthalpy change Δh at each heat exchanger.

The bond graph for the complete HVAC system has the same mathematical structure as a multi-domain electrical-mechanical coupled system — the same junction rules, the same power conservation, the same transfer function analysis.

`[STRUCTURAL CONNECTION]` — **the junction algebra is shared; the constitutive relations are not, and this is where HVAC gets hard.** The bond graph deliberately contains no physics: it says only that power is conserved at a junction and that effort/flow are conjugate. Everything domain-specific lives in the effort-flow relations attached to each element, and in an HVAC loop those relations are strongly coupled and strongly nonlinear:

Element	Realistic relation	Why the linear template is insufficient
Compressor	$\dot{W} = f(\dot{m}, P_{in}, P_{out})$	power rises steeply with pressure ratio; efficiency varies with operating point
Condenser / evaporator	$Q = UA\,\Delta T_{lm}$	UA changes with refrigerant phase and fouling; ΔT_{lm} is logarithmic
Expansion valve	$P_{in} \to P_{out}$ at constant h	throttle is genuinely nonlinear and is the loop's dominant nonlinearity
Refrigerant	two-phase Δh varies with quality	latent heat dominates; small quality changes move Δh a lot

So the *topology* is transferable and the *element values* are not. An HVAC control loop tuned as a linear second-order system will mispredict exactly where the interesting behaviour is: turndown, cycling, and frost recovery all live in the regions where the linear model breaks. Ch. 21 §21.0.2 treats this as the canonical case of a nonlinear loop needing a different representation.

Model Ledger — lumped HVAC system to a linear transfer function

Field	Content
Parent theory	Bond-graph power conservation (Ch. 17 §17.5); refrigerant thermodynamics (Ch. 10, Ch. 19 §19.5)
Reduction	Fixed operating point; linearise each effort-flow relation; $Bi \ll 1$ for each heat exchanger; quasi-steady refrigerant
Model	$H(s) = Q_{cooling}(s)/T_{space}(s)$ — a first-order-plus-delay loop gain
Assumptions	linearisation valid over the control band; UA constant; no frost; uniform room air; no cycling
Physics retained	power flow through the loop; the coupling between refrigerant mass flow and heat transfer
Physics neglected	compressor map nonlinearity; two-phase slide; frost accumulation; cycling/hysteresis; heat gains and losses from the envelope
Validity	small disturbances about a chosen operating point in one operating regime
Fails when	setpoint changes cross regime boundaries; frost forms; the compressor cycles; ambient conditions shift the operating point
Next model	piecewise-linear or gain-scheduled control; nonlinear simulation; two-phase refrigerant models

19.10 — Where ME Layer 3 Fails

Failure mode	Condition	What to use instead	Return to
Structural wave propagation	$f > v_P/10L$ (high-frequency)	Euler-Bernoulli beam PDE; FEM	Ch. 15, 16
Turbulent pipe flow	$Re > 2300$	Moody chart + empirical correlations; CFD	Ch. 15 §15.8
Thermal: $Bi > 0.1$	Significant temperature gradient	1D or 3D heat equation; FEM	Ch. 13, 16
Compressible flow ($M > 0.3$)	Density changes significantly	Isentropic flow relations; shock equations	Ch. 15
Nonlinear vibration (large amplitude)	Geometric or material nonlinearity	Nonlinear ODE; Duffing oscillator	Ch. 9 §9.10
Phase change in heat transfer	Boiling/condensation	Nucleate boiling correlations; CHF models	Ch. 10 §10.9
Fracture and fatigue	Stress intensity exceeds K_{Ic}	LEFM, Paris law	Ch. 15 §15.11
High-temperature creep	Stress relaxation (Arrhenius)	Norton power law; Larson-Miller parameter	Ch. 13, 14

19.11 — Summary

Concept	ME element	ω_0 / characteristic
Mass-spring-damper	m, k, b	$\omega_0 = \sqrt{k/m}$, $\zeta = b/2\sqrt{km}$
Torsional oscillator	I_r, k_r, b_r	$\omega_0 = \sqrt{k_r/I_r}$
Shaft critical speed	m_{rotor}, k_{shaft}	$\omega_c = \sqrt{k/m}$ (avoid in operation)
Gear train	TF with $n = N_2/N_1$	$I_{refl} = I_{load}/n^2$
Rankine efficiency	Steam cycle	$\eta = 1 - Q_C/Q_H \leq \eta_{Carnot}$

Concept	ME element	ω_0 / characteristic
Brayton efficiency	Gas turbine	$\eta = 1 - r_p^{-(\gamma-1)/\gamma}$
Otto efficiency	Gasoline engine	$\eta = 1 - r^{-(\gamma-1)}$
Thermal resistance	Conduction	$R_{th} = L/\kappa A$
Convection resistance	Newton's law	$R_{conv} = 1/hA$
Nusselt correlation	Turbulent pipe	$Nu = 0.023 Re^{0.8} Pr^{0.4}$
Hydraulic resistance	Poiseuille	$R_{hyd} = 8\mu L/\pi r^4$
Vibration transmissibility	Isolation	$T = \sqrt{(1+4\zeta^2 r^2)/((1-r^2)^2+4\zeta^2 r^2)}$
Acoustic SPL	Sound power	$L_p = 20\log_{10}(p_{rms}/20\mu\text{Pa})$
Reverberation time	Room acoustics	$RT_{60} = 0.161 V/A_{room}$

19.12 — Engineering Thread

Physics	Engineering system
Mass-spring-damper ω_0, ζ	Vibration isolation; NVH in vehicles; seismic design
Gear as TF $n = N_2/N_1$	Robot gearboxes; wind turbine pitch drives; automotive transmissions
Flywheel $E = \frac{1}{2} I_r \omega^2$	Grid-scale energy storage; UPS flywheel; kinetic energy recovery (F1)
Rankine cycle	Coal/nuclear/CSP power plants; combined cycle (Rankine + Brayton)
Brayton cycle	Aircraft jet engines; land-based gas turbines; combined heat and power
Otto/Diesel cycle	Automotive engines; gensets; marine diesels
Vapor compression refrigeration	HVAC; cold chain; industrial refrigeration
Heat exchanger NTU-ε	Shell-and-tube design; plate heat exchangers; recuperators
Hagen-Poiseuille R_{hyd}	Pipe sizing; hydraulic circuit design; microfluidics
Pump curve intersection	Pump selection; system design; variable speed drives
Vibration absorber TMD	Tall building sway control; bridge flutter suppression
Acoustic SPL + RT_{60}	Concert hall design; noise control; industrial hygiene

19.13 — Looking Ahead

Chapter 19 established the mechanical, thermal, and fluid instantiations of the Layer-3 template. The governing equations — mass-spring-damper, thermodynamic cycle, thermal circuit, hydraulic network — are all special cases of the universal template from Chapter 17.

Chapter 20 covers **CE and ChE Systems**: the civil structural, hydraulic infrastructure, and chemical process instantiations. Structural mechanics (from Ch. 15) becomes the stiffness matrix method; hydraulics (from Ch. 15, 17) becomes pipe network and open-channel flow; chemical engineering adds reaction kinetics (from Ch. 10 Boltzmann factor) and mass transfer.

Chapter 21 then provides the capstone: feedback and control, operating entirely at the level of the transfer function $H(s)$ — above and across all physical domains simultaneously.

End of Chapter 19.

Next: Chapter 20 — CE and ChE Systems: Structures, Hydraulics, and Process Engineering

CHAPTER 20

CE and ChE Systems: Structures, Hydraulics, and Process Engineering

The Structural, Hydraulic, and Chemical Instantiations of Layer 3

> *Terzaghi's consolidation equation and the 1D heat equation are the same equation. The NTU method for an absorption column and the NTU method for a heat exchanger are the same design framework. Darcy's law and Fick's law are the same constitutive shape arrived at independently. The Arrhenius factor and the Boltzmann factor share a statistical origin, not an equation.*
>
> Four different claims, and the difference between them matters. This chapter labels which is which rather than calling all of them "identical".

20.0 — Overview

Chapters 18 and 19 instantiated the Layer-3 template (Ch. 17) in the electrical and mechanical-thermal domains. This chapter completes the engineering coverage:

Civil Engineering (CE): Structural analysis, open-channel hydraulics, groundwater flow, soil consolidation.

Chemical Engineering (ChE): Mass and energy balances, reaction kinetics and reactor design, separation processes, mass transfer.

Every concept in this chapter traces to earlier physics. The last column states **what kind of trace** — see §20.0.1 for why the distinction is not pedantry.

Ch. 20 concept	Physics origin	Connection
Beam bending $EI\,d^4w/dx^4 = q$	Euler-Bernoulli (Ch. 16 §16.7.2); Navier equation (Ch. 15)	`[APPROXIMATION]` — slender beam, $L/r \gg 1$
Stiffness method $\mathbf{Ku} = \mathbf{F}$	FEM discretisation of the Navier equation (Ch. 16 §16.11)	`[DERIVATION]`
Terzaghi consolidation $\partial u_e/\partial t = c_v\,\partial^2 u_e/\partial z^2$	1D diffusion equation (Ch. 16 §16.8; Ch. 14 §14.8)	`[DERIVATION]` — same equation
Transient groundwater $S\,\partial h/\partial t = T\nabla^2 h$	Advection–diffusion / storage formulation (Ch. 16)	`[DERIVATION]` under stated storage assumptions
Darcy's law $q = -K\nabla h$	Linear transport law, same form as Fick (Ch. 13 §13.6)	`[STRUCTURAL CONNECTION]`
PFR design equation	1D advection–reaction PDE (Ch. 16)	`[DERIVATION]`

Ch. 20 concept	Physics origin	Connection
CSTR design equation	Lumped mass balance (Bridge C, Ch. 17)	`[APPROXIMATION]` — well-mixed is an assumption
Arrhenius $k = A\,e^{-E_a/RT}$	Boltzmann factor e^{-E/k_BT} (Ch. 10 §10.3.3)	`[STRUCTURAL CONNECTION]` — shared statistical origin
NTU for absorption columns	NTU for heat exchangers (Ch. 19 §19.6.3)	`[STRUCTURAL CONNECTION]` — shared design framework
Two-resistance mass transfer	Thermal resistances in series (Ch. 17, 19)	`[STRUCTURAL CONNECTION]`
Manning's equation	Navier–Stokes, empirically integrated (Ch. 15)	`[PHENOMENOLOGICAL]`
Rational method unit hydrograph	Linear filter / transfer function (Ch. 21 §21.4)	`[ANALOGY]`

20.0.1 — Four kinds of "same", and which one this chapter is allowed to claim

CE and ChE reuse the mathematics of physics constantly. But "the same mathematics" is not one claim — it is four, with different strength and different failure conditions. Collapsing them is how a physics text turns into a false analogy, so the distinction is stated once here and then applied.

Kind	What is actually shared	What is *not* shared	Can fail because…
Mathematical identity	The governing equation itself, term by term, once the assumptions are imposed	Nothing — the equations are the same object	The *assumptions* fail (1D, homogeneous, linear, incompressible)
Constitutive correspondence	The algebraic/functional form of a constitutive law, arrived at independently by different routes	The microscopic derivation, the closure, the coefficient values, the state variables	The closure model fails (turbulent, unsaturated, non-Newtonian, reactive)
Shared statistical origin	A statistical-mechanical structure (exponential activation, fluctuation–response)	The macroscopic law built on top of it	The statistical regime fails (non-equilibrium, non-ideal, many-body)
Shared dimensionless framework	The way dimensionless groups organise a design space	The driving forces, the equilibrium relations, the performance definition	The framework is applied outside its validated range

Worked instantiations, in the order they appear:

- **Mathematical identity** — Terzaghi consolidation (§20.4.2). Once the soil skeleton is treated as linear-elastic and the flow as 1D, the equation *is* the diffusion equation. The erfc solution, the T_v time factor, and the $T_v \approx (\pi/4)U^2$ parabola are not "analogues" of anything: they are the same solutions.
- **Constitutive correspondence** — Darcy's law (§20.4.1). Darcy's law is *not* a special case of Fick's law. It was measured, not derived, and the linear form emerges from pore-scale momentum balance plus an empirical closure. What it shares with Fick's law is the *shape* — linear in the gradient, flux proportional to a transport coefficient. The coefficient K is not a diffusivity and does not have units that make it one.

- **Shared statistical origin** — Arrhenius kinetics (§20.7.1). The Arrhenius factor and the Boltzmann factor are the same exponential because both come from the Maxwell–Boltzmann energy distribution. But $k(T)$ is a rate constant for a specific reaction mechanism, whereas e^{-E/k_BT} is a population ratio. Sharing an origin is not sharing an equation.
- **Shared dimensionless framework** — NTU/HTU (§20.9.2). The heat-exchanger and absorption-column methods are organised the same way, and the tabulated ε–NTU curves transfer usefully. But $NTU_{OG} = \int dy/(y^* - y)$ and $NTU = UA/C_{min}$ have *different driving forces* (composition deficit against a VLE equilibrium vs temperature deficit against a fixed inlet), and the capacity-rate ratio is mimicked, not the same quantity.

The rule used throughout this chapter: the word "identical" appears only for the first kind. Everything else is named for what it is.

20.1 — Structural Analysis: The Stiffness Method

20.1.1 — From PDE to Matrix: FEM for Structures

From Ch. 16 §16.11.2: the finite element method discretizes the domain into elements and assembles a global matrix equation. For linear elastic structures (Navier equation, Ch. 15 §15.9.2), the FEM gives:

$$\boxed{\mathbf{K}\mathbf{u} = \mathbf{F}}$$

where:

- $\mathbf{u}$: global displacement vector (degrees of freedom at nodes)
- $\mathbf{F}$: global force vector (applied loads and reactions)
- $\mathbf{K}$: global stiffness matrix (assembled from element stiffness matrices)

The stiffness matrix is the discrete version of the elastic modulus tensor from Ch. 15 §15.9.1 — the same physical law, now expressed as a matrix equation rather than a PDE.

Model Ledger — continuum elastodynamics to the stiffness method

Field	Content
Parent theory	Linear elasticity (Ch. 15 §15.9): $\sigma_{ij} = \lambda\varepsilon_{kk}\delta_{ij} + 2\mu_L\varepsilon_{ij}$, $\varepsilon_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i)$, $\varepsilon_{kk} = \nabla\cdot\mathbf{u}$
Reduction	static equilibrium; small strain; linear elastic material; prismatic geometry; conforming interpolation
Model	$\mathbf{K}\mathbf{u} = \mathbf{F}$, with $\mathbf{K}$ assembled from element matrices
Assumptions	$\varepsilon \ll 1$; $\sigma = D\varepsilon$ with constant D; no body force in the element; full continuity of u across elements
Physics retained	load path through stiffness; stress redistribution; equilibrium
Physics neglected	inertia (dynamics), geometric non-linearity, plasticity, fracture, contact, thermal strain
Validity	deflections $\lesssim L/100$; stresses below yield; mesh refined so element stiffness converges
Fails when	large deflection (buckling, P-Δ), yielding, or a singular/non-conforming element stiffness
Next model	nonlinear FEM; dynamic stiffness (modal analysis, Ch. 21); plastic $D(\varepsilon)$

> **Note on the constitutive form.** The isotropic law has two structurally distinct pieces, and both are load-bearing. The term $\lambda\varepsilon_{kk}\delta_{ij}$ is the *volumetric* response: it is multiplied by the trace $\varepsilon_{kk} = \nabla \cdot \mathbf{u}$, and it produces stress even in a pure hydrostatic strain where $\varepsilon_{ij} = 0$ for $i \neq j$. The factor of 2 in $2\mu_L\varepsilon_{ij}$ is fixed by the requirement that μ_L be the shear modulus, i.e. that $\sigma_{ij} = 2\mu_L\varepsilon_{ij}$ in pure shear where $\varepsilon_{kk} = 0$. Writing $\lambda\varepsilon_{ij}\delta_{ij}$ drops the trace dependence, and writing $\mu_L\varepsilon_{ij}$ without the 2 makes μ_L the wrong modulus by a factor of two. An element stiffness matrix built on a corrupted form of this equation will pass equilibrium checks and still give the wrong displacements.

20.1.2 — Bar Elements: Trusses

A truss member (bar) carries only axial force. For a bar of cross-sectional area A, elastic modulus E, length L, along the x-axis:

Element stiffness matrix (in local coordinates, 2 nodes × 1 DOF = 2×2):

$$\mathbf{k}^{(e)} = \frac{EA}{L}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

This is the spring stiffness from Ch. 17 §17.3.2 — a bar element is literally a spring with $k = EA/L$ (Hooke's law, Ch. 13 §13.8.2, now applied to a bar).

For a general truss member at angle θ to the global x-axis: transform the local stiffness matrix to global coordinates using the rotation matrix $\mathbf{T}(\theta)$ — giving a 4×4 element stiffness matrix in the global frame.

Assembly: Add element stiffness contributions to the global positions corresponding to their node DOFs. Apply displacement boundary conditions (strike rows/columns for zero-displacement nodes). Solve the reduced system.

Example: Three-bar truss Three members meeting at a central node (2 DOFs: u_x, u_y) with outer nodes pinned to the ground. Each member contributes to the 2×2 global stiffness matrix:

$$\mathbf{K} = \sum_{e} \mathbf{k}^{(e)}, \qquad \begin{pmatrix} K_{xx} & K_{xy} \\ K_{yx} & K_{yy} \end{pmatrix}\begin{pmatrix} u_x \\ u_y \end{pmatrix} = \begin{pmatrix} F_x \\ F_y \end{pmatrix}$$

Member forces: $N^{(e)} = \frac{EA}{L}(\Delta u_{axial})$ — axial stress $\sigma = N/A$.

20.1.3 — Beam Elements: Frames

A beam element carries transverse load, bending moment, and shear force. For an Euler-Bernoulli beam element with 2 nodes × 2 DOFs (transverse displacement w + rotation $\theta = dw/dx$) = 4 DOFs per element:

Element stiffness matrix (4×4, Hermite shape functions):

$$\mathbf{k}^{(e)} = \frac{EI}{L^3}\begin{pmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{pmatrix}$$

This matrix is the lumped version of the Euler-Bernoulli PDE $EI\, d^4w/dx^4 = q(x)$ (Ch. 16 §16.7.2) — exactly Bridge C applied to beam bending.

A frame element combines axial bar behavior + bending beam behavior in a 6-DOF element (3 per node: u_x, u_y, θ_z). This is the basis of commercial structural analysis software (SAP2000, ETABS, ANSYS, OpenSees).

20.2 — Beam Bending: Stress and Deflection

20.2.1 — The Euler-Bernoulli Beam Equation

For a slender beam with bending stiffness EI (Young's modulus × second moment of area) and distributed transverse load $q(x)$ [N/m]:

$$\boxed{EI\frac{d^4w}{dx^4} = q(x)}$$

This is a 4th-order ODE — a static limit of the flexural wave equation from Ch. 16 §16.7.2 (with $\partial_{tt}w = 0$). Integrating successively:

- $EI\,d^3w/dx^3 = V(x)$ — shear force, $V = \int q\,dx + C_1$
- $EI\,d^2w/dx^2 = M(x)$ — bending moment, $M = \int V\,dx + C_2$
- $dw/dx = \theta(x)$ — slope, $\theta = \frac{1}{EI}\int M\,dx + C_3$
- $w(x) = \int \theta\,dx + C_4$ — deflection

(sign convention here: $dV/dx = q$, $dM/dx = V$; the four constants are fixed by the boundary conditions.)

Bending stress at distance y from the neutral axis:

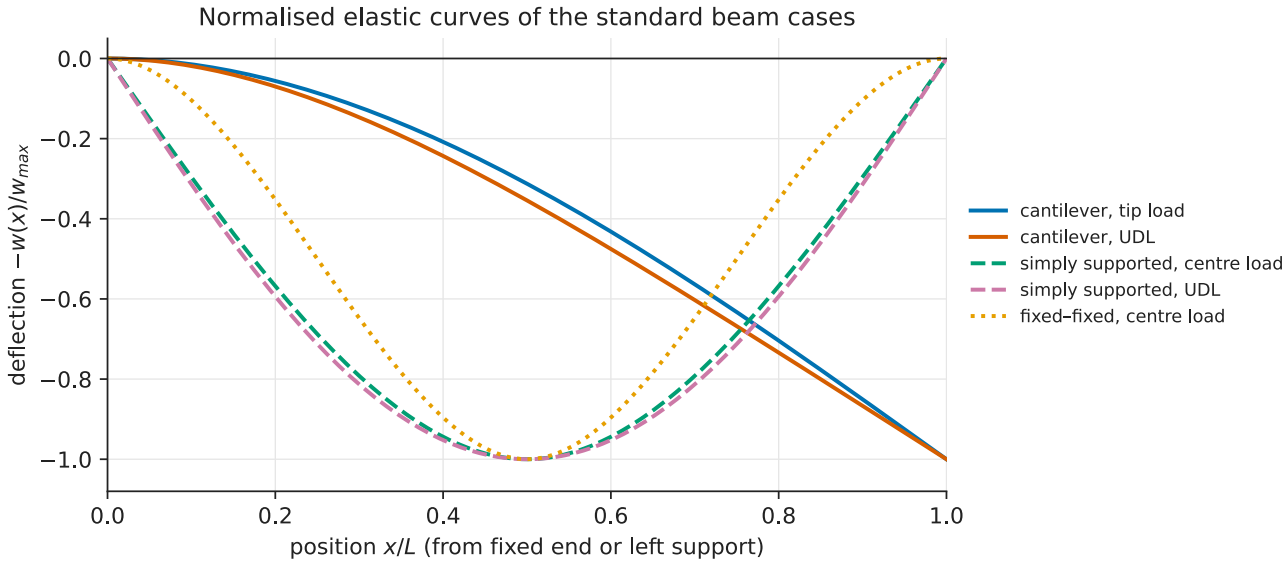
$$\sigma_x = -\frac{My}{I}$$

Maximum bending stress at $y = c$ (outermost fiber): $\sigma_{max} = Mc/I = M/S$ where $S = I/c$ is the **section modulus** — the key geometric quantity in beam design.

20.2.2 — Standard Beam Cases

Beam	Loading	Max deflection w_{max}	Max moment M_{max}
Cantilever, end load P	P at tip	$PL^3/3EI$	PL (at root)
Cantilever, UDL q	q uniform	$qL^4/8EI$	$qL^2/2$ (at root)
Simply-supported, centre load P	P at midspan	$PL^3/48EI$	$PL/4$ (at centre)
Simply-supported, UDL q	q uniform	$5qL^4/384EI$	$qL^2/8$ (at centre)
Fixed-fixed, centre load P	P at midspan	$PL^3/192EI$	$PL/8$ (at centre)

Design check: $\sigma_{max} = M_{max}c/I \le \sigma_{allowable}$ (strength) and $w_{max} \le L/360$ or $L/250$ (serviceability — deflection limit).



Elastic curves $w(x)/w_{max}$ from the closed-form Euler–Bernoulli solutions for the five tabulated cases, each normalised by its tabulated w_{max} (which the curves reproduce exactly); cantilevers peak at the free tip, the symmetric cases at midspan, and fixed ends flatten the curve to zero slope at the supports.

20.2.3 — Second Moment of Area

The second moment of area I for common cross-sections about the centroidal axis:

Section	I
Rectangle $b \times h$	$bh^3/12$
Circle radius r	$\pi r^4/4$
Hollow rectangle (outer $B \times H$, inner $b \times h$)	$(BH^3 - bh^3)/12$
I-beam (approx.)	$I_{flanges} + I_{web}$ (parallel axis theorem)

Parallel axis theorem: $I = I_{centroid} + Ad^2$ (shift axis by d from centroid, area A).

20.3 — Open-Channel Hydraulics

20.3.1 — Manning's Equation

For steady, uniform flow in an open channel (river, canal), the Navier-Stokes equation integrated over the cross-section gives:

$$\boxed{V = \frac{1}{n} R_h^{2/3} S_0^{1/2}}$$

where V is the average flow velocity, n is Manning's roughness coefficient, $R_h = A/P$ is the hydraulic radius (area/wetted perimeter), and S_0 is the channel bed slope.

Surface	Manning's n
Smooth concrete	0.011
Brick/masonry	0.015

Surface	Manning's n
Gravel channel	0.025–0.030
Earth channel (clean)	0.022
Natural river (clean)	0.025–0.035
Floodplain (heavy brush)	0.075–0.15

Volume flow rate: $Q = VA = \frac{1}{n} A R_h^{2/3} S_0^{1/2}$

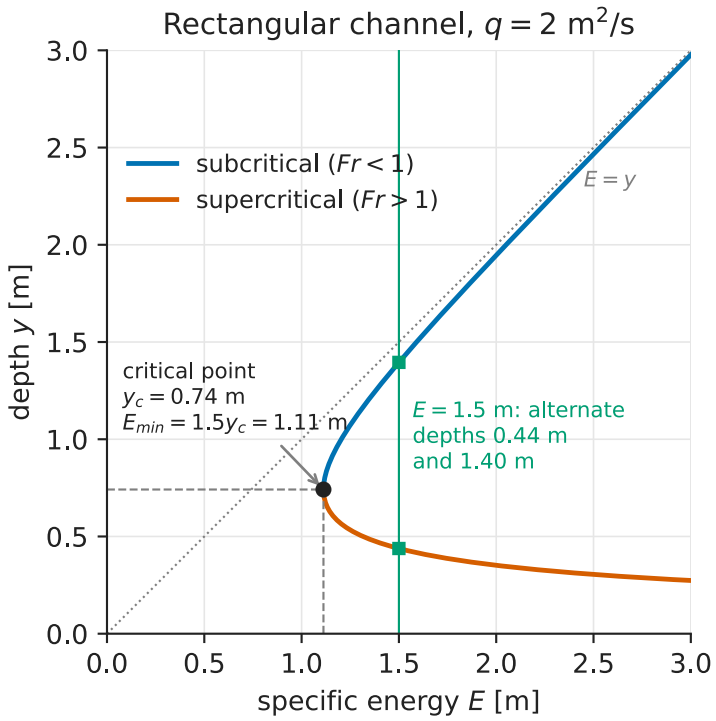
Connection to physics: Manning's equation is an empirical fit to the Navier-Stokes momentum equation with a turbulence model. The roughness n encodes the turbulent dissipation that Ch. 15 §15.8.2 showed cannot be derived analytically for open-channel turbulent flow.

20.3.2 — The Froude Number and Flow Regimes

Froude number $Fr = V/\sqrt{gD}$ (analogous to Mach number in compressible flow):

- $Fr < 1$: **subcritical (tranquil) flow** — disturbances propagate both upstream and downstream; controls from downstream
- $Fr = 1$: **critical flow** — minimum specific energy for a given flow rate
- $Fr > 1$: **supercritical (rapid) flow** — disturbances only travel downstream; controls from upstream

Specific energy: $E = y + \frac{Q^2}{2gA^2}$ where y is the flow depth. For a given Q, minimum E occurs at critical depth y_c (where $Fr = 1$).



Specific energy $E = y + q^2/(2gy^2)$ for a rectangular channel carrying $q = 2$ m²/s per unit width: the curve's nose is the critical point $y_c = (q^2/g)^{1/3} = 0.74$ m, $E_{min} = 1.5y_c = 1.11$ m, separating the subcritical (upper) and supercritical (lower) branches; any $E > E_{min}$, e.g. 1.5 m, admits two alternate depths.

Hydraulic jump: An abrupt transition from supercritical to subcritical flow (always increases depth, always dissipates energy). Sequent depths:

$$\frac{y_2}{y_1} = \frac{1}{2}\left(\sqrt{1 + 8Fr_1^2} - 1\right)$$

Energy loss: $\Delta E = (y_2 - y_1)^3 / 4y_1 y_2$. Used in spillway design to dissipate kinetic energy before the downstream channel.

20.4 — Groundwater and Soil: The Diffusion Equation in CE

20.4.1 — Darcy's Law: Ohm's Law for Groundwater

For water flowing through a saturated porous medium (soil, rock):

$$q = -K \nabla h$$

where q is the specific discharge (Darcy flux, [m/s]), K is the hydraulic conductivity [m/s], and h is the hydraulic head $h = z + p/\gamma_w$.

`[STRUCTURAL CONNECTION]` — **same constitutive shape as Fick's law, different derivation.** Fick's law (Ch. 13 §13.6) is a constitutive postulate. Darcy's law was *measured*, not derived: Darcy's 1856 sand experiment produced a linear relation between flux and head gradient, and the pore-scale justification — drag balance on a grain, closed over an unresolved pore geometry — is a different argument that arrives at the same algebraic form:

	Fick's law	Darcy's law
Driving force	chemical potential gradient $\nabla\mu$	hydraulic head gradient ∇h
Transport coefficient	diffusivity D [m²/s]	hydraulic conductivity K [m/s]
Driving variable	concentration	hydraulic head (length)
Status	constitutive postulate	empirical law, later rationalised
Fails when	activity coefficients matter	flow is turbulent, unsaturated, or reactive

Note the dimensional asymmetry: D carries the length that K does not. That single difference is why these are not the same equation and why groundwater flow is *not* "Fick's law with different letters". What is shared is the linear-gradient structure, which is why the same solvers, error-function solutions, and network analogies reappear.

Hydraulic conductivity values:

Material	K (m/s)
Gravel	10^{-2}–10^{-1}
Coarse sand	10^{-4}–10^{-2}
Fine sand	10^{-5}–10^{-4}
Silt	10^{-7}–10^{-5}
Clay	10^{-10}–10^{-7}

Groundwater flow equation (confined, transient):

$$S\frac{\partial h}{\partial t} = T\nabla^2 h + W$$

where S is the storage coefficient, $T = Kb$ is transmissivity, and W is the recharge rate. `[DERIVATION]` — given confined flow, constant S and T, and horizontal isotropic flow, this is the diffusion equation with $D = T/S$, so the same error-function and well-function solutions apply. The identity is exact *under those assumptions* and fails outside them: for unconfined flow h varies enough to matter in $T = Kb(h)$, and for unsaturated flow the saturation-dependent Richards equation is nonlinear.

20.4.2 — Terzaghi's Consolidation: Settlement by Diffusion

When a saturated clay layer is loaded (by a building, embankment, or fill), excess pore water pressure u_e is generated. The clay consolidates (settles) as water squeezes out under the excess pressure. Terzaghi's consolidation equation:

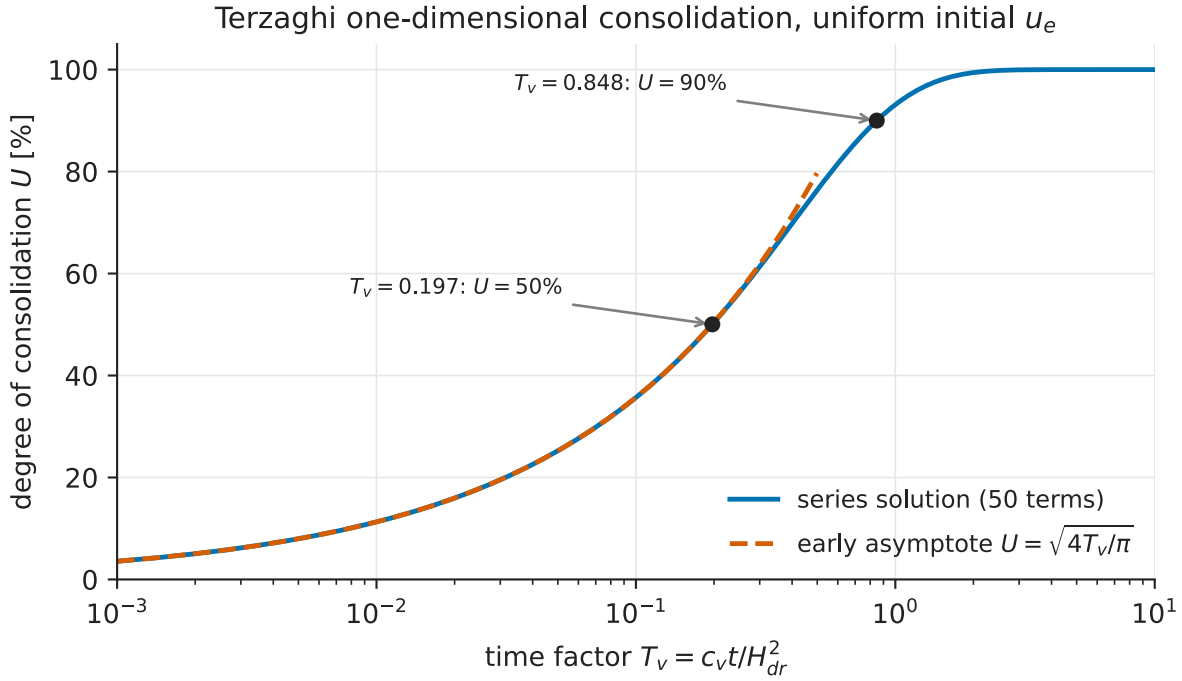
$$\boxed{\frac{\partial u_e}{\partial t} = c_v \frac{\partial^2 u_e}{\partial z^2}}$$

where $c_v = k(1+e_0)/\gamma_w m_v$ is the **coefficient of consolidation** (hydraulic conductivity k, initial void ratio e_0, water unit weight γ_w, volume compressibility m_v).

This is the 1D diffusion equation from Ch. 16 §16.8 and Ch. 14 §14.8. The solution for a clay layer of thickness H (drained top and bottom, so drainage path $H_{dr} = H/2$), initially at uniform excess pressure u_0:

$$U(t) = 1 - \frac{8}{\pi^2}\sum_{n=0}^{\infty}\frac{1}{(2n+1)^2}\exp\left(-\frac{(2n+1)^2\pi^2 T_v}{4}\right)$$

where $U(t) = \Delta H/\Delta H_{final}$ is the **degree of consolidation** and $T_v = c_v t/H_{dr}^2$ is the dimensionless time factor.



Average degree of consolidation U versus T_v (log axis) from the series above (50 terms), with the early-time asymptote $U = \sqrt{4T_v/\pi}$ (accurate to $U \approx 60\%$); notice $T_v = 0.197$ gives 50% and $T_v = 0.848$ gives 90%, the value used in the settlement example below.

`[DERIVATION]` **— this is a mathematical identity, not an analogy.** The Fourier-number series above, the step-load boundary condition, and the $T_v \approx (\pi/4)U^2$ parabola are the *same* solutions as the slab cooling problem of Ch. 14 §14.8, applied to a different field. Nothing is being reinterpreted; only the name of the field changes.

	Heat conduction (Ch. 14 §14.8)	Terzaghi consolidation
Field	temperature T	excess pore pressure u_e
Governing equation	$\partial T/\partial t = \alpha\, \partial^2 T/\partial z^2$	$\partial u_e/\partial t = c_v\, \partial^2 u_e/\partial z^2$
Transport coefficient	thermal diffusivity α	coefficient of consolidation c_v
Step boundary condition	surface suddenly cooled	load suddenly applied
Solution	Fourier series in Fo	same series in T_v
Early-time asymptote	$1 - \mathrm{erf}(z/2\sqrt{\alpha t})$	$T_v \approx (\pi/4)U^2$, i.e. $U \approx \sqrt{4T_v/\pi}$

Model Ledger — load step to degree of consolidation

Field	Content
Parent theory	Poroelasticity: Terzaghi's one-dimensional consolidation theory
Reduction	Linear elastic soil skeleton + incompressible pore water + 1D flow + constant c_v
Model	$\partial u_e/\partial t = c_v\, \partial^2 u_e/\partial z^2$ with step-load BCs
Assumptions	1D; saturated; linear skeleton (m_v constant); negligible self-weight; uniform c_v
Physics retained	transient pore-pressure dissipation driving settlement
Physics neglected	lateral drainage, skeleton non-linearity, overconsolidation, secondary compression (creep), air entrapment
Validity	normally consolidated, single load increment, $U \lesssim 90\%$, drainage path well defined
Fails when	load is removed (swelling, not consolidation), m_v varies with stress, or secondary compression dominates
Next model	nonlinear $m_v(\sigma')$; coupled 2D radial consolidation; creep superposition

Engineering application: A 6 m clay layer ($c_v = 10^{-3}$ cm²/s, $H_{dr} = 3$ m) achieves 90% consolidation at $T_v = 0.848$:

$$t_{90\%} = \frac{T_v H_{dr}^2}{c_v} = \frac{0.848 \times (300\ \mathrm{cm})^2}{10^{-3}\ \mathrm{cm^2/s}} = 7.6 \times 10^7\ \mathrm{s} \approx 2.4\ \mathrm{years}$$

Settlement continues for years — the engineer must design for long-term deformation, not just immediate elastic deflection.

20.5 — Hydrology: Rainfall to Runoff

20.5.1 — The Rational Method

For small urban catchments, peak runoff:

$$Q_{peak} = CiA$$

where C is the dimensionless runoff coefficient (0.1 for meadow, 0.95 for pavement), i is the rainfall intensity [m/s], and A is the catchment area [m²].

This is the lumped approximation of the full hydrological balance — the Bridge C operation applied to a catchment. The rational method assumes uniform rainfall, uniform catchment response, and that the storm duration equals the time of concentration t_c (time for water from the farthest point to reach the outlet).

20.5.2 — Linear Systems Theory in Hydrology: Unit Hydrograph

The unit hydrograph $u(t)$ is the response of a catchment to a unit depth of rainfall excess over unit time — exactly the **impulse response** (Green's function) of the hydrological system.

For any rainfall excess pattern $r(t)$, the direct runoff hydrograph:

$$Q(t) = \int_0^t r(\tau)\, u(t-\tau)\, d\tau$$

This is a **convolution integral** — the same operation as a linear time-invariant system in Ch. 18 §18.6.2. The hydrological system is treated as a linear filter (provided watershed response is truly linear — an approximation that holds for moderate rainfall events).

Clark's method: The unit hydrograph is modeled as a linear reservoir (storage $S = KQ$) followed by a channel routing — exactly an RC system in the hydraulic domain. The time-area method provides the input.

20.6 — Chemical Engineering: Mass and Energy Balances

20.6.1 — The Noether Foundation of Process Balances

Every process balance in ChE is a Noether conservation law (Ch. 1 §1.7) applied to a control volume:

Total mass balance (Noether: particle number conservation):

$$\frac{dm_{sys}}{dt} = \sum \dot{m}_{in} - \sum \dot{m}_{out}$$

Species mass balance for component A (with reaction):

$$\frac{dN_A}{dt} = \sum \dot{N}_{A,in} - \sum \dot{N}_{A,out} + r_A V$$

Energy balance (Noether: energy conservation, first law, Ch. 10):

$$\frac{dU_{sys}}{dt} = \dot{Q} - \dot{W}_s + \sum \dot{m}_{in} H_{in} - \sum \dot{m}_{out} H_{out}$$

where $H = U + pV/m = h$ is the specific enthalpy (the natural thermodynamic potential for constant-pressure processes).

The Kirchhoff analogy: These balance equations at steady state are:

- Mass balance: $\sum \dot{m}_k = 0$ at every process node (KCL for mass)
- Energy balance: determines the enthalpy change (KVL for energy)

A process flow diagram is a bond graph in the chemical domain.

20.6.2 — Degrees of Freedom

For a process unit with C components, S streams, and E equations:

$$DOF = S \times (C + 1) - E - specifications$$

where the $+1$ accounts for the stream flow rate (in addition to $C - 1$ mole fractions). A system with $DOF = 0$ is fully determined; $DOF > 0$ requires additional specifications (design variables).

This counting procedure is the chemical engineering analog of the Gibbs phase rule (Ch. 10 §10.9.3) — both trace to the mathematical structure of simultaneous equations.

20.7 — Chemical Reaction Kinetics

20.7.1 — The Arrhenius Rate Constant: Boltzmann Factor at Work

The temperature dependence of a reaction rate constant:

$$k_{rxn}(T) = A\, e^{-E_a/RT}$$

This is the Boltzmann factor from Ch. 10 §10.3.3. The activation energy E_a is the energy barrier that reactant molecules must exceed to reach the transition state. The fraction of molecules with enough energy is $e^{-E_a/k_B T}$ (from the Maxwell-Boltzmann distribution, Ch. 10 §10.8.2) — and this fraction sets the reaction rate.

`[STRUCTURAL CONNECTION]` — **shared statistical origin, not a shared equation.** Both exponentials come from the same place: the Boltzmann weight of a state at energy E relative to thermal energy $k_B T$. But they answer different questions.

	Boltzmann factor (Ch. 10)	Arrhenius factor (here)
Quantity	population ratio n_E/n_0	rate constant $k(T)$ [1/s]
E means	energy of a *state*	height of a *barrier*
Denominator	$k_B T$, per molecule	RT, per mole (hence the factor 1000)
What follows	equilibrium populations	a reaction rate, only after a mechanism is assumed
Needs a mechanism?	no	yes — the exponential is necessary, not sufficient

The gap between the second and third rows is where mechanism lives. A rate constant cannot be read off the Boltzmann factor; it is the Boltzmann factor multiplied by a prefactor that encodes collision frequency, steric orientation, entropy of activation, and — for a specific mechanism — the number of distinct reaction routes. Two reactions with the same E_a can differ in k by orders of magnitude.

Every thermally-activated process in engineering follows Arrhenius:

- Chemical reactions (this section)
- Solid-state diffusion (Ch. 14 §14.8.2: same formula!)
- Creep in metals (Ch. 19)
- Electrode reactions in batteries
- Crystal nucleation in processing

The pre-exponential A encodes collision frequency and steric factors — it comes from statistical mechanics (Ch. 10) and transition state theory.

Model Ledger — Arrhenius fit to a reactor rate constant

Field	Content
Parent theory	Maxwell-Boltzmann energy distribution; transition state theory
Reduction	Single rate-limiting step, dilute collision-controlled regime, $E_a \gg k_B T$
Model	$k(T) = A\exp(-E_a/RT)$, A taken as T-independent
Assumptions	one mechanism dominates; A constant over the fitted range; isothermal
Physics retained	thermal activation of a barrier
Physics neglected	entropy of activation, pressure dependence, non-Arrhenius curvature, falloff at high T, diffusion limitation
Validity	Arrhenius plot $ln\,k$ vs $1/T$ is linear over the fitted range
Fails when	plot curves (indicating a mechanism change or a diffusion-limited regime), or A must vary with T
Next model	modified Arrhenius / Eyring; explicit diffusion–reaction coupling

20.7.2 — Rate Laws and Reaction Orders

For the reaction $A + B \to C$:

$$-r_A = k_{rxn}[A]^m[B]^n \quad [\text{mol/L}\cdot\text{s}]$$

where m and n are the reaction orders (determined experimentally, not from stoichiometry in general). The overall rate expression must be determined from kinetic experiments — it is a Layer-2 constitutive relation, not derivable from Layer-0 without a detailed molecular mechanism.

Integrated rate laws for simple cases:

Order	$-r_A$	$C_A(t)$	**Half-life** $t_{1/2}$
Zero	k	$C_{A0} - kt$	$C_{A0}/2k$
First	kC_A	$C_{A0}e^{-kt}$	$\ln 2/k$
Second	kC_A^2	$1/(1/C_{A0} + kt)$	$1/kC_{A0}$

First-order kinetics: same exponential decay as radioactive decay (Ch. 6 §6.10.3). The nuclear decay rate constant λ is the reaction rate constant for the nuclear reaction — same Boltzmann physics, different energy scale.

20.8 — Reactor Design: CSTR and PFR

20.8.1 — The Three Ideal Reactor Types

Batch reactor: Closed, time-varying, perfectly mixed. Design equation (species balance):

$$N_{A0}\frac{dX_A}{dt} = -r_A(X_A)\cdot V$$

where $X_A = (N_{A0} - N_A)/N_{A0}$ is the fractional conversion. At constant volume:

$$\frac{dC_A}{dt} = r_A(C_A) \quad \Longrightarrow \quad t = C_{A0}\int_0^{X_A}\frac{dX_A}{-r_A}$$

CSTR (Continuous Stirred Tank Reactor): Perfect mixing → uniform concentration throughout the vessel = concentration at outlet. Steady-state species balance (input − output + generation = 0):

$$\dot{N}_{A,in} - \dot{N}_{A,out} + r_A V = 0 \quad \Longrightarrow \quad F_{A0} - F_{A0}(1 - X_A) + r_A V = 0$$

$$\boxed{V_{CSTR} = \frac{F_{A0}X_A}{-r_A(X_A)}}$$

The CSTR design equation is **algebraic** (not an ODE) because perfect mixing eliminates spatial gradients. It is the lumped (Bridge C) version of the reactor balance — one control volume, uniform conditions inside.

PFR (Plug Flow Reactor): No axial mixing; plug of fluid advances through the reactor. Species balance over differential volume element dV:

$$\frac{dF_A}{dV} = r_A \quad \Longrightarrow \quad \boxed{V_{PFR} = F_{A0}\int_0^{X_A}\frac{dX_A}{-r_A(X_A)}}$$

The PFR design equation is an **integral** over conversion. The PFR is the distributed-parameter version (Ch. 16: 1D convection-reaction PDE) of the CSTR lump.

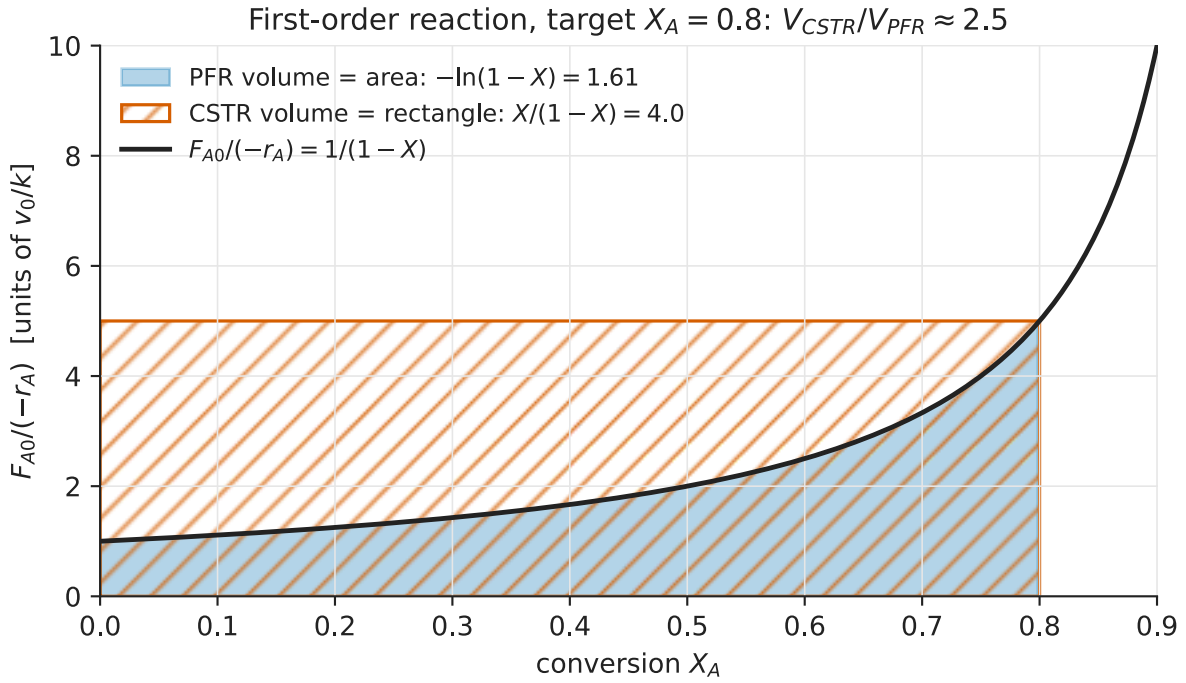
20.8.2 — Comparing CSTR and PFR: The Levenspiel Plot

For a positive-order reaction ($-r_A$ decreasing with X_A), the Levenspiel plot shows $1/(-r_A)$ vs. X_A:

- **PFR volume** = area under the curve from 0 to X_A
- **CSTR volume** = rectangle height $1/(-r_A)|_{X_{A,out}}$ × width X_A

Since $1/(-r_A)$ increases with X_A (rate decreases as reactant depletes), the rectangle is always larger than the area under the curve:

$$V_{CSTR} > V_{PFR} \quad \text{(for positive-order reactions at same conversion)}$$



Levenspiel plot $F_{A0}/(-r_A) = (v_0/k)/(1 - X_A)$ for a first-order reaction: to reach $X_A = 0.8$ the PFR needs the shaded area $-\ln(0.2) = 1.61\, v_0/k$ while the CSTR needs the full rectangle $0.8/0.2 = 4.0\, v_0/k$, about 2.5 times larger.

Engineering trade-off:

- PFR: more efficient (less volume needed); harder to control; temperature gradients; sensitive to flow disturbances
- CSTR: easier control; well-suited for exothermic reactions (good heat removal); larger volume; cascades of CSTRs approach PFR performance

Damköhler number:

$$Da = \frac{\tau \cdot (-r_A)|_{X_A=0}}{C_{A0}} = \frac{\text{reaction rate}}{\text{convective rate}}$$

$Da \ll 1$: reaction is slow (conversion control by residence time) $Da \gg 1$: reaction is fast (conversion control by equilibrium or transport)

20.9 — Separation Processes

20.9.1 — Distillation: Vapor-Liquid Equilibrium

Binary distillation separates two components with different volatilities. The vapor-liquid equilibrium (VLE) relationship:

$$y_A = \frac{\alpha_{AB} x_A}{1 + (\alpha_{AB} - 1) x_A}$$

where $\alpha_{AB} = p_A^{sat}/p_B^{sat}$ is the **relative volatility** (ratio of vapor pressures, both from the Clausius-Clapeyron equation, Ch. 10 §10.9.3).

McCabe-Thiele method (graphical design for a distillation column):

1. Plot VLE curve y vs. x
2. Draw **rectifying operating line** (above feed): $y = (L/V)x + x_D(1 - L/V)$, slope = L/V (internal reflux ratio)
3. Draw **stripping operating line** (below feed)
4. **Step off stages** between VLE curve and operating lines from the distillate composition x_D to the bottoms composition x_B
5. Number of steps = number of theoretical stages

Minimum reflux ratio R_{min}: when the operating line passes through a pinch point with the VLE curve — infinite stages needed. **Actual reflux:** $R = 1.2–1.5\, R_{min}$ (practical trade-off: more reflux = fewer stages = shorter column, but more energy).

20.9.2 — NTU-HTU Method: The Transfer Unit Approach

For continuous-contact equipment (packed columns for absorption, stripping, liquid-liquid extraction), the number of theoretical stages is replaced by **Transfer Units (NTU)** — the NTU concept used for heat exchangers (Ch. 19 §19.6.3), applied here to a mass-transfer driving force. The organisation of the method is shared; the equations are not (§20.9.2 below).

NTU (gas-phase basis):

$$NTU_{OG} = \int_{y_1}^{y_2} \frac{dy}{y^* - y}$$

where y^* is the equilibrium gas-phase composition at the current liquid composition, and y is the actual gas composition.

Height of a Transfer Unit (HTU):

$$HTU_{OG} = \frac{G}{K_{ya}}$$

where G is the gas molar flux, K_y is the overall mass transfer coefficient, and a is the interfacial area per unit volume of packing.

Packed column height:

$$Z = NTU_{OG} \times HTU_{OG}$$

The correspondence with heat exchangers:

- Heat exchanger: $NTU = UA/C_{min}$, $\varepsilon = f(NTU, C_r)$
- Absorption column: $NTU_{OG} = K_{ya} Z / G$, efficiency $= f(NTU, mG/L)$ where mG/L plays the role of C_r

`[STRUCTURAL CONNECTION]` — **shared dimensionless design framework, not the same equation.** What genuinely transfers is the *organising idea*: divide the required driving-force change by the local driving force, count the number of increments, multiply by the size of one increment. That is why the tabulated ε–NTU curves and the "how many transfer units, how tall per unit" workflow carry over intact.

What does **not** transfer, and is where a careless reading goes wrong:

	Heat exchanger	Absorption column
Driving force	ΔT against a fixed $T_{h,\text{in}}$	$y^* - y$ against a *moving* VLE equilibrium
Equilibrium	constant T_h (single phase)	$y^* = f(x, T, P)$ — composition-dependent
Capacity rate	C_{min} of one stream	mG/L — a transformed, not identical, capacity
Transport coefficient	U [W/m²K]	K_{ya} [mol/m²s] — area *and* driving-force dependent
Failure mode	one side phase-changes	flooding, channeling, mass-transfer limitation

Both are ultimately diffusive transport (Ch. 13 §13.10), but they are driven by *different currents* with *different coefficients* against *different equilibria*. The framework is shared; the equations are not interchangeable, and the column's K_{ya} cannot be measured with an NTU chart borrowed from a heat-exchanger correlation.

Model Ledger — packed column to NTU design

Field	Content
Parent theory	Two-film mass transfer (Ch. 13 §13.6, §20.10.1); countercurrent diffusion
Reduction	steady state; constant molar flow rates; linearised driving force; uniform packing
Model	$Z = NTU_{OG} \times HTU_{OG}$, $HTU_{OG} = G/K_{ya}$
Assumptions	dilute solute; negligible pressure drop; constant T, P; no radial gradients; dilute-phase operation
Physics retained	composition-driven diffusion and the countercurrent gradient
Physics neglected	pressure drop, heat effects, non-dilute VLE curvature, channeling, flooding
Validity	ε within the tabulated range; column below flooding limit
Fails when	flooding onset, K_{ya} varies with composition, or heat effects shift y^*
Next model	rigorous stagewise solve; non-isothermal VLE; packed-bed pressure-drop coupling

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20.10 — Mass Transfer: From Molecular to Engineering Scale

20.10.1 — Film Theory and Resistances

At a gas-liquid interface, the two-film theory assumes:

- Thin stagnant film on the gas side (thickness δ_G)
- Thin stagnant film on the liquid side (thickness δ_L)
- Equilibrium at the interface: $y_{Ai} = m\, x_{Ai}$ (Henry's law)

Fluxes through each film (Fick's law, Ch. 13):

$$N_A = k_G(y_A - y_{Ai}) = k_L(x_{Ai} - x_A)$$

where $k_G = D_{AG}/\delta_G$ and $k_L = D_{AL}/\delta_L$ are the individual film coefficients.

Overall resistance (same logic as thermal resistances in series, Ch. 14 §14.7.2):

$$\frac{1}{K_G} = \frac{1}{k_G} + \frac{m}{k_L}$$

where m is the Henry's law constant (equilibrium slope). This is the two-resistance model in series. The **algebra** is identical to the overall heat transfer coefficient of Ch. 19 §19.6.1 — $1/U = 1/h_1 + R_{wall} + 1/h_2$ — because series addition of resistances is one piece of algebra. The **resistances** are not the same physical quantities: k_G and k_L are mass-transfer coefficients set by molecular diffusivity, whereas h is set by convection, and m is an equilibrium slope that has no thermal counterpart. `[STRUCTURAL CONNECTION]`.

The chemical potential μ is the effort variable for mass transfer — diffusion runs from high μ to low μ, in the same sense as heat flows from high T to low T. `[ANALOGY]`: the two effort variables are not interchangeable, and the driving force is $-\nabla\mu$ rather than $-k\nabla T$. In the ideal case, $\Delta\mu = RT\ln(y/y^*)$ is the driving force for absorption.

20.10.2 — Dimensionless Groups for Mass Transfer

Dimensionless group	Formula	Analog in heat transfer
Schmidt number Sc	ν/D_{AB} (momentum/mass diffusivity)	Prandtl number $Pr = \nu/\alpha$
Sherwood number Sh	$k_m L/D_{AB}$ (convective/diffusive mass transfer)	Nusselt number $Nu = hL/\kappa$
Mass transfer coefficient	$k_m = Sh \cdot D_{AB}/L$	$h = Nu \cdot \kappa/L$

Chilton-Colburn analogy: For turbulent flow over a flat plate:

$$j_D = j_H = f/2$$

where $j_D = Sh/Re \cdot Sc^{1/3}$, $j_H = Nu/Re \cdot Pr^{1/3}$, and f is the friction factor. Heat and mass transfer have equal efficiency when $Pr \approx Sc$ — which occurs for gases ($Pr \approx Sc \approx 0.7$). Liquids differ ($Pr \gg Sc$ for most organic solvents).

20.11 — Coupled CE-ChE Systems: Multi-Physics at Layer 3

20.11.1 — Wastewater Treatment as a Multi-Domain Network

A wastewater treatment plant couples:

- **Hydraulic domain** (flow between unit processes): pipe network (§20.3)
- **Chemical domain** (biological oxygen demand, nutrient removal): CSTR kinetics (§20.8)
- **Mass transfer** (oxygen transfer to activated sludge): film theory (§20.10)
- **Thermal domain** (temperature affects microbial activity via Arrhenius, §20.7)

The entire plant is a **bond graph** (Ch. 17 §17.5) in the hydraulic + chemical domains, with energy flows connecting to the thermal domain. The Arrhenius coupling means this is a nonlinear system — temperature changes affect kinetics, which affects oxygen demand, which affects aeration energy, which affects temperature.

20.11.2 — The Reactor-Separator Network

Many chemical processes alternate between reaction (CSTR or PFR) and separation (distillation column). The recycle of unconverted reactant creates a **closed loop**:

```
Feed → [Reactor] → [Separator] → Product
         ↑                ↓
         └── Recycle ─────┘
```

The recycle creates positive feedback (higher conversion per pass → more product in separator → more recycle → higher reactor inlet concentration → ...). This is analyzed using the same closed-loop transfer function as Ch. 21 control theory:

$$X_{overall} = \frac{X_{pass}}{1 - R(1 - X_{pass})}$$

where R is the recycle ratio and X_{pass} is the single-pass conversion. The same feedback algebra (Ch. 21 preview) governs both feedback control systems and chemical recycle loops.

20.12 — Where CE/ChE Layer 3 Fails

Failure mode	Physical cause	Required model	Return to
Soil liquefaction	Dynamic loading, loss of pore pressure	Nonlinear soil dynamics, wave propagation	Ch. 15, 16
Turbulent open channels	$Re \gg 1$, no laminar model	CFD (RANS, LES)	Ch. 15 §15.8
Non-ideal CSTR (RTD)	Bypassing, dead zones	Residence time distribution analysis	Ch. 16 (dispersion PDE)
Reactive distillation	Reaction + separation coupled	Rate-based simulation	Ch. 10, 13, 14
Membrane transport	Non-Fickian at high pressure	Solution-diffusion model	Ch. 13 §13.6
Non-equilibrium separation	Stage efficiency < 1	Mass transfer rate model	Ch. 20 §20.10
Large deflection structures	Geometric nonlinearity	Nonlinear FEM	Ch. 15
Seismic structural response	Dynamic wave propagation	Time-history analysis (FEM + Ch. 16)	Ch. 16

20.13 — Summary

Domain	Effort	Flow	R	C	L	Conservation law
CE Structural	Force F	Velocity v	Damping b	Compliance $1/k$	Mass m	Newton's 2nd law
CE Hydraulic (pipe)	Pressure P	Flow Q	$8\mu L/\pi r^4$	Compressibility	Inertance $\rho L/A$	Mass conservation
CE Open channel	Head H	Flow Q	Manning resistance	—	—	Continuity
CE Groundwater	Head h	Flux q	$1/K$	Storativity S/T	—	Darcy + continuity

Domain	Effort	Flow	R	C	L	Conservation law
ChE Chemical	Chem. potential μ	Molar flow $\dot{n}$	Diffusion resistance	Tank capacity	—	Mole balance

Key PDEs that appear literally in CE/ChE Layer 3:

PDE	CE/ChE appearance	Chapter origin	Kind
Diffusion $\partial_t u = c_v\, \partial_{xx} u$	Terzaghi consolidation (clay settlement)	Ch. 13, 16	mathematical identity
Diffusion $\partial_t h = (T/S)\nabla^2 h$	Groundwater (confined, transient)	Ch. 13, 16	mathematical identity, under stated storage assumptions
Beam equation $EI\, \partial_{xxxx} w = q$	Euler-Bernoulli beam deflection	Ch. 16 §16.7.2	mathematical identity
Advection–reaction $\partial_V F_A = r_A$	PFR design equation	Ch. 16	mathematical identity
Linear gradient law $q = -K\nabla h$	Darcy's law	Ch. 13 §13.6	constitutive correspondence
$k = A\, e^{-E_a/RT}$	Arrhenius kinetics	Ch. 10 §10.3.3	shared statistical origin
$Z = NTU \times HTU$	Packed-column design	Ch. 19 §19.6.3	shared dimensionless framework

Only the first four rows earn the word "identical" — and each one earns it conditionally, on the assumptions named in its own section. The last three rows are the ones that get over-claimed in textbooks, and they are the ones this chapter now labels explicitly (§20.0.1).

20.14 — Engineering Thread

Physics	Engineering application
Stiffness method $\mathbf{Ku} = \mathbf{F}$	Structural analysis of buildings, bridges, towers (SAP2000, ETABS)
Beam bending $\sigma = My/I$	Member sizing; serviceability deflection checks; composite slab design
Manning's equation	River flood routing; culvert design; storm drain sizing
Hydraulic jump	Spillway energy dissipator design; canal protection
Darcy's law	Groundwater contamination; dewatering systems; dam seepage
Terzaghi consolidation	Foundation settlement prediction; construction scheduling for embankments
Rational method + unit hydrograph	Stormwater management; culvert design; detention basin sizing
Arrhenius rate constant	Reaction temperature control; catalyst selection; reactor sizing
CSTR/PFR design equation	Reactor volume sizing; conversion optimization; operating conditions
NTU-HTU for columns	Packed absorption tower height; distillation column design
Two-resistance model $1/K_G$	Gas scrubber design; oxygen transfer in bioreactors
Chilton-Colburn analogy	Mass transfer prediction from heat transfer data

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20.15 — Looking Ahead

Chapter 20 completes the Layer-3 engineering coverage across all four branches. The inventory of what Layer-3 models are available:

- **EEE (Ch. 18):** Circuits, semiconductor devices, signals, power electronics
- **ME (Ch. 19):** Machines, vibrations, thermodynamic cycles, heat exchangers, fluid systems
- **CE (Ch. 20):** Structural analysis, open-channel hydraulics, groundwater, soil settlement
- **ChE (Ch. 20):** Process balances, reaction kinetics, reactor design, separations, mass transfer

Chapter 21 closes the book with the capstone: **feedback and control**. For any system that admits an appropriate linear time-invariant representation, control theory operates at the level of the transfer function $H(s)$ — above all physical domains simultaneously. A PID controller does not know whether it is controlling temperature, position, flow rate, or pH; it sees the error signal and the process transfer function, and nothing about the underlying physics. That claim is bounded, and Chapter 21 §21.0.2 is explicit about the bound: nonlinear, time-varying, hybrid, distributed, stochastic, and delayed systems are not described by one $H(s)$, and each needs a different representation before the same design machinery applies. Chapter 21 brings everything from the layer map into a unified closing argument for the book's central claim: physics, in the right limits, becomes engineering.

End of Chapter 20.

Next: Chapter 21 — Feedback, Control, and the Cross-Branch Capstone

Feedback, Control, and the Cross-Branch Capstone

The Branch-Agnostic Mathematics of Layer 3

> *The art of control engineering is the art of designing systems that behave well in spite of uncertainty. This art is the same whether the system is a chemical reactor, a jet engine, a bridge, or a power grid.* — paraphrased from Åström and Wittenmark, *Feedback Systems*

21.0 — The Capstone

Chapters 18–20 instantiated the Layer-3 template in four physical domains. In each, the governing equation took the same form, the same ω_0 and ζ appeared, the same Bode plot methodology applied, and Thévenin/Norton equivalents transferred directly.

This chapter goes one abstraction higher. **Control theory operates on the system model, not on the physics that produced it.** A PID controller does not know whether it is regulating temperature, angular velocity, chemical concentration, or electrical current. It sees only:

$$e(s) = R(s) - Y(s) \quad \text{and} \quad U(s) = C(s)\,e(s)$$

21.0.1 — What the capstone claim is, precisely

Here is the claim this chapter defends, stated at the strength it can actually carry:

> **For systems that admit an appropriate linear time-invariant representation, the transfer function provides a branch-independent mathematical language for analysis and control.**

Four qualifications are load-bearing, and each is developed below:

1. **The transfer function is a representation of *linear time-invariant input–output* behaviour.** It is not a representation of physics, and it is not universal.
2. **Not every engineering system admits one.** Nonlinear, time-varying, hybrid, distributed-parameter, stochastic, and constrained systems all need other formulations. §21.0.2 and §21.10 give the full list and the correct tool for each.
3. **Linearisation is itself a Bridge C operation**, with its own validity regime — not a free simplification. It is tagged `[APPROXIMATION]` throughout this chapter, and its Ledger is in §21.0.3.
4. **Where the transfer function does apply, the unification is exact and branch-independent.** This half of the claim is not weaker for the caveats above. A PID loop does genuinely not know what domain it is in.

21.0.2 — The boundary of the transfer function

A transfer function $P(s) = Y(s)/U(s)$ exists, in the ordinary sense, only when the system is **linear** (a fixed operator on the signals), **time-invariant** (that operator does not change with time), and has a **well-defined input–output map with no internal initial-condition dependence**. Each requirement has a real engineering meaning, and each has a price when violated.

System class	Why $H(s)$ fails	Correct formulation	Where
Nonlinear	Superposition fails; the "operator" depends on the operating point	Linearise about an operating point (local $H(s)$), or work with the nonlinear model directly	§21.0.3, §21.10
Time-varying	P is a function of time; there is no single $P(s)$	Time-domain simulation; linear time-varying (state-transition) matrices $\Phi(t, \tau)$	§21.10
Hybrid	Continuous dynamics + discrete mode switches; no single operator	Hybrid automaton; switched systems; mode-dependent A, B	§21.10
Distributed parameter	Infinitely many degrees of freedom; $H(s)$ is a boundary-value problem, not a rational function	Discretise (FEM/FDM) then control the finite model, or H_∞/PDE control	§21.10, Ch. 17
Stochastic	The output is a random process; $Y(s)/U(s)$ is not a deterministic ratio	State-space with process/measurement noise; Kalman filter, LQG	§21.7.4, §21.10
Constrained	Feasible inputs are a subset of the domain; an unconstrained $H(s)$ has no meaning there	MPC, which optimises over the model subject to constraints	§21.9.3
Non-minimum phase (RHP zeros)	$H(s)$ exists, but the inverse does not	H_∞ loop shaping, waterbed-aware design	§21.6, §21.10
Pure time delay	$e^{-\theta s}$ is not a rational function of s	Padé approximation, Smith predictor, dead-time compensator	§21.8.2, §21.10
Nonlinear / chaotic	Sensitive dependence on initial conditions; no useful linearisation	Lyapunov methods, feedback linearisation, nonlinear observers, chaos control	§21.10

The correct statement of the capstone is therefore not *"all engineering problems reduce to transfer functions."* It is:

```
Layer 3 physical system
        │
        │  [APPROXIMATION] linearise about an operating point
        │  [APPROXIMATION] lump / discretise (Bridge C)
        ▼
   LTI plant model P(s)   ◄── only if the system is in the class
        │                       where this is defensible
        ▼
  transfer-function design: Bode, root locus, PID, margins
```

Everything in §§21.1–21.6 lives inside that box. Everything in §21.7 (state-space) is the formulation that *does* generalise to multi-input/multi-output, and §21.7.5 onward extends to stochastic and constrained cases. §21.10 is the honest accounting of what is left over.

21.0.3 — The linearisation step, tagged

Linearisation is the operation that makes the transfer function possible, and it is where most of the over-claiming in cross-domain control narratives happens. It belongs to Bridge C, and it is an approximation with a named regime:

> **[APPROXIMATION]** Linearisation. If the plant is $\dot{x} = f(x, u)$ with f smooth, then near an equilibrium x_0, u_0 with $f(x_0, u_0) = 0$, write $x = x_0 + \delta x$, $u = u_0 + \delta u$, and discard all terms of second and higher order in δx, δu. The result is $\delta\dot{x} = A\,\delta x + B\,\delta u$ with $A = \partial f/\partial x|_{x_0}$, $B = \partial f/\partial u|_{x_0}$.
>
> **Small parameter:** the ratio of the excursion to the local linearisation radius. **Valid when:** the signal stays inside the region where the linear model tracks the nonlinear one — typically a few percent of full scale, and always domain-specific. **Fails when:** the excursion leaves that region, at which point $P(s)$ is no longer the plant and the loop gains you designed are no longer the loop you have.

Model Ledger — LTI plant transfer function

Property	Description
Parent theory	The nonlinear (and usually distributed) Layer-3 model, e.g. Navier–Stokes + energy balance, or the CSTR material and energy balances (Ch. 20 §20.8)
Reduction	Bridge C: control-volume lumping, then linearisation about a nominal operating point
Model	P(s) = Y(s)/U(s), or the equivalent ẋ = Ax + Bu, y = Cx + Du
Assumptions	Dynamics linear over the signal range; coefficients time-invariant; initial conditions do not affect the input–output map; the operating point is an equilibrium and stays near it
Retained	First-order sensitivity of output to input about the operating point; the modes that matter locally
Neglected	All nonlinear coupling; drift of the operating point; unmodelled dynamics outside the linearisation region; measurement and process noise (unless added explicitly)
Valid when	Small-signal operation about a stable equilibrium; bandwidth low enough that the neglected higher-order terms stay small
Fails when	Large excursions, saturation, hysteresis, dead zones, sign changes, or a moving operating point
Next model	Gain scheduling (one local model per operating point), feedback linearisation, or the nonlinear model itself with Lyapunov / MPC design

This is the single most important caveat in the chapter, and it is the reason the chapter is not as aggressive as a first reading suggests. A transfer function is a *local* description of a system near a chosen point, obtained by two named approximations. The mathematics downstream of it is exact; the step that produced it is not.

21.1 — Feedback: The Core Concept

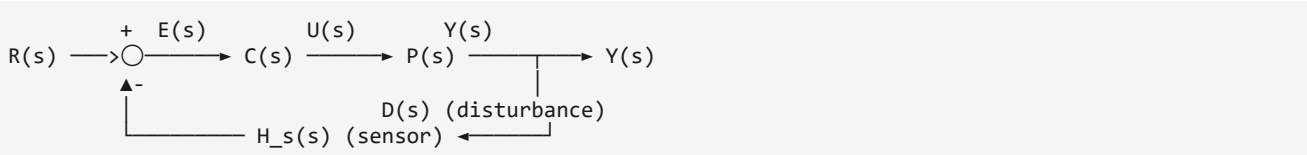
21.1.1 — Why Open-Loop Control Is Insufficient

An **open-loop** system applies a predetermined input and hopes the output follows. It fails when:

1. **Disturbances** enter the system (load changes, ambient temperature, etc.)
2. **Model uncertainty** means the plant $P(s)$ is not exactly known
3. **Nonlinearity** means the linear model is only locally valid

Example: A heater set to 50% power. If room temperature drops or a window opens, the controlled temperature drops. The controller has no information that anything has changed.

Closed-loop (feedback) control measures the output Y and adjusts the input U based on the error $E = R - Y$:



The **closed-loop transfer function** from reference R to output Y:

$$\boxed{T(s) = \frac{C(s)P(s)H_s(s)}{1 + C(s)P(s)H_s(s)}}$$

For ideal sensors ($H_s = 1$): $T(s) = CP/(1 + CP)$.

The **sensitivity function** $S(s) = 1/(1 + CP)$ characterizes how well the system rejects disturbances D entering at the plant input:

$$Y_{dist}(s) = \frac{P(s)}{1 + C(s)P(s)} D(s) = S(s)P(s)D(s)$$

Small S (high loop gain $|CP| \gg 1$) → strong disturbance rejection.

21.1.2 — The Fundamental Feedback Trade-Off

High gain ($|C(s)P(s)| \gg 1$):

- Good tracking: $Y \approx R$
- Good disturbance rejection: $Y_{dist} \approx 0$
- Sensitive to model uncertainty in $C(s)$
- Risk of instability

Low gain:

- Poor tracking and regulation
- Stable, robust to model errors

Control design is the art of achieving adequate performance while maintaining adequate stability margins — navigating this trade-off with mathematical precision.

21.2 — Block Diagram Algebra

21.2.1 — Reduction Rules

Any linear control system can be represented as a block diagram and reduced to a single transfer function using four rules:

Series: $H_{total}(s) = H_1(s) \cdot H_2(s)$

Parallel: $H_{total}(s) = H_1(s) + H_2(s)$

Negative feedback loop:

$$H_{total}(s) = \frac{G(s)}{1 + G(s)H(s)}$$

Moving a summing junction past a block G: replace the block on the branch being moved with $1/G$ (moving forward) or G (moving backward).

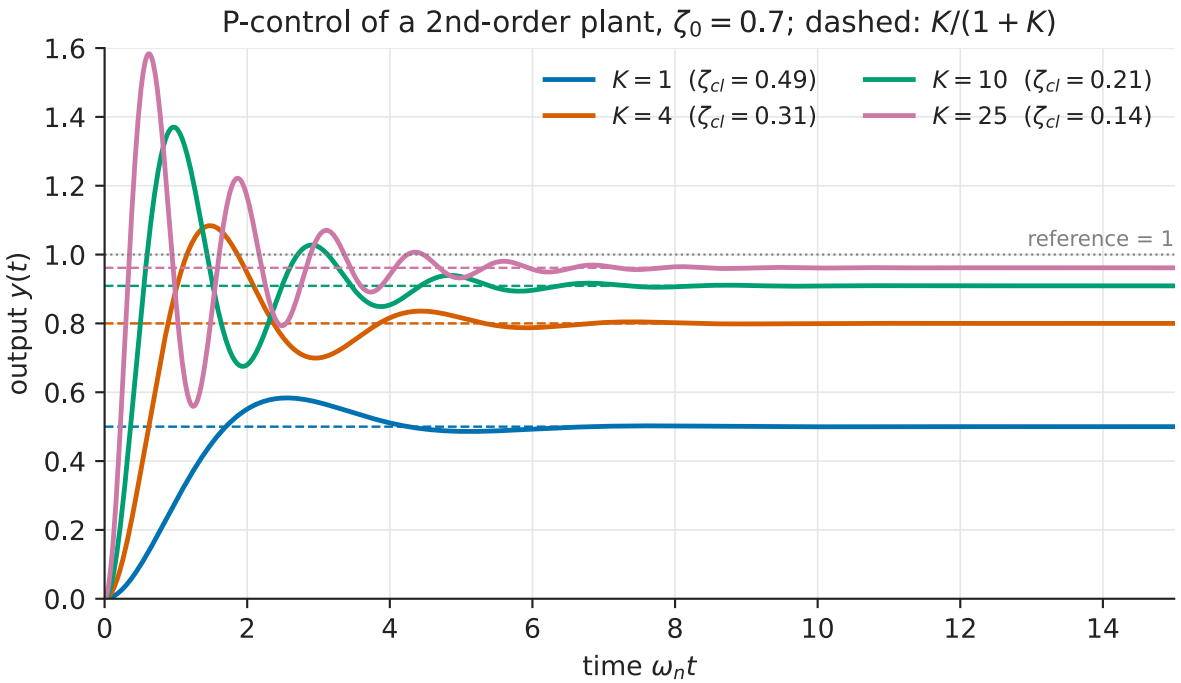
21.2.2 — Standard Second-Order Closed-Loop Form

For a proportional controller $C = K$ and a second-order plant $P(s) = \omega_n^2/(s^2 + 2\zeta_0\omega_n s + \omega_n^2)$:

$$T(s) = \frac{K\omega_n^2}{s^2 + 2\zeta_0\omega_n s + (1+K)\omega_n^2}$$

Closed-loop natural frequency: $\omega_{n,cl} = \omega_n\sqrt{1+K}$. Closed-loop damping: $\zeta_{cl} = \zeta_0/\sqrt{1+K}$.

Higher gain → higher bandwidth (faster) but lower damping (more oscillatory). This is the fundamental speed-stability trade-off, visible directly from the second-order template of Ch. 17 §17.7.



Unit-step responses of $T(s) = K\omega_n^2/[s^2 + 2\zeta_0\omega_n s + (1+K)\omega_n^2]$ with $\zeta_0 = 0.7$ for $K = 1, 4, 10, 25$ (time in units of $1/\omega_n$): raising K speeds the response and lifts the final value $K/(1+K)$ (dashed) towards the reference, but $\zeta_{cl} = \zeta_0/\sqrt{1+K}$ falls and the overshoot grows — and a proportional loop never removes

the steady-state error.

21.3 — Stability Theory

21.3.1 — BIBO Stability and Pole Locations

A system is **BIBO (Bounded-Input Bounded-Output) stable** iff all poles of the closed-loop transfer function $T(s)$ lie in the **open left half-plane** ($\text{Re}[s_k] < 0$).

From Ch. 17 §17.7.2: poles with $\text{Re}[s] < 0$ give decaying transients ($e^{\sigma t} \to 0$ as $t \to \infty$ for $\sigma < 0$). Any pole in the right half-plane gives a growing transient — the system diverges.

21.3.2 — Routh-Hurwitz Criterion

Given a characteristic polynomial:

$$\Delta(s) = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0$$

Form the **Routh array** by arranging coefficients:

$$\begin{array}{c|cccc} s^n & a_n & a_{n-2} & a_{n-4} & \cdots \\ s^{n-1} & a_{n-1} & a_{n-3} & a_{n-5} & \cdots \\ s^{n-2} & b_1 & b_2 & b_3 & \cdots \\ s^{n-3} & c_1 & c_2 & c_3 & \cdots \\ \vdots & \vdots & & & \\ s^0 & \star & & & \end{array}$$

where $b_1 = (a_{n-1}a_{n-2} - a_n a_{n-3})/a_{n-1}$, $b_2 = (a_{n-1}a_{n-4} - a_n a_{n-5})/a_{n-1}$, etc.

Routh-Hurwitz theorem: The number of closed-loop poles in the right half-plane equals the number of sign changes in the first column of the Routh array. For stability: all first-column entries must be positive.

Special cases:

- Zero in first column (non-zero row): replace with small $\epsilon > 0$ and proceed
- All-zero row: the polynomial has roots placed symmetrically about the origin (real pairs $\pm\sigma$, imaginary pairs $\pm j\omega$, or complex quadruplets); form the auxiliary polynomial from the row above, differentiate it to replace the zero row, and proceed — imaginary-axis pairs give marginal stability, the other cases instability

Example: Third-order system $\Delta(s) = s^3 + a_2 s^2 + a_1 s + a_0$

Routh array:

$$\begin{array}{c|cc} s^3 & 1 & a_1 \\ s^2 & a_2 & a_0 \\ s^1 & a_1 - a_0/a_2 & 0 \\ s^0 & a_0 & \end{array}$$

Stability conditions: $a_2 > 0$, $a_0 > 0$, and $a_1 a_2 > a_0$.

21.3.3 — Gain and Phase Margins: The Bode Stability Criterion

For the **open-loop transfer function** $L(s) = C(s)P(s)$, the Bode plot ($|L(j\omega)|$ and $\angle L(j\omega)$ vs. ω) reveals two critical frequencies:

- **Gain crossover frequency** ω_{gc}**:** where $|L(j\omega_{gc})| = 1$ (0 dB)
- **Phase crossover frequency** ω_{pc}**:** where $\angle L(j\omega_{pc}) = -180^\circ$

Phase margin (PM): How much additional phase lag before instability:

$$PM = 180^\circ + \angle L(j\omega_{gc})$$

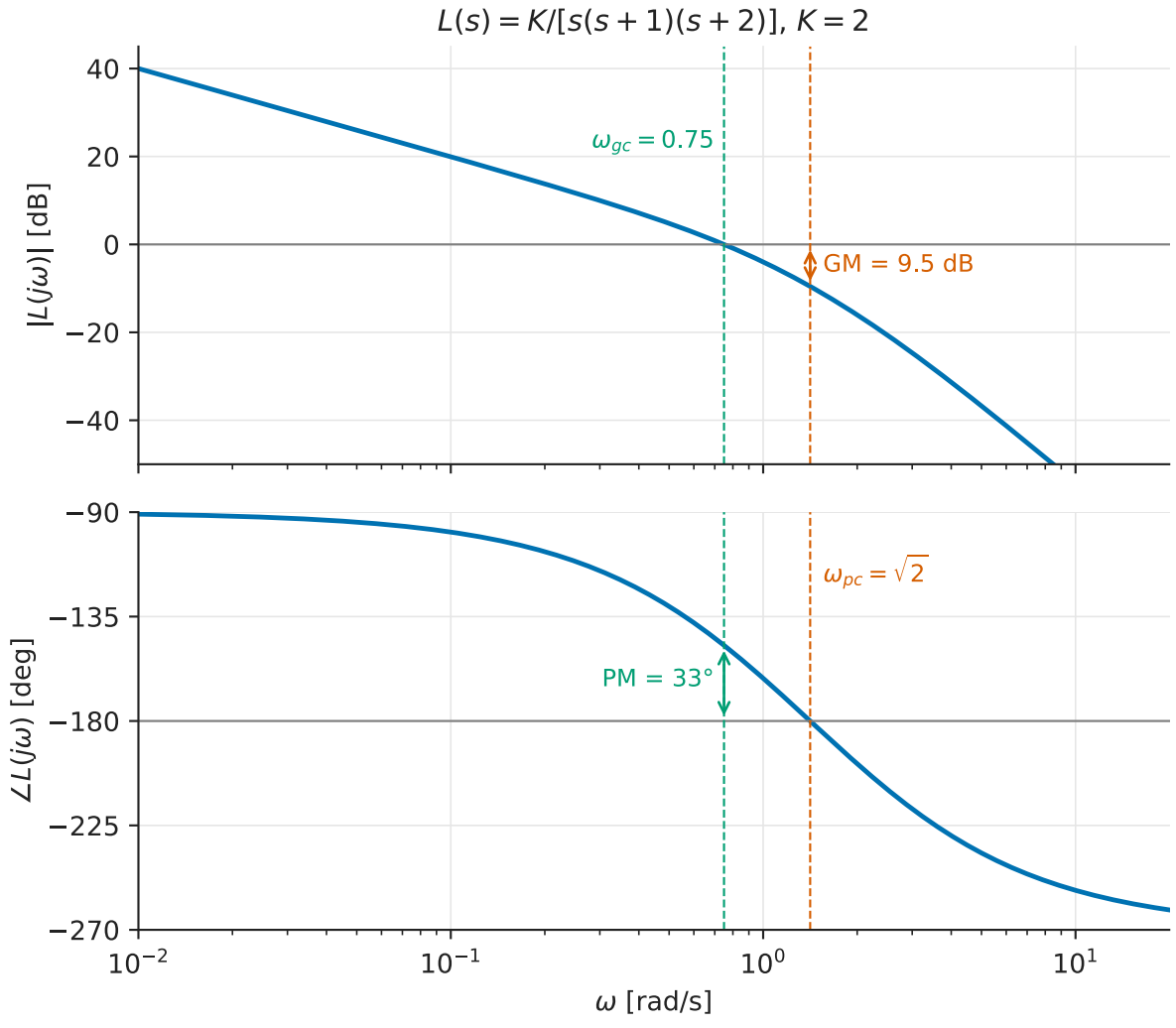
Gain margin (GM): How much the gain can be increased before instability:

$$GM = 20\log_{10}\left(\frac{1}{|L(j\omega_{pc})|}\right) \text{ dB}$$

Stability rules of thumb:

- $PM > 30^\circ$ (robust: $> 45^\circ$)
- $GM > 6$ dB (robust: > 12 dB)

Systems with adequate PM and GM are robust to model uncertainty — if the real $P(s)$ differs from the model, the system still remains stable.



Bode plot of $L(s) = K/[s(s+1)(s+2)]$ with $K = 2$: the gain crossover is at $\omega_{gc} \approx 0.75$ rad/s with PM $\approx 33^\circ$, and the phase crosses -180° at $\omega_{pc} = \sqrt{2}$ rad/s where $|L| = K/6$, giving GM $= 20\log_{10}(6/K) \approx 9.5$ dB — adequate by the rules of thumb above, though short of "robust".

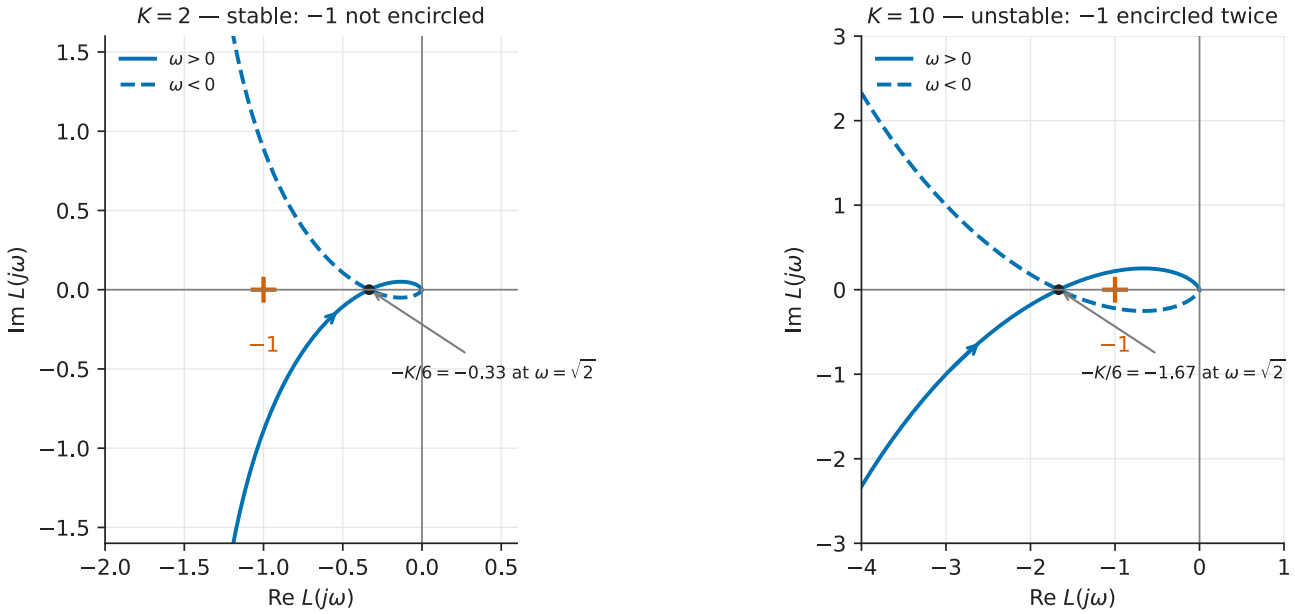
21.3.4 — The Nyquist Stability Criterion

The Nyquist criterion is exact (not approximate like Bode margin rules). Plot the polar (Nyquist) plot of $L(j\omega)$ for $\omega \in (-\infty, +\infty)$.

Nyquist theorem: The number of unstable closed-loop poles equals the number of open-loop unstable poles plus the number of clockwise encirclements of the critical point $(-1, 0)$ by the Nyquist plot.

For a stable open-loop plant: the closed-loop is stable iff the Nyquist plot does not encircle $(-1, 0)$.

Engineering insight: The $(-1, 0)$ point corresponds to gain = 1 AND phase = -180°. If $L = -1$, the feedback signal is exactly equal and opposite to the reference, causing sustained oscillation (marginal stability). Encircling it means the system can go unstable.



Nyquist plot of $L(j\omega) = K/[j\omega(j\omega+1)(j\omega+2)]$ (solid $\omega > 0$, dashed $\omega < 0$; the closing arc at infinity from the pole at $s = 0$ lies off-scale): the real-axis crossing at $\omega = \sqrt{2}$ is $-K/6$, so $K = 2$ leaves -1 unencircled (stable, GM $= 3\times$), while $K = 10$ crosses at -1.67 and encircles -1 twice clockwise (two right-half-plane closed-loop poles).

A note on the word "topological" here: [STRUCTURAL CONNECTION] The encirclement count is an integer that cannot change under a continuous deformation of the contour unless the contour crosses a pole or zero — in that sense it is a topological invariant of the map $L(j\omega)$. This is the same *kind* of mathematical object as a Chern number or the QCD θ-term's Pontryagin number (an integer classifying a map, insensitive to smooth deformation), but it is not derived from either of them, and lives on a completely different space (the complex plane traced out by an engineering transfer function, not a physical field configuration). See §21.11.2 for where this book makes that distinction explicit.

21.4 — The PID Controller

21.4.1 — The Three Terms

The **Proportional-Integral-Derivative (PID) controller** in time and frequency:

$$u(t) = \underbrace{K_p\, e(t)}_{\text{proportional}} + \underbrace{K_i \int_0^t e(\tau)\, d\tau}_{\text{integral}} + \underbrace{K_d \frac{de}{dt}}_{\text{derivative}}$$

$$C(s) = K_p + \frac{K_i}{s} + K_d s = K_p \left(1 + \frac{1}{T_i s} + T_d s\right)$$

where $T_i = K_p/K_i$ is the integral time constant and $T_d = K_d/K_p$ is the derivative time constant.

Each term's role:

Proportional (K_p): Reduces error in proportion to its magnitude. Cannot eliminate steady-state error for a constant disturbance (residual error = $1/(1 + K_p G_{DC})$).

Integral (K_i): Accumulates error over time; drives steady-state error to zero (because if $e \neq 0$, u keeps changing until it eliminates e). Adds a pole at $s = 0$ — one more integrator in the loop.

Derivative (K_d): Responds to the rate of change of error; anticipates future error and provides a damping-like effect. Sensitive to noise (high-frequency amplification from the s term) → always used with a low-pass filter in practice.

21.4.2 — Steady-State Error and System Type

The steady-state error for a unit step input depends on the **system type** — the number of open-loop integrators in $L(s)$:

System type	Integrators in $L(s)$	Error to step $r(t) = 1$	Error to ramp
Type 0	0	$1/(1 + K_{pos})$	∞
Type 1	1 (e.g., with I action)	0	$1/K_v$
Type 2	2	0	0

Here $K_{pos} = \lim_{s\to 0} L(s)$ is the position error constant (not the PID gain K_p) and $K_v = \lim_{s\to 0} sL(s)$ the velocity error constant.

Adding an integrator (I action in PID, or an integrating plant) makes the system Type 1 → zero steady-state error for constant inputs. This is the primary reason for the integral term in PID.

21.4.3 — Ziegler-Nichols Tuning

A purely empirical tuning method [PHENOMENOLOGICAL — requires no plant model]:

Step 1: Set $K_i = 0$, $K_d = 0$. Increase K_p until the output oscillates with constant amplitude → record the **ultimate gain** K_u and **ultimate period** T_u.

Step 2: Apply the tuning formulas:

Controller	K_p	T_i	T_d
P	$0.5K_u$	—	—
PI	$0.45K_u$	$T_u/1.2$	—
PID	$0.6K_u$	$T_u/2$	$T_u/8$

Z-N tuning gives PM $\approx 30^\circ$ — adequate but not robust. Fine-tune from the Z-N starting point using simulation or Bode analysis.

ITAE (Integral of Time-weighted Absolute Error) tuning gives better responses: minimizes $\int_0^\infty t|e(t)|\,dt$, prioritizing reduction of error at later times.

21.4.4 — PID in Every Engineering Domain

The controller transfer function $C(s) = K_p(1 + 1/T_i s + T_d s)$ is identical across all domains. Only the plant model $P(s)$ and the physical meaning of e, u, and y change:

Domain	Error e	Controller output u	Controlled variable y
EEE (temperature)	$T_{set} - T_{meas}$ [K]	Heater power [W]	Temperature [K]
EEE (motor speed)	$\omega_{set} - \omega_{meas}$ [rad/s]	Armature voltage [V]	Angular velocity [rad/s]
ME (position)	$x_{set} - x_{meas}$ [m]	Force or motor torque [N]	Position [m]
ME (pressure)	$P_{set} - P_{meas}$ [Pa]	Valve opening [%]	Pressure [Pa]
CE (water level)	$h_{set} - h_{meas}$ [m]	Pump speed [rpm]	Level [m]
ChE (pH)	$pH_{set} - pH_{meas}$	Acid/base flow [L/min]	pH
ChE (reactor T)	$T_{set} - T_{meas}$ [K]	Coolant flow [kg/s]	Reactor temperature [K]

The controller design method, the tuning procedure, the stability analysis — all identical regardless of the row. This is the book's central claim in its most concentrated form.

21.5 — Root Locus: Tracking Poles as Gain Varies

21.5.1 — The Root Locus Concept

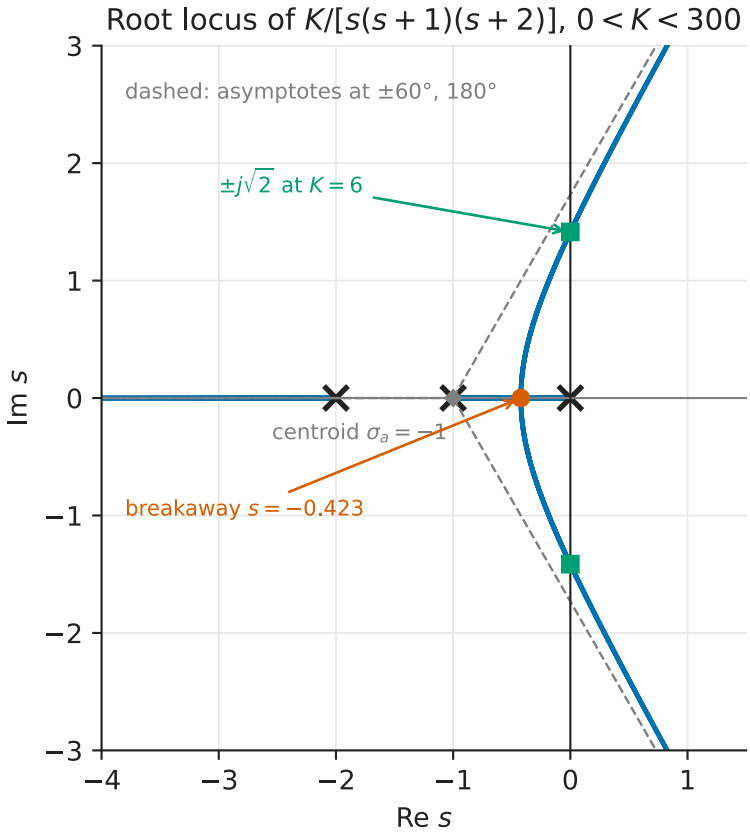
For a proportional controller $C = K$ with plant $P(s)$, the closed-loop characteristic equation: $1 + KP(s) = 0$, or $P(s) = -1/K$.

As K varies from 0 to ∞, the closed-loop poles trace the **root locus** — paths in the complex plane from the open-loop poles ($K = 0$) to the open-loop zeros ($K = \infty$) or to infinity.

Root locus construction rules (for $P(s)$ with n poles and m zeros, $n > m$):

1. **Number of branches:** n branches, one starting at each open-loop pole
2. **Start/end:** Branches start ($K = 0$) at open-loop poles; end ($K \to \infty$) at open-loop zeros (m branches) or ∞ (n-m branches)
3. **Real axis:** Locus exists on real axis to the left of an odd number of real poles and zeros
4. **Asymptotes:** $(n - m)$ asymptotes at angles $\theta_k = (2k + 1)180^\circ/(n - m)$, emanating from the centroid $\sigma_a = (\sum \text{poles} - \sum \text{zeros})/(n - m)$

5. **Breakaway/break-in:** Points where branches leave/enter the real axis, found from $dK/ds = 0$
6. **Imaginary axis crossing:** Found from Routh-Hurwitz (gives the gain K for marginal stability)



Closed-loop poles of $s^3 + 3s^2 + 2s + K = 0$ as K rises from 0 to 300, starting at the open-loop poles $0, -1, -2$: the branches from 0 and -1 meet and break away at $s = -1 + 1/\sqrt{3} \approx -0.423$, follow the $\pm 60^\circ$ asymptotes from the centroid $\sigma_a = -1$ (the third runs to $-\infty$ along 180°), and cross the imaginary axis at $\pm j\sqrt{2}$ when $K = 6$, the Routh–Hurwitz limit $a_1 a_2 > a_0$.

21.5.2 — Design by Root Locus

Design objective: Choose K (or a more complex $C(s)$) to place the dominant closed-loop poles at desired locations in the s-plane.

Desired pole locations from performance specs:

- Settling time $t_s \approx 4/\sigma$ where $\sigma = \zeta\omega_n = \text{Re}[s_{pole}]$
- Damping ratio $\zeta = \cos(\angle s_{pole})$, the angle measured from the negative real axis
- Natural frequency $\omega_n = |s_{pole}|$

Lead compensator $C(s) = K(s + z_c)/(s + p_c)$ with $p_c > z_c > 0$: adds phase lead (positive phase contribution) at the design frequency, allowing the root locus to pass through the desired pole locations.

Lag compensator $C(s) = K(s + z_c)/(s + p_c)$ with $z_c > p_c > 0$: adds a near-integrator, improving low-frequency gain and reducing steady-state error without significantly changing the high-frequency dynamics.

21.6 — Frequency Domain Design

21.6.1 — Shaping the Loop Transfer Function

The open-loop Bode plot of $L(j\omega) = C(j\omega)P(j\omega)$ directly reveals closed-loop performance:

- **Low frequency ($\omega < \omega_{gc}$):** High $|L|$ → good tracking and disturbance rejection. Slope should be -20 dB/decade or steeper.
- **Crossover region ($\omega \approx \omega_{gc}$):** Should have -20 dB/decade slope and PM $> 45^\circ$. Steeper slope → lower PM → more oscillatory.
- **High frequency ($\omega > \omega_{gc}$):** Roll off fast to reject noise and sensor noise.

The Bode integral (Bode's sensitivity integral):

$$\int_0^\infty \ln|S(j\omega)|\, d\omega = \pi \sum_k \mathrm{Re}[p_k]$$

where p_k are the open-loop unstable poles; the result assumes a stable closed loop and an open loop $L(s)$ of relative degree ≥ 2. If there are no unstable poles (open loop stable), the integral is zero — meaning **you cannot reduce sensitivity at some frequencies without increasing it at others.** Suppressing disturbances in one frequency range amplifies them in another. This is a fundamental constraint on what feedback can achieve.

21.6.2 — Lead-Lag Compensator Design

Phase lead compensator: $C(s) = K_c \dfrac{s+z}{s+p}$ with $p > z$

- Adds positive phase (phase lead) between z and p
- Maximum phase lead: $\phi_{max} = \arcsin\left(\dfrac{p-z}{p+z}\right)$ at $\omega_{max} = \sqrt{pz}$
- Use to increase phase margin by ϕ_{add} degrees: choose z, p to give $\phi_{max} \approx \phi_{add} + 5^\circ$ (5° extra for gain change)

Phase lag compensator: $C(s) = K_c \dfrac{s+z}{s+p}$ with $z > p$

- Increases low-frequency gain → reduces steady-state error
- Reduces phase margin → place pole/zero far below ω_{gc} to minimize PM loss
- Use $z/p =$ required low-frequency gain boost

Design procedure:

1. Plot uncompensated Bode; identify deficiencies in PM, GM, or steady-state error
2. Add lead for insufficient PM; add lag for insufficient steady-state accuracy
3. Iterate: a lead changes ω_{gc} and thus may affect the lag design

---

21.7 — State-Space Representation: Modern Control

21.7.1 — The State-Space Model

Any n-th order LTI system:

$$\boxed{\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u, \qquad y = \mathbf{C}\mathbf{x} + Du}$$

where $\mathbf{x} \in \mathbb{R}^n$ is the state vector, u is the input, y is the output.

Relationship to transfer function:

$$P(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + D$$

Eigenvalues of $\mathbf{A}$ = poles of $P(s)$ = natural frequencies of the system, for a minimal (controllable and observable) realization; otherwise some eigenvalues cancel and do not appear as poles.

Physical interpretation:

- State $\mathbf{x}$: minimum set of variables that fully characterizes the system's future (e.g., for RLC: $x_1 = V_C$, $x_2 = I_L$)
- Each element of $\mathbf{x}$ corresponds to an energy storage element (C or L)

21.7.2 — Controllability and Observability

Controllability: Can any state $\mathbf{x}$ be driven to zero in finite time by choosing $u(t)$?

$$\mathbf{W}_c = [\mathbf{B},\ \mathbf{A}\mathbf{B},\ \mathbf{A}^2\mathbf{B},\ \cdots,\ \mathbf{A}^{n-1}\mathbf{B}]$$

System is controllable iff $\text{rank}(\mathbf{W}_c) = n$.

Observability: Can the initial state $\mathbf{x}(0)$ be determined from the output history $y(0), y(1), \ldots$?

$$\mathbf{W}_o = \begin{bmatrix} \mathbf{C} \\ \mathbf{C}\mathbf{A} \\ \vdots \\ \mathbf{C}\mathbf{A}^{n-1} \end{bmatrix}$$

System is observable iff $\text{rank}(\mathbf{W}_o) = n$.

Engineering meaning:

- Uncontrollable modes: the input cannot reach certain states — physical modes (e.g., a disconnected circuit element) that the actuator cannot influence
- Unobservable modes: certain states cannot be inferred from the output — physical modes that don't affect the sensor

21.7.3 — State Feedback and Pole Placement

If the system is controllable, choose $u = -\mathbf{K}\mathbf{x}$ (full state feedback):

$$\dot{\mathbf{x}} = (\mathbf{A} - \mathbf{B}\mathbf{K})\mathbf{x}$$

The closed-loop poles are the eigenvalues of $(\mathbf{A} - \mathbf{BK})$. By **pole placement** (Ackermann's formula or direct design), choose $\mathbf{K}$ to put the closed-loop poles at any desired locations — exactly analogous to choosing K in root locus but for MIMO systems.

21.7.4 — The State Observer (Luenberger Observer)

When all states are not directly measurable (common in practice), estimate them:

$$\dot{\hat{\mathbf{x}}} = \mathbf{A}\hat{\mathbf{x}} + \mathbf{B}u + \mathbf{L}(y - \mathbf{C}\hat{\mathbf{x}})$$

The observer gain $\mathbf{L}$ drives $\hat{\mathbf{x}} \to \mathbf{x}$ asymptotically. Observer error dynamics: $\dot{\mathbf{e}} = (\mathbf{A} - \mathbf{LC})\mathbf{e} \to$ place observer poles (eigenvalues of $\mathbf{A} - \mathbf{LC}$) to the left of the controller poles (rule of thumb: 3–5× faster).

Separation principle: For linear systems, design the state feedback $\mathbf{K}$ and observer gain $\mathbf{L}$ independently — their combined performance is identical to if all states were measured directly.

21.7.5 — LQR: Optimal State Feedback

The **Linear-Quadratic Regulator (LQR)** chooses $\mathbf{K}$ to minimize:

$$J = \int_0^\infty (\mathbf{x}^T\mathbf{Q}\mathbf{x} + u^T\mathbf{R}u)\, dt$$

where $\mathbf{Q} \geq 0$ penalizes state deviation and $\mathbf{R} > 0$ penalizes control effort. Solution: $\mathbf{K} = \mathbf{R}^{-1}\mathbf{B}^T\mathbf{P}$ where $\mathbf{P}$ solves the algebraic Riccati equation:

$$\mathbf{A}^T\mathbf{P} + \mathbf{PA} - \mathbf{PBR}^{-1}\mathbf{B}^T\mathbf{P} + \mathbf{Q} = 0$$

The LQR automatically guarantees GM ≥ 6 dB and PM $\geq 60^\circ$ for single-input systems — built-in robustness from optimal design.

21.8 — Cross-Branch Worked Examples

Each example below follows the same six-step shape, and each records its validity limits:

```
starting theory  →  approximation  →  mathematical reduction
                 →  reduced model  →  engineering equation
                 →  validity limits
```

21.8.1 — Example 1: DC Motor Speed Control (EEE + ME)

Starting theory. A DC motor is a device that converts electrical energy to mechanical torque. From the electromagnetic side it is a two-port: voltage across the armature produces a current, and rotation produces a back-EMF. From the mechanical side it is a rotor inertia with viscous friction and a load.

Approximation. (a) Linear magnetics — the torque constant K_t and back-EMF constant K_e are constant, valid away from magnetic saturation and brush contact changes. (b) Lumped armature — the armature winding is treated as a single R_a–L_a branch, valid when the winding is electrically small compared with the EM wavelength it operates at. (c) Linearised mechanics — friction is taken as viscous, $T_f = b\,\omega$, valid for a fixed hydrodynamic/brush regime.

Mathematical reduction. Kirchhoff's voltage law on the armature plus Newton's second law on the rotor:

$$L_a \frac{di_a}{dt} = V_a - K_e\omega, \qquad J\frac{d\omega}{dt} = K_t i_a - b\omega$$

Reduced model. Eliminating i_a from the two equations gives a single second-order ODE in ω, whose Laplace transform is:

$$P(s) = \frac{\Omega(s)}{V_a(s)} = \frac{K_t}{(L_a s + R_a)(Js + b) + K_t K_e}$$

with electrical time constant $\tau_e = L_a/R_a$ and mechanical time constant $\tau_m = R_a J/(R_a b + K_t K_e)$ (the bare J/b, shortened by back-EMF damping).

Engineering equation. Further, for $\tau_e \ll \tau_m$:

$$P(s) \approx \frac{K_m}{\tau_m s + 1}, \qquad K_m = \frac{K_t}{R_a b + K_t K_e}$$

There is no free integrator in the speed transfer function: the back-EMF is internal speed feedback that gives a finite DC gain K_m (a constant voltage settles to a constant speed). The integrator appears only in the position transfer function $\Theta(s)/V_a(s) = P(s)/s$. What open-loop speed control cannot do is hold speed against load torque and parameter drift.

PID controller design:

1. Plot uncompensated Bode: flat at K_m, then -20 dB/dec beyond $1/\tau_m$ (-40 dB/dec beyond $1/\tau_e$)
2. Proportional only: ample PM, but a steady-state speed error of $1/(1 + K_p K_m)$
3. Add integral action (PI) to remove the offset; add lead/derivative only if PM is eroded — or tune PID directly with Z-N

With PID: The integral term handles steady-state error (speed offset under load); derivative term improves transient response (reduces settling time).

Model Ledger — DC motor speed plant

Property	Description
Parent theory	Two-port electromechanical coupling (Ch. 18 §18.x machines + Ch. 19 §19.x rotational dynamics)
Reduction	Lumped armature R_a–L_a; linear magnetics; linearised (viscous) friction; τ_e ≪ τ_m
Model	P(s) = K_m / (τ_m s + 1)
Assumptions	K_t, K_e constant (no saturation); armature electrically small; friction ∝ ω; no cogging; rigid coupling to a load that can be lumped into J and b
Retained	Back-EMF feedback (sets K_m, shortens τ_m); rotor inertia; electrical and mechanical time constants; load torque as a disturbance
Neglected	Magnetic saturation and hysteresis; brush contact dynamics and dead band; torque ripple from commutation; elastic shaft compliance; gearbox backlash; thermal derating of R_a
Valid when	Continuous operation near a nominal speed and duty cycle; τ_e ≪ τ_m; currents below the saturation knee

Property	Description
Fails when	Starting (large slip, $K_e\omega$ negligible, current-limited — the linear model is optimistic there); heavy load transients; high speed into field weakening; brushless/PMSM machines (different torque law, $T \propto i \rightarrow T \propto i_q$)
Next model	Full nonlinear ẋ = f(x,u); field-weakening control; gearbox as a separate compliant stage; PMSM/BLDC torque–current–flux maps

Cross-branch coupling. This motor drives a gear and load: the mechanical Layer 3 (Ch. 19) and electrical Layer 3 (Ch. 18) are coupled through the motor — exactly a gyrator (GY element) in the bond graph (Ch. 17 §17.5). The coupling is a `[STRUCTURAL CONNECTION]`: the motor is a power-converting element that maps a current-flow pair to a force-flow pair, and that mapping is the same object in Ch. 17's template regardless of the two domains involved.

21.8.2 — Example 2: Chemical Reactor Temperature Control (ChE)

Starting theory. A jacketed CSTR (Ch. 20 §20.8.1) with an exothermic reaction. Two coupled nonlinear ODEs: a material balance on the reacting species and an energy balance on the reactor and jacket.

Approximation. (a) Perfect mixing in both vessels — the well-mixed assumption that defines an ideal reactor. (b) Constant density and heat capacity. (c) Linearisation about a nominal operating point (the step in §21.0.3). (d) A pure transport delay inserted for the sensor and coolant piping.

Mathematical reduction. Linearising the two coupled balances and separating the dominant time constants yields a second-order-plus-delay plant:

$$P(s) \approx \frac{K_{process}}{(\tau_1 s + 1)(\tau_2 s + 1)} e^{-\theta s}$$

where the dead time $e^{-\theta s}$ represents transport delay in the temperature sensor or coolant piping — very common in chemical processes, where the delay is physical (a fluid parcel takes time to travel) and not an artefact.

Engineering equation. The controller-facing object is this $P(s)$; everything after it is domain-free.

Dead time and the Padé approximation:

$$e^{-\theta s} \approx \frac{1 - \theta s/2}{1 + \theta s/2}$$

Dead time adds phase lag proportional to $\theta\omega$ — it degrades PM severely at high frequencies and limits the achievable bandwidth to roughly $\omega_{max} \approx 1/\theta$.

IMC (Internal Model Control) tuning for dead-time processes:

$$C(s) = \frac{1}{P(s)} \cdot \frac{1/\lambda}{s + 1/\lambda}$$

where λ is a tuning parameter (closed-loop time constant). Gives inherently good robustness and explicit trade-off between performance (λ small) and robustness (λ large).

Model Ledger — CSTR temperature plant

Property	Description
Parent theory	Coupled nonlinear material + energy balances for an exothermic reactor (Ch. 20 §20.8)
Reduction	Perfect mixing; constant ρ, c_p; linearisation about a nominal steady state; transport delay lumped as e^{−θs}
Model	P(s) = K / [(τ1s+1)(τ2s+1)] · e^{−θs}
Assumptions	Both vessels well mixed; no spatial gradients; single reaction path; constant heat transfer coefficient; reactor is at steady state before the step; delay is pure, not a distributed-parameter diffusion
Retained	Thermal capacitance of vessel and jacket; jacket heat-transfer resistance; the Arrhenius temperature dependence *linearised* into K; sensor and piping delay
Neglected	Multiplicity and ignition/extinction (the true plant is nonlinear enough to have several steady states — linearising near the wrong one gives a stable-looking model of an unstable operation); reactant depletion; catalyst deactivation; jacket-side mixing; varying UA with flow
Valid when	Small excursions about a chosen stable steady state; $\theta\omega \ll 1$ below the bandwidth; single stable operating point in the region of interest
Fails when	Large setpoint steps that cross the ignition/extinction threshold; thermal runaway; batch operation; slow catalyst decay drifting the operating point
Next model	Nonlinear MPC on the full balances — the standard industrial answer for exothermic reactors, precisely because the linear model is local; gain scheduling across multiple steady states

This example is the clearest demonstration in the book that **the transfer function is a local object.** An exothermic CSTR can have three steady states. The linearisation is valid at each of them separately, with a different $P(s)$ at each, and one of them may be unstable. A single "the" transfer function for that reactor does not exist.

21.8.3 — Example 3: Active Vibration Control (ME/CE)

Starting theory. A distributed elastic structure. From Ch. 19 §19.3, the transverse displacement of a beam or floor plate obeys a PDE; the structure has an infinite set of modes, each with a natural frequency ω_n and a mode shape.

Approximation. (a) Modal truncation — keep the lowest n modes, discard the rest. (b) Lumped actuator and sensor (collocated pair). (c) For a single dominant mode, second-order approximation.

Mathematical reduction. Projecting the PDE onto the retained modes via the modal transformation gives a diagonal second-order system; retaining one mode and applying a co-located sensor/actuator pair gives an inertial plant:

$$P(s) = \frac{1/m}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Reduced model. One damped second-order oscillator. The other $n-1$ modes become the *unmodelled* dynamics — and in flexible structures those modes are the design constraint, not an afterthought.

Engineering equation. For collocated sensor–actuator pairs, the closed loop is unconditionally stable, and the controller adds damping to the targeted mode without exciting the others:

$$C_{PPF}(s) = \frac{g\omega_f^2}{s^2 + 2\zeta_f\omega_f s + \omega_f^2}$$

Positive Position Feedback (PPF): a control strategy specific to flexible structures — uses position feedback through a filter tuned to the target mode. When $\omega_f \approx \omega_n$, the PPF controller adds damping to the structural mode. The Bode analysis shows that PPF is unconditionally stable for collocated sensor-actuator pairs — a major advantage over direct velocity feedback in flexible structures where spillover modes can cause instability.

Model Ledger — single-mode structural plant

Property	Description
Parent theory	Euler–Bernoulli / plate PDE with ρA ∂²w/∂t² + EI ∂⁴w/∂x⁴ = q (Ch. 16 §16.7, Ch. 19 §19.3)
Reduction	Modal projection; truncation to the dominant mode; collocated sensor/actuator; linearised (small vibration)
Model	P(s) = (1/m) / (s² + 2ζω_n s + ω_n²)
Assumptions	Linear elasticity; small deflections; viscous (Rayleigh) damping; rigid sensor/actuator attachment; actuator authority sufficient over the mode; no structural nonlinearity (no yielding, no gap, no buckling)
Retained	The targeted mode's frequency, damping, and modal mass; the force-to-displacement path; the gain of the control path
Neglected	All higher modes (**spillover**); actuator dynamics (if the collocated pair is not truly collocated at high frequency, stability is lost); measurement noise amplification; control-induced structural nonlinearity
Valid when	Excitation is dominated by the targeted mode; the structure remains linear; the collocation assumption holds across the achieved bandwidth
Fails when	Broadband or unpredicted excitation; approaching yield, buckling, or a frictional joint; non-collocated actuator/sensor geometry; higher modes entering the control bandwidth
Next model	Multi-mode modal controller (LQG on the truncated modal model, with a stability check on the unmodelled modes); or collocation-aware H_∞ design

The bridge-zone point. This is Bridge C in its purest form: the structure is a *distributed* system (a PDE), and the control engineer has deliberately chosen a *lumped* model of it, discarding the rest of the spectrum. Ch. 19's failure-mode table and Ch. 16's Beam Ledger are the same content seen from two directions.

21.8.4 — The common shape

All three examples, plus the temperature oven and the distillation column in §21.13, have the identical skeleton:

Step	Motor speed	CSTR temperature	Structural mode
Parent theory	Electromechanical two-port	Nonlinear coupled balances	Elastic PDE
Bridge C reduction	Lumped R_a–L_a; lumped J, b	Perfect mixing; lumped thermal masses	Modal truncation
Linearisation	Constant K_t, K_e; viscous friction	About a chosen steady state	Small vibration
Reduced model	$K_m/(\tau_m s + 1)$	$K/[(1+\tau_1 s)(1+\tau_2 s)]e^{-\theta s}$	$(1/m)/(s^2 + 2\zeta\omega_n s + \omega_n^2)$

Step	Motor speed	CSTR temperature	Structural mode
Engineering equation	PID / PI on speed	IMC / PID on temperature	PPF / H_∞ on displacement
Validity limit	Saturation, field weakening	Ignition/extinction, multiplicity	Spillover, nonlinearity

The controller is identical across all three columns. That is the capstone claim, and it holds — within the LTI box drawn in §21.0.2, and no wider.

21.9 — Advanced Control Architectures

21.9.1 — Feedforward Control

When a disturbance $D(s)$ can be measured before it affects the output, add a **feedforward path** $C_{ff}(s)$:

$$U(s) = C_{ff}(s)D(s) + C_{fb}(s)E(s)$$

Set $C_{ff}(s) = -P_d(s)/P(s)$ (where P_d is the disturbance-to-output path) to cancel the disturbance perfectly.

Limitation: Feedforward requires an exact model. Feedback handles model uncertainty but is reactive. **The optimal strategy: feedback for robustness, feedforward for known disturbances** (e.g., known load changes in motor control, solar irradiance forecast for building HVAC).

21.9.2 — Cascade Control

Two nested feedback loops: a fast **inner loop** (e.g., current, flow, temperature) and a slow **outer loop** (e.g., speed, composition, pressure):

```
R ──► C_outer ──► R_inner ──► C_inner ──► Plant ──► Y ──► outer feedback
                              ↑──────────────────────────── inner feedback
```

The inner loop rejects fast disturbances; the outer loop maintains the set-point. Bandwidth: inner loop 3–5× faster than outer.

Common in: Electric drives (current loop → speed loop → position loop); distillation columns (composition control → temperature → heat duty); aircraft (roll rate → roll angle → lateral position).

21.9.3 — Model Predictive Control (MPC)

At each time step, solve an optimization over a **prediction horizon** N:

$$\min_{u_{0|t},\ldots,u_{N-1|t}} \sum_{k=0}^{N-1} \left[|x_{k|t} - x_{ref}|_Q^2 + |u_{k|t}|_R^2 \right]$$

subject to the plant model $x_{k+1} = \mathbf{A}x_k + \mathbf{B}u_k$ and constraints $u_{min} \le u \le u_{max}$, $x_{min} \le x \le x_{max}$.

Apply only the first control action; re-solve at the next time step (receding horizon).

Advantages: Handles multivariable systems and constraints explicitly — the dominant control method in oil refining, petrochemical, and power generation.

Computational cost: A quadratic program (QP) must be solved in real-time. For fast systems (millisecond sample rates), dedicated hardware is required.

21.10 — Where Control Theory Fails

This section is the counterpart to §21.0.2. Read together, they bound the capstone claim: §§21.1–21.6 are exact mathematics *inside* the LTI box, and everything in this table is a reason the system is not in the box.

Failure mode	Cause	Alternative
Large nonlinearity	Operating point changes; linearization invalid	Gain scheduling; feedback linearization; Lyapunov methods; nonlinear MPC
Time-varying coefficients	Aircraft in flight, turbine load ramps, drifting process conditions	Linear time-varying state-transition matrices $\Phi(t, \tau)$; gain scheduling; re-linearise and re-tune
Hybrid / switching dynamics	Discrete mode changes (contact, saturation, clutch, phase change)	Hybrid automaton; mode-dependent A and B with mode estimators; switched-system stability (common Lyapunov function)
Distributed parameter plant	PDE, not ODE; infinite-dimensional	Spatial discretization (FEM) + LQR; H_∞ control for PDEs
Pure time delay θ	Bandwidth limited to $\sim 1/\theta$; Padé inaccurate	Smith predictor; dead-time compensator
RHP zeros	Fundamental bandwidth limitation; non-minimum phase	H_∞ optimization; waterbed constraint aware design
RHP poles	Must be stabilized; robustness constraints	Requires careful loop shaping; stabilization bandwidth limits
Quantum systems	Measurement collapses the state	Quantum optimal control; open quantum systems theory
Stochastic systems with large noise	Deterministic control insufficient	Kalman filter + LQG; stochastic MPC
Constrained inputs / states	Feasible set is a subset of the domain; unconstrained $H(s)$ is undefined there	MPC, which optimises over the model subject to constraints
Chaotic systems	Sensitive dependence on initial conditions; linearisation radius collapses	Lyapunov methods; nonlinear observers; feedback linearisation; chaos control (where a controlled attractor exists)
Bode sensitivity integral	Cannot improve at all frequencies	Shapes trade-off; waterbed unavoidable

The point of this table is not defeat. It is that the Layer-3 unification delivered by the transfer function is a *conditional* result, and the condition is worth stating precisely: the system must admit an LTI input–output description. Where it does, the unification is exact and branch-independent — a PID loop genuinely does not know what domain it is in. Where it does not, the Layer-3 template still holds at the level of conservation laws and balance equations, and the correct next step is a different *representation* (state-space, stochastic, constrained, nonlinear), not a different physics. Chapters 18–20 remain valid in every row of this table.

The sensitivity integral (repeated from §21.6.1) is perhaps the most important theoretical limitation: for a stable closed loop with no RHP open-loop poles and an open loop of relative degree ≥ 2:

$$\int_0^\infty \ln |S(j\omega)|\, d\omega = 0$$

You cannot have $|S| < 1$ (good disturbance rejection) at some frequencies without having $|S| > 1$ (disturbance amplification) at others. This is a conservation law for the sensitivity function [ANALOGY — a conceptual echo of Heisenberg's uncertainty principle, not the same mathematics]: improving performance in one frequency band degrades it in another.

21.11 — The Cross-Branch Capstone: The Book's Central Claim

21.11.1 — The Complete Layer Map, Assembled

The full descent from the Standard Model action to engineering control systems:

```
LAYER 0: S = ∫√-g [(R−2Λ)/16πG + L_SM + F[…]] d⁴x
         │
         BRIDGE A (E ≪ mc², v ≪ c, single particle)
         │
LAYER 1: Quantum mechanics (wavefunctions, bands, nuclei, topology)
         Chapters 3–7
         │
         BRIDGE B (five simultaneous limits)
         B.a: ħ→0          → Classical mechanics (Ch. 9)
         B.b: N→∞          → Thermodynamics (Ch. 10)
         B.c: EM field limit → Maxwell equations (Ch. 11–12)
         B.d: Weak-field GR → Newton (Ch. 8)
         B.e: Kubo          → Transport laws (Ch. 13–14)
         │
LAYER 2: Classical continuum physics
         Navier-Stokes, Navier elasticity, wave equations, diffusion (Ch. 15–16)
         │
         BRIDGE C: Integrate PDE over control volume; L/λ ≪ 1
         │
LAYER 3: Engineering systems
         R/C/L in every domain; H(s); PID; Bode; root locus (Ch. 17–21)
```

What this diagram does and does not claim. Every arrow from L0 to L3 is a named reduction with a stated regime — that is the book's claim, and Chapters 3–17 defend it link by link. The final arrow, *L3 system → transfer function*, is a Bridge C operation like the others, and it is the one most often left implicit. It carries three assumptions at once: **lumping** ($L_{element} \ll \lambda_{field}$), **linearity** (small excursions about an equilibrium), and **time-invariance** (fixed coefficients). Drop any one and the transfer function is the wrong object — which is why §21.0.2 and §21.10 exist.

The correct terminal statement of the capstone is therefore:

> For systems that admit an appropriate LTI representation, the transfer function provides a branch-independent mathematical language for analysis and control — and where it does not, the same Layer-3 template survives in conservation-law and state-space form, pointing at a different *representation* rather than a different physics.

That is a narrower claim than "all engineering reduces to transfer functions." It is also the only version of the claim this book can actually defend, and the difference is not cosmetic: it is the difference between a framework and a slogan.

21.11.2 — The Five Cross-Layer Threads, Completed

Thread 1: Noether's theorem — from the global phase symmetry of charged matter (in a theory with local $U(1)$ gauge structure) to KCL at a circuit node, to mass balance at a process unit, to mole balance in a reactor. One mathematical idea, at every level. The full six-step chain, with each step named and tagged, is in `00 Map.md` §0.2 and §3.1 — it is not a one-line jump. (Link 1 is often compressed to "U(1) gauge symmetry ⟹ Noether"; the precise attribution — global symmetry carries the Noether charge, local gauge structure is associated with Noether's *second* theorem — is set out in `00 Map.md` §0.2.)

Thread 2: The Mexican Hat (Broken Symmetry) — Ch. 0's Higgs mechanism ($\sim$ 246 GeV), BCS superconductivity (Ch. 7, $\sim$ meV), and Landau phase transitions (Ch. 10) are independent physical instances of one mathematical template (an order parameter with a sign-changing coefficient), not a causal sequence running through the Higgs field. Engineering bistability and hysteresis (magnetic cores, structural snap-through, chemical multiplicity) are a further, separate instance of the same template. See `00 Map.md` §0.3 for the branching structure and the tagged connections.

Thread 3: The Action Principle — from Ch. 0 $\delta S = 0$ through Hamilton's principle in Ch. 1 through the PDE Euler-Lagrange equations in Ch. 15 through virtual work in structures to the Pontryagin minimum principle in optimal control (the continuous-time version of LQR).

Thread 4: Topology — [STRUCTURAL CONNECTION, not a causal chain] Three separate physical and mathematical settings in this book share the concept of a *topological invariant* — an integer-valued quantity that cannot change under a smooth deformation, only across a genuine qualitative transition. That shared mathematical skeleton is the point of this thread; reading it as one physical derivation running from particle physics to a control-loop diagram is exactly the mistake to avoid.

Topological invariants (general mathematical concept: classification of maps by winding/degree) —

The θ-term, the Chern number, and the Nyquist encirclement count are three independent physical questions answered by the same branch of mathematics. Recognizing the shared skeleton is valuable; treating it as a physical genealogy from particle physics to a control loop is not — that reading is explicitly disclaimed here.

Thread 5: Wave/Diffusion Dichotomy — from Ch. 0 hyperbolic/parabolic field equations through Ch. 16 unified PDE through filter design (transfer functions are dispersion relations, Ch. 18) to the bandwidth/stability trade-off in control.

21.11.3 — The One Unanswered Question

Throughout this book, a term has appeared in the action but has never been given content:

$$\mathcal{F}[g,\ \phi_?,\ \text{new fields},\ \text{topology},\ \text{anomalies},\ \ldots]$$

This placeholder represents:

- Dark matter (what is it? $\phi_?$ in the action?)
- Dark energy (a cosmological constant? a quintessence field?)

- Quantum gravity (how does the EH term quantize?)
- The strong CP problem (why is $\theta_{QCD} < 10^{-10}$?)
- The hierarchy problem (why is $m_{Higgs} \ll m_{Planck}$?)
- Matter-antimatter asymmetry (why is there more matter than antimatter?)
- The identity of the three generations (why e, μ, τ? Why three?)

These are not failures of the book's framework. They are failures of our current knowledge of $\mathcal{F}$. The framework itself is sound — the four-layer architecture, the five threads, the named approximations — all of this remains valid when $\mathcal{F}$ is discovered or constrained.

The $\mathcal{F}[\ldots]$ placeholder tells you exactly where the frontier is. Every future discovery in fundamental physics — a dark matter direct detection, a graviton, a proton decay event, a new particle at a future collider — will be described as a modification of the unknown sector. Engineers who understand this will recognize the new physics for what it is: a refinement of Layer 0, whose effects propagate down through Layers 1, 2, and 3 via the same bridges described in this book.

21.12 — Summary

The complete control toolbox, organized by method:

Method	What it does	When to use
Routh-Hurwitz	Algebraic stability test	Quick check; finding stability margins vs. gain
Bode plot + PM/GM	Frequency domain stability margins	Loop shaping; robust design
Nyquist criterion	Exact stability (handles unstable plants)	When Bode approximation is insufficient
Root locus	Pole locations vs. proportional gain	Single-loop design; visual insight
PID + Z-N tuning	Quick empirical design	No model available; standard loops
Lead-lag compensator	Bode-based loop shaping	PM/GM and steady-state error correction
State feedback (LQR)	Optimal multivariable control	All states measurable; known model
Observer (Luenberger)	State estimation	States not all measurable
MPC	Constraint handling; optimization	Multi-variable; input/output constraints
Feedforward	Known disturbance rejection	Measured disturbances available
Cascade	Fast inner loop disturbance rejection	Measurable intermediate variable

---

21.13 — Engineering Thread: Control Everywhere

System	Inner plant $P(s)$	Controller	Performance metric
Motor drive	$K_m/(\tau s + 1)$	PI speed + P current	Speed regulation, step response
Temperature oven	$K/(\tau s + 1)$	PID	Setpoint tracking, ±0.1°C
Chemical pH	Nonlinear, approximated	PID + feedforward	Disturbance rejection
Aircraft autopilot	6-DOF dynamics	LQR + feedforward	Stability, ride quality

System	Inner plant $P(s)$	Controller	Performance metric
Distillation column	Multivariable (RGA analysis)	MPC	Product purity, energy
Building HVAC	Thermal RC network	PID + cascade	Zone temperature, energy
Power grid frequency	Generator + load dynamics	Droop + AGC	60 Hz ± 0.1 Hz
Structural vibration	Modal second-order system	PPF or H∞	Vibration attenuation dB
Microprocessor V_core	Buck converter	Type III compensator	Load transient, ΔV
CNC machine axis	Mass + friction	Cascade P-velocity + PI-position	Contour error μm

---

21.14 — The Last Word

The book opened with a single equation:

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G} + \mathcal{L}_{SM} + \mathcal{F}(\phi_?, \ldots) \right]$$

It closes with a PID controller:

$$u(t) = K_p\, e(t) + K_i \int e\, dt + K_d\, \dot{e}$$

Between these two equations: six chapters of Layer-1 quantum mechanics, six chapters of Layer-2 classical physics, six chapters of Layer-3 engineering systems, and four explicit bridge operations that connect them.

The PID controller — implemented in a microcontroller costing less than a dollar, installed in every industrial process, every climate system, every motor drive — is the action principle of Ch. 0 viewed from far away, through four layers of named approximation.

And that closing equation is honest only with its caveats attached. The PID loop does not know what domain it is in, because the mathematics it operates on does not know either. But the transfer function that reaches it was obtained by lumping and linearising a physical system, and it describes that system only locally. The distance from the action to the controller is real; so are the four places where the approximation can be caught out. §21.0.2 and §21.10 say which.

Engineering is physics seen from far away. This book has shown you both ends of the telescope — and the four lenses in between, each with its own stated limits.

End of Chapter 21. End of the numbered chapters.

Next: Epilogue — The Unfinished Equation

EPILOGUE

The Unfinished Equation

What the Placeholder Knows, and What It Doesn't

> *Physics is like sex: sure, it may give some practical results, but that's not why we do it.* — Richard Feynman
>
> *We shall not cease from exploration, and the end of all our exploring will be to arrive where we started and know the place for the first time.* — T.S. Eliot, *Little Gidding*

E.1 — The Equation, One Final Time

Twenty-one chapters ago, the book opened with this:

$$S_{\text{eff}} = S_{EH} + S_{SM} + S_{\text{unknown}} = \int d^4x\,\sqrt{-g}\left[\frac{R}{16\pi G} + \mathcal{L}_{SM} + \mathcal{F}[g,\,\phi_?,\,\text{new fields},\,\text{topology},\,\ldots]\right]$$

Everything in this book — from hydrogen wavefunctions to Navier-Stokes to PID controllers to NTU heat exchanger methods — was derived from the first two terms.

The third term, $\mathcal{F}$, was never given content. It appeared in every restatement of the action. It was called out in Chapter 0 as a precisely-located frontier. It connected in Chapter 7 to the axion electrodynamics of topological insulators. It has silently governed whether any of the 19 free parameters of the Standard Model will ever be derived rather than measured.

This epilogue is about $\mathcal{F}$.

Not about what it is — we do not know. But about what we know about it, what it must contain, how we are searching for it, and what its discovery would mean for the four-layer framework this book has built.

E.2 — What We Know with Extraordinary Precision

The Standard Model Lagrangian $\mathcal{L}_{SM}$ and the Einstein-Hilbert term $R/16\pi G$ together describe:

- Every atomic spectrum ever measured, to better than one part in 10^{12} in the best cases (QED precision on the electron g-factor: theory and experiment agree to 12 significant figures)
- Every particle collision at every accelerator ever built, up to the TeV scale
- The orbits of the planets, to precision limited by our knowledge of the Sun's gravitational multipole moments
- Gravitational wave signals from merging black holes 10^9 light-years away
- The large-scale structure of the universe from its first 10^{-32} seconds to today

These are not approximate descriptions. They are among the most precisely confirmed scientific claims in all of human knowledge.

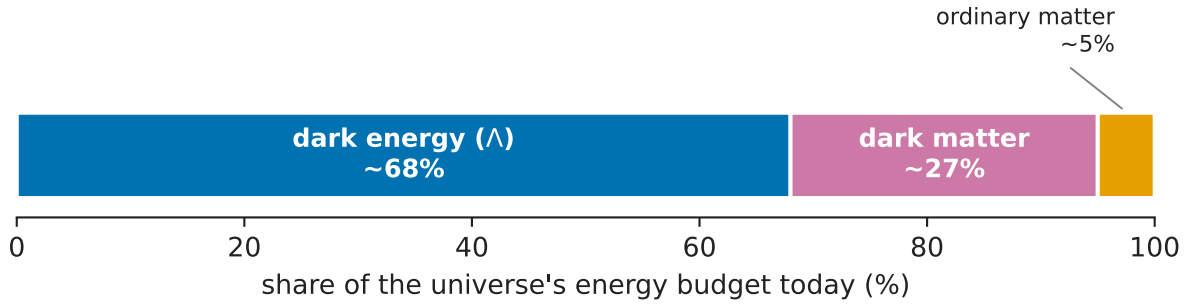
And yet they are incomplete. Not slightly incomplete. Not "correct to 99.9% and missing 0.1%." Fundamentally incomplete in ways that cannot be patched without new physics.

E.3 — The Inventory: What $\mathcal{F}$ Must Contain

E.3.1 — Dark Matter: 27% of the Universe's Energy Budget

The rotation curves of galaxies do not follow Kepler's law. Stars in the outer regions of spiral galaxies orbit too fast. Galaxy clusters bend light too much. The cosmic microwave background (CMB) temperature anisotropies require more gravitating matter than visible matter provides.

The required dark matter density: $\Omega_{DM}h^2 = 0.120 \pm 0.001$ (Planck 2018). The total visible matter: $\Omega_b h^2 = 0.022$. **Dark matter is 5.4 times more abundant than ordinary matter.**



Today's cosmic energy budget (Planck 2018, rounded): dark energy ~68%, dark matter ~27%, ordinary matter ~5%. Everything the Standard Model describes sits in the thin last slice.

Not one Standard Model particle can be dark matter:

- Neutrinos: too light, move too fast, would wash out the structures we observe
- The Higgs: unstable on cosmological timescales
- Protons and neutrons: the BBN (Big Bang Nucleosynthesis) abundance of helium-4 precisely constrains the total baryon content — ordinary matter is fully accounted for

$\mathcal{F}$ must contain at least one new stable, neutral, weakly-interacting particle. Candidates: WIMPs (Weakly Interacting Massive Particles), axions (which also solve the strong CP problem below), sterile neutrinos, primordial black holes. After 40 years of direct detection experiments — XENON, LUX, PandaX, LZ — nothing has been found. The parameter space is narrowing.

E.3.2 — Dark Energy: 68% of the Universe's Energy Budget

The universe's expansion is accelerating. The cosmological constant Λ is the simplest explanation, and it belongs **inside the known gravitational sector** as the $\frac{1}{16\pi G}\int d^4x\sqrt{-g}(-2\Lambda)$ term of S_{EH} (Ch. 0 §0.3) — it is not part of the unknown sector $\mathcal{F}$. What is unknown is not the term's place in the action but its **value and origin**. Why is it so small?

The measured value: $\Lambda \approx 10^{-52}\ \text{m}^{-2}$ → energy density $\rho_\Lambda \approx 10^{-9}\ \text{J/m}^3$.

The naive quantum field theory prediction: every quantum field contributes to the vacuum energy via zero-point fluctuations, giving:

$$\rho_{vac}^{QFT} \sim \frac{\hbar}{c} \int_0^{\Lambda_{UV}} \frac{k^3 \, dk}{(2\pi)^2} \sim \frac{\hbar \Lambda_{UV}^4}{c}$$

At the Planck scale ($\Lambda_{UV} = M_{Pl} c / \hbar$): $\rho_{vac}^{QFT} \sim 10^{111}$ J/m³.

The ratio: $\rho_\Lambda / \rho_{vac}^{QFT} \sim 10^{-120}$.

This is the most precise failure of theoretical prediction in the history of science. A discrepancy of 120 orders of magnitude between the measured cosmological constant and the naive expectation from quantum field theory.

Something beyond the minimal Λ term either generates the observed value dynamically (quintessence and related fields), or cancels the QFT contributions with extraordinary precision (fine-tuning, whose origin is unknown), or changes the framework entirely (modifications of gravity at cosmological scales, or the landscape of string theory with its many possible vacua, possibly selected anthropically). We do not know which. The **discrepancy is the open problem**; the term's location in the action is not.

E.3.3 — Quantum Gravity: The Incompatibility at the Planck Scale

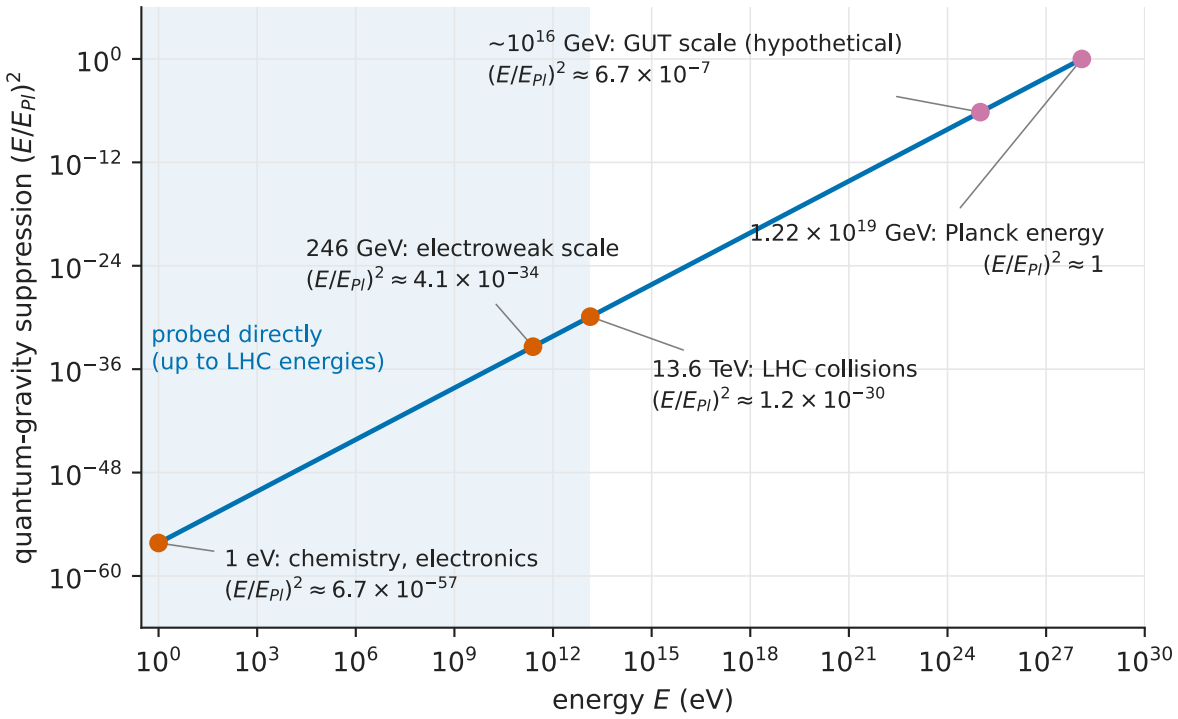
General relativity and quantum mechanics are the two most successful theories in physics. They are also incompatible.

GR describes curved spacetime as a dynamical classical field. QM describes matter and energy as quantum fields on a fixed background.

At energies approaching $E_{Pl} = \sqrt{\hbar c^5 / G} \approx 1.22 \times 10^{19}$ GeV (the Planck energy) and lengths approaching $\ell_{Pl} = \sqrt{\hbar G / c^3} \approx 1.6 \times 10^{-35}$ m, the metric itself must be treated as a quantum field — but no consistent theory of this currently exists.

String theory, loop quantum gravity, causal dynamical triangulations, asymptotic safety — these are candidates for the quantum gravity content of $\mathcal{F}$. None is confirmed. None makes unique, currently-testable predictions that distinguish it from the others.

The engineering implication is small but real: At every length scale relevant to human engineering ($\ell \gg \ell_{Pl}$), quantum gravity effects are suppressed by factors of $(E/E_{Pl})^2 \sim (1 \text{ eV}/10^{28} \text{ eV})^2 \to 0$. The four-layer framework of this book is valid to extraordinary precision in all engineering contexts. But at black hole horizons, at the Big Bang singularity, at the earliest moments of the universe — the framework has an edge.



The quantum-gravity suppression $(E/E_{Pl})^2$ on log axes from 1 eV through the electroweak scale (246 GeV), LHC collisions (13.6 TeV) and the hypothetical GUT scale (~10¹⁶ GeV) to the Planck energy (1.22×10¹⁹ GeV): at 1 eV the factor is ~10⁻⁵⁶, and even at the LHC it is ~10⁻³⁰.

E.3.4 — The Strong CP Problem and the Axion

The QCD θ-term in $\mathcal{L}_{SM}$ (Chapter 0 §0.9) would, for generic θ, generate a neutron electric dipole moment:

$$d_n \sim e \cdot \theta \cdot (10^{-3}\ \text{fm})$$

Experiment: $|d_n| < 1.8 \times 10^{-26}\ e \cdot \text{cm}$. This requires $|\theta| < 10^{-10}$.

Why is θ so small? There is no symmetry in the SM forcing this. This is the **strong CP problem**.

The most elegant solution: promote θ to a dynamical field (the **axion**, predicted by Peccei and Quinn in 1977). The axion rolls to zero, solving the strong CP problem. As a bonus, axions are dark matter candidates.

The connection to Chapter 7: the topological insulator effective action has $\theta = \pi$, not $\theta \approx 0$. This is not the QCD axion, but it is the same mathematical term — the θ-parameter of a U(1) gauge theory. Laboratory-scale experiments with topological insulators may provide insight into the physics of the QCD θ-term. **The condensed matter system in the laboratory is a controlled realization of the same topological structure that appears in the strong interaction.**

E.3.5 — Matter-Antimatter Asymmetry

The universe contains matter. Almost no antimatter. Yet the Big Bang should have produced equal amounts of both.

The Sakharov conditions for generating asymmetry:

1. Baryon number violation (must exist, otherwise no net baryon number can be generated)

2. C and CP violation (matter and antimatter must behave differently)
3. Departure from thermal equilibrium (otherwise any asymmetry would be washed out)

The SM has CP violation (the CKM phase δ_{CP}) but not enough — by approximately 10 orders of magnitude — to account for the observed asymmetry. $\mathcal{F}$ must contain additional CP violation sources.

E.3.6 — Neutrino Masses

The original Standard Model predicted massless neutrinos. This was confirmed to be wrong by the observation of neutrino oscillations (Super-Kamiokande, 1998; SNO, 2001) — neutrinos change flavor as they propagate, which is only possible if they have nonzero masses.

Current limits: $\sum m_\nu < 0.12$ eV (cosmological bound). Individual masses: not yet measured, only mass-squared differences.

The minimum modification: add right-handed neutrinos ν_R to $\mathcal{L}_{SM}$ (they are absent in the original) and give them Majorana masses via $\mathcal{F}$. The seesaw mechanism then generates very light left-handed neutrino masses: $m_\nu \sim m_{Dirac}^2 / M_R$ — the observed lightness is because $M_R \gg m_{Dirac}$.

The detection of neutrinoless double beta decay (which would confirm Majorana mass) is one of the primary experimental goals of current low-background underground experiments.

E.4 — What We Know About $\mathcal{F}$ Without Knowing $\mathcal{F}$

Even without knowing the content of $\mathcal{F}$, its form is severely constrained:

Mathematical constraints (tested and robust):

- Must be a Lorentz scalar — Lorentz invariance is tested to one part in 10^{23}
- Must be diffeomorphism-covariant — GR's equivalence principle is tested to 10^{-15}
- Must be gauge-invariant under $SU(3) \times SU(2) \times U(1)$ at energies < 100 GeV

Phenomenological constraints:

- Must reproduce all SM+GR predictions in all tested regimes (to their tested precision)
- Must not produce large flavor-changing neutral currents (FCNCs) — tightly constrained
- Must not violate baryon number at currently accessible energies (10^{31} year proton lifetime lower limit)
- Must contain a dark matter candidate with the right relic abundance

The Swampland: Recent work in string theory suggests that not all consistent effective field theories can be completed into a full quantum gravity theory. The "Swampland conjectures" constrain which potentials $V(\phi)$ are consistent with quantum gravity. These are not theorems — but they suggest that $\mathcal{F}$ is not arbitrary; the landscape of possibilities is smaller than it might appear.

E.5 — How $\mathcal{F}$'s Discovery Would Propagate Through the Layers

The four-layer framework of this book describes how $\mathcal{L}_{SM}$ and $R/16\pi G$ propagate down to engineering. The same framework tells us how the discovery of $\mathcal{F}$ would propagate:

Layer 0 → Layer 1: A new dark matter particle would add a new term to the Lagrangian. Its effects on atomic physics would be at the sub-10^{-30} level (since it interacts only weakly with SM matter). Bridge A would gain a small correction. Layer-1 quantum chemistry and band theory would be essentially unchanged — but the cosmological evolution of structure would be profoundly affected.

Layer 0 → Layer 2 via cosmology: Quantum gravity corrections to GR become important near the Big Bang. The inflationary model (a scalar field in $\mathcal{F}$ with a flat potential) determines the primordial density fluctuations that seed galaxy formation. Layer-2 astrophysics — large-scale structure, CMB — is the imprint of Layer-0 physics on the classical world.

Layer 0 → Layer 1 via nuclear: If $\mathcal{F}$ contains new particles that couple to quarks, nuclear binding energies and cross-sections could be slightly modified. But for any new particle with mass $m_{new} \gg 1$ MeV, the correction to nuclear physics goes as $(1\text{ MeV}/m_{new})^2$ — negligible for any particle above a few GeV.

The stability of engineering: The four-layer framework is robust to $\mathcal{F}$ because engineering operates at energy scales $E \ll E_{Pl}$ and length scales $L \gg \ell_{Pl}$. The corrections from any reasonable $\mathcal{F}$ to the physics of Chapters 3–21 are unmeasurably small in any engineering context.

But: The technologies that probe $\mathcal{F}$ are themselves engineering achievements — dark matter detectors, gravitational wave observatories, particle accelerators. **Engineering makes the experiments that constrain $\mathcal{F}$.** The boundary between "physics frontier" and "engineering frontier" runs through the laboratory, not through a diagram.

E.6 — The Unfinished Business of the Other Layers

$\mathcal{F}$ is not the only open question. Each layer has its own frontier:

Layer 1 — Quantum:

- **High-temperature superconductivity:** What is the pairing mechanism in cuprates and iron-based superconductors? The BCS theory (Ch. 7) works for conventional SCs, but $T_c = 133$ K in $HgBa_2Ca_2Cu_3O_8$ at ambient pressure has no agreed theoretical explanation after 35 years of study
- **Strongly correlated systems:** Mott insulators, heavy fermion systems, strange metals — where quasiparticles cease to exist and the Kubo framework (Ch. 13) breaks down
- **Topological quantum computing:** Can Majorana zero modes (Ch. 7) be braided to form fault-tolerant qubits? The first experiments are underway

Layer 2 — Classical continuum:

- **Turbulence:** The existence and smoothness of solutions to the Navier-Stokes equations is one of the seven Millennium Prize Problems ($1M prize unclaimed since 2000). We cannot prove that smooth initial conditions remain smooth — or that they do not. Turbulence itself remains phenomenologically described, not derived
- **The glass transition:** Why do liquids become amorphous solids (glasses) when cooled quickly? The structural relaxation time grows by 13 orders of magnitude near T_g with no obvious structural change. No first-principles theory explains this
- **Protein folding:** The native structure of a protein is determined by its amino acid sequence, but predicting it from first principles requires solving the Schrödinger equation for a $\sim 10^5$-atom system — computationally intractable from first principles. AlphaFold (2020) provides empirical accuracy without mechanistic understanding

Layer 3 — Engineering systems:

- **Aging infrastructure:** The degradation of bridges, pipelines, concrete structures, power grids involves multi-scale physics (fatigue at the nanoscale, corrosion at the atomic scale, load redistribution at the structural scale) that is not yet predictable from first principles
- **Resilience of complex networks:** Power grids, communication networks, financial systems exhibit cascading failures — the same topology as the Noether-current networks of Ch. 17, but with nonlinear feedbacks that produce catastrophic events with no simple early warning

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E.7 — The Book's Central Claim, Restated Precisely

This book made one central claim. It can now be stated precisely:

> **Many equations used in engineering practice — from KCL to Navier–Stokes to the Arrhenius rate law to the PID controller — can be connected to deeper physical theories through a sequence of controlled limits, constitutive assumptions, statistical closures, discretisations, and phenomenological models. Where a controlled derivation exists, this book shows it; where an empirical or constitutive model enters, the book marks that boundary explicitly.**

The chain that makes this possible is

```
fundamental theory → controlled approximation → effective theory
   → constitutive closure → discretisation → engineering model
```

and the bridges name the steps: (Bridge A) restriction to one particle below $m_e c^2$; (Bridge B.a) $\hbar \to 0$ classical limit; (Bridge B.b) $N \to \infty$ statistical limit; (Bridge B.c) classical field limit of U(1); (Bridge B.d) weak-field, slow-motion, quasi-static metric limit; (Bridge B.e) linear-response (Kubo) averaging; (Bridge C) spatial lumping.

Two qualifications are load-bearing, and the book's own apparatus exists to enforce them:

1. **Not every step is a derivation.** Several links are *closures* or *constitutive assumptions*, not consequences: the Drude relaxation time (§13.4.2), the Wiedemann–Franz ratio, Einstein's diffusion–mobility relation, engineering correlations, and the lumping criteria. These are labelled `[PHENOMENOLOGICAL]` or `[APPROXIMATION]` wherever they appear, because the distinction between "derived" and "assumed with a stated criterion" is the whole epistemic content of the book.
2. **Not every equation in engineering arrives this way.** The claim is *connectability*, not universal coverage. Many engineering relations are empirical correlations that no current theory derives; the book marks those as such rather than claiming descent for them.

The qualifier "the Standard Model action plus the Einstein–Hilbert term" is the honest version of the origin side. It acknowledges that $\mathcal{F}$ exists, that we do not know its content, and that its engineering consequences are unmeasurably small — but real. The framework is correct; it is simply incomplete at the level of the action itself.

E.8 — A Note on the 19 Free Parameters

The Standard Model Lagrangian contains 19 free parameters that must be measured rather than derived:

- 6 quark masses ($m_u, m_d, m_c, m_s, m_t, m_b$)
- 3 charged lepton masses (m_e, m_μ, m_τ)
- 3 gauge coupling constants (g_1, g_2, g_3 for U(1), SU(2), SU(3))
- 4 CKM parameters (3 mixing angles + 1 CP-violating phase)
- 2 Higgs sector parameters (m_H, $v = 246$ GeV)
- 1 QCD theta parameter (θ_{QCD})

(Plus 7 more if neutrino masses are included.)

Every material property that engineering depends on — the conductivity of copper, the strength of steel, the reactivity of chemical bonds, the melting points of alloys, the bandgap of silicon — ultimately traces to these 19 numbers.

Why these specific values? The answer is: we don't know. A theory of everything — a discovered $\mathcal{F}$ — should predict at least some of these as derived quantities. The fine structure constant $\alpha \approx 1/137$: why 137 and not 200? No one knows. It is one of the most famous unexplained numbers in physics.

The engineer who uses these constants every day is using the outputs of measurements that humanity has not yet theoretically explained. That is not a failure — it is an accurate description of where we stand.

E.9 — The Frontier, Right Now

The experimental program searching for $\mathcal{F}$ is the largest scientific enterprise in history:

At CERN (Large Hadron Collider): Proton-proton collisions at $\sqrt{s} = 13.6$ TeV probe $\mathcal{F}$ at momentum transfers up to a few TeV. The Higgs boson was found in 2012; nothing else beyond the SM has been found. The High-Luminosity LHC (HL-LHC), coming online in the late 2020s, will accumulate 20× more data.

Dark matter direct detection: The LZ detector (seven tonnes of liquid xenon underground in South Dakota) can detect WIMP-nucleus collisions with cross-sections down to 10^{-48} cm². The next generation (XLZD: 60–80 tonnes) will reach the "neutrino fog" — the irreducible background from solar neutrinos scattering off xenon nuclei.

Axion searches: ADMX, CASPEr, and many others search for axion-photon conversion in strong magnetic fields. The frequency scan covers $m_a \sim 1$–100 μeV (axion masses that could explain both the strong CP problem and dark matter).

Gravitational wave astronomy: LIGO/Virgo/KAGRA observe mergers of black holes and neutron stars. LISA (Laser Interferometer Space Antenna, planned for 2034) will observe mergers of supermassive black holes, stochastic backgrounds from the early universe, and potentially signals from cosmic strings — topological defects that appear in some extensions of $\mathcal{F}$.

CMB experiments: CMB-S4 (Stage 4 CMB experiment, 2030s) will measure the B-mode polarization of the CMB — a direct imprint of primordial gravitational waves from inflation. If detected, it would constrain the energy scale at which inflation occurred, and thus constrain $\mathcal{F}$'s inflationary sector.

Neutrino physics: DUNE (Deep Underground Neutrino Experiment) and Hyper-Kamiokande will measure the remaining neutrino mixing parameters and search for CP violation in the neutrino sector — a potential source of matter-antimatter asymmetry.

Each of these is an engineering achievement of the first order — systems whose sensitivity requirements push the absolute limits of known physics in mechanics, cryogenics, materials science, electronics, and data analysis. **The search for $\mathcal{F}$ is, simultaneously, the frontier of engineering.**

E.10 — The Last Equation

We end with the same equation we started with, annotated:

$$S = \int d^4x\,\sqrt{-g}\left[\underbrace{\frac{R}{16\pi G}}_{\substack{\text{gravity}\\ \text{GR}\to\text{Newton (Ch.8)}}} + \underbrace{\mathcal{L}_{SM}}_{\substack{\text{Standard Model}\\ \text{Chs. 3–7, 9–18}}} + \underbrace{\mathcal{F}(\phi_?,\ldots)}_{\substack{\text{dark matter}\\ \text{dark energy}\\ \text{quantum gravity}\\ \text{the frontier}}}\right]$$

The first term: understood, tested to extraordinary precision, gives Newton and gravitational waves.

The second term: understood, tested to part-per-trillion precision, gives all of atomic physics, all of band theory, all of chemistry, and — through the bridges of this book — all of engineering.

The third term: constrained but unknown. The most important and most humbling placeholder in science.

The book has shown you what the first two terms give when you descend through four layers of named approximation. The third term is the invitation to keep going — to learn more physics, to think more carefully, to do experiments more precise than any that have come before.

The equation is unfinished. So is the physics. So is the engineering.

The descent continues.

End of the Epilogue.

End of Engineering Physics: Top Down.

Acknowledgment

This book was written in the conviction that engineering students deserve the same honest account of physical reality that physics students receive — and that the honest account is also the most useful one.

The author thanks the tradition of physicists and engineers who insisted on showing their work: on naming approximations rather than hiding them, on deriving rather than postulating, on admitting uncertainty rather than projecting false confidence.

The framework equation will be updated when $\mathcal{F}$ is discovered. The rest of the book will not need to change.

"The most we can say is that if there is a Theory of Everything, it will appear in this form. Whether it exists, we do not know. That we do not know is not an embarrassment — it is the most precise statement we can make about the frontier of human knowledge."

— Chapter 0, this book

EXAMPLES — Six Worked Descents

Each file in this folder is a complete, self-contained worked example following one shape:

```
1. THEORY            what the starting equation is, and why it is exact
2. APPROXIMATION     every reduction named, typed, and costed
3. REDUCTION         the algebra, step by step, with no physics hidden
4. REDUCED MODEL     the engineering object that results
5. ENGINEERING EQN   a worked numerical problem and the design finding
6. VALIDITY LIMITS   a table of assumptions, failure conditions, and alternatives
```

Step 6 is not a formality. The book's central claim is that physics becomes engineering in the right limits, and an example that does not state its limits is not demonstrating the claim — it is asserting it.

#	File	Descent	Chapters
1	Dirac → Pauli	Relativistic spin-½ field theory → 2×2 magnetic Hamiltonian	Ch. 0, 3
2	Quantum → Thermodynamics	Single-particle QM + quantisation → Gibbs distribution, equipartition, $U, pV = Nk_BT$	Ch. 3, 6, 10
3	Maxwell → Circuit	Maxwell's equations → transmission line → R, C, KCL/KVL	Ch. 11, 16, 17, 18
4	Heat Equation → Thermal RC	Energy conservation + Fourier's law → thermal network — and why it has no inductors	Ch. 15, 16, 17, 19
5	Navier–Stokes → Fluid Network	Cauchy–Navier flow → pipe network, and why it is *not* a linear circuit	Ch. 13, 15, 16, 19, 20
6	Motor → Control System	Two coupled ODEs → second-order $G(s)$ → PI loop → saturation limit	Ch. 17, 18, 19, 21

What the set is designed to show

A descent is a chain of typed reductions, not a single leap. Every step carries one of the five connection tags — `[DERIVATION]`, `[APPROXIMATION]`, `[STRUCTURAL CONNECTION]`, `[ANALOGY]`, `[PHENOMENOLOGICAL]` — and the distinction is load-bearing, not decorative:

- Example 1 shows a `[DERIVATION]` (Euler–Lagrange) immediately followed by two `[APPROXIMATION]`s that produce a *more useful* model. Usefulness went up; truth did not.
- Example 3 shows that three of four steps are exact and the entire descent rests on one inequality, $L/\lambda \ll 1$.
- Example 4 shows a model that is a genuine circuit and genuinely not an electrical circuit: first order, no inertia, one empirical coefficient.
- Example 5 shows that "Ohm's law for pipes" is the same algebra but a *nonlinear* network, because inertia survives as a quadratic term.
- Example 6 shows the chain working end to end, and the final design finding coming from the physics that the transfer function discarded.

The tag `[PHENOMENOLOGICAL]` earns its place. Examples 4, 5, and 6 all end up depending on at least one coefficient that was fitted rather than derived (h, f, the friction curve). A book that claims everything reduces to first principles would have to hide this. Naming it is the honest move, and it is the boundary between physics and engineering practice.

Model Ledgers

Every descent is recorded in the **Model Ledger** format defined in `../METADATA/00 Map.md` — nine rows covering parent theory, reduction, model, assumptions, retained physics, neglected physics, validity, failure, and the next model. Ledgers appear in Chapters 3, 7, 13, 16, 17, 19, 20, and 21, and each example above points to the ledger for its own descent.

The placement rule is deliberate: a Ledger is a record of a *reduction*, so only chapters that actually reduce something carry one. Making the count uniform across all twenty-two chapters would dilute the convention into decoration.

Sources

Numerical constants and empirical correlations used in these examples are sourced in `../REFERENCES.md`, tiered as [MEASURED], [DERIVED / TEXTBOOK], or [SYNTHESIS].

Example 1 — Dirac → Pauli

A complete worked descent in six steps. Follows the shape every example in this folder uses: theory → approximation → mathematical reduction → reduced model → engineering equation → validity limits.

Source chapters: Ch. 0 (the Lagrangian) → Ch. 3 §§3.2–3.6 (Dirac, Foldy–Wouthuysen, Pauli)

Question answered: Where does the Pauli equation come from, and what was given up to get it?

Step 1 — Theory

The electron's dynamics start from a Lagrangian density. Keeping only the U(1)-coupled fermion sector and treating the gauge field classically:

$$\mathcal{L} = \bar{\psi}\left(i\hbar c\gamma^{\mu}D_{\mu} - m_e c^2\right)\psi, \qquad D_{\mu} = \partial_{\mu} + \frac{ie}{\hbar c}A_{\mu}$$

(Gaussian units and the conventions of Ch. 3: e is the signed charge, $e = -|e|$ for the electron, and the Dirac algebra uses the signature of Ch. 3 §3.3.1.) Vary with respect to $\bar{\psi}$ and the Euler–Lagrange equation is immediate:

$$i\hbar c\gamma^{\mu}D_{\mu}\psi - m_e c^2\psi = 0$$

Written with γ^0 on the left, this is the Dirac equation in Hamiltonian form. **This step is a derivation** — `[DERIVATION]` — and nothing has been discarded. Four-component spinor, antiparticles, relativistic dispersion: all present.

Status: exact within the assumed Lagrangian.

Step 2 — Approximation

Two independent things now have to go wrong before a 2×2 equation appears, and they are often conflated:

#	Reduction	Kind	Cost
A	Non-relativistic limit, $v/c \ll 1$	`[APPROXIMATION]`	antiparticle sector suppressed
B	Project onto the large component	`[APPROXIMATION]`	χ discarded; α^4 terms either kept or dropped

Reduction B is subtle. The Dirac spinor has large and small components, and they are *not* independent — χ is coupled to ϕ by $c\boldsymbol{\sigma}\cdot\hat{\boldsymbol{\pi}}$. You cannot simply set $\chi = 0$. The Foldy–Wouthuysen transformation is what makes the decoupling legitimate, and it is worth being precise about how it works, because the usual shorthand attaches the eigenvalues to the wrong operator.

The Dirac Hamiltonian splits by parity:

$$H = \underbrace{\beta m_e c^2 + \mathcal{E}}_{\text{even}} + \underbrace{\mathcal{O}}_{\text{odd}}, \qquad \mathcal{E} = eV, \qquad \mathcal{O} = c\boldsymbol{\alpha}\cdot\hat{\boldsymbol{\pi}}$$

Note carefully what is *not* in the odd operator. There is no $-e\boldsymbol{\sigma}\cdot\mathbf{E}$ term, and it is worth resisting the temptation to add one. The coupling to the field arrives entirely through $\hat{\boldsymbol{\pi}}$; the electric field never enters as an independent $\boldsymbol{\sigma}\cdot\mathbf{E}$ piece, because the minimal Dirac Hamiltonian's only odd operator is $c\boldsymbol{\alpha}\cdot\hat{\boldsymbol{\pi}}$ ($\mathbf{E}$ itself is gauge invariant; an extra $\boldsymbol{\sigma}\cdot\mathbf{E}$-type term would be a non-minimal, Pauli-type coupling added by hand). The familiar magnetic coupling is likewise not an *input* — it is **generated**, out of the $\mathbf{A}$ already sitting inside $\hat{\boldsymbol{\pi}}$, when that piece is iterated against the even Hamiltonian. That generation is the subject of step 4, and it is the reason $g=2$ comes out as a result rather than going in as an assumption.

The Foldy–Wouthuysen transformation removes the odd part, $U^{\dagger}\mathcal{O}U=0$, by exponentiating an odd generator, $U=\exp(-iS/\hbar)$, with S fixed order-by-order to cancel the odd terms. The first two orders of the resulting formal series are

$$H_{FW}=\beta m_e c^2+\mathcal{E}+\frac{\beta\mathcal{O}^2}{2m_e c^2}-\frac{1}{8m_e^2c^4}\big[\mathcal{O},[\mathcal{O},\mathcal{E}]\big]+\cdots$$

This is exactly where the exact/approximate distinction has to be drawn, because it is the whole point of this book. The transformation is **unitary at every order** — the exponential of a Hermitian generator is unitary, with no approximation anywhere. What is *not* exact is the **truncation**. The generator S is determined by an asymptotic expansion in $1/m_e$, each order fixed by cancelling the odd terms at that order; for a general external electromagnetic field there is no closed-form S, and hence no closed-form U that removes $\mathcal{O}$ in a single stroke. So H_{FW} above is a **truncated expansion**, valid through the order displayed, and the Hamiltonian in step 3 is truncated *again* on top of it. Unitarity is a statement about U and it is available; exactness would be a statement about H_{FW} and it is not.

Note also what the transformation does *not* do: it does not diagonalise $\mathcal{O}$. It annihilates it. That is the entire content of the trick — once the odd operator is gone, what remains is even, and an even operator is block diagonal in β.

The $\pm m_e c^2$ eigenvalues belong to the mass term $\beta m_e c^2$, not to $\mathcal{O}$. The odd operator has no such spectrum at all: $\mathcal{O}^2$ is assembled from $\hat{\boldsymbol{\pi}}^2$ and the $\mathbf{A}$-piece, and since $\mathcal{O}\sim m_e c\,\alpha$ it is suppressed by α^2 relative to $m_e^2c^2$. Carrying the positive/negative energy separation is the job of $\beta m_e c^2$, and the selection rule that follows is a projection onto a β block.

Projecting onto $\beta=+1$ and retaining the $1/m_e$ and $1/m_e^2$ terms of the series above produces the order-α^2 effective Hamiltonian of step 3. Keeping exactly those orders and discarding the rest is itself a truncation — the second one in this example, after the truncation already built into H_{FW}.

Step 3 — Mathematical reduction

Truncating the H_{FW} series above to the $1/m_e$ and $1/m_e^2$ terms — order α^2 and α^4 relative to $m_e c^2$ — gives the standard effective Hamiltonian:

$$\hat{H}_{FW}=m_e c^2+\frac{\hat{\pi}^2}{2m_e}+eV\underbrace{-\frac{e\hbar}{2m_e c}\boldsymbol{\sigma}\cdot\mathbf{B}}_{\text{Zeeman}}\underbrace{-\frac{\hat{\pi}^4}{8m_e^3c^2}}_{\text{relativistic mass}}\underbrace{-\frac{e\hbar}{4m_e^2c^2}\boldsymbol{\sigma}\cdot(\mathbf{E}\times\hat{\mathbf{p}})}_{\text{spin-orbit}}\underbrace{-\frac{e\hbar^2}{8m_e^2c^2}\nabla\cdot\mathbf{E}}_{\text{Darwin}}$$

where $\hat{\boldsymbol{\pi}}=\hat{\mathbf{p}}-e\mathbf{A}/c$.

Now truncate *again*, to reach the Pauli equation: drop the three α^4 terms and the rest-mass constant, keep the Zeeman term.

Step 4 — Reduced model

$$\boxed{i\hbar\frac{\partial\phi}{\partial t} = \left[\frac{(\hat{\mathbf{p}} - e\mathbf{A}/c)^2}{2m_e} - \frac{e\hbar}{2m_e c}\boldsymbol{\sigma}\cdot\mathbf{B} + eV\right]\phi, \qquad \phi = \begin{pmatrix}\phi_\uparrow \\ \phi_\downarrow\end{pmatrix}}$$

Two components, not four. Spin now lives in a 2×2 matrix, so spin-dependent physics becomes algebra.

Check that $g = 2$ was not put in by hand — from the right place. The obvious place to look is the orbital minimal-coupling term, and that is exactly where the check fails. In the Coulomb gauge ($\nabla\cdot\mathbf{A} = 0$):

$$\frac{1}{2m_e}\left(\hat{\mathbf{p}} - \frac{e}{c}\mathbf{A}\right)^2 = \frac{\hat{p}^2}{2m_e} - \frac{e}{m_e c}\mathbf{A}\cdot\hat{\mathbf{p}} + \frac{e^2}{2m_e c^2}\mathbf{A}^2$$

No $\hat{\mathbf{S}}$ appears anywhere in this — unsurprisingly, because the term is spin-free by construction. It also retains the diamagnetic A^2 piece. So a derivation that "finds" $g = 2$ by expanding this square has smuggled the spin term into an expression that cannot contain it.

The spin coupling comes from a *different* piece of the Dirac equation. The time-derivative half of the interaction, $-\frac{e}{c}\boldsymbol{\alpha}\cdot\mathbf{A}$, is an **odd** operator, so it belongs to $\mathcal{O}$ and not to the even kinetic term. Iterating that odd piece once in the Foldy–Wouthuysen reduction generates *both* magnetic couplings at once:

$$H_{\text{Zeeman}} = -\frac{e}{2m_e c}\left(\hat{\mathbf{L}} + g_e\,\hat{\mathbf{S}}\right)\cdot\mathbf{B} = +\frac{|e|}{2m_e c}\left(\hat{\mathbf{L}} + g_e\,\hat{\mathbf{S}}\right)\cdot\mathbf{B}\ (\text{electron}), \qquad g_e = 2$$

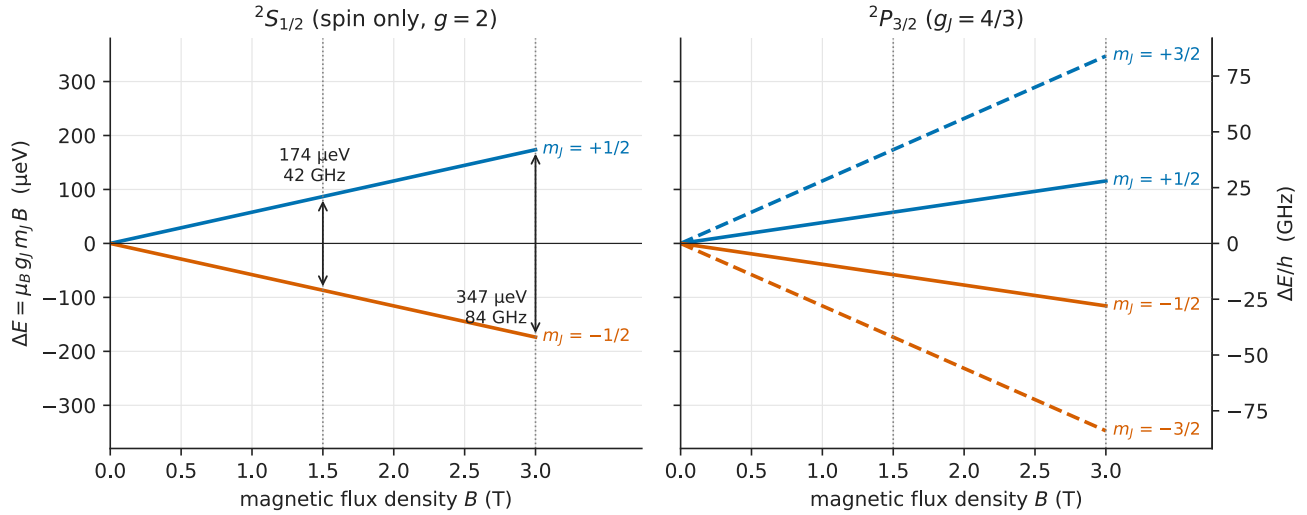
The orbital piece and the spin piece descend from the *same* operator $\boldsymbol{\alpha}\cdot\mathbf{A}$, and the algebra fixes their ratio at exactly 2. That ratio is the $g_e = 2$ of §3.4.2, and it is the same number as $-\frac{e\hbar}{2m_e c}\boldsymbol{\sigma}\cdot\mathbf{B}$ in step 3, since $2\hat{\mathbf{S}} = \hbar\boldsymbol{\sigma}$.

So $g = 2$ really does fall out — it was never inserted — but it falls out of the **odd** sector, and only after the projection onto the large component. The kinetic term is where a careless derivation goes looking for it and finds a spurious answer. That is the single best test of whether a descent is real: the new structure should appear whether or not you were looking for it, *and from the term that can actually produce it*.

Step 5 — Engineering equation

Expanding in $\hat{\mathbf{L}}$ on a hydrogen-like state, $\hat{\mathbf{L}} \to 0$, leaves the Zeeman splitting:

$$\Delta E = \mu_B\, g_J\, m_J\, B, \qquad \mu_B = \frac{|e|\hbar}{2m_e}\ (\text{SI}) = 9.274\times10^{-24}\ \text{J/T}\ \left(\text{Gaussian: } \frac{|e|\hbar}{2m_e c}\right), \qquad g_J = 1 + \frac{J(J+1}$$



Level shifts $\Delta E = \mu_B g_J m_J B$ for B = 0–3 T, for a spin-only $^2S_{1/2}$ level ($g = 2$, left) and a $^2P_{3/2}$ level ($g_J = 4/3$, right), with $\Delta E/h$ on the right axis ($\mu_B/h = 13.996$ GHz/T): the $g = 2$ splitting is 174 μeV (42 GHz) at 1.5 T and 347 μeV (84 GHz) at 3 T, linear in B.

Note the structure: the coefficient multiplying $m_J B$ is the **Bohr magneton**, and the reduction has produced it rather than assumed it. This is the equation worked through in full — weak field, strong field, Paschen–Back — in Ch. 5 §5.6, and it is the equation behind every NMR and MRI system.

Design consequence. A permanent magnet of flux density B across a sample splits spin levels by $2\mu_B B$. The engineering move — choose B to set the splitting you need — follows from the reduced model, and would not follow from the Dirac equation without all of steps 1–4.

Step 6 — Validity limits

Regime	Correct equation
$E \gtrsim m_e c^2$ (511 keV)	Dirac; pair creation is available
$E \ll m_e c^2$, spin observed	**Pauli**
$E \ll m_e c^2$, spin irrelevant	Schrödinger (Ch. 3 §3.7)
Binding energies, Lamb shift resolved	Dirac or full FW — Pauli is insufficient

The criterion is $v \ll c$, not "small". For hydrogen ground state $v/c = \alpha \approx 1/137$, so the relative kinetic-energy error is $\alpha^2 \approx 5 \times 10^{-5}$. For heavy nuclei and high-Z ions this stops being small and the Pauli equation is simply wrong.

What this example demonstrates. A working 2×2 model with a famously correct $g = 2$ was obtained by discarding the antiparticle sector and three correction terms. The discarded terms are recoverable — add them back and you have fine structure. "Valid in a regime" and "true" are different claims, and the ledger format separates them.

Model Ledger: see Ch. 3 §3.5.3 and §3.6 for the full nine-row ledgers with assumptions, neglected physics, and failure modes.

Example 2 — Quantum → Thermodynamics

A complete worked descent in six steps.

Source chapters: Ch. 3 (single-particle QM) → Ch. 6 (many-body, quantisation) → Ch. 10 §§10.2–10.6 (ensemble, Gibbs, entropy)

Question answered: Why do thermal physics and statistical mechanics follow from quantum mechanics at all, rather than being a separate layer?

Step 1 — Theory

A single quantum system is described by a state $|\psi\rangle$ evolving under $\hat{H}$. The Schrödinger equation is deterministic and time-reversible: run it backward and you recover the initial state exactly. Nothing in it produces thermodynamics.

Thermodynamics needs three things the Schrödinger equation does not have: **many** systems, a **large** number of degrees of freedom, and a **statistical statement** about typical rather than particular outcomes.

Step 2 — Approximation

Three approximations, in order of how much they cost:

#	Reduction	Kind	What it assumes away
A	N independent identical systems	`[APPROXIMATION]`	interactions between subsystems
B	Replace the microstate by an energy distribution	`[APPROXIMATION]`	which microstate is occupied
C	Equipartition: every quadratic degree of freedom carries $\frac{1}{2}k_BT$	`[APPROXIMATION]`	anharmonicity, non-quadratic potentials

Reduction A is the load-bearing one. **It is an approximation and it is a bad one** — which is precisely why its failure is thermodynamically interesting. It is good enough to derive the equipartition theorem and the ideal gas law, and bad enough that its corrections (van der Waals, phase transitions) are the whole of real thermodynamics.

Step 3 — Mathematical reduction

Quantise N independent one-particle levels $\epsilon_1, \ldots, \epsilon_N$. A microstate is an occupation assignment. The multiplicity of a configuration with n_i particles in level i:

$$W = \frac{N!}{\prod_i n_i!}$$

That is **all** statistical mechanics is at this stage: a counting statement about integers. Gibbs entropy is the logarithm:

$$S = k_B \ln W$$

Maximise S subject to fixed $N = \sum_i n_i$ and fixed $E = \sum_i n_i \epsilon_i$. The method of Lagrange multipliers gives the **Boltzmann distribution**:

$$\frac{n_i}{N} = \frac{e^{-\beta \epsilon_i}}{Z}, \qquad Z = \sum_i e^{-\beta \epsilon_i}, \qquad \beta = \frac{1}{k_B T}$$

Note what did *not* appear. There is no appeal to probability axioms, no assumption of randomness, and no time dynamics. Entropy maximisation under two constraints is enough. The second law is a statement about where the maximum is.

Step 4 — Reduced model

Equip the classical phase space with quantised levels and the distribution becomes the continuous Maxwell–Boltzmann form:

$$f(E) = \frac{2}{\sqrt{\pi}} \frac{1}{(k_B T)^{3/2}} \sqrt{E}\, e^{-E/k_B T}$$

From here, two results follow in a few lines each.

Equipartition. For each independent quadratic term $\tfrac{1}{2} a x^2$ in the Hamiltonian, $\langle x^2 \rangle = k_B T / a$, so $\langle E \rangle = \tfrac{1}{2} k_B T$ per term. With f quadratic terms per particle, $U = \tfrac{f}{2} N k_B T$.

Ideal gas. Substituting the level spacing $\epsilon_n = p^2 / 2m$ into Z and taking the continuum limit gives $pV = N k_B T$, with $U = \tfrac{3}{2} N k_B T$ (monatomic, $f = 3$).

The macro-thermodynamics follows by elimination: $p = -(\partial F / \partial V)_T$, $S = -(\partial F / \partial T)_V$, $F = U - TS$ — the Gibbs free energy. Every classical thermodynamic relation is a derivative of one scalar function.

Step 5 — Engineering equation

Worked problem. 1.00 mol of monatomic ideal argon, initial state 300 K and 1.00 atm, compressed adiabatically and reversibly to 5.00 atm. Find the final temperature, the work done on the gas, and ΔS.

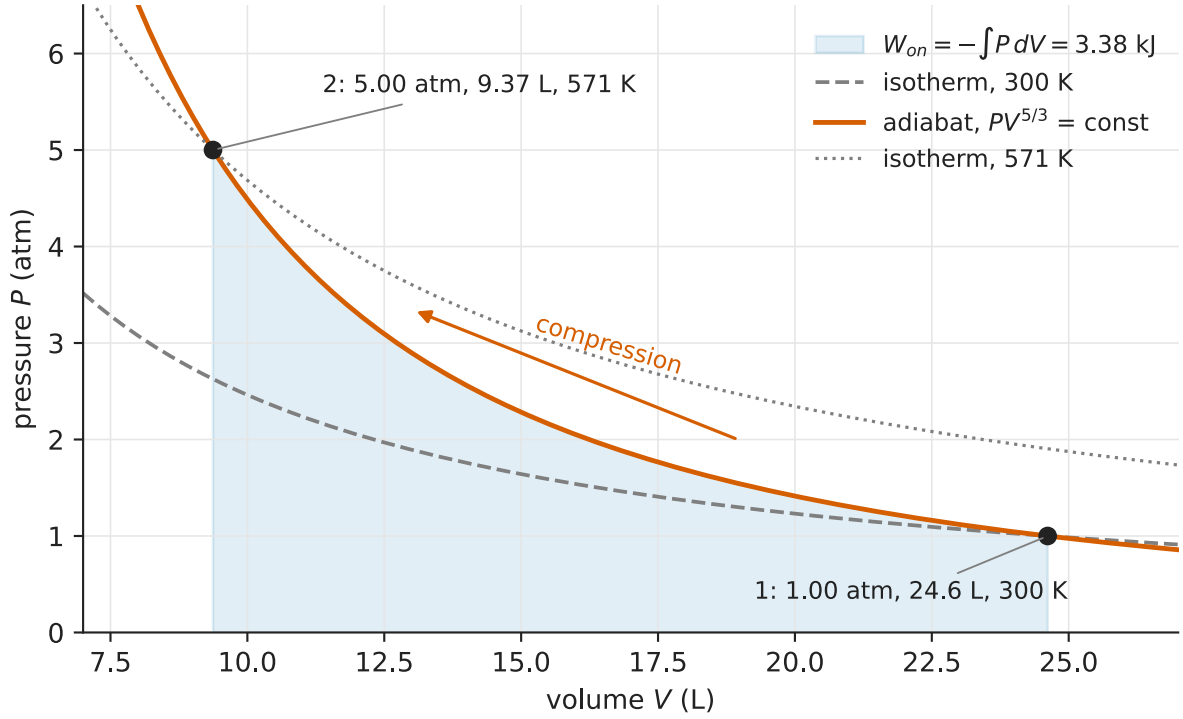
Reversible adiabatic means S is constant, and for an ideal gas $S = n C_V \ln T + n R \ln V + \text{const}$, so $T V^{\gamma - 1} = \text{const}$ with $\gamma = C_P / C_V = 5/3$. Equivalently $T_2 / T_1 = (P_2 / P_1)^{(\gamma - 1)/\gamma}$. Note the direction: this process is a **compression**, $P_2 > P_1$, so $T_2 > T_1$. (The reciprocal form $(P_1/P_2)^{(\gamma-1)/\gamma}$ is the *expansion* case, where $T_2 < T_1$ and the gas cools — it is the same equation read with $P_1 \leftrightarrow P_2$ swapped, and using it here would silently describe a different process.)

$$T_2 = T_1 \left(\frac{P_2}{P_1} \right)^{(\gamma-1)/\gamma} = 300 \left(\frac{5}{1} \right)^{0.4} = 300 \times 1.904 = 571 \text{ K}$$

$$W_{on} = \Delta U = n C_V (T_2 - T_1) = 1.00 \times 12.47 \times (571 - 300) = +3.38 \text{ kJ}$$

The work is **positive** because work is done *on* the gas, and the temperature rises. A compression that cooled the gas would be a physical contradiction.

$$\Delta S = nC_V \ln\frac{T_2}{T_1} + nR\ln\frac{V_2}{V_1} = 12.47(0.6438) + 8.314(-0.9657) = 0 \quad \text{(the two terms cancel)}$$



1.00 mol of monatomic argon compressed along the adiabat $PV^{5/3}$ = const from (1.00 atm, 24.6 L, 300 K) to (5.00 atm, 9.37 L, 571 K); the shaded area under the adiabat is the work done on the gas, 3.38 kJ, and the adiabat climbs from the 300 K isotherm to the 571 K one, which is the temperature rise.

Three engineering numbers, all from step 4, and the last one is a *check*: it is zero because "reversible adiabatic" means exactly that. If a compressor model returns $\Delta S \neq 0$, either the compression is irreversible or the model is wrong.

Design consequence. Compressing gas is how a compressor is specified, and the $T_2 = 571$ K outlet temperature is what forces intercooling. The whole engineering rationale for multi-stage compression with intercooling falls out of γ, which falls out of the equipartition count.

Step 6 — Validity limits

Assumption	Fails when	Consequence
Ideal gas, no interactions	High P, low T	van der Waals; condensation
Independent subsystems	N small, or interactions strong	quantum correlations, Bose/Fermi statistics
Boltzmann statistics	Degenerate gas, $n\lambda^3 \gtrsim 1$	Fermi–Dirac or Bose–Einstein; the same partition function, different counting
Equipartition	Anharmonic potentials, low T	frozen degrees of freedom; heat capacity falls below $\frac{f}{2}k_B$
Equilibrium	Isolated system, integrable dynamics	non-ergodicity; microcanonical preparation required

The Bose/Fermi caveat is the important one. Steps 1–4 assumed $W = N!/\prod n_i!$, which is *Maxwell–Boltzmann* counting. It is valid only when $\lambda = h/\sqrt{2\pi m k_B T} \ll$ mean interparticle spacing. When it fails, the counting changes — and so does the distribution, while the thermodynamics of $F = -k_B T \ln Z$ is unchanged. Same framework, different statistics. This is why Ch. 6 sits between quantum mechanics and statistical mechanics in the architecture rather than after it.

What this example demonstrates. Thermodynamics is not a separate foundation layered on top of quantum mechanics. It is a counting argument about a large-number limit, and the entire classical apparatus is recoverable from one constrained maximisation. The one place the story is genuinely incomplete is the non-equilibrium case, which is why Chapter 10 ends at equilibrium.

Model Ledger: thermodynamics is a *reduction* of microscopic counting, so its assumptions belong in a Ledger. See Ch. 10 for the ensemble and free-energy derivation these steps track.

Example 3 — Maxwell → Circuit

A complete worked descent in six steps.

Source chapters: Ch. 11 §§11.2–11.4 (Maxwell) → Ch. 16 §§16.3.2, 16.10 (telegraph equation, lumping) → Ch. 17 §§17.1–17.3 (R/C/L) → Ch. 18 (circuits)

Question answered: What exactly is thrown away when Maxwell's equations become a circuit diagram?

Step 1 — Theory

Maxwell's equations, in vacuum, in differential form:

$$\nabla \cdot \mathbf{E} = 0 \qquad \nabla \times \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \qquad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \qquad \nabla \cdot \mathbf{B} = 0$$

Take the curl of the Faraday–Ampère pair and use $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$. On a source-free region, $\mathbf{E}$ and $\mathbf{B}$ both satisfy the wave equation:

$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}, \qquad c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

Maxwell did not assume light is electromagnetic. It fell out.

Step 2 — Approximation

Circuit theory requires a **geometry** that Maxwell does not have. Three assumptions must be added before any of this becomes a diagram:

#	Assumption	Kind	Why it is needed
A	Quasi-static: $\ell \ll \lambda$, so $L/\lambda \ll 1$	`[APPROXIMATION]`	lets $\nabla \to$ a difference over a lump length
B	Lumped geometry: a single path, a single pair of conductors	`[APPROXIMATION]`	a circuit element has one voltage and one current
C	Linear, instantaneous medium: ϵ, μ constant	`[APPROXIMATION]`	makes $Q = CV$ and $\Phi = LI$ reversible, single-valued

Assumption A is the one that fails first, and its failure is named: beyond $L/\lambda \sim 1$ you have **transmission line** behaviour, where voltage is a function of position, and the correct object is the telegrapher's equations, not Kirchhoff's laws.

Step 3 — Mathematical reduction

Step 3a — the transmission line: an already-reduced distributed model. For a line with distributed (per-unit-length) parameters R', L', G', C':

$$\frac{\partial V}{\partial x} = -R'I - L'\frac{\partial I}{\partial t} \qquad \frac{\partial I}{\partial x} = -G'V - C'\frac{\partial V}{\partial t}$$

In the **lossless** case ($R' = G' = 0$), which is the form usually quoted:

$$\frac{\partial V}{\partial x} = -L'\frac{\partial I}{\partial t} \qquad \frac{\partial I}{\partial x} = -C'\frac{\partial V}{\partial t}, \qquad Z_0 = \sqrt{L'/C'}, \qquad v = \frac{1}{\sqrt{L'C'}}$$

Be precise about the epistemic status here, because "before lumping" is not the same as "exact." These are *not* Maxwell's equations verbatim with the transverse dimensions integrated out. Getting here has already cost something: the transverse field structure has been replaced by a single voltage and a single current (a single-mode, quasi-TEM idealisation), R', L', G', C' are being treated as constants lumped per unit length, and the conductor geometry is assumed benign enough that the line can be described by distributed scalars at all. Each of those is a modelling choice. What is exact is the *reduction* — once the line model is granted, the relations below are consequences of it, which is why the R and G terms already present here reappear in step 3c without any new assumption.

Step 3b — integrate over a segment of length ℓ. Exact, by the divergence theorem:

$$\ell L'\frac{dI}{dt} = V(x) - V(x+\ell) \qquad \ell C'\frac{dV}{dt} = I(x) - I(x+\ell)$$

Step 3c — invoke lumping. If $\ell \ll \lambda$, the spatial differences become derivatives scaled by ℓ:

$$L\frac{dI}{dt} + RI = V, \qquad C\frac{dV}{dt} + GV = I$$

Step 3b is an identity. **Step 3c is the approximation** — and it is the entire content of the descent. Note also that $L = L'\ell$, $C = C'\ell$, $R = \ell/(\sigma A)$: the lumped parameters are *definitions by correspondence*, not derived quantities.

Step 4 — Reduced model

Three constitutive laws, one per element type:

Element	Law	Energy	Physics
Resistor	$V = RI$	dissipated, I^2R	conductivity, scattering
Capacitor	$I = C\dot{V}$	stored, $\frac{1}{2}CV^2$	field energy, polarisation
Inductor	$V = L\dot{I}$	stored, $\frac{1}{2}LI^2$	magnetic field, inertia

KCL and KVL follow, and they are *not* extra postulates — but each is a specific conservation law, and KVL's identity in particular is usually mis-stated:

$$\sum_k I_k = 0 \qquad \text{(charge conservation: } \nabla\cdot\mathbf{J} = -\partial_t\rho \text{ integrated over a node)}$$

$$\sum_k V_k = 0 \qquad \text{(Faraday's law expressed in the lumped-element representation)}$$

KVL is a loop statement about E, not a statement about energy. It is Faraday's law $\oint \mathbf{E}\cdot d\mathbf{l} = -d\Phi_B/dt$, rewritten once every branch has been assigned a terminal voltage — which is the content of the chain below. Calling it "energy conservation" conflates two different laws: the conservation of energy in the electromagnetic field is **Poynting's theorem**,

$$\frac{\partial}{\partial t}\left(u_{\text{em}}\right) + \nabla \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}, \qquad \mathbf{S} = \frac{1}{\mu_0}\mathbf{E} \times \mathbf{B}$$

which is a *local, volumetric* statement involving the energy density u_{em} and the flux $\mathbf{S}$. KVL is an *integrated, topological* statement around a loop.

The two are related, but the relation runs through the circuit elements rather than through identity: once the three constitutive laws of step 4 are substituted, the resulting RC, RL and RLC equations do satisfy an energy balance — each element either stores ($\frac{1}{2}CV^2$, $\frac{1}{2}LI^2$) or dissipates (I^2R). That consistency is a *result* of the reduction, and a useful check on it. It is not what KVL states.

The correct chain is Faraday → flux linkage → induced EMF → KVL. Start from

$$\oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d\Phi_B}{dt}$$

In a lumped inductor, Φ_B is *proportional to the current* by construction — that proportionality is the definition of the inductance, $L = N\Phi_B/I$. Differentiating gives the terminal voltage of the inductor itself:

$$V_L = -\frac{d\Phi_B}{dt} = L\frac{dI}{dt}$$

(or $N\,d\Phi/dt$ for an N-turn coil). So the induced EMF **is** the inductor's contribution to the loop sum, and Kirchhoff's voltage law is simply what that loop integral becomes once every branch has been given a terminal voltage:

$$\oint \mathbf{E} \cdot d\mathbf{l} = \sum_k V_k \implies \sum_k V_k = 0$$

The thing to unlearn is the tempting shortcut — "KVL is the loop law with the time derivative dropped," i.e. $\nabla \times \mathbf{E} = -\partial_t \mathbf{B} = 0$. That is wrong, and wrong in a self-defeating way: setting $\partial_t \mathbf{B} = 0$ is precisely the condition that makes $V_L = L\dot{I} = 0$. You would be deleting the inductor in the act of deriving the law that has to accommodate it. The quasi-static approximation in circuit theory does **not** mean the magnetic field stops changing; it means the *radiation* and *displacement* effects can be neglected so that $\mathbf{B}$ is slaved to $\mathbf{I}$ by the constitutive relation rather than solved independently. The changing flux is not dropped — it is *accounted for*, in the inductor term.

Step 5 — Engineering equation

Worked problem. A 10 nF capacitor and a 2.2 kΩ resistor form an RC stage driving a 1 MΩ oscilloscope input. What is the −3 dB bandwidth, and what does the scope's loading do to it?

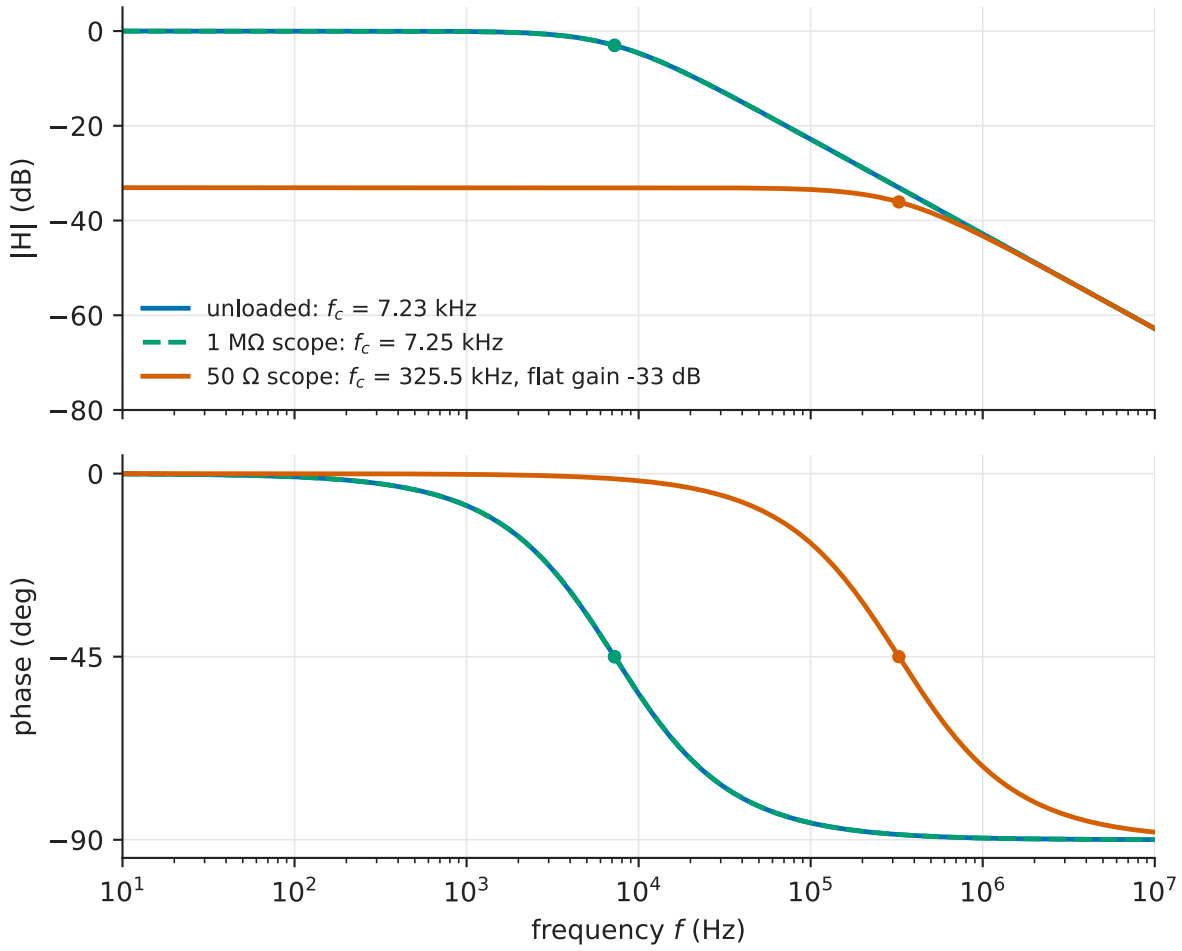
$$\tau = RC = 2.2 \times 10^3 \times 10 \times 10^{-9} = 22\ \mu\text{s}$$

$$f_{-3\text{dB}} = \frac{1}{2\pi\tau} = 7.2\ \text{kHz}$$

Now the loading. The scope presents 1 MΩ in parallel with the 2.2 kΩ:

$$R_{eq} = \frac{2.2 \times 10^3 \times 10^6}{2.2 \times 10^3 + 10^6} = 2.195\ \text{k}\Omega$$

A 0.2% change. Negligible — *because* 1 MΩ ≫ 2.2 kΩ. Swap in a 50 Ω scope input and $R_{eq} = \frac{2.2\times10^3\times50}{2.2\times10^3+50} = 48.89\ \Omega$. The source now sees a ~45:1 divider (2.2 kΩ into 50 Ω), so the signal reaching the scope is attenuated ~45-fold, and the time constant collapses from 22 µs to $48.89 \times 10 \times 10^{-9} = 0.49\ \mu$s, pushing $f_{-3\text{dB}} = \frac{1}{2\pi\times48.89\times10\times10^{-9}} = 325$ kHz — 45× *higher* than the designed corner. The design is no longer the circuit that was analysed. **The same design is correct or useless depending on a load the circuit theory never mentions**, because the load is outside the lumped element and only enters as a boundary condition.



$H = [R_L/(R + R_L)]/(1 + j\omega R_{eq}C)$ for R = 2.2 kΩ, C = 10 nF, from 10 Hz to 10 MHz: the 1 MΩ scope leaves the 7.2 kHz corner where it was, while the 50 Ω scope (Thevenin R_{eq} = 48.89 Ω) drops the passband to 50/2250 ≈ −33 dB and moves the corner to ≈325 kHz.

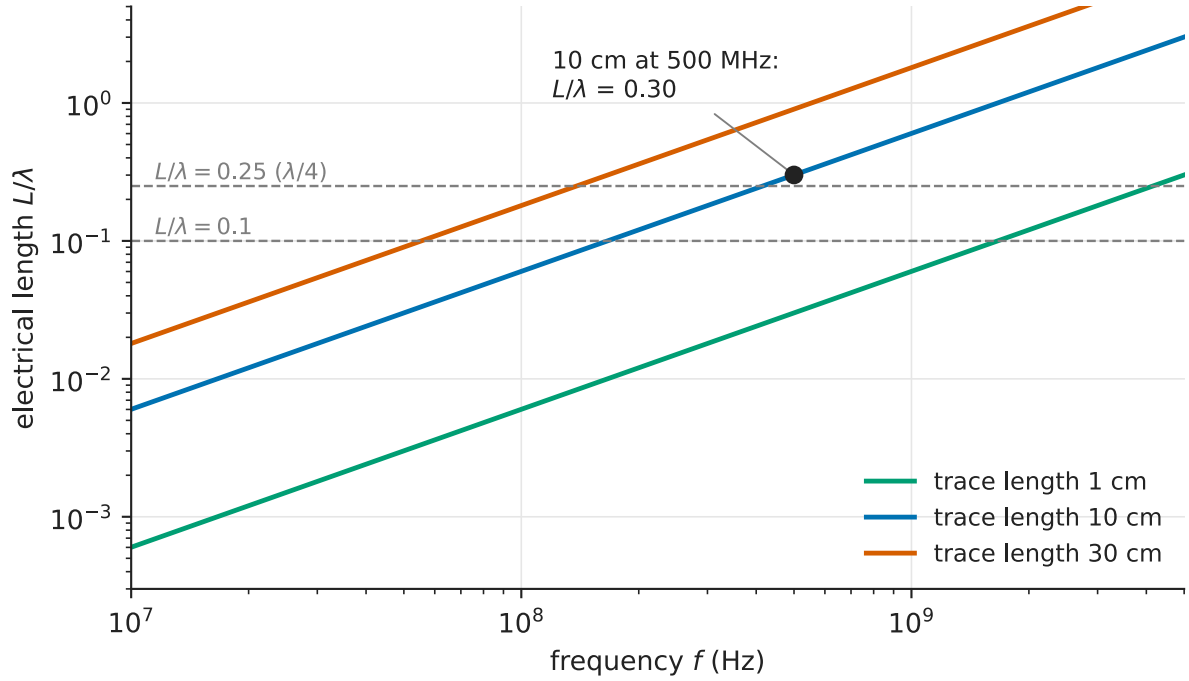
This is the practical meaning of the lumping criterion: it is not a statement about the circuit, it is a statement about the circuit
and its surroundings
.

Step 6 — Validity limits

Assumption	Fails when	Use instead
$L/\lambda \ll 1$	Trace or interconnect length $\gtrsim \lambda/10$	Transmission-line theory; scattering parameters

Assumption	Fails when	Use instead
Quasi-static ($V_L = L\dot{I}$ treated as lumped)	Fast edges, high dI/dt	Retain distributed L explicitly; transmission-line or full-wave treatment
Lumped geometry	Distributed sources, radiation, slots	Full-wave FEM (Ch. 16 §16.11)
Constant ϵ, μ	Non-linear dielectrics, ferroelectrics	Non-linear capacitance; hysteresis
$R \gg 0$	Superconductors, ideal conductors	$R = 0$ gives a genuinely different topology (flux quantisation, Ch. 7 §7.8)
Frequency-independent R	Skin effect, $f > R_{skin}$	$Z(\omega)$, complex permittivity

A worked confirmation of the first row: on FR-4 ($\epsilon_r \approx 4$, effective permittivity somewhat lower on microstrip, so $\epsilon_{eff} \approx 3$–3.5) a trace on 500 MHz has a guided wavelength of only $\lambda = c/(f\sqrt{\epsilon_{eff}}) \approx 0.32$–$0.35$ m, so $\lambda/4 \approx 8$–9 cm. A 10 cm trace therefore exceeds a quarter wavelength — $L/\lambda \approx 0.3$ — and the lumping criterion is already violated in ordinary FR-4 at ordinary signal rates. The "just wires" model is an approximation with a hard frequency ceiling, and that ceiling is the reason signal-integrity engineering exists as a discipline.



$L/\lambda = Lf\sqrt{\epsilon_{eff}}/c$ on FR-4 microstrip (ϵ_{eff} = 3.25) from 10 MHz to 5 GHz for 1, 10 and 30 cm traces, with the L/λ = 0.1 lumping threshold and the quarter-wave line: the 10 cm trace is already at $L/\lambda \approx$ 0.30 at 500 MHz.

What this example demonstrates. Three of the four steps — integrate, apply the divergence theorem, identify L and C — are exact. The entire descent rests on one inequality, $L/\lambda \ll 1$, and everything the lumped model cannot do (resonance, standing waves, radiation, propagation delay) is recovered the moment that inequality fails.

Model Ledger: see Ch. 16 §16.10 and Ch. 17 §17.1.1 for the nine-row ledgers recording this reduction's assumptions, neglected physics, and failure modes.

Example 4 — Heat Equation → Thermal RC Network

A complete worked descent in six steps.

Source chapters: Ch. 15 §15.7 (Navier–Stokes/energy) → Ch. 16 §16.2 (diffusion) → Ch. 17 §§17.1–17.3 (lumping, effort/flow) → Ch. 19 §19.6 (resistances, dimensionless numbers)

Question answered: A thermal network looks like a circuit diagram but has no inductors. What is missing, and why does it matter?

Step 1 — Theory

Start from energy conservation for a solid with internal energy u and heat flux $\mathbf{q}$:

$$\rho c_p \frac{\partial T}{\partial t} = -\nabla \cdot \mathbf{q}$$

Energy is the conserved quantity, not heat. Heat is not a substance that accumulates — it is energy *in transit*, and it has no independent conserved density, which is why the balance above is written in terms of internal energy u and flux $\mathbf{q}$ rather than some "heat content." This step is the **first law of thermodynamics applied locally** — a law of physics, not a definition. What it does *not* determine is $\mathbf{q}$, and closing that gap is the entire job of the next line.

For a continuum, the flux follows from irreversible thermodynamics — the Onsager form of Fourier's law (Ch. 13 §13.5.2):

$$\mathbf{q} = -k \nabla T$$

Substituting gives the heat equation:

$$\boxed{\rho c_p \frac{\partial T}{\partial t} = k \nabla^2 T \qquad \Longleftrightarrow \qquad \frac{\partial T}{\partial t} = \alpha \nabla^2 T, \quad \alpha = \frac{k}{\rho c_p}}$$

The epistemic status here is mixed, and the ledger depends on not blurring it. Energy conservation is a law. Fourier's law is a **constitutive law** — `[PHENOMENOLOGICAL]` in its macroscopic form. It can be *justified* as a linear-response limit near equilibrium (the Onsager route of Ch. 13), which is what earns it more than a bare empirical fit, but it is not an identity and it does not follow from conservation. The heat equation is therefore a **composition**:

$$\underbrace{\text{energy conservation}}_{\text{law}} + \underbrace{\text{Fourier constitutive law}}_{\text{constitutive}} \longrightarrow \text{heat equation}$$

not a pure derivation from conservation alone. α is a genuine material property with units $\mathrm{m^2/s}$ — a diffusivity — and the linear-response reading also gets its scaling right, but the constitutive step is where the physics is *supplied*, and it is the step that fails when k is anisotropic or strongly temperature-dependent, or when the linear-gradient ansatz breaks down.

Step 2 — Approximation

#	Reduction	Kind	Cost
A	Constant k, ρc_p	`[APPROXIMATION]`	no temperature-dependent properties, no phase change
B	One-dimensional conduction	`[APPROXIMATION]`	no radial or 3D loss
C	Lumped: $Bi = hL_c/k \ll 0.1$	`[APPROXIMATION]`	no internal gradient
D	Convection boundary: $q'' = h(T_s - T_\infty)$	`[PHENOMENOLOGICAL]`	h is empirical, not derived from first principles

Reduction D deserves attention: it is the one place a thermal model meets a purely empirical constitutive law. Newton's law of cooling has no microscopic derivation — h is a fitted number depending on geometry, orientation, and flow regime. Everything else in a thermal network is `[DERIVATION]` or `[APPROXIMATION]`; this is `[PHENOMENOLOGICAL]`, and it is usually the dominant uncertainty in a real thermal design.

Step 3 — Mathematical reduction

Step 3a — integrate over the body (exact):

$$\rho c_p V \frac{dT}{dt} = -\oint \mathbf{q} \cdot \hat{n}\, dA$$

Step 3b — split the boundary into convection, radiation, and contact terms:

$$\rho c_p V \frac{dT}{dt} = hA(T_\infty - \bar{T}) + \varepsilon\sigma A\left(T_{sur}^4 - \bar{T}^4\right) + \cdots$$

Step 3c — linearise radiation about the operating point (the small-ΔT approximation):

$$\varepsilon\sigma A(T_{sur}^4 - \bar{T}^4) \approx 4\varepsilon\sigma A\bar{T}^3\,(T_{sur} - \bar{T})$$

Step 3d — identify the network elements:

Physical term	Lumped element	Value
$\rho c_p V\,\dot{T}$	capacitance C	$C = \rho c_p V$ [J/K]
$hA\,(T_\infty - T)$	resistance R_{conv}	$R_{conv} = 1/hA$ [K/W]
$4\varepsilon\sigma A\bar{T}^3$	added resistance	$R_{rad} = 1/(4\varepsilon\sigma A\bar{T}^3)$
conduction through a wall	resistance	$R_{cond} = \delta/(kA)$

Steps 3a–3b are exact. Step 3c is an approximation that fails for large temperature swings. Step 3d is a definition.

Step 4 — Reduced model

The energy balance becomes a first-order ODE:

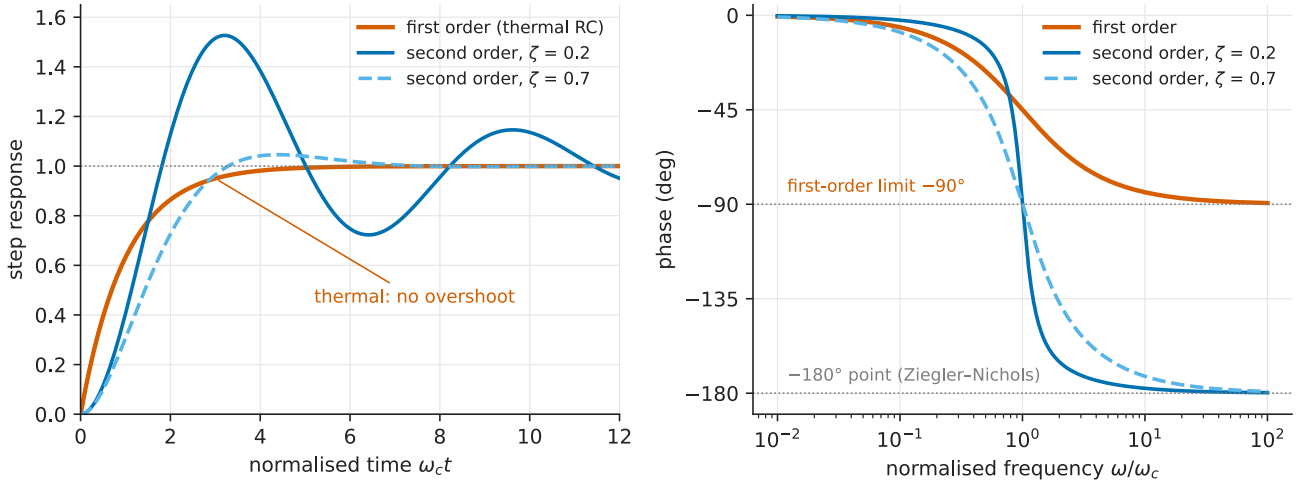
$$C\dot{T} + \frac{T - T_\infty}{R_{conv}} = \dot{Q}_{source}$$

Read as a circuit: a capacitor C to ground, a resistor R_{conv} to the ambient node, a current source injecting $\dot{Q}_{source}$.

The missing element. There is no L. Thermal energy has no inertial term because heat is not a conserved *quantity in motion* with momentum — it is energy being transferred, and the transfer rate is set by the resistance alone. So:

Domain	Order	Resonance?	Overshoot?
Electrical	2nd order	yes	yes
Mechanical	2nd order	yes	yes
Thermal	**1st order**	**no**	**no**

This is a design constraint, not a curiosity. A first-order loop has no phase lag beyond -90°, so a PID tuned on a mechanical second-order plant will be systematically mistuned on a thermal one — typically too aggressive, because the plant never rings. Ziegler–Nichols ultimate-gain tuning, which deliberately drives a plant to its -180° phase point, cannot even be applied: a first-order plant has no such point.



Left: normalised step responses of a first-order plant (the thermal RC) and second-order plants with ζ = 0.2 and 0.7, against $\omega_c t$; right: their phase against ω/ω_c. The first-order curve never overshoots and its phase never passes −90°, so it has no −180° point for Ziegler–Nichols tuning to find.

Step 5 — Engineering equation

Worked problem. A 100 mm cube of aluminium ($k = 205$ W/m·K, $\rho c_p = 2.90 \times 10^6$ J/m³·K) is cooled by forced air, $h = 50$ W/m²·K, on all six faces in ambient air at 20 °C. Initially at 80 °C. Find τ and the temperature after 60 s, and state whether the lumped model was justified.

Check the criterion first — this is the step most solutions skip:

$$L_c = \frac{V}{A} = \frac{10^{-3}\ \text{m}^3}{6 \times 10^{-2}\ \text{m}^2} = 1.667 \times 10^{-2}\ \text{m} = 16.67\ \text{mm}, \qquad Bi = \frac{hL_c}{k} = \frac{50 \times 0.01667}{205} = 4.07 \times 10^{-3}$$

$Bi \ll 0.1$: the lumped model is justified, with a large margin. (At $Bi = 0.1$ the internal temperature difference becomes non-negligible — about 5% centre-to-surface for a slab, shape-dependent and larger for other geometries — and lumping begins to fail.)

Capacitance and resistance:

$$C = \rho c_p V = 2.90 \times 10^{6} \times 10^{-3} = 2900 \text{ J/K}$$

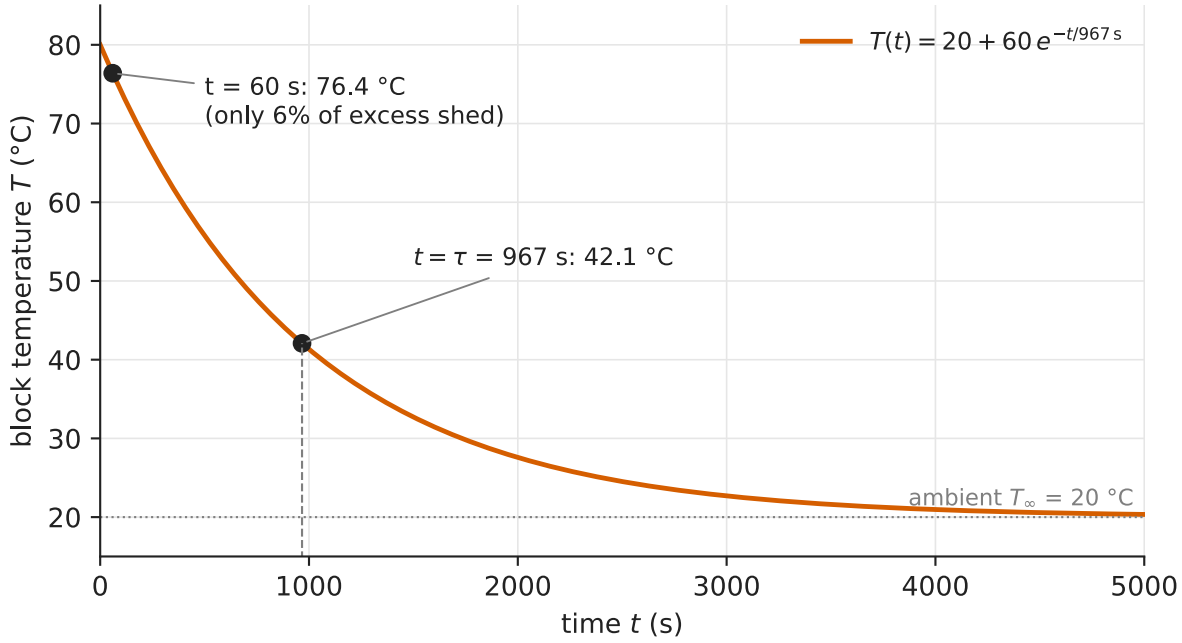
$$R = \frac{1}{hA} = \frac{1}{50 \times 6 \times 10^{-2}} = 0.333 \text{ K/W}$$

Time constant and transient:

$$\tau = RC = 0.3333 \times 2900 = 967 \text{ s} \approx 16 \text{ min}$$

$$T(t) = T_\infty + (T_0 - T_\infty)e^{-t/\tau}$$

$$T(60) = 20 + 60\,e^{-60/967} = 20 + 60 \times 0.940 = 76.4\ ^\circ\text{C}$$



$T(t) = 20 + 60\,e^{-t/967\,\text{s}}$ over 0–5000 s: after one time constant (967 s) the block is at 42.1 °C, while after the 60 s dwell it is still at 76.4 °C.

After one minute the block has shed only 6% of its excess temperature. **That is the design finding.** A 60 s dwell is thermally meaningless for this part, and any process timed to it will run out of equilibrium long before it starts.

Step 6 — Validity limits

Assumption	Fails when	Consequence
$Bi \ll 0.1$	$Bi \gtrsim 0.1$ — thin plates, low k, low h	internal gradients; surface $T \neq$ mean T
Constant h	flow regime changes, natural convection, boiling	h is itself a function of the answer
Constant properties	high T; phase change	conductivity falls; latent heat adds a plateau
Lumped radiation	ΔT large	T^4 cannot be linearised
Steady h, no radiation coupling	enclosure geometry matters	view factors

Assumption	Fails when	Consequence
Time-invariant α	transient conduction at short t	**see below**

The subtle one. Two different criteria get confused here, and they are opposites. For a **semi-infinite** solid — one whose depth greatly exceeds the conduction depth — the surface response never settles: the conduction front spreads as $\sqrt{\alpha t}$, the effective resistance behaves like $1/\sqrt{\alpha t}$, and there is no constant R in the lumped sense at all. Lumping requires the **opposite** limit. The body must be *small* compared with the diffusion length, so that heat has time to spread through it and it becomes nearly isothermal.

Here $\alpha = k/(\rho c_p) = 205/(2.90 \times 10^6) = 7.07 \times 10^{-5}$ m²/s, so over the thermal time constant $\tau = RC = 967$ s the diffusion length is $\sqrt{\alpha\tau} = \sqrt{7.07 \times 10^{-5} \times 967} = 0.26$ m — about five times the 50 mm half-width. In Fourier-number form, $Fo = \alpha\tau / L_c^2 = (7.07 \times 10^{-5} \times 967)/(0.01667)^2 = 246 \gg 1$. **The body is therefore well mixed internally, and the single-node RC model is self-consistent** — which is exactly what the $Bi \ll 0.1$ check already said. Bi compares surface resistance against internal resistance; Fo compares diffusion against the time available. Both are required, and together they are the lumped regime.

Had $\sqrt{\alpha t}$ come out *smaller* than L_c, the conclusion inverts: the lumped assumption fails, and the one-node model has to be replaced by a spatial discretisation — a semi-infinite or fin analysis, depending on geometry. Getting this one inequality backwards is the standard way a thermal RC model turns out to be wrong at the edge of its validity window.

What this example demonstrates. The thermal network is a real circuit — the same conservation laws, the same algebra, the same solution methods. It is also not an electrical circuit: it is first order, it has no inertia, and its single constitutive parameter (h) is empirical rather than derived. Recognising *which* engineering template applies is the skill the cross-branch identity is supposed to build.

Model Ledger: see Ch. 17 §17.1.1 and Ch. 19 §19.9.1 for the nine-row ledgers, including the HVAC example that shows why a lumped thermal model has a *time-dependent* validity window rather than a bandwidth.

Example 5 — Navier–Stokes → Fluid Network

A complete worked descent in six steps.

Source chapters: Ch. 13 §13.7 (Newtonian viscosity from Kubo) → Ch. 15 §§15.3–15.7 (stress tensor, continuum conservation, ideal and viscous flow, exact solutions) → Ch. 16 §16.10 (lumping) → Ch. 19 §§19.7.1–19.7.3 (hydraulic effort–flow pair, pipe networks, pump curves) → Ch. 20 §20.3 (open-channel hydraulics)

Question answered: How close is pipe-network analysis to Ohm's law, and which term breaks the analogy? The answer turns out to be friction, not inertia.

Step 1 — Theory

The incompressible Navier–Stokes equations, with a Newtonian stress tensor:

$$\rho\left(\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v}\cdot\nabla\mathbf{v}\right) = -\nabla p + \mu\nabla^2\mathbf{v} + \rho\mathbf{g}$$

$$\frac{\partial \rho}{\partial t} + \nabla\cdot(\rho\mathbf{v}) = 0, \qquad \nabla\cdot\mathbf{v} = 0$$

The second line is a *separate* conservation law — mass continuity — and is **not** a consequence of the momentum balance above. What *is* an approximation is the reduction to $\nabla\cdot\mathbf{v} = 0$, which requires continuity **together with** the assumption that ρ is constant: it is mass conservation *plus* an isochoric assumption, not momentum balance alone. Keeping all three statements visible keeps the distinctions honest, and it also flags the status of the constitutive and kinematic choices, because they are **not** the same kind of claim:

- **Newtonian viscosity** ($\tau = \mu\, d\mathbf{v}/dy$) — a **constitutive law**, valid `[PHENOMENOLOGICAL]`-to-`[DERIVATION]`: linear in the rate of strain over a range of shear rates, and for a Newtonian fluid the coefficient μ can indeed be obtained from the Kubo expression of Ch. 13. But what Kubo yields is a *microscopic expression for the coefficient*, not a derivation of the whole linear stress–rate law for arbitrary engineering fluids. That law still has to be postulated as constitutive. Water, air, and most liquids and gases satisfy it; blood, toothpaste, and polymer melts do not — and no coefficient fixes that.
- **Incompressibility** ($\nabla\cdot\mathbf{v} = 0$) — an `[APPROXIMATION]` / **model assumption**, a limit rather than a derivation. Constant density *along material trajectories* gives $D\rho/Dt = 0$, and it is mass conservation (above) that turns this into $\nabla\cdot\mathbf{v} = 0$. So the logic runs "assume the flow is nearly isochoric," not "the density is constant, therefore the continuity equation has been derived." It is a good approximation for liquids and low-Mach gases, and a poor one for cavitation, compressible transients, and detonations.

Step 2 — Approximation

#	Reduction	Kind	Typical validity
A	Steady, $\partial_t \to 0$	`[APPROXIMATION]`	quasi-steady network analysis

#	Reduction	Kind	Typical validity
B	Inertia, $\mathbf{v}\cdot\nabla\mathbf{v}\to 0$	`[APPROXIMATION]`	$Re \ll 1$
C	Unidirectional, fully developed flow	`[APPROXIMATION]`	straight pipe, away from fittings
D	Lumped per segment	`[APPROXIMATION]`	$L \gg D$
E	Empirical friction factor $f(Re, \varepsilon/D)$	`[PHENOMENOLOGICAL]`	Moody chart / Colebrook

Reduction B is the dangerous one. Dropping inertia is a *low*-Reynolds assumption, and a whole lot of real pipe flow is high-Reynolds. The resolution is that inertia can be retained in a lumped form — and, importantly, it comes out **linear**.

Inertia is linear. For uniform (plug) acceleration $a = \dot{v}$, the pressure force $\Delta p\, A$ balances the fluid's inertia $\rho A L a$:

$$\Delta p = \rho L \dot{v} = \frac{\rho L}{A}\dot{Q}, \qquad \boxed{\mathcal{I}_h = \frac{\rho L}{A}}$$

This is linear in $\dot{Q}$ and it is *exactly* the inductor law $V = L_{ind}\dot{I}$. Hydraulic inertance is a genuine energy-storing element, and the inductor analogy holds term for term.

Friction is what is nonlinear. The steady momentum balance, with friction retained, gives **Darcy–Weisbach**:

$$\boxed{\Delta p = f\frac{L}{D}\frac{\rho v^2}{2} + \rho g\,\Delta z} \qquad \text{or in head form} \qquad \boxed{\Delta h = f\frac{L}{D}\frac{v^2}{2g} + \Delta z}$$

The v^2 belongs to *this* term, and this term is quadratic drag, not inertia. It is worth being explicit about the difference, because conflating them is the classic error: an inertance whose drop went as v^2 would be dimensionally plausible but physically meaningless, and it would destroy the inductor correspondence that makes hydraulic networks tractable at all. Darcy–Weisbach itself contains no time derivative, so it says nothing about inertia either way.

The complete lumped pipe segment, for a transient, is the sum of the three separate contributions:

$$\Delta p = \underbrace{\frac{\rho L}{A}\dot{Q}}_{\text{inertance, linear}} + \underbrace{f\frac{L}{D}\frac{\rho Q^2}{2A^2}}_{\text{friction, quadratic, } f \text{ empirical}} + \underbrace{\rho g\,\Delta z}_{\text{gravity, static}}$$

So the hydraulic network is a linear RLC-like circuit *plus* one nonlinear, curve-fitted element. The nonlinearity is real, but it enters in one place only, and it is the empirical one.

Step 3 — Mathematical reduction

For a straight uniform pipe, assuming a velocity profile, integrating $\nabla\cdot\mathbf{v} = 0$ along the axis gives the volume flow:

$$Q = vA = v\frac{\pi D^2}{4}$$

and the momentum balance gives Darcy–Weisbach. Identify the effort/flow pair:

Role	Quantity	Units
Effort	pressure p (or head $h = p/\rho g$)	Pa, or m
Flow	volume flow Q	m³/s

For laminar flow, integrating the parabolic profile profile-for-profile (Hagen–Poiseuille) gives an *exact* linear law:

$$\Delta p = \frac{128 \mu L}{\pi D^4} Q \qquad \Longrightarrow \qquad R_h = \frac{\Delta p}{Q} = \frac{128 \mu L}{\pi D^4}$$

That D^4 is the reason pipe sizing is dominated by diameter rather than length. It is also why the laminar resistance is a clean `[DERIVATION]` and the turbulent one is not.

Step 4 — Reduced model

Network elements, exactly parallel to the electrical case:

Physical element	Hydraulic form	Electrical analog	Notes
Pipe, laminar	$R_h = \frac{128\mu L}{\pi D^4}$, $R_h = \Delta p / Q$	resistor	linear; exact, `[DERIVATION]`
Pipe, turbulent	$R_h = \frac{fL}{D}\frac{\rho v}{2}$, $f = f(Re, \varepsilon/D)$	nonlinear resistor	I^2R-like; f empirical
Fitting / valve	R_h	resistor	K-resistance; highly empirical
Fluid column in transit	$\Delta p = \frac{\rho L}{A}\dot{Q}$	**inductor**	inertance $\mathcal{I}_h = \rho L / A$; **linear**, energy-storing
Expansion vessel / tank	$C = dV/dp$	capacitor	compressibility storage
Pump	pressure source	voltage source	head–flow curve, not an ideal source

Note the asymmetry that trips people up: the **inertance row is linear** and the **friction row is not**. A pipe is therefore an RLC segment in the linear regime, and it is the turbulent f — not the inertia — that makes real networks nonlinear.

Node and loop rules are exact.

- **QCL** (flow sums to zero at a junction) is **mass conservation**, $\sum_k \dot{m}_k = 0$, which is the local continuity equation $\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0$ integrated over a control volume enclosing the junction. For constant density this reduces to $\boxed{\sum_k Q_k = 0}$, the form actually used in network analysis. (Note that $\partial_t \rho = 0$ on its own is *not* mass conservation — it is only the steady-density piece of it, and it drops the flux term that does the real work. Stated that way it loses exactly the claim being made.)
- **Loop rule** (head sums to zero around a loop) is a **hydraulic energy balance**, not Bernoulli. Bernoulli's ideal form $\frac{p}{\rho g} + \frac{v^2}{2g} + z = \text{const}$ requires a steady, inviscid flow along a single streamline with no losses and no machinery — and every one of those fails in the network above, which has friction, fittings, a pump, and turbulent sections. The working statement collects head gains and losses around the closed loop,

$$\sum h_{\text{pump}} - \sum h_{\text{loss}} = 0, \qquad h_f = f\frac{L}{D}\frac{v^2}{2g}$$

with h_f the distributed friction loss plus minor losses from fittings, valves and entrances. **Bernoulli is the frictionless, lossless, no-machinery limit of this energy equation** — a useful check, not the working rule. It is the *energy* bookkeeping that KVL tracks, and that is what supplies the analogy; streamline mechanics is a different and much narrower statement.

Both are identities given the assumptions; neither is an electrical coincidence.

Step 5 — Engineering equation

Worked problem. A 40 mm nominal steel pipe, 60 m long, carries water ($\nu = 1.0 \times 10^{-6}$ m²/s, $\varepsilon = 0.045$ mm) from a reservoir at the 30 m mark to a tank at the 10 m mark. Find the flow rate.

Available head: $\Delta h = 30 - 10 = 20$ m.

Iterate on Re because f depends on it. Guess $v = 1$ m/s: $Re = vD/\nu = 1 \times 0.04/10^{-6} = 4 \times 10^4$ (turbulent). $\varepsilon/D = 0.000045/0.04 = 1.125 \times 10^{-3}$; Colebrook gives $f \approx 0.0251$.

$$h_f = f\frac{L}{D}\frac{v^2}{2g} = 0.0251 \times \frac{60}{0.04} \times \frac{1}{19.62} = 37.7 \times 0.051 = 1.92 \text{ m}$$

That is far below the 20 m available, so the guess was too low. Try $v = 3$ m/s: $Re = 1.2 \times 10^5$, $f \approx 0.0223$,

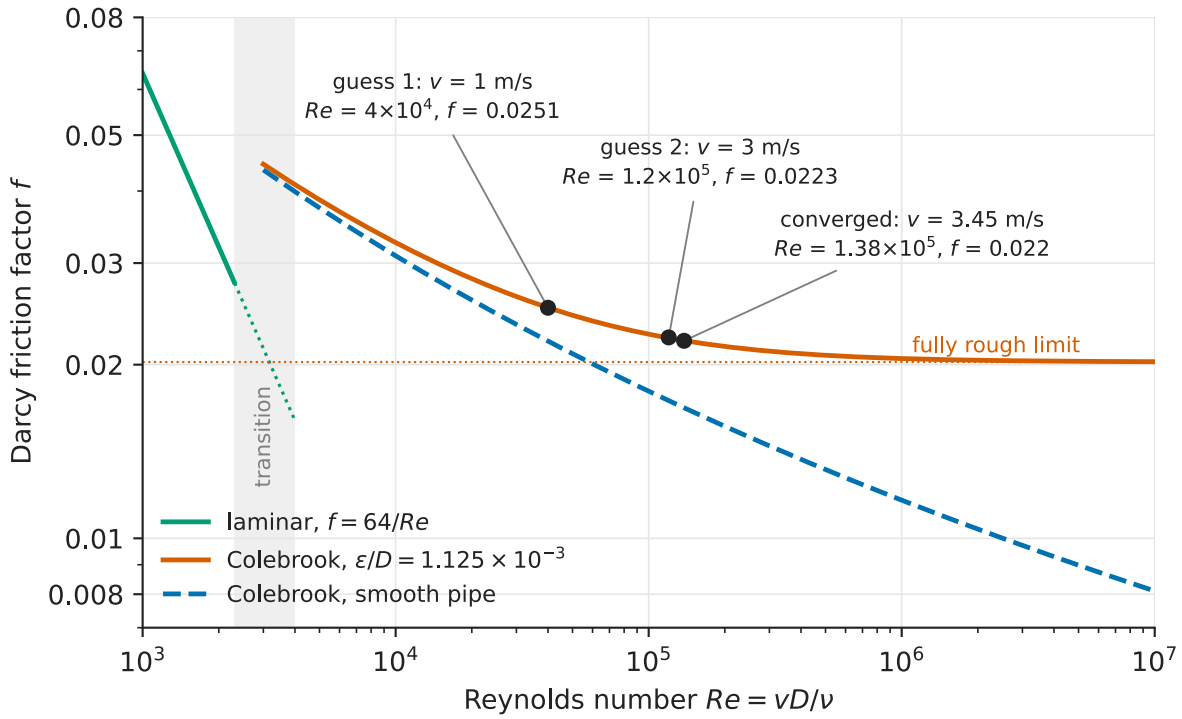
$$h_f = 0.0223 \times 1500 \times \frac{9}{19.62} = 33.4 \times 0.459 = 15.3 \text{ m}$$

Still below 20. Try $v = 3.45$ m/s: $Re = 1.38 \times 10^5$, $f \approx 0.0220$,

$$h_f = 0.0220 \times 1500 \times \frac{11.90}{19.62} = 33.0 \times 0.607 = 20.0 \text{ m}$$

Converged: $v \approx 3.45$ **m/s**, at $Re \approx 1.38 \times 10^5$ with $f \approx 0.0220$.

$$Q = vA = 3.45 \times \frac{\pi(0.04)^2}{4} = 3.45 \times 1.257 \times 10^{-3} = 4.33 \times 10^{-3} \text{ m}^3/\text{s} \approx 4.33 \text{ L/s}$$



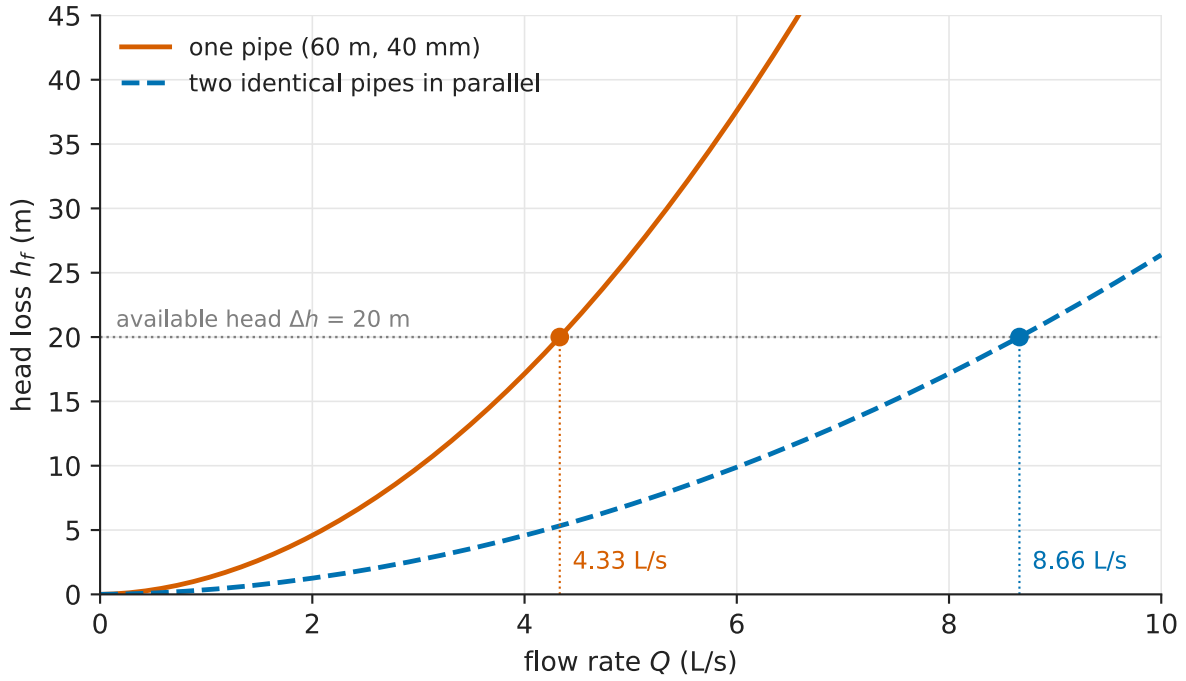
Colebrook $f(Re)$ for $\varepsilon/D = 1.125 \times 10^{-3}$ and for a smooth pipe, with laminar $f = 64/Re$, over Re = 10^3–10^7: the three iterates (4×10^4, 0.0251), (1.2×10^5, 0.0223) and (1.38×10^5, 0.0220) sit where the rough-pipe curve is already flattening toward its fully rough limit, which is why f barely moves between guesses.

A caution about the friction factor. It is worth being explicit about why the answer is sensitive to f, because the arithmetic is easy to get wrong in the direction that flatters the result. Colebrook at this roughness ($\varepsilon/D = 1.125 \times 10^{-3}$) returns $f \approx 0.022$ across the whole range $Re = 10^5$–$10^{5.5}$, and using a value near 0.018 — as a Moody-chart eyeball or a smooth-pipe habit might suggest — understates the loss by roughly 20% and inflates the flow by about 10%. The converged f is only weakly dependent on Re here precisely *because* the relative roughness dominates at these Reynolds numbers, which is exactly the regime the Colebrook form is built to handle.

The design finding. Head loss scales as v^2, so for a fixed pipe the flow rate grows only as the *square root* of the available head: $Q \propto \sqrt{\Delta h}$. That is the nonlinearity that matters, and it is why a pump curve has to be intersected with a system curve numerically.

What a second parallel pipe does — and does not — change. Adding an identical second pipe between the same two reservoirs **doubles the flow exactly**: $Q_{total} = 2Q_1$. Each parallel branch spans the same two nodes, so each sees the *same* head difference of 20 m, each runs at the same Re and the same f, and each independently carries $Q_1 = 4.33$ L/s. For this example that gives $Q_{total} \approx 8.66$ L/s. Head is *shared* between parallel branches, not divided among them; it is **series** elements that split the available head, each taking a share in proportion to its resistance.

(This is exact only while the branches really are in parallel between common nodes. If the two pipes share a common upstream run of finite resistance, that shared run develops a pressure drop as total flow rises, the effective head across each branch falls, and the gain is less than a factor of two. The doubling is the ideal limit, and stating it as "roughly double, but not quite" is wrong for the clean case and vague for the imperfect one.)



Head loss $h_f(Q)$ from Darcy–Weisbach with Colebrook f for the 60 m, 40 mm pipe, and for two identical pipes in parallel (each carrying $Q/2$ at the same head): the curves cross the 20 m available head at 4.33 L/s and 8.66 L/s, exactly a factor of two apart.

Step 6 — Validity limits

Assumption	Fails when	Use instead
Newtonian fluid	non-Newtonian rheology ($n \neq 1$)	power-law model; apparent viscosity
Incompressible	$M > 0.3$, cavitation	compressible flow; flashing
$Re < 2000$, Hagen–Poiseuille	$Re > 2300$	Colebrook/Moody; f empirical
Fully developed	$L/D < 20$, fittings	entrance lengths; K-factors
Inertia dropped	Re large, or transients of interest	**retained as the linear inertance $\rho L/A$; drop it only for slow, quasi-steady, low-Re analysis**
Lumped per segment	fittings, valves, short lines	Ch. 15 §15.8.2 — turbulence has no exact closed-form model
Single-phase	two-phase, flashing	two-phase flow correlations
f constant with Q	varying Re over the operating range	solve network iteratively

The f-versus-Re dependency is worth flagging as an engineering trap: because $f \propto Re^{-0.2}$ in turbulent flow, the head loss goes as $Q^{1.8}$, not Q^2. A pump curve intersecting a system curve needs a numerical solution, not algebra, and treating f as constant is a common source of wrong answers.

What this example demonstrates. Ohm's law and the Darcy pressure–flow relation have the same *network form* — an effort drop proportional to a flow variable in the linear regime — and that shared shape is not a coincidence of notation: both networks acquire it the same way, by combining conservation

laws with constitutive relations and then lumping. The *laws*, however, are not the same. Electrical KCL is charge conservation while its pipe counterpart is mass conservation; electrical KVL descends from Faraday's law while the hydraulic loop relation is a mechanical energy balance; $V = IR$ is an electrical constitutive law while $\Delta p = R_h Q$ is a hydraulic correlation fitted to real pipe behaviour. The defensible claim is therefore **structural equivalence without physical identity** — the network algebra transfers, the physics underneath it does not. The inertia is linear and maps cleanly onto an inductor; the nonlinearity enters through *friction*, and the one coefficient that governs it (f) is empirical. So a pipe network is circuit theory with a curve-fit in the resistive branch — closer to RLC than the usual "pipes are not circuits" refrain suggests, but different enough that the f dependence must be respected rather than assumed away.

Model Ledger: see Ch. 19 §19.9.1 for the nine-row ledger on a coupled thermofluid network, and Ch. 20 §20.4.1 for the four-way classification that keeps Darcy's law labelled as constitutive correspondence rather than mathematical identity — they are *not* the same thing, as Step 3 shows.

Example 6 — Motor → Control System

A complete worked descent in six steps. This is the book's capstone example in miniature: a physical machine, reduced to a transfer function, closed into a control loop, with the validity limits stated at every stage.

Source chapters: Ch. 17 §17.7 (second-order template) → Ch. 19 §19.2 (rotational mechanical systems) → Ch. 21 §§21.0.1–21.0.2 (what the capstone claim is, and the boundary of the transfer function), §§21.2–21.4 (block-diagram algebra, stability, PID), §21.8.1 (DC motor speed control)

Question answered: How do you actually design a motor speed controller, and where does the physics stop being trustworthy?

Step 1 — Theory

A brushed DC motor is two coupled physical laws and nothing else.

Electrical (Ohm's law for the armature, with the back-EMF written as a voltage *drop* in the direction that opposes the applied source — Lenz's law):

$$V_a = R_a I_a + L_a \frac{dI_a}{dt} + K_b \omega$$

The sign matters and is worth being explicit about. In motoring, ω and I_a have the same sign, so $K_b\omega$ is a drop in series with the applied voltage; it can never *add* energy. Writing it as $-K_b\omega$ on the right-hand side would make the back-EMF a source, and every sign downstream (the $K_t K_b$ term in the transfer-function denominator) would be wrong.

Mechanical (Newton's second law for the rotor, Ch. 19 §19.2.1):

$$J \frac{d\omega}{dt} + B\omega = K_t I_a$$

Symbol	Meaning	Typical value
V_a	applied armature voltage	24 V
R_a	armature resistance	0.5 Ω
L_a	armature inductance	5 mH
J	rotor + load inertia	2×10^{-4} kg·m²
B	viscous friction	2×10^{-5} N·m·s
K_t	torque constant	8×10^{-3} N·m/A
K_b	back-EMF constant	8×10^{-3} V·s/rad

Two first-order ODEs, one electrical state (I_a) and one mechanical state (ω). Nothing about control theory has entered yet.

Step 2 — Approximation

A DC motor is **not** a first-order plant, and treating it as one is the most common modelling error in motor control. There are two states.

#	Reduction	Kind	Cost
A	Neglect $L_a \dot{I}_a$	`[APPROXIMATION]`	valid only if $\omega_{cl}\tau_e \ll 1$ where $\tau_e = L_a/R_a$
B	Linearise $B\omega$ (already linear)	—	none
C	Treat K_t, K_b as constants	`[APPROXIMATION]`	ignores magnetic saturation
D	Lump load inertia into J	`[APPROXIMATION]`	ignores compliant shafts

Reduction A is the tempting one and it is usually wrong in the wrong direction. Neglecting L_a *deletes a state*, so the question is not "is the term small compared with the others" but "is the pole it creates outside the band I care about". That pole is the bare electrical pole, at $-1/\tau_e = -R_a/L_a$. It is negligible only when the **closed-loop** bandwidth sits far *below* it: $\omega_{cl}\tau_e \ll 1$. (The condition is $a \ll 1$, not $a \gg 1$: neglecting an inductance is justified when the frequency is small compared with the inductance's own cutoff, exactly as a low-pass filter passes signals below its corner. An inductance may only be dropped at frequencies well below R_a/L_a.) Here $\tau_e = L_a/R_a = 0.005/0.5 = 10$ ms, so $\omega_{cl} \ll 100$ rad/s.

The mechanical time constant is $\tau_m = J/R_t$ with $R_t = B + K_t K_b/R_a = 2\times10^{-5} + 1.28\times10^{-4} = 1.48\times10^{-4}$:

$$\tau_m = \frac{2\times10^{-4}}{1.48\times10^{-4}} = 1.35 \text{ s}$$

Since $\tau_e \ll \tau_m$, the two poles are widely separated and reducing to first order is defensible — but only once step 4 confirms where the poles actually sit, and only for a bandwidth well under 100 rad/s. **The reduction is valid over a range that depends on the controller you are about to design** — a fact that must be checked *after* the design, not before. Step 5 does exactly that, and step 6 states where it breaks.

Step 3 — Mathematical reduction

Take the Laplace transform with zero initial conditions.

Electrical: $(R_a + L_a s)I_a + K_b\Omega = V_a$, i.e. $(R_a + L_a s)I_a = V_a - K_b\Omega$

Mechanical: $(Js + B)\Omega - K_t I_a = 0$

Eliminate I_a. From the mechanical equation, $I_a = (Js+B)\Omega/K_t$. Substitute:

$$\left[(R_a + L_a s)(Js + B) + K_t K_b\right]\Omega = K_t V_a$$

The $K_t K_b$ enters with a **plus** sign, and it is worth seeing why. The mechanical side feeds the electrical side back: since $\Omega = \frac{K_t}{Js+B} I_a$, the induced voltage $K_b\Omega = \frac{K_t K_b}{Js+B} I_a$ is an impedance $K_t K_b/(Js+B)$ seen by the armature, in parallel with $R_a + L_a s$. A parallel impedance lowers the effective impedance, which shows up as an *increased* constant term in the denominator. The s-coefficient is untouched by the back-EMF, because the coupling feeds back through the mechanical state, which is already in the denominator. This is the standard DC motor reduction, and it is exact within the model.

$$\boxed{\frac{\Omega(s)}{V_a(s)} = \frac{K_t}{(L_a s + R_a)(Js + B) + K_t K_b} = \frac{K_t}{L_a J s^2 + (L_a B + R_a J)s + (R_a B + K_t K_b)}}$$

The middle coefficient is $L_a B + R_a J$. Each term pairs an electrical quantity with the mechanical quantity of the same role: L_a (electrical inertia) with B (rotational damping), and R_a (electrical resistance) with J (rotational inertia). Pairing them the other way round, $JB + L_a R_a$, is not a convention difference — it is dimensionally wrong (JB cannot be a coefficient of s in this polynomial) and it is the single most consequential error this example is here to prevent.

Step 4 — Reduced model

Normalise to the standard second-order form $s^2/\omega_n^2 + 2\zeta s/\omega_n + 1$:

Coefficients, with the values from step 1:

$$L_a J = 1 \times 10^{-6} \qquad L_a B + R_a J = \underbrace{1 \times 10^{-7}}_{L_a B} + \underbrace{1 \times 10^{-4}}_{R_a J} = 1.001 \times 10^{-4} \qquad R_a B + K_t K_b = 7.4 \times 10^{-5}$$

Note how lopsided that middle coefficient is: $R_a J = 10^{-4}$ dominates $L_a B = 10^{-7}$ by three orders of magnitude.

DC gain:

$$K = \frac{K_t}{R_a B + K_t K_b} = \frac{8 \times 10^{-3}}{7.4 \times 10^{-5}} = 108 \text{ rad/s per V}$$

At 24 V that predicts 2595 rad/s ≈ 24,800 rpm. **The real motor cannot do this** — it is torque-limited, not supply-limited. This is the first sign that the linear model's validity is bounded, and it is quantified in step 5.

Natural frequency and damping:

$$\omega_n^2 = \frac{R_a B + K_t K_b}{L_a J} = \frac{7.4 \times 10^{-5}}{1 \times 10^{-6}} = 74 \text{ s}^{-2} \qquad \omega_n = 8.60 \text{ rad/s} \quad (1.37 \text{ Hz})$$

$$\zeta = \frac{L_a B + R_a J}{2\sqrt{(R_a B + K_t K_b)\, L_a J}} = \frac{1.001 \times 10^{-4}}{2\sqrt{7.4 \times 10^{-5} \times 1 \times 10^{-6}}} = \frac{1.001 \times 10^{-4}}{2 \times 8.60 \times 10^{-6}} = 5.82$$

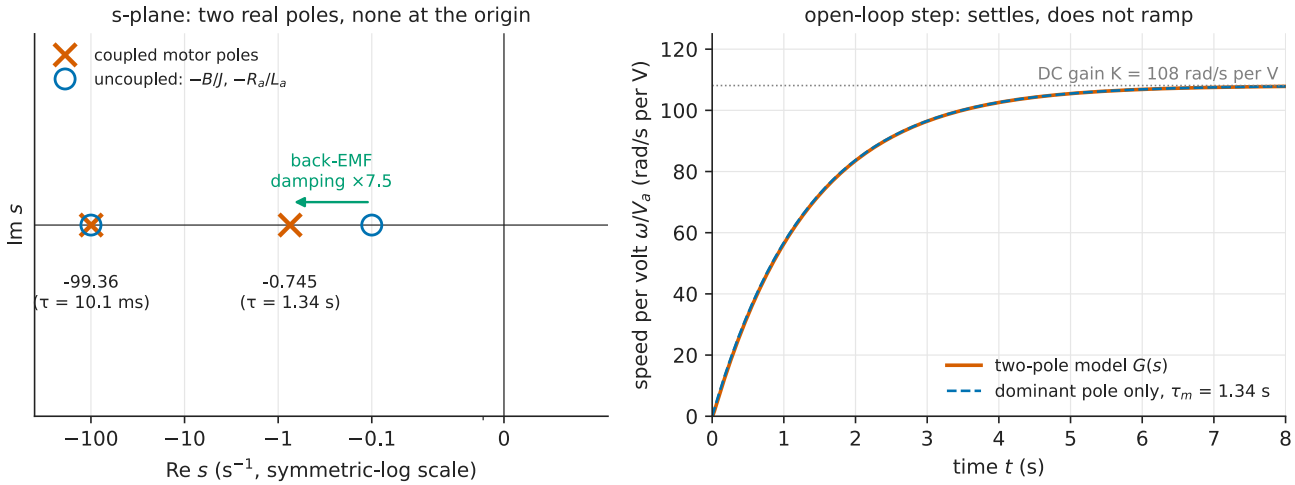
$\zeta = 5.82 \gg 1$**: the open-loop plant is strongly overdamped.** No resonance, no overshoot. This single number drives the entire control design.

$$\boxed{G(s) = \frac{108}{s^2/\omega_n^2 + 2(5.82)s/\omega_n + 1} = \frac{8000}{s^2 + 100.1\, s + 74} = \frac{8000}{(s + 0.745)(s + 99.36)}}$$

The two real poles are at $s = -0.745$ and $s = -99.36$ s⁻¹, i.e. time constants of 1.34 s and 10.1 ms. Two things are worth understanding here, because both are counter-intuitive:

- **The electrical pole is essentially where it was supposed to be.** $1/\tau_e = R_a/L_a = 100$ rad/s, and the actual fast pole is -99.36 s⁻¹. The back-EMF has barely moved it, exactly as step 3 predicts: the $K_t K_b$ coupling changes only the constant term, so it shifts ω_n and the DC gain but leaves the s-coefficient — and hence the fast pole — essentially alone. (A *closed-loop* design can move this pole; the open-loop plant cannot.)

- **The back-EMF's real job is to damp the mechanical side.** Without it, the mechanical pole would sit at $-B/J = -0.1$ s^{-1} ($\tau = 10$ s). With it, the pole is at -0.745 s^{-1} ($\tau = 1.34$ s) — a factor of 7.5 faster. The back-EMF behaves like an added resistance coupled through the motor constants K_t and K_b, and resistance is damping.



Left: poles of $G(s) = 8000/(s^2 + 100.1s + 74)$ at −0.745 and −99.36 s^{-1} (crosses) against the uncoupled references $-B/J = -0.1$ and $-R_a/L_a = -100$ (circles), on a symmetric-log axis; there is no pole at the origin. Right: the open-loop speed per volt settles at the DC gain of 108 rad/s per V, and the dominant-pole approximation is indistinguishable from the two-pole model on this time scale.

Step 5 — Engineering equation

Design a PI velocity loop. For a strongly overdamped plant, PI on velocity is the right structure. The plant's poles are far apart (0.745 vs 99.4 s^{-1}), so over any sane velocity-loop bandwidth it is well approximated by its dominant first-order pole:

$$G(s) \approx \frac{K}{1 + \tau_m s} \qquad K = 108 \text{ rad/s per V}, \quad \tau_m = 1.34 \text{ s}$$

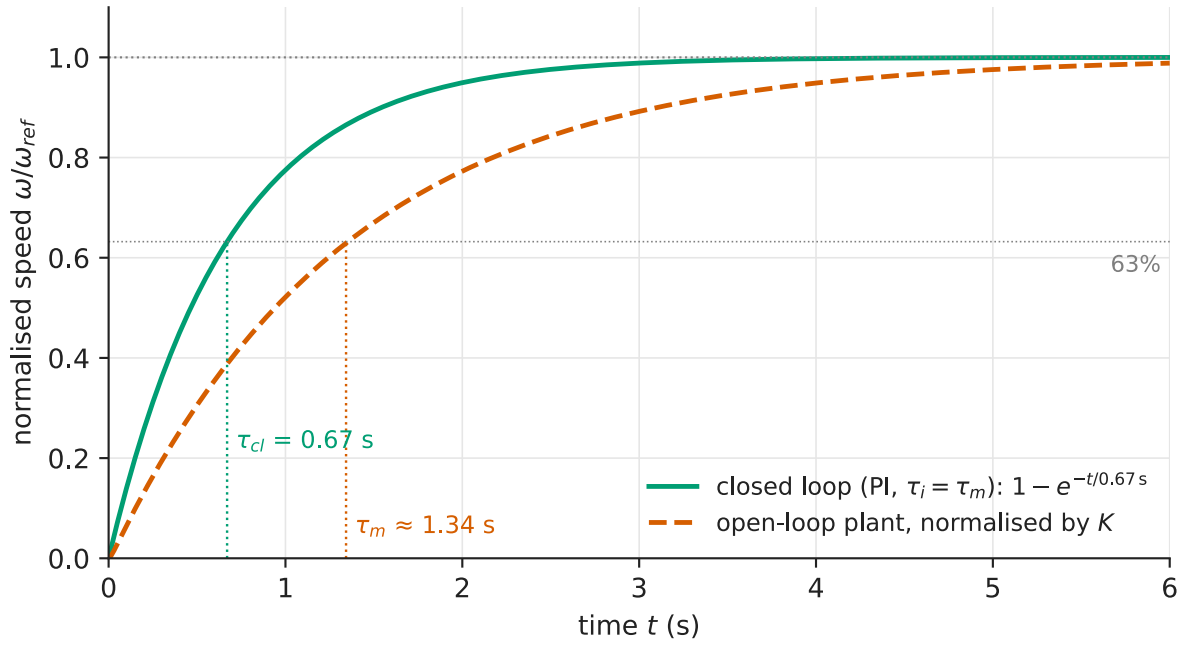
Take $C(s) = K_p\left(1 + \frac{1}{\tau_i s}\right)$ with unity feedback, and place the PI zero on the dominant plant pole, $\tau_i = \tau_m$. The $(1 + \tau_m s)$ then cancels exactly:

$$C(s)G(s) = \frac{K_p K(\tau_m s + 1)}{\tau_m s(1 + \tau_m s)} = \frac{K_p K}{\tau_m s} \qquad \Longrightarrow \qquad T(s) = \frac{K_p K}{\tau_m s + K_p K} = \frac{1}{1 + \tau_{cl} s}$$

That is a **pure first-order closed loop**: unity DC gain, no overshoot at all, and — because the integrator survives the cancellation — exactly zero steady-state error to a velocity step. Choosing a closed loop about twice as fast as the open loop, $\tau_{cl} = \tau_m/2 = 0.67$ s:

$$K_p = \frac{\tau_m}{K\,\tau_{cl}} = \frac{1.34}{108 \times 0.67} = 0.0185 \text{ V/(rad/s)} \qquad \tau_i = 1.34 \text{ s}$$

Now the check step 2 promised. The closed-loop bandwidth is $\sim 1/\tau_{cl} = 1.5$ rad/s, which sits a factor of 66 below the neglected electrical pole at 99.4 rad/s. So $\omega_{cl}\tau_e = 0.015 \ll 1$, reduction A is self-consistent, and the plant really was first order over this band. Had we demanded a closed loop ten times faster, the check would have failed and the second-order plant would have had to be kept.



Normalised speed after a step: the PI loop with $\tau_i = \tau_m$ gives the pure first-order response $1 - e^{-t/0.67\,\mathrm{s}}$, twice as fast as the open-loop plant ($\tau_m \approx 1.34$ s), with no overshoot and zero steady-state error.

Now check the real constraint. There are two candidate speed limits, and the point of the example is that they do not agree.

Supply-limited. At speed ω the steady-state current is whatever friction demands, $I_a = B\omega/K_t$, so the supply equation $V_a = R_a I_a + K_b\omega$ gives

$$\omega_{max} = \frac{V_a}{K_b + R_a B/K_t} = \frac{24}{8 \times 10^{-3} + 1.25 \times 10^{-3}} = 2595 \text{ rad/s}$$

which is just KV_a, the linear model's own prediction.

Torque-limited. Current is torque-limited: $T = K_t I_a$, and $I_{cont} = 4$ A gives $T_{max} = K_t I_{cont} = 0.032$ N·m. Setting friction torque equal to available torque:

$$\omega_{max} = \frac{K_t I_{cont}}{B} = \frac{0.032}{2 \times 10^{-5}} = 1600 \text{ rad/s} \approx 15{,}300 \text{ rpm}$$

Design finding. $1600 < 2595$, so the *torque* limit binds, not the supply — the friction-limited speed is 62% of what the linear transfer function promises. The diagnosis is visible in one number: at 1600 rad/s the motor draws its full 4 A and the friction torque is $B\omega = 0.032$ N·m, exactly the available torque, so there is nothing left to accelerate with. (For completeness, the 1600 rad/s operating point needs $V_a = R_a I + K_b\omega = 2 + 12.8 = 14.8$ V of the available 24 V, so supply is not close to binding either.) The linear model predicted 24,800 rpm; the machine delivers 15,300. A speed setpoint above 15,300 rpm integrates the velocity error until the controller saturates on current, and the loop then runs in open loop. Every commissioning procedure must check for this, because the symptom (controller wound up, plant unresponsive) looks like a control fault rather than a saturation limit.

Step 6 — Validity limits

Assumption	Fails when	Consequence
L_a neglected	$\omega_{cl} \gtrsim 1/\tau_e = 100$ rad/s (i.e. $\omega_{cl}\tau_e \gtrsim 1$)	neglected pole re-enters; current spike on step
Linear friction $B\omega$	Coulomb friction, stiction	dead band; limit cycle at low speed
No saturation	V_a or I_a limit reached	**integral windup**; the failure above
No magnetic saturation	I_a above rated	K_t falls; loop gain drops; heating
Rigid load	compliant coupling, backlash	torsional resonance; must be modelled (Ch. 19 §19.2.3)
Lumped J	gearbox ratio, reflected inertia	use reflected inertia at the motor shaft
No disturbance	load torque varying	steady-state error; needs feedforward or integral action

Anti-windup is not optional here. With an integral term and a saturating actuator, a setpoint beyond the achievable speed drives the integrator arbitrarily far. The standard remedies are back-calculation or clamping, and both must be specified before tuning, not after.

Anti-aliasing and current loop bandwidth. Because L_a was neglected, the model contains no information about the current loop. In practice the current loop is closed *inner* and *fast* (bandwidth $\gg$ velocity loop), which is what justifies the reduction retrospectively. If the current loop is slow, the velocity loop is fighting an unmodelled state.

What this example demonstrates. The full chain works and every step is defensible: physics → two coupled ODEs → second-order transfer function → closed-loop design → hardware numbers. And the final design finding came from the *physics* (friction vs available torque), not from the transfer function. That is the point of the book's architecture — the transfer function is a powerful tool, but it is valid on a bounded region, and the bounds are set by the physics that was reduced away.

Model Ledger: the full nine-row ledgers for this example are in Ch. 21 §21.8.1 (DC motor) and §21.8.2 (CSTR), which also work through a plant where the nonlinearity *is* the point.

Next: this example completes the six-step set. Return to `../README.md` for the full chapter architecture, or to Example 5 for a network whose constitutive law is empirical rather than derived.

Appendix A — Variational Calculus, Noether's Theorems, and Conservation Laws

This appendix supplies the complete proofs behind the variational machinery that the main text states or sketches. It proves the Euler–Lagrange equations with their boundary terms, the Legendre transform to Hamilton's equations, the basic Poisson-bracket results, **Noether's first theorem** for particles and fields (including the total-derivative term that the short proof in Ch. 1 §1.7.2 omits), the canonical and Hilbert stress–energy tensors and how they are related, **Noether's second theorem** for local gauge invariance, the chain from the continuity equation through the integral charge balance to Kirchhoff's current law (with every lumping assumption stated), and the Rayleigh dissipation function that carries the Lagrangian formalism over to R/C/L networks. The chapters that rely on it are Ch. 0 (the action and its symmetries), Ch. 1 §§1.5–1.7 and §1.9 (the machinery itself), Ch. 8 §8.1 (the metric definition of $T_{\mu\nu}$), Ch. 9 §§9.5–9.6 (Lagrangian and Hamiltonian mechanics as tools), Ch. 11 §11.5 and §11.12 (charge conservation and the Kirchhoff laws), and Ch. 17 §§17.3–17.4 (the R/C/L template and the node laws).

Nothing here is new physics. What the appendix adds is the bookkeeping: which assumption each step needs, and the point at which each result stops holding.

A.1 — Conventions

Assumptions used throughout.

1. **Signature.** The book's metric signature is $\eta_{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1)$ (Ch. 0 §0.2.2, Ch. 8 §8.2.1, Ch. 11 §11.2.1), with $x^\mu = (ct, \mathbf{r})$, so $\partial_0 = c^{-1}\partial_t$. Indices are raised and lowered with η (or g in curved spacetime); repeated indices are summed.
2. **Units.** SI. Electromagnetism follows Ch. 11: $A^\mu = (\phi/c,\ \mathbf{A})$, so $A_0 = -\phi/c$; $j^\mu = (c\rho,\ \mathbf{J})$; $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. In Gaussian units (as in Ch. 1 §1.6.1 and Ch. 3) replace $q\mathbf{A} \to q\mathbf{A}/c$.
3. **Smoothness.** All fields and paths are taken C^2 on the region of interest, and variations $\eta(t)$ or $\delta\phi(x)$ are smooth. Distributional sources (point charges, surface currents) are handled by the usual limiting arguments and are not treated separately.
4. **Charge convention.** A matter field of charge q (signed; $q = -e$ for the electron) transforms as $\psi \to e^{iq\alpha/\hbar}\psi$ with $A_\mu \to A_\mu + \partial_\mu\alpha$, and couples through $D_\mu = \partial_\mu - (iq/\hbar)A_\mu$. This matches the sign of the covariant derivative in Ch. 0 §0.5. With it, $-i\hbar D_k = p_k - qA_k$ and $i\hbar c\, D_0 = i\hbar\partial_t - q\phi$, so the nonrelativistic limit carries $(\mathbf{p} - q\mathbf{A})^2/2m + q\phi$.

A **functional** $F[\phi]$ assigns a number to a function. Its **functional derivative** $\delta F/\delta\phi(x)$ is defined by the first-order change under $\phi \to \phi + \delta\phi$ with $\delta\phi$ of compact support in the interior of the domain:

$$\delta F = \int dx\, \frac{\delta F}{\delta\phi(x)}\,\delta\phi(x) + \mathcal{O}(\delta\phi^2)$$

Boundary contributions are excluded from this definition on purpose. They are tracked separately in §A.2, because they are where boundary conditions come from.

A.2 — The Euler–Lagrange Equations, with Boundary Terms

A.2.1 — The fundamental lemma

Lemma. Let f be continuous on $[t_1, t_2]$. If $\int_{t_1}^{t_2} f(t)\,\eta(t)\,dt = 0$ for every smooth η that vanishes outside a closed subinterval of (t_1, t_2), then $f \equiv 0$ on $[t_1, t_2]$.

Proof. Suppose $f(t_0) > 0$ at some interior t_0. By continuity, $f > f(t_0)/2$ on some interval $(t_0 - \delta,\ t_0 + \delta)$. Choose η to be a smooth bump, positive on that interval and zero outside it. Then $\int f\eta\,dt > 0$, a contradiction. The case $f(t_0) < 0$ is the same with the sign reversed, and continuity extends $f = 0$ to the endpoints. □

The same argument works in d dimensions with bumps supported in a small ball, which is the version needed for fields.

A.2.2 — One degree of freedom, endpoints free or fixed

Assumptions. $L(q, \dot{q}, t)$ is C^2, and $S[q] = \int_{t_1}^{t_2} L\,dt$. The variation is $q \to q + \epsilon\eta$ with η smooth. At first we do **not** require $\eta(t_1) = \eta(t_2) = 0$.

Differentiate under the integral:

$$\left.\frac{dS[q+\epsilon\eta]}{d\epsilon}\right|_{\epsilon=0} = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q}\,\eta + \frac{\partial L}{\partial \dot{q}}\,\dot{\eta}\right) dt$$

Integrate the second term by parts and define $p \equiv \partial L/\partial\dot{q}$:

$$\delta S = \int_{t_1}^{t_2} \underbrace{\left(\frac{\partial L}{\partial q} - \frac{d}{dt}\frac{\partial L}{\partial \dot{q}}\right)}_{\equiv\,\mathsf{E}(t)} \eta\,dt \;+\; \Big[\,p\,\eta\,\Big]_{t_1}^{t_2}$$

This is the complete first variation. Three things follow from it.

1. **Interior equation.** Restricting to η of compact support in (t_1, t_2) kills the bracket. The lemma then gives $\mathsf{E}(t) = 0$, which is the Euler–Lagrange equation. In the notation of §A.1, $\delta S/\delta q(t) = \mathsf{E}(t)$.
2. **Fixed endpoints (Hamilton's principle, Ch. 1 §1.4.2).** If $q(t_1)$ and $q(t_2)$ are prescribed, then $\eta(t_1) = \eta(t_2) = 0$ and the bracket vanishes identically. Stationarity is then equivalent to $\mathsf{E} = 0$.
3. **Free endpoint (natural boundary condition).** If $q(t_2)$ is left free, then stationarity for all η requires $\mathsf{E} = 0$ *and* $p(t_2) = 0$. The boundary condition is not imposed. It is produced by the variational principle.

> **[DERIVATION]** Hamilton's principle with fixed endpoints is equivalent to $\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = 0$ for each i. This is Ch. 1 §1.5.1 with the boundary term kept explicit. The n-coordinate case follows by varying each q_i independently.

Total-derivative ambiguity. If $L' = L + dG(q,t)/dt$, then $S' = S + G(q(t_2), t_2) - G(q(t_1), t_1)$. With fixed endpoints the extra term does not vary, so L and L' give identical equations of motion. The canonical momentum does change, $p' = p + \partial G/\partial q$. The gauge transformation of a charged particle (§A.3.2) is the

standard example. This ambiguity is the reason Noether's theorem has to allow symmetries "up to a total derivative" (§A.4).

Stationary, not minimal. $\delta S = 0$ says nothing about the second variation. For $L = \frac{1}{2}m\dot{q}^2 - V(q)$ the action is a true local minimum only for time intervals shorter than the first conjugate point. For the harmonic oscillator that limit is half a period, π/ω. See Gelfand & Fomin, *Calculus of Variations*, for the Jacobi condition. Physics only ever uses stationarity.

A.2.3 — Higher derivatives: the Euler–Poisson equation

If $L = L(q, \dot{q}, \ddot{q}, t)$, two integrations by parts give

$$\delta S = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q} - \frac{d}{dt}\frac{\partial L}{\partial \dot{q}} + \frac{d^2}{dt^2}\frac{\partial L}{\partial \ddot{q}} \right) \eta \, dt + \left[\left(\frac{\partial L}{\partial \dot{q}} - \frac{d}{dt}\frac{\partial L}{\partial \ddot{q}} \right) \eta + \frac{\partial L}{\partial \ddot{q}}\,\dot{\eta} \right]_{t_1}^{t_2}$$

The interior equation is fourth order. The boundary term now needs two conditions at each end, one on η and one on $\dot{\eta}$. The static Euler–Bernoulli beam (Ch. 16) is this structure with x in place of t. The energy is $\int [\frac{1}{2}EI\,(w'')^2 - q\,w]\,dx$, the interior equation is $EI\,w'''' = q$, and the boundary bracket yields the familiar pairs: clamped ($w = w' = 0$), simply supported ($w = 0$ with $EI\,w'' = 0$ produced naturally), and free ($EI\,w'' = 0$ and $(EI\,w'')' = 0$, both natural).

A.2.4 — Fields

Assumptions. $S[\phi] = \int_\Omega d^4x\, \mathcal{L}(\phi_a, \partial_\mu \phi_a, x)$ over a spacetime region Ω with boundary $\partial\Omega$. In flat spacetime $d^4x = c\,dt\,d^3x$. The constant overall factor of c does not affect anything below.

$$\delta S = \int_\Omega d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \right] \delta\phi_a + \oint_{\partial\Omega} d\Sigma_\mu \, \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)}\, \delta\phi_a$$

The second term comes from Gauss's theorem in four dimensions. For variations of compact support the lemma gives the field Euler–Lagrange equation of Ch. 1 §1.9.2. The surface term supplies the boundary conditions of field problems in the same way as in §A.2.2: Dirichlet data where $\delta\phi = 0$ on the boundary, and natural (Neumann-type) conditions where ϕ is free.

Where the naive statement fails: gravity. The Einstein–Hilbert Lagrangian $\sqrt{-g}\,R$ contains second derivatives of $g_{\mu\nu}$. Its variation therefore leaves a boundary term in $\partial\delta g$ that does not vanish when only $\delta g_{\mu\nu}$ is fixed on $\partial\Omega$. A well-posed Dirichlet principle needs the Gibbons–Hawking–York boundary term $\frac{1}{8\pi G}\oint \sqrt{|h|}\,K$ added to S_{EH}. Here h is the induced metric on the boundary and K the trace of its extrinsic curvature (Wald, *General Relativity*, Appendix E). It does not change the field equations of Ch. 0 §0.3, which is why the chapters can omit it.

What was proved. Stationarity under variations of compact support is equivalent to the Euler–Lagrange equations. The boundary term decides which boundary conditions make the variational problem well posed. **Where it fails.** It fails for non-smooth extremals (corners need the Weierstrass–Erdmann conditions), for Lagrangians that are not differentiable in $\dot{q}$, and for systems whose equations of motion are not Euler–Lagrange equations of any Lagrangian. Dissipative systems are the important case of the last kind. They are handled in §A.10, and not by pretending they have an action.

A.3 — The Legendre Transform and Hamilton's Equations

A.3.1 — Regularity and the transform

Assumption (regularity). The Hessian $W_{ij} = \partial^2 L / \partial \dot{q}_i \partial \dot{q}_j$ is nonsingular. Then $p_i = \partial L / \partial \dot{q}_i (q, \dot{q}, t)$ can be solved locally for $\dot{q}_i(q, p, t)$ by the implicit function theorem. If L is strictly convex in $\dot{q}$, which covers any $L = \frac{1}{2}\dot{q}^{\mathsf{T}} M(q) \dot{q} + \dots$ with M positive definite, the inversion is global.

Define $H(q, p, t) = \sum_i p_i \dot{q}_i - L$, with $\dot{q} = \dot{q}(q, p, t)$ substituted. Take the differential and use the definition of p_i to cancel the $d\dot{q}_i$ terms:

$$dH = \sum_i \left(\dot{q}_i \, dp_i + p_i \, d\dot{q}_i - \frac{\partial L}{\partial q_i} dq_i - \frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i \right) - \frac{\partial L}{\partial t} dt = \sum_i \left(\dot{q}_i \, dp_i - \frac{\partial L}{\partial q_i} dq_i \right) - \frac{\partial L}{\partial t} dt$$

Reading off the partial derivatives gives three identities that hold **off shell**:

$$\frac{\partial H}{\partial p_i} = \dot{q}_i, \qquad \frac{\partial H}{\partial q_i} = -\frac{\partial L}{\partial q_i}, \qquad \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}$$

The Euler–Lagrange equation $\dot{p}_i = \partial L / \partial q_i$ turns the middle identity into $\dot{p}_i = -\partial H / \partial q_i$. That completes Hamilton's equations (Ch. 1 §1.6.3). Along a solution, $dH/dt = \partial H / \partial t = -\partial L / \partial t$, so H is conserved when L has no explicit time dependence. This is the content of Ch. 1 §1.7.3, here obtained without any symmetry language.

> **[DERIVATION]** For regular L, the n second-order Euler–Lagrange equations are equivalent to the $2n$ first-order Hamilton equations. Equivalently, Hamilton's equations are the Euler–Lagrange equations of the **phase-space action** $S_H[q, p] = \int (\sum_i p_i \dot{q}_i - H)\, dt$, varied independently in q (endpoints fixed) and p (endpoints free). Varying p_i gives $\dot{q}_i = \partial H / \partial p_i$, and varying q_i gives $\dot{p}_i = -\partial H / \partial q_i$.

H is not always the energy. H equals $T + V$ only when T is homogeneous quadratic in $\dot{q}$ and V is velocity independent. In a rotating frame, or with time-dependent constraints, H is conserved (when $\partial_t L = 0$) but is not $T + V$.

Where it fails: singular Lagrangians. If $\det W = 0$, some momenta are not independent and the equations $p_i = \partial L / \partial \dot{q}_i$ define **primary constraints**. Maxwell theory is the key example: $\mathcal{L}$ contains no $\partial_0 A_0$, so the momentum conjugate to A_0 vanishes identically. Preserving that constraint in time yields Gauss's law as a secondary constraint. This is the Hamiltonian face of the gauge redundancy treated in §A.8. The systematic method is Dirac's constraint analysis (Henneaux & Teitelboim, *Quantization of Gauge Systems*).

A.3.2 — The charged particle

From Ch. 9 §9.5.2, $L = \frac{1}{2} m \dot{\mathbf{r}}^2 - q\phi + q \dot{\mathbf{r}} \cdot \mathbf{A}$. The Hessian is $m\,1$, so L is regular, and

$$\mathbf{p} = m\dot{\mathbf{r}} + q\mathbf{A}, \qquad H = \mathbf{p} \cdot \dot{\mathbf{r}} - L = \frac{(\mathbf{p} - q\mathbf{A})^2}{2m} + q\phi$$

Under $\phi \to \phi - \partial_t \chi$, $\mathbf{A} \to \mathbf{A} + \nabla\chi$, the Lagrangian changes by $q(\partial_t\chi + \dot{\mathbf{r}} \cdot \nabla\chi) = q\, d\chi/dt$. That is a total derivative, so by §A.2.2 the motion is unchanged. The canonical momentum shifts by $q\nabla\chi$, while the **kinetic** momentum $\boldsymbol{\pi} = \mathbf{p} - q\mathbf{A} = m\dot{\mathbf{r}}$ is gauge invariant. This is the classical origin of minimal coupling $\hat{\mathbf{p}} \to \hat{\mathbf{p}} - q\mathbf{A}$ (Ch. 3 §3.9.4, where Gaussian units give $e\mathbf{A}/c$).

A.4 — Poisson Brackets and the Hamiltonian Form of Noether's Theorem

Definition (Ch. 1 §1.6.4):

$$\{f, g\} = \sum_i \left(\frac{\partial f}{\partial q_i}\frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i}\frac{\partial g}{\partial q_i} \right)$$

Properties, all verified by direct expansion: antisymmetry $\{f,g\} = -\{g,f\}$; bilinearity; the Leibniz rule $\{f, gh\} = \{f,g\}h + g\{f,h\}$; and the **Jacobi identity** $\{f,\{g,h\}\} + \{g,\{h,f\}\} + \{h,\{f,g\}\} = 0$. In the Jacobi identity every term contains a second derivative of exactly one of f, g, h, and these cancel in pairs because mixed partial derivatives commute. The canonical brackets are $\{q_i, p_j\} = \delta_{ij}$ and $\{q_i, q_j\} = \{p_i, p_j\} = 0$.

Equation of motion. By the chain rule and Hamilton's equations, for any $f(q,p,t)$ along a solution,

$$\frac{df}{dt} = \sum_i \left(\frac{\partial f}{\partial q_i}\dot{q}_i + \frac{\partial f}{\partial p_i}\dot{p}_i \right) + \frac{\partial f}{\partial t} = \{f, H\} + \frac{\partial f}{\partial t}$$

Generators. A phase-space function G generates the infinitesimal canonical transformation $\delta q_i = \epsilon\, \partial G/\partial p_i = \epsilon\{q_i, G\}$, $\delta p_i = -\epsilon\, \partial G/\partial q_i = \epsilon\{p_i, G\}$. To first order this preserves the canonical brackets, and it changes any function by $\delta f = \epsilon\{f, G\}$. In particular, $\delta H = \epsilon\{H, G\} = -\epsilon\{G, H\}$.

Theorem (Hamiltonian Noether, both directions). Let G have no explicit time dependence. Then

$$\frac{dG}{dt} = 0 \quad \Longleftrightarrow \quad \{G, H\} = 0 \quad \Longleftrightarrow \quad \delta_G H = 0$$

Proof. The first equivalence is the equation of motion above. The second follows from $\delta_G H = -\epsilon\{G, H\}$. □

So in Hamiltonian form the correspondence between symmetries and conserved quantities runs **both ways**: every conserved G generates a symmetry of H. The Lagrangian first theorem (§A.5) proves only the direction from symmetry to conservation.

Examples. $G = \mathbf{n}\cdot\mathbf{P}$ generates translations, $\delta\mathbf{r}_a = \epsilon\mathbf{n}$. $G = \mathbf{n}\cdot\mathbf{L}$ generates rotations, $\delta\mathbf{r}_a = \epsilon\, \mathbf{n}\times\mathbf{r}_a$. The angular momenta close into the rotation algebra $\{L_i, L_j\} = \epsilon_{ijk}L_k$. **Poisson's theorem** follows from the Jacobi identity: if $\{F,H\} = \{G,H\} = 0$, then $\{\{F,G\},H\} = 0$, so the bracket of two conserved quantities is conserved.

> **[STRUCTURAL CONNECTION]** The map $\{\,,\} \to (i\hbar)^{-1}[\,,]$ of Ch. 1 §1.6.4 preserves antisymmetry, bilinearity, Leibniz and Jacobi. It is **not** a derivation of quantum mechanics. The Groenewold–van Hove theorem shows that no linear map can send all polynomial phase-space functions to operators while turning every Poisson bracket into $(i\hbar)^{-1}$ times the commutator. The

> correspondence holds exactly only on a restricted class of observables, for example polynomials of degree two or less. Canonical quantization is a structural template, and it needs ordering choices beyond that.

A.5 — Noether's First Theorem in Mechanics

Assumptions.

1. The system has a regular Lagrangian $L(q, \dot{q}, t)$ (regularity is needed only for the Hamiltonian reading of the result).
2. There is a one-parameter family of transformations $q_i \to q_i + \epsilon K_i(q, \dot{q}, t) + \mathcal{O}(\epsilon^2)$, with ϵ a constant.
3. **Symmetry condition, off shell.** For every path $q(t)$, not just solutions, L changes by a total derivative:

$$\delta L \equiv \epsilon \sum_i \left(\frac{\partial L}{\partial q_i} K_i + \frac{\partial L}{\partial \dot{q}_i} \dot{K}_i \right) = \epsilon \, \frac{dF}{dt}$$

   for some function $F(q, \dot{q}, t)$.

Allowing $F \neq 0$ is essential, not decorative. Without it, time translation and Galilean boosts would not count as symmetries. The proof in Ch. 1 §1.7.2 sets $F = 0$, and Ch. 1 §1.7.3 then handles time translation by a separate argument. The proof here covers both cases at once.

Proof. Rewrite the left side using the product rule $\frac{\partial L}{\partial \dot{q}_i} \dot{K}_i = \frac{d}{dt}\big(p_i K_i\big) - \dot{p}_i K_i$:

$$\sum_i \left(\frac{\partial L}{\partial q_i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} \right) K_i + \frac{d}{dt} \sum_i p_i K_i = \frac{dF}{dt}$$

This is an **identity**, valid on every path. Define the Noether charge and the Euler–Lagrange expressions,

$$J \equiv \sum_i p_i K_i - F, \qquad \mathsf{E}_i \equiv \frac{\partial L}{\partial q_i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i}$$

The identity then reads

$$\boxed{\frac{dJ}{dt} = -\sum_i \mathsf{E}_i \, K_i}$$

On solutions $\mathsf{E}_i = 0$, so J is constant. □

> **[DERIVATION]** A continuous symmetry of L up to a total derivative implies the off-shell identity $\dot{J} = -\sum_i \mathsf{E}_i K_i$, hence on-shell conservation of $J = \sum_i p_i K_i - F$. Boundary conditions play no role. The time-endpoint argument of Ch. 1 §1.7.2 is equivalent when $F = 0$.

The standard cases. Each row is checked against the symmetry condition with the assumption it needs.

Symmetry	K_i	F	Conserved J	Requires
Time translation $q(t) \to q(t+\epsilon)$	$\dot{q}_i$	L (since $\delta L = \epsilon\, dL/dt$ when $\partial_t L = 0$)	$\sum p_i \dot{q}_i - L = H$	$\partial L/\partial t = 0$
Translation along $\mathbf{n}$	$\mathbf{n}$ on every $\mathbf{r}_a$	0	$\mathbf{n} \cdot \mathbf{P}$	V depends on differences $\mathbf{r}_a - \mathbf{r}_b$
Rotation about $\mathbf{n}$	$\mathbf{n} \times \mathbf{r}_a$	0	$\mathbf{n} \cdot \sum_a \mathbf{r}_a \times \mathbf{p}_a$	V depends on $\lvert\mathbf{r}_a - \mathbf{r}_b\rvert$ only
Galilean boost along $\mathbf{n}$	$t\,\mathbf{n}$ on every $\mathbf{r}_a$	$\mathbf{n} \cdot M\mathbf{X}$	$\mathbf{n} \cdot (\mathbf{P}t - M\mathbf{X})$	translation invariance plus $T = \sum \frac{1}{2} m_a \dot{\mathbf{r}}_a^2$

Here $M = \sum m_a$ and $\mathbf{X}$ is the centre of mass. The boost row is the **centre-of-mass theorem**, $M\mathbf{X} = \mathbf{P}t + \text{const}$. It is a genuine Noether charge with explicit time dependence, and it exists only because $F \neq 0$ is allowed. The rotation row reproduces Ch. 1 §1.7.3 with $\mathbf{n} = \hat{\mathbf{z}}$: $K = (-y, x, 0)$ gives $J = xp_y - yp_x = L_z$.

What was proved. Each continuous symmetry of the stated kind yields one conserved quantity. **Where it fails.** (i) Discrete symmetries such as parity and time reversal give selection rules, not conserved charges. (ii) A symmetry of the equations of motion that is not a symmetry of L up to a total derivative gives no Noether charge. (iii) With external time-dependent forces, $\partial_t L \neq 0$ and energy is not conserved. (iv) Dissipative forces are not in L at all (§A.10). (v) In quantum theory a classical Noether symmetry can fail to survive regularization. This is an **anomaly** (Ch. 0 §0.10), so classical conservation is a statement about the classical action.

A.6 — Noether's First Theorem for Fields

A.6.1 — The current and the charge

Assumptions. Take a flat spacetime Lagrangian density $\mathcal{L}(\phi_a, \partial_\mu \phi_a)$ and an infinitesimal transformation $\phi_a \to \phi_a + \epsilon\, \Delta\phi_a$ (a *form* variation at fixed x) such that, **off shell**,

$$\delta\mathcal{L} = \epsilon\, \partial_\mu K^\mu$$

for some $K^\mu(\phi, \partial\phi, x)$. Spacetime transformations are included by writing their effect as a form variation, for example $\Delta\phi = -a^\nu \partial_\nu \phi$ for a translation.

Proof. Expand $\delta\mathcal{L}$ and use the product rule on the derivative term:

$$\delta\mathcal{L} = \epsilon \left[\frac{\partial \mathcal{L}}{\partial \phi_a} \Delta\phi_a + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \partial_\mu \Delta\phi_a \right] = \epsilon \left[\mathsf{E}^a\, \Delta\phi_a + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \Delta\phi_a \right) \right]$$

Here $\mathsf{E}^a = \partial\mathcal{L}/\partial\phi_a - \partial_\mu\big(\partial\mathcal{L}/\partial(\partial_\mu\phi_a)\big)$. Equating to $\epsilon\, \partial_\mu K^\mu$ gives the off-shell identity

$$\boxed{\partial_\mu j^\mu = -\,\mathsf{E}^a\, \Delta\phi_a, \qquad j^\mu \equiv \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \Delta\phi_a - K^\mu}$$

On shell, $\partial_\mu j^\mu = 0$. $\square$

Ch. 1 §1.9.3 gives the special case $K^\mu = 0$.

From local to global. Write $\partial_\mu j^\mu = c^{-1} \partial_t j^0 + \nabla \cdot \mathbf{j}$ and define $Q(t) = c^{-1} \int_{\mathbb{R}^3} j^0\, d^3x$. Integrating over a large ball B_R and applying the divergence theorem gives

$$\frac{dQ_R}{dt} = -\oint_{\partial B_R} \mathbf{j}\cdot d\mathbf{A}$$

Here Q_R is the charge inside the ball. So Q is conserved **only if** the flux through the sphere at infinity vanishes, which needs $|\mathbf{j}|$ to fall faster than R^{-2}. Local conservation, $\partial_\mu j^\mu = 0$, is the stronger and more useful statement, because it holds region by region (§A.9).

Ambiguities. The current is not unique. (i) It can be rescaled by a constant. (ii) K^μ is defined only up to an identically conserved piece. (iii) For any antisymmetric $U^{\mu\nu} = -U^{\nu\mu}$, the current $j'^\mu = j^\mu + \partial_\nu U^{\mu\nu}$ is also conserved, because $\partial_\mu\partial_\nu U^{\mu\nu} \equiv 0$. Its charge differs by a surface integral at infinity. The Belinfante improvement (§A.7.3) and the gauge-theory "superpotential" (§A.8) are both instances of (iii).

A.6.2 — Internal U(1): the charge current of a matter field

Let ψ be a complex (or Dirac) field with $\mathcal{L}_m(\psi, \bar\psi, D_\mu\psi, \overline{D_\mu\psi})$ built only from the combinations ψ, $\bar\psi$ and the covariant derivative of §A.1. Under the **global** transformation $\psi \to e^{iq\epsilon/\hbar}\psi$, with A_μ held fixed, $\Delta\psi = (iq/\hbar)\psi$, $\Delta\bar\psi = -(iq/\hbar)\bar\psi$, and $\mathcal{L}_m$ is invariant with $K^\mu = 0$. The Noether current is

$$j^\mu_N = \frac{\partial\mathcal{L}_m}{\partial(\partial_\mu\psi)}\frac{iq}{\hbar}\psi - \frac{iq}{\hbar}\bar\psi\,\frac{\partial\mathcal{L}_m}{\partial(\partial_\mu\bar\psi)}$$

Identity: the Noether current is the source current. A_μ enters $\mathcal{L}_m$ only through $D_\mu\psi = \partial_\mu\psi - (iq/\hbar)A_\mu\psi$ and its conjugate. Hence $\partial\mathcal{L}_m/\partial A_\mu = -(iq/\hbar)\psi\,\partial\mathcal{L}_m/\partial(\partial_\mu\psi) + (iq/\hbar)\bar\psi\,\partial\mathcal{L}_m/\partial(\partial_\mu\bar\psi)$, so

$$J^\mu \equiv \frac{\partial\mathcal{L}_m}{\partial A_\mu} = -\,j^\mu_N$$

The overall sign is the convention ambiguity (i) above. What matters is that the current that sources Maxwell's equations (the $A_\mu j^\mu$ term of Ch. 11 §11.1) **is** the Noether current of the global phase symmetry. For the Dirac field, $\mathcal{L}_m = \bar\psi(i\hbar c\,\gamma^\mu D_\mu - mc^2)\psi$ with $\{\gamma^\mu,\gamma^\nu\} = -2\eta^{\mu\nu}$ in this signature, which keeps $\gamma^0 = \beta$ with $\beta^2 = 1$ (see Appendix B §B.1). It gives $J^\mu = qc\,\bar\psi\gamma^\mu\psi$, hence $\rho = J^0/c = q\,\psi^\dagger\psi$ and $\mathbf{J} = qc\,\psi^\dagger\boldsymbol{\alpha}\psi$.

> **[DERIVATION]** Global U(1) invariance of the matter Lagrangian implies, by Noether's first theorem, a current with $\partial_\mu J^\mu = 0$ whenever the **matter** equations of motion hold. For minimally coupled matter this current equals $\partial\mathcal{L}_m/\partial A_\mu$. Whether A_μ obeys its own equation is irrelevant to this statement (§A.8). Reading $\psi^\dagger\psi$ as a probability density is the Born postulate, not part of the theorem (Ch. 3 §3.9.2).

A.7 — Energy–Momentum: Canonical and Hilbert Tensors

A.7.1 — Translations and the canonical tensor

Assumption. $\mathcal{L}$ has no explicit x-dependence. Under $x^\mu \to x^\mu + a^\mu$ with constant a^μ, a field transported along is $\phi'(x) = \phi(x-a)$, so $\Delta\phi_a = -a^\nu\partial_\nu\phi_a$. Then $\delta\mathcal{L} = -a^\nu\partial_\nu\mathcal{L} = \partial_\mu(-a^\mu\mathcal{L})$, so $K^\mu = -a^\mu\mathcal{L}$. The Noether current of §A.6.1 is

$$j^\mu = -\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}a^\nu\partial_\nu\phi_a + a^\mu\mathcal{L} = a_\nu\,T^{\mu\nu}_{\text{can}}$$

with, in the book's signature,

$$\boxed{T^{\mu\nu}_{\text{can}} = -\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)}\,\partial^\nu \phi_a + \eta^{\mu\nu}\mathcal{L}, \qquad \partial_\mu T^{\mu\nu}_{\text{can}} = 0 \text{ on shell}}$$

There is one current for each of the four independent a^ν.

Check of signs. Let $\pi_a = \partial\mathcal{L}/\partial(\partial_t \phi_a)$, so that $\partial\mathcal{L}/\partial(\partial_0\phi_a) = c\,\pi_a$, and use $\partial^0 = -\partial_0$:

$$T^{00}_{\text{can}} = \pi_a \dot{\phi}_a - \mathcal{L} = \mathcal{H}, \qquad \frac{T^{0i}_{\text{can}}}{c} = -\pi_a\,\partial_i \phi_a$$

The first is the Hamiltonian density. The second is the momentum density, whose integral $-\int \pi_a \nabla\phi_a\, d^3x$ is the generator of spatial translations. The energy is $\int T^{00} d^3x$ and the momentum is $c^{-1}\int T^{0i} d^3x$.

> **Note on signature.** The form $T^{\mu\nu} = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\partial^\nu\phi - g^{\mu\nu}\mathcal{L}$ written in Ch. 1 §1.9.3 is the correct expression in the $(+,-,-,-)$ signature. In the book's $(-,+,+,+)$ signature both terms flip sign, as boxed above, so that $T^{00} = +\mathcal{H}$.

Example (real scalar). For $\mathcal{L} = -\frac{1}{2}\partial_\mu\varphi\,\partial^\mu\varphi - V(\varphi)$, we have $\partial\mathcal{L}/\partial(\partial_\mu\varphi) = -\partial^\mu\varphi$ and

$$T^{\mu\nu}_{\text{can}} = \partial^\mu\varphi\,\partial^\nu\varphi + \eta^{\mu\nu}\mathcal{L}, \qquad T^{00} = \tfrac{1}{2}c^{-2}\dot{\varphi}^2 + \tfrac{1}{2}(\nabla\varphi)^2 + V \geq 0 \;\text{ for } V \geq 0$$

The c^{-2} arises because $\partial^0\varphi = -c^{-1}\dot{\varphi}$. Here φ is normalized so that $\mathcal{L}$ has dimensions of energy density. This tensor is symmetric.

A.7.2 — Lorentz transformations and angular momentum

Under $x^\mu \to x^\mu + \omega^\mu{}_\nu x^\nu$ with $\omega_{\mu\nu} = -\omega_{\nu\mu}$, a scalar field has $\Delta\varphi = -\omega^\lambda{}_\nu x^\nu \partial_\lambda\varphi$. Because $\partial_\mu(\omega^\mu{}_\nu x^\nu) = \omega^\mu{}_\mu = 0$, $\delta\mathcal{L} = \partial_\mu(-\omega^\mu{}_\nu x^\nu \mathcal{L})$. The Noether current is $j^\mu = \omega_{\lambda\nu}x^\nu T^{\mu\lambda}_{\text{can}} = -\frac{1}{2}\omega_{\lambda\nu}M^{\mu\lambda\nu}$, with

$$M^{\mu\lambda\nu} = x^\lambda T^{\mu\nu} - x^\nu T^{\mu\lambda}, \qquad \partial_\mu M^{\mu\lambda\nu} = T^{\lambda\nu} - T^{\nu\lambda} \;\text{ (on shell)}$$

So conservation of angular momentum requires a **symmetric** energy–momentum tensor. For the scalar it holds automatically. The conserved charges are $J^{ij} = c^{-1}\int M^{0ij} d^3x = \int (x^i g^j - x^j g^i)\, d^3x$, with $\mathbf{g}$ the momentum density, which is the field version of $\mathbf{L} = \mathbf{r}\times\mathbf{p}$. The boost charges J^{0i} give the relativistic centre-of-energy theorem, the analogue of the Galilean row of §A.5.

For fields with spin, $\Delta\phi$ acquires an additional spin-matrix term. $T^{\mu\nu}_{\text{can}}$ is then not symmetric, and only orbital plus spin angular momentum is conserved.

A.7.3 — Belinfante improvement: the Maxwell example

For $\mathcal{L} = -\frac{1}{4\mu_0}F_{\alpha\beta}F^{\alpha\beta}$ in vacuum, $\partial\mathcal{L}/\partial(\partial_\mu A_\alpha) = -\mu_0^{-1}F^{\mu\alpha}$, so

$$T^{\mu\nu}_{\text{can}} = \frac{1}{\mu_0}F^{\mu\alpha}\partial^\nu A_\alpha + \eta^{\mu\nu}\mathcal{L}$$

This is neither symmetric nor gauge invariant. Write $\partial^\nu A_\alpha = F^\nu{}_\alpha + \partial_\alpha A^\nu$, and use the vacuum field equation $\partial_\alpha F^{\mu\alpha} = 0$ to rewrite $F^{\mu\alpha}\partial_\alpha A^\nu = \partial_\alpha(F^{\mu\alpha}A^\nu)$. Then

$$T^{\mu\nu}_{\text{can}} = \underbrace{\frac{1}{\mu_0}\left(F^{\mu\alpha}F^{\nu}{}_{\alpha} - \tfrac{1}{4}\eta^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}\right)}_{T^{\mu\nu}_{\text{B}}} + \partial_\lambda\left(\frac{1}{\mu_0}F^{\mu\lambda}A^{\nu}\right)$$

The added term has the form $\partial_\lambda B^{\lambda\mu\nu}$ with $B^{\lambda\mu\nu} = -\mu_0^{-1}F^{\mu\lambda}A^{\nu}$ antisymmetric in $\lambda\mu$. It is therefore identically divergence-free, it changes the four-momentum only by a surface term, and it can be dropped. The remainder T_{B} is symmetric, gauge invariant, and exactly the tensor of Ch. 11 §11.8.1. It gives $T^{00}_{\text{B}} = \frac{1}{2}(\epsilon_0 E^2 + B^2/\mu_0)$ and $T^{0i}_{\text{B}} = (\mathbf{E}\times\mathbf{B})^i/(\mu_0 c)$. The general Belinfante–Rosenfeld construction, Belinfante (1940) and Rosenfeld (1940), builds $B^{\lambda\mu\nu}$ from the spin current for any field. See Weinberg, *The Quantum Theory of Fields*, Vol. I (the chapter on the canonical formalism).

A.7.4 — The Hilbert tensor

Definition and sign convention. In curved spacetime with signature $(-,+,+,+)$:

$$\boxed{T_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta S_{\text{matter}}}{\delta g^{\mu\nu}}} \qquad \Longleftrightarrow \qquad \delta S_{\text{matter}} = -\tfrac{1}{2}\int d^4x\,\sqrt{-g}\,T_{\mu\nu}\,\delta g^{\mu\nu} = +\tfrac{1}{2}\int d^4x\,\sqrt{-g}\,T^{\mu\nu}\,\delta g_{\mu\nu}$$

The second form uses $\delta g^{\mu\nu} = -g^{\mu\alpha}g^{\nu\beta}\delta g_{\alpha\beta}$. This is the convention of Ch. 1 §1.9.3 and Ch. 8 §8.1. **The minus sign is tied to the signature.** In $(+,-,-,-)$ the same physical tensor is $+(2/\sqrt{-g})\,\delta S/\delta g^{\mu\nu}$. The check below fixes the sign by requiring positive energy density.

Two facts used repeatedly: $\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}\,g_{\mu\nu}\delta g^{\mu\nu}$ (Jacobi's formula for the determinant), and $g^{\mu\nu}$ appears explicitly in every contraction of lower-index derivatives.

Scalar check. $S = \int\sqrt{-g}\,[-\frac{1}{2}g^{\mu\nu}\partial_\mu\varphi\partial_\nu\varphi - V]$ gives $\delta S/\delta g^{\mu\nu} = \sqrt{-g}\,[-\frac{1}{2}\partial_\mu\varphi\partial_\nu\varphi - \frac{1}{2}g_{\mu\nu}\mathcal{L}]$. Hence

$$T_{\mu\nu} = \partial_\mu\varphi\,\partial_\nu\varphi + g_{\mu\nu}\mathcal{L}$$

In flat space this is **identical** to the canonical tensor of §A.7.1 with its indices lowered, and $T_{00} > 0$. With $+2/\sqrt{-g}$ in this signature, the energy would come out negative.

Maxwell check. $\delta(\sqrt{-g}\,g^{\mu\alpha}g^{\nu\beta}F_{\mu\nu}F_{\alpha\beta})/\delta g^{\mu\nu} = \sqrt{-g}\,(2F_{\mu\alpha}F_{\nu}{}^{\alpha} - \frac{1}{2}g_{\mu\nu}F^2)$. With the prefactor $-1/(4\mu_0)$ this gives $T_{\mu\nu} = \mu_0^{-1}(F_{\mu\alpha}F_{\nu}{}^{\alpha} - \frac{1}{4}g_{\mu\nu}F^2)$, which is **exactly the Belinfante tensor** T_{B}. No improvement term is needed: the metric variation produces the symmetric, gauge-invariant tensor directly.

Einstein's equations. Vary $S_{EH} + S_{\text{matter}}$ with S_{EH} as in Ch. 0 §0.3, using $\delta(\sqrt{-g}R) = \sqrt{-g}\,(R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu})\,\delta g^{\mu\nu}$ + (total derivative; §A.2.4) and $\delta(-2\Lambda\sqrt{-g}) = \Lambda\sqrt{-g}\,g_{\mu\nu}\delta g^{\mu\nu}$:

$$\frac{1}{16\pi G}\,(G_{\mu\nu} + \Lambda g_{\mu\nu}) - \frac{1}{2}T_{\mu\nu} = 0 \quad\Longrightarrow\quad G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G\,T_{\mu\nu}$$

This is the boxed equation of Ch. 0 §0.3 ($c = 1$). The factor -2 in the definition is chosen precisely so that this coefficient comes out as $8\pi G$ with positive energy. The unknown sector S_{unknown} contributes its own term of the same form wherever it depends on the metric. Nothing here assumes that contribution vanishes.

Covariant conservation from diffeomorphism invariance. Assumptions: S_{matter} is a scalar, and the matter fields obey their own equations of motion. Under an infinitesimal diffeomorphism generated by ξ^μ (compact support), $\delta g^{\mu\nu} = \mp(\nabla^\mu\xi^\nu + \nabla^\nu\xi^\mu)$, with the sign depending on active or passive convention. The matter-field variations multiply their Euler–Lagrange expressions and drop out on shell. Invariance then requires

$$0 = \delta S_{\text{matter}} = \mp\tfrac{1}{2}\int \sqrt{-g}\,(-T_{\mu\nu})(2\nabla^\mu \xi^\nu) = \mp\int \sqrt{-g}\,\xi^\nu\,\nabla^\mu T_{\mu\nu}$$

The last step integrates by parts, using $\nabla_\mu(\sqrt{-g}\,V^\mu) = \partial_\mu(\sqrt{-g}\,V^\mu)$. Since ξ is arbitrary,

$$\nabla^\mu T_{\mu\nu} = 0$$

This is an instance of **Noether's second theorem** (§A.8), with an arbitrary function $\xi^\mu(x)$ playing the role of the gauge parameter. It reproduces the ordering stated in Ch. 8 §8.1.

> **[DERIVATION]** $\nabla_\mu T^{\mu\nu} = 0$ follows from diffeomorphism invariance of S_{matter} together with the matter equations of motion. It is **not** in general a conservation law with a conserved integral, because ∇_μ contains connection terms. A conserved charge exists when the spacetime has a **Killing vector** k^ν ($\nabla_\mu k_\nu + \nabla_\nu k_\mu = 0$). Then $J^\mu = T^{\mu\nu}k_\nu$ satisfies $\nabla_\mu J^\mu = T^{\mu\nu}\nabla_\mu k_\nu = 0$, because a symmetric tensor contracted with an antisymmetric one vanishes. That is a genuine continuity equation. In flat space the ten Killing vectors of the Poincaré group return the energy, momentum, angular momentum and boost charges of §§A.7.1–A.7.2.

How the two tensors relate. In flat spacetime, for scalar, Maxwell, and (using a tetrad formalism) Dirac matter, the Hilbert tensor equals the Belinfante-improved canonical tensor, up to terms that vanish on shell. The scalar and Maxwell cases are proved above. The Dirac case needs a vierbein variation and is not done here (see a treatment of spinors in curved spacetime, e.g. Birrell & Davies, *Quantum Fields in Curved Space*, or the Belinfante–Rosenfeld papers). In general relativity only the Hilbert definition is available, as Ch. 8 §8.1 states.

A.8 — Noether's Second Theorem and Gauge Redundancy

Statement. Suppose S is invariant under a transformation that depends on arbitrary functions $\alpha(x)$ and their derivatives. Then the Euler–Lagrange expressions satisfy **identities**, one per arbitrary function, that hold whether or not the equations of motion do. Noether's 1918 paper (*Invariante Variationsprobleme*) contains both theorems. A modern account is Kosmann-Schwarzbach, *The Noether Theorems* (Springer, 2011).

A.8.1 — The U(1) identity

Assumptions. $S[A,\psi,\bar\psi] = \int d^4x\,[-\frac{1}{4\mu_0}F_{\mu\nu}F^{\mu\nu} + \mathcal{L}_m(\psi,\bar\psi,D\psi,\overline{D\psi})]$. The action is invariant under the **local** transformation

$$\delta A_\mu = \partial_\mu\alpha, \qquad \delta\psi = \frac{iq}{\hbar}\alpha\,\psi, \qquad \delta\bar\psi = -\frac{iq}{\hbar}\alpha\,\bar\psi$$

for arbitrary $\alpha(x)$ of compact support. Invariance holds because $F_{\mu\nu}$ is invariant and $D_\mu\psi \to e^{iq\alpha/\hbar}D_\mu\psi$.

Proof. For variations of compact support, §A.1 gives

$$0 = \delta S = \int d^4x\left[\frac{\delta S}{\delta A_\mu}\partial_\mu\alpha + \frac{iq}{\hbar}\alpha\left(\frac{\delta S}{\delta\psi}\psi - \bar\psi\frac{\delta S}{\delta\bar\psi}\right)\right] = \int d^4x\;\alpha\left[-\partial_\mu\frac{\delta S}{\delta A_\mu} + \frac{iq}{\hbar}\left(\frac{\delta S}{\delta\psi}\psi - \bar\psi\frac{\delta S}{\delta\bar\psi}\right)\right]$$

Since α is arbitrary, the fundamental lemma gives the **Noether identity**:

$$\boxed{\partial_\mu \frac{\delta S}{\delta A_\mu} \equiv \frac{iq}{\hbar}\left(\frac{\delta S}{\delta \psi}\psi - \bar{\psi}\frac{\delta S}{\delta \bar{\psi}}\right)}$$

This holds **off shell**. □

A.8.2 — What the identity does and does not say

Compute the left side explicitly. From $\mathcal{L}_{\text{Maxwell}}$, $\delta S/\delta A_\mu \supset -\partial_\nu\big(\partial\mathcal{L}/\partial(\partial_\nu A_\mu)\big) = \mu_0^{-1}\partial_\nu F^{\nu\mu}$. From $\mathcal{L}_m$, the contribution is J^μ of §A.6.2. So the Maxwell equation $\delta S/\delta A_\mu = 0$ reads

$$\partial_\nu F^{\mu\nu} = \mu_0 J^\mu \qquad \Longleftrightarrow \qquad \nabla\cdot\mathbf{E} = \rho/\epsilon_0,\;\; \nabla\times\mathbf{B} - c^{-2}\partial_t\mathbf{E} = \mu_0\mathbf{J}$$

The $\nu = 0$ component checks Gauss's law: $\partial_i F^{0i} = c^{-1}\nabla\cdot\mathbf{E}$ with $F^{0i} = E_i/c$ from Ch. 11 §11.2.1. **The index order matters.** $\partial_\nu F^{\nu\mu} = -\mu_0 J^\mu$ is the same equation.

Substituting into the identity, the Maxwell piece drops out by antisymmetry ($\partial_\mu\partial_\nu F^{\nu\mu} \equiv 0$), leaving

$$\partial_\mu J^\mu \equiv \frac{iq}{\hbar}\left(\frac{\delta S}{\delta \psi}\psi - \bar{\psi}\frac{\delta S}{\delta \bar{\psi}}\right)$$

Four consequences follow, and the chapters need each one separately.

1. **Charge conservation needs the matter equations.** $\partial_\mu J^\mu = 0$ holds when ψ is on shell, whatever A_μ is doing. This is the same conclusion as Noether's **first** theorem applied to the **global** subgroup (α = const; §A.6.2). Local invariance gives an identity. The global subgroup plus the matter equations gives the conservation law.
2. **Consistency of Maxwell's equations.** Taking ∂_μ of $\partial_\nu F^{\mu\nu} = \mu_0 J^\mu$ shows that a solution can exist only if $\partial_\mu J^\mu = 0$. This is the argument of Ch. 11 §11.5. It proves that **external** sources must be conserved for Maxwell's equations to be solvable, and it shows that the field equation and matter conservation are compatible. It is a consistency condition, not a derivation of charge conservation from the gauge field alone.
3. **The charge is a surface integral.** On the full shell, $J^\mu = \mu_0^{-1}\partial_\nu F^{\mu\nu}$ has the improper form $\partial_\nu U^{\mu\nu}$ of §A.6.1(iii). Hence $Q = \int \rho\, d^3x = \epsilon_0 \oint_{S^2_\infty} \mathbf{E}\cdot d\mathbf{A}$, which is Gauss's law. This is characteristic of second-theorem symmetries: the charges of the global subgroup of a gauge symmetry are measurable at infinity.
4. **Redundancy, not symmetry.** The identity means the field equations are not independent. One combination of them holds automatically, so the Cauchy data do not determine A_μ uniquely: A_μ and $A_\mu + \partial_\mu\alpha$ with α vanishing at $t = 0$ evolve from the same data. Physical predictions are therefore restricted to gauge-invariant quantities, and solving the equations requires gauge fixing (Ch. 11 §11.6). In Hamiltonian form the same fact appears as the primary constraint $\pi^0 = 0$ and the Gauss-law constraint (§A.3.1). A local gauge transformation maps a physical state to **the same** physical state. That is why it is called a redundancy, and why it carries no Noether charge of its own beyond the global one.

> **[DERIVATION]** Local U(1) invariance implies, by Noether's second theorem, the off-shell identity above. Together with the matter equations of motion, it implies $\partial_\mu J^\mu = 0$. Equivalently, the conservation law is the first theorem applied to the global constant-phase subgroup. The local

> invariance supplies an identity, the consistency of Maxwell's equations, and the surface-integral form of the charge, but on its own it does not supply a conserved charge. This is the distinction drawn in Ch. 17 §17.2.3 and required by MCP §8.

Which U(1). The symmetry relevant to circuits is $U(1)_{\text{em}}$, generated by $Q = T_3 + Y$ after electroweak symmetry breaking (Ch. 0 §0.4.2, §0.7). It is not $U(1)_Y$. The argument above applies verbatim to $U(1)_{\text{em}}$ with A_μ the photon field. The electromagnetic current is anomaly-free in the Standard Model, so the classical conservation law survives quantization.

The gravitational analogue. For S_{EH} the same theorem with diffeomorphism parameter ξ^μ gives the off-shell contracted Bianchi identity $\nabla^\mu G_{\mu\nu} \equiv 0$ (Ch. 8 §8.1). For the matter action it gives the result of §A.7.4.

A.9 — Continuity Equation → Integral Charge Balance → KCL

A.9.1 — From local to integral

Assumptions. (C1) $\partial_t \rho + \nabla \cdot \mathbf{J} = 0$ holds pointwise (§A.8; Ch. 11 §11.5). (C2) V is a fixed region with piecewise smooth boundary ∂V and outward normal. (C3) ρ and $\mathbf{J}$ are C^1 on a neighbourhood of $\bar{V}$. Surface charges and currents are included by the standard distributional limit.

Integrate over V, exchange d/dt with $\int_V$ (allowed because V is fixed and the integrand is C^1), and apply the divergence theorem:

$$\boxed{\frac{dQ_V}{dt} = -\oint_{\partial V} \mathbf{J} \cdot d\mathbf{A}, \qquad Q_V = \int_V \rho \, d^3x}$$

> **[DERIVATION]** The integral charge balance is mathematically equivalent to the local continuity equation for all fixed regions. The converse follows by shrinking V to a point. For a moving region, the Reynolds transport theorem adds the flux $\oint \rho \mathbf{u} \cdot d\mathbf{A}$ of the boundary velocity $\mathbf{u}$. The balance contains no approximation.

The exactly conserved current. Taking the divergence of the Ampère–Maxwell law gives $\nabla \cdot (\mathbf{J} + \epsilon_0 \partial_t \mathbf{E}) = 0$ **identically**, which is the integral balance with $Q_V = \epsilon_0 \oint \mathbf{E} \cdot d\mathbf{A}$ by Gauss's law. So

$$\oint_{\partial V} \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \cdot d\mathbf{A} = 0 \quad \text{exactly, for every closed surface}$$

The **total** current, conduction plus displacement, always obeys a Kirchhoff-type law. All the approximation in KCL lies in neglecting part of the displacement current.

A.9.2 — The lumping assumptions

A **node** is a region V enclosing a junction of conductors. Its surface ∂V cuts the conductors at terminal cross-sections $S_1, \ldots, S_N$, and elsewhere passes through insulator or vacuum.

- **(L1) No leakage.** Conduction current crosses ∂V only through the S_k, so $\sigma = 0$ on the rest of ∂V. Define the branch currents $I_k = \int_{S_k} \mathbf{J} \cdot d\mathbf{A}$ (counted outward).

- **(L2) Negligible stray displacement current.** $\left|\epsilon_0 \oint_{\partial V \setminus \cup S_k} \partial_t \mathbf{E} \cdot d\mathbf{A}\right| \ll \max_k |I_k|$. Equivalently, $|dQ_V/dt| \ll \max |I_k|$.
- **(L3) Intended storage is assigned to elements.** Charge that is *supposed* to accumulate, on capacitor plates, is not counted at the node. The node surface is drawn so that it does not cut a capacitor's dielectric. If it does, the displacement current through the gap is the capacitor's branch current, and including it there is exact by §A.9.1.
- **(L4) Well-defined branch current.** The current along each wire is the same at every cross-section between nodes. By the continuity equation in one dimension, $\partial_z I = -\partial_t \lambda$ with λ the charge per unit length, which requires the wire's own charging current to be negligible. This is again an electrical-smallness condition. Its breakdown is the transmission line (Ch. 17 §17.10).

The small parameter. Let the node have size ℓ, angular frequency ω, and node voltage V_n relative to its surroundings. Its stray capacitance is $C_s \sim \epsilon_0 \ell$, so $|dQ_V/dt| \sim \omega C_s |V_n|$. With $|V_n| \sim |Z|\,|I|$ for a characteristic impedance Z at the node, and $\epsilon_0 = 1/(Z_0 c)$ with $Z_0 = \sqrt{\mu_0/\epsilon_0} \approx 377\,\Omega$:

$$\varepsilon_{\mathrm{KCL}} \sim \omega C_s |Z| \sim \frac{\omega \ell}{c} \cdot \frac{|Z|}{Z_0}$$

> **[APPROXIMATION]** Under (L1)–(L4), $\sum_k I_k = -dQ_V/dt$ becomes $\sum_k I_k = 0$, with a relative error of order $\varepsilon_{\mathrm{KCL}} = (\omega\ell/c)(|Z|/Z_0)$. KCL therefore requires electrically small nodes ($\omega\ell/c \ll 1$, the criterion of Ch. 11 §11.12.1). It **also** requires node impedances that are not too large compared with Z_0. A high-impedance node in a picoampere circuit can violate KCL through stray capacitance even at low frequency, which is why guard rings exist. At DC with all fields static, $dQ_V/dt = 0$ exactly and KCL is exact under (L1).

The dual criterion for KVL. Faraday's law $\oint_{\mathcal{C}} \mathbf{E} \cdot d\mathbf{l} = -d\Phi_B/dt$ is exact. Lumping chooses the loop $\mathcal{C}$ to run, between element terminals, through a region where $\partial_t \mathbf{B}$ is negligible. There $\mathbf{E} \approx -\nabla\varphi$, the terminal voltages $v_k = \varphi(a_k) - \varphi(b_k)$ are path independent, and their sum around the loop vanishes identically. The time-varying flux is **not dropped**: it is confined inside the inductors. There Faraday's law, applied to the element's own winding, produces the branch relation $v_L = L\,dI/dt$ as part of the sum. The neglected piece is the stray flux through the loop outside the elements, $\omega L_s |I|$ with $L_s \sim \mu_0 \ell$. The small parameter is

$$\varepsilon_{\mathrm{KVL}} \sim \frac{\omega L_s}{|Z|} \sim \frac{\omega \ell}{c} \cdot \frac{Z_0}{|Z|}$$

> **[APPROXIMATION]** KVL in the form $\sum_k v_k = 0$, with inductive branch voltages included, holds to relative order $(\omega\ell/c)(Z_0/|Z|)$. KCL fails first at high impedance and KVL fails first at low impedance. Both require $\omega\ell/c \ll 1$. Energy bookkeeping in the lumped circuit is Poynting's theorem (Ch. 11 §11.8.2) integrated over the element volumes, not KVL. The full step-by-step proof of lumped KVL from Faraday's law is in Appendix E §E.10. This paragraph supplies only the small parameter dual to the KCL one.

What was proved. KCL is an exact consequence of the charge balance plus four stated lumping assumptions, with a named small parameter. **Where it fails.** It fails for stray capacitance at high-impedance nodes, for electrically long interconnects and radiation ($\omega\ell/c \gtrsim 0.1$), and for leakage through imperfect insulation. The replacement is the telegrapher's equations or full-wave analysis (Ch. 17 §17.10).

The same template gives the hydraulic junction law from mass conservation, with (L2) replaced by negligible junction compliance. The mechanical and thermal "node laws" of Ch. 17 §17.4.1 keep their storage terms ($m\ddot{x}$, $C_{th}\dot{T}$) and are balance laws, not $\sum f_k = 0$.

A.10 — Dissipation: The Rayleigh Function and the Bridge to R/C/L

A.10.1 — Why dissipation is not in the action

For a closed system with time-independent L, §A.5 makes H conserved. Linear friction $m\ddot{x} + b\dot{x} + kx = 0$ dissipates the physical energy $\frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2$, so it cannot come from a Lagrangian of the standard form $T - V$ with time-independent T and V. Non-standard Lagrangians do exist: for one degree of freedom the inverse problem of the calculus of variations always has a solution (Darboux), and time-independent ones exist for this equation. But their conserved energy function is then not the physical energy, and they do not generalize usefully to networks. Time-dependent constructions also exist. The Caldirola–Kanai Lagrangian $L = e^{\gamma t}\left(\frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2\right)$ gives $\frac{d}{dt}(e^{\gamma t}m\dot{x}) + e^{\gamma t}kx = 0$, that is $m\ddot{x} + \gamma m\dot{x} + kx = 0$. But its "Hamiltonian" is not the physical energy, and its canonical quantization does not describe a physically damped quantum oscillator. Physically, dissipation arises from coupling to unobserved degrees of freedom (Ch. 10, Ch. 13). The honest classical treatment adds it as a **generalized force**.

A.10.2 — The Lagrange–d'Alembert principle with a Rayleigh function

Assumptions.

- (D1) Conservative dynamics come from $L(q, \dot{q})$.
- (D2) Non-conservative forces are linear in velocities and derivable from a Rayleigh function

$$\mathcal{R}(q, \dot{q}) = \tfrac{1}{2}\sum_{ij} b_{ij}(q)\,\dot{q}_i\dot{q}_j, \qquad Q_i^{\text{diss}} = -\frac{\partial \mathcal{R}}{\partial \dot{q}_i}, \qquad b_{ij} = b_{ji}, \;\; b \succeq 0$$

- (D3) External drives are generalized forces $Q_i^{\text{ext}}(t)$.

The Lagrange–d'Alembert principle $\delta \int L\,dt + \int \sum_i (Q_i^{\text{diss}} + Q_i^{\text{ext}})\,\delta q_i\,dt = 0$, with fixed endpoints, gives by §A.2.2

$$\boxed{\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} + \frac{\partial \mathcal{R}}{\partial \dot{q}_i} = Q_i^{\text{ext}}}$$

This is **not** a stationary-action principle. It is the variational form of Newton's law with the extra forces included.

Energy theorem. Let $h = \sum_i \dot{q}_i\,\partial L/\partial \dot{q}_i - L$ (equal to H on the Legendre image). Differentiating along a solution and using the equation above:

$$\frac{dh}{dt} = \sum_i \dot{q}_i\left(\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i}\right) = \sum_i \dot{q}_i Q_i^{\text{ext}} - \sum_i \dot{q}_i \frac{\partial \mathcal{R}}{\partial \dot{q}_i} = P_{\text{in}} - 2\mathcal{R}$$

The last step is Euler's theorem for the degree-2 homogeneous function $\mathcal{R}$. Thus $2\mathcal{R} \geq 0$ is the dissipated power, and $b \succeq 0$ is **passivity**. An antisymmetric part of b_{ij} would do no work. Such gyroscopic or magnetic forces belong in L, as the $q\dot{\mathbf{r}} \cdot \mathbf{A}$ term does, not in $\mathcal{R}$.

A.10.3 — Circuits and the R/C/L template

Series RLC, mesh-charge coordinate q, $I = \dot{q}$.

$$L = \tfrac{1}{2}\mathsf{L}\,\dot{q}^2 - \frac{q^2}{2C}, \qquad \mathcal{R} = \tfrac{1}{2}R\,\dot{q}^2, \qquad Q^{\text{ext}} = V_s(t) \implies \mathsf{L}\ddot{q} + R\dot{q} + \frac{q}{C} = V_s(t)$$

This is KVL for the loop, with $h = \tfrac{1}{2}\mathsf{L}I^2 + q^2/2C$ and $dh/dt = V_s I - RI^2$.

Parallel RLC, node-flux coordinate Φ, $v = \dot{\Phi}$.

$$L = \tfrac{1}{2}C\,\dot{\Phi}^2 - \frac{\Phi^2}{2\mathsf{L}}, \qquad \mathcal{R} = \frac{\dot{\Phi}^2}{2R}, \qquad Q^{\text{ext}} = I_s(t) \implies C\dot{v} + \frac{v}{R} + \frac{1}{\mathsf{L}}\int v\,dt = I_s(t)$$

This is KCL for the node.

Here L is inductance, to avoid a clash with the Lagrangian L. The two choices are dual. In the mesh formulation magnetic energy is "kinetic". In the node formulation electric energy is "kinetic", which matches the field Lagrangian $\frac{\epsilon_0}{2}(E^2 - c^2B^2)$ of Ch. 11 §11.2.2, in which $\mathbf{E} = -\dot{\mathbf{A}} - \nabla\phi$ is a velocity.

Mechanical. $L = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2$ and $\mathcal{R} = \frac{1}{2}b\dot{x}^2$ give $m\ddot{x} + b\dot{x} + kx = F(t)$.

Template slot (Ch. 17 §17.3)	Lagrangian role	Electrical (mesh)	Mechanical
Inductor / inertia	kinetic term $\frac{1}{2}(\cdot)\dot{q}^2$	L	m
Capacitor / compliance	potential term $q^2/2(\cdot)$	C	$1/k$
Resistor / damper	Rayleigh $\frac{1}{2}(\cdot)\dot{q}^2$	R	b
Source	generalized force	V_s	F

> **[STRUCTURAL CONNECTION]** The electrical, mechanical and hydraulic second-order systems share the Lagrange–Rayleigh form $\mathsf{M}\ddot{q} + \mathsf{B}\dot{q} + \mathsf{K}q = Q$ because each has one quadratic kinetic store, one quadratic potential store and one linear dissipation. That is a shared mathematical structure, not a common derivation (Ch. 17 §17.7). The coefficients L and C follow from integrating the field energies over the element under the lumping assumptions of §A.9 (`[APPROXIMATION]`, small parameter $\omega\ell/c$). The coefficient R, and every b_{ij}, is a **constitutive** input: Ohm's law or viscosity, understood within linear response (Ch. 13). It is not a consequence of the action or of any conservation law.

> **[PHENOMENOLOGICAL]** The quadratic form of $\mathcal{R}$ encodes the assumption of **linear** dissipation, which holds near equilibrium. Coulomb friction, diode conduction, turbulent pipe losses ($\Delta p \propto Q^2$) and hysteresis do not fit a quadratic Rayleigh function. They need a non-quadratic dissipation potential or no potential at all. Symmetry of b_{ij} in multi-port linear dissipation is the Onsager reciprocity condition, which needs microscopic time-reversal arguments. It is assumed here, not proved.

What was proved. With dissipation added as a Rayleigh force, the Euler–Lagrange machinery reproduces the R/C/L network equations, and the energy theorem gives dissipated power $2\mathcal{R}$. **Where it fails.** It fails for nonlinear dissipation, for memory effects (frequency-dependent $R(\omega)$, dielectric relaxation), for distributed systems ($\omega\ell/c$ not small), and in quantum regimes where the bath must be modelled explicitly.

A.11 — Summary

Result	Statement	Assumptions	Used in
First variation	$\delta S = \int \mathsf{E}\,\eta\,dt + [p\,\eta]_{t_1}^{t_2}$; fields: plus $\oint d\Sigma_\mu\,\partial\mathcal{L}/\partial(\partial_\mu\phi)\,\delta\phi$	C^2 Lagrangian; smooth variations	Ch. 1 §1.5.1, §1.9.2
Euler–Lagrange	$\frac{d}{dt}\partial L/\partial\dot{q} - \partial L/\partial q = 0$	fixed endpoints, or natural BCs $p = 0$	Ch. 1 §1.5; Ch. 9 §9.5; Ch. 11 §11.3
Euler–Poisson	fourth-order interior equation, two BCs per end	$L(q, \dot{q}, \ddot{q})$	Ch. 16 (beam)
Total-derivative freedom	$L \to L + dG/dt$ leaves EOM, shifts p	fixed endpoints	Ch. 1 §1.6.1; Ch. 9 §9.5.2
Hamilton's equations	$\dot{q} = \partial H/\partial p$, $\dot{p} = -\partial H/\partial q$; $dH/dt = -\partial_t L$	regular Hessian	Ch. 1 §1.6; Ch. 9 §9.6
Poisson dynamics	$\dot{f} = \{f, H\} + \partial_t f$; conserved G iff $\{G, H\} = 0$	canonical coordinates	Ch. 1 §1.6.4; Ch. 9 §9.6.2
Noether I (mechanics)	$\dot{J} = -\sum \mathsf{E}_i K_i$, $J = \sum p_i K_i - F$	$\delta L = \epsilon\,dF/dt$ off shell	Ch. 1 §1.7
Noether I (fields)	$\partial_\mu j^\mu = -\mathsf{E}^a \Delta\phi_a$	$\delta\mathcal{L} = \epsilon\,\partial_\mu K^\mu$ off shell; falloff for global Q	Ch. 1 §1.9.3; Ch. 3 §3.9.2
Source = Noether current	$J^\mu = \partial\mathcal{L}_m/\partial A_\mu = -j_N^\mu$	minimal coupling	Ch. 11 §11.1, §11.5
Canonical $T^{\mu\nu}$	$-\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\partial^\nu\phi + \eta^{\mu\nu}\mathcal{L}$, with $T^{00} = \mathcal{H}$	translation invariance; $(-,+,+,+)$	Ch. 1 §1.9.3
Angular momentum	$\partial_\mu M^{\mu\lambda\nu} = T^{\lambda\nu} - T^{\nu\lambda}$	Lorentz invariance; symmetric T (or spin term)	Ch. 1 §1.7.3
Belinfante = Maxwell T	$T_{\text{can}} = T_{\text{B}} + \partial_\lambda B^{\lambda\mu\nu}$	vacuum Maxwell, on shell	Ch. 11 §11.8.1
Hilbert $T_{\mu\nu}$	$-(2/\sqrt{-g})\,\delta S_m/\delta g^{\mu\nu}$; equals improved canonical in flat space (scalar, Maxwell)	$(-,+,+,+)$ signature	Ch. 0 §0.3; Ch. 1 §1.9.3; Ch. 8 §8.1
Covariant conservation	$\nabla_\mu T^{\mu\nu} = 0$; conserved charge only with a Killing vector	diffeo invariance; matter on shell	Ch. 8 §8.1
Noether II (U(1))	$\partial_\mu\,\delta S/\delta A_\mu \equiv (iq/\hbar)(\dots)$ off shell; $\partial_\mu J^\mu = 0$ needs matter EOM	local U(1) invariance	Ch. 11 §11.5–11.6; Ch. 17 §17.2.3
Gauss charge	$Q = \epsilon_0 \oint \mathbf{E}\cdot d\mathbf{A}$	full shell	Ch. 11 §11.4
Integral charge balance	$dQ_V/dt = -\oint \mathbf{J}\cdot d\mathbf{A}$; total current exactly solenoidal	fixed V; C^1 fields	Ch. 11 §11.12.3; Ch. 17 §17.4.1
KCL	$\sum I_k = 0$, error $\sim (\omega\ell/c)(|Z|/Z_0)$	(L1)–(L4)	Ch. 11 §11.12.3; Ch. 17 §17.4; Ch. 18
KVL (lumped)	$\sum v_k = 0$ incl. $\mathsf{L}\,dI/dt$, error $\sim (\omega\ell/c)(Z_0/|Z|)$	flux confined to elements	Ch. 11 §11.12.2; Ch. 17 §17.4.2
Rayleigh–Lagrange	$\frac{d}{dt}\partial_{\dot{q}}L - \partial_q L + \partial_{\dot{q}}\mathcal{R} = Q^{\text{ext}}$; $\dot{h} = P_{\text{in}} - 2\mathcal{R}$	linear, symmetric, positive dissipation	Ch. 17 §17.3, §17.7, §17.9

References (standard texts; no claim here depends on a single page): Goldstein, Poole & Safko, *Classical Mechanics*, 3rd ed. (2002); Landau & Lifshitz, *Mechanics* and *The Classical Theory of Fields*; Gelfand & Fomin, *Calculus of Variations*; Arnold, *Mathematical Methods of Classical Mechanics*; Noether, "Invariante Variationsprobleme", *Nachr. Ges. Wiss. Göttingen* (1918); Kosmann-Schwarzbach, *The Noether Theorems* (2011); Weinberg, *The Quantum Theory of Fields*, Vol. I; Wald, *General Relativity*; Carroll, *Spacetime and Geometry*; Jackson, *Classical Electrodynamics*, 3rd ed.; Henneaux & Teitelboim, *Quantization of Gauge Systems* (1992).

End of Appendix A

Appendix B — Relativistic to Nonrelativistic Quantum Mechanics: Dirac, Foldy–Wouthuysen, Pauli

This appendix carries out in full the descent that Bridge A states. It starts from the Dirac equation with minimal coupling, separates the Hamiltonian into even and odd operators, and performs the Foldy–Wouthuysen (FW) transformation order by order to $(v/c)^2$. From the result it reads off five terms: the Pauli Hamiltonian with its Zeeman coupling and $g = 2$, the relativistic mass correction, the spin–orbit coupling (Thomas factor included automatically), and the Darwin term. Applied to hydrogen in first-order perturbation theory, these terms give the fine-structure formula to order α^4. The appendix then checks that formula against the expansion of the exact Dirac–Coulomb energy and closes with the Gordon decomposition, the cleanest statement of where $g = 2$ comes from. The chapters that rely on it are Ch. 3 §§3.2–3.6 (Bridge A itself), Ch. 4 §4.7 (the Bohr levels as the zeroth-order problem), Ch. 5 §§5.5–5.6 (spin–orbit, Zeeman, Landé), and Example 01.

The FW series derived here is **truncated**. The truncation error and the conditions under which the series fails altogether are stated explicitly in §B.11.

B.1 — Conventions and the Dirac Equation

Assumptions and conventions.

1. **Units: Gaussian**, as in Ch. 3. The fine-structure constant is $\alpha = e^2/\hbar c$ and the Bohr radius is $a_0 = \hbar^2/me^2$. To convert to SI (Ch. 4, Ch. 5), replace $e^2 \to e^2/4\pi\epsilon_0$, $q\mathbf{A}/c \to q\mathbf{A}$, $q\hbar/2mc \to q\hbar/2m$ in magnetic terms, and $\nabla\cdot\mathbf{E} = 4\pi\rho \to \rho/\epsilon_0$. Terms built from $\mathbf{E}$ and $1/c^2$ keep their form.
2. **Charge.** The particle has signed charge q. For the electron $q = -e$, with $e > 0$. The formulas of Ch. 3 §§3.2–3.5 are those below with q written as e, which is the signed-charge convention of Bjorken & Drell.
3. **Matrices.** Bold $\boldsymbol{\alpha}$ and β are the Dirac matrices. Plain α is always the fine-structure constant. $\boldsymbol{\Sigma} = \mathrm{diag}(\boldsymbol{\sigma}, \boldsymbol{\sigma})$.
4. **Signature for gamma matrices.** With $\gamma^0 = \beta$ and $\gamma^i = \beta\alpha_i$ one finds $(\gamma^0)^2 = 1$ and $(\gamma^i)^2 = -\beta^2\alpha_i^2 = -1$. So $\{\gamma^\mu, \gamma^\nu\} = 2\,\mathrm{diag}(1,-1,-1,-1)$. This is $2\eta^{\mu\nu}$ in the $(+,-,-,-)$ signature, which is the form Ch. 3 §3.3.1 writes, and $-2\eta^{\mu\nu}$ in the book's $(-,+,+,+)$ signature (Appendix A §A.1). The covariant formulas in §B.1 and §B.10 use $(+,-,-,-)$ and say so. Everything else is in Hamiltonian form, which does not depend on the signature.

The Dirac Hamiltonian. With Φ the scalar potential and $\mathbf{A}$ the vector potential:

$$i\hbar\,\partial_t\psi = H_D\psi, \qquad H_D = c\,\boldsymbol{\alpha}\cdot\boldsymbol{\pi} + \beta mc^2 + q\Phi, \qquad \boldsymbol{\pi} = \mathbf{p} - \frac{q}{c}\mathbf{A}, \quad \mathbf{p} = -i\hbar\nabla$$

In the Dirac representation (Ch. 3 §3.3.1):

$$\beta = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \boldsymbol{\alpha} = \begin{pmatrix} 0 & \boldsymbol{\sigma} \\ \boldsymbol{\sigma} & 0 \end{pmatrix}$$

The defining algebra, which holds in any representation, is

$$\alpha_i^2 = \beta^2 = 1, \qquad \{\alpha_i, \alpha_j\} = 2\delta_{ij}, \qquad \{\alpha_i, \beta\} = 0, \qquad \alpha_i^\dagger = \alpha_i, \;\; \beta^\dagger = \beta$$

Why this algebra. For a free particle,

$$(c\boldsymbol{\alpha}\cdot\mathbf{p} + \beta mc^2)^2 = c^2\alpha_i\alpha_j p_i p_j + mc^3\{\alpha_i,\beta\}p_i + \beta^2 m^2c^4 = c^2p^2 + m^2c^4$$

So every component obeys the Klein–Gordon dispersion relation $E^2 = c^2p^2 + m^2c^4$. The Clifford algebra is exactly the condition for a first-order Hamiltonian to square to the relativistic dispersion relation, and 4×4 is the smallest dimension in which four mutually anticommuting Hermitian matrices exist.

From the Lagrangian. In $(+,-,-,-)$ with $A^\mu = (\Phi, \mathbf{A})$, take $\mathcal{L} = \bar\psi\left(i\hbar c\,\gamma^\mu D_\mu - mc^2\right)\psi$ with $D_\mu = \partial_\mu + (iq/\hbar c)A_\mu$. The spatial components give $-i\hbar D_i = p^i - (q/c)A^i = \pi^i$ (using $A_i = -A^i$), and the time component gives $i\hbar c\, D_0 = i\hbar\partial_t - q\Phi$. Varying with respect to $\bar\psi$ and multiplying by γ^0 gives H_D exactly. Dividing by c, this is the equation of Ch. 3 §3.2.2 with $e \to q$.

> **[DERIVATION]** The Euler–Lagrange equation of the minimally coupled Dirac Lagrangian is the Hamiltonian equation $i\hbar\partial_t\psi = H_D\psi$. No approximation has been made.

Conserved current. Since H_D is Hermitian, $\partial_t(\psi^\dagger\psi) + \nabla\cdot(c\,\psi^\dagger\boldsymbol{\alpha}\psi) = 0$. Multiplied by q, this is the Noether current of the global phase symmetry (Appendix A §A.6.2): $\rho = q\psi^\dagger\psi$ and $\mathbf{J} = qc\,\psi^\dagger\boldsymbol{\alpha}\psi$. The density $\psi^\dagger\psi \geq 0$ is conserved. Reading it as a probability density is the Born postulate (Ch. 3 §3.9.2).

B.2 — Minimal Coupling: The Operator Identities

Gauge covariance. Under $\mathbf{A} \to \mathbf{A} + \nabla\chi$, $\Phi \to \Phi - c^{-1}\partial_t\chi$ and $\psi \to e^{iq\chi/\hbar c}\psi$, the operator $\boldsymbol{\pi}$ acting on ψ transforms covariantly: $\boldsymbol{\pi}\left(e^{iq\chi/\hbar c}\psi\right) = e^{iq\chi/\hbar c}\,\boldsymbol{\pi}_{\text{old}}\psi$. The combination $i\hbar\partial_t - q\Phi$ does the same. So the Dirac equation is gauge covariant, and the physical fields $\mathbf{E} = -\nabla\Phi - c^{-1}\partial_t\mathbf{A}$ and $\mathbf{B} = \nabla\times\mathbf{A}$ are gauge invariant.

Three commutators used throughout. Each follows from $[p_i, f(\mathbf{r})] = -i\hbar\,\partial_i f$.

$$[\pi_i,\; f(\mathbf{r})] = -i\hbar\,\partial_i f$$

$$[\pi_i, \pi_j] = -\frac{q}{c}\left([p_i, A_j] + [A_i, p_j]\right) = \frac{i\hbar q}{c}\left(\partial_i A_j - \partial_j A_i\right) = \frac{i\hbar q}{c}\,\epsilon_{ijk}B_k$$

$$(\boldsymbol{\pi}\times\boldsymbol{\pi})_k = \epsilon_{ijk}\pi_i\pi_j = \tfrac{1}{2}\epsilon_{ijk}[\pi_i,\pi_j] = \frac{i\hbar q}{c}B_k$$

Pauli identity, with operators. Let $\mathbf{a}$ and $\mathbf{b}$ be vector operators whose components commute with $\boldsymbol{\sigma}$, but not necessarily with each other. Using $\sigma_i\sigma_j = \delta_{ij} + i\epsilon_{ijk}\sigma_k$ and keeping the order a_ib_j:

$$(\boldsymbol{\sigma}\cdot\mathbf{a})(\boldsymbol{\sigma}\cdot\mathbf{b}) = \mathbf{a}\cdot\mathbf{b} + i\boldsymbol{\sigma}\cdot(\mathbf{a}\times\mathbf{b}), \qquad (\mathbf{a}\times\mathbf{b})_k \equiv \epsilon_{ijk}a_ib_j$$

Since $\alpha_i\alpha_j = \mathrm{diag}(\sigma_i\sigma_j,\, \sigma_i\sigma_j)$, the same identity holds with $\boldsymbol{\sigma} \to \boldsymbol{\alpha}$ on the left and $\boldsymbol{\sigma} \to \boldsymbol{\Sigma}$ on the right.

The key identity. Set $\mathbf{a} = \mathbf{b} = \boldsymbol{\pi}$:

$$\boxed{(\boldsymbol{\sigma}\cdot\boldsymbol{\pi})^2 = \boldsymbol{\pi}^2 - \frac{q\hbar}{c}\,\boldsymbol{\sigma}\cdot\mathbf{B}}$$

This one line is where the spin coupling and $g = 2$ come from (§B.6). The spin term arises only because π_i and π_j fail to commute when $\mathbf{B} \neq 0$.

B.3 — Even and Odd Operators

Definitions. An operator is **even** if it commutes with β. It is then block diagonal and does not mix upper and lower components. An operator is **odd** if it anticommutes with β. It is then block off-diagonal. Even·even and odd·odd products are even. Even·odd products are odd. The commutator of an odd operator with an even one is odd.

$$H_D = \beta mc^2 + \mathcal{E} + \mathcal{O}, \qquad \mathcal{E} = q\Phi, \qquad \mathcal{O} = c\,\boldsymbol{\alpha}\cdot\boldsymbol{\pi}, \qquad \beta\mathcal{E} = \mathcal{E}\beta, \quad \beta\mathcal{O} = -\mathcal{O}\beta$$

What is in $\mathcal{O}$, and what is not. The odd part of H_D is *exactly* $c\boldsymbol{\alpha}\cdot\boldsymbol{\pi}$. The electric field does **not** appear in $\mathcal{O}$, as $\boldsymbol{\sigma}\cdot\mathbf{E}$ or in any other form. The reason is that $\mathbf{E}$ does not appear in H_D at all: only the potentials Φ and $\mathbf{A}$ do. $\mathbf{E}$ first appears in §B.4 as the gauge-covariant combination generated when $\mathcal{O}$ fails to commute with $\mathcal{E}$ and with ∂_t:

$$[\mathcal{O}, \mathcal{E}] + i\hbar\dot{\mathcal{O}} = i\hbar qc\,\boldsymbol{\alpha}\cdot\mathbf{E}$$

To verify this, note $[c\alpha_k\pi_k,\, q\Phi] = -i\hbar qc\,\boldsymbol{\alpha}\cdot\nabla\Phi$ and $i\hbar\dot{\mathcal{O}} = i\hbar c\,\alpha_k\partial_t(-qA_k/c) = -i\hbar q\,\boldsymbol{\alpha}\cdot\partial_t\mathbf{A}$. Their sum is $-i\hbar qc\,\boldsymbol{\alpha}\cdot(\nabla\Phi + c^{-1}\partial_t\mathbf{A}) = i\hbar qc\,\boldsymbol{\alpha}\cdot\mathbf{E}$. This generated odd term carries the 4×4 matrix $\boldsymbol{\alpha}$, not $\boldsymbol{\sigma}$.

Adding an independent odd field term by hand would change the theory. An odd term proportional to $i\beta\boldsymbol{\alpha}\cdot\mathbf{E}$ is part of the Pauli anomalous-moment coupling, whose Hamiltonian form is proportional to $\beta(\boldsymbol{\Sigma}\cdot\mathbf{B} - i\boldsymbol{\alpha}\cdot\mathbf{E})$. That coupling shifts g away from 2. Minimal coupling excludes it by assumption.

Power counting. For a bound state of size r and speed v: $|\mathcal{O}| \sim cp \sim mc^2(v/c)$ and $|\mathcal{E}| \sim mv^2 = mc^2(v/c)^2$ (by the virial theorem), with $\hbar/r \sim p$. The FW expansion is an expansion in $\mathcal{O}/mc^2 \sim v/c$. For hydrogen-like ions, $v/c \sim Z\alpha$.

B.4 — The Foldy–Wouthuysen Transformation

B.4.1 — Transformation law for a time-dependent unitary

Let $\psi' = e^{iS}\psi$ with S Hermitian and possibly time dependent. Then

$$i\hbar\partial_t\psi' = e^{iS}\,i\hbar\partial_t\psi + i\hbar\,(\partial_t e^{iS})\psi = \left[e^{iS}He^{-iS} + i\hbar\,(\partial_t e^{iS})\,e^{-iS}\right]\psi'$$

so $H' = e^{iS}He^{-iS} + i\hbar(\partial_t e^{iS})e^{-iS}$. Unitarity is exact. Expanding with the Baker–Campbell–Hausdorff series:

$$H' = H + [iS, H] + \frac{1}{2!}[iS,[iS,H]] + \frac{1}{3!}[iS,[iS,[iS,H]]] + \frac{1}{4!}[iS,[iS,[iS,[iS,H]]]] + \cdots - \hbar\dot{S} - \frac{i\hbar}{2}[S,\dot{S}]$$

B.4.2 — First transformation

Write $M \equiv mc^2$ and choose

$$S = -\frac{i\beta\mathcal{O}}{2M}, \qquad iS = \frac{\beta\mathcal{O}}{2M}$$

S is Hermitian because $(\beta\mathcal{O})^\dagger = \mathcal{O}\beta = -\beta\mathcal{O}$. The building blocks follow from $\beta\mathcal{O}\beta = -\mathcal{O}$ and $[\beta, \mathcal{E}] = 0$:

$$[iS, \beta M] = \tfrac{1}{2}(\beta\mathcal{O}\beta - \mathcal{O}) = -\mathcal{O}, \qquad [iS, \mathcal{E}] = \frac{\beta}{2M}[\mathcal{O}, \mathcal{E}], \qquad [iS, \mathcal{O}] = \frac{\beta\mathcal{O}^2}{M}$$

The first identity is the design condition: the first commutator cancels the odd term. Denote the BCH terms by $X_n = \frac{1}{n}[iS, X_{n-1}]$ with $X_0 = H$, and keep everything through $1/M^2$, together with the kinetic $1/M^3$ term $\mathcal{O}^4/M^3$, which is of the same order as the $1/M^2$ terms for bound states.

$$X_1 = -\mathcal{O} + \frac{\beta}{2M}[\mathcal{O}, \mathcal{E}] + \frac{\beta\mathcal{O}^2}{M}$$

$$X_2 = \frac{1}{2}\left[\frac{\beta\mathcal{O}}{2M}, X_1\right] = -\frac{\beta\mathcal{O}^2}{2M} - \frac{\mathcal{O}^3}{2M^2} - \frac{1}{8M^2}[\mathcal{O}, [\mathcal{O}, \mathcal{E}]]$$

In the last term we used $\beta[\mathcal{O}, \mathcal{E}]\beta = -[\mathcal{O}, \mathcal{E}]$, so $\left[\beta\mathcal{O}, \beta[\mathcal{O}, \mathcal{E}]\right] = -[\mathcal{O}, [\mathcal{O}, \mathcal{E}]]$.

$$X_3 = \frac{\mathcal{O}^3}{6M^2} - \frac{\beta\mathcal{O}^4}{6M^3} + \ldots, \qquad X_4 = \frac{\beta\mathcal{O}^4}{24M^3} + \ldots$$

The time-derivative terms, with $\dot{S} = -i\beta\dot{\mathcal{O}}/2M$, are

$$-\hbar\dot{S} = \frac{i\hbar\beta\dot{\mathcal{O}}}{2M}, \qquad -\frac{i\hbar}{2}[S, \dot{S}] = -\frac{i\hbar}{8M^2}[\mathcal{O}, \dot{\mathcal{O}}]$$

The second uses $[\beta\mathcal{O}, \beta\dot{\mathcal{O}}] = -[\mathcal{O}, \dot{\mathcal{O}}]$. Collecting terms, with the coefficients $\frac{1}{M} - \frac{1}{2M} = \frac{1}{2M}$ for $\beta\mathcal{O}^2$, $-\frac{1}{6} + \frac{1}{24} = -\frac{1}{8}$ for $\beta\mathcal{O}^4/M^3$, and $-\frac{1}{2} + \frac{1}{6} = -\frac{1}{3}$ for $\mathcal{O}^3/M^2$:

$$H' = \beta M + \mathcal{E}' + \mathcal{O}'$$

$$\mathcal{E}' = \mathcal{E} + \beta\left(\frac{\mathcal{O}^2}{2M} - \frac{\mathcal{O}^4}{8M^3}\right) - \frac{1}{8M^2}\Big[\mathcal{O},\ [\mathcal{O}, \mathcal{E}] + i\hbar\dot{\mathcal{O}}\Big]$$

$$\mathcal{O}' = \frac{\beta}{2M}\Big([\mathcal{O}, \mathcal{E}] + i\hbar\dot{\mathcal{O}}\Big) - \frac{\mathcal{O}^3}{3M^2}$$

The original odd operator, of order $M(v/c)$, has been replaced by $\mathcal{O}'$, of order $M(v/c)^3$. Ch. 3 §3.5.2 states the same reduction.

B.4.3 — Second and third transformations

H' has the same structure as H, with $\mathcal{O} \to \mathcal{O}'$ and $\mathcal{E} \to \mathcal{E}'$. Repeat with $S' = -i\beta\mathcal{O}'/2M$:

$$H'' = \beta M + \mathcal{E}' + \underbrace{\frac{\beta}{2M}\Big([\mathcal{O}', \mathcal{E}'] + i\hbar\dot{\mathcal{O}}'\Big)}_{\mathcal{O}'' \sim 1/M^2} + \underbrace{\frac{\beta\mathcal{O}'^2}{2M} + \ldots}_{\text{even},\ \sim M(v/c)^6}$$

A third transformation with $S'' = -i\beta\mathcal{O}''/2M$ removes $\mathcal{O}''$ and leaves only even corrections of order $M(v/c)^6$. To the order retained, the result is

$$\boxed{H_{FW} = \beta\left(mc^2 + \frac{\mathcal{O}^2}{2mc^2} - \frac{\mathcal{O}^4}{8m^3c^6}\right) + \mathcal{E} - \frac{1}{8m^2c^4}\Big[\mathcal{O},\ [\mathcal{O}, \mathcal{E}] + i\hbar\dot{\mathcal{O}}\Big] + \mathcal{O}\big(mc^2(v/c)^6\big)}$$

This is the general result of Foldy & Wouthuysen (1950), in the form given by Bjorken & Drell, *Relativistic Quantum Mechanics*. Every term is even, so the Hamiltonian is block diagonal.

> **[APPROXIMATION]** Each FW step is an exact unitary transformation, but the generators are fixed order by order, and the sequence is stopped after removing odd terms through $1/M^2$. The small parameter is v/c (equivalently $|\mathcal{O}|/mc^2$). The boxed H_{FW} is exact through relative order $(v/c)^2$ beyond the Schrödinger terms and wrong at order $(v/c)^4$. It also requires the field conditions of §B.11.

B.4.4 — Check: the free particle, exactly

For $\Phi = 0$ and $\mathbf{A} = 0$, $\mathcal{O} = c\boldsymbol{\alpha}\cdot\mathbf{p}$ commutes with itself at all times and $\mathcal{O}^2 = c^2p^2$. Foldy and Wouthuysen found the exact transformation $e^{iS} = \exp(\beta\boldsymbol{\alpha}\cdot\hat{\mathbf{p}}\,\theta)$, with $\tan 2\theta = p/mc$. Since $(\beta\boldsymbol{\alpha}\cdot\hat{\mathbf{p}})^2 = -1$, this equals $\cos\theta + \beta\boldsymbol{\alpha}\cdot\hat{\mathbf{p}}\sin\theta$. It gives

$$H_{FW}^{\text{free}} = \beta\sqrt{c^2p^2 + m^2c^4} = \beta\left(mc^2 + \frac{p^2}{2m} - \frac{p^4}{8m^3c^2} + \dots\right)$$

This agrees term by term with the boxed series. For the free particle the expansion converges for $p < mc$. For a general external field no closed form exists, and the series is at best asymptotic.

B.5 — Evaluating H_{FW} for Electromagnetic Fields

Kinetic term. By §B.2 with $\boldsymbol{\alpha}\boldsymbol{\alpha} \to \boldsymbol{\Sigma}$,

$$\frac{\beta\mathcal{O}^2}{2mc^2} = \frac{\beta(\boldsymbol{\alpha}\cdot\boldsymbol{\pi})^2}{2m} = \beta\left(\frac{\boldsymbol{\pi}^2}{2m} - \frac{q\hbar}{2mc}\boldsymbol{\Sigma}\cdot\mathbf{B}\right)$$

$$-\frac{\beta\mathcal{O}^4}{8m^3c^6} = -\frac{\beta}{8m^3c^2}\left(\boldsymbol{\pi}^2 - \frac{q\hbar}{c}\boldsymbol{\Sigma}\cdot\mathbf{B}\right)^2 = -\beta\frac{\boldsymbol{\pi}^4}{8m^3c^2} + (\text{terms} \propto \mathbf{B})$$

The dropped $\mathbf{B}$-dependent pieces are relativistic corrections to the Zeeman term. Relative to it they are of order $(v/c)^2$, plus a term of order $\hbar|q|B/m^2c^3$. They are negligible for laboratory fields and are not retained in Ch. 3.

The commutator term. By §B.3, $[\mathcal{O}, \mathcal{E}] + i\hbar\dot{\mathcal{O}} = i\hbar qc\,\boldsymbol{\alpha}\cdot\mathbf{E}$, so

$$-\frac{1}{8m^2c^4}\Big[c\boldsymbol{\alpha}\cdot\boldsymbol{\pi},\ i\hbar qc\,\boldsymbol{\alpha}\cdot\mathbf{E}\Big] = -\frac{i\hbar q}{8m^2c^2}\left[\boldsymbol{\alpha}\cdot\boldsymbol{\pi},\ \boldsymbol{\alpha}\cdot\mathbf{E}\right]$$

Evaluate the commutator with the Pauli identity in $\boldsymbol{\Sigma}$ form, keeping operator order:

$$(\boldsymbol{\alpha}\cdot\boldsymbol{\pi})(\boldsymbol{\alpha}\cdot\mathbf{E}) = \boldsymbol{\pi}\cdot\mathbf{E} + i\boldsymbol{\Sigma}\cdot(\boldsymbol{\pi}\times\mathbf{E}), \qquad (\boldsymbol{\alpha}\cdot\mathbf{E})(\boldsymbol{\alpha}\cdot\boldsymbol{\pi}) = \mathbf{E}\cdot\boldsymbol{\pi} + i\boldsymbol{\Sigma}\cdot(\mathbf{E}\times\boldsymbol{\pi})$$

Two reorderings follow from $[\pi_i, E_j] = -i\hbar\partial_i E_j$:

$$\boldsymbol{\pi}\cdot\mathbf{E} - \mathbf{E}\cdot\boldsymbol{\pi} = -i\hbar\,\nabla\cdot\mathbf{E}, \qquad \boldsymbol{\pi}\times\mathbf{E} = -\mathbf{E}\times\boldsymbol{\pi} - i\hbar\,\nabla\times\mathbf{E}$$

Hence

$$\left[\boldsymbol{\alpha}\cdot\boldsymbol{\pi},\ \boldsymbol{\alpha}\cdot\mathbf{E}\right] = -i\hbar\,\nabla\cdot\mathbf{E} - 2i\,\boldsymbol{\Sigma}\cdot(\mathbf{E}\times\boldsymbol{\pi}) + \hbar\,\boldsymbol{\Sigma}\cdot(\nabla\times\mathbf{E})$$

Multiplying by $-i\hbar q/8m^2c^2$:

$$-\frac{q\hbar^2}{8m^2c^2}\nabla\cdot\mathbf{E} \;-\; \frac{q\hbar}{4m^2c^2}\boldsymbol{\Sigma}\cdot(\mathbf{E}\times\boldsymbol{\pi}) \;-\; \frac{iq\hbar^2}{8m^2c^2}\boldsymbol{\Sigma}\cdot(\nabla\times\mathbf{E})$$

The last two terms combine into a manifestly Hermitian operator, $-\frac{q\hbar}{8m^2c^2}\boldsymbol{\Sigma}\cdot(\mathbf{E}\times\boldsymbol{\pi}-\boldsymbol{\pi}\times\mathbf{E})$. Neither is Hermitian alone. For static fields, $\nabla\times\mathbf{E}=-c^{-1}\partial_t\mathbf{B}=0$, and the spin–orbit term reduces to its familiar form.

Projection onto the positive-energy block. H_{FW} is block diagonal. Its $\beta=+1$ block acts on two-component spinors, which are the large components in the FW basis, with $\boldsymbol{\Sigma}\to\boldsymbol{\sigma}$. The result is

$$\boxed{H_{FW}^{(+)}=mc^2+\frac{\boldsymbol{\pi}^2}{2m}+q\Phi\underbrace{-\frac{q\hbar}{2mc}\boldsymbol{\sigma}\cdot\mathbf{B}}_{\text{Zeeman}}\underbrace{-\frac{\boldsymbol{\pi}^4}{8m^3c^2}}_{\text{mass}}\underbrace{-\frac{q\hbar}{8m^2c^2}\boldsymbol{\sigma}\cdot(\mathbf{E}\times\boldsymbol{\pi}-\boldsymbol{\pi}\times\mathbf{E})}_{\text{spin–orbit}}\underbrace{-\frac{q\hbar^2}{8m^2c^2}\nabla\cdot\mathbf{E}}_{\text{Darwin}}}$$

In terms of the potential energy. For a static electric field write $V=q\Phi$, so that $q\mathbf{E}=-\nabla V$. The two electric terms become **independent of the sign of** q:

$$H_{SO}=\frac{\hbar}{4m^2c^2}\,\boldsymbol{\sigma}\cdot(\nabla V\times\mathbf{p})\xrightarrow{V=V(r)}\frac{1}{2m^2c^2}\,\frac{1}{r}\frac{dV}{dr}\,\mathbf{L}\cdot\mathbf{S},\qquad H_D=\frac{\hbar^2}{8m^2c^2}\,\nabla^2V$$

Here $\mathbf{S}=\hbar\boldsymbol{\sigma}/2$, and for a central field $\nabla V=(V'/r)\,\mathbf{r}$, so $\nabla V\times\mathbf{p}=(V'/r)\,\mathbf{L}$. The central-field form is the one used in Ch. 5 §5.5.

For the electron ($q=-e$): the Zeeman term is $+\frac{e\hbar}{2mc}\boldsymbol{\sigma}\cdot\mathbf{B}$, the spin–orbit term is $+\frac{e\hbar}{4m^2c^2}\boldsymbol{\sigma}\cdot(\mathbf{E}\times\mathbf{p})$ (static field), and the Darwin term is $+\frac{e\hbar^2}{8m^2c^2}\nabla\cdot\mathbf{E}$. In the Coulomb field of a nucleus, $V=-Ze^2/r$, so $V'>0$ and $\nabla^2V=4\pi Ze^2\,\delta^3(\mathbf{r})$. Both H_{SO} and H_D are then positive operators on the relevant states.

> **[DERIVATION]** Given the truncated FW Hamiltonian of §B.4.3, the five terms above follow by exact operator algebra, with no further approximation apart from dropping the $\mathbf{B}$-dependent parts of $\mathcal{O}^4$. **Convention check against Ch. 3 §3.5.3:** the Zeeman and spin–orbit terms there match the boxed expression with $q\to e$. The Darwin term as printed there carries the opposite sign, $+e\hbar^2\nabla\cdot\mathbf{E}/8m^2c^2$, which is the electron value with $e>0$. The V form above is free of this ambiguity.

B.6 — The Pauli Hamiltonian, the Zeeman Term, and $g=2$

Truncation. Drop the rest energy and the three terms of relative order $(v/c)^2$:

$$H_{\text{Pauli}}=\frac{\boldsymbol{\pi}^2}{2m}+q\Phi-\frac{q\hbar}{2mc}\boldsymbol{\sigma}\cdot\mathbf{B}=\frac{(\boldsymbol{\sigma}\cdot\boldsymbol{\pi})^2}{2m}+q\Phi$$

> **[APPROXIMATION]** Pauli from FW: discard terms of order $mc^2(v/c)^4$, that is, relative order $(v/c)^2\sim(Z\alpha)^2$ for bound states. Valid when fine structure need not be resolved (Ch. 3 §3.6 Ledger).

Orbital and spin moments. Expand $\boldsymbol{\pi}^2=p^2-\frac{q}{c}(\mathbf{p}\cdot\mathbf{A}+\mathbf{A}\cdot\mathbf{p})+\frac{q^2}{c^2}A^2$, with $\mathbf{p}\cdot\mathbf{A}=\mathbf{A}\cdot\mathbf{p}-i\hbar\nabla\cdot\mathbf{A}$. For a uniform field, choose $\mathbf{A}=\frac{1}{2}\mathbf{B}\times\mathbf{r}$. It satisfies $\nabla\cdot\mathbf{A}=0$, and $\mathbf{A}\cdot\mathbf{p}=\frac{1}{2}(\mathbf{B}\times\mathbf{r})\cdot\mathbf{p}=\frac{1}{2}\mathbf{B}\cdot\mathbf{L}$. Then

$$H_{\text{Pauli}}=\frac{p^2}{2m}+q\Phi-\frac{q}{2mc}\,(\mathbf{L}+2\mathbf{S})\cdot\mathbf{B}+\frac{q^2}{8mc^2}(\mathbf{B}\times\mathbf{r})^2$$

The magnetic moment operator is therefore $\boldsymbol{\mu} = \frac{q}{2mc}(g_L \mathbf{L} + g_S \mathbf{S})$ with $g_L = 1$ and $g_S = 2$. For the electron this is $+\frac{\mu_B}{\hbar}(\mathbf{L} + 2\mathbf{S}) \cdot \mathbf{B}$, with $\mu_B = e\hbar/2mc$ (Gaussian), which is $e\hbar/2m = 9.274 \times 10^{-24}$ J/T in SI. That is the Hamiltonian of Ch. 5 §5.6.1. The last term is the diamagnetic term. Relative to the Zeeman term it is of order B/B_{at}, with $B_{\text{at}} = \hbar/(ea_0^2) \approx 2.35 \times 10^5$ T in SI, so it is negligible for atoms in laboratory fields.

Where $g = 2$ comes from. The factor 2 is the coefficient in the identity $(\boldsymbol{\sigma} \cdot \boldsymbol{\pi})^2 = \boldsymbol{\pi}^2 - (q\hbar/c)\boldsymbol{\sigma} \cdot \mathbf{B}$. The FW kinetic term is $\mathcal{O}^2/2mc^2 = (\boldsymbol{\alpha} \cdot \boldsymbol{\pi})^2/2m$: the Dirac Hamiltonian is linear in $\boldsymbol{\pi}$, so its square necessarily produces the spin term with this coefficient. Three qualifications prevent an overclaim.

1. **Minimal coupling is an input.** A Pauli term $\propto \bar{\psi}\sigma^{\mu\nu}F_{\mu\nu}\psi$ is Lorentz and gauge invariant and would add an arbitrary anomalous moment. It is excluded from the fundamental Lagrangian because it has mass dimension five and is non-renormalizable. So $g = 2$ is the prediction of the **minimally coupled** Dirac equation, not of the Clifford algebra alone (compare Ch. 3 §3.4.2).
2. **It is not specifically relativistic.** Lévy-Leblond (1967) showed that a Galilean-invariant first-order linearization of the Schrödinger equation, minimally coupled, also gives $g = 2$. What forces $g = 2$ is linearity in $\boldsymbol{\pi}$, and the Dirac equation is the relativistic instance of that structure.
3. **QED corrects it.** Radiative corrections give $a_e = (g-2)/2 = \alpha/2\pi + \mathcal{O}(\alpha^2)$ (Schwinger, 1948), so that $g_e \approx 2.00231930$. This was discarded in Ch. 3 §3.1.2.

> **[DERIVATION]** Within the minimally coupled Dirac theory, $g_S = 2$ and $g_L = 1$ follow exactly from the operator identity of §B.2. The **[APPROXIMATION]** is only the truncation of the FW series that brings the Hamiltonian into Pauli form.

B.7 — The Three $(v/c)^2$ Terms

Relativistic mass correction. For the free particle (§B.4.4), $-p^4/8m^3c^2$ is the second term of $\sqrt{c^2p^2 + m^2c^4}$. It is the relativistic kinetic-energy correction. In a field, $\mathbf{p} \to \boldsymbol{\pi}$.

Spin–orbit coupling and the Thomas factor. The following semiclassical estimate serves as a check. In the particle's instantaneous rest frame there is, to first order in v/c, a magnetic field $\mathbf{B}' = -\frac{1}{c}\mathbf{v} \times \mathbf{E}$. A moment $\boldsymbol{\mu} = g\frac{q}{2mc}\mathbf{S}$ in that field has energy

$$-\boldsymbol{\mu} \cdot \mathbf{B}' = -\frac{gq}{2m^2c^2}\,\mathbf{S} \cdot (\mathbf{E} \times \mathbf{p})$$

With $g = 2$ this is **twice** the FW result $-\frac{q}{2m^2c^2}\mathbf{S} \cdot (\mathbf{E} \times \mathbf{p})$ (from §B.5 with $\hbar\boldsymbol{\sigma} = 2\mathbf{S}$). The discrepancy is the **Thomas precession**. The rest frame of an accelerated particle rotates relative to the lab at $\boldsymbol{\omega}_T \approx \frac{1}{2c^2}\mathbf{a} \times \mathbf{v}$. Transforming the spin energy to the lab frame replaces g by $g - 1$, so that $H_{SO} = -\frac{(g-1)q}{2m^2c^2}\mathbf{S} \cdot (\mathbf{E} \times \mathbf{p})$. See Jackson, *Classical Electrodynamics*, Ch. 11. For $g = 2$ the factor is 1, which is exactly the FW coefficient.

> **[STRUCTURAL CONNECTION]** The FW derivation and the semiclassical rest-frame-plus-Thomas argument are independent calculations that agree. In the FW derivation the factor ½ is not inserted. It is the coefficient $-1/8M^2$ of the double commutator, doubled by the two orderings in $\boldsymbol{\Sigma} \cdot (\mathbf{E} \times \boldsymbol{\pi} - \boldsymbol{\pi} \times \mathbf{E})$. The semiclassical argument shows **what that factor means**.

Darwin term. $H_D = \frac{\hbar^2}{8m^2c^2}\nabla^2 V$ has no nonrelativistic analogue and acts only where $\nabla^2 V \neq 0$. For a point Coulomb source that is the origin, so it shifts only s states.

> **[ANALOGY]** Here is a picture, not a derivation. If the electron's effective position fluctuates isotropically by $\delta\mathbf{r}$ with $\langle \delta r^2 \rangle$ of order the squared reduced Compton wavelength $(\hbar/mc)^2$, then $\langle V(\mathbf{r}+\delta\mathbf{r})\rangle \approx V + \frac{1}{6}\langle \delta r^2\rangle \nabla^2 V$. This matches the Darwin form if $\langle \delta r^2\rangle = \frac{3}{4}(\hbar/mc)^2$. The coefficient is not fixed by the picture. The precise statement is that the FW-basis position operator is a nonlocal transform of the Dirac position operator, smeared over $\sim \hbar/mc$ (the Newton–Wigner position).

B.8 — Hydrogen Fine Structure to Order α^4

B.8.1 — Setup

Assumptions.

- (H1) A point nucleus of charge Ze and infinite mass (no recoil and no reduced mass), so $V = -Ze^2/r$.
- (H2) $\mathbf{B} = 0$, and the fields are static.
- (H3) No hyperfine structure and no QED corrections.
- (H4) $(Z\alpha)^2 \ll 1$.

The Hamiltonian is $H = H_0 + H_1$, with $H_0 = p^2/2m + V$ and $E_n^{(0)} = -mc^2(Z\alpha)^2/2n^2$ (Ch. 4 §4.7.3 with $Z = 1$), and

$$H_1 = H_{\text{rel}} + H_{SO} + H_D = -\frac{p^4}{8m^3c^2} + \frac{Ze^2}{2m^2c^2}\frac{1}{r^3}\,\mathbf{L}\cdot\mathbf{S} + \frac{\pi\hbar^2 Ze^2}{2m^2c^2}\,\delta^3(\mathbf{r})$$

Why first-order perturbation theory in the $|n\,\ell\,j\,m_j\rangle$ basis is legitimate. The level n is $2n^2$-fold degenerate, so degenerate perturbation theory is required in principle. H_{rel} and H_D are spin independent and rotationally scalar. $H_{SO} \propto \xi(r)\,\mathbf{L}\cdot\mathbf{S}$ commutes with $\mathbf{L}^2$, $\mathbf{S}^2$, $\mathbf{J}^2$ and J_z. All three are parity even. Within the shell, states with equal (j, m_j) but different $\ell = j \pm \frac{1}{2}$ have opposite parity and cannot mix. So H_1 is already diagonal in $|n\,\ell\,j\,m_j\rangle$, and the first-order shifts are its diagonal expectation values. They are of order $(Z\alpha)^2 E_n^{(0)}$. Second-order shifts would be of order $(Z\alpha)^4 E_n^{(0)}$, which is the same order as the FW terms already discarded, so including them without the next FW order would be inconsistent.

B.8.2 — Radial expectation values

With $a \equiv a_0/Z$:

$$\left\langle \frac{1}{r} \right\rangle = \frac{1}{n^2 a}, \qquad \left\langle \frac{1}{r^2} \right\rangle = \frac{1}{n^3(\ell+\frac{1}{2})a^2}, \qquad \left\langle \frac{1}{r^3} \right\rangle = \frac{1}{n^3\,\ell(\ell+\frac{1}{2})(\ell+1)\,a^3} \;\; (\ell \geq 1), \qquad |\psi_{n00}(0)|^2 = \frac{1}{\pi n^3 a}$$

Proofs. (i) **Virial theorem.** For a $1/r$ potential, $\langle V\rangle = 2E_n^{(0)}$, which gives $\langle 1/r\rangle$. (ii) **Feynman–Hellmann in ℓ.** The radial Hamiltonian is $H_\ell = p_r^2/2m + \hbar^2\ell(\ell+1)/2mr^2 - Ze^2/r$, with $E = -Ze^2/(2a\,n^2)$ and $n = n_r + \ell + 1$. Treat ℓ as a continuous parameter at fixed n_r. Then $\partial E/\partial \ell = Ze^2/(a\,n^3)$ and $\partial H_\ell/\partial \ell = \hbar^2(2\ell+1)/2mr^2$. Equating them and using $me^2/\hbar^2 = 1/a_0$ gives $\langle 1/r^2\rangle$. (iii) **Radial commutator.** $\langle [H_\ell, p_r]\rangle = 0$ in a bound state implies $\langle dV_{\text{eff}}/dr\rangle = 0$ for $\ell \geq 1$, that is $\hbar^2\ell(\ell+$

$1)\langle r^{-3}\rangle/m = Ze^2\langle r^{-2}\rangle$, which gives $\langle 1/r^3\rangle$. For $\ell = 0$ the same identity acquires a boundary term at $r = 0$. It becomes $|\psi(0)|^2 = \frac{m}{2\pi\hbar^2}\langle dV/dr\rangle$, which with (ii) gives the last formula. A standard reference is Bethe & Salpeter, *Quantum Mechanics of One- and Two-Electron Atoms*.

It is convenient to note $\left(E_n^{(0)}\right)^2/mc^2 = mc^2(Z\alpha)^4/4n^4$.

B.8.3 — The three shifts

Mass correction. Write $p^4/4m^2 = (H_0 - V)^2$. The expectation value is computed as $\langle p^2\psi \,|\, p^2\psi\rangle$, using the hermiticity of p^2. That avoids the δ-function subtleties of applying p^4 to s states.

$$\Delta E_{\text{rel}} = -\frac{1}{2mc^2}\left[(E_n^{(0)})^2 - 2E_n^{(0)}\langle V\rangle + \langle V^2\rangle\right] = -\frac{(E_n^{(0)})^2}{2mc^2}\left[\frac{4n}{\ell+\frac{1}{2}} - 3\right]$$

This uses $\langle V\rangle = 2E_n^{(0)}$ and $\langle V^2\rangle = Z^2e^4\langle r^{-2}\rangle = 4(E_n^{(0)})^2\, n/(\ell+\frac{1}{2})$.

Spin–orbit ($\ell \geq 1$). $\langle \mathbf{L}\cdot\mathbf{S}\rangle = \frac{\hbar^2}{2}\left[j(j+1) - \ell(\ell+1) - \frac{3}{4}\right]$, which equals $\frac{\hbar^2}{2}\ell$ for $j = \ell + \frac{1}{2}$ and $-\frac{\hbar^2}{2}(\ell+1)$ for $j = \ell - \frac{1}{2}$. With $\hbar^2 = me^2a_0$:

$$\Delta E_{SO} = \frac{(E_n^{(0)})^2}{2mc^2}\cdot\frac{2n\left[j(j+1) - \ell(\ell+1) - \frac{3}{4}\right]}{\ell(\ell+\frac{1}{2})(\ell+1)} = \frac{(E_n^{(0)})^2}{2mc^2}\times\begin{cases}\dfrac{2n}{(\ell+\frac{1}{2})(\ell+1)} & j = \ell+\frac{1}{2}\\[2ex] -\dfrac{2n}{\ell(\ell+\frac{1}{2})} & j = \ell-\frac{1}{2}\end{cases}$$

For $\ell = 0$, $\mathbf{L}\cdot\mathbf{S} = 0$ exactly, while $\langle r^{-3}\rangle$ diverges. The product is assigned the value 0. The Darwin term then supplies the s-state shift.

Darwin ($\ell = 0$ only):

$$\Delta E_D = \frac{\pi\hbar^2 Ze^2}{2m^2c^2}|\psi_{n00}(0)|^2 = \frac{(E_n^{(0)})^2}{2mc^2}\cdot 4n\;\delta_{\ell 0}$$

B.8.4 — The sum

For $j = \ell + \frac{1}{2}$:

$$\frac{4n}{\ell+\frac{1}{2}}\left(1 - \frac{1}{2(\ell+1)}\right) = \frac{4n}{\ell+\frac{1}{2}}\cdot\frac{2\ell+1}{2(\ell+1)} = \frac{4n}{\ell+1} = \frac{4n}{j+\frac{1}{2}}$$

For $j = \ell - \frac{1}{2}$:

$$\frac{4n}{\ell+\frac{1}{2}}\left(1 + \frac{1}{2\ell}\right) = \frac{4n}{\ell} = \frac{4n}{j+\frac{1}{2}}$$

For $\ell = 0$ the Darwin term contributes $+4n$ and ΔE_{rel} contributes $-(8n-3)$, so the bracket is $3 - 4n = 3 - 4n/(j+\frac{1}{2})$ with $j = \frac{1}{2}$. The Darwin shift has exactly the value that the $j = \ell + \frac{1}{2}$ spin–orbit formula would give if continued to $\ell = 0$. All cases collapse to

$$\boxed{\Delta E_{nj} = \frac{(E_n^{(0)})^2}{2mc^2}\left[3 - \frac{4n}{j+\frac{1}{2}}\right] = -\frac{mc^2(Z\alpha)^4}{2n^4}\left(\frac{n}{j+\frac{1}{2}} - \frac{3}{4}\right)}$$

The shift depends on n and j only, not on ℓ. So $2S_{1/2}$ and $2P_{1/2}$ remain degenerate.

> **[DERIVATION]** The fine-structure formula follows by first-order perturbation theory from the truncated FW Hamiltonian under (H1)–(H4). It agrees with the α^4 term of the exact Dirac energy (§B.9), which is the check stated in Ch. 3 §3.4.4.

Numbers, hydrogen ($Z = 1$). $mc^2\alpha^4 = 1.449 \times 10^{-3}$ eV.

- Ground state: $\Delta E_{1,1/2} = -mc^2\alpha^4/8 = -1.81 \times 10^{-4}$ eV.
- $n = 2$: $\Delta E_{2,1/2} = -5mc^2\alpha^4/128$ and $\Delta E_{2,3/2} = -mc^2\alpha^4/128$, so the $2P_{3/2}$–$2P_{1/2}$ splitting is $mc^2\alpha^4/32 = 4.53 \times 10^{-5}$ eV ≈ 10.95 GHz. This is the value quoted in Ch. 5 §5.5.4. The measured interval is ≈ 10.97 GHz. The $\approx 0.2\%$ residual is dominated by QED, chiefly the anomalous moment, which scales the spin–orbit term by $\approx 1 + 2a_e$. The $2S_{1/2}$–$2P_{1/2}$ Lamb shift, ≈ 1.06 GHz, is pure QED and is absent here.

Where it fails. For large Z the neglected $(Z\alpha)^6$ terms grow (§B.9). Recoil enters at relative order m/M_N, and the leading part is captured by using the reduced mass in $E_n^{(0)}$. Hyperfine structure enters at order $\alpha^4(m/m_p)g_p$. Finite nuclear size modifies the s-state Darwin term. In many-electron atoms, V is a screened potential and the shifts must be recomputed with it (Ch. 5 §5.5).

B.9 — The Exact Dirac–Coulomb Energy and Its Expansion

Separation. With $V(r) = -Ze^2/r = -Z\alpha\hbar c/r$, the conserved quantities are $\mathbf{J}^2$, J_z, parity, and Dirac's operator $K = \beta(\boldsymbol{\Sigma}\cdot\mathbf{L} + \hbar)$. The eigenvalue of K is $-\hbar\kappa$ with $\kappa = \mp(j + \tfrac{1}{2})$ for $j = \ell \pm \tfrac{1}{2}$. Write $\psi = r^{-1}\big(G(r)\,\Omega_{\kappa m},\ iF(r)\,\Omega_{-\kappa m}\big)$, with spin–angular functions Ω. The radial equations are

$$\frac{dG}{dr} + \frac{\kappa}{r}G - \frac{E + mc^2 - V}{\hbar c}F = 0, \qquad \frac{dF}{dr} - \frac{\kappa}{r}F + \frac{E - mc^2 - V}{\hbar c}G = 0$$

Sign conventions for κ differ between texts. These equations are consistent with the stated κ: for $E \approx mc^2$, eliminating $F \approx \frac{\hbar}{2mc}(G' + \kappa G/r)$ gives $G'' - \frac{\kappa(\kappa+1)}{r^2}G + \frac{2m}{\hbar^2}(E - mc^2 - V)G = 0$. Since $\kappa(\kappa+1) = \ell(\ell+1)$ in both cases, this is the Schrödinger radial equation of Ch. 4 §4.7.

Behaviour at the origin. For $r \to 0$, $(E \pm mc^2 - V)/\hbar c \to Z\alpha/r$. Try $G \sim a\,r^s$ and $F \sim b\,r^s$:

$$(s + \kappa)\,a - Z\alpha\,b = 0, \qquad Z\alpha\,a + (s - \kappa)\,b = 0 \quad\Longrightarrow\quad s^2 = \kappa^2 - (Z\alpha)^2$$

The regular solution has $s = \sqrt{\kappa^2 - (Z\alpha)^2}$. It is real only for $Z\alpha < |\kappa|$.

Quantization. At large r the solutions decay as $e^{-\lambda r}$, with $\hbar c\lambda = \sqrt{m^2c^4 - E^2}$. Writing G and F as $r^s e^{-\lambda r}$ times power series, normalizability requires both series to terminate. That condition yields

$$E_{n_r\kappa} = mc^2\left[1 + \frac{(Z\alpha)^2}{\Big(n_r + \sqrt{\kappa^2 - (Z\alpha)^2}\Big)^2}\right]^{-1/2}, \qquad n_r = 0, 1, 2, \ldots$$

with $n_r = 0$ allowed only for $\kappa < 0$. The termination argument, a confluent-hypergeometric analysis, is lengthy and is **not reproduced here**. See Bjorken & Drell, *Relativistic Quantum Mechanics*; Berestetskii, Lifshitz & Pitaevskii, *Quantum Electrodynamics*; or Sakurai, *Advanced Quantum Mechanics*. With $n = n_r + |\kappa|$ and $|\kappa| = j + \tfrac{1}{2}$, this is the formula of Ch. 3 §3.4.4.

Expansion. Let $x = Z\alpha$ and $k = j + \frac{1}{2}$. Then $\sqrt{k^2 - x^2} = k - \frac{x^2}{2k} + \mathcal{O}(x^4)$, so $n_r + \sqrt{k^2 - x^2} = n - \frac{x^2}{2k} + \mathcal{O}(x^4)$, and

$$y \equiv \frac{x^2}{(n - x^2/2k)^2} = \frac{x^2}{n^2} + \frac{x^4}{n^3 k} + \mathcal{O}(x^6), \qquad (1+y)^{-1/2} = 1 - \frac{y}{2} + \frac{3y^2}{8} + \mathcal{O}(y^3)$$

$$\boxed{E_{nj} = mc^2 \left[1 - \frac{(Z\alpha)^2}{2n^2} - \frac{(Z\alpha)^4}{2n^4} \left(\frac{n}{j + \frac{1}{2}} - \frac{3}{4} \right) + \mathcal{O}\big((Z\alpha)^6\big) \right]}$$

The second term is the Bohr energy. The third is **identical** to the FW perturbative result of §B.8.4.

> **[DERIVATION]** The α^4 fine structure obtained from the truncated FW Hamiltonian agrees with the expansion of the exact Dirac eigenvalue. Two independent routes, an exact solution and a systematic expansion, give the same coefficient. Agreement at this order does not validate the FW series at higher orders, where it must be handled with care (§B.11).

Where the expansion fails: a heavy-atom example. For the $1s$ state, $E = mc^2\sqrt{1 - (Z\alpha)^2}$, so the binding energy is $mc^2\big(1 - \sqrt{1 - x^2}\big) = mc^2\big(\frac{x^2}{2} + \frac{x^4}{8} + \frac{x^6}{16} + \dots\big)$. For $Z = 82$, $x = 0.598$. The exact point-nucleus value is $0.1988\, mc^2 \approx 101.6$ keV, while the series truncated at x^4 gives $0.1951\, mc^2 \approx 99.7$ keV. The error is about 2%, and the x^6 term already accounts for most of it. As $Z\alpha \to 1$, $s \to 0$ for $|\kappa| = 1$ and the point-nucleus solution ceases to exist. With a finite nucleus the $1s$ level reaches $E = -mc^2$ near $Z \approx 170$. Beyond that point the vacuum becomes unstable to pair creation, and single-particle theory is meaningless (Ch. 3 §3.8).

B.10 — The Gordon Decomposition and $g = 2$

Assumptions. Natural units $\hbar = c = 1$ for this section, signature $(+,-,-,-)$, $D_\mu = \partial_\mu + iqA_\mu$, and $\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]$, so that $\gamma^\mu\gamma^\nu = \eta^{\mu\nu} - i\sigma^{\mu\nu}$ and $\gamma^\nu\gamma^\mu = \eta^{\mu\nu} + i\sigma^{\mu\nu}$. The field ψ satisfies $i\gamma^\nu D_\nu \psi = m\psi$, and its adjoint satisfies $-i\,(\partial_\nu - iqA_\nu)\bar\psi\,\gamma^\nu = m\bar\psi$.

Derivation. Write $\bar\psi\gamma^\mu\psi$ half with $\psi = m^{-1} i\gamma^\nu D_\nu \psi$ and half with the adjoint equation:

$$\bar\psi\gamma^\mu\psi = \frac{1}{2m}\Big[\bar\psi\gamma^\mu\gamma^\nu(iD_\nu\psi) + \big(-i\overline{D_\nu\psi}\big)\gamma^\nu\gamma^\mu\psi\Big]$$

Insert the two product formulas. The $\eta^{\mu\nu}$ pieces give $\frac{i}{2m}\big[\bar\psi D^\mu\psi - \overline{D^\mu\psi}\,\psi\big]$. The $\sigma^{\mu\nu}$ pieces give $\frac{1}{2m}\big[\bar\psi\sigma^{\mu\nu}D_\nu\psi + \overline{D_\nu\psi}\,\sigma^{\mu\nu}\psi\big]$. In the second bracket the gauge terms $\pm iqA_\nu$ cancel, leaving a total derivative:

$$\boxed{J^\mu = q\bar\psi\gamma^\mu\psi = \underbrace{\frac{iq}{2m}\Big[\bar\psi D^\mu\psi - \overline{D^\mu\psi}\,\psi\Big]}_{\text{convective}} + \underbrace{\frac{q}{2m}\,\partial_\nu\big(\bar\psi\sigma^{\mu\nu}\psi\big)}_{\text{spin (magnetization/polarization)}}}$$

For plane waves $\psi = u(p)e^{-ip\cdot x}$ and $\bar\psi = \bar u(p')e^{ip'\cdot x}$ with $A = 0$, this becomes the familiar

$$\bar u(p')\gamma^\mu u(p) = \bar u(p')\left[\frac{(p+p')^\mu}{2m} + \frac{i\sigma^{\mu\nu}(p'-p)_\nu}{2m}\right]u(p)$$

Reading off $g = 2$. In the Dirac representation $\sigma^{ij} = \epsilon_{ijk}\Sigma_k$ (for example, $\sigma^{12} = \frac{i}{2}[\gamma^1,\gamma^2] = \Sigma_3$) and $\sigma^{0i} = i\alpha_i$. The spatial part of the spin term is

$$\frac{q}{2m}\partial_j\left(\bar\psi\sigma^{ij}\psi\right)+\frac{q}{2m}\partial_0\left(\bar\psi\sigma^{i0}\psi\right)=\left(\nabla\times\mathbf{M}\right)^i+\partial_t P^i,\qquad \mathbf{M}=\frac{q}{2m}\bar\psi\boldsymbol{\Sigma}\psi$$

The polarization piece $\mathbf{P}$ involves $\bar\psi\boldsymbol{\alpha}\psi$, which mixes large and small components, and is suppressed by v/c. In the nonrelativistic limit, $\bar\psi\boldsymbol{\Sigma}\psi=\phi^\dagger\boldsymbol{\sigma}\phi-\chi^\dagger\boldsymbol{\sigma}\chi\to\phi^\dagger\boldsymbol{\sigma}\phi$. Restoring units, $\mathbf{M}=\frac{q\hbar}{2mc}\,\phi^\dagger\boldsymbol{\sigma}\phi=2\cdot\frac{q}{2mc}\,\phi^\dagger\mathbf{S}\phi$ in Gaussian units. A magnetization current $c\nabla\times\mathbf{M}$ couples to an external field through $-\frac{1}{c}\int\mathbf{J}\cdot\mathbf{A}\,d^3x=-\int\mathbf{M}\cdot\mathbf{B}\,d^3x$ after integrating by parts. That is a spin moment with $g_S=2$. The convective term carries the orbital current, with $g_L=1$.

> **[DERIVATION]** The Gordon decomposition is an exact identity on solutions of the minimally coupled Dirac equation. It shows that the Dirac current already contains a spin-magnetization current with gyromagnetic ratio $2\times q/2mc$. This is the same $g=2$ found in §B.6, obtained here without any FW truncation.

B.11 — Where the FW Series Fails

The FW Hamiltonian of §B.4.3 is a truncated asymptotic expansion. Each item below names the condition under which the truncation, or the expansion itself, breaks down.

1. **Speed.** The expansion parameter is v/c, which is $Z\alpha$ for bound states. The truncation error is of relative order $(Z\alpha)^2$ beyond the retained terms. It is negligible for light atoms, about 2% of the $1s$ binding energy at $Z=82$ (§B.9), and the expansion is useless as $Z\alpha\to1$. Use the exact Dirac equation, or Dirac–Fock and Dirac–Coulomb–Breit methods for many-electron atoms.
2. **Field strength.** The generated odd term $i\hbar qc\,\boldsymbol{\alpha}\cdot\mathbf{E}/2mc^2$ must be small compared with mc^2. This requires $|\mathbf{E}|\ll E_S=m^2c^3/(e\hbar)\approx1.3\times10^{18}$ V/m (SI), the Schwinger field. Similarly, $\hbar|q|B/mc\ll mc^2$ requires $B\ll m^2c^2/(e\hbar)\approx4.4\times10^9$ T (SI).
3. **Field variation.** Potentials must not change by $\gtrsim2mc^2$ over a reduced Compton wavelength $\hbar/mc=3.86\times10^{-13}$ m. Otherwise the upper and lower blocks cannot be separated perturbatively: this is the Klein-paradox regime.
4. **Frequency.** Time-dependent fields must have $\hbar\omega\ll mc^2$. The $\dot{\mathcal{O}}$ terms were expanded on this assumption.
5. **Pair creation.** When any of conditions 2–4 fails, electron–positron pairs can be created, the number of particles is not conserved, and no single-particle Hamiltonian is correct. QED is required (Ch. 3 §3.1.2, §3.8).
6. **Singular potentials.** For a point Coulomb potential, successive FW orders produce increasingly singular operators at $r=0$. The Darwin δ-function is the first. At order $(Z\alpha)^6$, naive term-by-term expectation values for s states are not finite, and the expansion must be reorganized, for example by regularizing with a finite nucleus. The α^4 results of §B.8 are unaffected.
7. **Picture change.** The FW transformation also changes the operators. Position, spin and contact densities in the FW basis differ from the Dirac ones at order $(v/c)^2$. Computing a property with a Dirac-basis operator and FW-basis states (or the reverse) introduces a "picture-change" error of the same order as the effects being computed.
8. **Closed forms are rare.** Exact FW transformations are known for the free particle (§B.4.4) and for a pure magnetic field (Case, 1954), where the result is $\beta\sqrt{m^2c^4+c^2(\boldsymbol{\sigma}\cdot\boldsymbol{\pi})^2}$. For combined electric and magnetic fields only the series exists.

B.12 — Summary

Result	Statement	Assumptions	Used in
Dirac Hamiltonian	$H_D = c\boldsymbol{\alpha}\cdot\boldsymbol{\pi} + \beta mc^2 + q\Phi$, $\boldsymbol{\pi} = \mathbf{p} - q\mathbf{A}/c$	minimal coupling; external classical field	Ch. 3 §3.2–3.3; Ex. 01 Step 1
Clifford algebra	$\{\alpha_i,\alpha_j\} = 2\delta_{ij}$, $\{\alpha_i,\beta\} = 0$, $\beta^2 = 1$; $\{\gamma^\mu,\gamma^\nu\} = 2\eta^{\mu\nu}_{(+---)}$	first-order H squaring to $c^2p^2 + m^2c^4$	Ch. 0 §0.5; Ch. 3 §3.3.1
Spin identity	$(\boldsymbol{\sigma}\cdot\boldsymbol{\pi})^2 = \boldsymbol{\pi}^2 - (q\hbar/c)\boldsymbol{\sigma}\cdot\mathbf{B}$	exact	Ch. 3 §3.4.2, §3.6
Odd operator	$\mathcal{O} = c\boldsymbol{\alpha}\cdot\boldsymbol{\pi}$ only; $\mathbf{E}$ enters via $[\mathcal{O},\mathcal{E}] + i\hbar\dot{\mathcal{O}} = i\hbar qc\,\boldsymbol{\alpha}\cdot\mathbf{E}$	minimal coupling	Ch. 3 §3.5.1; Ex. 01 Step 2
General FW	$\beta(mc^2 + \mathcal{O}^2/2mc^2 - \mathcal{O}^4/8m^3c^6) + \mathcal{E} - [\mathcal{O},[\mathcal{O},\mathcal{E}] + i\hbar\dot{\mathcal{O}}]/8m^2c^4$	truncated at $mc^2(v/c)^4$; §B.11 conditions	Ch. 3 §3.5.2–3.5.3
FW Hamiltonian (EM)	Zeeman, $-\boldsymbol{\pi}^4/8m^3c^2$, spin–orbit, Darwin (§B.5)	as above; static fields for the simple SO form	Ch. 3 §3.5.3; Ex. 01 Step 3
Pauli Hamiltonian	$(\boldsymbol{\sigma}\cdot\boldsymbol{\pi})^2/2m + q\Phi$; $\boldsymbol{\mu} = (q/2mc)(\mathbf{L} + 2\mathbf{S})$	drop $(v/c)^2$ terms	Ch. 3 §3.6; Ch. 5 §5.6; Ex. 01 Steps 4–5
$g = 2$	$g_S = 2$, $g_L = 1$ (FW and Gordon)	minimal coupling; no Pauli term; no QED	Ch. 3 §3.4.2; Ch. 5 §5.6
Spin–orbit	$\frac{1}{2m^2c^2}\frac{1}{r}\frac{dV}{dr}\mathbf{L}\cdot\mathbf{S}$; Thomas factor automatic	static central V	Ch. 5 §5.5
Darwin	$\frac{\hbar^2}{8m^2c^2}\nabla^2 V$; Coulomb: $\frac{\pi\hbar^2 Ze^2}{2m^2c^2}\delta^3(\mathbf{r})$	static V	Ch. 3 §3.5.3
Fine structure	$\Delta E_{nj} = -\frac{mc^2(Z\alpha)^4}{2n^4}\left(\frac{n}{j+\frac{1}{2}} - \frac{3}{4}\right)$	(H1)–(H4); first-order PT	Ch. 3 §3.4.4; Ch. 5 §5.5.4
Exact Dirac–Coulomb	$E = mc^2[1 + (Z\alpha)^2/(n_r + \sqrt{\kappa^2 - (Z\alpha)^2})^2]^{-1/2}$	point nucleus; $Z\alpha < \lvert\kappa\rvert$	Ch. 3 §3.4.4
Expansion check	exact energy $= mc^2[1 - (Z\alpha)^2/2n^2 +$ FW α^4 term $+ \mathcal{O}((Z\alpha)^6)]$	$Z\alpha \ll 1$	Ch. 3 §3.4.4, §3.5.3
Gordon decomposition	J^μ = convective + $\frac{q}{2m}\partial_\nu(\bar\psi\sigma^{\mu\nu}\psi)$	on-shell Dirac fields	Ch. 3 §3.4.2, §3.9.2
FW failure	$Z\alpha \to 1$; $E \gtrsim E_S$; $B \gtrsim 4.4\times10^9$ T; $\Delta V \gtrsim 2mc^2$ over $\hbar/mc$; $\hbar\omega \gtrsim mc^2$	—	Ch. 3 §3.5.3 and §3.8 Ledgers

References (standard texts; no claim depends on a single page): Foldy, L.L. & Wouthuysen, S.A., *Phys. Rev.* 78, 29 (1950) [R9]; Dirac, *Proc. R. Soc. A* 117, 610 (1928) [R7]; Bjorken & Drell, *Relativistic Quantum Mechanics* (1964); Sakurai, *Advanced Quantum Mechanics* (1967); Berestetskii, Lifshitz & Pitaevskii, *Quantum Electrodynamics* (Landau–Lifshitz Vol. 4); Bethe & Salpeter, *Quantum Mechanics of One- and Two-Electron Atoms* (1957); Thaller, *The Dirac Equation* (Springer, 1992); Jackson, *Classical Electrodynamics*, 3rd ed. (Thomas precession); Lévy-Leblond, *Commun. Math. Phys.* 6, 286 (1967); Case, *Phys. Rev.* 95, 1323 (1954); Schwinger, *Phys. Rev.* 73, 416 (1948).

End of Appendix B

Appendix C — From Quantum to Classical: Stationary Phase, WKB, and Ehrenfest

This appendix supplies the derivations behind two of the Bridge B limits. The first is Bridge B.a, the descent $\hbar \to 0$ from quantum to classical mechanics, which Ch. 1 §1.4.3 and §1.8.3–§1.8.4 introduce and Ch. 9 §9.1–§9.3 develops: why the classical path is the one with $\delta S = 0$, in what sense expectation values obey Newton's law, how the WKB wavefunction and its connection formulas are derived, and where the Bohr–Sommerfeld rule and the tunnelling exponent used in Ch. 4 §4.4 and Ch. 6 §6.10.1 come from. The second is Bridge B.d, the weak-field, slow-motion, quasi-static limit of general relativity that Ch. 8 §8.2–§8.4 and §8.8 use to recover Newtonian gravity and the GPS clock correction.

The two limits are grouped together because they have the same logical shape. In each, one dimensionless number is small ($\hbar/S$ in the first case, $|\Phi|/c^2 \sim v^2/c^2$ in the second). The reduced theory is the leading term in an expansion in that number, and the reduction fails exactly where that number stops being small. Neither limit is a theorem about all states or all spacetimes. Each is an asymptotic statement with stated hypotheses, and those hypotheses are written out below.

Conventions follow the chapters. The metric signature is $(-,+,+,+)$ with $x^0 = ct$. Wavefunctions satisfy $i\hbar\,\partial_t\psi = \hat{H}\psi$. The local wavenumber in a classically forbidden region is $\kappa(x) = \sqrt{2m(V-E)}/\hbar$, as in Ch. 4 §4.4.2 and Ch. 9 §9.3.2. Euler–Lagrange equations are derived in Ch. 1 §1.5 and in Appendix A; they are used here, not re-derived.

C.1 — Stationary Phase and the Classical Path

C.1.1 — What is being claimed

Ch. 1 §1.4.3 and Ch. 9 §9.2.2 state that as $\hbar \to 0$ the Feynman propagator is dominated by paths near the one with stationary action. This section separates that claim into a part that is a theorem and a part that is a heuristic.

The theorem is the finite-dimensional method of stationary phase.

The heuristic is applying that method to an integral over infinitely many variables (all paths), for which no Lebesgue measure exists.

The heuristic can be made rigorous in specific settings. Those settings are stated in §C.1.5.

C.1.2 — The one-dimensional stationary-phase lemma

Assumptions.

1. $\phi(x)$ is real and smooth.
2. $g(x)$ is smooth and has compact support (or decays fast enough that the boundary terms below vanish).
3. On the support of g, ϕ has exactly one stationary point x_0, with $\phi'(x_0) = 0$ and $\phi''(x_0) \neq 0$. A stationary point with $\phi''(x_0) \neq 0$ is called *nondegenerate*.

Consider

$$I(\hbar) = \int_{-\infty}^{\infty} g(x)\, e^{i\phi(x)/\hbar}\, dx.$$

Step 1: regions without a stationary point contribute only $O(\hbar^\infty)$. Where $\phi' \neq 0$ we can write $e^{i\phi/\hbar} = \frac{\hbar}{i\phi'}\frac{d}{dx}e^{i\phi/\hbar}$ and integrate by parts:

$$\int g\, e^{i\phi/\hbar}\, dx = -\frac{\hbar}{i}\int \frac{d}{dx}\left(\frac{g}{\phi'}\right) e^{i\phi/\hbar}\, dx.$$

The boundary terms vanish by assumption 2. Each further integration by parts gains another factor of $\hbar$, so the contribution from any region bounded away from x_0 is smaller than every power of $\hbar$. This is the precise form of "rapidly oscillating phases cancel".

Step 2: the neighbourhood of x_0. Near x_0 we have $\phi(x) = \phi(x_0) + \tfrac{1}{2}\phi''(x_0)(x-x_0)^2 + O((x-x_0)^3)$. The phase stays within order one of its stationary value over a window of width $|x - x_0| \sim \sqrt{\hbar/|\phi''(x_0)|}$, and that window shrinks to zero as $\hbar \to 0$. Over it, g can be replaced by $g(x_0)$, and the cubic term changes the phase by only $O(\hbar^{1/2})$. What remains is the Fresnel integral

$$\int_{-\infty}^{\infty} e^{iau^2/2\hbar}\, du = \sqrt{\frac{2\pi\hbar}{|a|}}\, e^{i\pi\,\mathrm{sgn}(a)/4}, \qquad a \neq 0.$$

Result.

$$I(\hbar) = g(x_0)\sqrt{\frac{2\pi\hbar}{|\phi''(x_0)|}}\, \exp\!\left(\frac{i\phi(x_0)}{\hbar} + \frac{i\pi}{4}\,\mathrm{sgn}\,\phi''(x_0)\right)\big[1 + O(\hbar)\big].$$

A full proof, which uses the Morse lemma to make Step 2 exact and controls the error to all orders, is given in Hörmander, *The Analysis of Linear Partial Differential Operators I*, §7.7.

In n dimensions with a nondegenerate stationary point $\mathbf{x}_0$ (Hessian $\phi_{ij} = \partial_i\partial_j\phi$ invertible), the same steps give

$$I(\hbar) \simeq g(\mathbf{x}_0)\, \frac{(2\pi\hbar)^{n/2}}{\sqrt{|\det \phi_{ij}|}}\, e^{i\phi(\mathbf{x}_0)/\hbar}\, e^{i\pi\,\mathrm{sgn}(\phi_{ij})/4},$$

where $\mathrm{sgn}(\phi_{ij})$ is the number of positive eigenvalues of the Hessian minus the number of negative ones.

> `[DERIVATION]` For finitely many integration variables, the dominance of the stationary point is a theorem, with corrections controlled by the small parameter $\hbar/|\phi''|\,\delta x^2$, where δx is the scale over which g and ϕ'' vary.

C.1.3 — The time-sliced path integral

The propagator for $\hat{H} = \hat{p}^2/2m + V(\hat{x})$ in one dimension is defined as the limit of a finite-dimensional integral. Divide $[t_a, t_b]$ into N steps of length $\varepsilon = (t_b - t_a)/N$ and fix $x_0 = x_a$, $x_N = x_b$. Then

$$K(x_b, t_b; x_a, t_a) = \lim_{N\to\infty} \left(\frac{m}{2\pi i\hbar\varepsilon}\right)^{N/2} \int \prod_{k=1}^{N-1} dx_k\; \exp\!\left(\frac{i}{\hbar} S_N[x_0, \ldots, x_N]\right),$$

$$S_N = \sum_{k=0}^{N-1} \varepsilon \left[\frac{m}{2} \left(\frac{x_{k+1} - x_k}{\varepsilon} \right)^2 - V(x_k) \right].$$

For fixed N this is an ordinary $(N-1)$-dimensional oscillatory integral, and §C.1.2 applies to it. Its stationarity condition $\partial S_N / \partial x_k = 0$ is

$$m \, \frac{x_{k+1} - 2x_k + x_{k-1}}{\varepsilon^2} = -V'(x_k), \qquad k = 1, \dots, N-1.$$

This is the finite-difference form of $m\ddot{x} = -V'(x)$. As $\varepsilon \to 0$ it becomes the Euler–Lagrange equation of Ch. 1 §1.5.2 with the fixed endpoints $x(t_a) = x_a$ and $x(t_b) = x_b$.

> [DERIVATION] For each finite N, the stationary point of the discretised action is the discretised classical trajectory.

Expanding the action to second order around the classical path $x_{cl}(t)$ gives a Gaussian integral over the fluctuations. Its value is the **Van Vleck–Pauli–Morette prefactor**:

$$K \simeq \left(\frac{1}{2\pi i\hbar} \left| -\frac{\partial^2 S_{cl}}{\partial x_b \, \partial x_a} \right| \right)^{1/2} e^{i S_{cl}(x_b, t_b; x_a, t_a)/\hbar - i\pi\nu/2},$$

where ν counts the conjugate (focal) points crossed by the classical path. This ν is the Maslov index; the same index reappears at turning points in §C.4. The prefactor is the quantity written A in Ch. 9 §9.2.2.

Check on the free particle. For $V = 0$, $S_{cl} = m(x_b - x_a)^2 / 2T$ with $T = t_b - t_a$, so $-\partial^2 S_{cl}/\partial x_b \partial x_a = m/T$ and the formula gives

$$K_{free} = \sqrt{\frac{m}{2\pi i\hbar T}} \, \exp\!\left(\frac{i m (x_b - x_a)^2}{2\hbar T} \right).$$

This is the *exact* free propagator. The semiclassical formula is exact for every action that is at most quadratic in the path, which includes the free particle, a uniform force, and the harmonic oscillator, because then the expansion to second order around x_{cl} leaves nothing out. This is the same structural reason that Ehrenfest's theorem is exact for these potentials (§C.2.3).

C.1.4 — Where the heuristic step lies

To go from §C.1.3 to "$\hbar \to 0$ selects the classical path", two limits must be exchanged: $N \to \infty$ (which defines the path integral) and $\hbar \to 0$ (which selects the stationary point). Three facts make this exchange more delicate than it looks.

1. **There is no measure on paths.** The factor $(m/2\pi i\hbar\varepsilon)^{N/2}$ diverges as $N \to \infty$. The object "$\mathcal{D}x(t)$" in Ch. 9 §9.2.1 is shorthand for the limiting procedure above, not a measure. (In imaginary time, the Wiener measure does exist; in real time it does not.)
2. **The typical paths are not smooth.** The paths that dominate the time-sliced integral at fixed $\hbar$ are continuous but nowhere differentiable: their increments scale as $|x_{k+1} - x_k| \sim \sqrt{\hbar\varepsilon/m}$. The classical path is an exceptional, smooth path. It dominates only after the oscillatory cancellation has been carried out.

3. **Uniqueness can fail.** There may be several classical paths between the same endpoints, for example in a harmonic oscillator over more than half a period. There may also be degenerate stationary points (caustics, where $\partial^2 S_{cl}/\partial x_b \partial x_a \to \infty$), or no real stationary path at all (tunnelling, §C.6).

What survives in general is therefore not "the particle takes the classical path". It is

$$K \simeq \sum_{\text{classical paths } j} A_j \, e^{iS_j/\hbar - i\pi\nu_j/2},$$

an interfering sum over classical paths, valid away from caustics. Classical mechanics proper, with a single trajectory, also requires the interference between different j to be washed out, either by averaging over initial conditions or by decoherence (Ch. 9 §9.4.2). Stationary phase alone does not provide that step.

C.1.5 — What makes it rigorous

For potentials V that are smooth with bounded derivatives of order two and higher, the time-sliced propagator converges for short times, and the semiclassical expansion around the classical path can be proved with controlled errors. Real-time Feynman integrals can also be defined as infinite-dimensional oscillatory integrals with a stationary-phase theorem of their own. See Albeverio, Høegh-Krohn and Mazzucchi, *Mathematical Theory of Feynman Path Integrals* (Lecture Notes in Mathematics 523, 2nd ed., Springer 2008), and Schulman, *Techniques and Applications of Path Integration* (Wiley 1981) for the physics-level treatment. A separate route to the same physical conclusion, which avoids path integrals altogether, is the coherent-state limit of Hepp, *Commun. Math. Phys.* **35**, 265 (1974): quantum correlation functions in suitably scaled coherent states converge to classical ones as $\hbar \to 0$.

> `[APPROXIMATION]` The small parameter is $\hbar/S_{cl}$, the ratio of $\hbar$ to the action of the relevant classical motion. The selection of the stationary path holds asymptotically in that parameter and fails at caustics, for classically forbidden processes, and at times long enough for the semiclassical sum to break down (§C.2.4).

What was proved. For every finite discretisation, the stationary point of the action is the discretised classical path, and its neighbourhood dominates by a theorem. In the continuum limit, the same conclusion holds asymptotically under the regularity conditions above, in the form of a sum over classical paths with Van Vleck amplitudes and Maslov phases. The conclusion fails where stationary points merge (caustics), where no real stationary point exists (tunnelling), and where interference between several classical paths is not suppressed.

C.2 — Ehrenfest's Theorem

C.2.1 — Proof

Assumptions.

1. $\hat{H}$ is self-adjoint.
2. $\psi(t)$ stays in a domain on which $\hat{H}\hat{A}$ and $\hat{A}\hat{H}$ are defined, so that the expectation values below are finite.
3. The state evolves by $i\hbar\,\partial_t|\psi\rangle = \hat{H}|\psi\rangle$.

Then $\partial_t\langle\psi| = (i/\hbar)\langle\psi|\hat{H}$, because $\hat{H}^\dagger = \hat{H}$. Differentiating $\langle\hat{A}\rangle = \langle\psi|\hat{A}|\psi\rangle$ gives

$$\frac{d\langle\hat{A}\rangle}{dt} = \frac{i}{\hbar}\langle\psi|\hat{H}\hat{A}|\psi\rangle - \frac{i}{\hbar}\langle\psi|\hat{A}\hat{H}|\psi\rangle + \left\langle\frac{\partial\hat{A}}{\partial t}\right\rangle = \frac{i}{\hbar}\langle[\hat{H},\hat{A}]\rangle + \left\langle\frac{\partial\hat{A}}{\partial t}\right\rangle.$$

This is the boxed result of Ch. 9 §9.1.1. Assumption 2 is not decorative. For an unbounded operator, the domain can fail to be preserved, and then the formal identity can fail. Standard examples involve wavefunctions with slowly decaying tails.

For $\hat{H} = \hat{p}^2/2m + V(\hat{x})$, apply the canonical commutator $[\hat{x},\hat{p}] = i\hbar$ and the identity $[f(\hat{x}),\hat{p}] = i\hbar f'(\hat{x})$. This gives $[\hat{H},\hat{x}] = -i\hbar\hat{p}/m$ and $[\hat{H},\hat{p}] = i\hbar V'(\hat{x})$, so

$$\frac{d\langle\hat{x}\rangle}{dt} = \frac{\langle\hat{p}\rangle}{m}, \qquad \frac{d\langle\hat{p}\rangle}{dt} = -\langle V'(\hat{x})\rangle, \qquad m\frac{d^2\langle\hat{x}\rangle}{dt^2} = \langle F(\hat{x})\rangle,$$

with $F = -V'$. These equations hold for *every* state and *every* potential.

> `[DERIVATION]` Ehrenfest's theorem is exact, given the Schrödinger equation and the domain conditions. It is not itself the classical limit.

C.2.2 — The gap between $\langle F(\hat{x})\rangle$ and $F(\langle\hat{x}\rangle)$

Write $\bar{x} = \langle\hat{x}\rangle$, $\delta\hat{x} = \hat{x} - \bar{x}$, and $\sigma_x^2 = \langle\delta\hat{x}^2\rangle$. For F analytic over the region where $|\psi|^2$ is appreciable, expand F about $\bar{x}$. The linear term averages to zero, so

$$\langle F(\hat{x})\rangle = F(\bar{x}) + \tfrac{1}{2}F''(\bar{x})\,\sigma_x^2 + \tfrac{1}{6}F'''(\bar{x})\,\langle\delta\hat{x}^3\rangle + \cdots$$

With $F = -V'$, the leading correction to Newton's law is

$$m\ddot{\bar{x}} = -V'(\bar{x}) - \tfrac{1}{2}V'''(\bar{x})\,\sigma_x^2 + \cdots, \qquad \text{equivalently} \qquad \langle V'\rangle = V'(\bar{x}) + \tfrac{1}{2}\sigma_x^2 V'''(\bar{x}) + \cdots$$

Note the sign: the correction to $\langle\nabla V\rangle$ is $+\frac{1}{2}\sigma_x^2\,\nabla^3 V$. The centroid obeys Newton's law with the force evaluated at the centroid provided

$$\sigma_x^2\,|V'''(\bar{x})| \ll |V'(\bar{x})| \qquad\Longleftrightarrow\qquad \sigma_x \ll L_V \equiv \sqrt{|V'/V'''|}.$$

This is the precise version of $\Delta x \ll L_V$ in Ch. 9 §9.1.3.

> `[APPROXIMATION]` The small parameter is σ_x/L_V, the width of the wavepacket divided by the length over which the force changes.

C.2.3 — When the gap is exactly zero, and when it is large

Exact cases. The equality $\langle F(\hat{x})\rangle = F(\langle\hat{x}\rangle)$ holds for every state if and only if F is affine, $F(x) = a + bx$, which means V is at most quadratic. The free particle, the uniform field, and the harmonic oscillator are the three examples. For them, the centroid of *any* wavefunction (including a cat state or a highly excited eigenstate) follows the classical trajectory exactly.

Following the classical trajectory does not make a state classical. A harmonic-oscillator energy eigenstate has $\langle \hat{x} \rangle = 0$ at all times, which is a correct classical trajectory (rest at the minimum), but the state is nothing like a classical particle at rest.

A failure case. In a symmetric double well, a packet that splits at the barrier into transmitted and reflected parts can have $\bar{x}$ near the barrier top, where no classical particle with that energy could be. There σ_x is comparable to the well separation, the expansion of §C.2.2 does not converge usefully, and $F(\bar{x})$ has nothing to do with $\langle F \rangle$.

Ehrenfest is necessary but not sufficient. A classical description requires the whole distribution to evolve classically, not only its first moment. The Wigner function $W(x, p, t)$ obeys the Moyal equation

$$\frac{\partial W}{\partial t} = -\frac{p}{m}\frac{\partial W}{\partial x} + V'(x)\frac{\partial W}{\partial p} - \frac{\hbar^2}{24}V'''(x)\frac{\partial^3 W}{\partial p^3} + O(\hbar^4).$$

Its first two terms are the classical Liouville equation (Ch. 9 §9.6.2).

> `[STRUCTURAL CONNECTION]` Quantum and classical phase-space dynamics differ by terms of order $\hbar^2 V'''$. These vanish identically for potentials that are at most quadratic, which is the same condition as the exact cases of Ehrenfest's theorem above.

C.2.4 — How long the classical description lasts

Even when $\sigma_x \ll L_V$ initially, wavepackets spread. For a free Gaussian of initial minimum-uncertainty width σ_0,

$$\sigma_x^2(t) = \sigma_0^2 + \left(\frac{\hbar t}{2m\sigma_0}\right)^2.$$

The Ehrenfest condition therefore holds only up to a time at which $\sigma_x(t)$ reaches L_V.

- In a regular (integrable) system, spreading is typically algebraic in t, and this time is long.
- In a chaotic system, the width grows like $e^{\lambda t}$ with Lyapunov exponent λ (Ch. 9 §9.10.2). The standard semiclassical estimate is then $t_E \sim \lambda^{-1}\ln(S/\hbar)$, where S is a typical action of the motion. This is only logarithmically long in $1/\hbar$; see Haake, *Quantum Signatures of Chaos* (Springer).

For macroscopic bodies the observed classicality on all practical time scales does not come from t_E alone. It comes from decoherence by the environment (Ch. 9 §9.4.2), which suppresses the interference terms that would otherwise make the evolution non-classical.

What was proved. The Ehrenfest equations hold exactly. Newton's law for the centroid follows when the packet is narrow compared with L_V, with leading error $\frac{1}{2}V'''\sigma_x^2$. The result is exact for potentials that are at most quadratic, and it fails when the packet splits or spreads to L_V, which happens on the Ehrenfest time t_E.

C.3 — The WKB Approximation to First Order

C.3.1 — The exact Riccati form

Assumptions.

1. The problem is one-dimensional and time-independent: $-\frac{\hbar^2}{2m}\psi'' + V(x)\psi = E\psi$.
2. V is smooth.

No approximation is made in this subsection. Write $\psi = e^{i\sigma(x)/\hbar}$ with σ complex. Then $\psi' = (i\sigma'/\hbar)\psi$ and $\psi'' = \left(i\sigma''/\hbar - \sigma'^2/\hbar^2\right)\psi$. The Schrödinger equation becomes, exactly,

$$\frac{\sigma'^2}{2m} - \frac{i\hbar}{2m}\sigma'' + V(x) - E = 0.$$

C.3.2 — Expansion in $\hbar$

Expand $\sigma = \sigma_0 + \hbar\sigma_1 + \hbar^2\sigma_2 + \cdots$ and collect powers of $\hbar$.

Order $\hbar^0$.

$$\sigma_0'^2 = 2m(E - V) \equiv p(x)^2 \quad \Longrightarrow \quad \sigma_0 = \pm\int^x p(x')\,dx'.$$

This is the time-independent Hamilton–Jacobi equation (Ch. 1 §1.8.2), with σ_0 playing the role of Hamilton's characteristic function.

Order $\hbar^1$.

$$2\sigma_0'\sigma_1' - i\sigma_0'' = 0 \quad \Longrightarrow \quad \sigma_1' = \frac{i}{2}\frac{\sigma_0''}{\sigma_0'} = \frac{i}{2}\frac{d}{dx}\ln p \quad \Longrightarrow \quad e^{i\sigma_1} = p^{-1/2}.$$

Keeping these two orders gives the WKB wavefunction of Ch. 1 §1.8.4 and Ch. 9 §9.3.2:

$$\psi_{WKB}(x) = \frac{C_+}{\sqrt{p(x)}}\,e^{\frac{i}{\hbar}\int^x p\,dx'} + \frac{C_-}{\sqrt{p(x)}}\,e^{-\frac{i}{\hbar}\int^x p\,dx'} \qquad (E > V),$$

$$\psi_{WKB}(x) = \frac{D_+}{\sqrt{\kappa(x)}}\,e^{\int^x \kappa\,dx'} + \frac{D_-}{\sqrt{\kappa(x)}}\,e^{-\int^x \kappa\,dx'} \qquad (E < V), \qquad \kappa = \frac{\sqrt{2m(V-E)}}{\hbar}.$$

In the forbidden region $p = i\hbar\kappa$, and the factor $|p|^{-1/2}$ is written $\kappa^{-1/2}$ up to a constant that is absorbed into $D_\pm$.

Two consistency checks.

1. The probability current of each travelling component is $j = |C_\pm|^2(\pm 1/m)$, independent of x. This is the time-independent form of the order-$\hbar$ continuity equation of Ch. 1 §1.8.3, $(A^2\sigma_0')' = 0$, which forces $A \propto p^{-1/2}$.
2. The density $|\psi|^2 \propto 1/p \propto 1/v$ is proportional to the time a classical particle spends in dx. This is the semiclassical correspondence for densities.

> [DERIVATION] Given the expansion, the WKB form follows at first order in $\hbar$. Whether the expansion is

accurate is a separate question, answered in §C.3.3.

C.3.3 — The validity criterion

Truncating after σ_1 is justified when each successive term in the expansion is small compared with the one before. The first such comparison is

$$\left|\frac{\hbar\sigma_1'}{\sigma_0'}\right| = \frac{\hbar}{2}\frac{|p'|}{p^2} \ll 1.$$

The order-$\hbar^2$ equation, $2\sigma_0'\sigma_2' + \sigma_1'^2 - i\sigma_1'' = 0$, gives $\hbar^2\sigma_2' \sim \hbar^2(p'^2/p^3 + p''/p^2)$. This is small under the same condition together with $\hbar|p''| \ll |p'|\,p$.

In terms of the reduced de Broglie wavelength $\tilde{\lambda} \equiv \lambda/2\pi = \hbar/p$,

$$\left|\frac{d\tilde{\lambda}}{dx}\right| = \frac{\hbar|p'|}{p^2} \ll 1 \qquad \Longleftrightarrow \qquad \frac{m\hbar\,|V'(x)|}{p(x)^3} \ll 1,$$

using $p' = -mV'/p$. Ch. 9 §9.3.3 writes this as $|d\lambda/dx| \ll 1$ with the full wavelength $\lambda = h/p$. That form differs by a factor of 2π, which does not change a "much less than" statement. The reduced-wavelength form is the one that appears directly in the expansion.

Physical reading. The second form says that the change in potential energy across one reduced wavelength, $|V'|\tilde{\lambda}$, must be small compared with the local kinetic energy p^2/m.

Where it fails. At a turning point $x = a$ where $V(a) = E$, we have $p^2 \approx 2m|V'(a)|\,|x-a|$, so

$$\frac{m\hbar|V'|}{p^3} \approx \frac{1}{2^{3/2}}\left(\frac{\ell}{|x-a|}\right)^{3/2}, \qquad \ell \equiv \left(\frac{\hbar^2}{2m|V'(a)|}\right)^{1/3}.$$

WKB therefore fails within a distance of order ℓ of every turning point. It is also unreliable wherever V changes appreciably within a wavelength, for example at a step or a hard wall. This last case is why the rectangular barrier of Ch. 4 §4.4.3 has a prefactor that WKB does not reproduce (§C.6.3).

`[APPROXIMATION]` The small parameter is $\hbar|p'|/p^2 = |d\tilde{\lambda}/dx|$. The approximation is valid where V is smooth on the scale of $\tilde{\lambda}$, and it fails within $\sim \ell$ of turning points and at discontinuities of V.

C.4 — Connection Formulas at a Linear Turning Point

C.4.1 — The local problem

Assumptions.

1. There is an isolated turning point at $x = a$ with $V'(a) \equiv F > 0$, so the allowed region is $x < a$ and the forbidden region is $x > a$.
2. V is linear over a region around a that is much wider than ℓ. If L is the scale over which V' changes, $L \sim |V'/V''|$, this requires $\ell \ll L$.

> [APPROXIMATION] Linearise $V(x) \approx E + F(x-a)$. The small parameter is ℓ/L, and it goes to zero as $\hbar \to 0$ because $\ell \propto \hbar^{2/3}$.

In the linear region the Schrödinger equation reads $\psi'' = (2mF/\hbar^2)(x-a)\psi$. Set $z = (x-a)/\ell$ with $\ell = (\hbar^2/2mF)^{1/3}$. This gives the **Airy equation**

$$\frac{d^2\psi}{dz^2} = z\,\psi,$$

whose two independent solutions are $\mathrm{Ai}(z)$ and $\mathrm{Bi}(z)$.

C.4.2 — Airy asymptotics

Let $\zeta = \frac{2}{3}|z|^{3/2}$. The standard asymptotic forms are (Abramowitz and Stegun, *Handbook of Mathematical Functions*, §10.4; or the NIST *Digital Library of Mathematical Functions*, §9.7):

$$\begin{aligned} z \to +\infty: &\qquad \mathrm{Ai}(z) \simeq \frac{e^{-\zeta}}{2\sqrt{\pi}\,z^{1/4}}, \qquad \mathrm{Bi}(z) \simeq \frac{e^{\zeta}}{\sqrt{\pi}\,z^{1/4}}, \\ z \to -\infty: &\qquad \mathrm{Ai}(z) \simeq \frac{\sin(\zeta+\pi/4)}{\sqrt{\pi}\,|z|^{1/4}}, \qquad \mathrm{Bi}(z) \simeq \frac{\cos(\zeta+\pi/4)}{\sqrt{\pi}\,|z|^{1/4}}. \end{aligned}$$

C.4.3 — Matching

In the linear region the WKB quantities are $\kappa(x) = z^{1/2}/\ell$ for $z > 0$ and $k(x) \equiv p/\hbar = |z|^{1/2}/\ell$ for $z < 0$. The phase integrals are

$$\int_a^x \kappa\,dx' = \int_0^z z'^{1/2}\,dz' = \tfrac{2}{3}z^{3/2} = \zeta, \qquad \int_x^a k\,dx' = \tfrac{2}{3}|z|^{3/2} = \zeta,$$

and the amplitude factors $\kappa^{-1/2}$ and $k^{-1/2}$ are both proportional to $|z|^{-1/4}$.

The WKB solutions are valid for $|x-a| \gg \ell$, and the linearisation is valid for $|x-a| \ll L$. When $\ell \ll L$ the two regions overlap, and in the overlap the WKB solutions must coincide with the Airy asymptotics above. Reading off coefficients:

- Ai has coefficient $1/2\sqrt{\pi}$ on the decaying side and $1/\sqrt{\pi}$ on the oscillating side, a ratio of 2.
- Bi has the same coefficient on both sides, a ratio of 1.

Hence:

$$\begin{aligned} \frac{2}{\sqrt{k}}\cos\!\left(\int_x^a k\,dx' - \frac{\pi}{4}\right) &\longleftarrow \frac{1}{\sqrt{\kappa}}\exp\!\left(-\int_a^x \kappa\,dx'\right), \\ \frac{1}{\sqrt{k}}\cos\!\left(\int_x^a k\,dx' + \frac{\pi}{4}\right) &\longleftrightarrow \frac{1}{\sqrt{\kappa}}\exp\!\left(+\int_a^x \kappa\,dx'\right). \end{aligned}$$

The first line uses $\sin(\zeta+\pi/4) = \cos(\zeta-\pi/4)$.

For a turning point b with the allowed region on the *right* (so $V'(b) < 0$), the same argument with $z = -(x-b)/\ell$ gives the mirror-image formulas:

$$\frac{1}{\sqrt{\kappa}}e^{-\int_x^b \kappa\,dx'} \longrightarrow \frac{2}{\sqrt{k}}\cos\!\left(\int_b^x k\,dx' - \frac{\pi}{4}\right), \qquad \frac{1}{\sqrt{\kappa}}e^{+\int_x^b \kappa\,dx'} \longleftrightarrow \frac{1}{\sqrt{k}}\cos\!\left(\int_b^x k\,dx' + \frac{\pi}{4}\right).$$

> [DERIVATION] Given the linearisation and the overlap condition $\ell \ll L$, the connection formulas follow from the exact Airy solution.

C.4.4 — Directionality and the meaning of $\pi/4$

The arrows matter. The decaying exponential connects reliably to the oscillatory solution: a pure Ai is fixed by the requirement of decay. The reverse inference is unsafe. A small error in the phase of the oscillating solution admixes some Bi, which grows exponentially into the forbidden region and swamps the decaying piece. Conversely, a subdominant decaying term cannot be resolved underneath a growing one. Statements that rely on keeping both exponentials in the same forbidden region therefore carry errors of the same order as the subdominant term (see §C.6.2).

The phase $\pi/4$ is the turning-point instance of the Maslov phase met in §C.1.3: a phase loss of $\pi/2$ for each reflection at a smooth turning point, split as $\pi/4$ on the way in and $\pi/4$ on the way out. The same result can be obtained without Airy functions by continuing the WKB solution around the turning point in the complex x plane at a distance $\gg \ell$. That is the Zwaan–Kemble method; see Berry and Mount, *Rep. Prog. Phys.* **35**, 315 (1972), for a review of both routes.

What was proved. For an isolated turning point at which V is linear over a width much larger than ℓ, the WKB solutions on the two sides are connected as above. The derivation fails when two turning points are closer than a few ℓ (near a barrier top or a well bottom), when $V'(a) = 0$ (a degenerate turning point), and at hard walls, where the phase loss is $\pi/2$ per wall instead of $\pi/4$ per side.

C.5 — Bohr–Sommerfeld Quantisation

C.5.1 — Derivation

Assumptions.

1. The bound state lives in a single well with two isolated linear turning points $x_1 < x_2$.
2. The state is classically forbidden outside $[x_1, x_2]$.
3. The connection formulas of §C.4 hold at both turning points.

Define $k = p/\hbar$, $\theta(x) = \int_{x_1}^{x} k\,dx'$, and $\Theta = \int_{x_1}^{x_2} k\,dx$.

From x_1. The allowed region is on the right of x_1, so the decaying solution in $x < x_1$ connects, by the mirror formula of §C.4.3, to

$$\psi \simeq \frac{2C_1}{\sqrt{k}} \cos\!\left(\theta(x) - \frac{\pi}{4}\right).$$

From x_2. The allowed region is on the left of x_2, so the decaying solution in $x > x_2$ connects to

$$\psi \simeq \frac{2C_2}{\sqrt{k}} \cos\!\left(\Theta - \theta(x) - \frac{\pi}{4}\right) = \frac{2C_2}{\sqrt{k}} \cos\!\left(\theta(x) - \Theta + \frac{\pi}{4}\right).$$

Matching. Both expressions describe the same function throughout the interior. Two cosines $\cos(\theta - \alpha)$ and $\cos(\theta - \beta)$ are proportional for all θ if and only if $\alpha - \beta = n\pi$. Here $\alpha = \pi/4$ and $\beta = \Theta - \pi/4$, so

$$\Theta - \frac{\pi}{2} = n\pi, \qquad C_2 = (-1)^n C_1.$$

Since $\Theta > 0$, $n = 0, 1, 2, \ldots$, and n is the number of nodes of ψ in the interior. Using $\oint p\,dq = 2\int_{x_1}^{x_2} p\,dx = 2\hbar\Theta$,

$$\boxed{\oint p\,dq = \left(n + \frac{1}{2}\right) h, \qquad n = 0, 1, 2, \ldots}$$

This is the result quoted in Ch. 1 §1.8.4 and Ch. 9 §9.3.4.

> [DERIVATION] The Bohr–Sommerfeld rule follows from first-order WKB plus the two linear-turning-point connections.

Level spacing and the classical period. Differentiating the rule with respect to E gives

$$\frac{d}{dE}\oint p\,dq = \oint \frac{dq}{\dot{q}} = T_{cl}(E).$$

So adjacent levels are separated by $\Delta E \simeq h/T_{cl} = \hbar\omega_{cl}$. Emission frequencies $\Delta E/\hbar$ between neighbouring levels therefore approach the classical orbital frequency, which is the correspondence principle of Ch. 4 §4.12 in quantitative form.

C.5.2 — Check 1: the harmonic oscillator

For $V = \frac{1}{2}m\omega^2 x^2$, the orbit at energy E is an ellipse in phase space with semi-axes $\sqrt{2mE}$ and $\sqrt{2E/m\omega^2}$. Its area is

$$\oint p\,dq = \pi\sqrt{2mE}\sqrt{\frac{2E}{m\omega^2}} = \frac{2\pi E}{\omega}.$$

Setting this equal to $(n + \frac{1}{2})h$ gives $E_n = (n + \frac{1}{2})\hbar\omega$, which is the exact spectrum of Ch. 4 §4.3.2. Exactness here is special. All higher-order WKB corrections to the quantisation condition vanish identically for the oscillator; see Dunham, *Phys. Rev.* **41**, 713 (1932). In general the first correction to the quantisation condition is of relative order $\hbar^2$, which for low-lying states means corrections of order $1/n^2$ or so.

C.5.3 — Check 2: the $\frac{1}{2}$ is not universal

The $\frac{1}{2}$ counts two *smooth* turning points, each contributing a phase of $\pi/4$ on each side. Writing the rule as $\oint p\,dq = 2\pi\hbar\,(n + \mu/4)$, μ is the Maslov index, with $\mu = 1$ for each smooth turning point and $\mu = 2$ for each hard wall (where $\psi = 0$ forces a phase loss of $\pi/2$ on each side).

- **Two hard walls** (the infinite square well of Ch. 4 §4.2): $\mu = 4$. With $p = \sqrt{2mE}$ constant, $\oint p\,dq = 2L\sqrt{2mE} = (n+1)h$, so $E = (n+1)^2\pi^2\hbar^2/2mL^2$. This is exact, with $n + 1$ the usual quantum number.
- **One wall and one smooth turning point** (the half-oscillator, $V = \frac{1}{2}m\omega^2x^2$ for $x > 0$ with a wall at $x = 0$): $\mu = 3$. Half the ellipse gives $\pi E/\omega = 2\pi\hbar(n + \frac{3}{4})$, so $E = (2n + \frac{3}{2})\hbar\omega$. These are exactly the odd-parity oscillator levels, which are the true spectrum of this problem.

What was proved. For a well bounded by two isolated linear turning points, $\oint p\,dq = (n + \frac{1}{2})h$ up to corrections of relative order $\hbar^2$. The rule is exact for the oscillator. The $\frac{1}{2}$ must be replaced according to the Maslov index when the boundary conditions change. The derivation fails for a well so shallow that the turning points lie within a few ℓ of each other, which typically means the lowest one or two levels of an anharmonic well. It also fails in more than one dimension when the motion is not separable or integrable: there the Einstein–Brillouin–Keller generalisation applies to integrable systems, and chaotic systems need trace formulas.

C.6 — The WKB Tunnelling Exponent

C.6.1 — Setup

Assumptions.

1. A smooth barrier has $V(x) > E$ for $x_1 < x < x_2$, with linear turning points at both ends.
2. The barrier is thick in the sense that $\Theta \equiv \int_{x_1}^{x_2} \kappa\,dx \gg 1$.
3. A wave is incident from the left, and there is only a transmitted wave for $x > x_2$.

C.6.2 — Derivation

Right region. Take the transmitted wave to be $\psi_{III} = \frac{F}{\sqrt{k}} \exp\left[i\left(\int_{x_2}^{x} k\,dx' + \frac{\pi}{4}\right)\right]$. The phase $\pi/4$ is chosen for convenience. Expanding $e^{iu} = \cos u + i \sin u$ and using the mirror formulas at x_2 in their Airy form, in which Bi pairs $\cos(\cdot + \pi/4)$ with the growing exponential and Ai pairs $2\sin(\cdot + \pi/4)$ with the decaying one:

$$\psi_{II} = \frac{F}{\sqrt{\kappa}}\left[e^{+\int_x^{x_2}\kappa\,dx'} + \frac{i}{2}e^{-\int_x^{x_2}\kappa\,dx'}\right] = \frac{F}{\sqrt{\kappa}}\left[e^{\Theta}e^{-\int_{x_1}^{x}\kappa\,dx'} + \frac{i}{2}e^{-\Theta}e^{+\int_{x_1}^{x}\kappa\,dx'}\right].$$

Left region. Connecting at x_1 with the formulas of §C.4.3 (with $a = x_1$), and setting $\chi = \int_x^{x_1} k\,dx'$ and $u = \chi + \pi/4$:

$$\psi_I = \frac{F}{\sqrt{k}}\left[2e^{\Theta}\sin u + \frac{i}{2}e^{-\Theta}\cos u\right] = \frac{F}{\sqrt{k}}\left[e^{iu}\left(-ie^{\Theta} + \tfrac{i}{4}e^{-\Theta}\right) + e^{-iu}\left(ie^{\Theta} + \tfrac{i}{4}e^{-\Theta}\right)\right].$$

Since χ decreases as x increases, e^{-iu} is the right-moving (incident) wave.

Transmission. All waves carry the same $k^{-1/2}$ normalisation, so the transmission probability is the ratio of squared amplitudes:

$$T = \frac{1}{\left(e^{\Theta} + \frac{1}{4}e^{-\Theta}\right)^2}, \qquad R = \frac{\left(e^{\Theta} - \frac{1}{4}e^{-\Theta}\right)^2}{\left(e^{\Theta} + \frac{1}{4}e^{-\Theta}\right)^2}, \qquad R + T = 1.$$

The last identity follows from $(a+b)^2 - (a-b)^2 = 4ab = 1$ with $a = e^{\Theta}$, $b = \frac{1}{4}e^{-\Theta}$, and it is a useful check on the algebra. However, the $e^{-\Theta}$ terms are of the same order as the error committed by carrying a subdominant exponential through the connection formulas (§C.4.4). The only result that is reliable to leading order is therefore

$$\boxed{T \simeq \exp\left(-2\int_{x_1}^{x_2}\kappa(x)\,dx\right) = \exp\left(-\frac{2}{\hbar}\int_{x_1}^{x_2}\sqrt{2m\big(V(x) - E\big)}\,dx\right)}$$

> [APPROXIMATION] The small parameters are $e^{-2\Theta}$ (the thick-barrier condition) and ℓ/L at each turning point (smoothness). Near the top of the barrier the two turning points merge, and the parabolic-barrier result $T = [1 + e^{2\Theta}]^{-1}$ replaces the formula above (Landau and Lifshitz, *Quantum Mechanics: Non-Relativistic Theory*, Ch. VII).

C.6.3 — Checks and uses

Rectangular barrier (Ch. 4 §4.4.3). Here $\Theta = \kappa d$, so the WKB exponent $e^{-2\kappa d}$ is the same as the exact thick-barrier result. The exact prefactor $16(E/V_0)(1 - E/V_0)$ is not reproduced, because the walls of a rectangular barrier violate the smoothness condition of §C.3.3. **The exponent is robust; the prefactor is not.**

Alpha decay (Ch. 6 §6.10.1). Take $V = B/r$ with $B = 2Ze^2/4\pi\varepsilon_0$, outer turning point $r_c = B/E$, inner boundary at the nuclear radius R, and $v = \sqrt{2E/m_\alpha}$. Then

$$G \equiv \int_R^{r_c} \kappa \, dr = \frac{\sqrt{2m_\alpha E}}{\hbar} r_c \left[\arccos\sqrt{\frac{R}{r_c}} - \sqrt{\frac{R}{r_c}\left(1 - \frac{R}{r_c}\right)} \right] \simeq \frac{\pi B}{\hbar v}\left(1 - \frac{4}{\pi}\sqrt{\frac{R}{r_c}}\right),$$

with $T_\alpha \simeq e^{-2G}$. In SI units the leading term is $\pi B/\hbar v = 2\pi Ze^2/(4\pi\varepsilon_0\hbar v)$, and $R/r_c = E/V_C(R)$. The radial problem with $\ell = 0$ reduces to this one-dimensional form, and the inner boundary at R is not a smooth turning point, which affects only the prefactor.

Field emission. For a triangular barrier $V - E = W - e\mathcal{E}x$ (work function W, applied field $\mathcal{E}$, no image force), $\Theta = \frac{2\sqrt{2m}\,W^{3/2}}{3\hbar e\mathcal{E}}$. This gives the Fowler–Nordheim exponent $T \sim \exp\left(-\frac{4\sqrt{2m}\,W^{3/2}}{3\hbar e\mathcal{E}}\right)$, which is the high-field counterpart of the thermionic and Schottky emission treated in Ch. 14 §14.6.

What was proved. For a smooth barrier with $\Theta \gg 1$, the tunnelling probability is $e^{-2\Theta}$ to leading exponential accuracy. The result fails near the top of the barrier, where the turning points merge. For sharp barriers it gives the right exponent but the wrong prefactor. For a non-separable barrier in several dimensions it must be replaced by the instanton (complex-time stationary-phase) method, which is the many-dimensional generalisation of §C.1 to imaginary time.

C.7 — General Relativity to Newton: the Weak-Field Limit

C.7.1 — Assumptions and the bookkeeping parameter

Ch. 8 §8.3 names three conditions. Here they are stated as a single expansion. There exist coordinates $x^\mu = (ct, \mathbf{x})$ in which:

1. **(W) Weak field.** $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ with $|h_{\mu\nu}| \ll 1$. Work to first order in h.
2. **(S) Slow motion.** Sources and test bodies move with $v \ll c$, and stresses are small: $|T^{ij}| \ll T^{00}$. For a perfect fluid this means $p \ll \rho c^2$.
3. **(Q) Quasi-static.** Time derivatives of the field are small compared with space derivatives: $|\partial_0 h| \sim (v/c)|\partial_i h|$.

For bound systems the virial theorem links (W) and (S): $|\Phi|/c^2 \sim v^2/c^2 \equiv \epsilon$. Terms are therefore ordered as follows: $h_{00} \sim \epsilon$, $T^{0i}/T^{00} \sim \epsilon^{1/2}$, $T^{ij}/T^{00} \sim \epsilon$, and $\partial_0/\partial_i \sim \epsilon^{1/2}$.

> [APPROXIMATION] Newtonian gravity is the leading order in ϵ. It requires all three assumptions at once.

C.7.2 — Geodesic equation to Newton's second law

The geodesic equation is

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\alpha\beta}\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau} = 0.$$

Under (S), $dx^0/d\tau = c\,\gamma \simeq c$ and $dx^i/d\tau \simeq v^i$. In the spatial equation, the three groups of terms scale as

$$\Gamma^i_{00}\,c^2 \;:\; \Gamma^i_{0j}\,c\,v^j \;:\; \Gamma^i_{jk}\,v^j v^k \;\sim\; 1 : \epsilon^{1/2} : \epsilon,$$

so only Γ^i_{00} survives at leading order.

Converting from τ to t adds a term $(d^2t/d\tau^2)\,dx^i/dt$, which involves $\Gamma^0_{00} = -\tfrac{1}{2}\partial_0 h_{00}$ and is suppressed by (Q). To first order in h,

$$\Gamma^i_{00} = \tfrac{1}{2}\eta^{ij}\left(2\partial_0 h_{0j} - \partial_j h_{00}\right) \xrightarrow{(Q)} -\tfrac{1}{2}\partial_i h_{00},$$

and therefore

$$\frac{d^2x^i}{dt^2} = -c^2\,\Gamma^i_{00} = +\frac{c^2}{2}\,\partial_i h_{00}.$$

Newton's law $\ddot{x}^i = -\partial_i\Phi$ is recovered if and only if $h_{00} = -2\Phi/c^2 + \text{const}$. Requiring $h_{00} \to 0$ where $\Phi \to 0$ (far from all sources) sets the constant to zero:

$$\boxed{g_{00} \simeq -\left(1 + \frac{2\Phi}{c^2}\right)}$$

Sign check. Ch. 8 §8.3.2 writes the intermediate step as $-c^2 \cdot \tfrac{1}{2}\eta^{ij}\partial_j h_{00}$. The correct expression carries the opposite sign, because $\Gamma^i_{00} = -\tfrac{1}{2}\partial_i h_{00}$. With $h_{00} = -2\Phi/c^2$, the result $\ddot{x}^i = -\partial_i\Phi$ follows as stated.

Slow test bodies probe only g_{00} at this order. The spatial components g_{ij} enter at relative order ϵ, and they are what doubles the light-deflection angle (Ch. 8 §8.7.2), because light has $v = c$ and so violates (S).

> [DERIVATION] Under (W), (S) and (Q), the geodesic equation reduces to $\ddot{\mathbf{x}} = -\nabla\Phi$ with $\Phi \equiv -c^2 h_{00}/2$.

C.7.3 — Einstein equations to Poisson's equation

Taking the trace of $G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$ gives $R = -(8\pi G/c^4)T$, and so the trace-reversed form

$$R_{\mu\nu} = \frac{8\pi G}{c^4}\left(T_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}T\right).$$

Only the 00 component is needed.

Matter side. For a perfect fluid in its rest frame to leading order, $T_{00} = \rho c^2$, $T_{ij} = p\,\delta_{ij}$, and $T = g^{\mu\nu}T_{\mu\nu} = -\rho c^2 + 3p$. Therefore

$$T_{00} - \tfrac{1}{2} g_{00} T = \rho c^2 + \tfrac{1}{2}\left(-\rho c^2 + 3p\right) = \tfrac{1}{2}\left(\rho c^2 + 3p\right) \xrightarrow{p \ll \rho c^2} \tfrac{1}{2}\rho c^2.$$

This is the factor that comes from T_{00} dominance. The trace reversal halves the source. Pressure contributes to the "active" gravitational mass density as $\rho + 3p/c^2$, and Newtonian gravity requires that contribution to be negligible: $p \ll \rho c^2$ is part of assumption (S).

Geometry side. The linearised Ricci tensor of Ch. 8 §8.2.2, with every ∂_0 dropped under (Q), gives

$$R^{(1)}_{00} = \tfrac{1}{2}\left(-\Box h_{00} - \partial_0\partial_0 h + 2\partial^\rho \partial_0 h_{0\rho}\right) \xrightarrow{(Q)} -\tfrac{1}{2}\nabla^2 h_{00}.$$

Combining.

$$-\tfrac{1}{2}\nabla^2 h_{00} = \frac{8\pi G}{c^4}\cdot\frac{\rho c^2}{2} \quad\Longrightarrow\quad \nabla^2\left(-\frac{c^2 h_{00}}{2}\right) = 4\pi G\rho \quad\Longrightarrow\quad \boxed{\nabla^2\Phi = 4\pi G\rho}$$

where $\Phi = -c^2 h_{00}/2$, exactly as identified in §C.7.2. This route needs no gauge choice, because the static R_{00} involves only h_{00}.

Cross-check with Ch. 8's gauge route. In the Lorenz gauge, the derivation of Ch. 8 §8.3.1 gives $\bar{h}_{00} = -4\Phi/c^2$ with the other components of $\bar{h}_{\mu\nu}$ of higher order. The trace is $\bar{h} = \eta^{\mu\nu}\bar{h}_{\mu\nu} = +4\Phi/c^2$, so $h = -\bar{h} = -4\Phi/c^2$. Then $h_{\mu\nu} = \bar{h}_{\mu\nu} - \tfrac{1}{2}\eta_{\mu\nu}\bar{h}$ gives

$$h_{00} = -\frac{4\Phi}{c^2} + \frac{2\Phi}{c^2} = -\frac{2\Phi}{c^2}, \qquad h_{ij} = -\frac{2\Phi}{c^2}\delta_{ij}.$$

This reproduces the metric of Ch. 8 §8.4, including the spatial factor $(1 - 2\Phi/c^2)$, and is consistent with the result above.

Dimensional remark. Ch. 8 §8.3 writes the dust tensor as $T^{\mu\nu} = \rho c^2 u^\mu u^\nu$. With $u^\mu = \gamma(c, \mathbf{v})$ carrying units of velocity, the dimensionally consistent form is $T^{\mu\nu} = \rho\, u^\mu u^\nu$. That form gives $T^{00} = \rho c^2$, which is what both derivations use.

> `[DERIVATION]` Under (W), (S) and (Q), the 00 Einstein equation reduces to Poisson's equation, with the factor $\tfrac{1}{2}$ from trace reversal cancelling against the factor 2 in $h_{00} = -2\Phi/c^2$.

What was proved, and where it fails. Newtonian gravity, meaning both the field equation and the equation of motion, is the leading term in $\epsilon \sim |\Phi|/c^2 \sim v^2/c^2$. It fails in four ways:

1. When $|\Phi|/c^2$ is not small (neutron stars and black holes; Ch. 8 §8.9).
2. When v is comparable to c for the test body (light deflection, where g_{ij} matters).
3. When sources move fast enough that g_{0i} (gravitomagnetism, frame dragging; Ch. 8 §8.7.3) or radiation (Ch. 8 §8.6) is needed.
4. When the pressure is relativistic.

The first corrections (perihelion advance and the Shapiro delay) are the post-Newtonian terms of order ϵ beyond this limit.

C.8 — The Leading GPS Clock-Rate Correction

C.8.1 — Rate of an orbiting clock

Assumptions.

1. The metric of §C.7.3, $ds^2 = -(1 + 2\Phi/c^2)c^2dt^2 + (1 - 2\Phi/c^2)|d\mathbf{x}|^2$, in an Earth-centred non-rotating frame.
2. The potential is the monopole $\Phi = -GM_\oplus/r$ (oblateness J_2 neglected).
3. The clock moves with coordinate speed v.

Proper time along the clock's worldline satisfies

$$\frac{d\tau}{dt} = \sqrt{1 + \frac{2\Phi}{c^2} - \left(1 - \frac{2\Phi}{c^2}\right)\frac{v^2}{c^2}} = 1 + \frac{\Phi}{c^2} - \frac{v^2}{2c^2} + O(\epsilon^2).$$

> [DERIVATION] Both the "gravitational" and the "velocity" terms of Ch. 8 §8.8.2–§8.8.3 come from one expansion of one metric. They are the two order-ϵ terms of the same quantity, not two separate effects added by hand.

For a satellite on a circular orbit of radius r, $v^2 = GM_\oplus/r$, so

$$\left.\frac{d\tau}{dt}\right|_{sat} = 1 - \frac{3GM_\oplus}{2rc^2}.$$

For a ground clock at radius $R_\oplus$ with speed v_g due to Earth's rotation,

$$\left.\frac{d\tau}{dt}\right|_{gnd} = 1 - \frac{GM_\oplus}{R_\oplus c^2} - \frac{v_g^2}{2c^2}.$$

C.8.2 — Numbers

Take $GM_\oplus = 3.986 \times 10^{14}\ \mathrm{m^3\,s^{-2}}$, $c^2 = 8.988 \times 10^{16}\ \mathrm{m^2\,s^{-2}}$, $R_\oplus = 6.371 \times 10^6$ m, and $r = 2.656 \times 10^7$ m (the GPS semi-major axis). Then:

Term	Value	Per day
$GM_\oplus/R_\oplus c^2 - GM_\oplus/rc^2$	$6.961 \times 10^{-10} - 1.670 \times 10^{-10} = 5.29 \times 10^{-10}$	$+45.7\,\mu$s
$-v_{sat}^2/2c^2$, with $v_{sat} = 3.874$ km/s	-8.35×10^{-11}	$-7.2\,\mu$s
$+v_g^2/2c^2$ at the equator ($v_g = 465$ m/s)	$+1.2 \times 10^{-12}$	$+0.10\,\mu$s
Net fractional rate, satellite minus ground	$\approx 4.47 \times 10^{-10}$	$\approx +38.6\,\mu$s

Omitting the ground-rotation term reproduces the $+38.5\,\mu$s/day and 4.46×10^{-10} of Ch. 8 §8.8.4. The operational system sets the satellite clocks' nominal 10.23 MHz to 10.22999999543 MHz, a fractional offset of 4.467×10^{-10}. Treating the ground clock on the rotating geoid, and including J_2 in the potential, closes the remaining gap. See Ashby, "Relativity in the Global Positioning System", *Living Rev. Relativ.* **6**, 1 (2003).

C.8.3 — The eccentricity term

For a Keplerian orbit with semi-major axis a, energy conservation gives $v^2/2 = GM_\oplus(1/r - 1/2a)$, so

$$\frac{d\tau}{dt} = 1 - \frac{3GM_\oplus}{2ac^2} - \frac{2GM_\oplus}{c^2}\left(\frac{1}{r} - \frac{1}{a}\right).$$

With $r = a(1 - e\cos E)$ and $dt = \sqrt{a^3/GM_\oplus}\,(1 - e\cos E)\,dE$, where E here is the eccentric anomaly, the last term integrates to

$$\Delta\tau_{periodic} = -\frac{2\sqrt{GM_\oplus a}}{c^2}\, e\, \sin E.$$

Its amplitude is about 23 ns for $e = 0.01$. Receivers apply this term in software; it is not absorbed by the factory frequency offset.

What was proved. The fractional rate offset of a GPS clock is the order-ϵ term in the proper-time expansion of the weak-field metric. That is $\approx 4.5 \times 10^{-10}$, of which the gravitational part dominates. Neglected here are signal-propagation effects (the Sagnac correction in the rotating frame and the Shapiro delay), J_2, and terms of order $\epsilon^2 \sim 10^{-19}$, all of which are treated in Ashby (2003).

C.9 — Summary

Result	Statement	Assumptions	Used in
Stationary-phase lemma	$\int g\, e^{i\phi/\hbar} \simeq g(x_0)\sqrt{2\pi\hbar/\lvert\phi''\rvert}\, e^{i\phi(x_0)/\hbar \pm i\pi/4}$	finite dimension; nondegenerate stationary point; smooth g, ϕ	Ch. 1 §1.4.3; Ch. 9 §9.2.2
Classical path from path integral	stationary point of S_N is the discretised Euler–Lagrange path; $K \simeq \sum_j A_j e^{iS_j/\hbar - i\pi\nu_j/2}$	$\hbar/S_{cl} \ll 1$; no caustics; heuristic exchange of $N \to \infty$ and $\hbar \to 0$ (rigorous under regularity conditions, §C.1.5)	Ch. 1 §1.4.3; Ch. 9 §9.2
Ehrenfest theorem	$d\langle\hat A\rangle/dt = (i/\hbar)\langle[\hat H, \hat A]\rangle + \langle\partial_t \hat A\rangle$; $m\ddot{\bar x} = \langle F\rangle$	self-adjoint $\hat H$; domain conditions	Ch. 9 §9.1
Centroid follows Newton	$\langle V'\rangle = V'(\bar x) + \tfrac{1}{2}\sigma_x^2 V''' + \cdots$	$\sigma_x \ll L_V$; $t \ll t_E$; exact for V at most quadratic	Ch. 9 §9.1.3
WKB wavefunction	$\psi \simeq Cp^{-1/2}e^{\pm i\int p\,dx/\hbar}$; $\kappa^{-1/2}e^{\pm\int\kappa\,dx}$	$\hbar\lvert p'\rvert/p^2 \ll 1$	Ch. 1 §1.8.4; Ch. 9 §9.3.2–§9.3.3
Connection formulas	$\frac{2}{\sqrt{k}}\cos(\int_x^a k - \frac{\pi}{4}) \leftarrow \frac{1}{\sqrt{\kappa}}e^{-\int_a^x\kappa}$	isolated linear turning point; $\ell \ll L$; direction matters	Ch. 9 §9.3.3
Bohr–Sommerfeld	$\oint p\,dq = (n + \tfrac{1}{2})h$; generally $2\pi\hbar(n + \mu/4)$	two smooth turning points; corrections $O(\hbar^2)$; exact for oscillator	Ch. 1 §1.8.4; Ch. 9 §9.3.4; Ch. 4 §4.3, §4.12
Tunnelling exponent	$T \simeq \exp(-2\int\kappa\,dx)$	$\Theta \gg 1$; smooth barrier; not near the top	Ch. 4 §4.4; Ch. 6 §6.10.1; Ch. 14 §14.6
Newtonian metric	$g_{00} \simeq -(1 + 2\Phi/c^2)$, $g_{ij} \simeq (1 - 2\Phi/c^2)\delta_{ij}$	weak field, slow motion, quasi-static	Ch. 8 §8.3.2, §8.4

Result	Statement	Assumptions	Used in
Poisson equation	$\nabla^2\Phi = 4\pi G\rho$ from the 00 trace-reversed Einstein equation	as above plus $p \ll \rho c^2$; source $\frac{1}{2}(\rho c^2 + 3p)$	Ch. 8 §8.3.1
GPS clock rate	$d\tau/dt \simeq 1 + \Phi/c^2 - v^2/2c^2$; net $\approx 4.5 \times 10^{-10}$, $\approx 38.5\,\mu$s/day	monopole potential; order ϵ; periodic eccentricity term separate	Ch. 8 §8.8

C.10 — References

Standard sources used in this appendix (see also `REFERENCES.md`):

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- J. L. Dunham, *Phys. Rev.* **41**, 713 (1932).
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- F. Haake, *Quantum Signatures of Chaos*, Springer.
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Appendix D — Statistical Mechanics and Linear Response: Ensembles, Quantum Statistics, Kubo, Green–Kubo

This appendix proves the statistical-mechanical results that Ch. 10 states, and the linear-response results that Ch. 13 sketches and Ch. 14 uses in engineering form. The first half (§D.1–§D.8) covers Bridge B.b: the canonical ensemble derived from a reservoir, thermodynamics from the partition function, equipartition and where it fails, the Fermi–Dirac and Bose–Einstein distributions, the classical limit, the Planck and Debye laws, and the Sommerfeld expansion. The second half (§D.9–§D.15) covers Bridge B.e: the Kubo formula derived from first-order perturbation theory, the fluctuation–dissipation theorem, the Green–Kubo expressions, the Drude conductivity under a relaxation-time closure, the Einstein and Wiedemann–Franz relations, and Onsager reciprocity under microscopic reversibility.

The appendix keeps one distinction in view throughout, as Ch. 13 §13.2.2 does. The Kubo formula is an exact statement *within linear response*. Each transport coefficient obtained from it (σ, κ, D, η) additionally needs a choice of operators, an order of limits, and in practice a closure such as a relaxation time. These laws are constitutive transport coefficients within appropriate linear-response regimes. No single calculation produces all of them.

Conventions follow Ch. 10 and Ch. 13. $\beta = 1/k_B T$. Electrons carry charge $-e$ with $e > 0$. The spin degeneracy is g_s (equal to 2 for electrons). Fourier transforms are $\tilde{X}(\omega) = \int X(t)\, e^{i\omega t}\, dt$, so the Drude form is $1/(1 - i\omega\tau)$. $\hat{J}$ is a total current, and $\hat{j} = \hat{J}/V$ is a current density (Ch. 13 §13.4.1).

D.1 — From the Microcanonical to the Canonical Ensemble

Ch. 10 §10.3 obtains $\hat{\rho} = e^{-\beta \hat{H}}/Z$ by maximising entropy at fixed mean energy. Here the same result is derived from a different starting point: a small system in contact with a large reservoir. The two routes agree; they rest on different postulates.

D.1.1 — Assumptions

1. **Isolation.** System S and reservoir R together are isolated, with total energy in a narrow shell $[E_{tot}, E_{tot} + \delta E]$.
2. **Weak coupling.** $\hat{H} = \hat{H}_S + \hat{H}_R + \hat{H}_{int}$, where $\hat{H}_{int}$ is strong enough to let energy flow between S and R but contributes negligibly to the energy itself. Each joint state then has energy $E_n + E_R$.
3. **Equal a priori probabilities.** Every joint microstate in the shell is equally likely. This is the microcanonical postulate. It is not derived here.
4. **Large reservoir.** R has many more degrees of freedom than S, so that the energies E_n of S that are actually occupied satisfy $E_n \ll E_{tot}$.

D.1.2 — Derivation

Let $\Omega_R(E)$ be the number of reservoir states in the shell at energy E. By assumption 3, the probability that S is in its state n is proportional to the number of reservoir states compatible with it:

$$p_n \propto \Omega_R(E_{tot} - E_n) = \exp\left[\frac{S_R(E_{tot} - E_n)}{k_B}\right], \qquad S_R \equiv k_B \ln \Omega_R.$$

Expand the reservoir entropy about E_{tot}:

$$\frac{S_R(E_{tot} - E_n)}{k_B} = \frac{S_R(E_{tot})}{k_B} - \beta E_n + \frac{1}{2}\frac{\partial \beta}{\partial E} E_n^2 + \cdots, \qquad \beta \equiv \frac{1}{k_B}\frac{\partial S_R}{\partial E}\bigg|_{E_{tot}}.$$

The first term is the same for every n and is absorbed into the normalisation.

> [APPROXIMATION] Drop the quadratic term. Using $\partial\beta/\partial E = -1/(k_B T^2 C_R)$, where C_R is the reservoir heat capacity, the ratio of the quadratic term to the linear one is $E_n/(2TC_R)$. That ratio is of order N_S/N_R for comparable materials, and it vanishes as the reservoir grows.

Normalising what remains gives

$$\boxed{p_n = \frac{e^{-\beta E_n}}{Z}, \qquad Z = \sum_n e^{-\beta E_n} = \operatorname{Tr} e^{-\beta \hat{H}_S}}$$

and β is fixed by the reservoir, not by the system. Two systems in contact with the same reservoir therefore share the same β, which is the statistical form of the zeroth law (Ch. 10 §10.5).

D.1.3 — Energy fluctuations and the equivalence of ensembles

In the canonical ensemble, $\langle H \rangle = -\partial_\beta \ln Z$ and $\langle H^2 \rangle - \langle H \rangle^2 = \partial_\beta^2 \ln Z = k_B T^2 C_V$. Both U and C_V are extensive, so

$$\frac{\Delta E}{U} = \frac{\sqrt{k_B T^2 C_V}}{U} \sim N^{-1/2}.$$

For macroscopic N the canonical energy distribution is therefore sharply peaked, and canonical and microcanonical averages of extensive quantities agree to relative order $N^{-1/2}$.

> [APPROXIMATION] The equivalence of ensembles holds in the thermodynamic limit ($N, V \to \infty$ at fixed N/V). It requires $\ln Z$ to be extensive, which needs short-range interactions, or Coulomb interactions with screening and stability. For the Coulomb case see Lieb and Lebowitz, *Adv. Math.* **9**, 316 (1972), and for the general theory Ruelle, *Statistical Mechanics: Rigorous Results* (1969).

Equivalence fails at first-order phase transitions (where the energy distribution is bimodal), for self-gravitating systems (where C_V can be negative in the microcanonical ensemble), and for small systems.

The thermodynamic limit is not the continuum limit. Taking $N \to \infty$ makes fluctuations of extensive quantities negligible. It does not produce smooth fields; that needs coarse-graining with $\ell_{micro} \ll L$ (Ch. 11).

What was proved. If equal a priori probabilities hold for the combined system and the reservoir is much larger than the system, the system's states are Boltzmann distributed, with corrections of relative order $E_n/(TC_R)$. The derivation fails for strongly coupled system–reservoir pairs, where $\hat{H}_{int}$ is not negligible and the system's own Hamiltonian is no longer well defined.

D.2 — Thermodynamic Potentials from the Partition Function

Assumptions. The canonical distribution of §D.1. Energy levels $E_n(V)$ depend on an external parameter (here the volume V).

Internal energy and entropy.

$$U = \sum_n p_n E_n = -\frac{\partial \ln Z}{\partial \beta}, \qquad S = -k_B \sum_n p_n \ln p_n = k_B \left(\ln Z + \beta U\right),$$

where the second form uses $\ln p_n = -\beta E_n - \ln Z$.

Free energy. $F \equiv U - TS = -k_B T \ln Z$.

First law, with heat and work separated. Write $dU = \sum_n E_n\, dp_n + \sum_n p_n\, dE_n$. The first sum is the heat term. Using $\sum_n dp_n = 0$,

$$dS = -k_B \sum_n dp_n \ln p_n = k_B \beta \sum_n E_n\, dp_n \quad \Longrightarrow \quad \sum_n E_n\, dp_n = T\, dS.$$

The second sum is the work term. With $dE_n = (\partial E_n/\partial V)\, dV$ it equals $-P\, dV$, where

$$P \equiv -\left\langle \frac{\partial \hat{H}}{\partial V} \right\rangle = k_B T \left.\frac{\partial \ln Z}{\partial V}\right|_T.$$

Hence $dU = T\, dS - P\, dV$ and $dF = -S\, dT - P\, dV$. This reproduces every entry of Ch. 10 §10.4, and $S = -\partial F/\partial T|_V$ follows directly.

> `[DERIVATION]` These identities are exact for the canonical ensemble. Identifying them with the macroscopic T, S and P of thermodynamics requires the thermodynamic limit of §D.1.3. Statistical mechanics supplies the microscopic interpretation; the four laws carry separate content and assumptions (Ch. 10 §10.5).

Grand canonical analogue. If the reservoir also exchanges particles, the same expansion as in §D.1.2 applied to $S_R(E_{tot} - E, N_{tot} - N)$ gives

$$p \propto e^{-\beta(E - \mu N)}, \qquad \mu \equiv -T \left.\frac{\partial S_R}{\partial N}\right|_E, \qquad \Xi = \mathrm{Tr}\, e^{-\beta(\hat{H} - \mu \hat{N})},$$

with $\Omega \equiv -k_B T \ln \Xi = F - \mu N$, $\langle N \rangle = -\partial \Omega/\partial \mu$, and $\Omega = -PV$ for a homogeneous system.

D.3 — Equipartition and Its Failure

D.3.1 — The generalised equipartition theorem

Assumptions.

1. The system is classical: phase-space variables x_i (coordinates and momenta) commute, and the canonical density is $\propto e^{-\beta H(x)}$.

2. Each x_i ranges over the whole real line, or over an interval at whose ends $e^{-\beta H}$ vanishes.

Integrating by parts in x_j,

$$\left\langle x_i \frac{\partial H}{\partial x_j} \right\rangle = \frac{1}{Z}\int d\Gamma\; x_i \frac{\partial H}{\partial x_j} e^{-\beta H} = -\frac{1}{\beta Z}\int d\Gamma\; x_i \frac{\partial}{\partial x_j} e^{-\beta H} = \frac{\delta_{ij}}{\beta Z}\int d\Gamma\; e^{-\beta H} = \delta_{ij} k_B T.$$

The boundary terms vanish by assumption 2. For a term $\frac{1}{2}\alpha q^2$ in H, choosing $x_i = x_j = q$ gives

$$\left\langle \tfrac{1}{2}\alpha q^2 \right\rangle = \tfrac{1}{2} k_B T,$$

which is the statement of Ch. 10 §10.7.2. For a potential $V \propto |q|^s$, the same identity gives $\langle V \rangle = k_B T / s$. So "$\frac{1}{2}k_B T$ per degree of freedom" is specific to quadratic terms.

> [DERIVATION] Equipartition is exact in classical statistical mechanics.

D.3.2 — Failure for quantised modes

For a quantum harmonic oscillator with levels $(n + \frac{1}{2})\hbar\omega$, the canonical sum is geometric:

$$Z = \frac{e^{-x/2}}{1 - e^{-x}}, \qquad x = \beta\hbar\omega, \qquad \langle E \rangle = \frac{\hbar\omega}{2} + \frac{\hbar\omega}{e^x - 1}, \qquad C = k_B \frac{x^2 e^x}{(e^x - 1)^2}.$$

High temperature, $x \ll 1$. Using $1/(e^x - 1) = 1/x - 1/2 + x/12 + O(x^3)$,

$$\langle E \rangle = k_B T + \frac{(\hbar\omega)^2}{12 k_B T} + \cdots$$

This recovers equipartition, the full $k_B T$ for a kinetic plus a potential quadratic term, with a quantum correction of relative size $x^2/12$.

Low temperature, $x \gg 1$. $C \simeq k_B x^2 e^{-x}$. The mode is frozen out, as in Ch. 10 §10.7.3.

The classical theorem fails here because the levels are discrete. Assumption 1 is violated: q and p do not commute, and the integral over phase space is not the trace.

> [APPROXIMATION] The small parameter is $\hbar\omega / k_B T$. For a general system, the classical partition function $Z_{cl} = (N!\, h^{3N})^{-1} \int d\Gamma\, e^{-\beta H}$ is the leading term of an expansion in $\hbar$. Its first correction is $F = F_{cl} + \frac{\hbar^2\beta^2}{24}\sum_i \frac{\langle (\nabla_i V)^2 \rangle_{cl}}{m_i} + O(\hbar^4)$, the Wigner–Kirkwood expansion (Landau and Lifshitz, *Statistical Physics, Part 1*, §33). The classical result requires the thermal wavelength to be small compared with the scale on which V varies.

What was proved. Equipartition is exact for quadratic terms in classical statistics. It is the high-temperature limit of the quantum result, and it fails for any mode with $\hbar\omega \gtrsim k_B T$.

D.4 — Fermi–Dirac and Bose–Einstein from the Grand Canonical Ensemble

D.4.1 — Assumptions

1. Grand canonical ensemble (§D.2).
2. Non-interacting particles: $\hat{H} = \sum_j \varepsilon_j \hat{n}_j$ and $\hat{N} = \sum_j \hat{n}_j$ in a basis of single-particle states j.
3. Occupation numbers: $n_j \in \{0, 1\}$ for fermions, and $n_j \in \{0, 1, 2, \dots\}$ for bosons.

The fermion restriction is not an extra postulate here. It follows from the anticommutation $\{\hat{c}_j, \hat{c}_j^\dagger\} = 1$, $\hat{c}_j^2 = 0$ (Ch. 0; Ch. 5), which gives $\hat{n}_j^2 = \hat{n}_j$.

D.4.2 — Factorisation and the distributions

Because $\hat{H} - \mu\hat{N} = \sum_j (\varepsilon_j - \mu)\hat{n}_j$ is a sum of commuting single-mode terms, the grand partition function factorises over modes:

$$\Xi = \prod_j \xi_j, \qquad \xi_j = \sum_n e^{-\beta(\varepsilon_j - \mu)n}.$$

Fermions.

$$\xi_j = 1 + e^{-\beta(\varepsilon_j - \mu)}, \qquad \bar{n}_j = -\frac{1}{\beta}\frac{\partial \ln \xi_j}{\partial \varepsilon_j} = \frac{1}{e^{\beta(\varepsilon_j - \mu)} + 1}.$$

Bosons. The geometric series converges only if $\mu < \varepsilon_j$ for every j, that is, $\mu < \varepsilon_{min}$. Then

$$\xi_j = \frac{1}{1 - e^{-\beta(\varepsilon_j - \mu)}}, \qquad \bar{n}_j = \frac{1}{e^{\beta(\varepsilon_j - \mu)} - 1}.$$

These are the boxed results of Ch. 10 §10.8.2.

> [DERIVATION] Given non-interacting modes and the grand canonical ensemble, the derivation is exact. In an interacting system the same functions describe quasiparticles only approximately; that is Fermi-liquid theory, which is not derived here.

D.4.3 — Fluctuations and the grand potential

Differentiating once more,

$$\langle \delta n_j^2 \rangle = -\frac{1}{\beta}\frac{\partial \bar{n}_j}{\partial \varepsilon_j} = \bar{n}_j(1 \mp \bar{n}_j),$$

with the upper sign for fermions and the lower for bosons. Fermion occupation fluctuations are suppressed relative to Poisson, which is the origin of sub-Poissonian shot noise in Ch. 13 §13.3.3; boson fluctuations are enhanced. Occupations of different modes are statistically independent: $\langle n_j n_{j'} \rangle = \bar{n}_j \bar{n}_{j'}$ for $j \neq j'$. This fact is used in §D.12.

The grand potential is $\Omega = \mp k_B T \sum_j \ln\left(1 \pm e^{-\beta(\varepsilon_j - \mu)}\right)$, again with the upper sign for fermions.

Photons, $\mu = 0$. Photon number is not conserved: the dynamics does not commute with $\hat{N}$. There is then no reservoir variable conjugate to N, and at fixed T and V the equilibrium value of N minimises F, so $\partial F/\partial N = \mu = 0$. The same holds for phonons.

What was proved. The Fermi–Dirac and Bose–Einstein distributions follow exactly for ideal quantum gases in the grand canonical ensemble. They fail as exact statements once interactions are present.

D.5 — The Classical (Maxwell–Boltzmann) Limit and $n\lambda^3 \ll 1$

D.5.1 — The condition

Both quantum distributions reduce to $\bar{n} \simeq e^{-\beta(\varepsilon-\mu)}$ when $e^{\beta(\varepsilon-\mu)} \gg 1$ for every occupied state. Since $\varepsilon \geq 0$, this requires the fugacity to be small: $z \equiv e^{\beta\mu} \ll 1$.

To relate z to the density, take an ideal gas in a box with $\varepsilon = \hbar^2 k^2/2m$ and degeneracy g_s:

$$n = g_s \int \frac{d^3k}{(2\pi)^3} \frac{1}{z^{-1}e^{\beta\varepsilon} \pm 1} = \frac{g_s}{\lambda_{th}^3}\left(z \mp \frac{z^2}{2^{3/2}} + \cdots\right), \qquad \lambda_{th} = \frac{h}{\sqrt{2\pi m k_B T}}.$$

The first term uses $\int \frac{d^3k}{(2\pi)^3} e^{-\beta\hbar^2k^2/2m} = (mk_BT/2\pi\hbar^2)^{3/2} = \lambda_{th}^{-3}$. To leading order, then, $z = n\lambda_{th}^3/g_s$, and

$$\boxed{z \ll 1 \quad \Longleftrightarrow \quad \frac{n\lambda_{th}^3}{g_s} \ll 1}$$

This is the criterion of Ch. 10 §10.7.1 and §10.8.2: the mean interparticle spacing $n^{-1/3}$ is large compared with λ_{th}.

D.5.2 — First quantum correction

The pressure is $P/k_BT = (g_s/\lambda_{th}^3)(z \mp z^2/2^{5/2} + \cdots)$. Eliminating z with $y \equiv n\lambda_{th}^3/g_s$ and $z = y \pm y^2/2^{3/2} + O(y^3)$ gives

$$P = nk_BT\left(1 \pm \frac{n\lambda_{th}^3}{2^{5/2}g_s} + \cdots\right).$$

The upper sign is for fermions. Fermi statistics acts like an effective repulsion, and Bose statistics like an effective attraction.

D.5.3 — Numbers

System	n (m^{-3})	λ_{th} at 300 K	$n\lambda_{th}^3$	Regime
N_2 gas at 1 atm	2.5×10^{25}	1.9×10^{-11} m	$\sim 2 \times 10^{-7}$	classical
Conduction electrons in Cu	8.5×10^{28}	4.3×10^{-9} m	$\sim 7 \times 10^{3}$	degenerate

The Gibbs factor emerges. In the regime $z \ll 1$, multiply occupied states are negligible. The quantum partition function then reduces to $Z = Z_1^N/N!$, so the Gibbs correction of Ch. 10 §10.7.1 is a *consequence* of quantum indistinguishability in this limit, not an extra rule.

> [APPROXIMATION] The small parameter is $n\lambda_{th}^3/g_s$. The approximation fails for degenerate electrons, for liquid helium, and for cold atomic gases near Bose–Einstein condensation, which for bosons occurs at $n\lambda_{th}^3 = \zeta(3/2) \approx 2.612$ per spin state.

D.6 — Planck's Law from Mode Counting

Assumptions.

1. Electromagnetic radiation in a cavity of volume V, in equilibrium with the walls at temperature T.
2. The field is a set of independent harmonic modes. This is exact for the free Maxwell field.
3. Two polarisations per wavevector.
4. Each mode is occupied with the Bose–Einstein distribution at $\mu = 0$ (§D.4.3).

Counting modes. With periodic boundary conditions, $\mathbf{k} = 2\pi\mathbf{n}/L$, so each mode occupies a volume $(2\pi)^3/V$ in k-space. The number of modes with angular frequency in $[\omega, \omega + d\omega]$, using $\omega = ck$, is

$$dN_{modes} = 2 \cdot \frac{V\, 4\pi k^2\, dk}{(2\pi)^3} = \frac{V\omega^2}{\pi^2 c^3}\, d\omega.$$

The large-volume mode density does not depend on the cavity shape or the boundary conditions. That is Weyl's law; see Courant and Hilbert, *Methods of Mathematical Physics*, Vol. 1, Ch. VI.

Energy density. Multiplying by the mean thermal energy per mode,

$$u(\omega, T)\, d\omega = \frac{\hbar\omega^3}{\pi^2 c^3} \frac{d\omega}{e^{\beta\hbar\omega} - 1}.$$

The zero-point energy $\tfrac{1}{2}\hbar\omega$ per mode has been dropped. It is T-independent and does not appear in thermal emission; its differences between geometries are the Casimir effect. Converting with $\omega = 2\pi\nu$ and $d\omega = 2\pi\, d\nu$ gives $\hbar(2\pi)^4\nu^3/\pi^2c^3 = 8\pi h\nu^3/c^3$, so

$$u(\nu, T) = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/k_BT} - 1},$$

which is the boxed result of Ch. 10 §10.8.4.

> [DERIVATION] Given the assumptions, Planck's law is exact. Real emitters depart from it through emissivity below 1 and through finite size, when the cavity is no longer much larger than the thermal wavelength hc/k_BT.

Integrals.

- Using $\int_0^\infty x^3/(e^x - 1)\, dx = \pi^4/15$, the total energy density is $U/V = \pi^2 k_B^4 T^4/(15\hbar^3c^3)$.
- The flux through a hole is $j = cU/4V = \sigma T^4$ with $\sigma = \pi^2 k_B^4/(60\hbar^3c^2) = 2\pi^5 k_B^4/(15h^3c^2)$.

Peaks. The peak depends on the variable used.

- Maximising $u(\nu)$ gives $x = 3(1 - e^{-x})$, so $h\nu_{max} = 2.821\, k_BT$.

- Maximising the per-wavelength density u_λ gives $x = 5(1 - e^{-x})$, so $hc/\lambda_{max} = 4.965\, k_B T$ and $\lambda_{max} T = 2.898 \times 10^{-3}$ m·K.

These are different statements, since $\lambda_{max} \neq c/\nu_{max}$. Ch. 10 §10.8.4 quotes them together.

D.7 — Debye Heat Capacity and the T^3 Law

Assumptions.

1. A harmonic crystal of N atoms in a monatomic Bravais lattice, with $3N$ normal modes (Ch. 6 §6.7.1; Ch. 9 §9.8.4).
2. Each mode is an independent quantum oscillator (§D.3.2).
3. The Debye model:
   - `[APPROXIMATION]` Replace the true dispersion by $\omega = v_s k$ for all three acoustic branches, with an averaged speed $3/v_s^3 = 1/v_L^3 + 2/v_T^3$.
   - Cut the spectrum off at ω_D, chosen so that the total number of modes is $3N$.

Density of states. Counting as in §D.6 with three branches, $g(\omega) = 3V\omega^2/2\pi^2 v_s^3$. The cutoff condition $\int_0^{\omega_D} g\, d\omega = 3N$ gives $\omega_D^3 = 6\pi^2 (N/V) v_s^3$, and therefore

$$g(\omega) = \frac{9N\omega^2}{\omega_D^3}.$$

Heat capacity. Each mode contributes the oscillator heat capacity of §D.3.2. With $x = \beta\hbar\omega$ and $T_D = \hbar\omega_D/k_B$,

$$C_V = \int_0^{\omega_D} g(\omega)\, k_B \frac{x^2 e^x}{(e^x - 1)^2}\, d\omega = 9Nk_B \left(\frac{T}{T_D}\right)^3 \int_0^{T_D/T} \frac{x^4 e^x}{(e^x - 1)^2}\, dx,$$

which is the result of Ch. 10 §10.7.3.

High-temperature limit. For $T \gg T_D$, the integrand is $\simeq x^2$, the integral is $\frac{1}{3}(T_D/T)^3$, and $C_V \to 3Nk_B$ (Dulong–Petit).

Low-temperature limit. For $T \ll T_D$, extend the upper limit to infinity. Integration by parts gives

$$\int_0^\infty \frac{x^4 e^x}{(e^x - 1)^2}\, dx = \left[-\frac{x^4}{e^x - 1}\right]_0^\infty + 4\int_0^\infty \frac{x^3}{e^x - 1}\, dx = \frac{4\pi^4}{15},$$

and hence

$$\boxed{C_V \simeq \frac{12\pi^4}{5} N k_B \left(\frac{T}{T_D}\right)^3, \qquad T \ll T_D}$$

What is robust and what is model. At $T \ll T_D$ only modes with $\hbar\omega \lesssim k_B T$ are excited. Those are long-wavelength acoustic phonons, for which linear dispersion is asymptotically exact. The T^3 law therefore does not depend on the cutoff or on the details of the dispersion: its coefficient involves only the sound speeds. The interpolation between the two limits is the model.

> [APPROXIMATION] The small parameter for the T^3 law is T/T_D, more precisely k_BT compared with the energy at which the dispersion departs from linear. Debye fails at intermediate T wherever the real $g(\omega)$ has van Hove structure or optical branches. In amorphous solids an additional low-T contribution appears that is not phononic.

D.8 — The Sommerfeld Expansion and $C_{el} \propto T$

D.8.1 — The expansion

Assumptions.

1. $H(\varepsilon)$ is smooth near μ on the scale of k_BT.
2. $H(\varepsilon)$ vanishes for $\varepsilon < 0$ (the band bottom) and grows at most polynomially.
3. The system is degenerate: $k_BT \ll \mu$.

Let f be the Fermi function and $K(\varepsilon) = \int_0^\varepsilon H\, d\varepsilon'$. Integration by parts gives

$$I = \int_0^\infty H(\varepsilon) f(\varepsilon)\, d\varepsilon = \int_0^\infty K(\varepsilon) \left(-\frac{\partial f}{\partial \varepsilon}\right) d\varepsilon.$$

The function $-\partial f/\partial\varepsilon$ is even in $\varepsilon - \mu$, integrates to 1, and has width of order k_BT. Expand K about μ and extend the lower limit to $-\infty$, at a cost of order $e^{-\beta\mu}$. The odd moments vanish. The second moment is

$$\int_{-\infty}^{\infty} y^2 \frac{e^y}{(e^y+1)^2}\, dy = 4\int_0^\infty \frac{y}{e^y+1}\, dy = 4 \cdot \frac{\pi^2}{12} = \frac{\pi^2}{3}.$$

Therefore

$$\boxed{\int_0^\infty H(\varepsilon) f(\varepsilon)\, d\varepsilon = \int_0^\mu H(\varepsilon)\, d\varepsilon + \frac{\pi^2}{6}(k_BT)^2 H'(\mu) + O\big((k_BT)^4 H'''\big)}$$

The expansion is asymptotic, not convergent: it misses terms of order $e^{-\beta\mu}$.

D.8.2 — Chemical potential and heat capacity

Let $g(\varepsilon)$ be the density of states per unit volume including spin, and n the electron density.

Chemical potential. Fixed $n = \int g f\, d\varepsilon$ requires $0 = (\mu - E_F) g(E_F) + \frac{\pi^2}{6}(k_BT)^2 g'(E_F)$, so

$$\mu = E_F - \frac{\pi^2}{6}(k_BT)^2 \frac{g'(E_F)}{g(E_F)}.$$

Energy. $u = \int \varepsilon g f\, d\varepsilon = u_0 + E_F g(E_F)(\mu - E_F) + \frac{\pi^2}{6}(k_BT)^2\,[g(E_F) + E_F g'(E_F)]$. Substituting $\mu - E_F$ cancels the g' terms, leaving

$$u = u_0 + \frac{\pi^2}{6}(k_BT)^2 g(E_F), \qquad \boxed{c_{el} = \frac{\partial u}{\partial T} = \frac{\pi^2}{3} k_B^2 T\, g(E_F)}$$

Free electrons. Here $g(E_F) = 3n/2E_F$. Multiplying by V for N electrons, $C_{el} = \frac{\pi^2}{2} N k_B\, T/T_F$, which is the Ch. 10 §10.8.3 result. In this case $g'/g = 1/2E_F$, so $\mu = E_F[1 - \frac{\pi^2}{12}(k_BT/E_F)^2]$.

> [APPROXIMATION] The small parameter is $k_B T / E_F$, about 4×10^{-3} for Cu at 300 K. The expansion fails when $g(\varepsilon)$ has structure on the scale of $k_B T$ (van Hove singularities, narrow d or f bands, band edges), in non-degenerate semiconductors, and as $T \to T_F$.

D.9 — The Kubo Formula from First-Order Perturbation Theory

D.9.1 — Assumptions

1. **Equilibrium in the past.** The unperturbed system is in equilibrium in the distant past: $\hat{\rho}(-\infty) = \hat{\rho}_0 = e^{-\beta \hat{H}_0}/Z$.
2. **Weak coupling to one classical field.** A perturbation $\hat{H}(t) = \hat{H}_0 - \hat{A}\, F(t)$ acts, with F a classical c-number field (as in Ch. 13 §13.1.1) switched on adiabatically, $F(t) \propto e^{\eta t}$ with $\eta \to 0^+$.
3. **Linearity.** Only the response linear in F is kept.

> [APPROXIMATION] The small parameter is the ratio of the field-induced change in the state to the equilibrium state over the time window of interest. It is not simply "F small". Absorbed energy heats the system at second order in F, so a truly steady dissipative state also requires coupling to a bath, or a limit in which $V \to \infty$ is taken before $t \to \infty$. This is the substance of van Kampen's objection to linear response (*Physica Norvegica* **5**, 279 (1971)). The standard answer is that the linear coefficient is well defined in these limits, even if the microscopic trajectories respond nonlinearly.

D.9.2 — Derivation

The von Neumann equation (Ch. 10 §10.1.2) is $i\hbar\, \partial_t \hat{\rho} = [\hat{H}(t), \hat{\rho}]$. In the interaction picture,

$$\hat{\rho}_I(t) = e^{i\hat{H}_0 t/\hbar}\, \hat{\rho}(t)\, e^{-i\hat{H}_0 t/\hbar}, \qquad \hat{X}_I(t) = e^{i\hat{H}_0 t/\hbar}\, \hat{X}\, e^{-i\hat{H}_0 t/\hbar},$$

the equation becomes $i\hbar\, \partial_t \hat{\rho}_I = [\hat{H}'_I(t), \hat{\rho}_I]$ with $\hat{H}'_I = -\hat{A}_I(t) F(t)$. Note that $\hat{\rho}_0$ is stationary, since it commutes with $\hat{H}_0$. Integrating once, and keeping only first order in F:

$$\hat{\rho}_I(t) = \hat{\rho}_0 + \frac{1}{i\hbar}\int_{-\infty}^{t} [\hat{H}'_I(t'), \hat{\rho}_0]\, dt' + O(F^2) = \hat{\rho}_0 + \frac{i}{\hbar}\int_{-\infty}^{t} F(t')\, [\hat{A}_I(t'), \hat{\rho}_0]\, dt' + O(F^2).$$

The response of an observable $\hat{B}$ is $\langle \hat{B} \rangle(t) = \mathrm{Tr}[\hat{\rho}(t)\hat{B}] = \mathrm{Tr}[\hat{\rho}_I(t)\hat{B}_I(t)]$. Use the cyclic identity

$$\mathrm{Tr}\big([\hat{X}, \hat{\rho}_0]\hat{Y}\big) = \mathrm{Tr}\big(\hat{\rho}_0 [\hat{Y}, \hat{X}]\big).$$

Then

$$\delta\langle \hat{B} \rangle(t) = \int_{-\infty}^{\infty} \chi_{BA}(t - t')\, F(t')\, dt', \qquad \boxed{\chi_{BA}(t) = \frac{i}{\hbar}\, \Theta(t)\, \big\langle [\hat{B}_I(t), \hat{A}_I(0)] \big\rangle_0}$$

The response depends only on $t - t'$ because $\hat{\rho}_0$ is stationary.

> [DERIVATION] This is the result of Ch. 13 §13.1.2, exact to first order in F.

Sign check. For a perturbation written $\hat{H}'$, the first-order correction is $\delta\hat{\rho}_I = -(i/\hbar)\int[\hat{H}'_I, \hat{\rho}_0]\,dt'$. Ch. 13 §13.2.1 writes it with $+i/\hbar$. That sign is right only after substituting $\hat{H}' = -\hat{A}F$; its final χ_{BA} is correct.

Causality. The factor $\Theta(t)$ makes $\tilde{\chi}(\omega) = \int_0^\infty \chi(t)\, e^{i\omega t - \eta t}\, dt$ analytic in the upper half of the complex ω plane. That analyticity gives the Kramers–Kronig relations between $\tilde{\chi}'$ and $\tilde{\chi}''$, where $\tilde{\chi} = \tilde{\chi}' + i\tilde{\chi}''$.

D.9.3 — The Kubo canonical form

For many applications it is useful to rewrite χ in terms of the time derivative of $\hat{A}$.

Kubo identity. Define $G(\beta) = e^{\beta\hat{H}_0}[\hat{A}, e^{-\beta\hat{H}_0}]$. Then $G(0) = 0$ and $dG/d\beta = e^{\beta\hat{H}_0}[\hat{H}_0, \hat{A}]e^{-\beta\hat{H}_0}$. Integrating, and using $[\hat{H}_0, \hat{A}] = -i\hbar\dot{\hat{A}}$ with $e^{\lambda\hat{H}_0}\dot{\hat{A}}e^{-\lambda\hat{H}_0} = \dot{\hat{A}}(-i\hbar\lambda)$,

$$[\hat{A}, \hat{\rho}_0] = -i\hbar\,\hat{\rho}_0 \int_0^\beta \dot{\hat{A}}(-i\hbar\lambda)\, d\lambda.$$

Substituting into $\langle[\hat{B}(t), \hat{A}]\rangle_0 = \mathrm{Tr}([\hat{A}, \hat{\rho}_0]\hat{B}(t))$ gives

$$\chi_{BA}(t) = \Theta(t)\int_0^\beta \left\langle \dot{\hat{A}}(-i\hbar\lambda)\,\hat{B}(t)\right\rangle_0 d\lambda \xrightarrow{\hbar\to 0} \beta\,\Theta(t)\left\langle \dot{A}(0)\,B(t)\right\rangle_0.$$

The classical limit on the right is the same result that one obtains by linearising the classical Liouville equation directly.

What was proved. To first order in a weak classical field, the response is the retarded commutator in the equilibrium state. Equivalently it is the Kubo canonical correlation of $\dot{\hat{A}}$ with $\hat{B}$, whose classical limit is $\beta\langle\dot{A}B(t)\rangle$. Nothing in this derivation specifies a transport law. It specifies how to compute one, once $\hat{A}$ and $\hat{B}$ are chosen.

D.10 — The Fluctuation–Dissipation Theorem

D.10.1 — Classical limit

Assumptions. Classical commuting variables, the classical Kubo form of §D.9.3 with $B = A$, and $\langle A\rangle_0 = 0$.

Let $C(t) = \langle A(0)A(t)\rangle_0$. Stationarity gives $\frac{d}{ds}\langle A(s)A(t+s)\rangle = 0$, hence $\langle\dot{A}(0)A(t)\rangle = -\dot{C}(t)$, and so

$$\chi_{AA}(t) = -\beta\,\Theta(t)\,\dot{C}(t).$$

Classically $C(t)$ is real and even, so $\dot{C}$ is odd. Therefore

$$\tilde{\chi}''(\omega) = -\beta\int_0^\infty \dot{C}\sin\omega t\, dt = -\frac{\beta}{2}\int_{-\infty}^\infty \dot{C}\sin\omega t\, dt = \frac{\beta\omega}{2}\int_{-\infty}^\infty C(t)\cos\omega t\, dt,$$

after integrating by parts. That is,

$$\boxed{\tilde{\chi}''_{AA}(\omega) = \frac{\omega}{2k_BT}\,S_A(\omega), \qquad S_A(\omega) = \int_{-\infty}^\infty C(t)\,e^{i\omega t}\,dt}$$

which is the classical boxed formula of Ch. 13 §13.3.1.

Static limit. At $\omega = 0$, $\tilde{\chi}_{AA}(0) = -\beta \int_0^\infty \dot{C}\, dt = \beta[C(0) - C(\infty)] = \beta\langle \delta A^2 \rangle$. A static susceptibility equals β times an equilibrium variance. The heat-capacity formula of Ch. 10 §10.4 has this structure.

D.10.2 — Why $\tilde{\chi}''$ is dissipation

Under $F = F_0 \cos \omega t$, the power absorbed is $\langle d\hat{H}/dt \rangle = -\langle \hat{A} \rangle \dot{F}$. Averaged over a period, only the out-of-phase part of $\langle \hat{A} \rangle$ survives:

$$\bar{P} = \frac{\omega F_0^2}{2}\, \tilde{\chi}''(\omega).$$

Passivity of an equilibrium state requires $\bar{P} \geq 0$. That is guaranteed by the spectral representation below.

D.10.3 — Quantum form, with the operator ordering explicit

Define the two orderings

$$S_>(\omega) = \int_{-\infty}^{\infty} \langle \hat{A}(t)\hat{A}(0) \rangle_0\, e^{i\omega t}\, dt, \qquad S_<(\omega) = \int_{-\infty}^{\infty} \langle \hat{A}(0)\hat{A}(t) \rangle_0\, e^{i\omega t}\, dt = S_>(-\omega).$$

The equilibrium state satisfies the KMS condition $\langle \hat{A}(t)\hat{B} \rangle_0 = \langle \hat{B}\hat{A}(t + i\hbar\beta) \rangle_0$. Shifting the contour by $i\hbar\beta$ gives the detailed-balance relation

$$S_>(-\omega) = e^{-\beta\hbar\omega} S_>(\omega).$$

From §D.9.2, $\tilde{\chi}''(\omega) = \frac{1}{2\hbar} \int \langle [\hat{A}(t), \hat{A}(0)] \rangle e^{i\omega t}\, dt = \frac{1}{2\hbar}\big[S_>(\omega) - S_>(-\omega)\big]$. Therefore

$$\tilde{\chi}''_{AA}(\omega) = \frac{1 - e^{-\beta\hbar\omega}}{2\hbar}\, S_>(\omega) = \frac{e^{\beta\hbar\omega} - 1}{2\hbar}\, S_<(\omega).$$

The spectral form $S_>(\omega) = 2\pi \sum_{nm} p_n |A_{nm}|^2\, \delta\big(\omega - (E_m - E_n)/\hbar\big) \geq 0$ shows that $\tilde{\chi}''(\omega) \geq 0$ for $\omega > 0$, as passivity requires.

Ordering note. Ch. 13 §13.3.1 defines S_A with the ordering $\langle \hat{A}(0)\hat{A}(t) \rangle$, which is $S_<$ here. Its prefactor $(1 - e^{-\beta\hbar\omega})/2\hbar$ and its detailed-balance relation $S_A(-\omega) = e^{-\beta\hbar\omega} S_A(\omega)$ hold for the opposite ordering, $S_>$, with the $e^{+i\omega t}$ convention. All orderings agree in the classical limit $\beta\hbar\omega \to 0$, and the symmetrised coth form quoted there is unaffected. See Kubo, *Rep. Prog. Phys.* **29**, 255 (1966), and Callen and Welton, *Phys. Rev.* **83**, 34 (1951).

> `[DERIVATION]` Given equilibrium (KMS) and linear response, the FDT is exact. The classical form is the limit $\hbar\omega \ll k_B T$. The FDT does not hold in non-equilibrium steady states, such as driven or aging systems, where violations of it are a diagnostic.

D.11 — Green–Kubo Expressions: Diffusion and Conductivity

D.11.1 — Diffusion from the velocity autocorrelation

Assumptions.

1. A tagged particle in an equilibrium, stationary environment.

2. The velocity autocorrelation $C_v(t) = \langle v_x(0) v_x(t) \rangle$ decays fast enough that $\int_0^\infty C_v\, dt$ and $\int_0^\infty t\, C_v\, dt$ converge.
3. At long times and long wavelengths, the tagged-particle density obeys the diffusion equation $\partial_t c = D \nabla^2 c$.

Assumption 3 is the hydrodynamic hypothesis, which supplies the definition of D.

From displacement to correlation. Since $x(t) - x(0) = \int_0^t v_x(s)\, ds$, stationarity gives

$$\langle [x(t) - x(0)]^2 \rangle = \int_0^t \!\! \int_0^t C_v(s - s')\, ds\, ds' = 2 \int_0^t (t - u)\, C_v(u)\, du \xrightarrow{t \gg \tau_c} 2t \int_0^\infty C_v\, du - 2 \int_0^\infty u\, C_v\, du.$$

From the diffusion equation. A point source spreading under the diffusion equation has $\langle x^2 \rangle = 2Dt$ per Cartesian direction. Matching the coefficients of t:

$$\boxed{D = \int_0^\infty \langle v_x(0) v_x(t) \rangle\, dt = \frac{1}{3} \int_0^\infty \langle \mathbf{v}(0) \cdot \mathbf{v}(t) \rangle\, dt}$$

which is the formula of Ch. 13 §13.6.1.

> `[APPROXIMATION]` The small parameters are τ_c / t, the correlation time compared with the observation time, and ℓ / L, the microscopic length compared with the length over which the concentration varies. Assumption 2 can fail. In fluids, C_v has a hydrodynamic long-time tail $\propto t^{-d/2}$ (Alder and Wainwright, *Phys. Rev. A* **1**, 18 (1970)). In $d = 3$ the tail is integrable and D exists. In $d = 2$ the integral diverges logarithmically, and a strictly two-dimensional fluid has no finite D.

Example: Langevin dynamics. For $m\dot{v} = -mv/\tau + \xi(t)$ with white noise ξ normalised so that $\langle v^2 \rangle = k_B T / m$ (equipartition, legitimate for a non-degenerate tracer), $C_v = (k_B T / m) e^{-|t|/\tau}$ and $D = k_B T \tau / m$. This is the example of Ch. 13 §13.6.1.

D.11.2 — Conductivity

Couple a uniform field to the total dipole: $\hat{H}' = -\hat{\mathbf{P}} \cdot \mathbf{E}(t)$ with $\hat{\mathbf{P}} = \sum_i q_i \hat{\mathbf{x}}_i$, so that $\dot{\hat{\mathbf{P}}} = \hat{\mathbf{J}}$, the total current. Take the response $\hat{j}_\alpha = \hat{J}_\alpha / V$. Then §D.9.3 gives

$$\sigma_{\alpha\beta}(\omega) = \frac{1}{V} \int_0^\infty dt\, e^{i\omega t - \eta t} \int_0^\beta d\lambda\, \big\langle \hat{J}_\beta(-i\hbar\lambda)\, \hat{J}_\alpha(t) \big\rangle_0.$$

Two remarks on this expression:

- **Index order.** $\sigma_{\alpha\beta}$ is the response of j_α to E_β, so the *driven* index β sits on the imaginary-time argument. Ch. 13 §13.4.1 places it the other way round. For an isotropic system, or for symmetric σ, this is immaterial. For the Hall response it is a transposition, $\sigma_{\alpha\beta} \leftrightarrow \sigma_{\beta\alpha}$, which by §D.15 is equivalent to $\mathbf{B} \to -\mathbf{B}$.
- **Dipole coupling.** $\hat{\mathbf{P}}$ is ill-defined under periodic boundary conditions. The careful derivation couples to a vector potential, with $\mathbf{E} = -\partial_t \mathbf{A}$. That produces a paramagnetic current–current term plus a diamagnetic term ne^2/m, which combine into the expression above. See Mahan, *Many-Particle Physics*, chapter on linear response.

DC limit. In the classical limit, the λ integral gives a factor β. For an isotropic system,

$$\sigma_{DC} = \frac{1}{3Vk_BT}\int_0^\infty \langle \hat{\mathbf{J}}(0)\cdot\hat{\mathbf{J}}(t)\rangle_0\, dt,$$

as in Ch. 13 §13.4.1. Quantum mechanically, the λ integral reduces to β exactly whenever $\hat{J}$ commutes with $\hat{H}_0$ (§D.12). Otherwise it reduces to β only in the classical limit, so the statement in Ch. 13 that "the $\omega = 0$ correlator is λ-independent" should be read as one of these two cases.

Other coefficients. The thermal conductivity $\kappa = (Vk_BT^2)^{-1}\int_0^\infty \langle \hat{J}_Q \hat{J}_Q(t)\rangle\, dt$ and the viscosity η have the same Green–Kubo structure. But ∇T is not a term in the Hamiltonian, so the derivation of §D.9 does not apply directly. Luttinger's device of a fictitious gravitational potential coupled to the energy density, *Phys. Rev.* **135**, A1505 (1964), or a hydrodynamic regression argument, supplies it. That derivation is out of scope here.

D.12 — The Drude Conductivity under the Relaxation-Time Assumption

D.12.1 — The exact free-gas correlator

Assumptions.

1. Non-interacting electrons with quadratic dispersion $\varepsilon_k = \hbar^2k^2/2m$.
2. The grand canonical ensemble.

Then $\hat{J}_x = -(e\hbar/m)\sum_k k_x \hat{n}_k$, where the sum includes spin. This operator commutes with $\hat{H}_0$, so it is conserved. Hence $\hat{J}_x(-i\hbar\lambda) = \hat{J}_x$, and the Kubo correlation reduces to $\beta\langle \hat{J}_x^2\rangle$.

Using the mode independence of §D.4.3, $\langle n_k n_{k'}\rangle = f_k f_{k'} + \delta_{kk'} f_k(1-f_k)$. The first term contributes $(\sum_k k_x f_k)^2 = 0$. Then:

- Use $f(1-f) = -k_BT\,\partial f/\partial\varepsilon$.
- Use $\partial f/\partial k_x = (\hbar^2 k_x/m)\,\partial f/\partial\varepsilon$.
- Integrate by parts in k_x.

The result is

$$\frac{\beta}{V}\langle \hat{J}_x^2\rangle = \frac{\beta e^2\hbar^2}{m^2V}\sum_k k_x^2 f_k(1-f_k) = \frac{e^2}{mV}\sum_k f_k = \frac{ne^2}{m}.$$

This holds **at any degeneracy**. The factor k_BT that appears is not equipartition. It comes from $f(1-f) = -k_BT\,\partial f/\partial\varepsilon$, which converts the sum into a Fermi-surface integral, exactly as in the Boltzmann route of Ch. 13 §13.4.2.

The exact Kubo result for the free gas is therefore

$$\sigma(\omega) = \frac{ne^2}{m}\int_0^\infty e^{i\omega t - \eta t}\, dt = \frac{ne^2}{m}\,\frac{i}{\omega + i\eta},$$

a pure Drude weight, with $\sigma_{DC} = \infty$ as $\eta \to 0$. This agrees with Ch. 13 §13.4.2, obstruction 1.

Note on the occupation correlator. Ch. 13 §13.4.2 writes $\langle \hat{n}_k \hat{n}_{k'}\rangle_0 = \delta_{kk'} f_k$. For $k \neq k'$ the correct value is $f_k f_{k'}$, and the connected part $\delta_{kk'} f_k(1-f_k)$ is precisely what enters the current correlator. That chapter's conclusion, that a closure is required, is unaffected.

D.12.2 — The relaxation-time closure

> [PHENOMENOLOGICAL] Assume momentum-relaxing collisions make the current correlator decay as $\langle \hat{J}_x; \hat{J}_x(t)\rangle = \langle \hat{J}_x; \hat{J}_x\rangle\, e^{-t/\tau}$, with a single τ and the equal-time value of §D.12.1 unchanged. Here $\langle \cdot\,;\cdot\rangle$ denotes the Kubo canonical correlation of §D.9.3.

Then

$$\boxed{\sigma(\omega) = \frac{ne^2}{m}\int_0^\infty e^{-t/\tau}e^{i\omega t}\,dt = \frac{\sigma_0}{1-i\omega\tau}, \qquad \sigma_0 = \frac{ne^2\tau}{m}}$$

The Boltzmann route of Ch. 13 §13.4.2 gives the same result with $\partial_t \delta f = -i\omega\,\delta f$ retained, since $(-i\omega + 1/\tau)\,\delta f$ replaces $\delta f/\tau$.

A check independent of τ. The real part satisfies

$$\int_0^\infty \mathrm{Re}\,\sigma\, d\omega = \sigma_0\frac{\pi}{2\tau} = \frac{\pi n e^2}{2m}$$

for any τ. This is the f-sum rule. Scattering redistributes spectral weight; it does not create or remove it.

> [APPROXIMATION] The closure requires well-defined quasiparticles, $\hbar/\tau \ll E_F$, a momentum-relaxing scattering mechanism, and τ independent of energy on the scale k_BT. It fails for:
>
> - interband transitions at optical frequencies;
> - the anomalous skin effect, where the mean free path $\ell = v_F\tau$ exceeds the field penetration depth;
> - ballistic samples with $\ell > L$ (Ch. 13 §13.11);
> - strongly correlated metals.
>
> Replacing m by an effective mass m^* is a band-structure correction, not part of the closure.

D.13 — The Einstein Relation

D.13.1 — From Kubo, for a non-degenerate particle

Assumptions. A classical tagged particle, with the perturbation $\hat{H}' = -xF$ from a constant force F. Then $\dot{A} = \dot{x} = v$, and the classical form of §D.9.3 gives the drift velocity

$$\langle v\rangle = F\beta\int_0^\infty \langle v(0)v(t)\rangle\, dt = \frac{D}{k_BT}F.$$

The second equality uses §D.11.1. So the force mobility is $\mu_F = \langle v\rangle/F = D/k_BT$. For charge q the electrical mobility is $\mu_{mob} = q\mu_F$, so

$$\boxed{D = \frac{k_BT}{q}\,\mu_{mob}}$$

as in Ch. 13 §13.6.2. The Kubo route makes explicit that this is the FDT applied to one particle: the same correlation function controls both spreading (D) and drift (μ).

D.13.2 — General (degenerate) form

Assumptions. Local equilibrium, and a single diffusion constant governing the particle flux $-D\nabla n$.

In equilibrium in an electrostatic potential ϕ, the electrochemical potential $\mu(n(\mathbf{x})) - e\phi(\mathbf{x})$ is constant. Hence

$$\nabla n = -e\frac{\partial n}{\partial \mu}\mathbf{E}.$$

The electrical current density is the drift term $\sigma\mathbf{E}$ plus the diffusive term $(-e)(-D\nabla n)$, and it must vanish in equilibrium:

$$\mathbf{j} = \sigma\mathbf{E} - e^2 D\frac{\partial n}{\partial \mu}\mathbf{E} = 0 \quad \Longrightarrow \quad \boxed{\sigma = e^2\frac{\partial n}{\partial \mu}D}$$

Limits.

- **Non-degenerate.** $\partial n/\partial\mu = n/k_BT$, which gives $D = \mu_{mob}k_BT/e$. This is the semiconductor relation used in Ch. 14 §14.5.1, with $V_T = k_BT/e$.
- **Degenerate.** $\partial n/\partial\mu = g(E_F)$, which gives $\sigma = e^2 g(E_F)D$ with $D = v_F^2\tau/3$.

For degenerate electrons the factor k_BT/e is replaced by $n/(e\,g(E_F)) = 2E_F/3e$ for free electrons. Using k_BT/e for metals or degenerate semiconductors is an error.

D.14 — Wiedemann–Franz and the Lorenz Number

D.14.1 — Assumptions

1. Linearised Boltzmann transport with a relaxation time $\tau(\varepsilon)$ that may depend on energy but is the same for charge and heat currents.
2. **Elastic scattering.** This is what makes assumption 1 legitimate: an elastic collision degrades a charge current and a heat current carried by the same electron equally.
3. Isotropy.
4. Degeneracy, $k_BT \ll E_F$, with $\tau(\varepsilon)$, $v(\varepsilon)$ and $g(\varepsilon)$ smooth on the scale k_BT.
5. Heat is carried by electrons only; phonon conduction is neglected.

D.14.2 — Transport integrals

The distribution. With a local equilibrium f_0 whose μ and T vary in space, an electric field $\mathbf{E}$, and $\mathbf{F} = -e\mathbf{E}$, the steady linearised Boltzmann equation gives

$$\delta f = \tau(\varepsilon)\left(-\frac{\partial f_0}{\partial\varepsilon}\right)\mathbf{v}\cdot\left[-e\mathcal{E} - (\varepsilon-\mu)\frac{\nabla T}{T}\right], \qquad \mathcal{E} \equiv \mathbf{E} + \frac{\nabla\mu}{e}.$$

This reduces to Ch. 13 §13.4.2 when $\nabla T = \nabla\mu = 0$.

The integrals. Define $\sigma(\varepsilon) = e^2\tau(\varepsilon)v(\varepsilon)^2 g(\varepsilon)/3$ and

$$\mathcal{L}^{(\alpha)} = \int d\varepsilon \left(-\frac{\partial f_0}{\partial \varepsilon}\right) (\varepsilon - \mu)^\alpha \, \sigma(\varepsilon).$$

The currents. The charge current $\mathbf{j} = -e\int \mathbf{v}\,\delta f$ and the heat current $\mathbf{j}_Q = \int (\varepsilon-\mu)\mathbf{v}\,\delta f$ are then

$$\mathbf{j} = \mathcal{L}^{(0)}\mathcal{E} + \frac{\mathcal{L}^{(1)}}{eT}\nabla T, \qquad \mathbf{j}_Q = -\frac{\mathcal{L}^{(1)}}{e}\mathcal{E} - \frac{\mathcal{L}^{(2)}}{e^2 T}\nabla T.$$

The coefficients. At $\mathbf{j} = 0$ the field is $\mathcal{E} = S\nabla T$. This gives

$$\sigma = \mathcal{L}^{(0)}, \qquad S = -\frac{\mathcal{L}^{(1)}}{eT\mathcal{L}^{(0)}}, \qquad \kappa = \frac{1}{e^2 T}\left[\mathcal{L}^{(2)} - \frac{(\mathcal{L}^{(1)})^2}{\mathcal{L}^{(0)}}\right].$$

D.14.3 — Sommerfeld evaluation

The moments of $-\partial f_0/\partial\varepsilon$ from §D.8 give

$$\mathcal{L}^{(0)} = \sigma(\mu) + O(T^2), \qquad \mathcal{L}^{(1)} = \frac{\pi^2}{3}(k_B T)^2 \sigma'(\mu), \qquad \mathcal{L}^{(2)} = \frac{\pi^2}{3}(k_B T)^2 \sigma(\mu).$$

The term $(\mathcal{L}^{(1)})^2/\mathcal{L}^{(0)}$ is $O(T^4)$, so

$$\boxed{\frac{\kappa}{\sigma T} = \frac{\pi^2}{3}\left(\frac{k_B}{e}\right)^2 = 2.443 \times 10^{-8}\ \mathrm{W\,\Omega\,K^{-2}}}$$

with corrections of relative order $(k_B T/E_F)^2$. This is Ch. 13 §13.5.2. The derivation needs neither parabolic dispersion nor energy-independent τ: $\sigma(\varepsilon)$ is arbitrary as long as it is smooth. What it needs is the *same* $\tau(\varepsilon)$ for both currents, which is assumption 2. Ch. 13 §13.5.2 lists "Galilean-invariant parabolic dispersion" among the requirements, which is stronger than necessary.

> `[APPROXIMATION]` Small parameters: $k_B T/E_F$, and the inelasticity of scattering. Inelastic small-angle electron–phonon scattering at intermediate T relaxes heat currents more efficiently than charge currents, pushing $\kappa/\sigma T$ below L_0. This is the regime table of Ch. 13 §13.5.2.

D.14.4 — By-product: the Mott formula

From the same integrals,

$$S = -\frac{\pi^2 k_B^2 T}{3e}\frac{\sigma'(\mu)}{\sigma(\mu)} = -\frac{\pi^2 k_B^2 T}{3eE_F}\left.\frac{d\ln\sigma}{d\ln\varepsilon}\right|_{E_F}.$$

For free electrons $\sigma(\varepsilon) \propto \varepsilon^{3/2}\tau(\varepsilon)$, so $d\ln\sigma/d\ln\varepsilon = 3/2 + d\ln\tau/d\ln\varepsilon$:

- Constant τ gives $S = -\pi^2 k_B^2 T/2eE_F$.
- Constant mean free path ($\tau \propto \varepsilon^{-1/2}$) gives $S = -\pi^2 k_B^2 T/3eE_F$, the form quoted in Ch. 13 §13.9.3.

Numerical check. At 300 K with $E_F = 7$ eV, these give $-1.6\,\mu$V/K and $-1.05\,\mu$V/K respectively. The measured thermopower of Cu at room temperature is about $+1.8\,\mu$V/K. The sign is positive because $\sigma'(\mu)/\sigma$ for copper's real Fermi surface and scattering is not the free-electron value. The Mott formula still holds; the free-electron input to it fails. Both the sign and the magnitude quoted in Ch. 13 §13.9.3 should therefore be treated with care.

D.15 — Onsager Reciprocity

D.15.1 — Statement

Let fluxes J_i and forces X_i be chosen so that the entropy production is $\dot{s} = \sum_i J_i X_i$. In the linear regime, $J_i = \sum_j L_{ij} X_j$. Then

$$\boxed{L_{ij}(\mathbf{B}) = \varepsilon_i \varepsilon_j \, L_{ji}(-\mathbf{B})}$$

where $\varepsilon_i = \pm 1$ is the time-reversal parity of the quantity whose rate is J_i, and $\mathbf{B}$ is any external magnetic field. A rotation rate $\boldsymbol{\Omega}$ is treated the same way as $\mathbf{B}$. Charge and heat currents have the same parity, so $\varepsilon_i \varepsilon_j = +1$ for thermoelectric coefficients.

D.15.2 — Assumptions

1. **Microscopic reversibility.** If $\Gamma(t)$ is a trajectory with field $\mathbf{B}$, then $\Theta\Gamma(-t)$ is a trajectory with field $-\mathbf{B}$, where Θ maps $(\mathbf{q}, \mathbf{p}) \to (\mathbf{q}, -\mathbf{p})$. For the flow map this reads $\varphi_t^{\mathbf{B}} \Theta = \Theta \, \varphi_{-t}^{-\mathbf{B}}$.
2. **Invariant equilibrium.** $\rho_0^{\mathbf{B}}(\Theta\Gamma) = \rho_0^{-\mathbf{B}}(\Gamma)$, and the phase-space measure is Θ-invariant.
3. **Definite parity.** $J_i(\Theta\Gamma) = \varepsilon_i J_i(\Gamma)$.
4. **Green–Kubo representation.** $L_{ij} = c \int_0^\infty \langle J_i(0) J_j(t) \rangle_0 \, dt$, with the same positive constant c (for example $1/V k_B$ for forces normalised as in §D.15.4) for every pair. For mechanical forces this is §D.9. For thermal forces it is the Luttinger or regression argument of §D.11.2, which is an additional assumption.
5. **Classical variables.** The quantum proof uses the Kubo canonical correlation and an antiunitary Θ, and goes through the same way (Kubo, *J. Phys. Soc. Jpn.* **12**, 570 (1957)).

D.15.3 — Proof

Let $C_{ij}(t; \mathbf{B}) = \int d\Gamma \, \rho_0^{\mathbf{B}}(\Gamma) \, J_i(\Gamma) \, J_j(\varphi_t^{\mathbf{B}} \Gamma)$. Substitute $\Gamma = \Theta\Gamma'$ and use assumptions 1–3:

$$C_{ij}(t; \mathbf{B}) = \int d\Gamma' \, \rho_0^{-\mathbf{B}}(\Gamma') \, \varepsilon_i J_i(\Gamma') \, J_j\big(\Theta \varphi_{-t}^{-\mathbf{B}} \Gamma'\big) = \varepsilon_i \varepsilon_j \, C_{ij}(-t; -\mathbf{B}).$$

Stationarity and commuting classical variables give $C_{ij}(-t) = \langle J_i(t) J_j(0) \rangle = C_{ji}(t)$. Hence $C_{ij}(t; \mathbf{B}) = \varepsilon_i \varepsilon_j \, C_{ji}(t; -\mathbf{B})$, and integrating over t with assumption 4 proves the statement.

> `[DERIVATION]` Reciprocity follows from microscopic reversibility plus a Green–Kubo representation. It is not a separate empirical law.

Ch. 13 §13.9.2 gives the argument without $\mathbf{B}$ and without the parities. With a magnetic field present, the correct statement is the Onsager–Casimir form above (Casimir, *Rev. Mod. Phys.* **17**, 343 (1945)), for example $\sigma_{xy}(\mathbf{B}) = \sigma_{yx}(-\mathbf{B})$ for the Hall effect. Onsager's original papers are *Phys. Rev.* **37**, 405 and **38**, 2265 (1931).

D.15.4 — The Kelvin relation as a check

The entropy production of §D.14 is $\dot{s} = \mathbf{j} \cdot \mathcal{E}/T + \mathbf{j}_Q \cdot (-\nabla T / T^2)$. With the conjugate forces $X_1 = \mathcal{E}/T$ and $X_2 = -\nabla T / T^2$, the currents of §D.14.2 become

$$L_{11} = T\mathcal{L}^{(0)}, \qquad L_{12} = -\frac{T\mathcal{L}^{(1)}}{e}, \qquad L_{21} = -\frac{T\mathcal{L}^{(1)}}{e}, \qquad L_{22} = \frac{T\mathcal{L}^{(2)}}{e^2},$$

so $L_{12} = L_{21}$ as required.

Kelvin relation. The Peltier coefficient at $\nabla T = 0$ is $\Pi = j_Q/j = -\mathcal{L}^{(1)}/e\mathcal{L}^{(0)} = TS$, which is the Kelvin relation of Ch. 13 §13.9.3.

Convention note. With these conjugate forces, $\sigma = L_{11}/T$ and $\kappa = (L_{22} - L_{12}L_{21}/L_{11})/T^2$. Ch. 13 §13.9.1 uses these same $1/T$ and $1/T^2$ factors, but defines its forces as $\mathbf{X}_E = \mathbf{E} - \nabla\mu/e$ and $\mathbf{X}_Q = -\nabla T/T$. Those forces correspond to different normalisations of L, so the factors and the forces there should be made consistent. The sign of the $\nabla\mu$ term also depends on the carrier charge: it is $+\nabla\mu/e$ for electrons of charge $-e$.

Where reciprocity fails. Outside linear response (for example, rectification in non-centrosymmetric conductors at second order). When Θ-invariance of the equilibrium state fails for reasons not captured by external fields, as in spontaneously magnetised systems, the order parameter must be reversed together with $\mathbf{B}$. And when the chosen fluxes and forces are not thermodynamically conjugate, the matrix of coefficients need not be symmetric.

D.16 — The Scope of Kubo, Stated Once

The derivations above support a precise version of the convergence claim of Ch. 13, which is *synthesis* in the sense of CONTRIBUTING §6.

1. **Exact within its domain.** Kubo gives every linear transport coefficient as an equilibrium correlation function. This is exact to first order in the driving field (§D.9).
2. **What turns it into a number.** Reaching a specific value of σ, κ, D or η requires several further inputs: the correct operator pair; the order of limits (the volume limit $V \to \infty$ before $\omega \to 0$); the separation of diamagnetic, contact and Drude-weight pieces (§D.12.1); for thermal forces, a further argument (§D.11.2); and in practice a closure such as τ (§D.12.2), which is `[PHENOMENOLOGICAL]`.
3. **What the laws are.** Ohm's, Fourier's, Fick's and Newton's viscosity laws are constitutive relations whose coefficients can be expressed in this framework, each in its own linear-response regime. No single calculation produces them all. Where the framework does not apply (ballistic transport, Ch. 13 §13.11; far from equilibrium), other descriptions such as Landauer's take over. The statement in Ch. 14 §14.7.1 that Fourier's and Ohm's laws are "the same Kubo result applied to different currents" is a `[STRUCTURAL CONNECTION]`: they are the same template, not the same calculation.

D.17 — Summary

Result	Statement	Assumptions	Used in
Canonical ensemble	$p_n = e^{-\beta E_n}/Z$, $\beta = \partial_E S_R/k_B$	equal a priori probabilities; weak coupling; $E_n/(TC_R) \to 0$	Ch. 10 §10.3
Thermodynamics from Z	$F = -k_BT \ln Z$; $T\,dS = \sum E_n\,dp_n$; $P = k_BT\,\partial_V \ln Z$	canonical ensemble; thermodynamic limit for identification	Ch. 10 §10.4

Result	Statement	Assumptions	Used in
Equipartition	$\langle x_i \partial_j H \rangle = \delta_{ij} k_B T$; $\frac{1}{2} k_B T$ per quadratic term	classical phase space; vanishing boundary terms	Ch. 10 §10.7.2
Its failure	$\langle E \rangle = k_B T + (\hbar\omega)^2/12 k_B T + \cdots$; frozen for $\hbar\omega \gg k_B T$	quantised oscillator	Ch. 10 §10.7.3
FD and BE	$\bar{n} = 1/(e^{\beta(\varepsilon-\mu)} \pm 1)$; $\langle \delta n^2 \rangle = \bar{n}(1 \mp \bar{n})$	ideal quantum gas, grand canonical; $\mu < \varepsilon_{min}$ for bosons	Ch. 10 §10.8.2
Classical limit	MB when $n\lambda_{th}^3/g_s \ll 1$; $P = nk_BT(1 \pm n\lambda_{th}^3/2^{5/2} g_s)$	ideal gas	Ch. 10 §10.7.1, §10.8.2
Planck law	$u(\nu) = (8\pi h\nu^3/c^3)/(e^{h\nu/k_BT} - 1)$	free field; 2 polarisations; $\mu = 0$; large cavity	Ch. 10 §10.8.4
Debye T^3	$C_V \simeq (12\pi^4/5) N k_B (T/T_D)^3$	harmonic crystal; $T \ll T_D$; coefficient set by sound speeds	Ch. 10 §10.7.3
Sommerfeld	$\int Hf = \int_0^\mu H + (\pi^2/6)(k_BT)^2 H'(\mu)$; $c_{el} = (\pi^2/3) k_B^2 T\, g(E_F)$	$k_BT \ll E_F$; smooth g	Ch. 10 §10.8.3
Kubo formula	$\chi_{BA}(t) = (i/\hbar)\Theta(t)\langle[\hat{B}(t), \hat{A}(0)]\rangle_0$	equilibrium past; first order in a classical field	Ch. 13 §13.1–§13.2
FDT	classical $\tilde{\chi}'' = (\omega/2k_BT) S$; quantum $\tilde{\chi}'' = (1 - e^{-\beta\hbar\omega}) S_>/2\hbar$	equilibrium (KMS); linear response	Ch. 13 §13.3
Green–Kubo D	$D = \frac{1}{3}\int_0^\infty \langle \mathbf{v}(0) \cdot \mathbf{v}(t) \rangle\, dt$	integrable VACF; hydrodynamic diffusion; $d = 3$	Ch. 13 §13.6.1
Green–Kubo σ	$\sigma_{\alpha\beta}(\omega) = V^{-1} \int dt\, e^{i\omega t} \int d\lambda\, \langle J_\beta(-i\hbar\lambda) J_\alpha(t) \rangle$	as Kubo; $V \to \infty$ before $\omega \to 0$	Ch. 13 §13.4.1
Drude	$\sigma(\omega) = \sigma_0/(1 - i\omega\tau)$, $\sigma_0 = ne^2\tau/m$; f-sum $\pi n e^2/2m$	exact free-gas weight plus single-τ closure	Ch. 13 §13.4.2; Ch. 14 §14.1
Einstein relation	$D = k_B T \mu_{mob}/q$ (non-degenerate); $\sigma = e^2(\partial n/\partial\mu) D$ (general)	linear response; local equilibrium	Ch. 13 §13.6.2; Ch. 14 §14.5
Wiedemann–Franz	$\kappa/\sigma T = \pi^2 k_B^2/3e^2$	elastic scattering (same $\tau(\varepsilon)$); $k_BT \ll E_F$; electrons only	Ch. 13 §13.5.2; Ch. 14 §14.4
Mott formula	$S = -(\pi^2 k_B^2 T/3e)\, \sigma'(\mu)/\sigma(\mu)$	as Wiedemann–Franz	Ch. 13 §13.9.3; Ch. 14 §14.3
Onsager–Casimir	$L_{ij}(\mathbf{B}) = \varepsilon_i \varepsilon_j L_{ji}(-\mathbf{B})$; Kelvin $\Pi = TS$	microscopic reversibility; Green–Kubo form; conjugate forces	Ch. 13 §13.9; Ch. 14 §14.3

D.18 — References

Standard sources used in this appendix (see also `REFERENCES.md`):

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Appendix E — Continuum Balance Laws and Their Reduction to Lumped Networks

This appendix supplies the proofs that the continuum and Bridge C chapters state, sketch, or use without derivation. It proves the Reynolds transport theorem; derives the local mass, momentum, and energy balances from their integral forms; proves the symmetry of the Cauchy stress from angular-momentum balance; derives Navier–Stokes from the Newtonian constitutive assumption; solves Hagen–Poiseuille flow exactly; derives the heat equation and the lumped-capacitance model, with an explicit error estimate in the Biot number; reduces Maxwell's equations to the telegrapher's equations and then to a lumped ladder; derives KVL from Faraday's law by lumping, with the inductive branch voltage kept; proves Poynting's theorem and the lumped port-power identity $P = v\,i$; derives the Euler–Bernoulli beam from three-dimensional elasticity and states what Timoshenko adds; and proves the Buckingham π theorem. The chapters that rely on these results are Ch. 11 (§11.8, §11.12), Ch. 15 (§15.3–§15.8), Ch. 16 (§16.2, §16.3, §16.7.2, §16.10), Ch. 17 (§17.1, §17.4, §17.10), Ch. 19 (§19.6, §19.7), and Ch. 20 (§20.2), together with the worked examples 03, 04, and 05.

Two distinctions organise everything that follows, and they are the distinctions the book insists on throughout. First, **balance laws are not constitutive laws.** Mass, momentum, angular momentum, energy, and charge balances hold for any continuum; Newtonian viscosity, Fourier conduction, Hooke elasticity, and Ohm conduction are closures that describe a particular class of material and can fail without any conservation law failing. Second, **lumping is an approximation with a small parameter**, never a free identity. Each section below names which steps are exact and which step discards something.

Notation. It matches Ch. 15: velocity $\mathbf{v}$, density ρ, Cauchy stress σ_{ij} with traction $t_i = \sigma_{ij} n_j$, strain-rate tensor $\dot{\varepsilon}_{ij} = \tfrac{1}{2}(\partial_i v_j + \partial_j v_i)$, material derivative $D/Dt = \partial_t + \mathbf{v}\cdot\nabla$, specific internal energy e, heat flux $\mathbf{q}$, volumetric heat source $\dot{Q}_{vol}$. Thermal conductivity is κ (as in Ch. 16 and Ch. 19; Example 04 writes k). Summation over repeated indices is implied.

E.1 — The Reynolds Transport Theorem

E.1.1 — Setting and assumptions

Label each material particle by its position $\mathbf{X}$ in a reference configuration at $t = 0$. The motion is a map $\mathbf{x} = \boldsymbol{\chi}(\mathbf{X}, t)$ with velocity $\mathbf{v}(\mathbf{x}, t) = \partial_t \boldsymbol{\chi}(\mathbf{X}, t)$ evaluated at $\mathbf{X} = \boldsymbol{\chi}^{-1}(\mathbf{x}, t)$. Assume:

1. $\boldsymbol{\chi}$ is continuously differentiable in $\mathbf{X}$ and t and invertible at each t — no fracture, cavitation, or interpenetration;
2. the deformation gradient $F_{iK} = \partial \chi_i / \partial X_K$ has $J = \det F > 0$;
3. the field $f(\mathbf{x}, t)$ being transported is continuously differentiable on a neighbourhood of the material volume $V_m(t) = \boldsymbol{\chi}(V_0, t)$.

The continuum hypothesis itself (Ch. 15 §15.1.1) is presupposed: the fields are coarse-grained averages over a volume much larger than the microscopic scale and much smaller than the scale on which the fields vary.

E.1.2 — Euler's expansion formula

Lemma. $\partial J/\partial t\,|_{\mathbf{X}} = J\,\nabla\cdot\mathbf{v}$.

Proof. Jacobi's formula for the derivative of a determinant gives $dJ = J\,\mathrm{tr}(F^{-1}dF)$. Differentiating F in time at fixed $\mathbf{X}$ and using the chain rule,

$$\frac{\partial F_{iK}}{\partial t} = \frac{\partial v_i}{\partial X_K} = \frac{\partial v_i}{\partial x_j}\frac{\partial x_j}{\partial X_K} = (\partial_j v_i)\,F_{jK}$$

so $\partial_t F = (\nabla\mathbf{v})F$ with $(\nabla\mathbf{v})_{ij} = \partial_j v_i$. Then $\mathrm{tr}(F^{-1}\partial_t F) = \mathrm{tr}(F^{-1}(\nabla\mathbf{v})F) = \mathrm{tr}(\nabla\mathbf{v}) = \nabla\cdot\mathbf{v}$, by cyclicity of the trace. $\square$

E.1.3 — The theorem

Theorem (Reynolds). Under E.1.1,

$$\frac{d}{dt}\int_{V_m(t)} f\,dV = \int_{V_m(t)}\left(\frac{\partial f}{\partial t} + \nabla\cdot(f\mathbf{v})\right)dV = \int_{V_m(t)}\frac{\partial f}{\partial t}\,dV + \oint_{\partial V_m(t)} f\,\mathbf{v}\cdot\hat{n}\,dA$$

Proof. Change variables to the fixed reference volume, $dV = J\,dV_0$:

$$\frac{d}{dt}\int_{V_m(t)} f\,dV = \frac{d}{dt}\int_{V_0} f(\boldsymbol{\chi}(\mathbf{X},t),t)\,J(\mathbf{X},t)\,dV_0$$

The domain V_0 no longer depends on t, so by assumption 3 the derivative passes inside the integral. The time derivative of f at fixed $\mathbf{X}$ is the material derivative $Df/Dt = \partial_t f + \mathbf{v}\cdot\nabla f$, and by the lemma $\partial_t J = J\nabla\cdot\mathbf{v}$. Therefore

$$\frac{d}{dt}\int_{V_m(t)} f\,dV = \int_{V_0}\left(\frac{Df}{Dt} + f\,\nabla\cdot\mathbf{v}\right)J\,dV_0 = \int_{V_m(t)}\left(\frac{\partial f}{\partial t} + \nabla\cdot(f\mathbf{v})\right)dV$$

using $\mathbf{v}\cdot\nabla f + f\nabla\cdot\mathbf{v} = \nabla\cdot(f\mathbf{v})$. The surface form follows from the divergence theorem. $\square$

> **[DERIVATION]** The Reynolds transport theorem is an identity of calculus given a smooth, invertible motion. No physics beyond the existence of a continuum velocity field enters.

E.1.4 — Arbitrary control volumes

Engineering balances are written over control volumes that are not material: a pipe section, a heat-exchanger shell, a tank with a moving free surface. Let $V_c(t)$ have boundary points moving with velocity $\mathbf{w}$. Applying the same argument with the flow map of $\mathbf{w}$ in place of $\mathbf{v}$,

$$\frac{d}{dt}\int_{V_c(t)} f\,dV = \int_{V_c}\frac{\partial f}{\partial t}\,dV + \oint_{\partial V_c} f\,\mathbf{w}\cdot\hat{n}\,dA$$

If f obeys a balance law $\partial_t f + \nabla\cdot(f\mathbf{v} + \mathbf{j}) = s$ (convective flux $f\mathbf{v}$, non-convective flux $\mathbf{j}$, production s), then eliminating $\partial_t f$ gives the **control-volume balance**

$$\frac{d}{dt}\int_{V_c(t)} f\,dV + \oint_{\partial V_c}\left[f(\mathbf{v}-\mathbf{w}) + \mathbf{j}\right]\cdot\hat{n}\,dA = \int_{V_c} s\,dV$$

This is the form behind Step 1 and Step 2 of the lumping procedure in Ch. 17 §17.1.1: it is exact, and it is the last exact statement before the lumping step.

E.1.5 — Where it fails

The theorem fails where the motion is not smooth. Across a shock, a contact discontinuity, or a phase boundary, f jumps, and the local differential form is replaced by jump conditions (Rankine–Hugoniot) derived from the same integral balance applied to a pillbox straddling the surface. The integral balance survives; only its localisation fails. It also fails, more fundamentally, when the continuum hypothesis does: at Knudsen numbers $Kn = \ell_{mfp}/L$ that are not small, there is no well-defined local velocity field to transport.

E.2 — From Integral to Local Balances: Mass and Linear Momentum

E.2.1 — The localisation lemma

Lemma. If g is continuous on a region Ω and $\int_V g\, dV = 0$ for every subvolume $V \subset \Omega$, then $g \equiv 0$ on Ω.

Proof. Suppose $g(\mathbf{x}_0) > 0$. By continuity there is a ball $B_\delta(\mathbf{x}_0)$ on which $g > g(\mathbf{x}_0)/2$, so $\int_{B_\delta} g\, dV > 0$, a contradiction; likewise for $g(\mathbf{x}_0) < 0$. □

The lemma is how every postulated integral balance becomes a PDE. It requires continuity of the integrand, which is exactly what fails at the discontinuities of E.1.5.

E.2.2 — Mass

Postulate. The mass of a material volume is constant: $\frac{d}{dt}\int_{V_m} \rho\, dV = 0$.

Reynolds with $f = \rho$ and localisation give

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \qquad \Longleftrightarrow \qquad \frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} = 0$$

which is Ch. 15 §15.4.1. Its status is as Ch. 15 states it: for a single species of fixed particle mass m it follows from particle-number conservation; it is not a consequence of a fundamental $U(1)$ in general.

Corollary (transport of specific quantities). For any specific (per-unit-mass) quantity ϕ,

$$\frac{d}{dt}\int_{V_m} \rho\phi\, dV = \int_{V_m} \rho \frac{D\phi}{Dt}\, dV$$

Proof. Reynolds gives the integrand $\partial_t(\rho\phi) + \nabla\cdot(\rho\phi\mathbf{v}) = \phi[\partial_t\rho + \nabla\cdot(\rho\mathbf{v})] + \rho[\partial_t\phi + \mathbf{v}\cdot\nabla\phi]$, and the first bracket vanishes by mass balance. □

E.2.3 — Linear momentum and the Cauchy tetrahedron

Postulate (Euler's first law). For every material volume,

$$\frac{d}{dt}\int_{V_m} \rho\mathbf{v}\, dV = \oint_{\partial V_m} \mathbf{t}^{(\hat{n})}\, dA + \int_{V_m} \mathbf{f}\, dV$$

where $\mathbf{t}^{(\hat{n})}(\mathbf{x}, t)$ is the surface force per unit area (traction) on a surface element with outward normal $\hat{n}$, and $\mathbf{f}$ is the body force per unit volume. This is Newton's second law applied to a continuum, and it is the place where the "short-range contact force" idealisation enters: the force exerted by the outside on the inside is assumed to be representable as a surface density depending only on position, time, and the local normal (Cauchy's postulate).

Theorem (Cauchy). The traction is linear in $\hat{n}$: $t_i^{(\hat{n})} = \sigma_{ij} n_j$ for a tensor field σ_{ij}.

Proof. Take a tetrahedron with three faces in the coordinate planes, through a point $\mathbf{x}_0$, and an oblique face with outward normal $\hat{n}$ (all $n_j > 0$) at distance h from $\mathbf{x}_0$. If the oblique face has area A, the coordinate face with outward normal $-\hat{e}_j$ has area $A n_j$, and the volume is $Ah/3$. Apply the momentum postulate, using the corollary of E.2.2 for the left side:

$$\int_{tet} \left(\rho \frac{D\mathbf{v}}{Dt} - \mathbf{f} \right) dV = \int_A \mathbf{t}^{(\hat{n})}\, dA + \sum_j \int_{An_j} \mathbf{t}^{(-\hat{e}_j)}\, dA$$

The left side is bounded by a constant times Ah (bounded integrand times volume). Divide by A and let $h \to 0$; by continuity of the traction in $\mathbf{x}$,

$$\mathbf{t}^{(\hat{n})} + \sum_j n_j\, \mathbf{t}^{(-\hat{e}_j)} = 0$$

Applying the same argument to a thin slab with faces $\pm\hat{n}$ gives Newton's third law in the form $\mathbf{t}^{(-\hat{n})} = -\mathbf{t}^{(\hat{n})}$. Hence $\mathbf{t}^{(\hat{n})} = \sum_j n_j \mathbf{t}^{(\hat{e}_j)}$. Defining $\sigma_{ij} = t_i^{(\hat{e}_j)}$ gives $t_i^{(\hat{n})} = \sigma_{ij} n_j$; normals with some $n_j < 0$ follow by reorienting the tetrahedron. □

> **[DERIVATION]** Cauchy's theorem follows from the momentum postulate plus continuity of the traction. The surface-to-volume scaling (A versus Ah) is the whole argument: inertia and body forces are volume effects and cannot balance an imbalance in surface forces on a vanishing tetrahedron.

E.2.4 — Cauchy's equation of motion

Substituting $t_i = \sigma_{ij} n_j$ into the postulate and using the divergence theorem, $\oint \sigma_{ij} n_j\, dA = \int \partial_j \sigma_{ij}\, dV$. Localisation gives

$$\rho \frac{Dv_i}{Dt} = \partial_j \sigma_{ij} + f_i$$

and, adding v_i times the mass balance, the conservation form

$$\frac{\partial (\rho v_i)}{\partial t} + \partial_j \left(\rho v_i v_j - \sigma_{ij} \right) = f_i$$

This is Ch. 15 §15.4.2. No constitutive information has been used: Cauchy's equation holds equally for water, steel, and blood. Its connection to translational symmetry (Noether) is the one Ch. 15 states; the derivation here starts from the integral postulate rather than from an action, which is how continuum mechanics is usually built and has the advantage of admitting dissipative stresses that no simple action produces.

E.3 — Symmetry of the Cauchy Stress from Angular Momentum

E.3.1 — Assumptions

The continuum is **non-polar**: (a) there are no body couples (torque per unit volume not arising as $\mathbf{x} \times \mathbf{f}$), (b) there are no couple stresses (torque per unit area not arising as $\mathbf{x} \times \mathbf{t}$), and (c) there is no intrinsic angular momentum density (spin) of the material elements.

E.3.2 — Proof

Postulate (Euler's second law). About a fixed origin,

$$\frac{d}{dt}\int_{V_m} \mathbf{x} \times \rho\mathbf{v}\, dV = \oint_{\partial V_m} \mathbf{x} \times \mathbf{t}\, dA + \int_{V_m} \mathbf{x} \times \mathbf{f}\, dV$$

In components, with the Levi-Civita symbol ϵ_{ijk}: the left side, by the corollary of E.2.2 with $\phi = \epsilon_{ijk} x_j v_k$ and $Dx_j/Dt = v_j$, is

$$\int \rho\, \epsilon_{ijk} \left(v_j v_k + x_j \frac{Dv_k}{Dt} \right) dV = \int \rho\, \epsilon_{ijk} x_j \frac{Dv_k}{Dt}\, dV$$

since $\epsilon_{ijk} v_j v_k = 0$. The surface term, by the divergence theorem, is

$$\oint \epsilon_{ijk} x_j \sigma_{kl} n_l\, dA = \int \epsilon_{ijk} \partial_l (x_j \sigma_{kl})\, dV = \int \epsilon_{ijk} \left(\delta_{jl}\sigma_{kl} + x_j \partial_l \sigma_{kl} \right) dV = \int \left(\epsilon_{ijk}\sigma_{kj} + \epsilon_{ijk} x_j \partial_l \sigma_{kl} \right) dV$$

Collecting, the postulate becomes

$$\int_{V_m} \epsilon_{ijk} x_j \left(\rho \frac{Dv_k}{Dt} - \partial_l \sigma_{kl} - f_k \right) dV = \int_{V_m} \epsilon_{ijk}\sigma_{kj}\, dV$$

The bracket on the left vanishes by Cauchy's equation (E.2.4). So $\int \epsilon_{ijk}\sigma_{kj}\, dV = 0$ for every volume, and localisation gives $\epsilon_{ijk}\sigma_{kj} = 0$, i.e. $\sigma_{23} = \sigma_{32}$, $\sigma_{31} = \sigma_{13}$, $\sigma_{12} = \sigma_{21}$. □

> **[DERIVATION]** Given linear-momentum balance, angular-momentum balance for a non-polar continuum is *equivalent* to $\sigma_{ij} = \sigma_{ji}$. It is not an independent equation of motion; its entire content is the symmetry of the stress. This is the result Ch. 15 §15.3.1 cites.

E.3.3 — Where it fails

Each assumption of E.3.1 can fail. Ferrofluids in a magnetic field experience a body couple $\mathbf{M} \times \mathbf{B}$, and their stress is not symmetric; polarised dielectrics in a field likewise ($\mathbf{P} \times \mathbf{E}$). Granular media, foams, liquid crystals, and lattice materials at scales comparable to their microstructure are modelled as Cosserat (micropolar) continua with couple stresses and independent microrotation. In all of these the antisymmetric part of σ_{ij} is balanced by the couple terms, and the six-component stress of Ch. 15 is incomplete.

E.4 — The Energy Balance

E.4.1 — Postulate

Postulate (first law for a material volume). With e the specific internal energy, $\mathbf{q}$ the heat flux (outward-positive), and $\dot{Q}_{vol}$ the volumetric heat supply,

$$\frac{d}{dt}\int_{V_m}\rho\left(e+\tfrac{1}{2}|\mathbf{v}|^2\right)dV=\oint_{\partial V_m}(\mathbf{t}\cdot\mathbf{v}-\mathbf{q}\cdot\hat{n})\,dA+\int_{V_m}\left(\mathbf{f}\cdot\mathbf{v}+\dot{Q}_{vol}\right)dV$$

The right side is the rate of work by surface and body forces plus the rate of heat supply.

E.4.2 — Local total-energy balance

Using the corollary of E.2.2, $\mathbf{t}\cdot\mathbf{v}=\sigma_{ij}n_jv_i$, the divergence theorem, and localisation:

$$\rho\frac{D}{Dt}\left(e+\tfrac{1}{2}|\mathbf{v}|^2\right)=\partial_j(\sigma_{ij}v_i)-\nabla\cdot\mathbf{q}+f_iv_i+\dot{Q}_{vol}$$

E.4.3 — Separating mechanical and thermal energy

Contract Cauchy's equation with v_i. This is not a new law; it is a consequence of momentum balance, the **mechanical energy theorem**:

$$\rho\frac{D}{Dt}\left(\tfrac{1}{2}|\mathbf{v}|^2\right)=v_i\partial_j\sigma_{ij}+f_iv_i$$

Subtracting from E.4.2, and using $\partial_j(\sigma_{ij}v_i)-v_i\partial_j\sigma_{ij}=\sigma_{ij}\partial_jv_i=\sigma_{ij}\dot{\varepsilon}_{ij}$ (the antisymmetric part of ∂_jv_i drops out because σ is symmetric, E.3):

$$\rho\frac{De}{Dt}=\sigma_{ij}\dot{\varepsilon}_{ij}-\nabla\cdot\mathbf{q}+\dot{Q}_{vol}$$

This is the equation of Ch. 15 §15.4.3. Now split the stress into pressure and viscous (deviatoric-plus-bulk) parts, $\sigma_{ij}=-p\,\delta_{ij}+\tau_{ij}$:

$$\sigma_{ij}\dot{\varepsilon}_{ij}=-p\,\nabla\cdot\mathbf{v}+\Phi,\qquad\Phi\equiv\tau_{ij}\dot{\varepsilon}_{ij}$$

The term $-p\nabla\cdot\mathbf{v}$ is the **reversible** work of compression: it is positive when the fluid is compressed and negative when it expands, and it is exchanged back and forth between kinetic and internal energy. Only Φ — the viscous dissipation — is sign-definite, and its sign ($\Phi\geq 0$) is a constraint that the second law places on the constitutive law (E.5.4), not a consequence of the energy balance. The full stress power $\sigma_{ij}\dot{\varepsilon}_{ij}$ is not "always positive"; that statement is true only of its viscous part.

> **[DERIVATION]** The internal-energy equation is the first law minus the mechanical energy theorem. The split into reversible compression work and dissipation is exact; the sign of the dissipation is a second-law constraint on the closure.

E.5 — The Newtonian Constitutive Law and the Navier–Stokes Equations

E.5.1 — What is being assumed

Cauchy's equation has more unknowns than equations: σ_{ij} needs a closure. For a Newtonian fluid we assume:

1. **Fluidity.** At rest the stress is isotropic, $\sigma_{ij} = -p\,\delta_{ij}$, with p the thermodynamic pressure.
2. **Linearity and locality.** The viscous stress τ_{ij} is a linear function of the velocity gradient $\partial_l v_k$ at the same point and time — no memory, no gradient-of-gradient terms.
3. **Frame indifference.** The viscous stress cannot depend on the antisymmetric part $W_{kl} = \frac{1}{2}(\partial_l v_k - \partial_k v_l)$, which describes local rigid rotation; a fluid in rigid rotation has no viscous stress.
4. **Isotropy.** The linear map has no preferred direction.

> **[PHENOMENOLOGICAL]** The Newtonian law is a constitutive assumption, not a consequence of any balance law. Its microscopic foundation, where it exists, is the linear-response (Green–Kubo) framework of Ch. 13 §13.7, which supplies μ as a transport coefficient in the regime of small gradients; it does not supply the linear form for polymer solutions, suspensions, or blood at low shear rate (Ch. 15 §15.12.2).

E.5.2 — The isotropic linear form

By assumptions 2–3, $\tau_{ij} = C_{ijkl}\dot{\varepsilon}_{kl}$. By isotropy, C_{ijkl} is an isotropic fourth-order tensor, and every such tensor has the form $a\,\delta_{ij}\delta_{kl} + b\,\delta_{ik}\delta_{jl} + c\,\delta_{il}\delta_{jk}$ (a classical representation theorem; a proof is in R. Aris, *Vectors, Tensors, and the Basic Equations of Fluid Mechanics*, Prentice-Hall 1962, reprinted Dover). Contracting with the symmetric $\dot{\varepsilon}_{kl}$, the b and c terms combine:

$$\tau_{ij} = 2\mu\,\dot{\varepsilon}_{ij} + \lambda\,\dot{\varepsilon}_{kk}\,\delta_{ij}$$

with $2\mu = b + c$ and $\lambda = a$. Symmetry of τ_{ij} — required by E.3 — is automatic. This is the relation of Ch. 15 §15.6.1, and the bulk viscosity is $\zeta_b = \lambda + \frac{2}{3}\mu$.

E.5.3 — Substitution into Cauchy's equation

For constant μ and λ,

$$\partial_j \tau_{ij} = \mu\,\partial_j(\partial_i v_j + \partial_j v_i) + \lambda\,\partial_i(\nabla\cdot\mathbf{v}) = \mu\nabla^2 v_i + (\mu+\lambda)\,\partial_i(\nabla\cdot\mathbf{v})$$

so Cauchy's equation becomes the **compressible Navier–Stokes equation**

$$\rho\frac{D\mathbf{v}}{Dt} = -\nabla p + \mu\nabla^2\mathbf{v} + (\mu+\lambda)\nabla(\nabla\cdot\mathbf{v}) + \mathbf{f}$$

> **[DERIVATION]** Navier–Stokes is Cauchy's equation with the Newtonian closure substituted, for spatially uniform μ, λ. If μ varies (temperature-dependent viscosity), the extra term $2(\partial_j\mu)\dot{\varepsilon}_{ij}$ must be kept.

> **[APPROXIMATION]** Incompressibility. If $\nabla \cdot \mathbf{v} = 0$, the bulk term vanishes regardless of λ, giving $\rho\, D\mathbf{v}/Dt = -\nabla p + \mu \nabla^2 \mathbf{v} + \rho \mathbf{g}$ (Ch. 15 §15.6.2). **Small parameter:** the relative density variation $\delta\rho/\rho$, which for steady isothermal flow scales as Ma^2 with $Ma = U/c_s$, and for unsteady flow also requires $L/(c_s T) \ll 1$ (acoustic transit time small compared with the flow time T). Thermal expansion gives a separate contribution $\beta \Delta T$; keeping it only in the buoyancy term is the Boussinesq approximation. **Fails when:** $Ma \gtrsim 0.3$, in acoustics, in water hammer (Ch. 19 §19.7), or with large heating.

In the incompressible case p is no longer a thermodynamic variable but a Lagrange multiplier enforcing $\nabla \cdot \mathbf{v} = 0$, determined by the Poisson equation obtained from the divergence of the momentum equation.

E.5.4 — The second-law constraint

Decompose $\dot{\varepsilon}_{ij} = \dot{\varepsilon}'_{ij} + \frac{1}{3}\dot{\varepsilon}_{kk}\delta_{ij}$ with $\dot{\varepsilon}'$ traceless. Then

$$\Phi = \tau_{ij}\dot{\varepsilon}_{ij} = 2\mu\, \dot{\varepsilon}_{ij}\dot{\varepsilon}_{ij} + \lambda (\dot{\varepsilon}_{kk})^2 = 2\mu\, \dot{\varepsilon}'_{ij}\dot{\varepsilon}'_{ij} + \left(\lambda + \tfrac{2}{3}\mu\right)(\dot{\varepsilon}_{kk})^2$$

since $\dot{\varepsilon}_{ij}\dot{\varepsilon}_{ij} = \dot{\varepsilon}'_{ij}\dot{\varepsilon}'_{ij} + \frac{1}{3}(\dot{\varepsilon}_{kk})^2$. The two terms are independent (one can choose pure shear or pure dilatation), so $\Phi \geq 0$ for all motions iff $\mu \geq 0$ and $\zeta_b = \lambda + \frac{2}{3}\mu \geq 0$. Stokes' hypothesis $\zeta_b = 0$ is the boundary case; its status is discussed in Ch. 15 §15.6.1.

E.6 — Hagen–Poiseuille Flow and the Hydraulic Resistance

E.6.1 — Assumptions

Incompressible Newtonian fluid with constant μ; circular pipe of radius R (diameter $D = 2R$), length L; rigid wall with no-slip; steady; **fully developed**, meaning $\mathbf{v} = v_z(r)\,\hat{z}$ with no radial or swirl component and no z-dependence; gravity absorbed into the modified pressure $P = p + \rho g\, h$ (h the height), so that body forces drop out.

E.6.2 — Exact reduction of Navier–Stokes

Continuity: $\nabla \cdot \mathbf{v} = \partial_z v_z = 0$ — satisfied identically. The convective term is $(\mathbf{v}\cdot\nabla)\mathbf{v} = v_z\, \partial_z v_z\, \hat{z} = 0$, so **the nonlinear term vanishes identically**, not approximately; this is why the solution is an exact solution of the full nonlinear equations. The radial and azimuthal momentum equations reduce to $\partial_r P = 0$ and $\partial_\theta P = 0$, so $P = P(z)$. The axial equation is

$$0 = -\frac{dP}{dz} + \mu \frac{1}{r}\frac{d}{dr}\left(r \frac{dv_z}{dr}\right)$$

The first term depends only on z, the second only on r; both must equal the same constant. Write $dP/dz = -G$ with $G = \Delta P / L > 0$ for flow in $+z$.

E.6.3 — Solution

Integrate once: $r\, dv_z/dr = -Gr^2/(2\mu) + c_1$. Regularity at $r = 0$ (finite shear) forces $c_1 = 0$. Integrate again and impose $v_z(R) = 0$:

$$v_z(r) = \frac{G}{4\mu}\left(R^2 - r^2\right)$$

The volume flow rate is

$$Q = \int_0^R v_z(r)\, 2\pi r\, dr = \frac{2\pi G}{4\mu}\left[\frac{R^4}{2} - \frac{R^4}{4}\right] = \frac{\pi G R^4}{8\mu}$$

so

$$\Delta P = \frac{8\mu L}{\pi R^4}\, Q, \qquad R_{hyd} = \frac{\Delta P}{Q} = \frac{8\mu L}{\pi R^4} = \frac{128\,\mu L}{\pi D^4}$$

using $R^4 = D^4/16$. The units check: $[\mu L/D^4] = \mathrm{Pa\cdot s\cdot m/m^4} = \mathrm{Pa\cdot s/m^3}$, i.e. pressure per volume flow rate.

Consistency checks. The mean velocity is $U = Q/\pi R^2 = GR^2/(8\mu)$ and the centreline velocity is $2U$. The wall shear stress is $\tau_w = \mu |dv_z/dr|_{r=R} = GR/2$; a force balance on the fluid cylinder, $\Delta P\, \pi R^2 = \tau_w\, 2\pi R L$, gives the same value independently. Writing $\Delta P = f_D (L/D)\tfrac{1}{2}\rho U^2$ with $G = 32\mu U/D^2$ gives the Darcy friction factor $f_D = 64/Re_D$, $Re_D = \rho U D/\mu$.

> **[DERIVATION]** Under E.6.1, $R_{hyd} = 128\mu L/(\pi D^4)$ is an exact consequence of Navier–Stokes. The constitutive input is the Newtonian law; the "R" element of the hydraulic network (Ch. 17 §17.3.1, Ch. 19 §19.7.1) is this number.

> **[STRUCTURAL CONNECTION]** R_{hyd} and the electrical $R = L/(\sigma A)$ share the structure "length over (transport coefficient × geometric factor)", because both are integrals of a linear flux–force law over a cross-section. The geometric factors differ (D^4 versus D^2) because the viscous flux is a momentum flux whose profile is set by a no-slip boundary condition, whereas the conduction current in a uniform wire is uniform. Same template, different physics.

E.6.4 — Where it fails

Assumption	Small parameter / condition	What happens when it fails
Laminar	$Re_D \lesssim 2300$ (empirical, `[PHENOMENOLOGICAL]`)	Transition to turbulence; ΔP becomes roughly $\propto Q^{1.75}$–Q^2 (Darcy–Weisbach, Ch. 19 §19.7.2). Hagen–Poiseuille flow is linearly stable at every Re investigated (no proof for all Re is known); transition is triggered by finite-amplitude disturbances, which is why Re_c depends on inlet quality
Fully developed	$L \gg L_e$, with $L_e \approx 0.06\, Re_D\, D$ for laminar flow (widely used correlation, `[PHENOMENOLOGICAL]`)	Developing profile; extra entrance pressure drop
Steady	Womersley number $Wo = R\sqrt{\omega/\nu} \ll 1$	Pulsatile flow; the profile lags the pressure; effective resistance and an inertance $\rho L/(\pi R^2)$ both matter
Incompressible, rigid wall	$L\omega/c \ll 1$	Pressure waves; the pipe becomes a hydraulic transmission line
No-slip	$Kn = \ell_{mfp}/D \ll 1$	Slip flow in microchannels and rarefied gases

Assumption	Small parameter / condition	What happens when it fails
Newtonian	constant μ	Shear-thinning fluids give a blunter profile and a different D-scaling

E.7 — The Heat Equation

E.7.1 — Assumptions

A solid at rest ($\mathbf{v} = 0$) or a fluid in which convection is ignored; constant density; caloric equation of state $de = c\,dT$ with constant specific heat c (for solids and liquids $c_p \approx c_v$ to within a few percent, which is why the book writes ρc_p); Fourier's law $\mathbf{q} = -\kappa \nabla T$.

E.7.2 — Derivation

With $\mathbf{v} = 0$, $\dot{\varepsilon}_{ij} = 0$ and the internal-energy equation of E.4.3 becomes $\rho\, \partial_t e = -\nabla \cdot \mathbf{q} + \dot{Q}_{vol}$. Substituting the caloric relation and Fourier's law,

$$\rho c \frac{\partial T}{\partial t} = \nabla \cdot (\kappa \nabla T) + \dot{Q}_{vol} \quad \xrightarrow{\kappa \text{ uniform}} \quad \frac{\partial T}{\partial t} = \alpha \nabla^2 T + \frac{\dot{Q}_{vol}}{\rho c}, \qquad \alpha = \frac{\kappa}{\rho c}$$

> **[DERIVATION]** Energy balance + caloric equation of state + Fourier closure → heat equation. Only the first is a conservation law.

> **[PHENOMENOLOGICAL]** Fourier's law is a constitutive, linear-response relation (Ch. 13 §13.5). It fails for ballistic phonon transport ($Kn_{phonon} = \ell_{ph}/L \not\ll 1$), at times shorter than the heat-flux relaxation time (where a hyperbolic Cattaneo-type closure is sometimes used), and in anisotropic media, where κ becomes a tensor.

For a moving fluid the same steps give $\rho c_p\, DT/Dt = \nabla \cdot (\kappa \nabla T) + \Phi + \dot{Q}_{vol}$ (plus a pressure-work term $\beta T\, Dp/Dt$ for compressible flow), which is the starting point of convective heat transfer.

E.7.3 — The convective boundary condition

At a solid surface exposed to a fluid at T_∞, the heat flux leaving the solid is written

$$-\kappa \left.\frac{\partial T}{\partial n}\right|_{surface} = h\,(T_s - T_\infty)$$

> **[PHENOMENOLOGICAL]** Newton's law of cooling with a heat-transfer coefficient h is not derived here: h comes from a solution of the fluid's own energy and momentum equations or, in practice, from correlations (Ch. 19 §19.6.2). The solid's heat equation and this Robin condition together are the problem whose lumped limit is E.8.

E.8 — Lumped Capacitance and the Biot Criterion

E.8.1 — The exact integral balance

Integrate the heat equation (no source) over the body volume V, with surface A, and use the Robin condition with uniform h:

$$\rho c V \frac{dT}{dt} = -\oint_A \mathbf{q}\cdot\hat{n}\, dA = -hA\left(\bar{T}_s - T_\infty\right)$$

where $\bar{T}$ is the volume average and $\bar{T}_s$ the surface average of T. **This is exact.** The lumped-capacitance model makes one replacement:

> **[APPROXIMATION]** $\bar{T}_s \approx \bar{T}$. Then $\rho c V\, d\bar{T}/dt = -hA(\bar{T} - T_\infty)$, with solution $\bar{T} - T_\infty = (T_0 - T_\infty)\, e^{-t/\tau}$ and
>
> $$\tau = \frac{\rho c V}{hA} = R_{conv}\, C_{th}, \qquad R_{conv} = \frac{1}{hA}, \qquad C_{th} = \rho c V$$
>
> **Small parameter:** the Biot number $Bi = hL_c/\kappa$ with $L_c = V/A$. **Regime:** $Bi \ll 1$, conventionally $Bi < 0.1$.

The rest of this section shows that the error of this replacement is $O(Bi)$ and computes the coefficient.

E.8.2 — Why Bi is the right parameter: scaling

Inside the body heat flows by conduction across a distance $\sim L_c$, a resistance $\sim L_c/(\kappa A)$; at the surface it leaves by convection, a resistance $1/(hA)$. The ratio is $Bi = hL_c/\kappa$. The surface flux $hA(T_s - T_\infty)$ must be carried through the interior by a gradient $\sim (T_{centre} - T_s)/L_c$, so

$$\frac{T_{centre} - T_s}{T_s - T_\infty} \sim \frac{hL_c}{\kappa} = Bi$$

Small Bi means the internal temperature differences are small compared with the temperature drop across the boundary layer, which is precisely the replacement $\bar{T}_s \approx \bar{T}$. The exact slab problem below confirms the scaling and fixes the coefficients.

E.8.3 — Exact solution for a slab

Take a slab of half-thickness L (so $L_c = V/A = L$), insulated by symmetry at its mid-plane and exposed to convection on both faces, initially at uniform T_0. Define $\theta = (T - T_\infty)/(T_0 - T_\infty)$, $\xi = x/L$, and the Fourier number $Fo = \alpha t/L^2$. The problem is

$$\frac{\partial\theta}{\partial Fo} = \frac{\partial^2\theta}{\partial\xi^2}, \qquad \left.\frac{\partial\theta}{\partial\xi}\right|_{\xi=0} = 0, \qquad \left.\frac{\partial\theta}{\partial\xi}\right|_{\xi=1} = -Bi\,\theta(1,Fo), \qquad \theta(\xi,0) = 1$$

Separation of variables (Ch. 16 §16.5) gives

$$\theta(\xi,Fo) = \sum_{n=1}^{\infty} C_n \cos(\lambda_n\xi)\, e^{-\lambda_n^2 Fo}, \qquad \lambda_n \tan\lambda_n = Bi, \qquad C_n = \frac{4\sin\lambda_n}{2\lambda_n + \sin 2\lambda_n}$$

where the eigenvalue condition is the Robin condition applied to $\cos\lambda\xi$, and C_n follows from orthogonality of $\{\cos\lambda_n\xi\}$ on $[0,1]$ (a Sturm–Liouville system, Ch. 16 §16.5.2). This classical solution is in H. S. Carslaw and J. C. Jaeger, *Conduction of Heat in Solids*, 2nd ed., Oxford 1959, Ch. III.

E.8.4 — Asymptotics as $Bi \to 0$

Fundamental mode. For small λ, $\tan\lambda = \lambda + \lambda^3/3 + O(\lambda^5)$, so $\lambda^2 + \lambda^4/3 + O(\lambda^6) = Bi$. Inverting,

$$\lambda_1^2 = Bi - \frac{Bi^2}{3} + O(Bi^3)$$

The lumped model predicts the decay rate t/τ with $\tau = \rho c L/h$, i.e. in dimensionless form $t/\tau = hL\,t/(\rho c L^2) = Bi\cdot Fo$: a decay exponent Bi in place of the exact λ_1^2. Therefore

$$\frac{\tau_{exact}}{\tau_{lump}} = \frac{Bi}{\lambda_1^2} = 1 + \frac{Bi}{3} + O(Bi^2)$$

Amplitude. With $\sin\lambda = \lambda - \lambda^3/6 + \ldots$ and $\sin 2\lambda = 2\lambda - \tfrac{4}{3}\lambda^3 + \ldots$,

$$C_1 = \frac{4\lambda_1(1-\lambda_1^2/6)}{4\lambda_1(1-\lambda_1^2/3)} + O(\lambda_1^4) = 1 + \frac{Bi}{6} + O(Bi^2)$$

Internal non-uniformity. In the fundamental mode $\theta(1)/\theta(0) = \cos\lambda_1 = 1 - Bi/2 + O(Bi^2)$: the surface–centre temperature difference is $Bi/2$ of the local excess temperature, in agreement with the scaling of E.8.2.

Higher modes. For $n \geq 2$ write $m = n-1$; then $\lambda_n = m\pi + \delta_m$ with $\tan\lambda_n = \tan\delta_m \approx \delta_m$, so the eigenvalue condition gives $m\pi\,\delta_m \approx Bi$, i.e. $\delta_m \approx Bi/(m\pi)$ to leading order. Hence $\sin\lambda_n \approx (-1)^m\delta_m$, $\sin 2\lambda_n \approx 2\delta_m$, and

$$C_n \approx \frac{2(-1)^m\,Bi}{m^2\pi^2}, \qquad \sum_{n\geq 2}|C_n| \approx \frac{2Bi}{\pi^2}\sum_{m\geq 1}\frac{1}{m^2} = \frac{Bi}{3}$$

As a check, $\theta(0,0) = 1$ requires $C_1 + \sum_{n\geq 2} C_n = 1$; the alternating sum is $\frac{2Bi}{\pi^2}\left(-\frac{\pi^2}{12}\right) = -\frac{Bi}{6}$, which exactly cancels the $+Bi/6$ in C_1. The higher modes decay at rates $\lambda_n^2 \geq \pi^2$ in units of α/L^2 — the diffusion time — and so are gone once $Fo \gtrsim 0.5$.

E.8.5 — The error bound

Combining, for all $\xi \in [0,1]$ and $Fo \geq 0$, to leading order in Bi:

$$\theta(\xi, Fo) = e^{-Bi\,Fo}\left[1 + Bi\left(\frac{1}{6} - \frac{\xi^2}{2}\right) + \frac{Bi^2\,Fo}{3} + O(Bi^2)\right] + R(\xi, Fo), \qquad |R| \leq \frac{Bi}{3}\,e^{-\pi^2 Fo}\,(1 + O(Bi))$$

Three statements follow.

1. **Uniform $O(Bi)$ accuracy on the relevant timescale.** For $t \lesssim$ a few τ (i.e. $Bi\,Fo = O(1)$), the term $Bi^2 Fo/3 = (Bi/3)(Bi\,Fo)$ is $O(Bi)$, so the lumped solution is accurate to relative error $O(Bi)$ everywhere in the body.
2. **Time-constant error.** The lumped τ is too short by the factor $1/(1 + Bi/3)$: about 3.4% at $Bi = 0.1$ (the exact root gives $\lambda_1^2 = 0.09675$, so $Bi/\lambda_1^2 = 1.034$).

3. **Long-time relative error grows.** For $t \gg \tau$ the relative error grows linearly as $(Bi/3)(t/\tau)$, although the absolute error is exponentially small. A lumped model used to predict *when* a body reaches 1% of its initial excess ($t \approx 4.6\,\tau$) carries a timing error of about $1.5\,Bi$ — small, but larger than $Bi/3$.

The numbers depend on shape. For a sphere of radius R_s ($L_c = R_s/3$, $Bi_R = hR_s/\kappa = 3Bi$), the eigenvalue condition is $1 - \lambda \cot \lambda = Bi_R$, giving $\lambda_1^2 = 3Bi_R(1 - Bi_R/5 + \ldots)$; the decay-rate error is then $3Bi/5$ (6% at $Bi = 0.1$) and the surface/centre ratio $\sin\lambda_1/\lambda_1 \approx 1 - \frac{3}{2}Bi$ (about 14% difference at $Bi = 0.1$). The criterion $Bi < 0.1$ is therefore a convention guaranteeing errors of a few to roughly fifteen percent, not a sharp threshold.

> **[APPROXIMATION]** Proved: for $Bi \to 0$ the exact conduction solution approaches the lumped exponential with $\tau = \rho c V/(hA)$, with relative error $O(Bi)$ for $t = O(\tau)$; the coefficient is $Bi/3$ in the time constant for a slab and $3Bi/5$ for a sphere.

E.8.6 — Where it fails

The lumped model fails (a) when $Bi \gtrsim 0.1$; (b) when h is non-uniform over the surface, since then even the exact balance of E.8.1 involves a weighted surface average; (c) when the forcing ($T_\infty(t)$ or a surface flux) varies on timescales shorter than the diffusion time L_c^2/α, because then the higher modes are continually re-excited — this is the time-window caveat of Ch. 16 §16.10.1, and it is independent of Bi; (d) with radiation, unless $h_{rad} \approx 4\epsilon\sigma T_m^3$ is linearised about a mean temperature (Ch. 19 §19.6.1) and included in Bi; (e) with internal generation that is non-uniform.

Model Ledger — conduction with a convective boundary to lumped thermal RC

Field	Content
Parent theory	Heat equation with Robin boundary condition (§E.7; Ch. 16 §16.2)
Reduction	Integrate over the body (exact); replace surface-average by volume-average temperature
Model	$\rho c V\, d\bar{T}/dt = -hA(\bar{T} - T_\infty)$, $\tau = \rho c V/(hA)$
Assumptions	uniform h; constant κ, ρc; no internal source; Fourier conduction; Newton cooling
Physics retained	total stored energy and the surface heat-loss rate
Physics neglected	internal gradients (fractional size $\approx Bi/2$ for a slab); higher spatial modes (total amplitude $\approx Bi/3$)
Validity	$Bi = hL_c/\kappa \ll 1$, and forcing slow compared with L_c^2/α
Fails when	$Bi \gtrsim 0.1$; rapid surface forcing; non-uniform h; strong radiation without linearisation
Next model	one-term (Heisler) eigenfunction solution; full heat equation; finite-volume thermal network (Ch. 16 §16.11.3)

E.9 — Maxwell's Equations to the Telegrapher's Equations

E.9.1 — Assumptions

A uniform two-conductor line along z (coaxial cable, twin lead, stripline), cross-section independent of z, embedded in a linear, isotropic, homogeneous dielectric with permittivity ϵ and permeability μ. Begin with perfect conductors and a lossless dielectric; losses are added in E.9.4. The mode of interest is **TEM**: $E_z = 0$, $B_z = 0$.

E.9.2 — The transverse problem is electrostatic and magnetostatic

Write $\nabla = \nabla_t + \hat{z}\,\partial_z$. The z-component of Faraday's law reads $(\nabla_t \times \mathbf{E}_t)\cdot\hat{z} = -\partial_t B_z = 0$. So in each transverse plane $\mathbf{E}_t$ is curl-free and $\mathbf{E}_t = -\nabla_t\phi(x,y;z,t)$. Gauss's law in the charge-free dielectric, with $E_z = 0$, gives $\nabla_t\cdot\mathbf{E}_t = 0$, so $\nabla_t^2\phi = 0$: in every plane, ϕ solves the two-dimensional electrostatic problem, with each perfect conductor an equipotential (tangential $\mathbf{E}$ vanishes there). Define the line voltage

$$V(z,t) = \phi_1 - \phi_2 = \int_1^2 \mathbf{E}_t\cdot d\mathbf{l}$$

along any path in the transverse plane from conductor 1 to conductor 2; path-independence follows from $\nabla_t\times\mathbf{E}_t = 0$. Similarly, the z-component of the Ampère–Maxwell law in the dielectric gives $\nabla_t\times\mathbf{H}_t = 0$ there, so the line current $I(z,t) = \oint \mathbf{H}_t\cdot d\mathbf{l}$ around conductor 1 is independent of the loop chosen. Because both transverse problems are linear and depend only on the cross-section geometry, the charge per unit length is $q' = C'V$ and the magnetic flux per unit length passing between the conductors is $\Phi' = L'I$, with C' and L' fixed by geometry and material.

E.9.3 — The two line equations

Faraday. Take a rectangular contour in a plane containing the z axis: from conductor 1 to conductor 2 at z, along conductor 2 to $z+\Delta z$, back to conductor 1, and along conductor 1 to z. The segments along perfect conductors contribute nothing ($E_{tan} = 0$). Faraday's law gives

$$V(z+\Delta z) - V(z) = -\frac{\partial}{\partial t}\left(\Phi'\Delta z\right) \quad\Longrightarrow\quad \frac{\partial V}{\partial z} = -L'\frac{\partial I}{\partial t}$$

(the sign of the flux being fixed by the contour orientation; with V and I defined as above and power flowing in $+z$ for $V, I > 0$, this is the standard sign).

Charge conservation. The continuity equation integrated over the surface of conductor 1 between z and $z+\Delta z$ gives $I(z) - I(z+\Delta z) = \partial_t(q'\Delta z)$, so

$$\frac{\partial I}{\partial z} = -C'\frac{\partial V}{\partial t}$$

For the exact TEM mode in a homogeneous medium, $L'C' = \mu\epsilon$, so the phase speed $1/\sqrt{L'C'}$ equals the plane-wave speed in the dielectric (D. M. Pozar, *Microwave Engineering*, 4th ed., Wiley 2012, Ch. 2–3).

E.9.4 — Losses and the quasi-TEM approximation

With finite conductivity, current in the conductors requires a small axial field E_z at their surfaces, and the mode is no longer exactly TEM. Treating E_z perturbatively, the contour of E.9.3 picks up a series term $R'I\,\Delta z$, where R' is the conductor resistance per unit length (frequency-dependent through the skin

effect). A slightly conducting dielectric carries a transverse conduction current $G'V$ per unit length. The result is the **telegrapher's equations** of Ch. 16 §16.3.2 and Example 03:

$$\frac{\partial V}{\partial z} = -R'I - L'\frac{\partial I}{\partial t}, \qquad \frac{\partial I}{\partial z} = -G'V - C'\frac{\partial V}{\partial t}$$

> **[APPROXIMATION]** Quasi-TEM. **Small parameters:** $|E_z|/|E_t|$, which is of order $R'/(\omega L')$ for conductor loss; for an inhomogeneous dielectric (microstrip, with field partly in air) the dispersion correction scales with $(d/\lambda)^2$, d the transverse dimension. **Fails when:** $d \gtrsim \lambda/2$ (higher-order waveguide modes propagate, Ch. 12), when losses are so large that $R' \sim \omega L'$ and the field is far from TEM, or near discontinuities, where evanescent higher-order modes are excited locally.

E.9.5 — The lumped limit $\ell \ll \lambda$

For a section of length ℓ, integrate the telegrapher's equations over $z \in [0, \ell]$. Exactly,

$$V(0) - V(\ell) = R'\ell\,\langle I\rangle + L'\ell\,\frac{d\langle I\rangle}{dt}, \qquad I(0) - I(\ell) = G'\ell\,\langle V\rangle + C'\ell\,\frac{d\langle V\rangle}{dt}$$

with $\langle\cdot\rangle$ the average over the section. The approximation is to replace $\langle I\rangle$ and $\langle V\rangle$ by terminal values. To quantify it, use the exact phasor solution. With $Z' = R' + j\omega L'$, $Y' = G' + j\omega C'$, $\gamma = \sqrt{Z'Y'}$, and $Z_0 = \sqrt{Z'/Y'}$, the section is the two-port

$$\begin{pmatrix} V(0) \\ I(0) \end{pmatrix} = \begin{pmatrix} \cosh\gamma\ell & Z_0\sinh\gamma\ell \\ Z_0^{-1}\sinh\gamma\ell & \cosh\gamma\ell \end{pmatrix} \begin{pmatrix} V(\ell) \\ I(\ell) \end{pmatrix}$$

Let $Z = Z'\ell$ and $Y = Y'\ell$, so that $(\gamma\ell)^2 = ZY$. Expanding for $|\gamma\ell| \ll 1$:

$$\cosh\gamma\ell = 1 + \frac{ZY}{2} + O\big((\gamma\ell)^4\big), \qquad Z_0\sinh\gamma\ell = Z\left(1 + \frac{ZY}{6} + \dots\right), \qquad \frac{\sinh\gamma\ell}{Z_0} = Y\left(1 + \frac{ZY}{6} + \dots\right)$$

The symmetric π-section (shunt $Y/2$, series Z, shunt $Y/2$) has transmission matrix with diagonal entries $1 + ZY/2$, off-diagonal entries Z and $Y(1 + ZY/4)$ — agreeing with the exact line to relative error $O(ZY) = O((\gamma\ell)^2)$. For a low-loss line $|\gamma\ell| \approx \omega\ell\sqrt{L'C'} = 2\pi\ell/\lambda$.

> **[APPROXIMATION]** Lumping a line section into $L = L'\ell$, $C = C'\ell$, $R = R'\ell$, $G = G'\ell$. **Small parameter:** ℓ/λ; error $O((2\pi\ell/\lambda)^2)$ per section. **Fails when:** $\ell \gtrsim \lambda/10$, the conventional engineering threshold: there $(\gamma\ell)^2 \approx 0.39$ and the π-section's off-diagonal entries err by about 3–7%; for a digital edge with rise time t_r, when the one-way delay ℓ/v is not small compared with t_r. A ladder of N sections converges to the line as $N \to \infty$ at fixed total length; that convergence, read backwards, is the descent from line to lumped circuit in Ch. 16 §16.10.2.

E.10 — Kirchhoff's Voltage Law from Faraday's Law by Lumping

E.10.1 — The lumping hypotheses

This section proves KVL as a consequence of Faraday's law plus explicit assumptions, and shows where the $d\Phi_B/dt$ term goes. It does **not** go to zero; it is relocated into the inductor branch voltages. The hypotheses are:

1. **Element boxes.** The circuit is a set of elements, each enclosed in a closed surface (its "box"), joined by wires outside the boxes. Each element has two terminals on its box surface.
2. **Flux confinement.** Outside the boxes, $\partial \mathbf{B}/\partial t = 0$ in the region swept by the circuit and its spanning surfaces. All time-varying magnetic flux lives inside inductor (and transformer) boxes.
3. **Charge confinement.** Outside the boxes, $\partial \rho/\partial t = 0$; time-varying charge lives on capacitor plates inside capacitor boxes. (Needed for KCL, E.10.4.)
4. **Ideal wires.** The wires outside the boxes are perfect conductors, so $\mathbf{E} \cdot d\mathbf{l} = 0$ along them (or have resistance assigned to a resistor box).
5. **Quasi-statics.** Circuit dimension $\ell \ll \lambda$, so that retardation across the circuit is negligible and radiation is negligible (E.10.5).

E.10.2 — Branch voltages are well defined

In the exterior region of hypothesis 2, $\nabla \times \mathbf{E} = 0$, so locally $\mathbf{E} = -\nabla\phi$. Define the branch voltage of element k with terminals a_k, b_k as

$$v_k = \int_{a_k}^{b_k} \mathbf{E} \cdot d\mathbf{l} = \phi(a_k) - \phi(b_k) \qquad \text{(path just outside the box)}$$

The integral is path-independent provided paths are deformable into one another within the exterior without crossing time-varying flux — which hypothesis 2 guarantees for paths that stay outside every box. This is exactly what a voltmeter with short leads routed away from the inductor measures; a voltmeter whose leads loop around a transformer core measures something else, and that is not a paradox but a violation of hypothesis 2.

E.10.3 — The loop law and the inductor's voltage

Loop law. Take any loop of the circuit graph. Construct a closed contour $\mathcal{C}$ that runs along the wires and, at each element, crosses from terminal to terminal along the exterior path used to define v_k. Choose a spanning surface for $\mathcal{C}$ lying entirely in the exterior — possible because each box is crossed only on its outside. Faraday's law on $\mathcal{C}$:

$$\oint_{\mathcal{C}} \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt}\int_S \mathbf{B} \cdot d\mathbf{S} = 0$$

by hypothesis 2. The wire segments contribute zero (hypothesis 4) and each element contributes $\pm v_k$ according to orientation. Hence

$$\sum_{k \in \text{loop}} \pm v_k = 0$$

Inductor branch relation. Now apply Faraday's law to a *different* contour: the one that runs from a to b **through the winding** of an inductor and returns from b to a through the exterior. Its spanning surface is pierced by the inductor's flux; the flux linkage through it is $\lambda = Li$ for a linear inductor, with i the winding current from a to b and orientations related by the right-hand rule. Inside a winding of resistance R_w, $\mathbf{E} = \mathbf{J}/\sigma$, so $\int_{winding} \mathbf{E} \cdot d\mathbf{l} = R_w i$. Faraday's law gives

$$R_w\, i + \int_b^a \mathbf{E} \cdot d\mathbf{l}\bigg|_{ext} = -\frac{d\lambda}{dt} \quad \Longrightarrow \quad R_w\, i - v_{ab} = -L\frac{di}{dt} \quad \Longrightarrow \quad v_{ab} = R_w\, i + L\frac{di}{dt}$$

This is where the $-d\Phi_B/dt$ went: the flux of the circuit lives inside the inductor, and Faraday's law applied to the contour *through* the inductor produces its branch relation. For coupled coils, $\lambda_j = \sum_k M_{jk} i_k$ and $v_j = \sum_k M_{jk}\, di_k/dt$.

> **[DERIVATION]** Under hypotheses 1–4, KVL in the form $\sum \pm v_k = 0$, with inductor branch voltages $v = L\, di/dt$ (plus winding resistance), follows from Faraday's law. KVL is not "Faraday's law with $d\Phi/dt$ dropped": the flux term is kept and assigned to the elements that hold the flux. This supersedes the phrasing in Ch. 11 §11.12.2 and Ch. 17 §17.4.2 where the two might be read as equivalent.

> **[APPROXIMATION]** The lumping hypotheses. **Small parameters:** stray flux through the circuit loop compared with the inductors' flux, $\Phi_{stray}/(Li)$; and ℓ/λ. **Fails when:** loop areas are large or fast edges couple stray flux (the "parasitic inductance" of a layout is precisely a violation of hypothesis 2 that is then repaired by adding an inductor box); when ℓ/λ is not small (E.9).

E.10.4 — KCL by the same method

Enclose a node by a closed surface lying in the exterior. Charge conservation gives $\oint \mathbf{J} \cdot \hat{n}\, dA = -dQ_{enc}/dt = 0$ by hypothesis 3, and $\mathbf{J}$ crosses the surface only in the wires, so $\sum i_k = 0$. For a capacitor, enclosing one plate gives $i = dq/dt$ with $q = Cv$ from the electrostatics of the box. Note that the total current $\mathbf{J} + \partial_t \mathbf{D}$ is always solenoidal; KCL for conduction currents alone is the statement that displacement current outside capacitor boxes is negligible — the electric counterpart of hypothesis 2.

E.10.5 — Radiation and retardation

Hypotheses 2 and 3 cannot hold exactly for a time-varying circuit, since a changing current always produces some changing field outside the boxes. The leading corrections are (i) retardation, of relative size $\omega\ell/c = 2\pi\ell/\lambda$, and (ii) radiation, which for an electrically small loop appears as a radiation resistance scaling as $(\ell/\lambda)^4$ (C. A. Balanis, *Antenna Theory*, 4th ed., Wiley 2016, Ch. 5). Both vanish as $\ell/\lambda \to 0$, which is the content of the criterion of Ch. 11 §11.12.1.

E.11 — Poynting's Theorem and Lumped Port Power

E.11.1 — Assumptions

Macroscopic Maxwell equations $\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$, $\nabla \times \mathbf{H} = \mathbf{J} + \partial_t \mathbf{D}$; media linear, non-dispersive, and time-invariant, $\mathbf{D} = \epsilon \mathbf{E}$, $\mathbf{B} = \mu \mathbf{H}$, with ϵ, μ symmetric tensors constant in time.

E.11.2 — Proof

The vector identity $\nabla \cdot (\mathbf{E} \times \mathbf{H}) = \mathbf{H} \cdot (\nabla \times \mathbf{E}) - \mathbf{E} \cdot (\nabla \times \mathbf{H})$ and the two curl equations give

$$\nabla \cdot (\mathbf{E} \times \mathbf{H}) = -\mathbf{H} \cdot \partial_t \mathbf{B} - \mathbf{E} \cdot \partial_t \mathbf{D} - \mathbf{J} \cdot \mathbf{E}$$

For symmetric, time-independent ϵ: $\mathbf{E} \cdot \partial_t (\epsilon \mathbf{E}) = \partial_t (\tfrac{1}{2} \mathbf{E} \cdot \epsilon \mathbf{E})$, and likewise for μ. Hence

$$\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}, \qquad u = \tfrac{1}{2} \left(\mathbf{E} \cdot \mathbf{D} + \mathbf{B} \cdot \mathbf{H} \right), \qquad \mathbf{S} = \mathbf{E} \times \mathbf{H}$$

and in vacuum this is Ch. 11 §11.8.2. In integral form over a fixed volume V,

$$-\oint_{\partial V} \mathbf{S} \cdot \hat{n} \, dA = \frac{d}{dt} \int_V u \, dV + \int_V \mathbf{J} \cdot \mathbf{E} \, dV$$

power flowing in equals the rate of increase of field energy plus the work done on charges (Joule heat σE^2 for an ohmic conductor).

> **[DERIVATION]** Poynting's theorem is an identity following from the two curl equations; the identification of u as an energy density uses the linear, non-dispersive constitutive relations. In dispersive or lossy-dielectric media the clean split fails and u must be redefined (J. D. Jackson, *Classical Electrodynamics*, 3rd ed., Wiley 1999, Ch. 6).

E.11.3 — Lumped port power $P = v\,i$

Enclose a two-terminal element in a surface Σ lying in the exterior region of E.10, crossed only by its two leads. On Σ, $\mathbf{E} = -\nabla \phi$ (hypothesis 2), so

$$\mathbf{E} \times \mathbf{H} = -\nabla \phi \times \mathbf{H} = -\nabla \times (\phi \mathbf{H}) + \phi \, \nabla \times \mathbf{H}$$

The surface integral of a curl over a closed surface vanishes, so

$$-\oint_{\Sigma} \mathbf{S} \cdot \hat{n} \, dA = -\oint_{\Sigma} \phi \, (\mathbf{J} + \partial_t \mathbf{D}) \cdot \hat{n} \, dA$$

Neglect the displacement current through Σ (the hypothesis 3 counterpart). $\mathbf{J}$ crosses Σ only within the two leads, at potentials ϕ_a, ϕ_b, carrying current i inward at a and outward at b. So

$$-\oint_{\Sigma} \mathbf{S} \cdot \hat{n} \, dA = \phi_a \, i - \phi_b \, i = v \, i$$

> **[DERIVATION]** Under the lumping hypotheses, the electromagnetic power delivered to an element through its box equals $v\,i$. This is the field-theoretic origin of the power-conjugate pairing of effort and flow (Ch. 17 §17.2) in the electrical domain. KVL itself is not an energy statement; Tellegen's

> theorem ($\sum v_k i_k = 0$ for any v satisfying KVL and any i satisfying KCL on the same graph) is the network form of energy balance, and it requires both laws.

E.12 — The Euler–Bernoulli Beam from Three-Dimensional Elasticity

E.12.1 — Assumptions

A straight prismatic beam along x, length L, cross-section A with characteristic transverse size h, symmetric about the x–y plane in which it bends. Linear isotropic elasticity with Young's modulus E and shear modulus G (Ch. 15 §15.9). The **slender-beam assumptions** are:

1. **Slenderness:** $h/L \ll 1$ (equivalently $r_g/L \ll 1$, $r_g = \sqrt{I/A}$ the radius of gyration);
2. **Small deflection:** $|w|/L \ll 1$ and $|w'| \ll 1$, so the problem is linear;
3. **Bernoulli–Kirchhoff kinematics:** plane cross-sections remain plane and normal to the deformed axis;
4. **Uniaxial stress:** $\sigma_{yy}, \sigma_{zz} \ll \sigma_{xx}$; lateral surfaces carry only the applied transverse load $q(x,t)$ per unit length.

E.12.2 — Kinematics, stress, and moment

Assumption 3 gives the displacement field $u_x = -y\,w'(x)$, $u_y = w(x)$, so the only significant strain is

$$\varepsilon_{xx} = \partial_x u_x = -y\,w''$$

Assumption 4 with Hooke's law gives $\sigma_{xx} = E\varepsilon_{xx} = -Ey\,w''$ (uniaxial stress, hence E and not the plane-strain modulus). Measuring y from the centroid makes the axial force $\int_A \sigma_{xx}\,dA$ vanish. Define the bending moment and shear force

$$M(x) = -\int_A y\,\sigma_{xx}\,dA = EI\,w'', \qquad V(x) = \int_A \sigma_{xy}\,dA, \qquad I = \int_A y^2\,dA$$

and so $\sigma_{xx} = -My/I$, matching Ch. 20 §20.2.1.

E.12.3 — Integrating Cauchy's equation over the cross-section

The beam equations are cross-sectional moments of Cauchy's equation (E.2.4). Statically and without body force, the x-component is $\partial_x\sigma_{xx} + \partial_y\sigma_{xy} + \partial_z\sigma_{xz} = 0$. Multiply by y and integrate over A; the lateral surface is traction-free in x, so integrating by parts gives $\int_A y(\partial_y\sigma_{xy} + \partial_z\sigma_{xz})\,dA = -\int_A \sigma_{xy}\,dA = -V$. Hence

$$-M' - V = 0 \quad\Longrightarrow\quad V = -M'$$

The y-component, $\partial_x\sigma_{yx} + \partial_y\sigma_{yy} + \partial_z\sigma_{yz} = 0$, integrated over A, gives $V' + q = 0$, since the boundary integral of $\sigma_{yy}n_y + \sigma_{yz}n_z$ around the cross-section's perimeter is the applied load per unit length. Combining, $M'' = q$ and

$$(EI\,w'')'' = q \quad \xrightarrow{EI \text{ uniform}} \quad EI\,\frac{d^4w}{dx^4} = q$$

Dynamics. Restoring $\rho\,\partial_t^2 u_y = \rho\ddot{w}$ in the y-equation adds $-\rho A\ddot{w}$ to the load, giving the bridge-zone equation of Ch. 16 §16.7.2 and Ch. 17 §17.10.1:

$$EI\,\frac{\partial^4 w}{\partial x^4} + \rho A\,\frac{\partial^2 w}{\partial t^2} = q(x,t)$$

The x-equation also acquires $\rho\,\partial_t^2 u_x = -\rho y\,\ddot{w}'$, whose y-moment adds the **rotary inertia** term $-\rho I\,\ddot{w}''$ (the Rayleigh beam). Euler–Bernoulli drops it.

E.12.4 — The internal inconsistency, and why it is harmless

Assumption 3 implies $\varepsilon_{xy} = 0$ and therefore $\sigma_{xy} = 0$ — yet the shear force V is nonzero. The resolution is that Euler–Bernoulli is the leading order of an asymptotic expansion in h/L: the shear stress is $O(h/L)$ smaller than the bending stress, its *strain* contribution to the deflection is $O((h/L)^2)$ smaller, and its value is recovered *a posteriori* from equilibrium (the Jourawski/Zhuravskii formula $\sigma_{xy} = VQ_A/(Ib)$). The kinematic hypothesis is a statement about which strains are negligible, not a claim that the shear stress is zero.

> **[APPROXIMATION]** Three-dimensional elasticity → Euler–Bernoulli beam. **Small parameters:** h/L (slenderness) and $|w|/L$ (linearity); in dynamics, kr_g with k the flexural wavenumber. Neglected: shear deformation (relative deflection error $O(Er_g^2/(\kappa_s G L^2))$), rotary inertia ($O((kr_g)^2)$), cross-section distortion (Poisson effects), and geometric nonlinearity. The status `[DERIVATION]` sometimes attached to the beam equation (Ch. 20 overview table) is accurate only for the reduction *given* these assumptions.

E.12.5 — What Timoshenko adds

Timoshenko (S. P. Timoshenko, *Philosophical Magazine* 41, 744–746, 1921) relaxes assumption 3: cross-sections remain plane but rotate by an independent angle $\varphi(x,t)$, so the shear strain $\gamma = w' - \varphi$ is retained, and rotary inertia is kept. With a shear coefficient κ_s correcting for the non-uniform shear distribution ($\kappa_s = 5/6$ is the usual value for a rectangle),

$$\rho A\,\ddot{w} = \kappa_s G A\,(w'' - \varphi') + q, \qquad \rho I\,\ddot{\varphi} = EI\,\varphi'' + \kappa_s G A\,(w' - \varphi)$$

Two consequences show the size of the correction. *Statics:* a cantilever with tip load P deflects by $PL^3/(3EI) + PL/(\kappa_s GA)$; the ratio of shear to bending terms is $3EI/(\kappa_s GAL^2) = 3(E/\kappa_s G)(r_g/L)^2$, which is $O((h/L)^2)$ and becomes significant for $L/h \lesssim 10$ or for sandwich and composite beams with small G/E. *Dynamics:* Euler–Bernoulli's dispersion relation $\omega^2 = (EI/\rho A)k^4$ (Ch. 16 §16.7.2) gives a phase velocity growing without bound as $\sqrt{\omega}$; Timoshenko's has two branches whose high-k speeds approach the finite values $\sqrt{E/\rho}$ and $\sqrt{\kappa_s G/\rho}$, so it corrects the unphysical infinite signal speed. The Timoshenko model is in turn an approximation to three-dimensional elastodynamics, accurate up to wavelengths comparable to h (K. F. Graff, *Wave Motion in Elastic Solids*, Ohio State University Press 1975, reprinted Dover).

E.13 — The Buckingham π Theorem

E.13.1 — Statement

Let a physical relation $f(q_1, \dots, q_n) = 0$ connect n positive dimensional quantities. Let there be k base dimensions (M, L, T, Θ, ...), and let the **dimension matrix** D be the $k \times n$ matrix whose column i lists the exponents of the base dimensions in q_i. Let $r = \operatorname{rank} \mathsf{D}$. Assume:

1. **Completeness:** the list $q_1, \dots, q_n$ contains every quantity on which the relation depends (including dimensional constants);
2. **Unit covariance:** the relation holds in every system of units obtained from one another by rescaling the base units.

Then the relation is equivalent to $F(\pi_1, \dots, \pi_{n-r}) = 0$ among $n - r$ independent dimensionless products $\pi_j = \prod_i q_i^{a_{ij}}$.

E.13.2 — Proof sketch

Counting. A product $\prod_i q_i^{a_i}$ is dimensionless iff its exponent vector $\mathbf{a} \in \mathbb{R}^n$ satisfies $\mathsf{D}\mathbf{a} = 0$. By the rank–nullity theorem, $\dim \ker \mathsf{D} = n - r$, so there are exactly $n - r$ independent dimensionless products; any other is a product of powers of these. (Because D has integer, in practice rational, entries, a basis of $\ker \mathsf{D}$ can be chosen with rational exponents.)

Reduction. Reorder so that $q_1, \dots, q_r$ have linearly independent dimension columns (the "repeating variables"). For each $j > r$, column j of D is a combination of the first r columns, so there are exponents with $\pi_{j-r} = q_j \prod_{i \le r} q_i^{-b_{ij}}$ dimensionless. Now rescale the base units by positive factors $\Lambda_1, \dots, \Lambda_k$; the numerical value of q_i changes by $\prod_m \Lambda_m^{\mathsf{D}_{mi}}$. Taking logarithms, the change of $(\ln q_1, \dots, \ln q_r)$ is $\mathsf{D}_{rep}^{T}(\ln \Lambda_1, \dots, \ln \Lambda_k)$, where D_{rep} is the $k \times r$ submatrix of rank r; its transpose maps $\mathbb{R}^k$ onto $\mathbb{R}^r$. So a unit system exists in which $q_1 = \cdots = q_r = 1$. In that system the π's equal $q_{r+1}, \dots, q_n$ numerically (dimensionless products are unit-invariant). By unit covariance the relation holds there:

$$f(1, \dots, 1, \pi_1, \dots, \pi_{n-r}) = 0 \equiv F(\pi_1, \dots, \pi_{n-r})$$

and since the π's are the same in every unit system, this relation holds in all of them. □

A careful treatment, including the role of the completeness assumption and of intermediate asymptotics, is G. I. Barenblatt, *Scaling, Self-Similarity, and Intermediate Asymptotics*, Cambridge University Press 1996, Ch. 1; the original is E. Buckingham, *Physical Review* 4, 345–376 (1914).

> **[DERIVATION]** The π theorem is a theorem of linear algebra plus unit covariance. Its physical content is entirely in assumption 1: a missing variable, or an irrelevant one included, produces a correct theorem about the wrong relation.

E.13.3 — Example: pipe flow, and what the theorem does not give

Take ΔP, L, D, U, ρ, μ: $n = 6$, base dimensions M, L, T with $r = 3$, so $n - r = 3$ groups, e.g. $\Delta P/(\rho U^2)$, L/D, $Re_D = \rho U D/\mu$. The theorem gives $\Delta P/(\rho U^2) = \Psi(L/D, Re_D)$ — nothing more. The further statement that $\Delta P \propto L$ for a long pipe is a *physical* input (fully developed flow, E.6), giving $\Delta P/(\rho U^2) =$

$(L/D)\,\frac{1}{2}f_D(Re_D)$; and the function $f_D = 64/Re_D$ comes only from solving Navier–Stokes (E.6.3) or, for turbulent flow, from experiment. The same reasoning applied to the dimensionless Navier–Stokes equation is how Re and Fr emerge in Ch. 15 §15.8.1.

E.14 — Summary

Result	Statement	Assumptions	Used in
Reynolds transport theorem	$\frac{d}{dt}\int_{V_m} f\,dV = \int \partial_t f\,dV + \oint f\,\mathbf{v}\cdot\hat{n}\,dA$	smooth invertible motion; C^1 field	Ch. 15 §15.4; Ch. 17 §17.1.1
Local mass balance	$\partial_t\rho + \nabla\cdot(\rho\mathbf{v}) = 0$	material mass constant; continuity of fields	Ch. 15 §15.4.1; Ch. 19 §19.7
Cauchy's theorem	$t_i = \sigma_{ij}n_j$	momentum postulate; contact forces as surface densities	Ch. 15 §15.3.1
Cauchy's equation	$\rho\,Dv_i/Dt = \partial_j\sigma_{ij} + f_i$	as above; any material	Ch. 15 §15.4.2
Stress symmetry	$\sigma_{ij} = \sigma_{ji}$	non-polar continuum (no body or couple stresses, no spin)	Ch. 15 §15.3.1
Internal-energy equation	$\rho\,De/Dt = -p\nabla\cdot\mathbf{v} + \Phi - \nabla\cdot\mathbf{q} + \dot{Q}_{vol}$	first law; stress symmetric	Ch. 15 §15.4.3
Navier–Stokes	$\rho\,D\mathbf{v}/Dt = -\nabla p + \mu\nabla^2\mathbf{v} + (\mu+\lambda)\nabla(\nabla\cdot\mathbf{v}) + \mathbf{f}$	Newtonian closure (constitutive); uniform μ, λ	Ch. 15 §15.6
Second-law constraint	$\mu \geq 0,\ \lambda + \frac{2}{3}\mu \geq 0$	$\Phi \geq 0$ for all motions	Ch. 15 §15.6.1
Hagen–Poiseuille	$R_{hyd} = 128\mu L/(\pi D^4)$, $f_D = 64/Re_D$	laminar, fully developed, steady, Newtonian, no-slip	Ch. 15 §15.7.1; Ch. 17 §17.3.1; Ch. 19 §19.7.1
Heat equation	$\rho c\,\partial_t T = \nabla\cdot(\kappa\nabla T) + \dot{Q}_{vol}$	energy balance + Fourier (constitutive) + $de = c\,dT$	Ch. 16 §16.2.2; Ch. 19 §19.6
Lumped capacitance	$\tau = \rho c V/(hA)$; $\tau_{exact}/\tau_{lump} = 1 + Bi/3 + O(Bi^2)$ (slab)	$Bi \ll 1$; uniform h; slow forcing	Ch. 16 §16.10.1; Ch. 17 §17.1; Ch. 19 §19.6.2
Telegrapher's equations	$\partial_z V = -R'I - L'\partial_t I$, $\partial_z I = -G'V - C'\partial_t V$	uniform line; quasi-TEM; $d \ll \lambda$	Ch. 16 §16.3.2; Ch. 17 §17.10
Line → lumped section	π-section matches exact two-port to $O((\gamma\ell)^2)$	$\ell \ll \lambda$	Ch. 16 §16.10.2; Ch. 17 §17.10.2
KVL with inductor branches	$\sum \pm v_k = 0$, $v_L = L\,di/dt$	flux confined to element boxes; ideal wires; $\ell \ll \lambda$	Ch. 11 §11.12.2; Ch. 17 §17.4.2; Ch. 18 §18.1.1
KCL	$\sum i_k = 0$, $i_C = C\,dv/dt$	charge confined to capacitor boxes	Ch. 11 §11.12.3; Ch. 17 §17.4.1
Poynting's theorem	$\partial_t u + \nabla\cdot\mathbf{S} = -\mathbf{J}\cdot\mathbf{E}$	linear, non-dispersive, time-invariant media	Ch. 11 §11.8.2
Lumped port power	$-\oint \mathbf{S}\cdot\hat{n}\,dA = v\,i$	lumping hypotheses on the box surface	Ch. 17 §17.2, §17.9
Euler–Bernoulli beam	$EI\,w'''' + \rho A\,\ddot{w} = q$	$h/L \ll 1$; small deflection; plane sections normal; uniaxial stress	Ch. 16 §16.7.2; Ch. 17 §17.10; Ch. 20 §20.1.3, §20.2

Result	Statement	Assumptions	Used in
Timoshenko correction	shear deflection ratio $3(E/\kappa_s G)(r_g/L)^2$; finite high-k wave speeds	plane sections, not normal; rotary inertia kept	Ch. 16 §16.7.2; Ch. 19 §19.10
Buckingham π	$n - r$ independent dimensionless groups	complete variable list; unit covariance	Ch. 15 §15.8.1; Ch. 19 §19.6.2; Ch. 20 §20.10.2

End of Appendix E

Appendix F — Linear Systems and Feedback: Laplace, Transfer Functions, Stability Proofs

This appendix proves the results of linear systems and classical control that the chapters state and use: the Laplace-transform properties (with the conditions of the final value theorem, which are routinely dropped), the state-space formula for the transfer function, the second-order step response and its metrics, the Bode asymptotes, the argument principle and the Nyquist criterion $Z = N + P$, the Routh–Hurwitz criterion with a worked example, the root-locus construction rules, steady-state error and system type, the Bode sensitivity integral, and the DC-motor transfer function. The chapters that rely on it are Ch. 17 (§17.6, §17.7), Ch. 18 (§18.6.3), and Ch. 21 (§21.2–§21.8), together with Example 06.

Scope. Everything here concerns **linear time-invariant (LTI), causal, finite-dimensional** systems, whose transfer functions are rational functions of s. The reduction of a physical plant to such a model is a separate step — lumping (Bridge C, Ch. 17 §17.1; Appendix E) followed by linearisation about an operating point (Ch. 21 §21.0.3) — and both are approximations with named regimes. The mathematics below is exact *for the model*; nothing in it repairs a model that has left its regime. Where a statement extends beyond rational functions (time delays, distributed plants) this is said explicitly. Ch. 21 §21.0.2 lists the system classes for which an ordinary transfer function does not exist.

Notation. It follows Ch. 21: plant $P(s)$, controller $C(s)$, loop transfer function $L(s) = C(s)P(s)$, sensitivity $S = 1/(1+L)$, complementary sensitivity $T = L/(1+L)$, state-space matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}, D$. The imaginary unit is j. Natural frequency and damping ratio are written ω_0, ζ as in Ch. 17 §17.7 (Ch. 21 writes ω_n; they are the same quantity). For the DC motor the notation of Example 06 is used (K_b, B); Ch. 21 §21.8.1 writes K_e, b for the same constants.

F.1 — The Laplace Transform: Properties Needed for Control

F.1.1 — Definition and convergence

For $f : [0, \infty) \to \mathbb{C}$ piecewise continuous and of **exponential order** — $|f(t)| \le M e^{\alpha t}$ for some M, α — the unilateral transform

$$F(s) = \mathcal{L}\{f\}(s) = \int_{0^-}^{\infty} f(t)\, e^{-st}\, dt$$

converges absolutely for $\operatorname{Re} s > \alpha$ and is analytic there. The lower limit 0^- lets impulses at $t = 0$ be included and makes initial conditions enter as $f(0^-)$, the state *before* any input is applied. Ch. 18 §18.6.3 introduces the bilateral transform; for causal signals the two coincide.

F.1.2 — Derivative and integral

Derivative. If f and f' are of exponential order, integration by parts gives, for $\operatorname{Re} s > \alpha$,

$$\mathcal{L}\{f'\} = \Big[f(t) e^{-st} \Big]_{0^-}^{\infty} + s \int_{0^-}^{\infty} f\, e^{-st}\, dt = sF(s) - f(0^-)$$

since $f(t)e^{-st} \to 0$ as $t \to \infty$. Iterating, $\mathcal{L}\{f''\} = s^2 F(s) - s f(0^-) - f'(0^-)$.

Integral. With $g(t) = \int_{0^-}^{t} f\,d\tau$, $g' = f$ and $g(0^-) = 0$, so $G(s) = F(s)/s$.

Delay. For $T \geq 0$, $\mathcal{L}\{f(t-T)u(t-T)\} = e^{-sT}F(s)$ — not a rational function, which is why a pure delay sits outside the rational-function machinery of this appendix (Ch. 21 §21.0.2).

F.1.3 — Convolution

Theorem. If f, g are causal and of exponential order, then $\mathcal{L}\{f * g\} = F(s)\,G(s)$ with $(f*g)(t) = \int_0^t f(\tau)\,g(t-\tau)\,d\tau$.

Proof. For $\operatorname{Re} s$ large enough the double integral converges absolutely, so Fubini's theorem permits exchanging the order:

$$\int_0^\infty e^{-st}\int_0^t f(\tau)g(t-\tau)\,d\tau\,dt = \int_0^\infty f(\tau)e^{-s\tau}\int_\tau^\infty g(t-\tau)e^{-s(t-\tau)}\,dt\,d\tau = F(s)\,G(s) \qquad \square$$

This is why an LTI system is characterised by one function: if $y = g * u$ with g the impulse response, then $Y = G\,U$. The ratio $G = Y/U$ is the transfer function, and it exists precisely because the system is linear (superposition) and time-invariant (a shifted input gives a shifted output, so the response is a convolution).

F.1.4 — The initial and final value theorems

Initial value theorem. If F is a strictly proper rational function, $f(0^+) = \lim_{s\to\infty} sF(s)$.

Final value theorem (FVT). Let F be rational. If **all poles of $sF(s)$ lie in the open left half-plane** — equivalently, F has at most a simple pole at $s = 0$ and every other pole has negative real part — then $\lim_{t\to\infty} f(t)$ exists and

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$$

Proof for rational F. Expand in partial fractions. A term c/s contributes the constant c to f and c to $\lim_{s\to0} sF$. Every other term has the form $c/(s-p)^k$ with $\operatorname{Re} p < 0$; it contributes $c\,t^{k-1}e^{pt}/(k-1)! \to 0$ to f and $\lim_{s\to0} s\,c/(s-p)^k = 0$ to sF. Both sides agree term by term. $\square$

The hypothesis is not decoration. Without it the formula returns a finite number for a signal that has no limit:

$F(s)$	$\lim_{s\to0} sF(s)$	**actual $f(t)$**	**why FVT fails**
$1/(s-1)$	0	$e^t \to \infty$	pole in the right half-plane
$\omega/(s^2+\omega^2)$	0	$\sin\omega t$, no limit	poles on the imaginary axis
$1/s^2$	∞	$t \to \infty$	double pole at the origin

In control, F is usually $E(s) = S(s)R(s)$, and the hypothesis reduces to **closed-loop stability** plus a reference whose transform has at most a simple pole at the origin (F.8). Ch. 18 §18.6.3 states the FVT without this condition; it should be read with it.

F.2 — State Space and the Transfer Function

F.2.1 — From state space to $G(s)$

Consider the LTI state-space model of Ch. 21 §21.7.1,

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u, \qquad y = \mathbf{C}\mathbf{x} + Du, \qquad \mathbf{x} \in \mathbb{R}^n$$

Transform with F.1.2: $s\mathbf{X}(s) - \mathbf{x}(0^-) = \mathbf{A}\mathbf{X}(s) + \mathbf{B}U(s)$. For s not an eigenvalue of $\mathbf{A}$, $s\mathbf{I} - \mathbf{A}$ is invertible and

$$Y(s) = \underbrace{\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{x}(0^-)}_{\text{zero-input response}} + \underbrace{\left[\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + D\right]}_{G(s)} U(s)$$

The transfer function is defined with $\mathbf{x}(0^-) = 0$:

$$\boxed{G(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + D}$$

> **[DERIVATION]** The formula is exact for LTI state-space models. The zero-input term is why Ch. 21 §21.0.2 lists "no internal initial-condition dependence" as a condition: the transfer function describes the forced response only.

In the time domain, $\mathcal{L}\{e^{\mathbf{A}t}\} = (s\mathbf{I} - \mathbf{A})^{-1}$, and the impulse response is $g(t) = \mathbf{C}e^{\mathbf{A}t}\mathbf{B}\,u(t) + D\,\delta(t)$; the full solution is $\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0) + \int_0^t e^{\mathbf{A}(t-\tau)}\mathbf{B}u(\tau)\,d\tau$.

F.2.2 — Properties

1. **Rationality and properness.** $(s\mathbf{I} - \mathbf{A})^{-1} = \operatorname{adj}(s\mathbf{I} - \mathbf{A}) / \det(s\mathbf{I} - \mathbf{A})$. The adjugate's entries are polynomials of degree at most $n-1$ and the determinant has degree n, so $\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}$ is strictly proper, and G is proper, strictly proper iff $D = 0$.
2. **Poles are eigenvalues — but not necessarily all of them.** Every pole of G is an eigenvalue of $\mathbf{A}$. The converse holds iff the realisation is **minimal** (controllable and observable); otherwise some eigenvalues cancel between numerator and denominator (T. Kailath, *Linear Systems*, Prentice-Hall 1980, Ch. 2 and Ch. 6). Ch. 21 §21.7.1's statement "eigenvalues of $\mathbf{A}$ = poles of $P(s)$" assumes minimality.
3. **Coordinate invariance.** Under $\mathbf{x} = \mathbf{T}\mathbf{z}$, $(\mathbf{A}, \mathbf{B}, \mathbf{C}, D) \to (\mathbf{T}^{-1}\mathbf{A}\mathbf{T}, \mathbf{T}^{-1}\mathbf{B}, \mathbf{C}\mathbf{T}, D)$ and G is unchanged, since $\mathbf{C}\mathbf{T}(s\mathbf{I} - \mathbf{T}^{-1}\mathbf{A}\mathbf{T})^{-1}\mathbf{T}^{-1}\mathbf{B} = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}$.

The practical danger of property 2: an unstable eigenvalue that cancels from G is a *hidden* mode. The input–output map is then bounded while an internal state grows. This is why closed-loop stability in F.5 is stated for the characteristic polynomial, not just for one transfer function.

F.2.3 — Example: the series RLC circuit

With states $x_1 = v_C$, $x_2 = i$, KVL (Appendix E §E.10) and the capacitor law give $C\dot{x}_1 = x_2$ and $L\dot{x}_2 = v_{in} - Rx_2 - x_1$:

$$\mathbf{A} = \begin{pmatrix} 0 & 1/C \\ -1/L & -R/L \end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix} 0 \\ 1/L \end{pmatrix}, \qquad \mathbf{C} = \begin{pmatrix} 1 & 0 \end{pmatrix}, \qquad D = 0$$

Then $\det(s\mathbf{I}-\mathbf{A}) = s^2 + (R/L)s + 1/(LC)$, and the first row of the adjugate is $(s + R/L,\ 1/C)$, so

$$G(s) = \frac{V_C}{V_{in}} = \frac{1/(LC)}{s^2 + (R/L)s + 1/(LC)} = \frac{\omega_0^2}{s^2 + 2\zeta\omega_0 s + \omega_0^2}$$

with $\omega_0 = 1/\sqrt{LC}$ and $\zeta = (R/2)\sqrt{C/L}$, the result of Ch. 17 §17.6.2.

F.2.4 — BIBO stability and pole locations

Theorem. A proper rational $G(s)$ is bounded-input bounded-output stable iff every pole has negative real part.

Proof sketch. (i) An LTI system $y = g * u$ is BIBO stable iff $\int_0^\infty |g(t)|\,dt < \infty$, plus a possible finite impulse at $t = 0$ from D. Sufficiency: $|y(t)| \le \sup|u| \int |g|$. Necessity: for fixed t_0 choose $u(\tau) = \operatorname{sgn} g(t_0 - \tau)$; then $y(t_0) = \int_0^{t_0} |g|$, which must stay bounded. (ii) By partial fractions, g is a finite sum of terms $t^k e^{pt}$; each is integrable iff $\operatorname{Re} p < 0$. $\square$

This is the statement of Ch. 21 §21.3.1. Poles on the imaginary axis, called "marginally stable" in Ch. 18 §18.6.3, are **not** BIBO stable: a bounded input at the pole frequency produces an unbounded (resonant) output.

F.3 — The Standard Second-Order Step Response

F.3.1 — Setting

$G(s) = \omega_0^2/(s^2 + 2\zeta\omega_0 s + \omega_0^2)$ with $\omega_0 > 0$, $\zeta > 0$, unit DC gain, no zeros, and a unit step input $U = 1/s$, zero initial state. The poles are $s_{1,2} = -\zeta\omega_0 \pm \omega_0\sqrt{\zeta^2 - 1}$ (Ch. 17 §17.7.2). Then $Y(s) = \omega_0^2/[s(s^2 + 2\zeta\omega_0 s + \omega_0^2)]$.

F.3.2 — Underdamped, $0 < \zeta < 1$

Let $\sigma = \zeta\omega_0$ and $\omega_d = \omega_0\sqrt{1-\zeta^2}$, so the denominator is $(s+\sigma)^2 + \omega_d^2$. Partial fractions:

$$Y(s) = \frac{1}{s} - \frac{s + 2\sigma}{(s+\sigma)^2 + \omega_d^2} = \frac{1}{s} - \frac{s+\sigma}{(s+\sigma)^2 + \omega_d^2} - \frac{\sigma}{\omega_d}\,\frac{\omega_d}{(s+\sigma)^2 + \omega_d^2}$$

(check: over the common denominator, $(s^2 + 2\sigma s + \omega_0^2) - s(s + 2\sigma) = \omega_0^2$, using $(s+\sigma)^2 + \omega_d^2 = s^2 + 2\sigma s + \omega_0^2$). Inverting with the frequency-shift rule $\mathcal{L}\{e^{-\sigma t} f(t)\} = F(s+\sigma)$,

$$y(t) = 1 - e^{-\sigma t}\left(\cos\omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_d t\right) = 1 - \frac{e^{-\sigma t}}{\sqrt{1-\zeta^2}}\sin(\omega_d t + \phi), \quad \cos\phi = \zeta$$

which is the bracket of Ch. 17 §17.7.3.

Peak time and overshoot. $y'(t)$ is the impulse response, $g(t) = \frac{\omega_0}{\sqrt{1-\zeta^2}} e^{-\sigma t}\sin\omega_d t$, which vanishes at $t = k\pi/\omega_d$. The first maximum is at $t_p = \pi/\omega_d$, where $\cos\omega_d t_p = -1$ and $\sin\omega_d t_p = 0$:

$$y(t_p) = 1 + e^{-\sigma\pi/\omega_d} \qquad \Longrightarrow \qquad \boxed{M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}}$$

> **[DERIVATION]** M_p is exact for the pure second-order system with no zeros. A zero, or a third pole that is not far to the left, changes it; for higher-order systems the formula is used under a dominant-pole approximation (small parameter: ratio of the dominant to the next pole's real part).

F.3.3 — Critically damped, $\zeta = 1$

$Y(s) = \omega_0^2/[s(s+\omega_0)^2] = 1/s - 1/(s+\omega_0) - \omega_0/(s+\omega_0)^2$, so

$$y(t) = 1 - e^{-\omega_0 t}\,(1 + \omega_0 t)$$

which rises monotonically: $y' = \omega_0^2 t e^{-\omega_0 t} \geq 0$.

F.3.4 — Overdamped, $\zeta > 1$

The poles $s_{1,2} = -\omega_0(\zeta \mp \sqrt{\zeta^2 - 1})$ are real and negative, with $s_1 s_2 = \omega_0^2$. Partial fractions give

$$y(t) = 1 + \frac{s_2 e^{s_1 t} - s_1 e^{s_2 t}}{s_1 - s_2}$$

(check: $y(0) = 0$ and $y'(0) = 0$, the latter because G has relative degree 2). There is no overshoot. For $\zeta \gg 1$ the slow pole is $s_1 \approx -\omega_0/(2\zeta)$ and dominates; the system behaves as first-order with time constant $2\zeta/\omega_0$.

F.3.5 — Settling time and rise time

The underdamped response lies within the envelope $1 \pm e^{-\sigma t}/\sqrt{1-\zeta^2}$. Requiring the envelope to be inside a 2% band gives

$$t_s = \frac{1}{\zeta\omega_0}\left[\ln 50 + \ln\frac{1}{\sqrt{1-\zeta^2}}\right] \approx \frac{3.9 + \ln(1/\sqrt{1-\zeta^2})}{\zeta\omega_0}$$

> **[APPROXIMATION]** $t_s \approx 4/(\zeta\omega_0)$. Three simplifications: $\ln 50 = 3.91$ rounded to 4; the factor $1/\sqrt{1-\zeta^2}$ dropped (adds 0.14 to the numerator at $\zeta = 0.5$, 0.34 at $\zeta = 0.7$); and the envelope used in place of the actual last crossing of the band. **Small parameter:** ζ^2, in the sense that the dropped logarithm is $\approx \zeta^2/2$. Numerically, the true 2% settling time is $11.2/\omega_0$ at $\zeta = 0.3$ (formula: 13.3), $8.1/\omega_0$ at $\zeta = 0.5$ (formula: 8.0), and $6.0/\omega_0$ at $\zeta = 0.7$ (formula: 5.7). **Fails when:** $\zeta \to 1$ or $\zeta > 1$, where the response is not oscillatory and settling is set by the slow real pole; for a 5% band the constant is $\ln 20 \approx 3$.

The rise time quoted in Ch. 17 §17.7.3, $t_r \approx 1.8/\omega_0$, is a curve fit rather than a derivation. The exact 10%–90% rise time is $1.32/\omega_0$ at $\zeta = 0.3$, $1.64/\omega_0$ at $\zeta = 0.5$, and $2.13/\omega_0$ at $\zeta = 0.7$; $1.8/\omega_0$ is a single number representative of the middle of this range (G. F. Franklin, J. D. Powell, A. Emami-Naeini, *Feedback Control of Dynamic Systems*, Pearson, any recent edition, Ch. 3).

F.4 — Bode Asymptotes

F.4.1 — Factorisation

A rational $G(j\omega)$ with real coefficients factors into gains, integrators $(j\omega)^{\pm 1}$, first-order factors $(1 + j\omega/p)^{\pm 1}$, and second-order factors $[1 + 2\zeta(j\omega/\omega_0) + (j\omega/\omega_0)^2]^{\pm 1}$. Because $\log|G_1 G_2| = \log|G_1| + \log|G_2|$ and $\angle G_1 G_2 = \angle G_1 + \angle G_2$, magnitude in decibels ($20\log_{10}|G|$) and phase are sums of the contributions of each factor. This additivity is the entire reason the Bode plot is drawn on logarithmic axes.

F.4.2 — First-order pole

For $G = 1/(1 + j\omega/p)$, $p > 0$:

$$20\log_{10}|G| = -10\log_{10}\left(1 + \omega^2/p^2\right) \to \begin{cases} 0 & \omega \ll p \\ -20\log_{10}(\omega/p) & \omega \gg p \end{cases}, \qquad \angle G = -\arctan(\omega/p)$$

The two asymptotes meet at the corner $\omega = p$, where the true value is $-10\log_{10} 2 = -3.01$ dB — the largest error of the asymptotic magnitude. The phase passes through -45° at the corner; the straight-line phase approximation from 0° at $p/10$ to -90° at $10p$ has maximum error $\arctan(0.1) = 5.7^\circ$ at its ends. A left-half-plane zero mirrors both plots. A right-half-plane zero, $1 - j\omega/z$, has the magnitude of a left-half-plane zero but the phase of a pole: the non-minimum-phase case of Ch. 21 §21.0.2.

F.4.3 — Integrator

$G = 1/(j\omega)$: a straight line at -20 dB/decade through 0 dB at $\omega = 1$, phase -90° at all frequencies.

F.4.4 — Second-order pole pair

With $u = \omega/\omega_0$,

$$|G|^2 = \frac{1}{(1-u^2)^2 + 4\zeta^2 u^2}$$

The asymptotes are 0 dB for $u \ll 1$ and $-40\log_{10} u$ for $u \gg 1$, meeting at $u = 1$, where the true value is $|G| = 1/(2\zeta)$. The error near the corner depends on ζ and is large for light damping. To find the resonant peak, minimise $Q(x) = (1-x)^2 + 4\zeta^2 x$ over $x = u^2$: $Q'(x) = -2(1-x) + 4\zeta^2 = 0$ gives $x = 1 - 2\zeta^2$, which is positive iff $\zeta < 1/\sqrt{2}$. Then $Q_{min} = 4\zeta^4 + 4\zeta^2(1 - 2\zeta^2) = 4\zeta^2(1-\zeta^2)$, so

$$\omega_r = \omega_0\sqrt{1 - 2\zeta^2}, \qquad M_r = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$$

which are the expressions of Ch. 17 §17.7.4. The phase goes from 0° to -180°, passing -90° at ω_0, increasingly abruptly as $\zeta \to 0$.

> **[APPROXIMATION]** Asymptotic Bode plot. **Small parameter:** distance from the corner frequencies in decades. Errors: 3 dB at a first-order corner; $|20\log_{10}(2\zeta)|$ at a second-order corner (more than 6 dB for $\zeta < 0.25$). **Fails when:** corners are closer than about a decade apart (contributions overlap) or a lightly damped pair is present.

For minimum-phase systems the phase is determined by the magnitude (Bode's gain–phase relation), so that a sustained slope of $-20n$ dB/decade implies a phase near $-90n^\circ$ (H. W. Bode, *Network Analysis and Feedback Amplifier Design*, Van Nostrand 1945). This is the basis of the loop-shaping rule in Ch. 21 §21.6.1 that the crossover slope should be about -20 dB/decade.

F.5 — The Argument Principle and the Nyquist Criterion

F.5.1 — The argument principle

Theorem. Let f be meromorphic on an open set containing a simple closed contour Γ and its interior, with no zeros or poles on Γ. If Γ is traversed once counter-clockwise, then

$$\frac{1}{2\pi j}\oint_{\Gamma}\frac{f'(s)}{f(s)}\,ds = Z - P$$

where Z and P count zeros and poles inside Γ with multiplicity. Equivalently, the image curve $f(\Gamma)$ winds $Z - P$ times counter-clockwise around the origin.

Proof sketch. Near a zero a of order m, $f(s) = (s-a)^m h(s)$ with h analytic and non-zero, so $f'/f = m/(s-a) + h'/h$: a simple pole with residue m. Near a pole of order m, the same computation gives residue $-m$. Elsewhere inside Γ, f'/f is analytic. The residue theorem gives $Z - P$. For the second form, note $f'/f = d(\log f)/ds$, so the integral equals $\frac{1}{2\pi j}$ times the change of $\log f = \ln|f| + j\arg f$ around Γ; $\ln|f|$ returns to its starting value, and the change of $\arg f$ is 2π times the winding number of $f(\Gamma)$ about 0. $\square$

If Γ is traversed **clockwise**, both signs flip: $f(\Gamma)$ encircles the origin $Z - P$ times **clockwise**. (Any standard complex-analysis text, e.g. L. V. Ahlfors, *Complex Analysis*, 3rd ed., McGraw-Hill 1979, Ch. 4.)

F.5.2 — The Nyquist contour

Let $L(s)$ be proper and rational. The **Nyquist contour** $\mathcal{D}$ runs up the imaginary axis from $-jR$ to $+jR$ and closes with a semicircle of radius $R \to \infty$ in the right half-plane; traversed in this direction it is clockwise and encloses the entire open right half-plane. Where L has poles on the imaginary axis (integrators, undamped resonances), $\mathcal{D}$ detours around them by small semicircles of radius $\epsilon \to 0$ into the right half-plane, so that those poles are counted as outside.

F.5.3 — Derivation of $Z = N + P$

Assume a unity-feedback loop with no cancellation of right-half-plane poles or zeros between C and P (otherwise a hidden unstable mode exists, F.2.2). Apply F.5.1 to $f(s) = 1 + L(s)$ on $\mathcal{D}$:

- The **zeros** of $1 + L$ are the closed-loop poles, since $T = L/(1+L)$ and $S = 1/(1+L)$ share the denominator $1 + L$ once common factors are absent. Let Z be the number in the right half-plane.
- The **poles** of $1 + L$ are the poles of L, the open-loop poles. Let P be the number in the right half-plane.
- The image $(1 + L)(\mathcal{D})$ encircles the origin exactly as $L(\mathcal{D})$ encircles -1. Let N be the number of **clockwise** encirclements of -1 by $L(\mathcal{D})$.

Since $\mathcal{D}$ is clockwise, F.5.1 gives $N = Z - P$, that is,

$$\boxed{Z = N + P}$$

The closed loop is stable iff $Z = 0$, i.e. iff $N = -P$: the Nyquist plot must encircle -1 **counter-clockwise** exactly P times. For an open-loop-stable L ($P = 0$) this reduces to "no encirclement of -1", which is the form in Ch. 21 §21.3.4.

The large semicircle. Because L is proper, $L(s) \to D_L$ (a constant, zero if strictly proper) as $|s| \to \infty$: the whole semicircle maps to one point, and in practice only $\omega \in (-\infty, \infty)$ needs plotting. Because L has real coefficients, $L(-j\omega) = \overline{L(j\omega)}$: the negative-frequency half is the mirror image of the positive half in the real axis.

The indentation at an integrator. If $L(s) \approx K_v/s$ near $s = 0$ with $K_v > 0$, the detour $s = \epsilon e^{j\theta}$, θ from $-\pi/2$ to $+\pi/2$, maps to $L \approx (K_v/\epsilon)e^{-j\theta}$: an arc of radius $K_v/\epsilon \to \infty$ swept **clockwise** through 180°, from $+90^\circ$ to -90°. This arc closes the plot at infinity and must be included when counting N.

> **[DERIVATION]** The Nyquist criterion is an exact consequence of the argument principle for rational L; it extends to L containing a pure delay $e^{-\theta s}$, because the argument principle needs only meromorphy, not rationality. It is a statement about the LTI model, and inherits every approximation that produced that model.

> **[STRUCTURAL CONNECTION]** The encirclement count is the winding number (degree) of a closed curve in $\mathbb{C} \setminus \{-1\}$ — the same mathematical object as the integer invariants of Ch. 7 in that it is unchanged by continuous deformation that avoids the singular point. The connection is mathematical, not a physical genealogy: the Nyquist plot is not a field configuration and nothing in it descends from band topology. Ch. 21 §21.3.4 tags this remark `[ANALOGY]`; the classification in this book's own taxonomy (two systems sharing one mathematical object) is closer to `[STRUCTURAL CONNECTION]`.

F.5.4 — Gain and phase margins

For an open-loop-stable L whose Nyquist plot crosses the negative real axis and the unit circle once each:

- **Phase crossover** ω_{pc}: $\angle L(j\omega_{pc}) = -180^\circ$. **Gain margin** $GM = 1/|L(j\omega_{pc})|$, i.e. $-20\log_{10}|L(j\omega_{pc})|$ dB: the factor by which the gain can rise before the plot passes through -1.
- **Gain crossover** ω_{gc}: $|L(j\omega_{gc})| = 1$. **Phase margin** $PM = 180^\circ + \angle L(j\omega_{gc})$: the extra phase lag that would bring the plot through -1.

These are the definitions of Ch. 21 §21.3.3. Their meaning comes from F.5.3: they measure the distance to the critical point along two particular directions. They can mislead when the plot crosses the unit circle or the real axis more than once, or for conditionally stable systems (stable only for an interval of gains). A direction-free measure is the **vector margin** $\min_\omega |1 + L(j\omega)| = 1/\max_\omega |S(j\omega)|$.

F.5.5 — Worked example: $L(s) = K/[s(s+1)(s+2)]$

$P = 0$; one integrator. The phase is $\angle L(j\omega) = -90^\circ - \arctan\omega - \arctan(\omega/2)$. It equals -180° when $\arctan\omega + \arctan(\omega/2) = 90^\circ$, i.e. when $\omega \cdot \omega/2 = 1$ (the tangent-addition denominator vanishes), so $\omega_{pc} = \sqrt{2}$ rad/s. There

$$|L(j\sqrt{2})| = \frac{K}{\sqrt{2}\cdot\sqrt{3}\cdot\sqrt{6}} = \frac{K}{6}$$

so $GM = 6/K$, and the loop is closed-loop stable iff $K < 6$. For $K = 1$: $GM = 6$ (15.6 dB), and solving $\omega\sqrt{\omega^2+1}\sqrt{\omega^2+4} = 1$ gives $\omega_{gc} = 0.446$ rad/s and $PM = 53.4^\circ$. F.6 recovers $K_u = 6, \omega = \sqrt{2}$ algebraically.

F.6 — The Routh–Hurwitz Criterion

F.6.1 — Statement

Let $\Delta(s) = a_n s^n + a_{n-1}s^{n-1} + \cdots + a_0$ with real coefficients and $a_n > 0$. Build the Routh array: row s^n holds $a_n, a_{n-2}, a_{n-4}, \ldots$; row s^{n-1} holds $a_{n-1}, a_{n-3}, \ldots$; each subsequent entry is

$$b_i = \frac{a_{n-1}a_{n-2i} - a_n a_{n-2i-1}}{a_{n-1}}$$

formed from the two rows above it, and so on down to row s^0 (missing coefficients are zero).

Theorem (Routh 1877, Hurwitz 1895). If no first-column entry is zero, the number of roots of Δ in the open right half-plane equals the number of sign changes in the first column. In particular Δ is **Hurwitz** (all roots in the open left half-plane) iff every first-column entry is positive.

F.6.2 — A necessary condition, proved

If Δ is Hurwitz with $a_n > 0$, every coefficient is positive. *Proof.* $\Delta = a_n \prod(s+\alpha_i)\prod(s^2+\beta_k s+\gamma_k)$ over real roots $-\alpha_i$ ($\alpha_i > 0$) and complex pairs with $\beta_k = -2\,\mathrm{Re}(\text{root}) > 0$, $\gamma_k = |\text{root}|^2 > 0$. A product of polynomials with positive coefficients has positive coefficients. □ The condition is not sufficient for $n \geq 3$: $s^3 + s^2 + s + 6 = (s+2)(s^2 - s + 3)$ has positive coefficients and two right-half-plane roots.

F.6.3 — Proof sketch

The idea is a reduction lemma. Write $\Delta(s) = r_1(s) + r_2(s)$ where r_1 collects the terms of degree $n, n-2, \ldots$ and r_2 those of degree $n-1, n-3, \ldots$ (the first two rows of the array). The third row is the polynomial $r_3 = r_1 - c\,s\,r_2$ with $c = a_n/a_{n-1}$, which removes the leading term. **Lemma:** if $c > 0$, then Δ is Hurwitz iff the degree-$(n-1)$ polynomial $r_2 + r_3$ — whose array is the original array with its first row deleted — is Hurwitz. Applied recursively, this reduces the question to a chain of positivity conditions on the first column. The lemma is proved through the Hermite–Biehler theorem: Δ is Hurwitz iff its even and odd parts have real, simple, interlacing roots on the imaginary axis with leading coefficients of the same sign — equivalently, iff $\arg\Delta(j\omega)$ increases monotonically by $n\pi/2$ as ω goes from 0 to ∞ (an argument-principle statement, F.5.1) — and the Routh step preserves interlacing. The count of right-half-plane roots by sign changes follows from Sturm-sequence (Cauchy-index) arguments. A complete proof is in F. R. Gantmacher, *The Theory of Matrices*, Vol. 2, Chelsea 1959, Ch. XV.

F.6.4 — Special cases

- **A zero in the first column, rest of row non-zero.** Replace it by a small $\epsilon > 0$, continue, and count sign changes as $\epsilon \to 0^+$. This is a perturbation argument: roots move continuously with coefficients, and the perturbation does not move any root across the imaginary axis for ϵ small enough unless one lies on it.

- **An entire row of zeros.** This happens iff Δ has a factor that is an **even polynomial**, the *auxiliary polynomial* formed from the row above. An even polynomial's roots are symmetric about the origin: they come as $\pm a$ (real), $\pm j\omega$ (imaginary), or quadruples $\pm\alpha \pm j\beta$. A zero row therefore signals roots symmetric about the origin, **not necessarily** imaginary-axis roots; Ch. 21 §21.3.2's gloss "the system has poles on the imaginary axis" covers only one of the three cases. The procedure replaces the zero row by the coefficients of the derivative of the auxiliary polynomial and continues; the roots of the auxiliary polynomial are then found directly.

F.6.5 — Third order

For $a_3 s^3 + a_2 s^2 + a_1 s + a_0$ with $a_3 > 0$, the array is

$$\begin{array}{c|cc} s^3 & a_3 & a_1 \\ s^2 & a_2 & a_0 \\ s^1 & (a_2 a_1 - a_3 a_0)/a_2 & 0 \\ s^0 & a_0 & \end{array}$$

so the polynomial is Hurwitz iff $a_2 > 0$, $a_0 > 0$, and $a_2 a_1 > a_3 a_0$; with $a_3 = 1$ these are the conditions of Ch. 21 §21.3.2.

F.6.6 — Worked example: $1 + K/[s(s+1)(s+2)] = 0$

The characteristic polynomial is $s(s+1)(s+2) + K = s^3 + 3s^2 + 2s + K$, with array

$$\begin{array}{c|cc} s^3 & 1 & 2 \\ s^2 & 3 & K \\ s^1 & (6-K)/3 & 0 \\ s^0 & K & \end{array}$$

Stability requires $K > 0$ and $6 - K > 0$: $0 < K < 6$, so the ultimate gain is $K_u = 6$. At $K = 6$ the s^1 row vanishes; the auxiliary polynomial from the s^2 row is $3s^2 + 6$, with roots $s = \pm j\sqrt{2}$. Indeed $s^3 + 3s^2 + 2s + 6 = (s+3)(s^2+2)$. The loop oscillates at $\omega_u = \sqrt{2}$ rad/s, period $T_u = 2\pi/\sqrt{2} = \pi\sqrt{2} \approx 4.44$ s — the quantities a Ziegler–Nichols experiment (Ch. 21 §21.4.3) would measure. For $K > 6$ the first column reads $1, 3, (6-K)/3 < 0, K > 0$: two sign changes, two right-half-plane poles. This agrees with F.5.5.

F.7 — Root-Locus Rules

F.7.1 — Angle and magnitude conditions

Let $L(s) = KG(s)$ with $G = N(s)/D(s)$, N and D monic of degrees $m < n$, open-loop zeros z_i and poles p_i, and $K \geq 0$. The closed-loop poles are the roots of $D(s) + KN(s) = 0$, i.e. $KG(s) = -1$. A point s is on the locus iff

$$\angle G(s) = \sum_i \angle(s - z_i) - \sum_i \angle(s - p_i) = (2k+1)\,180^\circ, \qquad K = \frac{1}{|G(s)|} = \frac{\prod |s - p_i|}{\prod |s - z_i|}$$

The angle condition determines the locus; the magnitude condition then gives the gain at each point.

F.7.2 — Derivation of the rules

1. n **branches, continuous in** K**.** $D + KN$ has degree n for every K, and the roots of a polynomial depend continuously on its coefficients.
2. **Start and end.** At $K = 0$ the roots are those of D — the open-loop poles. Dividing by K, $D/K + N = 0$; as $K \to \infty$, m roots approach the zeros of N, and the remaining $n - m$ go to infinity.
3. **Real axis.** For real s, each complex-conjugate pair of poles or zeros contributes angles that cancel. A real pole or zero to the right of s contributes 180°; one to the left contributes 0°. The angle condition holds iff the number of real poles plus zeros to the right of s is **odd**.
4. **Asymptotes and centroid.** Expand for large $|s|$: $D = s^n - (\sum p_i)s^{n-1} + \cdots$ and $N = s^m - (\sum z_i)s^{m-1} + \cdots$, so

$$\frac{D}{N} = s^{n-m} - \left(\sum p_i - \sum z_i\right) s^{n-m-1} + O(s^{n-m-2}) = (s - \sigma_a)^{n-m} + O(s^{n-m-2})$$

with $\sigma_a = (\sum p_i - \sum z_i)/(n-m)$, using $(s-\sigma_a)^q = s^q - q\sigma_a s^{q-1} + \cdots$. The characteristic equation $D/N = -K$ becomes $(s-\sigma_a)^{n-m} \approx -K$, so

$$s \approx \sigma_a + K^{1/(n-m)} e^{j(2k+1)\pi/(n-m)}, \qquad k = 0, \ldots, n-m-1$$

$n - m$ straight-line asymptotes radiating from the centroid σ_a at angles $(2k+1)180^\circ/(n-m)$, as in Ch. 21 §21.5.1. 5. **Breakaway and break-in points.** Where two branches meet on the real axis, s is a double root of $\Delta(s) = D + KN$: $\Delta(s) = 0$ and $\Delta'(s) = D' + KN' = 0$. Eliminating K ($K = -D/N = -D'/N'$) gives $D'N - DN' = 0$. Since along the real axis $K(s) = -D(s)/N(s)$ and $dK/ds = -(D'N - DN')/N^2$, this is $dK/ds = 0$. The condition is necessary, not sufficient: a candidate is accepted only if it lies on a real-axis segment of the locus and gives $K > 0$. 6. **Imaginary-axis crossings.** Found from Routh–Hurwitz (F.6) as the gain at which a row vanishes, or by substituting $s = j\omega$ into Δ and solving real and imaginary parts. 7. **Departure angles** from a complex pole p_ℓ follow from the angle condition at a point infinitesimally close to p_ℓ: $\theta_{dep} = 180^\circ + \sum \angle(p_\ell - z_i) - \sum_{i \neq \ell} \angle(p_\ell - p_i)$.

F.7.3 — Worked example: $G(s) = 1/[s(s+1)(s+2)]$

Here $n = 3$, $m = 0$, poles $0, -1, -2$.

- **Real-axis segments:** $[-1, 0]$ (one pole to the right) and $(-\infty, -2]$ (three poles to the right).
- **Asymptotes:** angles $60^\circ, 180^\circ, 300^\circ$; centroid $\sigma_a = (0 - 1 - 2)/3 = -1$.
- **Breakaway:** $K(s) = -s(s+1)(s+2) = -(s^3 + 3s^2 + 2s)$, so $dK/ds = -(3s^2 + 6s + 2) = 0$ gives

$$s = -1 \pm \frac{1}{\sqrt{3}} = -0.423 \text{ or } -1.577$$

The point -1.577 lies in $(-2, -1)$, which is not on the locus, and gives $K = -0.385 < 0$: rejected. The point $s = -0.423$ lies on $[-1, 0]$. With $u = s + 1 = 1/\sqrt{3}$, $s(s+1)(s+2) = u(u^2 - 1) = -2/(3\sqrt{3})$, so $K = 2/(3\sqrt{3}) \approx 0.385$. At this gain the closed-loop poles are a double pole at -0.423 and a third at $-3 + 2(0.423) = -2.155$ (their sum must equal -3, the negative of the s^2 coefficient).

- **Imaginary-axis crossing:** $K = 6$ at $s = \pm j\sqrt{2}$ (F.6.6), with the third pole at -3.

For $0 < K < 0.385$ all three closed-loop poles are real (overdamped); for $0.385 < K < 6$ the dominant pair is complex and moves toward the 60° asymptote, with damping decreasing as K rises; for $K > 6$ it is in the right half-plane. **Reading damping from pole angle:** a pole $-\zeta\omega_0 + j\omega_0\sqrt{1-\zeta^2}$ lies at distance

ω_0 from the origin, at angle θ from the **negative real axis** with $\cos\theta = \zeta$. Ch. 21 §21.5.2 measures this angle "from the imaginary axis"; measured from there, ζ is the *sine* of the angle.

F.8 — Steady-State Error and System Type

F.8.1 — Setting

Unity negative feedback, $E(s) = R(s) - Y(s) = S(s)R(s) = R(s)/[1 + L(s)]$. Write $L(s) = \frac{K\,\tilde{N}(s)}{s^q\,\tilde{D}(s)}$ with $\tilde{N}(0)$, $\tilde{D}(0)$ non-zero and normalised to 1; the integer $q \geq 0$ is the **system type**, the number of open-loop integrators. Assume the closed loop is stable, so that the FVT hypothesis of F.1.4 can hold.

F.8.2 — Error constants

Apply the FVT to $E(s)$ for references $R = 1/s$, $1/s^2$, $1/s^3$:

$$e_{step} = \lim_{s\to 0} \frac{1}{1 + L(s)} = \frac{1}{1 + K_{pos}}, \qquad e_{ramp} = \lim_{s\to 0} \frac{1}{s + sL(s)} = \frac{1}{K_v}, \qquad e_{parab} = \lim_{s\to 0} \frac{1}{s^2 + s^2 L(s)} = \frac{1}{K_a}$$

with $K_{pos} = \lim_{s\to 0} L(s)$, $K_v = \lim_{s\to 0} sL(s)$, $K_a = \lim_{s\to 0} s^2 L(s)$. (Ch. 21 §21.4.2 writes the position constant as K_p, which collides with the proportional gain of §21.4.1; it is written K_{pos} here.)

Type q	K_{pos}	K_v	K_a	e_{step}	e_{ramp}	e_{parab}
0	K	0	0	$1/(1+K)$	∞	∞
1	∞	K	0	0	$1/K$	∞
2	∞	∞	K	0	0	$1/K$

Where an entry is ∞, the FVT hypothesis fails for $E(s)$ — it has a double or higher pole at the origin — and the error grows without bound; the table entry is the correct conclusion, but it is reached by noting the failure, not by applying the formula.

> **[DERIVATION]** The error constants follow from the FVT under closed-loop stability. Without stability the FVT returns a finite number for a diverging error.

Disturbances. For a step disturbance entering at the plant input, the error is $-P/(1 + CP)$ times the disturbance, and its steady-state value is zero iff the integrator is in the **controller** C — an integrator in the plant downstream of the disturbance does not reject it. This is the elementary case of the internal model principle.

Example. For $L = K/[s(s+1)(s+2)]$ the type is 1, $K_v = \lim sL = K/2$, and $e_{ramp} = 2/K$. Since stability requires $K < 6$, the ramp error cannot be made smaller than $1/3$ with proportional control alone: a concrete instance of the gain–stability trade-off of Ch. 21 §21.1.2.

F.9 — The Bode Sensitivity Integral

F.9.1 — Statement

Theorem. Let $L(s)$ be rational with **relative degree at least 2**, let the closed loop be stable, and let $p_1, \dots, p_{N_p}$ be the open-loop poles in the open right half-plane. Then

$$\int_0^\infty \ln |S(j\omega)| \, d\omega = \pi \sum_{k=1}^{N_p} \operatorname{Re} p_k$$

For a stable open loop the right side is zero. This is the integral of Ch. 21 §21.6.1, which states it without the relative-degree and closed-loop-stability conditions; both are needed. (Bode 1945, as cited in F.4.4, for the stable case; J. S. Freudenberg and D. P. Looze, "Right half plane poles and zeros and design tradeoffs in feedback systems", *IEEE Transactions on Automatic Control* 30(6), 555–565, 1985, for unstable open loops; a textbook treatment is M. M. Seron, J. H. Braslavsky, and G. C. Goodwin, *Fundamental Limitations in Filtering and Control*, Springer 1997.)

F.9.2 — Proof for a stable open loop

With $P = 0$ and closed-loop stability, $S = 1/(1+L)$ has no poles in the closed right half-plane (closed-loop stability) and no zeros there (its zeros are the poles of L). So $\ln S$ has an analytic branch on the right half-plane and is continuous up to the imaginary axis. (If L has imaginary-axis poles, S has zeros there and $\ln|S| \to -\infty$ integrably; the standard indentation argument handles them.) By Cauchy's theorem, the integral of $\ln S$ around the Nyquist contour $\mathcal{D}$ of F.5.2 vanishes. Split it:

- **Imaginary axis**, $s = j\omega$, $ds = j\,d\omega$: the contribution is $j\int_{-R}^{R} \ln S(j\omega)\,d\omega$. Because $S(-j\omega) = \overline{S(j\omega)}$, $\ln|S|$ is even and $\arg S$ is odd, so this equals $2j\int_0^R \ln|S(j\omega)|\,d\omega$.
- **Semicircle**, $s = Re^{j\theta}$, θ from $\pi/2$ to $-\pi/2$: for large $|s|$, $L(s) \approx c/s^r$ with r the relative degree, so $\ln S = -\ln(1+L) = -L + O(L^2) \approx -c/s^r$. The arc has length πR and the integrand is $O(R^{-r})$, so the contribution is $O(R^{1-r}) \to 0$ when $r \geq 2$.

Hence $\int_0^\infty \ln|S(j\omega)|\,d\omega = 0$. $\square$

For relative degree exactly 1, the arc integral does not vanish: with $L \approx c/s$, $\int_{arc}(-c/s)\,ds = -c\int_{\pi/2}^{-\pi/2} j\,d\theta = j\pi c$, and the result becomes $\int_0^\infty \ln|S|\,d\omega = -\frac{\pi}{2}\lim_{s\to\infty} sL(s)$. For unstable open loops, S has zeros at the p_k; factoring out the Blaschke product $\prod (s-p_k)/(s+\bar{p}_k)$ and repeating the argument gives the $\pi\sum \operatorname{Re} p_k$ term (Freudenberg and Looze, cited above).

F.9.3 — Interpretation: the waterbed

$\ln|S| < 0$ where feedback attenuates disturbances and $\ln|S| > 0$ where it amplifies them. With zero total area, any band of attenuation must be paid for by a band of amplification — the **waterbed effect**. The theorem alone does not force a large peak, because the amplification could be spread thinly over an infinite band. What forces the peak is the combination with a bandwidth limit: actuator limits, sensor noise, and unmodelled high-frequency dynamics make $|L|$ small above some ω_{max}, so $\ln|S| \approx 0$ there, and the positive area must be concentrated below ω_{max}. Open-loop unstable poles make the budget strictly positive: they cost sensitivity whatever the controller.

> **[DERIVATION]** The integral is an exact consequence of analyticity for LTI rational loops. Its engineering force depends on a bandwidth constraint that is a modelling statement about the plant and actuators, not part of the theorem.

F.10 — The DC Motor Transfer Function

F.10.1 — Model and assumptions

The assumptions are those of Example 06 and Ch. 21 §21.8.1: linear magnetics (constant torque constant K_t and back-EMF constant K_b), a lumped armature R_a–L_a branch, viscous friction $B\omega$, a rigid rotor of inertia J, and a load torque T_L treated as a disturbance.

Electrical side. By the lumped KVL of Appendix E §E.10 — with the armature inductance carrying its own $L_a\, di_a/dt$ branch voltage, and the motional EMF $K_b\omega$ appearing as a branch voltage in series —

$$V_a = R_a I_a + L_a \frac{dI_a}{dt} + K_b\,\omega$$

Mechanical side. Newton's second law for the rotor:

$$J\frac{d\omega}{dt} + B\omega = K_t I_a - T_L$$

Energy consistency. The electrical power absorbed by the back-EMF is $K_b\omega I_a$ and the mechanical power produced is $K_t I_a \omega$; for ideal electromechanical conversion these are equal, so $K_t = K_b$ in SI units (N·m/A = V·s/rad). This is the gyrator of Ch. 17 §17.5 and Ch. 21 §21.8.1.

F.10.2 — Speed transfer function

With zero initial conditions and $T_L = 0$, $(L_a s + R_a) I_a = V_a - K_b\Omega$ and $(Js + B)\Omega = K_t I_a$. Eliminating $I_a = (Js+B)\Omega/K_t$:

$$\left[(L_a s + R_a)(Js + B) + K_t K_b\right]\Omega = K_t V_a$$

$$\boxed{\frac{\Omega(s)}{V_a(s)} = \frac{K_t}{(L_a s + R_a)(Js + B) + K_t K_b} = \frac{K_t}{L_a J s^2 + (L_a B + R_a J)s + (R_a B + K_t K_b)}}$$

This is Example 06's result. Three facts follow.

1. **No free integrator.** The denominator's constant term is $R_a B + K_t K_b > 0$, so $s = 0$ is not a pole. The DC gain is finite: $\Omega/V_a \to K_t/(R_a B + K_t K_b)$, and even with $B = 0$ it is $1/K_b$. A constant voltage produces a constant speed, not unbounded acceleration.
2. **The back-EMF removes, not creates, the integrator.** Without back-EMF and friction ($K_b = 0$, $B = 0$) the plant would be $K_t/[(L_a s + R_a)Js]$, which does contain an integrator: torque proportional to voltage, integrated by inertia. The $K_t K_b$ term is a velocity feedback internal to the motor; it adds damping (Example 06 shows it moving the mechanical pole from $-B/J$ to a much faster value).
3. **Stability.** All coefficients are positive and the polynomial is second order, so both poles are in the left half-plane (F.6.2 is sufficient at $n = 2$).

F.10.3 — Position transfer function

The shaft angle satisfies $\dot{\theta} = \omega$, so $\Theta = \Omega/s$ and

$$\frac{\Theta(s)}{V_a(s)} = \frac{K_t}{s\left[(L_a s + R_a)(Js + B) + K_t K_b\right]}$$

which has exactly one integrator. A position loop around this plant with proportional control is type 1 (zero steady-state error to a position step, F.8); a speed loop is type 0 and needs integral action for zero steady-state speed error.

F.10.4 — The reduced first-order speed model

> **[APPROXIMATION]** Neglect of armature inductance. Setting $L_a \to 0$ gives
>
> $$\frac{\Omega}{V_a} \approx \frac{K_m}{\tau_m s + 1}, \qquad K_m = \frac{K_t}{R_a B + K_t K_b}, \qquad \tau_m = \frac{R_a J}{R_a B + K_t K_b}$$
>
> **Small parameter:** τ_e/τ_m, with $\tau_e = L_a/R_a$. Formally, the second-order denominator has two real, widely separated poles near $-1/\tau_e$ and $-1/\tau_m$ when $4L_aJ(R_aB + K_tK_b) \ll (L_aB + R_aJ)^2$; dropping the fast pole is a singular-perturbation (dominant-pole) approximation. **Fails when:** the controller bandwidth approaches $1/\tau_e$, where the neglected electrical pole adds phase lag and the margins computed from the first-order model are optimistic.

With τ_m written as J/R_t, $R_t = B + K_tK_b/R_a$, this is the time constant of Example 06. The form $K_m/[s(\tau_m s + 1)]$ that appears in Ch. 21 §21.8.1 is the **position** plant Θ/V_a under the same approximation — not the speed plant. Ch. 21 §21.8.1 presents it as Ω/V_a and attributes its integrator to the back-EMF; by F.10.2 both statements are incorrect, and the $1/s$ belongs to the angle, $\dot{\theta} = \omega$.

F.11 — Summary

Result	Statement	Assumptions	Used in
Derivative rule	$\mathcal{L}\{f'\} = sF - f(0^-)$	f, f' of exponential order	Ch. 17 §17.6; Ch. 18 §18.6.3; Ch. 21 §21.7.1
Convolution	$\mathcal{L}\{f * g\} = FG$	causal, exponential order	Ch. 18 §18.6.3; Ch. 21 §21.2
Final value theorem	$f(\infty) = \lim_{s\to 0} sF(s)$	poles of sF in open LHP	Ch. 18 §18.6.3; Ch. 21 §21.4.2
State space → TF	$G = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + D$	LTI; zero initial state	Ch. 21 §21.7.1
Poles vs eigenvalues	poles $\subseteq$ eig($\mathbf{A}$), equal iff minimal	LTI	Ch. 21 §21.7.1–§21.7.2
BIBO stability	iff all poles have $\operatorname{Re} p < 0$	proper rational G	Ch. 18 §18.6.3; Ch. 21 §21.3.1
Second-order step	$y = 1 - e^{-\zeta\omega_0 t}(\cos\omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_d t)$	standard form, no zeros	Ch. 17 §17.7.3; Ch. 21 §21.2.2

Result	Statement	Assumptions	Used in
Overshoot	$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}$, $t_p = \pi/\omega_d$	$0 < \zeta < 1$, no zeros	Ch. 17 §17.7.3; Ch. 21 §21.5.2
Settling time	$t_s \approx 4/(\zeta\omega_0)$ (2% band)	envelope bound; ζ^2 small	Ch. 17 §17.7.3; Ch. 21 §21.5.2
Bode asymptotes	-20 dB/dec per pole; 3 dB corner error; $M_r = 1/(2\zeta\sqrt{1-\zeta^2})$	rational G; corners separated	Ch. 17 §17.6.2, §17.7.4; Ch. 21 §21.6
Argument principle	winding of $f(\Gamma)$ about $0 = Z - P$	f meromorphic, nothing on Γ	Ch. 21 §21.3.4
Nyquist criterion	$Z = N + P$ (N clockwise about -1)	LTI; no unstable cancellations	Ch. 21 §21.3.4
Margins	$GM = 1/\lvert L(j\omega_{pc})\rvert$, $PM = 180^\circ + \angle L(j\omega_{gc})$	single crossings; stable L	Ch. 21 §21.3.3, §21.6
Routh–Hurwitz	RHP roots = sign changes in first column	real coefficients; special cases handled	Ch. 21 §21.3.2, §21.5.1
Example $K/[s(s+1)(s+2)]$	$K_u = 6$, $\omega_u = \sqrt{2}$; breakaway -0.423 at $K = 0.385$	unity feedback, $K \geq 0$	Ch. 21 §21.3, §21.4.3, §21.5
Root-locus rules	angle/magnitude conditions; centroid $(\sum p - \sum z)/(n-m)$; $dK/ds = 0$	$m < n$, $K \geq 0$	Ch. 21 §21.5.1
Steady-state error	$1/(1+K_{pos})$, $1/K_v$, $1/K_a$ by type	closed-loop stable	Ch. 21 §21.4.1–§21.4.2
Bode sensitivity integral	$\int_0^\infty \ln\lvert S\rvert\, d\omega = \pi \sum \operatorname{Re} p_k$	relative degree ≥ 2; closed-loop stable	Ch. 21 §21.6.1, §21.10
DC motor speed	$\Omega/V_a = K_t/[(L_a s + R_a)(Js + B) + K_t K_b]$, no integrator	linear magnetics; lumped armature; viscous friction	Ch. 21 §21.8.1; Example 06
DC motor position	$\Theta/V_a = (\Omega/V_a)/s$, one integrator	as above	Ch. 21 §21.8.1

End of Appendix F

REFERENCES.md — Sources for the Load-Bearing Claims

Engineering Physics: Top Down — provenance for every claim that carries weight in the argument.

How to read this file

This file traces every load-bearing factual claim back to its source, chapter by chapter, in the order the book proceeds — Ch. 0 through Ch. 21 and the Epilogue — so the reference chain mirrors the derivation chain. A claim in Ch. 7 that echoes a structure introduced in Ch. 0 is filed under Ch. 7 and cross-references back to Ch. 0, exactly as the book's own "Revealed by" / "→ Ch. X" convention does.

Design principle: a citation is a liability as well as an asset. Attaching a reference to a claim that does not need one, or that the reference does not actually support, makes the book look better-sourced than it is. So every entry here is typed, and the typing carries information.

Evidence tiers

Tier	Meaning	How it is cited
[MEASURED]	An experimentally measured number or a discovery, traceable to one primary paper (or a small number of them)	Full citation with DOI where one exists
[DERIVED / TEXTBOOK]	Standard, uncontested physics, not tied to a single paper — the kind of thing that appears identically in every graduate textbook on the subject	Cited to a standard textbook (`T#` below), because that is the honest citation. No single paper "discovered" the textbook derivation of the Wiedemann–Franz ratio, even though von Klitzing's *measurement* of it is a paper
[THEORY]	An original theoretical result, prediction, or no-go theorem — traceable to one or a small number of primary papers, but not itself an observation	Full citation with DOI to the primary theory paper(s). Kept distinct from [DERIVED / TEXTBOOK]: a *named original result* (BCS, ABJ anomaly, Nielsen–Ninomiya, Higgs mechanism, Abrikosov vortices) is credited to its authors' paper, not to a textbook, and distinct from [MEASURED]: a theoretical prediction is not a measurement even when it is later confirmed
[SYNTHESIS]	The book's own argument or pedagogical framing, not a fact borrowed from the literature	Flagged explicitly rather than given a decorative citation. This is the tier that matters most: the book's central claims live here, and marking them is the point
[TARGET]	A forward-looking engineering target, design goal, or institutional projection — documented by the responsible body but *not* an observed result and not yet achieved	Cited to the project's own documentation. Kept distinct from [MEASURED]: a goal is not data, and must never be quoted as though the specification had been met

Numbered citation key

Each entry below carries an `[R#]` label and names the chapter(s) it serves, in the "Ch." column of the key table. **Chapter prose does not carry inline `[R#]` markers** — the manuscript cites by author and year in the sentence ("Pound and Rebka (1959)", "von Klitzing, 1980"), which reads better in continuous technical prose than a bracketed number would. The `[R#]` labels are therefore an index *into this file*, not a citation syntax the text depends on. When you add a source, give it the next free number, add it to the key table with its chapter coverage, and put the full entry in that chapter's section below.

Textbook references are numbered `[T#]` and collected in the Textbooks section.

#	Source	Ch.
R1	CODATA recommended values (NIST)	0, 4, 7, 10, 11
R2	Particle Data Group, *Review of Particle Physics*	0
R3	Englert & Brout; Higgs; Guralnik–Hagen–Kibble (1964)	0
R4	Wu et al., parity violation (1957)	0
R5	ACME Collaboration, electron EDM bound (2018)	0
R6	Georgi, Quinn & Weinberg, coupling unification (1974)	0
R7	Dirac, "The Quantum Theory of the Electron" (1928)	3
R8	Anderson, positron (1933)	3
R9	Foldy & Wouthuysen (1950)	3
R10	Rydberg (1890)	4
R11	Stern & Gerlach (1922)	5
R12	Lamb & Retherford (1947)	5
R13	BIPM, *SI Brochure*, 9th ed.	5, 7
R14	Bragg (1913)	6
R15	von Weizsäcker, semi-empirical mass formula (1935)	6
R16	Geiger & Nuttall (1911–12)	6
R17	ITER Organization, project goals	6
R18	Berry, quantal phase factors (1984)	7
R19	von Klitzing, Dorda & Pepper (1980)	7, 13
R20	Adler (1969); Bell & Jackiw (1969)	7
R21	Nielsen & Ninomiya (1981)	7
R22	Huang et al., chiral anomaly in TaAs (2015)	7
R23	Bardeen, Cooper & Schrieffer (1957)	7
R24	Deaver & Fairbank; Doll & Näbauer (1961)	7
R25	Josephson (1962)	7
R26	Abrikosov (1957)	7
R27	Pound & Rebka (1959)	8
R28	Everitt et al., Gravity Probe B (2011)	8

#	Source	Ch.
R29	Abbott et al., GW150914 (2016)	8
R30	Einstein, Mercury precession (1915)	8
R31	Dyson, Eddington & Davidson (1920); Shapiro et al. (2004)	8
R32	Ashby, relativity in GPS (2003)	8
R33	Zurek, decoherence (2003)	9
R34	Boltzmann, Stefan's law (1884)	10
R35	Wien, displacement law (1896)	10
R36	Landau, phase transitions (1937)	10
R37	Maxwell, dynamical theory of the EM field (1865)	11
R38	Kubo, linear response (1957)	13
R39	Wiedemann & Franz (1853)	13
R40	Mott & Jones, theory of metals and alloys (1936)	13
R41	Hall (1879)	14
R42	Richardson; Dushman (1921/1923)	14
R43	BIPM Resolution 1, 26th CGPM — 2019 SI redefinition	14
R44	IEC 60751:2008, Pt100 RTD	14
R45	Reynolds (1883)	15–20
R46	Kolmogorov (1941)	15–20
R47	Dittus & Boelter (1930)	19, 20
R48	Terzaghi, *Erdbaumechanik* (1925)	20
R49	Cooley & Tukey, FFT (1965)	18
R50	Kalman, general theory of control systems (1960)	20, 21

Chapter 0 — The Equation (Standard Model + Einstein–Hilbert action)

[MEASURED] [R1] Newton's gravitational constant, $G = 6.674 \times 10^{-11}\,\mathrm{N \cdot m^2/kg^2}$ Source: CODATA recommended values (NIST). This is the least precisely known fundamental constant (relative uncertainty ~2.2×10⁻⁵), a fact worth noting alongside the number itself. Link: https://physics.nist.gov/cgi-bin/cuu/Value?bg

[MEASURED] [R2] Weinberg angle, $\theta_W \approx 28.7^\circ$ **(**$\sin^2\theta_W \approx 0.231$**)** Source: Particle Data Group, *Review of Particle Physics* (current edition), Electroweak Model section. Link: https://pdg.lbl.gov/

[MEASURED] [R2] W and Z boson masses ($m_W \approx 80.4$ **GeV,** $m_Z \approx 91.2$ **GeV)** Source: Particle Data Group, *Review of Particle Physics*, Gauge and Higgs Boson Summary Table.

[DERIVED / TEXTBOOK] [R2] Higgs vacuum expectation value $v \approx 246$ **GeV** Derived from the measured Fermi constant G_F via $v = (\sqrt{2}G_F)^{-1/2}$; this is a standard electroweak-fit result, not an independent measurement. Tiered [DERIVED / TEXTBOOK] accordingly — the input G_F is [MEASURED] (see R1/R2), the VEV quoted here is a fit-derived quantity.

[MEASURED] [R3] Higgs boson discovery and mass ($m_h = 125.09$ GeV) Source: ATLAS Collaboration, "Observation of a new particle in the search for the Standard Model Higgs boson with the ATLAS detector at the LHC," *Phys. Lett. B* 716, 1–29 (2012). DOI: 10.1016/j.physletb.2012.08.020 Companion paper: CMS Collaboration, "Observation of a new boson at a mass of 125 GeV with the CMS experiment at the LHC," *Phys. Lett. B* 716, 30–61 (2012). DOI: 10.1016/j.physletb.2012.08.021 Combined ATLAS+CMS mass measurement, 125.09 ± 0.24 GeV: ATLAS+CMS Collaborations, *Phys. Rev. Lett.* 114, 191803 (2015). DOI: 10.1103/PhysRevLett.114.191803

[THEORY] [R3] Original Higgs-mechanism theory papers (1964)

- Englert, F. & Brout, R., "Broken Symmetry and the Mass of Gauge Vector Mesons," *Phys. Rev. Lett.* 13, 321–323 (1964). DOI: 10.1103/PhysRevLett.13.321
- Higgs, P.W., "Broken Symmetries, Massless Particles and Gauge Fields," *Phys. Lett.* 12, 132–133 (1964). DOI: 10.1016/0031-9163(64)91136-9
- Higgs, P.W., "Broken Symmetries and the Masses of Gauge Bosons," *Phys. Rev. Lett.* 13, 508–509 (1964). DOI: 10.1103/PhysRevLett.13.508
- Guralnik, G.S., Hagen, C.R., Kibble, T.W.B., "Global Conservation Laws and Massless Particles," *Phys. Rev. Lett.* 13, 585–587 (1964). DOI: 10.1103/PhysRevLett.13.585

[MEASURED] [R4] Wu et al. parity-violation experiment (1957) Wu, C.S., Ambler, E., Hayward, R.W., Hoppes, D.D., Hudson, R.P., "Experimental Test of Parity Conservation in Beta Decay," *Phys. Rev.* 105, 1413–1415 (1957). DOI: 10.1103/PhysRev.105.1413 This is the direct source for the book's claim that the cobalt-60 nucleus emitted electrons preferentially opposite to its spin.

[MEASURED] [R5] Electron electric dipole moment — experimental upper bound, $d_e < 1.8 \times 10^{-26}\ e \cdot \text{cm}$ Source: ACME Collaboration (Andreev, V. et al.), "Improved limit on the electric dipole moment of the electron," *Nature* 562, 355–360 (2018). DOI: 10.1038/s41586-018-0599-8 Tier note: this is an *experimental bound/limit* — a null-result exclusion, not a measured central value. [MEASURED] is retained because the limit is set by experiment, but the claim it supports is the inequality, never a value for d_e; the number must not be cited as though d_e had been resolved. The strong-CP bound $|\theta_{QCD}| < 10^{-10}$ derived from neutron EDM bounds is [DERIVED / TEXTBOOK] — see T8, Ch. 19.

[SYNTHESIS] [R6] Gauge coupling unification near 10^{16} GeV "suggesting a GUT" A theoretical extrapolation (running couplings under the MSSM nearly meet), not an experimental measurement — no GUT has been observed. The running-coupling calculation itself is standard: Georgi, H., Quinn, H.R., Weinberg, S., "Hierarchy of Interactions in Unified Gauge Theories," *Phys. Rev. Lett.* 33, 451 (1974), DOI: 10.1103/PhysRevLett.33.451. The word "suggesting" in the book's text is doing real work; this is not a confirmed discovery and must not be cited as one.

Chapter 1 — The Action Principle

No claim in this chapter relies on external experimental data. Lagrangian mechanics, Noether's theorem, and Hamilton–Jacobi theory are [DERIVED / TEXTBOOK] — see T1, T13.

Chapter 2 — Phenomena Catalogue

An index, not a source of new claims. Every phenomenon listed is sourced under the chapter where it is actually derived (see the "Revealed by → Ch. X" column in the book). No separate entries needed; cross-reference the chapter listed.

Chapter 3 — Bridge A: Dirac → Schrödinger

[THEORY] [R7] Anomalous magnetic moment $g_e = 2$ as a pre-1928 mystery, resolved by the Dirac equation Historical mystery: no single citable "discovery" — $g = 2$ emerged from spectroscopic anomalies (anomalous Zeeman effect) through the 1920s. The resolution is the theory paper: Dirac, P.A.M., "The Quantum Theory of the Electron," *Proc. R. Soc. Lond. A* 117, 610–624 (1928). DOI: 10.1098/rspa.1928.0023 Tier note: tiered [THEORY] because the cited source is Dirac's theoretical prediction of $g = 2$; the 1920s spectroscopic anomaly that motivated it is the empirical observation, the resolution is not.

[MEASURED] [R8] Discovery of the positron (Anderson, 1932/1933) Anderson, C.D., "The Positive Electron," *Phys. Rev.* 43, 491–494 (1933). DOI: 10.1103/PhysRev.43.491 The particle was photographed 2 August 1932; this is the full data paper. A short note appeared earlier in *Science* 76, 238 (1932).

[DERIVED / TEXTBOOK] [R9] Foldy–Wouthuysen transformation and its terms (Zeeman, spin-orbit, Darwin) Foldy, L.L. & Wouthuysen, S.A., "On the Dirac Theory of Spin 1/2 Particles and Its Non-Relativistic Limit," *Phys. Rev.* 78, 29–36 (1950). DOI: 10.1103/PhysRev.78.29

Chapter 4 — Schrödinger Equation, Hydrogen Atom

[MEASURED] [R10] Rydberg formula (empirical, 1888/1890) and Rydberg constant Rydberg, J.R., "On the Structure of the Line-Spectra of the Chemical Elements," *Phil. Mag.*, Series 5, Vol. 29, 331–337 (1890). Formula first presented in a paper read to the Lund Physiographic Society, 1888. Modern precision value $R_\infty = 1.0973731568539 \times 10^7\ \mathrm{m}^{-1}$: CODATA (NIST), one of the most precisely measured constants in physics. Link: https://physics.nist.gov/cgi-bin/cuu/Value?ryd

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Chapter 5 — Spin, Many-Electron Systems, Periodic Table

[MEASURED] [R11] Stern–Gerlach experiment (1922) Gerlach, W. & Stern, O., "Der experimentelle Nachweis der Richtungsquantelung im Magnetfeld," *Zeitschrift für Physik* 9, 349–352 (1922). DOI: 10.1007/BF01326983

[MEASURED] [R12] Lamb shift measurement Lamb, W.E. & Retherford, R.C., "Fine Structure of the Hydrogen Atom by a Microwave Method," *Phys. Rev.* 72, 241–243 (1947). DOI: 10.1103/PhysRev.72.241 The modern precision value of ~1057.8 MHz splitting comes from later refinements; the 1947 paper is the original discovery.

[DERIVED / TEXTBOOK] [R13] Cesium atomic clock definition of the SI second (9,192,631,770 Hz hyperfine transition) Source: BIPM, *The International System of Units (SI Brochure)*, 9th edition, definition of the second. Link: https://www.bipm.org/en/publications/si-brochure Tier note: the second is a DEFINITION (SI Brochure), fixed exactly, not a measured frequency — same class as the 2019 redefinition (R43). Tiered with the standards, not with measurements.

[MEASURED] [R1] Bohr magneton, $\mu_B = 9.274 \times 10^{-24}$ **J/T** Source: CODATA recommended values (NIST). Link: https://physics.nist.gov/cgi-bin/cuu/Value?mub

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Chapter 6 — Band Theory, Bonding, Nuclear Physics

[MEASURED] [R14] Bragg reflection condition Bragg, W.L., "The Diffraction of Short Electromagnetic Waves by a Crystal," *Proc. Camb. Phil. Soc.* 17, 43–57 (1913). Bragg's law itself, $n\lambda = 2d\sin\theta$; the book's application to electron-wave band gaps at the Brillouin zone boundary is [DERIVED / TEXTBOOK] — see T2, Ch. 7–9.

[DERIVED / TEXTBOOK] [R15] Semi-Empirical Mass Formula (Bethe–Weizsäcker) and its coefficients Weizsäcker, C.F. von, "Zur Theorie der Kernmassen," *Zeitschrift für Physik* 96, 431–458 (1935). DOI: 10.1007/BF01337700 Current best-fit coefficients ($a_V = 15.8$ MeV, $a_S = 18.3$ MeV, $a_C = 0.714$ MeV, $a_A = 23.2$ MeV, $\delta = \pm 12$ MeV): standard textbook values — see T4, Ch. 3. Binding-energy-per-nucleon data (peak at ^{56}Fe, 8.79 MeV): NNDC Atomic Mass Evaluation. Link: https://www.nndc.bnl.gov/

[DERIVED / TEXTBOOK] [R16] Geiger–Nuttall law Geiger, H. & Nuttall, J.M., "The ranges of the α particles from various radioactive substances," *Phil. Mag.* 22, 613–621 (1911); follow-up, *Phil. Mag.* 23, 439 (1912). Modern half-life data spanning ^{212}Po to ^{232}Th: NNDC.

[TARGET] [R17] ITER fusion energy gain target, $Q \geq 10$ Source: ITER Organization, official project documentation. Link: https://www.iter.org/sci/Goals Tier note: this is a design TARGET, not an achieved or measured value - $Q \geq 10$ is a forward-looking engineering goal, and the achieved gain to date is well below unity. Flagged so the projection is never cited as a measurement.

Chapter 7 — Topology, Quantum Hall Effect, Superconductivity

[THEORY] [R18] Berry phase Berry, M.V., "Quantal Phase Factors Accompanying Adiabatic Changes," *Proc. R. Soc. Lond. A* 392, 45–57 (1984). DOI: 10.1098/rspa.1984.0023 Tier note: tiered [THEORY] - Berry's 1984 adiabatic-phase paper is a theoretical result; experimental realisations (e.g. polarisation as a spin-1 Berry phase) are demonstrations, not the cited source.

[MEASURED] [R19] Von Klitzing's discovery of the (integer) quantum Hall effect (1980) Klitzing, K. von, Dorda, G., Pepper, M., "New Method for High-Accuracy Determination of the Fine-Structure Constant Based on Quantized Hall Resistance," *Phys. Rev. Lett.* 45, 494–497 (1980). DOI: 10.1103/PhysRevLett.45.494 Quantization to 1 part in 10^9 is repeatedly re-confirmed metrologically; see BIPM/NIST resistance-standard documentation for current figures.

[THEORY] [R20] Adler–Bell–Jackiw chiral anomaly (original theory papers, 1969)

- Adler, S.L., "Axial-Vector Vertex in Spinor Electrodynamics," *Phys. Rev.* 177, 2426–2438 (1969). DOI: 10.1103/PhysRev.177.2426
- Bell, J.S. & Jackiw, R., "A PCAC puzzle: $\pi^0 \to \gamma\gamma$ in the σ-model," *Nuovo Cimento A* 60, 47–61 (1969). DOI: 10.1007/BF02823296

[THEORY] [R21] Nielsen–Ninomiya (no-go) theorem for chiral fermions on a lattice Nielsen, H.B. & Ninomiya, M., "Absence of Neutrinos on a Lattice: I. Proof by Homotopy Theory," *Nucl. Phys. B* 185, 20–40 (1981). DOI: 10.1016/0550-3213(82)90011-6 Part II: *Nucl. Phys. B* 193, 173–194 (1981). DOI: 10.1016/0550-3213(81)90524-1

[MEASURED] [R22] Negative longitudinal magnetoresistance from the chiral anomaly, observed in TaAs (2015) Huang, X. et al., "Observation of the Chiral-Anomaly-Induced Negative Magnetoresistance in 3D Weyl Semimetal TaAs," *Phys. Rev. X* 5, 031023 (2015). DOI: 10.1103/PhysRevX.5.031023

[THEORY] [R23] BCS theory of superconductivity (1957) Bardeen, J., Cooper, L.N., Schrieffer, J.R., "Theory of Superconductivity," *Phys. Rev.* 108, 1175–1204 (1957). DOI: 10.1103/PhysRev.108.1175 The universal gap ratio $2\Delta(0)/k_B T_c = 3.528$ is a prediction of this paper, confirmed across conventional superconductors — [DERIVED / TEXTBOOK] for the confirming measurements; see T5, Ch. 3, for the standard compiled comparison table.

[MEASURED] [R24] Flux quantization in superconducting rings (1961, two independent groups)

- Deaver, B.S. & Fairbank, W.M., "Experimental Evidence for Quantized Flux in Superconducting Cylinders," *Phys. Rev. Lett.* 7, 43–46 (1961). DOI: 10.1103/PhysRevLett.7.43
- Doll, R. & Näbauer, M., "Experimental Proof of Magnetic Flux Quantization in a Superconducting Ring," *Phys. Rev. Lett.* 7, 51–52 (1961). DOI: 10.1103/PhysRevLett.7.51 Flux quantum $\Phi_0 = h/2e = 2.067833848 \times 10^{-15}$ Wb: CODATA (NIST). Link: https://physics.nist.gov/cgi-bin/cuu/Value?flxquhs2e

[THEORY] [R25] Josephson effect (theory, 1962) Josephson, B.D., "Possible new effects in superconductive tunnelling," *Phys. Lett.* 1, 251–253 (1962). DOI: 10.1016/0031-9163(62)91369-0 Josephson constant $K_J = 2e/h = 483597.8484\ldots \times 10^9$ Hz/V (exact since the 2019 SI redefinition): CODATA (NIST). Link: https://physics.nist.gov/cgi-bin/cuu/Value?kjos

[THEORY] [R26] Abrikosov vortices (theory, 1957; Nobel Prize 2003) Abrikosov, A.A., "On the Magnetic Properties of Superconductors of the Second Group," *Sov. Phys. JETP* 5, 1174–1182 (1957) [Zh. Eksp. Teor. Fiz. 32, 1442 (1957)].

[MEASURED] [R27] Quantized conductance of a point contact (1988) van Wees, B.J. et al., "Quantized Conductance of Point Contacts in a Two-Dimensional Electron Gas," *Phys. Rev. Lett.* 60, 848–850 (1988). DOI: 10.1103/PhysRevLett.60.848 Independently: Wharam, D.A. et al., *J. Phys. C* 21, L209 (1988).

Chapter 8 — Bridge B.d: General Relativity → Newton's Law

[MEASURED] [R28] Pound–Rebka gravitational redshift experiment (1959) Pound, R.V. & Rebka, G.A., "Gravitational Red-Shift in Nuclear Resonance," *Phys. Rev. Lett.* 3, 439–441 (1959). DOI: 10.1103/PhysRevLett.3.439 Measured $\Delta\nu/\nu \approx -2.5 \times 10^{-15}$ over 22.5 m, confirmed to ~10%; refined by

Pound & Snider, *Phys. Rev. Lett.* 13, 539 (1964), to ~1%.

[MEASURED] [R29] Gravity Probe B: frame dragging and geodetic precession (2011) Everitt, C.W.F. et al., "Gravity Probe B: Final Results of a Space Experiment to Test General Relativity," *Phys. Rev. Lett.* 106, 221101 (2011). DOI: 10.1103/PhysRevLett.106.221101 Measured frame-dragging drift: -37.2 ± 7.2 mas/yr vs. GR prediction -39.2 mas/yr — consistent with the book's 39 mas/yr figure within quoted precision.

[MEASURED] [R30] LIGO first direct detection of gravitational waves, GW150914 (2016) Abbott, B.P. et al. (LIGO Scientific and Virgo Collaborations), "Observation of Gravitational Waves from a Binary Black Hole Merger," *Phys. Rev. Lett.* 116, 061102 (2016). DOI: 10.1103/PhysRevLett.116.061102

[MEASURED] [R31] Mercury's anomalous perihelion precession (43″/century) explained by GR Original resolution: Einstein, A., "Erklärung der Perihelbewegung des Merkur aus der allgemeinen Relativitätstheorie," *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften*, 831–839 (1915). The anomaly was identified observationally by Le Verrier (1859) and refined by Newcomb; modern confirmation via radar ranging and ephemerides is [DERIVED / TEXTBOOK] — see T6, Ch. 40, or Will, C.M., "The Confrontation between General Relativity and Experiment," *Living Reviews in Relativity* 17, 4 (2014). DOI: 10.12942/lrr-2014-4 Tier note: mixed provenance, tiered [MEASURED] for the observed quantity. The perihelion anomaly (43"/century) is an empirical astronomical measurement (Le Verrier, Newcomb); Einstein's 1915 explanation is [THEORY]. The header tracks the measurement because that is the claim the book leans on, and the theoretical resolution is cited in the same entry.

[MEASURED] [R32] Light deflection by the Sun (1.75″) — Eddington 1919, modern VLBI confirmation Dyson, F.W., Eddington, A.S., Davidson, C., "A Determination of the Deflection of Light by the Sun's Gravitational Field, from Observations Made at the Total Eclipse of May 29, 1919," *Phil. Trans. R. Soc. A* 220, 291–333 (1920). DOI: 10.1098/rsta.1920.0009 Modern VLBI confirmation to 10^{-4}: Shapiro, S.S. et al., "Measurement of the Solar Gravitational Deflection of Radio Waves using Geodetic Very-Long-Baseline Interferometry Data, 1979–1999," *Phys. Rev. Lett.* 92, 121101 (2004). DOI: 10.1103/PhysRevLett.92.121101

[DERIVED / TEXTBOOK] [R33] GPS relativistic clock corrections (GR +45.7 μs/day, SR −7.2 μs/day, net +38.5 μs/day) Canonical citation for the full worked calculation: Ashby, N., "Relativity in the Global Positioning System," *Living Reviews in Relativity* 6, 1 (2003). DOI: 10.12942/lrr-2003-1 Every number in that section traces back to this review.

Chapter 9 — Bridge B.a: Quantum Mechanics → Classical Mechanics

No new external claims beyond what is already sourced above. Ehrenfest's theorem, WKB, and decoherence timescales are [DERIVED / TEXTBOOK] — see T3, Ch. 2 and 4; for decoherence, Zurek, W.H., "Decoherence, einselection, and the quantum origins of the classical," *Rev. Mod. Phys.* 75, 715 (2003). DOI: 10.1103/RevModPhys.75.715

Chapter 10 — Bridge B.b: Statistical Mechanics and Thermodynamics

[DERIVED / TEXTBOOK] [R34] Stefan–Boltzmann law and constant, $\sigma = 5.670 \times 10^{-8}\ \mathrm{W/m^2K^4}$ Original derivation: Boltzmann, L., "Ableitung des Stefan'schen Gesetzes," *Annalen der Physik* 258, 291–294 (1884). DOI: 10.1002/andp.18842580616 Current CODATA value:

https://physics.nist.gov/cgi-bin/cuu/Value?sigma

[DERIVED / TEXTBOOK] [R35] Wien displacement law constant, $b = 2.898 \times 10^{-3}$ **m·K** Wien, W., "Über die Energieverteilung im Emissionsspectrum eines schwarzen Körpers," *Annalen der Physik* 294, 662–669 (1896). DOI: 10.1002/andp.18962940803 Current CODATA value: https://physics.nist.gov/cgi-bin/cuu/Value?bwien

[DERIVED / TEXTBOOK] [R1] Copper Fermi energy / Fermi temperature ($E_F = 7.0$ **eV,** $T_F \approx 81{,}000$ **K)** Standard free-electron-model value — see T2, Table 2.1.

[SYNTHESIS] [R36] Framing of the Higgs mechanism, BCS superconductivity, and Landau theory as "independent instances of one mathematical template" This cross-thread claim (echoed in Ch. 21, Thread 2) is the book's own organising argument, not a claim from any single source. The three underlying physical theories are separately and correctly cited above (Ch. 0, Ch. 7, and Landau, L.D., "On the theory of phase transitions," *Zh. Eksp. Teor. Fiz.* 7, 19 and 627 (1937)) — but the *unification narrative itself* is the book's pedagogical synthesis and must be presented as such.

Chapter 11 — Bridge B.c: Maxwell's Equations from the U(1) Gauge Sector

[DERIVED / TEXTBOOK] [R37] Speed of light from $\mu_0\,\epsilon_0$**; Maxwell's identification of light as an EM wave** Maxwell, J.C., "A Dynamical Theory of the Electromagnetic Field," *Phil. Trans. R. Soc. Lond.* 155, 459–512 (1865). DOI: 10.1098/rstl.1865.0008 Modern value $c = 299{,}792{,}458$ m/s (exact, by SI definition since 1983): https://physics.nist.gov/cgi-bin/cuu/Value?c

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Chapter 12 — EM Waves, Optics, Photonics

Standard, uncontested optics — distributed Bragg reflectors, Fabry–Perot linewidths, the Kerr coefficient of silica fibre ($n_2 = 2.6 \times 10^{-20}$ m²/W): all [DERIVED / TEXTBOOK]. See T7 (optics); T9, Ch. 2 (Kerr coefficient value).

Chapter 13 — Kubo Formula → Generalized Transport Law

[THEORY] [R38] Kubo's linear-response theory (1957) Kubo, R., "Statistical-Mechanical Theory of Irreversible Processes. I," *J. Phys. Soc. Jpn.* 12, 570–586 (1957). DOI: 10.1143/JPSJ.12.570 Tier note: this is a theoretical framework — a general linear-response construction that *yields* correlation functions and fluctuation–dissipation relations — not a measurement. The experimentally measured quantities built on top of it (thermal conductivity, diffusion coefficients) are cited separately as [MEASURED].

[DERIVED / TEXTBOOK] [R39] Wiedemann–Franz law and the Lorenz number, $L_0 = 2.44 \times 10^{-8}\,\mathrm{W}\cdot\Omega\cdot\mathrm{K}^{-2}$ Original empirical law: Wiedemann, G. & Franz, R., "Ueber die Wärme-Leitungsfähigkeit der Metalle," *Annalen der Physik* 165, 497–531 (1853). DOI: 10.1002/andp.18531650802 Tier note: this entry carries two different provenances and is tiered by the number it headlines. The 1853 Wiedemann–Franz relation is *empirical* — a regularities-in-data observation with no microscopic basis at the time. The Lorenz number $L_0 = \pi^2 k_B^2/3e^2 = 2.44 \times 10^{-8}\,\mathrm{W}\cdot\Omega\,\mathrm{K}^{-2}$ quoted in the title is a *derivation*, exact only in the free-electron/Sommerfeld model — see T2, Ch. 13, and the Ch. 13 Model Ledger for the scattering assumptions this universality depends on.

[DERIVED / TEXTBOOK] [R40] Mott formula for thermopower Mott, N.F. & Jones, H., *The Theory of the Properties of Metals and Alloys*, Oxford (1936), Ch. 7. Standard modern reference: T2, Ch. 13, eq. 13.62.

[MEASURED] [R27] First direct measurement of conductance quantization (quantum point contacts, 1988) Cited under Ch. 7; the value $G_0 = 2e^2/h$ is the same constant referenced here.

Chapter 14 — Sensors, Semiconductor Devices, Thermal/Electrical Metrology

[MEASURED] [R41] Hall effect (original discovery, 1879) Hall, E.H., "On a New Action of the Magnet on Electric Currents," *American Journal of Mathematics* 2, 287–292 (1879). DOI: 10.2307/2369245

[DERIVED / TEXTBOOK] [R42] Richardson–Dushman thermionic emission equation and Richardson constant Richardson, O.W., "The Emission of Electricity from Hot Bodies," Longmans, Green & Co. (1921/1924 editions summarize his 1901–1911 work). Dushman, S., "Electron Emission from Metals as a Function of Temperature," *Phys. Rev.* 21, 623–636 (1923). DOI: 10.1103/PhysRev.21.623

[DERIVED / TEXTBOOK] [R43] SI redefinition of the second/kilogram/ampere via exact h, e (2019) Source: BIPM, Resolution 1 of the 26th CGPM (2018), "On the revision of the International System of Units (SI)," effective 20 May 2019. Link: https://www.bipm.org/en/measurement-units/rev-si/ Tier note: this is an authoritative *definitional standard*, not an experimental measurement paper — a metrological act of definition, not a result. The constants it fixes (h, e) are **exact by definition** after the redefinition, not fitted or measured values; tied to it, CODATA-derived uncertainties on formerly-measured constants became exact. Cited because the book's reported values must be pinned to a specific SI edition, not because a number was observed.

[DERIVED / TEXTBOOK] [R44] IEC 60751 Pt100 RTD coefficients (A, B, C) Source: IEC 60751:2008, "Industrial platinum resistance thermometers and platinum temperature sensors." Link: https://webstore.iec.ch/publication/3376

[DERIVED / TEXTBOOK] [GAP] Silicon dopant diffusion coefficients, minority-carrier lifetimes, solar-cell I_0/I_{SC} figures Standard semiconductor-device values. The textbook pointer originally given here (T10) is absent from the table above and its intended identity could not be recovered, so no source is cited rather than a guessed one. These are routine tabulated device parameters; pin a recognised semiconductor-device textbook and the relevant chapters on dopant diffusion and p-n junctions before press.

Chapters 15–20 — Continuum Engineering Physics (Fluids, Thermal, Structural, Geotechnical)

These chapters draw on established textbook-level engineering correlations rather than single discovery papers. Cite the textbook, not a paper, unless noted.

[DERIVED / TEXTBOOK] [R45] Reynolds number regime table (laminar/transitional/turbulent boundaries) Reynolds, O., "An Experimental Investigation of the Circumstances Which Determine Whether the Motion of Water Shall Be Direct or Sinuous...," *Phil. Trans. R. Soc. Lond.* 174, 935–982 (1883). DOI: 10.1098/rstl.1883.0029 Modern correlation tables: T11, Ch. 6.

[DERIVED / TEXTBOOK] [R46] Kolmogorov length scale, DNS intractability at high Re Kolmogorov, A.N., "The local structure of turbulence in incompressible viscous fluid for very large Reynolds numbers," *Dokl. Akad. Nauk SSSR* 30, 299–303 (1941).

[DERIVED / TEXTBOOK] [R47] Dittus–Boelter correlation (turbulent pipe-flow heat transfer) Dittus, F.W. & Boelter, L.M.K., University of California Publications in Engineering, Vol. 2, 443 (1930); reprinted *Int. Comm. Heat Mass Transfer* 12, 3–22 (1985). DOI: 10.1016/0735-1933(85)90003-X Standard modern reference: T12, Ch. 8.

[DERIVED / TEXTBOOK] [R48] Terzaghi consolidation theory ($t_{90\%}$ formula, hydraulic conductivity ranges by soil type) Terzaghi, K., "Erdbaumechanik auf bodenphysikalischer Grundlage," Deuticke, Vienna (1925). Standard modern reference: T16, Ch. 9 (Consolidation).

[DERIVED / TEXTBOOK] [R50] Separation principle (state feedback + observer design independence) Standard control result — see T17, Ch. 10, or Kalman, R.E., "On the general theory of control systems," *Proc. 1st IFAC Congress* (1960), for the original state-space formulation.

[DERIVED / TEXTBOOK] Manning's equation; Froude number; open-channel regime classification Empirical/derived engineering correlations — see T15, Ch. 4 (open-channel flow) and T11, Ch. 10.

[DERIVED / TEXTBOOK] Bond-graph formalism; effort-flow variable pairs Payne, H.T., "Analysis and Design of Elastic Systems," Academic Press; and the standard bond-graph treatment in Karnopp, D.C., Margolis, R.C., Rosenberg, R.C., *System Bond Graphs: Fundamentals and Applications*, Elsevier (2012) — see T14 for the effort/flow reference table.

Chapter 18 — Circuit Theory / Signal Processing Cross-Threads

[DERIVED / TEXTBOOK] [R49] Cooley–Tukey FFT algorithm (1965) Cooley, J.W. & Tukey, J.W., "An algorithm for the machine calculation of complex Fourier series," *Mathematics of Computation* 19, 297–301 (1965). DOI: 10.1090/S0025-5718-1965-0178586-1 The book's operation-count comparison (10^{12} vs. 2×10^7 for $N = 10^6$) follows directly from the $O(N \log N)$ complexity established in this paper. Tier note: tiered [DERIVED / TEXTBOOK] - the FFT is an algorithm (a mathematical/computational method with a complexity result), not a measured quantity; cited because the book's operation-count comparison rests on its $O(N \log N)$ analysis.

Chapter 21 — Synthesis, Cross-Layer Threads, Open Problems

[SYNTHESIS] "Thread 2: The Mexican Hat" — framing Higgs/BCS/Landau as independent instances of one template, not a causal chain Same flag as the Ch. 10 entry. Correctly cited components (Higgs mechanism, BCS, Landau) are sourced under their chapters; the unifying claim itself is the book's argument.

[SYNTHESIS] The Generalized Transport Law as the book's "convergence chapter" unifying Ohm, Fourier, Fick, and Newtonian viscosity under one Kubo-formula linear-response scheme, with Hooke's law entering as static response The individual transport laws and the Kubo formula are separately and correctly citable (see Ch. 13); their explicit unification under this framing is original pedagogical synthesis specific to this book. This is the most important claim in the book to keep honestly labelled, since it is the thesis the convergence chapter rests on. The claim must also be stated precisely: the four dissipative coefficients σ, κ, D, η share a linear-response framework in which the transport coefficient is expressed through an equilibrium current-correlation integral, subject to the relevant quantum/classical and closure assumptions. The elastic modulus C_{ijkl} is not a member of that family — it is a static, nondissipative response coefficient relating stress to strain, so Hooke's law is a constitutive relation arising from static response, not a fifth dissipative Green–Kubo transport coefficient. See the Ch. 13 §13.0 caveat on which members of the family use which part of the Kubo expression.

[SYNTHESIS] The four-way classification of CE/ChE correspondence in Ch. 20 (§20.0.1) The distinction between mathematical identity, constitutive correspondence, shared statistical origin, and shared dimensionless framework is the book's own contribution. The underlying results it sorts (Terzaghi consolidation, Darcy's law, Arrhenius kinetics, NTU methods) are textbook material — T16, T15, T18, T12 respectively.

[DERIVED / TEXTBOOK] Strong CP problem statement ($|\theta_{QCD}| < 10^{-10}$) See the Ch. 0 entry above (neutron/electron EDM bounds). Standard pedagogical reference: T8, Ch. 19, or T8b, §23.6.

Textbooks [T#]

Full citations for the [DERIVED / TEXTBOOK] tier. Chapter text cites these as `[T#]`.

Physics

#	Reference
T1	Goldstein, H., Poole, C., Safko, J., *Classical Mechanics*, 3rd ed., Addison-Wesley (2002)
T2	Ashcroft, N.W. & Mermin, N.D., *Solid State Physics*, Harcourt (1976)
T3	Sakurai, J.J. & Napolitano, J., *Modern Quantum Mechanics*, 3rd ed., Cambridge University Press (2020)
T4	Krane, K.S., *Introductory Nuclear Physics*, Wiley (1988)
T5	Tinkham, M., *Introduction to Superconductivity*, 2nd ed., Dover (2004)
T6	Misner, C.W., Thorne, K.S., Wheeler, J.A., *Gravitation*, W.H. Freeman (1973)
T7	Hecht, E., *Optics*, 5th ed., Pearson (2016)
T8	Peskin, M.E. & Schroeder, D.V., *An Introduction to Quantum Field Theory*, Westview Press (1995)
T8b	Weinberg, S., *The Quantum Theory of Fields, Vol. II*, Cambridge University Press (1996)
T9	Agrawal, G.P., *Nonlinear Fiber Optics*, 5th ed., Academic Press (2012)
T13	Pathria, R.K. & Beale, P.D., *Statistical Mechanics*, 4th ed., Academic Press (2021)
T19	Jackson, J.D., *Classical Electrodynamics*, 3rd ed., Wiley (1998)
T20	Griffiths, D.J., *Introduction to Quantum Mechanics*, 3rd ed., Cambridge University Press (2018)

Continuum mechanics and numerical methods

#	Reference
T21	Lai, J., Rubin, D., Krempl, E., *Introduction to Continuum Mechanics*, Cambridge University Press (2004)
T22	Spencer, P.A., *The Mechanics of Structures*, Cambridge University Press (1997)
T23	Marsden, J.E. & Hughes, T.J.R., *Mathematical Foundations of Elasticity*, Cambridge University Press (1999)
T24	Zienkiewicz, O.C., Taylor, R.L., Nitschke, P., *The Finite Element Method: Its Basis and Fundamentals*, 7th ed., Butterworth-Heinemann (2014)
T25	Bathe, K.-J., *Finite Element Procedures*, 2nd ed., Prentice Hall (2016)
T26	LeVeque, R.J., *Finite Difference Methods for Ordinary and Partial Differential Equations*, SIAM (2007)
T27	Versteeg, M.H.C. & Malalasekera, W., *An Introduction to Computational Fluid Dynamics*, 3rd ed., Prentice Hall (2010)

Fluids, heat, and mass transfer

#	Reference
T11	White, F.M., *Fluid Mechanics*, 8th ed., McGraw-Hill (2015)
T12	Incropera, F.P., DeWitt, D.P., Bergman, T.L., Lavine, A.S., *Fundamentals of Heat and Mass Transfer*, 8th ed., Wiley (2017)
T28	Geankoplis, T.J., *Transport Processes and Separation Process Principles*, 5th ed., Prentice Hall (2018)
T29	Welty, J.R., Wicks, C.J., Wilson, R., Rorrer, G.L., *Fundamentals of Momentum, Heat and Mass Transfer*, 5th ed., Wiley (2007)
T30	Çengel, Y.A. & Cimbala, J.M., *Fluid Mechanics: Fundamentals and Applications*, 4th ed., McGraw-Hill (2018)
T31	Landau, L.D. & Lifshitz, E.M., *Fluid Mechanics*, Course of Theoretical Physics Vol. 6, Pergamon (1987)

Structural, geotechnical, and chemical engineering

#	Reference
T14	Karnopp, D.C., Margolis, R.C., Rosenberg, R.C., *System Bond Graphs: Fundamentals and Applications*, Elsevier (2012)
T15	Das, B.M., *Principles of Geotechnical Engineering*, 9th ed., Cengage (2016)
T16	Das, B.M., *Principles of Foundation Engineering*, 9th ed., Cengage (2016)
T18	Felder, R.M., Rousseau, R.M., Bullard, L.G., *Elementary Principles of Chemical Processes*, 4th ed., Wiley (2019)
T32	Bird, R.B., Stewart, W.E., Lightfoot, E.N., *Transport Phenomena*, 2nd ed., Wiley (2006)
T33	Timoshenko, S.P. & Young, D.H., *Theory of Structures*, 2nd ed., McGraw-Hill (1965)
T34	Gere, J.M. & Goodno, B.J., *Mechanics of Materials*, 9th ed., Cengage (2018)
T35	Levenspiel, O., *Chemical Reaction Engineering*, 3rd ed., Wiley (1999)

Control theory

#	Reference
T17	Ogata, K., *Modern Control Engineering*, 5th ed., Prentice Hall (2010)

#	Reference
T36	Nise, N.S., *Control Systems Engineering*, 8th ed., Pearson (2019)
T37	Kuo, B.C., *Automatic Control Systems*, 10th ed., Wiley (2017)
T38	Åström, K.J. & Murray, R.M., *Feedback Systems*, 2nd ed., Princeton University Press (2021)
T39	Franklin, G.F., Powell, J.D., Emami, A., *Feedback Control of Dynamic Systems*, 8th ed., Pearson (2019)

Gaps and flags

Recorded honestly rather than papered over. These are known-unfinished items.

- **[GAP] Dangling identifier T10.** The Ch. 14 semiconductor-device entry (dopant diffusion coefficients, minority-carrier lifetimes, solar-cell I_0/I_{SC}) cited textbook "T10", but no T10 entry exists in the textbook key above — the table runs T1–T9, then T13 in the core block, with T11/T12 relocated to the fluids/heat block. The intended textbook could not be identified from the surrounding numbering (it is not sequential by subject). Rather than guess an entry, the in-text citation is flagged and no source is asserted. Resolve by adding a verified semiconductor-device text as T10 (or renumbering); do not leave a bare "T10" pointer in the table.
- **[GAP] Chapters 16, 17, 19, 20** contain further named correlations — acoustic cavitation and Rayleigh–Plesset dynamics, MEMS resonator Q-factors, the fissility parameter, geological dating precision — not individually traced to primary sources. These are almost certainly [DERIVED / TEXTBOOK], and the textbook key above now names the standard references for each domain (T21, T22, T27, T11, T12, T30, T4), but no page-level citation has been pinned. Flagged rather than manufactured.
- **[FLAG] CODATA/PDG links** point to live NIST/PDG reference pages rather than a fixed dataset year. These bodies update periodically (CODATA roughly every 4 years; PDG annually). A print bibliography must pin the specific edition actually used, or "current CODATA value" will silently drift.
- **[FLAG] Section numbers** were matched to chapters, not re-verified line-by-line against the manuscript after the Chapter 20, Chapter 7, and Chapter 13 rewrites changed section numbering. Re-pin against the chapter files before press.
- **[FLAG] The book's three synthesis claims** — the convergence thesis (Ch. 13), the Mexican-hat cross-thread (Ch. 10/21), and the four-way correspondence classification (Ch. 20) — are original to this work. They are flagged as [SYNTHESIS] here and must be flagged the same way wherever they are restated. A synthesis claim presented with a textbook citation attached is the failure mode this file exists to prevent.
- **[UNVERIFIED]** No entries currently carry this marker. If a new claim is added without a verified source, mark it rather than dropping it.

Cross-referenced from `METADATA/00 Map.md`*. Chapter text cites sources as* `[R#]` *and* `[T#]`*.*

Authoring *Engineering Physics: Top Down*

This file is the standard for how this book is written. It covers structure, notation, epistemic stance, and the review checklist. It is not a contribution workflow document

If anything here conflicts with `00 Map.md` or `README.md`, those two are authoritative and the conflict here is a bug in this file.

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1. The core principle: one chain, not a collection of essays

The spine is a single action, introduced in Chapter 0 and stated canonically in Ch. 0 §0.10:

```
S_eff = S_EH + S_SM + S_unknown
      = ∫ d⁴x √(−g) [ R/16πG + ℒ_SM + ℱ[g, ϕ?, new fields, topology, …] ]
```

Every major result must identify its physical or mathematical provenance. Where a genuine descent from this framework exists, show it and name the step. Where one does not, say so and label the result `[PHENOMENOLOGICAL]`, `[STRUCTURAL CONNECTION]`, `[ANALOGY]`, or a constitutive correspondence.

An earlier draft of this rule said *every* chapter must connect back to the master action. That was too strong, and it was wrong in a way that mattered: Ch. 13's engineering correlations, Ch. 19's thermofluid property fits, and Ch. 20's open-channel formulas are `[PHENOMENOLOGICAL]` or empirical. Forcing them into a descent would have manufactured derivations that do not exist. The book's own epistemic taxonomy already recognises this; the rule now matches it.

Concretely:

- Before writing a section, identify **which term of the master action it descends from and which named bridge gets you there** (Bridge A, Bridge B.a–B.e, Bridge C — see §7) — *or* identify why no such descent exists and tag it accordingly.
- When you introduce a new equation, state in prose what was discarded to get it and why discarding it was legitimate. The recurring question every chapter must answer somewhere: **"What did we throw away to get here, and why were we allowed to?"**
- When two results *look* structurally similar but are not the same derivation (the Higgs mechanism, Landau theory, ferromagnetism, superconductivity, and hysteresis all sharing a Mexican-hat potential), say so as an explicit `[STRUCTURAL CONNECTION]` thread. Do not imply a literal genealogy. **This distinction has caused real errors in review and must not recur.**
- Cross-reference by chapter and section number, always (§7). A chapter that does not cite where its starting equation came from is incomplete.

The five cross-layer threads to maintain: **Noether's theorem, symmetry breaking (Mexican-hat potential), the action principle, topology, and the wave/diffusion dichotomy**.

1.1 Four kinds of "same"

The single most common failure mode in a book like this is using one word — "identical", "the same", "is just" — for four different relationships. Chapter 20 §20.0.1 names them, and the distinction is used throughout:

Kind	Shared	Not shared	Chapter 20 example
Mathematical identity	the equation, term by term, once assumptions hold	nothing	Terzaghi consolidation ≡ 1D heat equation
Constitutive correspondence	the algebraic form, reached independently	derivation, closure, coefficients	Darcy's law ≅ Fick's law in form
Shared statistical origin	a statistical-mechanical structure	the macroscopic law built on it	Arrhenius ≅ Boltzmann factor
Shared dimensionless framework	how dimensionless groups organise design	driving forces, equilibria, performance	NTU for columns ≅ NTU for exchangers

"The word *identical* appears only for the first kind." The other three are named for what they are.

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2. Notation and the tagging system

Because "connect everything into one chain" invites overclaiming, every non-trivial connection between two results carries one of five tags:

- `[DERIVATION]` — a literal mathematical step follows from the previous equation via a named, checkable limit or substitution.
- `[APPROXIMATION]` — a term is dropped or a limit taken. **State the small parameter and the regime of validity.** An unlabelled approximation is a defect.
- `[STRUCTURAL CONNECTION]` — two results share mathematical form but one is **not** derived from the other.
- `[ANALOGY]` — a pedagogical comparison with no claimed formal equivalence.
- `[PHENOMENOLOGICAL]` — an empirical relation used as-is, not derived from the master action.

Use these consistently. Do not invent new tag categories without revising this section first.

2.1 Model Ledgers

Tags classify a **connection**. The **Model Ledger** classifies a **model**.

Wherever a chapter performs a major model transition — a PDE becoming an ODE, a field equation becoming a lumped circuit, a full model becoming a linearisation — it records the transition in a fixed nine-row table:

Field	Content
Parent theory	the equation it descends from, with chapter/section
Reduction	the specific approximations applied
Model	the resulting reduced equation
Assumptions	enumerated, including those inherited silently
Physics retained	what survives
Physics neglected	what does not — this row is the honest one
Validity	the regime where the model is trustworthy
Fails when	the specific condition, not "outside its domain"

Field	Content
Next model	what to use when it fails

Placement rule. One Ledger per major model transition — not one per equation, and not one per chapter. Chapters carrying Ledgers: **3, 7, 13, 16, 17, 19, 20, 21**. A chapter that only establishes, catalogues, or surveys has not reduced anything, and adding a Ledger to it would dilute the convention into decoration.

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3. Formatting rules

LaTeX must be portable. Display math uses `$$ ... $$`, inline math uses `$ ... $`.

1. **Markdown is the source of truth.** Never treat a word-processor export as canonical — round-trips have corrupted LaTeX before, silently turning underscores into asterisks.
2. **No `\[ \]`, no `\( \)`, no `equation`/`align` environments, no package-dependent macros, no `\newcommand`.** These break on import into most Markdown/LaTeX pipelines and render as literal source.
3. **Never number equations.** No `\tag{}`, no `\label{}`/`\ref{}`, no `(1)`, `(2)` style numbering. Reference equations in prose by chapter and section — *"the action from Ch. 0 §0.3"*, not *"Eq. (3)"*. Numbering is the single most common cause of broken rendering.
4. **Escape LaTeX specials even in obvious places.** A literal `%` inside a math span is a *comment* in LaTeX and will silently swallow the rest of the expression. Write `\%`. This has bitten the manuscript more than once: a quantity like `$\alpha^2 \approx 0.005\%$` lost its backslash and became an equation that rendered as nothing after the decimal point.
5. **Thin spaces are `\,` and they are not optional.** `C(s),e(s)` and `C(s)\,e(s)` are different expressions; so are `12.9\mu V` and `12.9\,\mu V`. Before submitting, scan your own new equations for bare `,` and `;` where a thin space belongs.
6. **Matrix row separators are `\\`.** A single `\` renders as garbage, and the failure is invisible in plain text — check `pmatrix`, `bmatrix`, `vmatrix`, and `cases` environments specifically.
7. **Verify rendering before considering the section done.** Paste the equations into a blank Markdown preview and confirm each one renders as intended.

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4. Chapter structure

Every chapter follows this skeleton:

- Chapter number and title
- Epigraph
- **§X.0 — Overview**: where we are, what this chapter does, what it assumes
- Numbered sections (`§X.1`, `§X.2`, …) with derivations, interpretation, tables
- **§X.N — Summary**: a table of results, each with its origin
- **§X.(N+1) — Engineering Thread**: physics results mapped to concrete engineering applications
- **§X.(N+2) — Looking Ahead**: what comes next and why it follows
- Closing: `End of Chapter X`, then `Next: Chapter Y — Title`

Do not skip the Engineering Thread or Looking Ahead sections. They are what make the single-chain structure legible, not optional polish.

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5. Prose style

- Full textbook prose. Paragraphs, not fragments. Not bullet skeletons.
- Tone: a professor who has thought carefully about why the standard bottom-up teaching order is pedagogically wrong, and is directly correcting it. Honest, direct, precise. Never breezy, never hand-wavy.
- **No bluffing.** If something is unknown or unresolved, say so. Never claim confirmation the physics does not support.
- Never write "it can be shown that" without either showing it or stating where it is shown.
- **Name every approximation**, with its small parameter and its regime.
- One consistent voice across all chapters. Register drift — one section at introductory level, the next in unexplained tensor notation — is a real defect.

---

6. Epistemic stance

The book does **not** claim that a Lagrangian Theory of Everything is proven or established. The correct framing, to be preserved in spirit wherever this comes up:

> "If a Theory of Everything exists, we have overwhelming reason to expect it will appear in this form."

Do not write language implying the master action has been confirmed. Do not present the hierarchy of approximations as a proven derivation chain where a structural connection is what is actually meant.

The same discipline applies to the book's own load-bearing claims. Three of them are **synthesis** rather than sourced fact, and must be labelled that way wherever they appear:

1. the convergence thesis (Ch. 13) — Ohm, Fourier, Fick, and viscosity within a common linear-response framework, with Hooke's law appearing as a static susceptibility (zero-frequency limit) rather than an ordinary transport coefficient
2. the Mexican-hat cross-thread (Ch. 10, Ch. 21) — Higgs, BCS, and Landau as instances of one template
3. the four-way correspondence classification (Ch. 20 §20.0.1)

These are the book's contribution. A synthesis claim wearing a textbook citation is the failure mode this rule exists to prevent. See `REFERENCES.md` for the tiering scheme.

---

7. Cross-chapter referencing

- Always by chapter and section: `Ch. 3 Bridge A`, `Ch. 0 §0.7`, `Ch. 8 §8.4`.
- **"will be developed in Ch. X"** forward; **"from Ch. X §Y.Z"** backward.
- The named bridges are fixed: **Bridge A** (Dirac → Schrödinger), **Bridge B.a** ($\hbar \to 0$), **Bridge B.b** ($N \to \infty$), remaining Bridge B sub-paths as defined in `00 Map.md`, and **Bridge C** (continuum → engineering limits).
- The phenomena catalogue (Ch. 2) uses **final** chapter numbers. If chapter numbering changes, update it in the same pass.

8. Design decisions — do not reverse these silently

- **Ch. 0 before Ch. 1.** Ch. 0 gives the full equation; Ch. 1 teaches the machinery that explains why it works. Deliberate.
- **The unknown sector `𝒮[…]` is never dropped.** The action is always written with all three sectors — S_{EH}, S_{SM}, S_{unknown} — or with an explicit statement about $\mathcal{F}$. Writing $S_{EH} + S_{SM}$ alone is a silent claim that the unknown sector is empty.
- **Never write the unknown sector as $f(\phi)R$.** That form asserts the missing physics is a scalar times the Ricci scalar, which nothing supports. $\mathcal{F}[\ldots]$ is a placeholder for an unknown *sector*, not a functional form. This supersedes earlier drafts that used `f(ϕ)`.
- **Pauli exclusion is a theorem, not a postulate**, derived from fermion-field anticommutation in Ch. 0 and carried into Ch. 5. Do not reintroduce it as an axiom.
- **The Born rule remains a foundational probabilistic postulate of standard quantum mechanics.** It is *not* derived in this book, and no design decision may be reversed to make it look as though it were. What U(1) symmetry and Noether's theorem *do* give (Ch. 3 §3.9.2) is a conserved current of $|\psi|^2$, so that probability conservation follows from charge conservation *once the Born postulate is adopted*. That arrow runs one way. Never write the reverse — "the Born rule is Noether's theorem" is false, and it is the single most common error in "everything is symmetry" treatments of QM.
- **The periodic table is derived** term by term from Aufbau, Hund's rule, and the four quantum numbers in Ch. 5.
- **Newton's law is presented as GR's weak-field limit.** Ch. 8 derives it; Ch. 9 assumes the reader has seen that.
- **Ch. 13 is the convergence chapter.** Do not fork discipline-specific derivations elsewhere that duplicate it.
- **Bridge zones are chapters in their own right** and must be as rigorous as any other chapter.
- **The architecture is frozen at Chapters 0–21 plus the Epilogue.** This is settled, not provisional. There is no Chapter 22.

---

9. Files and naming

- Chapters: `CHAPTERS/CHAPTER N.md`, N = 0…21.
- Metadata: `METADATA/00 Map.md`, `METADATA/EPILOGUE.md`.
- Provenance: `REFERENCES.md` (root — the single source of truth for sources; `METADATA/Reference.md` is a pointer, not a second copy).
- Worked examples: `EXAMPLES/`, each following the six-step shape documented in `EXAMPLES/README.md`.
- Landing page: `README.md`.
- Appendices: `APPENDIX/APPENDIX X.md`, X = A…F. An appendix holds the full derivation or proof of results a chapter states or sketches. It follows every rule in this file (tags, no numbering, stated assumptions, failure regime), links back to the chapters that use it, and ends with a summary table. It is supporting material, not a new chapter: the chapter count stays at 21.

- Figures: `FIGURES/<FOLDER>/<slug>.svg`, where `<FOLDER>` is `CH00`…`CH21`, `EX01`…`EX06`, `MAP` or `EPILOGUE`, and `<slug>` is kebab-case and names the content (`moody-colebrook`, not `fig3`). Every figure is drawn by `FIGURES/<FOLDER>/make_figures.py`, which imports the shared style in `FIGURES/_style.py`. Its parameters come from the text the figure sits in. No hand-drawn or pasted images, and no images that have no script. `python FIGURES/build_all.py` regenerates everything.
- Embed a figure with standard Markdown, not the Obsidian-only `![[...]]`, and put an italic caption directly below it:

  ```
  ![Colebrook friction factor against Reynolds number](../FIGURES/EX05/moody-colebrook.svg)
  *Darcy friction factor from the Colebrook equation for ε/D = 1.125×10⁻³, with
  the three iterates of Step 5 marked.*
  ```

  Never number figures or write "Figure 3" (same reason as §3 item 3). The caption states what is plotted, with which parameters, and what to notice. An AI agent reading the repo cannot see the SVG; it reads the caption and the script.
- A figure must agree with the text. If plotting exposes an error in the text, fix the text in the same pass.
- Any new top-level file requires updating `00 Map.md` and `README.md` in the same pass.

---

10. Review checklist

Before a section is considered done:

- [ ] Every connection to another result carries one of the five tags (§2)
- [ ] Every major model transition has a nine-row Model Ledger (§2.1)
- [ ] Every dropped term is named, with its small parameter and regime
- [ ] No unescaped `%`, no bare `,` or `;` where `\,` belongs, no single `\` in a matrix environment (§3, items 4–6)
- [ ] No equation numbering; all cross-references by chapter and section
- [ ] Synthesis claims labelled as synthesis (§6)
- [ ] Summary table gives each result its origin
- [ ] Engineering Thread and Looking Ahead present
- [ ] All equations render correctly in preview
- [ ] Every figure has a script, alt text and an italic caption, and it agrees with the numbers in the text (§9)
- [ ] `python mcp-server/smoke_test.py` passes (broken links and image embeds fail it)

---

11. Known open items

Not defects. Recorded so they are not rediscovered as bugs:

- Chapter-level §-number pinning in `REFERENCES.md` needs a re-verification pass after the Ch. 7, 13, and 20 rewrites.
- Several named engineering correlations in Ch. 16, 17, 19, 20 are domain-sourced to a textbook but not pinned to a page.
- Non-equilibrium statistical mechanics is out of scope; Ch. 10 ends at equilibrium.

- The taxonomy of *why* a `[STRUCTURAL CONNECTION]` exists is stated in Ch. 20 §20.0.1 but is not yet propagated as a standalone section.

MCP.md

Engineering Physics — AI/MCP Context Manifest

> Machine-readable project context for AI agents, LLMs, coding assistants, research assistants, and MCP-connected tools.
>
> Repository: `The-Studio-of-Sadiq/Engineering-Physics` Project: **ENGINEERING PHYSICS: TOP DOWN** Purpose: a rigorous top-down knowledge framework connecting fundamental physics to engineering models through explicit derivation, approximation, reduction, and model boundaries.

1. Agent Role

Act as a physics-aware research and authoring assistant, not an autonomous authority.

Responsibilities:

1. Understand the hierarchy of physical models.
2. Distinguish exact derivation, controlled approximation, structural connection, analogy, and phenomenological modelling.
3. Detect conceptual, mathematical, dimensional, numerical, citation, and consistency errors.
4. Preserve the project's top-down philosophy.
5. Prefer explicit assumptions and validity boundaries over compressed claims.
6. Never silently strengthen a scientific claim.
7. Mark uncertainty and identify what would verify it.
8. Treat the repository as an evolving scholarly manuscript.

AI/MCP functionality belongs in the authoring, validation, navigation, and research infrastructure around the knowledge base; AI is not part of the physics itself.

2. Core Thesis

> **Engineering is physics seen from far away.**

Engineering models are treated as reduced descriptions of deeper physical theories.

Preferred hierarchy:

```
Fundamental Framework
        ↓
Quantum / Relativistic Physics
        ↓
Classical / Statistical / Continuum Physics
        ↓
Reduced Constitutive Models
        ↓
Lumped / Distributed Engineering Models
        ↓
Engineering Systems and Control
```

Always ask: what parent model produced this equation, what assumptions reduced it, what was discarded, and when does the approximation fail?

3. Repository Structure

```
README.md
CONTRIBUTING.md
REFERENCES.md
MCP.md
LICENSE
.mcp.json              MCP client config (Claude Code) for mcp-server/
METADATA/
  00 Map.md
  EPILOGUE.md
CHAPTERS/
  CHAPTER 0.md
  ...
  CHAPTER 21.md
EXAMPLES/
  README.md
  01 Dirac-to-Pauli.md
  02 Quantum-to-Thermodynamics.md
  03 Maxwell-to-Circuit.md
  04 Heat-Equation-to-Thermal-RC.md
  05 Navier-Stokes-to-Fluid-Network.md
  06 Motor-to-Control-System.md
APPENDIX/
  APPENDIX A.md        variational calculus, Noether's theorems
  APPENDIX B.md        Dirac → Foldy–Wouthuysen → Pauli
  APPENDIX C.md        quantum → classical; GR → Newton
  APPENDIX D.md        statistical mechanics, Kubo, Green–Kubo
  APPENDIX E.md        continuum balance laws → lumped networks
  APPENDIX F.md        linear systems, feedback, stability proofs
FIGURES/
  _style.py            shared plot style
  build_all.py         regenerates every figure
  CH00/ … CH21/, EX01/ … EX06/, MAP/, EPILOGUE/
    make_figures.py    draws every SVG in its folder
    *.svg
mcp-server/
  server.py            deterministic MCP tooling layer (§16)
  smoke_test.py
```

Figures are SVGs and cannot be read as images. Each one is described by the italic caption under its embed, and exactly by the equation and parameters in its folder's `make_figures.py`. The MCP tool `get_figure_index` lists both.

Orientation order:

1. `README.md`
2. `METADATA/00 Map.md`
3. `CHAPTERS/CHAPTER 0.md`
4. Chapters 1–21
5. `METADATA/EPILOGUE.md`
6. `EXAMPLES/README.md`
7. Worked examples
8. `APPENDIX/` (full derivations, on demand)
9. `CONTRIBUTING.md`
10. `REFERENCES.md`

The Map is the architectural source of truth.

4. Conceptual Architecture

Layer 0 — Fundamental Framework

Standard Model, Einstein–Hilbert gravity, action principles, symmetries, conservation laws, spacetime/field theory, and an explicitly unknown frontier sector `\mathcal F`.

Do not claim current fundamental physics is complete.

Bridge A — Fundamental → Quantum / Atomic

Typical chain:

```
relativistic field theory
        ↓
single-particle / low-energy sector
        ↓
Dirac
        ↓
Foldy–Wouthuysen
        ↓
Pauli / Schrödinger + corrections
```

Layer 1 — Quantum / Atomic

Quantum mechanics, atoms, nuclei, scattering, condensed matter, bands, topology, effective quasiparticles.

Bridge B — Quantum → Classical / Statistical

Important reductions include:

- `\hbar → 0` → classical mechanics
- thermodynamic limit + statistical description → statistical mechanics/thermodynamics
- classical electromagnetic-field limit → Maxwell
- weak-field, slow-motion, quasi-static GR → Newtonian gravity
- linear-response/Kubo framework → transport coefficients
- ballistic transport → Landauer-type descriptions

Layer 2 — Classical Continuum & Statistical Physics

Classical mechanics, thermodynamics, statistical mechanics, electromagnetism, continuum mechanics, transport, waves, diffusion, fluid mechanics.

Bridge C — Continuum → Engineering

```
PDE / field description
        ↓
control-volume integration
        ↓
spatial discretization / lumping
        ↓
network or state-space model
        ↓
engineering system
```

Lumping is an approximation whose validity must be stated.

Layer 3 — Engineering Systems

Electrical, mechanical, thermal, civil, chemical/process, and control systems.

5. Epistemic Tags

Tag	Meaning
`[DERIVATION]`	Derived from explicit assumptions
`[APPROXIMATION]`	Controlled approximation, asymptotic limit, truncation, or reduction
`[STRUCTURAL CONNECTION]`	Same mathematical structure without claiming identical physics
`[ANALOGY]`	Useful analogy with different underlying mechanisms
`[PHENOMENOLOGICAL]`	Empirical/effective law or model
`[GAP]`	Connection/source not yet established
`[MEASURED]`	Genuine experimental/metrological result
`[THEORETICAL]`	Theoretical derivation or prediction
`[SYNTHESIS]`	Author's synthesis across sources

Never use `[DERIVATION]` merely because two equations look algebraically similar.

6. Four Kinds of Same

Classify cross-domain claims as:

1. **Exact identity** — same mathematical/physical statement under stated definitions.
2. **Controlled derivation** — one model follows from another under explicit limits/approximations.
3. **Structural equivalence** — same mathematical form, different physical interpretation.
4. **Analogy** — useful conceptual similarity, not physical equivalence.

When evidence permits multiple readings, use the weaker classification.

7. Model Ledger

For important models track:

```
Level
Parent theory
Origin
Approximation
Assumptions
Model type
Validity regime
Failure regime
Next reduced model
```

Example:

```
Parent theory:
  Navier–Stokes + continuity

Reduction:
  incompressible + Newtonian + fully developed pipe flow

Model:
  Hagen–Poiseuille relation

Validity:
  steady, laminar, incompressible, Newtonian,
  fully developed flow in a straight circular pipe,
  no-slip boundary condition

Failure:
  turbulence, compressibility, developing flow,
  non-Newtonian rheology, inappropriate geometry
```

Preserve model provenance when editing.

8. Critical Physics Rules

Do not overclaim

Avoid:

```
"all engineering laws are derived from the Standard Model"
"every engineering equation follows exactly from fundamental physics"
"all conserved quantities are effort-flow pairs"
"every physical theory has an action"
"every system has exactly R/C/L elements"
```

Prefer:

```
"can be connected through..."
"admits a reduction to..."
"shares a mathematical structure with..."
"under the stated assumptions..."
"provides a microscopic foundation for..."
```

Conservation vs constitutive law

Keep these distinct.

Examples:

- charge conservation → continuity equation → KCL after lumping
- mass conservation → continuity equation → hydraulic junction balance
- energy conservation → energy balance
- Ohm's law → constitutive relation
- Fourier's law → constitutive/linear-response approximation
- Darcy friction → empirical/correlation-based constitutive model
- Newtonian viscosity → constitutive assumption

Never turn a constitutive law into a conservation law.

Noether theorem

Distinguish:

- global continuous symmetries and Noether's first theorem,
- local gauge redundancy and Noether's second theorem,
- conserved currents,
- stress-energy definitions from metric variation.

Do not equate local gauge invariance with a conserved charge without specifying the Noether framework.

Electromagnetic U(1)

After electroweak symmetry breaking, `U(1)_em` is the relevant electromagnetic gauge symmetry.

Do not write `U(1)_Y → KCL`. Electromagnetic charge emerges from the electroweak structure.

KCL

Preferred chain:

```
electromagnetic charge symmetry / U(1)_em
        ↓
conserved electric current
        ↓
continuity equation
        ↓
integral charge balance
        ↓
lumped-node approximation
        ↓
Σ I_k = 0
```

KVL

Do not describe KVL as merely Faraday's law with the derivative removed.

Preferred:

```
Faraday's law
        ↓
lumped circuit approximation
        ↓
branch voltage relations, including inductive voltage
        ↓
circuit loop equation
```

Energy conservation is associated with electromagnetic energy balance/Poynting's theorem, not KVL alone.

Thermal networks

Engineering thermal resistance:

```
R_th = ΔT / Q̇
```

has units K/W.

Entropy-flow formulations use different power-conjugate variables. A relation such as `R ≈ T_0 R_th` is convention-dependent and belongs to a small-signal linearization about `T_0`.

Higgs mechanism

Do not claim Higgs coupling accounts for all ordinary matter mass.

- Higgs mechanism generates elementary fermion and weak-boson masses through the relevant couplings.
- Most visible mass of ordinary baryonic matter arises from QCD dynamics.
- The Higgs doublet has four real field components.
- `H†H = v²/2` describes an S^3 minimum manifold in four-real-dimensional field space.
- "Mexican hat" is a pedagogical picture, not literal one-dimensional field geometry.

Gravity

Do not use `R = 0 ⇒ flat spacetime`.

The Schwarzschild exterior has vanishing Ricci scalar while the full Riemann curvature is nonzero.

The Ricci scalar is one contraction of curvature, not the complete curvature.

Stress-energy

Distinguish canonical Noether stress-energy from the Hilbert tensor used as the gravitational source.

A useful convention is:

```
T_{μν} = -(2/√(-g)) δS_matter / δg^{μν}
```

Continuum mechanics

The thermodynamic limit is:

```
N,V → ∞ at fixed density
```

but this alone does not produce a continuum field. Continuum modelling also needs coarse-graining and scale separation such as `ℓ_micro << L_variation`.

Thermodynamics/statistical mechanics

Do not claim all four thermodynamic laws emerge from a single `N → ∞` operation. Statistical mechanics supplies microscopic foundations and statistical interpretations, but the four laws involve distinct definitions, assumptions, limits, and empirical content.

Kubo / transport

Kubo is a linear-response framework. Do not claim one calculation literally produces Ohm's, Fourier's, Fick's, and Newtonian-viscosity laws.

Prefer:

> These laws can be understood as constitutive transport coefficients within appropriate linear-response regimes.

Kubo is not restricted to diffusive transport; ballistic transport has different descriptions such as Landauer-type approaches.

Transfer functions

Ordinary transfer functions are associated with LTI input-output models. Do not apply them indiscriminately to nonlinear or time-varying systems.

Topology

- Weyl-semimetal quasiparticles are described by effective Weyl Hamiltonians; they are not literally Standard Model Weyl fermions.
- Topological-insulator and QCD θ terms can be structurally analogous without being the same physics.
- Topology → Nyquist is a shared mathematical/topological connection, not a direct physical genealogy.
- TR-protected helical-edge statements must specify elastic single-particle Kramers backscattering; interactions/inelastic/multiparticle processes require separate treatment.

9. Engineering Network Abstraction

Effort/flow and R/C/L-like abstractions are cross-domain modelling tools, not universal laws.

Examples:

```
Electrical:
  voltage ↔ current
  resistor ↔ capacitor ↔ inductor

Mechanical translational:
  force ↔ velocity
  damper ↔ mass ↔ spring

Thermal:
  temperature difference ↔ heat/entropy flow
  thermal resistance ↔ thermal capacitance
```

Important:

- Not every flow is a Noether current.
- Not every system has exactly three passive element types.
- The analogy depends on chosen variables and representation.
- Mechanical force–velocity is power-conjugate, not itself a conservation law.
- Sources, nonlinearities, constraints, and domain-specific elements may be required.

10. Worked Examples

01 — Dirac → Pauli

FW is unitary order-by-order, but the effective Hamiltonian is a truncated perturbative expansion for general external fields.

Use:

```
O = c α · π
```

Do not put `-e σ·E` directly into the odd operator. Electric spin-orbit and Darwin terms arise through the FW commutator expansion.

02 — Quantum → Thermodynamics

```
microscopic quantum states
→ statistical description
→ partition functions / thermodynamic potentials
→ macroscopic thermodynamics
```

Keep thermodynamic-limit assumptions explicit.

03 — Maxwell → Circuit

```
Maxwell equations
→ lumped/distributed approximation
→ circuit laws
→ RLC dynamics
```

Telegrapher equations are reduced distributed models and require appropriate geometry/mode/quasi-TEM assumptions.

04 — Heat equation → Thermal RC

```
heat equation
→ control-volume / lumped-capacitance approximation
→ thermal R + C
→ first-order RC-like response
```

Fourier's law is constitutive/linear-response, not a fundamental identity.

05 — Navier–Stokes → Fluid network

Continuity and momentum equations are separate.

```
∂ρ/∂t + ∇·(ρv) = 0
```

For constant-density incompressible flow:

```
∇·v = 0
```

Hagen–Poiseuille is exact only within its stated laminar-flow assumptions.

Hydraulic loop balance should not be called Bernoulli unless the ideal assumptions apply.

06 — Motor → Control

```
electromagnetic motor model
→ electromechanical coupling
→ state-space representation
→ transfer function where LTI assumptions apply
→ feedback/control
```

11. Numerical Audit Rules

For every numerical result:

1. Recalculate independently.
2. Check SI units.
3. Check significant figures.
4. Check intermediate quantities.
5. Check limiting behaviour.
6. Check empirical-correlation validity.
7. Distinguish exact algebra from numerical approximation.

Known checks, to be reverified against current source:

```
Example 3:
  R_eq ≈ 48.89 Ω
  f_-3dB ≈ 325 kHz
  500 MHz FR-4 wavelength ≈ 0.32–0.35 m
  quarter wavelength ≈ 8–9 cm

Example 4:
  L_c ≈ 16.67 mm
  Bi ≈ 4.07×10^-3
  C ≈ 2900 J/K
  R ≈ 0.333 K/W
  τ ≈ 967 s
  T(60 s) ≈ 76.4 °C

Example 5:
  v=1 m/s: Re ≈ 4×10^4, f ≈ 0.0251, h_f ≈ 1.92 m
  v=3 m/s: Re ≈ 1.2×10^5, f ≈ 0.0223, h_f ≈ 15.3 m
  v=3.45 m/s: Re ≈ 1.38×10^5, f ≈ 0.0220, h_f ≈ 20.0 m
  Q ≈ 4.33 L/s per pipe
  two identical parallel pipes ≈ 8.66 L/s
```

12. Reference Rules

`REFERENCES.md` is part of the scholarly infrastructure.

For every citation:

1. Verify the source actually supports the claim.
2. Do not label theoretical work `[MEASURED]`.
3. Distinguish measurement, theory, textbook derivation, standard/metrological definition, and synthesis.
4. Never fabricate missing references.
5. If support cannot be verified, use `[GAP]`.

6. Preserve original bibliographic identity.
7. Prefer primary literature for important historical scientific claims.

Examples of correct evidence classification:

- Adler–Bell–Jackiw anomaly → theoretical.
- Nielsen–Ninomiya theorem → theoretical.
- BCS theory → theoretical.
- Josephson 1962 → theoretical prediction.
- Abrikosov vortex theory → theoretical.
- Kubo → theoretical.
- Original Wiedemann–Franz observation → empirical/experimental.
- SI redefinition → metrological/legal standard, not a measurement paper.

13. Editing Protocol

Before editing:

1. Fetch the current source file.
2. Identify the exact target passage.
3. Re-read surrounding context.
4. Make the smallest necessary change.
5. Re-read the edited region.
6. Check Markdown tables/fences/equations.
7. Check cross-references.
8. Check whether the edit changes an epistemic claim elsewhere.
9. Report the change.

Do not edit from a rendered snippet when exact source is available.

Preserve existing Unicode/ASCII conventions unless deliberately standardizing them.

14. Validation

Structural

- chapter files exist
- chapter numbering intact
- internal links resolve
- anchors valid
- Markdown tables rectangular
- code fences balanced
- LaTeX delimiters balanced
- special characters correctly escaped
- no accidental encoding substitutions

Physics

- derivation chain valid
- assumptions visible

- approximation order correct
- conservation vs constitutive distinction preserved
- structural analogies not presented as identity
- validity boundaries retained
- empirical laws not relabelled as first-principles derivations

Numerical

- values recompute
- units match
- scaling laws valid in stated regime
- correlations used in intended domain

References

- strong claims supported
- evidence tiers correct
- citations support the statements using them
- unverified sources marked `[GAP]`

15. AI Audit Modes

`ORIENT`

Return project thesis, repository map, chapter hierarchy, bridges, and epistemic rules.

`AUDIT_PHYSICS`

Search for incorrect equations, dimensional errors, invalid derivations, missing assumptions, wrong limits, overclaims, conservation/constitutive confusion, and terminology errors.

Severity:

```
P0 — fundamental physics error
P1 — substantive conceptual/technical issue
P2 — qualification/clarity issue
P3 — stylistic/noncritical issue
```

`AUDIT_MATH`

Check algebra, dimensions, signs, factors, limiting cases, and notation.

`AUDIT_REFERENCES`

Check source existence, bibliographic accuracy, claim-source alignment, evidence tier, and missing citations.

`AUDIT_STRUCTURE`

Check links, anchors, headings, tables, fences, equation delimiters, numbering, and cross-file references.

`TRACE_MODEL`

For an engineering equation return:

```
engineering model
↓
reduction
↓
parent continuum model
↓
statistical/quantum foundation where justified
↓
assumptions
↓
failure boundary
```

Never invent intermediate theories.

`TRACE_CLAIM`

Return:

```
claim
evidence
epistemic tag
source
assumptions
confidence
possible stronger/weaker wording
```

`GRAPH`

Build a typed graph of theories, equations, models, approximations, assumptions, engineering domains, and references.

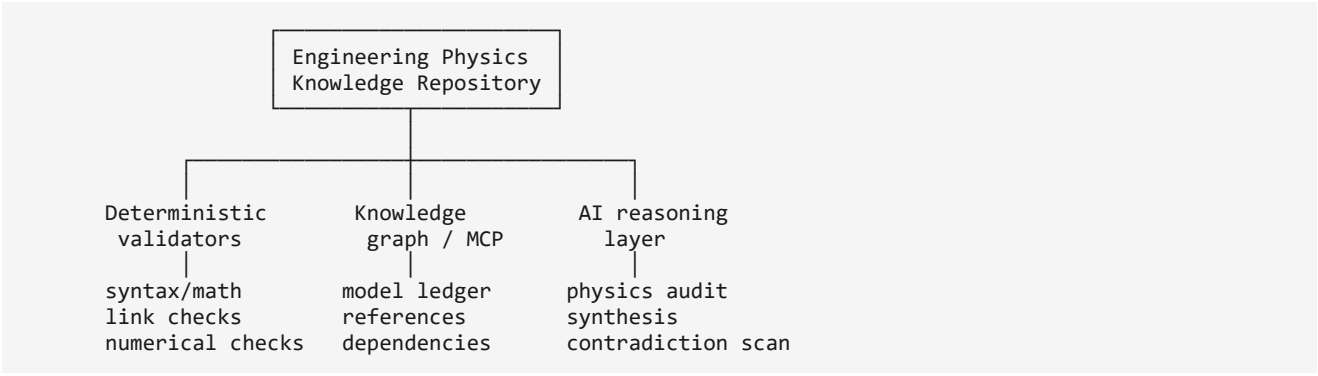
Allowed edge types:

```
DERIVES_FROM
APPROXIMATES
REDUCES_TO
STRUCTURALLY_MATCHES
ANALOGOUS_TO
SUPPORTED_BY
FAILS_AT
```

Do not represent every relationship as `DERIVES_FROM`.

16. MCP / Tooling Architecture

AI should operate as infrastructure around the repository:



Use deterministic tools for deterministic checks and LLM reasoning for conceptual synthesis.

17. Safe Write Policy

Preferred workflow:

```
READ
→ ANALYZE
→ PROPOSE DIFF
→ VALIDATE
→ HUMAN APPROVAL
→ WRITE
```

For potentially destructive changes:

- show the diff,
- identify affected files,
- identify downstream conceptual consequences,
- validate,
- obtain explicit approval,
- preserve Git history.

18. Source-of-Truth Priority

When information conflicts:

```
1. Current repository source files
2. Current references cited by the repository
3. Primary literature / authoritative textbooks
4. Established external documentation
5. AI inference
```

AI inference must not silently override verified source material.

If repository content conflicts with an older memory, trust the current repository.

19. High-Risk Audit Areas

Pay special attention to:

- Noether theorem claims
- gauge symmetry vs gauge redundancy

- Higgs mechanism and ordinary matter mass
- QCD contribution to baryonic mass
- GR → Newton limits
- thermodynamic/statistical limits
- continuum-limit language
- Kubo/linear-response claims
- transport laws
- R/C/L cross-domain analogies
- topology claims
- Foldy–Wouthuysen expansions
- electromagnetic circuit reductions
- fluid-network reductions
- claims of exactness
- claims that one law derives another
- reference evidence-tier metadata

20. Preferred Language

Prefer:

```
"under these assumptions"
"to leading order"
"in the weak-field limit"
"in the low-energy effective theory"
"within the lumped approximation"
"shares the same mathematical structure"
"is structurally analogous"
"provides a statistical foundation"
"can be reduced to"
"the model fails when"
```

Avoid unsupported absolutes:

```
"always"
"exactly"
"every"
"all"
"nothing but"
"the same physics"
"proves"
"fundamentally derives"
```

21. Project Identity

This is not merely a conventional engineering textbook, a collection of equations, or a claim that all engineering has already been derived from fundamental physics.

It is:

> A rigorous map of how physical theories become engineering models through symmetry, conservation, limiting procedures, constitutive assumptions, coarse-graining, discretization, and system reduction.

The core intellectual object is the **model hierarchy and its boundaries**.

22. Minimal Agent Instruction

```
You are assisting with Engineering Physics: Top Down
(The-Studio-of-Sadiq/Engineering-Physics).

Treat the repository as a rigorous hierarchy:
fundamental physics → quantum/statistical/classical physics →
continuum/effective models → engineering systems.

For every claimed connection, distinguish:
exact identity, controlled derivation, structural equivalence, and analogy.

Always state important assumptions, approximations, validity regimes,
and failure boundaries.

Do not confuse conservation laws with constitutive laws.
Do not overclaim that engineering equations are directly derived from
the Standard Model.
Do not treat Noether symmetry, gauge redundancy, and conserved currents
as interchangeable.
Do not treat Higgs coupling as the source of all ordinary matter mass.
Do not treat thermodynamic limit as identical to continuum limit.
Do not treat Kubo, transfer functions, R/C/L analogies, or topology
more strongly than their mathematical/physical scope permits.

Use the Model Ledger:
parent theory → origin → approximation → assumptions →
model → validity → failure → next model.

When editing:
read current source first, make minimal changes, preserve notation,
then validate structure, equations, links, references, and numerical values.

AI is authoring/research infrastructure, not part of the physical theory.
Prefer deterministic validation for deterministic problems and LLM reasoning
for conceptual synthesis and consistency auditing.
```

23. Machine-Readable Metadata

```yaml
project:
  name: "Engineering Physics: Top Down"
  repository: "The-Studio-of-Sadiq/Engineering-Physics"
  type: "scholarly knowledge repository"
  domains:
    - physics
    - engineering
    - mathematical_physics
    - systems
    - control
  philosophy: "Engineering is physics seen from far away."

architecture:
  layer_0: "fundamental framework"
  layer_1: "quantum and atomic physics"
  layer_2: "classical continuum and statistical physics"
  layer_3: "engineering systems"
  bridges:
    - "fundamental → quantum"
    - "quantum → classical/statistical"
    - "continuum → engineering"

epistemic_tags:
  - DERIVATION
  - APPROXIMATION
  - STRUCTURAL_CONNECTION
  - ANALOGY
  - PHENOMENOLOGICAL
  - GAP
  - MEASURED
  - THEORETICAL
  - SYNTHESIS

audit_severity:
  P0: "fundamental physics or mathematical error"
  P1: "substantive technical/conceptual issue"
  P2: "qualification or clarity issue"
  P3: "style/noncritical issue"

agent_policy:
  default_mode: "read-analyze-propose-validate"
  autonomous_writes: false
  preserve_git_history: true
  prefer_minimal_diffs: true
  never_invent_sources: true
  never_strengthen_claims_without_evidence: true

validation:
  structural: true
  mathematical: true
  dimensional: true
  numerical: true
  references: true
  conceptual: true
```

Engineering Physics — MCP Server

An MCP server for `The-Studio-of-Sadiq/Engineering-Physics`, implementing the **deterministic tooling layer** described in the repo's own `MCP.md` (section 16). By default it reads the **local working copy** it ships in, straight from disk on every call, so uncommitted edits are visible with no restart. It can instead read the live repo over GitHub's HTTP APIs (`EP_SOURCE=github`). Either way it runs mechanical checks and extraction that don't require physics judgment.

What it does, and deliberately does not do

`MCP.md` (section 15/16) splits work between a deterministic layer and an "AI reasoning layer." This server is the deterministic layer only:

Tool	MCP.md mode	What it actually does
`orient`	`ORIENT`	Returns README.md, MCP.md, 00 Map.md, CONTRIBUTING.md, and which source (local/GitHub) was read
`list_repo_structure`	—	Lists every file path in the repo (optionally filtered by prefix)
`get_file`	—	Returns one file's raw content (local mode refuses paths outside the repo)
`audit_structure`	`AUDIT_STRUCTURE`	Heading-level jumps, broken/relative links (files and folders), same-file anchor checks, **image embeds** (missing file, empty alt text, remote images, Obsidian-only `![[...]]`), unclosed code fences, unbalanced `$` / `$$` / `\(\)` / `\[\]` math delimiters, **numbered-equation LaTeX** (`\begin{equation}`, `\tag{}`, etc.) flagged as forbidden, mismatched table columns
`extract_model_ledger`	(section 7)	Parses every Model Ledger — both the Markdown-table form (`| Field | Content |` under a "Model Ledger — *name*" title) and the fenced `Field: value` form — into structured fields, verbatim
`get_epistemic_tag_index`	(section 5, feeds `GRAPH`)	Indexes every `[DERIVATION]` / `[APPROXIMATION]` /etc. tag with its nearest heading and an excerpt
`build_reduction_graph`	`GRAPH` (partial)	Builds `REDUCES_TO` and `FAILS_AT` edges strictly from Model Ledger fields
`trace_model`	`TRACE_MODEL`	Returns the ledger-stated chain for a named model: model → reduction → parent theory → assumptions → failure boundary
`get_figure_index`	—	Lists every embedded figure: path, whether the file exists, alt text, caption, nearest heading, and the `FIGURES/<FOLDER>/make_figures.py` script that draws it; also lists figure files nothing embeds

Figures are SVGs, which a model can't look at. `get_figure_index` returns each figure's caption, and `get_file` on its `make_figures.py` returns the exact equation and parameters plotted. Between them, that is the readable description of a figure.

Not implemented, on purpose: `AUDIT_PHYSICS`, `AUDIT_MATH`, `AUDIT_REFERENCES`, `TRACE_CLAIM`, and the `DERIVES_FROM` / `STRUCTURALLY_MATCHES` / `ANALOGOUS_TO` / `SUPPORTED_BY` edges of `GRAPH`. All of these require judging whether an equation, claim, or citation is *correct* — physics/mathematical/scholarly judgment

that `MCP.md` itself assigns to the calling LLM, not to tooling. Hard-coding that judgment here would be exactly the kind of silent overclaim `MCP.md` §8 warns against. Use `get_file` / `list_repo_structure` to pull the text, then reason over it directly — that's the intended split.

No write tools. There is no commit/PR/push capability. This matches `MCP.md` §17 (Safe Write Policy) and its own metadata: `autonomous_writes: false`. If/when you want write support, it should implement the full `READ → ANALYZE → PROPOSE DIFF → VALIDATE → HUMAN APPROVAL → WRITE` sequence `MCP.md` specifies, not a direct write tool — that's a deliberate gap, not an oversight, and worth a separate conversation before building.

Known caveats (mechanical checks, not opinions)

- **Anchors in *other* files aren't verified**, only that the target file exists — `audit_structure` says so explicitly in that case rather than silently passing.
- **A "heading skips a level" warning fires on a `#` title immediately followed by a `###` subtitle/tagline** (as in this repo's own `README.md`), which is a common and reasonable style choice, not necessarily an error — read these as prompts to check, not verdicts.
- **Ledgers written as plain aligned text** (no colon, no table — the illustrative block in `README.md`) are not parsed. Use the table form.
- **GitHub mode only:** the tree API is rate-limited to 60 requests/hour unauthenticated (raw file fetches are not); set `GITHUB_TOKEN` to raise it to 5000/hour. GitHub mode also caches in memory per process with no invalidation on push — restart to see upstream changes.

Setup

```
pip install -r requirements.txt
```

Requires `mcp>=2.0`, whose Python API renamed `FastMCP` to `MCPServer` (same decorator style: `@mcp.tool()`). If you have an older `mcp<2` pinned elsewhere, either upgrade it or change the two lines in `server.py`:

```
from mcp.server.mcpserver import MCPServer   # -> from mcp.server.fastmcp import FastMCP
mcp = MCPServer("engineering-physics")        # -> mcp = FastMCP("engineering-physics")
```

Configuration is by environment variable; `.env.example` lists them all. The server doesn't load a `.env` file itself — put the variables in your MCP client's `env` block (below) or your shell. With none set it reads the local working copy.

Run it directly to confirm it starts:

```
python server.py
```

(It will sit waiting for stdio MCP messages — that's expected; Ctrl+C to stop when just checking it starts cleanly.) On Windows use `python`, not `python3`, which is usually the Microsoft Store stub.

Wiring it into Claude Code

The repo root has a `.mcp.json`, so opening Claude Code **in the repo root** picks the server up automatically (it asks you to approve it the first time):

```
{
  "mcpServers": {
    "engineering-physics": {
      "command": "python",
      "args": ["mcp-server/server.py"],
      "env": { "EP_SOURCE": "local" }
    }
  }
}
```

Claude Desktop and other clients

Add the same entry to the client's config (e.g. `claude_desktop_config.json`), with an absolute path, since those clients don't start in the repo:

```
{
  "mcpServers": {
    "engineering-physics": {
      "command": "python",
      "args": ["/absolute/path/to/mcp-server/server.py"]
    }
  }
}
```

To read GitHub instead of the local copy, add `"env": { "EP_SOURCE": "github", "GITHUB_TOKEN": "..." }`.

Testing without a full MCP client

`smoke_test.py` calls every tool function directly (no stdio transport) and prints PASS/FAIL per check. It exits non-zero on any failure, so it works as a pre-commit check:

```
python smoke_test.py
```

It runs against the local working copy, so it needs no network. Run it with `EP_SOURCE=github` to exercise the GitHub path.